

Hi3518E V20X/Hi3516C V200 Economical HD IP Camera SoC

Data Sheet

Issue 02

Date 2017-06-12

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HiSilicon Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: http://www.hisilicon.com

Email: support@hisilicon.com

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About This Document.....1

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About This Document

Purpose

This document describes the features, logical structures, functions, operating modes, and related registers of each module of Hi3518E V20X/Hi3516C V200. This document also describes the interface timings and related parameters in diagrams. In addition, this document describes the pins, pin usages, performance parameters, and package dimension of Hi3518E V20X/Hi3516C V200 in detail.

Related Versions

The following table lists the product versions related to this document.

Product Name	Version
Hi3518E	V200
Hi3518E	V201
Hi3516C	V200

Intended Audience

This document is intended for:

- Design and maintenance personnel for electronics
- Sales personnel for electronic parts and components

Conventions

Symbol Conventions

The symbols that may be found in this document are defined as follows.

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Symbol	Description
 DANGER	Indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a hazard with a medium or low level of risk that, if not avoided, could result in minor or moderate injury.
 CAUTION	Indicates a potentially hazardous situation, which if not avoided, could result in equipment damage, data loss, performance degradation, or unexpected results.
 TIP	Indicates a tip that may help you solve a problem or save time.
 NOTE	Provides additional information to emphasize or supplement important points of the main text.

General Conventions

The general conventions that may be found in this document are defined as follows.

Convention	Description
Times New Roman	Normal paragraphs are in Times New Roman.
Boldface	Names of files, directories, folders, and users are in **boldface**. For example, log in as user **root**.
Italic	Book titles are in *italics*.
`Courier New`	Examples of information displayed on the screen are in `Courier New`.

Table Content Conventions

The table content conventions that may be found in this document are defined as follows.

Content	Description
–	The cell is blank.
*	The content in this cell is configurable.

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Notes

Register Attributes

The register attributes that may be found in this document are defined as follows.

Symbol	Description	Symbol	Description
RO	The register is read-only.	RW	The register is read/write.
RC	The register is cleared on a read.	WC	The register can be read. The register is cleared when 1 is written. The register keeps unchanged when 0 is written.

Reset Value Conventions

In the register definition tables:

- If the reset value (for the Reset row) of a bit is "?", the reset value is undefined.
- If the reset values of one or multiple bits are "?", the total reset value of a register is undefined and is marked as "-".

Numerical System

The expressions of data capacity, frequency, and data rate are described as follows.

Type	Symbol	Value
Data capacity (such as the RAM capacity)	K	1024
	M	1,048,576
	G	1,073,741,824
Frequency, data rate	k	1000
	M	1,000,000
	G	1,000,000,000

The expressions of addresses and data are described as follows.

Symbol	Example	Description
0x	0xFE04, 0x18	Address or data in hexadecimal
0b	0b000, 0b00 00000000	Data or sequence in binary (register description is excluded.)

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Symbol	Example	Description
X	00X, 1XX	In data expression, X indicates 0 or 1. For example, 00X indicates 000 or 001 and 1XX indicates 100, 101, 110, or 111.

Change History

Changes between document issues are cumulative. Therefore, the latest document issue contains all changes made in previous issues.

Issue 02 (2017-06-12)

This issue is the second official release, which incorporates the following changes:

Chapter 1 Product Description

Section 1.2.4 is modified.

Chapter 2 Hardware-- 192LD TFBGA

In section 2.2.3, table 2-26 is modified.

In section 2.5, table 2-56 is modified.

In section 2.8.5, table 2-94 and table 2-95 are modified.

In section 2.8.6, table 2-96 is modified.

In section 2.8.7, table 2-97 is modified.

In section 2.8.8, table 2-98 is modified.

Chapter 2 Hardware-- 273LD TFBGA

In section 2.2.3, table 2-27 is modified.

In section 2.5, table 2-57 is modified.

In section 2.8.6, table 2-92 and table 2-93 are modified.

In section 2.8.7, table 2-94 is modified.

In section 2.8.8, table 2-95 is modified.

In section 2.8.9, table 2-96 is modified.

Chapter 4 Memory Interfaces

Section 4.2.2 is modified.

Sections 4.2.3.6 and 4.2.5 are modified.

In section 4.2.8, FMC_CFG and FMC_INT are modified.

Chapter 6 Video Encoder

In section 6.2.2, "H.264 video encoding" is modified.

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Chapter 12 Peripherals

In section 12.1.5, I2C_SCL_HCNT, I2C_SCL_LCNT, and I2C_SDA_HOLD are modified.

Issue 01 (2016-10-28)

This issue is the first official release, which incorporates the following changes:

Chapter 2 Hardware-- 192LD TFBGA

In section 2.7.3, table 2-70 is modified.

Section 2.8.1 and section 2.7.6 are modified.

In section 2.8.2.2, table 2-90 is modified.

Chapter 2 Hardware-- 273LD TFBGA

In section 2.7.3, table 2-64 is modified.

Section 2.8.2 and section 2.7.6 are modified.

In section 2.8.3.2, table 2-88 is modified.

Chapter 3 System

In section 3.5.4.2, SC_CTRL is modified.

In section 3.5.5, MISC_CTRL56 is modified

Section 3.8.3 and section 3.8.4 are modified.

Chapter 5 ETH

In section 5.6.1, MDIO_RWCTRL is modified.

Chapter 9 Video Interfaces

In section 9.2.6, LCD_CTRL and LCD_SYNC_INV are added.

Chapter 11 Audio Interfaces

In section 11.1.6, I2S_CRG_CFG0_08 is added, and I2S_CRG_CFG1_08 is modified.

Chapter 12 Peripherals

In section 12.3.7, SPICR1 register is modified.

Issue 00B04 (2016-05-10)

This issue is the fourth draft release, which incorporates the following changes:

Chapter 2 Hardware-- 192LD TFBGA

Section 2.7 is modified

Chapter 2 Hardware-- 273LD TFBGA

Section 2.7.3, section 2.7.4, section 2.7.5, section 2.7.6, section 2.7.7, and section 2.7.8 are modified.

Section 2.9.1.3, table 2-84 is modified.

Chapter 3 System

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Section 3.2.7, the description of the [11:10] bit in PERI_CRG49 is modified.

Chapter 6 Video Encoder

Section 6.6.2 is modified.

Chapter 9 Video Interfaces

Section 9.1.3.1 is modified.

Chapter 12 Peripherals

Section 12.6.1 is modified.

Issue 00B03 (2016-03-14)

This issue is the third draft release, which incorporates the following changes:

Chapter 2 Hardware-- 192LD TFBGA

Section 2.7.3, table 2-67 is modified.

Section 2.7.6 and section 2.7.8 are modified.

Table 2-83 is added.

Chapter 2 Hardware-- 273LD TFBGA

Section 2.7.3, table 2-64 is modified.

Section 2.7.6 and section 2.7.8 are modified.

Table 2-84 is added.

Issue 00B02 (2015-12-28)

This issue is the second draft release, which incorporates the following changes:

Chapter 1 Product Description

The contents related in section 1.2 are modified.

Chapter 2 Hardware-- 192LD TFBGA

Section 2.2.3, section 2.7.1, section 2.7.2 and section 2.7.3 are modified.

In section 2.7.5, table 2-62 and table 2-63 are modified.

In section 2.9.3, table 2-74 is modified.

In section 2.9.7, the description of the SPI Timings is modified.

Chapter 2 Hardware-- 273LD TFBGA

Section 2.2.3, section 2.7.1, section 2.7.2 and section 2.7.3 are modified.

In section 2.7.5, table 2-63 and table 2-64 are modified.

In section 2.9.8, the description of the SPI Timings is modified.

Chapter 3 System

In section 3.4.1, table 3-8 is modified.

Chapter 6 Video Encoder

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Section 6.2.2 is modified.

Chapter 11 Audio Interfaces

In section 11.1.6, the description of the TX_BUFF_SADDR and TX_BUFF_SADDR are modified.

Issue 00B01 (2015-09-20)

This issue is the first draft release.

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1 Product Description

1.1 Application Scenarios

Hi3518E V20X/Hi3516C V200 is a new-generation SoC designed for the HD IP camera. It has an integrated new-generation ISP and H.264 encoder. It uses the optimized picture processing algorithm before encoding, advanced low-power technology, and low-power architecture design. These designs and functions enable Hi3518E V20X/Hi3516C V200 to feature industry-leading low bit rate, high picture quality, and low power consumption. It supports 90° or 270° rotation and lens distortion correction by using hardware, which meets requirements in various surveillance application scenarios. It also supports 3A algorithms, which allow customers to design various types of IP cameras including the movement of the integrated camera. Hi3518E V20X/Hi3516C V200 integrates the POR, RTC as well as audio CODEC and supports various sensor levels and clock outputs, which significantly reduces the EBOM costs for Hi3518E V20X/Hi3516C V200 HD IP camera. Similar to other HiSilicon DVR and NVR SDKs, Hi3518E V20X/Hi3516C V200 SDK supports rapid mass production and the system layout of IP cameras, DVRs, and NVRs.

CAUTION

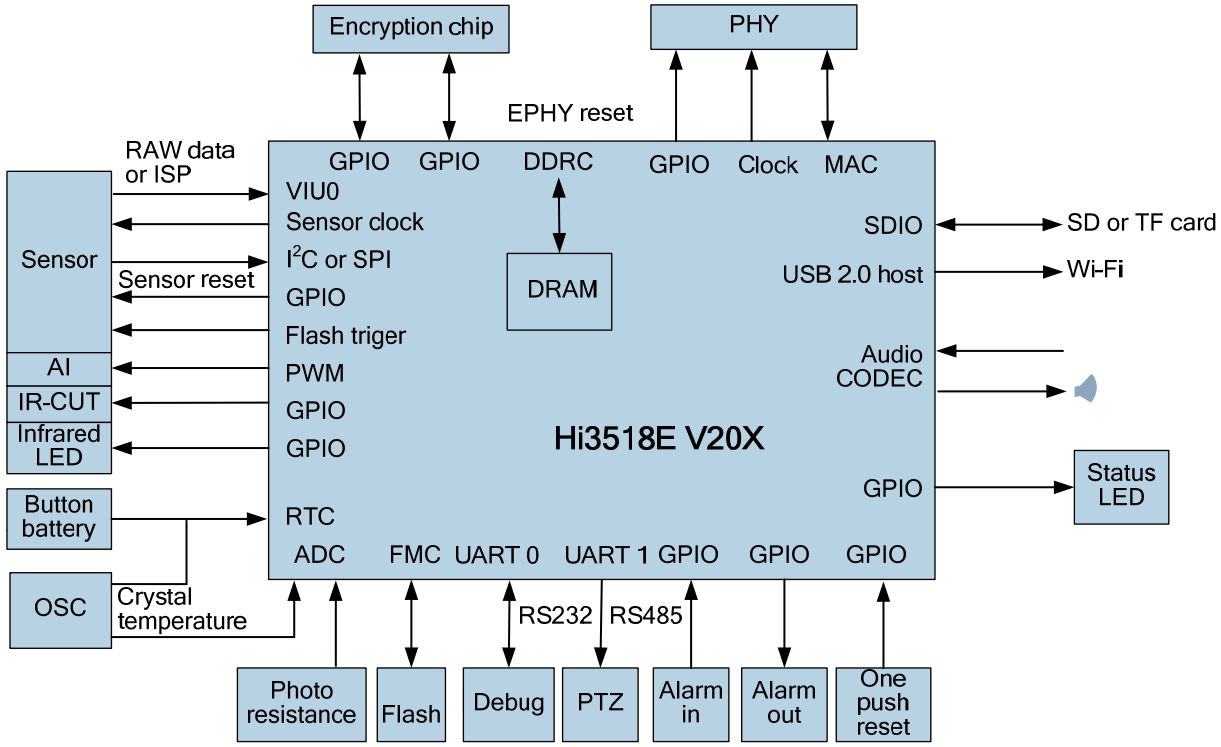
- In this document, Hi3518E V20X indicates Hi3518E V200 and Hi3518E V201.
- Unless otherwise specified, this document applies to Hi3518E V200, Hi3518E V201, and Hi3516C V200. The descriptions are provided using Hi3518E V20X as an example.

1.1.1 Hi3518E V20X Economical HD IP Camera Solution

Figure 1-1 shows the typical application scenario of Hi3518E V20X.

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Figure 1-1 Application block diagram of Hi3518E V20X

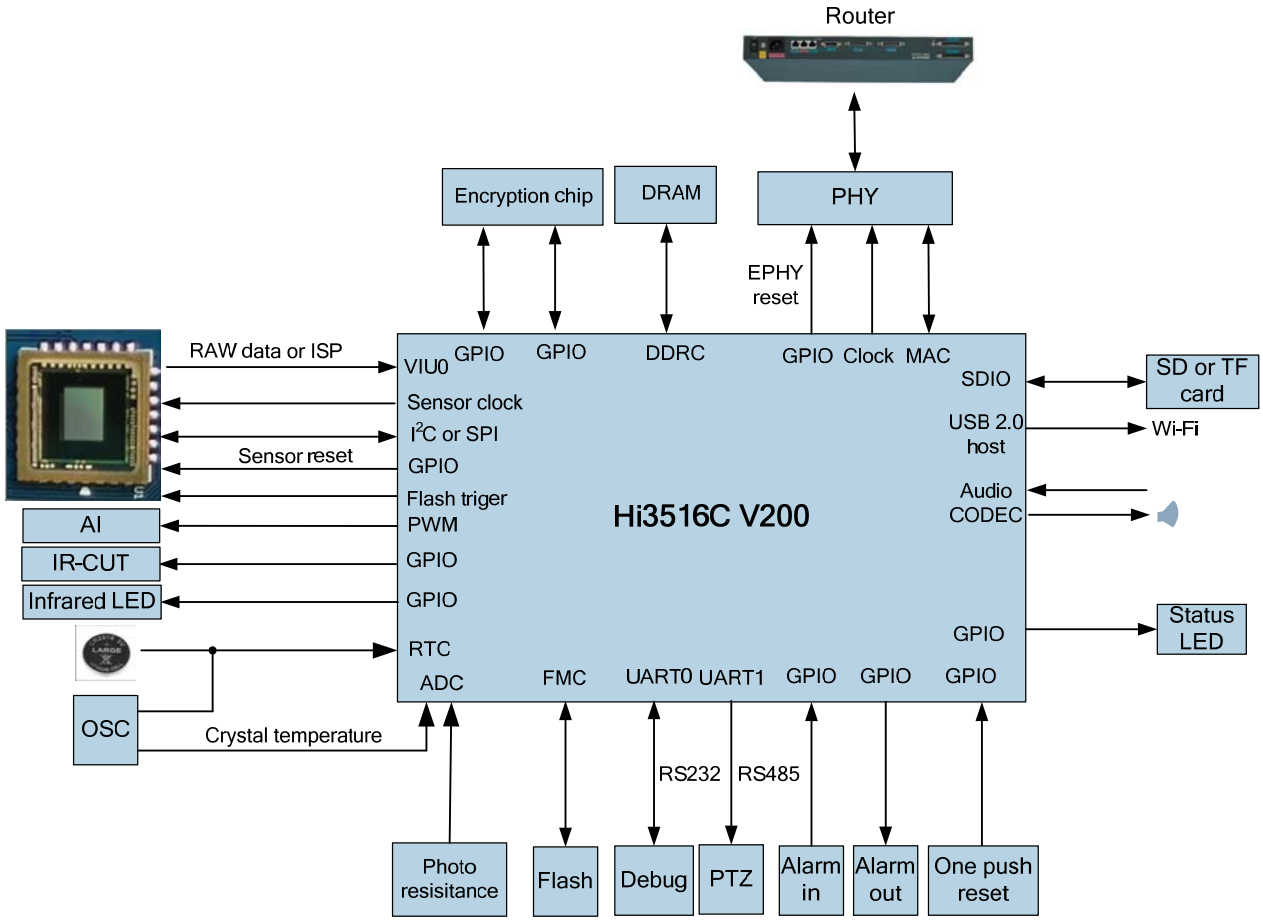


1.1.2 Hi3516C V200 Economical HD IP Camera Solution

Figure 1-2 shows the typical application scenario of the Hi3516C V200.

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Figure 1-2 Application block diagram of the Hi3516C V200



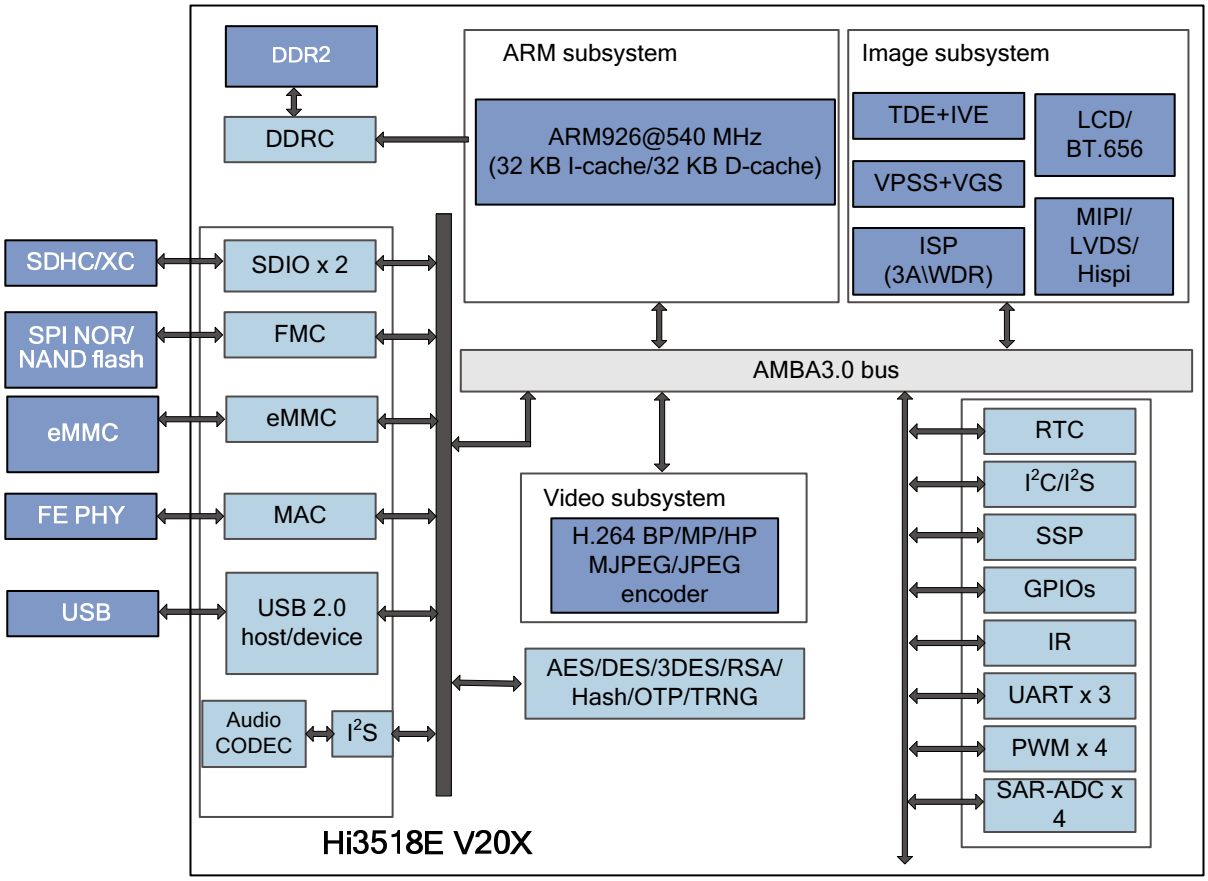
1.2 Architecture

1.2.1 Overview

Figure 1-3 and Figure 1-4 show the logical block diagrams Hi3518E V20X and Hi3516C V200 respectively.

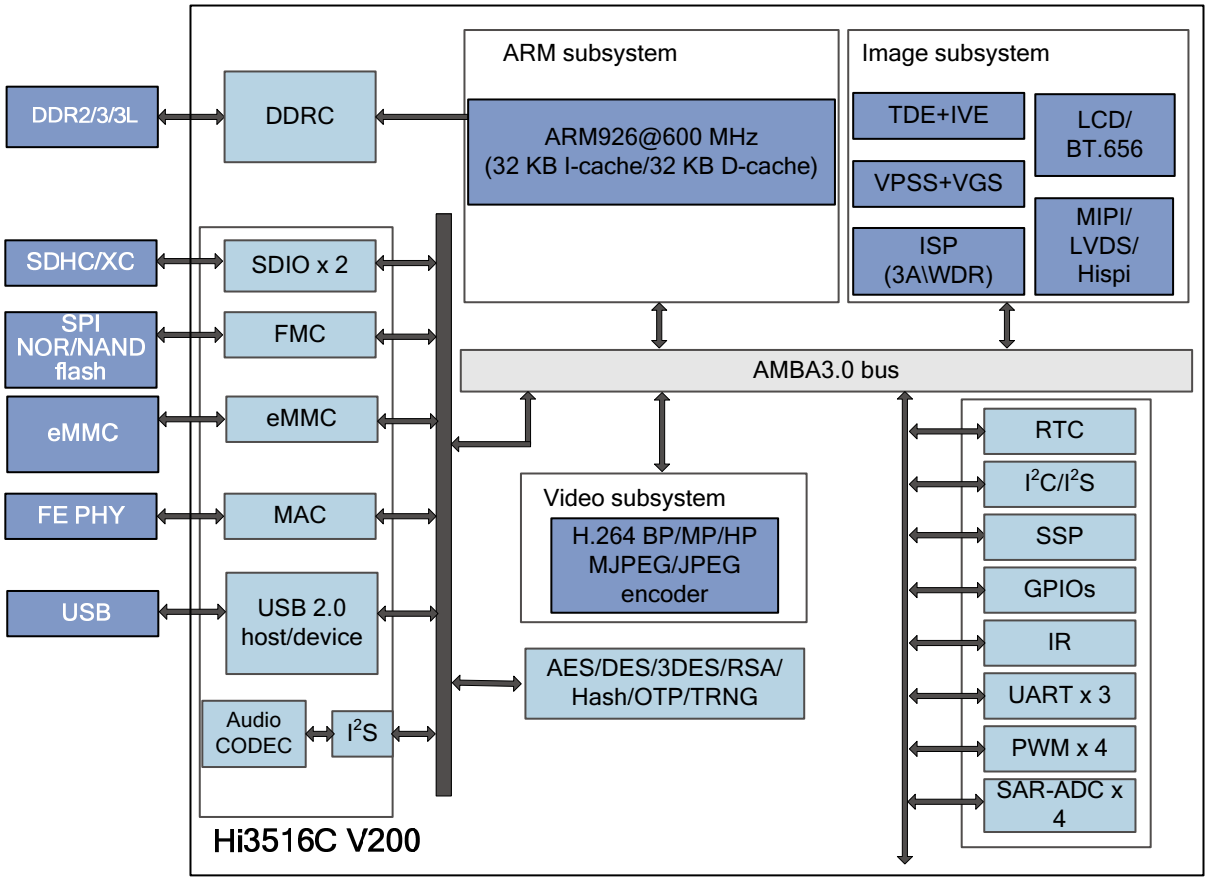
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Figure 1-3 Logical block diagram of Hi3518E V20X



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Figure 1-4 Logical block diagram of the Hi3516C V200



1.2.2 Processor Core

Hi3518E V20X: ARM926@540 MHz, 32 KB I-cache, 32 KB D-cache

Hi3516C V200: ARM926@600 MHz, 32 KB I-cache, 32 KB D-cache

1.2.3 Video Encoding

- H.264 main/high profile L4.0
- H.264 baseline encoding
- MJPEG/JPEG baseline encoding

1.2.4 Video Encoding Performance

- A maximum of 2-megapixel resolution for H.264 encoding
- Real-time H.264 & JPEG encoding of multiple streams:

  Hi3518E V200:

  – 720p@30 fps+VGA@30 fps+QVGA@30 fps+720p@1 fps JPEG snapshot

  – 1080p@20 fps+D1@20 fps+1080p@1 fps JPEG snapshot, maximum performance of 1080p@20 fps for a single channel

  Hi3518E V201:

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  - 720p@30 fps+QVGA@30 fps+720P @1 fpsJPEG snapshot
  - 960p@25 fps+D1@25 fps+960@1 fps JPEG snapshot

  Hi3516C V200:

  - 1080p@30 fps+VGA@30 fps+1080p@1 fps JPEG snapshot
- 2 megapixels@5 fps JPEG snapshot
- CBR or VBR with the output bit rate ranging from 2 kbit/s to 100 Mbit/s
- Encoding frame rate ranging from 1/16 fps to 30 fps
- Encoding of eight ROIs
- OSD overlaying of eight regions before encoding

1.2.5 Intelligent Video Analysis

Integrated IVE, supporting various intelligent analysis applications such as motion detection, perimeter defense, and video diagnosis.

1.2.6 Video and Graphics Processing

- Video pre-processing, including 3D denoising, image enhancement, and edge enhancement
- Anti-flicker for output videos and graphics
- 1/15x to 16x video scaling
- 1/2x to 2x graphics scaling
- OSD overlaying of eight regions before encoding
- Hardware graphics overlaying for videos at the video layer and graphics layer 1 during post-processing

1.2.7 ISP

- 2x2 Pattern RGB-IR sensor
- Adjustable 3A (AE, AF, and AWB) functions
- Highlight compensation, backlight compensation, gamma correction, and color enhancement
- Defect pixel correction, denoising, and digital image stabilization
- Anti-fog
- Lens distortion correction (not supported by Hi3518E V201)
- Picture rotation by 90° or 270° (not supported by Hi3518E V201)
- Mirroring and flipping
- Build-in WDR and tone mapping
- ISP tuning tools for the PC

1.2.8 Audio Encoding/Decoding

- Voice encoding/decoding complying with multiple protocols by using software
- G.711, ADPCM, and G.726 protocols supported
- Echo cancellation, noise suppression, and automatic gain

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1.2.9 Security Engine

- AES, DES, 3DES, and RSA encryption/decryption algorithms implemented by hardware
- Hash digest tamper proofing implemented by hardware
- Integrated 512-bit OTP storage space and hardware random number generator

1.2.10 Video Interfaces

- VI interfaces
  - 8-, 10-, 12-, or 14-bit RGB Bayer/RGB-IR inputs, maximum clock frequency of 100 MHz
  - BT.601, BT.656, and BT.1120 VI interfaces
  - 4-Lane MIPI/HiSPI/LVDS VI interfaces
  - Compatibility with mainstream HD CMOS sensors provided by Sony, Aptina, OmniVision, and Panasonic
  - Various sensor levels supported
  - Programmable sensor clock output
  - Maximum input resolution of 2 (1920*1080) megapixels
- VO interfaces
  - One BT.656 VO interface supporting 8-bit serial LCD outputs

1.2.11 Audio Interfaces

- Integrated audio CODEC that supports 16-bit audio inputs and outputs
- Mono-channel differential microphone inputs for reducing the background noises
- I^2S input
- Single-end input

1.2.12 Peripheral Interfaces

- POR
- One integrated high-precision RTC
- One 4-channel SAR ADC
- Three UART interfaces
- IR interfaces, I^2C interfaces, SPI master interfaces, and GPIO interfaces
- Four PWM interfaces
- Two SDIO interfaces, one of which supports SD 3.0
- One USB 2.0 interface that supports the host/device mode
- RMII in 10/100 Mbit/s full-duplex or half-duplex mode, TSO network acceleration, and PHY clock output

1.2.13 External Memory Interfaces

- DDR interface

  Hi3518E V20X supports only the DDR2 SDRAM interface.
  - Hi3518E V200: embedded 16-bit DDR2 SDRAM with the capacity of 512 Mbits

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  - Hi3518E V201: embedded 16-bit DDR2 SDRAM with the capacity of 256 Mbits
  - Maximum frequency of 360 MHz

  Hi3516C V200:
  - 16-bit DDR2/DDR3/DDR3L interface
  - Maximum frequency of 400 MHz
  - Maximum capacity of 4 Gbits
- SPI NOR flash interface

  1-, 2-, or 4-bit SPI NOR flash interface
- SPI NAND flash interface

  Maximum capacity of 4 Gbits
- eMMC 5.0 interface

  Maximum capacity of 64 GB
- Booting from the SPI NOR flash, SPI NAND flash, or eMMC

1.2.14 SDK

- Linux 3.4-based SDK
- High-performance H.264 PC decoding library

1.2.15 Physical Specifications

- Power consumption
  - Typical power consumption of 700 mW (including DDR2 SDRAM)
  - Multi-level power saving mode
- Operating voltages
  - 1.1 V core voltage
  - 3.3 V I/O voltage and 3.8 V margin voltage
- Package
  - Hi3518E V20X: body size of 10 mm x 10 mm (0.39 in. x 0.39 in.), 0.65 mm (0.03 in.) ball pitch, TFBGA RoHS package with 192 pins
  - Hi3516C V200: body size of 12 mm x 12 mm (0.47 in. x 0.47 in.), 0.65 mm (0.03 in.) ball pitch, TFBGA RoHS package with 273 pins

1.3 Boot Modes

Hi3518E V20X/Hi3516C V200 can boot from:

- External eMMC
- External SPI flash
- BOOTROM

During POR, the boot mode of Hi3518E V20X/Hi3516C V200 depends on the values of the BOOTROM_SEL and BOOT_SEL signals. Table 1-1 describes the mapping between signal values and boot modes.

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Table 1-1 Mapping between signal values and boot modes

BOOTROM_SEL	BOOT_SEL	Boot Mode
1	X	Booting from the BOOTROM
0	0	Booting from an external SPI flash
0	1	Booting from an external eMMC

NOTE

- The BOOTROM_SEL signal is multiplexed with the external pin MDCK.
- The BOOT_SEL signal is multiplexed with the external pin EPHY_RST.

1.3.1 Booting from the BOOTROM

When Hi3518E V20X/Hi3516C V200 boots from the BOOTROM, the serial port communication mechanism starts, the communication between the serial port and the related software running on the PC is set up, and then the boot program is downloaded. For details, see the *HiBurn User Guide*. If the serial port communication times out, the level of BOOT_SEL is checked. If BOOT_SEL is 0, Hi3518E V20X/Hi3516C V200 boots from the SPI flash; if BOOT_SEL is 1, Hi3518E V20X/Hi3516C V200 boots from the eMMC.

1.3.2 Booting from the eMMC

When Hi3518E V20X/Hi3516C V200 boots from the eMMC, the external memory is a eMMC. In this case, you must select the boot memory by configuring the levels of the pins BOOTROM_SEL (multiplexed with the external pin MDCK) and BOOT_SEL (multiplexed with the external pin EPHY_RST). When BOOTROM_SEL is 0 and BOOT_SEL is 1, Hi3518E V20X/Hi3516C V200 can boot from the eMMC mounted on the eMMC interface.

Table 1-2 eMMC boot mode

SFC_EMMC_BOOT_MODE	eMMC Boot Mode
0	4-wire boot mode
1	8-wire boot mode

1.3.3 Booting from the SPI Flash

When Hi3518E V20X/Hi3516C V200 boots from the SPI flash, the external memory is an SPI flash. In this case, you must select the boot memory by configuring the levels of the pins BOOTROM_SEL (multiplexed with the external pin MDCK) and BOOT_SEL (multiplexed with the external pin EPHY_RST). When BOOTROM_SEL is 0 and BOOT_SEL is 0, Hi3518E V20X/Hi3516C V200 can boot from the SPI flash mounted on the SFC interface.

During POR, the SPI flash boot mode of Hi3518E V20X/Hi3516C V200 depends on the values of the SFC_EMMC_BOOT_MODE and SFC_DEVICE_MODE signals. Table 1-3 describes the mapping between signal values and SPI flash boot modes.

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Table 1-3 Mapping between signal values and SPI flash boot modes

SFC_DEVICE_MODE	SFC_EMMC_BOOT_MODE	SPI Flash Boot Mode
0	0	SPI NOR flash, 3-byte address mode
0	1	SPI NOR flash, 4-byte address mode
1	0	SPI NAND flash, 1-wire boot mode
1	1	SPI NAND flash, 4-wire boot mode

NOTE

- The SFC_DEVICE_MODE signal is multiplexed with the external pin SHUTTER_TRIG.
- The SFC_EMMC_BOOT_MODE signal is multiplexed with the external pin FLASH_TRIG.

1.4 Address Space Mapping

Table 1-4 describes the address space mapping.

Table 1-4 Address space mapping

Start Address	End Address	Function	Capacity	Remarks
0xA000_0000	0xFFFF_FFFF	Reserved	-	-
0x8000_0000	0x9FFF_FFFF	Address space of the connected external DDR	512 MB	Hi3518E V200 supports 64 MB address space, whereas Hi3518E V201 supports 32 MB address space.
0x5C00_0000	0x7FFF_FFFF	Reserved	-	-
0x5800_0000	0x5BFF_FFFF	Storage space of the SPI flash	64 MB	-
0x5400_0000	0x57FF_FFFF	Reserved	64 MB	-
0x5000_0000	0x53FF_FFFF	Reserved	64 MB	-
0x2070_0000	0x4FFF_FFFF	Reserved	-	-
0x206F_0000	0x206F_FFFF	Reserved	64 KB	-
0x206E_0000	0x206E_FFFF	Reserved	64 KB	-
0x206D_0000	0x206D_FFFF	DDR test module	64 KB	-
0x206C_0000	0x206C_FFFF	Reserved	64 KB	-
0x206B_0000	0x206B_FFFF	Reserved	-	-

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Start Address	End Address	Function	Capacity	Remarks
0x206A_0000	0x206A_FFFF	IVE register	64 KB	-
0x2069_0000	0x2069_FFFF	Reserved	64 KB	-
0x2068_0000	0x2068_FFFF	MIPI register	64 KB	-
0x2067_0000	0x2067_FFFF	Reserved	-	-
0x2066_0000	0x2066_FFFF	JPGE register	64 KB	-
0x2065_0000	0x2065_FFFF	AIAO register	64 KB	-
0x2064_0000	0x2064_FFFF	Reserved	64 KB	-
0x2063_0000	0x2063_FFFF	VGS register	64 KB	-
0x2062_0000	0x2062_FFFF	AVC register	64 KB	-
0x2061_0000	0x2061_FFFF	TDE register	64 KB	-
0x2060_0000	0x2060_FFFF	VPSS register	64 KB	-
0x205D_0000	0x205F_FFFF	Reserved	-	-
0x205C_0000	0x205C_FFFF	VDP register	64 KB	-
0x2058_0000	0x205B_FFFF	VICAP register	256 KB	-
0x2029_0000	0x2057_FFFF	Reserved	-	-
0x2028_0000	0x2028_FFFF	TRNG register	64 KB	-
0x2027_0000	0x2027_FFFF	PMC register	64 KB	-
0x2026_0000	0x2026_FFFF	RSA register	64 KB	-
0x2025_0000	0x2025_FFFF	I^2C2 register	64 KB	-
0x2024_0000	0x2024_FFFF	I^2C1 register	64 KB	-
0x2023_0000	0x2023_FFFF	Reserved	64 KB	-
0x2022_0000	0x2022_FFFF	Reserved	64 KB	-
0x2021_0000	0x2021_FFFF	Reserved	64 KB	-
0x2020_0000	0x2020_FFFF	Reserved	64 KB	-
0x201F_0000	0x201F_FFFF	Reserved	64 KB	-
0x201E_0000	0x201E_FFFF	Reserved	64 KB	-
0x201D_0000	0x201D_FFFF	Reserved	64 KB	-
0x201C_0000	0x201C_FFFF	GPIO8 register	64 KB	-
0x201B_0000	0x201B_FFFF	GPIO7 register	64 KB	-
0x201A_0000	0x201A_FFFF	GPIO6 register	64 KB	-
0x2019_0000	0x2019_FFFF	GPIO5 register	64 KB	-

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Start Address	End Address	Function	Capacity	Remarks
0x2018_0000	0x2018_FFFF	GPIO4 register	64 KB	-
0x2017_0000	0x2017_FFFF	GPIO3 register	64 KB	-
0x2016_0000	0x2016_FFFF	GPIO2 register	64 KB	-
0x2015_0000	0x2015_FFFF	GPIO1 register	64 KB	-
0x2014_0000	0x2014_FFFF	GPIO0 register	64 KB	-
0x2013_0000	0x2013_FFFF	PWM register	64 KB	-
0x2012_0000	0x2012_FFFF	MISC register	64 KB	-
0x2011_0000	0x2011_FFFF	DDRC/DDR PHY register	64 KB	-
0x2010_0800	0x2010_FFFF	cipher_hash_key_ctrl register	62 KB	
0x2010_0000	0x2010_07FF	Reserved	2 KB	-
0x200F_0800	0x200F_FFFF	IO CTR register	62 KB	-
0x200F_0000	0x200F_07FF	IO MUX register	2 KB	-
0x200E_0000	0x200E_FFFF	SPI1 register	64 KB	-
0x200D_0000	0x200D_FFFF	I^2C0 register	64 KB	-
0x200C_0000	0x200C_FFFF	SPI0 register	64 KB	-
0x200B_0000	0x200B_FFFF	SAR_ADC register	64 KB	-
0x200A_0000	0x200A_FFFF	UART2 register	64 KB	-
0x2009_0000	0x2009_FFFF	UART1 register	64 KB	-
0x2008_0000	0x2008_FFFF	UART0 register	64 KB	-
0x2007_0000	0x2007_FFFF	IR register	64 KB	-
0x2006_0000	0x2006_FFFF	RTC register	64 KB	-
0x2005_0000	0x2005_FFFF	SYS_CTRL register	64 KB	-
0x2004_0000	0x2004_FFFF	WDG register	64 KB	-
0x2003_0000	0x2003_FFFF	CRG register	64 KB	-
0x2002_0000	0x2002_FFFF	Reserved	64 KB	-
0x2001_0000	0x2001_FFFF	Timer 2/Timer 3 register	64 KB	-
0x2000_0000	0x2000_FFFF	Timer 0/Timer 1 register	64 KB	-
0x100E_0000	0x1FFF_FFFF	Reserved	-	-
0x100D_0000	0x100D_FFFF	Interrupt control register	64 KB	-
0x100C_0000	0x100C_FFFF	Cipher register	64 KB	-
0x100B_0000	0x100B_FFFF	USB EHCI register	64 KB	-

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Start Address	End Address	Function	Capacity	Remarks
0x100A_0000	0x100A_FFFF	USB OHCI register	64 KB	-
0x1009_0000	0x1009_FFFF	ETH register	64 KB	-
0x1008_0000	0x1008_FFFF	USB device register	64 KB	-
0x1007_0000	0x1007_FFFF	HASH register	64 KB	-
0x1006_0000	0x1006_FFFF	DMAC register	64 KB	-
0x1005_0000	0x1005_FFFF	Reserved	64 KB	-
0x1004_0000	0x1004_FFFF	Reserved	64 KB	-
0x1003_0000	0x1003_FFFF	SDIO1 register	64 KB	-
0x1002_0000	0x1002_FFFF	SDIO0 register	64 KB	
0x1001_0000	0x1001_FFFF	FMC control register	64 KB	-
0x1000_0000	0x1000_FFFF	Reserved	64 KB	-
0x0402_0000	0x0FFF_FFFF	Reserved	-	-
0x0401_0000	0x0401_FFFF	On-chip RAM	64 KB	-
0x0400_0000	0x0400_FFFF	BOOTROM	64 KB	-
0x0000_0000	0x03FF_FFFF	Memory selected after address remapping is cleared. The address points to the on-chip RAM after remapping is cleared.	64 MB	The boot address space varies according to the setting of BOOT_SEL and BOOTROOM_SEL. {BOOT_SEL, BOOTROOM_SEL}: 00: storage space of the SPI flash Other values: BOOTROM

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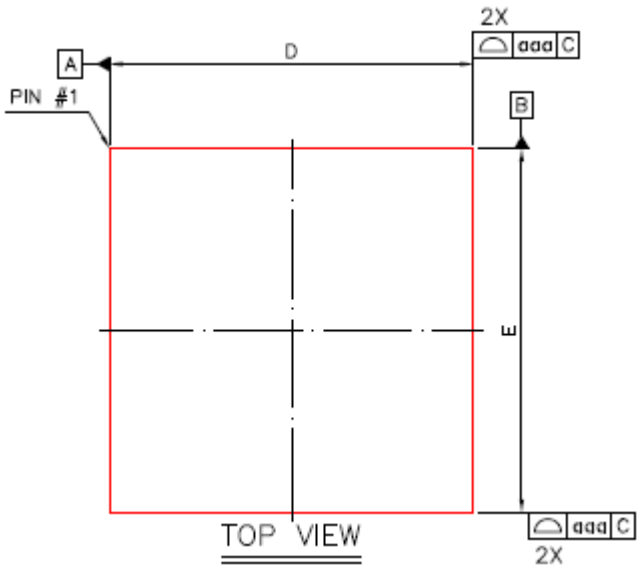
2 Hardware

2.1 Package and Pinout

2.1.1 Package

Hi3518E V20X uses the lead thin & fine-pitch ball grid array (LD TFBGA) package. It has 192 pins, its body size is 10 mm x 10 mm (0.39 in. x 0.39 in.), and its ball pitch is 0.65 mm (0.03 in.). Figure 2-1 to Figure 2-4 show package views. Figure 2-5 shows package dimensions.

Figure 2-1 Top view



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Figure 2-2 Bottom view

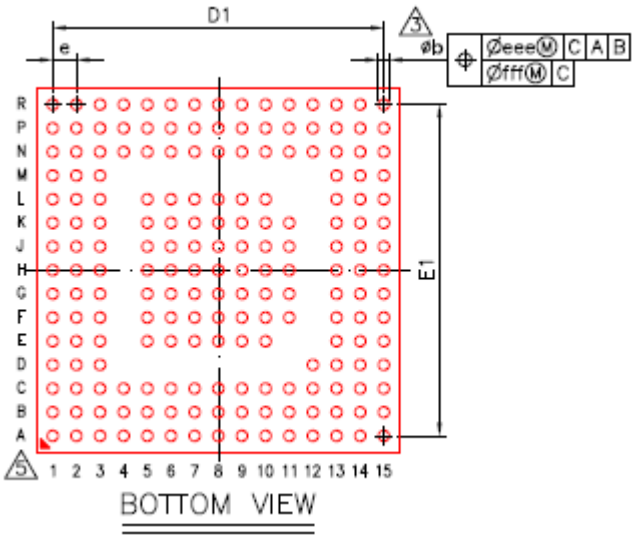


Figure 2-3 Side view

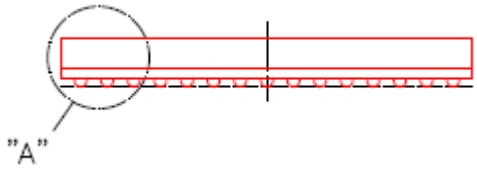
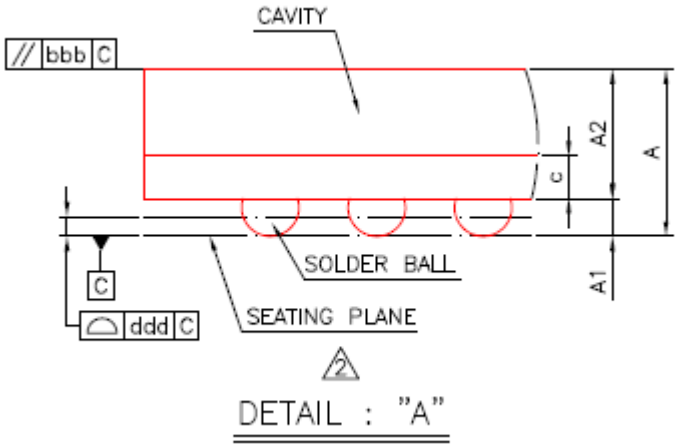


Figure 2-4 Enlarged view of detail "A"



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Figure 2-5 Package dimensions

Symbol	Dimension in mm			Dimension in inch		
	MIN	NOM	MAX	MIN	NOM	MAX
A	1.10	1.17	1.24	0.043	0.046	0.049
A1	0.16	0.21	0.26	0.006	0.008	0.010
A2	0.91	0.96	1.01	0.036	0.038	0.040
c	0.22	0.26	0.30	0.009	0.010	0.012
D	9.90	10.00	10.10	0.390	0.394	0.398
E	9.90	10.00	10.10	0.390	0.394	0.398
D1	---	9.10	---	---	0.358	---
E1	---	9.10	---	---	0.358	---
e	---	0.65	---	---	0.026	---
b	0.25	0.30	0.35	0.010	0.012	0.014
aaa	0.15			0.006		
bbb	0.10			0.004		
ddd	0.08			0.003		
eee	0.15			0.006		
fff	0.05			0.002		
MD/ME	15/15					

NOTE :
1. CONTROLLING DIMENSION : MILLIMETER.
2. PRIMARY DATUM C AND SEATING PLANE ARE DEFINED BY THE SPHERICAL CROWNS OF THE SOLDER BALLS.
3. DIMENSION b IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER, PARALLEL TO PRIMARY DATUM C.
4. SPECIAL CHARACTERISTICS C CLASS: bbb,ddd
5. THE PATTERN OF PIN 1 FIDUCIAL IS FOR REFERENCE ONLY.
6. REFERENCE DOCUMENT : JEDEC PUBLICATION 95 DESIGN GUIDE 4.5

2.1.2 Pinout

Hi3518E V20X has 192 pins. Table 2-1 lists the pin quantity of Hi3518E V20X by type.

Table 2-1 Pin quantity of Hi3518E V20X by type

Pin Type	Quantity
Input/Output (I/O)	101
Digital power	23
Digital ground (GND)	41
Others/Analog power	7
Others/Analog GND	11
Double data rate (DDR) reference power	9
Total	192

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Pin Map

Figure 2-6 to Figure 2-9 show pin maps of Hi3518E V20X.

Figure 2-6 Pin map part 1 of Hi3518E V20X (A1−G7)

	1	2	3	4	5	6	7
A	VSS	UART0_TXD	JTAG_TDI	JTAG_TCK	VSS	VDDIO_DDR	VSS
B	RTC_XIN	RTC_XOUT	UART0_RXD	JTAG_TRSTN	JTAG_TMS	VDDIO_DDR	VDDIO_CK_DDR
C	VSS	AVDD_BAT	TEST_MODE	JTAG_TDO	JTAG_EN	VSS	AVDD_DDRPLL
D	XOUT	XIN	AVSS_RTC				
E	VSS	PWM1	AVDD11_PLL		VDD	VDD	VDD
F	PWM3	PWM2	AVSS_PLL		VSS	VSS	VSS
G	PWM0	I2C2_SDA	AVDD33_PLL		VSS	VSS	VSS

Figure 2-7 Pin map part 2 of Hi3518E V20X (A8−G15)

8	9	10	11	12	13	14	15
VDDIO_DDR	DDR_VREF	SPI1_SCLK	SPI1_SDI	EPHY_CLK	RMII_RXD0	RMII_CLK	VSS
VSS	VDDIO_DDR	VSS	SPI1_CSN1	VSS	RMII_RXD1	RMII_RX_DV	RMII_TXD1
VDDIO_DDR	VDDIO_DDR	SPI1_SDO	SPI1_CSN0	MDCK	RMII_TXD0	RMII_TX_EN	EPHY_RST
				MDIO	UART1_CTSN	UART1_RXD	UART1_RTSN
VDD	VDD	DVDD33			UART2_TXD	UART1_TXD	VSS
VSS	VSS	VSS	VDD		UART2_RXD	SFC_CLK	SFC_HOLD_IO3
VSS	VSS	VSS	DVDDIO_RMII		SYS_RSTN_OUT	SFC_SIO_IO0	SFC_CSN0

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Figure 2-8 Pin map part 3 of Hi3518E V20X (H1−R7)

H	I2C2_SCL	GPIO0_2	AVDD33_RTC		DVDD33	VSS	VSS
J	GPIO0_1	GPIO0_0	DVDD33		VDD	VSS	VSS
K	SAR_ADC_CH0	SAR_ADC_CH1	VSS		VDD	VDD	VSS
L	SAR_ADC_CH2	VSS	SAR_ADC_CH3		DVDD3318_SENSOR	VDD	VDD
M	SPI0_SCLK	SPI0_SDO	SENSOR_RSTN				
N	SPI0_CSN	SPI0_SDI	FLASH_TRIG	AVDD3318_MIPI	AVSS3318_MIPI	GPIO1_0	GPIO1_2
P	SENSOR_CLK	SHUTTER_TRIG	MIPI_D3P	MIPI_D1P	MIPI_CKP	MIPI_D0P	MIPI_D2P
R	VSS	AVSS3318_MIPI	MIPI_D3M	MIPI_D1M	MIPI_CKM	MIPI_D0M	MIPI_D2M

Figure 2-9 Pin map part 4 of Hi3518E V20X (H8−R15)

VSS	VSS	VSS	DVDD3318_EMMC		SFC_CSN1	SFC_SOI_IO1	SFC_WP_IO2
VSS	VSS	VSS	VSS		DVDD18_SDIO	SDIO0_CDATA2	SDIO0_CDATA3
VSS	VSS	AVSS_AC	VDD		DVDD33	SDIO0_VOUT	SDIO0_CCMD
VDD	DVDD3318_VI	AVDD33_AC			SDIO0_CWPR	VSS	SDIO0_CCLK_OUT
					AVSS_AC	SDIO0_CDATA1	SDIO0_CDATA0
GPIO1_5	GPIO1_4	AVDD33_USB	AVSS_USB	USB_REXT	AC_VREF	SDIO0_CARD_POWER_EN	SDIO0_CARD_DETECT
GPIO1_6	GPIO1_3	USB_OVRCUR	USB_DP	AVSS_USB	AC_INL	AC_OUTL	AC_OUTR
AVSS3318_MIPI	GPIO1_1	USB_PWREN	USB_DM	AVSS_AC	AC_INR	AC_MICBIAS	AVSS_AC

Pin Arrangement

Table 2-2 lists the pins of Hi3518E V20X based on their positions.

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Table 2-2 Pin arrangement

Position	Pin Name	Position	Pin Name	Position	Pin Name
A1	VSS	F1	PWM3	K15	SDIO0_CCMD
A2	UART0_TXD	F2	PWM2	L1	SAR_ADC_CH2
A3	JTAG_TDI	F3	AVSS_PLL	L2	VSS
A4	JTAG_TCK	F5	VSS	L3	SAR_ADC_CH3
A5	VSS	F6	VSS	L5	DVDD3318_SENSOR
A6	VDDIO_DDR	F7	VSS	L6	VDD
A7	VSS	F8	VSS	L7	VDD
A8	VDDIO_DDR	F9	VSS	L8	VDD
A9	DDR_VREF	F10	VSS	L9	DVDD3318_VI
A10	SPI1_SCLK	F11	VDD	L10	AVDD33_AC
A11	SPI1_SDI	F13	UART2_RXD	L13	SDIO0_CWPR
A12	EPHY_CLK	F14	SFC_CLK	L14	VSS
A13	RMII_RXD0	F15	SFC_HOLD_IO3	L15	SDIO0_CCLK_OUT
A14	RMII_CLK	G1	PWM0	M1	SPI0_SCLK
A15	VSS	G2	I2C2_SDA	M2	SPI0_SDO
B1	RTC_XIN	G3	AVDD33_PLL	M3	SENSOR_RSTN
B2	RTC_XOUT	G5	VSS	M13	AVSS_AC
B3	UART0_RXD	G6	VSS	M14	SDIO0_CDATA1
B4	JTAG_TRSTN	G7	VSS	M15	SDIO0_CDATA0
B5	JTAG_TMS	G8	VSS	N1	SPI0_CSN
B6	VDDIO_DDR	G9	VSS	N2	SPI0_SDI
B7	VDDIO_CK_DDR	G10	VSS	N3	FLASH_TRIG
B8	VSS	G11	DVDDIO_RMII	N4	AVDD3318_MIPI
B9	VDDIO_DDR	G13	SYS_RSTN_OUT	N5	AVSS3318_MIPI
B10	VSS	G14	SFC_SIO_IO0	N6	GPIO1_0
B11	SPI1_CSN1	G15	SFC_CSN0	N7	GPIO1_2
B12	VSS	H1	I2C2_SCL	N8	GPIO1_5
B13	RMII_RXD1	H2	GPIO0_2	N9	GPIO1_4
B14	RMII_RX_DV	H3	AVDD33_RTC	N10	AVDD33_USB
B15	RMII_TXD1	H5	DVDD33	N11	AVSS_USB

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Position	Pin Name	Position	Pin Name	Position	Pin Name
C1	VSS	H6	VSS	N12	USB_REXT
C2	AVDD_BAT	H7	VSS	N13	AC_VREF
C3	TEST_MODE	H8	VSS	N14	SDIO0_CARD_POWER_EN
C4	JTAG_TDO	H9	VSS	N15	SDIO0_CARD_DETECT
C5	JTAG_EN	H10	VSS	P1	SENSOR_CLK
C6	VSS	H11	DVDD3318_EMMC	P2	SHUTTER_TRIG
C7	AVDD_DDRPLL	H13	SFC_CSN1	P3	MIPI_D3P
C8	VDDIO_DDR	H14	SFC_SOI_IO1	P4	MIPI_D1P
C9	VDDIO_DDR	H15	SFC_WP_IO2	P5	MIPI_CKP
C10	SPI1_SDO	J1	GPIO0_1	P6	MIPI_D0P
C11	SPI1_CSN0	J2	GPIO0_0	P7	MIPI_D2P
C12	MDCK	J3	DVDD33	P8	GPIO1_6
C13	RMII_TXD0	J5	VDD	P9	GPIO1_3
C14	RMII_TX_EN	J6	VSS	P10	USB_OVRCUR
C15	EPHY_RST	J7	VSS	P11	USB_DP
D1	XOUT	J8	VSS	P12	AVSS_USB
D2	XIN	J9	VSS	P13	AC_INL
D3	AVSS_RTC	J10	VSS	P14	AC_OUTL
D12	MDIO	J11	VSS	P15	AC_OUTR
D13	UART1_CTSN	J13	DVDD18_SDIO	R1	VSS
D14	UART1_RXD	J14	SDIO0_CDATA2	R2	AVSS3318_MIPI
D15	UART1_RTSN	J15	SDIO0_CDATA3	R3	MIPI_D3M
E1	VSS	K1	SAR_ADC_CH0	R4	MIPI_D1M
E2	PWM1	K2	SAR_ADC_CH1	R5	MIPI_CKM
E3	AVDD11_PLL	K3	VSS	R6	MIPI_D0M
E5	VDD	K5	VDD	R7	MIPI_D2M
E6	VDD	K6	VDD	R8	AVSS3318_MIPI
E7	VDD	K7	VSS	R9	GPIO1_1
E8	VDD	K8	VSS	R10	USB_PWREN
E9	VDD	K9	VSS	R11	USB_DM

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Position	Pin Name	Position	Pin Name	Position	Pin Name
E10	DVDD33	K10	AVSS_AC	R12	AVSS_AC
E13	UART2_TXD	K11	VDD	R13	AC_INR
E14	UART1_TXD	K13	DVDD33	R14	AC_MICBIAS
E15	VSS	K14	SDIO0_VOUT	R15	AVSS_AC

2.2 Pin Description

2.2.1 Pin Types

Table 2-3 describes the I/O pin types.

Table 2-3 I/O pin types

I/O	Description
I	Input signal
I_{PD}	Input signal, internal pull-down
I_{PU}	Input signal, internal pull-up
I_S	Input signal with a Schmitt trigger
I_{SPD}	Input signal with a Schmitt trigger, internal pull-down
I_{SPU}	Input signal with a Schmitt trigger, internal pull-up
O	Output signal
O_{OD}	Output open drain (OD)
I/O	Bidirectional (input/output) signal
I_{PD}/O	Bidirectional signal, input pull-down
I_{PU}/O	Bidirectional signal, input pull-up
I_{SPU}/O	Bidirectional signal with a Schmitt trigger, input pull-up
I_{SPD}/O	Bidirectional signal with a Schmitt trigger, input pull-down
I_{PD}/O_{OD}	Bidirectional signal, input pull-down and output OD
I_{PU}/O_{OD}	Bidirectional signal, input pull-up and output OD
I_S/O	Bidirectional signal, input with a Schmitt trigger
I_S/O_{OD}	Bidirectional signal, input with a Schmitt trigger and output OD
CIN	Crystal oscillator input

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I/O	Description
COUT	Crystal oscillator output
P	Power supply
G	GND

2.2.2 Pin Details

ACODEC Pins

Table 2-4 describes audio coder/decoder (ACODEC) pins.

Table 2-4 ACODEC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
P13	AC_INL	I	N/A	N/A	Audio-left channel input or positive end of the audio differential input
R13	AC_INR	I	N/A	N/A	Audio-right channel input or negative end of the audio differential input
R14	AC_MICBIAS	I/O	N/A	2.1	Audio input microphone bias
P14	AC_OUTL	O	N/A	N/A	Audio-left channel output
P15	AC_OUTR	O	N/A	N/A	Audio-right channel output
N13	AC_VREF	P	N/A	1.65	Audio reference source. This pin connects to a 10 µF external capacitor that connects to a 100 nF ceramic capacitor with low equivalent series resistance (ESR) in parallel. The 100 nF capacitor should be placed as close as possible to this pin.
L10	AVDD33_AC	P	N/A	3.3	Analog power of the audio interface
K10 M13 R12 R15	AVSS_AC	G	N/A	N/A	Analog GND of the audio interface

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EPHY Pins

Table 2-5 describes external physical layer (EPHY) pins.

Table 2-5 EPHY pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
A12	EPHY_CLK	I/O	Configurable (for details, see the PADCTRL_REG13 register)	1.8/3.3	**Function 0: GPIO3_0** General-purpose input/output (GPIO) **Function 1: EPHY_CLK** Working clock of the EPHY **Function 3: VO_DATA1** Video output (VO) data 1 **Function 4: SDIO1_CDATA2** Secure digital input/output 1 (SDIO1) data 2
C15	EPHY_RST	I_{PD}/O	Configurable (for details, see the PADCTRL_REG11 register)	1.8/3.3	**Function 0: GPIO2_7** GPIO **Function 1: EPHY_RST** EPHY reset **Function 2: BOOT_SEL** Boot enable 0: booting from the serial peripheral interface (SPI) flash 1: booting from the embedded multimedia card (eMMC) **Function 3: VO_HS** VO horizontal sync (HS) signal **Function 4: SDIO1_CARD_POWER_EN** SDIO1 power enable signal. The value 1 indicates power-on.

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Sensor Pins

Table 2-6 describes sensor pins.

Table 2-6 Sensor pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
N3	FLASH_TRIG	I_{PD}/O	Configurable (for details, see the PADCTRL_REG2 register)	3.3	**Function 0: GPIO0_6** GPIO **Function 1: FLASH_TRIG** Camera flash control **Function 2: SFC_EMMC_BOOT_MODE** Boot address mode of the SPI NOR flash when BOOT_SEL and SFC_DEVICE_MODE are 0 0: 3-byte address mode 1: 4-byte address mode Boot mode of the SPI NAND flash when BOOT_SEL is 0 and SFC_DEVICE_MODE is 1 0: 1-wire boot mode 1: 4-wire boot mode eMMC boot mode when BOOT_SEL is 1 0: 4-wire boot mode 1: 8-wire boot mode **Function 3: SPI1_CSN1** SPI1 chip select 1 (CS1) **Function 4: VI_VS_BT1120** Video input (VI) vertical sync (VS) signal
P1	SENSOR_CLK	I/O	Configurable (for details, see the PADCTRL_REG0 register)	3.3/1.8	**Function 0: GPIO0_4** GPIO **Function 1: SENSOR_CLK**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					Sensor working clock
M3	SENSOR_RSTN	I/O	Configurable (for details, see the PADCTRL_REG1 register)	3.3/1.8	**Function 0: SENSOR_RSTN** Sensor reset signal **Function 1: GPIO0_5** GPIO
P2	SHUTTER_TRIG	I_{PD}/O	Configurable (for details, see the PADCTRL_REG3 register)	3.3	**Function 0: GPIO0_7** GPIO **Function 1: SHUTTER_TRIG** Shutter control **Function 2: SFC_DEVICE_MODE** SPI flash from which the system boots (that is, type of the flash memory connected to SFC_CSN0) 0: SPI NOR flash 1: SPI NAND flash **Function 4: VI_HS_BT1120** VI HS signal

GPIO Pins

Table 2-7 describes GPIO pins.

Table 2-7 GPIO pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
J2	GPIO0_0	I/O	Configurable (for details, see the PADCTRL_REG32 register)	3.3	**Function 0: GPIO0_0** GPIO **Function 1: IR_IN** Infrared (IR) input **Function 2: TEMPER_DQ** Temperature measurement. It is used to communicate with the external

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					temperature measurement chip.
J1	GPIO0_1	I/O	Configurable (for details, see the PADCTRL_REG33 register)	3.3	**Function 0: GPIO0_1** GPIO **Function 2: TEMPER_DQ** Temperature measurement. It is used to communicate with the external temperature measurement chip.
H2	GPIO0_2	I/O	Configurable (for details, see the PADCTRL_REG34 register)	3.3	**Function 0: GPIO0_2** GPIO **Function 2: TEMPER_DQ** Temperature measurement. It is used to communicate with the external temperature measurement chip.
N6	GPIO1_0	I/O	Configurable (for details, see the PADCTRL_REG35 register)	3.3/1.8	**Function 0: GPIO1_0** GPIO **Function 1: VI_DATA13** VI data 13 **Function 3: I2S_BCLK_TX** Transmit (TX) clock of the inter-IC sound (I^2S) or pulse code modulation (PCM) interface **Function 4: PWM0** Pulse-width modulation (PWM) output 0
R9	GPIO1_1	I/O	Configurable (for details, see the PADCTRL_REG36 register)	3.3/1.8	**Function 0: GPIO1_1** GPIO **Function 1: VI_DATA10** VI data 10 **Function 3: I2S_SD_TX** TX data of the I^2S or PCM interface **Function 4: UART1_TXD** Universal asynchronous receiver transmitter 1

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					(UART1) TX data
N7	GPIO1_2	I/O	Configurable (for details, see the PADCTRL_REG37 register)	3.3/1.8	**Function 0: GPIO1_2** GPIO **Function 1: VI_DATA12** VI data 12 **Function 3: I2S_MCLK** Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC. **Function 4: UART1_CTSN** UART1 modem status input, active low
P9	GPIO1_3	I/O	Configurable (for details, see the PADCTRL_REG38 register)	3.3/1.8	**Function 0: GPIO1_3** GPIO **Function 1: VI_DATA11** VI data 11 **Function 3: I2S_WS_TX** I^2S TX audio channel select signal or PCM TX VS signal **Function 4: UART2_RXD** UART2 receive (RX) data
N9	GPIO1_4	I/O	Configurable (for details, see the PADCTRL_REG39 register)	3.3/1.8	**Function 0: GPIO1_4** GPIO **Function 1: VI_DATA15** VI data 15 **Function 2: VI_VS_SEN** VI VS signal **Function 3: I2S_WS_RX** I^2S RX audio channel select signal or PCM TX VS signal **Function 4: UART1_RXD** UART1 RX data
N8	GPIO1_5	I/O	Configurable (for details, see the	3.3/1.8	**Function 0: GPIO1_5** GPIO

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
			PADCTRL_REG40 register)		**Function 1: VI_DATA14** VI data 14 **Function 2: VI_HS_SEN** VI HS signal **Function 3: I2S_BCLK_RX** RX clock of the I²S or PCM interface **Function 4: UART1_RTSN** UART1 modem status output, active low
P8	GPIO1_6	I/O	Configurable (for details, see the PADCTRL_REG41 register)	3.3/1.8	**Function 0: GPIO1_6** GPIO **Function 1: VI_DATA9** VI data 9 **Function 3: I2S_SD_RX** RX data of the I²S or PCM interface **Function 4: UART2_TXD** UART2 TX data

I²C2 Pins

Table 2-8 describes inter-integrated circuit 2 (I²C2) pins.

Table 2-8 I²C2 pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
H1	I2C2_SCL	I/O	Configurable (for details, see the PADCTRL_REG25 register)	3.3	**Function 0: GPIO4_4** GPIO **Function 1: I2C2_SCL** I²C2 bus clock
G2	I2C2_SDA	I/O	Configurable (for details, see the PADCTRL_REG24 register)	3.3	**Function 0: GPIO4_3** GPIO **Function 1: I2C2_SDA** I²C2 bus data/address

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JTAG Pins

Table 2-9 describes Joint Test Action Group (JTAG) pins.

Table 2-9 JTAG pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
C5	JTAG_EN	I_{PD}	N/A	3.3	JTAG pin enable 0: disabled 1: enabled
A4	JTAG_TCK	I/O	Configurable (for details, see the PADCTRL_REG65 register)	3.3	**Function 0: GPIO6_6** GPIO **Function 1: JTAG_TCK** JTAG clock signal **Function 2: SPI1_SCLK** SPI1 clock signal **Function 3: I2S_WS_TX** I^2S TX audio channel select signal or PCM TX VS signal **Function 4: I2C1_SCL** I^2C1 bus clock
A3	JTAG_TDI	I/O	Configurable (for details, see the PADCTRL_REG68 register)	3.3	**Function 0: GPIO7_1** GPIO **Function 1: JTAG_TDI** JTAG input data **Function 2: SPI1_SDI** SPI1 input data **Function 3: I2S_BCLK_TX** TX clock of the I^2S or PCM interface
C4	JTAG_TDO	I/O	Configurable (for details, see the PADCTRL_REG67 register)	3.3	**Function 0: GPIO7_0** GPIO **Function 1: JTAG_TDO** JTAG output data **Function 2: SPI1_SDO** SPI1 output data **Function 3: I2S_SD_RX**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					RX data of the I²S or PCM interface **Function 4: I2C1_SDA** I²C1 bus data/address
B5	JTAG_TMS	I/O	Configurable (for details, see the PADCTRL_REG66 register)	3.3	**Function 0: GPIO6_7** GPIO **Function 1: JTAG_TMS** JTAG mode select input **Function 2: SPI1_CSN0** SPI1 CS0 **Function 3: I2S_SD_TX** TX data of the I²S or PCM interface
B4	JTAG_TRSTN	I/O	Configurable (for details, see the PADCTRL_REG64 register)	3.3	**Function 0: GPIO6_5** GPIO **Function 1: JTAG_TRSTN** JTAG reset signal **Function 2: SPI1_CSN1** SPI1 CS1 **Function 3: I2S_MCLK** Main clock of the I²S or PCM interface. It can act as the working clock of the ACODEC.

NOTE

Functions 0, 2, and 3 are valid only when the input of the JTAG_EN pin is 0, and function 1 is valid only when the input of the JTAG_EN pin is 1.

MDIO Pins

Table 2-10 describes management data input/output (MDIO) pins.

Table 2-10 MDIO pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
C12	MDCK	I_{PD}/O	Configurable (for details, see the	1.8/3.3	**Function 0: GPIO3_1** GPIO

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
			PADCTRL_REG14 register)		**Function 1: MDCK** MDIO interface clock **Function 2: BOOTROM_SEL** BOOTROM boot enable 0: booting from other space 1: booting from the BOOTROM **Function 3: VO_DATA6** VO data 6
D12	MDIO	I/O	Configurable (for details, see the PADCTRL_REG15 register)	1.8/3.3	**Function 0: GPIO3_2** GPIO **Function 1: MDIO** Data signal of the MDIO interface **Function 3: VO_DATA4** VO data 4

MIPI Pins

The mobile industry processor interface (MIPI) of Hi3518E V20X supports the differential and single-ended modes. Table 2-11 describes the functions of MIPI pins in the two modes.

Table 2-11 Functions of MIPI pins in two modes

Pin Name	Function in Differential Mode	Function in Single-Ended Mode
MIPI_CKM	MIPI differential clock	VI_DATA3
MIPI_CKP	MIPI differential clock	VI_DATA4
MIPI_D0M	MIPI differential data 0	VI_DATA5
MIPI_D0P	MIPI differential data 0	VI_DATA6
MIPI_D1M	MIPI differential data 1	VI_ DATA1
MIPI_D1P	MIPI differential data 1	VI_ DATA2
MIPI_D2M	MIPI differential data 2	VI_DATA7
MIPI_D2P	MIPI differential data 2	VI_DATA8
MIPI_D3M	MIPI differential data 3	VI_CLK
MIPI_D3P	MIPI differential data 3	VI_DATA0

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Table 2-12 describes MIPI pins.

Table 2-12 MIPI pins

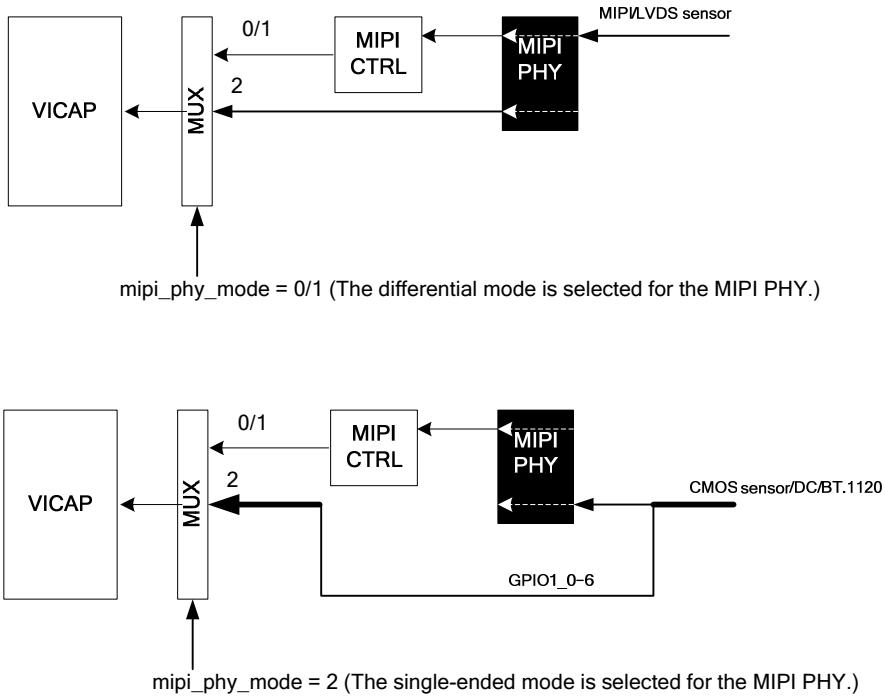
Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
R5	MIPI_CKM	I	N/A	3.3/1.8	VI_DATA3 in single-ended mode or differential clock of MIPI
P5	MIPI_CKP	I	N/A	3.3/1.8	VI_DATA4 in single-ended mode or differential clock of MIPI
R6	MIPI_D0M	I	N/A	3.3/1.8	VI_DATA5 in single-ended mode or differential data 0 of MIPI
P6	MIPI_D0P	I	N/A	3.3/1.8	VI_DATA6 in single-ended mode or differential data 0 of MIPI
R4	MIPI_D1M	I	N/A	3.3/1.8	VI_DATA1 in single-ended mode or differential data 1 of MIPI
P4	MIPI_D1P	I	N/A	3.3/1.8	VI_DATA2 in single-ended mode or differential data 1 of MIPI
R7	MIPI_D2M	I	N/A	3.3/1.8	VI_DATA7 in single-ended mode or differential data 2 of MIPI
P7	MIPI_D2P	I	N/A	3.3/1.8	VI_DATA8 in single-ended mode or differential data 2 of MIPI
R3	MIPI_D3M	I	N/A	3.3/1.8	VI_CLK in single-ended mode or differential data 3 of MIPI
P3	MIPI_D3P	I	N/A	3.3/1.8	VI_DATA0 in single-ended mode or differential data 3 of MIPI
N4	AVDD3318_MIPI	P	N/A	3.3/1.8	Analog power of the MIPI PHY
N5 R2 R8	AVSS3318_MIPI	G	N/A	N/A	Analog GND of the MIPI PHY

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CAUTION

As the video capture (VICAP) supports only one video input, the video input paths of the MIPI PHY and VI pin are mutually exclusive (dual sensor inputs are not supported). The mipi_phy_mode register determines the video input of the VICAP. See Figure 2-10.

Figure 2-10 Video input paths of the MIPI PHY and VI pin



PWM Pins

Table 2-13 describes PWM pins.

Table 2-13 PWM pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
G1	PWM0	I/O	Configurable (for details, see the PADCTRL_REG 70 register)	3.3	**Function 0: PMC_PWM** Power management controller (PMC) PWM output **Function 1: GPIO7_2** GPIO **Function 2: PWM0** PWM output 0

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
E2	PWM1	I/O	Configurable (for details, see the PADCTRL_REG71 register)	3.3	**Function 0: PWM1** PWM output 1 **Function 1: GPIO7_3** GPIO
F2	PWM2	I/O	Configurable (for details, see the PADCTRL_REG72 register)	3.3	**Function 0: PWM2** PWM output 2 **Function 1: GPIO7_4** GPIO **Function 2: SPI1_CSN1** SPI1 CS1
F1	PWM3	I/O	Configurable (for details, see the PADCTRL_REG73 register)	3.3	**Function 0: PWM3** PWM output 3 **Function 1: GPIO7_5** GPIO

RMII Pins

Table 2-14 describes reduced media independent interface (RMII) pins.

Table 2-14 RMII pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
A14	RMII_CLK	I/O	Configurable (for details, see the PADCTRL_REG4 register)	1.8/3.3	**Function 0: GPIO2_0** GPIO **Function 1: RMII_CLK** RMII clock **Function 3: VO_CLK** VO clock signal **Function 4: SDIO1_CCLK_OUT** SDIO1 output working clock for the card
B14	RMII_RX_DV	I/O	Configurable (for details, see the PADCTRL_REG8 register)	1.8/3.3	**Function 0: GPIO2_4** GPIO **Function 1: RMII_RX_DV** RX data validity and carrier sense signal **Function 3: VO_DATA7**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					VO data 7 **Function 4: SDIO1_CDATA0** SDIO1 data 0
A13	RMII_RXD0	I/O	Configurable (for details, see the PADCTRL_REG9 register)	1.8/3.3	**Function 0: GPIO2_5** GPIO **Function 1: RMII_RXD0** RX data 0 **Function 3: VO_DATA2** VO data 2 **Function 4: SDIO1_CDATA3** SDIO1 data 3
B13	RMII_RXD1	I/O	Configurable (for details, see the PADCTRL_REG10 register)	1.8/3.3	**Function 0: GPIO2_6** GPIO **Function 1: RMII_RXD1** RX data 1 **Function 3: VO_DATA3** VO data 3 **Function 4: SDIO1_CCMD** SDIO1 command
C14	RMII_TX_EN	I/O	Configurable (for details, see the PADCTRL_REG5 register)	1.8/3.3	**Function 0: GPIO2_1** GPIO **Function 1: RMII_TX_EN** TX enable **Function 3: VO_VS** VO VS signal **Function 4: SDIO1_CARD_DETECT** SDIO1 card detection signal, active low
C13	RMII_TXD0	I/O	Configurable (for details, see the PADCTRL_REG6 register)	1.8/3.3	**Function 0: GPIO2_2** GPIO **Function 1: RMII_TXD0** TX data 0 **Function 3: VO_DATA5** VO data 5 **Function 4: SDIO1_CWPR** SDIO1 card write protection

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					detection
B15	RMII_TXD1	I/O	Configurable (for details, see the PADCTRL_REG7 register)	1.8/3.3	**Function 0: GPIO2_3** GPIO **Function 1: RMII_TXD1** TX data 1 **Function 3: VO_DE** VO data validity signal **Function 4: SDIO1_CDATA1** SDIO1 data 1

RTC Pins

Table 2-15 describes real-time clock (RTC) pins.

Table 2-15 RTC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
B1	RTC_XIN	CIN	N/A	N/A	RTC crystal input
B2	RTC_XOUT	COUT	N/A	N/A	RTC crystal drive output
C2	AVDD_BAT	P	N/A	3.3	RTC external battery power
D3	AVSS_RTC	G	N/A	N/A	RTC analog GND
H3	AVDD_RTC	G	N/A	3.3	RTC analog power

SAR_ADC Pins

Table 2-16 describes successive approximation register analog-to-digital converter (SAR_ADC) pins.

Table 2-16 SAR_ADC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
K1	SAR_ADC	I/O	Configurable (for details, see the	3.3	**Function 0:**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
	_CH0		PADCTRL_REG74 register)		**SAR_ADC_CH0** LSADC channel 0 **Function 1: GPIO7_6** GPIO
K2	SAR_ADC_CH1	I/O	Configurable (for details, see the PADCTRL_REG75 register)	3.3	**Function 0: SAR_ADC_CH1** LSADC channel 1 **Function 1: GPIO7_7** GPIO
L1	SAR_ADC_CH2	I/O	Configurable (for details, see the PADCTRL_REG76 register)	3.3	**Function 0: SAR_ADC_CH2** LSADC channel 2 **Function 1: GPIO8_0** GPIO
L3	SAR_ADC_CH3	I/O	Configurable (for details, see the PADCTRL_REG77 register)	3.3	**Function 0: SAR_ADC_CH3** LSADC channel 3 **Function 1: GPIO8_1** GPIO

SDIO Pins

Table 2-17 describes SDIO pins.

Table 2-17 SDIO pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
N15	SDIO0_CARD_DETECT	I/O	Configurable (for details, see the PADCTRL_REG43 register)	3.3	**Function 0: GPIO4_7** GPIO **Function 1: SDIO0_CARD_DETECT** SDIO0 card detection signal, active low
N14	SDIO0_CARD_POWER_EN	I/O	Configurable (for details, see the PADCTRL_REG42 register)	3.3	**Function 0: GPIO1_7** GPIO **Function 1: SDIO0_CARD_POWER_EN** SDIO0 power enable signal.

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					The value 1 indicates power-on.
L15	SDIO0_CCLK_OUT	I/O	Configurable (for details, see the PADCTRL_REG45 register)	3.3/1.8	**Function 0: GPIO5_1** GPIO **Function 1: SDIO0_CCLK_OUT** SDIO0 output working clock for the card
K15	SDIO0_CCMD	I/O	Configurable (for details, see the PADCTRL_REG46 register)	3.3/1.8	**Function 0: GPIO5_2** GPIO **Function 1: SDIO0_CCMD** SDIO0 command
M15	SDIO0_CDATA0	I/O	Configurable (for details, see the PADCTRL_REG47 register)	3.3/1.8	**Function 0: GPIO5_3** GPIO **Function 1: SDIO0_CDATA0** SDIO0 data 0
M14	SDIO0_CDATA1	I/O	Configurable (for details, see the PADCTRL_REG48 register)	3.3/1.8	**Function 0: TEST_CLK** Test clock **Function 1: SDIO0_CDATA1** SDIO0 data 1 **Function 2: GPIO5_4** GPIO
J14	SDIO0_CDATA2	I/O	Configurable (for details, see the PADCTRL_REG49 register)	3.3/1.8	**Function 0: GPIO5_5** GPIO **Function 1: SDIO0_CDATA2** SDIO0 data 2
J15	SDIO0_CDATA3	I/O	Configurable (for details, see the PADCTRL_REG50 register)	3.3/1.8	**Function 0: GPIO5_6** GPIO **Function 1: SDIO0_CDATA3** SDIO0 data 3
L13	SDIO0_CWPR	I/O	Configurable (for details, see the PADCTRL_REG44 register)	3.3	**Function 0: GPIO5_0** GPIO **Function 1: SDIO0_CWPR** SDIO0 card write protection detection

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
K14	SDIO0_VOUT	P	N/A	3.3/1.8	Power output of the pull-up resistor of SDIO0

SFC Pins

Table 2-18 describes SPI flash controller (SFC) pins.

Table 2-18 SFC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
F14	SFC_CLK	O	Configurable (for details, see the PADCTRL_REG51 register)	3.3/1.8	**Function 0: SFC_CLK** Clock signal transmitted to the SPI flash **Function 1: EMMC_CLK** eMMC clock
G15	SFC_CSN0	O	Configurable (for details, see the PADCTRL_REG56 register)	3.3/1.8	**Function 0: SFC_CSN0** CS0 of the SPI flash for mounting the SPI flash from which the system boots, active low **Function 1: EMMC_RST_N** eMMC reset
H13	SFC_CSN1	O	Configurable (for details, see the PADCTRL_REG57 register)	3.3/1.8	**Function 0: SFC_CSN1** CS1 of the SPI flash, active low **Function 1: EMMC_POWER_EN** eMMC power enable signal. The value 1 indicates power-on.
F15	SFC_HOLD_IO3	I/O	Configurable (for details, see the PADCTRL_REG55 register)	3.3/1.8	**Function 0: SFC_HOLD_IO3** Hold function in standard SPI mode, active low Hold function in dual-SPI mode, active low Data I/O signal 3 in quad-SPI mode **Function 1: EMMC_DATA0** eMMC data 0

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
G14	SFC_SIO_IO0	I/O	Configurable (for details, see the PADCTRL_REG52 register)	3.3/1.8	**Function 0: SFC_SIO_IO0** Data output signal in standard SPI mode Data I/O signal 0 in dual-SPI mode Data I/O signal 0 in quad-SPI mode **Function 1: EMMC_DATA4** eMMC data 4
H14	SFC_SOI_IO1	I/O	Configurable (for details, see the PADCTRL_REG53 register)	3.3/1.8	**Function 0: SFC_SOI_IO1** Data input signal in standard SPI mode Data I/O signal 1 in dual-SPI mode Data I/O signal 1 in quad-SPI mode **Function 1: EMMC_DATA3** eMMC data 3
H15	SFC_WP_IO2	I/O	Configurable (for details, see the PADCTRL_REG54 register)	3.3/1.8	**Function 0: SFC_WP_IO2** Write protection function in standard SPI mode, active low Write protection function in dual-SPI mode, active low Data I/O signal 2 in quad-SPI mode **Function 1: EMMC_CMD** eMMC command/response

SPI0 Pins

Table 2-19 describes SPI0 pins.

Table 2-19 SPI0 pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
N1	SPI0_CSN	I/O	Configurable (for details, see the PADCTRL_REG19 register)	3.3/1.8	**Function 0: GPIO3_6** GPIO **Function 1: SPI0_CSN** SPI0 CS signal

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
M1	SPI0_SCLK	I/O	Configurable (for details, see the PADCTRL_REG16 register)	3.3/1.8	**Function 0: GPIO3_3** GPIO **Function 1: SPI0_SCLK** SPI0 clock signal **Function 2: I2C0_SCL** I^2C0 bus clock
N2	SPI0_SDI	I/O	Configurable (for details, see the PADCTRL_REG18 register)	3.3/1.8	**Function 0: GPIO3_5** GPIO **Function 1: SPI0_SDI** SPI0 input data
M2	SPI0_SDO	I/O	Configurable (for details, see the PADCTRL_REG17 register)	3.3/1.8	**Function 0: GPIO3_4** GPIO **Function 1: SPI0_SDO** SPI0 output data **Function 2: I2C0_SDA** I^2C0 bus data/address

SPI1 Pins

Table 2-20 describes SPI1 pins.

Table 2-20 SPI1 pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
C11	SPI1_CSN0	I/O	Configurable (for details, see the PADCTRL_REG23 register)	3.3	**Function 0: GPIO4_2** GPIO **Function 1: SPI1_CSN0** SPI1 CS0
B11	SPI1_CSN1	I/O	Configurable (for details, see the PADCTRL_REG12 register)	3.3	**Function 0: GPIO0_3** GPIO **Function 1: SPI1_CSN1** SPI1 CS1 **Function 3: VO_DATA0** VO data 0

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
A10	SPI1_SCLK	I/O	Configurable (for details, see the PADCTRL_REG20 register)	3.3	**Function 0: GPIO3_7** GPIO **Function 1: SPI1_SCLK** SPI1 clock signal **Function 2: I2C1_SCL** I^2C1 bus clock
A11	SPI1_SDI	I/O	Configurable (for details, see the PADCTRL_REG22 register)	3.3	**Function 0: GPIO4_1** GPIO **Function 1: SPI1_SDI** SPI1 input data
C10	SPI1_SDO	I/O	Configurable (for details, see the PADCTRL_REG21 register)	3.3	**Function 0: GPIO4_0** GPIO **Function 1: SPI1_SDO** SPI1 output data **Function 2: I2C1_SDA** I^2C1 bus data/address

SYS Pins

Table 2-21 describes system (SYS) pins.

Table 2-21 SYS pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
G13	SYS_RSTN_OUT	O	N/A	1.8/3.3	System reset output 0: reset 1: deassert reset
C3	TEST_MODE	I_{PD}	N/A	3.3	Mode select 0: functional mode 1: test mode

UART Pins

Table 2-22 describes UART pins.

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Table 2-22 UART pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
B3	UART0_RXD	I_{PU}	Configurable (for details, see the PADCTRL_REG26 register)	3.3	UART0 RX data
A2	UART0_TXD	O	Configurable (for details, see the PADCTRL_REG27 register)	3.3	UART0 TX data
D13	UART1_CTSN	I/O	Configurable (for details, see the PADCTRL_REG60 register)	3.3/1.8	**Function 0: GPIO6_1** GPIO **Function 1: EMMC_DATA7** eMMC data 7 **Function 2: I2S_MCLK** Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC. **Function 3: UART1_CTSN** UART1 modem status input, active low
D15	UART1_RTSN	I/O	Configurable (for details, see the PADCTRL_REG58 register)	3.3/1.8	**Function 0: GPIO5_7** GPIO **Function 1: EMMC_DATA6** eMMC data 6 **Function 2: I2S_SD_TX** TX data of the I^2S or PCM interface **Function 3: UART1_RTSN** UART1 modem status output, active low
D14	UART1_RXD	I/O	Configurable (for details, see the PADCTRL_REG59 register)	3.3/1.8	**Function 0: GPIO6_0** GPIO **Function 1: EMMC_DATA5** eMMC data 5 **Function 2: I2S_WS_TX** I^2S TX audio channel select signal or PCM TX VS signal **Function 3: UART1_RXD**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					UART1 RX data
E14	UART1_TXD	I/O	Configurable (for details, see the PADCTRL_REG61 register)	3.3/1.8	**Function 0: GPIO6_2** GPIO **Function 1: EMMC_DS** eMMC DATA_STROBE **Function 2: I2S_SD_RX** RX data of the I^2S or PCM interface **Function 3: UART1_TXD** UART1 TX data
F13	UART2_RXD	I/O	Configurable (for details, see the PADCTRL_REG62 register)	3.3/1.8	**Function 0: GPIO6_3** GPIO **Function 1: EMMC_DATA1** eMMC data 1 **Function 3: UART2_RXD** UART2 RX data
E13	UART2_TXD	I/O	Configurable (for details, see the PADCTRL_REG63 register)	3.3/1.8	**Function 0: GPIO6_4** GPIO **Function 1: EMMC_DATA2** eMMC data 2 **Function 2: I2S_BCLK_TX** TX clock of the I^2S or PCM interface **Function 3: UART2_TXD** UART2 TX data

USB Pins

Table 2-23 describes USB pins.

Table 2-23 USB pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
R11	USB_DM	I/O	N/A	N/A	USB D− signal. In high-speed mode, the maximum voltage of this port is 400 mV. In full-speed or low-speed mode, the voltage of this port is 3.3 V.

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
P11	USB_DP	I/O	N/A	N/A	USB D+ signal. In high-speed mode, the maximum voltage of this port is 400 mV. In full-speed or low-speed mode, the voltage of this port is 3.3 V.
P10	USB_OVRCUR	I/O	Configurable (for details, see the PADCTRL_REG30register)	3.3	**Function 0: GPIO4_5** GPIO **Function 1: USB_OVRCUR** USB port overcurrent indicator. The level polarity can be specified by configuring the peripheral control register MISC_CTRL30 bit[17].
R10	USB_PWREN	I/O	Configurable (for details, see the PADCTRL_REG31register)	3.3	**Function 0: GPIO4_6** GPIO **Function 1: USB_PWREN** Power control output pin of the USB port. The level polarity can be specified by configuring the peripheral control register MISC_CTRL30 bit[16].
N12	USB_REXT	I/O	N/A	N/A	USB pin for connecting to an external reference resistor. The 135 Ω resistor is recommended.
N10	AVDD33_USB	P	N/A	3.3	USB analog power
N11 P12	AVSS_USB	G	N/A	N/A	USB analog GND

OSC Pins

Table 2-24 describes oscillator (OSC) pins.

Table 2-24 OSC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
D2	XIN	CIN	N/A	N/A	24 MHz system crystal input
D1	XOUT	COUT	N/A	N/A	24 MHz system crystal drive output

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Other Power Pins and GND Pins

Table 2-25 describes other power pins and GND pins.

Table 2-25 Power pins and GND pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
C7	AVDD_DDRPLL	P	N/A	3.3	Analog power of the DDR phase-locked loop (PLL)
A9	DDR_VREF	P	N/A	0.9	DDR reference power
B7	VDDIO_CK_DDR	P	N/A	1.8	I/O power for the DDR clock
A6, A8, B6, B9, C8, C9	VDDIO_DDR	P	N/A	1.8	Other I/O power for the DDR
L5	DVDD3318_SENSOR	P	N/A	3.3/1.8	Sensor I/O power
L9	DVDD3318_VI	P	N/A	3.3/1.8	VI I/O power
G11	DVDDIO_RMII	P	N/A	3.3/1.8	RMII I/O power
E5, E6, E7, E8, E9, F11, J5, K5, K6, K11, L6, L7, L8	VDD	P	N/A	1.1	Core working power
A1, A5, A7, A15, B8, B10, B12, C1, C6, E1, E15, F5, F6, F7, F8, F9, F10, G5, G6, G7, G8, G9, G10, H6, H7, H8, H9, H10, J6, J7, J8, J9, J10, J11, K3, K7, K8, K9, L14, L2, R1	VSS	G	N/A	N/A	Digital GND
E10, H5, J3, K13	DVDD33	P	N/A	3.3	Digital I/O power

2.2.3 Default Reset Status of Digital Pins

Table 2-26 describes the default reset status of digital pins.

NOTE

The level states are described as follows:

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- 0: Output the low level.
- 1: Output the high level.
- Z: high impedance
- PU: input state with internal pull-up
- PD: input state with internal pull-down
- Toggle: Level jump occurs.

For details about the specific resistance of the pull-up/pull-down resistors in the chip, see Table 2-72 and Table 2-73.

Table 2-26 Default reset status of digital pins

PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
SENSOR_CLK	GPIO0_4	Z	N/A	Z	I	-	-
SENSOR_RSTN	SENSOR_RSTN	PD	N/A	0	O	R_{PD2}	Fixed pull-down in the chip
FLASH_TRIG	GPIO0_6	PD	N/A	PD	I	R_{PD5}	Fixed pull-down in the chip
SHUTTER_TRIG	GPIO0_7	PD	N/A	PD	I	R_{PD5}	Fixed pull-down in the chip
RMII_CLK	GPIO2_0	Z	N/A	Z	I	-	-
RMII_TX_EN	GPIO2_1	Z	N/A	Z	I	-	-
RMII_TXD0	GPIO2_2	Z	N/A	Z	I	-	-
RMII_TXD1	GPIO2_3	Z	N/A	Z	I	-	-
RMII_RX_DV	GPIO2_4	Z	N/A	Z	I	-	-
RMII_RXD0	GPIO2_5	Z	N/A	Z	I	-	-
RMII_RXD1	GPIO2_6	Z	N/A	Z	I	-	-
EPHY_RST	GPIO2_7	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip
EPHY_CLK	GPIO3_0	Z	N/A	Z	I	-	-
MDCK	GPIO3_1	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
MDIO	GPIO3_2	Z	N/A	Z	I	-	-
SPI0_SCLK	GPIO3_3	Z	N/A	Z	I	-	-
SPI0_SDO	GPIO3_4	Z	N/A	Z	I	-	-
SPI0_SDI	GPIO3_5	Z	N/A	Z	I	-	-
SPI0_CSN	GPIO3_6	Z	N/A	Z	I	-	-
SPI1_SCLK	GPIO3_7	Z	N/A	Z	I	-	-
SPI1_SDO	GPIO4_0	Z	N/A	Z	I	-	-
SPI1_SDI	GPIO4_1	Z	N/A	Z	I	-	-
SPI1_CSN0	GPIO4_2	Z	N/A	Z	I	-	-
SPI1_CSN1	GPIO0_3	Z	N/A	Z	I	-	-
I2C2_SDA	GPIO4_3	Z	N/A	Z	I	-	-
I2C2_SCL	GPIO4_4	Z	N/A	Z	I	-	-
UART0_RXD	UART0_RXD	Z	N/A	PU	I	R_{PU1}	
UART0_TXD	UART0_TXD	Z	N/A	1	O	-	-
SYS_RSTN_OUT	SYS_RSTN_OUT	PD	N/A	0	O	R_{PD2}	Fixed pull-down in the chip
TEST_MODE	TEST_MODE	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip
USB_OVRCUR	GPIO4_5	Z	N/A	Z	I	-	-
USB_PWREN	GPIO4_6	Z	N/A	Z	I	-	-
GPIO0_0	GPIO0_0	PD	N/A	Z	I	R_{PD2}	-
GPIO0_1	GPIO0_1	PD	N/A	Z	I	R_{PD2}	-
GPIO0_2	GPIO0_2	PD	N/A	Z	I	R_{PD2}	-
GPIO1_0	GPIO1_0	Z	N/A	Z	I	-	-

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
GPIO1_1	GPIO1_1	Z	N/A	Z	I	-	-
GPIO1_2	GPIO1_2	Z	N/A	Z	I	-	-
GPIO1_3	GPIO1_3	Z	N/A	Z	I	-	-
GPIO1_4	GPIO1_4	Z	N/A	Z	I	-	-
GPIO1_5	GPIO1_5	Z	N/A	Z	I	-	-
GPIO1_6	GPIO1_6	Z	N/A	Z	I	-	-
SDIO0_CARD_POWER_EN	GPIO1_7	Z	N/A	Z	I	-	-
SDIO0_CARD_DETECT	GPIO4_7	Z	N/A	Z	I	-	-
SDIO0_CWPR	GPIO5_0	Z	N/A	Z	I	-	-
SDIO0_CCLK_OUT	GPIO5_1	Z	N/A	Z	I	-	-
SDIO0_CCMD	GPIO5_2	Z	N/A	Z	I	-	-
SDIO0_CDATA0	GPIO5_3	Z	N/A	Z	I	-	-
SDIO0_CDATA1	TEST_CLK	Z	N/A	Toggle	O	-	Test clock
SDIO0_CDATA2	GPIO5_5	Z	N/A	Z	I	-	-
SDIO0_CDATA3	GPIO5_6	Z	N/A	Z	I	-	-
JTAG_TRSTN	GPIO6_5	Z	N/A	Z	I	-	-
JTAG_TCK	GPIO6_6	Z	N/A	Z	I	-	-
JTAG_TMS	GPIO6_7	Z	N/A	Z	I	-	-
JTAG_TDO	GPIO7_0	Z	N/A	Z	I	-	-
JTAG_TDI	GPIO7_1	Z	N/A	Z	I	-	-
JTAG_EN	JTAG_EN	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
PWM0	PMC_PWM	Z	N/A	0	O	-	-
PWM1	PWM1	Z	N/A	0	O	-	-
PWM2	PWM2	Z	N/A	0	O	-	-
PWM3	PWM3	Z	N/A	0	O	-	-
SAR_ADC_CH0	SAR_ADC_CH0	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SAR_ADC_CH1	SAR_ADC_CH1	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SAR_ADC_CH2	SAR_ADC_CH2	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SAR_ADC_CH3	SAR_ADC_CH3	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_CLK	SFC_CLK	Z	N/A	0	O	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
SFC_SIO_IO0	SFC_SIO_IO0	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_SOI_IO1	SFC_SOI_IO1	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_WP_IO2	SFC_WP_IO2	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_HOLD_IO3	SFC_HOLD_IO3	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_CSN0	SFC_CSN0	PU	N/A	1	O	R_{PU2}	Fixed pull-up in the chip
SFC_CSN1	SFC_CSN1	PU	N/A	1	O	R_{PU2}	Fixed pull-up in the chip
UART1_RTSN	GPIO5_7	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART1_RXD	GPIO6_0	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
UART1_CTSN	GPIO6_1	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART1_TXD	GPIO6_2	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART2_RXD	GPIO6_3	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART2_TXD	GPIO6_4	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default

2.3 Pin Multiplexing Control Registers

2.3.1 Register Summary

Table 2-27 describes pin multiplexing control registers.

Table 2-27 Summary of pin multiplexing control registers (base address: 0x200F_0000)

Offset Address	Register	Description	Page
0x000	muxctrl_reg0	Multiplexing control register for the SENSOR_CLK pin	2-43

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Offset Address	Register	Description	Page
0x004	muxctrl_reg1	Multiplexing control register for the SENSOR_RSTN pin	2-44
0x008	muxctrl_reg2	Multiplexing control register for the FLASH_TRIG pin	2-44
0x00C	muxctrl_reg3	Multiplexing control register for the SHUTTER_TRIG pin	2-45
0x010	muxctrl_reg4	Multiplexing control register for the RMII_CLK pin	2-45
0x014	muxctrl_reg5	Multiplexing control register for the RMII_TX_EN pin	2-46
0x018	muxctrl_reg6	Multiplexing control register for the RMII_TXD0 pin	2-46
0x01C	muxctrl_reg7	Multiplexing control register for the RMII_TXD1 pin	2-47
0x020	muxctrl_reg8	Multiplexing control register for the RMII_RX_DV pin	2-47
0x024	muxctrl_reg9	Multiplexing control register for the RMII_RXD0 pin	2-48
0x028	muxctrl_reg10	Multiplexing control register for the RMII_RXD1 pin	2-48
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Offset Address	Register	Description	Page
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2.3.2 Register Description

muxctrl_reg0

muxctrl_reg0 is a multiplexing control register for the SENSOR_CLK pin.

Offset Address	Register Name	Total Reset Value
0x000	muxctrl_reg0	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg0
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg0	Multiplexing control for the SENSOR_CLK pin 0: GPIO0_4 1: SENSOR_CLK

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muxctrl_reg1

muxctrl_reg1 is a multiplexing control register for the SENSOR_RSTN pin.

Offset Address	Register Name	Total Reset Value
0x004	muxctrl_reg1	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg1
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg1	Multiplexing control for the SENSOR_RSTN pin 0: SENSOR_RSTN 1: GPIO0_5

muxctrl_reg2

muxctrl_reg2 is a multiplexing control register for the FLASH_TRIG pin.

Offset Address	Register Name	Total Reset Value
0x008	muxctrl_reg2	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg2
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg2	Multiplexing control for the FLASH_TRIG pin 000: GPIO0_6 001: FLASH_TRIG 010: SFC_EMMC_BOOT_MODE 011: SPI1_CSN1 100: VI_VS_BT1120 Other values: reserved

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muxctrl_reg3

muxctrl_reg3 is a multiplexing control register for the SHUTTER_TRIG pin.

Offset Address	Register Name	Total Reset Value
0x00C	muxctrl_reg3	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg3	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg3	Multiplexing control for the SHUTTER_TRIG pin 000: GPIO0_7 001: SHUTTER_TRIG 010: SFC_DEVICE_MODE 100: VI_HS_BT1120 Other values: reserved

muxctrl_reg4

muxctrl_reg4 is a multiplexing control register for the RMII_CLK pin.

Offset Address	Register Name	Total Reset Value
0x010	muxctrl_reg4	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg4	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg4	Multiplexing control for the RMII_CLK pin 000: GPIO2_0 001: RMII_CLK

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			011: VO_CLK 100: SDIO1_CCLK_OUT Other values: reserved

muxctrl_reg5

muxctrl_reg5 is a multiplexing control register for the RMII_TX_EN pin.

Offset Address	Register Name	Total Reset Value
0x014	muxctrl_reg5	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg5	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg5	Multiplexing control for the RMII_TX_EN pin 000: GPIO2_1 001: RMII_TX_EN 011: VO_VS 100: SDIO1_CARD_DETECT Other values: reserved

muxctrl_reg6

muxctrl_reg6 is a multiplexing control register for the RMII_TXD0 pin.

Offset Address	Register Name	Total Reset Value
0x018	muxctrl_reg6	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg6	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved

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[2:0]	RW	muxctrl_reg6	Multiplexing control for the RMII_TXD0 pin 000: GPIO2_2 001: RMII_TXD0 011: VO_DATA5 100: SDIO1_CWPR Other values: reserved

muxctrl_reg7

muxctrl_reg7 is a multiplexing control register for the RMII_TXD1 pin.

Offset Address	Register Name	Total Reset Value
0x01C	muxctrl_reg7	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg7	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg7	Multiplexing control for the RMII_TXD1 pin 000: GPIO2_3 001: RMII_TXD1 011: VO_DE 100: SDIO1_CDATA1 Other values: reserved

muxctrl_reg8

muxctrl_reg8 is a multiplexing control register for the RMII_RX_DV pin.

Offset Address	Register Name	Total Reset Value
0x020	muxctrl_reg8	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg8	

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg8	Multiplexing control for the RMII_RX_DV pin 000: GPIO2_4 001: RMII_RX_DV 011: VO_DATA7 100: SDIO1_CDATA0 Other values: reserved

muxctrl_reg9

muxctrl_reg9 is a multiplexing control register for the RMII_RXD0 pin.

Offset Address	Register Name	Total Reset Value
0x024	muxctrl_reg9	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg9	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg9	Multiplexing control for the RMII_RXD0 pin 000: GPIO2_5 001: RMII_RXD0 011: VO_DATA2 100: SDIO1_CDATA3 Other values: reserved

muxctrl_reg10

muxctrl_reg10 is a multiplexing control register for the RMII_RXD1 pin.

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Offset Address	Register Name	Total Reset Value
0x028	muxctrl_reg10	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg10	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg10	Multiplexing control for the RMII_RXD1 pin 000: GPIO2_6 001: RMII_RXD1 011: VO_DATA3 100: SDIO1_CCMD Other values: reserved

muxctrl_reg11

muxctrl_reg11 is a multiplexing control register for the EPHY_RST pin.

Offset Address	Register Name	Total Reset Value
0x02C	muxctrl_reg11	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg11	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg11	Multiplexing control for the EPHY_RST pin 000: GPIO2_7 001: EPHY_RST 010: BOOT_SEL 011: VO_HS 100: SDIO1_CARD_POWER_EN Other values: reserved

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muxctrl_reg12

muxctrl_reg12 is a multiplexing control register for the SPI1_CSN1 pin.

Offset Address	Register Name	Total Reset Value
0x030	muxctrl_reg12	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg12
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg12	Multiplexing control for the SPI1_CSN1 pin 00: GPIO0_3 01: SPI1_CSN1 11: VO_DATA0 Other values: reserved

muxctrl_reg13

muxctrl_reg13 is a multiplexing control register for the EPHY_CLK pin.

Offset Address	Register Name	Total Reset Value
0x034	muxctrl_reg13	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg13
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg13	Multiplexing control for the EPHY_CLK pin 000: GPIO3_0 001: EPHY_CLK 011: VO_DATA1 100: SDIO1_CDATA2 Other values: reserved

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muxctrl_reg14

muxctrl_reg14 is a multiplexing control register for the MDCK pin.

Offset Address	Register Name	Total Reset Value
0x038	muxctrl_reg14	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg14
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg14	Multiplexing control for the MDCK pin 00: GPIO3_1 01: MDCK 10: BOOTROM_SEL 11: VO_DATA6

muxctrl_reg15

muxctrl_reg15 is a multiplexing control register for the MDIO pin.

Offset Address	Register Name	Total Reset Value
0x03C	muxctrl_reg15	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg15
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg15	Multiplexing control for the MDIO pin 00: GPIO3_2 01: MDIO 11: VO_DATA4

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			Other values: reserved

muxctrl_reg16

muxctrl_reg16 is a multiplexing control register for the SPI0_SCLK pin.

Offset Address	Register Name	Total Reset Value
0x040	muxctrl_reg16	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg16	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg16	Multiplexing control for the SPI0_SCLK pin 00: GPIO3_3 01: SPI0_SCLK 10: I2C0_SCL Other values: reserved

muxctrl_reg17

muxctrl_reg17 is a multiplexing control register for the SPI0_SDO pin.

Offset Address	Register Name	Total Reset Value
0x044	muxctrl_reg17	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg17	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg17	Multiplexing control for the SPI0_SDO pin 00: GPIO3_4 01: SPI0_SDO

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			10: I2C0_SDA Other values: reserved

muxctrl_reg18

muxctrl_reg18 is a multiplexing control register for the SPI0_SDI pin.

Offset Address	Register Name	Total Reset Value
0x048	muxctrl_reg18	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg18
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg18	Multiplexing control for the SPI0_SDI pin 0: GPIO3_5 1: SPI0_SDI

muxctrl_reg19

muxctrl_reg19 is a multiplexing control register for the SPI0_CSN pin.

Offset Address	Register Name	Total Reset Value
0x04C	muxctrl_reg19	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg19
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg19	Multiplexing control for the SPI0_CSN pin 0: GPIO3_6 1: SPI0_CSN

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muxctrl_reg20

muxctrl_reg20 is a multiplexing control register for the SPI1_SCLK pin.

Offset Address	Register Name	Total Reset Value
0x050	muxctrl_reg20	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg20	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg20	Multiplexing control for the SPI1_SCLK pin 00: GPIO3_7 01: SPI1_SCLK 10: I2C1_SCL Other values: reserved

muxctrl_reg21

muxctrl_reg21 is a multiplexing control register for the SPI1_SDO pin.

Offset Address	Register Name	Total Reset Value
0x054	muxctrl_reg21	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg21	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg21	Multiplexing control for the SPI1_SDO pin 00: GPIO4_0 01: SPI1_SDO 10: I2C1_SDA

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			Other values: reserved

muxctrl_reg22

muxctrl_reg22 is a multiplexing control register for the SPI1_SDI pin.

Offset Address	Register Name	Total Reset Value
0x058	muxctrl_reg22	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg22
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg22	Multiplexing control for the SPI1_SDI pin 0: GPIO4_1 1: SPI1_SDI

muxctrl_reg23

muxctrl_reg23 is a multiplexing control register for the SPI1_CSN0 pin.

Offset Address	Register Name	Total Reset Value
0x05C	muxctrl_reg23	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg23
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg23	Multiplexing control for the SPI1_CSN0 pin 0: GPIO4_2 1: SPI1_CSN0

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muxctrl_reg24

muxctrl_reg24 is a multiplexing control register for the I2C2_SDA pin.

Offset Address	Register Name	Total Reset Value
0x060	muxctrl_reg24	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg24
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg24	Multiplexing control for the I2C2_SDA pin 0: GPIO4_3 1: I2C2_SDA

muxctrl_reg25

muxctrl_reg25 is a multiplexing control register for the I2C2_SCL pin.

Offset Address	Register Name	Total Reset Value
0x064	muxctrl_reg25	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg25
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg25	Multiplexing control for the I2C2_SCL pin 0: GPIO4_4 1: I2C2_SCL

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muxctrl_reg26

muxctrl_reg26 is a multiplexing control register for the USB_OVRCUR pin.

Offset Address	Register Name	Total Reset Value
0x068	muxctrl_reg26	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg26
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg26	Multiplexing control for the USB_OVRCUR pin 0: GPIO4_5 1: USB_OVRCUR

muxctrl_reg27

muxctrl_reg27 is a multiplexing control register for the USB_PWREN pin.

Offset Address	Register Name	Total Reset Value
0x06C	muxctrl_reg27	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg27
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg27	Multiplexing control for the USB_PWREN pin 0: GPIO4_6 1: USB_PWREN

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muxctrl_reg28

muxctrl_reg28 is a multiplexing control register for the GPIO0_0 pin.

Offset Address	Register Name	Total Reset Value
0x070	muxctrl_reg28	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg28
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg28	Multiplexing control for the GPIO0_0 pin 00: GPIO0_0 01: IR_IN 10: TEMPER_DQ Other values: reserved

muxctrl_reg29

muxctrl_reg29 is a multiplexing control register for the GPIO0_1 pin.

Offset Address	Register Name	Total Reset Value
0x074	muxctrl_reg29	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg29
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg29	Multiplexing control for the GPIO0_1 pin 00: GPIO0_1 10: TEMPER_DQ Other values: reserved

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muxctrl_reg30

muxctrl_reg30 is a multiplexing control register for the GPIO0_2 pin.

Offset Address	Register Name	Total Reset Value
0x078	muxctrl_reg30	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg30
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg30	Multiplexing control for the GPIO0_2 pin 00: GPIO0_2 10: TEMPER_DQ Other values: reserved

muxctrl_reg31

muxctrl_reg31 is a multiplexing control register for the GPIO1_0 pin.

Offset Address	Register Name	Total Reset Value
0x07C	muxctrl_reg31	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg31
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg31	Multiplexing control for the GPIO1_0 pin 000: GPIO1_0 001: VI_DATA13 011: I2S_BCLK_TX 100: PWM0 Other values: reserved

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muxctrl_reg32

muxctrl_reg32 is a multiplexing control register for the GPIO1_1 pin.

Offset Address	Register Name	Total Reset Value
0x080	muxctrl_reg32	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg32	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg32	Multiplexing control for the GPIO1_1 pin 000: GPIO1_1 001: VI_DATA10 011: I2S_SD_TX 100: UART1_TXD Other values: reserved

muxctrl_reg33

muxctrl_reg33 is a multiplexing control register for the GPIO1_2 pin.

Offset Address	Register Name	Total Reset Value
0x084	muxctrl_reg33	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg33	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg33	Multiplexing control for the GPIO1_2 pin 000: GPIO1_2 001: VI_DATA12 011: I2S_MCLK 100: UART1_CTSN

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			Other values: reserved

muxctrl_reg34

muxctrl_reg34 is a multiplexing control register for the GPIO1_3 pin.

Offset Address	Register Name	Total Reset Value
0x088	muxctrl_reg34	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg34	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg34	Multiplexing control for the GPIO1_3 pin 000: GPIO1_3 001: VI_DATA11 011: I2S_WS_TX 100: UART2_RXD Other values: reserved

muxctrl_reg35

muxctrl_reg35 is a multiplexing control register for the GPIO1_4 pin.

Offset Address	Register Name	Total Reset Value
0x08C	muxctrl_reg35	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg35	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg35	Multiplexing control for the GPIO1_4 pin 000: GPIO1_4

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			001: VI_DATA15 010: VI_VS_SEN 011: I2S_WS_RX 100: UART1_RXD Other values: reserved

muxctrl_reg36

muxctrl_reg36 is a multiplexing control register for the GPIO1_5 pin.

Offset Address	Register Name	Total Reset Value
0x090	muxctrl_reg36	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg36	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg36	Multiplexing control for the GPIO1_5 pin 000: GPIO1_5 001: VI_DATA14 010: VI_HS_SEN 011: I2S_BCLK_RX 100: UART1_RTSN Other values: reserved

muxctrl_reg37

muxctrl_reg37 is a multiplexing control register for the GPIO1_6 pin.

Offset Address	Register Name	Total Reset Value
0x094	muxctrl_reg37	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg37	

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg37	Multiplexing control for the GPIO1_6 pin 000: GPIO1_6 001: VI_DATA9 011: I2S_SD_RX 100: UART2_TXD Other values: reserved

muxctrl_reg38

muxctrl_reg38 is a multiplexing control register for the SDIO0_CARD_POWER_EN pin.

Offset Address	Register Name	Total Reset Value
0x098	muxctrl_reg38	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg38
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg38	Multiplexing control for the SDIO0_CARD_POWER_EN pin 0: GPIO1_7 1: SDIO0_CARD_POWER_EN

muxctrl_reg39

muxctrl_reg39 is a multiplexing control register for the SDIO0_CARD_DETECT pin.

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Offset Address	Register Name	Total Reset Value
0x09C	muxctrl_reg39	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg39
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg39	Multiplexing control for the SDIO0_CARD_DETECT pin 0: GPIO4_7 1: SDIO0_CARD_DETECT

muxctrl_reg40

muxctrl_reg40 is a multiplexing control register for the SDIO0_CWPR pin.

Offset Address	Register Name	Total Reset Value
0x0A0	muxctrl_reg40	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg40
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg40	Multiplexing control for the SDIO0_CWPR pin 0: GPIO5_0 1: SDIO0_CWPR

muxctrl_reg41

muxctrl_reg41 is a multiplexing control register for the SDIO0_CCLK_OUT pin.

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Offset Address	Register Name	Total Reset Value
0x0A4	muxctrl_reg41	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg41
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg41	Multiplexing control for the SDIO0_CCLK_OUT pin 0: GPIO5_1 1: SDIO0_CCLK_OUT

muxctrl_reg42

muxctrl_reg42 is a multiplexing control register for the SDIO0_CCMD pin.

Offset Address	Register Name	Total Reset Value
0x0A8	muxctrl_reg42	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg42
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg42	Multiplexing control for the SDIO0_CCMD pin 0: GPIO5_2 1: SDIO0_CCMD

muxctrl_reg43

muxctrl_reg43 is a multiplexing control register for the SDIO0_CDATA0 pin.

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Offset Address	Register Name	Total Reset Value
0x0AC	muxctrl_reg43	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg43
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg43	Multiplexing control for the SDIO0_CDATA0 pin 0: GPIO5_3 1: SDIO0_CDATA0

muxctrl_reg44

muxctrl_reg44 is a multiplexing control register for the SDIO0_CDATA1 pin.

Offset Address	Register Name	Total Reset Value
0x0B0	muxctrl_reg44	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg44
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg44	Multiplexing control for the SDIO0_CDATA1 pin 00: TEST_CLK 01: SDIO0_CDATA1 10: GPIO5_4 Other values: reserved

muxctrl_reg45

muxctrl_reg45 is a multiplexing control register for the SDIO0_CDATA2 pin.

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Offset Address	Register Name	Total Reset Value
0x0B4	muxctrl_reg45	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg45
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg45	Multiplexing control for the SDIO0_CDATA2 pin 0: GPIO5_5 1: SDIO0_CDATA2

muxctrl_reg46

muxctrl_reg46 is a multiplexing control register for the SDIO0_CDATA3 pin.

Offset Address	Register Name	Total Reset Value
0x0B8	muxctrl_reg46	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg46
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg46	Multiplexing control for the SDIO0_CDATA3 pin 0: GPIO5_6 1: SDIO0_CDATA3

muxctrl_reg47

muxctrl_reg47 is a multiplexing control register for the UART1_RTSN pin.

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Offset Address	Register Name	Total Reset Value
0x0BC	muxctrl_reg47	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg47
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg47	Multiplexing control for the UART1_RTSN pin 00: GPIO5_7 01: EMMC_DATA6 10: I2S_SD_TX 11: UART1_RTSN

muxctrl_reg48

muxctrl_reg48 is a multiplexing control register for the UART1_RXD pin.

Offset Address	Register Name	Total Reset Value
0x0C0	muxctrl_reg48	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg48
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg48	Multiplexing control for the UART1_RXD pin 00: GPIO6_0 01: EMMC_DATA5 10: I2S_WS_TX 11: UART1_RXD

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muxctrl_reg49

muxctrl_reg49 is a multiplexing control register for the UART1_CTSN pin.

Offset Address	Register Name	Total Reset Value
0x0C4	muxctrl_reg49	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg49
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg49	Multiplexing control for the UART1_CTSN pin 00: GPIO6_1 01: EMMC_DATA7 10: I2S_MCLK 11: UART1_CTSN

muxctrl_reg50

muxctrl_reg50 is a multiplexing control register for the UART1_TXD pin.

Offset Address	Register Name	Total Reset Value
0x0C8	muxctrl_reg50	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg50
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg50	Multiplexing control for the UART1_TXD pin 00: GPIO6_2 01: EMMC_DS 10: I2S_SD_RX 11: UART1_TXD

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muxctrl_reg51

muxctrl_reg51 is a multiplexing control register for the UART2_RXD pin.

Offset Address	Register Name	Total Reset Value
0x0CC	muxctrl_reg51	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg51	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg51	Multiplexing control for the UART2_RXD pin 00: GPIO6_3 01: EMMC_DATA1 11: UART2_RXD Other values: reserved

muxctrl_reg52

muxctrl_reg52 is a multiplexing control register for the UART2_TXD pin.

Offset Address	Register Name	Total Reset Value
0x0D0	muxctrl_reg52	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg52	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg52	Multiplexing control for the UART2_TXD pin 00: GPIO6_4 01: EMMC_DATA2 10: I2S_BCLK_TX 11: UART2_TXD

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muxctrl_reg53

muxctrl_reg53 is a multiplexing control register for the JTAG_TRSTN pin.

Offset Address	Register Name	Total Reset Value
0x0D4	muxctrl_reg53	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg53
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg53	Multiplexing control for the JTAG_TRSTN pin 00: GPIO6_5 01: JTAG_TRSTN 10: SPI1_CSN1 11: I2S_MCLK

muxctrl_reg54

muxctrl_reg54 is a multiplexing control register for the JTAG_TCK pin.

Offset Address	Register Name	Total Reset Value
0x0D8	muxctrl_reg54	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg54
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg54	Multiplexing control for the JTAG_TCK pin 000: GPIO6_6 001: JTAG_TCK 010: SPI1_SCLK 011: I2S_WS_TX 100: I2C1_SCL

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			Other values: reserved

muxctrl_reg55

muxctrl_reg55 is a multiplexing control register for the JTAG_TMS pin.

Offset Address	Register Name	Total Reset Value
0x0DC	muxctrl_reg55	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg55	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg55	Multiplexing control for the JTAG_TMS pin 00: GPIO6_7 01: JTAG_TMS 10: SPI1_CSN0 11: I2S_SD_TX

muxctrl_reg56

muxctrl_reg56 is a multiplexing control register for the JTAG_TDO pin.

Offset Address	Register Name	Total Reset Value
0x0E0	muxctrl_reg56	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg56	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg56	Multiplexing control for the JTAG_TDO pin 000: GPIO7_0 001: JTAG_TDO

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			010: SPI1_SDO 011: I2S_SD_RX 100: I2C1_SDA Other values: reserved

muxctrl_reg57

muxctrl_reg57 is a multiplexing control register for the JTAG_TDI pin.

Offset Address	Register Name	Total Reset Value
0x0E4	muxctrl_reg57	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg57	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg57	Multiplexing control for the JTAG_TDI pin 00: GPIO7_1 01: JTAG_TDI 10: SPI1_SDI 11: I2S_BCLK_TX

muxctrl_reg58

muxctrl_reg58 is a multiplexing control register for the PWM0 pin.

Offset Address	Register Name	Total Reset Value
0x0E8	muxctrl_reg58	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg58	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved

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[1:0]	RW	muxctrl_reg58	Multiplexing control for the PWM0 pin 00: PMC_PWM 01: GPIO7_2 10: PWM0 Other values: reserved

muxctrl_reg59

muxctrl_reg59 is a multiplexing control register for the PWM1 pin.

Offset Address	Register Name	Total Reset Value
0x0EC	muxctrl_reg59	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg59
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg59	Multiplexing control for the PWM1 pin 0: PWM1 1: GPIO7_3

muxctrl_reg60

muxctrl_reg60 is a multiplexing control register for the PWM2 pin.

Offset Address	Register Name	Total Reset Value
0x0F0	muxctrl_reg60	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg60
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved

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[1:0]	RW	muxctrl_reg60	Multiplexing control for the PWM2 pin 00: PWM2 01: GPIO7_4 10: SPI1_CSN1 Other values: reserved

muxctrl_reg61

muxctrl_reg61 is a multiplexing control register for the PWM3 pin.

Offset Address	Register Name	Total Reset Value
0x0F4	muxctrl_reg61	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg61	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg61	Multiplexing control for the PWM3 pin 0: PWM3 1: GPIO7_5

muxctrl_reg62

muxctrl_reg62 is a multiplexing control register for the SAR_ADC_CH0 pin.

Offset Address	Register Name	Total Reset Value
0x0F8	muxctrl_reg62	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg62	

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg62	Multiplexing control for the SAR_ADC_CH0 pin 0: SAR_ADC_CH0 1: GPIO7_6

muxctrl_reg63

muxctrl_reg63 is a multiplexing control register for the SAR_ADC_CH1 pin.

Offset Address	Register Name	Total Reset Value
0x0FC	muxctrl_reg63	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg63
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg63	Multiplexing control for the SAR_ADC_CH1 pin 0: SAR_ADC_CH1 1: GPIO7_7

muxctrl_reg64

muxctrl_reg64 is a multiplexing control register for the SAR_ADC_CH2 pin.

Offset Address	Register Name	Total Reset Value
0x100	muxctrl_reg64	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg64

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg64	Multiplexing control for the SAR_ADC_CH2 pin 0: SAR_ADC_CH2 1: GPIO8_0

muxctrl_reg65

muxctrl_reg65 is a multiplexing control register for the SAR_ADC_CH3 pin.

Offset Address	Register Name	Total Reset Value
0x104	muxctrl_reg65	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg65
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg65	Multiplexing control for the SAR_ADC_CH3 pin 0: SAR_ADC_CH3 1: GPIO8_1

2.4 Software Multiplexed Pins

Sensor

Table 2-28 lists software multiplexed pins of the sensor.

Table 2-28 Software multiplexed pins of the sensor

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signals 2–4
P1	SENSOR_CLK	muxctrl_reg0	GPIO0_4	SENSOR_CLK	2–4: N/A

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Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signals 2–4
M3	SENSOR_RSTN	muxctrl_reg1	SENSOR_RSTN	GPIO0_5	2–4: N/A
N3	FLASH_TRIG	muxctrl_reg2	GPIO0_6	FLASH_TRIG	2: SFC_EMMC_BOOT_MODE 3: SPI1_CSN1 4: VI_VS_BT1120
P2	SHUTTER_TRIG	muxctrl_reg3	GPIO0_7	SHUTTER_TRIG	2: SFC_DEVICE_MODE 3: - 4: VI_HS_BT1120

Table 2-29 describes software multiplexed signals of the sensor.

Table 2-29 Description of software multiplexed signals of the sensor

Signal	Direction	Description
FLASH_TRIG	O	Camera flash control
GPIO0_4	I/O	GPIO
GPIO0_5	I/O	GPIO
GPIO0_6	I/O	GPIO
GPIO0_7	I/O	GPIO
SENSOR_CLK	O	Sensor working clock
SENSOR_RSTN	O	Sensor reset signal
SFC_DEVICE_MODE	I	SPI flash from which the system boots (that is, type of the flash memory connected to SFC_CSN0) 0: SPI NOR flash 1: SPI NAND flash
SFC_EMMC_BOOT_MODE	I	Boot address mode of the SPI NOR flash when BOOT_SEL and SFC_DEVICE_MODE are 0 0: 3-byte address mode 1: 4-byte address mode Boot mode of the SPI NAND flash when BOOT_SEL is 0 and SFC_DEVICE_MODE is 1 0: 1-wire boot mode

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Signal	Direction	Description
		1: 4-wire boot mode eMMC boot mode when BOOT_SEL is 1 0: 4-wire boot mode 1: 8-wire boot mode
SHUTTER_TRIG	O	Shutter control
SPI1_CSN1	O	SPI1 CS1
VI_HS_SEN	I	VI HS signal
VI_VS_SEN	I	VI VS signal

RMII

Table 2-30 lists software multiplexed pins of the RMII.

Table 2-30 Software multiplexed pins of the RMII

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
A14	RMII_CLK	muxctrl_reg4	GPIO2_0	RMII_CLK	VO_CLK	SDIO1_CCLK_OUT
C14	RMII_TX_EN	muxctrl_reg5	GPIO2_1	RMII_TX_EN	VO_VS	SDIO1_CARD_DETECT
C13	RMII_TXD0	muxctrl_reg6	GPIO2_2	RMII_TXD0	VO_DATA5	SDIO1_CWPR
B15	RMII_TXD1	muxctrl_reg7	GPIO2_3	RMII_TXD1	VO_DE	SDIO1_CDATA1
B14	RMII_RX_DV	muxctrl_reg8	GPIO2_4	RMII_RX_DV	VO_DATA7	SDIO1_CDATA0
A13	RMII_RXD0	muxctrl_reg9	GPIO2_5	RMII_RXD0	VO_DATA2	SDIO1_CDATA3
B13	RMII_RXD1	muxctrl_reg10	GPIO2_6	RMII_RXD1	VO_DATA3	SDIO1_CCMD

Table 2-31 describes software multiplexed signals of the RMII.

Table 2-31 Description of software multiplexed signals of the RMII

Signal	Direction	Description
GPIO2_0	I/O	GPIO

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Signal	Direction	Description
GPIO2_1	I/O	GPIO
GPIO2_2	I/O	GPIO
GPIO2_3	I/O	GPIO
GPIO2_4	I/O	GPIO
GPIO2_5	I/O	GPIO
GPIO2_6	I/O	GPIO
RMII_CLK	I/O	RMII clock
RMII_RX_DV	I	RX data validity and carrier sense signal
RMII_RXD0	I	RX data 0
RMII_RXD1	I	RX data 1
RMII_TX_EN	O	TX enable
RMII_TXD0	O	TX data 0
RMII_TXD1	O	TX data 1
SDIO1_CARD_DETECT	I	SDIO1 card detection signal, active low
SDIO1_CCLK_OUT	O	SDIO1 output working clock for the card
SDIO1_CCMD	I/O	SDIO1 command
SDIO1_CDATA0	I/O	SDIO1 data 0
SDIO1_CDATA1	I/O	SDIO1 data 1
SDIO1_CDATA3	I/O	SDIO1 data 3
SDIO1_CWPR	I	SDIO1 card write protection detection
VO_CLK	O	VO clock signal
VO_DATA2	O	VO data 2
VO_DATA3	O	VO data 3
VO_DATA5	O	VO data 5
VO_DATA7	O	VO data 7
VO_DE	O	VO data validity signal
VO_VS	O	VO VS signal

EPHY

Table 2-32 lists software multiplexed pins of the EPHY.

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Table 2-32 Software multiplexed pins of the EPHY

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
C15	EPHY_RST	muxctrl_reg11	GPIO2_7	EPHY_RST	BOOT_SEL	VO_HS	SDIO1_CARD_POWER_EN
A12	EPHY_CLK	muxctrl_reg13	GPIO3_0	EPHY_CLK	VO_DATA1	SDIO1_CDATA2	-

Table 2-33 describes software multiplexed signals of the EPHY.

Table 2-33 Description of software multiplexed signals of the EPHY

Signal	Direction	Description
BOOT_SEL	I	Boot enable 0: booting from the SPI flash 1: booting from the eMMC
EPHY_CLK	O	EPHY working clock
EPHY_RST	O	EPHY reset
GPIO2_7	I/O	GPIO
GPIO3_0	I/O	GPIO
SDIO1_CARD_POWER_EN	O	SDIO1 power enable signal. The value 1 indicates power-on.
SDIO1_CDATA2	I/O	SDIO1 data 2
VO_DATA1	O	VO data 1
VO_HS	O	VO HS signal

SPI1

Table 2-34 lists software multiplexed pins of SPI1.

Table 2-34 Software multiplexed pins of SPI1

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
B11	SPI1_CSN1	muxctrl_reg12	GPIO0_3	SPI1_CSN1	VO_DATA0
A10	SPI1_SCLK	muxctrl_reg20	GPIO3_7	SPI1_SCLK	I2C1_SCL

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Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
C10	SPI1_SDO	muxctrl_reg21	GPIO4_0	SPI1_SDO	I2C1_SDA
A11	SPI1_SDI	muxctrl_reg22	GPIO4_1	SPI1_SDI	N/A
C11	SPI1_CSN0	muxctrl_reg23	GPIO4_2	SPI1_CSN0	N/A

Table 2-35 describes software multiplexed signals of SPI1.

Table 2-35 Description of software multiplexed signals of SPI1

Signal	Direction	Description
GPIO0_3	I/O	GPIO
GPIO3_7	I/O	GPIO
GPIO4_0	I/O	GPIO
GPIO4_1	I/O	GPIO
GPIO4_2	I/O	GPIO
I2C1_SCL	I/O	I^2C1 bus clock
I2C1_SDA	I/O	I^2C1 bus data/address
SPI1_CSN0	O	SPI1 CS0
SPI1_CSN1	O	SPI1 CS1
SPI1_SCLK	O	SPI1 clock signal
SPI1_SDI	I	SPI1 input data
SPI1_SDO	O	SPI1 output data
VO_DATA0	O	VO data 0

MDIO

Table 2-36 lists software multiplexed pins of the MDIO.

Table 2-36 Software multiplexed pins of the MDIO

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
C12	MDCK	muxctrl_reg14	GPIO3_1	MDCK	BOOTROM_SEL	VO_DATA6
D12	MDIO	muxctrl_reg15	GPIO3_2	MDIO	VO_DATA4	N/A

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Table 2-37 describes software multiplexed signals of the MDIO.

Table 2-37 Description of software multiplexed signals of the MDIO

Signal	Direction	Description
BOOTROM_SEL	I	BOOTROM enable 0: booting from other space 1: booting from BOOTROM
GPIO3_1	I/O	GPIO
GPIO3_2	I/O	GPIO
MDCK	O	MDIO interface clock
MDIO	I/O	Data signal of the MDIO interface
VO_DATA4	O	VO data 4
VO_DATA6	O	VO data 6

SPI0

Table 2-38 lists software multiplexed pins of SPI0.

Table 2-38 Software multiplexed pins of SPI0

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
M1	SPI0_SCLK	muxctrl_reg16	GPIO3_3	SPI0_SCLK	I2C0_SCL
M2	SPI0_SDO	muxctrl_reg17	GPIO3_4	SPI0_SDO	I2C0_SDA
N2	SPI0_SDI	muxctrl_reg18	GPIO3_5	SPI0_SDI	N/A
N1	SPI0_CSN	muxctrl_reg19	GPIO3_6	SPI0_CSN	N/A

Table 2-39 describes software multiplexed signals of SPI0.

Table 2-39 Description of software multiplexed signals of SPI0

Signal	Direction	Description
GPIO3_3	I/O	GPIO
GPIO3_4	I/O	GPIO

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Signal	Direction	Description
GPIO3_5	I/O	GPIO
GPIO3_6	I/O	GPIO
I2C0_SCL	I/O	I^2C0 bus clock
I2C0_SDA	I/O	I^2C0 bus data/address
SPI0_CSN	O	SPI0 CS signal
SPI0_SCLK	O	SPI0 clock signal
SPI0_SDI	I	SPI0 input data
SPI0_SDO	O	SPI0 output data

I^2C2

Table 2-40 lists software multiplexed pins of I^2C2.

Table 2-40 Software multiplexed pins of I^2C2

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1
G2	I2C2_SDA	muxctrl_reg24	GPIO4_3	I2C2_SDA
H1	I2C2_SCL	muxctrl_reg25	GPIO4_4	I2C2_SCL

Table 2-41 describes software multiplexed signals of I^2C2.

Table 2-41 Description of software multiplexed signals of I^2C2

Signal	Direction	Description
GPIO4_3	I/O	GPIO
GPIO4_4	I/O	GPIO
I2C2_SCL	I/O	I^2C2 bus clock
I2C2_SDA	I/O	I^2C2 bus data/address

USB

Table 2-42 lists software multiplexed pins of the USB.

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Table 2-42 Software multiplexed pins of the USB

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1
P10	USB_OVRCUR	muxctrl_reg26	GPIO4_5	USB_OVRCUR
R10	USB_PWREN	muxctrl_reg27	GPIO4_6	USB_PWREN

Table 2-43 describes software multiplexed signals of the USB.

Table 2-43 Description of software multiplexed signals of the USB

Signal	Direction	Description
GPIO4_5	I/O	GPIO
GPIO4_6	I/O	GPIO
USB_OVRCUR	I	USB port overcurrent indicator. The level polarity can be specified by configuring the peripheral control register MISC_CTRL30 bit[17].
USB_PWREN	O	Power control output pin of the USB port. The level polarity can be specified by configuring the peripheral control register MISC_CTRL30 bit[16].

GPIO

Table 2-44 lists software multiplexed pins of the GPIO.

Table 2-44 Software multiplexed pins of the GPIO

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
J2	GPIO0_0	muxctrl_reg28	GPIO0_0	IR_IN	TEMPER_DQ	N/A	N/A
J1	GPIO0_1	muxctrl_reg29	GPIO0_1	TEMPER_DQ	N/A	N/A	N/A
H2	GPIO0_2	muxctrl_reg30	GPIO0_2	TEMPER_DQ	N/A	N/A	N/A
N6	GPIO1_0	muxctrl_reg31	GPIO1_0	VI_DATA13	I2S_BCLK_TX	PWM0	N/A
R9	GPIO1_1	muxctrl_reg32	GPIO1_1	VI_DATA10	I2S_SD_TX	UART1_TXD	N/A
N7	GPIO1	muxctrl_reg33	GPIO1_2	VI_DATA12	I2S_MCLK	UART1_CT	N/A

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Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
	_2					SN	
P9	GPIO1_3	muxctrl_reg34	GPIO1_3	VI_DATA11	I2S_WS_TX	UART2_RXD	N/A
N9	GPIO1_4	muxctrl_reg35	GPIO1_4	VI_DATA15	VI_VS_SEN	I2S_WS_RX	UART1_RXD
N8	GPIO1_5	muxctrl_reg36	GPIO1_5	VI_DATA14	VI_HS_SEN	I2S_BCLK_RX	UART1_RTSN
P8	GPIO1_6	muxctrl_reg37	GPIO1_6	VI_DATA9	I2S_SD_RX	UART2_TXD	N/A

Table 2-45 describes software multiplexed signals of the GPIO.

Table 2-45 Description of software multiplexed signals of the GPIO

Signal	Direction	Description
GPIO0_0	I/O	GPIO
GPIO0_1	I/O	GPIO
GPIO0_2	I/O	GPIO
GPIO1_0	I/O	GPIO
GPIO1_1	I/O	GPIO
GPIO1_2	I/O	GPIO
GPIO1_3	I/O	GPIO
GPIO1_4	I/O	GPIO
GPIO1_5	I/O	GPIO
GPIO1_6	I/O	GPIO
I2S_BCLK_RX	I/O	RX clock of the I^2S or PCM interface
I2S_BCLK_TX	I/O	TX clock of the I^2S or PCM interface
I2S_MCLK	O	Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC.
I2S_SD_RX	I	RX data of the I^2S or PCM interface
I2S_SD_TX	O	TX data of the I^2S or PCM interface
I2S_WS_RX	I/O	I^2S RX audio channel select signal or PCM TX VS signal

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Signal	Direction	Description
I2S_WS_TX	I/O	I^2S TX audio channel select signal or PCM TX VS signal
IR_IN	I	IR input
PWM0	O	PWM output 0
TEMPER_DQ	I/O	Temperature measurement. It is used to communicate with the external temperature measurement chip.
UART1_CTSN	I	UART1 modem status input, active low
UART1_RTSN	O	UART1 modem status output, active low
UART1_RXD	I	UART1 RX data
UART1_TXD	O	UART1 TX data
UART2_RXD	I	UART2 RX data
UART2_TXD	O	UART2 TX data
VI_DATA10	I	VI data 10
VI_DATA11	I	VI data 11
VI_DATA12	I	VI data 12
VI_DATA13	I	VI data 13
VI_DATA14	I	VI data 14
VI_DATA15	I	VI data 15
VI_DATA9	I	VI data 9
VI_HS_SEN	I	VI HS signal
VI_VS_SEN	I	VI VS signal

SDIO

Table 2-46 lists software multiplexed pins of the SDIO.

Table 2-46 Software multiplexed pins of the SDIO

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
N14	SDIO0_CARD_POWER_EN	muxctrl_reg38	GPIO1_7	SDIO0_CARD_POWER_EN	N/A
N15	SDIO0_CARD_DETECT	muxctrl_reg39	GPIO4_7	SDIO0_CARD_DETECT	N/A

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Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
L13	SDIO0_CWPR	muxctrl_reg40	GPIO5_0	SDIO0_CWPR	N/A
L15	SDIO0_CCLK_OUT	muxctrl_reg41	GPIO5_1	SDIO0_CCLK_OUT	N/A
K15	SDIO0_CCMD	muxctrl_reg42	GPIO5_2	SDIO0_CCMD	N/A
M15	SDIO0_CDATA0	muxctrl_reg43	GPIO5_3	SDIO0_CDATA0	N/A
M14	SDIO0_CDATA1	muxctrl_reg44	TEST_CLK	SDIO0_CDATA1	GPIO5_4
J14	SDIO0_CDATA2	muxctrl_reg45	GPIO5_5	SDIO0_CDATA2	N/A
J15	SDIO0_CDATA3	muxctrl_reg46	GPIO5_6	SDIO0_CDATA3	N/A

Table 2-47 describes software multiplexed signals of the SDIO.

Table 2-47 Description of software multiplexed signals of the SDIO

Signal	Direction	Description
GPIO1_7	I/O	GPIO
GPIO4_7	I/O	GPIO
GPIO5_0	I/O	GPIO
GPIO5_1	I/O	GPIO
GPIO5_2	I/O	GPIO
GPIO5_3	I/O	GPIO
GPIO5_4	I/O	GPIO
GPIO5_5	I/O	GPIO
GPIO5_6	I/O	GPIO
SDIO0_CARD_DETECT	I	SDIO0 card detection signal, active low
SDIO0_CARD_POWER_EN	O	SDIO0 power enable signal. The value 1 indicates power-on.
SDIO0_CCLK_OUT	O	SDIO0 output working clock for the card
SDIO0_CCMD	I/O	SDIO0 command
SDIO0_CDATA0	I/O	SDIO0 data 0
SDIO0_CDATA1	I/O	SDIO0 data 1
SDIO0_CDATA2	I/O	SDIO0 data 2

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Signal	Direction	Description
SDIO0_CDATA3	I/O	SDIO0 data 3
SDIO0_CWPR	I	SDIO0 card write protection detection
TEST_CLK	O	Test clock

UART

Table 2-48 lists software multiplexed pins of the UART.

Table 2-48 Software multiplexed pins of the UART

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
D15	UART1_RTSN	muxctrl_reg47	GPIO5_7	EMMC_DATA6	I2S_SD_TX	UART1_RTSN
D14	UART1_RXD	muxctrl_reg48	GPIO6_0	EMMC_DATA5	I2S_WS_TX	UART1_RXD
D13	UART1_CTSN	muxctrl_reg49	GPIO6_1	EMMC_DATA7	I2S_MCLK	UART1_CTSN
E14	UART1_TXD	muxctrl_reg50	GPIO6_2	EMMC_DS	I2S_SD_RX	UART1_TXD
F13	UART2_RXD	muxctrl_reg51	GPIO6_3	EMMC_DATA1	UART2_RXD	-
E13	UART2_TXD	muxctrl_reg52	GPIO6_4	EMMC_DATA2	I2S_BCLK_TX	UART2_TXD

Table 2-49 describes software multiplexed signals of the UART.

Table 2-49 Description of software multiplexed signals of the UART

Signal	Direction	Description
EMMC_DATA1	I/O	eMMC data 1
EMMC_DATA2	I/O	eMMC data 2
EMMC_DATA5	I/O	eMMC data 5
EMMC_DATA6	I/O	eMMC data 6
EMMC_DATA7	I/O	eMMC data 7
EMMC_DS	I	eMMC DATA_STROBE

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Signal	Direction	Description
GPIO5_7	I/O	GPIO
GPIO6_0	I/O	GPIO
GPIO6_1	I/O	GPIO
GPIO6_2	I/O	GPIO
GPIO6_3	I/O	GPIO
GPIO6_4	I/O	GPIO
I2S_BCLK_TX	I/O	TX clock of the I^2S or PCM interface
I2S_MCLK	O	Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC.
I2S_SD_RX	I	RX data of the I^2S or PCM interface
I2S_SD_TX	O	TX data of the I^2S or PCM interface
I2S_WS_TX	I/O	I^2S TX audio channel select signal or PCM TX VS signal
UART1_CTSN	I	UART1 modem status input, active low
UART1_RTSN	O	UART1 modem status output, active low
UART1_RXD	I	UART1 RX data
UART1_TXD	O	UART1 TX data
UART2_RXD	I	UART2 RX data
UART2_TXD	O	UART2 TX data

JTAG

Table 2-50 lists software multiplexed pins of the JTAG.

Table 2-50 Software multiplexed pins of the JTAG

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
B4	JTAG_TRSTN	muxctrl_reg53	GPIO6_5	JTAG_TRSTN	SPI1_CSN1	I2S_MCLK	N/A
A4	JTAG_TCK	muxctrl_reg54	GPIO6_6	JTAG_TCK	SPI1_SCLK	I2S_WS_TX	I2C1_SCL
B5	JTAG_TMS	muxctrl_reg55	GPIO6_7	JTAG_TMS	SPI1_CSN0	I2S_SD_TX	N/A

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Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
C4	JTAG_TDO	muxctrl_reg56	GPIO7_0	JTAG_TDO	SPI1_SDO	I2S_SD_RX	I2C1_SDA
A3	JTAG_TDI	muxctrl_reg57	GPIO7_1	JTAG_TDI	SPI1_SDI	I2S_BCLK_TX	N/A

Table 2-51 describes software multiplexed signals of the JTAG.

Table 2-51 Description of software multiplexed signals of the JTAG

Signal	Direction	Description
GPIO6_5	I/O	GPIO
GPIO6_6	I/O	GPIO
GPIO6_7	I/O	GPIO
GPIO7_0	I/O	GPIO
GPIO7_1	I/O	GPIO
I2C1_SCL	I/O	I^2C1 bus clock
I2C1_SDA	I/O	I^2C1 bus data/address
I2S_BCLK_TX	I/O	TX clock of the I^2S or PCM interface
I2S_MCLK	O	Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC.
I2S_SD_RX	I	RX data of the I^2S or PCM interface
I2S_SD_TX	O	TX data of the I^2S or PCM interface
I2S_WS_TX	I/O	I^2S TX audio channel select signal or PCM TX VS signal
JTAG_TCK	I	JTAG clock signal
JTAG_TDI	I	JTAG input data
JTAG_TDO	O	JTAG output data
JTAG_TMS	I	JTAG mode select input
JTAG_TRSTN	I	JTAG reset signal
SPI1_CSN0	O	SPI1 CS0
SPI1_CSN1	O	SPI1 CS1
SPI1_SCLK	O	SPI1 clock signal
SPI1_SDI	I	SPI1 input data

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Signal	Direction	Description
SPI1_SDO	O	SPI1 output data

PWM

Table 2-52 lists software multiplexed pins of the PWM.

Table 2-52 Software multiplexed pins of the PWM

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
G1	PWM0	muxctrl_reg58	PMC_PWM	GPIO7_2	PWM0
E2	PWM1	muxctrl_reg59	PWM1	GPIO7_3	N/A
F2	PWM2	muxctrl_reg60	PWM2	GPIO7_4	SPI1_CSN1
F1	PWM3	muxctrl_reg61	PWM3	GPIO7_5	N/A

Table 2-53 describes software multiplexed signals of the PWM.

Table 2-53 Description of software multiplexed signals of the PWM

Signal	Direction	Description
GPIO7_2	I/O	GPIO
GPIO7_3	I/O	GPIO
GPIO7_4	I/O	GPIO
GPIO7_5	I/O	GPIO
PMC_PWM	O	PMC PWM output
PWM0	O	PWM output 0
PWM1	O	PWM output 1
PWM2	O	PWM output 2
PWM3	O	PWM output 3
SPI1_CSN1	O	SPI1 CS1

SAR_ADC

Table 2-54 lists software multiplexed pins of the SAR_ADC.

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Table 2-54 Software multiplexed pins of the SAR_ADC

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1
K1	SAR_ADC_CH0	muxctrl_reg62	SAR_ADC_CH0	GPIO7_6
K2	SAR_ADC_CH1	muxctrl_reg63	SAR_ADC_CH1	GPIO7_7
L1	SAR_ADC_CH2	muxctrl_reg64	SAR_ADC_CH2	GPIO8_0
L3	SAR_ADC_CH3	muxctrl_reg65	SAR_ADC_CH3	GPIO8_1

Table 2-55 describes software multiplexed signals of the SAR_ADC.

Table 2-55 Description of software multiplexed signals of the SAR_ADC

Signal	Direction	Description
GPIO7_6	I/O	GPIO
GPIO7_7	I/O	GPIO
GPIO8_0	I/O	GPIO
GPIO8_1	I/O	GPIO
SAR_ADC_CH0	I	LSADC channel 0
SAR_ADC_CH1	I	LSADC channel 1
SAR_ADC_CH2	I	LSADC channel 2
SAR_ADC_CH3	I	LSADC channel 3

2.5 Hardware Multiplexed Pins

SFC Pins

Table 2-56 lists hardware multiplexed pins of the SFC.

Table 2-56 Hardware multiplexed pins of the SFC

Pin	Pad Signal	Multiplexed Signal 1 (BOOT_SEL == 1'b1)
F14	SFC_CLK	EMMC_CLK
G14	SFC_SIO_IO0	EMMC_DATA4

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Pin	Pad Signal	Multiplexed Signal 1 (BOOT_SEL == 1'b1)
H14	SFC_SOI_IO1	EMMC_DATA3
H15	SFC_WP_IO2	EMMC_CMD
F15	SFC_HOLD_IO3	EMMC_DATA0
G15	SFC_CSN0	EMMC_RST_N
H13	SFC_CSN1	EMMC_POWER_EN

Table 2-57 describes hardware multiplexed signals of the SFC.

Table 2-57 Description of hardware multiplexed signals of the SFC

Signal	Direction	Description
EMMC_CLK	O	eMMC clock
EMMC_DATA4	I/O	eMMC data 4
EMMC_DATA3	I/O	eMMC data 3
EMMC_CMD	I/O	eMMC command/response
EMMC_DATA0	I/O	eMMC data 0
EMMC_RST_N	O	eMMC reset
EMMC_POWER_EN	O	eMMC power enable signal. The value 1 indicates power-on.

2.6 Pin Drive Capability Registers

2.6.1 Register Summary

Table 2-58 describes pin drive capability registers.

Table 2-58 Summary of pin drive capability registers (base address: 0x200F_0800)

Offset Address	Register	Description	Page
0x000	PADCTRL_REG0	Control register for the SENSOR_CLK functional pin	2-98
0x004	PADCTRL_REG1	Control register for the SENSOR_RSTN functional pin	2-99

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Offset Address	Register	Description	Page
0x008	PADCTRL_REG2	Control register for the FLASH_TRIG functional pin	2-100
0x00C	PADCTRL_REG3	Control register for the SHUTTER_TRIG functional pin	2-101
0x010	PADCTRL_REG4	Control register for the RMII_CLK functional pin	2-102
0x014	PADCTRL_REG5	Control register for the RMII_TX_EN functional pin	2-103
0x018	PADCTRL_REG6	Control register for the RMII_TXD0 functional pin	2-103
0x01C	PADCTRL_REG7	Control register for the RMII_TXD1 functional pin	2-104
0x020	PADCTRL_REG8	Control register for the RMII_RX_DV functional pin	2-105
0x024	PADCTRL_REG9	Control register for the RMII_RXD0 functional pin	2-106
0x028	PADCTRL_REG10	Control register for the RMII_RXD1 functional pin	2-106
0x02C	PADCTRL_REG11	Control register for the EPHY_RST functional pin	2-107
0x030	PADCTRL_REG12	Control register for the SPI1_CSN1 functional pin	2-108
0x034	PADCTRL_REG13	Control register for the EPHY_CLK functional pin	2-109
0x038	PADCTRL_REG14	Control register for the MDCK functional pin	2-110
0x03C	PADCTRL_REG15	Control register for the MDIO functional pin	2-110
0x040	PADCTRL_REG16	Control register for the SPI0_SCLK functional pin	2-111
0x044	PADCTRL_REG17	Control register for the SPI0_SDO functional pin	2-112
0x048	PADCTRL_REG18	Control register for the SPI0_SDI functional pin	2-113
0x04C	PADCTRL_REG19	Control register for the SPI0_CSN functional pin	2-113
0x050	PADCTRL_REG20	Control register for the SPI1_SCLK functional pin	2-114

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Offset Address	Register	Description	Page
0x054	PADCTRL_REG21	Control register for the SPI1_SDO functional pin	2-115
0x058	PADCTRL_REG22	Control register for the SPI1_SDI functional pin	2-115
0x05C	PADCTRL_REG23	Control register for the SPI1_CSN0 functional pin	2-116
0x060	PADCTRL_REG24	Control register for the I2C2_SDA functional pin	2-117
0x064	PADCTRL_REG25	Control register for the I2C2_SCL functional pin	2-118
0x068	PADCTRL_REG26	Control register for the UART0_RXD functional pin	2-118
0x06C	PADCTRL_REG27	Control register for the UART0_TXD functional pin	2-119
0x078	PADCTRL_REG30	Control register for the USB_OVRCUR functional pin	2-120
0x07C	PADCTRL_REG31	Control register for the USB_PWREN functional pin	2-121
0x080	PADCTRL_REG32	Control register for the GPIO0_0 functional pin	2-121
0x084	PADCTRL_REG33	Control register for the GPIO0_1 functional pin	2-122
0x088	PADCTRL_REG34	Control register for the GPIO0_2 functional pin	2-123
0x08C	PADCTRL_REG35	Control register for the GPIO1_0 functional pin	2-124
0x090	PADCTRL_REG36	Control register for the GPIO1_1 functional pin	2-124
0x094	PADCTRL_REG37	Control register for the GPIO1_2 functional pin	2-125
0x098	PADCTRL_REG38	Control register for the GPIO1_3 functional pin	2-126
0x09C	PADCTRL_REG39	Control register for the GPIO1_4 functional pin	2-127
0x0A0	PADCTRL_REG40	Control register for the GPIO1_5 functional pin	2-127
0x0A4	PADCTRL_REG41	Control register for the GPIO1_6 functional pin	2-128

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Offset Address	Register	Description	Page
0x0A8	PADCTRL_REG42	Control register for the SDIO0_CARD_POWER_EN functional pin	2-129
0x0AC	PADCTRL_REG43	Control register for the SDIO0_CARD_DETECT functional pin	2-130
0x0B0	PADCTRL_REG44	Control register for the SDIO0_CWPR functional pin	2-130
0x0B4	PADCTRL_REG45	Control register for the SDIO0_CCLK_OUT functional pin	2-131
0x0B8	PADCTRL_REG46	Control register for the SDIO0_CCMD functional pin	2-132
0x0BC	PADCTRL_REG47	Control register for the SDIO0_CDATA0 functional pin	2-133
0x0C0	PADCTRL_REG48	Control register for the SDIO0_CDATA1 functional pin	2-134
0x0C4	PADCTRL_REG49	Control register for the SDIO0_CDATA2 functional pin	2-135
0x0C8	PADCTRL_REG50	Control register for the SDIO0_CDATA3 functional pin	2-136
0x0CC	PADCTRL_REG51	Control register for the SFC_CLK functional pin	2-137
0x0D0	PADCTRL_REG52	Control register for the SFC_SIO_IO0 functional pin	2-138
0x0D4	PADCTRL_REG53	Control register for the SFC_SOI_IO1 functional pin	2-139
0x0D8	PADCTRL_REG54	Control register for the SFC_WP_IO2 functional pin	2-140
0x0DC	PADCTRL_REG55	Control register for the SFC_HOLD_IO3 functional pin	2-141
0x0E0	PADCTRL_REG56	Control register for the SFC_CSN0 functional pin	2-142
0x0E4	PADCTRL_REG57	Control register for the SFC_CSN1 functional pin	2-143
0x0E8	PADCTRL_REG58	Control register for the UART1_RTSN functional pin	2-143
0x0EC	PADCTRL_REG59	Control register for the UART1_RXD functional pin	2-144
0x0F0	PADCTRL_REG60	Control register for the UART1_CTSN functional pin	2-145

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Offset Address	Register	Description	Page
0x0F4	PADCTRL_REG61	Control register for the UART1_TXD functional pin	2-146
0x0F8	PADCTRL_REG62	Control register for the UART2_RXD functional pin	2-147
0x0FC	PADCTRL_REG63	Control register for the UART2_TXD functional pin	2-148
0x100	PADCTRL_REG64	Control register for the JTAG_TRSTN functional pin	2-149
0x104	PADCTRL_REG65	Control register for the JTAG_TCK functional pin	2-150
0x108	PADCTRL_REG66	Control register for the JTAG_TMS functional pin	2-151
0x10C	PADCTRL_REG67	Control register for the JTAG_TDO functional pin	2-152
0x110	PADCTRL_REG68	Control register for the JTAG_TDI functional pin	2-152
0x118	PADCTRL_REG70	Control register for the PWM0 functional pin	2-153
0x11C	PADCTRL_REG71	Control register for the PWM1 functional pin	2-154
0x120	PADCTRL_REG72	Control register for the PWM2 functional pin	2-155
0x124	PADCTRL_REG73	Control register for the PWM3 functional pin	2-155
0x128	PADCTRL_REG74	Control register for the SAR_ADC_CH0 functional pin	2-156
0x12C	PADCTRL_REG75	Control register for the SAR_ADC_CH1 functional pin	2-157
0x130	PADCTRL_REG76	Control register for the SAR_ADC_CH2 functional pin	2-158
0x134	PADCTRL_REG77	Control register for the SAR_ADC_CH3 functional pin	2-159

2.6.2 Register Description

PADCTRL_REG0

PADCTRL_REG0 is a control register for the SENSOR_CLK functional pin.

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Offset Address	Register Name	Total Reset Value
0x000	PADCTRL_REG0	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg0_sr	padctrl_reg0_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg0_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6:4]	RW	padctrl_reg0_ds	Drive capability of the SENSOR_CLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG1

PADCTRL_REG1 is a control register for the SENSOR_RSTN functional pin.

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Offset Address	Register Name	Total Reset Value
0x004	PADCTRL_REG1	0x0000_0090

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg1_sr	reserved	padctrl_reg1_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	0	0 1	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg1_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg1_ds	Drive capability of the SENSOR_RSTN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG2

PADCTRL_REG2 is a control register for the FLASH_TRIG functional pin.

Offset Address	Register Name	Total Reset Value
0x008	PADCTRL_REG2	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg2_sl	reserved	padctrl_reg2_ds	reserved

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg2_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg2_ds	Drive capability of the FLASH_TRIG pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG3

PADCTRL_REG3 is a control register for the SHUTTER_TRIG functional pin.

Offset Address	Register Name	Total Reset Value
0x00C	PADCTRL_REG3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg3_sl	reserved	padctrl_reg3_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg3_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg3_ds	Drive capability of the SHUTTER_TRIG pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG4

PADCTRL_REG4 is a control register for the RMII_CLK functional pin.

Offset Address	Register Name	Total Reset Value
0x010	PADCTRL_REG4	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg4_sr	padctrl_reg4_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg4_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6:4]	RW	padctrl_reg4_ds	Drive capability of the RMII_CLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG5

PADCTRL_REG5 is a control register for the RMII_TX_EN functional pin.

Offset Address	Register Name	Total Reset Value
0x014	PADCTRL_REG5	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg5_sr	reserved	padctrl_reg5_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg5_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg5_ds	Drive capability of the RMII_TX_EN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG6

PADCTRL_REG6 is a control register for the RMII_TXD0 functional pin.

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Offset Address	Register Name	Total Reset Value
0x018	PADCTRL_REG6	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg6_sr	reserved	padctrl_reg6_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg6_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg6_ds	Drive capability of the RMII_TXD0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG7

PADCTRL_REG7 is a control register for the RMII_TXD1 functional pin.

Offset Address	Register Name	Total Reset Value
0x01C	PADCTRL_REG7	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg7_sr	reserved	padctrl_reg7_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg7_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg7_ds	Drive capability of the RMII_TXD1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG8

PADCTRL_REG8 is a control register for the RMII_RX_DV functional pin.

Offset Address	Register Name	Total Reset Value
0x020	PADCTRL_REG8	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg8_sr	reserved	padctrl_reg8_ds0		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg8_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg8_ds0	Drive capability of the RMII_RX_DV pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG9

PADCTRL_REG9 is a control register for the RMII_RXD0 functional pin.

Offset Address	Register Name	Total Reset Value
0x024	PADCTRL_REG9	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg9_sr	reserved	padctrl_reg9_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg9_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg9_ds	Drive capability of the RMII_RXD0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG10

PADCTRL_REG10 is a control register for the RMII_RXD1 functional pin.

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Offset Address	Register Name	Total Reset Value
0x028	PADCTRL_REG10	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg10_sr	reserved	padctrl_reg10_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg10_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg10_ds	Drive capability of the RMII_RXD1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG11

PADCTRL_REG11 is a control register for the EPHY_RST functional pin.

Offset Address	Register Name	Total Reset Value
0x02C	PADCTRL_REG11	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg11_sr	reserved	padctrl_reg11_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg11_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg11_ds	Drive capability of the EPHY_RST pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG12

PADCTRL_REG12 is a control register for the SPI1_CSN1 functional pin.

Offset Address	Register Name	Total Reset Value
0x030	PADCTRL_REG12	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg12_sr	reserved	padctrl_reg12_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg12_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg12_ds	Drive capability of the SPI1_CSN1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG13

PADCTRL_REG13 is a control register for the EPHY_CLK functional pin.

Offset Address	Register Name	Total Reset Value
0x034	PADCTRL_REG13	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg12_sr	padctrl_reg12_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg12_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6:4]	RW	padctrl_reg12_ds	Drive capability of the EPHY_CLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG14

PADCTRL_REG14 is a control register for the MDCK functional pin.

Offset Address	Register Name	Total Reset Value
0x038	PADCTRL_REG14	0x0000_0080

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg14_sr	reserved	padctrl_reg14_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg14_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg14_ds	Drive capability of the MDCK pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG15

PADCTRL_REG15 is a control register for the MDIO functional pin.

Offset Address	Register Name	Total Reset Value
0x03C	PADCTRL_REG15	0x0000_0090

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg15_sr	reserved	padctrl_reg15_ds	reserved

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg15_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg15_ds	Drive capability of the MDIO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG16

PADCTRL_REG16 is a control register for the SPI0_SCLK functional pin.

Offset Address	Register Name	Total Reset Value
0x040	PADCTRL_REG16	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg16_sr	padctrl_reg16_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg16_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg16_ds	Drive capability of the SPI0_SCLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG17

PADCTRL_REG17 is a control register for the SPI0_SDO functional pin.

Offset Address	Register Name	Total Reset Value
0x044	PADCTRL_REG17	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg17_sr	reserved	padctrl_reg17_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg17_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg17_ds	Drive capability of the SPI0_SDO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG18

PADCTRL_REG18 is a control register for the SPI0_SDI functional pin.

Offset Address	Register Name	Total Reset Value
0x048	PADCTRL_REG18	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg17_sr	reserved	padctrl_reg17_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg17_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg17_ds	Drive capability of the SPI0_SDI pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG19

PADCTRL_REG19 is a control register for the SPI0_CSN functional pin.

Offset Address	Register Name	Total Reset Value
0x04C	PADCTRL_REG19	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg19_sr	reserved	padctrl_reg19_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg19_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg19_ds	Drive capability of the SPI0_CSN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG20

PADCTRL_REG20 is a control register for the SPI1_SCLK functional pin.

Offset Address	Register Name	Total Reset Value
0x050	PADCTRL_REG20	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg20_sr	reserved	padctrl_reg20_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg20_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg20_ds	Drive capability of the SPI1_SCLK pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG21

PADCTRL_REG21 is a control register for the SPI1_SDO functional pin.

Offset Address	Register Name	Total Reset Value
0x054	PADCTRL_REG21	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg21_sr	reserved	padctrl_reg21_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg21_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg21_ds	Drive capability of the SPI1_SDO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG22

PADCTRL_REG22 is a control register for the SPI1_SDI functional pin.

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Offset Address	Register Name	Total Reset Value
0x058	PADCTRL_REG22	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg22_sr	reserved	padctrl_reg22_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg22_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg22_ds	Drive capability of the SPI1_SDI pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG23

PADCTRL_REG23 is a control register for the SPI1_CSN0 functional pin.

Offset Address	Register Name	Total Reset Value
0x05C	PADCTRL_REG23	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg23_sr	reserved	padctrl_reg23_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg23_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg23_ds	Drive capability of the SPI1_CSN0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG24

PADCTRL_REG24 is a control register for the I2C2_SDA functional pin.

Offset Address	Register Name	Total Reset Value
0x060	PADCTRL_REG24	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg24_sl	reserved	padctrl_reg24_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg24_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg24_ds	Drive capability of the I2C2_SDA pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG25

PADCTRL_REG25 is a control register for the I2C2_SCL functional pin.

Offset Address	Register Name	Total Reset Value
0x064	PADCTRL_REG25	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg25_sr	reserved	padctrl_reg25_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg25_sr	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg25_ds	Drive capability of the I2C2_SCL pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG26

PADCTRL_REG26 is a control register for the UART0_RXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x068	PADCTRL_REG26	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg26_sr	reserved	padctrl_reg26_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg26_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg26_ds	Drive capability of the UART0_RXD pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG27

PADCTRL_REG27 is a control register for the UART0_TXD functional pin.

Offset Address	Register Name	Total Reset Value
0x06C	PADCTRL_REG27	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg27_sr	reserved	padctrl_reg27_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved

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[7]	RW	padctrl_reg27_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg27_ds	Drive capability of the UART0_TXD pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG30

PADCTRL_REG30 is a control register for the USB_OVRCUR functional pin.

Offset Address	Register Name	Total Reset Value
0x078	PADCTRL_REG30	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg30_sr	reserved	padctrl_reg30_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg30_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg30_ds	Drive capability of the USB_OVRCUR pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG31

PADCTRL_REG31 is a control register for the USB_PWREN functional pin.

Offset Address	Register Name	Total Reset Value
0x07C	PADCTRL_REG31	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg31_sr	reserved	padctrl_reg31_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg31_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg31_ds	Drive capability of the USB_PWREN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG32

PADCTRL_REG32 is a control register for the GPIO0_0 functional pin.

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Offset Address	Register Name	Total Reset Value
0x080	PADCTRL_REG32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg32_sl	reserved	padctrl_reg32_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg32_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg32_ds	Drive capability of the GPIO0_0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG33

PADCTRL_REG33 is a control register for the GPIO0_1 functional pin.

Offset Address	Register Name	Total Reset Value
0x084	PADCTRL_REG33	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg33_sl	reserved	padctrl_reg33_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg33_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg33_ds	Drive capability of the GPIO0_1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG34

PADCTRL_REG34 is a control register for the GPIO0_2 functional pin.

Offset Address	Register Name	Total Reset Value
0x088	PADCTRL_REG34	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg34_sl	reserved	padctrl_reg34_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg34_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg34_ds	Drive capability of the GPIO0_2 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG35

PADCTRL_REG35 is a control register for the GPIO1_0 functional pin.

Offset Address	Register Name	Total Reset Value
0x08C	PADCTRL_REG35	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg35_sr	reserved	padctrl_reg35_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg35_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg35_ds	Drive capability of the GPIO1_0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG36

PADCTRL_REG36 is a control register for the GPIO1_1 functional pin.

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Offset Address	Register Name	Total Reset Value
0x090	PADCTRL_REG36	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg36_sr	reserved	padctrl_reg36_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg36_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg36_ds	Drive capability of the GPIO1_1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG37

PADCTRL_REG37 is a control register for the GPIO1_2 functional pin.

Offset Address	Register Name	Total Reset Value
0x094	PADCTRL_REG37	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg37_sr	reserved	padctrl_reg37_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg37_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg37_ds	Drive capability of the GPIO1_2 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG38

PADCTRL_REG38 is a control register for the GPIO1_3 functional pin.

Offset Address	Register Name	Total Reset Value
0x098	PADCTRL_REG38	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg38_sr	reserved	padctrl_reg38_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg38_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg38_ds	Drive capability of the GPIO1_3 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG39

PADCTRL_REG39 is a control register for the GPIO1_4 functional pin.

Offset Address	Register Name	Total Reset Value
0x09C	PADCTRL_REG39	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg39_sr	reserved	padctrl_reg39_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg39_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg39_ds	Drive capability of the GPIO1_4 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG40

PADCTRL_REG40 is a control register for the GPIO1_5 functional pin.

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Offset Address	Register Name	Total Reset Value
0x0A0	PADCTRL_REG40	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg40_sr	reserved	padctrl_reg40_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg40_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg40_ds	Drive capability of the GPIO1_5 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG41

PADCTRL_REG41 is a control register for the GPIO1_6 functional pin.

Offset Address	Register Name	Total Reset Value
0x0A4	PADCTRL_REG41	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg41_sr	reserved	padctrl_reg41_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg41_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg41_ds	Drive capability of the GPIO1_6 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG42

PADCTRL_REG42 is a control register for the SDIO0_CARD_POWER_EN functional pin.

Offset Address	Register Name	Total Reset Value
0x0A8	PADCTRL_REG42	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg42_sr	reserved	padctrl_reg42_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg42_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg42_ds	Drive capability of the SDIO0_CARD_POWER_EN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG43

PADCTRL_REG43 is a control register for the SDIO0_CARD_DETECT functional pin.

Offset Address	Register Name	Total Reset Value
0x0AC	PADCTRL_REG43	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg43_sr	reserved	padctrl_reg43_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg43_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg43_ds	Drive capability of the SDIO0_CARD_DETECT pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG44

PADCTRL_REG44 is a control register for the SDIO0_CWPR functional pin.

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Offset Address	Register Name	Total Reset Value
0x0B0	PADCTRL_REG44	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg44_sr	reserved	padctrl_reg44_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg44_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg44_ds	Drive capability of the SDIO0_CWPR pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG45

PADCTRL_REG45 is a control register for the SDIO0_CCLK_OUT functional pin.

Offset Address	Register Name	Total Reset Value
0x0B4	PADCTRL_REG45	0x0000_00D0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg45_sr	padctrl_reg45_ds			reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg45_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6:4]	RW	padctrl_reg45_ds	Drive capability of the SDIO0_CCLK_OUT pin 000: 18 mA 001: 16 mA 010: 14 mA 011: 12 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG46

PADCTRL_REG46 is a control register for the SDIO0_CCMD functional pin.

Offset Address	Register Name	Total Reset Value
0x0B8	PADCTRL_REG46	0x0000_00E0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg46_sr	padctrl_reg46_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg46_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg46_ds	Drive capability of the SDIO0_CCMD pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG47

PADCTRL_REG47 is a control register for the SDIO0_CDATA0 functional pin.

Offset Address	Register Name	Total Reset Value
0x0BC	PADCTRL_REG47	0x0000_00E0

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6 5 4	3 2 1 0
Name	reserved						padctrl_reg47_sr	padctrl_reg47_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	1 1 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg47_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg47_ds	Drive capability of the SDIO0_CDATA0 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG48

PADCTRL_REG48 is a control register for the SDIO0_CDATA1 functional pin.

Offset Address	Register Name	Total Reset Value
0x0C0	PADCTRL_REG48	0x0000_00E0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg48_sr	padctrl_reg48_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg48_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg48_ds	Drive capability of the SDIO0_CDATA1 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG49

PADCTRL_REG49 is a control register for the SDIO0_CDATA2 functional pin.

Offset Address	Register Name	Total Reset Value
0x0C4	PADCTRL_REG49	0x0000_00E0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg49_sr	padctrl_reg49_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg49_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg49_ds	Drive capability of the SDIO0_CDATA2 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG50

PADCTRL_REG50 is a control register for the SDIO0_CDATA3 functional pin.

Offset Address	Register Name	Total Reset Value
0x0C8	PADCTRL_REG50	0x0000_00E0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg50_sr	padctrl_reg50_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg50_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg50_ds	Drive capability of the SDIO0_CDATA3 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG51

PADCTRL_REG51 is a control register for the SFC_CLK functional pin.

Offset Address	Register Name	Total Reset Value
0x0CC	PADCTRL_REG51	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg51_st	padctrl_reg51_pd	padctrl_reg51_pu	reserved	padctrl_reg51_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg51_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg51_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg51_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg51_ds	Drive capability of the SFC_CLK pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG52

PADCTRL_REG52 is a control register for the SFC_SIO_IO0 functional pin.

Offset Address	Register Name	Total Reset Value
0x0D0	PADCTRL_REG52	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg52_st	padctrl_reg52_pd	padctrl_reg52_pu	reserved	padctrl_reg52_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg52_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg52_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg52_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg52_ds	Drive capability of the SFC_SIO_IO0 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG53

PADCTRL_REG53 is a control register for the SFC_SOI_IO1 functional pin.

Offset Address	Register Name	Total Reset Value
0x0D4	PADCTRL_REG53	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg53_st	padctrl_reg53_pd	padctrl_reg53_pu	reserved			padctrl_reg53_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg53_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg53_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg53_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg53_ds	Drive capability of the SFC_SOI_IO1 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG54

PADCTRL_REG54 is a control register for the SFC_WP_IO2 functional pin.

Offset Address	Register Name	Total Reset Value
0x0D8	PADCTRL_REG54	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg54_st	padctrl_reg54_pd	padctrl_reg54_pu	reserved			padctrl_reg54_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg54_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg54_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg54_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg54_ds	Drive capability of the SFC_WP_IO2 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG55

PADCTRL_REG55 is a control register for the SFC_HOLD_IO3 functional pin.

Offset Address	Register Name	Total Reset Value
0x0DC	PADCTRL_REG55	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg55_st	padctrl_reg55_pd	padctrl_reg55_pu	reserved			padctrl_reg55_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg55_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg55_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg55_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg55_ds	Drive capability of the SFC_HOLD_IO3 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG56

PADCTRL_REG56 is a control register for the SFC_CSN0 functional pin.

Offset Address	Register Name	Total Reset Value
0x0E0	PADCTRL_REG56	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg56_sl	reserved	padctrl_reg56_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg56_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg56_ds	Drive capability of the SFC_CSN0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG57

PADCTRL_REG57 is a control register for the SFC_CSN1 functional pin.

Offset Address	Register Name	Total Reset Value
0x0E4	PADCTRL_REG57	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg57_sl	reserved	padctrl_reg57_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg57_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg57_ds	Drive capability of the SFC_CSN1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG58

PADCTRL_REG58 is a control register for the UART1_RTSN functional pin.

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Offset Address	Register Name	Total Reset Value
0x0E8	PADCTRL_REG58	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg58_st	padctrl_reg58_pd	padctrl_reg58_pu	reserved				padctrl_reg58_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg58_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg58_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg58_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg58_ds	Drive capability of the UART1_RTSN pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG59

PADCTRL_REG59 is a control register for the UART1_RXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0EC	PADCTRL_REG59	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg59_st	padctrl_reg59_pd	padctrl_reg59_pu	reserved				padctrl_reg59_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg59_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg59_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg59_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg59_ds	Drive capability of the UART1_RXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG60

PADCTRL_REG60 is a control register for the UART1_CTSN functional pin.

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Offset Address	Register Name	Total Reset Value
0x0F0	PADCTRL_REG60	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg60_st	padctrl_reg60_pd	padctrl_reg60_pu	reserved			padctrl_reg60_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg60_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg60_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg60_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg60_ds	Drive capability of the UART1_CTSN pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG61

PADCTRL_REG61 is a control register for the UART1_TXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0F4	PADCTRL_REG61	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg61_st	padctrl_reg61_pd	padctrl_reg61_pu	reserved				padctrl_reg61_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg61_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg61_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg61_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg61_ds	Drive capability of the UART1_TXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG62

PADCTRL_REG62 is a control register for the UART2_RXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0F8	PADCTRL_REG62	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg62_st	padctrl_reg62_pd	padctrl_reg62_pu	reserved				padctrl_reg62_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg62_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg62_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg62_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg62_ds	Drive capability of the UART2_RXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG63

PADCTRL_REG63 is a control register for the UART2_TXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0FC	PADCTRL_REG63	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg63_st	padctrl_reg63_pd	padctrl_reg63_pu	reserved				padctrl_reg63_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg63_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg63_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg63_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg63_ds	Drive capability of the UART2_TXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG64

PADCTRL_REG64 is a control register for the JTAG_TRSTN functional pin.

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Offset Address	Register Name	Total Reset Value
0x100	PADCTRL_REG64	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg64_sr	reserved	padctrl_reg64_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg64_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg64_ds	Drive capability of the JTAG_TRSTN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG65

PADCTRL_REG65 is a control register for the JTAG_TCK functional pin.

Offset Address	Register Name	Total Reset Value
0x104	PADCTRL_REG65	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg65_sr	reserved	padctrl_reg65_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg65_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg65_ds	Drive capability of the JTAG_TCK pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG66

PADCTRL_REG66 is a control register for the JTAG_TMS functional pin.

Offset Address	Register Name	Total Reset Value
0x108	PADCTRL_REG66	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg66_sr	reserved	padctrl_reg66_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg66_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg66_ds	Drive capability of the JTAG_TMS pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG67

PADCTRL_REG67 is a control register for the JTAG_TDO functional pin.

Offset Address	Register Name	Total Reset Value
0x10C	PADCTRL_REG67	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg67_sr	reserved	padctrl_reg67_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg67_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg67_ds	Drive capability of the JTAG_TDO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG68

PADCTRL_REG68 is a control register for the JTAG_TDI functional pin.

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Offset Address	Register Name	Total Reset Value
0x110	PADCTRL_REG68	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg68_sr	reserved	padctrl_reg68_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg68_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg68_ds	Drive capability of the JTAG_TDI pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG70

PADCTRL_REG70 is a control register for the PWM0 functional pin.

Offset Address	Register Name	Total Reset Value
0x118	PADCTRL_REG70	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg70_sl	reserved	padctrl_reg70_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg70_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg70_ds	Drive capability of the PWM0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG71

PADCTRL_REG71 is a control register for the PWM1 functional pin.

Offset Address	Register Name	Total Reset Value
0x11C	PADCTRL_REG71	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg71_sl	reserved	padctrl_reg71_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg71_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg71_ds	Drive capability of the PWM1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG72

PADCTRL_REG72 is a control register for the PWM2 functional pin.

Offset Address	Register Name	Total Reset Value
0x120	PADCTRL_REG72	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg72_sl	reserved	padctrl_reg72_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg72_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg72_ds	Drive capability of the PWM2 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG73

PADCTRL_REG73 is a control register for the PWM3 functional pin.

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Offset Address	Register Name	Total Reset Value
0x124	PADCTRL_REG73	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg73_sl	reserved	padctrl_reg73_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg73_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg73_ds	Drive capability of the PWM3 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG74

PADCTRL_REG74 is a control register for the SAR_ADC_CH0 functional pin.

Offset Address	Register Name	Total Reset Value
0x128	PADCTRL_REG74	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg94_st	padctrl_reg94_pd	padctrl_reg94_pu	padctrl_reg94_sl	reserved	padctrl_reg94_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg94_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg94_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg94_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg94_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg94_ds	Drive capability of the SAR_ADC_CH0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG75

PADCTRL_REG75 is a control register for the SAR_ADC_CH1 functional pin.

Offset Address	Register Name	Total Reset Value
0x12C	PADCTRL_REG75	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg95_st	padctrl_reg95_pd	padctrl_reg95_pu	padctrl_reg95_sl	reserved	padctrl_reg95_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg95_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg95_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg95_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg95_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg95_ds	Drive capability of the SAR_ADC_CH1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG76

PADCTRL_REG76 is a control register for the SAR_ADC_CH2 functional pin.

Offset Address	Register Name	Total Reset Value
0x130	PADCTRL_REG76	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg96_st	padctrl_reg96_pd	padctrl_reg96_pu	padctrl_reg96_sl	reserved	padctrl_reg96_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg96_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg96_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg96_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg96_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg96_ds	Drive capability of the SAR_ADC_CH2 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG77

PADCTRL_REG77 is a control register for the SAR_ADC_CH3 functional pin.

Offset Address	Register Name	Total Reset Value
0x134	PADCTRL_REG77	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg97_st	padctrl_reg97_pd	padctrl_reg97_pu	padctrl_reg97_sl	reserved	padctrl_reg97_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg97_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg97_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg97_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg97_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg97_ds	Drive capability of the SAR_ADC_CH3 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

2.7 Electrical Specifications

2.7.1 Power Consumption Parameters

Power Consumption Parameters of Hi3518E V200

Table 2-59 to Table 2-64 describe power consumption parameters of Hi3518E V200.

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CAUTION

- The values of power consumption parameters are provided based on typical application scenarios.
- The maximum values are the tested maximum average power consumption within a specific time period, and cannot reflect the transient maximum power consumption when the system is working. Therefore, design board power supplies by following the *Hi3518E V20X/Hi3516C V200 Hardware Design User Guide*.

Table 2-59 Power consumption parameters when Hi3518E V200 does not use the SVB technology

Parameter	Description	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	209	308	mW
3.3 V power	3.3 V I/O power consumption	115	120	mW
1.8 V power	Power consumption of the DDR interface	182	195	mW
	Other power consumption of other 1.8 V I/O interfaces	54	60	mW

Table 2-60 Power consumption parameters when Hi3518E V200 uses the SVB technology

Parameter	Description	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	179	250	mW
3.3 V power	3.3 V I/O power consumption	115	120	mW
1.8 V power	Power consumption of the DDR interface	193	203	mW
	Other power consumption of other 1.8 V I/O interfaces	54	60	mW

Table 2-61 Power consumption benefit parameters when Hi3518E V200 uses the SVB technology

Typ	Max
~19 mW	~50 mW

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NOTE

The typical application scenario related to power consumption parameters in Table 2-59 to Table 2-61 is as follows: ARM9@540 MHz+720p@30 ISP+3DNR+1-channel video graphics subsystem (VGS) (only one of the rotation, LDC, and scaling functions is selected)+(720p@30 fps+VGA@30 fps+QVGA@30 fps) triple-stream H.264 encoding.

Table 2-62 Power consumption parameters when Hi3518E V200 does not use the SVB technology

Parameter	Description	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	215.5	310	mW
3.3 V power	3.3 V I/O power consumption	122	124	mW
1.8 V power	Power consumption of the DDR interface	186.7	202	mW
	Other power consumption of other 1.8 V I/O interfaces	54	60	mW

Table 2-63 Power consumption parameters when Hi3518E V200 uses the SVB technology

Parameter	Description	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	182	258	mW
3.3 V power	3.3 V I/O power consumption	122	124	mW
1.8 V power	Power consumption of the DDR interface	194	205	mW
	Other power consumption of other 1.8 V I/O interfaces	54	60	mW

Table 2-64 Power consumption benefit parameters when Hi3518E V200 uses the SVB technology

Typ	Max
~26.2 mW	~ 52 mW

NOTE

The typical application scenario related to power consumption parameters in Table 2-62 to Table 2-64 is as follows: ARM9@600 MHz+720p@30 ISP+3DNR+1-channel video graphics subsystem (VGS) (only one of the rotation, LDC, and scaling functions is selected)+(720p@30 fps+VGA@30 fps+QVGA@30 fps) triple-stream H.264 encoding.

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Power Consumption Parameters of Hi3518E V201

Table 2-65 to Table 2-67 describe power consumption parameters of Hi3518E V201.

CAUTION

- The values of power consumption parameters are provided based on typical application scenarios.
- The maximum values are the tested maximum average power consumption within a specific time period, and cannot reflect the transient maximum power consumption when the system is working. Therefore, design board power supplies by following the *Hi3518E V20X/Hi3516C V200 Hardware Design User Guide*.

Table 2-65 Power consumption parameters when Hi3518E V201 does not use the SVB technology

Symbol	Parameter	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	207	304	mW
3.3 V power	3.3 V I/O power consumption	102	106	mW
1.8 V power	Power consumption of the DDR interface	130	149	mW
	Other power consumption of other 1.8 V I/O interfaces	54	60	mW

Table 2-66 Power consumption parameters when Hi3518E V201 uses the SVB technology

Symbol	Parameter	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	175	244	mW
3.3 V power	3.3 V I/O power consumption	102	106	mW
1.8 V power	Power consumption of the DDR interface	133	156	mW
	Other power consumption of other 1.8 V I/O interfaces	54	60	mW

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Table 2-67 Power consumption benefit parameters when Hi3518E V201 uses the SVB technology

Typ	Max
~29 mW	~53 mW

NOTE

The typical application scenario of Hi3518E V201 is as follows: ARM9@540 MHz+720p@30 image signal processor (ISP) processing+3DNR+720p@30 fps single-stream H.264 encoding.

2.7.2 Temperature and Thermal Resistance Parameters

Table 2-68 and Table 2-69 describe temperature and thermal resistance parameters.

NOTE

- The thermal resistance is provided in compliance with the JEDEC JESD51-2 standard. The actual system design and environment may be different.
- The chip junction temperature is proportional to the chip power consumption. Ensure that the junction temperature is appropriate to match power supplies.
- Design heat dissipation by following the *Hi3518E V20X/Hi3516C V200 Hardware Design User Guide*.
- The chip can be stored in the sealed packaging bag for up to 12 months when the temperature is below 40ºC (104ºF) and the relative humidity (RH) is below 90%.
- After the packaging bag is opened, the following conditions must be met before the component is used in the reflow soldering process or other high-temperature processes:

  a. The process is complete within 168 hours at 30ºC (86ºF) or lower and at most 60% RH.

  b. The component is stored at the RH lower than 10%.
- The soldering temperature curve is based on the J-STD-020D.1 standard.

Table 2-68 Temperature and thermal resistance parameters of the Hi3518E V200

Parameter	Symbol	Min	Typ	Max	Unit
Ambient temperature	T_A	0	N/A	70	ºC
Rated temperature	T_{JMAX}	-20	N/A	105	ºC
Junction-to-ambient thermal resistance	θ_{JA}	N/A	34	N/A	ºC/W
Junction-to-board thermal resistance	θ_{JB}	N/A	16.9	N/A	ºC/W
Junction-to-case thermal resistance	θ_{JC}	N/A	12.7	N/A	ºC/W

Table 2-69 Temperature and thermal resistance parameters of the Hi3518E V201

Parameter	Symbol	Min	Typ	Max	Unit
Ambient temperature	T_A	0	N/A	70	ºC
Rated temperature	T_{JMAX}	-20	N/A	105	ºC

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Parameter	Symbol	Min	Typ	Max	Unit
Junction-to-ambient thermal resistance	θ_{JA}	N/A	34.5	N/A	ºC/W
Junction-to-board thermal resistance	θ_{JB}	N/A	16.5	N/A	ºC/W
Junction-to-case thermal resistance	θ_{JC}	N/A	14.1	N/A	ºC/W

CAUTION

The ambient temperature is used in chip operating conditions, while the junction temperature only serves as a reference during thermal simulation.

2.7.3 Operating Conditions

Table 2-70 and Table 2-71 describes operating conditions.

CAUTION

The prerequisite for using the VDD voltage range (when SVB is used) listed in Table 2-70 is as follows: the customer's board must use the SVB circuit, and the impedance and capacitance parameters of the SVB circuit must completely copy the related RC parameters of Hi3518EV20XDMEB VER.A.

Table 2-70 Operating conditions for the SVB related power supply

Symbol	Description	Voltage range			Unit
		Min	Typ	Max	
VDD	Internal core power (when SVB is not used)	1.045	1.1	1.155	V
	Internal core power (when SVB is used)	1.02	-	1.05	V
AVDD11_PLL	PLL core analog power (when SVB is not used)	1.045	1.1	1.155	V
	PLL core analog power (when SVB is used)	1.02	-	1.05	V

Table 2-71 Operating conditions for the power supply under normal voltage

Symbol	Description	Min	Typ	Max	Unit
DVDD33	I/O power	2.97	3.3	3.63	V

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Symbol	Description	Min	Typ	Max	Unit
DVDD3318_VI	VI I/O power	1.62	1.8	1.98	V
	VI I/O power	2.97	3.3	3.63	V
DVDD3318_SENSOR	Sensor, SPI0 I/O power	1.62	1.8	1.98	V
	Sensor, SPI0 I/O power	2.97	3.3	3.63	V
DVDD3318_EMMC	SFC, UART1/EMMC I/O power	1.62	1.8	1.98	V
	SFC, UART1/EMMC I/O power	2.97	3.3	3.63	V
DVDDIO_RMII	RMII I/O power	1.62	1.8	1.98	V
	RMII I/O power	2.97	3.3	3.63	V
DVDD18_SDIO	1.8 V SDIO I/O power	1.62	1.8	1.98	V
VDDIO_DDR	DDR2 interface power	1.7	1.8	1.9	V
VDDIO_CK_DDR	DDR2 clock interface power	1.7	1.8	1.9	V
AVDD_DDRPLL	DDR2 PLL3V3 analog power	2.97	3.3	3.63	V
AVDD33_PLL	3.3 V PLL analog power	2.97	3.3	3.63	V
AVDD33_AC	3.3 V ACODEC analog power	3	3.3	3.6	V
AVDD33_USB	3.3 V USB analog power	3	3.3	3.6	V
AVDD3318_MIPI	MIPI analog power	1.71	1.8	1.89	V
	3.3 V power when the MIPI signal is multiplexed as a single-ended digital signal	2.97	3.3	3.63	V
	1.8 V power when the MIPI signal is multiplexed as a single-ended digital signal	1.62	1.8	1.98	V

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Symbol	Description	Min	Typ	Max	Unit
AVDD_BAT	RTC battery power	1.6	3.3	3.6	V
AVDD33_RTC	RTC analog power	2.97	3.3	3.63	V

2.7.4 Power-On and Power-Off Sequences

For details about power-on and power-off sequences, see section 1.2.5 "requirements on the power-on and power-off timing" in the *Hi3518E V20X/Hi3516C V200 Hardware Design User Guide*.

2.7.5 DC and AC Electrical Specifications

Table 2-72 and Table 2-73 describe direct current (DC) electrical specifications.

Table 2-72 DC electrical specifications (DVDD33/DVDD3318_VI/DVDD3318_SENSOR/DVDD3318_EMMC/DVDDIO_RMII = 3.3 V, incompatible with the 5 V input voltage)

Symbol	Description	Min	Type	Max	Unit	Remarks
DVDD33	Interface voltage	2.97	3.3	3.63	V	N/A
V_{IH}	High-level input voltage	2.0	N/A	DVDD33 + 0.3 V	V	The interface is incompatible with the 5 V input voltage.
V_{IL}	Low-level input voltage	–0.3	N/A	0.8	V	N/A
I_L	Input leakage current	N/A	N/A	±10	μA	N/A
I_{OZ}	Tristate output leakage current	N/A	N/A	±10	μA	N/A
V_{OH}	High-level output voltage	2.4	N/A	N/A	V	N/A
V_{OL}	Low-level output voltage	N/A	N/A	0.4	V	N/A
R_{PU1}	Internal pull-up impedance	33	37	41	kΩ	N/A
R_{PU2}	Internal pull-up impedance	7	8	9	kΩ	N/A
R_{PU3}	Internal pull-up impedance	32	36	40	kΩ	N/A
R_{PU4}	Internal pull-up impedance	46	51	56	kΩ	N/A
R_{PD1}	Internal pull-down impedance	36	40	44	kΩ	N/A

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Symbol	Description	Min	Type	Max	Unit	Remarks
R_{PD2}	Internal pull-down impedance	7	8	9	kΩ	N/A
R_{PD3}	Internal pull-down impedance	36	40	44	kΩ	N/A
R_{PD4}	Internal pull-down impedance	50	51	56	kΩ	N/A
R_{PD5}	Internal pull-down impedance	36	40	44	kΩ	N/A

Table 2-73 DC electrical specifications (DVDD18_SDIO/DVDD3318/DVDDIO_RMII = 1.8 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
DVDD18	Interface voltage	1.62	1.8	1.98	V	N/A
V_{IH}	High-level input voltage	1.17	N/A	DVDD18+ 0.3V	V	N/A
V_{IL}	Low-level input voltage	–0.3	N/A	0.63	V	N/A
I_L	Input leakage current	N/A	N/A	±10	μA	N/A
I_{OZ}	Tristate output leakage current	N/A	N/A	±10	μA	N/A
V_{OH}	High-level output voltage	1.35	N/A	N/A	V	N/A
V_{OL}	Low-level output voltage	N/A	N/A	0.45	V	N/A
R_{PU1}	Internal pull-up impedance	74	82	90	kΩ	N/A
R_{PU2}	Internal pull-up impedance	8	9	10	kΩ	N/A
R_{PU3}	Internal pull-up impedance	76	85	94	kΩ	N/A
R_{PU4}	Internal pull-up impedance	50	51	56	kΩ	N/A
R_{PD1}	Internal pull-down impedance	77	86	95	kΩ	N/A
R_{PD2}	Internal pull-down impedance	8	9	10	kΩ	N/A
R_{PD3}	Internal pull-down impedance	77	86	95	kΩ	N/A
R_{PD4}	Internal pull-down impedance	46	51	56	kΩ	N/A
R_{PD5}	Internal pull-down impedance	77	86	95	kΩ	N/A

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Table 2-74 describes DC electrical specifications in DDR2 SSTL18 mode.

Table 2-74 DC electrical specifications in DDR2 SSTL18 mode (VDDIO_DDR = 1.8 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
VDDIO_DDR	Interface voltage	1.7	1.8	1.9	V	N/A
VREF	Reference voltage of the DDR interface	0.49 x VDDIO_DDR	0.5x VDDIO_DDR	0.51x VDDIO_DDR	V	(0.49–0.51) x VDDIO_DDR
VTT	Termination voltage	VREF – 0.04	VREF	VREF + 0.04	V	N/A
$V_{IH(DC)}$	High-level input voltage	VREF + 0.125	N/A	VDDIO_DDR + 0.3	V	N/A
$V_{IL(DC)}$	Low-level input voltage	–0.3	N/A	VREF – 0.125	V	N/A
V_{OH}	High-level output voltage	VTT + 0.603	N/A	N/A	V	N/A
V_{OL}	Low-level output voltage	N/A	N/A	VTT – 0.603	V	N/A
I_{OH}	High-level output current	–13.4	N/A	N/A	mA	N/A
I_{OL}	Low-level output current	13.4	N/A	N/A	mA	N/A

Table 2-75 describes alternating current (AC) electrical specifications in DDR2 SSTL18 mode.

Table 2-75 AC electrical specifications in DDR2 SSTL18 mode (VDDIO_DDR = 1.8 V)

Symbol	Description	533–1600 Mbit/s		Unit	Remarks
		Min	Max		
$V_{IH(AC)}$	AC high-level input voltage	VREF + 0.2	N/A	V	N/A
$V_{IL(AC)}$	AC low-level input voltage	N/A	VREF–0.2	V	N/A

2.7.6 MIPI/LVDS RX Electrical Parameters

Table 2-76 describes low-voltage differential signaling (LVDS) differential DC electrical parameters.

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Table 2-76 LVDS differential DC electrical parameters

Symbol	Description		Min	Typ	Max	Unit
VIDTH (SL)	Differential input threshold voltage |(VP—VM)|	Sub-LVDS	70	-	-	mV
VIDTH (HS)		HiSPi (SLVS)	70	-	-	
VIDTH (HiVCM)		HiSPi (HiVCM)	100	-	-	
VIDTH (DP)		D-PHY HS	70	-	-	
VIDTH (LV)		LVDS	100	-	-	
VIDTH (ML)		Mini-LVDS	100	-	-	
VCM (SL)	Common mode voltage range (VP+VM)/2	Sub-LVDS	0.5	0.9	1.3	V
VCM (HS)		HiSPi (SLVS)	0.07	0.2	0.35	
VCM (HiVCM)		HiSPi (HiVCM)	0.66	0.90	1.17	
VCM (DP)		D-PHY HS	0.07	0.2	0.33	
VCM (LV)		LVDS	0.925	1.2	1.475	
VCM (ML)		Mini-LVDS	1.025	1.2	1.375	
VISVR (SL)	Single-ended input voltage range VP,VM	Sub-LVDS	0.4	-	1.4	V
VCM (HS)		HiSPi (SLVS)	-0.04	-	0.49	
VCM (HiVCM)		HiSPi (HiVCM)	0.55	-	1.35	
VCM (DP)		D-PHY HS	-0.04	-	0.46	
VCM (LV)		LVDS	0	-	1.8	
VCM (ML)		Mini-LVDS	0.825	-	1.575	
ZID (SL)	Internal termination resister value	Sub-LVDS	80	100	120	Ω
ZID (HS)		HiSPi (SLVS)			125	
ZID (HiVCM)		HiSPi (HiVCM)			125	
ZID (LV)		LVDS			120	
ZID (ML)		Mini-LVDS			120	
ZID (DP)		D-PHY HS			125	

Table 2-77 to Table 2-79 describe the MIPI parameters.

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Table 2-77 MIPI high-speed (HS) DC parameters

Symbol	Description	Min	Typ	Max	Unit
VTERM-EN	Single-ended threshold for HS termination enable	-	-	450	mV

Table 2-78 MIPI HS AC parameters

Symbol	Description	Min	Typ	Max	Unit
ΔVCMRX (HF)	Common-mode interface beyond 450 MHz	-	-	100	-
ΔVCMRX (LF)	Common-mode interface 50 MHz−450 MHz	-50	-	50	mV
CCM	Common-mode termination	-	-	60	pF

Table 2-79 MIPI low-power (LP) DC parameters

Symbol	Description	Min	Typ	Max	Unit
VIHLP	Logic 1 input voltage	880	-	-	mV
VILLP	Logic 0 input voltage	-	-	550	
VHYST	Input hysteresis	25	-	-	

2.7.7 SDIO Electrical Specifications

Table 2-80 describes the SDIO electrical specifications (3.3 V).

Table 2-80 SDIO electrical specifications (3.3 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
VDDIO	Power supply voltage	2.97	3.3	3.6	V	N/A
VOH	Output high level	2.4	N/A	N/A	V	
VOL	Output low level	N/A	N/A	0.4	V	
VIH	Input high level	2.0	N/A	VDD IO+ 0.3	V	
VIL	Input low level	-0.3	N/A	0.8	V	
N/A	Power-on duration	N/A	N/A	250	Ms	0 V to VDDIO

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Table 2-81 describes the SDIO electrical specifications (1.8 V).

Table 2-81 SDIO electrical specifications (1.8 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
VDDIO	Power supply voltage	1.62	1.8	1.98	V	N/A
VOH	Output high level	1.35	N/A	N/A	V	
VOL	Output low level	N/A	N/A	0.45	V	
VIH	Input high level	1.17	N/A	VDDIO+0.3	V	
VIL	Input low level	-0.3	N/A	0.63	V	
N/A	Power-on duration	N/A	N/A	250	ms	0 V to VDDIO

2.7.8 Audio CODEC Electrical Parameters

Table 2-82 to Table 2-85 describe the electrical parameters of the audio CODEC.

Table 2-82 Overall specifications

Description	Min	Typ	Max	Unit	Remarks
Analog circuit power AVDD	3	3.3	3.6	V	Relative to AGND
VREF	-	AVDD/2		V	Relative to AGND

Table 2-83 Major DAC specifications

Description	Min	Typ	Max	Unit	Remarks
Output amplitude at full scale	-	0.875	-	Vrms	Maximum output signal swing

Table 2-84 Major ADC specifications

Description	Min	Typ	Max	Unit	Remarks
Maximum input amplitude	-	1	-	Vrms	Maximum input signal swing of the ADC

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Table 2-85 Major MICBIAS specifications

Description	Min	Typ	Max	Unit	Remarks
Bias voltage	-	2.1 x AVDD/3.3	-	V	MIC bias voltage
Maximum output current	-	-	3	mA	-

Table 2-86 Major MICPGA specifications

Description	Min	Typ	Max	Unit	Remarks
Input impedance	5	-	12	kΩ	MICPGA input impedance

2.8 Interface Timings

2.8.1 SFC Interface Timings

Figure 2-11 shows the SFC output timing.

Figure 2-11 SFC output timing

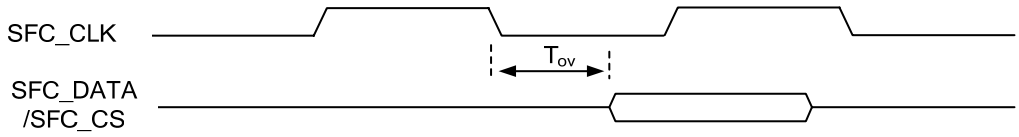


Table 2-87 describes the parameters of the SFC output timing.

Table 2-87 Parameters of the SFC output timing

Parameter	Symbol	Min	Typ	Max	Unit
Clock cycle of SFCCLK	T	10.1	N/A	83.2	ns
Output data signal delay	T_{ov}	-3	N/A	3	ns
Output CS signal delay	T_{ov}	-3	N/A	3	ns

Figure 2-12 shows the SFC input timing.

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Figure 2-12 SFC input timing

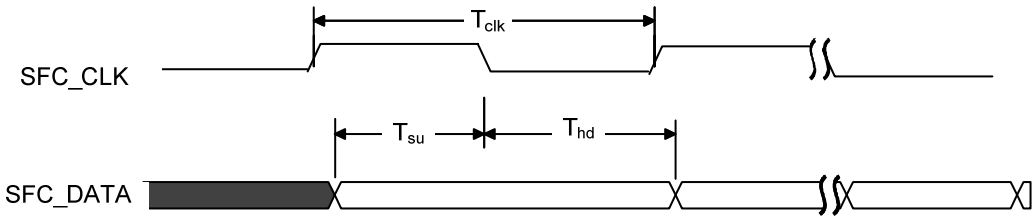


Table 2-88 describes the parameters of the SFC input timing.

Table 2-88 Parameters of the SFC input timing

Parameter	Symbol	Min	Typ	Max	Unit
Clock cycle of SFCCLK	T_{clk}	10.1	N/A	83.2	ns
Input signal setup time	T_{su}	2	N/A	N/A	ns
Input signal hold time	T_{hd}	0.1	N/A	N/A	ns

2.8.2 Ethernet MAC Interface Timings

RMII Timings

Figure 2-13 shows the 100 Mbit/s RX timing of the RMII.

Figure 2-13 100 Mbit/s RX timing of the RMII

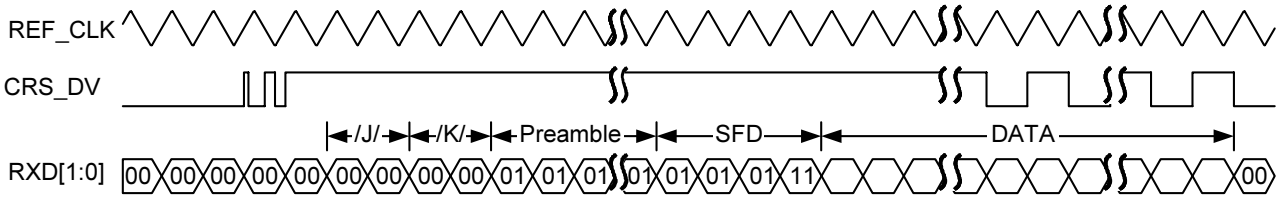


Figure 2-14 shows the 100 Mbit/s TX timing of the RMII.

Figure 2-14 100 Mbit/s TX timing of the RMII

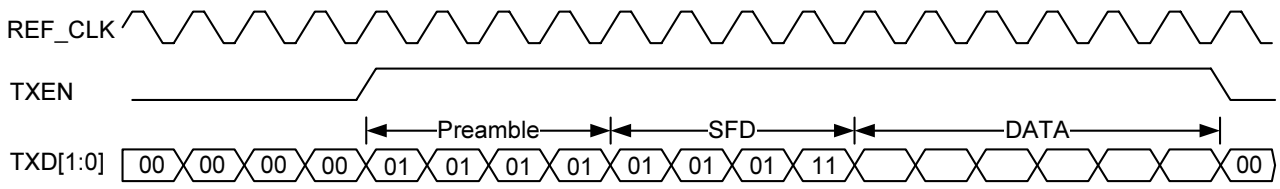


Figure 2-15 shows the 10 Mbit/s RX timing of the RMII.

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Figure 2-15 10 Mbit/s RX timing of the RMII

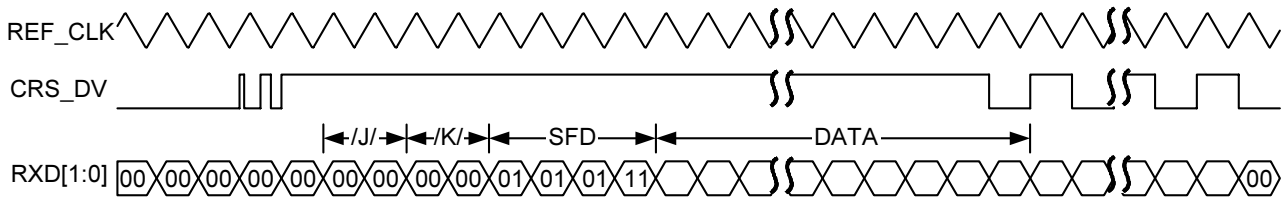


Figure 2-16 shows the 10 Mbit/s TX timing of the RMII.

Figure 2-16 10 Mbit/s TX timing of the RMII

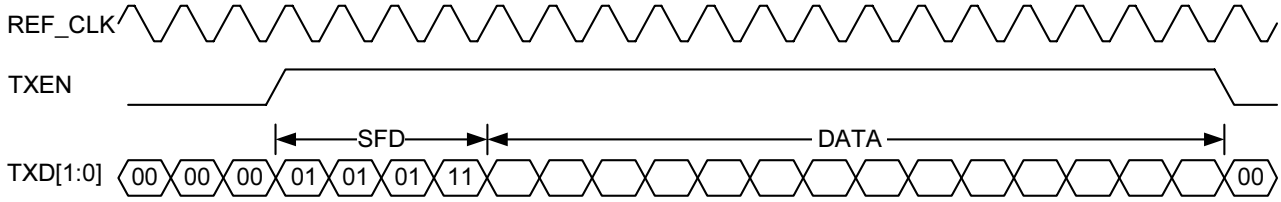


Figure 2-17 shows the RMII timing parameters.

Figure 2-17 RMII timing parameters

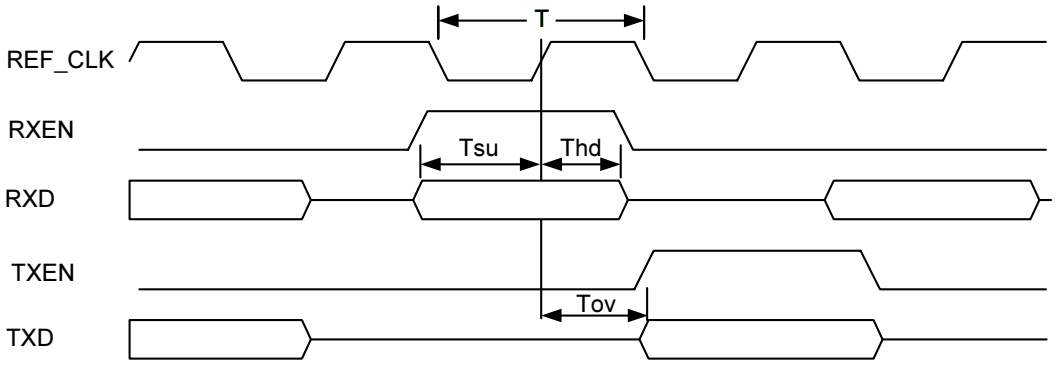


Table 2-89 describes the RMII timing parameters.

Table 2-89 RMII timing parameters

Parameter	Symbol	Signal	Min	Max	Unit
RMII clock cycle	T	REF_CLK	20	20	ns
RMII signal setup time	Tsu (RX)	CRS_DV/RXD[1:0]	4	N/A	ns
RMII signal hold time	Thd (RX)	CRS_DV/RXD[1:0]	2	N/A	ns
RMII output signal delay	Tov (TX)	TXEN/TXD[1:0]	3	16	ns

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2.8.2.2 MDIO Interface Timings

Figure 2-18 shows the read timing of the MDIO interface.

Figure 2-18 Read timing of the MDIO interface

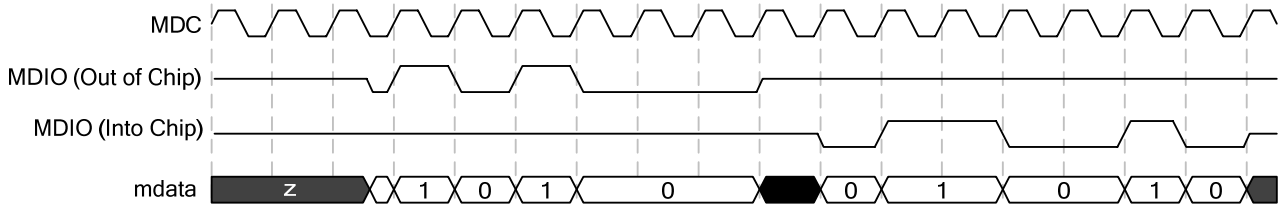


Figure 2-19 shows the write timing of the MDIO interface.

Figure 2-19 Write timing of the MDIO interface

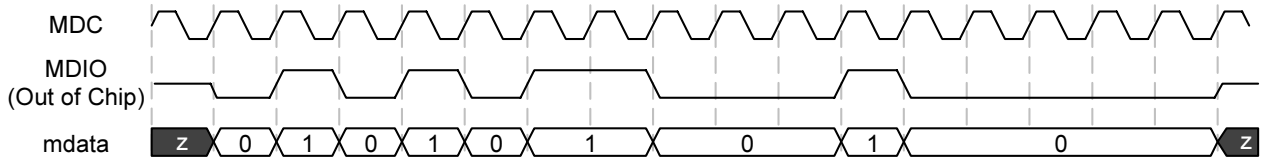


Figure 2-20 shows the RX timing parameters of the MDIO interface.

Figure 2-20 RX timing parameters of the MDIO interface

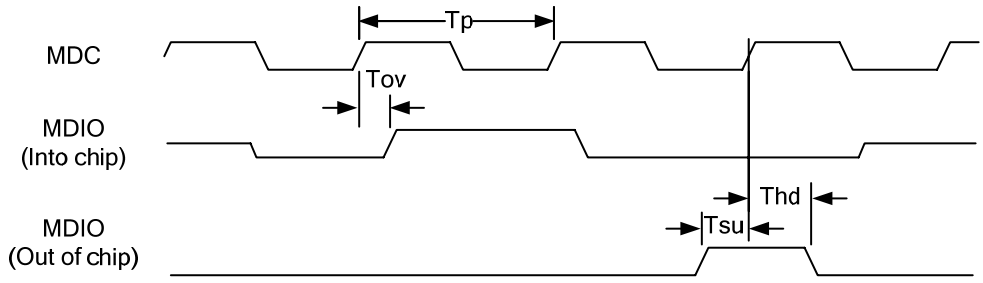


Table 2-90 describes the timing parameters of the MDIO interface.

Table 2-90 Timing parameters of the MDIO interface

Parameter	Symbol	Signal	Min	Max	Unit
MDIO data RX delay	Tov	MDIO	0	375	ns
MDIO clock cycle	Tp	MDCK	500	64000	ns
MDIO data TX setup time	Tsu	MDIO	10	N/A	ns
MDIO data TX hold time	Thd	MDIO	10	N/A	ns

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2.8.3 VI Interface Timing

Figure 2-21 shows the VI interface timing in CMOS mode.

Figure 2-21 VI interface timing in CMOS mode

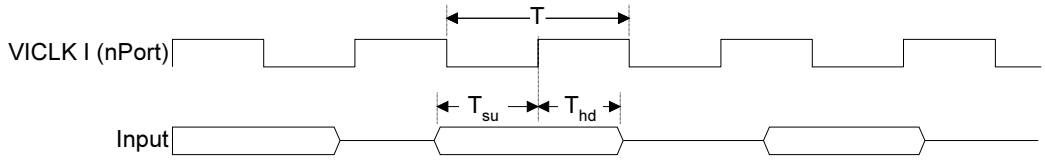


Table 2-91 describes the VI interface timing parameters.

Table 2-91 VI interface timing parameters

Parameter	Symbol	Min	Typ	Max	Unit
VICLK clock cycle	T	6.73	N/A	N/A	ns
Input signal setup time	T_{su}	2.5	N/A	N/A	ns
Input signal hold time	T_{hd}	2.0	N/A	N/A	ns

2.8.4 VO Interface Timing

Figure 2-22 shows the timing of the BT.656 interface.

Figure 2-22 Timing of the BT.656 interface

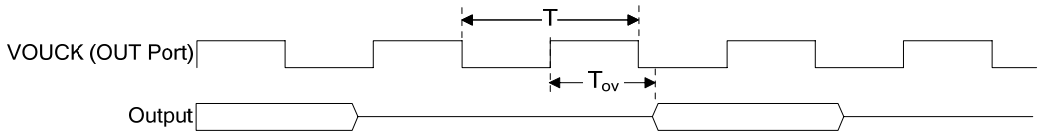


Table 2-92 describes the timing parameters of the BT.656 interface.

Table 2-92 Timing parameters of the BT.656 interface

Parameter	Symbol	Min	Typ	Max	Unit
VOUCLK clock cycle	T	13.46	13.48	37	ns
Output signal delay	T_{ov}	T/2-1.5	N/A	T/2+1.5	ns

Figure 2-23 shows the liquid crystal display (LCD) interface timing.

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Figure 2-23 LCD interface timing

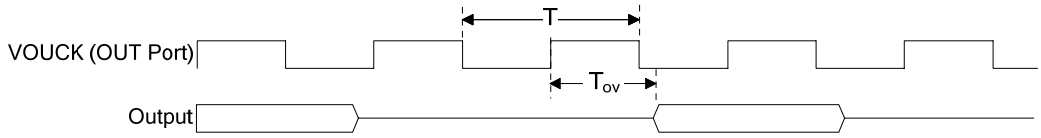


Table 2-93 describes the timing parameters of the LCD interface.

Table 2-93 Timing parameters of the LCD interface

Parameter	Symbol	Min	Typ	Max	Unit
VOUCLK clock cycle	T	37	N/A	500	ns
Output signal delay	T_{ov}	T/2 − 1.5	N/A	T/2 + 1.5	ns

2.8.5 AIAO Interface Timings

2.8.5.1 I²S Interface Timing

Figure 2-24 shows the RX timing of the I²S interface.

Figure 2-24 RX timing of the I²S interface

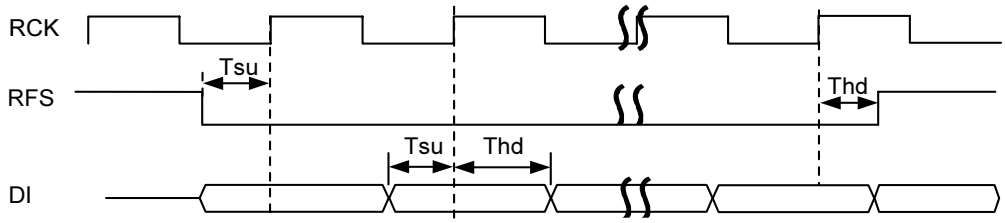


Figure 2-25 shows the TX timing of the I²S interface.

Figure 2-25 TX timing of the I²S interface

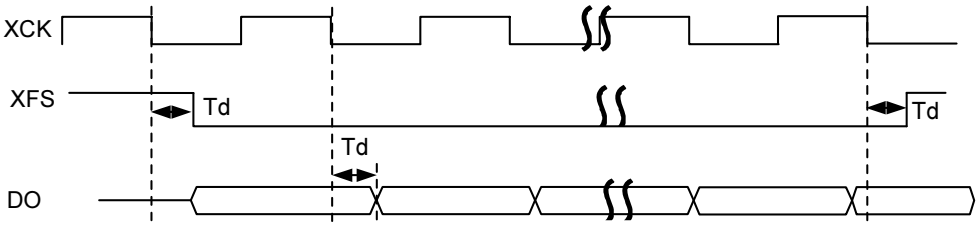


Table 2-94 describes the timing parameters of the I²S interface.

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Table 2-94 Timing parameters of the I²S interface

Parameter	Symbol	Min	Typ	Max	Unit
Input signal setup time	T_{su}	10	N/A	N/A	ns
Input signal hold time	T_{hd}	10	N/A	N/A	ns
Output signal delay	T_d	-10	N/A	10	ns

2.8.5.2 PCM Interface Timings

Figure 2-26 shows the RX timing of the PCM interface.

Figure 2-26 RX timing of the PCM interface

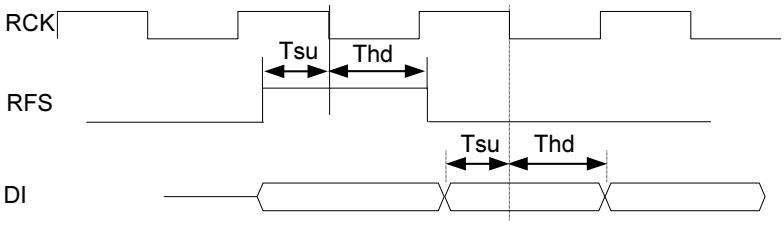


Figure 2-27 shows the TX timing of the PCM interface

Figure 2-27 TX timing of the PCM interface

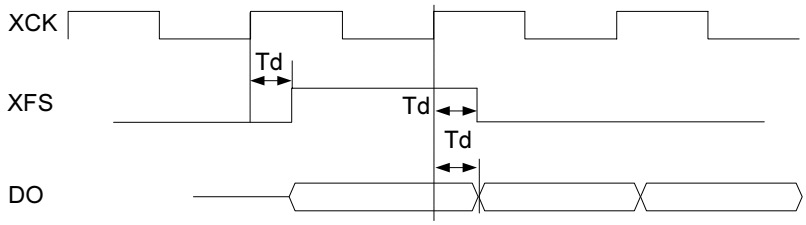


Table 2-95 describes the timing parameters of the PCM interface.

Table 2-95 Timing parameters of the PCM interface

Parameter	Symbol	Min	Typ	Max	Unit
Input signal setup time	T_{su}	10	N/A	N/A	ns
Input signal hold time	T_{hd}	10	N/A	N/A	ns
Output signal delay	T_d	-10	N/A	10	ns

2.8.6 I²C Interface Timing

Figure 2-28 shows the I²C transfer timing.

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Figure 2-28 I^2C transfer timing

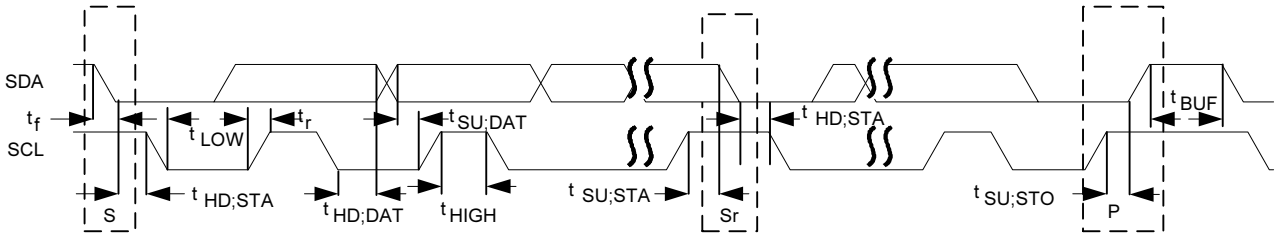


Table 2-96 describes the timing parameters of the I^2C interface.

Table 2-96 Timing parameters of the I^2C interface

Parameter	Symbol	Standard Mode		Fast Mode		Unit
		Min	Max	Min	Max	
Serial clock (SCL) frequency	f_{SCL}	0	100	0	400	kHz
Start hold time	$t_{HD;STA}$	4.0	N/A	0.6	N/A	µs
SCL low-level cycle	t_{LOW}	4.7	N/A	1.3	N/A	µs
SCL high-level cycle	t_{HIGH}	4.0	N/A	0.6	N/A	µs
Start setup time	$t_{SU;STA}$	4.7	N/A	0.6	N/A	µs
Data hold time	$t_{HD;DAT}$	0	3.45	0	0.9	µs
Data setup time	$t_{SU;DAT}$	250	N/A	100	N/A	ns
Serial data (SDA) and SCL rising time	t_r	N/A	1000	$20 + 0.1C_b$	300	ns
SDA and SCL falling time	t_f	N/A	300	$20 + 0.1C_b$	300	ns
End setup time	$t_{SU;STO}$	4.0	N/A	0.6	N/A	µs
Bus release time from start to end	t_{BUF}	4.7	N/A	1.3	N/A	µs
Bus load	C_b	N/A	400	N/A	400	pF

2.8.7 SPI Timings

NOTE

In Figure Figure 2-29 to Figure 2-31, the conventions are as follows:

- MSB = most significant bit
- LSB = least significant bit
- SPI_CK(0): spo = 0
- SPI_CK(1): spo = 1

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Figure 2-29 shows the SPI clock (SPICK) timing.

Figure 2-29 SPICK timing

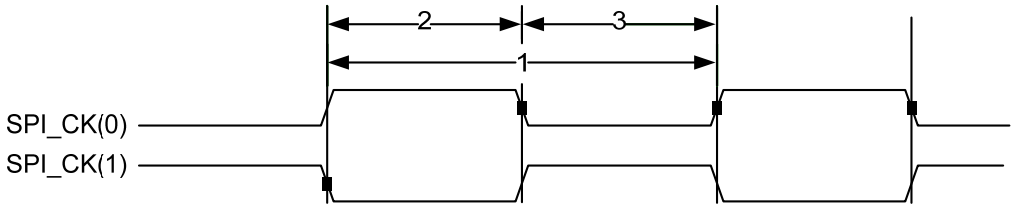


Figure 2-30 and Figure 2-31 show the SPI timings in master mode.

Figure 2-30 SPI timing in master mode (sph = 1)

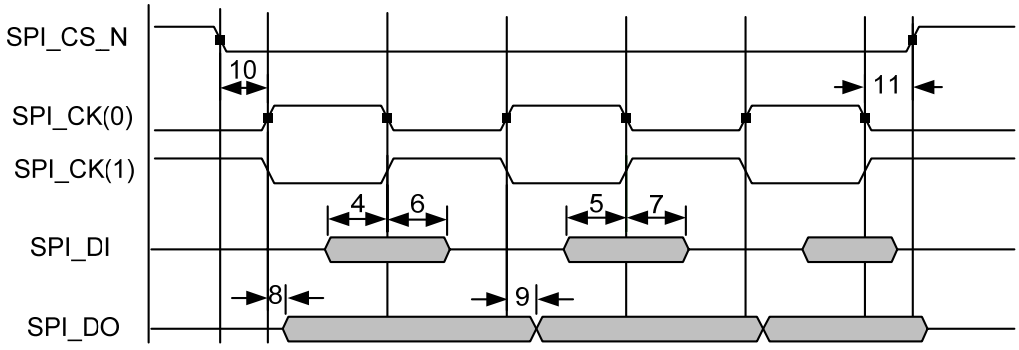


Figure 2-31 SPI timing in master mode (sph = 0)

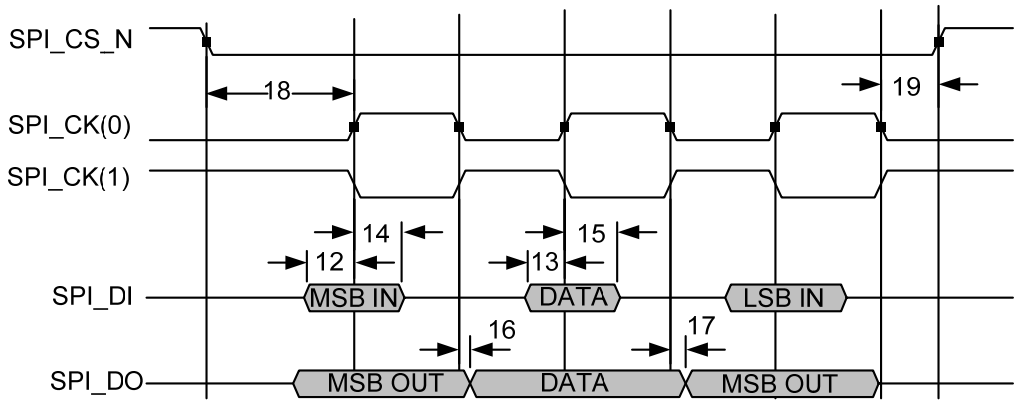


Table 2-97 describes the SPI timing parameters.

Table 2-97 SPI timing parameters

No.	Parameter	Symbol	Min	Typ	Max	Unit	Remark
1	Cycle time, SPI_CK	tc	2000/Fsspclk	N/A	65024000/Fsspclk	ns	Fsspclk unit: MHz

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No.	Parameter	Symbol	Min	Typ	Max	Unit	Remark
2	Pulse duration, SPI_CK high (all master modes)	tw1	N/A	1/2 tc	N/A	ns	
3	Pulse duration. SPI_CK low (all master modes)	tw2	N/A	1/2 tc	N/A	ns	
4	Setup time, SPI_DI (input) valid before SPI_CK (output) falling edge	tsu1	1/4 tc	N/A	N/A	ns	
5	Setup time, SPI_DI (in put) valid before SPICK (output) rising edge	tsu2	1/4 tc	N/A	N/A	ns	
6	Hold time, SPI_DI (input) valid after SPI_CK (output) falling edge	th1	1/4 tc	N/A	N/A	ns	
7	Hold time, SPI_DI (input) valid after SPI_CK (output) rising edge	th2	1/4 tc	N/A	N/A	ns	
8	Delay time, SPI_CK (output) rising edge to SPI_DO (output) transition	td1	0	N/A	N/A	ns	
9	Delay time, SPI_CK (output) falling edge to SPI_DO (output) transition	td2	0	N/A	N/A	ns	
10	Delay time, SPI_CS_N (output) falling edge to first SPI_CK (output) rising or falling edge	td3	N/A	tc	N/A	ns	
11	Delay time, SPI_CK (output) rising or falling edge to SPI_CS_N (output) rising edge	td4	N/A	1/2 tc	N/A	ns	
12	Setup time, SPI_DI (input) valid before SPI_CK (output) rising edge	tsu3	1/4 tc	N/A	N/A	ns	
13	Setup time, SPI_DI (in put) valid before SPI_CK (output) falling edge	tsu4	1/4 tc	N/A	N/A	ns	
14	Hold time, SPI_DI (input) valid after SPI_CK (output) rising edge	th3	1/4 tc	N/A	N/A	ns	
15	Hold time, SPI_DI (input) valid after SPI_CK (output) falling edge	th4	1/4 tc	N/A	N/A	ns	

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No.	Parameter	Symbol	Min	Typ	Max	Unit	Remark
16	Delay time, SPI_CK (output) falling edge to SPI_DO (output) transition	td5	0	N/A	N/A	ns	
17	Delay time, SPI_CK (output) rising edge to SPI_DO (output) transition	td6	0	N/A	N/A	ns	
18	Delay time, SPI_CS_N (output) falling edge to first SPI_CK (output) rising or falling edge	td7	N/A	1/2 tc	N/A	ns	
19	Delay time, SPI_CK (output) rising or falling edge to SPI_CS_N (output) rising edge	td8	N/A	tc	N/A	ns	

2.8.8 MIPI RX Timings

Table 2-98 describes the MIPI RX timing parameters.

Table 2-98 MIPI RX timing parameters

Symbol	Description	Min	Tye	Max	Unit
F_{MAX}	Data rate	N/A	N/A	1G	bit/s
T_{PERIOD}	Differential clock cycle	2	N/A	N/A	ns
$DUTY_{clk}$	Differential clock duty cycle	45	N/A	55	%
T_{SET}	Differential clock setup time	0.15 x UI	N/A	N/A	ns
T_{HD}	Differential clock hold time	0.15 x UI	N/A	N/A	ns

NOTE

$UI = T_{PERIOD}/2$

2.8.9 SDIO/MMC Interface Timings

Figure 2-32 shows the single-edge data input and output timings of the SDIO/MMC interface.

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Figure 2-32 Single-edge data input and output timings of the SDIO/MMC interface

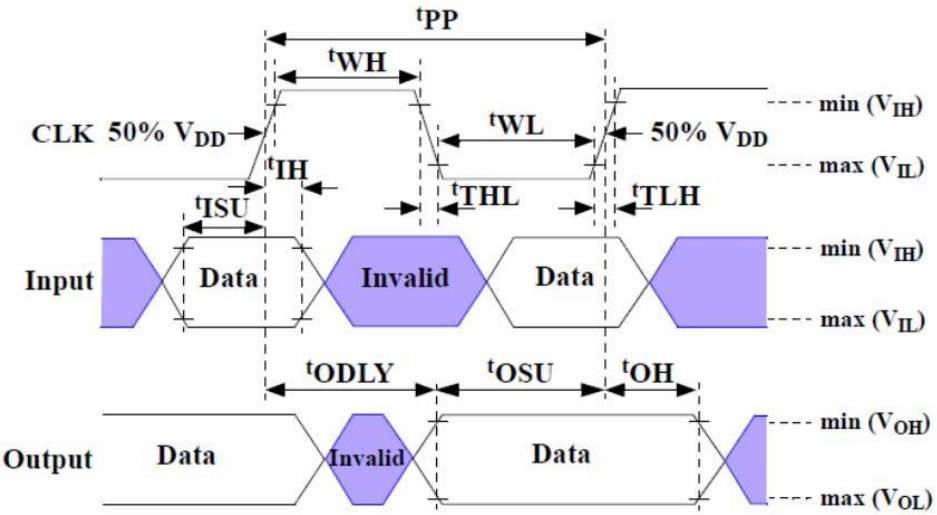


Figure 2-33 shows the dual-edge data input and output timings of the SDIO/MMC interface.

Figure 2-33 Dual-edge data input and output timings of the SDIO/MMC interface

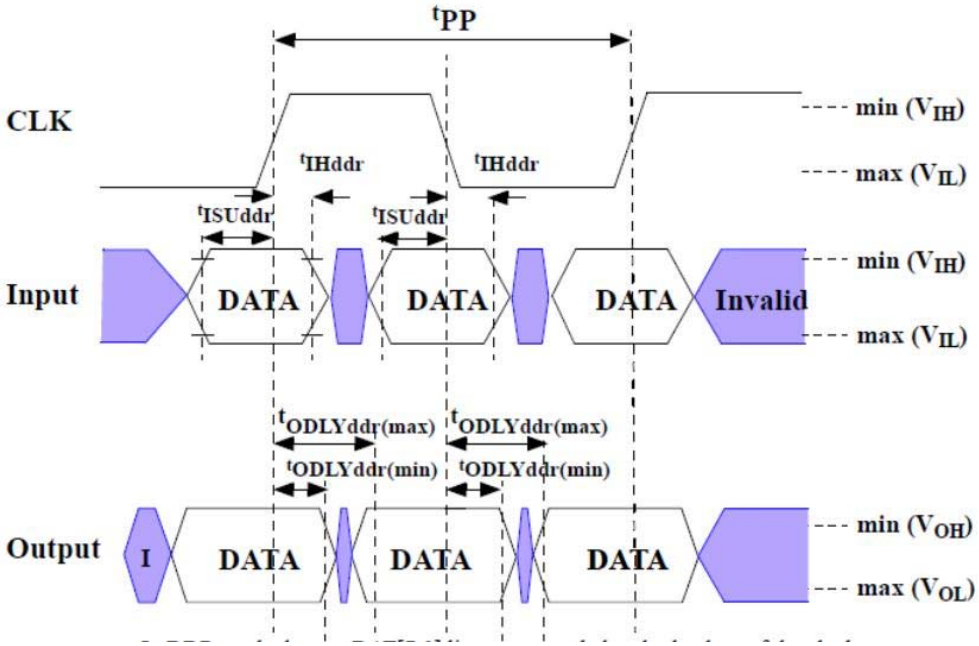


Table 2-99 describes the timing parameters of the SDIO/MMC interface.

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Table 2-99 Timing parameters of the SDIO/MMC interface

Speed Mode	Maximum Frequency (MHz)/Cycle (ns)		Minimum Hold Time	Minimum Setup Time	Maximum Output Delay of the Card	High Level Duration of the Card Clock
MMC_HS400	100 MHz	10 ns	0.4 ns	0.4 ns	N/A	(0.45−0.55) clock cycle
MMC_HS200	100 MHz	10 ns	0.8ns	1.4 ns	N/A	
MMC DDR (CMD line)	50 MHz	20 ns	3.0 ns	3.0 ns	13.7 ns	
MMC DDR (DAT line)	50 MHz	20 ns	2.5 ns	2.5 ns	7.0 ns	
MMC_HS	50 MHz	20 ns	3.0 ns	3.0 ns	13.7 ns	
SD_SDR104	100 MHz	10 ns	0.8 ns	1.4 ns	10.0 ns	
SD_SDR50	100 MHz	10 ns	0.8 ns	3.0 ns	7.5 ns	
SD_DDR50 (CMD line)	50 MHz	20 ns	0.8 ns	6.0 ns	13.7 ns	
SD_DDR50 (DAT line)	50 MHz	20 ns	0.8 ns	3.0 ns	7 ns	
SD_SDR25	50 MHz	20 ns	2.0 ns	6.0 ns	14 ns	
SD_SDR12	25 MHz	40 ns	5.0 ns	5.0 ns	14 ns	
SD_HS	50 MHz	20 ns	2.0 ns	6.0 ns	14 ns	
SD_DS	25 MHz	40 ns	5.0 ns	5.0 ns	14 ns	
Identification mode	400 kHz	2.5 μs	5.0 ns	5.0 ns	50 ns	

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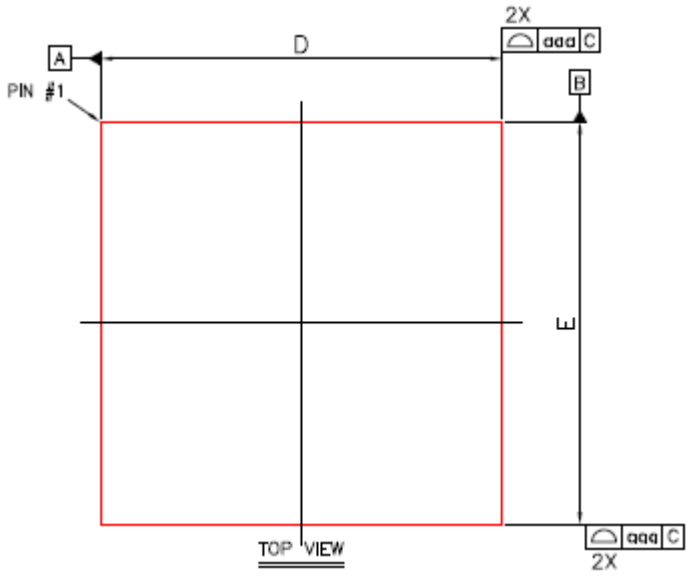
2 Hardware

2.1 Package and Pinout

2.1.1 Package

Hi3516C V200 uses the lead thin & fine-pitch ball grid array (LD TFBGA) package. It has 273 pins, its body size is 12 mm x 12 mm (0.47 in. x 0.47 in.), and its ball pitch is 0.65 mm (0.03 in.). Figure 2-1 to Figure 2-4 show package views. Figure 2-5 shows package dimensions.

Figure 2-1 Top view



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Figure 2-2 Bottom view

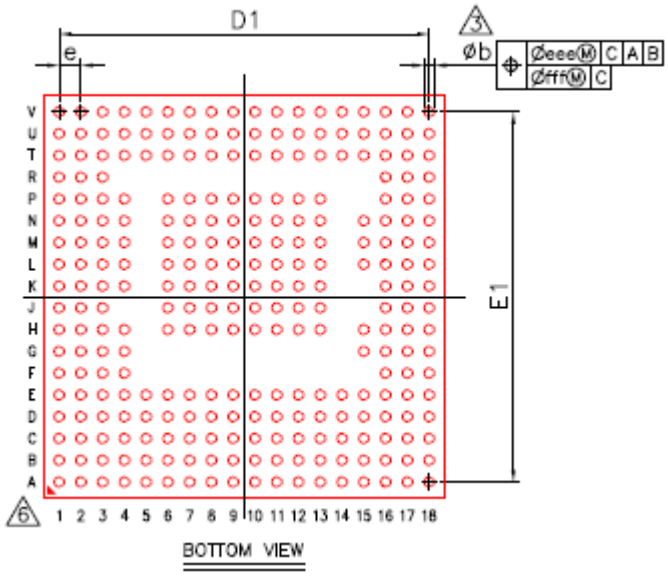


Figure 2-3 Side view

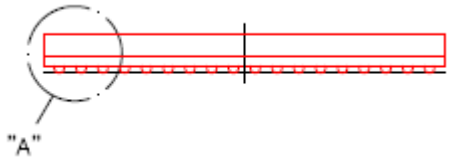
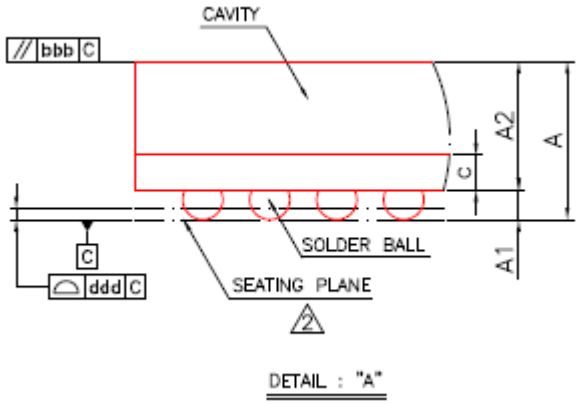


Figure 2-4 Enlarged view of detail "A"



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Figure 2-5 Package dimensions

Symbol	Dimension In mm			Dimension In Inch		
	MIN	NOM	MAX	MIN	NOM	MAX
A	1.10	1.17	1.24	0.043	0.046	0.049
A1	0.16	0.21	0.26	0.006	0.008	0.010
A2	0.91	0.96	1.01	0.036	0.038	0.040
c	0.22	0.26	0.30	0.009	0.010	0.012
D	11.90	12.00	12.10	0.469	0.472	0.476
E	11.90	12.00	12.10	0.469	0.472	0.476
D1	---	11.05	---	---	0.435	---
E1	---	11.05	---	---	0.435	---
e	---	0.65	---	---	0.026	---
b	0.25	0.30	0.35	0.010	0.012	0.014
aaa	0.15			0.006		
bbb	0.10			0.004		
ddd	0.08			0.003		
eee	0.15			0.006		
fff	0.08			0.003		
MD/ME	18/18					

NOTE :
1. CONTROLLING DIMENSION : MILLIMETER.
2. PRIMARY DATUM C AND SEATING PLANE ARE DEFINED BY THE SPHERICAL CROWNS OF THE SOLDER BALLS.
3. DIMENSION b IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER, PARALLEL TO PRIMARY DATUM C.
4. REFERANCE DOCUMENT : JEDEC PUBLICATION 95 DESIGN GUIDE 4.5
5. SPECIAL CHARACTERISTICS C CLASS: bbb, ddd
6. THE PATTERN OF PIN 1 FIDUCIAL IS FOR REFERENCE ONLY.

2.1.2 Pinout

Hi3516C V200 has 273 pins. Table 2-1 lists the pin quantity of Hi3516C V200 by type.

Table 2-1 Pin quantity of Hi3516C V200 by type

Pin Type	Quantity
Input/Output (I/O)	151
Digital power	23
Digital ground (GND)	72
Others/Analog power	7
Others/Analog GND	10
Double data rate (DDR) reference power	10
Total	273

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Pin Map

Figure 2-6 to Figure 2-9 show pin maps of Hi3516C V200.

Figure 2-6 Pin map part 1 of Hi3516C V200 (A1−J9)

	1	2	3	4	5	6	7	8	9
A	VSS	DDR_A11	VSS	DDR_A6	DDR_A4	DDR_BA1	DDR_CLK_P	VSS	DDR_DQ7
B	DDR_A7	DDR_A1	DDR_A14	DDR_A8	VSS	DDR_A10	DDR_CLK_N	VSS	DDR_DQ5
C	DDR_A13	VSS	DDR_A9	DDR_BA2	DDR_A12	DDR_WE_N	DDR_CAS_N	DDR_CKE	DDR_DQ4
D	DDR_RESET_N	DDR_A2	DDR_A0	VSS	DDR_CS_N	DDR_ODT	VSS	DDR_RAS_N	VSS
E	DDR_A5	VSS	DDR_A3	DDR_BA0	VSS	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_CK_DDR
F	DDR_ZQ	JTAG_TDO	JTAG_EN	JTAG_TMS					
G	JTAG_TCK	JTAG_TDI	JTAG_TRSTN	TEST_MODE					
H	VSS	AVDD_BAT	UART0_RXD	UART0_TXD		DVDD33	VDD	VDD	VSS
J	RTC_XIN	RTC_XOUT	AVSS_RTC			VSS	VSS	VSS	VSS

Figure 2-7 Pin map part 2 of Hi3516C V200 (A10−J18)

10	11	12	13	14	15	16	17	18
DDR_DQ1	VSS	DDR_DQS0_P	DDR_DQ8	DDR_DQ10	DDR_DQ14	DDR_DQ12	DDR_DQS1_P	VSS
DDR_DQ3	VSS	DDR_DQS0_N	DDR_DM0	VSS	DDR_DQ11	VSS	DDR_DQS1_N	VSS
DDR_DQ6	DDR_DQ2	DDR_DQ0	DDR_DM1	DDR_DQ9	DDR_DQ13	DDR_DQ15	SPI1_SCLK	SPI1_SDI
VSS	VSS	VSS	VSS	VSS	SPI1_CSN0	SPI1_SDO	SPI1_CSN1	EPHY_CLK
VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VSS	RMII_RXD0	VSS	RMII_CLK
						RMII_RXD1	RMII_RX_DV	RMII_TXD1
					VSS	RMII_TXD0	RMII_TX_EN	VSS
AVDD_DDRPLL	VDD	VDD	DVDD33		DVDDIO_RMII	MDIO	MDCK	EPHY_RST
VSS	VSS	VSS	VSS			UART1_CTSN	UART1_RXD	UART1_RTSN

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Figure 2-8 Pin map part 3 of Hi3516C V200 (K1−V9)

K	VSS	VSS	PWM2	AVDD33_PLL		AVDD11_PLL	VSS	VSS	VSS
L	XIN	XOUT	PWM3	AVDD33_RTC		AVSS_PLL	VSS	VSS	VSS
M	VSS	PWM0	PWM1	DVDD33		VDD	VSS	VSS	VSS
N	I2C2_SDA	I2C2_SCL	GPIO0_2	VSS		VDD	VSS	VSS	VSS
P	SAR_ADC_CH0	SAR_ADC_CH1	GPIO0_0	GPIO0_1		DVDD3318_SENS	VDD	VDD	DVDD3318_VI
R	SAR_ADC_CH2	VSS	SAR_ADC_CH3						
T	SPI0_SDO	SPI0_SDI	SENSOR_RSTN	SHUTTER_TRIG	AVDD3318_MIPI	AVSS3318_MIPI	GPIO1_0	GPIO1_2	GPIO1_5
U	SPI0_SCLK	SPI0_CSN	FLASH_TRIG	MIPI_D3P	MIPI_D1P	MIPI_CKP	MIPI_D0P	MIPI_D2P	GPIO1_6
V	VSS	SENSOR_CLK	AVSS3318_MIPI	MIPI_D3M	MIPI_D1M	MIPI_CKM	MIPI_D0M	MIPI_D2M	AVSS3318_MIPI

Figure 2-9 Pin map part 4 of Hi3516C V200 (K10−V18)

VSS	VSS	VSS	VDD			UART2_TXD	UART1_TXD	VSS
VSS	VSS	VSS	VDD		DVDD3318_EMMC	UART2_RXD	SFC_CLK	SFC_HOLD_IO3
VSS	VSS	VSS	VDD		VSS	SYS_RSTN_OUT	SFC_SIO_IO0	SFC_CSN0
VSS	VSS	VSS	VSS		DVDD18_SDIO	SFC_CSN1	SFC_SOI_IO1	SFC_WP_IO2
VDD	VDD	VSS	DVDD33			SDIO0_VOUT	SDIO0_CDATA2	SDIO0_CDATA3
						SDIO0_CARD_DE	SDIO0_CCLK_OU	SDIO0_CCMD
GPIO1_4	AVDD33_USB	AVSS_USB	USB_REXT	AVDD33_AC	AVSS_AC	SDIO0_CARD_PO	SDIO0_CDATA0	VSS
GPIO1_3	USB_PWREN	USB_DP	AVSS_AC	AC_INL	AC_MICBIAS	AC_OUTL	SDIO0_CWPR	SDIO0_CDATA1
GPIO1_1	USB_OVRCUR	USB_DM	AVSS_AC	AC_INR	AC_VREF	AC_OUTR	AVSS_AC	VSS

Pin Arrangement

Table 2-2 lists the pins of Hi3516C V200 based on their positions.

Table 2-2 Pin arrangement

Position	Pin Name	Position	Pin Name	Position	Pin Name
A1	VSS	F2	JTAG_TDO	N1	I2C2_SDA
A2	DDR_A11	F3	JTAG_EN	N2	I2C2_SCL

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Position	Pin Name	Position	Pin Name	Position	Pin Name
A3	VSS	F4	JTAG_TMS	N3	GPIO0_2
A4	DDR_A6	F16	RMII_RXD1	N4	VSS
A5	DDR_A4	F17	RMII_RX_DV	N6	VDD
A6	DDR_BA1	F18	RMII_TXD1	N7	VSS
A7	DDR_CLK_P	G1	JTAG_TCK	N8	VSS
A8	VSS	G2	JTAG_TDI	N9	VSS
A9	DDR_DQ7	G3	JTAG_TRSTN	N10	VSS
A10	DDR_DQ1	G4	TEST_MODE	N11	VSS
A11	VSS	G15	VSS	N12	VSS
A12	DDR_DQS0_P	G16	RMII_TXD0	N13	VSS
A13	DDR_DQ8	G17	RMII_TX_EN	N15	DVDD18_SDIO
A14	DDR_DQ10	G18	VSS	N16	SFC_CSN1
A15	DDR_DQ14	H1	VSS	N17	SFC_SOI_IO1
A16	DDR_DQ12	H2	AVDD_BAT	N18	SFC_WP_IO2
A17	DDR_DQS1_P	H3	UART0_RXD	P1	SAR_ADC_CH0
A18	VSS	H4	UART0_TXD	P2	SAR_ADC_CH1
B1	DDR_A7	H6	DVDD33	P3	GPIO0_0
B2	DDR_A1	H7	VDD	P4	GPIO0_1
B3	DDR_A14	H8	VDD	P6	DVDD3318_SENSOR
B4	DDR_A8	H9	VSS	P7	VDD
B5	VSS	H10	AVDD_DDRPLL	P8	VDD
B6	DDR_A10	H11	VDD	P9	DVDD3318_VI
B7	DDR_CLK_N	H12	VDD	P10	VDD
B8	VSS	H13	DVDD33	P11	VDD
B9	DDR_DQ5	H15	DVDDIO_RMII	P12	VSS
B10	DDR_DQ3	H16	MDIO	P13	DVDD33
B11	VSS	H17	MDCK	P16	SDIO0_VOUT
B12	DDR_DQS0_N	H18	EPHY_RST	P17	SDIO0_CDATA2
B13	DDR_DM0	J1	RTC_XIN	P18	SDIO0_CDATA3
B14	VSS	J2	RTC_XOUT	R1	SAR_ADC_CH2
B15	DDR_DQ11	J3	AVSS_RTC	R2	VSS

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Position	Pin Name	Position	Pin Name	Position	Pin Name
B16	VSS	J6	VSS	R3	SAR_ADC_CH3
B17	DDR_DQS1_N	J7	VSS	R16	SDIO0_CARD_DETECT
B18	VSS	J8	VSS	R17	SDIO0_CCLK_OUT
C1	DDR_A13	J9	VSS	R18	SDIO0_CCMD
C2	VSS	J10	VSS	T1	SPI0_SDO
C3	DDR_A9	J11	VSS	T2	SPI0_SDI
C4	DDR_BA2	J12	VSS	T3	SENSOR_RSTN
C5	DDR_A12	J13	VSS	T4	SHUTTER_TRIG
C6	DDR_WE_N	J16	UART1_CTSN	T5	AVDD3318_MIPI
C7	DDR_CAS_N	J17	UART1_RXD	T6	AVSS3318_MIPI
C8	DDR_CKE	J18	UART1_RTSN	T7	GPIO1_0
C9	DDR_DQ4	K1	VSS	T8	GPIO1_2
C10	DDR_DQ6	K2	VSS	T9	GPIO1_5
C11	DDR_DQ2	K3	PWM2	T10	GPIO1_4
C12	DDR_DQ0	K4	AVDD33_PLL	T11	AVDD33_USB
C13	DDR_DM1	K6	AVDD11_PLL	T12	AVSS_USB
C14	DDR_DQ9	K7	VSS	T13	USB_REXT
C15	DDR_DQ13	K8	VSS	T14	AVDD33_AC
C16	DDR_DQ15	K9	VSS	T15	AVSS_AC
C17	SPI1_SCLK	K10	VSS	T16	SDIO0_CARD_POWER_EN
C18	SPI1_SDI	K11	VSS	T17	SDIO0_CDATA0
D1	DDR_RESET_N	K12	VSS	T18	VSS
D2	DDR_A2	K13	VDD	U1	SPI0_SCLK
D3	DDR_A0	K16	UART2_TXD	U2	SPI0_CSN
D4	VSS	K17	UART1_TXD	U3	FLASH_TRIG
D5	DDR_CS_N	K18	VSS	U4	MIPI_D3P
D6	DDR_ODT	L1	XIN	U5	MIPI_D1P
D7	VSS	L2	XOUT	U6	MIPI_CKP
D8	DDR_RAS_N	L3	PWM3	U7	MIPI_D0P
D9	VSS	L4	AVDD33_RTC	U8	MIPI_D2P

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Position	Pin Name	Position	Pin Name	Position	Pin Name
D10	VSS	L6	AVSS_PLL	U9	GPIO1_6
D11	VSS	L7	VSS	U10	GPIO1_3
D12	VSS	L8	VSS	U11	USB_PWREN
D13	VSS	L9	VSS	U12	USB_DP
D14	VSS	L10	VSS	U13	AVSS_AC
D15	SPI1_CSN0	L11	VSS	U14	AC_INL
D16	SPI1_SDO	L12	VSS	U15	AC_MICBIAS
D17	SPI1_CSN1	L13	VDD	U16	AC_OUTL
D18	EPHY_CLK	L15	DVDD3318_EMMC	U17	SDIO0_CWPR
E1	DDR_A5	L16	UART2_RXD	U18	SDIO0_CDATA1
E2	VSS	L17	SFC_CLK	V1	VSS
E3	DDR_A3	L18	SFC_HOLD_IO3	V2	SENSOR_CLK
E4	DDR_BA0	M1	VSS	V3	AVSS3318_MIPI
E5	VSS	M2	PWM0	V4	MIPI_D3M
E6	VDDIO_DDR	M3	PWM1	V5	MIPI_D1M
E7	VDDIO_DDR	M4	DVDD33	V6	MIPI_CKM
E8	VDDIO_DDR	M6	VDD	V7	MIPI_D0M
E9	VDDIO_CK_DDR	M7	VSS	V8	MIPI_D2M
E10	VDDIO_DDR	M8	VSS	V9	AVSS3318_MIPI
E11	VDDIO_DDR	M9	VSS	V10	GPIO1_1
E12	VDDIO_DDR	M10	VSS	V11	USB_OVRCUR
E13	VDDIO_DDR	M11	VSS	V12	USB_DM
E14	VDDIO_DDR	M12	VSS	V13	AVSS_AC
E15	VSS	M13	VDD	V14	AC_INR
E16	RMII_RXD0	M15	VSS	V15	AC_VREF
E17	VSS	M16	SYS_RSTN_OUT	V16	AC_OUTR
E18	RMII_CLK	M17	SFC_SIO_IO0	V17	AVSS_AC
F1	DDR_ZQ	M18	SFC_CSN0	V18	VSS

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2.2 Pin Description

2.2.1 Pin Types

Table 2-3 describes the I/O pin types.

Table 2-3 I/O pin types

I/O	Description
I	Input signal
I_{PD}	Input signal, internal pull-down
I_{PU}	Input signal, internal pull-up
I_S	Input signal with a Schmitt trigger
I_{SPD}	Input signal with a Schmitt trigger, internal pull-down
I_{SPU}	Input signal with a Schmitt trigger, internal pull-up
O	Output signal
O_{OD}	Output open drain (OD)
I/O	Bidirectional (input/output) signal
I_{PD}/O	Bidirectional signal, input pull-down
I_{PU}/O	Bidirectional signal, input pull-up
I_{SPU}/O	Bidirectional signal with a Schmitt trigger, input pull-up
I_{SPD}/O	Bidirectional signal with a Schmitt trigger, input pull-down
I_{PD}/O_{OD}	Bidirectional signal, input pull-down and output OD
I_{PU}/O_{OD}	Bidirectional signal, input pull-up and output OD
I_S/O	Bidirectional signal, input with a Schmitt trigger
I_S/O_{OD}	Bidirectional signal, input with a Schmitt trigger and output OD
CIN	Crystal oscillator input
COUT	Crystal oscillator output
P	Power supply
G	GND

2.2.2 Pin Details

ACODEC Pins

Table 2-4 describes audio coder/decoder (ACODEC) pins.

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Table 2-4 ACODEC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
U14	AC_INL	I	N/A	N/A	Audio-left channel input or positive end of the audio differential input
V14	AC_INR	I	N/A	N/A	Audio-right channel input or negative end of the audio differential input
U15	AC_MICBIAS	I/O	N/A	2.1	Audio input microphone bias
U16	AC_OUTL	O	N/A	N/A	Audio-left channel output
V16	AC_OUTR	O	N/A	N/A	Audio-right channel output
V15	AC_VREF	P	N/A	1.65	Audio reference source. This pin connects to a 10 µF external capacitor that connects to a 100 nF ceramic capacitor with low equivalent series resistance (ESR) in parallel. The 100 nF capacitor should be placed as close as possible to this pin.
T14	AVDD33_AC	P	N/A	3.3	Analog power of the audio interface
T15 U13 V13 V17	AVSS_AC	G	N/A	N/A	Analog GND of the audio interface

DDR Pins

Table 2-5 describes DDR pins.

Table 2-5 DDR pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
D3	DDR_A0	O	N/A	1.35/1.5/1.8	Address signal 0 of the DDR synchronous dynamic random access memory (SDRAM)

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
B2	DDR_A1	O	N/A	1.35/1.5/1.8	Address signal 1 of the DDR SDRAM
D2	DDR_A2	O	N/A	1.35/1.5/1.8	Address signal 2 of the DDR SDRAM
E3	DDR_A3	O	N/A	1.35/1.5/1.8	Address signal 3 of the DDR SDRAM
A5	DDR_A4	O	N/A	1.35/1.5/1.8	Address signal 4 of the DDR SDRAM
E1	DDR_A5	O	N/A	1.35/1.5/1.8	Address signal 5 of the DDR SDRAM
A4	DDR_A6	O	N/A	1.35/1.5/1.8	Address signal 6 of the DDR SDRAM
B1	DDR_A7	O	N/A	1.35/1.5/1.8	Address signal 7 of the DDR SDRAM
B4	DDR_A8	O	N/A	1.35/1.5/1.8	Address signal 8 of the DDR SDRAM
C3	DDR_A9	O	N/A	1.35/1.5/1.8	Address signal 9 of the DDR SDRAM
B6	DDR_A10	O	N/A	1.35/1.5/1.8	Address signal 10 of the DDR SDRAM
A2	DDR_A11	O	N/A	1.35/1.5/1.8	Address signal 11 of the DDR SDRAM
C5	DDR_A12	O	N/A	1.35/1.5/1.8	Address signal 12 of the DDR SDRAM
C1	DDR_A13	O	N/A	1.35/1.5/1.8	Address signal 13 of the DDR SDRAM
B3	DDR_A14	O	N/A	1.35/1.5/1.8	Address signal 14 of the DDR SDRAM
E4	DDR_BA0	O	N/A	1.35/1.5/1.8	Bank address signal 0 of the DDR SDRAM
A6	DDR_BA1	O	N/A	1.35/1.5/1.8	Bank address signal 1 of the DDR SDRAM
C4	DDR_BA2	O	N/A	1.35/1.5/1.8	Bank address signal 2 of the DDR SDRAM
C7	DDR_CAS_N	O	N/A	1.35/1.5/1.8	Column address select signal of the DDR SDRAM
C8	DDR_CKE	O	N/A	1.35/1.5/1.8	Clock enable signal of the DDR SDRAM

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
B7	DDR_CLK_N	O	N/A	1.35/1.5/1.8	Negative differential clock of the DDR SDRAM
A7	DDR_CLK_P	O	N/A	1.35/1.5/1.8	Positive differential clock of the DDR SDRAM
D5	DDR_CS_N	O	N/A	1.35/1.5/1.8	DDR chip select (CS) signal
B13	DDR_DM0	O	N/A	1.35/1.5/1.8	Data mask signal 0 of the DDR SDRAM
C13	DDR_DM1	O	N/A	1.35/1.5/1.8	Data mask signal 1 of the DDR SDRAM
C12	DDR_DQ0	I/O	N/A	1.35/1.5/1.8	Data line 0 of the DDR SDRAM
A10	DDR_DQ1	I/O	N/A	1.35/1.5/1.8	Data line 1 of the DDR SDRAM
C11	DDR_DQ2	I/O	N/A	1.35/1.5/1.8	Data line 2 of the DDR SDRAM
B10	DDR_DQ3	I/O	N/A	1.35/1.5/1.8	Data line 3 of the DDR SDRAM
C9	DDR_DQ4	I/O	N/A	1.35/1.5/1.8	Data line 4 of the DDR SDRAM
B9	DDR_DQ5	I/O	N/A	1.35/1.5/1.8	Data line 5 of the DDR SDRAM
C10	DDR_DQ6	I/O	N/A	1.35/1.5/1.8	Data line 6 of the DDR SDRAM
A9	DDR_DQ7	I/O	N/A	1.35/1.5/1.8	Data line 7 of the DDR SDRAM
A13	DDR_DQ8	I/O	N/A	1.35/1.5/1.8	Data line 8 of the DDR SDRAM
C14	DDR_DQ9	I/O	N/A	1.35/1.5/1.8	Data line 9 of the DDR SDRAM
A14	DDR_DQ10	I/O	N/A	1.35/1.5/1.8	Data line 10 of the DDR SDRAM
B15	DDR_DQ11	I/O	N/A	1.35/1.5/1.8	Data line 11 of the DDR SDRAM
A16	DDR_DQ12	I/O	N/A	1.35/1.5/1.8	Data line 12 of the DDR SDRAM
C15	DDR_DQ13	I/O	N/A	1.35/1.5/1.8	Data line 13 of the DDR SDRAM
A15	DDR_DQ14	I/O	N/A	1.35/1.5/1.8	Data line 14 of the DDR SDRAM
C16	DDR_DQ15	I/O	N/A	1.35/1.5/1.8	Data line 15 of the DDR SDRAM
B12	DDR_DQS0_N	I/O	N/A	1.35/1.5/1.8	Negative data strobe (DQS) signal 0 of the DDR, corresponding to DQ[7:0]

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
A12	DDR_DQS0_P	I/O	N/A	1.35/1.5/1.8	Positive DQS signal 0 of the DDR, corresponding to DQ[7:0]
B17	DDR_DQS1_N	I/O	N/A	1.35/1.5/1.8	Negative DQS signal 1 of the DDR, corresponding to DQ[15:8]
A17	DDR_DQS1_P	I/O	N/A	1.35/1.5/1.8	Positive DQS signal 1 of the DDR, corresponding to DQ[15:8]
D6	DDR_ODT	I/O	N/A	1.35/1.5/1.8	DDR pin connected to an external matched reference resistor
D8	DDR_RAS_N	O	N/A	1.35/1.5/1.8	Row address select signal of the DDR SDRAM
D1	DDR_RESET_N	O	N/A	1.35/1.5/1.8	Reset signal of the DDR SDRAM
C6	DDR_WE_N	O	N/A	1.35/1.5/1.8	Write enable signal of the DDR SDRAM
F1	DDR_ZQ	I/O	N/A	N/A	Pin connected to an external reference resistor for calibrating the on-chip DDR I/O ZQ. This pin is different from the ZQ pin of the DDR.
H10	AVDD_DDRPLL	P	N/A	3.3	Analog power of the DDR phase-locked loop (PLL)
E9	VDDIO_CK_DDR	P	N/A	1.35/1.5/1.8	I/O power for the DDR clock
E6 E7 E8 E10 E11 E12 E13 E14	VDDIO_DDR	P	N/A	1.35/1.5/1.8	Other I/O power for the DDR

EPHY Pins

Table 2-6 describes external physical layer (EPHY) pins.

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Table 2-6 EPHY pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
D18	EPHY_CLK	I/O	Configurable (for details, see the PADCTRL_REG13 register)	1.8/3.3	**Function 0: GPIO3_0** General-purpose input/output (GPIO) **Function 1: EPHY_CLK** Working clock of the EPHY **Function 3: VO_DATA1** Video output (VO) data 1 **Function 4: SDIO1_CDATA2** Secure digital input/output 1 (SDIO1) data 2
H18	EPHY_RST	I_{PD}/O	Configurable (for details, see the PADCTRL_REG11 register)	1.8/3.3	**Function 0: GPIO2_7** GPIO **Function 1: EPHY_RST** EPHY reset **Function 2: BOOT_SEL** Boot enable 0: booting from the serial peripheral interface (SPI) flash 1: booting from the embedded multimedia card (eMMC) **Function 3: VO_HS** VO horizontal sync (HS) signal **Function 4: SDIO1_CARD_POWER_EN** SDIO1 power enable signal. The value 1 indicates power-on.

Sensor Pins

Table 2-7 describes sensor pins.

Table 2-7 Sensor pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
U3	FLASH_TRIG	I_{PD}/O	Configurable (for details, see the PADCTRL_REG2 register)	3.3	**Function 0: GPIO0_6** GPIO **Function 1: FLASH_TRIG** Camera flash control

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					Function 2: SFC_EMMC_BOOT_MODE Boot address mode of the SPI NOR flash when BOOT_SEL and SFC_DEVICE_MODE are 0 0: 3-byte address mode 1: 4-byte address mode Boot mode of the SPI NAND flash when BOOT_SEL is 0 and SFC_DEVICE_MODE is 1 0: 1-wire boot mode 1: 4-wire boot mode eMMC boot mode when BOOT_SEL is 1 0: 4-wire boot mode 1: 8-wire boot mode **Function 3: SPI1_CSN1** SPI1 CS1 **Function 4: VI_VS_BT1120** Video input (VI) vertical sync (VS) signal
V2	SENSOR_CLK	I/O	Configurable (for details, see the PADCTRL_REG0 register)	3.3/1.8	**Function 0: GPIO0_4** GPIO **Function 1: SENSOR_CLK** Sensor working clock
T3	SENSOR_RSTN	I/O	Configurable (for details, see the PADCTRL_REG1 register)	3.3/1.8	**Function 0: SENSOR_RSTN** Sensor reset signal **Function 1: GPIO0_5** GPIO
T4	SHUTTER_TRIG	I_{PD}/O	Configurable (for details, see the PADCTRL_REG3 register)	3.3	**Function 0: GPIO0_7** GPIO **Function 1: SHUTTER_TRIG** Shutter control **Function 2: SFC_DEVICE_MODE**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					SPI flash from which the system boots (that is, type of the flash memory connected to SFC_CSN0) 0: SPI NOR flash 1: SPI NAND flash **Function 4: VI_HS_BT1120** VI HS signal

GPIO Pins

Table 2-8 describes GPIO pins.

Table 2-8 GPIO pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
P3	GPIO0_0	I/O	Configurable (for details, see the PADCTRL_REG32 register)	3.3	**Function 0: GPIO0_0** GPIO **Function 1: IR_IN** Infrared (IR) input **Function 2: TEMPER_DQ** Temperature measurement. It is used to communicate with the external temperature measurement chip.
P4	GPIO0_1	I/O	Configurable (for details, see the PADCTRL_REG33 register)	3.3	**Function 0: GPIO0_1** GPIO **Function 2: TEMPER_DQ** Temperature measurement. It is used to communicate with the external temperature measurement chip.
N3	GPIO0_2	I/O	Configurable (for details, see the PADCTRL_REG34 register)	3.3	**Function 0: GPIO0_2** GPIO **Function 2: TEMPER_DQ** Temperature measurement. It is used to communicate with the external temperature measurement

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					chip.
T7	GPIO1_0	I/O	Configurable (for details, see the PADCTRL_REG35 register)	3.3/1.8	**Function 0: GPIO1_0** GPIO **Function 1: VI_DATA13** VI data 13 **Function 3: I2S_BCLK_TX** Transmit (TX) clock of the inter-IC sound (I^2S) or pulse code modulation (PCM) interface **Function 4: PWM0** Pulse-width modulation (PWM) output 0
V10	GPIO1_1	I/O	Configurable (for details, see the PADCTRL_REG36 register)	3.3/1.8	**Function 0: GPIO1_1** GPIO **Function 1: VI_DATA10** VI data 10 **Function 3: I2S_SD_TX** TX data of the I^2S or PCM interface **Function 4: UART1_TXD** Universal asynchronous receiver transmitter 1 (UART1) TX data
T8	GPIO1_2	I/O	Configurable (for details, see the PADCTRL_REG37 register)	3.3/1.8	**Function 0: GPIO1_2** GPIO **Function 1: VI_DATA12** VI data 12 **Function 3: I2S_MCLK** Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC. **Function 4: UART1_CTSN** UART1 modem status input, active low
U10	GPIO1_3	I/O	Configurable (for details, see the PADCTRL_REG38 register)	3.3/1.8	**Function 0: GPIO1_3** GPIO **Function 1: VI_DATA11**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					VI data 11 **Function 3: I2S_WS_TX** I^2S TX audio channel select signal or PCM TX VS signal **Function 4: UART2_RXD** UART2 receive (RX) data
T10	GPIO1_4	I/O	Configurable (for details, see the PADCTRL_REG39 register)	3.3/1.8	**Function 0: GPIO1_4** GPIO **Function 1: VI_DATA15** VI data 15 **Function 2: VI_VS_SEN** VI VS signal **Function 3: I2S_WS_RX** I^2S RX audio channel select signal or PCM TX VS signal **Function 4: UART1_RXD** UART1 RX data
T9	GPIO1_5	I/O	Configurable (for details, see the PADCTRL_REG40 register)	3.3/1.8	**Function 0: GPIO1_5** GPIO **Function 1: VI_DATA14** VI data 14 **Function 2: VI_HS_SEN** VI HS signal **Function 3: I2S_BCLK_RX** RX clock of the I^2S or PCM interface **Function 4: UART1_RTSN** UART1 modem status output, active low
U9	GPIO1_6	I/O	Configurable (for details, see the PADCTRL_REG41 register)	3.3/1.8	**Function 0: GPIO1_6** GPIO **Function 1: VI_DATA9** VI data 9 **Function 3: I2S_SD_RX** RX data of the I^2S or PCM interface **Function 4: UART2_TXD** UART2 TX data

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I²C2 Pins

Table 2-9 describes inter-integrated circuit 2 (I^2C2) pins.

Table 2-9 I^2C2 pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
N2	I2C2_SCL	I/O	Configurable (for details, see the PADCTRL_REG25 register)	3.3	**Function 0: GPIO4_4** GPIO **Function 1: I2C2_SCL** I^2C2 bus clock
N1	I2C2_SDA	I/O	Configurable (for details, see the PADCTRL_REG24 register)	3.3	**Function 0: GPIO4_3** GPIO **Function 1: I2C2_SDA** I^2C2 bus data/address

JTAG Pins

Table 2-10 describes Joint Test Action Group (JTAG) pins.

Table 2-10 JTAG pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
F3	JTAG_EN	I_{PD}	-	3.3	JTAG pin enable 0: disabled 1: enabled
G1	JTAG_TCK	I/O	Configurable (for details, see the PADCTRL_REG65 register)	3.3	**Function 0: GPIO6_6** GPIO **Function 1: JTAG_TCK** JTAG clock signal **Function 2: SPI1_SCLK** SPI1 clock signal **Function 3: I2S_WS_TX** I^2S TX audio channel select signal or PCM TX VS signal **Function 4: I2C1_SCL** I^2C1 bus clock
G2	JTAG_TDI	I/O	Configurable (for details, see the	3.3	**Function 0: GPIO7_1**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
			PADCTRL_REG68 register)		GPIO **Function 1: JTAG_TDI** JTAG input data **Function 2: SPI1_SDI** SPI1 input data **Function 3: I2S_BCLK_TX** TX clock of the I^2S or PCM interface
F2	JTAG_TDO	I/O	Configurable (for details, see the PADCTRL_REG67 register)	3.3	**Function 0: GPIO7_0** GPIO **Function 1: JTAG_TDO** JTAG output data **Function 2: SPI1_SDO** SPI1 output data **Function 3: I2S_SD_RX** RX data of the I^2S or PCM interface **Function 4: I2C1_SDA** I^2C1 bus data/address
F4	JTAG_TMS	I/O	Configurable (for details, see the PADCTRL_REG66 register)	3.3	**Function 0: GPIO6_7** GPIO **Function 1: JTAG_TMS** JTAG mode select input **Function 2: SPI1_CSN0** SPI1 CS0 **Function 3: I2S_SD_TX** TX data of the I^2S or PCM interface
G3	JTAG_TRSTN	I/O	Configurable (for details, see the PADCTRL_REG64 register)	3.3	**Function 0: GPIO6_5** GPIO **Function 1: JTAG_TRSTN** JTAG reset signal **Function 2: SPI1_CSN1** SPI1 CS1 **Function 3: I2S_MCLK** Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC.

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NOTE

Functions 0, 2, and 3 are valid only when the input of the JTAG_EN pin is 0, and function 1 is valid only when the input of the JTAG_EN pin is 1.

MDIO Pins

Table 2-11 describes management data input/output (MDIO) pins.

Table 2-11 MDIO pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
H17	MDCK	I_{PD}/O	Configurable (for details, see the PADCTRL_REG14 register)	1.8/3.3	**Function 0: GPIO3_1** GPIO **Function 1: MDCK** MDIO interface clock **Function 2: BOOTROM_SEL** BOOTROM boot enable 0: booting from other space 1: booting from the BOOTROM **Function 3: VO_DATA6** VO data 6
H16	MDIO	I/O	Configurable (for details, see the PADCTRL_REG15 register)	1.8/3.3	**Function 0: GPIO3_2** GPIO **Function 1: MDIO** Data signal of the MDIO interface **Function 3: VO_DATA4** VO data 4

MIPI Pins

The mobile industry processor interface (MIPI) of Hi3516C V200 supports the differential and single-ended modes. Table 2-12 describes the functions of MIPI pins in the two modes.

Table 2-12 Functions of MIPI pins in two modes

Pin Name	Function in Differential Mode	Function in Single-Ended Mode
MIPI_CKM	MIPI differential clock	VI_DATA3
MIPI_CKP	MIPI differential clock	VI_DATA4

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Pin Name	Function in Differential Mode	Function in Single-Ended Mode
MIPI_D0M	MIPI differential data 0	VI_DATA5
MIPI_D0P	MIPI differential data 0	VI_DATA6
MIPI_D1M	MIPI differential data 1	VI_ DATA1
MIPI_D1P	MIPI differential data 1	VI_ DATA2
MIPI_D2M	MIPI differential data 2	VI_DATA7
MIPI_D2P	MIPI differential data 2	VI_DATA8
MIPI_D3M	MIPI differential data 3	VI_CLK
MIPI_D3P	MIPI differential data 3	VI_DATA0

Table 2-13 describes MIPI pins.

Table 2-13 MIPI pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
V6	MIPI_CKM	I	N/A	3.3/1.8	VI_DATA3 in single-ended mode or differential clock of MIPI
U6	MIPI_CKP	I	N/A	3.3/1.8	VI_DATA4 in single-ended mode or differential clock of MIPI
V7	MIPI_D0M	I	N/A	3.3/1.8	VI_DATA5 in single-ended mode or differential data 0 of MIPI
U7	MIPI_D0P	I	N/A	3.3/1.8	VI_DATA6 in single-ended mode or differential data 0 of MIPI
V5	MIPI_D1M	I	N/A	3.3/1.8	VI_DATA1 in single-ended mode or differential data 1 of MIPI
U5	MIPI_D1P	I	N/A	3.3/1.8	VI_DATA2 in single-ended mode or differential data 1 of MIPI
V8	MIPI_D2M	I	N/A	3.3/1.8	VI_DATA7 in single-ended mode or differential data 2 of MIPI

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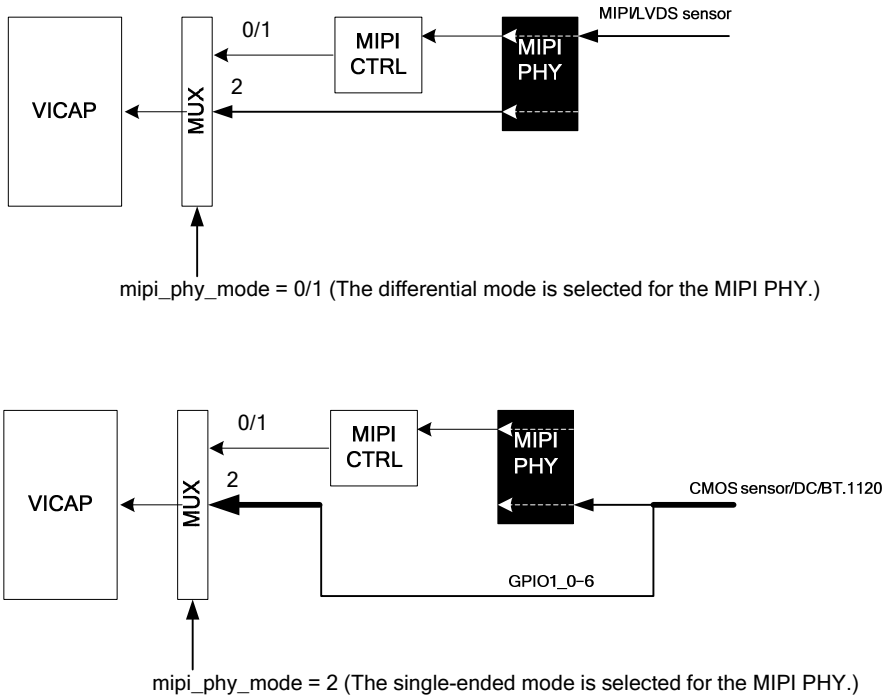
Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
U8	MIPI_D2P	I	N/A	3.3/1.8	VI_DATA8 in single-ended mode or differential data 2 of MIPI
V4	MIPI_D3M	I	N/A	3.3/1.8	VI_CLK in single-ended mode or differential data 3 of MIPI
U4	MIPI_D3P	I	N/A	3.3/1.8	VI_DATA0 in single-ended mode or differential data 3 of MIPI
T5	AVDD3318_MIPI	P	N/A	3.3/1.8	Analog power of the MIPI PHY
T6 V3 V9	AVSS3318_MIPI	G	N/A	N/A	Analog GND of the MIPI PHY

CAUTION

As the video capture (VICAP) supports only one video input, the video input paths of the MIPI PHY and VI pin are mutually exclusive (dual sensor inputs are not supported). The mipi_phy_mode register determines the video input of the VICAP. See Figure 2-10.

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Figure 2-10 Video input paths of the MIPI PHY and VI pin



PWM Pins

Table 2-14 describes PWM pins.

Table 2-14 PWM pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
M2	PWM0	I/O	Configurable (for details, see the PADCTRL_REG70 register)	3.3	**Function 0: PMC_PWM** Power management controller (PMC) PWM output **Function 1: GPIO7_2** GPIO **Function 2: PWM0** PWM output 0
M3	PWM1	I/O	Configurable (for details, see the PADCTRL_REG71 register)	3.3	**Function 0: PWM1** PWM output 1 **Function 1: GPIO7_3** GPIO
K3	PWM2	I/O	Configurable (for details, see the PADCTRL_REG72	3.3	**Function 0: PWM2** PWM output 2

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
			register)		**Function 1: GPIO7_4** GPIO **Function 2: SPI1_CSN1** SPI1 CS1
L3	PWM3	I/O	Configurable (for details, see the PADCTRL_REG73 register)	3.3	**Function 0: PWM3** PWM output 3 **Function 1: GPIO7_5** GPIO

RMII Pins

Table 2-15 describes reduced media independent interface (RMII) pins.

Table 2-15 RMII pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
E18	RMII_CLK	I/O	Configurable (for details, see the PADCTRL_REG4 register)	1.8/3.3	**Function 0: GPIO2_0** GPIO **Function 1: RMII_CLK** RMII clock **Function 3: VO_CLK** VO clock signal **Function 4: SDIO1_CCLK_OUT** SDIO1 output working clock for the card
F17	RMII_RX_DV	I/O	Configurable (for details, see the PADCTRL_REG8 register)	1.8/3.3	**Function 0: GPIO2_4** GPIO **Function 1: RMII_RX_DV** RX data validity and carrier sense signal **Function 3: VO_DATA7** VO data 7 **Function 4: SDIO1_CDATA0** SDIO1 data 0
E16	RMII_RXD0	I/O	Configurable (for details, see the PADCTRL_REG9 register)	1.8/3.3	**Function 0: GPIO2_5** GPIO **Function 1: RMII_RXD0**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					RX data 0 **Function 3: VO_DATA2** VO data 2 **Function 4: SDIO1_CDATA3** SDIO1 data 3
F16	RMII_RXD1	I/O	Configurable (for details, see the PADCTRL_REG10 register)	1.8/3.3	**Function 0: GPIO2_6** GPIO **Function 1: RMII_RXD1** RX data 1 **Function 3: VO_DATA3** VO data 3 **Function 4: SDIO1_CCMD** SDIO1 command
G17	RMII_TX_EN	I/O	Configurable (for details, see the PADCTRL_REG5 register)	1.8/3.3	**Function 0: GPIO2_1** GPIO **Function 1: RMII_TX_EN** TX enable **Function 3: VO_VS** VO VS signal **Function 4: SDIO1_CARD_DETECT** SDIO1 card detection signal, active low
G16	RMII_TXD0	I/O	Configurable (for details, see the PADCTRL_REG6 register)	1.8/3.3	**Function 0: GPIO2_2** GPIO **Function 1: RMII_TXD0** TX data 0 **Function 3: VO_DATA5** VO data 5 **Function 4: SDIO1_CWPR** SDIO1 card write protection detection
F18	RMII_TXD1	I/O	Configurable (for details, see the PADCTRL_REG7 register)	1.8/3.3	**Function 0: GPIO2_3** GPIO **Function 1: RMII_TXD1** TX data 1 **Function 3: VO_DE** VO data validity signal **Function 4: SDIO1_CDATA1**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					SDIO1 data 1

RTC Pins

Table 2-16 describes real-time clock (RTC) pins.

Table 2-16 RTC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
J1	RTC_XIN	CIN	N/A	N/A	RTC crystal input
J2	RTC_XOUT	COUT	N/A	N/A	RTC crystal drive output
H2	AVDD_BAT	P	N/A	3.3	RTC external battery power
L4	AVDD33_RTC	P	N/A	3.3	RTC analog power
J3	AVSS_RTC	G	N/A	N/A	RTC analog GND

SAR_ADC Pins

Table 2-17 describes successive approximation register analog-to-digital converter (SAR_ADC) pins.

Table 2-17 SAR_ADC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
P1	SAR_ADC_CH0	I/O	Configurable (for details, see the PADCTRL_REG74 register)	3.3	**Function 0: SAR_ADC_CH0** LSADC channel 0 **Function 1: GPIO7_6** GPIO
P2	SAR_ADC_CH1	I/O	Configurable (for details, see the PADCTRL_REG75 register)	3.3	**Function 0: SAR_ADC_CH1** LSADC channel 1 **Function 1: GPIO7_7** GPIO

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
R1	SAR_ADC_CH2	I/O	Configurable (for details, see the PADCTRL_REG76 register)	3.3	**Function 0: SAR_ADC_CH2** LSADC channel 2 **Function 1: GPIO8_0** GPIO
R3	SAR_ADC_CH3	I/O	Configurable (for details, see the PADCTRL_REG77 register)	3.3	**Function 0: SAR_ADC_CH3** LSADC channel 3 **Function 1: GPIO8_1** GPIO

SDIO Pins

Table 2-18 describes SDIO pins.

Table 2-18 SDIO pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
R16	SDIO0_CARD_DETECT	I/O	Configurable (for details, see the PADCTRL_REG43 register)	3.3	**Function 0: GPIO4_7** GPIO **Function 1: SDIO0_CARD_DETECT** SDIO0 card detection signal, active low
T16	SDIO0_CARD_POWER_EN	I/O	Configurable (for details, see the PADCTRL_REG42 register)	3.3	**Function 0: GPIO1_7** GPIO **Function 1: SDIO0_CARD_POWER_EN** SDIO0 power enable signal. The value 1 indicates power-on.
R17	SDIO0_CCLK_OUT	I/O	Configurable (for details, see the PADCTRL_REG45 register)	3.3/1.8	**Function 0: GPIO5_1** GPIO **Function 1: SDIO0_CCLK_OUT** SDIO0 output working clock for the card

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
R18	SDIO0_CCMD	I/O	Configurable (for details, see the PADCTRL_REG46 register)	3.3/1.8	**Function 0: GPIO5_2** GPIO **Function 1: SDIO0_CCMD** SDIO0 command
T17	SDIO0_CDATA0	I/O	Configurable (for details, see the PADCTRL_REG47 register)	3.3/1.8	**Function 0: GPIO5_3** GPIO **Function 1: SDIO0_CDATA0** SDIO0 data 0
U18	SDIO0_CDATA1	I/O	Configurable (for details, see the PADCTRL_REG48 register)	3.3/1.8	**Function 0: TEST_CLK** Test clock **Function 1: SDIO0_CDATA1** SDIO0 data 1 **Function 2: GPIO5_4** GPIO
P17	SDIO0_CDATA2	I/O	Configurable (for details, see the PADCTRL_REG49 register)	3.3/1.8	**Function 0: GPIO5_5** GPIO **Function 1: SDIO0_CDATA2** SDIO0 data 2
P18	SDIO0_CDATA3	I/O	Configurable (for details, see the PADCTRL_REG50 register)	3.3/1.8	**Function 0: GPIO5_6** GPIO **Function 1: SDIO0_CDATA3** SDIO0 data 3
U17	SDIO0_CWPR	I/O	Configurable (for details, see the PADCTRL_REG44 register)	3.3	**Function 0: GPIO5_0** GPIO **Function 1: SDIO0_CWPR** SDIO0 card write protection detection
P16	SDIO0_VOUT	P	N/A	3.3/1.8	Power output of the pull-up resistor of SDIO0

SFC Pins

Table 2-19 describes SPI flash controller (SFC) pins.

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Table 2-19 SFC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
L17	SFC_CLK	O	Configurable (for details, see the PADCTRL_REG51 register)	3.3/1.8	**Function 0: SFC_CLK** Clock signal transmitted to the SPI flash **Function 1: EMMC_CLK** eMMC clock
M18	SFC_CSN0	O	Configurable (for details, see the PADCTRL_REG56 register)	3.3/1.8	**Function 0: SFC_CSN0** CS0 of the SPI flash for mounting the SPI flash from which the system boots, active low **Function 1: EMMC_RST_N** eMMC reset
N16	SFC_CSN1	O	Configurable (for details, see the PADCTRL_REG57 register)	3.3/1.8	**Function 0: SFC_CSN1** CS1 of the SPI flash, active low **Function 1: EMMC_POWER_EN** eMMC power enable signal. The value 1 indicates power-on.
L18	SFC_HOLD_IO3	I/O	Configurable (for details, see the PADCTRL_REG55 register)	3.3/1.8	**Function 0: SFC_HOLD_IO3** Hold function in standard SPI mode, active low Hold function in dual-SPI mode, active low Data I/O signal 3 in quad-SPI mode **Function 1: EMMC_DATA0** eMMC data 0
M17	SFC_SIO_IO0	I/O	Configurable (for details, see the PADCTRL_REG52 register)	3.3/1.8	**Function 0: SFC_SIO_IO0** Data output signal in standard SPI mode Data I/O signal 0 in dual-SPI mode Data I/O signal 0 in quad-SPI mode **Function 1: EMMC_DATA4** eMMC data 4
N17	SFC_SOI_IO1	I/O	Configurable (for details, see the PADCTRL_REG53 register)	3.3/1.8	**Function 0: SFC_SOI_IO1** Data input signal in standard SPI mode Data I/O signal 1 in dual-SPI

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					mode Data I/O signal 1 in quad-SPI mode **Function 1: EMMC_DATA3** eMMC data 3
N18	SFC_WP_IO2	I/O	Configurable (for details, see the PADCTRL_REG54 register)	3.3/1.8	**Function 0: SFC_WP_IO2** Write protection function in standard SPI mode, active low Write protection function in dual-SPI mode, active low Data I/O signal 2 in quad-SPI mode **Function 1: EMMC_CMD** eMMC command/response

SPI0 Pins

Table 2-20 describes SPI0 pins.

Table 2-20 SPI0 pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
U2	SPI0_CSN	I/O	Configurable (for details, see the PADCTRL_REG19 register)	3.3/1.8	**Function 0: GPIO3_6** GPIO **Function 1: SPI0_CSN** SPI0 CS signal
U1	SPI0_SCLK	I/O	Configurable (for details, see the PADCTRL_REG16 register)	3.3/1.8	**Function 0: GPIO3_3** GPIO **Function 1: SPI0_SCLK** SPI0 clock signal **Function 2: I2C0_SCL** I^2C0 bus clock
T2	SPI0_SDI	I/O	Configurable (for details, see the PADCTRL_REG18 register)	3.3/1.8	**Function 0: GPIO3_5** GPIO **Function 1: SPI0_SDI** SPI0 input data
T1	SPI0_SDO	I/O	Configurable (for details, see the PADCTRL_REG17	3.3/1.8	**Function 0: GPIO3_4** GPIO

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
			register)		**Function 1: SPI0_SDO** SPI0 output data **Function 2: I2C0_SDA** I^2C0 bus data/address

SPI1 Pins

Table 2-21 describes SPI1 pins.

Table 2-21 SPI1 pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
D15	SPI1_CSN0	I/O	Configurable (for details, see the PADCTRL_REG23 register)	3.3	**Function 0: GPIO4_2** GPIO **Function 1: SPI1_CSN0** SPI1 CS0
D17	SPI1_CSN1	I/O	Configurable (for details, see the PADCTRL_REG12 register)	3.3	**Function 0: GPIO0_3** GPIO **Function 1: SPI1_CSN1** SPI1 CS1 **Function 3: VO_DATA0** VO data 0
C17	SPI1_SCLK	I/O	Configurable (for details, see the PADCTRL_REG20 register)	3.3	**Function 0: GPIO3_7** GPIO **Function 1: SPI1_SCLK** SPI1 clock signal **Function 2: I2C1_SCL** I^2C1 bus clock
C18	SPI1_SDI	I/O	Configurable (for details, see the PADCTRL_REG22 register)	3.3	**Function 0: GPIO4_1** GPIO **Function 1: SPI1_SDI** SPI1 input data
D16	SPI1_SDO	I/O	Configurable (for details, see the PADCTRL_REG21 register)	3.3	**Function 0: GPIO4_0** GPIO **Function 1: SPI1_SDO** SPI1 output data **Function 2: I2C1_SDA**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					I^2C1 bus data/address

SYS Pins

Table 2-22 describes system (SYS) pins.

Table 2-22 SYS pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
M16	SYS_RSTN_OUT	O	N/A	1.8/3.3	System reset output 0: reset 1: deassert reset
G4	TEST_MODE	I_{PD}	N/A	3.3	Mode select 0: functional mode 1: test mode

UART Pins

Table 2-23 describes UART pins.

Table 2-23 UART pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
H3	UART0_RXD	I_{PU}	Configurable (for details, see the PADCTRL_REG26 register)	3.3	UART0 RX data
H4	UART0_TXD	O	Configurable (for details, see the PADCTRL_REG27 register)	3.3	UART0 TX data
J16	UART1_CTSN	I/O	Configurable (for details, see the PADCTRL_REG60 register)	3.3/1.8	**Function 0: GPIO6_1** GPIO **Function 1: EMMC_DATA7** eMMC data 7 **Function 2: I2S_MCLK**

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					Main clock of the I^{2}S or PCM interface. It can act as the working clock of the ACODEC. **Function 3: UART1_CTSN** UART1 modem status input, active low
J18	UART1_RTSN	I/O	Configurable (for details, see the PADCTRL_REG58 register)	3.3/1.8	**Function 0: GPIO5_7** GPIO **Function 1: EMMC_DATA6** eMMC data 6 **Function 2: I2S_SD_TX** TX data of the I^{2}S or PCM interface **Function 3: UART1_RTSN** UART1 modem status output, active low
J17	UART1_RXD	I/O	Configurable (for details, see the PADCTRL_REG59 register)	3.3/1.8	**Function 0: GPIO6_0** GPIO **Function 1: EMMC_DATA5** eMMC data 5 **Function 2: I2S_WS_TX** I^{2}S TX audio channel select signal or PCM TX VS signal **Function 3: UART1_RXD** UART1 RX data
K17	UART1_TXD	I/O	Configurable (for details, see the PADCTRL_REG61 register)	3.3/1.8	**Function 0: GPIO6_2** GPIO **Function 1: EMMC_DS** eMMC DATA_STROBE **Function 2: I2S_SD_RX** RX data of the I^{2}S or PCM interface **Function 3: UART1_TXD** UART1 TX data
L16	UART2_RXD	I/O	Configurable (for details, see the PADCTRL_REG62	3.3/1.8	**Function 0: GPIO6_3** GPIO

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
			register)		**Function 1: EMMC_DATA1** eMMC data 1 **Function 3: UART2_RXD** UART2 RX data
K16	UART2_TXD	I/O	Configurable (for details, see the PADCTRL_REG63 register)	3.3/1.8	**Function 0: GPIO6_4** GPIO **Function 1: EMMC_DATA2** eMMC data 2 **Function 2: I2S_BCLK_TX** TX clock of the I^2S or PCM interface **Function 3: UART2_TXD** UART2 TX data

USB Pins

Table 2-24 describes USB pins.

Table 2-24 USB pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
V12	USB_DM	I/O	N/A	N/A	USB D− signal. In high-speed mode, the maximum voltage of this port is 400 mV. In full-speed or low-speed mode, the voltage of this port is 3.3 V.
U12	USB_DP	I/O	N/A	N/A	USB D+ signal. In high-speed mode, the maximum voltage of this port is 400 mV. In full-speed or low-speed mode, the voltage of this port is 3.3 V.
V11	USB_OVRCUR	I/O	Configurable (for details, see the PADCTRL_REG30 register)	3.3	**Function 0: GPIO4_5** GPIO **Function 1: USB_OVRCUR** USB port overcurrent indicator. The level polarity

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
					can be specified by configuring the peripheral control register MISC_CTRL30 bit[17].
U11	USB_PWREN	I/O	Configurable (for details, see the PADCTRL_REG31 register)	3.3	**Function 0: GPIO4_6** GPIO **Function 1: USB_PWREN** Power control output pin of the USB port. The level polarity can be specified by configuring the peripheral control register MISC_CTRL30 bit[16].
T13	USB_REXT	I/O	N/A	N/A	USB pin for connecting to an external reference resistor. The 135 Ω resistor is recommended.
T11	AVDD33_USB	P	N/A	3.3	USB analog power
T12	AVSS_USB	G	N/A	N/A	USB analog GND

OSC Pins

Table 2-25 describes oscillator (OSC) pins.

Table 2-25 OSC pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
L1	XIN	CIN	N/A	N/A	24 MHz system crystal input
L2	XOUT	COUT	N/A	N/A	24 MHz system crystal drive output

Other Power Pins and GND Pins

Table 2-26 describes other power pins and GND pins.

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Table 2-26 Power pins and GND pins

Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
K6	AVDD11_PLL	P	N/A	1.1	1.1 V PLL analog power
K4	AVDD33_PLL	P	N/A	3.3	3.3 V PLL analog power
L6	AVSS_PLL	G	N/A	N/A	PLL analog GND
H6, H13, M4, P13	DVDD33	P	N/A	3.3	Digital I/O power
N15	DVDD18_SDIO	P	N/A	1.8	1.8 V input power of the SDIO0 power switch
L15	DVDD3318_EMMC	P	N/A	3.3/1.8	eMMC I/O power
P6	DVDD3318_SENSOR	P	N/A	3.3/1.8	Sensor I/O power
P9	DVDD3318_VI	P	N/A	3.3/1.8	VI I/O power
H15	DVDDIO_RMII	P	N/A	3.3/1.8	RMII I/O power
H7, H8, H11, H12, K13, L13, M6, M13, N6, P7, P8, P10, P11	VDD	P	N/A	1.1	Core working power
A1, A3, A8, A11, A18, B5, B8, B11, B14, B16, B18, C2, D4, D7, D9, D10, D11, D12, D13, D14, E2, E5, E15, E17, G15, G18, H1, H9, J6, J7, J8, J9, J10, J11, J12, J13, K1, K2, K7, K8, K9, K10, K11, K12, K18, L7, L8, L9, L10, L11, L12, M1, M7, M8, M9, M10, M11, M12, M15, N4,	VSS	G	N/A	N/A	Digital GND

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Position	Pin Name	Type	Drive Current (mA)	Voltage (V)	Description
N7, N8, N9, N10, N11, N12, N13, P12, R2, T18, V1, V18					

2.2.3 Default Reset Status of Digital Pins

Table 2-27 describes the default reset status of digital pins.

NOTE

The level states are described as follows:

- 0: Output the low level.
- 1: Output the high level.
- Z: high impedance
- PU: input state with internal pull-up
- PD: input state with internal pull-down
- Toggle: Level jump occurs.

For details about the specific resistance of the pull-up/pull-down resistors in the chip, see Table 2-66 and Table 2-67.

Table 2-27 Default reset status of digital pins

PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
SENSOR_CLK	GPIO0_4	Z	N/A	Z	I	-	-
SENSOR_RSTN	SENSOR_RSTN	PD	N/A	0	O	R_{PD2}	Fixed pull-down in the chip
FLASH_TRIG	GPIO0_6	PD	N/A	PD	I	R_{PD5}	Fixed pull-down in the chip
SHUTTER_TRIG	GPIO0_7	PD	N/A	PD	I	R_{PD5}	Fixed pull-down in the chip
RMII_CLK	GPIO2_0	Z	N/A	Z	I	-	-

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
RMII_TX_EN	GPIO2_1	Z	N/A	Z	I	-	-
RMII_TXD0	GPIO2_2	Z	N/A	Z	I	-	-
RMII_TXD1	GPIO2_3	Z	N/A	Z	I	-	-
RMII_RX_DV	GPIO2_4	Z	N/A	Z	I	-	-
RMII_RXD0	GPIO2_5	Z	N/A	Z	I	-	-
RMII_RXD1	GPIO2_6	Z	N/A	Z	I	-	-
EPHY_RST	GPIO2_7	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip
EPHY_CLK	GPIO3_0	Z	N/A	Z	I	-	-
MDCK	GPIO3_1	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip
MDIO	GPIO3_2	Z	N/A	Z	I	-	-
SPI0_SCLK	GPIO3_3	Z	N/A	Z	I	-	-
SPI0_SDO	GPIO3_4	Z	N/A	Z	I	-	-
SPI0_SDI	GPIO3_5	Z	N/A	Z	I	-	-
SPI0_CSN	GPIO3_6	Z	N/A	Z	I	-	-
SPI1_SCLK	GPIO3_7	Z	N/A	Z	I	-	-
SPI1_SDO	GPIO4_0	Z	N/A	Z	I	-	-
SPI1_SDI	GPIO4_1	Z	N/A	Z	I	-	-
SPI1_CSN0	GPIO4_2	Z	N/A	Z	I	-	-
SPI1_CSN1	GPIO0_3	Z	N/A	Z	I	-	-
I2C2_SDA	GPIO4_3	Z	N/A	Z	I	-	-
I2C2_SCL	GPIO4_4	Z	N/A	Z	I	-	-
UART0_RXD	UART0_RXD	Z	N/A	PU	I	R_{PU1}	
UART0_TXD	UART0_TXD	Z	N/A	1	O	-	-

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
SYS_RSTN_OUT	SYS_RSTN_OUT	PD	N/A	0	O	R_{PD2}	Fixed pull-down in the chip
TEST_MODE	TEST_MODE	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip
USB_OVRCUR	GPIO4_5	Z	N/A	Z	I	-	-
USB_PWREN	GPIO4_6	Z	N/A	Z	I	-	-
GPIO0_0	GPIO0_0	PD	N/A	Z	I	R_{PD2}	-
GPIO0_1	GPIO0_1	PD	N/A	Z	I	R_{PD2}	-
GPIO0_2	GPIO0_2	PD	N/A	Z	I	R_{PD2}	-
GPIO1_0	GPIO1_0	Z	N/A	Z	I	-	-
GPIO1_1	GPIO1_1	Z	N/A	Z	I	-	-
GPIO1_2	GPIO1_2	Z	N/A	Z	I	-	-
GPIO1_3	GPIO1_3	Z	N/A	Z	I	-	-
GPIO1_4	GPIO1_4	Z	N/A	Z	I	-	-
GPIO1_5	GPIO1_5	Z	N/A	Z	I	-	-
GPIO1_6	GPIO1_6	Z	N/A	Z	I	-	-
SDIO0_CARD_POWER_EN	GPIO1_7	Z	N/A	Z	I	-	-
SDIO0_CARD_DETECT	GPIO4_7	Z	N/A	Z	I	-	-
SDIO0_CWPR	GPIO5_0	Z	N/A	Z	I	-	-
SDIO0_CCLK_OUT	GPIO5_1	Z	N/A	Z	I	-	-
SDIO0_CCMD	GPIO5_2	Z	N/A	Z	I	-	-
SDIO0_CDATA0	GPIO5_3	Z	N/A	Z	I	-	-
SDIO0_CDATA1	TEST_CLK	Z	N/A	Toggle	O	-	Test clock

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
SDIO0_CDATA2	GPIO5_5	Z	N/A	Z	I	-	-
SDIO0_CDATA3	GPIO5_6	Z	N/A	Z	I	-	-
JTAG_TRSTN	GPIO6_5	Z	N/A	Z	I	-	-
JTAG_TCK	GPIO6_6	Z	N/A	Z	I	-	-
JTAG_TMS	GPIO6_7	Z	N/A	Z	I	-	-
JTAG_TDO	GPIO7_0	Z	N/A	Z	I	-	-
JTAG_TDI	GPIO7_1	Z	N/A	Z	I	-	-
JTAG_EN	JTAG_EN	Z	N/A	PD	I	R_{PD1}	Fixed pull-down in the chip
PWM0	PMC_PWM	Z	N/A	0	O	-	-
PWM1	PWM1	Z	N/A	0	O	-	-
PWM2	PWM2	Z	N/A	0	O	-	-
PWM3	PWM3	Z	N/A	0	O	-	-
SAR_ADC_CH0	SAR_ADC_CH0	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SAR_ADC_CH1	SAR_ADC_CH1	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SAR_ADC_CH2	SAR_ADC_CH2	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
SAR_ADC_CH3	SAR_ADC_CH3	Z	N/A	Z	I	R_{PU3}/R_{PD3}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_CLK	SFC_CLK	Z	N/A	0	O	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_SIO_IO0	SFC_SIO_IO0	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_SOI_IO1	SFC_SOI_IO1	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_WP_IO2	SFC_WP_IO2	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_HOLD_IO3	SFC_HOLD_IO3	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
SFC_CSN0	SFC_CSN0	PU	N/A	1	O	R_{PU2}	Fixed pull-up in the chip

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PAD	Default Function	Level Status*				Resistance of Pull-up/Pull-down Resistor in the Chip**	Description
		I/O Is Powered on but the Core Is not Powered on	I/O Type	Both the I/O and Core Are Powered on and Before Chip Reset Is Deasserted	I/O Type		
SFC_CSN1	SFC_CSN1	PU	N/A	1	O	R_{PU2}	Fixed pull-up in the chip
UART1_RTSN	GPIO5_7	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART1_RXD	GPIO6_0	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART1_CTSN	GPIO6_1	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART1_TXD	GPIO6_2	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART2_RXD	GPIO6_3	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default
UART2_TXD	GPIO6_4	Z	N/A	Z	I	R_{PU4}/R_{PD4}	Configurable pull-up/pull-down register, and no pull-up/pull-down by default

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2.3 Pin Multiplexing Control Registers

2.3.1 Register Summary

Table 2-28 describes pin multiplexing control registers.

Table 2-28 Summary of pin multiplexing control registers (base address: 0x200F_0000)

Offset Address	Register	Description	Page
0x000	muxctrl_reg0	Multiplexing control register for the SENSOR_CLK pin	2-47
0x004	muxctrl_reg1	Multiplexing control register for the SENSOR_RSTN pin	2-48
0x008	muxctrl_reg2	Multiplexing control register for the FLASH_TRIG pin	2-48
0x00C	muxctrl_reg3	Multiplexing control register for the SHUTTER_TRIG pin	2-49
0x010	muxctrl_reg4	Multiplexing control register for the RMII_CLK pin	2-49
0x014	muxctrl_reg5	Multiplexing control register for the RMII_TX_EN pin	2-50
0x018	muxctrl_reg6	Multiplexing control register for the RMII_TXD0 pin	2-50
0x01C	muxctrl_reg7	Multiplexing control register for the RMII_TXD1 pin	2-51
0x020	muxctrl_reg8	Multiplexing control register for the RMII_RX_DV pin	2-51
0x024	muxctrl_reg9	Multiplexing control register for the RMII_RXD0 pin	2-52
0x028	muxctrl_reg10	Multiplexing control register for the RMII_RXD1 pin	2-53
0x02C	muxctrl_reg11	Multiplexing control register for the EPHY_RST pin	2-53
0x030	muxctrl_reg12	Multiplexing control register for the SPI1_CSN1 pin	2-54
0x034	muxctrl_reg13	Multiplexing control register for the EPHY_CLK pin	2-54
0x038	muxctrl_reg14	Multiplexing control register for the MDCK pin	2-55
0x03C	muxctrl_reg15	Multiplexing control register for the MDIO pin	2-55
0x040	muxctrl_reg16	Multiplexing control register for the	2-56

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Offset Address	Register	Description	Page
		SPI0_SCLK pin	
0x044	muxctrl_reg17	Multiplexing control register for the SPI0_SDO pin	2-56
0x048	muxctrl_reg18	Multiplexing control register for the SPI0_SDI pin	2-57
0x04C	muxctrl_reg19	Multiplexing control register for the SPI0_CSN pin	2-57
0x050	muxctrl_reg20	Multiplexing control register for the SPI1_SCLK pin	2-58
0x054	muxctrl_reg21	Multiplexing control register for the SPI1_SDO pin	2-58
0x058	muxctrl_reg22	Multiplexing control register for the SPI1_SDI pin	2-59
0x05C	muxctrl_reg23	Multiplexing control register for the SPI1_CSN0 pin	2-59
0x060	muxctrl_reg24	Multiplexing control register for the I2C2_SDA pin	2-60
0x064	muxctrl_reg25	Multiplexing control register for the I2C2_SCL pin	2-60
0x068	muxctrl_reg26	Multiplexing control register for the USB_OVRCUR pin	2-61
0x06C	muxctrl_reg27	Multiplexing control register for the USB_PWREN pin	2-61
0x070	muxctrl_reg28	Multiplexing control register for the GPIO0_0 pin	2-62
0x074	muxctrl_reg29	Multiplexing control register for the GPIO0_1 pin	2-62
0x078	muxctrl_reg30	Multiplexing control register for the GPIO0_2 pin	2-63
0x07C	muxctrl_reg31	Multiplexing control register for the GPIO1_0 pin	2-63
0x080	muxctrl_reg32	Multiplexing control register for the GPIO1_1 pin	2-64
0x084	muxctrl_reg33	Multiplexing control register for the GPIO1_2 pin	2-65
0x088	muxctrl_reg34	Multiplexing control register for the GPIO1_3 pin	2-65
0x08C	muxctrl_reg35	Multiplexing control register for the GPIO1_4	2-66

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Offset Address	Register	Description	Page
		pin	
0x090	muxctrl_reg36	Multiplexing control register for the GPIO1_5 pin	2-66
0x094	muxctrl_reg37	Multiplexing control register for the GPIO1_6 pin	2-67
0x098	muxctrl_reg38	Multiplexing control register for the SDIO0_CARD_POWER_EN pin	2-67
0x09C	muxctrl_reg39	Multiplexing control register for the SDIO0_CARD_DETECT pin	2-68
0x0A0	muxctrl_reg40	Multiplexing control register for the SDIO0_CWPR pin	2-68
0x0A4	muxctrl_reg41	Multiplexing control register for the SDIO0_CCLK_OUT pin	2-69
0x0A8	muxctrl_reg42	Multiplexing control register for the SDIO0_CCMD pin	2-69
0x0AC	muxctrl_reg43	Multiplexing control register for the SDIO0_CDATA0 pin	2-70
0x0B0	muxctrl_reg44	Multiplexing control register for the SDIO0_CDATA1 pin	2-70
0x0B4	muxctrl_reg45	Multiplexing control register for the SDIO0_CDATA2 pin	2-71
0x0B8	muxctrl_reg46	Multiplexing control register for the SDIO0_CDATA3 pin	2-71
0x0BC	muxctrl_reg47	Multiplexing control register for the UART1_RTSN pin	2-72
0x0C0	muxctrl_reg48	Multiplexing control register for the UART1_RXD pin	2-72
0x0C4	muxctrl_reg49	Multiplexing control register for the UART1_CTSN pin	2-73
0x0C8	muxctrl_reg50	Multiplexing control register for the UART1_TXD pin	2-74
0x0CC	muxctrl_reg51	Multiplexing control register for the UART2_RXD pin	2-74
0x0D0	muxctrl_reg52	Multiplexing control register for the UART2_TXD pin	2-75
0x0D4	muxctrl_reg53	Multiplexing control register for the JTAG_TRSTN pin	2-75
0x0D8	muxctrl_reg54	Multiplexing control register for the	2-76

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Offset Address	Register	Description	Page
		JTAG_TCK pin	
0x0DC	muxctrl_reg55	Multiplexing control register for the JTAG_TMS pin	2-76
0x0E0	muxctrl_reg56	Multiplexing control register for the JTAG_TDO pin	2-77
0x0E4	muxctrl_reg57	Multiplexing control register for the JTAG_TDI pin	2-77
0x0E8	muxctrl_reg58	Multiplexing control register for the PWM0 pin	2-78
0x0EC	muxctrl_reg59	Multiplexing control register for the PWM1 pin	2-78
0x0F0	muxctrl_reg60	Multiplexing control register for the PWM2 pin	2-79
0x0F4	muxctrl_reg61	Multiplexing control register for the PWM3 pin	2-79
0x0F8	muxctrl_reg62	Multiplexing control register for the SAR_ADC_CH0 pin	2-80
0x0FC	muxctrl_reg63	Multiplexing control register for the SAR_ADC_CH1 pin	2-80
0x100	muxctrl_reg64	Multiplexing control register for the SAR_ADC_CH2 pin	2-81
0x104	muxctrl_reg65	Multiplexing control register for the SAR_ADC_CH3 pin	2-81

2.3.2 Register Description

muxctrl_reg0

muxctrl_reg0 is a multiplexing control register for the SENSOR_CLK pin.

Offset Address	Register Name	Total Reset Value
0x000	muxctrl_reg0	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg0
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg0	Multiplexing control for the SENSOR_CLK pin

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			0: GPIO0_4 1: SENSOR_CLK

muxctrl_reg1

muxctrl_reg1 is a multiplexing control register for the SENSOR_RSTN pin.

Offset Address	Register Name	Total Reset Value
0x004	muxctrl_reg1	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg1
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg1	Multiplexing control for the SENSOR_RSTN pin 0: SENSOR_RSTN 1: GPIO0_5

muxctrl_reg2

muxctrl_reg2 is a multiplexing control register for the FLASH_TRIG pin.

Offset Address	Register Name	Total Reset Value
0x008	muxctrl_reg2	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg2	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg2	Multiplexing control for the FLASH_TRIG pin 000: GPIO0_6 001: FLASH_TRIG 010: SFC_EMMC_BOOT_MODE

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			011: SPI1_CSN1 100: VI_VS_BT1120 Other values: reserved

muxctrl_reg3

muxctrl_reg3 is a multiplexing control register for the SHUTTER_TRIG pin.

Offset Address	Register Name	Total Reset Value
0x00C	muxctrl_reg3	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg3	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg3	Multiplexing control for the SHUTTER_TRIG pin 000: GPIO0_7 001: SHUTTER_TRIG 010: SFC_DEVICE_MODE 100: VI_HS_BT1120 Other values: reserved

muxctrl_reg4

muxctrl_reg4 is a multiplexing control register for the RMII_CLK pin.

Offset Address	Register Name	Total Reset Value
0x010	muxctrl_reg4	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg4	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved

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[2:0]	RW	muxctrl_reg4	Multiplexing control for the RMII_CLK pin 000: GPIO2_0 001: RMII_CLK 011: VO_CLK 100: SDIO1_CCLK_OUT Other values: reserved

muxctrl_reg5

muxctrl_reg5 is a multiplexing control register for the RMII_TX_EN pin.

Offset Address	Register Name	Total Reset Value
0x014	muxctrl_reg5	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												muxctrl_reg5			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg5	Multiplexing control for the RMII_TX_EN pin 000: GPIO2_1 001: RMII_TX_EN 011: VO_VS 100: SDIO1_CARD_DETECT Other values: reserved

muxctrl_reg6

muxctrl_reg6 is a multiplexing control register for the RMII_TXD0 pin.

Offset Address	Register Name	Total Reset Value
0x018	muxctrl_reg6	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												muxctrl_reg6			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg6	Multiplexing control for the RMII_TXD0 pin 000: GPIO2_2 001: RMII_TXD0 011: VO_DATA5 100: SDIO1_CWPR Other values: reserved

muxctrl_reg7

muxctrl_reg7 is a multiplexing control register for the RMII_TXD1 pin.

Offset Address	Register Name	Total Reset Value
0x01C	muxctrl_reg7	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg7	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg7	Multiplexing control for the RMII_TXD1 pin 000: GPIO2_3 001: RMII_TXD1 011: VO_DE 100: SDIO1_CDATA1 Other values: reserved

muxctrl_reg8

muxctrl_reg8 is a multiplexing control register for the RMII_RX_DV pin.

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Offset Address	Register Name	Total Reset Value
0x020	muxctrl_reg8	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg8	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg8	Multiplexing control for the RMII_RX_DV pin 000: GPIO2_4 001: RMII_RX_DV 011: VO_DATA7 100: SDIO1_CDATA0 Other values: reserved

muxctrl_reg9

muxctrl_reg9 is a multiplexing control register for the RMII_RXD0 pin.

Offset Address	Register Name	Total Reset Value
0x024	muxctrl_reg9	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg9	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg9	Multiplexing control for the RMII_RXD0 pin 000: GPIO2_5 001: RMII_RXD0 011: VO_DATA2 100: SDIO1_CDATA3 Other values: reserved

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muxctrl_reg10

muxctrl_reg10 is a multiplexing control register for the RMII_RXD1 pin.

Offset Address	Register Name	Total Reset Value
0x028	muxctrl_reg10	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg10
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg10	Multiplexing control for the RMII_RXD1 pin 000: GPIO2_6 001: RMII_RXD1 011: VO_DATA3 100: SDIO1_CCMD Other values: reserved

muxctrl_reg11

muxctrl_reg11 is a multiplexing control register for the EPHY_RST pin.

Offset Address	Register Name	Total Reset Value
0x02C	muxctrl_reg11	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg11
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg11	Multiplexing control for the EPHY_RST pin 000: GPIO2_7 001: EPHY_RST 010: BOOT_SEL 011: VO_HS

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			100: SDIO1_CARD_POWER_EN Other values: reserved

muxctrl_reg12

muxctrl_reg12 is a multiplexing control register for the SPI1_CSN1 pin.

Offset Address	Register Name	Total Reset Value
0x030	muxctrl_reg12	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg12	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg12	Multiplexing control for the SPI1_CSN1 pin 00: GPIO0_3 01: SPI1_CSN1 11: VO_DATA0 Other values: reserved

muxctrl_reg13

muxctrl_reg13 is a multiplexing control register for the EPHY_CLK pin.

Offset Address	Register Name	Total Reset Value
0x034	muxctrl_reg13	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg13	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg13	Multiplexing control for the EPHY_CLK pin 000: GPIO3_0

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			001: EPHY_CLK 011: VO_DATA1 100: SDIO1_CDATA2 Other values: reserved

muxctrl_reg14

muxctrl_reg14 is a multiplexing control register for the MDCK pin.

Offset Address	Register Name	Total Reset Value
0x038	muxctrl_reg14	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg14	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg14	Multiplexing control for the MDCK pin 00: GPIO3_1 01: MDCK 10: BOOTROM_SEL 11: VO_DATA6

muxctrl_reg15

muxctrl_reg15 is a multiplexing control register for the MDIO pin.

Offset Address	Register Name	Total Reset Value
0x03C	muxctrl_reg15	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg15	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved

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[1:0]	RW	muxctrl_reg15	Multiplexing control for the MDIO pin 00: GPIO3_2 01: MDIO 11: VO_DATA4 Other values: reserved

muxctrl_reg16

muxctrl_reg16 is a multiplexing control register for the SPI0_SCLK pin.

Offset Address	Register Name	Total Reset Value
0x040	muxctrl_reg16	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg16	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg16	Multiplexing control for the SPI0_SCLK pin 00: GPIO3_3 01: SPI0_SCLK 10: I2C0_SCL Other values: reserved

muxctrl_reg17

muxctrl_reg17 is a multiplexing control register for the SPI0_SDO pin.

Offset Address	Register Name	Total Reset Value
0x044	muxctrl_reg17	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg17	

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg17	Multiplexing control for the SPI0_SDO pin 00: GPIO3_4 01: SPI0_SDO 10: I2C0_SDA Other values: reserved

muxctrl_reg18

muxctrl_reg18 is a multiplexing control register for the SPI0_SDI pin.

Offset Address	Register Name	Total Reset Value
0x048	muxctrl_reg18	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg18
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg18	Multiplexing control for the SPI0_SDI pin 0: GPIO3_5 1: SPI0_SDI

muxctrl_reg19

muxctrl_reg19 is a multiplexing control register for the SPI0_CSN pin.

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Offset Address	Register Name	Total Reset Value
0x04C	muxctrl_reg19	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg19
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg19	Multiplexing control for the SPI0_CSN pin 0: GPIO3_6 1: SPI0_CSN

muxctrl_reg20

muxctrl_reg20 is a multiplexing control register for the SPI1_SCLK pin.

Offset Address	Register Name	Total Reset Value
0x050	muxctrl_reg20	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg20
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg20	Multiplexing control for the SPI1_SCLK pin 00: GPIO3_7 01: SPI1_SCLK 10: I2C1_SCL Other values: reserved

muxctrl_reg21

muxctrl_reg21 is a multiplexing control register for the SPI1_SDO pin.

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Offset Address	Register Name	Total Reset Value
0x054	muxctrl_reg21	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg21	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg21	Multiplexing control for the SPI1_SDO pin 00: GPIO4_0 01: SPI1_SDO 10: I2C1_SDA Other values: reserved

muxctrl_reg22

muxctrl_reg22 is a multiplexing control register for the SPI1_SDI pin.

Offset Address	Register Name	Total Reset Value
0x058	muxctrl_reg22	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg22
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg22	Multiplexing control for the SPI1_SDI pin 0: GPIO4_1 1: SPI1_SDI

muxctrl_reg23

muxctrl_reg23 is a multiplexing control register for the SPI1_CSN0 pin.

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Offset Address	Register Name	Total Reset Value
0x05C	muxctrl_reg23	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg23
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg23	Multiplexing control for the SPI1_CSN0 pin 0: GPIO4_2 1: SPI1_CSN0

muxctrl_reg24

muxctrl_reg24 is a multiplexing control register for the I2C2_SDA pin.

Offset Address	Register Name	Total Reset Value
0x060	muxctrl_reg24	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg24
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg24	Multiplexing control for the I2C2_SDA pin 0: GPIO4_3 1: I2C2_SDA

muxctrl_reg25

muxctrl_reg25 is a multiplexing control register for the I2C2_SCL pin.

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Offset Address	Register Name	Total Reset Value
0x064	muxctrl_reg25	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg25
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg25	Multiplexing control for the I2C2_SCL pin 0: GPIO4_4 1: I2C2_SCL

muxctrl_reg26

muxctrl_reg26 is a multiplexing control register for the USB_OVRCUR pin.

Offset Address	Register Name	Total Reset Value
0x068	muxctrl_reg26	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg26
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg26	Multiplexing control for the USB_OVRCUR pin 0: GPIO4_5 1: USB_OVRCUR

muxctrl_reg27

muxctrl_reg27 is a multiplexing control register for the USB_PWREN pin.

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Offset Address	Register Name	Total Reset Value
0x06C	muxctrl_reg27	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg27
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg27	Multiplexing control for the USB_PWREN pin 0: GPIO4_6 1: USB_PWREN

muxctrl_reg28

muxctrl_reg28 is a multiplexing control register for the GPIO0_0 pin.

Offset Address	Register Name	Total Reset Value
0x070	muxctrl_reg28	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg28
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg28	Multiplexing control for the GPIO0_0 pin 00: GPIO0_0 01: IR_IN 10: TEMPER_DQ Other values: reserved

muxctrl_reg29

muxctrl_reg29 is a multiplexing control register for the GPIO0_1 pin.

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Offset Address	Register Name	Total Reset Value
0x074	muxctrl_reg29	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg29	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg29	Multiplexing control for the GPIO0_1 pin 00: GPIO0_1 10: TEMPER_DQ Other values: reserved

muxctrl_reg30

muxctrl_reg30 is a multiplexing control register for the GPIO0_2 pin.

Offset Address	Register Name	Total Reset Value
0x078	muxctrl_reg30	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg30	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg30	Multiplexing control for the GPIO0_2 pin 00: GPIO0_2 10: TEMPER_DQ Other values: reserved

muxctrl_reg31

muxctrl_reg31 is a multiplexing control register for the GPIO1_0 pin.

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Offset Address	Register Name	Total Reset Value
0x07C	muxctrl_reg31	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg31	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg31	Multiplexing control for the GPIO1_0 pin 000: GPIO1_0 001: VI_DATA13 011: I2S_BCLK_TX 100: PWM0 Other values: reserved

muxctrl_reg32

muxctrl_reg32 is a multiplexing control register for the GPIO1_1 pin.

Offset Address	Register Name	Total Reset Value
0x080	muxctrl_reg32	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg32	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg32	Multiplexing control for the GPIO1_1 pin 000: GPIO1_1 001: VI_DATA10 011: I2S_SD_TX 100: UART1_TXD Other values: reserved

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muxctrl_reg33

muxctrl_reg33 is a multiplexing control register for the GPIO1_2 pin.

Offset Address	Register Name	Total Reset Value
0x084	muxctrl_reg33	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg33	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg33	Multiplexing control for the GPIO1_2 pin 000: GPIO1_2 001: VI_DATA12 011: I2S_MCLK 100: UART1_CTSN Other values: reserved

muxctrl_reg34

muxctrl_reg34 is a multiplexing control register for the GPIO1_3 pin.

Offset Address	Register Name	Total Reset Value
0x088	muxctrl_reg34	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg34	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg34	Multiplexing control for the GPIO1_3 pin 000: GPIO1_3 001: VI_DATA11 011: I2S_WS_TX 100: UART2_RXD

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			Other values: reserved

muxctrl_reg35

muxctrl_reg35 is a multiplexing control register for the GPIO1_4 pin.

Offset Address	Register Name	Total Reset Value
0x08C	muxctrl_reg35	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg35	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg35	Multiplexing control for the GPIO1_4 pin 000: GPIO1_4 001: VI_DATA15 010: VI_VS_SEN 011: I2S_WS_RX 100: UART1_RXD Other values: reserved

muxctrl_reg36

muxctrl_reg36 is a multiplexing control register for the GPIO1_5 pin.

Offset Address	Register Name	Total Reset Value
0x090	muxctrl_reg36	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg36	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg36	Multiplexing control for the GPIO1_5 pin

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			000: GPIO1_5 001: VI_DATA14 010: VI_HS_SEN 011: I2S_BCLK_RX 100: UART1_RTSN Other values: reserved

muxctrl_reg37

muxctrl_reg37 is a multiplexing control register for the GPIO1_6 pin.

Offset Address	Register Name	Total Reset Value
0x094	muxctrl_reg37	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg37	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg37	Multiplexing control for the GPIO1_6 pin 000: GPIO1_6 001: VI_DATA9 011: I2S_SD_RX 100: UART2_TXD Other values: reserved

muxctrl_reg38

muxctrl_reg38 is a multiplexing control register for the SDIO0_CARD_POWER_EN pin.

Offset Address	Register Name	Total Reset Value
0x098	muxctrl_reg38	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg38

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg38	Multiplexing control for the SDIO0_CARD_POWER_EN pin 0: GPIO1_7 1: SDIO0_CARD_POWER_EN

muxctrl_reg39

muxctrl_reg39 is a multiplexing control register for the SDIO0_CARD_DETECT pin.

Offset Address	Register Name	Total Reset Value
0x09C	muxctrl_reg39	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg39
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg39	Multiplexing control for the SDIO0_CARD_DETECT pin 0: GPIO4_7 1: SDIO0_CARD_DETECT

muxctrl_reg40

muxctrl_reg40 is a multiplexing control register for the SDIO0_CWPR pin.

Offset Address	Register Name	Total Reset Value
0x0A0	muxctrl_reg40	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg40

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg40	Multiplexing control for the SDIO0_CWPR pin 0: GPIO5_0 1: SDIO0_CWPR

muxctrl_reg41

muxctrl_reg41 is a multiplexing control register for the SDIO0_CCLK_OUT pin.

Offset Address	Register Name	Total Reset Value
0x0A4	muxctrl_reg41	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg41
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg41	Multiplexing control for the SDIO0_CCLK_OUT pin 0: GPIO5_1 1: SDIO0_CCLK_OUT

muxctrl_reg42

muxctrl_reg42 is a multiplexing control register for the SDIO0_CCMD pin.

Offset Address	Register Name	Total Reset Value
0x0A8	muxctrl_reg42	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg42

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg42	Multiplexing control for the SDIO0_CCMD pin 0: GPIO5_2 1: SDIO0_CCMD

muxctrl_reg43

muxctrl_reg43 is a multiplexing control register for the SDIO0_CDATA0 pin.

Offset Address	Register Name	Total Reset Value
0x0AC	muxctrl_reg43	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg43
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg43	Multiplexing control for the SDIO0_CDATA0 pin 0: GPIO5_3 1: SDIO0_CDATA0

muxctrl_reg44

muxctrl_reg44 is a multiplexing control register for the SDIO0_CDATA1 pin.

Offset Address	Register Name	Total Reset Value
0x0B0	muxctrl_reg44	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg44

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg44	Multiplexing control for the SDIO0_CDATA1 pin 00: TEST_CLK 01: SDIO0_CDATA1 10: GPIO5_4 Other values: reserved

muxctrl_reg45

muxctrl_reg45 is a multiplexing control register for the SDIO0_CDATA2 pin.

Offset Address	Register Name	Total Reset Value
0x0B4	muxctrl_reg45	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg45
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg45	Multiplexing control for the SDIO0_CDATA2 pin 0: GPIO5_5 1: SDIO0_CDATA2

muxctrl_reg46

muxctrl_reg46 is a multiplexing control register for the SDIO0_CDATA3 pin.

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Offset Address	Register Name	Total Reset Value
0x0B8	muxctrl_reg46	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg46
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg46	Multiplexing control for the SDIO0_CDATA3 pin 0: GPIO5_6 1: SDIO0_CDATA3

muxctrl_reg47

muxctrl_reg47 is a multiplexing control register for the UART1_RTSN pin.

Offset Address	Register Name	Total Reset Value
0x0BC	muxctrl_reg47	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg47
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg47	Multiplexing control for the UART1_RTSN pin 00: GPIO5_7 01: EMMC_DATA6 10: I2S_SD_TX 11: UART1_RTSN

muxctrl_reg48

muxctrl_reg48 is a multiplexing control register for the UART1_RXD pin.

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Offset Address	Register Name	Total Reset Value
0x0C0	muxctrl_reg48	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg48
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg48	Multiplexing control for the UART1_RXD pin 00: GPIO6_0 01: EMMC_DATA5 10: I2S_WS_TX 11: UART1_RXD

muxctrl_reg49

muxctrl_reg49 is a multiplexing control register for the UART1_CTSN pin.

Offset Address	Register Name	Total Reset Value
0x0C4	muxctrl_reg49	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								muxctrl_reg49
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg49	Multiplexing control for the UART1_CTSN pin 00: GPIO6_1 01: EMMC_DATA7 10: I2S_MCLK 11: UART1_CTSN

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muxctrl_reg50

muxctrl_reg50 is a multiplexing control register for the UART1_TXD pin.

Offset Address	Register Name	Total Reset Value
0x0C8	muxctrl_reg50	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg50
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg50	Multiplexing control for the UART1_TXD pin 00: GPIO6_2 01: EMMC_DS 10: I2S_SD_RX 11: UART1_TXD

muxctrl_reg51

muxctrl_reg51 is a multiplexing control register for the UART2_RXD pin.

Offset Address	Register Name	Total Reset Value
0x0CC	muxctrl_reg51	0x00000000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							muxctrl_reg51
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg51	Multiplexing control for the UART2_RXD pin 00: GPIO6_3 01: EMMC_DATA1 11: UART2_RXD Other values: reserved

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muxctrl_reg52

muxctrl_reg52 is a multiplexing control register for the UART2_TXD pin.

Offset Address	Register Name	Total Reset Value
0x0D0	muxctrl_reg52	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg52	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg52	Multiplexing control for the UART2_TXD pin 00: GPIO6_4 01: EMMC_DATA2 10: I2S_BCLK_TX 11: UART2_TXD

muxctrl_reg53

muxctrl_reg53 is a multiplexing control register for the JTAG_TRSTN pin.

Offset Address	Register Name	Total Reset Value
0x0D4	muxctrl_reg53	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg53	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg53	Multiplexing control for the JTAG_TRSTN pin 00: GPIO6_5 01: JTAG_TRSTN 10: SPI1_CSN1 11: I2S_MCLK

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muxctrl_reg54

muxctrl_reg54 is a multiplexing control register for the JTAG_TCK pin.

Offset Address	Register Name	Total Reset Value
0x0D8	muxctrl_reg54	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg54	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg54	Multiplexing control for the JTAG_TCK pin 000: GPIO6_6 001: JTAG_TCK 010: SPI1_SCLK 011: I2S_WS_TX 100: I2C1_SCL Other values: reserved

muxctrl_reg55

muxctrl_reg55 is a multiplexing control register for the JTAG_TMS pin.

Offset Address	Register Name	Total Reset Value
0x0DC	muxctrl_reg55	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg55	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg55	Multiplexing control for the JTAG_TMS pin 00: GPIO6_7 01: JTAG_TMS 10: SPI1_CSN0

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			11: I2S_SD_TX

muxctrl_reg56

muxctrl_reg56 is a multiplexing control register for the JTAG_TDO pin.

Offset Address	Register Name	Total Reset Value
0x0E0	muxctrl_reg56	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg56	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	muxctrl_reg56	Multiplexing control for the JTAG_TDO pin 000: GPIO7_0 001: JTAG_TDO 010: SPI1_SDO 011: I2S_SD_RX 100: I2C1_SDA Other values: reserved

muxctrl_reg57

muxctrl_reg57 is a multiplexing control register for the JTAG_TDI pin.

Offset Address	Register Name	Total Reset Value
0x0E4	muxctrl_reg57	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg57	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg57	Multiplexing control for the JTAG_TDI pin

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			00: GPIO7_1 01: JTAG_TDI 10: SPI1_SDI 11: I2S_BCLK_TX

muxctrl_reg58

muxctrl_reg58 is a multiplexing control register for the PWM0 pin.

Offset Address	Register Name	Total Reset Value
0x0E8	muxctrl_reg58	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg58	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg58	Multiplexing control for the PWM0 pin 00: PMC_PWM 01: GPIO7_2 10: PWM0 Other values: reserved

muxctrl_reg59

muxctrl_reg59 is a multiplexing control register for the PWM1 pin.

Offset Address	Register Name	Total Reset Value
0x0EC	muxctrl_reg59	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg59

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg59	Multiplexing control for the PWM1 pin 0: PWM1 1: GPIO7_3

muxctrl_reg60

muxctrl_reg60 is a multiplexing control register for the PWM2 pin.

Offset Address	Register Name	Total Reset Value
0x0F0	muxctrl_reg60	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg60	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	muxctrl_reg60	Multiplexing control for the PWM2 pin 00: PWM2 01: GPIO7_4 10: SPI1_CSN1 Other values: reserved

muxctrl_reg61

muxctrl_reg61 is a multiplexing control register for the PWM3 pin.

Offset Address	Register Name	Total Reset Value
0x0F4	muxctrl_reg61	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														muxctrl_reg61	

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg61	Multiplexing control for the PWM3 pin 0: PWM3 1: GPIO7_5

muxctrl_reg62

muxctrl_reg62 is a multiplexing control register for the SAR_ADC_CH0 pin.

Offset Address	Register Name	Total Reset Value
0x0F8	muxctrl_reg62	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg62
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg62	Multiplexing control for the SAR_ADC_CH0 pin 0: SAR_ADC_CH0 1: GPIO7_6

muxctrl_reg63

muxctrl_reg63 is a multiplexing control register for the SAR_ADC_CH1 pin.

Offset Address	Register Name	Total Reset Value
0x0FC	muxctrl_reg63	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg63

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg63	Multiplexing control for the SAR_ADC_CH1 pin 0: SAR_ADC_CH1 1: GPIO7_7

muxctrl_reg64

muxctrl_reg64 is a multiplexing control register for the SAR_ADC_CH2 pin.

Offset Address	Register Name	Total Reset Value
0x100	muxctrl_reg64	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg64
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg64	Multiplexing control for the SAR_ADC_CH2 pin 0: SAR_ADC_CH2 1: GPIO8_0

muxctrl_reg65

muxctrl_reg65 is a multiplexing control register for the SAR_ADC_CH3 pin.

Offset Address	Register Name	Total Reset Value
0x104	muxctrl_reg65	0x00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															muxctrl_reg65

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	muxctrl_reg65	Multiplexing control for the SAR_ADC_CH3 pin 0: SAR_ADC_CH3 1: GPIO8_1

2.4 Software Multiplexed Pins

Sensor

Table 2-29 lists software multiplexed pins of the sensor.

Table 2-29 Software multiplexed pins of the sensor

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signals 2−4
V2	SENSOR_CLK	muxctrl_reg0	GPIO0_4	SENSOR_CLK	2−4: N/A
T3	SENSOR_RSTN	muxctrl_reg1	SENSOR_RSTN	GPIO0_5	2−4: N/A
U3	FLASH_TRIG	muxctrl_reg2	GPIO0_6	FLASH_TRIG	2: SFC_EMMC_BOOT_MODE 3: SPI1_CSN1 4: VI_VS_BT1120
T4	SHUTTER_TRIG	muxctrl_reg3	GPIO0_7	SHUTTER_TRIG	2: SFC_DEVICE_MODE 3: - 4: VI_HS_BT1120

Table 2-30 describes software multiplexed signals of the sensor.

Table 2-30 Description of software multiplexed signals of the sensor

Signal	Direction	Description
FLASH_TRIG	O	Camera flash control
GPIO0_4	I/O	GPIO

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Signal	Direction	Description
GPIO0_5	I/O	GPIO
GPIO0_6	I/O	GPIO
GPIO0_7	I/O	GPIO
SENSOR_CLK	O	Sensor working clock
SENSOR_RSTN	O	Sensor reset signal
SFC_DEVICE_MODE	I	SPI flash from which the system boots (that is, type of the flash memory connected to SFC_CSN0) 0: SPI NOR flash 1: SPI NAND flash
SFC_EMMC_BOOT_MODE	I	Boot address mode of the SPI NOR flash when BOOT_SEL and SFC_DEVICE_MODE are 0 0: 3-byte address mode 1: 4-byte address mode Boot mode of the SPI NAND flash when BOOT_SEL is 0 and SFC_DEVICE_MODE is 1 0: 1-wire boot mode 1: 4-wire boot mode eMMC boot mode when BOOT_SEL is 1 0: 4-wire boot mode 1: 8-wire boot mode
SHUTTER_TRIG	O	Shutter control
SPI1_CSN1	O	SPI1 CS1
VI_HS_BT1120	I	VI HS signal
VI_VS_BT1120	I	VI VS signal

RMII

Table 2-31 lists software multiplexed pins of the RMII.

Table 2-31 Software multiplexed pins of the RMII

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
E18	RMII_CLK	muxctrl_reg4	GPIO2_0	RMII_CLK	VO_CLK	SDIO1_CCLK_OUT

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Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
G17	RMII_TX_EN	muxctrl_reg5	GPIO2_1	RMII_TX_EN	VO_VS	SDIO1_CARD_DETECT
G16	RMII_TXD0	muxctrl_reg6	GPIO2_2	RMII_TXD0	VO_DATA5	SDIO1_CWPR
F18	RMII_TXD1	muxctrl_reg7	GPIO2_3	RMII_TXD1	VO_DE	SDIO1_CDATA1
F17	RMII_RX_DV	muxctrl_reg8	GPIO2_4	RMII_RX_DV	VO_DATA7	SDIO1_CDATA0
E16	RMII_RXD0	muxctrl_reg9	GPIO2_5	RMII_RXD0	VO_DATA2	SDIO1_CDATA3
F16	RMII_RXD1	muxctrl_reg10	GPIO2_6	RMII_RXD1	VO_DATA3	SDIO1_CCMD

Table 2-32 describes software multiplexed signals of the RMII.

Table 2-32 Description of software multiplexed signals of the RMII

Signal	Direction	Description
GPIO2_0	I/O	GPIO
GPIO2_1	I/O	GPIO
GPIO2_2	I/O	GPIO
GPIO2_3	I/O	GPIO
GPIO2_4	I/O	GPIO
GPIO2_5	I/O	GPIO
GPIO2_6	I/O	GPIO
RMII_CLK	I/O	RMII clock
RMII_RX_DV	I	RX data validity and carrier sense signal
RMII_RXD0	I	RX data 0
RMII_RXD1	I	RX data 1
RMII_TX_EN	O	TX enable
RMII_TXD0	O	TX data 0
RMII_TXD1	O	TX data 1
SDIO1_CARD_DETECT	I	SDIO1 card detection signal, active low
SDIO1_CCLK_OUT	O	SDIO1 output working clock for the card

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Signal	Direction	Description
SDIO1_CCMD	I/O	SDIO1 command
SDIO1_CDATA0	I/O	SDIO1 data 0
SDIO1_CDATA1	I/O	SDIO1 data 1
SDIO1_CDATA3	I/O	SDIO1 data 3
SDIO1_CWPR	I	SDIO1 card write protection detection
VO_CLK	O	VO clock signal
VO_DATA2	O	VO data 2
VO_DATA3	O	VO data 3
VO_DATA5	O	VO data 5
VO_DATA7	O	VO data 7
VO_DE	O	VO data validity signal
VO_VS	O	VO VS signal

EPHY

Table 2-33 lists software multiplexed pins of the EPHY.

Table 2-33 Software multiplexed pins of the EPHY

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
H18	EPHY_RST	muxctrl_reg11	GPIO2_7	EPHY_RST	BOOT_SEL	VO_HS	SDIO1_CARD_POWER_EN
D18	EPHY_CLK	muxctrl_reg13	GPIO3_0	EPHY_CLK	VO_DATA1	SDIO1_CDATA2	N/A

Table 2-34 describes software multiplexed signals of the EPHY.

Table 2-34 Description of software multiplexed signals of the EPHY

Signal	Direction	Description
BOOT_SEL	I	Boot enable 0: booting from the SPI flash 1: booting from the eMMC
EPHY_CLK	O	EPHY working clock

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Signal	Direction	Description
EPHY_RST	O	EPHY reset
GPIO2_7	I/O	GPIO
GPIO3_0	I/O	GPIO
SDIO1_CARD_POWER_EN	O	SDIO1 power enable signal. The value 1 indicates power-on.
SDIO1_CDATA2	I/O	SDIO1 data 2
VO_DATA1	O	VO data 1
VO_HS	O	VO HS signal

SPI1

Table 2-35 lists software multiplexed pins of SPI1.

Table 2-35 Software multiplexed pins of SPI1

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
D17	SPI1_CSN1	muxctrl_reg12	GPIO0_3	SPI1_CSN1	VO_DATA0
C17	SPI1_SCLK	muxctrl_reg20	GPIO3_7	SPI1_SCLK	I2C1_SCL
D16	SPI1_SDO	muxctrl_reg21	GPIO4_0	SPI1_SDO	I2C1_SDA
C18	SPI1_SDI	muxctrl_reg22	GPIO4_1	SPI1_SDI	N/A
D15	SPI1_CSN0	muxctrl_reg23	GPIO4_2	SPI1_CSN0	N/A

Table 2-36 describes software multiplexed signals of SPI1.

Table 2-36 Description of software multiplexed signals of SPI1

Signal	Direction	Description
GPIO0_3	I/O	GPIO
GPIO3_7	I/O	GPIO
GPIO4_0	I/O	GPIO
GPIO4_1	I/O	GPIO
GPIO4_2	I/O	GPIO

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Signal	Direction	Description
I2C1_SCL	I/O	I^2C1 bus clock
I2C1_SDA	I/O	I^2C1 bus data/address
SPI1_CSN0	O	SPI1 CS0
SPI1_CSN1	O	SPI1 CS1
SPI1_SCLK	O	SPI1 clock signal
SPI1_SDI	I	SPI1 input data
SPI1_SDO	O	SPI1 output data
VO_DATA0	O	VO data 0

MDIO

Table 2-37 lists software multiplexed pins of the MDIO.

Table 2-37 Software multiplexed pins of the MDIO

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
H17	MDCK	muxctrl_reg14	GPIO3_1	MDCK	BOOTROM_SEL	VO_DATA6
H16	MDIO	muxctrl_reg15	GPIO3_2	MDIO	VO_DATA4	N/A

Table 2-38 describes software multiplexed signals of the MDIO.

Table 2-38 Description of software multiplexed signals of the MDIO

Signal	Direction	Description
BOOTROM_SEL	I	BOOTROM enable 0: booting from other space 1: booting from BOOTROM
GPIO3_1	I/O	GPIO
GPIO3_2	I/O	GPIO
MDCK	O	MDIO interface clock
MDIO	I/O	Data signal of the MDIO interface
VO_DATA4	O	VO data 4
VO_DATA6	O	VO data 6

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SPI0

Table 2-39 lists software multiplexed pins of SPI0.

Table 2-39 Software multiplexed pins of SPI0

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
U1	SPI0_SCLK	muxctrl_reg16	GPIO3_3	SPI0_SCLK	I2C0_SCL
T1	SPI0_SDO	muxctrl_reg17	GPIO3_4	SPI0_SDO	I2C0_SDA
T2	SPI0_SDI	muxctrl_reg18	GPIO3_5	SPI0_SDI	N/A
U2	SPI0_CSN	muxctrl_reg19	GPIO3_6	SPI0_CSN	N/A

Table 2-40 describes software multiplexed signals of SPI0.

Table 2-40 Description of software multiplexed signals of SPI0

Signal	Direction	Description
GPIO3_3	I/O	GPIO
GPIO3_4	I/O	GPIO
GPIO3_5	I/O	GPIO
GPIO3_6	I/O	GPIO
I2C0_SCL	I/O	I^2C0 bus clock
I2C0_SDA	I/O	I^2C0 bus data/address
SPI0_CSN	O	SPI0 CS signal
SPI0_SCLK	O	SPI0 clock signal
SPI0_SDI	I	SPI0 input data
SPI0_SDO	O	SPI0 output data

I^2C2

Table 2-41 lists software multiplexed pins of I^2C2.

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Table 2-41 Software multiplexed pins of I²C2

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1
N1	I2C2_SDA	muxctrl_reg24	GPIO4_3	I2C2_SDA
N2	I2C2_SCL	muxctrl_reg25	GPIO4_4	I2C2_SCL

Table 2-42 describes software multiplexed signals of I²C2.

Table 2-42 Description of software multiplexed signals of I²C2

Signal	Direction	Description
GPIO4_3	I/O	GPIO
GPIO4_4	I/O	GPIO
I2C2_SCL	I/O	I²C2 bus clock
I2C2_SDA	I/O	I²C2 bus data/address

USB

Table 2-43 lists software multiplexed pins of the USB.

Table 2-43 Software multiplexed pins of the USB

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1
V11	USB_OVRCUR	muxctrl_reg26	GPIO4_5	USB_OVRCUR
U11	USB_PWREN	muxctrl_reg27	GPIO4_6	USB_PWREN

Table 2-44 describes software multiplexed signals of the USB.

Table 2-44 Description of software multiplexed signals of the USB

Signal	Direction	Description
GPIO4_5	I/O	GPIO
GPIO4_6	I/O	GPIO
USB_OVRCUR	I	USB port overcurrent indicator. The level polarity can be specified by configuring the peripheral control register MISC_CTRL30 bit[17].
USB_PWREN	O	Power control output pin of the USB port. The level

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Signal	Direction	Description
		polarity can be specified by configuring the peripheral control register MISC_CTRL30 bit[16].

GPIO

Table 2-45 lists software multiplexed pins of the GPIO.

Table 2-45 Software multiplexed pins of the GPIO

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
P3	GPIO0_0	muxctrl_reg28	GPIO0_0	IR_IN	TEMPER_DQ	N/A	N/A
P4	GPIO0_1	muxctrl_reg29	GPIO0_1	TEMPER_DQ	N/A	N/A	N/A
N3	GPIO0_2	muxctrl_reg30	GPIO0_2	TEMPER_DQ	N/A	N/A	N/A
T7	GPIO1_0	muxctrl_reg31	GPIO1_0	VI_DATA13	I2S_BCLK_TX	PWM0	N/A
V10	GPIO1_1	muxctrl_reg32	GPIO1_1	VI_DATA10	I2S_SD_TX	UART1_TXD	N/A
T8	GPIO1_2	muxctrl_reg33	GPIO1_2	VI_DATA12	I2S_MCLK	UART1_CTSN	N/A
U10	GPIO1_3	muxctrl_reg34	GPIO1_3	VI_DATA11	I2S_WS_TX	UART2_RXD	N/A
T10	GPIO1_4	muxctrl_reg35	GPIO1_4	VI_DATA15	VI_VS_SEN	I2S_WS_RX	UART1_RXD
T9	GPIO1_5	muxctrl_reg36	GPIO1_5	VI_DATA14	VI_HS_SEN	I2S_BCLK_RX	UART1_RTSN
U9	GPIO1_6	muxctrl_reg37	GPIO1_6	VI_DATA9	I2S_SD_RX	UART2_TXD	N/A

Table 2-46 describes software multiplexed signals of the GPIO.

Table 2-46 Description of software multiplexed signals of the GPIO

Signal	Direction	Description
GPIO0_0	I/O	GPIO
GPIO0_1	I/O	GPIO

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Signal	Direction	Description
GPIO0_2	I/O	GPIO
GPIO1_0	I/O	GPIO
GPIO1_1	I/O	GPIO
GPIO1_2	I/O	GPIO
GPIO1_3	I/O	GPIO
GPIO1_4	I/O	GPIO
GPIO1_5	I/O	GPIO
GPIO1_6	I/O	GPIO
I2S_BCLK_RX	I/O	RX clock of the I^2S or PCM interface
I2S_BCLK_TX	I/O	TX clock of the I^2S or PCM interface
I2S_MCLK	O	Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC.
I2S_SD_RX	I	RX data of the I^2S or PCM interface
I2S_SD_TX	O	TX data of the I^2S or PCM interface
I2S_WS_RX	I/O	I^2S RX audio channel select signal or PCM TX VS signal
I2S_WS_TX	I/O	I^2S TX audio channel select signal or PCM TX VS signal
IR_IN	I	IR input
PWM0	O	PWM output 0
TEMPER_DQ	I/O	Temperature measurement. It is used to communicate with the external temperature measurement chip.
UART1_CTSN	I	UART1 modem status input, active low
UART1_RTSN	O	UART1 modem status output, active low
UART1_RXD	I	UART1 RX data
UART1_TXD	O	UART1 TX data
UART2_RXD	I	UART2 RX data
UART2_TXD	O	UART2 TX data
VI_DATA10	I	VI data 10
VI_DATA11	I	VI data 11
VI_DATA12	I	VI data 12
VI_DATA13	I	VI data 13

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Signal	Direction	Description
VI_DATA14	I	VI data 14
VI_DATA15	I	VI data 15
VI_DATA9	I	VI data 9
VI_HS_SEN	I	VI HS signal
VI_VS_SEN	I	VI VS signal

SDIO

Table 2-47 lists software multiplexed pins of the SDIO.

Table 2-47 Software multiplexed pins of the SDIO

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
T16	SDIO0_CARD_POWER_EN	muxctrl_reg38	GPIO1_7	SDIO0_CARD_POWER_EN	N/A
R16	SDIO0_CARD_DETECT	muxctrl_reg39	GPIO4_7	SDIO0_CARD_DETECT	N/A
U17	SDIO0_CWPR	muxctrl_reg40	GPIO5_0	SDIO0_CWPR	N/A
R17	SDIO0_CCLK_OUT	muxctrl_reg41	GPIO5_1	SDIO0_CCLK_OUT	N/A
R18	SDIO0_CCMD	muxctrl_reg42	GPIO5_2	SDIO0_CCMD	N/A
T17	SDIO0_CDATA0	muxctrl_reg43	GPIO5_3	SDIO0_CDATA0	N/A
U18	SDIO0_CDATA1	muxctrl_reg44	TEST_CLK	SDIO0_CDATA1	GPIO5_4
P17	SDIO0_CDATA2	muxctrl_reg45	GPIO5_5	SDIO0_CDATA2	N/A
P18	SDIO0_CDATA3	muxctrl_reg46	GPIO5_6	SDIO0_CDATA3	N/A

Table 2-48 describes software multiplexed signals of the SDIO.

Table 2-48 Description of software multiplexed signals of the SDIO

Signal	Direction	Description
GPIO1_7	I/O	GPIO
GPIO4_7	I/O	GPIO

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Signal	Direction	Description
GPIO5_0	I/O	GPIO
GPIO5_1	I/O	GPIO
GPIO5_2	I/O	GPIO
GPIO5_3	I/O	GPIO
GPIO5_4	I/O	GPIO
GPIO5_5	I/O	GPIO
GPIO5_6	I/O	GPIO
SDIO0_CARD_DETECT	I	SDIO0 card detection signal, active low
SDIO0_CARD_POWER_EN	O	SDIO0 power enable signal. The value 1 indicates power-on.
SDIO0_CCLK_OUT	O	SDIO0 output working clock for the card
SDIO0_CCMD	I/O	SDIO0 command
SDIO0_CDATA0	I/O	SDIO0 data 0
SDIO0_CDATA1	I/O	SDIO0 data 1
SDIO0_CDATA2	I/O	SDIO0 data 2
SDIO0_CDATA3	I/O	SDIO0 data 3
SDIO0_CWPR	I	SDIO0 card write protection detection
TEST_CLK	O	Test clock

UART

Table 2-49 lists software multiplexed pins of the UART.

Table 2-49 Software multiplexed pins of the UART

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
J18	UART1_RTSN	muxctrl_reg47	GPIO5_7	EMMC_DATA6	I2S_SD_TX	UART1_RTSN
J17	UART1_RXD	muxctrl_reg48	GPIO6_0	EMMC_DATA5	I2S_WS_TX	UART1_RXD
J16	UART1_CTSN	muxctrl_reg49	GPIO6_1	EMMC_DATA7	I2S_MCLK	UART1_CTSN
K17	UART1_TXD	muxctrl_reg50	GPIO6_2	EMMC_DS	I2S_SD_RX	UART1_TXD

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Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3
L16	UART2_RXD	muxctrl_reg51	GPIO6_3	EMMC_DATA1	UART2_RXD	N/A
K16	UART2_TXD	muxctrl_reg52	GPIO6_4	EMMC_DATA2	I2S_BCLK_TX	UART2_TXD

Table 2-50 describes software multiplexed signals of the UART.

Table 2-50 Description of software multiplexed signals of the UART

Signal	Direction	Description
EMMC_DATA1	I/O	eMMC data 1
EMMC_DATA2	I/O	eMMC data 2
EMMC_DATA5	I/O	eMMC data 5
EMMC_DATA6	I/O	eMMC data 6
EMMC_DATA7	I/O	eMMC data 7
EMMC_DS	I	eMMC DATA_STROBE
GPIO5_7	I/O	GPIO
GPIO6_0	I/O	GPIO
GPIO6_1	I/O	GPIO
GPIO6_2	I/O	GPIO
GPIO6_3	I/O	GPIO
GPIO6_4	I/O	GPIO
I2S_BCLK_TX	I/O	TX clock of the I^2S or PCM interface
I2S_MCLK	O	Main clock of the I^2S or PCM interface. It can act as the working clock of the ACODEC.
I2S_SD_RX	I	RX data of the I^2S or PCM interface
I2S_SD_TX	O	TX data of the I^2S or PCM interface
I2S_WS_TX	I/O	I^2S TX audio channel select signal or PCM TX VS signal
UART1_CTSN	I	UART1 modem status input, active low
UART1_RTSN	O	UART1 modem status output, active low
UART1_RXD	I	UART1 RX data
UART1_TXD	O	UART1 TX data

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Signal	Direction	Description
UART2_RXD	I	UART2 RX data
UART2_TXD	O	UART2 TX data

JTAG

Table 2-51 lists software multiplexed pins of the JTAG.

Table 2-51 Software multiplexed pins of the JTAG

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2	Multiplexed Signal 3	Multiplexed Signal 4
G3	JTAG_TRSTN	muxctrl_reg53	GPIO6_5	JTAG_TRSTN	SPI1_CSN1	I2S_MCLK	N/A
G1	JTAG_TCK	muxctrl_reg54	GPIO6_6	JTAG_TCK	SPI1_SCLK	I2S_WS_TX	I2C1_SCL
F4	JTAG_TMS	muxctrl_reg55	GPIO6_7	JTAG_TMS	SPI1_CSN0	I2S_SD_TX	N/A
F2	JTAG_TDO	muxctrl_reg56	GPIO7_0	JTAG_TDO	SPI1_SDO	I2S_SD_RX	I2C1_SDA
G2	JTAG_TDI	muxctrl_reg57	GPIO7_1	JTAG_TDI	SPI1_SDI	I2S_BCLK_TX	N/A

Table 2-52 describes software multiplexed signals of the JTAG.

Table 2-52 Description of software multiplexed signals of the JTAG

Signal	Direction	Description
GPIO6_5	I/O	GPIO
GPIO6_6	I/O	GPIO
GPIO6_7	I/O	GPIO
GPIO7_0	I/O	GPIO
GPIO7_1	I/O	GPIO
I2C1_SCL	I/O	I^2C1 bus clock
I2C1_SDA	I/O	I^2C1 bus data/address
I2S_BCLK_TX	I/O	TX clock of the I^2S or PCM interface
I2S_MCLK	O	Main clock of the I^2S or PCM interface. It can act as

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Signal	Direction	Description
		the working clock of the ACODEC.
I2S_SD_RX	I	RX data of the I^2S or PCM interface
I2S_SD_TX	O	TX data of the I^2S or PCM interface
I2S_WS_TX	I/O	I^2S TX audio channel select signal or PCM TX VS signal
JTAG_TCK	I	JTAG clock signal
JTAG_TDI	I	JTAG input data
JTAG_TDO	O	JTAG output data
JTAG_TMS	I	JTAG mode select input
JTAG_TRSTN	I	JTAG reset signal
SPI1_CSN0	O	SPI1 CS0
SPI1_CSN1	O	SPI1 CS1
SPI1_SCLK	O	SPI1 clock signal
SPI1_SDI	I	SPI1 input data
SPI1_SDO	O	SPI1 output data

PWM

Table 2-53 lists software multiplexed pins of the PWM.

Table 2-53 Software multiplexed pins of the PWM

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1	Multiplexed Signal 2
M2	PWM0	muxctrl_reg58	PMC_PWM	GPIO7_2	PWM0
M3	PWM1	muxctrl_reg59	PWM1	GPIO7_3	N/A
K3	PWM2	muxctrl_reg60	PWM2	GPIO7_4	SPI1_CSN1
L3	PWM3	muxctrl_reg61	PWM3	GPIO7_5	N/A

Table 2-54 describes software multiplexed signals of the PWM.

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Table 2-54 Description of software multiplexed signals of the PWM

Signal	Direction	Description
GPIO7_2	I/O	GPIO
GPIO7_3	I/O	GPIO
GPIO7_4	I/O	GPIO
GPIO7_5	I/O	GPIO
PMC_PWM	O	PMC PWM output
PWM0	O	PWM output 0
PWM1	O	PWM output 1
PWM2	O	PWM output 2
PWM3	O	PWM output 3
SPI1_CSN1	O	SPI1 CS1

SAR_ADC

Table 2-55 lists software multiplexed pins of the SAR_ADC.

Table 2-55 Software multiplexed pins of the SAR_ADC

Pin	Pad Signal	Multiplexing Control Register	Multiplexed Signal 0	Multiplexed Signal 1
P1	SAR_ADC_CH0	muxctrl_reg62	SAR_ADC_CH0	GPIO7_6
P2	SAR_ADC_CH1	muxctrl_reg63	SAR_ADC_CH1	GPIO7_7
R1	SAR_ADC_CH2	muxctrl_reg64	SAR_ADC_CH2	GPIO8_0
R3	SAR_ADC_CH3	muxctrl_reg65	SAR_ADC_CH3	GPIO8_1

Table 2-56 describes software multiplexed signals of the SAR_ADC.

Table 2-56 Description of software multiplexed signals of the SAR_ADC

Signal	Direction	Description
GPIO7_6	I/O	GPIO
GPIO7_7	I/O	GPIO
GPIO8_0	I/O	GPIO
GPIO8_1	I/O	GPIO

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Signal	Direction	Description
SAR_ADC_CH0	I	LSADC channel 0
SAR_ADC_CH1	I	LSADC channel 1
SAR_ADC_CH2	I	LSADC channel 2
SAR_ADC_CH3	I	LSADC channel 3

2.5 Hardware Multiplexed Pins

SFC Pins

Table 2-57 lists hardware multiplexed pins of the SFC.

Table 2-57 Hardware multiplexed pins of the SFC

Pin	Pad Signal	Multiplexed Signal 1 (BOOT_SEL == 1'b1)
L17	SFC_CLK	EMMC_CLK
M17	SFC_SIO_IO0	EMMC_DATA4
N17	SFC_SOI_IO1	EMMC_DATA3
N18	SFC_WP_IO2	EMMC_CMD
L18	SFC_HOLD_IO3	EMMC_DATA0
M18	SFC_CSN0	EMMC_RST_N
N16	SFC_CSN1	EMMC_POWER_EN

Table 2-58 describes hardware multiplexed signals of the SFC.

Table 2-58 Description of hardware multiplexed signals of the SFC

Signal	Direction	Description
EMMC_CLK	O	eMMC clock
EMMC_DATA4	I/O	eMMC data 4
EMMC_DATA3	I/O	eMMC data 3
EMMC_CMD	I/O	eMMC command/response
EMMC_DATA0	I/O	eMMC data 0

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Signal	Direction	Description
EMMC_RST_N	O	eMMC reset
EMMC_POWER_EN	O	eMMC power enable signal. The value 1 indicates power-on.

2.6 Pin Drive Capability Registers

2.6.1 Register Summary

Table 2-59 describes pin drive capability registers.

Table 2-59 Summary of pin drive capability registers (base address: 0x200F_0800)

Offset Address	Register	Description	Page
0x000	PADCTRL_REG0	Control register for the SENSOR_CLK functional pin	2-103
0x004	PADCTRL_REG1	Control register for the SENSOR_RSTN functional pin	2-104
0x008	PADCTRL_REG2	Control register for the FLASH_TRIG functional pin	2-105
0x00C	PADCTRL_REG3	Control register for the SHUTTER_TRIG functional pin	2-105
0x010	PADCTRL_REG4	Control register for the RMII_CLK functional pin	2-106
0x014	PADCTRL_REG5	Control register for the RMII_TX_EN functional pin	2-107
0x018	PADCTRL_REG6	Control register for the RMII_TXD0 functional pin	2-108
0x01C	PADCTRL_REG7	Control register for the RMII_TXD1 functional pin	2-108
0x020	PADCTRL_REG8	Control register for the RMII_RX_DV functional pin	2-109
0x024	PADCTRL_REG9	Control register for the RMII_RXD0 functional pin	2-110
0x028	PADCTRL_REG10	Control register for the RMII_RXD1 functional pin	2-111
0x02C	PADCTRL_REG11	Control register for the EPHY_RST functional pin	2-111

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Offset Address	Register	Description	Page
0x030	PADCTRL_REG12	Control register for the SPI1_CSN1 functional pin	2-112
0x034	PADCTRL_REG13	Control register for the EPHY_CLK functional pin	2-113
0x038	PADCTRL_REG14	Control register for the MDCK functional pin	2-114
0x03C	PADCTRL_REG15	Control register for the MDIO functional pin	2-115
0x040	PADCTRL_REG16	Control register for the SPI0_SCLK functional pin	2-115
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0x094	PADCTRL_REG37	Control register for the GPIO1_2 functional pin	2-130
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0x0BC	PADCTRL_REG47	Control register for the SDIO0_CDATA0 functional pin	2-138
0x0C0	PADCTRL_REG48	Control register for the SDIO0_CDATA1 functional pin	2-139
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0x0E8	PADCTRL_REG58	Control register for the UART1_RTSN functional pin	2-148
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0x0F4	PADCTRL_REG61	Control register for the UART1_TXD functional pin	2-151
0x0F8	PADCTRL_REG62	Control register for the UART2_RXD functional pin	2-152
0x0FC	PADCTRL_REG63	Control register for the UART2_TXD functional pin	2-153
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0x108	PADCTRL_REG66	Control register for the JTAG_TMS functional pin	2-156
0x10C	PADCTRL_REG67	Control register for the JTAG_TDO functional pin	2-157
0x110	PADCTRL_REG68	Control register for the JTAG_TDI functional pin	2-157
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Offset Address	Register	Description	Page
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0x124	PADCTRL_REG73	Control register for the PWM3 functional pin	2-160
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0x12C	PADCTRL_REG75	Control register for the SAR_ADC_CH1 functional pin	2-162
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2.6.2 Register Description

PADCTRL_REG0

PADCTRL_REG0 is a control register for the SENSOR_CLK functional pin.

Offset Address	Register Name	Total Reset Value
0x000	PADCTRL_REG0	0x0000_0040

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6 5 4	3 2 1 0
Name	reserved						padctrl_reg0_sr	padctrl_reg0_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	1 0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg0_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg0_ds	Drive capability of the SENSOR_CLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG1

PADCTRL_REG1 is a control register for the SENSOR_RSTN functional pin.

Offset Address	Register Name	Total Reset Value
0x004	PADCTRL_REG1	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg1_sr	reserved	padctrl_reg1_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg1_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg1_ds	Drive capability of the SENSOR_RSTN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG2

PADCTRL_REG2 is a control register for the FLASH_TRIG functional pin.

Offset Address	Register Name	Total Reset Value
0x008	PADCTRL_REG2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg2_sl	reserved	padctrl_reg2_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg2_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg2_ds	Drive capability of the FLASH_TRIG pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG3

PADCTRL_REG3 is a control register for the SHUTTER_TRIG functional pin.

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Offset Address	Register Name	Total Reset Value
0x00C	PADCTRL_REG3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg3_sl	reserved	padctrl_reg3_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg3_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg3_ds	Drive capability of the SHUTTER_TRIG pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG4

PADCTRL_REG4 is a control register for the RMII_CLK functional pin.

Offset Address	Register Name	Total Reset Value
0x010	PADCTRL_REG4	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg4_sr	padctrl_reg4_ds			reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg4_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6:4]	RW	padctrl_reg4_ds	Drive capability of the RMII_CLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG5

PADCTRL_REG5 is a control register for the RMII_TX_EN functional pin.

Offset Address	Register Name	Total Reset Value
0x014	PADCTRL_REG5	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg5_sr	reserved	padctrl_reg5_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg5_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg5_ds	Drive capability of the RMII_TX_EN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG6

PADCTRL_REG6 is a control register for the RMII_TXD0 functional pin.

Offset Address	Register Name	Total Reset Value
0x018	PADCTRL_REG6	0x0000_0080

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg6_sr	reserved	padctrl_reg6_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg6_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg6_ds	Drive capability of the RMII_TXD0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG7

PADCTRL_REG7 is a control register for the RMII_TXD1 functional pin.

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Offset Address	Register Name	Total Reset Value
0x01C	PADCTRL_REG7	0x0000_0080

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg7_sr	reserved	padctrl_reg7_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg7_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg7_ds	Drive capability of the RMII_TXD1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG8

PADCTRL_REG8 is a control register for the RMII_RX_DV functional pin.

Offset Address	Register Name	Total Reset Value
0x020	PADCTRL_REG8	0x0000_0080

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg8_sr	reserved	padctrl_reg8_ds0	reserved

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg8_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg8_ds0	Drive capability of the RMII_RX_DV pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG9

PADCTRL_REG9 is a control register for the RMII_RXD0 functional pin.

Offset Address	Register Name	Total Reset Value
0x024	PADCTRL_REG9	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg9_sr	reserved	padctrl_reg9_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg9_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg9_ds	Drive capability of the RMII_RXD0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG10

PADCTRL_REG10 is a control register for the RMII_RXD1 functional pin.

Offset Address	Register Name	Total Reset Value
0x028	PADCTRL_REG10	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg10_sr	reserved	padctrl_reg10_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg10_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg10_ds	Drive capability of the RMII_RXD1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG11

PADCTRL_REG11 is a control register for the EPHY_RST functional pin.

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Offset Address	Register Name	Total Reset Value
0x02C	PADCTRL_REG11	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg11_sr	reserved	padctrl_reg11_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg11_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg11_ds	Drive capability of the EPHY_RST pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG12

PADCTRL_REG12 is a control register for the SPI1_CSN1 functional pin.

Offset Address	Register Name	Total Reset Value
0x030	PADCTRL_REG12	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg12_sr	reserved	padctrl_reg12_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg12_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg12_ds	Drive capability of the SPI1_CSN1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG13

PADCTRL_REG13 is a control register for the EPHY_CLK functional pin.

Offset Address	Register Name	Total Reset Value
0x034	PADCTRL_REG13	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg12_sr	padctrl_reg12_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg12_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg12_ds	Drive capability of the EPHY_CLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG14

PADCTRL_REG14 is a control register for the MDCK functional pin.

Offset Address	Register Name	Total Reset Value
0x038	PADCTRL_REG14	0x0000_0080

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg14_sr	reserved	padctrl_reg14_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg14_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg14_ds	Drive capability of the MDCK pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG15

PADCTRL_REG15 is a control register for the MDIO functional pin.

Offset Address	Register Name	Total Reset Value
0x03C	PADCTRL_REG15	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg15_sr	reserved	padctrl_reg15_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg15_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg15_ds	Drive capability of the MDIO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG16

PADCTRL_REG16 is a control register for the SPI0_SCLK functional pin.

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Offset Address	Register Name	Total Reset Value
0x040	PADCTRL_REG16	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg16_sr	padctrl_reg16_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg16_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6:4]	RW	padctrl_reg16_ds	Drive capability of the SPI0_SCLK pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG17

PADCTRL_REG17 is a control register for the SPI0_SDO functional pin.

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Offset Address	Register Name	Total Reset Value
0x044	PADCTRL_REG17	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg17_sr	reserved	padctrl_reg17_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg17_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg17_ds	Drive capability of the SPI0_SDO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG18

PADCTRL_REG18 is a control register for the SPI0_SDI functional pin.

Offset Address	Register Name	Total Reset Value
0x048	PADCTRL_REG18	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg17_sr	reserved	padctrl_reg17_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg17_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg17_ds	Drive capability of the SPI0_SDI pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG19

PADCTRL_REG19 is a control register for the SPI0_CSN functional pin.

Offset Address	Register Name	Total Reset Value
0x04C	PADCTRL_REG19	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg19_sr	reserved	padctrl_reg19_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg19_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg19_ds	Drive capability of the SPI0_CSN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG20

PADCTRL_REG20 is a control register for the SPI1_SCLK functional pin.

Offset Address	Register Name	Total Reset Value
0x050	PADCTRL_REG20	0x0000_0090

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg20_sr	reserved	padctrl_reg20_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg20_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg20_ds	Drive capability of the SPI1_SCLK pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG21

PADCTRL_REG21 is a control register for the SPI1_SDO functional pin.

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Offset Address	Register Name	Total Reset Value
0x054	PADCTRL_REG21	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg21_sr	reserved	padctrl_reg21_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg21_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg21_ds	Drive capability of the SPI1_SDO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG22

PADCTRL_REG22 is a control register for the SPI1_SDI functional pin.

Offset Address	Register Name	Total Reset Value
0x058	PADCTRL_REG22	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg22_sr	reserved	padctrl_reg22_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg22_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg22_ds	Drive capability of the SPI1_SDI pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG23

PADCTRL_REG23 is a control register for the SPI1_CSN0 functional pin.

Offset Address	Register Name	Total Reset Value
0x05C	PADCTRL_REG23	0x0000_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg23_sr	reserved	padctrl_reg23_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg23_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg23_ds	Drive capability of the SPI1_CSN0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG24

PADCTRL_REG24 is a control register for the I2C2_SDA functional pin.

Offset Address	Register Name	Total Reset Value
0x060	PADCTRL_REG24	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg24_sl	reserved	padctrl_reg24_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg24_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg24_ds	Drive capability of the I2C2_SDA pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG25

PADCTRL_REG25 is a control register for the I2C2_SCL functional pin.

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Offset Address	Register Name	Total Reset Value
0x064	PADCTRL_REG25	0x0000_0010

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg25_sr	reserved	padctrl_reg25_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0 1	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg25_sr	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg25_ds	Drive capability of the I2C2_SCL pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG26

PADCTRL_REG26 is a control register for the UART0_RXD functional pin.

Offset Address	Register Name	Total Reset Value
0x068	PADCTRL_REG26	0x0000_00B0

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg26_sr	reserved	padctrl_reg26_ds	reserved

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg26_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg26_ds	Drive capability of the UART0_RXD pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG27

PADCTRL_REG27 is a control register for the UART0_TXD functional pin.

Offset Address	Register Name	Total Reset Value
0x06C	PADCTRL_REG27	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg27_sr	reserved	padctrl_reg27_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg27_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg27_ds	Drive capability of the UART0_TXD pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG30

PADCTRL_REG30 is a control register for the USB_OVRCUR functional pin.

Offset Address	Register Name	Total Reset Value
0x078	PADCTRL_REG30	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg30_sr	reserved	padctrl_reg30_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg30_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg30_ds	Drive capability of the USB_OVRCUR pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG31

PADCTRL_REG31 is a control register for the USB_PWREN functional pin.

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Offset Address	Register Name	Total Reset Value
0x07C	PADCTRL_REG31	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg31_sr	reserved	padctrl_reg31_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg31_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg31_ds	Drive capability of the USB_PWREN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG32

PADCTRL_REG32 is a control register for the GPIO0_0 functional pin.

Offset Address	Register Name	Total Reset Value
0x080	PADCTRL_REG32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg32_sl	reserved	padctrl_reg32_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg32_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg32_ds	Drive capability of the GPIO0_0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG33

PADCTRL_REG33 is a control register for the GPIO0_1 functional pin.

Offset Address	Register Name	Total Reset Value
0x084	PADCTRL_REG33	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg33_sl	reserved	padctrl_reg33_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg33_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg33_ds	Drive capability of the GPIO0_1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG34

PADCTRL_REG34 is a control register for the GPIO0_2 functional pin.

Offset Address	Register Name	Total Reset Value
0x088	PADCTRL_REG34	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg34_sl	reserved	padctrl_reg34_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg34_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg34_ds	Drive capability of the GPIO0_2 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG35

PADCTRL_REG35 is a control register for the GPIO1_0 functional pin.

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Offset Address	Register Name	Total Reset Value
0x08C	PADCTRL_REG35	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg35_sr	reserved	padctrl_reg35_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg35_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg35_ds	Drive capability of the GPIO1_0 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG36

PADCTRL_REG36 is a control register for the GPIO1_1 functional pin.

Offset Address	Register Name	Total Reset Value
0x090	PADCTRL_REG36	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg36_sr	reserved	padctrl_reg36_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg36_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg36_ds	Drive capability of the GPIO1_1 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG37

PADCTRL_REG37 is a control register for the GPIO1_2 functional pin.

Offset Address	Register Name	Total Reset Value
0x094	PADCTRL_REG37	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg37_sr	reserved	padctrl_reg37_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg37_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg37_ds	Drive capability of the GPIO1_2 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG38

PADCTRL_REG38 is a control register for the GPIO1_3 functional pin.

Offset Address	Register Name	Total Reset Value
0x098	PADCTRL_REG38	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg38_sr	reserved	padctrl_reg38_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg38_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg38_ds	Drive capability of the GPIO1_3 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG39

PADCTRL_REG39 is a control register for the GPIO1_4 functional pin.

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Offset Address	Register Name	Total Reset Value
0x09C	PADCTRL_REG39	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg39_sr	reserved	padctrl_reg39_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg39_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg39_ds	Drive capability of the GPIO1_4 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG40

PADCTRL_REG40 is a control register for the GPIO1_5 functional pin.

Offset Address	Register Name	Total Reset Value
0x0A0	PADCTRL_REG40	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg40_sr	reserved	padctrl_reg40_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg40_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg40_ds	Drive capability of the GPIO1_5 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG41

PADCTRL_REG41 is a control register for the GPIO1_6 functional pin.

Offset Address	Register Name	Total Reset Value
0x0A4	PADCTRL_REG41	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg41_sr	reserved	padctrl_reg41_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg41_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg41_ds	Drive capability of the GPIO1_6 pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG42

PADCTRL_REG42 is a control register for the SDIO0_CARD_POWER_EN functional pin.

Offset Address	Register Name	Total Reset Value
0x0A8	PADCTRL_REG42	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg42_sr	reserved	padctrl_reg42_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg42_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg42_ds	Drive capability of the SDIO0_CARD_POWER_EN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG43

PADCTRL_REG43 is a control register for the SDIO0_CARD_DETECT functional pin.

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Offset Address	Register Name	Total Reset Value
0x0AC	PADCTRL_REG43	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg43_sr	reserved	padctrl_reg43_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg43_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg43_ds	Drive capability of the SDIO0_CARD_DETECT pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG44

PADCTRL_REG44 is a control register for the SDIO0_CWPR functional pin.

Offset Address	Register Name	Total Reset Value
0x0B0	PADCTRL_REG44	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg44_sr	reserved	padctrl_reg44_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg44_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg44_ds	Drive capability of the SDIO0_CWPR pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG45

PADCTRL_REG45 is a control register for the SDIO0_CCLK_OUT functional pin.

Offset Address	Register Name	Total Reset Value
0x0B4	PADCTRL_REG45	0x0000_00D0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg45_sr	padctrl_reg45_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg45_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg45_ds	Drive capability of the SDIO0_CCLK_OUT pin 000: 18 mA 001: 16 mA 010: 14 mA 011: 12 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG46

PADCTRL_REG46 is a control register for the SDIO0_CCMD functional pin.

Offset Address	Register Name	Total Reset Value
0x0B8	PADCTRL_REG46	0x0000_00E0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg46_sr	padctrl_reg46_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg46_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg46_ds	Drive capability of the SDIO0_CCMD pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG47

PADCTRL_REG47 is a control register for the SDIO0_CDATA0 functional pin.

Offset Address	Register Name	Total Reset Value
0x0BC	PADCTRL_REG47	0x0000_00E0

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6 5 4	3 2 1 0
Name	reserved						padctrl_reg47_sr	padctrl_reg47_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	1 1 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg47_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg47_ds	Drive capability of the SDIO0_CDATA0 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG48

PADCTRL_REG48 is a control register for the SDIO0_CDATA1 functional pin.

Offset Address	Register Name	Total Reset Value
0x0C0	PADCTRL_REG48	0x0000_00E0

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6 5 4	3 2 1 0
Name	reserved						padctrl_reg48_sr	padctrl_reg48_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	1 1 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg48_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg48_ds	Drive capability of the SDIO0_CDATA1 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG49

PADCTRL_REG49 is a control register for the SDIO0_CDATA2 functional pin.

Offset Address	Register Name	Total Reset Value
0x0C4	PADCTRL_REG49	0x0000_00E0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg49_sr	padctrl_reg49_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg49_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg49_ds	Drive capability of the SDIO0_CDATA2 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG50

PADCTRL_REG50 is a control register for the SDIO0_CDATA3 functional pin.

Offset Address	Register Name	Total Reset Value
0x0C8	PADCTRL_REG50	0x0000_00E0

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6 5 4	3 2 1 0
Name	reserved						padctrl_reg50_sr	padctrl_reg50_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1	1 1 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg50_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge

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[6:4]	RW	padctrl_reg50_ds	Drive capability of the SDIO0_CDATA3 pin 000: 12 mA 001: 11 mA 010: 9 mA 011: 8 mA 100: 5 mA 101: 4 mA 110: 2 mA 111: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG51

PADCTRL_REG51 is a control register for the SFC_CLK functional pin.

Offset Address	Register Name	Total Reset Value
0x0CC	PADCTRL_REG51	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg51_st	padctrl_reg51_pd	padctrl_reg51_pu	reserved	padctrl_reg51_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg51_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg51_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg51_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg51_ds	Drive capability of the SFC_CLK pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG52

PADCTRL_REG52 is a control register for the SFC_SIO_IO0 functional pin.

Offset Address	Register Name	Total Reset Value
0x0D0	PADCTRL_REG52	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg52_st	padctrl_reg52_pd	padctrl_reg52_pu	reserved	padctrl_reg52_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg52_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg52_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg52_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg52_ds	Drive capability of the SFC_SIO_IO0 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG53

PADCTRL_REG53 is a control register for the SFC_SOI_IO1 functional pin.

Offset Address	Register Name	Total Reset Value
0x0D4	PADCTRL_REG53	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg53_st	padctrl_reg53_pd	padctrl_reg53_pu	reserved	padctrl_reg53_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg53_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg53_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg53_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg53_ds	Drive capability of the SFC_SOI_IO1 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG54

PADCTRL_REG54 is a control register for the SFC_WP_IO2 functional pin.

Offset Address	Register Name	Total Reset Value
0x0D8	PADCTRL_REG54	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg54_st	padctrl_reg54_pd	padctrl_reg54_pu	reserved			padctrl_reg54_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg54_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg54_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg54_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg54_ds	Drive capability of the SFC_WP_IO2 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG55

PADCTRL_REG55 is a control register for the SFC_HOLD_IO3 functional pin.

Offset Address	Register Name	Total Reset Value
0x0DC	PADCTRL_REG55	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg55_st	padctrl_reg55_pd	padctrl_reg55_pu	reserved			padctrl_reg55_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg55_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg55_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg55_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved

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[6:4]	RW	padctrl_reg55_ds	Drive capability of the SFC_HOLD_IO3 pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG56

PADCTRL_REG56 is a control register for the SFC_CSN0 functional pin.

Offset Address	Register Name	Total Reset Value
0x0E0	PADCTRL_REG56	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg56_sl	reserved	padctrl_reg56_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg56_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg56_ds	Drive capability of the SFC_CSN0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

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PADCTRL_REG57

PADCTRL_REG57 is a control register for the SFC_CSN1 functional pin.

Offset Address	Register Name	Total Reset Value
0x0E4	PADCTRL_REG57	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg57_sl	reserved	padctrl_reg57_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg57_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg57_ds	Drive capability of the SFC_CSN1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG58

PADCTRL_REG58 is a control register for the UART1_RTSN functional pin.

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Offset Address	Register Name	Total Reset Value
0x0E8	PADCTRL_REG58	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg58_st	padctrl_reg58_pd	padctrl_reg58_pu	reserved				padctrl_reg58_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg58_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg58_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg58_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg58_ds	Drive capability of the UART1_RTSN pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG59

PADCTRL_REG59 is a control register for the UART1_RXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0EC	PADCTRL_REG59	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg59_st	padctrl_reg59_pd	padctrl_reg59_pu	reserved				padctrl_reg59_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg59_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg59_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg59_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg59_ds	Drive capability of the UART1_RXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG60

PADCTRL_REG60 is a control register for the UART1_CTSN functional pin.

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Offset Address	Register Name	Total Reset Value
0x0F0	PADCTRL_REG60	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg60_st	padctrl_reg60_pd	padctrl_reg60_pu	reserved			padctrl_reg60_ds			reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg60_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg60_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg60_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg60_ds	Drive capability of the UART1_CTSN pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG61

PADCTRL_REG61 is a control register for the UART1_TXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0F4	PADCTRL_REG61	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg61_st	padctrl_reg61_pd	padctrl_reg61_pu	reserved				padctrl_reg61_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg61_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg61_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg61_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg61_ds	Drive capability of the UART1_TXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG62

PADCTRL_REG62 is a control register for the UART2_RXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0F8	PADCTRL_REG62	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg62_st	padctrl_reg62_pd	padctrl_reg62_pu	reserved				padctrl_reg62_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg62_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg62_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg62_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg62_ds	Drive capability of the UART2_RXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG63

PADCTRL_REG63 is a control register for the UART2_TXD functional pin.

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Offset Address	Register Name	Total Reset Value
0x0FC	PADCTRL_REG63	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg63_st	padctrl_reg63_pd	padctrl_reg63_pu	reserved				padctrl_reg63_ds				reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg63_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg63_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg63_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RO	reserved	Reserved
[6:4]	RW	padctrl_reg63_ds	Drive capability of the UART2_TXD pin 000: 1 mA 001: 2 mA 010: 4 mA 011: 5 mA 100: 12 mA 101: 14 mA 110: 16 mA 111: 18 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG64

PADCTRL_REG64 is a control register for the JTAG_TRSTN functional pin.

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Offset Address	Register Name	Total Reset Value
0x100	PADCTRL_REG64	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg64_sr	reserved	padctrl_reg64_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg64_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg64_ds	Drive capability of the JTAG_TRSTN pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG65

PADCTRL_REG65 is a control register for the JTAG_TCK functional pin.

Offset Address	Register Name	Total Reset Value
0x104	PADCTRL_REG65	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg65_sr	reserved	padctrl_reg65_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg65_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg65_ds	Drive capability of the JTAG_TCK pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG66

PADCTRL_REG66 is a control register for the JTAG_TMS functional pin.

Offset Address	Register Name	Total Reset Value
0x108	PADCTRL_REG66	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg66_sr	reserved	padctrl_reg66_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg66_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg66_ds	Drive capability of the JTAG_TMS pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG67

PADCTRL_REG67 is a control register for the JTAG_TDO functional pin.

Offset Address	Register Name	Total Reset Value
0x10C	PADCTRL_REG67	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg67_sr	reserved	padctrl_reg67_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg67_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg67_ds	Drive capability of the JTAG_TDO pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG68

PADCTRL_REG68 is a control register for the JTAG_TDI functional pin.

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Offset Address	Register Name	Total Reset Value
0x110	PADCTRL_REG68	0x0000_00B0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg68_sr	reserved	padctrl_reg68_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg68_sr	Rising/Falling speed control of the output signal 0: fast edge 1: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg68_ds	Drive capability of the JTAG_TDI pin 00: 4 mA 01: 3 mA 10: 2 mA 11: 1 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG70

PADCTRL_REG70 is a control register for the PWM0 functional pin.

Offset Address	Register Name	Total Reset Value
0x118	PADCTRL_REG70	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg70_sl	reserved	padctrl_reg70_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg70_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg70_ds	Drive capability of the PWM0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG71

PADCTRL_REG71 is a control register for the PWM1 functional pin.

Offset Address	Register Name	Total Reset Value
0x11C	PADCTRL_REG71	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg71_sl	reserved	padctrl_reg71_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg71_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved

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[5:4]	RW	padctrl_reg71_ds	Drive capability of the PWM1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG72

PADCTRL_REG72 is a control register for the PWM2 functional pin.

Offset Address	Register Name	Total Reset Value
0x120	PADCTRL_REG72	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						padctrl_reg72_sl	reserved	padctrl_reg72_ds	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg72_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg72_ds	Drive capability of the PWM2 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG73

PADCTRL_REG73 is a control register for the PWM3 functional pin.

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Offset Address	Register Name	Total Reset Value
0x124	PADCTRL_REG73	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								padctrl_reg73_sl	reserved	padctrl_reg73_ds		reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	padctrl_reg73_sl	Rising/Falling speed control of the output signal 1: fast edge 0: slow edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg73_ds	Drive capability of the PWM3 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG74

PADCTRL_REG74 is a control register for the SAR_ADC_CH0 functional pin.

Offset Address	Register Name	Total Reset Value
0x128	PADCTRL_REG74	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg94_st	padctrl_reg94_pd	padctrl_reg94_pu	padctrl_reg94_sl	reserved	padctrl_reg94_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg94_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg94_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg94_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg94_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg94_ds	Drive capability of the SAR_ADC_CH0 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG75

PADCTRL_REG75 is a control register for the SAR_ADC_CH1 functional pin.

Offset Address	Register Name	Total Reset Value
0x12C	PADCTRL_REG75	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg95_st	padctrl_reg95_pd	padctrl_reg95_pu	padctrl_reg95_sl	reserved	padctrl_reg95_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg95_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg95_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg95_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg95_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg95_ds	Drive capability of the SAR_ADC_CH1 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG76

PADCTRL_REG76 is a control register for the SAR_ADC_CH2 functional pin.

Offset Address	Register Name	Total Reset Value
0x130	PADCTRL_REG76	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg96_st	padctrl_reg96_pd	padctrl_reg96_pu	padctrl_reg96_sl	reserved	padctrl_reg96_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg96_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg96_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg96_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg96_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg96_ds	Drive capability of the SAR_ADC_CH2 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

PADCTRL_REG77

PADCTRL_REG77 is a control register for the SAR_ADC_CH3 functional pin.

Offset Address	Register Name	Total Reset Value
0x134	PADCTRL_REG77	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					padctrl_reg97_st	padctrl_reg97_pd	padctrl_reg97_pu	padctrl_reg97_sl	reserved	padctrl_reg97_ds		reserved			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	padctrl_reg97_st	Input Schmitt control 0: no Schmitt input 1: Schmitt input
[9]	RW	padctrl_reg97_pd	Input pull-down control 0: no pull-down 1: pull-down
[8]	RW	padctrl_reg97_pu	Input pull-up control 0: no pull-up 1: pull-up
[7]	RW	padctrl_reg97_sl	Rising/Falling speed control of the output signal 0: slow edge 1: fast edge
[6]	RO	reserved	Reserved
[5:4]	RW	padctrl_reg97_ds	Drive capability of the SAR_ADC_CH3 pin 00: 1 mA 01: 2 mA 10: 3 mA 11: 4 mA
[3:0]	RO	reserved	Reserved

2.7 Electrical Specifications

2.7.1 Power Consumption Parameters

Table 2-60 to Table 2-62 describe power consumption parameters.

CAUTION

- The values of power consumption parameters are provided based on typical application scenarios.

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- The maximum values are the tested maximum average power consumption within a specific time period, and cannot reflect the transient maximum power consumption when the system is working. Therefore, design board power supplies by following the *Hi3518E V20X/Hi3516C V200 Hardware Design User Guide*.

Table 2-60 Power consumption parameters when Hi3516C V200 does not use the SVB technology

Parameter	Description	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	282	370	mW
1.8 V I/O power	1.8 V module power	54	60	mW
3.3 V power	Power consumption of the 3.3 V I/O power	122	125	mW
1.5 V power	Power consumption of the DDR interface	69	74	mW

Table 2-61 Power consumption parameters when Hi3516C V200 uses the SVB technology

Parameter	Description	Typ	Max	Unit
VDD core power	Power consumption of the VDD core power domain	259	310	mW
1.8 V I/O power	1.8 V module power	54	60	mW
3.3 V power	Power consumption of the 3.3 V I/O power	122	125	mW
1.5 V power	Power consumption of the DDR interface	70	75	mW

Table 2-62 Power consumption benefit parameters when Hi3516C V200 uses the SVB technology

Typ	Max
~22 mW	~ 59 mW

NOTE

The typical application scenario of Hi3516C V200 is as follows: ARM9@600 MHz+1080p@30 image signal processor (ISP) processing+3DNR+1-channel video graphics subsystem (VGS) (only one of the rotation, LDC, and scaling functions is selected)+(1080p@30 fps+VGA@30 fps) dual-stream H.264 encoding.

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2.7.2 Temperature and Thermal Resistance Parameters

Table 2-63 describes temperature and thermal resistance parameters.

NOTE

- The thermal resistance is provided in compliance with the JEDEC JESD51-2 standard. The actual system design and environment may be different.
- The chip junction temperature is proportional to the chip power consumption. Ensure that the junction temperature is appropriate to match power supplies.
- Design heat dissipation by following the *Hi3516C V200 Hardware Design User Guide*.
- The chip can be stored in the sealed packaging bag for up to 12 months when the temperature is below 40ºC (104ºF) and the relative humidity (RH) is below 90%.
- After the packaging bag is opened, the following conditions must be met before the component is used in the reflow soldering process or other high-temperature processes:

  a. The process is complete within 168 hours at 30ºC (86ºF) or lower and at most 60% RH.

  b. The component is stored at the RH lower than 10%.
- The soldering temperature curve is based on the J-STD-020D.1 standard.

Table 2-63 Temperature and thermal resistance parameters

Parameter	Symbol	Min	Typ	Max	Unit
Ambient temperature	T_A	0	N/A	70	ºC
Rated temperature	T_{JMAX}	−20	N/A	105	ºC
Junction-to-ambient thermal resistance	θ_{JA}	N/A	29	N/A	ºC/W
Junction-to-board thermal resistance	θ_{JB}	N/A	15.1	N/A	ºC/W
Junction-to-case thermal resistance	θ_{JC}	N/A	8.8	N/A	ºC/W

CAUTION

The ambient temperature is used in chip operating conditions, while the junction temperature only serves as a reference during thermal simulation.

2.7.3 Operating Conditions

Table 2-64 describes operating conditions.

CAUTION

The prerequisite for using the VDD voltage range (when SVB is used) listed in Table 2-64 is as follows: the customer's board must use the SVB circuit, and the impedance and capacitance parameters of the SVB circuit must completely copy the related RC parameters of Hi3516C V200DMEB VER.A.

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Table 2-64 Operating conditions for the SVB related power supply

Symbol	Description	Voltage range			Unit
		Min	Typ	Max	
VDD	Internal core power (when SVB is not used)	1.045	1.1	1.155	V
	Internal core power (when SVB is used)	1.06	1.06	1.06	V
AVDD11_PLL	PLL core analog power (when SVB is used)	1.045	1.1	1.155	V
	PLL core analog power (when SVB is used)	1.06	1.06	1.06	V

Table 2-65 Operating conditions for the power supply under normal voltage

Symbol	Description	Min	Typ	Max	Unit
DVDD33	I/O power	2.97	3.3	3.63	V
DVDD3318_VI	VI I/O power	1.62	1.8	1.98	V
	VI I/O power	2.97	3.3	3.63	V
DVDD3318_SENSOR	SENSOR, SPI0 I/O power	1.62	1.8	1.98	V
	SENSOR, SPI0 I/O power	2.97	3.3	3.63	V
DVDD3318_EMMC	SFC, UART1/EMMC I/O power	1.62	1.8	1.98	V
	SFC, UART1/EMMC I/O power	2.97	3.3	3.63	V
DVDDIO_RMII	RMII I/O power	1.62	1.8	1.98	V
	RMII I/O power	2.97	3.3	3.63	V
DVDD18_SDIO	SDIO 1.8V I/O power	1.62	1.8	1.98	V
VDDIO_DDR	DDR3L interface power	1.282	1.35	1.418	V
	DDR3 interface power	1.425	1.5	1.575	V
VDDIO_CK_DDR	DDR3/3L clock interface power	1.425/1.323	1.5/1.35	1.575/1.425	V

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Symbol	Description	Min	Typ	Max	Unit
AVDD_DDRPLL	DDR3/3L PLL3.3V analog power	2.97	3.3	3.63	V
AVDD33_PLL	PLL 3.3V analog power	2.97	3.3	3.63	V
AVDD33_AC	ACODEC 3.3V analog power	3	3.3	3.6	V
AVDD33_USB	USB 3.3V analog power	3	3.3	3.6	V
AVDD3318_MIPI	MIPI analog power	1.71	1.8	1.89	V
	3.3 V power when the MIPI signal is multiplexed as a single-ended digital signal	2.97	3.3	3.63	V
	1.8 V power when the MIPI signal is multiplexed as a single-ended digital signal	1.62	1.8	1.98	V
AVDD_BAT	RTC battery power	1.6	3.3	3.6	V
AVDD33_RTC	RTC analog power	2.97	3.3	3.63	V

2.7.4 Power-On and Power-Off Sequences

For details about power-on and power-off sequences, see section 1.2.5 "requirements on the power-on and power-off timing" in the *Hi3518E V20X/Hi3516C V200 Hardware Design User Guide*.

2.7.5 DC and AC Electrical Specifications

Table 2-66 and Table 2-67 describe direct current (DC) electrical specifications.

Table 2-66 DC electrical specifications (DVDD33/DVDD3318_VI/DVDD3318_SENSOR/DVDD3318_EMMC/DVDDIO_RMII = 3.3 V, incompatible with the 5 V input voltage)

Symbol	Description	Min	Typ	Max	Unit	Remarks
DVDD33	Interface voltage	2.97	3.3	3.63	V	-
V_{IH}	High-level input voltage	2.0	N/A	DVDD33 + 0.3 V	V	The interface is incompatible with the 5 V input voltage.

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Symbol	Description	Min	Typ	Max	Unit	Remarks
V_{IL}	Low-level input voltage	–0.3	N/A	0.8	V	N/A
I_L	Input leakage current	N/A	N/A	±10	μA	N/A
I_{OZ}	Tristate output leakage current	N/A	N/A	±10	μA	N/A
V_{OH}	High-level output voltage	2.4	N/A	N/A	V	N/A
V_{OL}	Low-level output voltage	N/A	N/A	0.4	V	N/A
R_{PU1}	Internal pull-up impedance	33	37	41	kΩ	N/A
R_{PU2}	Internal pull-up impedance	7	8	9	kΩ	N/A
R_{PU3}	Internal pull-up impedance	32	36	40	kΩ	N/A
R_{PU4}	Internal pull-up impedance	46	51	56	kΩ	N/A
R_{PD1}	Internal pull-down impedance	36	40	44	kΩ	N/A
R_{PD2}	Internal pull-down impedance	7	8	9	kΩ	N/A
R_{PD3}	Internal pull-down impedance	36	40	44	kΩ	N/A
R_{PD4}	Internal pull-down impedance	50	51	56	kΩ	N/A
R_{PD5}	Internal pull-down impedance	36	40	44	kΩ	N/A

Table 2-67 DC electrical specifications (DVDD18_SDIO/DVDD3318/DVDDIO_RMII = 1.8 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
DVDD18	Interface voltage	1.62	1.8	1.98	V	N/A
V_{IH}	High-level input voltage	1.17	N/A	DVDD18 + 0.3 V	V	N/A
V_{IL}	Low-level input voltage	–0.3	N/A	0.63	V	N/A
I_L	Input leakage current	N/A	N/A	±10	μA	N/A

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Symbol	Description	Min	Typ	Max	Unit	Remarks
I_{OZ}	Tristate output leakage current	N/A	N/A	±10	μA	N/A
V_{OH}	High-level output voltage	1.35	N/A	N/A	V	N/A
V_{OL}	Low-level output voltage	N/A	N/A	0.45	V	N/A
R_{PU1}	Internal pull-up impedance	74	82	90	kΩ	N/A
R_{PU2}	Internal pull-up impedance	8	9	10	kΩ	N/A
R_{PU3}	Internal pull-up impedance	76	85	94	kΩ	N/A
R_{PU4}	Internal pull-up impedance	50	51	56	kΩ	N/A
R_{PD1}	Internal pull-down impedance	77	86	95	kΩ	N/A
R_{PD2}	Internal pull-down impedance	8	9	10	kΩ	N/A
R_{PD3}	Internal pull-down impedance	77	86	95	kΩ	N/A
R_{PD4}	Internal pull-down impedance	46	51	56	kΩ	N/A
R_{PD5}	Internal pull-down impedance	77	86	95	kΩ	N/A

Table 2-68 describes DC electrical specifications in DDR3 SSTL15 mode.

Table 2-68 DC electrical specifications in DDR3 SSTL15 mode (VDDIO_DDR = 1.5 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
VDDIO_DDR	Interface voltage	1.425	1.5	1.575	V	N/A
Vref	Reference voltage	0.49 x VDDIO_DDR	0.5 x VDDIO_DDR	0.51 x VDDIO_DDR	V	(0.49~0.51) x VDDIO_DDR
$V_{IH(DC)}$	High-level input voltage	Vref+0.1	N/A	VDDIO_DDR +0.3	V	N/A
$V_{IL(DC)}$	Low-level input voltage	–0.3	N/A	Vref–0.1	V	N/A

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Symbol	Description	Min	Typ	Max	Unit	Remarks
V_{OH}	High-level output voltage	0.8 x VDDIO_DDR	N/A	(1+0.1) x VDDIO_DDR	V	The driver is configurable.
V_{OL}	Low-level output voltage	0	N/A	0.2 x VDDIO_DDR	V	The driver is configurable.
I_{OH}	High-level output current	N/A	10.50	10.83	mA	The DDR drive impedance is 34 Ω, and RTT is 60.
I_{OL}	Low-level output current	N/A	10.50	10.83	mA	The DDR drive impedance is 34 Ω, and RTT is 60.
Output impedance	N/A	34	N/A	80	Ω	N/A

Table 2-69 describes alternating current (AC) electrical specifications in DDR3 SSTL15 mode.

Table 2-69 AC electrical specifications in DDR3 SSTL15 mode (VDDIO_DDR = 1.5 V)

Symbol	Parameter	Min	Max	Unit	Remarks
$V_{IH(AC)}$	High-level input voltage	Vref + 0.15	VDDIO_DDR+0.3	V	N/A
$V_{IL(AC)}$	Low-level input voltage	-	Vref–0.15	V	N/A

Table 2-70 describes DC electrical specifications in DDR3L SSTL135 mode.

Table 2-70 DC electrical specifications in DDR3L SSTL135 mode (VDDIO_DDR = 1.35 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
VDDIO_DDR	Interface voltage	1.283	1.35	1.45	V	N/A
VREF	Reference voltage of the DDR interface	0.49 x VDDIO_DDR	0.5 x VDDIO_DDR	0.51 x VDDIO_DDR	V	N/A
$V_{IH(DC)}$	High-level input voltage	0.49 x VDDIO_DDR + 0.125	N/A	VDDIO_DDR + 0.3	V	N/A
$V_{IL(DC)}$	Low-level input voltage	–0.3	N/A	0.51 x VDDIO_DDR − 0.125	V	N/A

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Symbol	Description	Min	Typ	Max	Unit	Remarks
V_{OH}	High-level output voltage	VDDIO_DDR − 0.28	N/A	N/A	V	N/A
V_{OL}	Low-level output voltage	N/A	N/A	VDDIO_DDR + 0.28	V	N/A
I_{OH}	High-level output current	N/A	9.42	9.85	mA	The DDR drive impedance is 34 Ω, and RTT is 60.
I_{OL}	Low-level output current	N/A	9.42	9.85	mA	The DDR drive impedance is 34 Ω, and RTT is 60.

Table 2-71 describes AC electrical specifications in DDR3L SSTL135 mode.

Table 2-71 AC electrical specifications in DDR3L SSTL135 mode (VDDIO_DDR = 1.35 V)

Symbol	Description	533−1600 Mbit/s		Unit	Remarks
		Min	Max		
$V_{IH(AC)}$	AC high-level input voltage	0.49 x VDDIO_DDR + 0.25	N/A	V	N/A
$V_{IL(AC)}$	AC low-level input voltage	N/A	0.51 x VDDIO_DDR– 0.25	V	N/A

2.7.6 MIPI/LVDS RX Electrical Parameters

Table 2-72 describes low-voltage differential signaling (LVDS) differential DC electrical parameters.

Table 2-72 LVDS differential DC electrical parameters

Symbol	Description		Min	Typ	Max	Unit
VIDTH (SL)	Differential input threshold voltage |(VP—VM)|	Sub-LVDS	70	-	-	mV
VIDTH (HS)		HiSPi (SLVS)	70	-	-	
VIDTH (HiVCM)		HiSPi (HiVCM)	100	-	-	
VIDTH (DP)		D-PHY HS	70	-	-	
VIDTH (LV)		LVDS	100	-		
VIDTH (ML)		Mini-LVDS	100	-	-	

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Symbol	Description		Min	Typ	Max	Unit
VCM (SL)	Common mode voltage range (VP+VM)/2	Sub-LVDS	0.5	0.9	1.3	V
VCM (HS)		HiSPi (SLVS)	0.07	0.2	0.35	
VCM (HiVCM)		HiSPi (HiVCM)	0.66	0.90	1.17	
VCM (DP)		D-PHY HS	0.07	0.2	0.33	
VCM (LV)		LVDS	0.925	1.2	1.475	
VCM (ML)		Mini-LVDS	1.025	1.2	1.375	
VISVR (SL)	Single-ended input voltage range VP, VM	Sub-LVDS	0.4	-	1.4	V
VCM (HS)		HiSPi (SLVS)	-0.04	-	0.49	
VCM (HiVCM)		HiSPi (HiVCM)	0.55	-	1.35	
VCM (DP)		D-PHY HS	-0.04	-	0.46	
VCM (LV)		LVDS	0	-	1.8	
VCM (ML)		Mini-LVDS	0.825	-	1.575	
ZID (SL)	Internal termination resister value	Sub-LVDS	80	100	120	Ω
ZID (HS)		HiSPi (SLVS)			125	
ZID (HiVCM)		HiSPi (HiVCM)			125	
ZID (LV)		LVDS			120	
ZID (ML)		Mini-LVDS			120	
ZID (DP)		D-PHY HS			125	

Table 2-73 to Table 2-75 describe the MIPI parameters.

Table 2-73 MIPI high-speed (HS) DC parameters

Symbol	Description	Min	Typ	Max	Unit
VTERM-EN	Single-ended threshold for HS termination enable	-	-	450	mV

Table 2-74 MIPI HS AC parameters

Symbol	Description	Min	Typ	Max	Unit
ΔVCMRX (HF)	Common-mode interface beyond 450 MHz	-	-	100	-

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Symbol	Description	Min	Typ	Max	Unit
ΔVCMRX (LF)	Common-mode interface 50 MHz–450 MHz	-50	-	50	mV
CCM	Common-mode termination	-	-	60	pF

Table 2-75 MIPI low-power (LP) DC parameters

Symbol	Description	Min	Typ	Max	Unit
VIHLP	Logic 1 input voltage	880	-	-	mV
VILLP	Logic 0 input voltage	-	-	550	
VHYST	Input hysteresis	25	-	-	

2.7.7 SDIO Electrical Specifications

Table 2-76 describes the SDIO electrical specifications (3.3 V).

Table 2-76 SDIO electrical specifications (3.3 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
VDDIO	Power supply voltage	2.97	3.3	3.6	V	N/A
VOH	Output high level	2.4	N/A	N/A	V	
VOL	Output low level	N/A	N/A	0.4	V	
VIH	Input high level	2.0	N/A	VDD IO+ 0.3	V	
VIL	Input low level	-0.3	N/A	0.8	V	
N/A	Power-on duration	N/A	N/A	250	Ms	0 V to VDDIO

Table 2-77 describes the SDIO electrical specifications (1.8 V).

Table 2-77 SDIO electrical specifications (1.8 V)

Symbol	Description	Min	Typ	Max	Unit	Remarks
VDDIO	Power supply voltage	1.62	1.8	1.98	V	
VOH	Output high level	1.35	N/A	N/A	V	
VOL	Output low level	N/A	N/A	0.45	V	

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Symbol	Description	Min	Typ	Max	Unit	Remarks
VIH	Input high level	1.17	N/A	VDDIO+0.3	V	
VIL	Input low level	-0.3	N/A	0.63	V	
N/A	Power-on duration	N/A	N/A	250	ms	0 V to VDDIO

2.7.8 Audio CODEC Electrical Parameters

Table 2-78 to Table 2-81 describe the electrical parameters of the audio CODEC.

Table 2-78 Overall specifications

Description	Min	Typ	Max	Unit	Remarks
Analog circuit power AVDD	3	3.3	3.6	V	Relative to AGND
VREF	-	AVDD/2		V	Relative to AGND

Table 2-79 Major DAC specifications

Description	Min	Typ	Max	Unit	Remarks
Output amplitude at full scale	-	0.875	-	Vrms	Maximum output signal swing

Table 2-80 Major ADC specifications

Description	Min	Typ	Max	Unit	Remarks
Maximum input amplitude	-	1	-	Vrms	Maximum input signal swing of the ADC

Table 2-81 Major MICBIAS specifications

Description	Min	Typ	Max	Unit	Remarks
Bias voltage	-	2.1 x AVDD/3.3	-	V	MIC bias voltage
Maximum output current	-	-	3	mA	-

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Table 2-82 Major MICPGA specifications

Description	Min	Typ	Max	Unit	Remarks
Input voltage range	-	1	-	Vrms	Maximum input signal swing
Input impedance	5	-	12	kΩ	MICPGA input impedance

2.8 Interface Timings

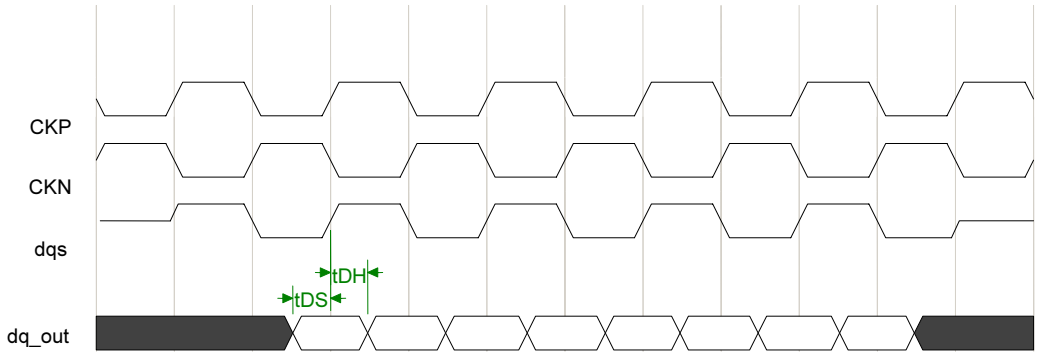
2.8.1 DDR Interface Timings

2.8.1.1 Write Timings

Write Timings of dqs_out Relative to dq_out

In the write timing of dqs_out relative to dq_out, the major parameters are tDS and tDH.

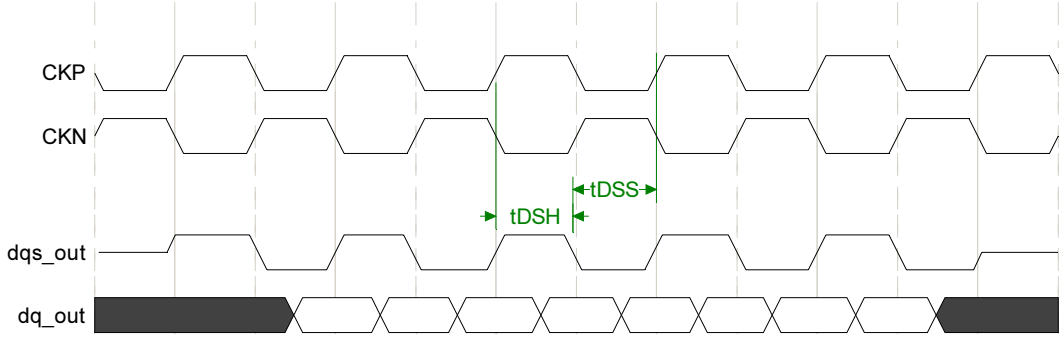
Figure 2-11 Write timing of dqs_out relative to dq_out for the DDR3



Write Timings of dqs_out Relative to CK

Figure 2-12 shows the write timing of dqs_out relative to CK for the DDR3.

Figure 2-12 Write timing of dqs_out relative to CK for the DDR3

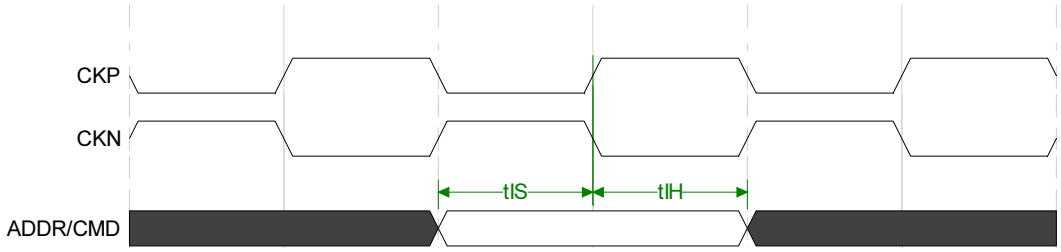


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Write Timing of CMD/ADDR Relative to CK

Figure 2-13 shows the write timing of CMD/ADDR relative to CK.

Figure 2-13 Write timing of CMD/ADDR relative to CK



2.8.1.2 Read Timings

Read Timing of CMD/ADDR Relative to CK

The read timing of CMD/ADDR relative to CK is the same as the "Write Timing of CMD/ADDR Relative to CK".

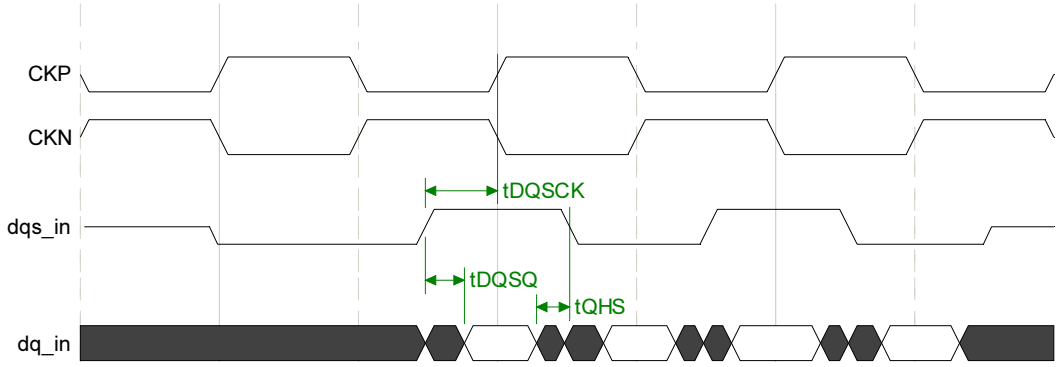
Read Timings of dqs_in Relative to dq_in

The read timings of dqs_in relative to dq_in are classified into the DDRn SDRAM output timing, dqs_in timing on the DDR PHY side, and dq_in timing on the DDR PHY side.

For the DDR SDRAM output timing, the phases of DQS and CK are the same in the ideal condition; however, there is a tDQSCK skew between DQS and CK. The value of tDQSCK is 0.35 ns. tDQSQ is the jitter of the last valid DQ relative to DQS and its value is 0.2 ns; tQHS is the jitter of the first valid DQ relative to DQS and its value is 0.3 ns.

Figure 2-14 shows the output timing of the DDRn SDRAM.

Figure 2-14 Output timing of the DDRn SDRAM



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2.8.1.3 Timing Parameters

The timings of the DDR interface comply with the JEDEC standards including JESD79-2E and JESD79-3B standards. All the timings in this document are output on the DDR PHY side.

The Hi3516C V200 is based on the timing parameters of the DDR3-1600 SDRAMs.

Table 2-83 and Table 2-84 describe the clock parameters of the DDR3-1600 SDRAM.

Table 2-83 DDR3 clock parameters

Parameter	Typ	Unit
DDR clock frequency	800.00	MHz
PLL jitter	0.200	ns
PLL duty ratio	47.000	%
Clock skew	0.100	ns

Table 2-84 Parameters for the DDR3-1600 SDRAM

Parameter	Symbol	Typ	Unit
Setup time, DQS falling edge to DDR clock	tDSS	0.18	tCK
Hold time, DQS falling edge to DDR clock	tDSH	0.18	tCK
Setup time, DQ/DM to DQS	tDS	0.010	ns
Hold time, DQ/DM to DQS	tDH	0.045	ns
Skew between DQS and DQ	tDQSQ	0.100	ns
Setup time, ADDR/CMD to DDR clock	tIS	0.170	ns
Hold time, ADDR/CMD to DDR clock	tIH	0.120	ns
Skew of DQS (output) to DDR clock	tDQSCK	0.225	ns

2.8.2 SFC Interface Timings

Figure 2-15 shows the SFC output timing.

Figure 2-15 SFC output timing

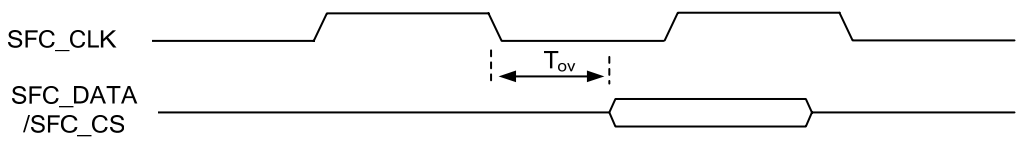


Table 2-85 describes the parameters of the SFC output timing.

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Table 2-85 Parameters of the SFC output timing

Parameter	Symbol	Min	Typ	Max	Unit
Clock cycle of SFCCLK	T	10.1	N/A	83.2	ns
Output data signal delay	T_{ov}	-3	N/A	3	ns
Output CS signal delay	T_{ov}	-3	N/A	3	ns

Figure 2-16 shows the SFC input timing.

Figure 2-16 SFC input timing

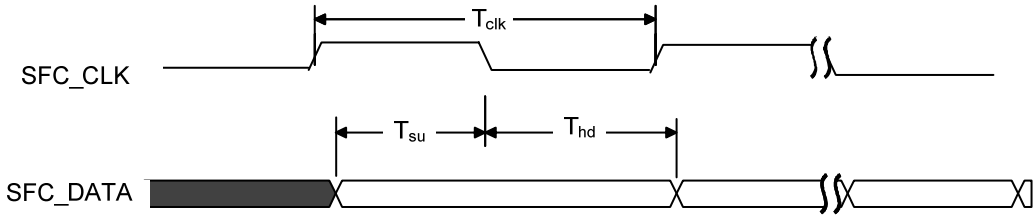


Table 2-86 describes the parameters of the SFC input timing.

Table 2-86 Parameters of the SFC input timing

Parameter	Symbol	Min	Typ	Max	Unit
Clock cycle of SFCCLK	T_{clk}	10.1	N/A	83.2	ns
Input signal setup time	T_{su}	2	N/A	N/A	ns
Input signal hold time	T_{hd}	0.1	N/A	N/A	ns

2.8.3 Ethernet MAC Interface Timings

2.8.3.1 RMII Timings

Figure 2-17 shows the 100 Mbit/s RX timing of the RMII.

Figure 2-17 100 Mbit/s RX timing of the RMII

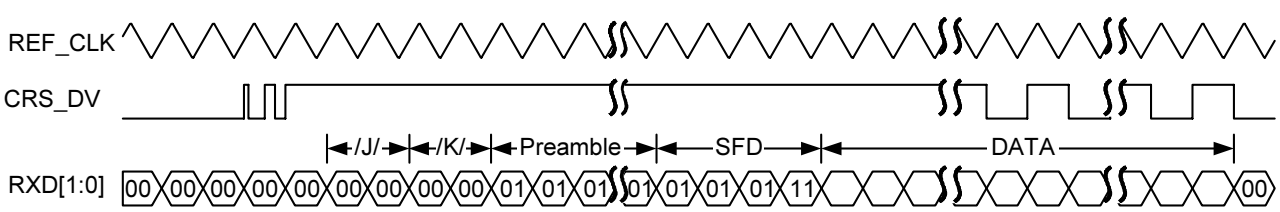


Figure 2-18 shows the 100 Mbit/s TX timing of the RMII.

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Figure 2-18 100 Mbit/s TX timing of the RMII

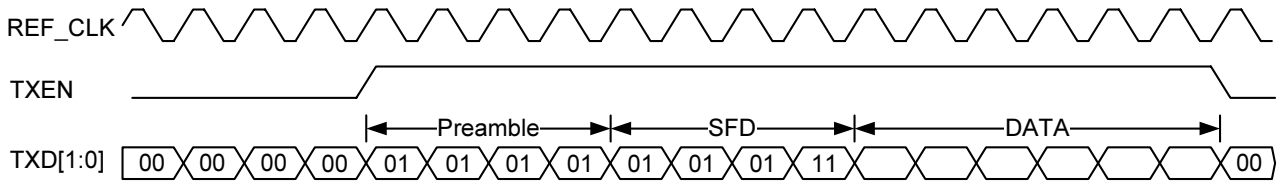


Figure 2-19 shows the 10 Mbit/s RX timing of the RMII.

Figure 2-19 10 Mbit/s RX timing of the RMII

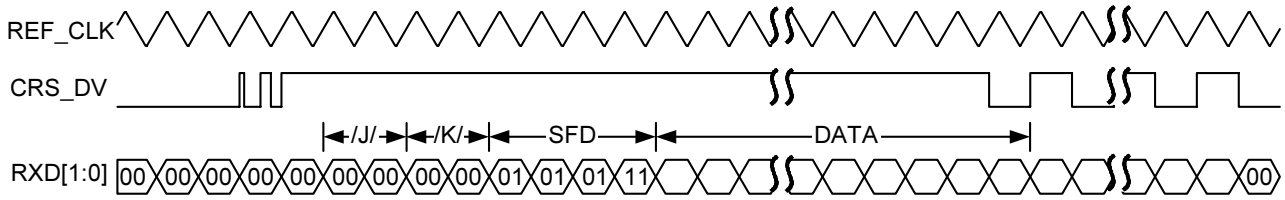


Figure 2-20 shows the 10 Mbit/s TX timing of the RMII.

Figure 2-20 10 Mbit/s TX timing of the RMII

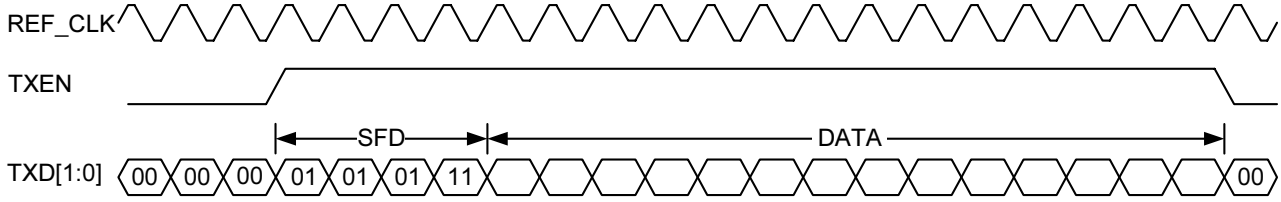


Figure 2-21 shows the RMII timing parameters.

Figure 2-21 RMII timing parameters

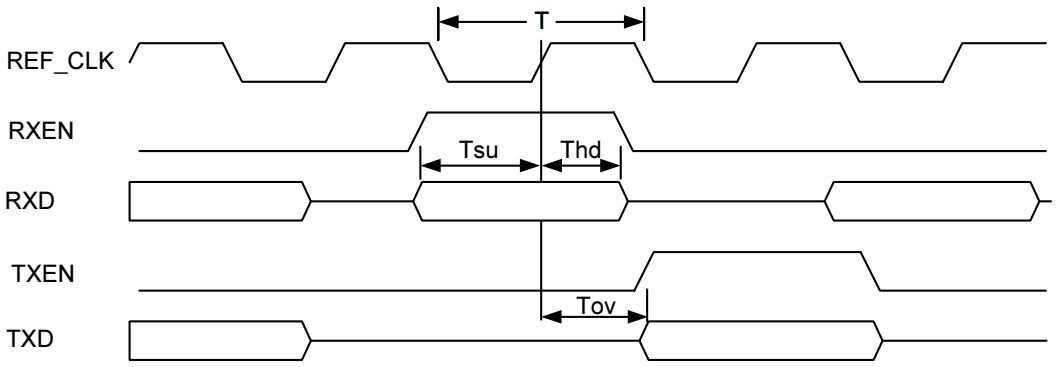


Table 2-87 describes the RMII timing parameters.

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Table 2-87 RMII timing parameters

Parameter	Symbol	Signal	Min	Max	Unit
RMII clock cycle	T	REF_CLK	20	20	ns
RMII signal setup time	Tsu (RX)	CRS_DV/RXD[1:0]	4	N/A	ns
RMII signal hold time	Thd (RX)	CRS_DV/RXD[1:0]	2	N/A	ns
RMII output signal delay	Tov (TX)	TXEN/TXD[1:0]	3	16	ns

2.8.3.2 MDIO Interface Timings

Figure 2-22 shows the read timing of the MDIO interface.

Figure 2-22 Read timing of the MDIO interface

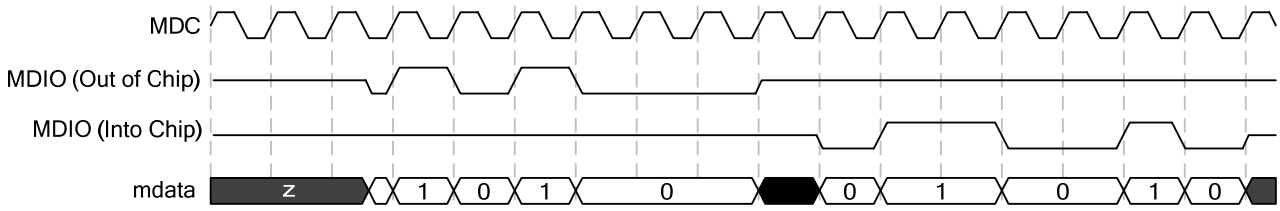


Figure 2-23 shows the write timing of the MDIO interface.

Figure 2-23 Write timing of the MDIO interface

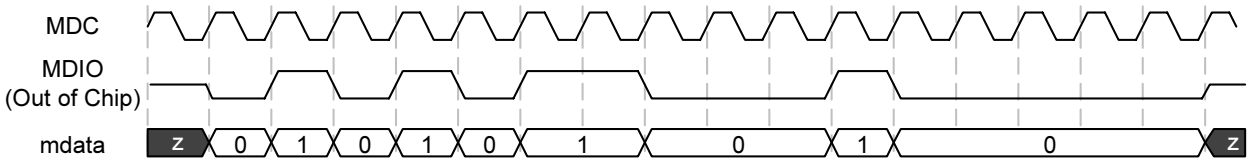
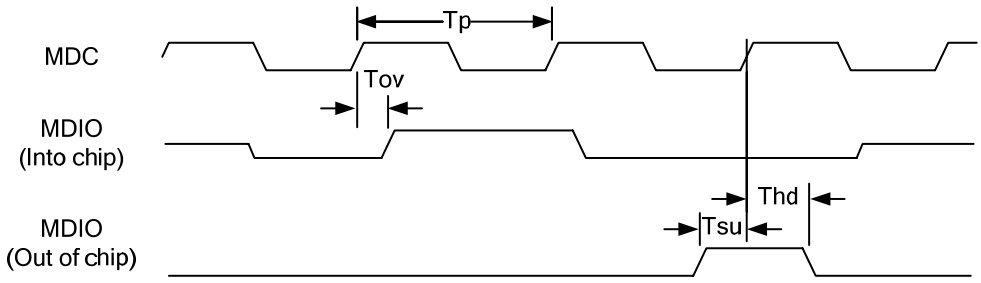


Figure 2-24 shows the RX timing parameters of the MDIO interface.

Figure 2-24 RX timing parameters of the MDIO interface



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Table 2-88 describes the timing parameters of the MDIO interface.

Table 2-88 Timing parameters of the MDIO interface

Parameter	Symbol	Signal	Min	Max	Unit
MDIO data RX delay	Tov	MDIO	0	375	ns
MDIO clock cycle	Tp	MDCK	500	64000	ns
MDIO data TX setup time	Tsu	MDIO	10	N/A	ns
MDIO data TX hold time	Thd	MDIO	10	N/A	ns

2.8.4 VI Interface Timing

Figure 2-25 shows the VI interface timing in CMOS mode.

Figure 2-25 VI interface timing in CMOS mode

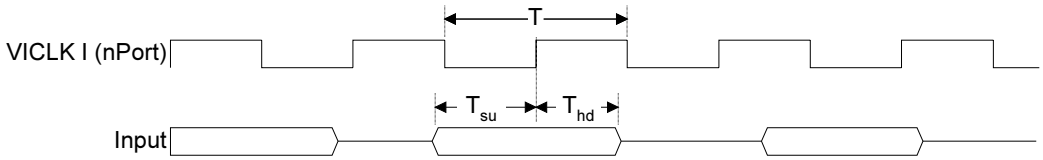


Table 2-89 describes the VI interface timing parameters.

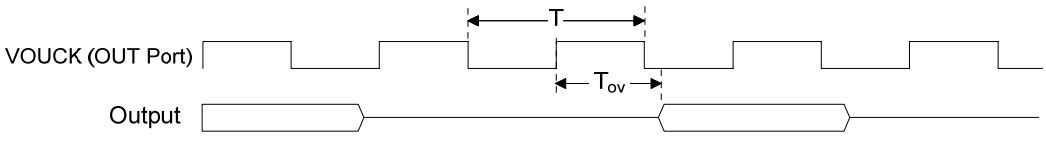
Table 2-89 VI interface timing parameters

Parameter	Symbol	Min	Typ	Max	Unit
VICLK clock cycle	T	6.73	N/A	N/A	ns
Input signal setup time	T_{su}	2.5	N/A	N/A	ns
Input signal hold time	T_{hd}	2.0	N/A	N/A	ns

2.8.5 VO Interface Timings

Figure 2-26 shows the BT.656 interface timing.

Figure 2-26 BT.656 interface timing



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Table 2-90 describes the timing parameters of the BT.656 interface.

Table 2-90 Timing parameters of the BT.656 interface

Parameter	Symbol	Min	Typ	Max	Unit
VOUCLK clock cycle	T	13.46	13.48	37	ns
Output signal delay	T_{ov}	T/2-1.5	N/A	T/2+1.5	ns

Figure 2-27 shows the liquid crystal display (LCD) interface timing.

Figure 2-27 LCD interface timing

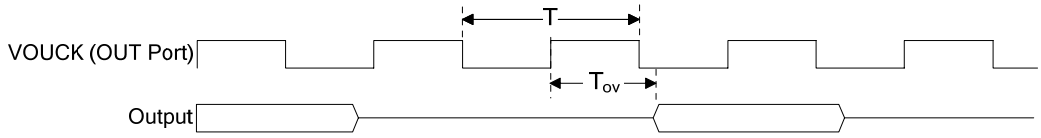


Table 2-91 describes the timing parameters of the LCD interface.

Table 2-91 Timing parameters of the LCD interface

Parameter	Symbol	Min	Typ	Max	Unit
VOUCLK clock cycle	T	37	N/A	500	ns
Output signal delay	T_{ov}	T/2 − 1.5	N/A	T/2 + 1.5	ns

2.8.6 AIAO Interface Timings

2.8.6.1 I²S Interface Timing

Figure 2-28 shows the RX timing of the I^2S interface.

Figure 2-28 RX timing of the I^2S interface

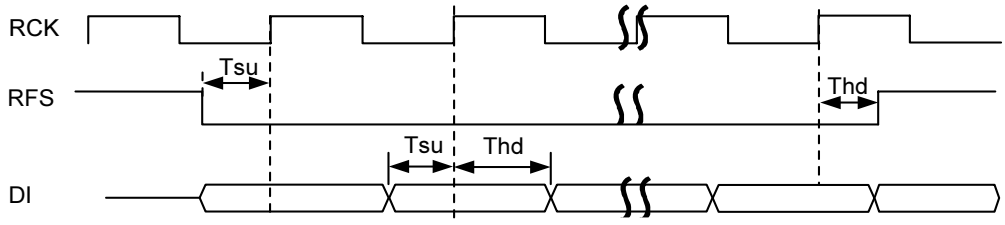


Figure 2-29 shows the TX timing of the I^2S interface.

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Figure 2-29 TX timing of the I²S interface

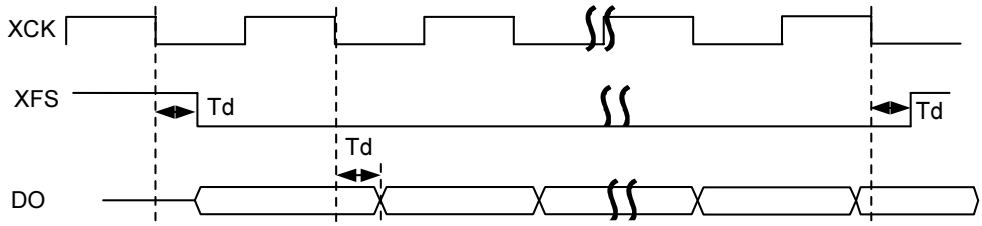


Table 2-92 describes the timing parameters of the I²S interface.

Table 2-92 Timing parameters of the I²S interface

Parameter	Symbol	Min	Typ	Max	Unit
Input signal setup time	T_{su}	10	N/A	N/A	ns
Input signal hold time	T_{hd}	10	N/A	N/A	ns
Output signal delay	T_d	-10	N/A	10	ns

2.8.6.2 PCM Interface Timings

Figure 2-30 shows the RX timing of the PCM interface.

Figure 2-30 RX timing of the PCM interface

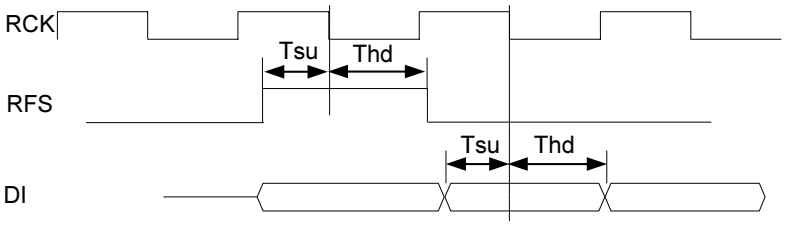


Figure 2-31 shows the TX timing of the PCM interface

Figure 2-31 TX timing of the PCM interface

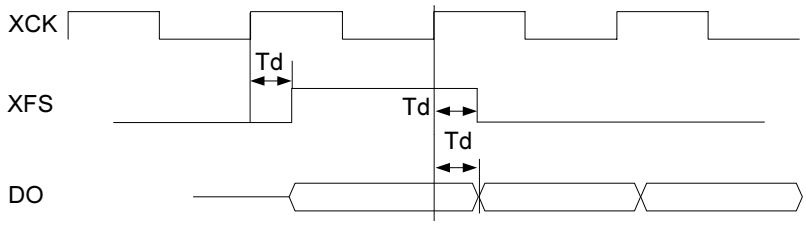


Table 2-93 describes the timing parameters of the PCM interface.

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Table 2-93 Timing parameters of the PCM interface

Parameter	Symbol	Min	Typ	Max	Unit
Input signal setup time	T_{su}	10	N/A	N/A	ns
Input signal hold time	T_{hd}	10	N/A	N/A	ns
Output signal delay	T_d	-10	N/A	10	ns

2.8.7 I²C Interface Timing

Figure 2-32 shows the I²C transfer timing.

Figure 2-32 I²C transfer timing

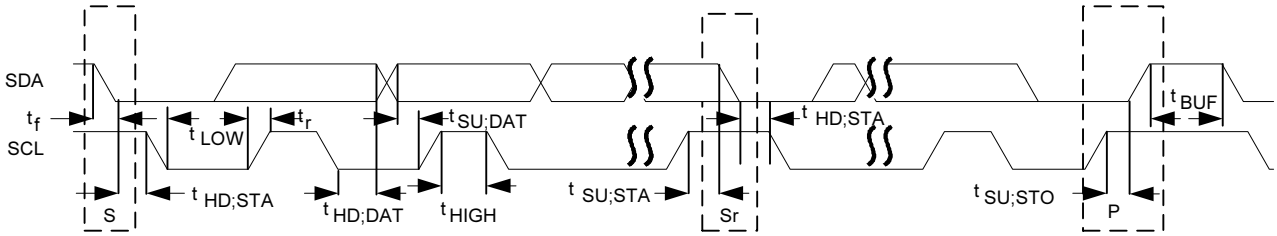


Table 2-94 describes the timing parameters of the I²C interface.

Table 2-94 Timing parameters of the I²C interface

Parameter	Symbol	Standard Mode		Fast Mode		Unit
		Min	Max	Min	Max	
Serial clock (SCL) frequency	f_{SCL}	0	100	0	400	kHz
Start hold time	$t_{HD;STA}$	4.0	N/A	0.6	N/A	µs
SCL low-level cycle	t_{LOW}	4.7	N/A	1.3	N/A	µs
SCL high-level cycle	t_{HIGH}	4.0	N/A	0.6	N/A	µs
Start setup time	$t_{SU;STA}$	4.7	N/A	0.6	N/A	µs
Data hold time	$t_{HD;DAT}$	0	3.45	0	0.9	µs
Data setup time	$t_{SU;DAT}$	250	N/A	100	N/A	ns
Serial data (SDA) and SCL rising time	t_r	N/A	1000	$20 + 0.1C_b$	300	ns
SDA and SCL falling time	t_f	N/A	300	$20 + 0.1C_b$	300	ns
End setup time	$t_{SU;STO}$	4.0	N/A	0.6	N/A	µs
Bus release time from start to	t_{BUF}	4.7	N/A	1.3	N/A	µs

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Parameter	Symbol	Standard Mode		Fast Mode		Unit
		Min	Max	Min	Max	
end						
Bus load	C_b	N/A	400	N/A	400	pF

2.8.8 SPI Timings

NOTE

In Figure Figure 2-33 to Figure 2-35, the conventions are as follows:

- MSB = most significant bit
- LSB = least significant bit
- SPI_CK(0):spo = 0
- SPI_CK(1):spo = 1

Figure 2-33 shows the SPI clock (SPICK) timing.

Figure 2-33 SPICK timing

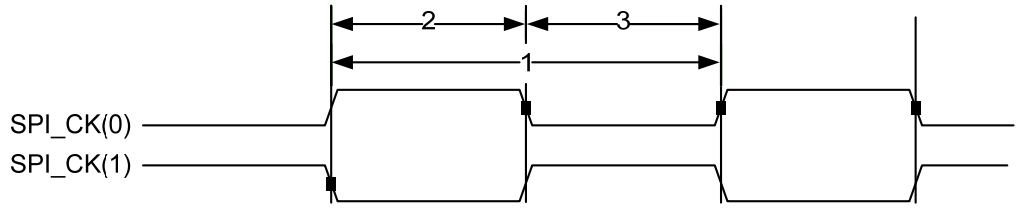
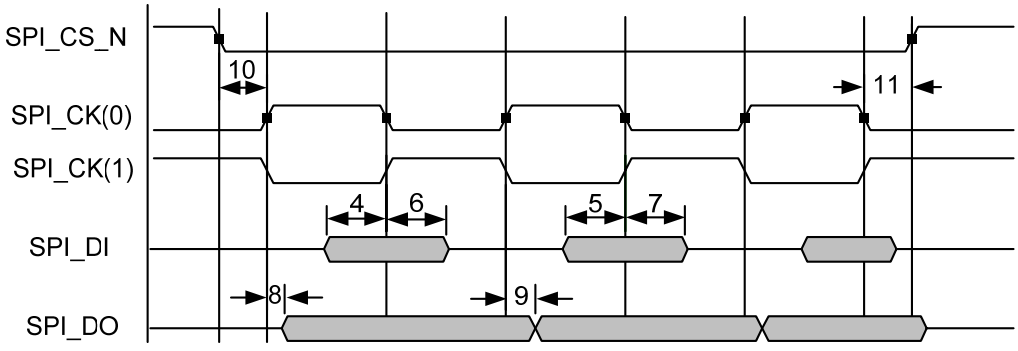


Figure 2-34 and Figure 2-35 show the SPI timings in master mode.

Figure 2-34 SPI timing in master mode (sph = 1)



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Figure 2-35 SPI timing in master mode (sph = 0)

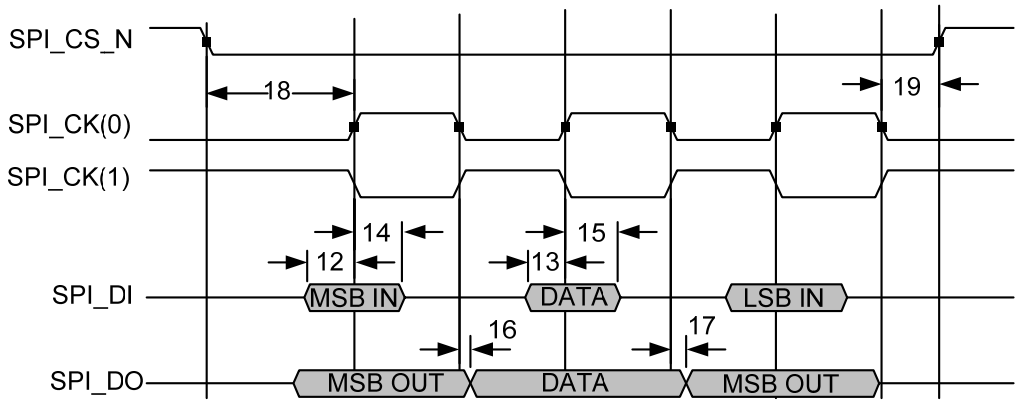


Table 2-95 describes the SPI timing parameters.

Table 2-95 SPI timing parameters

No.	Parameter	Symbol	Min	Typ	Max	Unit	Remark
1	Cycle time, SPI_CK	tc	2000/Fsspclk	N/A	65024000/Fsspclk	ns	Fsspclk unit: MHz
2	Pulse duration, SPI_CK high (all master modes)	tw1	N/A	1/2 tc	N/A	ns	
3	Pulse duration. SPI_CK low (all master modes)	tw2	N/A	1/2 tc	N/A	ns	
4	Setup time, SPI_DI (input) valid before SPI_CK (output) falling edge	tsu1	1/4 tc	N/A	N/A	ns	
5	Setup time, SPI_DI (in put) valid before SPICK (output) rising edge	tsu2	1/4 tc	N/A	N/A	ns	
6	Hold time, SPI_DI (input) valid after SPI_CK (output) falling edge	th1	1/4 tc	N/A	N/A	ns	
7	Hold time, SPI_DI (input) valid after SPI_CK (output) rising edge	th2	1/4 tc	N/A	N/A	ns	
8	Delay time, SPI_CK (output) rising edge to SPI_DO (output) transition	td1	0	N/A	N/A	ns	
9	Delay time, SPI_CK (output) falling edge to SPI_DO (output) transition	td2	0	N/A	N/A	ns	
10	Delay time, SPI_CS_N (output) falling edge to first	td3	N/A	tc	N/A	ns	

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No.	Parameter	Symbol	Min	Typ	Max	Unit	Remark
	SPI_CK (output) rising or falling edge						
11	Delay time, SPI_CK (output) rising or falling edge to SPI_CS_N (output) rising edge	td4	N/A	1/2 tc	N/A	ns	
12	Setup time, SPI_DI (input) valid before SPI_CK (output) rising edge	tsu3	1/4 tc	N/A	N/A	ns	
13	Setup time, SPI_DI (in put) valid before SPI_CK (output) falling edge	tsu4	1/4 tc	N/A	N/A	ns	
14	Hold time, SPI_DI (input) valid after SPI_CK (output) rising edge	th3	1/4 tc	N/A	N/A	ns	
15	Hold time, SPI_DI (input) valid after SPI_CK (output) falling edge	th4	1/4 tc	N/A	N/A	ns	
16	Delay time, SPI_CK (output) falling edge to SPI_DO (output) transition	td5	0	N/A	N/A	ns	
17	Delay time, SPI_CK (output) rising edge to SPI_DO (output) transition	td6	0	N/A	N/A	ns	
18	Delay time, SPI_CS_N (output) falling edge to first SPI_CK (output) rising or falling edge	td7	N/A	1/2 tc	N/A	ns	
19	Delay time, SPI_CK (output) rising or falling edge to SPI_CS_N (output) rising edge	td8	N/A	tc	N/A	ns	

2.8.9 MIPI RX Timings

Table 2-96 describes the MIPI RX timing parameters.

Table 2-96 MIPI RX timing parameters

Symbol	Description	Min	Tye	Max	Unit
F_{MAX}	Data rate	N/A	N/A	1G	bit/s
T_{PERIOD}	Differential clock cycle	2	N/A	N/A	ns
$DUTY_{clk}$	Differential clock duty cycle	45	N/A	55	%

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Symbol	Description	Min	Tye	Max	Unit
T_{SET}	Differential clock setup time	0.15 x UI	N/A	N/A	ns
T_{HD}	Differential clock hold time	0.15 x UI	N/A	N/A	ns

NOTE

UI = $T_{PERIOD}/2$

2.8.10 SDIO/MMC Interface Timings

Figure 2-36 shows the single-edge data input and output timings of the SDIO/MMC interface.

Figure 2-36 Single-edge data input and output timings of the SDIO/MMC interface

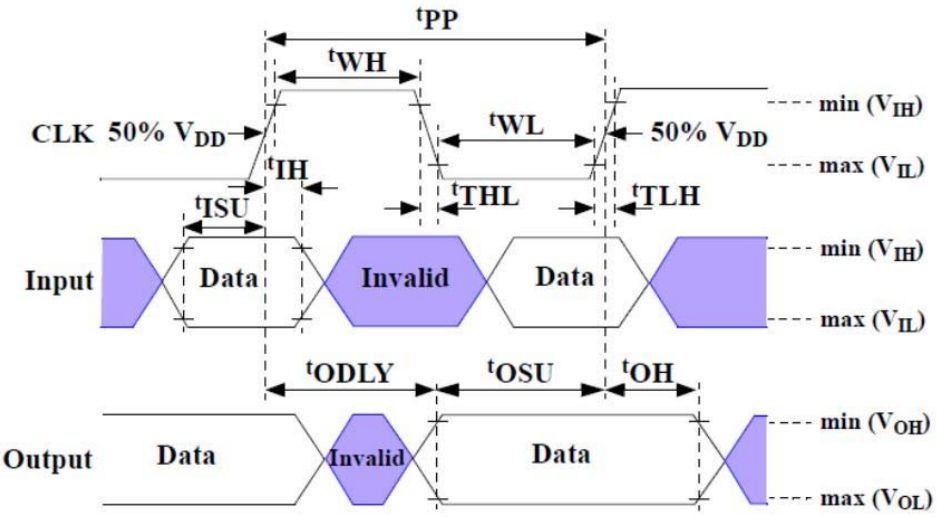


Figure 2-37 shows the dual-edge data input and output timings of the SDIO/MMC interface.

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Figure 2-37 Dual-edge data input and output timings of the SDIO/MMC interface

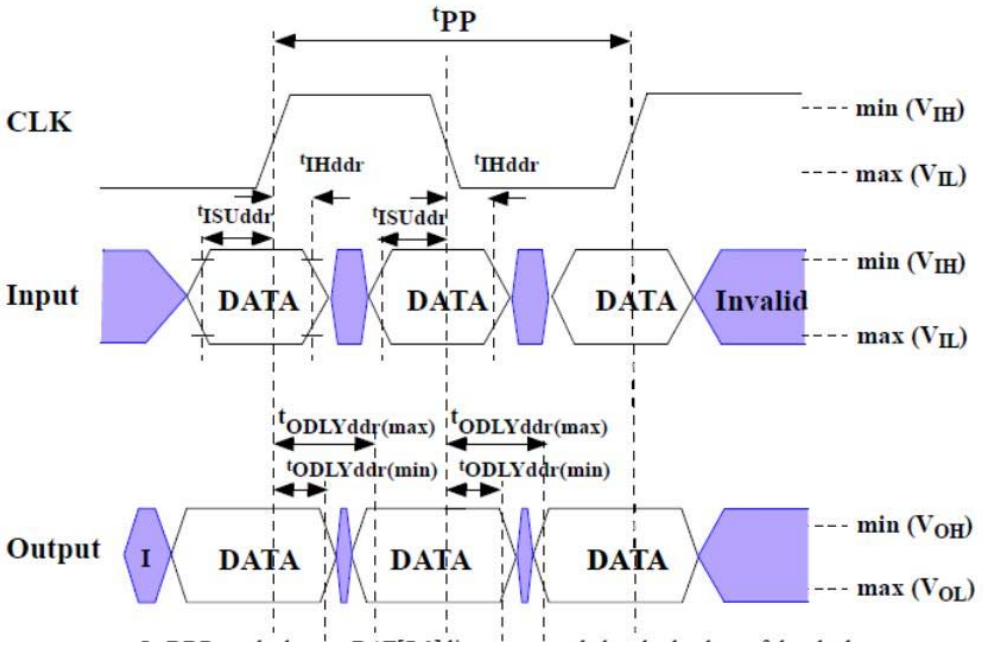


Table 2-97 describes the timing parameters of the SDIO/MMC interface.

Table 2-97 Timing parameters of the SDIO/MMC interface

Speed Mode	Maximum Frequency (MHz)/Cycle (ns)		Minimum Hold Time	Minimum Setup Time	Maximum Output Delay of the Card	High Level Duration of the Card Clock
MMC_HS400	100 MHz	10 ns	0.4 ns	0.4 ns	N/A	(0.45−0.55) clock cycle
MMC_HS200	100 MHz	10 ns	0.8 ns	1.4 ns	N/A	
MMC DDR (CMD line)	50 MHz	20 ns	3.0 ns	3.0 ns	13.7 ns	
MMC DDR (DAT line)	50 MHz	20 ns	2.5 ns	2.5 ns	7.0 ns	
MMC_HS	50 MHz	20 ns	3.0 ns	3.0 ns	13.7 ns	
SD_SDR104	100 MHz	10 ns	0.8 ns	1.4 ns	10.0 ns	
SD_SDR50	100 MHz	10 ns	0.8 ns	3.0 ns	7.5 ns	
SD_DDR50 (CMD line)	50 MHz	20 ns	0.8 ns	6.0 ns	13.7 ns	
SD_DDR50 (DAT line)	50 MHz	20 ns	0.8 ns	3.0 ns	7 ns	
SD_SDR25	50 MHz	20 ns	2.0 ns	6.0 ns	14 ns	

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Speed Mode	Maximum Frequency (MHz)/Cycle (ns)		Minimum Hold Time	Minimum Setup Time	Maximum Output Delay of the Card	High Level Duration of the Card Clock
SD_SDR12	25 MHz	40 ns	5.0 ns	5.0 ns	14 ns	
SD_HS	50 MHz	20 ns	2.0 ns	6.0 ns	14 ns	
SD_DS	25 MHz	40 ns	5.0 ns	5.0 ns	14 ns	
Identification mode	400 kHz	2.5 μs	5.0 ns	5.0 ns	50 ns	

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3 System

3.1 Reset

3.1.1 Overview

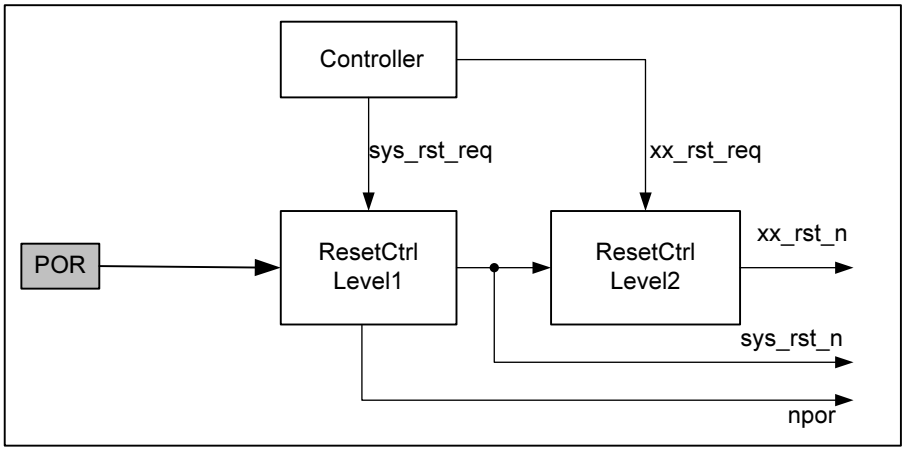
The reset management module resets the entire chip and all functional modules in a unified manner as follows:

- Manages and controls power-on reset.
- Controls the system soft reset and the separate soft reset of each functional module.
- Synchronizes reset signals to the clock domain corresponding to each module.
- Generates reset signals for each internal functional module.

3.1.2 Reset Control

Figure 3-1 shows the diagram of controlling reset signals.

Figure 3-1 Diagram of controlling reset signals



NOTE

- POR: internal power-on reset (POR) module
- sys_rst_req: global soft reset request signal. This signal is derived from the system controller.

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- xx_rst_req: separate soft reset request signal of each submodule. This signal is derived from the clock and reset generator (CRG) system controller.
- xx_rst_n, sys_rst_n, and npor: reset signals.

Table 3-1 Types of reset signals

Type	Generation Mode	Function
Global hard reset signal (npor)	Derived from the internal POR module	Globally resets the entire.
Global soft reset signal (syn_rst_n)	Derived from the global soft reset register of the software configuration system controller.	Globally resets all the modules of the Hi3518E V20X/Hi3516C V200, excluding the clock reset circuits, test circuits, and registers that do not support soft reset
Submodule reset signal (xx_rst_n)	Derived from the submodule reset control register of the CRG system controller.	Separately resets each submodule of the Hi3518E V20X/Hi3516C V200.

3.1.3 Reset Configuration

Power-On Reset

To implement power-on reset, the following conditions must be met:

- The internal POR module generates a low-level pulse, and the low pulse duration is greater than 12 XIN crystal clock cycles.
- The clock input by the XIN pin of the crystal oscillator clock works properly.

System Reset

The system is reset in either of the following ways:

- Power-on reset
- Global soft reset, controlled by the system controller

Soft Reset

Soft reset control is implemented by configuring the corresponding CRG controller. For details about configurations, see the description of the reset register for each module.

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CAUTION

- After a system soft reset request is sent, the circuit reset is deasserted after at least 360 system clock cycles.
- The separate soft reset of each module is not automatically deasserted. For example, if a module is reset after 1 is written to the related bit, the reset of this module is deasserted only when the related bit is set to 0.

3.2 Clock

3.2.1 Overview

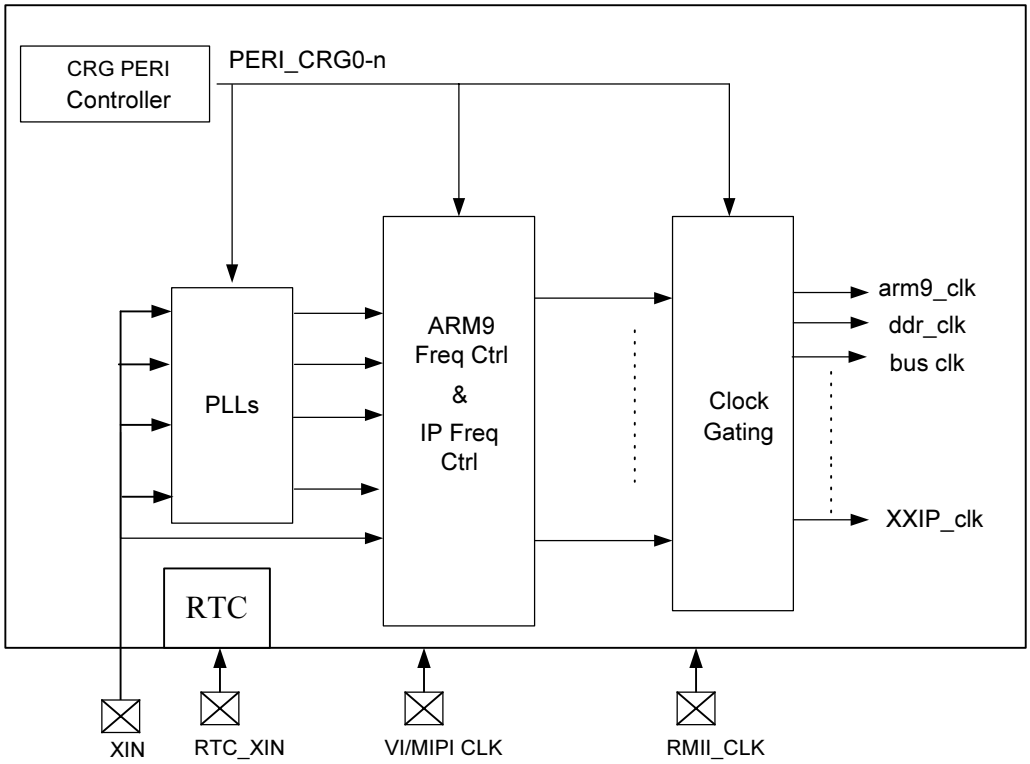
The clock and reset generator (CRG) manages clock input, clock generation, and clock control in a unified manner as follows:

- Manages and controls clock inputs
- Divides and controls clock frequencies
- Generates working clocks for each module

3.2.2 Clock Control Block Diagram

Figure 3-2 shows the functional block diagram of the CRG.

Figure 3-2 Functional block diagram of the CRG



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NOTE

XIN is the PLL input clock and is always connected to a 24 MHz crystal, whereas RTC_XIN is the RTC input clock and is always connected to a 32.768 kHz crystal.

3.2.3 Clock Distribution

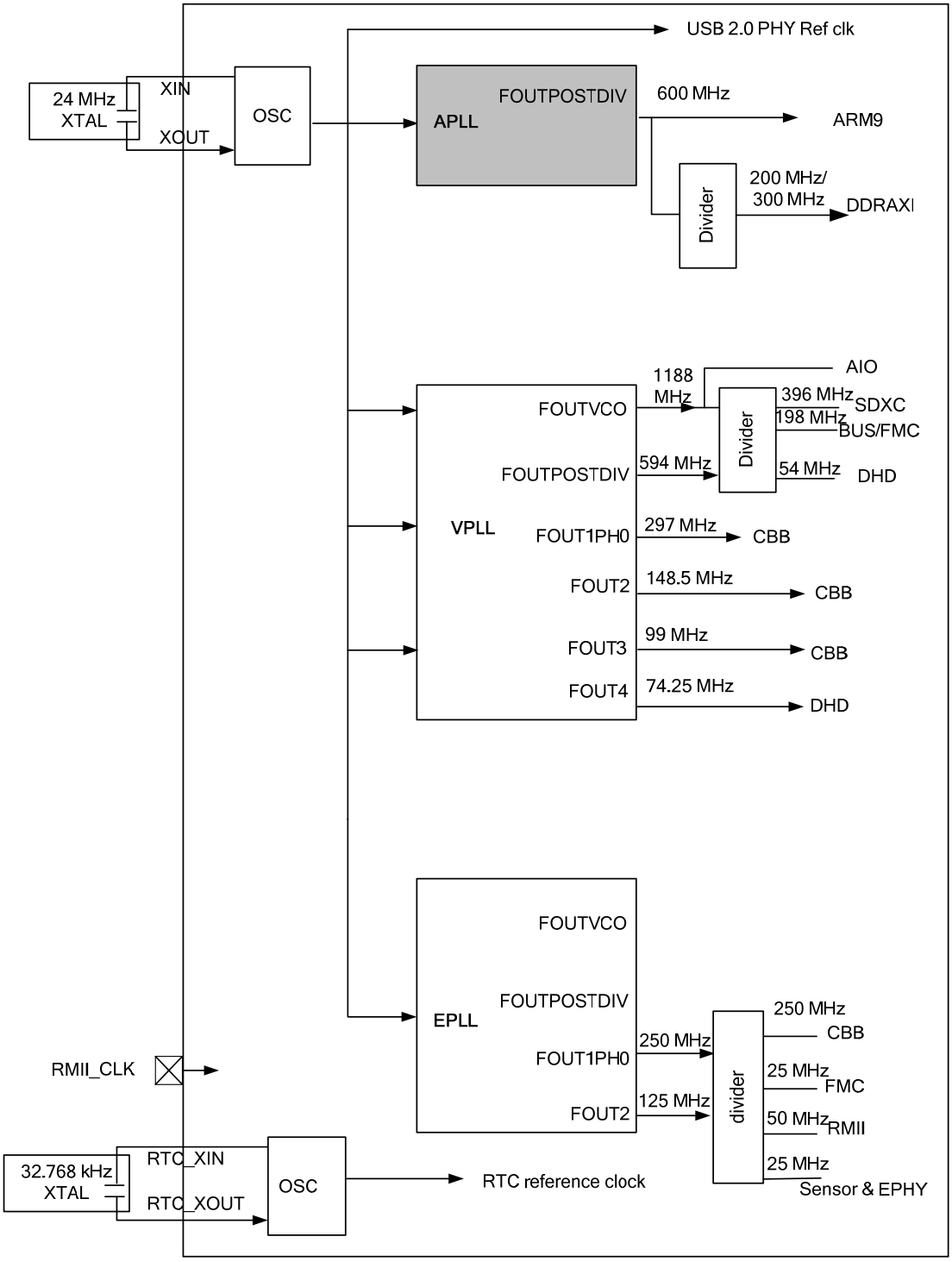
The CRG configures, controls, and manages the internal PLLs and the input clocks from pins. It also generates the clocks required by each module. Figure 3-3 shows the clock distribution diagram.

NOTE

The APLL in gray can be configured by programming, and the configurations of other PLLs are fixed. If a module has multiple clock sources, the default clock source is specified.

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Figure 3-3 Clock distribution diagram



3.2.4 PLL Configuration

The Hi3518E V20X/Hi3516C V200 has three internal PLLs. Each PLL uses two configuration registers. See Table 3-2.

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Table 3-2 Configuration registers corresponding to PLLs

PLL	Configuration Register 0	Configuration Register 1
APLL	PERI_CRG0	PERI_CRG1
VPLL	PERI_CRG4	PERI_CRG5
EPLL	PERI_CRG8	PERI_CRG9

All PLLs use the input crystal oscillator clock of the XIN pin as the input clock. For details on how to calculate PLL output frequencies, see Table 3-3.

Table 3-3 Methods of calculating PLL output frequencies

PLL Pin	Formula	Remarks
FREF	PLL input reference clock	The input clock must be 24 MHz.
FOUTVCO	FREF x (fbdiv + frac/2^24)/refdiv	PLL working frequency. It must be greater than 600 MHz but less than or equal to 2.4 GHz.
FOUTPOSTDIV	FOUTVCO/(pstdiv1 x pstdiv2)	None
FOUT1ph0	FOUTVCO/(pstdiv1 x pstdiv2 x 2)	None
FOUT2	FOUTVCO/(pstdiv1 x pstdiv2 x 4)	None
FOUT3	FOUTVCO/(pstdiv1 x pstdiv2 x 6)	None
FOUT4	FOUTVCO/(pstdiv1 x pstdiv2 x 8)	None

NOTE

- fbdiv: integral frequency multiplier
- frac: decimal frequency multiplier
- refdiv: frequency divider of the reference clock
- pstdiv1: level-1 output frequency divider
- pstdiv2: level-2 output frequency divider

For details about the frequency multiplier and divider of each PLL, see Table 3-1.

The following uses APLL as an example. If APLL outputs the 600 MHz FOUTPOSTDIV clock to the CPU module, the configured values of the register are calculated as follows:

- If postdiv2 is 1 and postdiv1 is 2, the value of FOUTVCO is 1200 MHz based on the following formula:

  FOUTOISTDIV = FOUTVCO/(pstdiv1 x pstdiv2)
- If refdiv is 2, fbdiv and frac can be calculated based on the following formula:

  FOUTVCO = FREF x (fbdiv + frac/2^24)/refdiv = 12 x (fbdiv + frac/2^24)/1 = 1200 MHz

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That is, fbdiv = 100 and frac = 000000

3.2.5 Frequency Configurations

Frequency Configurations of ARM9/DDR/Bus Clocks

Table 3-4 describes the frequency configurations of ARM9/DDR/bus clocks.

Table 3-4 Frequency configurations of ARM9/DDR/bus clocks

Signal	Configuration Register
cpu_sc_sel	You can control this signal by configuring PERI_CRG12 bit[8].
cpuclk_loaden	You can control this signal by configuring PERI_CRG10 bit[17].
cpuclk_skipcfg	You can control this signal by configuring PERI_CRG10 bit[16:12].
cpu_clk_div	You can control this signal by configuring PERI_CRG12 bit[5:4].
bus_sc_sel	You can control this signal by configuring PERI_CRG12 bit[0].

Table 3-5 shows the mapping between system controller modes and clock states.

Table 3-5 Mapping between system controller modes and clock states

System Controller Mode	Enable State of the 24 MHz Crystal Oscillator	Enable State of the APLL	System Clock State
Normal	Enabled	Enabled	The working clocks of the ARM subsystem are derived from the PLL.
Slow	Enabled	Disabled	The working clocks of the ARM subsystem are derived from the 24 MHz crystal oscillator.

Frequency Configurations of Module Clocks

Table 3-6 describes the frequency configurations of module clocks.

Table 3-6 Frequency configurations of module clocks

Signal	Configuration Register
VDP clock configurations	
lcd_cksel	PERI_CRG13 bit[20]
lcd_cken	PERI_CRG13 bit[18]

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Signal	Configuration Register
hd_lcd_cksel	PERI_CRG13 bit[17]
vo_out_cksel	PERI_CRG13 bit[14]
hd_cksel	PERI_CRG13 bit[13:12]
vo_out_pctrl	PERI_CRG13 bit[2]
VI_MIPI clock configurations	
sensor_cksel	PERI_CRG11 bit[18:16]
vi0_sc_sel	PERI_CRG11 bit[2]
phy_cmos_lat	PERI_CRG11 bit[7:6]
phy_hs_lat	PERI_CRG11 bit[5:4]
vi0_pctrl	PERI_CRG11 bit[1]
Advanced video coding (AVC) clock configurations	
avcclk_loaden	PERI_CRG16 bit[21]
avcclk_skipcfg	PERI_CRG16 bit[20:16]
VPSS clock configurations	
vpssclk_loaden	PERI_CRG18 bit[9]
vpssclk_skipcfg	PERI_CRG18 bit[8:4]
TDE clock configurations	
tdeclk_loaden	PERI_CRG22 bit[9]
tdeclk_skipcfg	PERI_CRG22 bit[8:4]
VGS clock configurations	
vgsclk_loaden	PERI_CRG23 bit[9]
vgsclk_skipcfg	PERI_CRG23 bit[8:4]
JPGE clock configurations	
jpgeclk_loaden	PERI_CRG24 bit[9]
tdeclk_skipcfg	PERI_CRG24 bit[8:4]
LCD clock configurations	
lcd_mclk_div	PERI_CRG26 bit[26:0]
FMC clock configurations	
sfc_cksel	PERI_CRG48 bit[3:2]
Secure digital extended capacity (SDXC) clock configurations	
sdxc0_cksel	PERI_CRG49 bit[11:10]

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Signal	Configuration Register
sdxc1_cksel	PERI_CRG49 bit[3:2]
Pulse-width modulation (PWM)/UART clock configurations	
pwm_cksel	PERI_CRG14 bit[3:2]
uart_cksel	PERI_CRG57 bit[20:19]
ETH clock configurations	
ethcore_cksel	PERI_CRG59 bit[7]
ethphy_cksel	PERI_CRG59 bit[5:4]
eth_rmiick_sel	PERI_CRG59 bit[2]

Precautions

Take the following precautions when configuring clocks:

- By default, the ARM9 working clock is in crystal oscillator mode after power-on. That is, the crystal oscillator clock input by the XIN pin is selected.
- If the frequency of the PLL is changed, the PLL can output a stable clock 0.1 ms later. The frequency of the PLL can be changed only when the system runs in slow mode.
- If the PLL output clock is not stable, the system mode cannot be switched to PLL mode. You can view the PLL lock bit to check whether the PLL is locked. The status of the PLL lock bit can be obtained by reading PERI_CRG58 bit[2:0].

3.2.6 Register Summary

Table 3-7 describes CRG registers.

Table 3-7 Summary of CRG registers (base address: 0x2003_0000)

Offset Address	Register	Description	Page
0x0000	PERI_CRG0	APLL configuration register 0	3-11
0x0004	PERI_CRG1	APLL configuration register 1	3-11
0x0010	PERI_CRG4	VPLL configuration register 0	3-13
0x0014	PERI_CRG5	VPLL configuration register 1	3-13
0x0020	PERI_CRG8	EPLL configuration register 0	3-15
0x0024	PERI_CRG9	EPLL configuration register 1	3-15
0x0028	PERI_CRG10	ARM9 frequency mode and reset configuration register	3-17
0x002C	PERI_CRG11	VICAP&MIPI CTRL and sensor clock and reset configuration register	3-17

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Offset Address	Register	Description	Page
0x0030	PERI_CRG12	System-on-chip (SoC) clock selection register	3-19
0x0034	PERI_CRG13	VOU clock and reset control register	3-20
0x0038	PERI_CRG14	PWM clock and reset control register	3-22
0x0040	PERI_CRG16	AVC clock and soft reset control register	3-23
0x0048	PERI_CRG18	VPSS clock and reset control register	3-24
0x0058	PERI_CRG22	TDE clock and soft reset control register	3-25
0x005C	PERI_CRG23	VGS clock and soft reset control register	3-26
0x0060	PERI_CRG24	JPGE clock and soft reset control register	3-27
0x0068	PERI_CRG26	LCD clock control register	3-28
0x006C	PERI_CRG27	IVE&HASH clock and soft reset control register	3-29
0x007C	PERI_CRG31	SAR_ADC&cipher clock and soft reset control register	3-29
0x008C	PERI_CRG35	AIAO clock and soft reset control register	3-30
0x00B8	PERI_CRG46	USB clock and soft reset control register	3-31
0x00C0	PERI_CRG48	FMC clock and soft reset control register	3-32
0x00C4	PERI_CRG49	SDIO0/1 clock and soft reset control register	3-32
0x00D4	PERI_CRG53	GZIP clock and soft reset control register	3-34
0x00D8	PERI_CRG54	DMA/DDRTEST clock and soft reset control register	3-35
0x00DC	PERI_CRG55	SoC SW feedback indicator register	3-36
0x00E0	PERI_CRG56	GZIP reset status register	3-36
0x00E4	PERI_CRG57	Soft reset control register for other CRG interface modules	3-37
0x00E8	PERI_CRG58	CRG status register	3-39
0x00EC	PERI_CRG59	ETH clock and soft reset control register	3-39
0x0100	PERI_CRG64	RSA clock and soft reset control register	3-40
0x0104	PERI_CRG65	Register reserved for debugging	3-41
0x0108	PERI_CRG66	APLL spread spectrum configuration register	3-42
0x0110	PERI_CRG68	VPLL spread spectrum configuration register	3-43

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Offset Address	Register	Description	Page
0x0118	PERI_CRG70	EPLL spread spectrum configuration register	3-44

3.2.7 Register Description

PERI_CRG0

PERI_CRG0 is APLL configuration register 0.

Offset Address	Register Name	Total Reset Value
0x0000	PERI_CRG0	0x1200_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		apll_postdiv2		reserved		apll_postdiv1		apll_frac																							
Reset	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:28]	RW	apll_postdiv2	Level-2 output divider of the APLL
[27]	RO	reserved	Reserved
[26:24]	RW	apll_postdiv1	Level-1 output divider of the APLL
[23:0]	RW	apll_frac	Decimal divider of the APLL

PERI_CRG1

PERI_CRG1 is APLL configuration register 1.

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Offset Address	Register Name	Total Reset Value
0x0004	PERI_CRG1	0x0100_2064

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					apll_bypass	apll_dacpd	apll_dsmpd	apll_pd	apll_foutvcopd	apll_postdivpd	apll_fout4phasepd	reserved		apll_refdiv						apll_fbdiv											
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	1	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RW	apll_bypass	APLL clock frequency division bypass 0: not bypassed 1: bypassed
[25]	RW	apll_dacpd	APLL test signal control 0: normal operating mode 1: power-down mode
[24]	RW	apll_dsmpd	APLL decimal division control 0: decimal frequency-division mode 1: integral frequency-division mode
[23]	RW	apll_pd	APLL power-down control 0: normal operating mode 1: power-down mode
[22]	RW	apll_foutvcopd	APLL VCO output power-down control 0: normal output clock 1: no output clock
[21]	RW	apll_postdivpd	APLL POSTDIV output power-down control 0: normal clock output 1: no output clock
[20]	RW	apll_fout4phasepd	APLL FOUT output power-down control 0: normal output clock 1: no output clock
[19:18]	RW	reserved	Reserved
[17:12]	RW	apll_refdiv	Frequency divider of the APLL reference clock
[11:0]	RW	apll_fbdiv	Integral frequency multiplier of the APLL

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PERI_CRG4

PERI_CRG4 is VPLL configuration register 0.

Offset Address	Register Name	Total Reset Value
0x0010	PERI_CRG4	0x1200_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	vpll_postdiv2			reserved	vpll_postdiv1			vpll_frac																							
Reset	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:28]	RW	vpll_postdiv2	Level-2 output divider of VPLL
[27]	RO	reserved	Reserved
[26:24]	RW	vpll_postdiv1	Level-1 output divider of VPLL
[23:0]	RW	vpll_frac	Decimal divider of VPLL

PERI_CRG5

PERI_CRG5 is VPLL configuration register 1.

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Offset Address	Register Name	Total Reset Value
0x0014	PERI_CRG5	0x0100_2063

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					vpll0_bypass	vpll0_dacpd	vpll0_dsmpd	vpll0_pd	vpll0_foutvcopd	vpll0_postdivpd	vpll0_fout4phasepd	reserved		vpll0_refdiv						vpll0_fbdiv												
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	0	1	1

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RW	vpll_bypass	VPLL clock frequency division bypass 0: not bypassed 1: bypassed
[25]	RW	vpll_dacpd	VPLL test signal control 0: normal operating mode 1: power-down mode
[24]	RW	vpll_dsmpd	VPLL decimal division control 0: decimal frequency-division mode 1: integral frequency-division mode
[23]	RW	vpll_pd	VPLL power-down control 0: normal operating mode 1: power-down mode
[22]	RW	vpll_foutvcopd	VPLL VCO output power-down control 0: normal output clock 1: no output clock
[21]	RW	vpll_postdivpd	VPLL POSTDIV output power-down control 0: normal output clock 1: no output clock
[20]	RW	vpll_fout4phasepd	VPLL FOUT output power-down control 0: normal output clock 1: no output clock
[19:18]	RO	reserved	Reserved
[17:12]	RW	vpll_refdiv	Frequency divider of the VPLL reference clock
[11:0]	RW	vpll_fbdiv	Integral frequency multiplier of VPLL

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PERI_CRG8

PERI_CRG8 is EPLL configuration register 0.

Offset Address	Register Name	Total Reset Value
0x0020	PERI_CRG8	0x1200_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	epll_postdiv2			reserved	epll_postdiv1			epll_frac																							
Reset	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:28]	RW	epll_postdiv2	Level-2 output divider of the EPLL
[27]	RO	reserved	Reserved
[26:24]	RW	epll_postdiv1	Level-1 output divider of the EPLL
[23:0]	RW	epll_frac	Decimal divider of the EPLL

PERI_CRG9

PERI_CRG9 is EPLL configuration register 1.

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Offset Address	Register Name	Total Reset Value
0x0024	PERI_CRG9	0x0100_307D

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					epll_bypass	epll_dacpd	epll_dsmpd	epll_pd	epll_foutvcopd	epll_postdivpd	epll_fout4phasepd	reserved		epll_refdiv						epll_fbdiv											
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1	1	1	1	0	1

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RW	epll_bypass	EPLL clock frequency division bypass 0: not bypassed 1: bypassed
[25]	RW	epll_dacpd	EPLL test signal control 0: normal operating mode 1: power-down mode
[24]	RW	epll_dsmpd	EPLL decimal division control 0: decimal frequency-division mode 1: integral frequency-division mode
[23]	RW	epll_pd	EPLL power-down control 0: normal operating mode 1: power-down mode
[22]	RW	epll_foutvcopd	EPLL VCO output power-down control 0: normal output clock 1: no output clock
[21]	RW	epll_postdivpd	EPLL POSTDIV output power-down control 0: normal output clock 1: no output clock
[20]	RW	epll_fout4phasepd	EPLL FOUT output power-down control 0: normal output clock 1: no output clock
[19:18]	RO	reserved	Reserved
[17:12]	RW	epll_refdiv	Frequency divider of the EPLL reference clock
[11:0]	RW	epll_fbdiv	Integral frequency multiplier of the EPLL

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PERI_CRG10

PERI_CRG10 is an ARM9 frequency mode and reset configuration register.

Offset Address	Register Name	Total Reset Value
0x0028	PERI_CRG10	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18	17	16	15 14 13 12	11 10 9 8	7 6 5	4	3 2 1 0
Name	reserved			reserved	cpuclk_loaden		cpuclk_skipcfg	reserved		arm_core_srst_req	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0	0		0 0 0 0	0 0 0 0	0 0 0	0	0 0 0 0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:18]	RO	reserved	Reserved
[17]	RW	cpuclk_loaden	Skip configuration enable for the CPU clock To change the skip configuration, perform the following steps: 1. Write a new skip configuration value. 2. Write 0 to loaden. 3. Write 1 to loaden.
[16:12]	RW	cpuclk_skipcfg	Skip configuration of the CPU clock N: *N*-beat clocks are disabled every 32-beat CPU clocks.
[11:6]	RO	reserved	Reserved
[5]	RW	arm_core_srst_req	ARM core soft reset request 0: deassert reset 1: reset
[4:0]	RO	reserved	Reserved

PERI_CRG11

PERI_CRG11 is a VICAP&MIPI CTRL and sensor clock and reset configuration register.

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Offset Address	Register Name	Total Reset Value
0x002C	PERI_CRG11	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved											sensor_srst_req	sensor_cken	sensor_cksel			reserved	mipi_cken	isp_cfg_srst_req	isp_core_srst_req	mipi_hrst_req	mipi_srst_req	vi_hrst_req	vi0_srst_req	phy_cmos_lat		phy_hs_lat		reserved	vi0_sc_sel	vi0_pctrl	vi0_cken
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20]	RW	sensor_srst_req	Sensor soft reset request 0: deassert reset 1: reset
[19]	RW	sensor_cken	Sensor CLKOUT clock gating 0: disabled 1: enabled
[18:16]	RW	sensor_cksel	Sensor clkout clock 000: 74.25 MHz 001: 37.125 MHz 010: 54 MHz 011: 27 MHz 1X0: 24 MHz 1X1: 25 MHz
[15]	RO	reserved	Reserved.
[14]	RW	mipi_cken	MIPI PIX clock gating 0: The clock is disabled. 1: The clock is enabled.
[13]	RW	isp_cfg_srst_req	ISP CFG soft reset request 0: deassert reset 1: reset
[12]	RW	isp_core_srst_req	ISP core soft reset request 0: deassert reset 1: reset
[11]	RO	reserved	Reserved

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[10]	RW	mipi_srst_req	MIPI Ctrl soft reset request 0: deassert reset 1: reset
[9]	RW	vi_hrst_req	VI bus soft reset request 0: deassert reset 1: reset
[8]	RW	vi0_srst_req	VI CH0 soft reset request 0: deassert reset 1: reset
[7:6]	RW	phy_cmos_lat	Interface clock latency in MIPI PHY CMOS mode n: *n* steps are added based on the default value (500 ps per step).
[5:4]	RW	phy_hs_lat	Interface clock latency in MIPI PHY HS mode n: *n* steps are added based on the default value (500 ps per step).
[3]	RO	reserved	Reserved
[2]	RW	vi0_sc_sel	VI CH0 clock select 0: 99 MHz 1: 148.5 MHz
[1]	RW	vi0_pctrl	Phase of the VI CH0 interface clock 0 (default): normal phase 1: inverted phase
[0]	RW	vi0_cken	VI CH0 clock gating 0: disabled 1: enabled

PERI_CRG12

PERI_CRG12 is an SoC clock selection register.

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Offset Address	Register Name	Total Reset Value
0x0030	PERI_CRG12	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved											mem_adjust_bypass	reserved											cpu_sc_sel	reserved		cpu_clk_div		reserved			bus_sc_sel
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20]	RW	mem_adjust_bypass	mem adjustability select 0: adjustable 1: not adjustable
[19:9]	RO	reserved	Reserved
[8]	RW	cpu_sc_sel	CPU clock select 0: 24 MHz 1: APLL_POSTDIV (configurable frequency of 540, 600, 660, or 750 MHz)
[7:6]	RO	reserved	Reserved
[5:4]	RW	cpu_clk_div	DDR bus clock 10: CPU clock after being divided by 3 Other values: reserved
[3:1]	RO	reserved	Reserved
[0]	RW	bus_sc_sel	Bus clock select 0: 24 MHz 1: 198 MHz

PERI_CRG13

PERI_CRG13 is a VOU clock and reset control register.

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Offset Address	Register Name	Total Reset Value
0x0034	PERI_CRG13	0x0000_0005

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved											lcd_cksel	reserved	lcd_cken	hd_lcd_cksel	reserved	vo_out_cksel	hd_cksel			reserved			vou_hd_cken	vou_acken	vo_out_cken	vou_pcken	vou_ppc_cken	vou_cfg_cken	reserved	vo_out_pctrl	reserved	vo_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20]	RW	lcd_cksel	LCD frequency division clock select 0: LCD clock after being divided by 3 1: LCD clock after being divided by 4
[19]	RO	reserved	Reserved
[18]	RW	lcd_cken	LCD working clock gating 0: disabled 1: enabled
[17]	RW	hd_lcd_cksel	HD_LCD working clock select 0: The HD frequency division clock is used as the DHD1 channel clock. 1: The LCD frequency division clock is used as the DHD1 channel clock.
[16:15]	RO	reserved	Reserved
[14]	RW	vo_out_cksel	VO OUT_CLK frequency select 0: The HD clock is used as the VDP output associated clock. 1: The LCD clock is used as the VDP output associated clock.
[13:12]	RW	hd_cksel	DHD frequency bit[12] 0: The HD clock frequency is 27 MHz. 1: The HD clock frequency is 74.25 MHz. bit[13] 0: HD frequency division clock with the clock divider of 2 1: HD frequency division clock with the clock divider of 1
[11:10]	RO	reserved	Reserved

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[9]	RW	vou_hd_cken	VOU HD clock gating register 0: disabled 1: enabled
[8]	RW	vou_acken	VOU AXI bus clock gating register 0: disabled 1: enabled
[7]	RW	vo_out_cken	VO_CLKOUT clock gating register 0: disabled 1: enabled
[6]	RW	vou_pcken	VO APB clock gating register 0: disabled 1: enabled
[5]	RW	vou_ppc_cken	VO PPC clock gating register 0: disabled 1: enabled
[4]	RW	vou_cfg_cken	VOU CFG clock gating (internally configured) register 0: disabled 1: enabled
[3]	RO	reserved	Reserved
[2]	RW	vo_out_pctrl	Phase of the output associated clock of the VOU HD 0: normal phase 1 (default): inverted phase
[1]	RO	reserved	Reserved
[0]	RW	vo_srst_req	VOU soft reset request 0: deassert reset 1: reset

PERI_CRG14

PERI_CRG14 is a PWM clock and reset control register.

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Offset Address	Register Name	Total Reset Value
0x0038	PERI_CRG14	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												pwm_cksel		pwm_cken	pwm_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:2]	RW	pwm_cksel	PWM clock select 00: 3 MHz 01: 50 MHz 1x: 24 MHz
[1]	RW	pwm_cken	PWM clock gating register 0: disabled 1: enabled
[0]	RW	pwm_srst_req	PWM soft reset request 0: deassert reset 1: reset

PERI_CRG16

PERI_CRG16 is an AVC clock and reset control register.

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Offset Address	Register Name	Total Reset Value
0x0040	PERI_CRG16	0x0000_3000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										avcclk_loaden	avcclk_skipcfg					reserved		avc_cken	avc_srst_req	reserved											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	avcclk_loaden	Skip configuration enable for the AVC clock To change the skip configuration, perform the following steps: 1. Write a new skip configuration value. 2. Write 0 to loaden. 3. Write 1 to loaden.
[20:16]	RW	avcclk_skipcfg	Skip configuration of the AVC clock N: *N*-beat clocks are disabled every 32-beat AVC clocks. The following example assumes that the AVC working source clock is 250 MHz. If avcclk_skipcfg is set to 00, the AVC works in full-speed mode; if avcclk_skipcfg is set to 01, the AVC works in speed-down mode. The actual working frequency is calculated as follows: 250 x 31/32 = 242 MHz This rule applies to other configurations.
[15:14]	RO	reserved	Reserved
[13]	RW	avc_cken	AVC clock gating 0: disabled 1: enabled
[12]	RW	avc_srst_req	AVC soft reset request 0: deassert reset. 1: reset
[11:0]	RO	reserved	Reserved

PERI_CRG18

PERI_CRG18 is a VPSS clock and reset control register.

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Offset Address	Register Name	Total Reset Value
0x0048	PERI_CRG18	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						vpsclk_loaden		vpsclk_skipcfg					reserved		vps_cken	vps_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	vpsclk_loaden	Skip configuration enable for the VPSS clock To change the skip configuration, perform the following steps: 1. Write a new skip configuration value. 2. Write 0 to loaden. 3. Write 1 to loaden.
[8:4]	RW	vpsclk_skipcfg	Skip configuration of the VPSS clock N: *N*-beat clocks are disabled every 32-beat VPS clocks.
[3:2]	RO	reserved	Reserved
[1]	RW	vps_cken	VPSS clock gating register 0: disabled 1: enabled
[0]	RW	vps_srst_req	VPSS soft reset request 0: deassert reset 1: reset

PERI_CRG22

PERI_CRG22 is a TDE clock and soft reset control register.

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Offset Address	Register Name	Total Reset Value
0x0058	PERI_CRG22	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						tdeclk_loaden	tdeclk_skipcfg					reserved		tde_cken	tde_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	tdeclk_loaden	Skip configuration enable for the TDE clock To change the skip configuration, perform the following steps: 1. Write a new skip configuration value. 2. Write 0 to loaden. 3. Write 1 to loaden.
[8:4]	RW	tdeclk_skipcfg	Skip configuration of the TDE clock N: *N*-beat clocks are disabled every 32-beat TDE clocks. The following example assumes that the TDE working source clock is 150 MHz. If tdeclk_skipcfg is set to 00, the TDE works in full-speed mode; if tdeclk_skipcfg is set to 01, the TDE works in speed-down mode. The actual working frequency is calculated as follows: 150 x 31/32 = 145 MHz This rule applies to other configurations.
[3:2]	RO	reserved	Reserved
[1]	RW	tde_cken	TDE clock gating register 0: disabled 1: enabled
[0]	RW	tde_srst_req	TDE soft reset request 0: deassert reset 1: reset

PERI_CRG23

PERI_CRG23 is a VGS clock and reset control register.

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Offset Address	Register Name	Total Reset Value
0x005C	PERI_CRG23	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						vgsclk_loaden		vgsclk_skipcfg					reserved		vgs_cken	vgs_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	vgsclk_loaden	Skip configuration enable for the VGS clock To change the skip configuration, perform the following steps: 1. Write a new skip configuration value. 2. Write 0 to loaden. 3. Write 1 to loaden.
[8:4]	RW	vgsclk_skipcfg	Skip configuration of the VGS clock N: *N*-beat clocks are disabled every 32-beat VGS clocks.
[3:2]	RO	reserved	Reserved
[1]	RW	vgs_cken	VGS clock gating register 0: disabled 1: enabled
[0]	RW	vgs_srst_req	VGS soft reset request 0: deassert reset 1: reset

PERI_CRG24

PERI_CRG24 is a JPGE clock and soft reset control register.

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Offset Address	Register Name	Total Reset Value
0x0060	PERI_CRG24	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						jpgeclk_loaden	jpgeclk_skipcfg					reserved		jpge_cken	jpge_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	jpgeclk_loaden	Skip configuration enable for the JPGE clock To change the skip configuration, perform the following steps: 1. Write a new skip configuration value. 2. Write 0 to loaden. 3. Write 1 to loaden.
[8:4]	RW	jpgeclk_skipcfg	Skip configuration of the JPGE clock N: *N*-beat clocks are disabled every 32-beat TDE clocks.
[3:2]	RO	reserved	Reserved
[1]	RW	jpge_cken	JPGE clock gating 0: disabled 1: enabled
[0]	RW	jpge_srst_req	JPGE soft reset request 0: deassert reset 1: reset

PERI_CRG26

PERI_CRG26 is an LCD clock control register.

Offset Address	Register Name	Total Reset Value
0x0068	PERI_CRG26	0x0015_E4C3

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				lcd_mclk_div																											
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	1	1	0	0	1	0	0	1	1	0	0	0	0	1	1

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved

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[26:0]	RO	lcd_mclk_div	LCD clock, configurable

PERI_CRG27

PERI_CRG27 is an IVE&HASH clock and soft reset control register.

Offset Address	Register Name	Total Reset Value
0x006C	PERI_CRG27	0x0000_000F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												hash_cken	hash_srst_req	ive_cken	ive_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RW	hash_cken	Hash clock gating 0: disabled 1: enabled
[2]	RW	hash_srst_req	Hash soft reset request 0: deassert reset 1: reset
[1]	RW	ive_cken	IVE clock gating 0: disabled 1: enabled
[0]	RW	ive_srst_req	IVE soft reset request 0: deassert reset 1: reset

PERI_CRG31

PERI_CRG31 is an SAR_ADC&cipher clock and soft reset control register.

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Offset Address	Register Name	Total Reset Value
0x007C	PERI_CRG31	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													sar_adc_srst_req	cipher_cken	cipher_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	sar_adc_srst_req	SAR ADC soft reset request 0: deassert reset 1: reset
[1]	RW	cipher_cken	Cipher clock gating 0: disabled 1: enabled
[0]	RW	cipher_srst_req	Cipher soft reset request 0: deassert reset 1: reset

PERI_CRG35

PERI_CRG35 is an AIAO clock and reset control register.

Offset Address	Register Name	Total Reset Value
0x008C	PERI_CRG35	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														aiao_cken	aiao_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved

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[1]	RW	aiao_cken	AIAO clock gating 0: disabled 1: enabled
[0]	RW	aiao_srst_req	AIAO soft reset request 0: deassert reset 1: reset

PERI_CRG46

PERI_CRG46 is a USB2.0 clock and soft reset control register.

Offset Address	Register Name	Total Reset Value
0x00B8	PERI_CRG46	0x0000_00A7

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																							usb_device_ctrl_srst_req	usb_cken	reserved	usb_ctrl_utmi0_req	reserved			usb_phy_port0_treq	usb_phy_req	usb_hrst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:9]	RO	reserved	Reserved
[8]	RW	usb_device_ctrl_srst_req	USB2.0 device controller soft reset request 0: deassert reset 1: reset
[7]	RW	usb_cken	Clock gating of the USB2.0 PHY reference clock 0: disabled 1: enabled
[6]	RO	reserved	Reserved
[5]	RW	usb_ctrl_utmi0_req	USB2.0 controller port 0 soft reset request 0: deassert reset 1: reset
[4:3]	RO	reserved	Reserved
[2]	RW	usb_phy_port0_treq	USB2.0 PHY port 0 soft reset request 0: deassert reset 1: reset

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[1]	RW	usb_phy_req	USB2.0 PHY soft reset request 0: deassert reset 1: reset
[0]	RW	usb_hrst_req	USB2.0 controller bus soft reset request 0: deassert reset 1: reset

PERI_CRG48

PERI_CRG48 is an FMC clock and soft reset control register.

Offset Address	Register Name	Total Reset Value
0x00C0	PERI_CRG48	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												sfc_cksel		sfc_cken	sfc_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:2]	RW	sfc_cksel	FMC 2X clock source select 00: 24 MHz 01: 148.5 MHz 10: 125 MHz 11: 198 MHz
[1]	RW	sfc_cken	FMC clock gating 0: disabled 1: enabled
[0]	RW	sfc_srst_req	FMC soft reset request 0: deassert reset 1: reset

PERI_CRG49

PERI_CRG49 is an SDIO0/1 clock and soft reset control register.

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Offset Address	Register Name	Total Reset Value
0x00C4	PERI_CRG49	0x0000_0202

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sdxc0_dll_srst_req	sdxc0_dll_cken	reserved	reserved	sdxc0_cksel		sdxc0_cken	sdxc0_srst_req	reserved		reserved	reserved	sdxc1_cksel		sdxc1_cken	sdxc1_srst_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RW	sdxc0_dll_srst_req	SDIO0 DLL soft reset request 0: deassert reset 1: reset
[14]	RW	sdxc0_dll_cken	SDIO0 DLL clock gating 0: disabled 1: enabled
[13]	RO	reserved	Reserved
[12]	RO	reserved	Reserved
[11:10]	RW	sdxc0_cksel	SDIO0 working clock select 00: 99 MHz 01: 74.25 MHz 10: 49.5 MHz 11: reserved
[9]	RW	sdxc0_cken	SDIO0 clock gating 0: disabled 1: enabled
[8]	RW	sdxc0_srst_req	SDIO0 soft reset request 0: deassert reset 1: reset
[7:6]	RO	reserved	Reserved
[5]	RO	reserved	Reserved
[4]	RO	reserved	Reserved

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[3:2]	RW	sdxc1_cksel	SDIO1 working clock select 00: 49.5 MHz 01: reserved 10: reserved 11: reserved
[1]	RW	sdxc1_cken	SDIO1 clock gating 0: disabled 1: enabled
[0]	RW	sdxc1_srst_req	SDIO1 soft reset request 0: deassert reset 1: reset

PERI_CRG53

PERI_CRG53 is a GZIP clock and soft reset control register.

Offset Address	Register Name	Total Reset Value
0x00D4	PERI_CRG53	0x0000_0007

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				reserved				gzip_srst_req	reserved				gzip_pcken	gzip_acken	gzip_cken	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10:8]	RO	reserved	Reserved
[7]	RW	gzip_srst_req	GZIP soft reset request 0: deassert reset 1: reset
[6:3]	RO	reserved	Reserved
[2]	RW	gzip_pcken	GZIP APB clock gating 0: disabled 1: enabled

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[1]	RW	gzip_acken	GZIP AXI clock gating 0: disabled 1: enabled
[0]	RW	gzip_cken	GZIP clock gating 0: disabled 1: enabled

PERI_CRG54

PERI_CRG54 is a DMA/DDRTEST clock and soft reset control register.

Offset Address	Register Name	Total Reset Value
0x00D8	PERI_CRG54	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										dma_cken	dma_srst_req	ddrtest_cken	ddrtest_srst_req	reserved	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5]	RW	dma_cken	DMA clock gating 0: disabled 1: enabled
[4]	RW	dma_srst_req	DMA soft reset request 0: deassert reset 1: reset
[3]	RW	ddrtest_cken	DDRTEST clock gating 0: disabled 1: enabled
[2]	RW	ddrtest_srst_req	DDRTEST soft reset request 0: deassert reset 1: reset
[1]	RO	reserved	Reserved
[0]	RO	reserved	Reserved

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PERI_CRG55

PERI_CRG55 is an SoC SW feedback indicator register.

Offset Address	Register Name	Total Reset Value
0x00DC	PERI_CRG55	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										reserved	clk_cpu_seled	reserved		clk_bus_seled	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5]	RO	reserved	Reserved
[4]	RO	clk_cpu_seled	CPU switch completion indicator
[3:2]	RO	reserved	Reserved
[1]	RO	clk_bus_seled	System bus switch completion indicator
[0]	RO	reserved	Reserved

PERI_CRG56

PERI_CRG56 is a GZIP reset status register.

Offset Address	Register Name	Total Reset Value
0x00E0	PERI_CRG56	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															gzip_rst_ok
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	gzip_rst_ok	GZIP reset status

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PERI_CRG57

PERI_CRG57 is a soft reset control register for other CRG interface modules.

Offset Address	Register Name	Total Reset Value
0x00E4	PERI_CRG57	0x040B_F001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				i2c_cksel		i2c2_srst_req	i2c1_srst_req	i2c0_srst_req	reserved			uart_cksel		reserved		uart2_cken	uart1_cken	uart0_cken	ssp1_cken	ssp0_cken	ir_cken	reserved		uart2_srst_req	uart1_srst_req	uart0_srst_req	ssp1_srst_req	ssp0_srst_req	ir_srst_req	reserved		rtc_srst_req	reserved	test_clk_en	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:26]	RW	i2c_cksel	I^2C clock source select 00: APB clock 01: 24 MHz clock 1x: 6 MHz clock
[25]	RW	i2c2_srst_req	I^2C2 soft reset request 0: deassert reset 1: reset
[24]	RW	i2c1_srst_req	I^2C1 soft reset request 0: deassert reset 1: reset
[23]	RW	i2c0_srst_req	I^2C0 soft reset request 0: deassert reset 1: reset
[22:21]	RW	reserved	Reserved
[20:19]	RW	uart_cksel	UART clock select 00: APB clock 01: 24 MHz clock 1x: 6 MHz clock
[18]	RW	reserved	Reserved
[17]	RW	uart2_cken	UART2 clock gating 0: disabled 1: enabled

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[16]	RW	uart1_cken	UART1 clock gating 0: disabled 1: enabled
[15]	RW	uart0_cken	UART0 clock gating 0: disabled 1: enabled
[14]	RW	ssp1_cken	SSP1 clock gating 0: disabled 1: enabled
[13]	RW	ssp0_cken	SSP0 clock gating 0: disabled 1: enabled
[12]	RW	ir_cken	IR clock gating 0: disabled 1: enabled
[11:10]	RO	reserved	Reserved
[9]	RW	uart2_srst_req	UART2 soft reset request 0: deassert reset 1: reset
[8]	RW	uart1_srst_req	UART1 soft reset request 0: deassert reset 1: reset
[7]	RW	uart0_srst_req	UART0 soft reset request 0: deassert reset 1: reset
[6]	RW	ssp1_srst_req	SSP1 soft reset request 0: deassert reset 1: reset
[5]	RW	ssp0_srst_req	SSP0 soft reset request 0: deassert reset 1: reset
[4]	RW	ir_srst_req	IR soft reset request 0: deassert reset 1: reset
[3]	RO	reserved	Reserved

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[2]	RW	rtc_srst_req	t_cap soft reset request 0: deassert reset 1: reset
[1]	RO	reserved	Reserved
[0]	RW	test_clk_en	Test clock enable 0: All test clocks are disabled. 1: All test clocks are enabled.

PERI_CRG58

PERI_CRG58 is a CRG status register.

Offset Address	Register Name	Total Reset Value
0x00E8	PERI_CRG58	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													epll_lock	vpll_lock	apll_lock
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RO	epll_lock	EPLL lock status 0: unlocked 1: locked
[1]	RO	vpll_lock	VPLL lock status 0: unlocked 1: locked
[0]	RO	apll_lock	APLL lock status 0: unlocked 1: locked

PERI_CRG59

PERI_CRG59 is an ETH clock and soft reset control register.

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Offset Address	Register Name	Total Reset Value
0x00EC	PERI_CRG59	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								ethcore_cksel	reserved	ethphy_cksel		ext_fephy_srst_req	eth_rmiick_sel	eth_cken	hrst_eth_s
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:8]	RW	reserved	Reserved
[7]	RW	ethcore_cksel	ETH clock source select 0: 99 MHz 1: 54 MHz
[6]	RW	reserved	Reserved
[5:4]	RW	ethphy_cksel	ETH PHY output clock source select 00: 25 MHz 01: 24 MHz 1x: 27 MHz
[3]	RW	ext_fephy_srst_req	EPHY soft reset request 0: deassert reset 1: reset
[2]	RW	eth_rmiick_sel	ETH RMII clock source select 0: internal CRG clock 1: PAD input clock
[1]	RW	eth_cken	ETH clock gating 0: The clock is disabled. 1: The clock is enabled.
[0]	RW	hrst_eth_s	ETH soft reset request 0: deassert reset 1: reset

PERI_CRG64

PERI_CRG64 is an RSA clock and soft reset control register.

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Offset Address	Register Name	Total Reset Value
0x0100	PERI_CRG64	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										rsa_cken	rsa_srst_req	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5]	RW	rsa_cken	RSA clock gating 0: The clock is disabled. 1: The clock is enabled.
[4]	RW	rsa_srst_req	RSA soft reset request 0: deassert reset 1: reset
[3:0]	RW	reserved	Reserved

PERI_CRG65

PERI_CRG65 is a register reserved for debugging.

Offset Address	Register Name	Total Reset Value
0x0104	PERI_CRG65	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																eco_reserv															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved

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[15:0]	RW	eco_reserv	Bit [5]: clock select for the FMC MUX I0 end 0: 24 MHz 1: 99 MHz Bit[3]: VGS source clock select 0: 148.5 MHz 1: 198 MHz Bit[2:1]: AVC source clock select 00: 148.5 MHz 01: 99 MHz 1x: 198 MHz Bit[0]: VPSS source clock select 0: 99 MHz 1: 148.5 MHz

PERI_CRG66

PERI_CRG66 is an APLL spread spectrum configuration register.

Offset Address	Register Name	Total Reset Value
0x0108	PERI_CRG66	0x0000_0004

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				apll_ssmod_ctrl											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bits	Access	Name	Description
[31:12]	RW	reserved	Reserved
[11:0]	RW	apll_ssmod_ctrl	ssmod divval[11:8]: SSMOD divval control ssmod spread[6:4]: SSMOD spread control 000: 0 001: 0.049% 010: 0.098% 011: 0.195% 100: 0.391% 101: 0.781% 110: 1.563% 111: 3.125% ssmod downspread[3]: SSMOD downspread control 0: central spread spectrum 1: down spread spectrum ssmod_disable[2]: SSMOD disable control

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			0: enabled 1: disabled ssmod_rst_req[1]: SSMOD reset control 0: not reset 1: reset ssmod_cken[0]: SSMOD clock gating 0 (default): disabled 1: enabled

PERI_CRG68

PERI_CRG68 is a VPLL spread spectrum configuration register.

Offset Address	Register Name	Total Reset Value
0x0110	PERI_CRG68	0x0000_0004

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				vpll_ssmod_ctrl											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bits	Access	Name	Description
[31:12]	RW	reserved	Reserved
[11:0]	RW	vpll_ssmod_ctrl	ssmod divval[11:8]: SSMOD divval control ssmod spread[6:4]: SSMOD spread control 000: 0 001: 0.049% 010: 0.098% 011: 0.195% 100: 0.391% 101: 0.781% 110: 1.563% 111: 3.125% ssmod downspread[3]: SSMOD downspread control 0: central spread spectrum 1: down spread spectrum ssmod_disable[2]: SSMOD disable control 0: enabled 1: disabled ssmod_rst_req[1]: SSMOD reset control 0: not reset 1: reset

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			ssmod_cken[0]: SSMOD clock gating 0 (default): disabled 1: enabled

PERI_CRG70

PERI_CRG70 is an EPLL spread spectrum configuration register.

Offset Address	Register Name	Total Reset Value
0x0118	PERI_CRG70	0x0000_0004

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				epll_ssmod_ctrl											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bits	Access	Name	Description
[31:12]	RW	reserved	Reserved
[11:0]	RW	epll_ssmod_ctrl	ssmod divval[11:8]: SSMOD divval control ssmod spread[6:4]: SSMOD spread control 000: 0 001: 0.049% 010: 0.098% 011: 0.195% 100: 0.391% 101: 0.781% 110: 1.563% 111: 3.125% ssmod downspread[3]: SSMOD downspread control 0: central spread spectrum 1: down spread spectrum ssmod_disable[2]: SSMOD disable control 0: enabled 1: disabled ssmod_rst_req[1]: SSMOD reset control 0: not reset 1: reset ssmod_cken[0]: SSMOD clock gating 0: disabled (default) 1: enabled

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3.3 Processor Subsystem

Hi3518E V20X/Hi3516C V200 has an embedded processor ARM926EJ-S, which has the following features:

- 32-bit ARM v5TEJ and 5-stage pipeline, compatible with the 32-bit ARM and the 16-bit thumb instruction set
- Embedded enhanced digital signal processing (DSP) instructions
- Java
- Independent 32 KB I-cache, 32 KB D-cache, and 4-way set-associative cache. The cache line size is 32 bytes. The D-cache supports configurable write-back and write-through operations.
- Caches that support the configurable pseudo-random or round-robin replacement algorithm.
- Independent instruction and data bus interfaces. The ratio of the working frequency of the bus clock to that of the system clock ARM926EJ-S is 1:1. The two clocks have the same phase.
- Memory management unit (MMU) that supports multiple open operating systems, such as VxWorks, Linux, WindowCE, and PalmOS.
- Little-endian mode
- Fast interrupt requests (FIQs) and interrupt requests (IRQs)
- Joint Test Action Group (JTAG) debugging interface
- Dynamic and static power management

Maximum working frequency of 540 MHz for Hi3518E V20X and maximum working frequency of 600 MHz for Hi3516C V200

3.4 Interrupt System

3.4.1 Interrupt Source Allocation

Table 3-8 describes the allocation of ARM9 interrupt sources.

Table 3-8 Allocation of ARM9 interrupt sources

ARM9	Interrupt Source
0	Global soft interrupt
1	WatchDog interrupt
2	RTC/Temper_Cap interrupt
3	Timer0/Timer1
4	Timer2/Timer3
5	UART0
6	SSP0

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ARM9	Interrupt Source
7	SSP1
8	GZIP/SDIO1
9	AIAO
10	USB_DEV
11	FMC
12	ETH
13	CIPHER
14	DMAC
15	USB_EHCI
16	USB_OHCI
17	VPSS
18	SDIO0
19	IR/SAR_ADC
20	$I^2C0/I^2C1/I^2C2$
21	IVE
22	VICAP
23	VDP
24	AVC
25	UART2
26	JPGE
27	TDE
28	MIPI
29	VGS
30	UART1
31	GPIO0−GPIO8

3.4.2 Register Description

Table 3-9 describes interrupt control registers.

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Table 3-9 Summary of interrupt control registers (base address: 0x100D_0000)

Offset Address	Register	Description	Page
0x000	INT_IRQSTATUS	IRQ interrupt status register	3-47
0x004	INT_FIQSTATUS	FIQ interrupt status register	3-47
0x008	INT_RAWINTR	Raw interrupt status register	3-48
0x00C	INT_INTSELECT	Interrupt source select register	3-48
0x010	INT_INTENABLE	Interrupt enable register	3-49
0x014	INT_INTENCLEAR	Interrupt enable clear register	3-49
0x018	INT_SOFTINT	Software interrupt register	3-50
0x01C	INT_SOFTINTCLEAR	Software interrupt clear register	3-50
0x020	INT_PROTECTION	Protection enable register	3-51

3.4.3 Register Description

INT_IRQSTATUS

INT_IRQSTATUS is an IRQ interrupt status register. The 32 bits correspond to 32 interrupt sources. For details, see Table 3-8.

Offset Address	Register Name	Total Reset Value
0x000	INT_IRQSTATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	irqstatus																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	irqstatus	Status of the IRQ interrupt source 0: No interrupt is generated. 1: An IRQ interrupt is generated and sent to the processor.

INT_FIQSTATUS

INT_FIQSTATUS is an FIQ interrupt status register. The 32 bits correspond to 32 interrupt sources.

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Offset Address	Register Name	Total Reset Value
0x004	INT_FIQSTATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fiqstatus																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	fiqstatus	Status of the FIQ interrupt source 0: No interrupt is generated. 1: An FIQ interrupt is generated and sent to the processor.

INT_RAWINTR

INT_RAWINTR is a raw interrupt status register. It shows the status of raw interrupt requests and the status of the software interrupts that are generated by configuring INT_SOFTINT. The 32 bits correspond to 32 interrupt sources.

Offset Address	Register Name	Total Reset Value
0x008	INT_RAWINTR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rawinterrupt																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	rawinterrupt	Status of the raw interrupt source request 0: No interrupt is generated. 1: An interrupt is generated.

INT_INTSELECT

INT_INTSELECT is an interrupt source select register. It determines whether the selected interrupt source generates the IRQ interrupt or FIQ interrupt. The 32 bits correspond to 32 interrupt sources.

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Offset Address	Register Name	Total Reset Value
0x00C	INT_INTSELECT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	intselect																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	intselect	Interrupt request type of the interrupt source 0: IRQ interrupt 1: FIQ interrupt

INT_INTENABLE

INT_INTENABLE is an interrupt enable register. It is used to enable interrupt request lines. During reset, the value of INT_INTENABLE is changed to 0x0000_0000. As a result, all interrupt sources are masked. The 32 bits correspond to 32 interrupt sources.

Offset Address	Register Name	Total Reset Value
0x010	INT_INTENABLE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	intenable																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	intenable	The mask status of each interrupt source is returned when this register is read. 0: masked 1: not masked The interrupt source is enabled by bit when this register is written. 0: The current value of the corresponding bit is not affected. 1: The corresponding bit is set to 1 to enable the corresponding interrupt request.

INT_INTENCLEAR

INT_INTENCLEAR is an interrupt enable clear register. It is used to clear the corresponding bits of INT_INTENABLE. INT_INTENCLEAR is write-only and has no default reset value.

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Offset Address	Register Name	Total Reset Value
0x014	INT_INTENCLEAR	-

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	intenableclear																															
Reset	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?

Bits	Access	Name	Description
[31:0]	WO	intenableclear	Mask of the interrupt source corresponding to INT_INTENABLE 0: The current value of the corresponding bit of INT_INTENABLE is not affected. 1: The corresponding bit of INT_INTENABLE is cleared and the corresponding interrupt request is masked.

INT_SOFTINT

INT_SOFTINT is a software interrupt register. It controls whether the interrupt source input line generates software interrupts by using software. The software interrupts can be cleared by writing INT_SOFTINTCLEAR after the interrupt service routine (ISR) ends.

Offset Address	Register Name	Total Reset Value
0x018	INT_SOFTINT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	softint																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	softint	Generation of a raw software interrupt on a specified interrupt source 0: The current value of the corresponding bit is not affected. 1: The corresponding bit is set to 1 and a software interrupt request is generated.

INT_SOFTINTCLEAR

INT_SOFTINTCLEAR is a software interrupt clear register. It is used to clear the corresponding bit of INT_SOFTINT. INT_SOFTINTCLEAR is write-only and has no default reset value.

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Offset Address	Register Name	Total Reset Value
0x01C	INT_SOFTINTCLEAR	-

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	softintclear																															
Reset	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?

Bits	Access	Name	Description
[31:0]	WO	softintclear	Mask of the interrupt request corresponding to INT_SOFTINT 0: The corresponding bit of INT_SOFTINT is not affected. 1: The corresponding bit of INT_SOFTINT is cleared and the corresponding interrupt request is masked.

INT_PROTECTION

INT_PROTECTION is a protection enable register. It is used to enable or disable the access to protected registers.

CAUTION

- This register is cleared during reset and the INT registers can be accessed in user mode or privileged mode.
- When the CPU fails to generate the correct protection information (HPROT), the INT registers can be accessed in user mode after this register is reset.

Offset Address	Register Name	Total Reset Value
0x020	INT_PROTECTION	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															protection
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	protection	Register access protection enable 0: disabled. The CPU can access INT registers in user mode or privileged mode. 1: enabled. The CPU can access INT registers only in privileged mode.

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3.5 System Controller

3.5.1 Overview

The system controller manages important system functions and configures some functions of the peripherals.

3.5.2 Features

The system controller has the following features:

- Controls the system address remap and monitors its status.
- Provides general peripheral registers
- Provides write protection for key registers.
- Provides chip identification (ID) registers.

3.5.3 Function Description

Soft Reset

The system controller can soft-reset the entire chip or some modules.

After the global soft reset register SC_SYSRES is configured, the system controller transmits a reset request to the on-chip reset module for setting the Hi3518E V20X/Hi3516C V200.

System Address Remap Control

For details, see section 1.4 "Address Space Mapping."

Write Protection for Key Registers

To prevent the entire system from being affected by misoperations on the system controller, the system controller provides write protection for the key configuration register: global soft reset control register SC_SYSRES. You must configure SC_LOCKEN to enable the write permission before writing to SC_SYSRES. After the write operation is complete, you need to disable the write permission by configuring SC_LOCKEN, ensuring that SC_SYSRES is not incorrectly written by software.

NOTE

By default, write protection is not enabled for key registers after reset. You are advised to enable write protection by configuring SC_LOCKEN when the system starts.

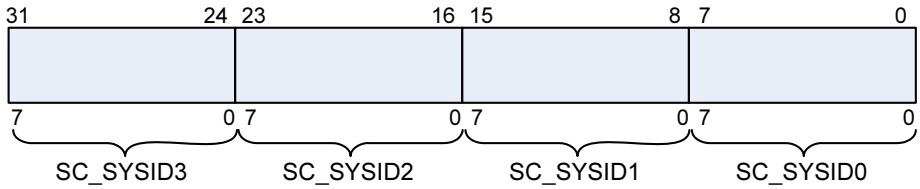
System ID Register of the Chip

The system controller provides a chip ID register SC_SYSID. This register is a virtual 32-bit read-only register. It consists of four 8-bit ID registers:

SCSYSID3, SCSYSID2, SCSYSID1, and SCSYSID0. After the values of these four registers are read and combined, the value of SC_SYSID, namely, 0x3518_E200, is obtained. Figure 3-4 shows the bit allocation of chip ID registers.

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Figure 3-4 Bit allocation of chip ID registers



3.5.4 System Controller Registers

3.5.4.1 Register Summary

Table 3-10 describes system controller registers.

Table 3-10 Summary of system controller registers (base address: 0x2005_0000)

Offset Address	Register	Description	Page
0x000	SC_CTRL	System control register	3-53
0x004	SC_SYSRES	System soft reset register	3-55
0x001C	SOLFINT	Software interrupt register	3-56
0x0044	SC_LOCKEN	Key register lock register	3-56
0x008C	SYSSTAT	System status register (PLL_LOCK)	3-57
0xEE0	SCSYSID0	Chip ID register 0	3-58
0xEE4	SCSYSID1	Chip ID register 1	3-58
0xEE8	SCSYSID2	Chip ID register 2	3-59
0xEEC	SCSYSID3	Chip ID register 3	3-59

3.5.4.2 Register Description

SC_CTRL

SC_CTRL is a system control register. It is used to specify the operations to be performed by the system.

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CAUTION

Write protection for this register can be enabled by configuring SC_LOCKEN. This register can be written only when write protection is disabled.

Offset Address	Register Name	Total Reset Value
0x000	SC_CTRL	0x0000_0202

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved									timeren3ov	reserved	timeren2ov	reserved	timeren1ov	reserved	timeren0ov	reserved	reserved					remapstat	remapclear	reserved								
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	1	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22]	RW	timeren3ov	Timer 3 count clock select 0: 3 MHz clock 1: bus clock
[21]	RO	reserved	Reserved
[20]	RW	timeren2ov	Timer 2 count clock select 0: 3 MHz clock 1: bus clock
[19]	RW	reserved	Reserved
[18]	RW	timeren1ov	Timer 1 count clock select 0: 3 MHz clock 1: bus clock
[17]	RW	reserved	Reserved
[16]	RW	timeren0ov	Timer 0 count clock select 0: 3 MHz clock 1: bus clock
[15]	RW	reserved	Reserved
[14:10]	RO	reserved	Reserved. Reading this field returns 0, and writing to this field has no effect.
[9]	RO	remapstat	Address remap status 0: The address is not remapped. 1: The address is remapped. The BOOTROM, or SFC CS0 is remapped to address 0.

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[8]	RW	remapclear	Address remap clear 0: The remap status is retained. 1: The remap status is cleared. For details about the address mapping relationships before and after remapping is cleared, see section 1.3 "Boot Mode" and section 1.3.4 "Address Space Mapping" in chapter 1.
[7:0]	RO	reserved	Reserved

SC_SYSRES

SC_SYSRES is a system soft reset register. When a value is written to this register, the system controller sends a system soft reset request to the reset module. Then the reset module resets the system.

CAUTION

Write protection for this register can be enabled by configuring SC_LOCKEN. This register can be written only when write protection is disabled.

Offset Address	Register Name	Total Reset Value
0x004	SC_SYSRES	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	softresreq																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:0]	WO	softresreq	Writing any value to this register soft-resets the system.

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SOLFINT

SOLFINT is a software interrupt register.

Offset Address	Register Name	Total Reset Value
0x001C	SOLFINT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															software_int
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	software_int	Software interrupt 0: No interrupt is generated. 1: An interrupt is generated.

SC_LOCKEN

SC_LOCKEN is a key register lock registers.

Offset Address	Register Name	Total Reset Value
0x0044	SC_LOCKEN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	scper_lockl																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

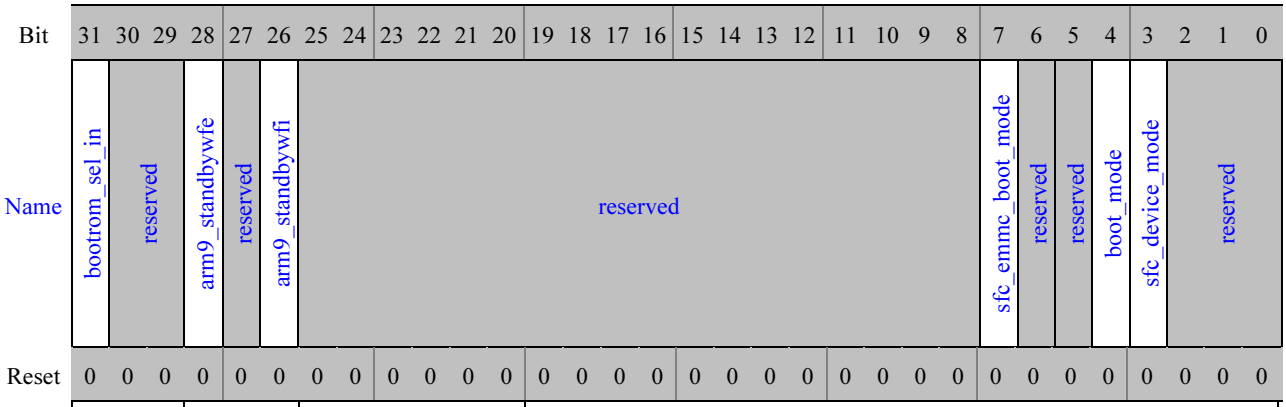
Bits	Access	Name	Description
[31:0]	RW	scper_lockl	Register for locking key system control registers. The key registers include SC_CTRL and SYSSTAT. When 0x1ACC_E551 is written to SC_LOCKEN, the write permission for all registers is enabled; when any other value is written to SC_LOCKEN, the write permission is disabled. Reading this register returns the lock status but not the written value. 0x0000_0000: The write permission is enabled (unlocked). 0x0000_0001: The write permission is disabled (locked).

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SYSSTAT

SYSSTAT is a system status register (PLL_LOCK).

Offset Address	Register Name	Total Reset Value
0x008C	SYSSTAT	0x0000_0000



Bits	Access	Name	Description
[31]	RO	bootrom_sel_in	Whether the system boots from the BOOTROM 0: no 1: yes
[30:29]	RO	reserved	Reserved
[28]	RO	arm9_standbywfe	Whether the ARM9 processor is in wait for event (WFE) status 0: no 1: yes
[27]	RO	reserved	Reserved
[26]	RO	arm9_standbywfi	Whether the ARM9 processor is in wait for interrupt (WFI) status 0: no 1: yes
[25:8]	RO	reserved	Reserved
[7]	RO	sfc_emmc_boot_mode	Boot address mode of the SPI NOR flash when boot_mode and sfc_device_mode are 0 0: 3-byte address mode 1: 4-byte address mode Boot mode of the SPI NAND flash when boot_mode is 0 and sfc_device_mode is 1 0: 1-wire boot mode 1: 4-wire boot mode Embedded multimedia card (eMMC) boot mode when boot_mode is 1 0: 4-wire boot mode 1: 8-wire boot mode

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[6]	RO	reserved	Reserved
[5]	RO	reserved	Reserved
[4]	RO	boot_mode	Chip boot mode 0: The chip boots from the SPI flash. 1: The chip boots from the eMMC.
[3]	RW	sfc_device_mode	SPI flash select 0: SPI NOR flash 1: SPI NAND flash
[2:0]	RW	reserved	Reserved

SCSYSID0

SCSYSID0 is chip ID register 0.

Offset Address	Register Name	Total Reset Value
0xEE0	SCSYSID0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	chipid								reserved																sysid0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	chipid	Version 01: Hi3516CV200 02: Hi3518EV200 03: Hi3518EV201 Others values: reserved
[23:8]	RO	reserved	Reserved
[7:0]	RO	sysid0	Reading this register returns 0x00.

SCSYSID1

SCSYSID1 is chip ID register 1.

Offset Address	Register Name	Total Reset Value
0xEE4	SCSYSID1	0xE2

Bit	7	6	5	4	3	2	1	0
Name	sysid1							

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Reset	1	1	1	0	0	0	1	0

Bits	Access	Name	Description
[7:0]	RO	sysid1	Reading this register returns 0xE2.

SCSYSID2

SCSYSID2 is chip ID register 2.

Offset Address	Register Name	Total Reset Value
0xEE8	SCSYSID2	0x18

Bit	7	6	5	4	3	2	1	0
Name	sysid2							
Reset	0	0	0	1	1	0	0	0

Bits	Access	Name	Description
[7:0]	RO	sysid2	Reading this register returns 0x18.

SCSYSID3

SCSYSID3 is chip ID register 3.

Offset Address	Register Name	Total Reset Value
0xEEC	SCSYSID3	0x35

Bit	7	6	5	4	3	2	1	0
Name	sysid3							
Reset	0	0	1	1	0	1	0	1

Bits	Access	Name	Description
[7:0]	RO	sysid3	Reading this register returns 0x35.

3.5.5 Peripheral Control Registers

3.5.5.1 Register Summary

Table 3-11 describes peripheral control registers.

Table 3-11 Summary of peripheral control registers (base address: 0x2012_0000)

Offset Address	Register	Description	Page
0x0004	MISC_CTRL1	MISC register 1	3-61

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Offset Address	Register	Description	Page
0x0008	MISC_CTRL2	MISC register 2	3-63
0x000C	MISC_CTRL3	MISC register 3	3-63
0x0010	MISC_CTRL4	MISC register 4	3-64
0x0018	MISC_CTRL6	MISC register 6	3-65
0x001C	MISC_CTRL7	MISC register 7	3-66
0x0020	MISC_CTRL8	MISC register 8	3-66
0x0024	MISC_CTRL9	MISC register 9	3-67
0x002C	MISC_CTRL11	MISC register 11	3-68
0x0038	MISC_CTRL14	MISC register 14	3-68
0x003C	MISC_CTRL15	MISC register 15	3-69
0x0040	MISC_CTRL16	MISC register 16	3-71
0x0048	MISC_CTRL18	MISC register 18	3-72
0x0058	MISC_CTRL22	MISC register 22	3-72
0x005C	MISC_CTRL23	MISC register 23	3-73
0x0060	MISC_CTRL24	MISC register 24	3-74
0x0064	MISC_CTRL25	MISC register 25	3-74
0x0068	MISC_CTRL26	MISC register 26	3-75
0x006C	MISC_CTRL27	MISC register 27	3-75
0x0078	MISC_CTRL30	MISC register 30	3-76
0x00C4	MISC_CTRL49	Audio CODEC (ACODEC) control register 6	3-78
0x00C8	MISC_CTRL50	ACODEC control register 0	3-79
0x00CC	MISC_CTRL51	ACODEC control register 1	3-81
0x00D0	MISC_CTRL52	ACODEC control register 2	3-82
0x00D4	MISC_CTRL53	ACODEC control register 3	3-84
0x00D8	MISC_CTRL54	ACODEC control register 4	3-85
0x00E0	MISC_CTRL56	MISC register 56	3-86
0x00F0	MISC_CTRL 60	MISC register 60	3-87
0x00F4	MISC_CTRL 61	MISC register 61	3-88
0x0140	MISC_CTRL 80	MISC register 80	3-89
0x0144	MISC_CTRL 81	MISC register 81	3-89

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3.5.5.2 Register Description

MISC_CTRL1

MISC_CTRL1 is MISC register 1. This register cannot be soft reset.

Offset Address	Register Name	Total Reset Value
0x0004	MISC_CTRL1	0x0010_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	vicap_vpss_online	reserved	reserved	ssp1_cs_sel		reserved				mipi_phy_mode		reserved							ive_gzip_mst_sel	reserved	spi1_cs1_ctrl	spi1_cs0_ctrl	spi0_cs0_ctrl	uart1_rts_ctrl	bootrom_pgen	reserved	bootram1_ck_gt_en	bootram0_ck_gt_en	tde_ddrt_mst_sel	sdio1_card_det_mode	sdio0_card_det_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30]	RW	vicap_vpss_online	VICAP and VPSS mode 0: offline mode 1: online mode
[29]	RO	reserved	Reserved
[28]	RO	reserved	Reserved
[27:26]	RW	ssp1_cs_sel	SSP1 CS select 00: CS0 01: CS1 1x: reserved
[25:22]	RO	reserved	Reserved
[21:20]	RW	mipi_phy_mode	MIPI PHY mode 00: MIPI mode 01: LVDS mode 10: CMOS mode 11: bypass mode
[19:13]	RO	reserved	Reserved

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[12]	RW	ive_gzip_mst_sel	IVE/GZIP access select through media1 AXI 0: GZIP 1: IVE
[11]	RO	reserved	Reserved
[10]	RW	spi1_cs1_ctrl	Reverse control for the spi1_cs1 signal 0: not inverted 1: inverted
[9]	RW	spi1_cs0_ctrl	Reverse control for the spi1_cs0 signal 0: not inverted 1: inverted
[8]	RW	spi0_cs0_ctrl	Reverse control for the spi0_cs0 signal 0: not inverted 1: inverted
[7]	RW	uart1_rts_ctrl	Reverse control for the uart1_rts signal 0: not inverted 1: inverted
[6]	RW	bootrom_pgen	BOOTROM power supply control 0: powered on 1: powered off
[5]	RO	reserved	Reserved
[4]	RW	bootram1_ck_gt_en	BOOTRAM1 clock enable 0: enabled 1: disabled
[3]	RW	bootram0_ck_gt_en	BOOTRAM0 clock enable 0: enabled 1: disabled
[2]	RW	tde_ddrt_mst_sel	MDU/DDRT access select through media1 AXI 0: TDE 1: DDRT
[1]	RW	sdio1_card_det_mode	Reverse control for the SDIO1 card detection signal 0: not inverted 1: inverted
[0]	RW	sdio0_card_det_mode	Reverse control for the SDIO0 card detection signal 0: not inverted 1: inverted

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MISC_CTRL2

MISC_CTRL2 is MISC register 2. This register cannot be soft reset.

Offset Address	Register Name	Total Reset Value
0x0008	MISC_CTRL2	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														arm9_mem_dly	
Reset	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	0	0	1	0	1	0	0	0	1	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	arm9_mem_dly	ARM9 mem speed adjustment control

MISC_CTRL3

MISC_CTRL3 is MISC register 3. This register cannot be soft reset.

Offset Address	Register Name	Total Reset Value
0x000C	MISC_CTRL3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														powerswitch_sel_mux	powerswitch_sel
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	powerswitch_sel_mux	Power switch source select 0: SDIO0 controller 1: powerswitch_sel register

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[0]	RW	powerswitch_sel	Power switch select (valid only when powerswitch_sel_mux is 1) 0: 3.3 V 1: 1.8 V

MISC_CTRL4

MISC_CTRL4 is MISC register 4. This register cannot be soft reset.

Offset Address	Register Name	Total Reset Value
0x0010	MISC_CTRL4	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	avc_ret1n_bypass	reserved																							sdio1_ret1n	sdio0_ret1n	reserved	usb_ret1n	reserved	cipher_ret1n	bootram_ret1n	
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

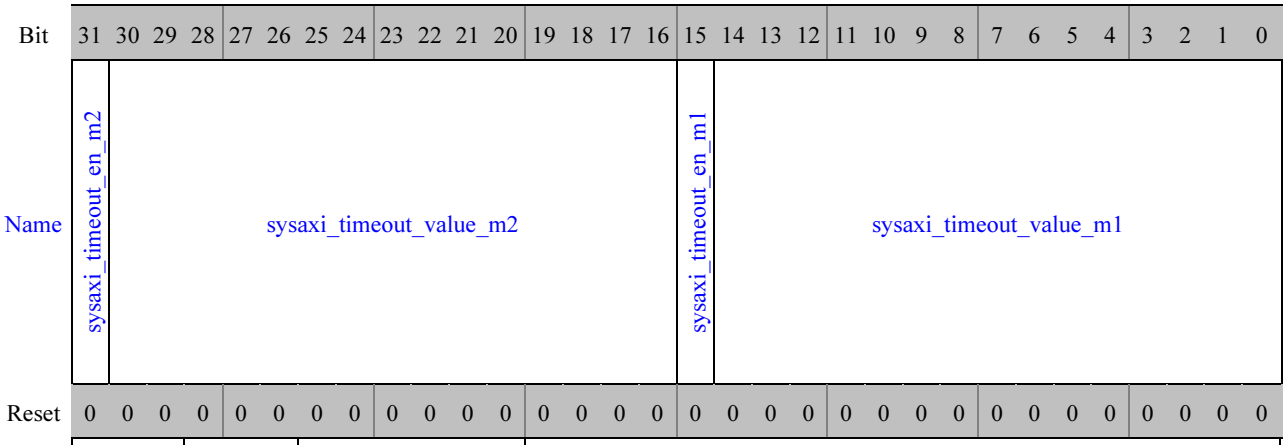
Bits	Access	Name	Description
[31]	RW	avc_ret1n_bypass	AVC automatic mem ret1n function bypass 0: enabled 1: bypassed
[30:7]	RO	reserved	Reserved
[6]	RW	sdio1_ret1n	mem ret1n signal in the SDIO1 CTRL. Set this signal to 1 when the SDIO1 CTRL is enabled or set this signal to 0 when the SDIO1 CTRL is disabled to reduce mem power consumption.
[5]	RW	sdio0_ret1n	mem ret1n signal in the SDIO0 CTRL. Set this signal to 1 when the SDIO0 CTRL is enabled or set this signal to 0 when the SDIO0 CTRL is disabled to reduce mem power consumption.
[4]	RO	reserved	Reserved
[3]	RW	usb_ret1n	mem ret1n signal in the USB CTRL. Set this signal to 1 when the USB CTRL is enabled or set this signal to 0 when the USB CTRL is disabled to reduce mem power consumption.
[2]	RO	reserved	Reserved
[1]	RW	cipher_ret1n	mem ret1n signal in the cipher. Set this signal to 1 when the cipher is enabled or set this signal to 0 when the cipher is disabled to reduce mem power consumption.
[0]	RW	bootram_ret1n	mem ret1n signal in the BOOTROM. Set this signal to 1 when the BOOTROM is enabled or set this signal to 0 when the BOOTROM is disabled to reduce mem power consumption.

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MISC_CTRL6

MISC_CTRL6 is MISC register 6.

Offset Address	Register Name	Total Reset Value
0x0018	MISC_CTRL6	0x0000_0000



Bits	Access	Name	Description
[31]	RW	sysaxi_timeout_en_m2	Timeout count enable for the SYS AXI bus M2 port (AHB0) 0: disabled 1: enabled
[30:16]	RW	sysaxi_timeout_value_m2	Timeout count value of the SYS AXI bus M2 port (AHB0) Count value = sysaxi_timeout_value_m2 x 2
[15]	RW	sysaxi_timeout_en_m1	Timeout count enable for the SYS AXI bus M1 port (CPU) 0: disabled 1: enabled
[14:0]	RW	sysaxi_timeout_value_m1	Timeout count value of the SYS AXI bus M1 port (CPU) Count value = sysaxi_timeout_value_m1 x 2

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MISC_CTRL7

MISC_CTRL7 is MISC register 7.

Offset Address	Register Name	Total Reset Value
0x001C	MISC_CTRL7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sysaxi_timeout_en_m3	sysaxi_timeout_value_m3															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RW	sysaxi_timeout_en_m3	Timeout count enable for the SYS AXI bus M3 port (AHB1) 0: disabled 1: enabled
[14:0]	RW	sysaxi_timeout_value_m3	Timeout count value of the SYS AXI bus M3 port (AHB1) Count value = sysaxi_timeout_value_m3 x 2

MISC_CTRL8

MISC_CTRL8 is MISC register 8.

Offset Address	Register Name	Total Reset Value
0x0020	MISC_CTRL8	0x0000_0012

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					sysaxi_port3_pri	reserved		sysaxi_port2_pri				reserved		sysaxi_port1_pri	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved

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[10:8]	RW	sysaxi_port3_pri	Priority of the SYS AXI bus M3 port (AHB1) The value 3 indicates the highest priority.
[7]	RO	reserved	Reserved
[6:4]	RW	sysaxi_port2_pri	Priority of the SYS AXI bus M2 port (AHB0) The value 3 indicates the highest priority.
[3]	RO	reserved	Reserved
[2:0]	RW	sysaxi_port1_pri	Priority of the SYS AXI bus M1 port (CPU) The value 3 indicates the highest priority.

MISC_CTRL9

MISC_CTRL9 is MISC register 9.

Offset Address	Register Name	Total Reset Value
0x0024	MISC_CTRL9	0x0000_0123

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			sysaxi_slave_priority_s4				reserved				sysaxi_slave_priority_s3				reserved				sysaxi_slave_priority_s2				reserved				sysaxi_slave_priority_s1			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14:12]	RW	sysaxi_slave_priority_s4	Bus access priority of the SYS AXI s4 (mddrc) port The value 3 indicates the highest priority.
[11]	RO	reserved	Reserved
[10:8]	RW	sysaxi_slave_priority_s3	Bus access priority of the SYS AXI s3 (mediaapb) port The value 3 indicates the highest priority.
[7]	RO	reserved	Reserved
[6:4]	RW	sysaxi_slave_priority_s2	Bus access priority of the SYS AXI s2 (sysapb) port The value 3 indicates the highest priority.
[3]	RO	reserved	Reserved

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[2:0]	RW	sysaxi_slave_priority_s1	Bus access priority of the SYS AXI s1 (ahb_cfg) port The value 3 indicates the highest priority.

MISC_CTRL11

MISC_CTRL11 is MISC register 11.

Offset Address	Register Name	Total Reset Value
0x002C	MISC_CTRL11	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	media0axi_timeout_en_m2	media0axi_timeout_value_m2															media0axi_timeout_en_m1	media0axi_timeout_value_m1														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	media0axi_timeout_en_m2	Timeout count enable for the MDA0 AXI bus M2 port (VGS) 0: disabled 1: enabled
[30:16]	RW	media0axi_timeout_value_m2	Timeout count value of the MDA0 AXI bus M2 port (VGS) Count value = media0axi_timeout_value_m2 x 2
[15]	RW	media0axi_timeout_en_m1	Timeout count enable for the MDA0 AXI bus M1 port (AVC) 0: disabled 1: enabled
[14:0]	RW	media0axi_timeout_value_m1	Timeout count value of the MDA0 AXI bus M1 port (AVC) Count value = media0axi_timeout_value_m1 x 2

MISC_CTRL14

MISC_CTRL14 is MISC register 14.

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Offset Address	Register Name	Total Reset Value
0x0038	MISC_CTRL14	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									media0axi_port2_pri			reserved	media0axi_port1_pri		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6:4]	RW	media0axi_port2_pri	Priority of the media 0 AXI bus M2 port (VGS) The value 1 indicates the highest priority.
[3]	RO	reserved	Reserved
[2:0]	RW	media0axi_port1_pri	Priority of the media 0 AXI bus M1 port (AVC) The value 1 indicates the highest priority.

MISC_CTRL15

MISC_CTRL15 is MISC register 15.

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Offset Address	Register Name	Total Reset Value
0x003C	MISC_CTRL15	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	media1axi_timeout_en_m2	media1axi_timeout_value_m2															media1axi_timeout_en_m1	media1axi_timeout_value_m1														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	media1axi_timeout_en_m2	Timeout count enable for the media 1 AXI bus M2 port (JPGE) 0: disabled 1: enabled
[30:16]	RW	media1axi_timeout_value_m2	Timeout count value of the media 1 AXI bus M2 port (JPGE) Count value = media1axi_timeout_value_m2 x 2
[15]	RW	media1axi_timeout_en_m1	Timeout count enable for the media 1 AXI bus M1 port (TDE_DDRT) 0: disabled 1: enabled
[14:0]	RW	media1axi_timeout_value_m1	Timeout count value of the media 1 AXI bus M1 port (TDE_DDRT) Count value = media1axi_timeout_value_m1 x 2

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MISC_CTRL16

MISC_CTRL16 is MISC register 16.

Offset Address	Register Name	Total Reset Value
0x0040	MISC_CTRL16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	media1axi_timeout_en_m4				media1axi_timeout_value_m4												media1axi_timeout_en_m3				media1axi_timeout_value_m3											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	media1axi_timeout_en_m4	Timeout count enable for the media 1 AXI bus M4 port (IVE_GZIP) 0: disabled 1: enabled
[30:16]	RW	media1axi_timeout_value_m4	Timeout count value of the media 1 AXI bus M4 port (IVE_GZIP) Count value = media1axi_timeout_value_m4 x 2
[15]	RW	media1axi_timeout_en_m3	Timeout count enable for the media 1 AXI bus M3 port (AIAO) 0: disabled 1: enabled
[14:0]	RW	media1axi_timeout_value_m3	Timeout count value of the media 1 AXI bus M3 port (AIAO) Count value = media1axi_timeout_value_m3 x 2

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MISC_CTRL18

MISC_CTRL18 is MISC register 18.

Offset Address	Register Name	Total Reset Value
0x0048	MISC_CTRL 18	0x0000_0123

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																	media1axi_port4_pri			reserved			media1axi_port3_pri			reserved			media1axi_port2_pri			reserved			media1axi_port1_pri		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14:12]	RW	media1axi_port4_pri	Priority of the media 1 AXI bus M4 port (IVE_GZIP) The value 3 indicates the highest priority.
[11]	RO	reserved	Reserved
[10:8]	RW	media1axi_port3_pri	Priority of the media 1 AXI bus M3 port (AIAO) The value 3 indicates the highest priority.
[7]	RO	reserved	Reserved
[6:4]	RW	media1axi_port2_pri	Priority of the media 1 AXI bus M2 port (JPGE) The value 3 indicates the highest priority.
[3]	RO	reserved	Reserved
[2:0]	RW	media1axi_port1_pri	Priority of the media 1 AXI bus M1 port (TD_DDRTE) The value 3 indicates the highest priority.

MISC_CTRL22

MISC_CTRL22 is MISC register 22.

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Offset Address	Register Name	Total Reset Value
0x0058	MISC_CTRL22	0x7654_3210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				hash_qosmap				ive_gzip_qosmap				aio_qosmap				jpge_qosmap				tde_ddrt_qosmap				vicap_qosmap				vdp_qosmap			
Reset	0	1	1	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:24]	RW	hash_qosmap	QoS value of the hash in the MDDRC
[23:20]	RW	ive_gzip_qosmap	QoS value of the ive_gzip in the MDDRC
[19:16]	RW	aio_qosmap	QoS value of the AIAO in the MDDRC
[15:12]	RW	jpge_qosmap	QoS value of the JPGE in the MDDRC
[11:8]	RW	tde_ddrt_qosmap	QoS value of the tde_ddrt in the MDDRC
[7:4]	RW	vicap_qosmap	QoS value of the VICAP in the MDDRC
[3:0]	RW	vdp_qosmap	QoS value of the VDP in the MDDRC

MISC_CTRL23

MISC_CTRL23 is MISC register 23.

Offset Address	Register Name	Total Reset Value
0x005C	MISC_CTRL23	0x7654_3210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								fmc_qosmap				sdio1_qosmap				sdio0_qosmap				cpu_qosmap				vpss_qosmap				vgs_qosmap			
Reset	0	1	1	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:20]	RW	fmc_qosmap	QoS value of the FMC in the MDDRC
[19:16]	RW	sdio1_qosmap	QoS value of SDIO1 in the MDDRC
[15:12]	RW	sdio0_qosmap	QoS value of SDIO0 in the MDDRC
[11:8]	RW	cpu_qosmap	QoS value of the CPU in the MDDRC
[7:4]	RW	vpss_qosmap	QoS value of the VPSS in the MDDRC
[3:0]	RW	vgs_qosmap	QoS value of the VGS in the MDDRC

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MISC_CTRL24

MISC_CTRL24 is MISC register 24.

Offset Address	Register Name	Total Reset Value
0x0060	MISC_CTRL24	0x7654_3210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								avc_qosmap				usb_qosmap				cipher_qosmap				dma2_qosmap				dma1_qosmap				eth_qosmap			
Reset	0	1	1	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:20]	RW	avc_qosmap	QoS value of the AVC in the MDDRC
[19:16]	RW	usb_qosmap	QoS value of the USB in the MDDRC
[15:12]	RW	cipher_qosmap	QoS value of the cipher in the MDDRC
[11:8]	RW	dma2_qosmap	QoS value of DMA2 in the MDDRC
[7:4]	RW	dma1_qosmap	QoS value of DMA1 in the MDDRC
[3:0]	RW	eth_qosmap	QoS value of the ETH in the MDDRC

MISC_CTRL25

MISC_CTRL25 is MISC register 25.

Offset Address	Register Name	Total Reset Value
0x0064	MISC_CTRL25	0x3210_3210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				hash_to				ive_gzip_to				aio_to				jpge_to				tde_ddrt_to				vicap_to				vdp_to			
Reset	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:24]	RW	hash_to	Timeout select for the hash in the MDDRC
[23:20]	RW	ive_gzip_to	Timeout select for the ive_gzip in the MDDRC
[19:16]	RW	aio_to	Timeout select for the AIAO in the MDDRC
[15:12]	RW	jpge_to	Timeout select for the JPGE in the MDDRC
[11:8]	RW	tde_ddrt_to	Timeout select for the tde_ddrt in the MDDRC
[7:4]	RW	vicap_to	Timeout select for the VICAP in the MDDRC
[3:0]	RW	vdp_to	Timeout select for the VDP in the MDDRC

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MISC_CTRL26

MISC_CTRL26 is MISC register 26.

Offset Address	Register Name	Total Reset Value
0x0068	MISC_CTRL26	0x3210_3210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								fmc_to				sdio1_to				sdio0_to				cpu_to				vpss_to				vgs_to			
Reset	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:20]	RW	fmc_to	Timeout select for the FMC in the MDDRC
[19:16]	RW	sdio1_to	Timeout select for SDIO1 in the MDDRC
[15:12]	RW	sdio0_to	Timeout select for SDIO0 in the MDDRC
[11:8]	RW	cpu_to	Timeout select for the CPU in the MDDRC
[7:4]	RW	vpss_to	Timeout select for the VPSS in the MDDRC
[3:0]	RW	vgs_to	Timeout select for the VGS in the MDDRC

MISC_CTRL27

MISC_CTRL27 is MISC register 27.

Offset Address	Register Name	Total Reset Value
0x006C	MISC_CTRL27	0x3210_3210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								avc_to				usb_to				cipher_to				dma2_to				dma1_to				eth_to			
Reset	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:20]	RW	avc_to	Timeout select for the AVC in the MDDRC
[19:16]	RW	usb_to	Timeout select for the USB in the MDDRC
[15:12]	RW	cipher_to	Timeout select for the cipher in the MDDRC
[11:8]	RW	dma2_to	Timeout select for DMA2 in the MDDRC
[7:4]	RW	dma1_to	Timeout select for DMA1 in the MDDRC

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[3:0]	RW	eth_to	Timeout select for the ETH in the MDDRC

MISC_CTRL30

MISC_CTRL30 is MISC register 30.

Offset Address	Register Name	Total Reset Value
0x0078	MISC_CTRL30	0x0C03_13A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	usb_chipid	reserved			usb2_phy_dmpulldown	usb2_phy_dppulldown	ss_scaledown_mode		reserved	ss_hubsetup_min_i	reserved		usbovr_p_ctrl	usbpwr_pctrl			reserved	port_ovr_en	reserved	port_pwr_en	reserved		ss_ena_incr16_i	ss_ena_incr8_i	ss_ena_incr4_i	ss_ena_incr_align_i	ss_autoppd_on_overcur_en_i	reserved		app_start_clk_i	ohci_susp_lgcy_i	ss_word_if_i
Reset	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	0	1	1	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	usb_chipid	Host and device switching control 0: host 1: device
[30:28]	RW	reserved	Reserved
[27]	RW	usb2_phy_dmpulldown	D− pull-down resistor enable 0: disabled 1: enabled
[26]	RW	usb2_phy_dppulldown	D+ pull-down resistor enable 0: disabled 1: enabled
[25:24]	RW	ss_scaledown_mode	Scale-down mode 00: Scale-down is disabled for all timings, and the actual timings are used. 01: Scale-down is enabled for all timings except the suspended and resumed timings in device mode. 10: Scale-down is enabled only for the suspended and resumed timings in device mode. 11: Scale-down timings of bit 0 and bit 1 are enabled.
[23]	RO	reserved	Reserved

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[22]	RW	ss_hubsetup_min_i	Number of idle cycles after the full-speed preamble packet 0: 5 idle full-speed cycles 1: 4 full-speed idle cycles
[21:18]	RO	reserved	Reserved
[17]	RW	usbovr_p_ctrl	Over-current protection polarity 0: active low 1: active high
[16]	RW	usbpwr_pctrl	USB PHY power shutdown polarity 0: active low 1: active high
[15]	RO	reserved	Reserved
[14]	RW	port_ovr_en	Over-current protection enable for the USB PHY 0: disabled 1: enabled
[13]	RO	reserved	Reserved
[12]	RW	port_pwr_en	USB PHY power shutdown control 0: The power is shut down. 1: The controller power output is enabled.
[11:10]	RO	reserved	Reserved
[9]	RW	ss_ena_incr16_i	AHB burst 16 enable 0: disabled 1 (default): enabled
[8]	RW	ss_ena_incr8_i	AHB burst 8 enable 0: disabled 1 (default): enabled
[7]	RW	ss_ena_incr4_i	AHB burst 4 enable 0: disabled 1 (default): enabled
[6]	RW	ss_ena_incr_align_i	Burst alignment enable 0 (default): disabled 1: enabled
[5]	RW	ss_autoppd_on_overcur_en_i	Automatic port power shutdown enable during over-current 0: disabled 1 (default): enabled
[4:3]	RO	reserved	Reserved

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[2]	RW	app_start_clk_i	Open host controller interface (OHCI) clock control 0 (default): The OHCI works normally. 1: The OHCI clock is enabled in suspend mode.
[1]	RW	ohci_susp_lgcy_i	Strap input signal when the OHCI is suspended
[0]	RW	ss_word_if_i	Data bit width select for the UTMI interface 0 (default): 8 bits 1: 16 bits

MISC_CTRL49

MISC_CTRL49 is ACODEC control register 6.

Offset Address	Register Name	Total Reset Value
0x00C4	MISC_CTRL49	0x0000_005B

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4	3 2 1 0
Name	reserved						pdb_ctcm_ibias	pd_ctcm	rx_chnll_sel	rx_chnlr_sel
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	1	0 1	1 0 1 1

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	pdb_ctcm_ibias	This bit is set to 1 in normal mode.
[6]	RW	pd_ctcm	This bit is set to 0 in normal mode.
[5:3]	RW	rx_chnll_sel	RXPGA_L input signal select 011: The AC_INL pin is selected as the single-ended input of the audio-left channel. 100: The AC_INL and AC_INR pins are selected as the positive end and negative end of the differential input for the audio-left channel respectively. Other values: reserved
[2:0]	RW	rx_chnlr_sel	RXPGA_R input signal select 011: The AC_INR pin is selected as the single-ended input of the audio-right channel. Other values: reserved

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MISC_CTRL50

MISC_CTRL50 is ACODEC control register 0.

Offset Address	Register Name	Total Reset Value
0x00C8	MISC_CTRL50	0xFCCC_8000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	pd_adcl	pd_adcr	pd_dacl	pd_dacr	pd_micbias	pd_vref	reserved	reserved	mute_dacl	mute_dacr	reserved	reserved	mute_adcl	mute_adcr	mclk_ana_sel	reserved	pd_ibias	gain_micl			gainboostl	reserved	reserved		gain_micr				gainboostr	reserved		
Reset	1	1	1	1	1	1	0	0	1	1	0	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	pd_adcl	ADCL power-off control 0: normal mode 1: powered off
[30]	RW	pd_adcr	ADCR power-off control. This bit needs to be set to 1 when the differential input is selected. 0: normal mode 1: powered off
[29]	RW	pd_dacl	DACL power-off control 0: normal mode 1: powered off
[28]	RW	pd_dacr	DACR power-off control 0: normal mode 1: powered off
[27]	RW	pd_micbias	Microphone bias circuit power-off control 0: normal mode 1: powered off
[26]	RW	pd_vref	Reference voltage power-off control 0: normal mode 1: powered off
[25]	RO	reserved	Reserved
[24]	RO	reserved	Reserved
[23]	RW	mute_dacl	DACL analog mute 0: normal mode 1: muted

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[22]	RW	mute_dacr	DACR analog mute 0: normal mode 1: muted
[21:20]	RO	reserved	Reserved
[19]	RW	mute_adcl	ADCL analog mute 0: normal mode 1: muted
[18]	RW	mute_adcr	ADCR analog mute 0: normal mode 1: muted
[17]	RW	mclk_ana_sel	This bit is set to 1 in normal mode.
[16]	RO	reserved	Reserved
[15]	RW	pd_ibias	Global IBIAS power-off control 0: normal mode 1: powered off
[14:10]	RW	gain_micl	AC_INL input gain control 00000: 0 dB 00001: 2 dB 00010: 4 dB 00011: 6 dB 00100: 8 dB 00101: 10 dB 00110: 12 dB 00111: 14 dB 01000: 16 dB 01001: 18 dB 01010: 20 dB 01011: 22 dB 01100: 24 dB 01101: 26 dB 01110: 28 dB 01111: 30 dB 10000: −1.5 dB
[9]	RW	gainboostl	AC_INL gain boost control 0: 0 dB 1: 20 dB
[8]	RO	reserved	Reserved
[7]	RO	reserved	Reserved

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[6:2]	RW	gain_micr	AC_INR input gain control 00000: 0 dB 00001: 2 dB 00010: 4 dB 00011: 6 dB 00100: 8 dB 00101: 10 dB 00110: 12 dB 00111: 14 dB 01000: 16 dB 01001: 18 dB 01010: 20 dB 01011: 22 dB 01100: 24 dB 01101: 26 dB 01110: 28 dB 01111: 30 dB 10000: −1.5 dB
[1]	RW	gainboostr	AC_INR gain boost control 0: 0 dB 1: 20 dB
[0]	RO	reserved	Reserved

MISC_CTRL51

MISC_CTRL51 is ACODEC control register 1.

Offset Address	Register Name	Total Reset Value
0x00CC	MISC_CTRL51	0x00F3_5A4A

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dacl_rst_n	dacr_rst_n	adcl_rst_n	adcr_rst_n	dacl_en	dacr_en	adcl_en	adcr_en	reserved				i2s1_fs_sel				reserved															rst
Reset	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1	0	1	0	1	1	0	1	0	0	1	0	0	1	0	1	0

Bits	Access	Name	Description
[31]	RW	dacl_rst_n	This bit is set to 1 in normal mode.
[30]	RW	dacr_rst_n	This bit is set to 1 in normal mode.
[29]	RW	adcl_rst_n	This bit is set to 1 in normal mode.

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[28]	RW	adcr_rst_n	This bit is set to 1 in normal mode.
[27]	RW	dacl_en	DACL enable 0: disabled 1: enabled
[26]	RW	dacr_en	DACR enable 0: disabled 1: enabled
[25]	RW	adcl_en	ADCL enable 0: disabled 1: enabled
[24]	RW	adcr_en	ADCR enable 0: disabled 1: enabled
[23:18]	RO	reserved	Reserved
[17:13]	RW	i2s1_fs_sel	Sampling rate of I^2S channel 11000: mclk/512/2, selected when the MCLK is 1024 FS clocks 11001: mclk/256/2, selected when the MCLK is 512 FS clocks 11010: mclk/128/2, selected when the MCLK is 256 FS clocks 11011: mclk/64/2, selected when the MCLK is 128 FS clocks 111xx: mclk/32/2, selected when the MCLK is 64 FS clocks
[12:1]	RO	reserved	Reserved
[0]	RW	rst	Analog circuit reset 0: normal mode 1: reset

MISC_CTRL52

MISC_CTRL52 is ACODEC control register 2.

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Offset Address	Register Name	Total Reset Value
0x00D0	MISC_CTRL52	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	smutel	smuter	sunmutel	sunmuter	dacvu	mutel_rate		muter_rate	reserved																							dacr_lrsel
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31]	RW	smutel	DACL soft mute control 0: disabled 1: enabled
[30]	RW	smuter	DACR soft mute control 0: disabled 1: enabled
[29]	RW	sunmutel	DACL soft mute control 0: disabled 1: enabled
[28]	RW	sunmuter	DACR soft mute control 0: disabled 1: enabled
[27]	RW	dacvu	DAC volume update control 0: not updated 1: updated
[26:25]	RW	mutel_rate	DACL soft mute rate control 00: fs/2 01: fs/8 10: fs/32 11: fs/64
[24:23]	RW	muter_rate	DACR soft mute rate control 00: fs/2 01: fs/8 10: fs/32 11: fs/64
[22:0]	RO	reserved	Reserved
[0]	RW	dacr_lrsel	This bit is set to 1 in normal mode.

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MISC_CTRL53

MISC_CTRL53 is ACODEC control register 3.

Offset Address	Register Name	Total Reset Value
0x00D4	MISC_CTRL53	0x0606_2424

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dacl_mute	dacl_vol							dacr_mute	dacr_vol							reserved															
Reset	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1	0	0	0	1	0	0	1	0	0	0	0	1	0	0	1	0	0

Bits	Access	Name	Description
[31]	RW	dacl_mute	DACL digital mute control 0: normal mode 1: muted
[30:24]	RW	dacl_vol	DACL digital volume control, You are advised not to use this field, but to use the digital volume adjustment function in the AIAO module to control the digital volume output. The volume is calculated as follows: (6 − dacl_vol) dB 0x00: 6 dB 0x01: 5 dB 0x02: 4 dB … 0x7E: −120 dB When dacl_vol is 0x7F, the DACL is muted.
[23]	RW	dacr_mute	DACR digital mute control 0: normal mode 1: muted
[22:16]	RW	dacr_vol	DACR digital volume control, You are advised not to use this field, but to use the digital volume adjustment function in the AIAO module to control the digital volume output. The volume is calculated as follows: (6 − dacr_vol) dB 0x00: 6 dB 0x01: 5 dB 0x02: 4 dB … 0x7E: −120 dB When dacr_vol is 0x7F, the DACR is muted.
[15:0]	RO	reserved	Reserved

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MISC_CTRL54

MISC_CTRL54 is ACODEC control register 4.

Offset Address	Register Name	Total Reset Value
0x00D8	MISC_CTRL54	0x1E1E_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	adcl_mute	adcl_vol							adcr_mute	adcr_vol							adcl_hpf_en	adcr_hpf_en	reserved													adcr_lrsel
Reset	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31]	RW	adcl_mute	ADCL digital mute control 0: unmuted 1: muted
[30:24]	RW	adcl_vol	ADCL volume control, You are advised not to use this field, but to use the digital volume adjustment function in the AIAO module to control the digital volume output. The volume is calculated as follows: (30 − adcl_vol) dB 00: 30 dB 01: 29 dB 02: 28 dB … 7E: −96 dB 7F: −97 dB
[23]	RW	adcr_mute	ADCR digital mute control 0: unmuted 1: muted
[22:16]	RW	adcr_vol	ADCR volume control, You are advised not to use this field, but to use the digital volume adjustment function in the AIAO module to control the digital volume output. The volume is calculated as follows: (30 − adcr_vol) dB 00: 30 dB 01: 29 dB 02: 28 dB … 7E: −96 dB 7F: −97 dB

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[15]	RW	adcl_hpf_en	ADCL high-pass filter enable 0: disabled 1: enabled
[14]	RW	adcr_hpf_en	ADCR high-pass filter enable 0: disabled 1: enabled
[13:1]	RO	reserved	Reserved
[0]	RW	adcr_lrsel	This bit is set to 1 in normal mode.

MISC_CTRL56

MISC_CTRL56 is MISC register 56.

Offset Address	Register Name	Total Reset Value
0x00E0	MISC_CTRL56	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												i2s_pad_mclk_sel	reserved	i2s_pad_enable	audio_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RW	i2s_pad_mclk_sel	mclk selection control for the chip I^2S interface. 0: aiao_i2s_rx_mclk 1: aiao_i2s_tx_mclk
[2]	RO	reserved	Reserved
[1]	RW	i2s_pad_enable	Enable for the channel between the I^2S interface of the chip and AIAO 0: The channel between the I^2S interface of the chip and AIAO is disabled, and the I^2S interface of the chip has no output. 1: The channel between the I^2S interface of the chip and AIAO is enabled, and the I^2S interface of the chip connects to the I^2S interface of the AIAO. Note: i2s_pad_enable and audio_enable cannot be 1 at the same time.

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[0]	RW	audio_enable	Enable for the channel between the I^2S interface of the internal audio CODEC and AIAO 0: The channel between the I^2S interface of the internal audio CODEC and AIAO is disabled, and the I^2S interface of the internal audio CODEC has no output. 1: The channel between the I^2S interface of the internal audio CODEC and AIAO is enabled, and the I^2S interface of internal audio CODEC connects to the I^2S interface of the AIAO. Note: i2s_pad_enable and audio_enable cannot be 1 at the same time.

MISC_CTRL60

MISC_CTRL60 is MISC register 60.

Offset Address	Register Name	Total Reset Value
0x00F0	MISC_CTRL 60	0x0002_2202

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										core_mem_ras_emaw		reserved	core_mem_ras_emab			reserved	core_mem_rft_emab			reserved	core_mem_rft_emaa			reserved	core_mem_rfs_emaw			reserved	core_mem_rfs_ema		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:20]	RW	core_mem_ras_emaw	emaw speed adjustment control signal of ras_mem at the core layer
[19]	RO	reserved	Reserved
[18:16]	RW	core_mem_ras_emab	ema speed adjustment control signal of ras_mem at the core layer
[15]	RO	reserved	Reserved
[14:12]	RW	core_mem_rft_emab	emab speed adjustment control signal of rft_mem at the core layer
[11]	RO	reserved	Reserved
[10:8]	RW	core_mem_rft_emaa	emaa speed adjustment control signal of rft_mem at the core layer
[7:6]	RO	reserved	Reserved

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[5:4]	RW	core_mem_rfs_emaw	emaw speed adjustment control signal of rfs_mem at the core layer
[3]	RO	reserved	Reserved
[2:0]	RW	core_mem_rfs_ema	ema speed adjustment control signal of rfs_mem at the core layer

MISC_CTRL61

MISC_CTRL61 is MISC register 61.

Offset Address	Register Name	Total Reset Value
0x00F4	MISC_CTRL 61	0x0002_0202

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												core_mem_ros_ema				reserved				core_mem_rad_emawb				reserved				core_mem_rad_emab			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0

Bit	7	6	5	4	3	2	1	0
Name	reserved				core_mem_rad_emawa			
Reset	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:19]	RO	reserved	Reserved
[18:16]	RW	core_mem_ros_ema	ema speed adjustment control signal of ros_mem at the core layer
[15:14]	RO	reserved	Reserved
[13:12]	RW	core_mem_rad_emawb	emawb speed adjustment control signal of rad_mem at the core layer
[11]	RO	reserved	Reserved
[10:8]	RW	core_mem_rad_emab	emab speed adjustment control signal of rad_mem at the core layer
[7:6]	RO	reserved	Reserved
[5:4]	RW	core_mem_rad_emawa	emawa speed adjustment control signal of rad_mem at the core layer
[3]	RO	reserved	Reserved
[2:0]	RW	core_mem_rad_emaa	emaa speed adjustment control signal of rad_mem at the core layer

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MISC_CTRL80

MISC_CTRL80 is MISC register 80.

Offset Address	Register Name	Total Reset Value
0x0140	MISC_CTRL 80	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														emmc_dll_dsel		emmc_dll_ssel								emmc_dll_mode	emmc_dll_stop	emmc_dll_bypass	emmc_dll_slave_en	emmc_dll_tune			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17:16]	RW	emmc_dll_dsel	DLL dsel control signal of the eMMC interface
[15:8]	RW	emmc_dll_ssel	DLL ssel control signal of the eMMC interface
[7]	RW	emmc_dll_mode	DLL mode control signal of the eMMC interface
[6]	RW	emmc_dll_stop	DLL stop control signal of the eMMC interface
[5]	RW	emmc_dll_bypass	DLL bypass control signal of the eMMC interface
[4]	RW	emmc_dll_slave_en	DLL slave_en control signal of the eMMC interface
[3:0]	RW	emmc_dll_tune	DLL tune control signal of the eMMC interface

MISC_CTRL81

MISC_CTRL81 is MISC register 81.

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Offset Address	Register Name	Total Reset Value
0x0144	MISC_CTRL 81	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					emmc_dll_lock	emmc_dll_ready	emmc_dll_overflow	emmc_dll_mdly_tap							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RO	emmc_dll_lock	emmc_dll_lock
[9]	RO	emmc_dll_ready	emmc_dll_ready
[8]	RO	emmc_dll_overflow	emmc_dll_overflow
[7:0]	RO	emmc_dll_mdly_tap	emmc_dll_mdly_tap

3.6 DMA Controller

3.6.1 Overview

The direction memory access (DMA) operation is a high-speed data transfer operation. It supports data read/write between peripherals and memories without using the CPU. The direction memory access controller (DMAC) directly transfers data between a memory and a peripheral, between peripherals, or between memories. This avoids the CPU intervention and reduces the interrupt handling overhead of the CPU.

3.6.2 Features

The DMAC has the following features:

- Transfers data in 8-bit, 16-bit, or 32-bit mode.
- Provides four DMA channels. Each channel can be configured to support unidirectional transfer.
- Provides two 32-bit master bus interfaces for data transfer.
- Supports the DMA requests controlled through software.
- Programs the DMA burst size.
- Configures the source address and the destination address as automatic incremental or decremented addresses during DMA transfer.

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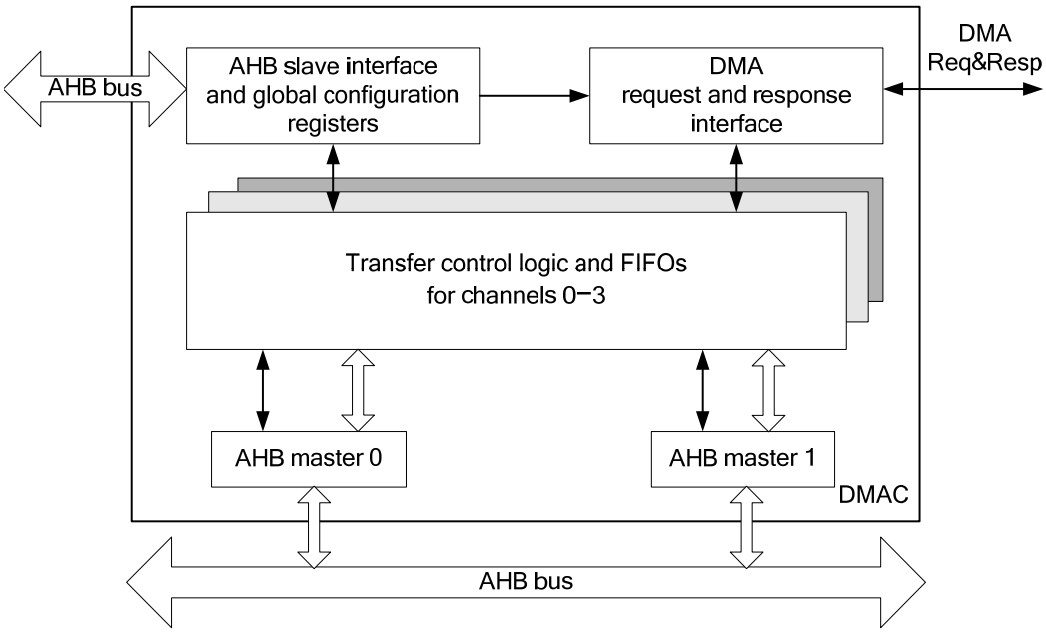
- Supports DMA transfer with the linked list.
- Supports the DMAC flow control.

3.6.3 Function Description

Functional Block Diagram

Figure 3-5 shows the functional block diagram of the DMAC.

Figure 3-5 Functional block diagram of the DMAC



NOTE

- The priorities of DMAC channels are fixed. DMA channel 0 has the highest priority; whereas channel 3 has the lowest priority. When the DMA requests from two peripherals are valid simultaneously, the channel with the higher priority starts data transfer first.
- DMA channel 0 and DMA channel 1 have one 4 x 32-bit FIFO each, and DMA channel 2 and DMA channel 3 have one 16 x 32-bit FIFO each.

Each DMA channel has a group of transfer control logic and one FIFO. The transfer control logic automatically performs the following operations:

Step 1 Read data from the source address specified by the software.

Step 2 Buffer data in the FIFO.

Step 3 Fetch data from the FIFO.

Step 4 Write the data to the destination address specified by the software.

----End

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Workflow

The workflow of the DMAC is as follows:

Step 1 The software selects one DMA channel for DMA transfer, configures the following items, and enables the channel:

- Source address
- Destination address
- Header pointer of the linked list
- Amount of the transferred data
- Source and destination peripheral request signal numbers
- Masters used at the source and destination ends of the channel.

After the channel is enabled, the DMAC starts to check the activities of the DMA request lines of the source peripheral and destination device connected to this channel.

Step 2 The source device transmits a DMA request to the DMAC. If the source device is a memory, the DMAC considers that the DMA request is always valid by default.

Step 3 The DMAC channel responds to the DMA request of the source device. Then the DAMC reads data from the source device and stores it in the internal FIFO of the channel.

Step 4 The destination device transmits a DMA request to the DMAC. If the destination device is a memory, the DMAC considers that the DMA request is always valid by default.

Step 5 The DMA channel responds to the DMA request of the destination device. Then the DMAC fetches data from the internal FIFO of the channel and writes it to the destination device.

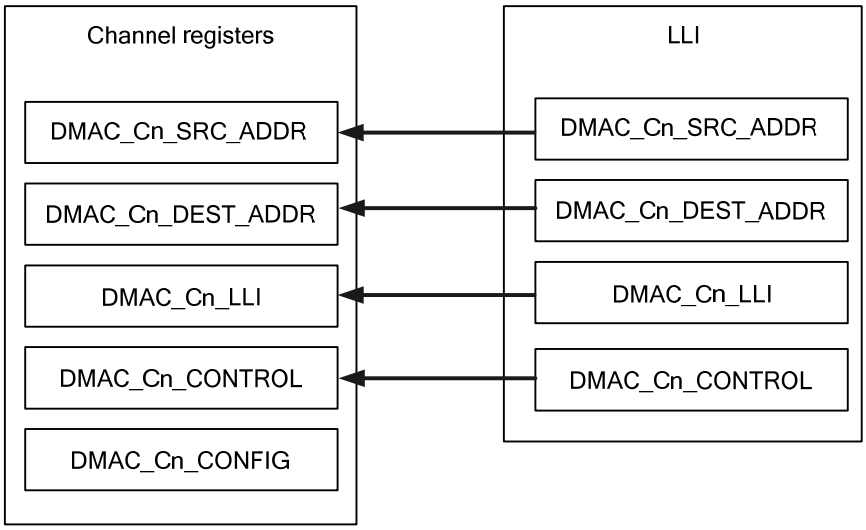
Step 6 The steps 2 and step 3 as well as step 4 and 5 may be performed concurrently, because the source and destination devices may transmit DMA requests to the DMAC at the same time. When the FIFO overrun or underrun occurs on the DMA channel, the DMAC blocks the DMA requests of the source device or destination device until the FIFO is full or empty. When the DMAC interacts with the source device and destination device for several times, step 2 to step 5 are performed repeatedly until the specified data is completely transferred and a maskable transfer terminal interrupt is sent. If the value of DMAC_Cn_LLI is not 0, read linked list item (LLI) nodes by considering the register value as an address, load the read values to DMAC_Cn_SRC_ADDR, DMAC_Cn_DEST_ADDR, DMAC_Cn_LLI, and DMAC_Cn_CONTROL in sequence (see Figure 3-5), and then go to step 2. If the value of DMAC_Cn_LLI is 0, the current DMA transfer is stopped. In this case, the channel is disabled automatically and the transfer ends.

----End

Figure 3-6 illustrates how to update channel registers through the LLI.

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Figure 3-6 Updating channel registers through the LLI



Connection Between the DMA and Peripherals

The peripherals initiate data transfer by transmitting DMA request signals to the DMAC.

The DMAC provides the following two DMA request signals for each peripheral:

- DMACBREQ

  Burst transfer request signal. It triggers a burst transfer and the burst size is preconfigured.
- DMACSREQ

  Single transfer request signal. It triggers a single transfer. That is, the DMAC reads a data segment from a peripheral or writes a data segment to a peripheral.

The DMAC provides a request clear signal DMACLR.

This signal is sent to each peripheral by the DMAC as a response to the DMA request signal of each peripheral.

DMAC Request Signals

Table 3-12 describes the mapping between DMAC hardware request signals and devices.

Table 3-12 DMAC hardware request signals and corresponding peripheral requests

DMAC Hardware Request Signal No.	Device Request
0	DMA request of the I^2C0 RX channel
1	DMA request of the I^2C0 TX channel
2	DMA request of the I^2C1 RX channel
3	DMA request of the I^2C1 TX channel

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DMAC Hardware Request Signal No.	Device Request
4	DMA request of the UART0 RX channel
5	DMA request of the UART0 TX channel
6	DMA request of the UART1 RX channel
7	DMA request of the UART1 TX channel
8	DMA request of the UART2 RX channel
9	DMA request of the UART2 TX channel
10	DMA request of the I^2C2 RX channel
11	DMA request of the I^2C2 TX channel
12	DMA request of the SSP1 RX channel
13	DMA request of the SSP1 TX channel
14	DMA request of the SSP0 RX channel
15	DMA request of the SSP0 TX channel

The source and destination requests of each DMA channel are configured by software. For example, DMA request 4 is the request of the UART0 RX channel. To transmit the UART0 RX data by using channel 3, you must configure DMA request 4 as the source request of channel 3.

As memories do not provide DMA request signals, when a memory is used for DMA transfer, the DMAC considers that the DMA request of the memory is always valid by default. In addition, an idle cycle is inserted after each bus operation during the DMAC transfer on channel 2 or channel 3. In this way, the master with a higher priority channel can transfer data on the bus first. Therefore, to prevent other channels from waiting for the bus for a long time, you are advised to transmit data from memory to memory using channel 2 or channel 3.

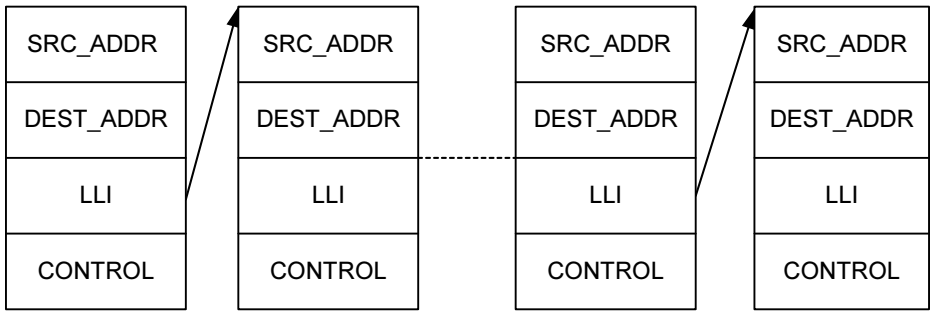
DMA Linked List

The data structure of the DMAC LLI node is as follows:

- Channel register DMAC_Cn_SRC_ADDR, for setting the start address for the source device
- Channel register DMAC_Cn_DEST_ADDR, for setting the start address for the destination device
- Channel register DMAC_Cn_LLI, for setting the address for the next node
- Channel register DMAC_Cn_CONTROL, for setting the master, data width, burst size, address increment, and transfer size of the source device and destination device

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Figure 3-7 Structure of DMAC LLIs



CAUTION

The LLI field value must be less than or equal to 0xFFFF_FFF0. Otherwise, the address is wrapped around to 0x0000_0000 during a 4-word burst transfer. As a result, the data structure of LLI nodes cannot be stored in a consecutive address area.

If the LLI field is set to 0, the current node is at the end of the linked list. In this case, the corresponding channel is disabled after data blocks corresponding to the current node are transferred.

3.6.4 Operating Mode

Clock Gating

In the following cases, the DMAC and DMAC clock can be disabled using the software to reduce power consumption:

- All DMA channels are idle and there is no DMA transfer request.
- DMAC_Cn_CONFIG[ch_en] is set to 0 and all the DMA channels are disabled.

To disable the DMAC clock, perform the following steps:

Step 1 Write 0 to DMAC_Cn_CONFIG[ch_en] to disable DMAC channels.

Step 2 Write 0 to DMAC_CONFIG [dmac_enable] to disable the DMAC.

Step 3 Write 0 to PERI_CRG54 [dma_cken] to disable DMAC bus clock gating. Then the DMAC clock is disabled.

Step 4 Enable the clock and the DMAC again when the DMAC is required for data transfer.

----End

Initialization

To initialize the DMAC, perform the following steps:

Step 1 Write to DMAC_CONFIG to set the endianness of DMAC master 1 and DMAC master 2, and write 1 to DMAC_CONFIG[dmac_enable] to enable the DMAC.

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Step 2 Write 1 to all the bits of DMAC_INT_ERR_CLR and DMAC_INT_TC_CLR to clear all interrupts.

Step 3 Write 0 to the corresponding bits of DMAC_SYNC to set the DMA request signal groups to be synchronized.

Step 4 Configure and disable channels in sequence. You can disable all the channels by writing 0 to DMAC_Cn_CONFIG[ch_en] of each channel.

----End

Enabling a Channel

After the DMAC is initialized, the DMAC can transmit data only when a DMAC channel is configured and enabled. To enable a DMA channel, perform the following steps:

Step 1 Read DMAC_ENABLED_CHNS to search for idle channels and select one.

Step 2 Write 1 to the corresponding bits of DMAC_INT_ERR_CLR and DMAC_INT_TC_CLR to clear the interrupt status of the selected channel.

Step 3 Write to DMAC_Cn_SRC_ADDR to set the access start address for the source device.

Step 4 Write to DMAC_Cn_DEST_ADDR to set the access start address for the destination device.

Step 5 Write to DMAC_Cn_LLI to set the linked list information. If the channel is used for single-block data transfer, set DMAC_Cn_LLI to 0. If the channel is used for linked list data transfer, set DMAC_Cn_LLI to the linked list header pointer.

Step 6 Write to DMAC_Cn_CONTROL to set the master, data width, burst size, address increment, and transfer size of the source device and destination device.

Step 7 Write to DMAC_Cn_CONFIG to set the DMA request signal, flow control mode, and interrupt mask of this channel.

Step 8 Write 1 to DMAC_Cn_CONFIG[ch_en] to enable this channel.

----End

Usage of DMAC_Cn_CONTROL

The DMAC_Cn_CONTROL register contains the control information about the DMA channels, such as the transfer size, burst size, and transfer bit width.

Before a channel is enabled, its corresponding register must be programmed by using software. After the channel is enabled, the register value is updated when being loaded from an LLI node after a complete data block is transferred.

If a channel is active, no valid information is obtained when this register is read. This is because that after software obtains the register value, the value changes during data transfer. As a result, the register can be read after the channel stops data transfer.

Table 3-13 lists the mapping between the value of dbsize or sbsize of DMAC_Cn_CONTROL and the burst length.

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Table 3-13 Mapping between the value of dbsize or sbsize and the burst length

Value of bsize or sbSize	Burst Length
000	1
001	4
010	8
011	16
100	32
101	64
110	128
111	256

Table 3-14 describes mapping between the value of dwidth or swidth of DMAC_Cn_CONTROL and the transfer data width.

Table 3-14 Mapping between the value of dwidth or swidth and the transfer bit width

Value of swidth or dwidth	Transfer Bit Width
000	Byte (8 bits)
001	Halfword (16 bits)
010	Word (32 bits)
011	Reserved
100	Reserved
101	Reserved
110	Reserved
111	Reserved

Note the following when configuring DMAC_Cn_CONTROL:

- When the transfer bit width of the source device is smaller than that of the destination device, the product of the transfer bit width and transfer size of the source device must be an integral multiple of the transfer bit width of the destination device. Otherwise, data retention and data loss occur in the FIFO.
- swidth and dwidth fields cannot be set to undefined bit widths.
- If the transfer size field is set to 0 and the DMAC is a flow controller, the DMAC does not transfer data. In this case, the programmer needs to disable the DMA channel and reprogram it.
- Never conduct common write/read tests on the DMAC_Cn_CONTROL register, because the transfer size field is different from the common register field whose written value and read value may be the same. During the write operation, this field serves as a control

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register, because it determines the number of data segments transferred by the DMAC. During the read operation, this field serves as a status register, because it returns the number (in the unit of the bit width of the source device) of the remaining data segments to be transferred.

- When the transfer size field is set to a value greater than the depth of the FIFO (peripheral FIFO but not the DMAC FIFO) of the source device or destination device, the mode of DMAC source address or destination address must be set to non-incremental mode. Otherwise, the peripheral FIFO may overflow.

The bus access information is provided for the source device or destination device over the master interface signals during data transfer. Such information is related to the bits DMAC_Cn_CONTROL[prot_stat] and DMAC_Cn_CONFIG[ch_lock] that are configured by programming channel registers. Table 3-15 describes the three protection bits of the prot field of DMAC_Cn_CONTROL.

Table 3-15 Definitions of the prot_stat field of DMAC_Cn_CONTROL

Bit	Description	Purpose
[2]	Cacheable or non-cacheable	Indicates whether the access is cacheable. 0: non-cacheable 1: cacheable For example, this bit can notify an advanced microcontroller bus architecture (AMBA) bridge of the following information: When finding the first read operation of the 8-digit burst, this bridge can originate one 8-digit burst read operation on the destination bus directly instead of transmitting the read operations on the source bus to the destination bus one by one. This bit controls the output of the bus signal HPROT[3].
[1]	Bufferable or non-bufferable	Indicates whether the access is bufferable. 0: non-bufferable 1: bufferable For example, this bit is used to notify an AMBA bridge that the write operation on the source bus can be complete without waiting. That is, the operation can be performed even when the bridge does not arbitrate the operation to the destination bus and the slave device does not receive data completely. This bit controls the output of the bus signal HPROT[2].
[0]	Privileged or User	Access mode. 0: user mode 1: privileged mode This bit controls the output of the bus signal HPROT[1].

NOTE

AMBA: advanced microcontroller bus architecture

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Usage of DMAC_Cn_CONFIG

Table 3-16 describes the flow controllers and transfer types corresponding to the flow_cntrl field of DMAC_Cn_CONFIG.

Table 3-16 Flow controllers and transfer types corresponding to the flow_cntrl field

Bit Value	Transfer Mode	Controller
000	Memory to memory	DMAC
001	Memory to peripheral	DMAC
010	Peripheral to memory	DMAC
011	Source device to destination device	DMAC
100	Source device to destination device	Destination device
101	Memory to peripheral	Destination device
110	Peripheral to memory	Source device
111	Source device to destination device	Source device

Interrupt Handling

When data transfer is complete or an error occurs during data transfer, interrupts are reported to the interrupt controller. An interrupt is handled as follows:

Step 1 Read DMAC_INT_STAT to find the channel that transmits an interrupt request. When multiple channels initiate interrupt requests at the same time, the interrupt request with the highest priority is handled first.

Step 2 Read DMAC_INT_TC_STAT to query the value of the selected bit to check whether the interrupt sent by the corresponding channel is a transfer terminal interrupt. If the value is 1, the interrupt is a transfer terminal interrupt. In this case, go to step 4; otherwise, go to step 3.

Step 3 Read DMAC_INT_ERR_STAT to query the value of the selected bit to check whether the interrupt sent by the corresponding channel is an error interrupt. If the selected bit is 1, the interrupt is an error interrupt. In this case, go to step 5; otherwise, end the operation.

Step 4 Handle the transfer terminal interrupt as follows:

1. Write 1 to the selected bit of DMAC_INT_TC_CLR to clear the interrupt status of the corresponding channel.
2. Fetch or use up the data buffered in the memory. If necessary (for example, you need to create a buffer in the memory), configure and enable the channel again.
3. End interrupt handling.

Step 5 Handle the error interrupt as follows:

1. Write 1 to the selected bit of DMAC_INT_ERR_CLR to clear the interrupt status of the corresponding channel.
2. Provide the error information. If necessary, configure and enable the channel again.
3. End interrupt handling.

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----End

3.6.5 Register Summary

NOTE

The *n* in the offset addresses for DMA registers indicates the DMA channel and its value range is 0−3.

Table 3-17 describes the DMAC registers.

Table 3-17 Summary of the DMAC registers (base address: 0x1006_0000)

Offset Address	Register	Description	Page
0x0000	DMAC_INT_STAT	DMAC interrupt status register	3-101
0x0004	DMAC_INT_TC_STAT	DMAC transfer terminal interrupt status register	3-102
0x0008	DMAC_INT_TC_CLR	DMAC transfer terminal interrupt clear register	3-102
0x000C	DMAC_INT_ERR_STAT	DMAC error interrupt status register	3-104
0x0010	DMAC_INT_ERR_CLR	DMAC error interrupt clear register	3-105
0x0014	DMAC_RAW_INT_TC_STAT	DMAC raw transfer terminal interrupt status register	3-106
0x0018	DMAC_RAW_INT_ERR_STAT	DMAC raw error interrupt status register	3-107
0x001C	DMAC_ENABLED_CHNS	DMAC channel enable status register	3-108
0x0020	DMAC_SOFT_BREQ	DMAC software burst transfer request register	3-108
0x0024	DMAC_SOFT_SREQ	DMAC software single transfer request register	3-109
0x0028	DMAC_SOFT_LBREQ	DMAC software last burst request register	3-110
0x002C	DMAC_SOFT_LSREQ	DMAC software last single request register	3-110
0x0030	DMAC_CONFIG	DMAC configuration register	3-111
0x0034	DMAC_SYNC	DMAC request line sync enable register	3-112
0x0100 + *n* x 0x20	DMAC_Cn_SRC_ADDR	Source address register of DMA channel *n* (*n* = 0−3)	3-112
0x0104 + *n* x 0x20	DMAC_Cn_DEST_ADDR	Destination address register of DMA channel *n* (*n* = 0−3)	3-113

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Offset Address	Register	Description	Page
0x0108 + *n* x 0x20	DMAC_Cn_LLI	LLI information register of DMA channel *n* (*n* = 0−3)	3-113
0x010C + *n* x 0x20	DMAC_Cn_CONTROL	Control register of DMA channel *n* (*n* = 0−3)	3-114
0x110 + *n* x 0x20	DMAC_Cn_CONFIG	Configuration register of DMA channel *n* (*n* = 0−3)	3-117

3.6.6 Register Description

DMAC_INT_STAT

DMAC_INT_STAT is an interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0000	DMAC_INT_STAT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3	2	1	0
Name	reserved							ch3_int_stat	ch2_int_stat	ch1_int_stat	ch0_int_stat
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RO	ch3_int_stat	Masked interrupt status of channel 3 0: No interrupt is generated. 1: A transfer error interrupt or transfer terminal interrupt is generated.
[2]	RO	ch2_int_stat	Masked interrupt status of channel 2 0: No interrupt is generated. 1: A transfer error interrupt or transfer terminal interrupt is generated.
[1]	RO	ch1_int_stat	Masked interrupt status of channel 1 0: No interrupt is generated. 1: A transfer error interrupt or transfer terminal interrupt is generated.

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[0]	RO	ch0_int_stat	Masked interrupt status of channel 0 0: No interrupt is generated. 1: A transfer error interrupt or transfer terminal interrupt is generated.

DMAC_INT_TC_STAT

DMAC_INT_TC_STAT is a DMAC transfer terminal interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0004	DMAC_INT_TC_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ch3_int_tc_stat		ch2_int_tc_stat		ch1_int_tc_stat		ch0_int_tc_stat	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RO	ch3_int_tc_stat	Status of the masked transfer terminal interrupt of channel 3 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	ch2_int_tc_stat	Status of the masked transfer terminal interrupt of channel 2 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	ch1_int_tc_stat	Status of the masked transfer terminal interrupt of channel 1 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	ch0_int_tc_stat	Status of the masked transfer terminal interrupt of channel 0 0: No interrupt is generated. 1: An interrupt is generated.

DMAC_INT_TC_CLR

DMAC_INT_TC_CLR is a DMAC transfer terminal interrupt clear register.

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Offset Address	Register Name	Total Reset Value
0x0008	DMAC_INT_TC_CLR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ch3_int_tc_clr	ch2_int_tc_clr	ch1_int_tc_clr	ch0_int_tc_clr
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	WO	ch3_int_tc_clr	Transfer terminal interrupt clear for channel 3 0: not cleared 1: cleared
[2]	WO	ch2_int_tc_clr	Transfer terminal interrupt clear for channel 2 0: not cleared 1: cleared
[1]	WO	ch1_int_tc_clr	Transfer terminal interrupt clear for channel 1 0: not cleared 1: cleared
[0]	WO	ch0_int_tc_clr	Transfer terminal interrupt clear for channel 0 0: not cleared 1: cleared

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DMAC_INT_ERR_STAT

DMAC_INT_ERR_STAT is a DMAC error interrupt status register.

Offset Address	Register Name	Total Reset Value
0x000C	DMAC_INT_ERR_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ch3_int_err_stat		ch2_int_err_stat		ch1_int_err_stat		ch0_int_err_stat	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RO	ch3_int_err_stat	Status of the masked error interrupt of channel 3 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	ch2_int_err_stat	Status of the masked error interrupt of channel 2 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	ch1_int_err_stat	Status of the masked error interrupt of channel 1 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	ch0_int_err_stat	Status of the masked error interrupt of channel 0 0: No interrupt is generated. 1: An interrupt is generated.

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DMAC_INT_ERR_CLR

DMAC_INT_ERR_CLR is a DMAC transfer error interrupt clear register.

Offset Address	Register Name	Total Reset Value
0x0010	DMAC_INT_ERR_CLR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ch3_int_err_clr		ch2_int_err_clr		ch1_int_err_clr		ch0_int_err_clr	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	WO	ch3_int_err_clr	Error interrupt clear for channel 3 0: not cleared 1: cleared
[2]	WO	ch2_int_err_clr	Error interrupt clear for channel 2 0: not cleared 1: cleared
[1]	WO	ch1_int_err_clr	Error interrupt clear for channel 1 0: not cleared 1: cleared
[0]	WO	ch0_int_err_clr	Error interrupt clear for channel 0 0: not cleared 1: cleared

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DMAC_RAW_INT_TC_STAT

DMAC_RAW_INT_TC_STAT is a DMAC raw transfer terminal interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0014	DMAC_RAW_INT_TC_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ch3_raw_int_tc		ch2_raw_int_tc		ch1_raw_int_tc		ch0_raw_int_tc	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RO	ch3_raw_int_tc	Status of the raw transfer terminal interrupt of channel 3 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	ch2_raw_int_tc	Status of the raw transfer terminal interrupt of channel 2 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	ch1_raw_int_tc	Status of the raw transfer terminal interrupt of channel 1 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	ch0_raw_int_tc	Status of the raw transfer terminal interrupt of channel 0 0: No interrupt is generated. 1: An interrupt is generated.

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DMAC_RAW_INT_ERR_STAT

DMAC_RAW_INT_ERR_STAT is a DMAC raw error interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0018	DMAC_RAW_INT_ERR_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ch3_raw_int_err		ch2_raw_int_err		ch1_raw_int_err		ch0_raw_int_err	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RO	ch3_raw_int_err	Status of the raw error interrupt of channel 3 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	ch2_raw_int_err	Status of the raw error interrupt of channel 2 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	ch1_raw_int_err	Status of the raw error interrupt of channel 1 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	ch0_raw_int_err	Status of the raw error interrupt of channel 0 0: No interrupt is generated. 1: An interrupt is generated.

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DMAC_ENABLED_CHNS

DMAC_ENABLED_CHNS is a DMAC channel enable status register.

Offset Address	Register Name	Total Reset Value
0x001C	DMAC_ENABLED_CHNS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ch3_enabled	ch2_enabled	ch1_enabled	ch0_enabled
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RO	ch3_enabled	Channel 3 enable 0: disabled 1: enabled
[2]	RO	ch2_enabled	Channel 2 enable 0: disabled 1: enabled
[1]	RO	ch1_enabled	Channel 1 enable 0: disabled 1: enabled
[0]	RO	ch0_enabled	Channel 0 enable 0: disabled 1: enabled

DMAC_SOFT_BREQ

DMAC_SOFT_BREQ is a DMAC software burst transfer request register.

Software controls the generation of a DMA burst transfer request by using this register.

When this register is read, the device that is requesting the DMA burst transfer can be queried. This register and any peripheral can generate a DMA request each.

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Offset Address	Register Name	Total Reset Value
0x0020	DMAC_SOFT_BREQ	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																soft_breq															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	soft_breq	Controls whether to generate a DMA burst transfer request When this register is written: 0: no effect 1: A DMA burst transfer request is generated. When the transfer is complete, the corresponding bit is cleared. When this register is read: 0: The peripheral corresponding to the request signal DMACBREQ[15:0] does not send any DMA burst request. 1: The peripheral corresponding to the request signal DMACBREQ[15:0] is requesting the DMA burst transfer.

DMAC_SOFT_SREQ

DMAC_SOFT_SREQ is a DMAC software single transfer request register.

Software controls the generation of DMA single transfer requests by using this register.

When this register is read, the device that is requesting the DMA single transfer can be queried. This register and any of the 16 DMA request input signals of the DMAC can generate a DMA request each.

Offset Address	Register Name	Total Reset Value
0x0024	DMAC_SOFT_SREQ	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																soft_sreq															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	soft_sreq	Controls whether to generate a DMA single transfer request When this register is written: 0: no effect 1: A DMA single transfer request is generated. When the transfer is complete, the corresponding bit is cleared.

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			When this register is read: 0: The peripheral corresponding to the request signal DMACSREQ[15:0] does not send a DMA single request. 1: The peripheral corresponding to the request signal DMACSREQ[15:0] is requesting the DMA single transfer.

DMAC_SOFT_LBREQ

DMAC_SOFT_LBREQ is a DMAC software last burst request register.

Software controls the generation of the DMA last burst transfer requests by using this register.

Offset Address	Register Name	Total Reset Value
0x0028	DMAC_SOFT_LBREQ	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																soft_lbreq															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	soft_lbreq	Last burst request issued by the software 0: no effect 1: A DMA last burst transfer request is generated. When the transfer is complete, the corresponding bit is cleared.

DMAC_SOFT_LSREQ

DMAC_SOFT_LSREQ is a DMAC software last single transfer request register.

Software controls the generation of the DMA last single transfer requests by using this register.

Offset Address	Register Name	Total Reset Value
0x002C	DMAC_SOFT_LSREQ	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																soft_lsreq															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved

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[15:0]	RW	soft_lsreq	Last single transfer request issued by software 0: no effect 1: A DMA last single transfer request is generated. When the transfer is complete, the corresponding bit is cleared.

DMAC_CONFIG

DMAC_CONFIG is a DMAC configuration register.

Offset Address	Register Name	Total Reset Value
0x0030	DMAC_CONFIG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													m2_endianness	m1_endianness	dmac_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	m2_endianness	Byte endianness of master 2 0: little endian 1: big endian
[1]	RW	m1_endianness	Byte endianness of master 1 0: little endian 1: big endian
[0]	RW	dmac_enable	DMA enable 0: disabled 1: enabled

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DMAC_SYNC

DMAC_SYNC is a DMAC request line sync enable register.

Offset Address	Register Name	Total Reset Value
0x0034	DMAC_SYNC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																dmac_sync															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	dmac_sync	Controls whether to synchronize the request signals 0: The sync logic provided for the DMA request signals of the corresponding peripheral is enabled. 1: The sync logic provided for the DMA request signals of the corresponding peripheral is disabled.

DMAC_Cn_SRC_ADDR

DMAC_Cn_SRC_ADDR is a source address register of DMA channel *n* (*n* = 0−3).

Its offset address is 0x100 + *n* x 0x20. The value of *n* ranges from 0 to 3. The values 0–3 correspond to DMA channels 0–3.

Before a channel is enabled, its corresponding register must be programmed by using software. After the channel is enabled, the register is updated in any of the following cases:

- The source address is incremented.
- A complete data block is transferred and then loaded from LLI nodes.
- If a channel is active, no valid information is obtained when this register is read. This is because that after software obtains the register value, the value changes during data transfer. Therefore, the register is read after the channel stops data transfer. At this time, the read value is the last source address read by the DMAC.

Offset Address	Register Name	Total Reset Value
0x0100 + *n* x 0x20 (*n* = 0−3)	DMAC_Cn_SRC_ADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	src_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	src_addr	DMA source address

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DMAC_Cn_DEST_ADDR

DMAC_Cn_DEST_ADDR is a destination address register of DMA channel *n* (*n* = 0−3).

Its offset address is 0x104 + *n* x 0x20. The value of *n* ranges from 0 to 3. The values 0–3 correspond to DMA channels 0–3.

This register contains the destination address for the data to be transferred. Before a channel is enabled, its corresponding register must be programmed by using software. After the channel is enabled, the register is updated in any of the following cases:

- Destination address increment.
- A complete data block is transferred and then loaded from LLI nodes.
- If a channel is active, no valid information is obtained when this register is read. This is because that after software obtains the register value, the value changes during data transfer. Therefore, the register is read after the channel stops data transfer. At this time, the read value is the last destination address written by the DMAC.

Offset Address	Register Name	Total Reset Value
0x0104 + *n* x 0x20 (*n* = 0−3)	DMAC_Cn_DEST_ADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dest_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dest_addr	DMA destination address

DMAC_Cn_LLI

DMAC_Cn_LLI is an LLI information register of DMA channel *n* (*n* = 0−3).

Its offset address is 0x108 + *n* x 0x20. The value of *n* ranges from 0 to 3. The values 0–3 correspond to DMA channels 0–3. For details, see section "DMA Linked List."

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Offset Address	Register Name	Total Reset Value
0x0108 + *n* x 0x20 (*n* = 0–3)	DMAC_Cn_LLI	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ll_item																														reserved	ll_master
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RW	ll_item	The bit[31:2] in the next LLI node address and the address bit[1:0] are set to 0. A linked list address must be 4-byte aligned.
[1]	RW	reserved	Reserved. This bit value must be 0 during write operations, and this bit must be masked during read operations.
[0]	RW	ll_master	Master for loading the next LLI node 0: master 1 1: master 2

DMAC_Cn_CONTROL

DMAC_Cn_CONTROL is a control register of DMA channel *n* (*n* = 0–3).

Its offset address is 0x10C + n x 0x20. The value of *n* ranges from 0 to 3. The values 0–3 correspond to DMA channels 0–3.

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Offset Address: 0x010C + *n* x 0x20 (*n* = 0–3)

Register Name: DMAC_Cn_CONTROL

Total Reset Value: 0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	int_tc_enable	prot_stat			dest_incr	src_incr	dest_select	src_select	dwidth				swidth				dbsize				sbsize				trans_size							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	int_tc_enable	Transfer terminal interrupt enable. This bit determines whether the current LLI node triggers a transfer terminal interrupt. 0: do not trigger 1: trigger
[30:28]	RW	prot_stat	HPROT[2:0] access protection signal transmitted by the master
[27]	RW	dest_incr	Destination address increment 0: The destination address is not incremented 1: The destination address is incremented once after a data segment is transferred If the destination device is a peripheral, the destination address is not incremented. If the destination device is a memory, the destination address is incremented.
[26]	RW	src_incr	Source address increment 0: The source address is not incremented 1: The source address is incremented once after a data segment is transferred If the source device is a peripheral, the source address is not incremented. If the source device is a memory, the source address is incremented.
[25]	RW	dest_select	Master for accessing the destination device 0: master 1 1: master 2
[24]	RW	src_select	Master for accessing the source device 0: master 1 1: master 2

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[23:21]	RW	dwidth	Transfer bit width of the destination device The transfer bit width is invalid if it is greater than the bit width of the master. The data widths of the destination and source devices can be different. The hardware automatically packs and unpacks the data. For the mapping between the value of dwidth and the bit width, see Table 3-14.
[20:18]	RW	swidth	Transfer bit width of the source device The transfer bit width is invalid if it is greater than the bit width of the master. The data widths of the destination and source devices can be different. The hardware automatically packs and unpacks the data. For the mapping between the value of swidth and the bit width, see Table 3-14.
[17:15]	RW	dbsize	Burst size of the destination device It indicates the number of data segments to be transferred by the destination device in a burst transfer, that is, the number of transferred data segments when DMACCxBREQ is valid. This value must be set to a burst size supported by the destination device. If the destination device is a memory, the value is set to the storage area size beyond the storage address boundary. For the mapping between the value of dbsize and the transfer length, see Table 3-13.
[14:12]	RW	sbsize	Burst size of the source device It indicates the amount of data to be transferred by the source device in a burst transfer, that is, the amount of transferred data when DMACCxBREQ is valid. The value must be set to a burst size supported by the source device. If the source device is a memory, the value is set to the storage area size beyond the storage address boundary. For the mapping between the value of sbsize and the transfer length, see Table 3-13.
[11:0]	RW	trans_size	The DMA transfer size can be configured by writing to this register only when the DMAC is a flow controller. This field indicates the amount of data to be transferred by the source device. When this register is read, the amount of data transferred through the bus connected to the destination device is obtained. If a channel is active, no valid information is obtained when this register is read. This is because that after software obtains the register value, the value changes during data transfer. Therefore, the register is read after the channel is enabled and data transfer stops.

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DMAC_Cn_CONFIG

DMAC_Cn_CONFIG is a configuration register of channel *n* (*n* = 0−3).

Its offset address is 0x110 + *n* x 0x20. The value of *n* ranges from 0 to 3. The values 0–3 correspond to DMA channels 0–3.

This register is not updated when a new LLI node is loaded.

Offset Address	Register Name	Total Reset Value
0x110 + *n* x 0x20 (*n* = 0−3)	DMAC_Cn_CONFIG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved													ch_halt	ch_active	ch_lock	tc_int_msk	err_int_msk	flow_ctrl		reserved		dest_periph		reserved		src_periph		ch_en			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:19]	RW	reserved	Reserved This bit value must be 0 during write operations, and this bit must be masked during read operations.
[18]	RW	ch_halt	Halt bit 0: The DMA request is allowed. 1: The subsequent DMA requests are ignored and the contents in the channel FIFO are completely transmitted. This bit can disable a DMA channel without data loss by working with the Active bit and the Channel Enable bit.
[17]	RW	ch_active	Active bit 0: There is no data in the channel FIFO. 1: There is data in the channel FIFO. This bit can disable a DMA channel without data loss by working with the Halt bit and Channel Enable bit.
[16]	RW	ch_lock	Lock bit 0: Lock transfer on the bus is disabled. 1: Lock transfer on the bus is enabled.
[15]	RW	tc_int_msk	Transfer terminal interrupt mask 0: The transfer terminal interrupts of the channel are masked. 1: The transfer terminal interrupts of the channel are not masked.

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[14]	RW	err_int_msk	Transfer error interrupt mask 0: The error interrupts of the channel are masked. 1: The error interrupts of the channel are not masked.
[13:11]	RW	flow_ctrl	Flow control and transfer type This field specifies the flow controller and transfer type. The flow controller can be the DMAC, source device, or destination device. The transfer type can be memory to peripheral, peripheral to memory, peripheral to peripheral, or memory to memory. For details, see Table 3-16.
[10]	RW	reserved	Reserved This bit value must be 0 during write operations, and this bit must be masked during read operations.
[9:6]	RW	dest_periph	Destination device. This field is used to select a peripheral request signal as the request signal for the DMA destination device of the channel. If the destination device for DMA transfer is a memory, this field is ignored.
[5]	RW	reserved	Reserved This bit value must be 0 during write operations, and this bit must be masked during read operations.
[4:1]	RW	src_periph	Source device. This field is used to select a peripheral request signal as the request signal for the DMA source device of the channel. If the source device for DMA transfer is a memory, this field is ignored.
[0]	RW	ch_en	Channel enable. The current status of the channel can be queried by reading this field or DMAC_ENABLED_CHNS. 0: disabled 1: enabled Clearing this bit can disable a channel. When this bit is cleared, the current bus transfer continues until the data transfer is complete. Then, the channel is disabled and the remaining data in the FIFO is lost. When the last LLI node is transferred or an error occurs during transfer, the channel is also disabled and this bit is cleared. If you want to disable a channel without data loss, the Halt bit must be set to 1 so the subsequent DMA requests are ignored by the channel. After this, the Active bit must be polled until its value becomes 0, indicating that there is no data in the channel FIFO. At this time, the Enable bit can be cleared. Before enabling a channel by setting this bit to 1, you must reinitialize the channel; otherwise, unexpected results may occur. When a channel is disabled by writing DMAC_Cn_CONFIG [ch_en], DMAC_Cn_CONFIG [ch_en] can be reset again only after the corresponding bit of DMAC_ENABLED_CHNS is 0. This is because the channel is not disabled immediately after the Channel Enable bit is cleared. The running delay during a bus

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			burst operation also needs to be considered.

3.7 Timer

3.7.1 Overview

The timer module implements the timing and counting functions. It not only serves as the system clock of the operating system, but also can be used by applications for timing and counting. The Hi3518E V20X/Hi3516C V200 has four timers.

3.7.2 Features

Timer has the following features:

- Provides 16-bit or 32-bit down counter that has a programmable 8-bit prescaler.
- Provides a configurable count clock, that is, the clock can serve as the 49.5 MHz bus clock or 3 MHz crystal oscillator clock.
- Supports three count modes: free-running mode, periodic mode, and one-shot mode.
- Loads the initial value through either of the following registers: TIMERx_LOAD and
- TIMERx_BGLOAD.
- Reads the current count value at any time.
- Generates an interrupt when the count value is decreased to 0.

3.7.3 Function Description

The timer is a 32-bit or 16-bit configurable down counter. The counter value is decremented by 1 on each rising edge of the count clock. When the count value reaches 0, the timer generates an interrupt.

The timer supports three count modes:

- Free-running mode

  The timer counts continuously. When the count value reaches 0, the timer wraps its value around to the maximum value automatically and then continues to count. When the count length is 32 bits, the maximum value is 0xFFFF_FFFF. When the count length is 16 bits, the maximum value is 0xFFFF. In free-running mode, the count value is decremented immediately from the loaded value. When the value reaches 0, the value is wrapped around to the maximum value.
- Periodic mode

  The timer counts continuously. When the count value reaches 0, the timer loads an initial value from

  TIMERx_BGLOAD again and then continues to count.
- One-shot mode

  The initial value is loaded to the timer. When the count value of the timer reaches 0, the timer stops counting. The timer starts to count again only when a new value is loaded and the timer is enabled.

Each timer has a prescaler that divides the frequency of the working clock of each timer by 1, 16, or 256. In this way, flexible frequencies of the count clock are provided. An initial value is loaded to the timer as follows:

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- An initial value can be loaded by writing TIMERx_LOAD. When the timer works, if a value is written to TIMERx_LOAD, the timer recounts starting from this value immediately. This method is applicable to all count modes.
- The count cycle in periodic mode can be set by writing
- TIMERx_BGLOAD. The current count value of the timer is not affected immediately when TIMERx_BGLOAD is written. Instead, the timer continues to count until the count value reaches 0. Then the timer loads the new value of
- TIMERx_BGLOAD and starts to count.

3.7.4 Operating Mode

Initialization

The timer must be initialized when the system is initialized. To initialize timer*X* (*X* ranges from 0 to 3), do as follows:

Step 1 Write to TIMERx_LOAD to load an initial value to the timer.

Step 2 When the timer is required to work in periodic mode and the count cycle is different from the initial value loaded to the timer, write to TIMERx_BGLOAD to set the count cycle of the timer.

Step 3 Configure the SC_CTRL register of the system controller to set the reference clock of the clock enable signal of the timer.

Step 4 Write to TIMERx_CONTROL to set the count mode, counter length, prescaling factor, and interrupt mask of the timer, and then enable the timer to count.

----End

Interrupt Processing

The timer is used to generate interrupts periodically. Therefore, interrupt processing is a process of activating and waiting the timing interrupt. To process an interrupt, do as follows:

Step 1 Configure TIMERx_INTCLR to clear the interrupt of the timer.

Step 2 Activate the processes of waiting for the interrupt and execute the process.

Step 3 When all the processes of waiting for the interrupt are complete or the wait interrupt is in hibernate state, resume the interrupt and continue to execute the interrupted program.

----End

Clock Selection

Each timer has two optional count clocks. The following sections describe how to select a clock by taking timer0 as an example.

To select the bus clock as the count clock, do as follows:

Step 1 Set SC_CTRL [timeren0ov] to 1.

Step 2 Initialize the timer and start to count.

----End

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To select the 3 MHz crystal oscillator clock as the count clock, do as follows:

Step 1 Set SC_CTRL [timeren0sel] to 0.

Step 2 Initialize the timer and start to count.

----End

3.7.5 Register Summary

The timer module consists of four timers and each timer involves a group of registers. The four groups of registers have the same features except that their base addresses are different. The details about their base addresses are as follows:

- The base address of timer 0: 0x2000_0000.
- The base address of timer 1: 0x2000_0020.
- The base address of timer 2: 0x2001_0000.
- The base address of timer 3: 0x2001_0020.

NOTE

The value of *X* in timerX ranges from 0 to 3. The registers for timer 0 to timer 3 are the same. In this section, timer 0 registers are described as an example.

Table 3-18 Summary of timer registers

Offset Address	Register	Description	Page
0x000	TIMERx_LOAD	Initial count value register	3-121
0x004	TIMERx_VALUE	Current count value register	3-122
0x008	TIMERx_CONTROL	Control register	3-122
0x00C	TIMERx_INTCLR	Interrupt clear register	3-124
0x010	TIMERx_RIS	Raw interrupt status register	3-124
0x014	TIMERx_MIS	Masked interrupt status register	3-125
0x018	TIMERx_BGLOAD	Initial count value register in periodic mode	3-125

3.7.6 Register Description

TIMERx_LOAD

TIMERx_LOAD is an initial count value register. It is used to set the initial count value of each timer. Each timer (timers 0–3) has one such register.

NOTE

- The minimum valid value written to TIMERx_LOAD is 1.
- When the value 0 is written to TIMERx_LOAD, the dual-timer module generates an interrupt immediately.

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If values are written to TIMERx_BGLOAD and TIMERx_LOAD before the rising edge of TIMCLK enabled by TIMCLKENx reaches, the count value of the next rising edge of TIMCLK is changed to the value written to TIMERx_LOAD. As the value of

TIMERx_BGLOAD changes when data is written to TIMERx_LOAD, the value returned after TIMERx_BGLOAD is read is the latest value that is written to TIMERx_LOAD and TIMERx_BGLOAD. When the timer works in periodic mode and the count value decreases to 0, the initial value is loaded from TIMERx_BGLOAD to continue counting.

Offset Address	Register Name	Total Reset Value
0x000	TIMER0_LOAD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timer0_load																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	timer0_load	Initial count value of timer 0

TIMERx_VALUE

TIMERx_VALUE is a current count value register. It shows the current value of the counter that is decremented. Each timer (timers 0–3) has one such register.

After a value is written to TIMERx_LOAD, TIMERx_VALUE immediately shows the newly loaded value of the counter in the PCLK domain without waiting for the clock edge of TIMCLK enabled by TIMCLKENx.

NOTE

When a timer is in 16-bit mode, the 16 upper bits of the 32-bit TIMERx_VALUE are not set to 0 automatically. If the timer is switched from 32-bit mode to 16-bit mode and no data is written to TIMERx_LOAD, the upper 16 bits of TIMERx_VALUE may be non-zero.

Offset Address	Register Name	Total Reset Value
0x004	TIMER0_VALUE	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timer0_value																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:0]	RO	timer0_value	Current count value of timer 0 that is decremented

TIMERx_CONTROL

TIMERx_CONTROL is a control register. Each timer (timers 0–3) has one such register.

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NOTE

When the periodic mode is selected, TIMERx_CONTROL[timermode] must be set to 1 and TIMERx_CONTROL[oneshot] must be set to 0.

Offset Address	Register Name	Total Reset Value
0x008	TIMER0_CONTROL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								timeren	timermode	intenable	reserved	timerpre		timersize	oneshot
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	timeren	Timer enable 0: disabled 1: enabled
[6]	RW	timermode	Timer count mode 0: free-running mode 1: periodic mode
[5]	RW	intenable	TIMERx_RIS interrupt mask 0: masked 1: not masked
[4]	RO	reserved	Reserved
[3:2]	RW	timerpre	Prescaling factor configuration 00: no prescaling. That is, the clock frequency of the timer is divided by 1 01: 4-level prescaling. That is, the clock frequency of the timer is divided by 16 10: 8-level prescaling. That is, the clock frequency of the timer is divided by 256 11: undefined. If the bits are set to 11, 8-level prescaling is considered. That is, the clock frequency of the timer is divided by 256.
[1]	RW	timersize	Counter select 0: 16-bit counter 1: 32-bit counter
[0]	RW	oneshot	Count mode select 0: periodic mode or free-running mode 1: one-shot mode

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TIMERx_INTCLR

TIMERx_INTCLR is the interrupt clear register. The interrupt status of a counter is cleared after any operation is performed on this register. Each timer (timers 0–3) has one such register.

CAUTION

This register is a write-only register. The timer clears interrupts when any value is written to this register. In addition, no value is recorded in this register and no default reset value is defined.

Offset Address	Register Name	Total Reset Value
0x00C	TIMER0_INTCLR	-

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timer0_intclr																															
Reset	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?

Bits	Access	Name	Description
[31:0]	WO	timer0_intclr	Writing this register clears the output interrupt of timer 0.

TIMERx_RIS

TIMERx_RIS is a raw interrupt status register. Each timer (timers 0–3) has one such register.

Offset Address	Register Name	Total Reset Value
0x010	TIMER0_RIS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															timer0ris
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved. Writing this register has no effect and reading this register returns 0.
[0]	RO	timer0ris	Raw interrupt status of timer 0 0: No interrupt is generated. 1: An interrupt is generated.

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TIMERx_MIS

TIMERx_MIS is a masked interrupt status register. Each timer (timers 0–3) has one such register.

Offset Address	Register Name	Total Reset Value
0x014	TIMER0_MIS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															timer0mis
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	timer0mis	Masked interrupt status of timer 0 0: The interrupt is invalid. 1: The interrupt is valid.

TIMERx_BGLOAD

TIMERx_BGLOAD is an initial count value register in periodic mode. Each timer (timers 0–3) has one such register.

The TIMERx_BGLOAD register contains the initial count value of the timer. This register is used to reload an initial count value when the count value of the timer reaches 0 in periodic mode.

In addition, this register provides another method of accessing TIMERx_LOAD. The difference is that after a value is written to TIMERx_BGLOAD, the timer does not count starting from the input value immediately.

Offset Address	Register Name	Total Reset Value
0x018	TIMER0_BGLOAD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timer0bgload																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	timer0bgload	Initial count value of timer 0 Note: This register differs from TIMERx_LOAD. For details, see the descriptions of TIMERx_LOAD.

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3.8 Watchdog

3.8.1 Overview

The watchdog is used to transmit a reset signal to reset the entire system within a period after an exception occurs in the system.

3.8.2 Features

The watchdog has the following features:

- Provides a 32-bit internal down counter.
- Supports the configurable timeout interval, namely, initial count value.
- Locks registers to avoid any modification to them.
- Supports the generation of timeout interrupts.
- Supports the generation of reset signals.
- Supports the debugging mode.

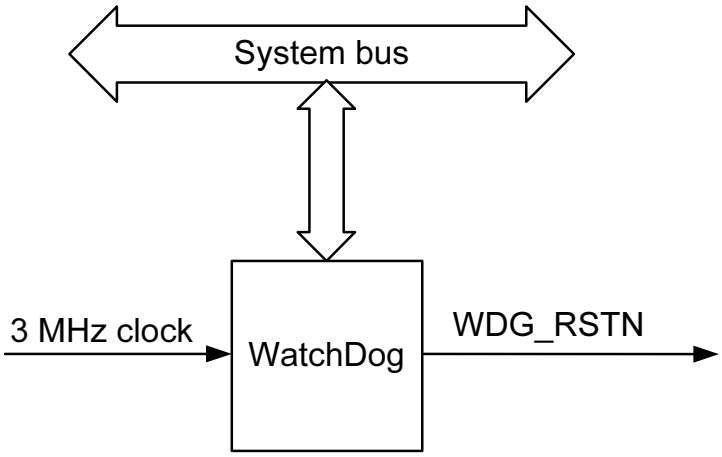
3.8.3 Function Description

Application Block Diagram

The system configures the parameter values of watchdog registers by using the system bus. The watchdog transmits interrupt requests to the system periodically. When the system does not respond to the interrupt requests (such as the suspend case), the watchdog transmits the WDG_RSTN reset signal to reset the system. In this way, the system running status is monitored.

Figure 3-8 shows the application block diagram of the watchdog.

Figure 3-8 Application block diagram of the watchdog



Function Principle

The watchdog works based on a 32-bit down counter. The initial value is loaded by WDG_LOAD. When the watchdog clock is enabled, the count value is decremented by 1 on

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the rising edge of each count clock. When the count value reaches 0, the watchdog generates an interrupt. On the next rising edge of the count clock, the counter reloads the initial value from WDG_LOAD and continues to count in decremental mode.

If the count value of the counter reaches 0 for the second time but the CPU does not clear the watchdog interrupt, the watchdog transmits the reset signal WDG_RSTN and the counter stops counting.

You can enable or disable the watchdog by configuring WDG_CONTROL as required. That is, you can control the watchdog whether to generate interrupts and reset signals.

- When the interrupt generation function is disabled, the watchdog counter stops counting.
- When the interrupt generation function is enabled again, the watchdog counter counts starting from the preset value of WDG_LOAD instead of the last count value. Before an interrupt is generated, the initial value can be reloaded.

The count clock of the watchdog can be a crystal oscillator clock or a bus clock so different count time ranges are available.

By configuring WDG_LOCK, you can disable the operation of writing to the internal registers of the watchdog.

- Writing 0x1ACC_E551 to WDG_LOCK to enable the write permission for all the registers of the watchdog.
- Writing any other values to WDG_LOCK to disable the write permission for all the registers of the watchdog except WDG_LOCK.

This feature avoids modifications to the watchdog registers by software. Therefore, the watchdog operation is not terminated by mistake by software when an exception occurs.

In debugging mode, the watchdog is disabled automatically to avoid the intervention to the normal debugging.

3.8.4 Operating Mode

Configuring the Frequency of the Count Clock

The watchdog counting clock is a 3 MHz.

The count time T_{WDG} of the watchdog is calculated as follows: $T_{WDG} = Value_{WDG_LOAD} \times (1/f_{clk})$

NOTE

The definition of each parameter in the preceding formula is as follows:

- T_{WDG} indicates the count time of the watchdog.
- $Value_{WDG_LOAD}$ indicates the initial count value of the watchdog.
- f_{clk} indicates the frequency of the watchdog count clock.

The counting time of the watchdog counting clock ranges from 0s to 1400s.

Initializing the System

The watchdog counter stops counting after the system power-on reset. Before the system is initialized, the watchdog must be initialized and enabled. To initialize the watchdog, perform the following steps:

Step 1 Write to WDG_LOAD to set the initial count value.

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Step 2 Write to WDG_CONTROL to enable the interrupt mask function and start the watchdog counter.

Step 3 Write to WDG_LOCK to lock the watchdog to avoid the watchdog settings being modified by the software by mistake.

----End

Processing an Interrupt

After an interrupt is received from the watchdog, the interrupt must be cleared in time and the initial count value must be reloaded to the watchdog to restart counting. A watchdog interrupt is processed as follows:

Step 1 Write 0x1ACC_E551 to WDG_LOCK to unlock the watchdog.

Step 2 Write to WDG_INTCLR to clear the watchdog interrupt and load the initial count value to the watchdog to restart counting.

Step 3 Write any other values rather than 0x1ACC_E551 to WDG_LOCK to lock the watchdog.

----End

Disabling the Watchdog

You can control the status of the watchdog by writing 0 or 1 to WDG_CONTROL[inten].

- 0: disabled
- 1: enabled

3.8.5 Register Summary

Table 3-19 describes watchdog registers.

Table 3-19 Summary of watchdog registers (base address: 0x2004_0000)

Offset Address	Register	Description	Page
0x0000	WDG_LOAD	Initial count value register	3-129
0x0004	WDG_VALUE	Current count value register	3-129
0x0008	WDG_CONTROL	Control register	3-129
0x000C	WDG_INTCLR	Interrupt clear register	3-130
0x0010	WDG_RIS	Raw interrupt register	3-130
0x0014	WDG_MIS	Masked interrupt status register	3-131
0x0C00	WDG_LOCK	Lock register	3-131

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3.8.6 Register Description

WDG_LOAD

WDG_LOAD is an initial count value register. It is used to configure the initial count value of the internal counter of the watchdog.

Offset Address	Register Name	Total Reset Value
0x0000	WDG_LOAD	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdg_load																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:0]	RW	wdg_load	Initial count value of the watchdog counter

WDG_VALUE

WDG_VALUE is a current count value register It is used to read the current count value of the internal counter of the watchdog.

Offset Address	Register Name	Total Reset Value
0x0004	WDG_VALUE	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdogvalue																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:0]	RO	wdogvalue	Current count value of the watchdog counter

WDG_CONTROL

WDG_CONTROL is a control register. It is used to enable or disable the watchdog and control the interrupt and reset functions of the watchdog.

Offset Address	Register Name	Total Reset Value
0x0008	WDG_CONTROL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														resen	inten

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	resen	Output enable of the watchdog reset signal 0: disabled 1: enabled
[0]	RW	inten	Output enable of the watchdog interrupt signal 0: The counter stops counting, the current count value remains unchanged, and the watchdog is disabled. 1: The counter, interrupt and watchdog are enabled.

WDG_INTCLR

WDG_INTCLR is an interrupt clear register. It is used to clear watchdog interrupts so the watchdog can reload an initial value for counting. This register is write-only. When a value is written to this register, the watchdog interrupts are cleared. No written value is recorded in this register and no default reset value is defined.

Offset Address	Register Name	Total Reset Value
0x000C	WDG_INTCLR	-

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdg_intclr																															
Reset	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?

Bits	Access	Name	Description
[31:0]	WO	wdg_intclr	Writing any value to this register clears the watchdog interrupts and enables the watchdog to reload an initial count value from WDG_LOAD to restart counting.

WDG_RIS

WDG_RIS is a raw interrupt status register. It shows the raw interrupt status of the watchdog.

Offset Address	Register Name	Total Reset Value
0x0010	WDG_RIS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															wdogris

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	wdogris	Status of the raw interrupts of the watchdog. When the count value reaches 0, this bit is set to 1. 0: No interrupt is generated. 1: An interrupt is generated.

WDG_MIS

WDG_MIS is a masked interrupt status register. It shows the masked interrupt status of the watchdog.

Offset Address	Register Name	Total Reset Value
0x0014	WDG_MIS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															wdogmis
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	wdogmis	Status of the masked interrupts of the watchdog 0: No interrupt is generated or the interrupt is masked. 1: An interrupt is generated.

WDG_LOCK

WDG_LOCK is a lock register. It is used to control the write and read permissions for the watchdog registers.

Offset Address	Register Name	Total Reset Value
0x0C00	WDG_LOCK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdg_lock																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	wdg_lock	Writing 0x1ACC_E551 to this register enables the write

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			permission for all the registers. Writing other values disables the write permission for all the registers. When this register is read, the lock status rather than the written value of this register is returned. 0x0000_0000: The write permission is available (unlocked). 0x0000_0001: The write permission is unavailable (locked).

3.9 RTC

3.9.1 Overview

The RTC is used to display the time in real time and periodically generate alarms.

3.9.2 Features

The RTC has the following features:

- Provides a 40-bit counter. Of the 40 bits, 16 bits are for counting days, five bits are for counting hours, six bits are for counting minutes, six bits are for counting seconds, and the other seven bits are for counting 10 ms.
- Provides a 100 Hz count clock.
- Supports the configurable initial count value.
- Supports the configurable count comparison value.
- Generates the timeout interrupt.
- Supports soft reset.
- Supports configurable frequency division parameters.
- Stores user data in a 64-bit user register.

3.9.3 Function Description

The RTC has a 40-bit up counter for counting days, hours, minutes, seconds, and 10 ms. The initial values are loaded from RTC_LR_10MS, RTC_LR_S, RTC_LR_M, RTC_LR_H, RTC_LR_D_L, and RTC_LR_D_H. This section assumes that the counter is divided into the day counter, hour counter, minute counter, second counter, and 10 ms counter. When the count value is equal to the values of RTC_MR_10MS, RTC_MR_S, RTC_MR_M, RTC_MR_H, RTC_MR_D_L, and RTC_MR_D_H, the RTC generates an interrupt. Then, the counter continues to count in incremental mode on the rising edge of the next count clock.

By configuring RTC_IMSC, you can allow or forbid the RTC to generate interrupts. Note the following two cases:

- When the function of generating interrupts is disabled, the RTC counter continues to count in incremental mode and no interrupts are generated. RTC_MSC_INT shows the status of masked interrupts and RTC_RAW_INT shows the status of raw interrupts.
- When the function of generating interrupts is enabled, the RTC counter still counts in incremental mode. When the count value is equal to the values of RTC_MR_10MS,

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RTC_MR_S, RTC_MR_M, RTC_MR_H, RTC_MR_D_L, and RTC_MR_D_H, the RTC generates an interrupt.

The RTC uses a 100 Hz count clock and a 16-bit day counter. The value of the day counter can be used to reckon the specific year, month, and day.

3.9.4 Operating Mode

3.9.4.1 Count Clock Frequency

The RTC uses a 100 Hz count clock. The maximum RTC count time (T_{RTC}) is calculated as follows:

$T_{RTC} = 2^{16} = 65536$ days

NOTE

T_{RTC} is the RTC count time.

3.9.4.2 Soft Reset

The RTC can be separately soft-reset by configuring RTC_POR_N. After soft reset, the value of each RTC configuration register is restored to its default value. Therefore, these registers must be initialized again.

To soft-reset the RTC, perform the following steps:

Step 1 Write 0 to RTC_POR_N.

Step 2 Wait 30 ms.

----End

3.9.4.3 Initializing the RTC

The system needs to initialize the RTC when the RTC is powered on for the first time. The initialization process is as follows:

Step 1 Configure RTC_POR_N to reset the RTC.

Step 2 Wait 30 ms.

Step 3 Configure RTC_IMSC to set the interrupt mask bit of the RTC.

Step 4 Configure RTC_MR_10MS, RTC_MR_S, RTC_MR_M, RTC_MR_H, RTC_MR_D_L, and RTC_MR_D_H to set the RTC match value.

Step 5 Configure RTC_MR_10MS, RTC_MR_S, RTC_MR_M, RTC_MR_H, RTC_MR_D_L, and RTC_MR_D_H to set the initial count value of the RTC.

Step 6 Set RTC_LORD to 1 to load the initial count value to the RTC counter.

Step 7 Wait 5 ms.

Step 8 Based on the 100 Hz count clock, the RTC counts starting from the values of RTC_MR_10MS, RTC_MR_S, RTC_MR_M, RTC_MR_H, RTC_MR_D_L, and RTC_MR_D_H. When the count value is equal to the values of RTC_MR_10MS, RTC_MR_S, RTC_MR_M, RTC_MR_H, RTC_MR_D_L, and RTC_MR_D_H, the RTC determines whether to generate an interrupt based on the settings of RTC_IMSC.

----End

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3.9.4.4 Handling Interrupts

If the system receives an interrupt from the RTC, the configured time is reached. Then user-defined operations can be performed. The RTC counter, however, still counts in incremental mode. To clear an RTC interrupt, set RTC_INT_CLR to 1. To continue to configure time, write a new match value to RTC_MR_10MS, RTC_MR_S, RTC_MR_M, RTC_MR_H, RTC_MR_D_L, and RTC_MR_D_H.

3.9.4.5 Accessing RTC Registers

Internal RTC registers are located in the RTC module, not on the bus. The RTC registers on the bus are used only for accessing internal RTC registers.

To write to internal RTC registers, perform the following steps:

Step 1 Configure SPI_CLK_DIV.

This example assumes that the clock is 120 MHz and the expected SPI clock is 12 MHz. The configured value of SPI_CLK_DIV is calculated as follows: (120/12)/2 – 1 = 4 = 0x04. If you have configured SPI_CLK_DIV and do not want to change the SPI clock frequency, skip this step.

Step 2 Read SPI_RW bit[31] until it is 0.

Step 3 Configure SPI_RW.

The internal offset address for RTC_MR_10MS is 0x06. If you want to write 0x10 to RTC_MR_10MS, write 0x01060010 to SPI_RW. That is, spi_start is 1, spi_rw is 0, spi_add is 0x06, and spi_wdata is 0x10.

----End

To read internal RTC registers, perform the following steps:

Step 1 Configure SPI_CLK_DIV.

This step assumes that the clock is 120 MHz and the expected SPI clock is 12 MHz. The configured value of SPI_CLK_DIV is calculated as follows: (120/12)/2 – 1 = 4 = 0x04. If you have configured SPI_CLK_DIV and do not want to change the SPI clock frequency, skip this step.

Step 2 Read SPI_RW bit[31] until it is 0.

Step 3 Configure SPI_RW.

The internal offset address for RTC_MR_10MS is 0x06. If you want to read SPI_RW, write 0x01860000 to SPI_RW. That is, spi_start is 1, spi_rw is 0, and spi_add is 0x06.

Step 4 Read SPI_RW bit[31] until it is 0. SPI_RW [15:8] is the value that is returned after RTC_MR_10MS is read.

----End

3.9.5 Register Summary

Table 3-20 describes RTC registers.

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Table 3-20 Summary of RTC registers (base address: 0x2006_0000)

Offset Address	Register	Description	Page
0x0000	SPI_CLK_DIV	SPI clock divider register	3-137
0x0004	SPI_RW	SPI read/write register	3-138
0x0080	CONVER_T	External temperature sensor DS1820 (DS18B20) measurement control register	3-138
0x0084	CRC_EN	External temperature sensor DS1820 (DS18B20) measurement CRC check enable register	3-139
0x0088	INT_MASK	External temperature sensor DS1820 (DS18B20) measurement interrupt mask register	3-139
0x008C	INT_CLEAR	External temperature sensor DS1820 (DS18B20) measurement interrupt clear register	3-140
0x0090	BUSY	External temperature sensor DS1820 (DS18B20) measurement status register	3-140
0x0094	INT_RAW	External temperature sensor DS1820 (DS18B20) measurement interrupt status register	3-141
0x0098	INT_TCAP	External temperature sensor DS1820 (DS18B20) measurement interrupt status register	3-141
0x009C	T_VALUE	External temperature sensor DS1820 (DS18B20) measurement value register	3-142
0x00A0	FILTER_NUM	Filter glitch width configuration register	3-142

Table 3-21 describes the internal RTC registers.

Table 3-21 Summary of internal RTC registers (base address: 0x00)

Offset Address	Register	Description	Page
0x00	RTC_10MS_COUNT	Count value register for the RTC 10 ms counter	3-143
0x01	RTC_S_COUNT	Count value register for the RTC second counter	3-143
0x02	RTC_M_COUNT	Count value register for the RTC minute counter	3-144
0x03	RTC_H_COUNT	Count value register for the RTC	3-144

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Offset Address	Register	Description	Page
		hour counter	
0x04	RTC_D_COUNT_L	Lower-8-bit count value register for the RTC day counter	3-145
0x05	RTC_D_COUNT_H	Upper-8-bit count value register for the RTC day counter	3-145
0x06	RTC_MR_10MS	Match value register for the RTC 10 ms counter	3-146
0x07	RTC_MR_S	Match value register for the RTC second counter	3-146
0x08	RTC_MR_M	Match value register for the RTC minute counter	3-147
0x09	RTC_MR_H	Match value register for the RTC hour counter	3-147
0x0A	RTC_MR_D_L	Lower-8-bit match value register for the RTC day counter	3-148
0x0B	RTC_MR_D_H	Upper-8-bit match value register for the RTC day counter	3-148
0x0C	RTC_LR_10MS	Configured value register for the RTC 10 ms counter	3-148
0x0D	RTC_LR_S	Configured value register for the RTC second counter	3-149
0x0E	RTC_LR_M	Configured value register for the RTC minute counter	3-149
0x0F	RTC_LR_H	Configured value register for the RTC hour counter	3-150
0x10	RTC_LR_D_L	Lower-8-bit configured value register for the RTC day counter	3-150
0x11	RTC_LR_D_H	Upper-8-bit configured value register for the RTC day counter	3-150
0x12	RTC_LORD	RTC configured value loading enable register	3-151
0x13	RTC_IMSC	RTC interrupt enable register	3-152
0x14	RTC_INT_CLR	RTC interrupt clear register	3-152
0x15	RTC_MSC_INT	RTC mask interrupt status register.	3-153
0x16	RTC_RAW_INT	RTC raw interrupt status register	3-153
0x17	RTC_CLK	RTC output clock select register	3-154
0x18	RTC_POR_N	RTC reset control register	3-154

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Offset Address	Register	Description	Page
0x51	SDM_COEF_OUSIDE_H	Upper 4-bit external frequency divider register	3-155
0x52	SDM_COEF_OUSIDE_L	Lower 8-bit external frequency divider register	3-155
0x53	USER_REGISTER1	64-bit user register 1	3-155
0x54	USER_REGISTER2	64-bit user register 2	3-156
0x55	USER_REGISTER3	64-bit user register 3	3-156
0x56	USER_REGISTER4	64-bit user register 4	3-156
0x57	USER_REGISTER5	64-bit user register 5	3-157
0x58	USER_REGISTER6	64-bit user register 6	3-157
0x59	USER_REGISTER7	64-bit user register 7	3-157
0x5A	USER_REGISTER8	64-bit user register 8	3-158

3.9.6 RTC Register Description

SPI_CLK_DIV

SPI_CLK_DIV is an SPI clock divider register.

Offset Address	Register Name	Total Reset Value
0x0000	SPI_CLK_DIV	0x0000_003B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								spi_clk_div							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	spi_clk_div	SPI clock divider. The SPI clock frequency cannot be higher than 20 MHz. The value 12 MHz is recommended. The field value ranges from 1 to 255. The value of spi_clk_div is used to define the SPI RX and RX bit rates. The formula is as follows: FSPICLK = FAPBCLK/[2 x (spi_clk_div + 1)]. FAPBCLK is the clock frequency. If the clock is 120 MHz and the expected SPI clock is 12 MHz, the configured value of spi_clk_div is calculated as follows: (120/12)/2 – 1 = 4

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SPI_RW

SPI_RW is an SPI read/write register.

Offset Address	Register Name	Total Reset Value
0x0004	SPI_RW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	spi_busy	reserved						spi_start	spi_rw	spi_add							spi_rdata								spi_wdata							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	spi_busy	SPI read/write status indicator 0: The SPI is idle, and a new read/write operation can be initiated over the SPI. 1: A read/write operation is being performed over the SPI, and no new operation is allowed.
[30:25]	RO	reserved	Reserved
[24]	W1_PULSE	spi_start	SPI read/write start. Writing 1 automatically clears this field. Writing has no effect when spi_busy is 1. That is, no new SPI operation is started before the previous read/write operation is complete. If a start request is sent, hardware ignores this request.
[23]	RW	spi_rw	SPI operation type 0: write 1: read
[22:16]	RW	spi_add	SPI operation address The address range is 0–127.
[15:8]	RO	spi_rdata	Data read from the SPI
[7:0]	RW	spi_wdata	Data to be written to the SPI

CONVER_T

CONVER_T is an external temperature sensor DS1820 (DS18B20) measurement control register.

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Offset Address	Register Name	Total Reset Value
0x0080	CONVER_T	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															conver_t
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	conver_t	External temperature sensor DS1820 (DS18B20) measurement start. This field must be set to 1. 1: start. Hardware automatically sets this field to 0 when the interrupt is cleared.

CRC_EN

CRC_EN is an external temperature sensor DS1820 (DS18B20) measurement CRC check enable register.

Offset Address	Register Name	Total Reset Value
0x0084	CRC_EN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															crc_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	crc_en	External temperature sensor DS1820 (DS18B20) measurement CRC check enable 0: disabled 1: enabled

INT_MASK

INT_MASK is an external temperature sensor DS1820 (DS18B20) measurement interrupt mask register.

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Offset Address	Register Name	Total Reset Value
0x0088	INT_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															int_mask
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	int_mask	External temperature sensor DS1820 (DS18B20) measurement interrupt mask 0: not masked 1: masked

INT_CLEAR

INT_CLEAR is an external temperature sensor DS1820 (DS18B20) measurement interrupt clear register.

Offset Address	Register Name	Total Reset Value
0x008C	INT_CLEAR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															int_clear
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	int_clear	External temperature sensor DS1820 (DS18B20) measurement interrupt clear Writing 1 clears the interrupt. Hardware automatically sets this field to 0 after the interrupt is cleared.

BUSY

BUSY is an external temperature sensor DS1820 (DS18B20) measurement status register.

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Offset Address	Register Name	Total Reset Value
0x0090	BUSY	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															busy
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	busy	External temperature sensor DS1820 (DS18B20) measurement status 0: ready 1: busy

INT_RAW

INT_RAW is an external temperature sensor DS1820 (DS18B20) raw measurement interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0094	INT_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														int_err	get_tmprt_int
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RO	int_err	Error interrupt
[0]	RO	get_tmprt_int	External temperature sensor DS1820 (DS18B20) measurement completion interrupt

INT_TCAP

INT_TCAP is an external temperature sensor DS1820 (DS18B20) measurement interrupt status register.

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Offset Address	Register Name	Total Reset Value
0x0098	INT_TCAP	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															int_tcap
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	int_tcap	Masked interrupt status 0: No interrupt is generated. 1: An interrupt is generated.

T_VALUE

T_VALUE is an external temperature sensor DS1820 (DS18B20) measurement value register.

Offset Address	Register Name	Total Reset Value
0x009C	T_VALUE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				t_value											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RO	t_value	Temperature measured by the external temperature sensor DS1820 (DS18B20)

FILTER_NUM

FILTER_NUM is a filter glitch width configuration register.

Offset Address	Register Name	Total Reset Value
0x00A0	FILTER_NUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												filter_num			

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	filter_num	Filter glitch width for inputs. The glitch with the width of (*N* + 1) APB clocks is filtered.

3.9.7 Internal Register Description

RTC_10MS_COUNT

RTC_10MS_COUNT is a count value register for the RTC 10 ms counter.

Offset Address	Register Name	Total Reset Value
0x00	RTC_10MS_COUNT	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved	rtc_10ms_count						
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7]	RO	reserved	Reserved
[6:0]	RO	rtc_10ms_count	RTC 10 ms counter value. It indicates the currently counted time. Its unit is 10 ms. The value range is 0–99.

RTC_S_COUNT

RTC_S_COUNT is a count value register for the RTC second counter.

Offset Address	Register Name	Total Reset Value
0x01	RTC_S_COUNT	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved		rtc_s_count					

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Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:6]	RO	reserved	Reserved
[5:0]	RO	rtc_s_count	RTC second counter value. It indicates the currently counted seconds. The value range is 0–59.

RTC_M_COUNT

RTC_M_COUNT is a count value register for the RTC minute counter.

Offset Address	Register Name	Total Reset Value
0x02	RTC_M_COUNT	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved			rtc_m_count				
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:6]	RO	reserved	Reserved
[5:0]	RO	rtc_m_count	RTC minute counter value. It indicates the currently counted minutes. The value range is 0–59.

RTC_H_COUNT

RTC_H_COUNT is a count value register for the RTC hour counter.

Offset Address	Register Name	Total Reset Value
0x03	RTC_H_COUNT	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved			rtc_h_count				
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:5]	RO	reserved	Reserved

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[4:0]	RO	rtc_h_count	RTC hour counter value. It indicates the currently counted hours. The value range is 0–23.

RTC_D_COUNT_L

RTC_D_COUNT_L is a lower-8-bit count value register for the RTC day counter.

Offset Address	Register Name	Total Reset Value
0x04	RTC_D_COUNT_L	0x00

Bit	7	6	5	4	3	2	1	0
Name	rtc_d_count_l							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RO	rtc_d_count_l	Lower eight bits of the RTC day counter value. This field works with RTC_D_COUNT_H to indicate the currently counted days. The day range is 0–65535.

RTC_D_COUNT_H

RTC_D_COUNT_H is an upper-8-bit count value register for the RTC day counter.

Offset Address	Register Name	Total Reset Value
0x05	RTC_D_COUNT_H	0x00

Bit	7	6	5	4	3	2	1	0
Name	rtc_d_count_h							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RO	rtc_d_count_h	Upper eight bits of the RTC day counter value. This field works with RTC_D_COUNT_L to indicate the currently counted day. The day range is 0–65535.

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RTC_MR_10MS

RTC_MR_10MS is a match value register for the RTC 10 ms counter.

Offset Address	Register Name	Total Reset Value
0x06	RTC_MR_10MS	0x7F

Bit	7	6	5	4	3	2	1	0
Name	reserved	rtc_mr_10ms						
Reset	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[7]	RO	reserved	Reserved
[6:0]	RW	rtc_mr_10ms	Match value of the RTC 10 ms counter The value range is 0–99.

RTC_MR_S

RTC_MR_S is a match value register for the RTC second counter.

Offset Address	Register Name	Total Reset Value
0x07	RTC_MR_S	0x3F

Bit	7	6	5	4	3	2	1	0
Name	reserved		rtc_mr_s					
Reset	0	0	1	1	1	1	1	1

Bits	Access	Name	Description
[7:6]	RO	reserved	Reserved
[5:0]	RW	rtc_mr_s	Match value of the RTC second counter The value range is 0–59.

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RTC_MR_M

RTC_MR_M is a match value register for the RTC minute counter.

Offset Address	Register Name	Total Reset Value
0x08	RTC_MR_M	0x3F

Bit	7	6	5	4	3	2	1	0
Name	reserved			rtc_mr_m				
Reset	0	0	1	1	1	1	1	1

Bits	Access	Name	Description
[7:6]	RO	reserved	Reserved
[5:0]	RW	rtc_mr_m	Match value of the RTC minute counter The value range is 0–59.

RTC_MR_H

RTC_MR_H is a match value register for the RTC hour counter.

Offset Address	Register Name	Total Reset Value
0x09	RTC_MR_H	0x1F

Bit	7	6	5	4	3	2	1	0
Name	reserved			rtc_mr_h				
Reset	0	0	0	1	1	1	1	1

Bits	Access	Name	Description
[7:5]	RO	reserved	Reserved
[4:0]	RW	rtc_mr_h	Match value of the RTC hour counter The value range is 0–23.

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RTC_MR_D_L

RTC_MR_D_L is a lower-8-bit match value register for the RTC day counter.

Offset Address	Register Name	Total Reset Value
0x0A	RTC_MR_D_L	0xFF

Bit	7	6	5	4	3	2	1	0
Name	rtc_mr_d_l							
Reset	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[7:0]	RW	rtc_mr_d_l	Lower eight bits of the match value of the RTC day counter. This field works with RTC_MR_D_H to indicate the matched day. The day range is 0–65535.

RTC_MR_D_H

RTC_MR_D_H is an upper-8-bit match value register for the RTC day counter.

Offset Address	Register Name	Total Reset Value
0x0B	RTC_MR_D_H	0xFF

Bit	7	6	5	4	3	2	1	0
Name	rtc_mr_d_h							
Reset	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[7:0]	RW	rtc_mr_d_h	Upper eight bits of the match value of the RTC day counter. This field works with RTC_MR_D_L to indicate the matched day. The day range is 0–65535.

RTC_LR_10MS

RTC_LR_10MS is a configured value register for the RTC 10 ms counter.

Offset Address	Register Name	Total Reset Value
0x0C	RTC_LR_10MS	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved	rtc_lr_10ms						

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Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7]	RO	reserved	Reserved
[6:0]	RW	rtc_lr_10ms	Configured value of the RTC 10 ms counter The value range is 0–99.

RTC_LR_S

RTC_LR_S is a configured value register for the RTC second counter.

Offset Address	Register Name	Total Reset Value
0x0D	RTC_LR_S	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved		rtc_lr_s					
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:6]	RO	reserved	Reserved
[5:0]	RW	rtc_lr_s	Configured value of the RTC second counter The value range is 0–59.

RTC_LR_M

RTC_LR_M is a configured value register for the RTC minute counter.

Offset Address	Register Name	Total Reset Value
0x0E	RTC_LR_M	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved		rtc_lr_m					
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:6]	RO	reserved	Reserved

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[5:0]	RW	rtc_lr_m	Configured value of the RTC minute counter The value range is 0–59.

RTC_LR_H

RTC_LR_H is a configured value register for the RTC hour counter.

Offset Address	Register Name	Total Reset Value
0x0F	RTC_LR_H	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved			rtc_lr_h				
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:5]	RO	reserved	Reserved
[4:0]	RW	rtc_lr_h	Configured value of the RTC hour counter The value range is 0–23.

RTC_LR_D_L

RTC_LR_D_L is a lower-8-bit configured value register for the RTC day counter.

Offset Address	Register Name	Total Reset Value
0x10	RTC_LR_D_L	0x00

Bit	7	6	5	4	3	2	1	0
Name	rtc_lr_d_l							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	rtc_lr_d_l	Lower eight bits of the configured value of the RTC day counter. This field works with RTC_LR_D_H to indicate the configured day. The day range is 0–65535.

RTC_LR_D_H

RTC_LR_D_H is an upper-8-bit configured value register for the RTC day counter.

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Offset Address	Register Name	Total Reset Value
0x11	RTC_LR_D_H	0x00

Bit	7	6	5	4	3	2	1	0
Name	rtc_lr_d_h							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	rtc_lr_d_h	Upper eight bits of the configured value of the RTC day counter. This field works with RTC_LR_D_L to indicate the configured day. The day range is 0–65535.

RTC_LORD

RTC_LORD is an RTC configured value loading enable register.

Offset Address	Register Name	Total Reset Value
0x12	RTC_LORD	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved					rtc_lock_bypass	rtc_lock	rtc_load
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:3]	RO	reserved	Reserved
[2]	RW	rtc_lock_bypass	RTC lock enable 0: enabled. The RTC count value (0x00−0x05) is updated only after the RTC is successfully locked. 1: disabled. The RTC count value (0x00−0x05) is updated in real time.
[1]	RW	rtc_lock	RTC lock. If software writes 1, hardware is automatically cleared after the RTC is successfully locked. Note: This field is valid only when rtc_lock_bypass is 0.
[0]	RW	rtc_load	RTC configured value loading enable. When the field is enabled, the RTC configured value will be loaded to the RTC accumulator. If software writes 1 to load the configured value, hardware will automatically set this field to 0 after successful loading.

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RTC_IMSC

RTC_IMSC is an RTC interrupt enable register.

Offset Address	Register Name	Total Reset Value
0x13	RTC_IMSC	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved							rtc_imsc_time
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:1]	RO	reserved	Reserved
[0]	RW	rtc_imsc_time	RTC timing interrupt enable 0: disabled 1: enabled

RTC_INT_CLR

RTC_INT_CLR is an RTC interrupt clear register.

Offset Address	Register Name	Total Reset Value
0x14	RTC_INT_CLR	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved							rtc_int_clr
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:1]	RO	reserved	Reserved
[0]	WO	rtc_int_clr	RTC interrupt clear register. Writing any value by software clears the interrupt. Reading back the value has no effect.

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RTC_MSC_INT

RTC_MSC_INT is an RTC mask interrupt status register.

Offset Address	Register Name	Total Reset Value
0x15	RTC_MSC_INT	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved							mask_int_time
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:1]	RO	reserved	Reserved
[0]	RO	mask_int_time	Status of the masked RTC timing interrupt

RTC_RAW_INT

RTC_RAW_INT is an RTC raw interrupt status register.

Offset Address	Register Name	Total Reset Value
0x16	RTC_RAW_INT	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved							raw_int_time
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:1]	RO	reserved	Reserved
[0]	RO	raw_int_time	Status of the raw RTC timing interrupt

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RTC_CLK

RTC_CLK is an RTC output clock select register.

Offset Address	Register Name	Total Reset Value
0x17	RTC_CLK	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved						clk_out_sel	
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:2]	RO	reserved	Reserved
[1:0]	RW	clk_out_sel	Output test clock of the RTC 00: crystal oscillator clock 01: corrected 100 Hz clock 1x: 1 Hz clock

RTC_POR_N

RTC_POR_N is an RTC reset control register.

Offset Address	Register Name	Total Reset Value
0x18	RTC_POR_N	0x01

Bit	7	6	5	4	3	2	1	0
Name	reserved							rtc_por_n
Reset	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[7:1]	RO	reserved	Reserved
[0]	RW	rtc_por_n	RTC reset. This field is automatically set to 1 after the RTC is successfully reset. 0: reset

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SDM_COEF_OUSIDE_H

SDM_COEF_OUSIDE_H is an upper 4-bit external frequency divider register.

Offset Address	Register Name	Total Reset Value
0x51	SDM_COEF_OUSIDE_H	0x8

Bit	7	6	5	4	3	2	1	0
Name	Reserved				Sdm_coef_ouside_h			
Reset	0	0	0	0	1	0	0	0

Bits	Access	Name	Description
[7:4]	RO	reserved	Reserved
[3:0]	RW	Sdm_coef_ouside_h	Upper four bits of the frequency divider

SDM_COEF_OUSIDE_L

SDM_COEF_OUSIDE_L is a lower 8-bit external frequency divider register.

Offset Address	Register Name	Total Reset Value
0x52	SDM_COEF_OUSIDE_L	0x1B

Bit	7	6	5	4	3	2	1	0
Name	sdm_coef_ouside_l							
Reset	0	0	0	1	1	0	1	1

Bits	Access	Name	Description
[7:0]	RW	sdm_coef_ouside_l	Lower eight bits of the frequency divider Note: When clock divider registers (addresses of 0x51 and 0x52) are read or written, you need to read or write to the upper four bits and then the lower eight bits continuously. The clock divider is calculated as follows: (f − 32700) x 30.52. f indicates the oscillation frequency of the external crystal and ranges from 32700 to 32799.

USER_REGISTER1

USER_REGISTER1 is 64-bit user register 1.

Offset Address	Register Name	Total Reset Value
0x53	USER_REGISTER1	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register1							

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Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register1	64-bit user register 1 corresponding to bit[7:0]

USER_REGISTER2

USER_REGISTER2 is 64-bit user register 2.

Offset Address	Register Name	Total Reset Value
0x54	USER_REGISTER2	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register2							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register2	64-bit user register 2 corresponding to bit[15:8]

USER_REGISTER3

USER_REGISTER3 is 64-bit user register 3.

Offset Address	Register Name	Total Reset Value
0x55	USER_REGISTER3	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register3							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register3	64-bit user register 3 corresponding to bit[23:16]

USER_REGISTER4

USER_REGISTER4 is 64-bit user register 4.

Offset Address	Register Name	Total Reset Value
0x56	USER_REGISTER4	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register4							

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Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register4	64-bit user register 4 corresponding to bit[31:24]

USER_REGISTER5

USER_REGISTER5 is 64-bit user register 5.

Offset Address	Register Name	Total Reset Value
0x57	USER_REGISTER5	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register5							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register5	64-bit user register 5 corresponding to bit[39:32]

USER_REGISTER6

USER_REGISTER6 is 64-bit user register 6.

Offset Address	Register Name	Total Reset Value
0x58	USER_REGISTER6	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register6							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register6	64-bit user register 6 corresponding to bit[47:40]

USER_REGISTER7

USER_REGISTER7 is 64-bit user register 7.

Offset Address	Register Name	Total Reset Value
0x59	USER_REGISTER7	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register7							

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Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register7	64-bit user register 7 corresponding to bit[55:48]

USER_REGISTER8

USER_REGISTER8 is 64-bit user register 8.

Offset Address	Register Name	Total Reset Value
0x5A	USER_REGISTER8	0x0

Bit	7	6	5	4	3	2	1	0
Name	user_register8							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	user_register8	64-bit user register 8 corresponding to bit[63:56]

3.10 Power Management and Low-Power Mode Control

3.10.1 Overview

In low-power mode, the power consumption of the chip is reduced effectively. The Hi3518E V20X/Hi3516C V200 reduces its power consumption in the following low-power control modes:

- Clock gating and clock frequency adjustment

  The clock disabling function is used to disable unnecessary clocks to reduce the power consumption of the chip. In addition, the frequency of the system working clock can be adjusted. That is, when the function requirement is met, you can adjust the clock frequency to reduce the power consumption of the chip.
- Module low-power control

  When a module is idle, it can be disabled or switched to low-power mode to reduce the power consumption.
- DDR low-power control

  The power consumption of the DDRC and related pins can be controlled dynamically. You can enable this function to reduce the power consumption of the chip. You can also enable the self-refresh function of the DDR to reduce the power consumption of the entire product.
- Adaptive voltage scaling (AVS) based on core timing monitoring

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3.10.2 Clock Gating and Clock Frequency Adjustment

The system supports clock gating of each module. When a module is idle, its clock can be disabled to reduce chip power consumption. For details about the process, see the description in section "clock gating" of each module.

The system can adjust its working frequency to reduce chip power consumption. To adjust the system working frequency, perform the following steps:

- Disable the service module to prevent it from accessing the DDR.
- Enable the system to run in the off-chip memory.

When DDRC_SREFCTRL[sr_seq] is set to a valid value, the DDRC forces the DDR*n* SDRAM to enter the self-refresh mode.

Step 1 Set PERI_CRG12 to select the clock sources for the SYS AXI, DDRC, and CPU.

Step 2 Wait and configure DDRC_SREF_CTRL to enable the DDRC to exit the self-refresh mode based on the configuration requirements of the DDRC.

Step 3 The program starts to run in the DDR.

----End

The working frequencies of some modules can also be adjusted separately. This reduces the power consumption of the system further. For details about the clock source of each module, see section 3.2.3 "Clock Distribution".

3.10.3 Module Low-Power Control

Most modules including USB 2.0 host/Device, and PLL modules support low-power operating modes. For details, see the descriptions of the system controller, and clocks.

- Low-power control for the USB 2.0 host/Device module:

  If the USB 2.0 host is not used, it can be disabled to switch the USB 2.0 host clock by setting PERI_CRG46 bit[7] to 0.
- The PLL also supports the low-power mode. If the PLL is not used, it can be disabled to switch the system to low-power mode.
  - If the APLL is not used, you can set PERI_CRG1 bit[23] to 1 to disable it. Then the APLL enters the low-power mode.
  - If VPLL is not used, you can set PERI_CRG5 bit[23] to 1 to disable it. Then VPLL enters the low-power mode.
  - If the EPLL is not used, you can set PERI_CRG9 bit[23] to 1 to disable it. Then the EPLL enters the low-power mode.

3.10.4 DDR Low-Power Control

For details about the low-power control mode of the DDRC, see the description of "Configuring the Low-Power Mode" in section 4.1.4 "Operating Mode."

3.10.5 Core AVS

The AVS technology further reduces power consumption by dynamically adjusting the voltage in real time based on the chip technology, temperature, and circuit timings. The system obtains the chip PVT information by using the power management controller (PMC) and

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implements automatic voltage scaling by using the PWM output signal in the PMC. For details, see section 3.11 "PMC."

3.11 PMC

3.11.1 Function Description

The PMC provides basic information about power consumption management of the chip and controls the power consumption.

- PWM output

  The PMC provides a programmable PWM module. The peripheral PWM voltage scaling circuit can be used to adjust the power supply voltage of the chip through the PWM outputs of the chip.
- High performance monitor (HPM) control

  The HPM can be used to obtain the comprehensive PVT information about the chip. The reference power supply voltage scaling solution can be developed based on the PVT information.
- Hardware AVS

  After the hardware AVS function is enabled, the chip obtains the HPM monitoring value periodically. The PWM output duty cycle is adjusted based on the configured policy to implement adaptive adjustment of the power supply voltage.
- Internal temperature detection

  The chip has an integrated temperature sensor for obtaining the internal chip temperature and providing temperature information for temperature protection.

3.11.2 Operating Mode

PWM Output

The PWM module in the PMC has a 24 MHz working clock, a 16-bit cycle counter, and a 16-bit duty cycle counter. The count cycle and duty cycle of the PWM output can be specified by configuring the PWM-related registers in the PMC.

To configure the count cycle and duty cycle of the PWM output, perform the following steps:

Step 1 Calculate the number of PWM count cycles and the count value of the high level based on the required PWM output frequency and duty cycle.

The number of count cycles is calculated as follows:

$$pwm_period = (24000000/Freq) - 1$$

The count value of the high level is calculated as follows:

$$pwm_duty = (24000000/Freq) * duty - 1$$

Step 2 Configure the PWM count cycle and high level count value register.

Step 3 Enable the PWM output.

----End

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For example, the voltage of the core power supply is controlled by the PWM0 signal. Perform the following steps if the required PWM0 output frequency is 200 kHz and the duty cycle is 75%:

Step 1 Calculate the required number of count cycles and count value of the high level.

Pwm_period = (24000000/200000) − 1 = 119

Pwm_duty = (24000000/200000) x 0.75 – 1 = 89

Step 2 Configure the parameters of the PWM corresponding to the core power supply by setting PERI_PMC0 bit[15:0] to **0x77** and PERI_PMC0 bit[31:16] to **0x59**.

Step 3 Enable the PWM corresponding to the core power supply by setting PERI_PMC4 bit[0] to **1**.

----End

HPM Control

The chip has only one core HPM for obtaining the PVT information about the chip.

The source clock of the HPM reference clock is the CPU working clock. You are advised to set core_hpm_div to an appropriate value to ensure that the HPM monitoring value is precise and the frequency of the HPM reference clock is close to 50 MHz.

The following describes the HPM configuration procedure by taking the core HPM monitoring value as an example:

Step 1 Set the HPM capture mode.

The HPM has two capture modes: single capture mode and cyclic capture mode. For details, see the description of PERI_PMC10.

If the cyclic capture mode is used, the capture cycle needs to be configured. The capture cycle T (in ms) is calculated as follows: T = (N x 2048 + M x 16)/1000 (N = PERI_PMC13 bit[31:24], M = PERI_PMC31 bit[7:0]).

Step 2 Calculate the value of core_hpm_div based on the CPU frequency, deassert reset, and start the HPM capture.

For example, if the CPU frequency is 600 MHz, core_hpm_div is calculates as follows: core_hpm_div = (600 MHz/50 MHz) – 1 = 11. In this case, set PERI_PMC10 to 0x0400000B and then 0x0500000B. Then enable the cyclic HPM capture mode.

Step 3 Wait until core_hpm_pc_valid is 1. Then software reads the HPM monitoring value.

Step 4 In single capture mode, only HPM original code pattern 0 recorded in PERI_PMC11 bit[9:0] is valid.

In cyclic capture mode, the latest four HPM original code patterns (patterns 0−3) are recorded in PERI_PMC11 bit[9:0], PERI_PMC11 bit[21:12], PERI_PMC12 bit[9:0], and PERI_PMC12 bit[21:12]. The latest recorded value is HPM original code pattern 0.

----End

Hardware AVS

The chip automatically adjusts the PWM output duty cycle based on the HPM monitoring value by using the hardware AVS function of the PMC to implement automatic power supply voltage scaling.

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The start process of the hardware AVS function is as follows:

Step 1 Specify the hardware AVS calculation mode by configuring PERI_PMC5 bit[2:1].

Step 2 Specify the voltage scaling step of hardware AVS by configuring PERI_PMC5 bit[31:16].

Step 3 Enable hardware AVS by setting PERI_PMC5 bit[0] to 1.

----End

Internal Temperature Detection

The chip has an integrated temperature sensor and the detected temperature ranges from −40ºC (−40ºF) to +140ºC (284ºF). To enable the temperature sensor to collect data, perform the following steps:

Step 1 Set the capture mode of the temperature sensor.

The temperature sensor captures the temperature in two modes: single capture mode and cyclic capture mode. For details, see the description of PERI_PMC68.

If the cyclic capture mode is used, the capture cycle needs to be configured. The capture cycle (in ms) is calculated as follows: Capture cycle = tsensor_monitor_period x 2.

Step 2 Enable the temperature sensor to start collecting the temperature code.

Step 3 Software reads the temperature code captured by the temperature sensor.

In single capture mode, only temperature code 0 recorded in PERI_PMC69 bit[7:0] is valid.

In cyclic capture mode, the latest eight temperature codes (codes 0−7) are recorded in PERI_PMC69 bit[31:0] and PERI_PMC70 bit[31:0]. The latest recorded temperature data is temperature code 0.

Step 4 Calculate the corresponding temperature value based on the recorded temperature code.

The temperature value T (in ºC) is calculated as follows: T = (tsensor_temp_code x 180/256) − 40.

----End

3.11.3 PMC Register Summary

Table 3-22 describes PMC registers.

Table 3-22 Summary of PMC registers (base address: 0x2027_0000)

Offset Address	Register	Description	Page
0x0000	PERI_PMC0	PMC register 0	3-163
0x0010	PERI_PMC4	PMC register 4	3-163
0x0014	PERI_PMC5	PMC register 5	3-164
0x0028	PERI_PMC10	Core HPM control register 1	3-165
0x002C	PERI_PMC11	Core HPM status register 1	3-166

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Offset Address	Register	Description	Page
0x0030	PERI_PMC12	Core HPM status register 2	3-167
0x0034	PERI_PMC13	Core HPM control register 2	3-167
0x007C	PERI_PMC31	HPM control register (fine 2)	3-168
0x110	PERI_PMC68	PMC register 68	3-168
0x0114	PERI_PMC69	PMC register 69	3-169
0x0118	PERI_PMC70	PMC register 70	3-170
0x011C	PERI_PMC71	PMC register 71	3-170

3.11.4 PMC Register Description

PERI_PMC0

PERI_PMC0 is PMC register 0.

Offset Address	Register Name	Total Reset Value
0x0000	PERI_PMC0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	core_pwm_duty																core_pwm_period															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	core_pwm_duty	Number of PWM high-level beats for the core AVS output. If the number is greater than or equal to the number of PWM cycles, the output level is always high. The value cannot be 0. If the configured value is 0, the value 1 is used.
[15:0]	RW	core_pwm_period	Number of PWM cycles for the core AVS output

PERI_PMC4

PERI_PMC4 is PMC register 4.

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Offset Address	Register Name	Total Reset Value
0x0010	PERI_PMC4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														core_pwm_inv	core_pwm_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	core_pwm_inv	PWM phase control for the core AVS output 0: positive-phase output 1: negative-phase output
[0]	RW	core_pwm_enable	PWM enable for the core AVS output 0: disabled 1: enabled

PERI_PMC5

PERI_PMC5 is PMC register 5.

Offset Address	Register Name	Total Reset Value
0x0014	PERI_PMC5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	core_avs_fsm_inc_pwm_step								core_avs_fsm_dec_pwm_step								reserved														core_avs_hpm_fsm_calc_mode	core_hpm_fsm_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	core_avs_fsm_inc_pwm_step	Automatic increase step of the core AVS FSM PWM

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[23:16]	RW	core_avs_fsm_dec_pwm_step	Automatic decrease step of the core AVS FSM PWM
[15:3]	RO	reserved	Reserved
[2:1]	RW	core_avs_hpm_fsm_calc_mode	Calculation mode of the core AVS HPM FSM 00: average value of four samplings 01: average value of eight samplings 10: average value of 16 samplings 11: average value of 32 samplings
[0]	RW	core_hpm_fsm_en	Core AVS HPM FSM enable 0: disabled 1: enabled

PERI_PMC10

PERI_PMC10 is core HPM control register 1.

Offset Address	Register Name	Total Reset Value
0x0028	PERI_PMC10	0x0000_000A

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				core_hpm_srst_req	core_hpm_monitor_en	reserved	core_hpm_en	reserved				core_hpm_offset								reserved		core_hpm_shift		reserved				core_hpm_div			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27]	RW	core_hpm_srst_req	Core HPM reset request, active high 0: deassert reset 1: reset
[26]	RW	core_hpm_monitor_en	Core HPM cyclic monitoring enable 0: disabled 1: enabled
[25]	RO	reserved	Reserved

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[24]	RW	core_hpm_en	Core HPM measurement enable 0: The value retains 0 before a process is started. 1: A frequency modulation process is started.
[23:22]	WO	reserved	Reserved
[21:12]	RW	core_hpm_offset	Core HPM offset
[11:10]	RO	reserved	Reserved
[9:8]	RW	core_hpm_shift	Core HPM shift
[7:6]	RO	reserved	Reserved
[5:0]	RW	core_hpm_div	Divider of the core HPM reference clock *n*: (*n* + 1) frequency division (*n* must be greater than 0)

PERI_PMC11

PERI_PMC11 is core HPM status register 1.

Offset Address	Register Name	Total Reset Value
0x002C	PERI_PMC11	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved						core_hpm_up_warning	core_hpm_low_warning	reserved				core_hpm_pc_record1								reserved	core_hpm_pc_valid	core_hpm_pc_record0									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25]	RO	core_hpm_up_warning	Upper limit for the core HPM warning
[24]	RO	core_hpm_low_warning	Lower limit for the core HPM warning
[23:22]	RO	reserved	Reserved
[21:12]	RO	core_hpm_pc_record1	Core HPM original code pattern 1
[11]	RO	reserved	Reserved

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[10]	RO	core_hpm_pc_valid	Core HPM output validity indicator 0: invalid 1: valid
[9:0]	RO	core_hpm_pc_record0	Core HPM original code pattern 0

PERI_PMC12

PERI_PMC12 is core HPM status register 2.

Offset Address	Register Name	Total Reset Value
0x0030	PERI_PMC12	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				core_hpm_rcc				reserved				core_hpm_pc_record3								reserved				core_hpm_pc_record2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:24]	RO	core_hpm_rcc	Core HPM output with the RCC code
[23:22]	RO	reserved	Reserved
[21:12]	RO	core_hpm_pc_record3	Core HPM original code pattern 3
[11:10]	RO	reserved	Reserved
[9:0]	RO	core_hpm_pc_record2	Core HPM original code pattern 2

PERI_PMC13

PERI_PMC13 is core HPM control register 2.

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Offset Address	Register Name	Total Reset Value
0x0034	PERI_PMC13	0x0100_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	core_hpm_monitor_period								reserved				core_hpm_lowlimit								reserved				core_hpm_uplimit							
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	core_hpm_monitor_period	Core HPM cyclic monitoring period If the configured value is *N*, the monitoring period (T) is as follows: T = *N* x 2048/1000 ms The maximum monitoring interval is 522 ms, and the minimum interval is 2 ms.
[23:22]	RO	reserved	Reserved
[21:12]	RW	core_hpm_lowlimit	Core HPM lower limit
[11:10]	RO	reserved	Reserved
[9:0]	RW	core_hpm_uplimit	Core HPM upper limit

PERI_PMC31

PERI_PMC31 is an HPM control register (fine 2).

Offset Address	Register Name	Total Reset Value
0x007C	PERI_PMC31	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								core_hpm_monitor_period							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	core_hpm_monitor_period	Parameter for fine tuning the core HPM cyclic monitoring cycle

PERI_PMC68

PERI_PMC68 is PMC register 68.

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Offset Address	Register Name	Total Reset Value
0x110	PERI_PMC68	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	tsensor_en	tsensor_monitor_en	reserved	reserved				tsensor_monitor_period								tsensor_temp_uplimit								tsensor_temp_lowlimit							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30]	RW	tsensor_en	Temperature sensor enable 0: disabled 1: enabled
[29]	RW	tsensor_monitor_en	Cyclic temperature monitoring enable 0: single mode 1: cyclic mode
[28]	RO	reserved	Reserved
[27:24]	RO	reserved	Reserved
[23:16]	RW	tsensor_monitor_period	Cyclic temperature monitoring cycle. The timing reference is 2 ms.
[15:8]	RW	tsensor_temp_uplimit	Temperature overflow value
[7:0]	RW	tsensor_temp_lowlimit	Temperature underflow value

PERI_PMC69

PERI_PMC69 is PMC register 69.

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Offset Address	Register Name	Total Reset Value
0x0114	PERI_PMC69	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tsensor_temp_code3								tsensor_temp_code2								tsensor_temp_code1								tsensor_temp_code0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	tsensor_temp_code3	Temperature record value 3
[23:16]	RO	tsensor_temp_code2	Temperature record value 2
[15:8]	RO	tsensor_temp_code1	Temperature record value 1
[7:0]	RO	tsensor_temp_code0	Temperature record value 0

PERI_PMC70

PERI_PMC70 is PMC register 70.

Offset Address	Register Name	Total Reset Value
0x0118	PERI_PMC70	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tsensor_temp_code7								tsensor_temp_code6								tsensor_temp_code5								tsensor_temp_code4							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	tsensor_temp_code7	Temperature record value 7
[23:16]	RO	tsensor_temp_code6	Temperature record value 6
[15:8]	RO	tsensor_temp_code5	Temperature record value 5
[7:0]	RO	tsensor_temp_code4	Temperature record value 4

PERI_PMC71

PERI_PMC71 is PMC register 71.

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Offset Address	Register Name	Total Reset Value
0x011C	PERI_PMC71	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														tsensor_low_warning	tsensor_up_warning
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RO	tsensor_low_warning	Temperature underflow alarm, active high
[0]	RO	tsensor_up_warning	Temperature overflow alarm, active high

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4 Memory Interfaces

4.1 DDRC

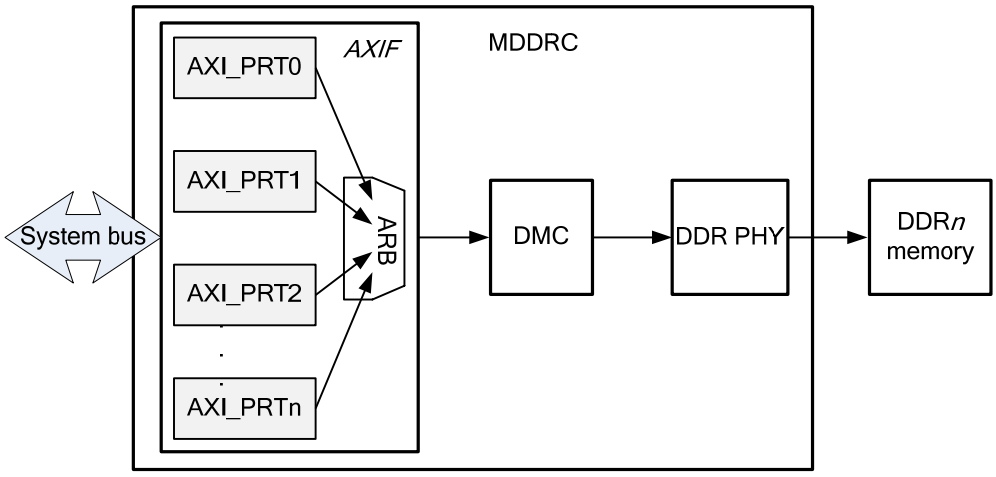
4.1.1 Overview

The DDR2/3/3L SDRAM controller (DDRC) controls access to the DDR2/3/3L SDRAM. Figure 4-1 shows the DDRC structure.

NOTE

Hi3518E V200/Hi3518E V201 has an embedded DDR2 SDRAM. Hi3516C V200 supports the DDR2/3/3L SDRAM. Unless otherwise specified, DDR2/3/3L SDRAM is represented by DDR*n* SDRAM in the following sections.

Figure 4-1 DDRC structure



4.1.2 Features

The DDRC has the following features:

- One DDRn SDRAM chip select (CS)
- 16-bit DDR*n* SDRAM data bus

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- Maximum 512 Mbits storage space for Hi3518E V200; maximum 256 Mbits storage space for Hi3518E V201; maximum 4 Gbits storage space for Hi3516C V200
- Maximum 360 MHz working frequency and 720 Mbit/s data rate of the DDR2 SDRAM for Hi3518E V200/Hi3518E V201; maximum 400 MHz working frequency and 800 Mbit/s data rate of the DDR*n* SDRAM and for Hi3516C V200
- Various DDRn SDRAM low-power modes, such as the power-down and self-refresh modes
- Burst8 transfer mode for the DDR*n* SDRAM

4.1.3 Function Description

4.1.3.1 Application Block Diagram

By using the DDRC, the master devices of the system-on-chip (SoC) such as the CPU can access the external DDR*n* SDRAM. It supports the DDR2 SDRAM and DDR3 SDRAM complying with the JEDEC (JESD79) standard after the timing parameter registers of the DDRC are configured by using the CPU.

Table 4-1 lists the DDR3 SDRAMs supported by the DDRC. The descriptions in Table 4-1 are based on the working frequencies of DDR3 SDRAMs. The restrictions such as the capacity are not taken into consideration.

Table 4-1 DDR3 SDRAMs supported by the DDRC

Vendor	400 MHz	533 MHz
JESD79 (DDR3 standard)	DDR3-800 DDR3-1066 DDR3-1333 DDR3-1600	DDR3-1066 DDR3-1333 DDR3-1600

Table 4-2 lists the mainstream DDR2 SDRAMs supported by the DDRC. The descriptions in Table 4-2 are based on the working frequencies of DDR2 SDRAMs. The restrictions such as the capacity are not taken into consideration.

Table 4-2 DDR2 SDRAMs supported by the DDRC

Vendor	333 MHz	400 MHz	533 MHz
JESD79 (DDR2 Standard)	DDR2-667 DDR2-800	DDR2-800	None
Micron	3 DDR2-667 3E DDR2-667 25 DDR2-800 25E DDR2-800	25 DDR2-800 25E DDR2-800	187E DDR2-1066
ELPIDA	6E DDR2-667 6C DDR2-667 8E DDR2-800	8E DDR2-800	None

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Vendor	333 MHz	400 MHz	533 MHz
Hynix	Y4 DDR2-667 Y5 DDR2-667 S5 DDR2-800 S6 DDR2-800	S5 DDR2-800 S6 DDR2-800	None
Samsung	E6 DDR2-667 E7 DDR2-800	E7 DDR2-800	None

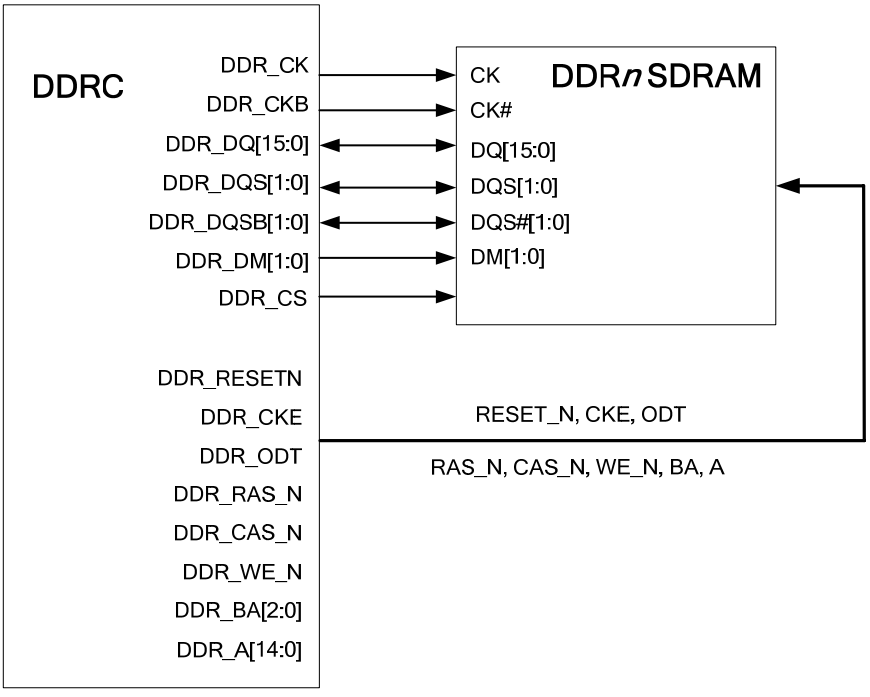
NOTE

For details about the supported component types, see the JEDEC standard and the component data sheet.

The DDRC supports the 16-bit DDR SDRAM. It can connect to one 16-bit DDR*n* SDRAM or two 8-bit DDR*n* SDRAMs.

Figure 4-2 shows the connection between the DDRC and a 16-bit DDR*n* SDRAM.

Figure 4-2 Connecting the DDRC to a 16-bit DDR*n* SDRAM



NOTE

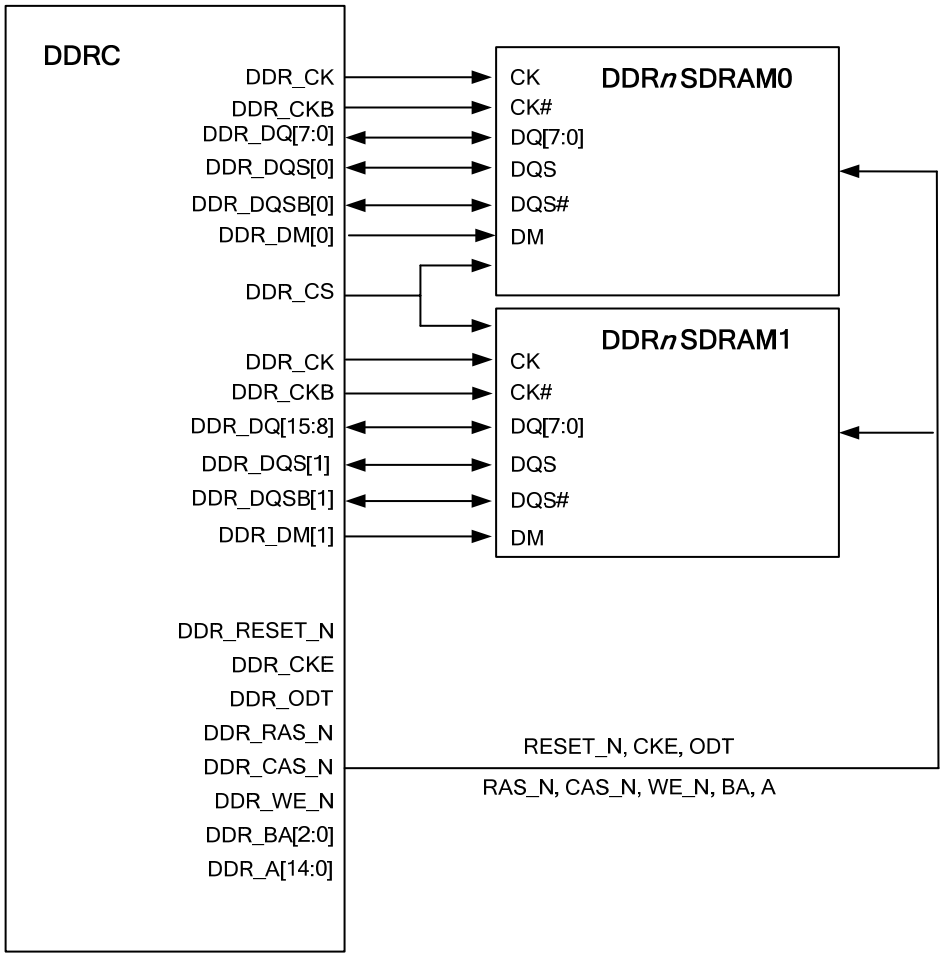
- The DDR*n* SDRAM is a 16-bit memory.

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- The DDRC command control signals (DDR_CS, DDR_CKE, DDR_ODT, DDR_RESET_N, DDR_RAS_N, DDR_CAS_N, DDR_WE_N, DDR_BA, and DDR_A) connect to the command control signals of the DDR*n* SDRAM respectively. The DDRC command control bus uses the one-drive-one mode. Note that the bit widths of DDR_A and DDR_BA vary according to the capacity of the DDR*n* SDRAM.
- If the DDR2 SDRAM is used, the DDR_RESET signal does not need to be connected.

Figure 4-3 shows the connection between the DDRC and two 8-bit DDR*n* SDRAMs.

Figure 4-3 Connecting the DDRC to two 8-bit DDR*n* SDRAMs



NOTE

- The DDR*n* SDRAM*x* is an 8-bit memory.
- The DDRC command control signals (DDR_CS, DDR_CKE, DDR_ODT, DDR_RESET_N, DDR_RAS_N, DDR_CAS_N, DDR_WE_N, DDR_BA, and DDR_A) connect to the command control signals of the DDR*n* SDRAM*x* respectively. The DDRC command control bus uses the one-drive-two mode (DDR_CS is one-drive-one). Note that the bit widths of DDR_A and DDR_BA vary according to the capacity of the DDR*n* SDRAM.
- If the DDR2 SDRAM is used, the DDR_RESET signal does not need to be connected.

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4.1.3.2 Function Implementation

As the timings of the DDRC interface comply with the JESD79 standard, the DDRC can access (read or write) data in the DDRn SDRAM and control the status of the DDRn SDRAM (including automatic refresh and low power control) by sending the command words of the DDRn SDRAM.

Command Truth Value Table

The DDRC can read and write to the DDRn SDRAM and control command words. Table 4-3 lists the command truth values of the DDRC. For details, see the JEDEC standard and component data sheet.

Table 4-3 Command truth values of the DDRC

Function	DDRn_CKE	DDRn_CSN	DDRn_RASN	DDRn_CASN	DDRn_WEN	DDRn_ADR			DDRn_BA
						15:11	AP(10)	9:0	
DESELECT	H	H	X	X	X	X	X	X	X
ACTIVE	H	L	L	H	H	V	V	V	V
READ	H	L	H	L	H	V	V	V	V
WRITE	H	L	H	L	L	V	V	V	V
PRECHARGE	H	L	L	H	L	V	L	V	BA
PRECHARGE ALL	H	L	L	H	L	V	H	V	V
AUTO REFRESH	H	L	L	L	H	V	V	V	V
SELF REFRESH ENTRY	H->L	L	L	L	H	V	V	V	V
SELF REFRESH EXIT	L->H	L	H	H	H	V	V	V	V
MODE REGISTER SET	H	L	L	L	L	V	V	V	V
ZQCL	H	L	H	H	L	X	H	X	X
ZQCS	H	L	H	H	L	X	L	X	X

NOTE

- H indicates high level; L indicates low level; V indicates valid; X indicates ignored.
- **ZQ Calibration Long** (ZQCL): It is used to start ZQ calibration on the DDR3 SDRAM when the DDR3 SDRAM is initialized during power-on. This command does not apply to the DDR2 SDRAM.
- **ZQ Calibration Short** (ZQCS): It is used to start ZQ calibration on the DDR3 SDRAM when the ambient environment is changed. This command does not apply to the DDR2 SDRAM.

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Auto Refresh

When DDRC_CFG_TIMING2[taref] is set to a non-zero value, the DDRC refreshes the DDRn SDRAM by automatically generating a periodical auto refresh command. At ambient temperature, the DDRn SDRAM is required to implement 8,192 auto-refresh operations within 64 ms. That is, the auto-refresh cycle is 7.8 µs. The relationship between the value of DDRC_CFG_TIMING2[taref] (taref) and the auto-refresh cycle T (T = 7.8 µs) is as follows:

T ≥ taref x (16 x DDR clock cycle)

When DDRC_CFG_TIMING2[taref] is configured, the internal counter of the DDRC automatically loads the taref value and then counts in decremented mode. When the count value reaches 0, the DDRC initiates an auto-refresh operation and the counter reloads the taref value to count.

Low-Power Management

The DDRC supports two modes of low-power management: common low-power mode and auto-refresh low-power mode.

When DDRC_CFG_PD[pd_en] is set to 1 to enable the SDRAM low-power mode, and the system is idle (the DDR is not read or written through the DDRC bus interface for a period), the DDRn SDRAM enters the common low-power mode automatically.

To switch the system mode to standby mode, you can force the DDRn SDRAM to enter the auto-refresh mode by setting DDRC_CTRL_SREF[sref_req] to 1. In this mode, the power consumption of the DDRn SDRAM is minimized, data in the DDRn SDRAM is retained, but the system cannot access the DDRn SDRAM.

Arbitration Mechanism

The DDRC uses the priority scheduling algorithm and flow control algorithm.

The priority mapping mode can be selected by configuring AXI_QOSCFG0n[pri_map_mode]. If AXI_QOSCFG0n[pri_map_mode] is set to 1, the system bus signal is used as the QoS priority. If AXI_QOSCFG0n[pri_map_mode] is set to 0, AXI_QOSCFG0n[id_map] needs to be configured to select some bits of the ID for QoS configuration. The DDRC adds priority attributes to the read/write commands sent to the bus by configuring AXI_WRPRIn and AXI_RDPRI, and then schedules the commands based on the priority attributes to implement highly effective access to the DDRn SDRAM.

The DDRC adds response timeout attributes to commands sent to the bus by setting AXI_WRTOUT and AXI_RDTOUT to non-zero values. The mapping mode of the timeout value can be selected by configuring AXI_QOSCFG0n[tout_map_mode]. If AXI_QOSCFG0n[tout_map_mode] is set to 1, the 2 bit select signals of the system bus are used the address to search for one of the four preset timeout values. If AXI_QOSCFG0n[tout_map_mode] is set to 0, AXI_QOSCFG0n[tout_id_map] needs to be configured to select 2 bit signals of the ID as the address to search for one of the four preset timeout values. Then the bus command response timeout is ensured based on the response timeout priority scheduling algorithm.

The DDRC adds the flow control attributes to bus interfaces by configuring DDRC_CFG_PERF[flux_en] and AXI_FLUX0n[flux_cfg0]. The DDRC allocates traffic to each bus interface to ensure sufficient bandwidth of each interface when the traffic is heavy. The commands with the response timeout attributes are not controlled after the timeout period elapse.

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Traffic Statistics and Command Latency Statistics

The DDRC supports traffic statistics. The interface read and write traffic statistics can be collected to determine whether dynamic frequency scaling (DFS) is required, that is, whether to increase or reduce the DDRC frequency. The DDRC can collect the read and write traffic statistics of a specific ID, or the overall read and write traffic statistics. The DDRC can also collect DDR interface usage statistics. The statistic counter supports the continuous counts and one-time count. When counting continuously, the counter is a non-saturating counter, and it is wrapped when the maximum count is reached to facilitate the continuous count. Therefore, the system needs to read the count value before the counter is wrapped. When counting for only one time, the counter stops counting when the statistic time elapses. The system can obtain the instantaneous traffic and latency by using this function.

The DDRC can collect command latency statistics, including the maximum latency of the read and write access to a specific ID and the accumulative latency statistics.

The process for collecting statistics is as follows:

Step 1 Set the statistic mode to continuous triggering or one-time triggering by configuring DDRC_CFG_PERF.perf_mode. If the one-time triggering is selected, set the perf_prd field to configure the statistic cycle.

Step 2 Set the ID for which statistics are to be collected by configuring DDRC_CFG_STAID.

Step 3 Set the mask value of the ID by configuring DDRC_CFG_STAIDMSK. The DDRC determines whether to collect statistics of the current access based on sta_idmask and the accessed ID. Statistics of multiple IDs can be collected by using this method.

Step 4 Enable the statistic function by configuring DDRC_CTRL_PERF[perf_en]. For the one-time count, counting is complete when DDRC_CTRL_PERF[perf_en] restores to 0. For the continuous count, the software needs to write 0 to DDRC_CTRL_PERF[perf_en] to stop counting.

Step 5 Observe the collected statistics by using DDRC_HIS_FLUX_WR, DDRC_HIS_FLUX_RD, DDRC_HIS_FLUX_WCMD, DDRC_HIS_FLUX_RCMD, DDRC_HIS_FLUXID_WR, DDRC_HIS_FLUXID_RD, DDRC_HIS_FLUXID_WCMD, DDRC_HIS_FLUXID_RCMD, DDRC_HIS_WLATCNT0, DDRC_HIS_WLATCNT1, DDRC_HIS_RLATCNT0, DDRC_HIS_RLATCNT1, and DDRC_HIS_INHERE_RLAT_CNT.

----End

Address Mapping

The DDRC can convert the access address for the system bus into that for the DDRn SDRAM. You can set the address mapping mode to row-bank-column (RBC) by configuring DDRC_CFG_RNKVOL[mem_map]. Currently only the RBC mode is supported. Then you can set the SDRAM row and column address bit width by configuring DDRC_CFG_RNKVOL[mem_row] and DDRC_CFG_RNKVOL[mem_col]. The DDRC then converts the system bus address into the DDR3 SDRAM address based on the address mapping algorithm.

The following describes the mapping algorithms for the system bus address and the address for the DDRn SDRAM by using the RBC mode as an example. Assume that the system bus address is BUSADR[31:0], the valid address is BUSADR[m – 1:0], and the address for the DDRn SDRAM is DDRADR[14:0]. When DDRADR[14:0] serves as the row address, its valid address is DDRROW[x – 1:0]; when DDRADR[13:0] serves as the column address, its valid address is DDRCOL[y – 1:0]. In addition, the bank address for the DDR is DDRBA[z –

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1:0] and the width of the storage data bus of the DDRC is DW. In this case, the address mappings are as follows:

- When DDRC_CFG_RNKVOL[mem_map] is set to 2'b00, the RBC mapping mode is as follows:

  BUSADR[m − 1:0] = {DDRROW[x − 1:0], DDRBA[z − 1:0], DDRCOL[y − 1:0], DW{b0}}

  In the preceding expressions, the following condition is met: m = x + y + z + DW. DW is 1 when the DDR*n* data with is 16 bits.
- When DDRC_CFG_RNKVOL[mem_map] is set to 2'b00 and A10 acts as the AP function bit of the DDR, the mapping between the system bus addresses and the DDRn SDRAM addresses is shown in Table 4-4.

Table 4-4 describes the address mapping in RBC mode.

Table 4-4 DDRC address mapping

Memory Type	Row Address Width	Column Address Width	DDR BA			Row/Column Address	DDR ADR							
Mbit x bw			2	1	0		14	13	12	11	10/AP	9	8	7:0
DDR2: 256 Mbits, 4 banks														
32 x 16	13	9	-	11	10	Row address	-	-	24	23	22	21	20	19:12
						Column address	-	-	-	-	AP		9	8:1
DDR2: 512 Mbits, 4 banks														
32 x 16	13	10	-	12	11	Row address	-	-	25	24	23	22	21	20:13
						Column address	-	-	-	-	AP	10	9	8:1
DDR*n*: 1 Gbits, 8 banks														
64 x 16	13	10	13	12	11	Row address			26	25	24	23	22	21:14
						Column address		-	-	-	AP	10	9	8:1
DDR*n*: 2 Gbits, 8 banks														
128 x 16	14	10	13	12	11	Row address		27	26	25	24	23	22	21:14
						Column		-	-	-	AP	10	9	8:1

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Memory Type	Row Address Width	Column Address Width	DDR BA			Row/ Column Address	DDR ADR							
Mbit x bw			2	1	0		14	13	12	11	10/ AP	9	8	7:0
						address								
DDR*n*: 4 Gbits, 8 banks														
256 x 16	15	10	13	12	11	Row address	28	27	26	25	24	23	22	21:14
						Column address		-	-	-	AP	10	9	8:1

4.1.4 Operating Mode

4.1.4.1 Clock Gating

After the system enters the low-power mode, the working clock of the DDRC can be disabled. Before the system restores to the normal mode, the working clock of the DDRC must be enabled.

To disable the working clock after the DDRC enters the low-power mode, perform the following steps:

Step 1 Set DDRC_CTRL_SREF[sref_req] to 1 to request to enter the self-refresh mode.

Step 2 Query DDRC_CURR_FUNC[in_sref] until its value is 1. It indicates that the DDRC enters the low-power mode.

Step 3 Disable the DDRC clock.

Step 4 Enter the low-power mode.

----End

To enable the working clock after the DDRC exits the low-power mode, perform the following steps:

Step 1 Enable the DDRC clock when the system runs in normal mode.

Step 2 Set PLLCTRL[31:0] to 0x2 to power down the PLL.

Step 3 Wait 1 μs, and set PLLCTRL[31:0] to 0x0 to enable the PLL to exit the power-down mode.

Step 4 Set PHYCTRL1 to 0x780078 to reset the PLL.

Step 5 Wait 1 μs and set PHYCTRL1 to 0x700070 to deassert the reset on the PLL.

Step 6 Set PHYINITCTRL to 0x0000_000B.

Step 7 Query PHYINITCTRL[init_en] until its value is 0, which indicates that PLL initialization and impedance calibration are complete.

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Step 8 Set DDRC_CTRL_SREF[sref_req] to 0 to request to exit the self-refresh mode.

Step 9 Query DDRC_CURR_FUNC[in_sref] until its value is 0, which indicates that the DDRC exits the low-power mode.

Step 10 Set DDRC_CTRL_SREF[sref_done] to 1.

Step 11 The system accesses the DDR properly.

----End

4.1.4.2 Soft Reset

The DDRC does not support separate soft reset. It can be reset only by a global soft reset. After reset, the DDRn DRAM must be reinitialized according to the following initialization processes.

4.1.4.3 Initializing the DDR2 SDRAM

After power-on, the system can access the DDR2 SDRAM only when the DDR2 SDRAM is initialized. Before initializing the DDR3 SDRAM, note the following:

- Power on the DDR2 SDRAM by following the standard power-on process provided by the component vendor.
- Initialize the DDR2 SDRAM after the system runs in normal mode.

The following describes the procedure for initializing the DDR subsystem when the ratio of the working frequency of the DDRC to that of the DDR*n* SDRAM is 1:2.

Step 1 Set DDRC_CFG_SREF to 0x8100 to back-press the DDRC commands when the DDR enters the auto-refresh mode.

Step 2 Set DDRC_CFG_AREF to 0x0 to disable auto-refresh during initialization.

Step 3 Set DDRC_CFG_PD to disable the power-down function during initialization.

Step 4 Set DDRC_CFG_WORKMODE to 0x11002011.

Step 5 Set DDRC_CFG_DDRMODE to 0x25 to set the DDR bit width to 16 bits, DDR type to DDR2, and DDRC burst length to BL4.

Step 6 Set DDRC_CFG_RNKVOL to 0x22.

NOTE

The row and column bit widths and the number of banks vary according to DDR2 SDRAMs. Change the value of DDRC_CFG_RNKVOL based on the used DDR2 SDRAM.

Step 7 Set DDRC_CFG_TIMING0−DDRC_CFG_TIMING5 to configure the DDR2 timing parameters.

Step 8 Set DDRC_CFG_ODT to 0x1 to enable ODT on write or disable ODT on read.

Step 9 Configure the PHY timing parameters by setting the related registers PHYTIMER0, PLLTIMER, DLYMEASCTRL, and MISC.

Step 10 Set DRAMCFG to 0x21 to set the DDR type to DDR2.

NOTE

Set DRAMCFG[ma2t] to 1 only when the frequency ratio is 1:2.

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Step 11 Set ACPHYCTL4 to 0xFFFF0770.

NOTE

ACPHYCTL4 is a register related to 2T timing. It needs to be configured only when the frequency ratio is 1:2. The row and column bit widths and the number of banks vary according to DDR2 SDRAMs. Change the value of ACPHYCTL4 based on the used DDR2 SDRAM.

Step 12 Set DRAMTIMER0–DRAMTIMER4 to configure the DDR timing parameters.

Step 13 Set ODTCR to 0x1 to ensure consistency with the DDRC.

Step 14 Set the DDR2 mode registers MODEREG01 and MODEREG23.

Step 15 Set PHYINITCTRL to start the PHY for initialization.

Step 16 Software queries PHYINITCTRL[init_en] until its value is 0, indicating that the initialization is complete.

Step 17 Set DDRC_CFG_TIMING2 to enable auto-refresh.

Step 18 Set DDRC_CTRL_SREF to 0x2 to enable the DDRC to exit the self-refresh mode.

Step 19 Software queries DDRC_CURR_FUNC[in_sref] until its value is 0, indicating that the DDRC exits self-refresh.

Step 20 Set DDRC_CTRL_SREF[sref_done] to 1.

----End

After the preceding steps are complete, the DDR2 SDRAM works properly.

NOTE

The register values may vary according to DDR2 SDRAMs, but the procedure is the same.

4.1.4.4 Initializing the DDR3 SDRAM

After power-on, the system can access the DDR3 SDRAM only when the DDR3 SDRAM is initialized. Before initializing the DDR3 SDRAM, note the following:

- Power on the DDR3 SDRAM by following the JEDEC standard. That is, you must power on VDD, VDDQ, VREF, and VTT in sequence.
- Initialize the DDR3 SDRAM after the system enters normal mode.

The following describes the procedure for initializing the DDR subsystem when the ratio of the working frequency of the DDRC to that of the DDR*n* SDRAM is 1:2.

Step 1 Set DDRC_CFG_SREF to 0x8100 to back-press the DDRC commands when the DDR enters the auto-refresh mode.

Step 2 Set DDRC_CFG_AREF to 0x0 to disable auto-refresh during DDR initialization.

Step 3 Set DDRC_CFG_PD to 0x0 to disable the power-down function during initialization.

Step 4 Set DDRC_CFG_WORKMODE to 0x11002011.

Step 5 Set DDRC_CFG_DDRMODE to 0x16 to set the DDR bit width to 16 bits, DDR type to DDR3, and DDRC burst length to BL4.

Step 6 Set DDRC_CFG_RNKVOL to 0x122.

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NOTE

The row and column bit widths vary according to DDRs. Change the value of DDRC_CFG_RNKVOL based on the used DDR.

Step 7 Set DDRC_CFG_TIMING0–DDRC_CFG_TIMING5 to configure the DDR timing parameters.

Step 8 Set DDRC_CFG_ODT to 0x1 to enable ODT on write or disable ODT on read.

Step 9 Configure the PHY timing parameters by setting the related registers PHYTIMER0, PLLTIMER, DLYMEASCTRL, and MISC.

Step 10 Set DRAMCFG to 0x22 to set the DDR type to DDR3.

NOTE

Set DRAMCFG[ma2t] to 1 only when the frequency ratio is 1:2.

Step 11 (Optional) Set ACPHYCTL4 to 0xFFFF0770 only when the frequency ratio to 1:2.

NOTE

ACPHYCTL4 is a register related to the 2T timing. Configure this register only when the frequency ratio to 1:2. The row and column bit widths and the number of banks vary according to DDR2 SDRAMs. Change the value of ACPHYCTL4 based on the used DDR2 SDRAM.

Step 12 Set DRAMTIMER0–DRAMTIMER4 to configure the DDR timing parameters.

Step 13 Set ODTCR to 0x1 to ensure consistency with the DDRC.

Step 14 Set the DDR3 mode registers MODEREG01 and MODEREG23.

Step 15 Set PHYINITCTRL to start the PHY for initialization.

Step 16 Software queries PHYINITCTRL[init_en] until its value is 0, indicating that the initialization is complete.

Step 17 Set DDRC_CFG_TIMING2 to enable auto-refresh.

Step 18 Set DDRC_CTRL_SREF to 0x2 to enable the DDRC to exit self-refresh.

Step 19 Software queries DDRC_CURR_FUNC[in_sref] until its value is 0, indicating that the DDRC exits self-refresh.

Step 20 Set DDRC_CTRL_SREF[sref_done] to 1.

----End

After the preceding steps are complete, the DDR3 SDRAM works properly.

NOTE

The register values may vary according to DDR SDRAMs, but the procedure is the same.

4.1.4.5 Low-Power Configuration

The DDRC supports two types of low-power modes for the DDRn SDRAM: DDRn SDRAM power-down mode and DDRn SDRAM self-refresh mode.

When DDRC_CFG_PD[pd_en] and DDRC_CFG_PD[pd_prd] are set to valid values, the DDRC automatically forces the DDRn SDRAM to enter the low-power mode if the system is idle. When DDRC_CFG_PD[pd_en] is set to 1 and the DDRC does not access

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the DDRn SDRAM within DDRC_CFG_PD[pd_prd] bus clock cycles, the DDRC forces the DDRn SDRAM to enter the low-power mode.

When DDRC_CTRL_SREF[sr_req] is set to 1, the DDRC forces the DDRn SDRAM to enter the self-refresh mode without responding to the bus requests after completing the current access operation.

4.1.5 AXI Registers of the DDRC

4.1.5.1 Register Summary

Table 4-5 describes AXI registers.

NOTE

The variable **ports** in the offset addresses indicates the number of AXI ports and ranges from 0 to 6.

Table 4-5 Summary of EMI registers (base address: 0x2011_0000)

Offset Address	Register	Description	Page
0x000	AXI_CONFIG	DDRC function configuration register	4-14
0x004	AXI_CKGCFG	DDRC clock gating configuration register	4-16
0x01C	AXI_STATUS	DDRC interface status register	4-17
0x020	AXI_CHCFG0	MDDRC channel configuration register 0	4-18
0x024	AXI_CHCFG1	MDDRC channel configuration register 1	4-19
0x028	AXI_CHCFG2	MDDRC channel configuration register 2	4-21
0x02C	AXI_CHCFG3	MDDRC channel configuration register 3	4-23
0x030	AXI_CHCFG4	MDDRC channel configuration register 4	4-25
0x034	AXI_CHCFG5	MDDRC channel configuration register 5	4-26
0x038	AXI_LPCFG	MDDRC low-power zone configuration register	4-27
0x040 + 0x4 x *ports*	AXI_QOSCFG0	DDRC QoS configuration register 0	4-28
0x080 + 0x4 x *ports*	AXI_QOSCFG1	DDRC QoS configuration register 1	4-29
0x0C0 + 0x4 x *ports*	AXI_WRPRI	DDRC write command priority configuration register	4-30

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Offset Address	Register	Description	Page
0x100 + 0x4 x *ports*	AXI_RDPRI	DDRC read command priority configuration register	4-31
0x140 + 0x4 x *ports*	AXI_WRTOUT	DDRC write command timeout configuration register	4-33
0x180 + 0x4 x *ports*	AXI_RDTOUT	DDRC read command timeout configuration register	4-33
0x200 + 0x4 x *ports*	AXI_OSTDCFG0	DDRC port command outstanding configuration register 0	4-34
0x240	AXI_OSTDCFG1	DDRC port command outstanding configuration register 1	4-35
0x244	AXI_OSTDCFG2	DDRC port command outstanding configuration register 2	4-35
0x248	AXI_OSTDSTATUS	DDRC channel command outstanding status register	4-38
0x280 + 0x4 x *ports*	AXI_FLUX0	DDRC port bandwidth traffic control configuration register for channel 0	4-38

4.1.5.2 Register Description

AXI_CONFIG

AXI_CONFIG is a DDRC function configuration register.

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Offset Address	Register Name	Total Reset Value
0x000	AXI_CONFIG	0x0006_000C

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved													wr_rcv_mode	exclusive_en	rd_wrap_split_en	reserved												bank_width			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0

Bits	Access	Name	Description
[31:19]	RO	reserved	Reserved
[18]	RW	wr_rcv_mode	Mode of receiving the write command over the DDRC AXI port 0: The write command is received directly. 1: The write command is received only after the expected written data arrives.
[17]	RW	exclusive_en	Exclusive command enable 0: disabled 1: enabled
[16]	RW	rd_wrap_split_en	WRAP read command split enable 0: not split 1: split
[15:4]	RO	reserved	Reserved
[3:0]	RW	bank_width	Bank bit width 1000: 256 bits 1001: 512 bits 1010: 1 KB 1011: 2 KB 1100: 4 KB 1101: 8 KB 1110: 16 KB 1111: 32 KB Other values: reserved For example, if the column is 10 for a 32-bit DDR, the bank width is calculated as follows: bank_width = (32/8) x (2^10) = 4 KB

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AXI_CKGCFG

AXI_CKGCFG is a DDRC clock gating configuration register.

Offset Address	Register Name	Total Reset Value
0x004	AXI_CKGCFG	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved			sta_ckg_pub0	reserved			sta_ckg_dmc0	reserved			sta_ckg_apb	reserved			dyn_ckg_rdr	reserved							dyn_ckg_axi6	dyn_ckg_axi5	dyn_ckg_axi4	dyn_ckg_axi3	dyn_ckg_axi2	dyn_ckg_axi1	dyn_ckg_axi0
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:29]	RW	reserved	Reserved
[28]	RW	sta_ckg_pub0	Static clock gating for PUB0 0: disabled 1: enabled
[27:25]	RW	reserved	Reserved
[24]	RW	sta_ckg_dmc0	Dynamic clock gating for DMC0 0: disabled 1: enabled
[23:21]	RO	reserved	Reserved
[20]	RW	sta_ckg_apb	Static clock gating for the register configuration module 0: disabled 1: enabled
[19:17]	RO	reserved	Reserved
[16]	RW	dyn_ckg_rdr	Dynamic clock gating for the REORDER module 0: The clock is always enabled. 1: Clock gating is automatic when the module is idle.
[15:7]	RW	reserved	Reserved
[6]	RW	dyn_ckg_axi6	Dynamic clock gating for AXI port 6 0: The clock is always enabled. 1: Clock gating is automatic when the AXI module is idle.
[5]	RW	dyn_ckg_axi5	Dynamic clock gating for AXI port 5 0: The clock is always enabled. 1: Clock gating is automatic when the AXI module is idle.

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[4]	RW	dyn_ckg_axi4	Dynamic clock gating for AXI port 4 0: The clock is always enabled. 1: Clock gating is automatic when the AXI module is idle.
[3]	RW	dyn_ckg_axi3	Dynamic clock gating for AXI port 3 0: The clock is always enabled. 1: Clock gating is automatic when the AXI module is idle.
[2]	RW	dyn_ckg_axi2	Dynamic clock gating for AXI port 2 0: The clock is always enabled. 1: Clock gating is automatic when the AXI module is idle.
[1]	RW	dyn_ckg_axi1	Dynamic clock gating for AXI port 1 0: The clock is always enabled. 1: Clock gating is automatic when the AXI module is idle.
[0]	RW	dyn_ckg_axi0	Dynamic clock gating for AXI port 0 0: The clock is always enabled. 1: Clock gating is automatic when the AXI module is idle.

AXI_STATUS

AXI_STATUS is a DDRC interface status register.

Offset Address	Register Name	Total Reset Value
0x01C	AXI_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									axist6	axist5	axist4	axist3	axist2	axist1	axist0
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6]	RO	axist6	Status of controller bus interface 6 0: The interface is idle. 1: A command is being executed.
[5]	RO	axist5	Status of controller bus interface 5 0: The interface is idle. 1: A command is being executed.
[4]	RO	axist4	Status of controller bus interface 4 0: The interface is idle. 1: A command is being executed.

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[3]	RO	axist3	Status of controller bus interface 3 0: The interface is idle. 1: A command is being executed.
[2]	RO	axist2	Status of controller bus interface 2 0: The interface is idle. 1: A command is being executed.
[1]	RO	axist1	Status of controller bus interface 1 0: The interface is idle. 1: A command is being executed.
[0]	RO	axist0	Status of controller bus interface 0 0: The interface is idle. 1: A command is being executed.

AXI_CHCFG0

AXI_CHCCFG0 is MDDRC channel configuration register 0.

Offset Address	Register Name	Total Reset Value
0x020	AXI_CHCFG0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														ch_addr_msk_idx	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RW	reserved	Reserved

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[2:0]	RW	ch_addr_msk_idx	Sequence number of mask for the address space index Four consecutive bits in the address are obtained for searching the channel configuration table. 000: addr[ADDR_WIDTH − 1: ADDR_WIDTH − 4] 001: addr[ADDR_WIDTH − 2: ADDR_WIDTH − 5] 010: addr[ADDR_WIDTH − 3: ADDR_WIDTH − 6] 011: addr[ADDR_WIDTH − 4: ADDR_WIDTH − 7] 100: addr[ADDR_WIDTH − 5: ADDR_WIDTH − 8] 101: addr[ADDR_WIDTH − 6: ADDR_WIDTH − 9] 110: addr[ADDR_WIDTH − 7: ADDR_WIDTH − 10] 111: addr[ADDR_WIDTH − 8: ADDR_WIDTH − 11] ADDR_WIDTH: bit width of the system address (32 bits to 40 bits) Prevent upper-bit addresses from being wrapped. For example, when the address bit width is 32 bits, if it is set to 3, addr[31:29] must be fixed.

AXI_CHCFG1

AXI_CHCCFG1 is MDDRC channel configuration register 1.

Offset Address	Register Name	Total Reset Value
0x024	AXI_CHCFG1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ch_start15		ch_start14		ch_start13		ch_start12		ch_start11		ch_start10		ch_start9		ch_start8		ch_start7		ch_start6		ch_start5		ch_start4		ch_start3		ch_start2		ch_start1		ch_start0	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RW	ch_start15	Channel configuration table, start channel of address space 15 0−3 correspond to channels 0−3.
[29:28]	RW	ch_start14	Channel configuration table, start channel of address space 14 0−3 correspond to channels 0−3.
[27:26]	RW	ch_start13	Channel configuration table, start channel of address space 13 0−3 correspond to channels 0−3.
[25:24]	RW	ch_start12	Channel configuration table, start channel of address space 12 0−3 correspond to channels 0−3.
[23:22]	RW	ch_start11	Channel configuration table, start channel of address space 11 0−3 correspond to channels 0−3.

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[21:20]	RW	ch_start10	Channel configuration table, start channel of address space 10 0−3 correspond to channels 0−3.
[19:18]	RW	ch_start9	Channel configuration table, start channel of address space 9 0−3 correspond to channels 0−3.
[17:16]	RW	ch_start8	Channel configuration table, start channel of address space 8 0−3 correspond to channels 0−3.
[15:14]	RW	ch_start7	Channel configuration table, start channel of address space 7 0−3 correspond to channels 0−3.
[13:12]	RW	ch_start6	Channel configuration table, start channel of address space 6 0−3 correspond to channels 0−3.
[11:10]	RW	ch_start5	Channel configuration table, start channel of address space 5 0−3 correspond to channels 0−3.
[9:8]	RW	ch_start4	Channel configuration table, start channel of address space 4 0−3 correspond to channels 0−3.
[7:6]	RW	ch_start3	Channel configuration table, start channel of address space 3 0−3 correspond to channels 0−3.
[5:4]	RW	ch_start2	Channel configuration table, start channel of address space 2 0−3 correspond to channels 0−3.
[3:2]	RW	ch_start1	Channel configuration table, start channel of address space 1 0−3 correspond to channels 0−3.
[1:0]	RW	ch_start0	Channel configuration table, start channel of address space 0 0−3 correspond to channels 0−3.

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AXI_CHCFG2

AXI_CHCCFG2 is MDDRC channel configuration register 2.

Offset Address	Register Name	Total Reset Value
0x028	AXI_CHCFG2	0x0000_0000

Bit	31 30	29 28	27 26	25 24	23 22	21 20	19 18	17 16	15 14	13 12	11 10	9 8	7 6	5 4	3 2	1 0
Name	ch_mode15	ch_mode14	ch_mode13	ch_mode12	ch_mode11	ch_mode10	ch_mode9	ch_mode8	ch_mode7	ch_mode6	ch_mode5	ch_mode4	ch_mode3	ch_mode2	ch_mode1	ch_mode0
Reset	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0	0 0

Bits	Access	Name	Description
[31:30]	RW	ch_mode15	Channel configuration table, channel mode of address space 15 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[29:28]	RW	ch_mode14	Channel configuration table, channel mode of address space 14 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[27:26]	RW	ch_mode13	Channel configuration table, channel mode of address space 13 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[25:24]	RW	ch_mode12	Channel configuration table, channel mode of address space 12 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[23:22]	RW	ch_mode11	Channel configuration table, channel mode of address space 11 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved

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[21:20]	RW	ch_mode10	Channel configuration table, channel mode of address space 10 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[19:18]	RW	ch_mode9	Channel configuration table, channel mode of address space 9 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[17:16]	RW	ch_mode8	Channel configuration table, channel mode of address space 8 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[15:14]	RW	ch_mode7	Channel configuration table, channel mode of address space 7 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[13:12]	RW	ch_mode6	Channel configuration table, channel mode of address space 6 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[11:10]	RW	ch_mode5	Channel configuration table, channel mode of address space 5 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[9:8]	RW	ch_mode4	Channel configuration table, channel mode of address space 4 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved

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[7:6]	RW	ch_mode3	Channel configuration table, channel mode of address space 3 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[5:4]	RW	ch_mode2	Channel configuration table, channel mode of address space 2 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[3:2]	RW	ch_mode1	Channel configuration table, channel mode of address space 1 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved
[1:0]	RW	ch_mode0	Channel configuration table, channel mode of address space 0 00: address-independent 01: reserved 10: dual-channel-interleaved 11: quad-channel-interleaved

AXI_CHCFG3

AXI_CHCCFG3 is MDDRC channel configuration register 3.

Offset Address	Register Name	Total Reset Value
0x02C	AXI_CHCFG3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ch_invld15	ch_invld14	ch_invld13	ch_invld12	ch_invld11	ch_invld10	ch_invld9	ch_invld8	ch_invld7	ch_invld6	ch_invld5	ch_invld4	ch_invld3	ch_invld2	ch_invld1	ch_invld0
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	reserved	Reserved
[15]	RW	ch_invld15	Channel configuration table, address space 15 mask enable 0: The current address space is valid. 1: The current address space is invalid.

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[14]	RW	ch_invld14	Channel configuration table, address space 14 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[13]	RW	ch_invld13	Channel configuration table, address space 13 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[12]	RW	ch_invld12	Channel configuration table, address space 12 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[11]	RW	ch_invld11	Channel configuration table, address space 11 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[10]	RW	ch_invld10	Channel configuration table, address space 10 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[9]	RW	ch_invld9	Channel configuration table, address space 9 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[8]	RW	ch_invld8	Channel configuration table, address space 8 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[7]	RW	ch_invld7	Channel configuration table, address space 7 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[6]	RW	ch_invld6	Channel configuration table, address space 6 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[5]	RW	ch_invld5	Channel configuration table, address space 5 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[4]	RW	ch_invld4	Channel configuration table, address space 4 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[3]	RW	ch_invld3	Channel configuration table, address space 3 mask enable 0: The current address space is valid. 1: The current address space is invalid.

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[2]	RW	ch_invld2	Channel configuration table, address space 2 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[1]	RW	ch_invld1	Channel configuration table, address space 1 mask enable 0: The current address space is valid. 1: The current address space is invalid.
[0]	RW	ch_invld0	Channel configuration table, address space 0 mask enable 0: The current address space is valid. 1: The current address space is invalid.

AXI_CHCFG4

AXI_CHCCFG4 is MDDRC channel configuration register 4.

Offset Address	Register Name	Total Reset Value
0x030	AXI_CHCFG4	0x7654_3210

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	ch_sect7	ch_sect6	ch_sect5	ch_sect4	ch_sect3	ch_sect2	ch_sect1	ch_sect0
Reset	0 1 1 1	0 1 1 0	0 1 0 1	0 1 0 0	0 0 1 1	0 0 1 0	0 0 0 1	0 0 0 0

Bits	Access	Name	Description
[31:28]	RW	ch_sect7	Channel configuration table, position of the channel section to which address space 7 is mapped
[27:24]	RW	ch_sect6	Channel configuration table, position of the channel section to which address space 6 is mapped
[23:20]	RW	ch_sect5	Channel configuration table, position of the channel section to which address space 5 is mapped
[19:16]	RW	ch_sect4	Channel configuration table, position of the channel section to which address space 4 is mapped
[15:12]	RW	ch_sect3	Channel configuration table, position of the channel section to which address space 3 is mapped
[11:8]	RW	ch_sect2	Channel configuration table, position of the channel section to which address space 2 is mapped
[7:4]	RW	ch_sect1	Channel configuration table, position of the channel section to which address space 1 is mapped
[3:0]	RW	ch_sect0	Channel configuration table, position of the channel section to which address space 0 is mapped

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AXI_CHCFG5

AXI_CHCCFG5 is MDDRC channel configuration register 5.

Offset Address	Register Name	Total Reset Value
0x034	AXI_CHCFG5	0xFEDC_BA98

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	ch_sect15	ch_sect14	ch_sect13	ch_sect12	ch_sect11	ch_sect10	ch_sect9	ch_sect8
Reset	1 1 1 1	1 1 1 0	1 1 0 1	1 1 0 0	1 0 1 1	1 0 1 0	1 0 0 1	1 0 0 0

Bits	Access	Name	Description
[31:28]	RW	ch_sect15	Channel configuration table, position of the channel section to which address space 15 is mapped
[27:24]	RW	ch_sect14	Channel configuration table, position of the channel section to which address space 14 is mapped
[23:20]	RW	ch_sect13	Channel configuration table, position of the channel section to which address space 13 is mapped
[19:16]	RW	ch_sect12	Channel configuration table, position of the channel section to which address space 12 is mapped
[15:12]	RW	ch_sect11	Channel configuration table, position of the channel section to which address space 11 is mapped
[11:8]	RW	ch_sect10	Channel configuration table, position of the channel section to which address space 10 is mapped
[7:4]	RW	ch_sect9	Channel configuration table, position of the channel section to which address space 9 is mapped
[3:0]	RW	ch_sect8	Channel configuration table, position of the channel section to which address space 8 is mapped

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AXI_LPCFG

AXI_LPCFG is a low-power zone configuration register of the MDDRC.

Offset Address	Register Name	Total Reset Value
0x038	AXI_LPCFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lp_zone_size																reserved															lp_zone_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lp_zone_size	Size of the low-power zone The zone size (in MB) equals lp_zone_size plus 1 and increases by the multiple of 2. 0x0000: 1 MB 0x0001: 2 MB 0x0003: 4 MB 0x0007: 8 MB ... If the address space where the low-power zone is located is dual-/quad-channel interleaved, the size of the low-power zone cannot exceed half or a quarter of the address space.
[15:1]	RW	reserved	Reserved
[0]	RW	lp_zone_en	Low-power zone enable 0: disabled 1: enabled

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AXI_QOSCFG0

AXI_QOSCFG0 is DDRC QoS configuration register 0.

Offset Address	Register Name	Total Reset Value
0x040 + 0x4 x *ports* (*ports* = 0–6)	AXI_QOSCFG0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	pri_deliver_en	push_deliver_en	reserved					tout_map_mode	tout_id_map								reserved			pri_map_mode	pri_id_map											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	pri_deliver_en	Port priority deliver enable 0: disabled 1: enabled
[30]	RW	push_deliver_en	Push priority deliver enable 0: disabled 1: enabled
[29:25]	RO	reserved	Reserved
[24]	RW	tout_map_mode	Timeout mapping mode of the read/write command 0: mapping mode specified by tout_id_map 1: mapping mode specified by the command associated signal wr(rd)_tout_sel
[23:16]	RW	tout_id_map	Two bits of the read/write command ID for timeout configuration id_map[7:4]: bit[1] for ID mapping id_map[3:0]: bit[0] for ID mapping For example, if ID_MAP is set to 0x32, {ID[3], ID[2]} of the read/write command ID is used for ID mapping and priority configuration. Note that the lower four bits of the read/write command ID indicate the AXI port ID. That is, read/write command ID = {Bus ID, 4-bit AXI port ID}.

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[15:13]	RO	reserved	Reserved
[12]	RW	pri_map_mode	Priority mapping mode of the read/write command 0: mapping mode specified by pri_id_map 1: QoS mapping mode specified by using the AxQOS[2:0] signal of the AXI4 read/write command
[11:0]	RW	pri_id_map	Three bits of the read/write command ID for QoS configuration pri_id_map[11:8]: bit[2] for ID mapping pri_id_map[7:4]: bit[1] for IP mapping pri_id_map[3:0]: bit[0] for ID mapping For example, if ID_MAP is set to 0x320, {ID[3], ID[2], ID[0]} of the read/write command ID is used for ID mapping and priority configuration. Note that the lower four bits of the read/write command ID indicate the AXI port ID. That is, read/write command ID = {Bus ID, 4-bit AXI port ID}.

AXI_QOSCFG1

AXI_QOSCFG1 is a DDRC QoS configuration register 1.

Offset Address	Register Name	Total Reset Value
0x080 + 0x4 x *ports* (*ports* = 0–6)	AXI_QOSCFG1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								wr_pri_apt				wr_age_prd				reserved								rd_pri_apt				rd_age_prd			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:20]	RW	wr_pri_apt	Write command priority adaptation cycle 0x0: disabled 0x1−0xF: (*n* x 16) clock cycles
[19:16]	RW	wr_age_prd	Write command aging cycle 0x0: disabled 0x1−0xF: (*n* x 16) clock cycles
[15:8]	RO	reserved	Reserved
[7:4]	RW	rd_pri_apt	Read command priority adaptation cycle 0x0: disabled 0x1−0xF: (*n* x 16) clock cycles

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[3:0]	RW	rd_age_prd	Read command aging cycle 0x0: disabled 0x1−0xF: (*n* x 16) clock cycles

AXI_WRPRI

AXI_WRPRI is a DDRC write command priority configuration register.

Offset Address	Register Name	Total Reset Value
0x0C0 + 0x4 x *ports* (*ports* = 0–6)	AXI_WRPRI	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	wr_pri7			reserved	wr_pri6			reserved	wr_pri5			reserved	wr_pri4			reserved	wr_pri3			reserved	wr_pri2			reserved	wr_pri1			reserved	wr_pri0		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:28]	RW	wr_pri7	Priority when the write command ID mapping is 7 000: highest priority ... 111: lowest priority
[27]	RO	reserved	Reserved
[26:24]	RW	wr_pri6	Priority when the write command ID mapping is 6 000: highest priority ... 111: lowest priority
[23]	RO	reserved	Reserved
[22:20]	RW	wr_pri5	Priority when the write command ID mapping is 5 000: highest priority ... 111: lowest priority
[19]	RO	reserved	Reserved
[18:16]	RW	wr_pri4	Priority when the write command ID mapping is 4 000: highest priority ... 111: lowest priority

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[15]	RO	reserved	Reserved
[14:12]	RW	wr_pri3	Priority when the write command ID mapping is 3 000: highest priority ... 111: lowest priority
[11]	RO	reserved	Reserved
[10:8]	RW	wr_pri2	Priority when the write command ID mapping is 2 000: highest priority ... 111: lowest priority
[7]	RO	reserved	Reserved
[6:4]	RW	wr_pri1	Priority when the write command ID mapping is 1 000: highest priority ... 111: lowest priority
[3]	RO	reserved	Reserved
[2:0]	RW	wr_pri0	Priority when the write command ID mapping is 0 000: highest priority ... 111: lowest priority

AXI_RDPRI

AXI_RDPRI is a DDRC read command priority configuration register.

Offset Address	Register Name	Total Reset Value
0x100 + 0x4 x *ports* (*ports* = 0–6)	AXI_RDPRI	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	rd_pri7			reserved	rd_pri6			reserved	rd_pri5			reserved	rd_pri4			reserved	rd_pri3			reserved	rd_pri2			reserved	rd_pri1			reserved	rd_pri0		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved

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[30:28]	RW	rd_pri7	Priority when the read command ID mapping is 7 000: highest priority ... 111: lowest priority
[27]	RO	reserved	Reserved
[26:24]	RW	rd_pri6	Priority when the read command ID mapping is 6 000: highest priority ... 111: lowest priority
[23]	RO	reserved	Reserved
[22:20]	RW	rd_pri5	Priority when the read command ID mapping is 5 000: highest priority ... 111: lowest priority
[19]	RO	reserved	Reserved
[18:16]	RW	rd_pri4	Priority when the read command ID mapping is 4 000: highest priority ... 111: lowest priority
[15]	RO	reserved	Reserved
[14:12]	RW	rd_pri3	Priority when the read command ID mapping is 3 000: highest priority ... 111: lowest priority
[11]	RO	reserved	Reserved
[10:8]	RW	rd_pri2	Priority when the read command ID mapping is 2 000: highest priority ... 111: lowest priority
[7]	RO	reserved	Reserved
[6:4]	RW	rd_pri1	Priority when the read command ID mapping is 1 000: highest priority ... 111: lowest priority
[3]	RO	reserved	Reserved

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[2:0]	RW	rd_pri0	Priority when the read command ID mapping is 0 000: highest priority ... 111: lowest priority

AXI_WRTOUT

AXI_WRTOUT is a DDRC write command timeout configuration register.

Offset Address	Register Name	Total Reset Value
0x140 + 0x4 x *ports* (*ports* = 0–6)	AXI_WRTOUT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wr_tout3								wr_tout2								wr_tout1								wr_tout0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	wr_tout3	Write command timeout period (level 3) The configuration mode is the same as that of wr_tout0.
[23:16]	RW	wr_tout2	Write command timeout period (level 2) The configuration mode is the same as that of wr_tout0.
[15:8]	RW	wr_tout1	Write command timeout period (level 1) The configuration mode is the same as that of wr_tout0.
[7:0]	RW	wr_tout0	Write command timeout period (level 0) 0x0: disabled 0x01–0xFF: (*n* x 4) clock cycles **NOTE** When the timeout period is set to 8 bits, the actual count value is 10 bit. That is, the lower two bits are fixed at 0.

AXI_RDTOUT

AXI_RDTOUT is a DDRC read command timeout configuration register.

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Offset Address	Register Name	Total Reset Value
0x180 + 0x4 x *ports* (*ports* = 0–6)	AXI_RDTOUT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rd_tout3								rd_tout2								rd_tout1								rd_tout0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	rd_tout3	Read command timeout period (level 3) The configuration mode is the same as that of rd_tout0.
[23:16]	RW	rd_tout2	Read command timeout period (level 2) The configuration mode is the same as that of rd_tout0.
[15:8]	RW	rd_tout1	Read command timeout period (level 1) The configuration mode is the same as that of rd_tout0.
[7:0]	RW	rd_tout0	Read command timeout period (level 0) 0x0: disabled 0x01–0xFF: (*n* x 4) clock cycles **NOTE** When the timeout period is set to 8 bits, the actual count value is 10 bit. That is, the lower two bits are fixed at 0.

AXI_OSTDCFG0

AXI_OSTDCFG0 is DDRC port command outstanding configuration register 0.

Offset Address	Register Name	Total Reset Value
0x200 + 0x4 x *ports* (*ports* = 0–6)	AXI_OSTDCFG0	0x1F1F_1F1F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												port_ostd_lvl0			
Reset	0	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	port_ostd_lvl0	OSTD threshold of the command transmitted from the port to channel 0 (used with the AXI_OSTDCFG2 register)

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AXI_OSTDCFG1

AXI_OSTDCFG1 is DDRC port command outstanding configuration register 1.

Offset Address	Register Name	Total Reset Value
0x240	AXI_OSTDCFG1	0x1F1F_1F1F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												dmc_ostd_lvl0			
Reset	0	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	dmc_ostd_lvl0	Accumulated OSTD threshold of commands transmitted from all ports to channel 0 (used with the AXI_OSTDCFG2 register)

AXI_OSTDCFG2

AXI_OSTDCFG2 is DDRC port command outstanding configuration register 2.

Offset Address	Register Name	Total Reset Value
0x244	AXI_OSTDCFG2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																		ostd_mode6		ostd_mode5		ostd_mode4		ostd_mode3		ostd_mode2		ostd_mode1		ostd_mode0	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved

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[13:12]	RW	ostd_mode6	OSTD mode of port 6 00: The OSTD statistics of the current port are disabled. 01: green port. The mode is controlled by channel. When the OSTD of the current port reaches port_ostd_lvl, the port is blocked. 10: non-green port mode 0. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl, the current port is blocked. 11: non-green port mode 1. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl or the OSTD of the current port reaches port_ostd_lvl, the current port is blocked.
[11:10]	RW	ostd_mode5	OSTD mode of port 5 00: The OSTD statistics of the current port are disabled. 01: green port. The mode is controlled by channel. When the OSTD of the current port reaches port_ostd_lvl, the port is blocked. 10: non-green port mode 0. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl, the current port is blocked. 11: non-green port mode 1. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl or the OSTD of the current port reaches port_ostd_lvl, the current port is blocked.
[9:8]	RW	ostd_mode4	OSTD mode of port 4 00: The OSTD statistics of the current port are disabled. 01: green port. The mode is controlled by channel. When the OSTD of the current port reaches port_ostd_lvl, the port is blocked. 10: non-green port mode 0. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl, the current port is blocked. 11: non-green port mode 1. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl or the OSTD of the current port reaches port_ostd_lvl, the current port is blocked.

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[7:6]	RW	ostd_mode3	OSTD mode of port 3 00: The OSTD statistics of the current port are disabled. 01: green port. The mode is controlled by channel. When the OSTD of the current port reaches port_ostd_lvl, the port is blocked. 10: non-green port mode 0. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl, the current port is blocked. 11: non-green port mode 1. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl or the OSTD of the current port reaches port_ostd_lvl, the current port is blocked.
[5:4]	RW	ostd_mode2	OSTD mode of port 2 00: The OSTD statistics of the current port are disabled. 01: green port. The mode is controlled by channel. When the OSTD of the current port reaches port_ostd_lvl, the port is blocked. 10: non-green port mode 0. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl, the current port is blocked. 11: non-green port mode 1. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl or the OSTD of the current port reaches port_ostd_lvl, the current port is blocked.
[3:2]	RW	ostd_mode1	OSTD mode of port 1 00: The OSTD statistics of the current port are disabled. 01: green port. The mode is controlled by channel. When the OSTD of the current port reaches port_ostd_lvl, the port is blocked. 10: non-green port mode 0. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl, the current port is blocked. 11: non-green port mode 1. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl or the OSTD of the current port reaches port_ostd_lvl, the current port is blocked.

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[1:0]	RW	ostd_mode0	OSTD mode of port 0 00: The OSTD statistics of the current port are disabled. 01: green port. The mode is controlled by channel. When the OSTD of the current port reaches port_ostd_lvl, the port is blocked. 10: non-green port mode 0. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl, the current port is blocked. 11: non-green port mode 1. The mode is controlled by channel. If the accumulated OSTD of all ports reaches dmc_ostd_lvl or the OSTD of the current port reaches port_ostd_lvl, the current port is blocked.

AXI_OSTDSTATUS

AXI_OSTDSTATUS is a DDRC channel command outstanding status register.

Offset Address	Register Name	Total Reset Value
0x248	AXI_OSTDSTATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											ostd_st0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RO	ostd_st0	Command OSTD status of channel 0. The statistics are valid when at least a port is in non-green port mode.

AXI_FLUX0

AXI_FLUX0 is a DDRC port bandwidth traffic control configuration register for channel 0.

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Offset Address	Register Name	Total Reset Value
0x280 + 0x4 x *ports* (*ports* = 0–6)	AXI_FLUX0	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved		flux_port_en0 / flux_ovfl_en0	reserved	flux_ovfl_lvl0	reserved	flux_cfg0	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	flux_port_en0	AXI interface DDRC traffic statistics enable 0: disabled 1: enabled
[20]	RW	flux_ovfl_en0	AXI interface traffic overflow allow enable 0: disabled 1: enabled When this field is set to 1, the bandwidth of the AXI interface can exceed the configured traffic if the AXI interface traffic exceeds the bandwidth limit, there is no AXI interface with traffic overflow, and a command request is initiated.
[19:17]	RO	reserved	Reserved
[16:12]	RW	flux_ovfl_lvl0	AXI interface traffic overflow threshold 0x0–0x10: DMC threshold for allowing traffic overflow Other values: reserved When the AXI interface traffic exceeds the configured bandwidth of flux, flux_ovfl is 1, and the number of commands to be processed in the DMC is less than the configured threshold, transmission is allowed. Otherwise, no arbitration is performed.
[11:10]	RO	reserved	Reserved
[9:0]	RW	flux_cfg0	Allowed bandwidth of the AXI interface 0x0–0x3FF: ratio of the maximum DDR bandwidth to the total bandwidth accessed by the AXI interface. The total bandwidth is 1024. The configured value is a ratio of the allowed bandwidth to the total bandwidth. Other values: reserved

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4.1.6 DMC Registers of the DDRC

4.1.6.1 Register Summary

Table 4-6 describes DMC registers.

Table 4-6 DMC registers (base address: 0x2011_1000)

Offset Address	Register	Description	Page
0x000	DDRC_CTRL_SREF	DDRC self-refresh control register	4-44
0x004	DDRC_CTRL_INIT	DDRC initialization control register	4-44
0x00C	DDRC_CTRL_SFC	DDRC software configuration command start register	4-45
0x010	DDRC_CTRL_PERF	DDRC performance statistics control register	4-45
0x020	DDRC_CFG_SREF	DDR self-refresh configuration register	4-46
0x028	DDRC_CFG_PD	DDR power-down status register	4-47
0x02C	DDRC_CFG_AREF	DDRC auto-refresh mode register	4-48
0x040	DDRC_CFG_WORKMODE	DDRC operating mode register	4-49
0x050	DDRC_CFG_DDRMODE	DDR operating mode register	4-50
0x060	DDRC_CFG_RNKVOL	DDRC-controlled DDR capacity configuration register	4-52
0x070	DDRC_CFG_EMRS01	DDR mode register 0 and mode register 1	4-54
0x074	DDRC_CFG_EMRS23	DDR mode register 2 and mode register 3	4-54
0x080	DDRC_CFG_TIMING0	DDRC timing parameter register 0	4-54
0x084	DDRC_CFG_TIMING1	DDRC timing parameter register 1	4-55
0x88	DDRC_CFG_TIMING2	DDRC timing parameter register 2	4-57
0x8C	DDRC_CFG_TIMING3	DDRC timing parameter register 3	4-58

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Offset Address	Register	Description	Page
0x090	DDRC_CFG_TIMING4	DDRC timing parameter register 4	4-59
0x094	DDRC_CFG_TIMING5	DDRC timing parameter register 5	4-60
0x098	DDRC_CFG_TIMING6	DDRC timing parameter register 6	4-61
0x0A0	DDRC_CFG_NXT_TIMING 0	DDRC timing parameter register 0 for switching the frequency	4-62
0x0A4	DDRC_CFG_NXT_TIMING 1	DDRC timing parameter register 1 for switching the frequency	4-63
0x0A8	DDRC_CFG_NXT_TIMING 2	DDRC timing parameter register 2 for switching the frequency	4-64
0x0AC	DDRC_CFG_NXT_TIMING 3	DDRC timing parameter register 3 for switching the frequency	4-65
0x0B0	DDRC_CFG_NXT_TIMING 4	DDRC timing parameter register 4 for switching the frequency	4-66
0x0B4	DDRC_CFG_NXT_TIMING 5	DDRC timing parameter register 5 for switching the frequency	4-67
0x0B8	DDRC_CFG_NXT_TIMING 6	DDRC timing parameter register 6 for switching the frequency	4-68
0x0BC	DDRC_CFG_BLDATA	DDRC pre-received write data configuration register	4-68
0x0C0	DDRC_CFG_ODT	DDR ODT feature configuration register	4-69
0x0C4	DDRC_CFG_DMCLVL	DDRC command queue depth threshold configuration register	4-70
0x200	DDRC_CFG_DDRPHY	DDR I/O configuration register	4-70
0x210	DDRC_CFG_SFC	DDRC software DDR command attribute register	4-71
0x214	DDRC_CFG_SFC_ADDR0	Read/Write memory address register 0 for the software configuration module	4-72

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Offset Address	Register	Description	Page
0x218	DDRC_CFG_SFC_ADDR1	Read/Write memory address register 1 for the software configuration module	4-72
0x21C	DDRC_CFG_SFC_WDATA0	Write data register 0 for the software configuration module	4-73
0x220	DDRC_CFG_SFC_WDATA1	Write data register 1 for the software configuration module	4-73
0x224	DDRC_CFG_SFC_WDATA2	Write data register 2 for the software configuration module	4-73
0x228	DDRC_CFG_SFC_WDATA3	Write data register 3 for the software configuration module	4-74
0x270	DDRC_CFG_PERF	DDRC performance statistics mode register	4-74
0x274	DDRC_CFG_STAID	DDRC performance statistics command ID register	4-75
0x278	DDRC_CFG_STAIDMSK	DDR performance statistics command ID mask register	4-76
0x280	DDRC_INTMSK	DDRC interrupt mask register	4-76
0x284	DDRC_RINT	DDRC raw interrupt register	4-77
0x288	DDRC_INTSTS	DDRC interrupt status register	4-78
0x290	DDRC_CURR_STATUS	DDRC status register	4-78
0x294	DDRC_CURR_FUNC	DDRC FUNC module status register	4-79
0x2A0	DDRC_CURR_EXECST	DDRC command state machine status register	4-80
0x2A4	DDRC_CURR_WGFIFOST	DDRC write data FIFO status register	4-81
0x380	DDRC_HIS_FLUX_WR	DDRC all write command traffic statistics register	4-81
0x384	DDRC_HIS_FLUX_RD	DDRC all read command traffic statistics register	4-82
0x388	DDRC_HIS_FLUX_WCMD	DDRC all write command count register	4-82
0x38C	DDRC_HIS_FLUX_RCMD	DDRC all read command	4-83

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Offset Address	Register	Description	Page
		count register	
0x390	DDRC_HIS_FLUXID_WR	DDRC specified ID write traffic statistics registers	4-83
0x394	DDRC_HIS_FLUXID_RD	DDRC specified ID read traffic statistics registers	4-84
0x0398	DDRC_HIS_FLUXID_WCMD	DDRC all ID write command count register	4-84
0x039C	DDRC_HIS_FLUXID_RCMD	DDRC all ID read command count register	4-84
0x3A0	DDRC_HIS_WLATCNT0	DDRC specified ID write command latency statistics register 0	4-85
0x3A4	DDRC_HIS_WLATCNT1	DDRC specified ID write command latency statistics register 1	4-85
0x3A8	DDRC_HIS_RLATCNT0	DDRC specified ID read command latency statistics register 0	4-86
0x3AC	DDRC_HIS_RLATCNT1	DDRC specified ID read command latency statistics register 1	4-86
0x3B0	DDRC_HIS_INHERE_RLAT_CNT	Read channel inherent latency register	4-87
0x3BC	DDRC_HIS_CMD_SUM	DMC accumulated command count register	4-87
0x4B4	DDRC_HIS_SFC_RDATA3	Software configuration read data register 3	4-88
0x4B8	DDRC_HIS_SFC_RDATA4	SFC read data register	4-88
0x4BC	DDRC_HIS_SFC_RDATA5	SFC read data register	4-88
0x4C0	DDRC_HIS_SFC_RDATA6	SFC read data register	4-89
0x4C4	DDRC_HIS_SFC_RDATA7	SFC read data register	4-89
0x4C8	DDRC_HIS_SFC_RDATA_ECC	SFC read data error correcting code (ECC) register	4-90

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4.1.6.2 Register Description

DDRC_CTRL_SREF

DDRC_CTRL_SREF is a DDRC self-refresh control register.

Offset Address	Register Name	Total Reset Value
0x000	DDRC_CTRL_SREF	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														sref_done	sref_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	sref_done	DDR PHY self-refresh done 1: The transition from 0 to 1 indicates that the DDR PHY completes all required operations after exiting the self-refresh status and the DMC can accept new requests.
[0]	RW	sref_req	SDRAM self-refresh request 0: exit the self-refresh status 1: enter the self-refresh status

DDRC_CTRL_INIT

DDRC_CTRL_INIT is a DDRC initialization control register.

Offset Address	Register Name	Total Reset Value
0x004	DDRC_CTRL_INIT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														dfi_init_start	init_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved

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[1]	RW	dfi_init_start	DFI initialization start 0: normal state 1: DFI interface initialization is started, and the PHY implements related control processes. This bit is automatically cleared by hardware after initialization is complete.
[0]	RW	init_req	Initialization enable 0: Initialization is complete or initialization is performed properly. 1: The SDRAM starts to be initialized.

DDRC_CTRL_SFC

DDRC_CTRL_SFC is a DDRC software configuration command start register.

Offset Address	Register Name	Total Reset Value
0x00C	DDRC_CTRL_SFC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															cmd_req
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	cmd_req	Request of executing the configuration command of the DDRC 0: The command is not executed or the parameter is automatically cleared after the command is executed. 1: The command is requested to be executed.

DDRC_CTRL_PERF

DDRC_CTRL_PERF is a DDRC performance statistics enable register.

Offset Address	Register Name	Total Reset Value
0x010	DDRC_CTRL_PERF	0x0014_F000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															perf_en

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Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	perf_en	Performance statistics mode 0: disabled 1: enabled **NOTE** When perf_mode is 0 and this bit is enabled, the performance statistics register starts cyclic counting. When perf_mode is 1, this bit is automatically cleared after counting is complete.

DDRC_CFG_SREF

DDRC_CFG_SREF is a DDRC self-refresh configuration register.

Offset Address	Register Name	Total Reset Value
0x020	DDRC_CFG_SREF	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sref_arefnum				reserved			clk_switch	reserved			sref_odis	reserved			sref_cc
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:12]	RW	sref_arefnum	Number of self-refresh operations initiated after DDRn SDRAM exits the self-refresh status during the DFS process 0x0: No self-refresh operation is initiated. 0x1–0xF: *n*
[11:9]	RO	reserved	Reserved
[8]	RW	clk_switch	DDRC low-power clock switch. Whether to back press AXI interface command when the DDRC enters the low-power status (DDR self-refresh). 0: The interface commands are not back pressed, and an error is returned directly. 1: The interface commands are back pressed, and the original command is executed after the clock is switched.
[7:5]	RO	reserved	Reserved

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[4]	RW	sref_odis	Output disable for the DDR command and data I/O in self-refresh mode. 0: enabled 1: disabled **NOTE** The configuration is a static configuration. It is recommended that this bit be set to 1 to disable the output of the DDR command and data I/O after the DDR enters the self-refresh status. It is recommended that this bit be set to 0 to enable the output of the DDR command and data I/O before the DDR exits the self-refresh status.
[3:1]	RO	reserved	Reserved
[0]	RW	sref_cc	SDRAM clock control in self-refresh mode 0: The SDRAM clock is enabled. 1: The SDRAM clock is disabled.

DDRC_CFG_PD

DDRC_CFG_PD is a DDR power-down status register.

Offset Address	Register Name	Total Reset Value
0x028	DDRC_CFG_PD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								t_clk_cke				reserved			pd_ac	reserved			pd_cc	pd_prd								reserved			pd_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:20]	RW	t_clk_cke	Relationship between CKL and CKE 0x0–0x7: The delay relative to CKE when the PHY disables the clock is related to the DDR PHY. For example, it is set to 5 for SNPS G2 MULTIPHY or 0 for SNPS 32PHY.
[19:17]	RW	reserved	Reserved
[16]	RW	pd_ac	SDRAM address command dynamic disable in power-down mode 0: enabled 1: disabled **NOTE** This field is valid when pd_en is enabled. The control pins exclude CKE, ODT, CSN, and RESET_N.

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[15:13]	RO	reserved	Reserved
[12]	RW	pd_cc	SDRAM clock control in power-down mode 0: The SDRAM clock is enabled. 1: The SDRAM clock is disabled. Note: This field is valid only when the LPDDR SDRAM or LPDDR2 SDRAM is used.
[11:4]	RW	pd_prd	SDRAM power-down cycle. When the DDRC does not receive any command in consecutive pd_prd cycles, it forces the SDRAM to enter the power-down status. When a command is received, the DDRC forces the SDRAM to exit the power-down status. 0x00: not to enter the power-down status 0x01−0xFF: *n* clock cycles **NOTE** This field is valid only when pd_en is 1.
[3:1]	RO	reserved	Reserved
[0]	RW	pd_en	Automatic power-down enable for the SDRAM 0: disabled 1: enabled

DDRC_CFG_AREF

DDRC_CFG_AREF is a DDRC auto-refresh mode register.

Offset Address	Register Name	Total Reset Value
0x02C	DDRC_CFG_AREF	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				aref_alarm_num		reserved	aref_alarm_en	reserved	aref_mode
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved

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[15:8]	RW	aref_alarm_num	The auto-refresh command is forced to be transmitted if a specific number of auto-refresh commands cannot be transmitted. 0x0–0xFF: lack of (*n* + 1) auto-refresh commands Note that only one auto-refresh command is considered when *n* is 255 because the upper bits are lost after the bits of the 8-bit counter are carried.
[7:5]	RO	reserved	Reserved
[4]	RW	aref_alarm_en	Auto-refresh loss alarm enable (AREF function) 0: disabled 1: enabled
[3:2]	RO	reserved	Reserved
[1:0]	RW	aref_mode	Auto-refresh mode 00: An auto-refresh operation is performed every one tREFI cycle. 01: Three auto-refresh operations are performed every three tREFI cycles. 10: Four auto-refresh operations are performed every five tREFI cycles. 11: Eight auto-refresh operations are performed every nine tREFI cycles.

DDRC_CFG_WORKMODE

DDRC_CFG_WORKMODE is a DDRC operating mode register.

Offset Address	Register Name	Total Reset Value
0x040	DDRC_CFG_WORKMODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																		hdr_mode	read_mode	reserved			wrap_en	reserved			apre_en	func_clkon	data_clkon	cmd_clkon	clk_ratio
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13]	RW	hdr_mode	DFI interface mode select 0: SDR mode 1: HDR mode Only the HDR mode is supported. Therefore, this bit is fixed at 1.

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[12]	RW	read_mode	DDRC read mode 0: associated read mode 1: delay read mode The associated read mode is a mode in which the DDRC samples data based on the data valid signal from the PHY. The delay read mode is a mode in which the DDRC samples the data from the PHY after the internal delay of the DDRC. **NOTE** This field must be set to 0 when DDRC_DTRCTRL[train_mode] is set to 0.
[11:9]	RO	reserved	Reserved
[8]	RW	wrap_en	Wrap command optimization enable 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved
[4]	RW	apre_en	Auto-precharge enable 0: disabled 1: enabled
[3]	RW	func_clkon	Functional module clock enable 0: The clock is internally controlled. 1: The clock is forcibly enabled.
[2]	RW	data_clkon	Data channel clock enable 0: The clock is internally controlled. 1: The clock is forcibly enabled.
[1]	RW	cmd_clkon	Command channel clock enable 0: The clock is internally controlled. 1: The clock is forcibly enabled.
[0]	RW	clk_ratio	Operating mode of the DDRC 0: The ratio of the DDRC frequency to the PHY frequency is 1:1. 1: The ratio of the DDRC frequency to the PHY frequency is 1:2. Only the HDR mode is supported. Therefore, this bit is fixed at 1.

DDRC_CFG_DDRMODE

DDRC_CFG_DDRMODE is a DDR operating mode register.

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Offset Address	Register Name	Total Reset Value
0x050	DDRC_CFG_DDRMODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										rank		reserved		odt_on	zqc_en	reserved			bc_en	reserved			brstlen	reserved			mem_width	reserved		dram_type	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:20]	RW	rank	DDRC rank 00: 1 rank 01: 2 ranks 10: 3 ranks 11: 4 ranks
[19:18]	RO	reserved	Reserved
[17]	RW	odt_on	Whether the ODT signal output to the SDRAM is fixed 0: The signal is controlled by the DDRC. 1: The output is fixed at the wodt configuration of rank 0.
[16]	RW	zqc_en	DDR3 SDRAM ZQ enable 0: disabled 1: enabled **NOTE** This field is valid only for the DDR3 SDRAM or LPDDR2 SDRAM and its default value is 0.
[15:13]	RO	reserved	Reserved
[12]	RW	bc_en	DDR3 burst chop mode 0: disabled 1: enabled **NOTE** This field is valid only for the DDR3 SDRAM.
[11:9]	RO	reserved	Reserved

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[8]	RW	brstlen	Burst length of the DDRC 0: BL4 1: BL8 The burst mode of the DDR2 SDRAM can be set to burst4 or burst8 mode. The burst mode of the DDR3 SDRAM must be burst8 mode. **NOTE** When DDRC_CFG_WORKMODE[clk_ratio] is 1, this bit must be set to 0 (BL4) for the DDR3 SDRAM, which indicates the burst8 mode.
[7:6]	RO	reserved	Reserved
[5:4]	RW	mem_width	Bit width of the storage data bus 00: 8 bits 01: 16 bits 10: 32 bits 11: 64 bits
[3]	RO	reserved	Reserved
[2:0]	RW	dram_type	External memory type 000: reserved 001: LPDDR 010: LPDDR2 011: LPDDR3 100: DDR 101: DDR2 110: DDR3 111: DDR4

DDRC_CFG_RNKVOL

DDRC_CFG_RNKVOL is a DDRC-controlled DDR capacity configuration register.

Offset Address	Register Name	Total Reset Value
0x060	DDRC_CFG_RNKVOL	0x0000_0022

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				mem_map	reserved	mem_bank	reserved	mem_row	reserved	mem_col

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:12]	RW	mem_map	Address translation mode of the SDRAM 00: {Rank, Row, Ba, Col, DW} = AXI_Address 01: {Rank, Ba, Row, Col, DW}= AXI_Address 10: {Rank, Row, Ba, Col, cs, Col, DW}= AXI_Address 11: {Rank, Ba, Row, Col, cs, Col, DW}= AXI_Address This parameter can be set to 10 or 11 only when DDRC_CFG_AREF is valid. When there are multiple ranks, the configuration must be the same for each rank.
[11:9]	RO	reserved	Reserved
[8]	RW	mem_bank	Number of banks of a single SDRAM 0: 4 banks 1: 8 banks
[7]	RO	reserved	Reserved
[6:4]	RW	mem_row	Bit width of the row address of a single SDRAM 000: 11 bits 001: 12 bits 010: 13 bits 011: 14 bits 100: 15 bits 101: 16 bits Other values: reserved
[3]	RO	reserved	Reserved
[2:0]	RW	mem_col	Bit width of the column address of a single SDRAM 000: 8 bits 001: 9 bits 010: 10 bits 011: 11 bits 100: 12 bits Other values: reserved **NOTE** The DDRC does not support the DDR whose col address is less than 8. That is, the 32-bit LPDDR2 with the capacity of 64 Mbits is not supported.

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DDRC_CFG_EMRS01

DDRC_CFG_EMRS01 is a configuration register for DDR mode register 0 and mode register 1.

Offset Address	Register Name	Total Reset Value
0x070	DDRC_CFG_EMRS01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	emrs1																mrs															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	emrs1	DDR*n* SDRAM extended mode register 1
[15:0]	RW	mrs	DDR*n* SDRAM mode register

DDRC_CFG_EMRS23

DDRC_CFG_EMRS23 is a configuration register for DDR mode register 2 and mode register 3.

Offset Address	Register Name	Total Reset Value
0x074	DDRC_CFG_EMRS23	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	emrs3																emrs2															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	emrs3	DDR*n* SDRAM extended mode register 3
[15:0]	RW	emrs2	DDR*n* SDRAM extended mode register 2

DDRC_CFG_TIMING0

DDRC_CFG_TIMING0 is DDRC timing parameter register 0.

Timing parameters in the register are obtained from DDR user manuals.

- If DDRC_CFG_WORKMODE[clk_ration] is 1, convert the time specified in the DDR user manuals into the DDR clock cycle and then divide it by 2. If the value has a remainder, add 1 to the value.
- If DDRC_CFG_WORKMODE[clk_ration] is 0, convert the time specified in the DDR user manuals into the DDR clock cycle and then use it directly.

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Offset Address	Register Name	Total Reset Value
0x080	DDRC_CFG_TIMING0	0xFFFF_FF3F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tmrd				trrd				trp				trcd				trc								reserved				tras			
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	1

Bits	Access	Name	Description
[31:28]	RW	tmrd	Number of wait cycles for the load mode register (LMR) command. The value is the larger value between tMOD and tMRD for the DDR2 or LPDDR2 and is tMRD for the DDR3. 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles
[27:24]	RW	trrd	Number of wait cycles from ACT bank A to ACT bank B 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles
[23:19]	RW	trp	Number of wait cycles for the disable command (PRE period) 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles
[18:14]	RW	trcd	Number of wait cycles from the ACT bank command to the read or write command 0x0–0x3: 3 clock cycles 0x4–0xF: *n* clock cycles
[13:8]	RW	trc	Number of wait cycles from an ACT bank command to the next ACT bank command 0x00–0x01: 1 clock cycle 0x02–0x3F: *n* clock cycles
[7:6]	RO	reserved	Reserved
[5:0]	RW	tras	Number of wait cycles from the ACT bank command to the disable command (PRE) 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles

DDRC_CFG_TIMING1

DDRC_CFG_TIMING1 is DDRC timing parameter register 1.

Timing parameters in the register are obtained from DDR user manuals.

- If DDRC_CFG_WORKMODE[clk_ration] is 1, convert the time specified in the DDR user manuals into the DDR clock cycle and then divide it by 2. If the value has a remainder, add 1 to the value.

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- If DDRC_CFG_WORKMODE[clk_ration] is 0, convert the time specified in the DDR user manuals into the DDR clock cycle and then use it directly.

Offset Address	Register Name	Total Reset Value
0x084	DDRC_CFG_TIMING1	0xFF24_51FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tsre								trtw				twl				tcl				reserved				trfc							
Reset	1	1	1	1	1	1	1	1	0	0	1	0	0	1	0	0	0	1	0	1	0	0	0	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:24]	RW	tsre	Number of wait cycles from the self-refresh exit command to the read command 0x0: reserved 0x01–0xFF: (*n* x 4) clock cycles This field is set to the larger value between tXSDLL and tXS for the DDR3 SDRAM.
[23:20]	RW	trtw	Delay from the last read data command to the first write data command 0x0–0x1: 1 clock cycle 0x2–0xF: (*n* + 1) clock cycles **NOTE** In DDR2/DDR3 mode, the field value depends on the latency of the board, package, and I/O; in LPDDR/LPDDR2/LPDDR3 mode, the field value depends on the latency of the board, package, and I/O and tdqsckmax.
[19:16]	RW	twl	Number of wait cycles from the write command to the write data command 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles For example, 0x3 indicates three clock cycles. The clock cycles are obtained from DDR user manuals. You can directly configure this field based on the clock cycles.
[15:12]	RW	tcl	Column address strobe latency (CL) from the read command to the read data operation 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles The clock cycles are obtained from DDR user manuals. You can directly configure this field based on the clock cycles.
[11:9]	RO	reserved	Reserved

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[8:0]	RW	trfc	Number of wait cycles for the AREF period or AREF to the ACT command. The field value is set to the maximum value of max{trfc, tzqcs}. 0x00: reserved 0x01–0x1FF: *n* clock cycles

DDRC_CFG_TIMING2

DDRC_CFG_TIMING2 is DDRC timing parameter register 2.

Timing parameters in the register are obtained from DDR user manuals.

- If DDRC_CFG_WORKMODE[clk_ration] is 1, convert the time specified in the DDR user manuals into the DDR clock cycle and then divide it by 2. If the value has a remainder, add 1 to the value.
- If DDRC_CFG_WORKMODE[clk_ration] is 0, convert the time specified in the DDR user manuals into the DDR clock cycle and then use it directly.

Offset Address	Register Name	Total Reset Value
0x88	DDRC_CFG_TIMING2	0xF303_F000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tcke				twtr				reserved								tfaw				reserved	taref										
Reset	1	1	1	1	0	0	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	tcke	Minimum cycle of retaining the self-refresh mode 0x0: reserved 0x1–0xF: *n* clock cycles The field value must be the maximum value among tCKESR, tCKSRE, tCKSRX, and tCKE.
[27:24]	RW	twtr	Number of wait cycles for the last write data to the write-to-read command 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles For example, the value 0x3 indicates three clock cycles.
[23:18]	RO	reserved	Reserved
[17:12]	RW	tfaw	Number of clock cycles for four consecutive activation commands 0x00–0x3F: *n* clock cycles For example, the value 0x14 indicates 20 clock cycles.
[11]	RO	reserved	Reserved

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[10:0]	RW	taref	Number of auto-refresh cycles 0x000: forbidden 0x001–0x7FF: The auto-refresh cycle of the SDRAM is (16 x *n*) clock cycles. For example, the value 0x008 indicates 128 (16 x 8) clock cycles. The interval tREFI is 7800/16/tclk. Tclk is the running cycle of the SDRAM. When DDRC_CFG_AREF[aref_mode] is 1, the interval must be set to 8 x tREFI.

DDRC_CFG_TIMING3

DDRC_CFG_TIMING3 is DDRC timing parameter register 3.

Timing parameters in the register are obtained from DDR user manuals.

- If DDRC_CFG_WORKMODE[clk_ration] is 1, convert the time specified in the DDR user manuals into the DDR clock cycle and then divide it by 2. If the value has a remainder, add 1 to the value.
- If DDRC_CFG_WORKMODE[clk_ration] is 0, convert the time specified in the DDR user manuals into the DDR clock cycle and then use it directly.

Offset Address	Register Name	Total Reset Value
0x8C	DDRC_CFG_TIMING3	0xFFDF_F0F2

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tzq_prd												tzqinit								taond				txard				trtp			
Reset	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	0	0	1	0

Bits	Access	Name	Description
[31:22]	RW	tzq_prd	Number of ZQCS command cycles 0x000: The ZQCS command is forbidden. 0x001–0x3FF: (*n* x 128) AREF cycles The number of ZQCS command cycles is equal to (*n* x 128) taref clock cycles.
[21:12]	RW	tzqinit	Number of ZQ initialization delay cycles 0x0–0x1FF: (*n* + 1) clock cycles The field value must be the larger value between tZQINIT and tDLLK.
[11:8]	RW	taond	Number of ODT enable/disable cycles In DDR2 mode (taond/taofd): 0x0: 2/2.5 0x1: 3/3.5 0x2: 4/4.5 0x3: 5/5.5

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			Other values: reserved In DDR3 mode, this value is set to tWL − 1. In 1:2 mode, this value needs to be set to tWL − 2. However, if the PHY has been separately processed, the value tWL − 1 is used. In LPDDR3 mode, the value is set to tODTon(max)/tCK.
[7:4]	RW	txard	Number of wait cycles of exiting the DDR low-power mode 0x0–0xF: *n* clock cycles (in decimal) For example, 0x7 indicates seven clock cycles. The field value is the maximum value among tXP, tXARD, tXARDS, and tXS. In DDR3 mode, the field value is the larger value between tXP and tCKE.
[3:0]	RW	trtp	Wait delay from the read command to the disable command 000–010: 2 clock cycles 011–111: *n* clock cycles Trtp is calculated as follows: Trtp = AL + BL/2 + Max (trtp, 2) – 2 **NOTE** For DDR2, Trtp is greater than or equal to 2; for DDR3, Trtp is greater than or equal to 4.

DDRC_CFG_TIMING4

DDRC_CFG_TIMING4 is DDRC timing parameter register 4.

Timing parameters in the register are obtained from DDR user manuals.

- If DDRC_CFG_WORKMODE[clk_ration] is 1, convert the time specified in the DDR user manuals into the DDR clock cycle and then divide it by 2. If the value has a remainder, add 1 to the value.
- If DDRC_CFG_WORKMODE[clk_ration] is 0, convert the time specified in the DDR user manuals into the DDR clock cycle and then use it directly.

Offset Address	Register Name	Total Reset Value
0x090	DDRC_CFG_TIMING4	0x01FF_2000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								tmod				twlo				reserved				twldqsen				reserved				twlmrd			
Reset	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	1	0		0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved

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[24:20]	RW	tmod	Delay from the MRS command to ODT and ZQCL validity 0x0–0x1: 1 clock cycle 0x2–0x1F: *n* clock cycles
[19:16]	RW	twlo	DDR3 write level status delay 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles The parameter is equal to twlo + twloe.
[15:14]	RO	reserved	Reserved
[13:8]	RW	twldqsen	DDR3 write level start delay 0x0 and 0x1: one clock cycle 0x2−0x3F: *n* clock cycles
[7:6]	RO	reserved	Reserved
[5:0]	RW	twlmrd	Delay of the initial valid DDR3 write level DQS 0x0 and 0x1: one clock cycle 0x2−0x3F: n clock cycles

DDRC_CFG_TIMING5

DDRC_CFG_TIMING5 is DDRC timing parameter register 5.

Timing parameters in the register are obtained from DDR user manuals.

- If DDRC_CFG_WORKMODE[clk_ration] is 1, convert the time specified in the DDR user manuals into the DDR clock cycle and then divide it by 2. If the value has a remainder, add 1 to the value.
- If DDRC_CFG_WORKMODE[clk_ration] is 0, convert the time specified in the DDR user manuals into the DDR clock cycle and then use it directly.

Offset Address	Register Name	Total Reset Value
0x094	DDRC_CFG_TIMING5	0x1113_FF1F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												trnk2rnk				tzqcs								reserved				twr			
Reset	0	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	1	1	1	1

Bits	Access	Name	Description
[31:20]	RW	reserved	Reserved

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[19:16]	RW	trnk2rnk	Rank-to-rank delay 0000–1111: *n* clock cycles Note: In DDR2/DDR3 mode, the field value depends on the latency of the board, package, and I/O; in LPDDR/LPDDR2/LPDDR3 mode, the field value depends on the latency of the board, package, and I/O, and tdqsckmax.
[15:8]	RW	tzqcs	ZQCS calibration delay period 0x0–0xFF: (*n* + 1) clock cycles The field value must be greater than or equal to 10 due to the DMC design.
[7:5]	RO	reserved	Reserved
[4:0]	RW	twr	Number of wait cycles of write recovery 0x0–0x1: 1 clock cycle 0x2–0x1F: *n* clock cycles **NOTE** When DFS is required, tWR must be set based on the maximum chip frequency that may be used. tWR cannot be changed when the DDR frequency changes.

DDRC_CFG_TIMING6

DDRC_CFG_TIMING6 is DDRC timing parameter register 6.

Timing parameters in the register are obtained from DDR user manuals.

- If DDRC_CFG_WORKMODE[clk_ration] is 1, convert the time specified in the DDR user manuals into the DDR clock cycle and then divide it by 2. If the value has a remainder, add 1 to the value.
- If DDRC_CFG_WORKMODE[clk_ration] is 0, convert the time specified in the DDR user manuals into the DDR clock cycle and then use it directly.

Offset Address	Register Name	Total Reset Value
0x098	DDRC_CFG_TIMING6	0x0000_00FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								tcksrx				tcksre			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:4]	RW	tcksrx	tCKSRX for the DDR3 or DDR4, that is, number of advanced valid beats before the DDR exits the self-refresh status 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles

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[3:0]	RW	tcksre	tCKSRE for the DDR3 or DDR4, that is, number of beats to be retained after the DDR enters the self-refresh status 0x0–0x1: 1 clock cycle 0x2–0xF: *n* clock cycles

DDRC_CFG_NXT_TIMING0

DDRC_CFG_NXT_TIMING0 is DDRC timing parameter register 0 for switching the frequency.

Offset Address	Register Name	Total Reset Value
0x0A0	DDRC_CFG_NXT_TIMING0	0xFFFF_FF3F

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6	5 4	3 2 1 0
Name	tmrd	trrd	trp	trcd	trc		reserved	tras	
Reset	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	0 0	1 1	1 1 1 1

Bits	Access	Name	Description
[31:28]	RW	tmrd	Number of wait cycles for the loading command of the loading mode register (LMR) 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles
[27:24]	RW	trrd	Number of wait cycles from ACT bank A to ACT bank B 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles
[23:19]	RW	trp	Number of wait cycles for the disable command (PRE period) 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles
[18:14]	RW	trcd	Number of wait cycles from the ACT bank command to the read or write command 0x0−0x3: three clock cycles 0x4−0xF: *n* clock cycles
[13:8]	RW	trc	Number of wait cycles from an ACT bank command to the next ACT bank command 0x00–0x01: one clock cycle 0x02−0x3F: *n* clock cycles
[7:6]	RO	reserved	Reserved

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[5:0]	RW	tras	Number of wait cycles from the ACT bank command to the disable command (PRE) 0x00 − 0x01: one clock cycle 0x02−0x3F: *n* clock cycles

DDRC_CFG_NXT_TIMING1

DDRC_CFG_NXT_TIMING1 is DDRC timing parameter register 1 for switching the frequency.

Offset Address	Register Name	Total Reset Value
0x0A4	DDRC_CFG_NXT_TIMING1	0xFF24_51FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tsre								trtw				twl				tcl				reserved				trfc							
Reset	1	1	1	1	1	1	1	1	0	0	1	0	0	1	0	0	0	1	0	1	0	0	0	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:24]	RW	tsre	Number of wait cycles from the self-refresh exit command to the read command 0x0: one clock cycle 0x01–0xFF: *n* x 4 clock cycles When the DDR3 SDRAM is used, the value is set to tXSDLL.
[23:20]	RW	trtw	Delay from the last read data command to the first write data command 0x0 and 0x1: one clock cycle 0x2−0xF: *n* + 1 clock cycles
[19:16]	RW	twl	Number of wait cycles from the write command to the data write operation 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles For example, the value 0x3 indicates three clock cycles. Note: In DDR2 mode, tw1 is set to tcl − 1 and the condition (twl − taond ≥ 1) must be met.
[15:12]	RW	tcl	Column address strobe latency (CL) from the read command to the data read operation 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles
[11:9]	RO	reserved	Reserved

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[8:0]	RW	trfc	Number of wait cycles for the AREF period or AREF to the ACT command. The field value is set to the maximum value of max{trfc, tzqcs}. 0x00−0x01: one clock cycle 0x02−0xFF: *n* clock cycles

DDRC_CFG_NXT_TIMING2

DDRC_CFG_NXT_TIMING2 is DDRC timing parameter register 2 for switching the frequency.

Offset Address	Register Name	Total Reset Value
0x0A8	DDRC_CFG_NXT_TIMING2	0xF303_F000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tcke				twtr				reserved								tfaw				reserved	taref										
Reset	1	1	1	1	0	0	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	tcke	Minimum cycle of retaining the low-power mode 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles The field value must be the maximum value among tCKESR, tCKSRE, tCKSRX, and tCKE.
[27:24]	RW	twtr	Number of wait cycles for the last data write operation to the write-to-read command 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles For example, the value 0x3 indicates three clock cycles.
[23:18]	RO	reserved	Reserved
[17:12]	RW	tfaw	Number of clock cycles for four consecutive activation commands 0x00−0x3F: *n* clock cycles For example, the value 0x14 indicates 20 clock cycles.
[11]	RO	reserved	Reserved

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[10:0]	RW	taref	Number of auto-refresh cycles 0x000: auto-refresh disabled 0x001–0x7FF: The auto-refresh cycle of the SDRAM is 16 x n clock cycles. For example, the value 0x008 indicates 128 (16 x 8) clock cycles. The interval tREF is 7800/16/tclk. Tclk is the running cycle of the SDRAM.

DDRC_CFG_NXT_TIMING3

DDRC_CFG_NXT_TIMING3 is DDRC timing parameter register 3 for switching the frequency.

Offset Address	Register Name	Total Reset Value
0x0AC	DDRC_CFG_NXT_TIMING3	0xFFDF_F0F2

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tzq_prd												tzqinit								taond				txard				trtp			
Reset	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	0	0	1	0

Bits	Access	Name	Description
[31:22]	RW	tzq_prd	Number of ZQCS command cycles 0x000: The ZQCS command is disabled. 0x001−0x3FF: *n* x 128 AREF cycles The time of ZQCS command cycles is equal to n x 128 taref clock cycles.
[21:12]	RW	tzqinit	Number of ZQ initialization delay cycles 0x0–0x3ff: (*n* + 1) clock cycles The value needs to be set to the larger one between tZQINIT and tDLLK.
[11:8]	RW	taond	Number of ODT enable/disable cycles In DDR2 mode (taond/taofd): 0x0: 2/2.5 0x1: 3/3.5 0x2: 4/4.5 0x3: 5/5.5 Other values: reserved In DDR3 mode, the value is set to tWL − 1. In 1:2 mode, the value needs to be set to tWL – 2. However, if the PHY has been separately processed, the value tWL − 1 is used. In LPDDR3 mode, the value is set to tODTon(max)/tCK.
[7:4]	RW	txard	Number of wait cycles of exiting the DDR low-power mode

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			0x0–0xF: *n* clock cycles. *n* indicates a decimal value. For example, 0x7 indicates seven clock cycles. The field value is the maximum value among tXP, tXARD, tXARDS, and tXS. In DDR3 mode, when the register is set to tXS, txard only needs to be set to an equivalent clock cycle of 10 ns.
[3:0]	RW	trtp	Wait delay from the read command to the disable command 0000−0010: two clock cycles 0011–1111: *n* clock cycles Trtp = AL + BL/2 + Max (trtp, 2) – 2

DDRC_CFG_NXT_TIMING4

DDRC_CFG_NXT_TIMING4 is DDRC timing parameter register 4 for switching the frequency.

Offset Address	Register Name	Total Reset Value
0x0B0	DDRC_CFG_NXT_TIMING4	0x01FF_2000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								tmod				twlo				reserved				twldqsen				reserved				twlmrd			
Reset	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RW	reserved	Reserved
[24:20]	RW	tmod	Delay from the MRS command to ODT valid 0x0 and 0x1: one clock cycle 0x02−0x1F: *n* clock cycles
[19:16]	RW	twlo	DDR3 write level status delay parameter 0x0 and 0x1: one clock cycle 0x2−0xF: *n* clock cycles The parameter is equal to (twlo + twloe).
[15:14]	RO	reserved	Reserved
[13:8]	RW	twldqsen	DDR3 write level start delay 0x0 and 0x1: one clock cycle 0x2−0x3F: *n* clock cycles
[7:6]	RO	reserved	Reserved

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[5:0]	RW	twlmrd	Delay of the initial valid DDR3 write level DQS 0x0 and 0x1: one clock cycle 0x2−0x3F: *n* clock cycles

DDRC_CFG_NXT_TIMING5

DDRC_CFG_NXT_TIMING5 is DDRC timing parameter register 5 for switching the frequency.

Offset Address	Register Name	Total Reset Value
0x0B4	DDRC_CFG_NXT_TIMING5	0x1113_FF1F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												trnk2rnk				tzqcs								reserved				twr			
Reset	0	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	1	1	1	1

Bits	Access	Name	Description
[31:20]	RW	reserved	Reserved
[19:16]	RW	trnk2rnk	Rank-to-rank delay 0000–1111: *n* clock cycles
[15:8]	RW	tzqcs	ZQCS calibration delay period 0x0−0xFF: (*n* + 1) clock cycles
[7:5]	RW	reserved	Reserved
[4:0]	RW	twr	Number of wait cycles of write recovery 0x0 and 0x1: one clock cycle 0x2−0x1F: *n* clock cycles **NOTE** When DFS is required, tWR must be set based on the maximum chip frequency that may be used. tWR cannot be changed when the DDR frequency changes.

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DDRC_CFG_NXT_TIMING6

DDRC_CFG_NXT_TIMING6 is DDRC timing parameter register 6 for switching the frequency.

Offset Address	Register Name	Total Reset Value
0x0B8	DDRC_CFG_NXT_TIMING6	0x0000_00FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				todt_sft				tcksrx				tcksre			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:12]	RW	reserved	Reserved
[11:8]	RW	todt_sft	ODT delay control in LPDDR3 mode, for compensating the delay of the DDR PHY for ODT IO of POP LPDDR3 0x0–0xF: *n* clock cycles ODT delay = tWL − taond + todt_sft. Note that the maximum ODT delay cannot exceed 0xF.
[7:4]	RW	tcksrx	tCKSRX for the DDR3 or DDR4, that is, number of advanced valid beats before the DDR exits the self-refresh status 0x0 and 0x1: one clock cycle 0x2–0xF: *n* clock cycles
[3:0]	RW	tcksre	tCKSRE for the DDR3 or DDR4, that is, number of beats to be retained after the DDR enters the self-refresh status 0x0 and 0x1: one clock cycle 0x2–0xF: *n* clock cycles

DDRC_CFG_BLDATA

DDRC_CFG_BLDATA is a DDRC pre-received write data configuration register.

Offset Address	Register Name	Total Reset Value
0x0BC	DDRC_CFG_BLDATA	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												bl_data			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	bl_data	Number of DMC data segments corresponding to each DDR command in the current mode 0x0–0xF: *n* data segments

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DDRC_CFG_ODT

DDRC_CFG_ODT is a DDR ODT feature configuration register.

Offset Address	Register Name	Total Reset Value
0x0C0	DDRC_CFG_ODT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	rodt3	rodt2	rodt1	rodt0	wodt3	wodt2	wodt1	wodt0
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RW	rodt3	Read ODT enable for rank 3 0: disabled 1: enabled
[27:24]	RW	rodt2	Read ODT enable for rank 2 0: disabled 1: enabled
[23:20]	RW	rodt1	Read ODT enable for rank 1 0: disabled 1: enabled
[19:16]	RW	rodt0	Read ODT enable for rank 0 0: disabled 1: enabled
[15:12]	RW	wodt3	Write ODT enable for rank 3 0: disabled 1: enabled
[11:8]	RW	wodt2	Write ODT enable for rank 2 0: disabled 1: enabled
[7:4]	RW	wodt1	Write ODT enable for rank 1 0: disabled 1: enabled
[3:0]	RW	wodt0	Write ODT enable for rank 0 0: disabled 1: enabled

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DDRC_CFG_DMCLVL

DDRC_CFG_DMCLVL is a DDRC command queue depth threshold configuration register.

Offset Address	Register Name	Total Reset Value
0x0C4	DDRC_CFG_DMCLVL	0x0000_010C

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												que_level			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	que_level	Depth of the command register FIFO in the DMC 0x1−0x10: depth of *n* commands Other values: reserved

DDRC_CFG_DDRPHY

DDRC_CFG_DDRPHY is a DDR IO configuration register.

Offset Address	Register Name	Total Reset Value
0x200	DDRC_CFG_DDRPHY	0x0000_1000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			phy_upden	trdlat				reserved				phy_zqen	reserved		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	phy_upden	dfi_phyupd_en enable in response to the DDR PHY 0: disabled 1: enabled
[11:8]	RW	trdlat	Inherent delay of the DDR PHY 0x0–0xF: (*n* + 1) clock cycles The value needs to be set to 8 when synopsys3/2phy is used.
[7:5]	RW	reserved	Reserved

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[4]	RW	phy_zqen	DDRPHY dfi_ctrlupd_req enable 0: disabled 1: enabled
[3:0]	RO	reserved	Reserved

DDRC_CFG_SFC

DDRC_CFG_SFC is a DDRC software DDR command attribute register.

Offset Address	Register Name	Total Reset Value
0x210	DDRC_CFG_SFC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cmd_mrs																cmd_ma								cmd_rank				reserved	cmd_type		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	cmd_mrs	Value of the DDR mode register when the LMR command is configured. This field is valid for the lower eight bits of the LPDDR2.
[15:8]	RW	cmd_ma	Address for the mode register to be configured when the LMR command is configured in LPDDR mode
[7:4]	RW	cmd_rank	DDR rank for executing commands 0: The configuration command is not executed. 1: The configuration command is executed. **NOTE** Each bit controls a rank. For example, cmd_rank[0] control DDR rank 0.
[3]	RO	reserved	Reserved
[2:0]	RW	cmd_type	DDR command configuration 000: enter the deep power-down status 001: exit the deep power-down status 010: LMR command 011: ZQCL 100: WRITE command 101: read command 110: precharge all command 111: read MRS command

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DDRC_CFG_SFC_ADDR0

DDRC_CFG_SFC_ADDR0 is read/write memory address register 0 for the software configuration module.

Offset Address	Register Name	Total Reset Value
0x214	DDRC_CFG_SFC_ADDR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sfc_row																sfc_col															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	sfc_row	Row address for the memory that is read or written by the software configuration module
[15:0]	RW	sfc_col	Column address for the memory that is read or written by the software configuration module **NOTE** The DDRC access address must be aligned based on the current DMC bit width. For example, if the current DMC bit width is 128 bits: When the 64-/72-bit external DDR is connected, the DMC addresses are accessed based on {sfc_col[15:1], 1'b0}. When the 32-/36-bit external DDR is connected, the DMC addresses are accessed based on {sfc_col[15:2]0.2'b0}.

DDRC_CFG_SFC_ADDR1

DDRC_CFG_SFC_ADDR1 is read/write memory address register 1 for the software configuration module.

Offset Address	Register Name	Total Reset Value
0x218	DDRC_CFG_SFC_ADDR1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														sfc_bank	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	sfc_bank	Bank address for the memory that is read or written by the software configuration module

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DDRC_CFG_SFC_WDATA0

DDRC_CFG_SFC_WDATA0 is write data configuration register 0 for the software configuration module.

Offset Address	Register Name	Total Reset Value
0x21C	DDRC_CFG_SFC_WDATA0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdata0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	wdata0	Bits 127−96 of the written data of the software configuration module

DDRC_CFG_SFC_WDATA1

DDRC_CFG_SFC_WDATA1 is write data configuration register 1 for the software configuration module.

Offset Address	Register Name	Total Reset Value
0x220	DDRC_CFG_SFC_WDATA1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdata1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	wdata1	Bits 95−64 of the written data of the software configuration module

DDRC_CFG_SFC_WDATA2

DDRC_CFG_SFC_WDATA2 is write data configuration register 2 for the software configuration module.

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Offset Address	Register Name	Total Reset Value
0x224	DDRC_CFG_SFC_WDATA2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdata2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	wdata2	Bits 63−32 of the written data of the software configuration module

DDRC_CFG_SFC_WDATA3

DDRC_CFG_SFC_WDATA3 is write data configuration register 3 for the software configuration module.

Offset Address	Register Name	Total Reset Value
0x228	DDRC_CFG_SFC_WDATA3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wdata3																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	wdata3	Bits 31−0 of the written data of the software configuration module

DDRC_CFG_PERF

DDRC_CFG_PERF is a DDRC performance statistics mode register.

Offset Address	Register Name	Total Reset Value
0x270	DDRC_CFG_PERF	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		flux_en	perf_mode	perf_prd																											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved

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[29]	RW	flux_en	DMC traffic monitoring enable 0: disabled 1: enabled When traffic monitoring is enabled, the statistics module provides the number of cycles occupied by the DDR interface by ID to each request port. The traffic can be controlled by using this field and the port traffic setting function.
[28]	RW	perf_mode	Performance statistical mode 0: continuous trigger mode. The performance counter continues to count. This ensures no overflow within 1s in continuous count mode. 1: single trigger mode. When the performance statistics time reaches perf_prd, the count value is retained and counting stops. **NOTE** After the count value overflows, the value is retained.
[27:0]	RW	perf_prd	Performance statistics cycle 0x0–0x1: invalid 0x2–0xFFFFFFF: statistics cycles The actual statistics cycle is perf_prd x 16 x tclk (tclk is the bus clock cycle of the DDRC) **NOTE** This field is valid only when perf_mode is 1. When perf_mode is 0, the performance counter keeps on counting.

DDRC_CFG_STAID

DDRC_CFG_STAID is a DDRC performance statistics command ID register.

Offset Address	Register Name	Total Reset Value
0x274	DDRC_CFG_STAID	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												sta_id																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:0]	RW	sta_id	Performance statistics by a specified ID. This field works with sta_idmask.

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DDRC_CFG_STAIDMSK

DDRC_CFG_STAIDMSK is a DDR performance statistics command ID mask register.

Offset Address	Register Name	Total Reset Value
0x278	DDRC_CFG_STAIDMSK	0x0000_1FFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				sta_idmask											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved **NOTE** This field must be set to 0.
[12:0]	RW	sta_idmask	Specified ID mask The DDRC performance statistics register collects statistics only for specified ID commands. Cmd_id&sta_idmask = sta_id

DDRC_INTMSK

DDRC_INTMSK is a DDRC interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x280	DDRC_INTMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																aref_alarm_int_mask	reserved			rdtimeout_int_mask	reserved							sref_err_int_mask	reserved				flux_int_mask	serr_int_mask
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RW	aref_alarm_int_mask	DDR AREF command error interrupt mask enable 0: enabled 1: disabled
[15:13]	RO	reserved	Reserved

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[12]	RW	rdtimeout_int_mask	DDR PHY read data timeout interrupt mask enable 0: enabled 1: disabled
[11:6]	RO	reserved	Reserved
[5]	RW	sref_err_int_mask	Interface command error interrupt enable in DDR self-refresh mode 0: enabled 1: disabled
[4:2]	RO	reserved	Reserved
[1]	RW	flux_int_mask	DDR FLUX statistics period reach interrupt mask enable 0: enabled 1: disabled
[0]	RW	serr_int_mask	DDR ECC single-bit error interrupt mask enable 0: enabled 1: disabled

DDRC_RINT

DDRC_RINT is a DDRC raw interrupt register.

Offset Address	Register Name	Total Reset Value
0x284	DDRC_RINT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															aref_alarm_rint	reserved			rdtimeout_rint	reserved							sref_err_rint	reserved				flux_rint	serr_rint
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	INT_WC	aref_alarm_rint	DDR AREF command error raw interrupt. Writing 1 clears the interrupt.
[15:13]	RO	reserved	Reserved
[12]	INT_WC	rdtimeout_rint	DDR PHY read data timeout raw interrupt. Writing 1 clears the interrupt.
[11:6]	RO	reserved	Reserved

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[5]	INT_WC	sref_err_rint	Self-refresh interface command access interrupt. Writing 1 clears the interrupt.
[4:2]	RO	reserved	Reserved
[1]	RW	flux_rint	DDR FLUX statistics period reach interrupt. Writing 1 clears the interrupt.
[0]	INT_WC	serr_rint	DDR ECC single-bit error raw interrupt. Writing 1 clears the interrupt.

DDRC_INTSTS

DDRC_INTSTS is a DDRC interrupt status register.

Offset Address	Register Name	Total Reset Value
0x288	DDRC_INTSTS	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved			aref_alarm_intsts	reserved	rdtimeout_intsts	reserved	sref_err_intsts	reserved	flux_intsts	serr_intsts
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	INT	aref_alarm_intsts	DDR AREF command error interrupt
[15:13]	RO	reserved	Reserved
[12]	INT	rdtimeout_intsts	DDR PHY read data timeout interrupt
[11:6]	RO	reserved	Reserved
[5]	INT	sref_err_intsts	Self-refresh interface command access interrupt
[4:2]	RO	reserved	Reserved
[1]	RW	flux_intsts	DDR FLUX statistics period reach interrupt
[0]	INT	serr_intsts	DDR ECC single-bit error interrupt

DDRC_CURR_STATUS

DDRC_CURR_STATUS is a DDRC status register.

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Offset Address	Register Name	Total Reset Value
0x290	DDRC_CURR_STATUS	0x0000_0101

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																							busy_func	reserved			busy_dmc	reserved			busy
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:9]	RO	reserved	Reserved
[8]	RO	busy_func	Status of the DDRC FUNC module. This field is in the SREF status during reset. 0: idle 1: A command is being processed.
[7:5]	RO	reserved	Reserved
[4]	RO	busy_dmc	Status of the DDRC DMC 0: idle 1: A command is being processed.
[3:1]	RO	reserved	Reserved
[0]	RO	busy	Overall status of the DDRC 0: idle 1: A command is being processed.

DDRC_CURR_FUNC

DDRC_CURR_FUNC is a DDRC FUNC module status register.

Offset Address	Register Name	Total Reset Value
0x294	DDRC_CURR_FUNC	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				in_phyupd	reserved														in_sfc	in_pd				reserved			dfi_init_complete	in_init	reserved			in_sref

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28]	RO	in_phyupd	DDRC PHY update status 0: idle 1: The PHY is being updated.
[27:13]	RO	reserved	Reserved
[12]	RO	in_sfc	DDRC SFC status 0: idle 1: An SFC command is being processed.
[11:8]	RO	in_pd	DDRC power-down status 0: normal 0: power-down status Each bit represents a rank.
[7:6]	RO	reserved	Reserved
[5]	RO	dfi_init_complete	DFI initialization status 0: initializing 1: normal operating mode
[4]	RO	in_init	DDRC initialization status 0: normal 1: initializing
[3:1]	RO	reserved	Reserved
[0]	RO	in_sref	DDR self-refresh status 0: normal 1: self-refresh

DDRC_CURR_EXECST

DDRC_CURR_EXECST is a DDRC command state machine status register.

Offset Address	Register Name	Total Reset Value
0x2A0	DDRC_CURR_EXECST	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dmc_ct																dmc_cv															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	dmc_ct	Controller command type 0: read command 1: write command
[15:0]	RO	dmc_cv	Controller command validity flag 0: invalid 1: valid

DDRC_CURR_WGFIFOST

DDRC_CURR_WGFIFOST is a DDRC write data FIFO status register.

Offset Address	Register Name	Total Reset Value
0x2A4	DDRC_CURR_WGFIFOST	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															wgntfifo_e
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	wgntfifo_e	WGNT_FIFO empty or full status. The empty status indicates that all written data is received, and the non-empty status indicates that the written data is not completely received. 0: not empty 1: empty

DDRC_HIS_FLUX_WR

DDRC_HIS_FLUX_WR is a DDRC all write command traffic statistics register.

Offset Address	Register Name	Total Reset Value
0x380	DDRC_HIS_FLUX_WR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flux_wr																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	flux_wr	Write traffic statistics of all masters with the IDs of the DDRC. The counting is performed in the valid statistics cycle. The unit is the DMC bit width. When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_FLUX_RD

DDRC_HIS_FLUX_RD is a DDRC all read command traffic statistics register.

Offset Address	Register Name	Total Reset Value
0x384	DDRC_HIS_FLUX_RD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flux_rd																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	flux_rd	Read traffic statistics of all masters with the IDs of the DDRC. The counting is performed in the valid statistics cycle. The unit is the DMC bit width. When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_FLUX_WCMD

DDRC_HIS_FLUX_WCMD is a DDRC all write command count register.

Offset Address	Register Name	Total Reset Value
0x388	DDRC_HIS_FLUX_WCMD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flux_wr_cmd																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	flux_wr_cmd	Write command count of the masters with all IDs of the DDRC. The counting is performed in the valid statistics cycle.

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			When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_FLUX_RCMD

DDRC_HIS_FLUX_RCMD is a DDRC all read command count register.

Offset Address	Register Name	Total Reset Value
0x38C	DDRC_HIS_FLUX_RCMD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flux_rd_cmd																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	flux_rd_cmd	Read command count of the masters with all IDs of the DDRC. The counting is performed in the valid statistics cycle. When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_FLUXID_WR

DDRC_HIS_FLUXID_WR is a DDRC specified ID write traffic statistics registers.

Offset Address	Register Name	Total Reset Value
0x390	DDRC_HIS_FLUXID_WR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fluxid_wr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	fluxid_wr	Write traffic statistics of the master with the specified ID of the DDRC. The counting is performed in the valid statistics cycle. The unit is the DMC bit width. When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

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DDRC_HIS_FLUXID_RD

DDRC_HIS_FLUXID_RD is a DDRC specified ID read traffic statistics registers.

Offset Address	Register Name	Total Reset Value
0x394	DDRC_HIS_FLUXID_RD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fluxid_rd																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	fluxid_rd	Read traffic statistics of the master with the specified ID of the DDRC. The counting is performed in the valid statistics cycle. The unit is the DMC bit width. When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_FLUXID_WCMD

DDRC_HIS_FLUXID_WCMD is a DDRC all ID write command count register.

Offset Address	Register Name	Total Reset Value
0x0398	DDRC_HIS_FLUXID_WCMD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fluxid_wr_cmd																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	fluxid_wr_cmd	Read traffic statistics of the master based on specified IDs. The counting is performed in the valid statistics cycle. The unit is the DMC bit width. When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_FLUXID_RCMD

DDRC_HIS_FLUXID_RCMD is a DDRC all ID read command count register.

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Offset Address	Register Name	Total Reset Value
0x039C	DDRC_HIS_FLUXID_RCMD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fluxid_rd_cmd																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	fluxid_rd_cmd	Read command count of the master based on specified IDs. The counting is performed in the valid statistics cycle. When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_WLATCNT0

DDRC_HIS_WLATCNT0 is DDRC specified ID write command latency statistics register 0.

Offset Address	Register Name	Total Reset Value
0x3A0	DDRC_HIS_WLATCNT0	0x0000_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wlatcnt_max																wlatcnt_min															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:16]	RO	wlatcnt_max	Maximum latency of the write command with the specified ID of the DDRC This field is cleared when the next statistics start.
[15:0]	RO	wlatcnt_min	Minimum latency of the write command with the specified ID of the DDRC. The maximum value is retained if overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_WLATCNT1

DDRC_HIS_WLATCNT1 is DDRC specified ID write command latency statistics register 1.

Offset Address	Register Name	Total Reset Value
0x3A4	DDRC_HIS_WLATCNT1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wlatcnt_all																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	wlatcnt_all	Accumulated latency of the write command with the specified ID in the statistics cycle (the results of the lower four bits are ignored) When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start. **NOTE** Software can obtain the average latency of the write command with the specified ID by dividing wlatcnt_all by fluxid_wr_cmd.

DDRC_HIS_RLATCNT0

DDRC_HIS_RLATCNT0 is a DDRC specified ID read command latency statistics register 0.

Offset Address	Register Name	Total Reset Value
0x3A8	DDRC_HIS_RLATCNT0	0x0000_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rlatcnt_max																rlatcnt_min															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:16]	RO	rlatcnt_max	Maximum latency of the read command with the specified ID of the DDRC The actual maximum read latency is rlatcnt_max plus inhere_rlatcnt. The maximum value is retained if overflow occurs. This field is cleared when the next statistics start.
[15:0]	RO	rlatcnt_min	Minimum latency of the read command with the specified ID of the DDRC Note that the actual minimum read latency is rlatcnt_min plus inhere_rlatcnt. The maximum value is retained if overflow occurs. This field is cleared when the next statistics start.

DDRC_HIS_RLATCNT1

DDRC_HIS_RLATCNT1 is DDRC specified ID read command latency statistics register 1.

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Offset Address	Register Name	Total Reset Value
0x3AC	DDRC_HIS_RLATCNT1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rlatcnt_all																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	rlatcnt_all	Accumulated latency of the read command with the specified ID in the statistics cycle (the results of the lower four bits are ignored) When perf_mode is 0, the value is wrapped when overflow occurs. When perf_mode is 1, the value is retained when overflow occurs. This field is cleared when the next statistics start. **NOTE** Software can obtain the average latency of the read command with the specified ID by using rlatcnt_all/fluxid_rd_cmd+inhere_rlatcnt.

DDRC_HIS_INHERE_RLAT_CNT

DDRC_HIS_INHERE_RLAT_CNT is a read channel inherent latency register.

Offset Address	Register Name	Total Reset Value
0x3B0	DDRC_HIS_INHERE_RLAT_CNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																inhere_rlatcnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	inhere_rlatcnt	Inherent latency of the read data channel for the DDRC and PHY. The actual latency can be obtained only by using this register and rlatcnt_min, rlatcnt_max, rlatcnt_all, and fluxid_rd_cmd.

DDRC_HIS_CMD_SUM

DDRC_HIS_CMD_SUM is a DDRC accumulated command count register.

Offset Address	Register Name	Total Reset Value
0x3BC	DDRC_HIS_CMD_SUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dmc_cmd_sum																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dmc_cmd_sum	Accumulated value of the commands that are temporarily stored in the DMC in the cycle. The value is wrapped if overflow occurs.

DDRC_HIS_SFC_RDATA3

DDRC_HIS_SFC_RDATA3 is software configuration read data register 3.

Offset Address	Register Name	Total Reset Value
0x4B4	DDRC_HIS_SFC_RDATA3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rdata3																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rdata3	Bits 159−128 of the data read by the SFC

DDRC_HIS_SFC_RDATA4

DDRC_HIS_SFC_RDATA4 is an SFC read data register.

Offset Address	Register Name	Total Reset Value
0x4B8	DDRC_HIS_SFC_RDATA4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rdata4																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rdata4	Bits 127−96 of the data read by the SFC

DDRC_HIS_SFC_RDATA5

DDRC_HIS_SFC_RDATA5 is an SFC read data register.

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Offset Address	Register Name	Total Reset Value
0x4BC	DDRC_HIS_SFC_RDATA5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rdata5																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rdata5	Bits 95−64 of the data read by the SFC

DDRC_HIS_SFC_RDATA6

DDRC_HIS_SFC_RDATA6 is an SFC read data register.

Offset Address	Register Name	Total Reset Value
0x4C0	DDRC_HIS_SFC_RDATA6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rdata6																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rdata6	Bits 63−32 of the data read by the SFC

DDRC_HIS_SFC_RDATA7

DDRC_HIS_SFC_RDATA7 is an SFC read data register.

Offset Address	Register Name	Total Reset Value
0x4C4	DDRC_HIS_SFC_RDATA7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rdata7																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rdata7	Bits 31−0 of the data read by the SFC The value of bits 7−0 is the readback value of the LPDDR2 MRS register in RD_MRS mode.

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DDRC_HIS_SFC_RDATA_ECC

DDRC_HIS_SFC_RDATA_ECC is an SFC read data ECC register.

Offset Address	Register Name	Total Reset Value
0x4C8	DDRC_HIS_SFC_RDATA_ECC	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	rdata_ecc3		rdata_ecc2		rdata_ecc1		rdata_ecc0	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:24]	RW	rdata_ecc3	8-bit ECC for bits 255−192 of the data read by the SFC
[23:16]	RW	rdata_ecc2	8-bit ECC for bits 191−128 of the data read by the SFC
[15:8]	RW	rdata_ecc1	8-bit ECC for bits 127−64 of the data read by the SFC
[7:0]	RW	rdata_ecc0	8-bit ECC for bits 63−0 of the data read by the SFC

4.1.7 DDR PHY Registers

4.1.7.1 Register Summary

NOTE

The variable *n* in the offset addresses for DDR PHY registers indicates the data block ID and ranges from 0 to 3.

Table 4-7 describes DDR PHY registers.

Table 4-7 Summary of DDR PHY registers (base address: 0x2011_8000)

Offset Address	Register	Description	Page
0x000	REVISION	PHY control block version register	4-96
0x004	PHYINITCTRL	PHY initialization control register	4-96
0x008	PHYINITSTATUS	PHY status register	4-99
0x00C	PHYCLKGATED	PHY clock gating control register	4-100
0x010	PHYTIMER0	PHY timing parameter register	4-101
0x014	PHYTIMER1	DRAM CKE low initialization register	4-102
0x018	PLLCTRL	Timing parameter register for the PLL in both address/command and data block	4-102
0x01C	PLLTIMER	PLL address/command and data block control register	4-103

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Offset Address	Register	Description	Page
0x020	DLYMEASCTRL	Read delay line delay measurement control register	4-104
0x024	IMPCTRL	ZQ calibration control register	4-105
0x028	IMPSTATUS	ZQ calibration result register	4-105
0x02C	DRAMCFG	DRAM system configuration register	4-107
0x030	DRAMTIMER0	DRAM timing parameter register 0	4-108
0x034	DRAMTIMER1	DRAM timing parameter register 1	4-109
0x038	DRAMTIMER2	DRAM timing parameter register 2	4-110
0x03C	DRAMTIMER3	DRAM timing parameter register 3	4-110
0x040	DRAMTIMER4	DRAM write leveling timing parameter register	4-111
0x044	ODTCR	Rank ODT control register while reading or writing to particular ranks	4-111
0x048	TRAINCTRL0	Training control register	4-112
0x04C	RANKEN	Training rank control register	4-114
0x0050	TRAINMADDR	Data training memory start address register	4-115
0x0054	BISTCTRL	Built-in self test (BIST) behavior register	4-115
0x0058	BISTDATA0	DRAM loopback test BIST data register 0	4-117
0x005C	BISTDATA1	DRAM loopback test BIST data register 1	4-117
0x0060	BISTSTATUS	BIST test result and status register	4-118
0x0064	MODEREG01	Mode register content register 1	4-119
0x0068	MODEREG23	Mode register content register 2	4-119
0x006C	DETPATTERN	Write/Read DET pattern register	4-120
0x0070	MISC	Miscellaneous control register	4-121
0x0074	RNK2RNK	Rank-to-rank delay control register	4-123
0x0078	PHYCTRL0	PHY control register 0	4-123
0x007C	PHYDBG	PHY debug register	4-124
0x0080	RETCTRL0	Retraining control register	4-124
0x0084	DMSEL	DM swap selection register	4-125
0x0088	IOCTL	I/O control register	4-126
0x008C	DQSSEL	PHY DQ swap register	4-126

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Offset Address	Register	Description	Page
0x0090	PHYCTRL1	AC PHY control register	4-127
0x0094	DXNCKCTRL	PLL phase selection register	4-128
0x0098	PHYPLLCTRL_AC	PHY PLL control register 1	4-128
0x009C	PHYPLLCTRL_DX	PHY PLL control register 2	4-129
0x00A0	SWTMODE	S/W training mode register	4-130
0x00A4	SWTWLDQS	Write DQS issue register during S/W write leveling training	4-130
0x00A8	SWTRLT	S/W training result register	4-131
0x00AC	PHYCTRL2	PHY control register 2	4-131
0x00B0	IOCTL1	PHY I/O control register 1	4-132
0x00B4	IOCTL2	PHY I/O control register 2	4-132
0x00B8	BISTTIMER	Code update period register during BIST running	4-133
0x00D0	TRAINCTRL1	Data training control register 1	4-134
0x00D4	TRAINCTRL2	Data training control register 2	4-135
0x00D8	REGBANKCTRL	Register bank control register	4-136
0x00DC	TRAINCTRL3	Data training control register 3	4-136
0x100	ACBISTCTRL0	Comparison control register 0 while BIST activated	4-138
0x104	ACBISTCTRL1	Comparison control register 1 while BIST activated	4-139
0x108	ACBISTSTS0	BIST test result and status register 0	4-140
0x10C	ACBISTSTS1	BIST test result and status register 1	4-141
0x120	ACCMDBDL0	AC command bit delay line configuration register 0	4-141
0x124	ACCMDBDL1	AC command bit delay line configuration register 1	4-141
0x128	ACCMDBDL2	AC command bit delay line configuration register 2	4-142
0x12C	ACCMDBDL3	AC command bit delay line configuration register 3	4-142
0x130	ACCMDBDL4	AC command bit delay line configuration register 4	4-143

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Offset Address	Register	Description	Page
0x134	ACCMDBDL5	AC command bit delay line configuration register 5	4-143
0x138	ACCMDBDL6	AC command bit delay line configuration register 6	4-144
0x13C	ACCMDBDL7	AC command bit delay line configuration register 7	4-144
0x140	ACADDRBDL0	AC address bit delay line configuration register 0	4-145
0x144	ACADDRBDL1	AC address bit delay line configuration register 1	4-145
0x148	ACADDRBDL2	AC address bit delay line configuration register 2	4-146
0x14C	ACADDRBDL3	AC address bit delay line configuration register 3	4-146
0x150	ACADDRBDL4	AC address bit delay line configuration register 4	4-147
0x154	ACADDRBDL5	AC address bit delay line configuration register 5	4-147
0x158	ACADDRBDL6	AC address bit delay line configuration register 6	4-148
0x15C	ACADDRBDL7	AC address bit delay line configuration register 7	4-148
0x160	ACADDRBDL8	AC address bit delay line configuration register 8	4-149
0x164	ACADDRBDL9	AC address bit delay line configuration register 9	4-149
0x168	ACCLKBDL	AC clock bit delay line configuration register	4-150
0x170	ACPHYCTL0	AC block PHY control register 0	4-150
0x174	ACPHYCTL1	AC block PHY control register 1	4-151
0x178	ACPHYCTL2	AC block PHY control register 2	4-152
0x17C	ACPHYCTL3	AC block PHY control register 3	4-153
0x180	ACPHYCTL4	AC block PHY control register 4	4-154
0x184	ACPHYCTL5	AC block PHY control register 5	4-155
0x188	ACPHYCTL6	AC block PHY control register 6	4-156
0x018C	ACPHYCTL7	AC block PHY control register 7	4-156

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Offset Address	Register	Description	Page
0x0190	ACDEBUG	PHY debugging register	4-157
0x0194	ACPHYRSVDC	AC block PHY reserved control pin register 1	4-158
0x0198	ACPHYRSVDS	AC block PHY reserved control pin register 2	4-158
0x019C	ACPHYCTL8	AC block PHY control register 8	4-159
0x1A0	ACCMDBDL8	AC command bit delay line configuration register 8	4-159
0x1A4	ACCMDBDL9	AC command bit delay line configuration register 9	4-160
0x1A8	ACCMDBDL10	AC command bit delay line configuration register 10	4-160
0x1AC	ACCMDBDL11	AC command bit delay line configuration register 11	4-161
0x01B0	ACPHYDCC	AC block PHY DCC control register	4-161
0x01B4	ACPHYCTL9	AC block PHY control register 9	4-162
0x01C0	CATTIMER0	CA training timer 0 register	4-163
0x01C4	CATTIMER1	CA training timer 1 register	4-164
0x01C8	CATCONFIG	CA training configuration register	4-164
0x01CC	CATRESULT	CA training result register (for debugging)	4-166
0x01D0	PHYDQRESULT	SW CA training DQ result (from PHY) register	4-166
0x01D4	ADDRPHBOUND	CA training address phase boundary register	4-166
0x01D8	SWCATPATTERN_P	SW CA training pattern register for the positive CK edge	4-168
0x01DC	SWCATPATTERN_N	SW CA training pattern register for the negative CK edge	4-168
0x01E4	LPCTRL	Low-power control register	4-168
0x200 + *n* x 0x80	DXNBISTCTRL	Loopback data comparison control register during BIST of data blocks	4-170
0x204 + *n* x 0x80	DXNBISTSTS	BIST test result and status register	4-171
0x208 + *n* x 0x80	DXNCTRL	Data block control register	4-172

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Offset Address	Register	Description	Page
0x210 + *n* x 0x80	DXNWDQNBDL0	Data block bit delay configuration register 0	4-172
0x214 + *n* x 0x80	DXNWDQNBDL1	Data block bit delay configuration register 1	4-173
0x218 + *n* x 0x80	DXNWDQNBDL2	Data block bit delay configuration register 2	4-174
0x21C + *n* x 0x80	DXNRDQNBDL0	Data block bit delay configuration register 3	4-174
0x220 + *n* x 0x80	DXNRDQNBDL1	Data block bit delay configuration register 4	4-175
0x224 + *n* x 0x80	DXNRDQNBDL2	Data block bit delay configuration register 5	4-176
0x22C + *n* x 0x80	DXNRDQSDLY	Local delay line control register	4-176
0x230 + *n* x 0x80	DXNWDQSDLY	Write leveling DQS delay control register	4-177
0x234 + *n* x 0x80	DXNWDQDLY	Write leveling delay line control register	4-177
0x238 + *n* x 0x80	DXNWLSL	Extra system latency addition control register	4-178
0x23C + *n* x 0x80 + *m* x 0x400	DXNRDQSGDLY	Local delay line control register	4-178
0x240 + *n* x 0x80	DXPHYCTRL	DX PHY control register	4-179
0x248 + *n* x 0x80	DXNGDS	Latch enable register	4-180
0x24C + *n* x 0x80	DXNCLKBDL	Data block clock bit delay line configuration register	4-180
0x250 + *n* x 0x80	DXNRDBOUND	Read data eye boundary register	4-181
0x254 + *n* x 0x80	DXNWRBOUND	Write data eye boundary register	4-181
0x258 + *n* x 0x80	DXNOEBDL	Output enable delay line control register	4-182
0x260 + *n* x 0x80	DXNMISCCTRL0	Data block PHY miscellaneous control register 0	4-182

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Offset Address	Register	Description	Page
0x264 + *n* x 0x80	DXNMISCCTRL1	Data block PHY miscellaneous control register 1	4-183
0x268 + *n* x 0x80	DXDEBUG0	Data block PHY debug signal register 0	4-184
0x26C + *n* x 0x80	DXDEBUG1	Data block PHY debug signal register 1	4-185
0x270 + *n* x 0x80	DXPHYRSVD	Data block PHY reserved control pin register	4-185
0x274 + *n* x 0x80	DXNMISCCTRL2	Data block PHY miscellaneous control register 2	4-186
0x278 + *n* x 0x80	DXDEBUGCONFIG	Data block PHY debugging control register	4-187
0x27C + *n* x 0x80	DXNDCC	Data block PHY DCC control register	4-187

4.1.7.2 Register Description

REVISION

REVISION is a PHY control block version register.

Offset Address	Register Name	Total Reset Value
0x000	REVISION	0x0000_0012

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																major								minor							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RO	major	Major version of the PHY control block
[7:0]	RO	minor	Minor version of the PHY control block

PHYINITCTRL

PHYINITCTRL is a PHY initialization control register.

NOTE

Writing to this register has no effect if init_en is 1.

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Offset Address	Register Name	Total Reset Value
0x004	PHYINITCTRL	0x0000_000E

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ctl_cke_bypass	reserved															phyconn_rst	pack_rst	phy_rst	dram_rst	cat_en	dram_init_en	wdet_en	rdet_en	wl2_en	gdst_en	gt_en	wl_en	zcal_en	dlymeas_en	pll_init_en	init_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0

Bits	Access	Name	Description
[31]	RW	ctl_cke_bypass	PACK CKE bypass function enable 1'b1: CKE is controlled by MC even when PACK controls the DFI interface. 1'b0: CKE is controlled by PACK when PACK controls the DFI interface.
[30:16]	RO	reserved	Reserved
[15]	RW	phyconn_rst	PHY counter reset This field is used to set the PHY counter reset signal. To issue a reset signal to PHY counter, this field should be set to 1. To end the reset signal to PHY counter, this field should be cleared. Note that the reset signal connects to the A/C block counter and the counter of all of the data blocks.
[14]	RW	pack_rst	PACK reset This field is used to issue reset to all the digital circuits of the PACK including the register file. If this field is set to 1, all of the digital parts (including control registers) of the PHY will be set to default value. The reset will be automatically executed, and this bit will be cleared after reset is complete.
[13]	RW	phy_rst	PHY reset This field is used to set PHY reset signal. To issue a reset signal to the PHY, this field should be set to 1. To end the reset signal to the PHY, this field should be cleared. Note that the reset signal connects to the A/C block and all of the data blocks.
[12]	RW	dram_rst	DRAM reset This field is used to initiate the DRAM reset sequence. If this field is set to 1, RESET# and CKE will be pulled low for t_dram_rst. After t_dram_rst, the CKE RESET# is pulled to high while keeping CKE at low for t_cke_low. After t_cke_low, CKE is pulled to high, and after t_cke_high, this field is cleared automatically. For LPDDR2/LPDDR3, only CKE will be active during DRAM reset sequence.

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[11]	RW	cat_en	HW CA training enable Writing 1 to this field starts the HW CA training sequence. This field will be cleared after CA training is complete. Note that the CA training applies only to LPDDR3.
[10]	RW	dram_init_en	DRAM initialization enable This field is used to control whether the internal DRAM initialization sequence is executed after PHY initialization is complete. For LPDDR2/LPDDR3, this field will complete the initial sequence including tINIT3−tINIT5 & tZQC.
[9]	RW	wdet_en	Write data eye training enable Writing 1 to this field starts the write data eye training sequence. This field will be cleared after the write data eye training is complete.
[8]	RW	rdet_en	Read data eye training enable Writing 1 to this field starts the read data eye training sequence. This field will be cleared after the read data eye training is complete.
[7]	RW	wl2_en	Second write leveling enable Writing 1 to this field starts write latency adjustment. This field will be cleared after adjustment is complete. Note that write leveling adjustment applies only to DDR3.
[6]	RW	gdst_en	PHY read data latch train enable Writing 1 to this field starts the read data latch position training. This field will be cleared after write leveling is complete.
[5]	RW	gt_en	Gate training enable Writing 1 to this field starts the gate training sequence. This field will be cleared after the gate training is complete.
[4]	RW	wl_en	Write leveling enable Writing 1 to this field starts the write leveling sequence. This field will be cleared after write leveling is complete. Note that write leveling applies only to DDR3.
[3]	RW	zcal_en	Impedance calibration enable This field is used to specify whether impedance calibration is executed. This bit will be cleared after impedance calibration is complete.
[2]	RW	dlymeas_en	Delay measurement enable This field is used to enable initial delay measurement of the activated read delay line. If it is set to 1, delay measurement is enabled.
[1]	RW	pll_init_en	PLL initialization enable If this field is used to activate the PLL initialization. If it is set to 1, the PLL is reset and waiting for a period of time for PLL to be locked.

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[0]	RW	init_en	PHY initialization enable Writing 1 to this field triggers the PHY initialization sequence. The exact initialization executed is specified in bits 1−15 of this register. If PHY initialization is complete, this bit will be cleared. Note that writing 0 to this field has no effect.

PHYINITSTATUS

PHYINITSTATUS is a PHY status register.

Offset Address	Register Name	Total Reset Value
0x008	PHYINITSTATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					cat_err	wdet_err	rdet_err	wl2_err	gdst_err	gt_err	wl_err	zcal_err	dlymeas_err	pll_lock_err	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RWC	cat_err	CA training error This field indicates whether an error occurs during CA training. Writing 1 to this field clears the error status.
[9]	RWC	wdet_err	Write data eye calibration error This field indicates whether an error occurs during write data eye calibration. Writing 1 to this field clears the error status.
[8]	RWC	rdet_err	Read data eye calibration error This field indicates whether an error occurs during read data eye calibration. Writing 1 to this field clears the error status.
[7]	RWC	wl2_err	Write leveling adjustment error This field indicates whether an error occurs during write leveling adjustment. Writing 1 to this field clears the error status.
[6]	RWC	gdst_err	PHY read data latch train error This field indicates whether an error occurs during read data latch training. Writing 1 to this field clears the error status.
[5]	RWC	gt_err	Gate training error This field indicates whether an error occurs during gate training. Writing 1 to this field clears the error status.

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[4]	RWC	wl_err	Write leveling error This field indicates whether an error occurs during write leveling. Writing 1 to this field clears the error status.
[3]	RWC	zcal_err	Impedance calibration error This field indicates whether an error occurs during impedance calibration. Writing 1 to this field clears the error status.
[2]	RWC	dlymeas_err	Delay measurement fail This field indicates whether one of the initial delay measurements fails. Writing 1 to this field clears the error status.
[1]	RWC	pll_lock_err	PLL locked fail This field indicates whether one of the PLLs within the PHY fails to assert locked flag after t_pll_lock timer. Writing 1 to this field clears the error status.
[0]	RO	reserved	Reserved

PHYCLKGATED

PHYCLKGATED is a PHY clock gating control register.

Offset Address	Register Name	Total Reset Value
0x000C	PHYCLKGATED	0x0000_0400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	byp_pll_lock	reserved																			dx_phgated_men	dx_phclkgated	dx_phgated_en	ck_phy_clkgated	ac_phy_clkgated	reserved						dx_phy_clkgated			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	byp_pll_lock	PLL lock bypass. The user can assert this field to bypass the PLL lock signal when re-locking the PLL in the bypass mode (DFS) or when the PLL lock signal cannot be used. 0: use the PLL lock to gate the PHY clocks to prevent clock glitch 1: bypass the PLL lock (do not use the PLL lock signal to gate the PHY clocks)
[30:13]	RO	reserved	Reserved

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[12]	RW	dx_phgated_men	PHY phase clock gated manual control enable. This field takes effect only when dx_phgated_en is 1'b0. If dx_phgated_en is high and this field is high, the user can use dx_phclkgated to manually control the clock gating of the PHY phase clock. 0: disabled 1: enabled
[11]	RW	dx_phclkgated	PHY phase clock gated control. This field takes effect only when dx_phgated_en is 1'b0. If dx_phgated_en is '1b1, this field is used, and the gated function is automatically controlled by PACK FSM. 0: The clock is disabled. 1: The clock is enabled.
[10]	RW	dx_phgated_en	PHY phase clock gated function (automatically controlled by PACK FSM) enable 0: disabled 1: enabled
[9]	RW	ck_phy_clkgated	Dedicated clock gated control for the output CK and related circuits 0: The clock is gated. 1: The clock is enabled. **NOTE** This field is automatically set to 1 when the PLL is locked without error.
[8]	RW	ac_phy_clkgated	Clock gated for the circuits in AC block except CK and CK related circuits 0: The clock is gated. 1: The clock is enabled. **NOTE** This field is automatically set to 1 when the PLL is locked without error.
[7:4]	RO	reserved	Reserved
[3:0]	RW	dx_phy_clkgated	Clock gated for circuits in data blocks 0: The clock is gated. 1: The clock is enabled. **NOTE** This field is automatically set to 1 when the PLL is locked without error.

PHYTIMER0

PHYTIMER0 is a PHY timing parameter register. The timing parameters are required by the PHY.

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Offset Address	Register Name	Total Reset Value
0x010	PHYTIMER0	0x2710_000F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	t_dram_cke_high																				reserved				t_dram_reset							
Reset	0	0	1	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

Bits	Access	Name	Description
[31:12]	RW	t_dram_cke_high	DRAM CKE high initialization This field specifies the number of PCLK clock cycles of the first DRAM command from CKE pulled to high (LPDDR2/LPDDR3 tINIT3).
[11:8]	RO	reserved	Reserved
[7:0]	RW	t_dram_reset	DDR3 DRAM reset. This field is used to specify the time issuing reset signaling to the DRAM (number of PCLK clock cycles).

PHYTIMER1

PHYTIMER1 is a DRAM CKE low initialization register.

Offset Address	Register Name	Total Reset Value
0x014	PHYTIMER1	0x0000_2000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																t_dram_cke_low															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	t_dram_cke_low	DRAM CKE low initialization This field specifies the number of PCLK clock cycles during which CKE should be kept low after power is reset and the clock is stable.

PLLCTRL

PLLCTRL is a timing parameter register for the PLL in both address/command and data block.

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Offset Address	Register Name	Total Reset Value
0x018	PLLCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										pll_freq_range		reserved		pll_pwdn			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5:4]	RW	pll_freq_range	PLL frequency range select This field specifies the frequency input range of the PLLs.
[3]	RO	reserved	Reserved
[2:0]	RW	pll_pwdn	PLL power down This field is used to set the power-down status of the PLL input. If this field is set to 1, the PLL power-down pin is activated. Writing 0 to this field will remove the deactivated PLL power down pin. Bit[2] for AC PLL Bit[1] for DX1 (DATA1) PLL Bit[0] for DX0 (DATA0) PLL

PLLTIMER

PLLTIMER is a PLL address/command and data block control register.

Offset Address	Register Name	Total Reset Value
0x01C	PLLTIMER	0x4E20_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	t_pll_lock																reserved								t_pll_rst							
Reset	0	1	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	t_pll_lock	PLL lock time This field specifies the number of PCLK clock cycles waiting for the PLL to be locked.
[15:8]	RO	reserved	Reserved

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[7:0]	RW	t_pll_rst	PLL reset time This field specifies the number of PCLK clock cycles to issue reset to the PLL.

DLYMEASCTRL

DLYMEASCTRL is a read delay line delay measurement control register.

Offset Address	Register Name	Total Reset Value
0x020	DLYMEASCTRL	0x0000_0471

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																dly_track_limit								reserved	phy_autoref_en	dynamic_dqsen	dynamic_dqsgen	dynamic_dqsgth		dly_track_type	dly_meas_type
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	1	0	0	0	1

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14:8]	RW	dly_track_limit	Delay tracking limit This field specifies the maximum number of accumulated taps allowed while dynamic tracking.
[7]	RO	reserved	Reserved
[6]	RW	phy_autoref_en	PHY auto-refresh tracking enable
[5]	RW	dynamic_dqsen	Dynamic DQS tracking enable This field should be set to high during delay measurement.
[4]	RW	dynamic_dqsgen	Dynamic DQSG tracking enable
[3:2]	RW	dynamic_dqsgth	Dynamic DQSG update threshold This field specifies the number of taps accumulated before the RDQSG delay line is updated. 0: update every 4 times 1: update every 2 times 2 or 3: update each time
[1]	RW	dly_track_type	Delay tracking type This field specifies whether priority updated from the PHY is used during dynamic tracking. If this field is set to 1, the priority updated from the PHY is used.

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[0]	RW	dly_meas_type	Delay measurement type This field specifies whether the delay measurement is for one clock cycle or half a clock cycle. If this field is set to 1, a full clock cycle is measured.

IMPCTRL

IMPCTRL is a ZQ calibration control register.

Offset Address	Register Name	Total Reset Value
0x024	IMPCTRL	0x8000_8020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	zcfuzzy_en	reserved															zcomp_num				reserved								zcomp_rsp_dly			
Reset	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	zcfuzzy_en	ZQ comparator result selection by weighted method. When this function is disabled, the precision is improved but ZQ comparison may fail. When this function is enabled, ZQ comparison is successful. 0: disabled 1: enabled (default)
[30:16]	RO	reserved	Reserved
[15:12]	RW	zcomp_num	ZQ comparator iteration time This field specifies the iteration time of zcomp changing detection.
[11:6]	RO	reserved	Reserved
[5:0]	RW	zcomp_rsp_dly	ZQ comparator response time This field specifies the delay of the zcomp result after zctrl is used.

IMPSTATUS

IMPSTATUS is a ZQ calibration status register.

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Offset Address	Register Name	Total Reset Value
0x028	IMPSTATUS	0x0020_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	zcode_pdrv_cal							pdrv_cal_result	reserved	zcode_pdrv						reserved	zcode_ndrv_cal							ndrv_cal_result	reserved	zcode_ndrv					
Reset	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:24]	RO	zcode_pdrv_cal	P-drive impedance calibration code This field shows the p-drive calibration code during impedance calibration.
[23]	RO	pdrv_cal_result	P-drive calibration result This field specifies the ZQ calibration result. The valid values are as follows: 0: The calibration is complete without error. 1: The calibration is complete with an error. Writing 1 to this field clears this field.
[22]	RO	reserved	Reserved
[21:16]	RW	zcode_pdrv	P-drive impedance control code This field is used to specify the p-drive impedance control code. This register is updated by the user or is updated each time the ZQ calibration is complete.
[15]	RO	reserved	Reserved
[14:8]	RO	zcode_ndrv_cal	N-drive impedance calibration code This field shows the n-driver calibration code during impedance calibration.
[7]	RO	ndrv_cal_result	N-drive calibration result This field specifies the ZQ calibration result. The valid values are as follows: 0: The calibration is complete without error. 1: The calibration is complete with an error. Writing 1 to this field clears this field.
[6]	RO	reserved	Reserved

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[5:0]	RW	zcode_ndrv	N-drive impedance control code This field is used to specify the n-drive impedance control code. This register is updated by the user or is updated each time the ZQ calibration is complete.

DRAMCFG

DRAMCFG is a DRAM system configuration register.

Offset Address	Register Name	Total Reset Value
0x02C	DRAMCFG	0x0000_0405

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					lpddr23_zqc_en	lpddr23_mrw1617_en	lpddr23_pre_en	reserved	maddr_mir	ma2t	nosra	reserved			dcfg_type
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	lpddr23_zqc_en	LPDDR2/LPDDR3 initialization ZQC (init) enable 0: disabled 1: issue ZQC(init) commands during initialization sequence rank by rank (default).
[9]	RW	lpddr23_mrw1617_en	LPDDR2/LPDDR3 initialization MRW16/17 enable 0: disabled (default) 1: issue MRW16 then MRW17 commands during initialization sequence
[8]	RW	lpddr23_pre_en	LPDDR2/LPDDR3 initialization precharge all enable 0: disabled (default) 1: issue the precharge all command before the reset command during initialization sequence
[7]	RO	reserved	Address mirroring This bit should be set to 0 in single rank configuration
[6]	RW	maddr_mir	Address mirroring This bit should be set to 0 in single rank configuration

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[5]	RW	ma2t	Two cycles on address/command This field specifies whether the 2T timing should be used by the PHY controller that internally generates the DRAM command.
[4]	RW	nosra	No simultaneous rank access This field specifies whether simultaneously rank access in the same clock cycle is allowed. If it set to 1, the rank access will be taken in turn. Note that this field can be enabled only when the number of ranks is 2 or 4. This field should be set to 0 in single rank configuration.
[3]	RO	reserved	Reserved
[2:0]	RW	dcfg_type	DRAM type selection This field specifies the DDR RAM type. The valid values are as follows: 010: DDR3 101: LPDDR2/LPDDR3 Other values: reserved

DRAMTIMER0

DRAMTIMER0 is a DRAM timing parameter register.

NOTE

The command delay specified in this register applies only to the command issued by the PHY controller internally.

Offset Address	Register Name	Total Reset Value
0x030	DRAMTIMER0	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	t_rc	t_rrd	t_ras	t_rcd	t_rp	t_wtr	t_rtp	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:26]	RW	t_rc	Activated to activated on same bank command delay
[25:22]	RW	t_rrd	Activated to activated on different bank command delay
[21:16]	RW	t_ras	Activated to precharge command delay
[15:12]	RW	t_rcd	Activated to read/write delay
[11:8]	RW	t_rp	Precharge command period The unit is two clock cycles.
[7:4]	RW	t_wtr	Write to read command delay

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[3:0]	RW	t_rtp	Read to precharge command delay

DRAMTIMER1

Offset Address	Register Name	Total Reset Value
0x034	DRAMTIMER1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		t_rtw						t_rfc								reserved	t_faw							t_mod				t_mrd			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:25]	RW	t_rtw	Read to write command delay This field specifies whether to insert extra cycles between read and write commands to compensate tdqsck. This value should be 3 for the common condition when DDR3 is used. This value should be 6 for the common condition when LPDDR2/LPDDR3 is used.
[24:16]	RW	t_rfc	Refresh to refresh command delay
[15]	RO	reserved	Reserved
[14:9]	RW	t_faw	Four-bank activated period. This field applies only to the 8-bank device.
[8:4]	RW	t_mod	Load mode register to non load mode register command delay. This field is used as max (t_mod, t_mrd) in DDR3, max (t_mrw, tmrd) in LPDDR3, and t_mrw in LPDDR2.
[3:0]	RW	t_mrd	Load mode register to load mode register command delay (not used)

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DRAMTIMER2

Offset Address	Register Name	Total Reset Value
0x038	DRAMTIMER2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	t_ccd	reserved	t_dllk										t_cke				reserved	t_xp							t_xs							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	t_ccd	Read to read and write to write command delay 0 = BL/2 (for DDR3, this value should be 4) 1 = BL/2 + 1 (for DDR3, this value should be 5)
[30]	RO	reserved	Reserved
[29:20]	RW	t_dllk	DLL lock time
[19:16]	RW	t_cke	CKE minimum pulse width
[15]	RO	reserved	Reserved
[14:10]	RW	t_xp	Power down exit delay
[9:0]	RW	t_xs	Self refresh exit delay

DRAMTIMER3

Offset Address	Register Name	Total Reset Value
0x03C	DRAMTIMER3	0x0010_0200

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	t_wr				t_wl				t_init5														t_zcal									
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	t_wr	Write recovery cycle
[27:24]	RW	t_wl	Write latency (AL + CWL)
[23:10]	RW	t_init5	LPDDR device initialization time This field specifies the time during which the LPDDR sets the DAI bit to 1.

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[9:0]	RW	t_zcal	ZQ calibration to command delay This field specifies the number of clock cycles to wait after initial ZQ calibration. For DDR3 ZQ normal operation full calibration, this value is divided by 2. For DDR3 ZQ initialization calibration, this value is divided by 4. For LPDDR2/3, this field specifies the value of tZQINIT during initialization.

DRAMTIMER4

DRAMTIMER4 is a DRAM write leveling timing parameter register. This register applies only to the DDR3/LPDDR3 SDRAM.

Offset Address	Register Name	Total Reset Value
0x040	DRAMTIMER4	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5 4 3 2 1 0
Name	reserved	t_odton	reserved		t_odt	t_wlo	reserved		t_wlmrd
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0		0 0 0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:24]	RW	t_odton	ODT turn-on latency (number of cycles from write command to assert of ODT)
[23:16]	RO	reserved	Reserved
[15:12]	RW	t_odt	ODT turn-on cycles
[11:7]	RW	t_wlo	Write leveling output delay This field specifies the write leveling output delay in cycles (dficlk cycle).
[6]	RO	reserved	Reserved
[5:0]	RW	t_wlmrd	Write leveling mode programmed (load mode register) to first DQS/DQS# rising edge delay (dficlk cycle)

ODTCR

ODTCR is a rank ODT control register while reading or writing to particular ranks.

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Offset Address	Register Name	Total Reset Value
0x044	ODTCR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																wodt_rank3				wodt_rank2				wodt_rank1				wodt_rank0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:12]	RW	wodt_rank3	Write rank 3 ODT control This field specifies the ODT issued while write on rank 3. The valid bit width is the number of ranks.
[11:8]	RW	wodt_rank2	Write rank 2 ODT control This field specifies the ODT issued while write on rank 2. The valid bit width is the number of ranks.
[7:4]	RW	wodt_rank1	Write rank 1 ODT control This field specifies the ODT issued while write on rank 1. The valid bit width is the number of ranks.
[3:0]	RW	wodt_rank0	Write rank 0 ODT control This field specifies the ODT issued while write on rank 0. The valid bit width is the number of ranks.

TRAINCTRL0

TRAINCTRL0 is a training control register.

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Offset Address	Register Name	Total Reset Value
0x048	TRAINCTRL0	0xD044_1080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	skip_eye_opening	dtrerrstop	gtds_rspdly						dqsg_cstep		dqsgrt_chk						dqsgrt_shft				dqsgsl_mrgn				dtr_rank							
Reset	1	1	0	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	skip_eye_opening	Skip eye opening of data eye training This field specifies whether the eye opening should be skipped. If this field is set to 1, the eye opening before data eye training is skipped. If this field is set to 0, the eye opening is executed before data eye training.
[30]	RW	dtrerrstop	Data training error stop This field controls whether the data training is terminated if an error occurs. The data training is stopped if this bit is set to 1. If this bit set to 0, the training will continue even an error occurs. The error status is recorded no matter when this field is set to 0 or 1.
[29:24]	RW	gtds_rspdly	Gate/Read latch training response delay This field specifies the waiting time to check the response from PHY while gate and read latch training is performed (dficlk cycle).
[23:22]	RW	dqsg_cstep	Gate training coarse tuning step (the unit is phase). The recommended values are 0x0−0x2. 2'd0: step = 1 2'd1: step = 2 (default) 2'd2: step = 3 2'd3: step = 4
[21:16]	RW	dqsgrt_chk	Gate retraining check window This field specifies the delta value to be checked during gate retraining check. The delta is added/subtracted to/from the read DQS gate. This position is checked to determine whether gating phase adjustment is required. The phase movement is the configured value multiple by 2. Note that 3'b000 is reserved and should not be used. The recommended values are 0x1−0x3.

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[15:10]	RW	dqsgrt_shft	Gate retraining shift window This field specifies the value to be shifted if gate phase adjustment is required after gate retraining checks. The phase movement is the configured value multiple by 2. Note that 3'b000 is reserved and should not be used. The recommended value is dqsgrt_chk.
[9:4]	RW	dqsgsl_mrgn	Read DQS gate margin This field specifies the margin to be reserved after the DQS gate training. The number of phases specified in this field is decreased to create DQS gating margin. The valid values are as follows: 6'b00100: 1/4 cycle 6'b01000: 1/2 cycle (recommended) 6'b01100: 3/4 cycle 6'b10000: 1 cycle Other values: reserved
[3:0]	RW	dtr_rank	Training rank This field specifies the rank used for training. After the training, this field is used to select the source of rank specific settings.

RANKEN

RANKEN is a training rank control register.

Offset Address	Register Name	Total Reset Value
0x04C	RANKEN	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rank_en															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	rank_en	Training rank enable This field specifies whether the dtr_rank is enabled. 0: disabled 1: enabled (default) Bit 0 indicates rank 0. Bit 1 indicates rank 1. ... Bit 15 indicates rank 15. The maximum number of ranks is specified by the system.

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TRAINMADDR

TRAINMADDR is a data training memory start address register. The minimum space required for the data training is 64 bytes, and the allocated memory must be 64-byte-aligned.

Offset Address	Register Name	Total Reset Value
0x0050	TRAINMADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dtrow																dtcol												reserved		dtbank	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	dtrow	Data training row address This field is used to configure the memory row address required for the data training.
[15:4]	RW	dtcol	Data training column address This field is used to configure the column address required for the data training. The field should be 0.
[3]	RO	reserved	Reserved
[2:0]	RW	dtbank	Data training bank address This field is used to configure the memory bank address required for the data training.

BISTCTRL

BISTCTRL is a BIST behavior register.

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Offset Address	Register Name	Total Reset Value
0x0054	BISTCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								bist_fail_stop								reserved						bist_pat		reserved		bist_mode			reserved		bist_op
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	bist_fail_stop	BIST stop error count This field specifies the number of failures at which the BIST should be stopped. If this field is set to 0x00, the BIST will run infinite until the BIST operation is stopped or reset. If a non-zero value is specified, the BIST is stopped while the accumulated errors reach the count specified in this field.
[15:10]	RO	reserved	Reserved
[9:8]	RW	bist_pat	BIST pattern generation This field specifies the BIST pattern generated for test. Note that bist_pat can be changed only when bist_done is 1. 00: walking 0 01: walking 1 10: LFSR-based random generation 11: user programmed
[7]	RO	reserved	Reserved
[6:4]	RW	bist_mode	BIST mode select This field specifies the BIST mode. The valid values are as follows: 000: pattern generated only 100: loopback before output buffer 101: loopback after output buffer 110: DRAM read/write mode Other values: reserved Note that bist_mode can be changed only when bist_done is 1. When the data loopback mode is selected, swapwl_en/swap_en must be set to all zeros (no swap).
[3:2]	RO	reserved	Reserved

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[1:0]	RW	bist_op	During Reg write, this field is used for selecting the BIST operation. The valid values are as follows: 00: BIST stop 01: BIST run 10: BIST reset 11: reserved Writing the Reset clears all the status registers, and this field is cleared to BIST Stop. During Reg read, this field is for BIST active status (only bit 0 is valid, and bit1 is read as 1'b0). 0: BIST inactive 1: BIST active

BISTDATA0

BISTDATA0 is DRAM loopback test BIST data register 0.

NOTE

This register is valid only when bist_pat is set to "User Defined."

Offset Address	Register Name	Total Reset Value
0x0058	BISTDATA0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bist_data_b3								bist_data_b2								bist_data_b1								bist_data_b0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	bist_data_b3	BIST data byte 3 This field is used as the fourth beat of burst during BIST test
[23:16]	RW	bist_data_b2	BIST data byte 2 This field is used as the third beat of burst during BIST test.
[15:8]	RW	bist_data_b1	BIST data byte 1 This field is used as the second beat of burst during BIST test.
[7:0]	RW	bist_data_b0	BIST data byte 0 This field is used as the first beat of burst during BIST test.

BISTDATA1

BISTDATA1 is DRAM loopback test BIST data register 1.

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Offset Address	Register Name	Total Reset Value
0x005C	BISTDATA1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bist_data_b7								bist_data_b6								bist_data_b5								bist_data_b4							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	bist_data_b7	BIST data byte 7 This field is used as the eighth beat of burst during BIST test
[23:16]	RW	bist_data_b6	BIST data byte 6 This field is used as the sixth beat of burst during BIST test.
[15:8]	RW	bist_data_b5	BIST data byte 5 This field is used as the fifth beat of burst during BIST test.
[7:0]	RW	bist_data_b4	BIST data byte 4 This field is used as the fourth beat of burst during BIST test.

BISTSTATUS

BISTSTATUS is a BIST test result and status register.

Offset Address	Register Name	Total Reset Value
0x0060	BISTSTATUS	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													bist_dxerr	bist_acerr	bist_done
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RO	bist_dxerr	BIST error on DX blocks This bit shows whether any error occurs on the DXn during BIST test.
[1]	RO	bist_acerr	BIST error on AC block This bit shows whether any error occurs on the AC during BIST test.

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[0]	RO	bist_done	BIST done This field shows whether the BIST is complete. Note that if the BIST Stop Error Count field (bist_fail_stop) is set to 0, this field is never asserted.

MODEREG01

MODEREG01 is mode register content register 1.

Offset Address	Register Name	Total Reset Value
0x0064	MODEREG01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mr1																mr0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	mr1	Mode register 1 For DDR/DDR2/DDR3/LPDDR, this field specifies the content of extend mode register 1. For LPDDR2/LPDDR3, this field defines the content of MR3/MR9 (mode register). bit[31:24]: MR9 OP bit[23:16]: MR3 OP
[15:0]	RW	mr0	Mode register 0 For DDR/DDR2/DDR3/LPDDR, this field specifies the content of mode register 0. For LPDDR2/LPDDR3, this field defines the mode registers MR1/MR2. bit[15:8]: MR2 OP bit[7:0]: MR1 OP

MODEREG23

MODEREG23 is mode register content register 2.

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Offset Address	Register Name	Total Reset Value
0x0068	MODEREG23	0xFF0A_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mr3																mr2															
Reset	1	1	1	1	1	1	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	mr3	Mode register 3 For DDR/DDR2/DDR3, this field specifies extend mode register 3. For LPDDR2/LPDDR3, this field is for the ZQ commands. bit[31:24]: OP bit[23:16]: MA The ZQ commands are issued rank by rank after tINIT5. bit[31:24]: MR10 OP bit[23:16]: MR10 MA
[15:0]	RW	mr2	Mode register 2 For DDR/DDR2/DDR3, this field specifies extend mode register 2. For LPDDR2/LPDDR3, this field specifies MR16/MR17. bit[15:8]: MR17 OP bit[7:0]: MR16 OP

DETPATTERN

DETPATTERN is a write/read DET pattern register.

NOTE

These fields are used to fine-tune the training procedure. You are advised not to modify them.

Offset Address	Register Name	Total Reset Value
0x006C	DETPATTERN	0x00FF_AA55

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								det_patFF								det_patAA								det_pat55							
Reset	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	0	1	0	1	0	1	0	0	1	0	1	0	1	0	1

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	det_patFF	DET pattern 2
[15:8]	RW	det_patAA	DET pattern 1

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[7:0]	RW	det_pat55	DET pattern 0

MISC

MISC is a miscellaneous control register.

Offset Address	Register Name	Total Reset Value
0x0070	MISC	0x0008_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	swapdfibyte_en		swapwl_en			swapdm_en		swapaddr_en		swap_en		cfg_dlyupd	addr_delay	swap_en_msb	addr_toggle	scramb_en	reserved				cfg_rl				reserved				cfg_wl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RW	swapdfibyte_en	Swap DFI byte lane 00: no swap 01: swap
[29:27]	RW	swapwl_en	Swap WL/CA result 000: no swap 001: swap 1 010: swap 2 011: swap 3 100: swap 4 Note: When the data BIST loopback mode is selected, this field must be set to 0x0.
[26:25]	RW	swapdm_en	Swap DM bus 00: no swap 01: 16-bit T2 swap
[24:23]	RW	swapaddr_en	Swap address bus 00: no swap 01: 32-bit flyby-4, flyby-2 swap 10: 16-bit T-2 swap

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[22:21]	RW	swap_en	Swap DQ bus bit[1:0] 000: no swap 001: swap 1 010: swap 2 011: swap 3 100: swap 4 **NOTE** When the data BIST loopback mode is selected, this field must be set to 0x0.
[20]	RW	cfg_dlyupd	Transparent mode for the delay update module. When this bit is set to 1, the delay setting in registers are applied to the PHY immediately.
[19]	RW	addr_delay	Add 1-T delay on address/command This bit must be set to 1 when CWL is 5.
[18]	RW	swap_en_msb	Swap DQ bus bit[2] 000: no swap 001: swap 1 010: swap 2 011: swap 3 100: swap 4 **NOTE** When the data BIST loopback mode is selected, this field must be set to 0x0.
[17]	RW	addr_toggle	Toggle address bus every clock cycle This bit is set to 1 only for the burn-in test.
[16]	RW	scramb_en	Data scrambler enable Set this field to 1 to enable data scramble function
[15:12]	RO	reserved	Reserved
[11:7]	RW	cfg_rl	PHY read latency It defines the number of cycles from read command to assert of rddata_en and is equal to CL + AL − 2 (for LPDDR2/3, the field should be RL-2).
[6:5]	RO	reserved	Reserved
[4:0]	RW	cfg_wl	PHY write latency It defines the number of cycles from write command to assert of wrdata_en and is equal to CWL + AL − 1 (for LPDDR2/3, the field should be WL + 1). If the DFI clock ratio is 1:2, cfg_wl must be greater than or equal to 2.

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RNK2RNK

RNK2RNK is a rank-to-rank delay control register.

Offset Address	Register Name	Total Reset Value
0x0074	RNK2RNK	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved			trnkwtr	trnkrtw	trnkrtr	trnkwtw	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:15]	RW	trnkwtr	Delay from write to read command targeted to different ranks
[14:10]	RW	trnkrtw	Delay from read to write command targeted to different ranks
[9:5]	RW	trnkrtr	Delay from read to read command targeted to different ranks
[4:0]	RW	trnkwtw	Delay from write to write command targeted to different ranks

PHYCTRL0

PHYCTRL0 is PHY control register 0.

NOTE

Register in this field are connected to PHY interface directly

Offset Address	Register Name	Total Reset Value
0x0078	PHYCTRL0	0x0000_3980

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14	13	12	11	10 9 8	7	6 5 4	3 2 1	0
Name	reserved					resetoen	csoen	ckeoen	ckoen	cmdoen	reserved	lp_ck_sel	sel_pos_rx	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0	1	1	1	0 0 1	1	0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13]	RW	resetoen	Set this bit to the default value.
[12]	RW	csoen	Set this bit to the default value.
[11]	RW	ckeoen	Set this bit to the default value.
[10:8]	RW	ckoen	Set this field to the default value.
[7]	RW	cmdoen	Set this bit to the default value.

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[6:5]	RO	reserved	Set this field to the default value.
[4:3]	RW	lp_ck_sel	Set this field to the default value.
[2]	RW	sel_pos_rx	PHY output trigger edge select 0: falling edge 1: rising edge
[1:0]	RO	reserved	Reserved

PHYDBG

PHYDBG is a PHY debug register.

Offset Address	Register Name	Total Reset Value
0x007C	PHYDBG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								dqsb_gated_error								reserved								dqs_gated_error							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RO	dqsb_gated_error	DQSN gated error flag This error flag can be cleared only by issuing PHY reset.
[15:8]	RO	reserved	Reserved
[7:0]	RO	dqs_gated_error	DQS gated error flag This error flag can be cleared only by issuing PHY reset.

RETCTRL0

RETCTRL0 is a retraining control register.

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Offset Address	Register Name	Total Reset Value
0x0080	RETCTRL0	0x0001_2400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	retrain_thrd																reserved	gtrten	retrain_en	phyupd_resp												
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	retrain_thrd	Retraining threshold The PHY counts the number of AREFs issued by MC. When the number reaches the threshold, the retraining operation is triggered to be started. Only a non-zero value is valid.
[15]	RO	reserved	Reserved
[14]	RW	gtrten	Gate retraining active This field controls whether the gate retraining is executed during retraining on-going. This field should be asserted only after the initial gate training is complete.
[13]	RW	retrain_en	PHY retraining enable This field controls whether the PHY is enabled to request memory controller for retraining.
[12:0]	RW	phyupd_resp	PHY update response latency When the PHY asserts dfi_phyupd_req to MC, MC should assert dfi_phyupd_ack no later than the number of cycles configured by the response latency.

DMSEL

DMSEL is a DM swap selection register.

Offset Address	Register Name	Total Reset Value
0x0084	DMSEL	0x0000_000A

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												dxctl_dmswap_sel			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	dxctl_dmswap_sel	bit[3:2]: DM1 swap select bit[1:0]: DM0 swap select

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IOCTL

IOCTL is an I/O control register.

Offset Address	Register Name	Total Reset Value
0x0088	IOCTL	0x4BFF_F801

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	odtsel_dqs				odtsel				ronsel_ck				ronsel_ac2t				ronsel_ac1t				ronsel_dqs				ronsel				reserved								aciopldn	dxiopldn	diff_dqs_en	
Reset	0	1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:29]	RW	odtsel_dqs	Set this field to the default value.
[28:26]	RW	odtsel	Set this field to the default value.
[25:23]	RW	ronsel_ck	Set this field to the default value.
[22:20]	RW	ronsel_ac2t	Set this field to the default value.
[19:17]	RW	ronsel_ac1t	Set this field to the default value.
[16:14]	RW	ronsel_dqs	Set this field to the default value.
[13:11]	RW	ronsel	Set this field to the default value.
[10]	RO	reserved	Set this bit to the default value.
[9:3]	RO	reserved	Reserved
[2]	RW	aciopldn	Set this bit to the default value.
[1]	RW	dxiopldn	Set this bit to the default value.
[0]	RW	diff_dqs_en	Differential DQS enable

DQSSEL

DQSSEL is a PHY DQ swap register.

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Offset Address	Register Name	Total Reset Value
0x008C	DQSSEL	0xFF00_FF00

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dqswap_sel																															
Reset	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dqswap_sel	DQ swap in PHY Each bit determines the relative DQS with DQ. 0: relative to DQS0 1: relative to DQS1

PHYCTRL1

PHYCTRL1 is an AC PHY control register.

NOTE

Register in this field are connected to PHY interface directly

Offset Address	Register Name	Total Reset Value
0x0090	PHYCTRL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	ac_pll_cal	ac_pll_calm	ac_pll_calt	ac_pll_lt	ac_pll_sp			ac_pll_cpi				ac_pll_init	ac_pll_testen	ac_pll_testpem	ac_pll_thrm	reserved																	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30]	RW	ac_pll_cal	The value 0x0 is recommended.
[29]	RW	ac_pll_calm	The value 0x0 is recommended.
[28]	RW	ac_pll_calt	The value 0x0 is recommended.
[27:26]	RW	ac_pll_lt	The value 0x0 is recommended.
[25:23]	RW	ac_pll_sp	The value 0x2 is recommended.
[22:20]	RW	ac_pll_cpi	The value 0x4 is recommended.
[19]	RW	ac_pll_init	The value 0x0 is recommended.
[18]	RW	ac_pll_testen	The value 0x0 is recommended.

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[17]	RW	ac_pll_testpem	The value 0x0 is recommended.
[16]	RW	ac_pll_thrm	The value 0x0 is recommended.
[15:0]	RO	reserved	Reserved

DXNCKCTRL

DXNCKCTRL is a PLL phase selection (within the PHY) register. You are advised not to modify this register.

Offset Address	Register Name	Total Reset Value
0x0094	DXNCKCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										dx_ckd2				dx_ck0p_mclk			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5:3]	RW	dx_ckd2	Set this field to the default value.
[2:0]	RW	dx_ck0p_mclk	Set this field to the default value.

PHYPLLCTRL_AC

PHYPLLCTRL_AC is PHY PLL control register 1.

NOTE

Register in this field are connected to PHY interface directly

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Offset Address	Register Name	Total Reset Value
0x0098	PHYPLLCTRL_AC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ac_pll_lock	reserved																							ac_pll_test							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	ac_pll_lock	AC PLL lock signal 0: not locked 1: locked
[30:8]	RO	reserved	Reserved
[7:0]	RW	ac_pll_test	AC PLL test mode 0: normal mode 1: test mode

PHYPLLCTRL_DX

PHYPLLCTRL_DX is PHY PLL control register 2.

Offset Address	Register Name	Total Reset Value
0x009C	PHYPLLCTRL_DX	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dx_pll_lock				reserved												dx_pll_test															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	dx_pll_lock	ND: number of data PHY blocks DX PLL lock signals for each byte lane 0: not locked 1: locked
[27:16]	RO	reserved	Reserved
[15:0]	RW	dx_pll_test	PLL test for 16-bit data block bit[7:0]: 16-bit block 0 bit[15:8]: 16-bit block 1

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SWTMODE

SWTMODE is an S/W training mode register.

Offset Address	Register Name	Total Reset Value
0x00A0	SWTMODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wl_wdqs_gated	reserved																													sw_gtmode	sw_wlmode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	wl_wdqs_gated	Write leveling WDQS strobe disable when the related PH/BDL is tuned to prevent glitch 0: not gated 1: gated (disable WDQS strobe)
[30:2]	RO	reserved	Reserved
[1]	RW	sw_gtmode	Set this bit to 1 when performing S/W gate training or GDS training.
[0]	RW	sw_wlmode	Set this bit to 1 when performing S/W write leveling training.

SWTWLDQS

SWTWLDQS is a write DQS issue register during S/W write leveling training.

Offset Address	Register Name	Total Reset Value
0x00A4	SWTWLDQS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															sw_wl_dqs
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	WO	sw_wl_dqs	Writing 1 to this bit once issues one DQS pulse from the PHY to DRAM.

SWTRLT

SWTRLT is an S/W training result register.

Offset Address	Register Name	Total Reset Value
0x00A8	SWTRLT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								gds_result								gt_result								wl_dq_result							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RO	gds_result	Each bit corresponds to one byte lane.
[15:8]	RO	gt_result	Each bit corresponds to one byte lane.
[7:0]	RO	wl_dq_result	Each bit corresponds to one byte lane.

PHYCTRL2

PHYCTRL2 is PHY control register 2.

NOTE

Register in this field are connected to PHY interface directly

Offset Address	Register Name	Total Reset Value
0x00AC	PHYCTRL2	0x0000_000B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											phy_dyn_pupden	phy_dummypad_use	phy_odten_gated	phy_gatetdc	phy_dqsgerror
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	RW	phy_dyn_pupden	DATA16BIT_blk.reg_dynamic_PUPDEN_16b control

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[3]	RW	phy_dummypad_use	DATA16BIT_blk.reg_dummypad_use control
[2]	RW	phy_odten_gated	DATA16BIT_blk.reg_ODTEN_Gated control
[1]	RW	phy_gatetdc	DATA16BIT_blk.BUFRESETCONTN_gated_TDC control
[0]	RW	phy_dqsgerror	DATA16BIT_blk.BUFRESETCONTN_gated_DQSGERROR control

IOCTL1

IOCTL1 is I/O PHY control register 1.

NOTE

Register in this field are connected to PHY interface directly

Offset Address	Register Name	Total Reset Value
0x00B0	IOCTL1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	ioctl_squeach_pd	reserved		ioctl_genvref_pd	ioctl_genvref_ie	ioctl_popmode	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:24]	RW	ioctl_squeach_pd	Set this field to the default value.
[23:18]	RO	reserved	Reserved
[17:12]	RW	ioctl_genvref_pd	Set this field to the default value.
[11:6]	RW	ioctl_genvref_ie	Set this field to the default value.
[5:0]	RW	ioctl_popmode	Set this field to the default value.

IOCTL2

IOCTL2 is I/O PHY control register 2.

NOTE

Register in this field are connected to PHY interface directly

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Offset Address	Register Name	Total Reset Value
0x00B4	IOCTL2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		ioctl_genvref_refsel_2						ioctl_genvref_refsel_1				ioctl_genvref_refsel_0				ioctl_genvref_rangel_1				ioctl_genvref_rangel_0											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:24]	RW	ioctl_genvref_refsel_2	Set this field to the default value.
[23:18]	RW	ioctl_genvref_refsel_1	Set this field to the default value.
[17:12]	RW	ioctl_genvref_refsel_0	Set this field to the default value.
[11:6]	RW	ioctl_genvref_rangel_1	Set this field to the default value.
[5:0]	RW	ioctl_genvref_rangel_0	Set this field to the default value.

BISTTIMER

BISTTIMER is a code update period register during BIST running.

Offset Address	Register Name	Total Reset Value
0x00B8	BISTTIMER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																bstupdt_timer															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved

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[15:0]	RW	bstupdt_timer	BIST update code timer This field controls the code update period. If this field is programmed to a non-zero value, the BIST is temporarily stopped, and the internal delay code is updated to the PHY when the timer reaches the configured value. The BIST will continue if the code update is complete. If this field is programmed to zero, the update code mechanism is disabled.

TRAINCTRL1

TRAINCTRL1 is data training control register 1.

Offset Address	Register Name	Total Reset Value
0x00D0	TRAINCTRL1	0xF036_F026

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rdet_lbs_en	rdet_ds_en	rdet_rbs_en	rdet_av_en	reserved	rdet_rb_backtap							rdet_samp_num				wdet_lbs_en	wdet_ds_en	wdet_rbs_en	wrdet_av_en	reserved				wdet_rb_backtap				wdet_samp_num			
Reset	1	1	1	1	0	0	0	0	0	0	1	1	0	1	1	0	1	1	1	1	0	0	0	0	0	0	1	0	0	1	1	0

Bits	Access	Name	Description
[31]	RW	rdet_lbs_en	Read DET left boundary search enable
[30]	RW	rdet_ds_en	Read DET deskew enable
[29]	RW	rdet_rbs_en	Read DET right boundary search enable
[28]	RW	rdet_av_en	Read DET average stage enable
[27]	RO	reserved	Reserved
[26:20]	RW	rdet_rb_backtap	Read DET right boundary back tap. This field is used to offset (+) the rdqsbdl before the right boundary search is started to overcome meta-sable zone.
[19:16]	RW	rdet_samp_num	Read DET sample number control. The multiple sample is used to overcome the meta situations. Valid value: 1−15
[15]	RW	wdet_lbs_en	Write DET left boundary search enable
[14]	RW	wdet_ds_en	Write DET deskew enable
[13]	RW	wdet_rbs_en	Write DET right boundary search enable
[12]	RW	wrdet_av_en	Write DET average stage enable

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[11:8]	RO	reserved	Reserved
[7:4]	RW	wdet_rb_backtap	Write DET right boundary back tap. This field is used to offset (−) the wdqphase before the right boundary search is started to overcome meta-sable zone.
[3:0]	RW	wdet_samp_num	Write DET sample number control. The multiple sample is used to overcome the meta situations. Valid value: 1−15

TRAINCTRL2

TRAINCTRL2 is data training control register 2.

Offset Address	Register Name	Total Reset Value
0x00D4	TRAINCTRL2	0x0000_CC55

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rdet_method_sel	mrr_pat_order			mrr_bl4	reserved		gds_automrg_en	mrr_pat_mask								mrr_pat_B								mrr_pat_A							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1	0	0	0	1	0	1	0	1	0	1

Bits	Access	Name	Description
[31]	RW	rdet_method_sel	Read DET method select 0: write-then-read 1: MRR32/40 (only for LPDDR3)
[30:28]	RW	mrr_pat_order	MRR pattern order. If MRR is used in the read DET, PACK will issue three MRR commands. The order is indicated by bit 2 -> bit 1 -> bit 0. 0: MRR32 1: MRR40
[27]	RW	mrr_bl4	MRR burst 4 indicator. If MRR is used in the read DET, PACK needs to know whether MRR burst is 4 (LPDDR2) or 8 (LPDDR3). 0: MRR32/40 burst length = 8 (default) 1: MRR32/40 burst length = 4
[26:25]	RO	reserved	Reserved
[24]	RW	gds_automrg_en	GDS training/retraining auto tuning enable 0: disable GDS auto margin tuning (will increase GDS when GDS training/retraining is complete) 1: enable GDS auto margin tuning (default)

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[23:16]	RW	mrr_pat_mask	MRR pattern mask 0: not masked 1: masked (does not compare data)
[15:8]	RW	mrr_pat_B	MRR40 pattern. This field should comply with the LPDDR2/LPDDR3 specifications.
[7:0]	RW	mrr_pat_A	MRR32 pattern. This field should comply with the LPDDR2/LPDDR3 specifications.

REGBANKCTRL

REGBANKCTRL is register bank control register.

Offset Address	Register Name	Total Reset Value
0x00D8	REGBANKCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														op_sel	wr_sel
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	op_sel	Register bank select related to DFS 0: bank 0 for PHY operating 1: bank 1 for PHY operating
[0]	RW	wr_sel	Register bank select related to DFS 0: bank 0 while APB read/write 1: bank 1 while APB read/write

TRAINCTRL3

TRAINCTRL3 is data training control register 3.

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Offset Address	Register Name	Total Reset Value
0x00DC	TRAINCTRL3	0x8002_0300

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	exd_gtres_en	reserved														gdsrt_backtap	reserved				rpat_cnt_num_open				reserved				rpat_cnt_num			
Reset	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	exd_gtres_en	Extended gate result enable. When this field is set as high, the PACK will use additional information (EN_DQSx_regread) to help extend the gate result in the "fine tune stage of gate training" and "gate retraining." 0: disabled 1: enabled (default)
[30:19]	RO	reserved	Reserved
[18:16]	RW	gdsrt_backtap	GDS retraining back tap. This field controls the number of back taps during GDS retraining. The value range is 1−7. The recommended value range is 1−2.
[15:12]	RO	reserved	Reserved
[11:8]	RW	rpat_cnt_num_open	RDET pattern length control for eye open. This field is used to set the pattern length for eye open. The same RDET pattern will be extended as the following settings (field value + 1): 0x0: pattern length = 1 0x1: pattern length = 2 ... 0xF: pattern length = 16
[7:4]	RO	reserved	Reserved
[3:0]	RW	rpat_cnt_num	RDET pattern length control. This field is used to set the pattern length. The same RDET pattern will be extended as the following settings (field value + 1): 0x0: pattern length = 1 0x1: pattern length = 2 ... 0xF: pattern length = 16

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ACBISTCTRL0

ACBISTCTRL0 is comparison control register 0 while BIST activated.

Offset Address	Register Name	Total Reset Value
0x100	ACBISTCTRL0	0x4000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ac_rdffsel			aclpbk_tg	acbist_en	rst_cmp_mask	cs_cmp_mask				ras_cmp_mask	cas_cmp_mask	we_cmp_mask	ba_cmp_mask			addr_cmp_mask															
Reset	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RW	ac_rdffsel	AC loopback read latch select
[28]	RW	aclpbk_tg	AC loopback toggle control. The user should not modify this bit. It is related to the analog PHY design. 0: 28LPP PHY (default) 1: 40PHY
[27]	RW	acbist_en	BIST enable This field specifies whether the AC block is included for loopback test while bist_op is set to BIST Enable.
[26]	RW	rst_cmp_mask	RST loopback compare mask This field controls whether the comparison of the loopback data on the RST bit is disabled. 0: not compared 1: compared
[25:22]	RW	cs_cmp_mask	CS loopback compare mask This field controls whether the comparison of the loopback data on the CS bits is disabled. 0: not compared 1: compared
[21]	RW	ras_cmp_mask	RAS loopback compare mask This field controls whether the comparison of the loopback data on RAS is disabled. 0: not compared 1: compared

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[20]	RW	cas_cmp_mask	CAS loopback compare mask This field controls whether the comparison of the loopback data on CAS is disabled. 0: not compared 1: compared
[19]	RW	we_cmp_mask	Write enable loopback compare mask This field controls whether the comparison of the loopback data on WE is disabled. 0: not compared 1: compared
[18:16]	RW	ba_cmp_mask	Bank address loopback compare mask This field controls whether the comparison of the loopback data on bank address bits is disabled. 0: not compared 1: compared
[15:0]	RW	addr_cmp_mask	Memory address loopback compare mask This field controls whether the comparison of the loopback data on address bits is disabled. 0: not compared 1: compared

ACBISTCTRL1

ACBISTCTRL1 is comparison control register 1 while BIST activated.

Offset Address	Register Name	Total Reset Value
0x104	ACBISTCTRL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ac_lfsr_seed								cke_cmp_mask				odt_cmp_mask			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	ac_lfsr_seed	LFSR seed value This field specifies the initial seed value of the LFSR generator to this byte lane.

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[7:4]	RW	cke_cmp_mask	CKE loopback compare mask This field controls whether the comparison of the loopback data on CKE bits is disabled. 0: not compared 1: compared
[3:0]	RW	odt_cmp_mask	ODT loopback compare mask This field controls whether the comparison of the loopback data on ODT bits is disabled. 0: not compared 1: compared

ACBISTSTS0

ACBISTSTS0 is BIST test result and status register 0.

NOTE

This register is read-only, and can only be reset by setting bist_op to BIST Reset.

Offset Address	Register Name	Total Reset Value
0x108	ACBISTSTS0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					rst_bist_err	cs_bist_err				ras_bist_err	cas_bist_err	we_bist_err	ba_bist_err			addr_bist_err															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RO	rst_bist_err	BIST error on RST
[25:22]	RO	cs_bist_err	BIST error on CS
[21]	RO	ras_bist_err	BIST error on RAS
[20]	RO	cas_bist_err	BIST error on CAS
[19]	RO	we_bist_err	BIST error on write enable
[18:16]	RO	ba_bist_err	BIST error on bank address
[15:0]	RO	addr_bist_err	BIST error on address bus

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ACBISTSTS1

ACBISTSTS1 is BIST test result and status register 1.

NOTE

This register is read-only, and can only be reset by setting bist_op to BIST Reset.

Offset Address	Register Name	Total Reset Value
0x10C	ACBISTSTS1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved						cke_bist_err	odt_bist_err
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:4]	RO	cke_bist_err	BIST error on CKE
[3:0]	RO	odt_bist_err	BIST error on ODT

ACCMDBDL0

ACCMDBDL0 is AC command bit delay line configuration register 0.

Offset Address	Register Name	Total Reset Value
0x120	ACCMDBDL0	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved			odt1_bdl	reserved			odt0_bdl
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	odt1_bdl	Bit delay line of ODT1
[15:5]	RO	reserved	Reserved
[4:0]	RW	odt0_bdl	Bit delay line of ODT0

ACCMDBDL1

ACCMDBDL1 is AC command bit delay line configuration register 1.

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Offset Address	Register Name	Total Reset Value
0x124	ACCMDBDL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												odt3_bdl				reserved												odt2_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	odt3_bdl	Bit delay line of ODT3
[15:5]	RO	reserved	Reserved
[4:0]	RW	odt2_bdl	Bit delay line of ODT2

ACCMDBDL2

ACCMDBDL2 is AC command bit delay line configuration register 2.

Offset Address	Register Name	Total Reset Value
0x128	ACCMDBDL2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cs1_bdl				reserved												cs0_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	cs1_bdl	Bit delay line of CS1
[15:5]	RO	reserved	Reserved
[4:0]	RW	cs0_bdl	Bit delay line of CS0

ACCMDBDL3

ACCMDBDL3 is AC command bit delay line configuration register 3.

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Offset Address	Register Name	Total Reset Value
0x12C	ACCMDBDL3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cs3_bdl				reserved												cs2_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	cs3_bdl	Bit delay line of CS3
[15:5]	RO	reserved	Reserved
[4:0]	RW	cs2_bdl	Bit delay line of CS2

ACCMDBDL4

ACCMDBDL4 is AC command bit delay line configuration register 4.

Offset Address	Register Name	Total Reset Value
0x130	ACCMDBDL4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cke1_bdl				reserved												cke0_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	cke1_bdl	Bit delay line of CKE1
[15:5]	RO	reserved	Reserved
[4:0]	RW	cke0_bdl	Bit delay line of CKE0

ACCMDBDL5

ACCMDBDL5 is AC command bit delay line configuration register 5.

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Offset Address	Register Name	Total Reset Value
0x134	ACCMDBDL5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cke3_bdl				reserved												cke2_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	cke3_bdl	Bit delay line of CKE3
[15:5]	RO	reserved	Reserved
[4:0]	RW	cke2_bdl	Bit delay line of CKE2

ACCMDBDL6

ACCMDBDL6 is AC command bit delay line configuration register 6.

Offset Address	Register Name	Total Reset Value
0x138	ACCMDBDL6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cas_bdl				reserved												we_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	cas_bdl	Bit delay line of CAS
[15:5]	RO	reserved	Reserved
[4:0]	RW	we_bdl	Bit delay line of WE

ACCMDBDL7

ACCMDBDL7 is AC command bit delay line configuration register 7.

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Offset Address	Register Name	Total Reset Value
0x13C	ACCMDBDL7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												resetn_bdl				reserved												ras_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	resetn_bdl	Bit delay line of RESETN
[15:5]	RO	reserved	Reserved
[4:0]	RW	ras_bdl	Bit delay line of RAS

ACADDRBDL0

ACADDRBDL0 is AC address bit delay line configuration register 0.

Offset Address	Register Name	Total Reset Value
0x140	ACADDRBDL0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr1_bdl				reserved												addr0_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr1_bdl	Bit delay line of ADDR1
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr0_bdl	Bit delay line of ADDR0

ACADDRBDL1

ACADDRBDL1 is AC address bit delay line configuration register 1.

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Offset Address	Register Name	Total Reset Value
0x144	ACADDRBDL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr3_bdl				reserved												addr2_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr3_bdl	Bit delay line of ADDR3
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr2_bdl	Bit delay line of ADDR2

ACADDRBDL2

ACADDRBDL2 is AC address bit delay line configuration register 2.

Offset Address	Register Name	Total Reset Value
0x148	ACADDRBDL2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr5_bdl				reserved												addr4_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr5_bdl	Bit delay line of ADDR5
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr4_bdl	Bit delay line of ADDR4

ACADDRBDL3

ACADDRBDL3 is AC address bit delay line configuration register 3.

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Offset Address	Register Name	Total Reset Value
0x14C	ACADDRBDL3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr7_bdl				reserved												addr6_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr7_bdl	Bit delay line of ADDR7
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr6_bdl	Bit delay line of ADDR6

ACADDRBDL4

ACADDRBDL4 is AC address bit delay line configuration register 4.

Offset Address	Register Name	Total Reset Value
0x150	ACADDRBDL4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr9_bdl				reserved												addr8_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr9_bdl	Bit delay line of ADDR9
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr8_bdl	Bit delay line of ADDR8

ACADDRBDL5

ACADDRBDL5 is AC address bit delay line configuration register 5.

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Offset Address	Register Name	Total Reset Value
0x154	ACADDRBDL5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr11_bdl				reserved												addr10_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr11_bdl	Bit delay line of ADDR11
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr10_bdl	Bit delay line of ADDR10

ACADDRBDL6

ACADDRBDL6 is AC address bit delay line configuration register 6.

Offset Address	Register Name	Total Reset Value
0x158	ACADDRBDL6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr13_bdl				reserved												addr12_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr13_bdl	Bit delay line of ADDR13
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr12_bdl	Bit delay line of ADDR12

ACADDRBDL7

ACADDRBDL7 is AC address bit delay line configuration register 7.

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Offset Address	Register Name	Total Reset Value
0x15C	ACADDRBDL7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												addr15_bdl				reserved												addr14_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	addr15_bdl	Bit delay line of ADDR15
[15:5]	RO	reserved	Reserved
[4:0]	RW	addr14_bdl	Bit delay line of ADDR14

ACADDRBDL8

ACADDRBDL8 is AC address bit delay line configuration register 8.

Offset Address	Register Name	Total Reset Value
0x160	ACADDRBDL8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												ba1_bdl				reserved												ba0_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	ba1_bdl	Bit delay line of BA1
[15:5]	RO	reserved	Reserved
[4:0]	RW	ba0_bdl	Bit delay line of BA0

ACADDRBDL9

ACADDRBDL9 is AC address bit delay line configuration register 9.

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Offset Address	Register Name	Total Reset Value
0x164	ACADDRBDL9	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ba2_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	ba2_bdl	Bit delay line of BA2

ACCLKBDL

ACCLKBDL is an AC clock bit delay line configuration register.

Offset Address	Register Name	Total Reset Value
0x168	ACCLKBDL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				fbclk_bdl				reserved				refclk_bdl				reserved				dramclk1_bdl				reserved				dramclk0_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:24]	RW	fbclk_bdl	Bit delay line of FBCLK
[23:21]	RO	reserved	Reserved
[20:16]	RW	refclk_bdl	Bit delay line of REFCLK
[15:13]	RO	reserved	Reserved
[12:8]	RW	dramclk1_bdl	Set this field to default value
[7:5]	RO	reserved	Reserved
[4:0]	RW	dramclk0_bdl	Set this field to default value

ACPHYCTL0

ACPHYCTL0 is AC block PHY control register 0. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x170	ACPHYCTL0	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									bufphyclk_div2	lvdqclkdiv2	dramclk_l	dramclk_h	margin_a1	margin_a0	enshift_a0
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6]	RW	bufphyclk_div2	PHY control
[5]	RW	lvdqclkdiv2	PHY control
[4]	RW	dramclk_l	PHY control
[3]	RW	dramclk_h	PHY control
[2]	RW	margin_a1	PHY control
[1]	RW	margin_a0	PHY control
[0]	RW	enshift_a0	PHY control

ACPHYCTL1

ACPHYCTL1 is AC block PHY control register 1. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x174	ACPHYCTL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dram1sel_addr																reserved				dram1sel_ba				reserved	dram1sel_we	dram1sel_cas	dram1sel_ras	dram1sel_odt	dram1sel_cke	dram1sel_cs	dram1sel_reset
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	dram1sel_addr	PHY control
[15:11]	RO	reserved	Reserved
[10:8]	RW	dram1sel_ba	PHY control
[7]	RO	reserved	Reserved
[6]	RW	dram1sel_we	PHY control
[5]	RW	dram1sel_cas	PHY control
[4]	RW	dram1sel_ras	PHY control
[3]	RW	dram1sel_odt	PHY control
[2]	RW	dram1sel_cke	PHY control
[1]	RW	dram1sel_cs	PHY control
[0]	RW	dram1sel_reset	PHY control

ACPHYCTL2

ACPHYCTL2 is AC block PHY control register 2. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x178	ACPHYCTL2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sdrsel_addr																reserved				sdrsel_ba				reserved	sdrsel_we	sdrsel_cas	sdrsel_ras	sdrsel_odt	sdrsel_cke	sdrsel_cs	sdrsel_reset
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	sdrsel_addr	PHY control
[15:11]	RO	reserved	Reserved
[10:8]	RW	sdrsel_ba	PHY control
[7]	RO	reserved	Reserved
[6]	RW	sdrsel_we	PHY control
[5]	RW	sdrsel_cas	PHY control
[4]	RW	sdrsel_ras	PHY control
[3]	RW	sdrsel_odt	PHY control
[2]	RW	sdrsel_cke	PHY control
[1]	RW	sdrsel_cs	PHY control
[0]	RW	sdrsel_reset	PHY control

ACPHYCTL3

ACPHYCTL3 is AC block PHY control register 3. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x17C	ACPHYCTL3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	posedge_addr																reserved				posedge_ba				reserved	posedge_we	posedge_cas	posedge_ras	posedge_odt	posedge_cke	posedge_cs	posedge_reset
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	posedge_addr	PHY control
[15:11]	RO	reserved	Reserved
[10:8]	RW	posedge_ba	PHY control
[7]	RO	reserved	Reserved
[6]	RW	posedge_we	PHY control
[5]	RW	posedge_cas	PHY control
[4]	RW	posedge_ras	PHY control
[3]	RW	posedge_odt	PHY control
[2]	RW	posedge_cke	PHY control
[1]	RW	posedge_cs	PHY control
[0]	RW	posedge_reset	PHY control

ACPHYCTL4

ACPHYCTL4 is AC block PHY control register 4. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x180	ACPHYCTL4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timing2T_addr																reserved				timing2T_ba				reserved	timing2T_we	timing2T_cas	timing2T_ras	timing2T_odt	timing2T_cke	timing2T_cs	timing2T_reset
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	timing2T_addr	PHY control
[15:11]	RO	reserved	Reserved
[10:8]	RW	timing2T_ba	PHY control
[7]	RO	reserved	Reserved
[6]	RW	timing2T_we	PHY control
[5]	RW	timing2T_cas	PHY control
[4]	RW	timing2T_ras	PHY control
[3]	RW	timing2T_odt	PHY control
[2]	RW	timing2T_cke	PHY control
[1]	RW	timing2T_cs	PHY control
[0]	RW	timing2T_reset	PHY control

ACPHYCTL5

ACPHYCTL5 is AC block PHY control register 5. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x184	ACPHYCTL5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				clk1x_cmd2t_ctl2				clk1x_mclk				clk1x_cmd1t				clk1x_cmd2t				clk1x_cmd2t_ctl				clk1x_dramclk1				clk1x_dramclk0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:24]	RW	clk1x_cmd2t_ctl2	PHY control

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[23:20]	RW	clk1x_mclk	PHY control
[19:16]	RW	clk1x_cmd1t	PHY control
[15:12]	RW	clk1x_cmd2t	PHY control
[11:8]	RW	clk1x_cmd2t_ctl	PHY control
[7:4]	RW	clk1x_dramclk1	PHY control
[3:0]	RW	clk1x_dramclk0	PHY control

ACPHYCTL6

ACPHYCTL6 is AC block PHY control register 6. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x188	ACPHYCTL6	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	clk2x_cmd2t_ctl2	clk2x_mclk	clk2x_cmd1t	clk2x_cmd2t	clk2x_cmd2t_ctl	clk2x_dramclk1	clk2x_dramclk0
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:24]	RW	clk2x_cmd2t_ctl2	PHY control
[23:20]	RW	clk2x_mclk	PHY control
[19:16]	RW	clk2x_cmd1t	PHY control
[15:12]	RW	clk2x_cmd2t	PHY control
[11:8]	RW	clk2x_cmd2t_ctl	PHY control
[7:4]	RW	clk2x_dramclk1	PHY control
[3:0]	RW	clk2x_dramclk0	PHY control

ACPHYCTL7

ACPHYCTL7 is AC block PHY control register 7. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x018C	ACPHYCTL7	0x0080_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					halft_cmd2tctl2	halft_dramclk0	halft_dramclk1	halft_cmd1t	reserved	halft_cmd2tctl	ck5p_cmd2tc				reserved	ck3p_cmd1t			ck2p_dclk1			ck1p_dclk0				ck0p_mclk				ckd2_mclk			
Reset	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RW	halft_cmd2tctl2	Set this bit to the default value.
[25]	RW	halft_dramclk0	Set this bit to the default value.
[24]	RW	halft_dramclk1	Set this bit to the default value.
[23]	RW	halft_cmd1t	Set this bit to the default value.
[22]	RO	reserved	Reserved
[21]	RW	halft_cmd2tctl	Set this bit to the default value.
[20:18]	RW	ck5p_cmd2tc	Set this field to the default value.
[17:15]	RO	reserved	Reserved
[14:12]	RW	ck3p_cmd1t	Set this field to the default value.
[11:9]	RW	ck2p_dclk1	Configure DRAM CK1 clock phase from PLL1 tap: 1/16 T.
[8:6]	RW	ck1p_dclk0	Configure DRAM CK0 clock phase from PLL1 tap: 1/16 T.
[5:3]	RW	ck0p_mclk	Set this field to the default value.
[2:0]	RW	ckd2_mclk	Set this field to the default value.

ACDEBUG

ACDEBUG is a PHY debugging register.

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Offset Address	Register Name	Total Reset Value
0x0190	ACDEBUG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												acdbg_config				reserved							DESKEW_regread	reserved				ac_rdcnt			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:16]	RW	acdbg_config	Set this field to the default value.
[15:9]	RO	reserved	Reserved
[8]	RO	DESKEW_regread	Set this bit to the default value.
[7:3]	RO	reserved	Reserved
[2:0]	RO	ac_rdcnt	Set this field to the default value.

ACPHYRSVDC

ACPHYRSVDC is AC block PHY reserved control pin register 1. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x0194	ACPHYRSVDC	0xFFFF_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ac_rsv_control																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	ac_rsv_control	Reserved

ACPHYRSVDS

ACPHYRSVDS is AC block PHY reserved control pin register 2. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x0198	ACPHYRSVDS	0xFFFF_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ac_rsv_status																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	ac_rsv_status	Reserved

ACPHYCTL8

ACPHYCTL8 is AC block PHY control register 8. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x019C	ACPHYCTL8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												ac_ck12p_reserve			ac_ck11p_reserve			ac_ck10p_reserve			ac_ck9p_reserve			ac_ck8p_reserve			ac_ck7p_reserve			ac_ck6p_reserve		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:18]	RW	ac_ck12p_reserve	Set this field to the default value.
[17:15]	RW	ac_ck11p_reserve	Set this field to the default value.
[14:12]	RW	ac_ck10p_reserve	Set this field to the default value.
[11:9]	RW	ac_ck9p_reserve	Set this field to the default value.
[8:6]	RW	ac_ck8p_reserve	Set this field to the default value.
[5:3]	RW	ac_ck7p_reserve	Set this field to the default value.
[2:0]	RW	ac_ck6p_reserve	Set this field to the default value.

ACCMDBDL8

ACCMDBDL8 is AC command bit delay line configuration register 8.

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Offset Address	Register Name	Total Reset Value
0x1A0	ACCMDBDL8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												odt1_bdl1				reserved												odt1_bdl0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	odt1_bdl1	Set this field to the default value.
[15:5]	RO	reserved	Reserved
[4:0]	RW	odt1_bdl0	Set this field to the default value.

ACCMDBDL9

ACCMDBDL9 is AC command bit delay line configuration register 9.

Offset Address	Register Name	Total Reset Value
0x1A4	ACCMDBDL9	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												odt1_bdl3				reserved												odt1_bdl2			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	odt1_bdl3	Set this field to the default value.
[15:5]	RO	reserved	Reserved
[4:0]	RW	odt1_bdl2	Set this field to the default value.

ACCMDBDL10

ACCMDBDL10 is AC command bit delay line configuration register 10.

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Offset Address	Register Name	Total Reset Value
0x1A8	ACCMDBDL10	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cs1_bdl1				reserved												cs1_bdl0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	cs1_bdl1	Set this field to the default value.
[15:5]	RO	reserved	Reserved
[4:0]	RW	cs1_bdl0	Set this field to the default value.

ACCMDBDL11

ACCMDBDL11 is AC command bit delay line configuration register 11.

Offset Address	Register Name	Total Reset Value
0x1AC	ACCMDBDL11	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cs1_bdl3				reserved												cs1_bdl2			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	cs1_bdl3	Set this field to the default value.
[15:5]	RO	reserved	Reserved
[4:0]	RW	cs1_bdl2	Set this field to the default value.

ACPHYDCC

ACPHYDCC is an AC block PHY DCC control register. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x01B0	ACPHYDCC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												acdcc_mclk				reserved	acdcc_dqs0			reserved	acdcc_dqs1			reserved	acdcc_dq0			reserved	acdcc_dq1		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:19]	RO	reserved	Reserved
[18:16]	RW	acdcc_mclk	Set this field to the default value.
[15]	RO	reserved	Reserved
[14:12]	RW	acdcc_dqs0	Set this field to the default value.
[11]	RO	reserved	Reserved
[10:8]	RW	acdcc_dqs1	Set this field to the default value.
[7]	RO	reserved	Reserved
[6:4]	RW	acdcc_dq0	Set this field to the default value.
[3]	RO	reserved	Reserved
[2:0]	RW	acdcc_dq1	Set this field to the default value.

ACPHYCTL9

ACPHYCTL9 is AC block PHY control register 9. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x01B4	ACPHYCTL9	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				actiming2t_odt1	actiming2t_cke1	actiming2t_cs1	acdram1sel_odt1	acdram1sel_cke1	acdram1sel_cs1	acposedge_odt1	acposedge_cke1	acposedge_cs1	acsdrsel_odt1	acsdrsel_cke1	acsdrsel_cs1
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved

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[11]	RW	actiming2t_odt1	Set this bit to the default value.
[10]	RW	actiming2t_cke1	Set this bit to the default value.
[9]	RW	actiming2t_cs1	Set this bit to the default value.
[8]	RW	acdram1sel_odt1	Set this bit to the default value.
[7]	RW	acdram1sel_cke1	Set this bit to the default value.
[6]	RW	acdram1sel_cs1	Set this bit to the default value.
[5]	RW	acposedge_odt1	Set this bit to the default value.
[4]	RW	acposedge_cke1	Set this bit to the default value.
[3]	RW	acposedge_cs1	Set this bit to the default value.
[2]	RW	acsdrsel_odt1	Set this bit to the default value.
[1]	RW	acsdrsel_cke1	Set this bit to the default value.
[0]	RW	acsdrsel_cs1	Set this bit to the default value.

CATTIMER0

CATTIMER0 is a CA training timer 0 register.

Offset Address	Register Name	Total Reset Value
0x01C0	CATTIMER0	0x0B17_0B0B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				tcackeh				reserved				tcacd				reserved				tcaent				reserved				tcackel			
Reset	0	0	0	0	1	0	1	1	0	0	0	1	0	1	1	1	0	0	0	0	1	0	1	1	0	0	0	0	1	0	1	1

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:24]	RW	tcackeh	tCACKEH (dficlk cycle)
[23:22]	RO	reserved	Reserved
[21:16]	RW	tcacd	tCACD (dficlk cycle)
[15:14]	RO	reserved	Reserved
[13:8]	RW	tcaent	tCAENT (dficlk cycle)
[7:6]	RO	reserved	Reserved
[5:0]	RW	tcackel	tCACKEL (dficlk cycle)

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CATTIMER1

CATTIMER1 is CA training timer 1 register.

Offset Address	Register Name	Total Reset Value
0x01C4	CATTIMER1	0x0000_040B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				tmrz				reserved				tcaext			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	1	1

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	tmrz	tMRZ (dficlk cycle)
[7:6]	RO	reserved	Reserved
[5:0]	RW	tcaext	tCAEXT (dficlk cycle)

CATCONFIG

CATCONFIG is a CA training configuration register.

Offset Address	Register Name	Total Reset Value
0x01C8	CATCONFIG	0x0008_2606

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sw_cat_en	sw_cat_dqvalid	sw_cat_cke_low	sw_cat_cke_high	sw_cat_strobe	sw_cat_mrw41	sw_cat_mrw48	sw_cat_mrw42	reserved				cat_openeye_en	cat_restore_en	cat_phydq_sel	reserved	cat_rb_backtap				ca_samp_num_ph				reserved				ca_samp_num_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	1	1	0	0	0	0	0	0	1	1	0

Bits	Access	Name	Description
[31]	RW	sw_cat_en	SW CA training enable When this field is asserted, PACK will enter the SW CA training state. After the SW CA training is complete, this bit should be deasserted by firmware. 0: disabled (default) 1: enabled

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[30]	RW	sw_cat_dqvalid	SW CA training. This field indicates whether the CA result is ready. This bit is cleared when sw_cat_strobe is asserted. Then after the timing tCACD, this bit is asserted. 0: not valid 1: valid
[29]	RW	sw_cat_cke_low	SW CA training. This field is used to request the CKE to be low and it cleared by HW.
[28]	RW	sw_cat_cke_high	SW CA training. This field is used to request the CKE to be high and is cleared by HW.
[27]	RW	sw_cat_strobe	SW CA training. This field is used to issue the predefined SW CA pattern and is cleared by HW.
[26]	RW	sw_cat_mrw41	SW CA training. This field is used to issue the MRW41 command and is cleared by HW.
[25]	RW	sw_cat_mrw48	SW CA training. This field is used to issue the MRW48 command and is cleared by HW.
[24]	RW	sw_cat_mrw42	SW CA training. This field is used to issue the MRW42 command and is cleared by HW.
[23:21]	RO	reserved	Reserved
[20]	RW	cat_openeye_en	HW CA training open eye enable 0: disabled (default) 1: enabled
[19]	RW	cat_restore_en	HW CA training restore enable. If this field is set to high, the CA training will restore the CA phase before the second CA training (mapping stage) starts. This field is recommended to set to high. 0: disabled 1: enabled (default)
[18]	RW	cat_cat_phydq_sel	HW CA training PHY DQ CA result select 0: LSB (default) 1: MSB
[17:16]	RO	reserved	Reserved
[15:12]	RW	cat_rb_backtap	HW CA training right boundary back tap. This field is used to offset the addrph_a before the right boundary search is started to overcome meta-sable zone.
[11:8]	RW	ca_samp_num_ph	HW CA training sample count for CA phase Value range: 1−15
[7:4]	RO	reserved	Reserved
[3:0]	RW	ca_samp_num_bdl	HW CA training sample count for CA BDL Value range: 1−15

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CATRESULT

CATRESULT is a CA training result register (for debugging).

Offset Address	Register Name	Total Reset Value
0x01CC	CATRESULT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						cat_reault									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:0]	RO	cat_reault	CA training. HW CA training result for each address bit during each HW CA training state. This field is meaningful only after the HW CA training is complete and is only for debugging. 0: fail 1: success

PHYDQRESULT

PHYDQRESULT is a SW CA training DQ result (from PHY) register.

Offset Address	Register Name	Total Reset Value
0x01D0	PHYDQRESULT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	phy_dq_result																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	phy_dq_result	SW CA training. SW CA training latched results from the DRAM device. For the LPDDR3 CA training, the meaningful bits are [15:0] (mapped to DQ[15:0]). PACK and PHY will not latch these bits, so these bits are those you obtained on DQ of the DRAM device.

ADDRPHBOUND

ADDRPHBOUND is a CA training address phase boundary register.

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Offset Address	Register Name	Total Reset Value
0x01D4	ADDRPHBOUND	0x0008_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				addrph_a_ori				reserved				addrph_a				reserved				addrph_a_left				reserved				addrph_a_right			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:24]	RO	addrph_a_ori	HW CA training. This field records the address phase before the HW CA training (for debugging).
[23:21]	RO	reserved	Reserved
[20:16]	RW	addrph_a	HW CA training. This field records the left boundary of the address phase after the HW CA training (for debugging).
[15:13]	RO	reserved	Reserved
[12:8]	RO	addrph_a_left	HW CA training. This field records the left boundary of the address phase after the HW CA training (for debugging).
[7:5]	RO	reserved	Reserved
[4:0]	RO	addrph_a_right	HW CA training. This field records the right boundary of the address phase after the HW CA training (for debugging).

SWCATPATTERN_P

SWCATPATTERN_P is an SW CA training pattern register for the positive CK edge.

Offset Address	Register Name	Total Reset Value
0x01D8	SWCATPATTERN_P	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								sw_cat_pat_p							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:0]	RW	sw_cat_pat_p	SW CA training pattern for the positive CK edge. You can fill any pattern.

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SWCATPATTERN_N

SWCATPATTERN_N is an SW CA training pattern register for the negative CK edge.

Offset Address	Register Name	Total Reset Value
0x01DC	SWCATPATTERN_N	0x0000_03FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						sw_cat_pat_n									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:0]	RW	sw_cat_pat_n	SW CA training pattern for the negative CK edge. You can fill any pattern.

LPCTRL

LPCTRL is a low-power control register.

Offset Address	Register Name	Total Reset Value
0x01E4	LPCTRL	0xFE80_8688

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gcken_auto_en	gcken_dfi_main	gcken_dinit	gcken_dtrain	gcken_retrain	gcken_acbist	gcken_dxbist	t_wakeup_thrd_lv2				lp_cmdoen_disable	lp_ac_phy_clkgated_disable	lp_dx_phy_clkgated_disable	lp_dx_phy_phgated_disable	lp_pll_powerdown_disable	t_resp				t_wakeup_thrd_lv3				t_dram_clk_disable				t_dram_clk_enable			
Reset	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	0	0	0	1	0	0	0

Bits	Access	Name	Description
[31]	RW	gcken_auto_en	Auto gated clock control enable. If this field is set to high, HW automatically/dynamically controls the PACK high-level clock gating. 0: off 1: on (default)

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[30]	RW	gcken_dfi_main	Manual gating control, main DFI clock gating switch 0: clock off 1: clock on (default) When this field is set to low (clock off), some control registers that use the dfi_main clock are affected.
[29]	RW	gcken_dinit	Manual gating control, DRAM initialization clock gating 0: clock off 1: clock on (default)
[28]	RW	gcken_dtrain	Manual gating control, data training clock gating 0: clock off 1: clock on (default)
[27]	RW	gcken_retrain	Manual gating control, retraining clock gating 0: clock off 1: clock on (default)
[26]	RW	gcken_acbist	Manual gating control, AC BIST clock gating 0: clock off 1: clock on (default)
[25]	RW	gcken_dxbist	Manual gating control, data BIST clock gating 0: clock off 1: clock on (default)
[24:21]	RW	t_wakeup_thrd_lv2	DFI low-power control interface timing threshold ≥ wakeup_thrd_lv2: PHY clock gating, disable AC output enable ≥ wakeup_thrd_lv3: PHY clock gating, disable AC output enable, PHY PLL disable **NOTE** wakeup_thrd_lv3 should ≥ wakeup_thrd_lv2
[20]	RW	lp_cmdoen_disable	LP CMDOEN function disable when low-power assert. If this function is enabled, the I/O output of CA is floating to reduce I/O power. 0: LP-related function enabled (default) 1: LP-related function disabled
[19]	RW	lp_ac_phy_clkgated_disable	LP AC PHY clock gating disable when low-power assert 0: LP-related function enabled (default) 1: LP-related function disabled
[18]	RW	lp_dx_phy_clkgated_disable	LP DX PHY clock gating disable when low-power assert 0: LP-related function enabled (default) 1: LP-related function disabled

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[17]	RW	lp_dx_phy_phgated_disable	DX PHY clock phase gating disable when low-power assert 0: LP-related function enabled (default) 1: LP-related function disabled
[16]	RW	lp_pll_powerdown_disable	LP PLL disable function disable when low-power assert 0: LP-related function enabled (default) 1: LP-related function disabled
[15:12]	RW	t_resp	DFI low-power control interface timing parameter. The field value must be smaller than DFI tlp_resp.
[11:8]	RW	t_wakeup_thrd_lv3	DFI low-power control interface timing threshold ≥ wakeup_thrd_lv2: PHY clock gating, disable AC output enable ≥ wakeup_thrd_lv3: PHY clock gating, disable AC output enable, PHY PLL power down Note: wakeup_thrd_lv3 should ≥ wakeup_thrd_lv2
[7:4]	RW	t_dram_clk_disable	DFI DRAM clock disable timing parameter. This field is used to count down before DRAM CK is disabled.
[3:0]	RW	t_dram_clk_enable	DFI DRAM clock enable timing parameter. This field is used to count down before DRAM CK is enabled.

DXNBISTCTRL

DXNBISTCTRL is a loopback data comparison control register during BIST of data blocks.

Offset Address	Register Name	Total Reset Value
0x200 + *n* x 0x80 (*n* = 0–3)	DXNBISTCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lfsr_seed								reserved							dqm_cmp_mask	dq_cmp_mask								reserved							bist_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	lfsr_seed	LFSR seed value This field specifies the initial seed value of the LFSR generator to this byte lane.
[23:17]	RO	reserved	Reserved

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[16]	RW	dqm_cmp_mask	DQM loopback compare mask This field controls whether the comparison of the loopback data on DQM bits is disabled. 0: not compared 1: compared
[15:8]	RW	dq_cmp_mask	DQ loopback compare mask This field controls whether the comparison of the loopback data on DQ bits is disabled. 0: not compared 1: compared
[7:1]	RO	reserved	Reserved
[0]	RW	bist_en	BIST enable This field specifies whether the data block is included for the loopback test while bist_op is set to BIST Enable.

DXNBISTSTS

DXNBISTSTS is a BIST test result and status register. Note that this register is read-only, and can be reset only by setting bist_op to BIST Reset.

Offset Address	Register Name	Total Reset Value
0x204 + *n* x 0x80 (*n* = 0–3)	DXNBISTSTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															dqm_cmp_err	dq_cmp_ferr								dq_cmp_rerr							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RO	dqm_cmp_err	BIST error on DQM
[15:8]	RO	dq_cmp_ferr	BIST error on DQ at the falling edge of DQS
[7:0]	RO	dq_cmp_rerr	BIST error on DQ at the rising edge of DQS

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DXNCTRL

DXNCTRL is a data block control register.

Offset Address	Register Name	Total Reset Value
0x208 + *n* x 0x80 (*n* = 0–3)	DXNCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														dm_dis	bl_dis
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	dm_dis	Data mask disable This field specifies whether the corresponding write data mask (DM) is enabled. When this field is set to 1, the corresponding DM is not used and calibrated during training.
[0]	RW	bl_dis	Byte lane disable This field specifies whether the corresponding byte lane is in use. If the corresponding byte lane is not used, the PLL/DDL is powered down, and the output driver is turned off.

DXNWDQNBDL0

DXNWDQNBDL0 is data block bit delay configuration register 0.

Offset Address	Register Name	Total Reset Value
0x210 + *n* x 0x80 (*n* = 0–3)	DXNWDQNBDL0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				wdq3bdl				reserved				wdq2bdl				reserved				wdq1bdl				reserved				wdq0bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:24]	RW	wdq3bdl	Delay select of WDQ3 This field specifies the delay value of the bit delay line on the write path of DQ3.

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[23:21]	RO	reserved	Reserved
[20:16]	RW	wdq2bdl	Delay select of WDQ2 This field specifies the delay value of the bit delay line on the write path of DQ2.
[15:13]	RO	reserved	Reserved
[12:8]	RW	wdq1bdl	Delay select of WDQ1 This field specifies the delay value of the bit delay line on the write path of DQ1.
[7:5]	RO	reserved	Reserved
[4:0]	RW	wdq0bdl	Delay select of WDQ0 This field specifies the delay value of the bit delay line on the write path of DQ0.

DXNWDQNBDL1

DXNWDQNBDL1 is data block bit delay configuration register 1.

Offset Address	Register Name	Total Reset Value
0x214 + *n* x 0x80 (*n* = 0–3)	DXNWDQNBDL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				wdq7bdl				reserved				wdq6bdl				reserved				wdq5bdl				reserved				wdq4bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:24]	RW	wdq7bdl	Delay select of WDQ7 This field specifies the delay value of the bit delay line on the write path of DQ3.
[23:21]	RO	reserved	Reserved
[20:16]	RW	wdq6bdl	Delay select of WDQ6 This field specifies the delay value of the bit delay line on the write path of DQ2.
[15:13]	RO	reserved	Reserved
[12:8]	RW	wdq5bdl	Delay select of WDQ5 This field specifies the delay value of the bit delay line on the write path of DQ1.

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[7:5]	RO	reserved	Reserved
[4:0]	RW	wdq4bdl	Delay select of WDQ4 This field specifies the delay value of the bit delay line on the write path of DQ0.

DXNWDQNBDL2

DXNWDQNBDL2 is data block bit delay configuration register 2.

Offset Address	Register Name	Total Reset Value
0x218 + *n* x 0x80 (*n* = 0–3)	DXNWDQNBDL2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											wdmbdl				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	wdmbdl	Delay select of WDQM This field specifies the delay value of the bit delay line on DQM.

DXNRDQNBDL0

DXNRDQNBDL0 is data block bit delay configuration register 4.

Offset Address	Register Name	Total Reset Value
0x21C + *n* x 0x80 (*n* = 0–3)	DXNRDQNBDL0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				rdq3bdl				reserved				rdq2bdl				reserved				rdq1bdl				reserved				rdq0bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:24]	RW	rdq3bdl	Delay select of RDQ3 This field specifies the delay value of the bit delay line on the write path of DQ3.

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[23:21]	RO	reserved	Reserved
[20:16]	RW	rdq2bdl	Delay select of RDQ2 This field specifies the delay value of the bit delay line on the write path of DQ2.
[15:13]	RO	reserved	Reserved
[12:8]	RW	rdq1bdl	Delay select of RDQ1 This field specifies the delay value of the bit delay line on the write path of DQ1.
[7:5]	RO	reserved	Reserved
[4:0]	RW	rdq0bdl	Delay select of RDQ0 This field specifies the delay value of the bit delay line on the write path of DQ0.

DXNRDQNBDL1

DXNRDQNBDL1 is data block bit delay configuration register 5.

Offset Address	Register Name	Total Reset Value
0x220 + *n* x 0x80 (*n* = 0–3)	DXNRDQNBDL1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	rdq7bdl	reserved	rdq6bdl	reserved	rdq5bdl	reserved	rdq4bdl
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:24]	RW	rdq7bdl	Delay select of RDQ7 This field specifies the delay value of the bit delay line on the write path of DQ3.
[23:21]	RO	reserved	Reserved
[20:16]	RW	rdq6bdl	Delay select of RDQ6 This field specifies the delay value of the bit delay line on the write path of DQ2.
[15:13]	RO	reserved	Reserved
[12:8]	RW	rdq5bdl	Delay select of RDQ5 This field specifies the delay value of the bit delay line on the write path of DQ1.

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[7:5]	RO	reserved	Reserved
[4:0]	RW	rdq4bdl	Delay select of RDQ4 This field specifies the delay value of the bit delay line on the write path of DQ0.

DXNRDQNBDL2

DXNRDQNBDL2 is data block bit delay configuration register 6.

Offset Address	Register Name	Total Reset Value
0x224 + *n* x 0x80 (*n* = 0–3)	DXNRDQNBDL2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											rdmbdl				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	rdmbdl	Delay select of RDQM This field specifies the delay value of the bit delay line on loopback read path of DQM.

DXNRDQSDLY

DXNRDQSDLY is a local delay line control register.

Offset Address	Register Name	Total Reset Value
0x22C + *n* x 0x80 (*n* = 0–3)	DXNRDQSDLY	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved									rdqscyc							reserved												rdqsbdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22:16]	RW	rdqscyc	Read DQS clock phase select This field specifies the number of taps in one clock cycle for the 1-T delay measurement type or the number of taps in half of clock cycle for the half-T delay measurement type. This field is used to create 90 degree delays for RDQS.

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[15:7]	RO	reserved	Reserved
[6:0]	RW	rdqsbdl	Read DQS delay select This field specifies the phase shift of the read DQS to create 90 degree delays.

DXNWDQSDLY

DXNWDQSDLY is a write leveling DQS delay control register.

Offset Address	Register Name	Total Reset Value
0x230 + *n* x 0x80 (*n* = 0–3)	DXNWDQSDLY	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				wdqsphase				reserved				wdqsbdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:8]	RW	wdqsphase	Write DQS clock phase select
[7:5]	RO	reserved	Reserved
[4:0]	RW	wdqsbdl	Write leveling DQS delay select This field specifies the delay value of delay applied on WDQS for write leveling.

DXNWDQDLY

DXNWDQDLY is a write leveling delay line control register.

Offset Address	Register Name	Total Reset Value
0x234 + *n* x 0x80 (*n* = 0–3)	DXNWDQDLY	0x0000_0C00

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				wdqphase				reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved

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[12:8]	RW	wdqphase	Write data eye clock phase select
[7:0]	RO	reserved	Reserved

DXNWLSL

DXNWLSL is an extra system latency addition control register.

Offset Address	Register Name	Total Reset Value
0x238 + *n* x 0x80 (*n* = 0–3)	DXNWLSL	0x0001_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														wlsl		reserved															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17:16]	RW	wlsl	Write leveling system latency This field is used to adjust the write latency after write leveling.
[15:0]	RO	reserved	Reserved

DXNRDQSGDLY

DXNRDQSGDLY is a local delay line control register.

Offset Address	Register Name	Total Reset Value
0x23C + *n* x 0x80 + *m* x 0x400 (*n* = 0–3) (*m* = 0–1)	DXNRDQSGDLY	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved			org_rdqsgph				reserved				rdqsgtxbdl				reserved				rdqsgphase				reserved				rdqsgbdl				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved

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[29:24]	RO	org_rdqsgph	Original read DQS gate phase before HW gate training post-update stage (for debugging)
[23:21]	RO	reserved	Reserved
[20:16]	RW	rdqsgtxbdl	Read DQS gate delay select This field specifies the delay value of the read gating delay before DQS dummy PAD.
[15:14]	RO	reserved	Reserved
[13:8]	RW	rdqsgphase	Read DQS gating clock phase select
[7:6]	RO	reserved	Reserved
[5:0]	RW	rdqsgbdl	Read DQS gating delay select This field specifies the delay value of the read gating delay after DQS dummy PAD.

DXPHYCTRL

DXPHYCTRL is a DX PHY control register.

Offset Address	Register Name	Total Reset Value
0x240 + *n* x 0x80 (*n* = 0–3)	DXPHYCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																dx_pll_cal	dx_pll_calm	dx_pll_calt	dx_pll_lt			dx_pll_sp				dx_pll_cpi				dx_pll_init	dx_pll_testen	dx_pll_testpem	dx_pll_thrm
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14]	RW	dx_pll_cal	The value 0x0 is recommended.
[13]	RW	dx_pll_calm	The value 0x0 is recommended.
[12]	RW	dx_pll_calt	The value 0x0 is recommended.
[11:10]	RW	dx_pll_lt	The value 0x0 is recommended.
[9:7]	RW	dx_pll_sp	The value 0x2 is recommended.
[6:4]	RW	dx_pll_cpi	The value 0x4 is recommended.

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[3]	RW	dx_pll_init	The value 0x0 is recommended.
[2]	RW	dx_pll_testen	The value 0x0 is recommended.
[1]	RW	dx_pll_testpem	The value 0x0 is recommended.
[0]	RW	dx_pll_thrm	The value 0x0 is recommended.

DXNGDS

DXNGDS is a latch enable register. It is used to control the latch enable within the PHY to obtain stable data.

Offset Address	Register Name	Total Reset Value
0x248 + *n* x 0x80 (*n* = 0–3)	DXNGDS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														gds	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	gds	Rank 0 read data latch delay This field is used as a reference stable region for the PHY to latch the read data.

DXNCLKBDL

DXNCLKBDL is a data block clock bit delay line configuration register.

Offset Address	Register Name	Total Reset Value
0x24C + *n* x 0x80 (*n* = 0–3)	DXNCLKBDL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												dx_refclk_bdl				reserved												dx_fbclk_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	dx_refclk_bdl	Bit delay line of REFCLK
[15:5]	RO	reserved	Reserved

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[4:0]	RW	dx_fbclk_bdl	Bit delay line of FBCLK

DXNRDBOUND

DXNRDBOUND is a read data eye boundary register. This register indicates the left and right boundaries of RDQSQDL of the data eye.

Offset Address	Register Name	Total Reset Value
0x250 + *n* x 0x80 (*n* = 0–3)	DXNRDBOUND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												rdqsbdl_left				reserved												rdqsbdl_right			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22:16]	RO	rdqsbdl_left	Left boundary of read data eye
[15:7]	RO	reserved	Reserved
[6:0]	RO	rdqsbdl_right	Right boundary of read data eye

DXNWRBOUND

DXNWRBOUND is a write data eye boundary register. This register indicates the left and right boundaries of WDQSQDL of the data eye.

Offset Address	Register Name	Total Reset Value
0x254 + *n* x 0x80 (*n* = 0–3)	DXNWRBOUND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												wdqphase_left				reserved												wdqphase_right			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RO	wdqphase_left	Left boundary of the write data eye
[15:5]	RO	reserved	Reserved
[4:0]	RO	wdqphase_right	Right boundary of the write data eye

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DXNOEBDL

DXNOEBDL is an output enable delay line control register. This register specifies the delay line value of the output enable delay line within the PHY data block.

Offset Address	Register Name	Total Reset Value
0x258 + *n* x 0x80 (*n* = 0–3)	DXNOEBDL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												wdqsoe_bdl				reserved												oen_bdl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	wdqsoe_bdl	Write DQS output enable delay control
[15:5]	RO	reserved	Reserved
[4:0]	RW	oen_bdl	DQ output enable delay control

DXNMISCCTRL0

DXNMISCCTRL0 is data block PHY miscellaneous control register 0. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x260 + *n* x 0x80 (*n* = 0–3)	DXNMISCCTRL0	0x0002_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ptrgated_en	enshift_a0	margin_a0	bufphyclkdiv2	lvdqclkdiv2	dqsgatedla	dx_rxp_2nd_dm	dx_rxn_2nd_dm	dx_dqs_h	dx_dqs_l	dx_margin_a1	rxp_2nd_dq					rxn_2nd_dq												
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RW	ptrgated_en	Set this bit to the default value.
[25]	RW	enshift_a0	Set this bit to the default value.

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[24]	RW	margin_a0	After this field is modified, "phyconn_rst" is required to update the latest setting. If the dly_track_type=1 tracking type is required, this field should be set to 0x1 before initial trainings and dly_track_type should be set to 0x1 after initial trainings.
[23]	RW	bufphyclkdiv2	Set this bit to the default value.
[22]	RW	lvdqclkdiv2	Set this bit to the default value.
[21]	RW	dqsgatedla	Set this bit to the default value.
[20]	RW	dx_rxp_2nd_dm	After this field is modified, "phyconn_rst" is required to update the latest setting. If the dly_track_type=1 tracking type is required, this field should be set to 0x1 before initial trainings and dly_track_type should be set to 0x1 after initial trainings.
[19]	RW	dx_rxn_2nd_dm	After this field is modified, "phyconn_rst" is required to update the latest setting. If the dly_track_type=1 tracking type is required, this field should be set to 0x1 before initial trainings and dly_track_type should be set to 0x1 after initial trainings.
[18]	RW	dx_dqs_h	Set this bit to the default value.
[17]	RW	dx_dqs_l	Set this bit to the default value.
[16]	RW	dx_margin_a1	After this field is modified, "phyconn_rst" is required to update the latest setting. If the dly_track_type=1 tracking type is required, this field should be set to 0x1 before initial trainings and dly_track_type should be set to 0x1 after initial trainings.
[15:8]	RW	rxp_2nd_dq	After this field is modified, "phyconn_rst" is required to update the latest setting. When the dly_track_type=1 tracking type is required, this field should be set to 0xFF before initial trainings and dly_track_type should be set to 0x1 after initial trainings.
[7:0]	RW	rxn_2nd_dq	After this field is modified, "phyconn_rst" is required to update the latest setting. When the dly_track_type=1 tracking type is required, this field should be set to 0xFF before initial trainings and dly_track_type should be set to 0x1 after initial trainings.

DXNMISCCTRL1

DXNMISCCTRL1 is data block PHY miscellaneous control register 1. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x264 + *n* x 0x80 (*n* = 0–3)	DXNMISCCTRL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dqclk1x				dqclk2x				dqsgclk1x				dqsgclk2x				dqsclk1x				dqsclk2x				mclk1x				mclk2x			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	dqclk1x	Set this field to the default value.
[27:24]	RW	dqclk2x	Set this field to the default value.
[23:20]	RW	dqsgclk1x	Set this field to the default value.
[19:16]	RW	dqsgclk2x	Set this field to the default value.
[15:12]	RW	dqsclk1x	Set this field to the default value.
[11:8]	RW	dqsclk2x	Set this field to the default value.
[7:4]	RW	mclk1x	Set this field to the default value.
[3:0]	RW	mclk2x	Set this field to the default value.

DXDEBUG0

DXDEBUG0 is data block PHY debug signal register 0.

Offset Address	Register Name	Total Reset Value
0x268 + *n* x 0x80 (*n* = 0–3)	DXDEBUG0	0x0000_5415

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																	s0a	s0b	s1a	reserved											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0	0	0	1	0	1	0	1

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14]	RO	s0a	Set this bit to the default value.
[13]	RO	s0b	Set this bit to the default value.
[12]	RO	s1a	Set this bit to the default value.
[11:0]	RO	reserved	Reserved

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DXDEBUG1

DXDEBUG1 is data block PHY debug signal register 1.

Offset Address	Register Name	Total Reset Value
0x26C + *n* x 0x80 (*n* = 0–3)	DXDEBUG1	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								dqs0								dqs0b								dqs0_gold							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:24]	RO	reserved	Debug signal. Note that the field may have unknown values in the simulation.
[23:0]	RO	dqs0	Reserved
[15:8]	RO	dqs0b	Reserved
[7:0]	RO	dqs0_gold	Reserved

DXPHYRSVD

DXPHYRSVD is a data block PHY reserved control pin register. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x270 + *n* x 0x80 (*n* = 0–3)	DXPHYRSVD	0x0000_FF00

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dx_rsv_status																dx_rsv_control															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	dx_rsv_status	Reserved
[15:0]	RW	dx_rsv_control	Reserved The followings are for (n = 0, 2): Bit 1 (reg_DQSG_extend_2T): 0/1: extend DQSG 1T/2T dfi_clk, when reg_DQSG_extend_en is 1 Bit 0 (reg_DQSG_extend_en): 0/1: disable/enable the DQSG extend function

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DXNMISCCTRL2

DXNMISCCTRL2 is data block PHY miscellaneous control register 2. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x274 + *n* x 0x80 (*n* = 0–3)	DXNMISCCTRL2	0x0000_026F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					reg_OE_extend1T_en	reg_dqsg_TX_2path	reg_squeach_en	reg_dqsglat1T_en	reg_sel_halfT_gated_16b	reg_dynamic_PUPDEN_16b	reg_dummypad_use	reg_ODTEN_gated	reg_GDS_r1T_sel_16b	BUFRESETCONTN_gated_TDC	BUFRESETCONTN_gated_DQSGERROR
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	1	1	1	1

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	reg_OE_extend1T_en	Set this bit to the default value.
[9]	RW	reg_dqsg_TX_2path	Set this bit to the default value.
[8]	RW	reg_squeach_en	Set this bit to the default value.
[7]	RW	reg_dqsglat1T_en	Set this bit to the default value.
[6]	RW	reg_sel_halfT_gated_16b	Set this bit to the default value.
[5]	RW	reg_dynamic_PUPDEN_16b	Set this bit to the default value.
[4]	RW	reg_dummypad_use	Set this bit to the default value.
[3]	RW	reg_ODTEN_gated	Set this bit to the default value.
[2]	RW	reg_GDS_r1T_sel_16b	Set this bit to the default value.
[1]	RW	BUFRESETCONTN_gated_TDC	Set this bit to the default value.

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[0]	RW	BUFRESETCONTN_gated_DQSGERROR	Set this bit to the default value.

DXDEBUGCONFIG

DXDEBUGCONFIG is a data block PHY debugging control register. This register is used to control the PHY and should not be modified.

Offset Address	Register Name	Total Reset Value
0x278 + *n* x 0x80 (*n* = 0–3)	DXDEBUGCONFIG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											DESKEW_regread	reg_dbg_prepost_sel	reg_dbg_pllclk_sel	reg_dbg_REFFB_predly	reg_dbg_REFFB_postdly
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	RO	DESKEW_regread	Reserved
[3]	RW	reg_dbg_prepost_sel	Set this bit to the default value.
[2]	RW	reg_dbg_pllclk_sel	Set this bit to the default value.
[1]	RW	reg_dbg_REFFB_predly	Set this bit to the default value.
[0]	RW	reg_dbg_REFFB_postdly	Set this bit to the default value.

DXNDCC

DXNDCC is a data block PHY DCC control register. This register is used to control the PHY and should not be modified.

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Offset Address	Register Name	Total Reset Value
0x27C + *n* x 0x80 (*n* = 0–3)	DXNDCC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				dxctl_mclk_dcc				dxctl_dqs_dcc				dxctl_dqsg_dcc				dxctl_dq_dcc			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:9]	RW	dxctl_mclk_dcc	bit[11]: dly_sel_xxx_DCC_16b bit[10]: ctrlC_xxx_DCC_16b bit[9]: ctrl2_xxx_DCC_16b
[8:6]	RW	dxctl_dqs_dcc	bit[8]: dly_sel_xxx_DCC_16b bit[7]: ctrlC_xxx_DCC_16b bit[6]: ctrl2_xxx_DCC_16b
[5:3]	RW	dxctl_dqsg_dcc	bit[5]: dly_sel_xxx_DCC_16b bit[4]: ctrlC_xxx_DCC_16b bit[3]: ctrl2_xxx_DCC_16b
[2:0]	RW	dxctl_dq_dcc	bit[2]: dly_sel_xxx_DCC_16b bit[1]: ctrlC_xxx_DCC_16b bit[0]: ctrl2_xxx_DCC_16b

4.2 Flash Memory Controller

4.2.1 Overview

The flash memory controller (FMC) provides memory controller interfaces for connecting to external SPI NAND flash or SPI NOR flash to access data.

4.2.2 Features

The FMC has the following features:

- Provides 4 KB+256 bytes on-chip buffer for improving the read speed.
- Supports two CSs that can be connected to two SPI NOR flash memories, two SPI NAND flash memories, or one SPI NAND flash memory and one SPI NOR flash memory.

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- Supports the SPI NOR flash with 3-byte or 4-byte address.
- Support the SPI NAND flash with 2 KB or 4 KB page size.
- Supports five types of SPIs, including the standard SPI, dual-output/dual-input SPI, quad-output/quad-input SPI, dual I/O SPI, and quad I/O SPI.
- Boots directly from the SPI NOR flash or SPI NAND flash and supports at most 1 MB boot space when booting from CS0.
- Supports the function of skipping bad blocks during booting from the SPI NAND flash. The FMC automatically searches for good blocks for booting.
- Supports 1-wire and 4-wire boot modes for the SPI NAND flash and only 1-wire boot mode for the SPI NOR flash
- Supports the adaptive boot mode for the SPI NAND flash. The configurations of ecc_type and the page size are automatically configured by logic.
- Supports 8 bits/1 KB and 24 bits/1 KB BCH code error correcting code (ECC) check and error correction for the SPI NAND flash.
- Supports the query mode and interrupt report mode and reports the read/write interrupt, erase interrupt, programming completion interrupt, DMA programming timeout failure interrupt, and ECC check error interrupt.
- Supports the read and write operations in internal DMA mode for the SPI NAND flash. You can write only the entire page and read the entire page or the control information (read only the OOB).
- Supports data transfer in internal DMA mode for the SPI NOR flash. The length of the data transferred each time is configurable.
- Supports the read and write operations in transparent transfer (ECC0) mode. That is, data is directly transferred with unprocessed data storage structure.
- Supports the low-power mode. The unused modules can be disabled.

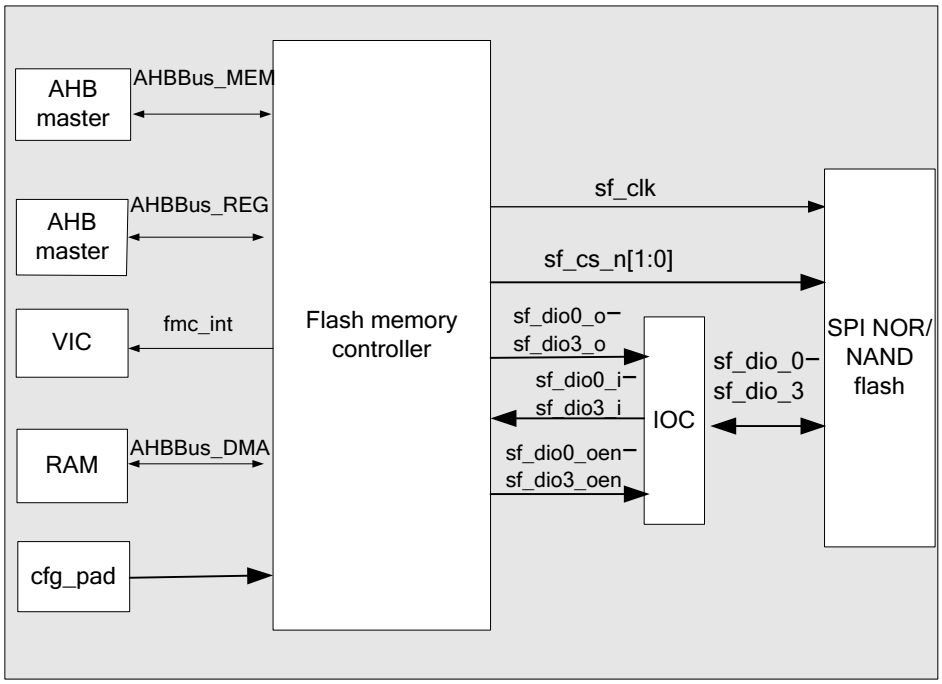
4.2.3 Function Description

4.2.3.1 Interface Block Diagram

The Hi3518E V20X or Hi3516C V200 provides two CSs that can connect to the SPI NOR flash or SPI NAND flash. Figure 4-4 shows the block diagram of FMC interfaces.

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Figure 4-4 Block diagram of FMC interfaces



In Figure 4-4, the pin names such as sf_clk indicate the names of pins in the FMC. Table 4-8 describes the mapping between the names of FMC pins and the names of external pins. In the following sections, the pin names related to interface description indicate the names of FMC pins.

Table 4-8 Mapping between the names of FMC pins and the names of chip pins

FMC Pin Name	Chip Pin Name
sf_clk	SFC_CLK
sf_dio_0	SFC_DIO
sf_dio_1	SFC_DOI
sf_dio_2	SFC_WP_IO2
sf_dio_3	SFC_HOLD_IO3
sf_cs_n[0]	SFC_CS0N
sf_cs_n[1]	SFC_CS1N

4.2.3.2 Interfaces

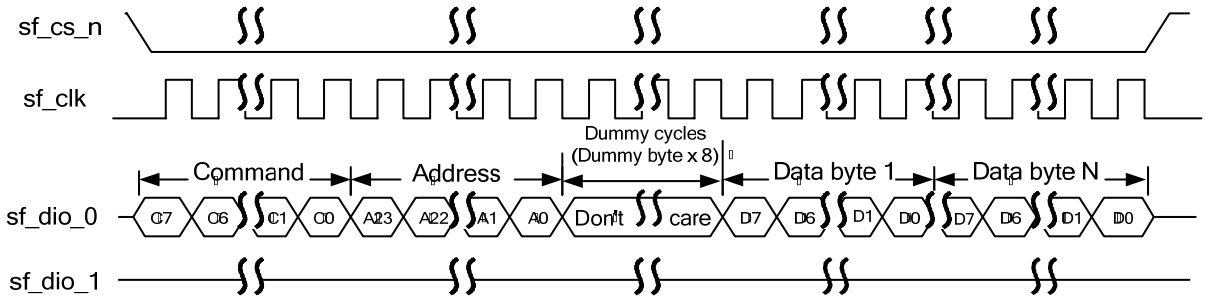
The FMC supports five types of SPIs, including the standard SPI, dual-output/dual-input SPI, quad-output/quad-input SPI, dual I/O SPI, and quad I/O SPI.

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Standard SPI Mode

The standard SPI is connected to a 1-bit data input line and a 1-bit data output line. Figure 4-5 shows the write timing of the standard SPI.

Figure 4-5 Write timing of the standard SPI

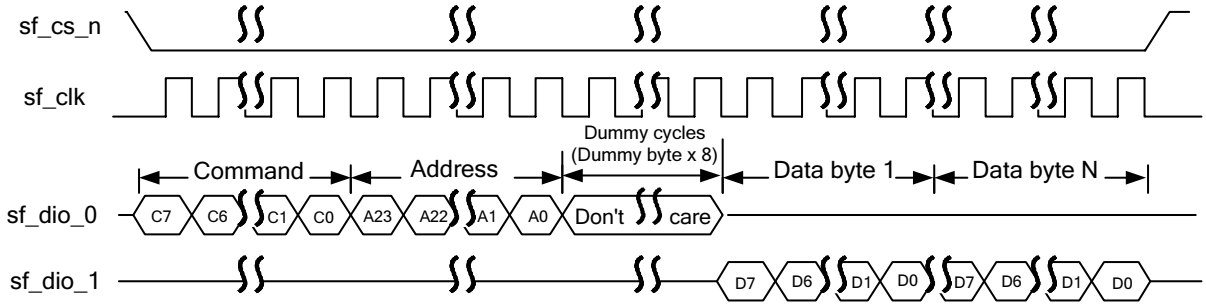


Note the following:

- The command cycles, address cycles, and dummy cycles are output in the unit of a single bit in serial mode through the sf_dio_0 line.
- The data bytes are output in the unit of a single bit in serial mode through the sf_dio_0 line.

Figure 4-6 shows the read timing of the standard SPI.

Figure 4-6 Read timing of the standard SPI



Note the following:

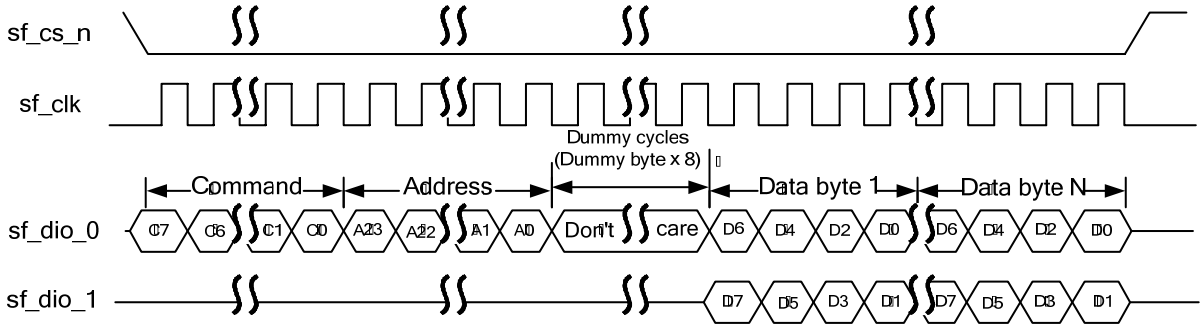
- The command cycles, address cycles, and dummy cycles are output in the unit of a single bit in serial mode through the sf_dio_0 line.
- The data bytes are input in the unit of a single bit in serial mode through the sf_dio_1 line.

Dual-Output/Dual-Input SPI

Figure 4-7 shows the timing of the dual-output/dual-input SPI.

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Figure 4-7 Timing of the dual-output/dual-input SPI



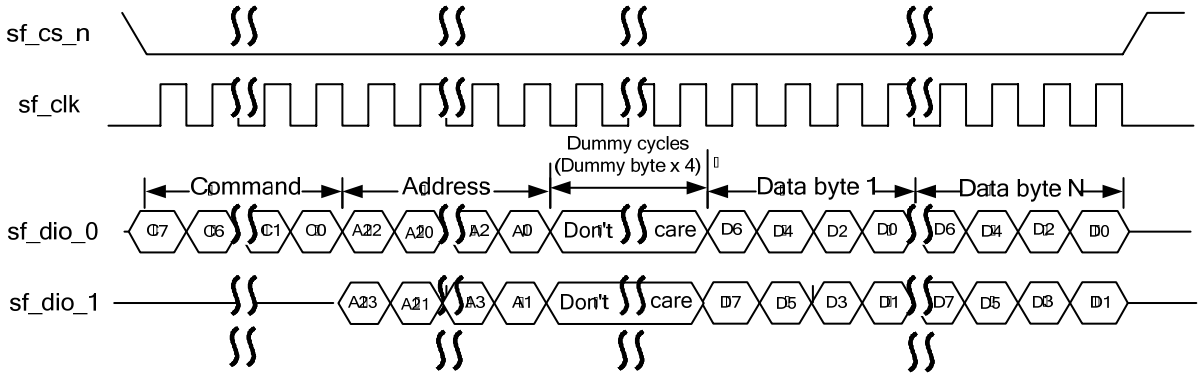
Note the following:

- The command cycles, address cycles, and dummy cycles are output in the unit of a single bit in serial mode through the sf_dio_0 line.
- The data bytes are output (written) or input (read) in the unit of two bits through the sf_dio_0 or sf_dio_1 line.

Dual I/O SPI

Figure 4-8 shows the timing of the dual I/O SPI.

Figure 4-8 Timing of the dual I/O SPI



Note the following:

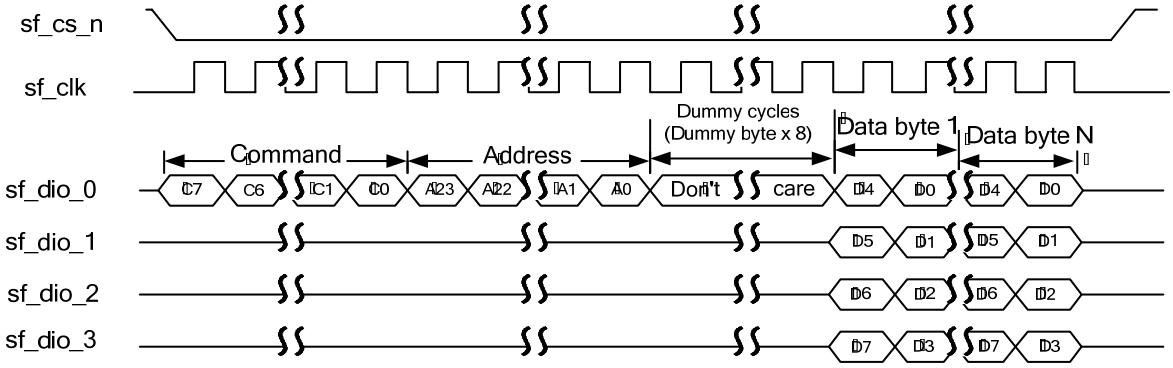
- The command cycles are output in the unit of a single bit in serial mode through the sf_dio_0 line.
- The address cycles and dummy cycles are output in the unit of two bits in serial mode through the sf_dio_0 or sf_dio_1 line.
- The data bytes are output (written) or input (read) in the unit of two bits through the sf_dio_0 or sf_dio_1 line.

Quad-Output/Quad-Input SPI

Figure 4-9 shows the timing of the quad-output/quad-input SPI.

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Figure 4-9 Timing of the quad-output/quad-input SPI



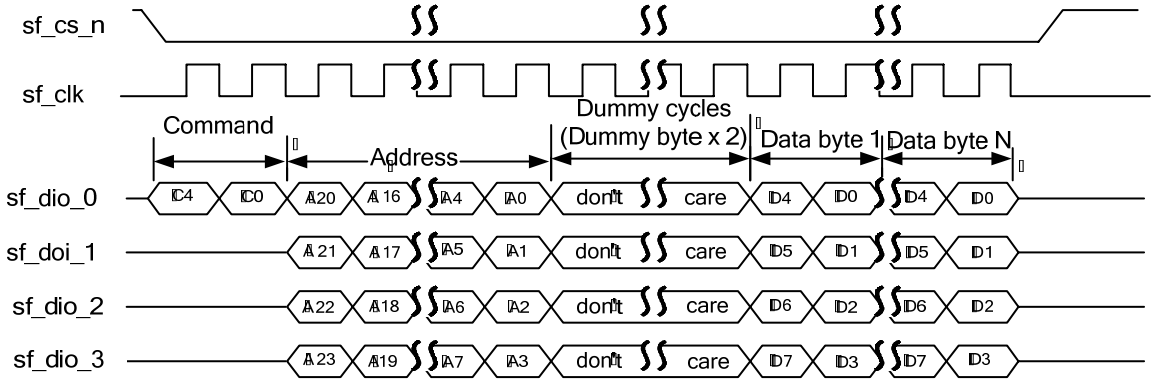
Note the following:

- The command cycles, address cycles, and dummy cycles are output in the unit of a single bit in serial mode through the sf_dio_0 line.
- The data bytes are output (written) or input (read) in the unit of four bits through the sf_dio_0, sf_dio_1, sf_dio_2, or sf_dio_3 line.

Quad I/O SPI

Figure 4-10 shows the timing of the quad I/O SPI.

Figure 4-10 Timing of the quad I/O SPI



Note the following:

- The command cycles are output in the unit of a single bit in serial mode through the sf_dio_0 line.
- The address cycles and dummy cycles are output in the unit of four bits in serial mode through the sf_dio_0, sf_dio_1, sf_dio_2, or sf_dio_3 line.
- The data bytes are output (written) or input (read) in the unit of four bits through the sf_dio_0, sf_dio_1, sf_dio_2, or sf_dio_3 line.

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4.2.3.3 Boot Function

The FMC is in boot mode by default. The chip can boot by reading data directly from the flash memory. The CPU can directly read the data stored in the address ranging from 0x00_0000 to 0x0F_FFFF. The size of the entire address space is 1 MB.

For the SPI NAND flash, physical block 0 must be a good block. For other physical blocks, the bad blocks are skipped and the good blocks are read during the boot process. If there are five consecutive bad blocks, the boot fails. In other cases, the boot is successful even if there are bad blocks.

The chip boots and automatically adapts to the page size and required ECC of the SPI NAND flash based on ECC_TYPE and PAGE_SIZE.

4.2.3.4 Operations in Register Mode

The registers related to the operation command and address are configured by using software. Then the corresponding command is issued by configuring the FMC_OP register. The FMC issues a command to the flash memory according to the value configured by software. If data needs to be transferred to the flash memory, the internal buffer is used.

Operations such as reading ID, setting feature, and erasing are performed in this mode.

In register mode, all the operations related to the flash memory can be combined, and commands related to the address and data transfer can be issued separately.

4.2.3.5 Operations in Internal DMA Mode

The FMC can perform read and write operations in internal DMA mode for improving the access speed. In internal DMA mode, the FMC can directly access the DDR through the bus.

- DMA write operation: For the SPI NOR flash, data with any length can be transferred from any address of the DDR and written to any address of the flash memory. For the SPI NAND flash, the write operation is performed only by page.
- DMA read operation: For the SPI NOR flash, data with any length can be transferred from any address of the flash memory and written to any address of the DDR. For the SPI NAND flash, the entire page is read or only the OOB is read.
- Only-OOB read operation: When the software requires only software management information such as the bad block flag and empty block flag, only the control information needs to be read. In this case, only the OOB is read in DMA mode.

4.2.3.6 ECC Check

The FMC supports ECC check and error correction for the SPI NAND flash. Two types of ECCs are supported, including 8 bits/1 KB ECC and 24 bits/1 KB ECC. The 8 bits/1 KB ECC indicates that errors occur in at most eight bits of the checked 1 KB data can be corrected.

The OOB information is added to the data with ECC protection, and then the ECC is added. The error correction algorithm runs by the error correction unit (1 KB). For example, the ECC is calculated for (DATA+OOB). If the page size is 2 KB, data in each error correction unit is (DATA+OOB)/2; if the page size is 4 KB, data in each error correction unit is (DATA+OOB)/4.

- The OOB is part of the software management information. For details, see the (BB+CTRL) part in the data structure.
- DATA indicates the real data. DATA indicates 2048-byte data when the page size is 2 KB and 4096-byte data when the page size is 4 KB.

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During the boot process, only the 8-bit or 24-bit ECC is supported, while the flash memory requires 4 bits/512 bytes or 8 bits/512 bytes ECC (the error correction unit is 512 bytes in the memory manual).

When the maximum error correction capability is exceeded, the uncorrectable error interrupt is reported. The FMC supports the alarm interrupt. When the number of error bits during one error correction operation is greater than or equal to the configured error threshold (FMC_ERR_THD), an error alarm interrupt is reported. If error bits occur in one or more error correction units and the number of error bits in each error correction unit is less than the error threshold and uncorrectable error value, the correctable error interrupt flag FMC_INT[err_val_int] is changed to 1.

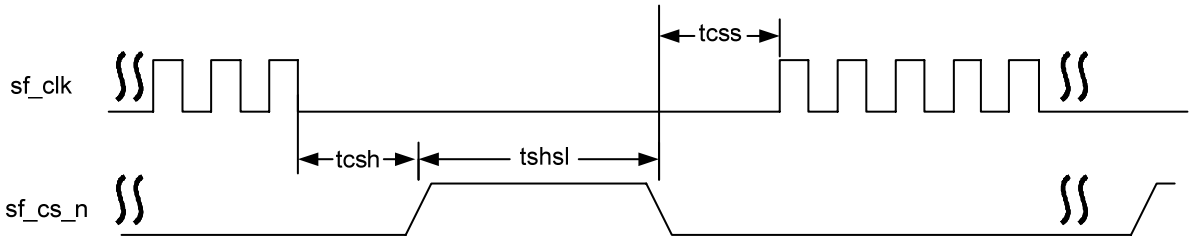
When the data on a page is read, if uncorrectable errors occur in one error correction unit, the uncorrectable error interrupt is reported. If an error alarm interrupt is reported for an error correction unit, the error correction alarm status is reported. If the number of error correction bits is less than the error threshold, the correctable error interrupt status is reported.

The FMC does not support ECC check for the SPI NOR flash. If FMC_CFG[ecc_type] is 0, the SPI NAND flash works in non-ECC check mode and the FMC transfers data directly from the flash without data structure processing.

4.2.3.7 SPI Timings

Figure 4-11 shows the SPI timings and related parameters.

Figure 4-11 SPI output timing



NOTE

- The timings are configured in the TIMING_SPI_CFG register.
- tcsh indicates the CS hold time.
- tcss indicates the CS setup time.
- tshsl indicates the CS deselect time.

4.2.3.8 SPI NAND Flash Address

Table 4-9 describes the address allocation of the SPI NAND flash. The first and second bytes indicate the column address, whereas the third, fourth, and fifth bytes indicate the row address.

Table 4-9 Address allocation of the SPI NAND flash

Byte	Bit 0	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
1st byte	A0	A1	A2	A3	A4	A5	A6	A7
2nd byte	A8	A9	A10	A11	A12*	A13*	0	0

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Byte	Bit 0	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
3rd byte	A12	A13	A14	A15	A16	A17	A18	A19
4th byte	A20	A21	A22	A23	A24	A25	A26	A27
5th byte*	A28*	A29*	0	0	0	0	0	0

NOTE

- Bits A0−A11 are configured as the valid column address when the page size is 2 KB. Bits A0−A12 are configured as the valid column address when the page size is 4 KB.
- Bit A12 (2 KB page size) and bit A13 (4 KB page size) indicate the plane address only for Micron flash memories. Other vendors do not have the concept of plane address.
- Whether bits A28 and A29 are required depends on the flash memory. If the two bits are not used, the outputs are 0.

When the read and write commands of the SPI NAND flash are issued, the column and row addresses are configured based on the following operations:

- For the write operation, the column address is configured during the loading process and the row address is configured during the programming process.
- For the read operation, the row address is configured when the page is read to cache and the column address is configured during the read operation.

In internal DMA mode, the address command is issued by the FMC. FMC_ADDRL and FMC_ADDRH are configured by software based on the operation address. The configuration values of FMC_ADDRL and FMC_ADDRH are 1st byte−4th byte and 5th byte respectively.

NOTE

- If the page size is 2 KB, bit A12 indicates the plane address. Other meanings of bit A12 are not supported; otherwise, the results of the read and write operations will be affected.
- If the page size is 4 KB, bit A13 indicates the plane address. Other meanings of bit A13 are not supported; otherwise, the results of the read and write operations will be affected.

4.2.4 Working Process

4.2.4.1 Initialization

The initialization process is as follows:

Step 1 Configure the TIMING_SPI_CFG register according to the flash memory if the timing parameters need to be configured.

Step 2 Configure the interface type, ecc_type and page size of the flash memory in the FMC configuration register (FMC_CFG) according to the manual of the connected flash.

Step 3 Switch the address mode of the SPI NOR flash to the 4-byte address mode if the default address mode is 3-byte address mode. For details, see section 4.2.4.4 "Process of Changing the Address Mode of the SPI NOR Flash."

----End

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4.2.4.2 Process of Reading or Configuring Component Registers

To read the IDs of the component registers or configure component registers by configuring FMC_OP, perform the following steps:

Step 1 Set ecc_type of the FMC_CFG register to 0.

Step 2 Write the expected operation data from the memory access start address for a register write operation (for example, configure the flash memory configuration register).

Step 3 Configure the operation command, operation address, and number of data segments to be read or written as required in FMC_CMD, FMC_ADDRL, and FMC_DATA_NUM respectively.

Step 4 Configure FMC_OP_CFG based on the component commands that are issued by configuring FMC_OP.

Step 5 Configure FMC_OP to issue commands. For details about the configuration value, see the description of FMC_OP.

Step 6 Check whether the operation is complete by querying of FMC_OP bit[0] in query mode and FMC_INT[op_done_init] in interrupt mode. If FMC_OP bit[0] or FMC_INT[op_done_init] is 1, the operation is complete.

Step 7 Read the value of the component register from the buffer that stores the read results in step 6.

----End

4.2.4.3 Process of Reading the Component Status

To read the component status, perform the following steps:

Step 1 Set FMC_OP[read_status_en] and FMC_OP[reg_op_start] to 1 to issue the component status read command.

Step 2 Store the read result in FMC_STATUS.

----End

4.2.4.4 Process of Changing the Address Mode of the SPI NOR Flash

The SPI NOR flash supports the 3-byte and 4-byte address modes. You can specify the default address mode by pulling up or pulling down the corresponding pin, and dynamically change the address mode by configuring registers after the chip boots.

If the default address mode is the 3-byte address mode and the address mode of the flash memory is 4-byte address mode, perform the following steps to change the address mode after the chip boots:

Step 1 Ensure that the operations on the SPI NOR flash are complete.

Step 2 Configure the related registers to issue the command for changing the address mode of the flash memory to 4-byte address mode in register mode according to the flash memory requirements.

Step 3 Set FMC_CFG[spi_nor_addr_mode] to 1 to change the address mode of the SPI NOR flash from the 3-byte address mode to 4-byte address mode.

----End

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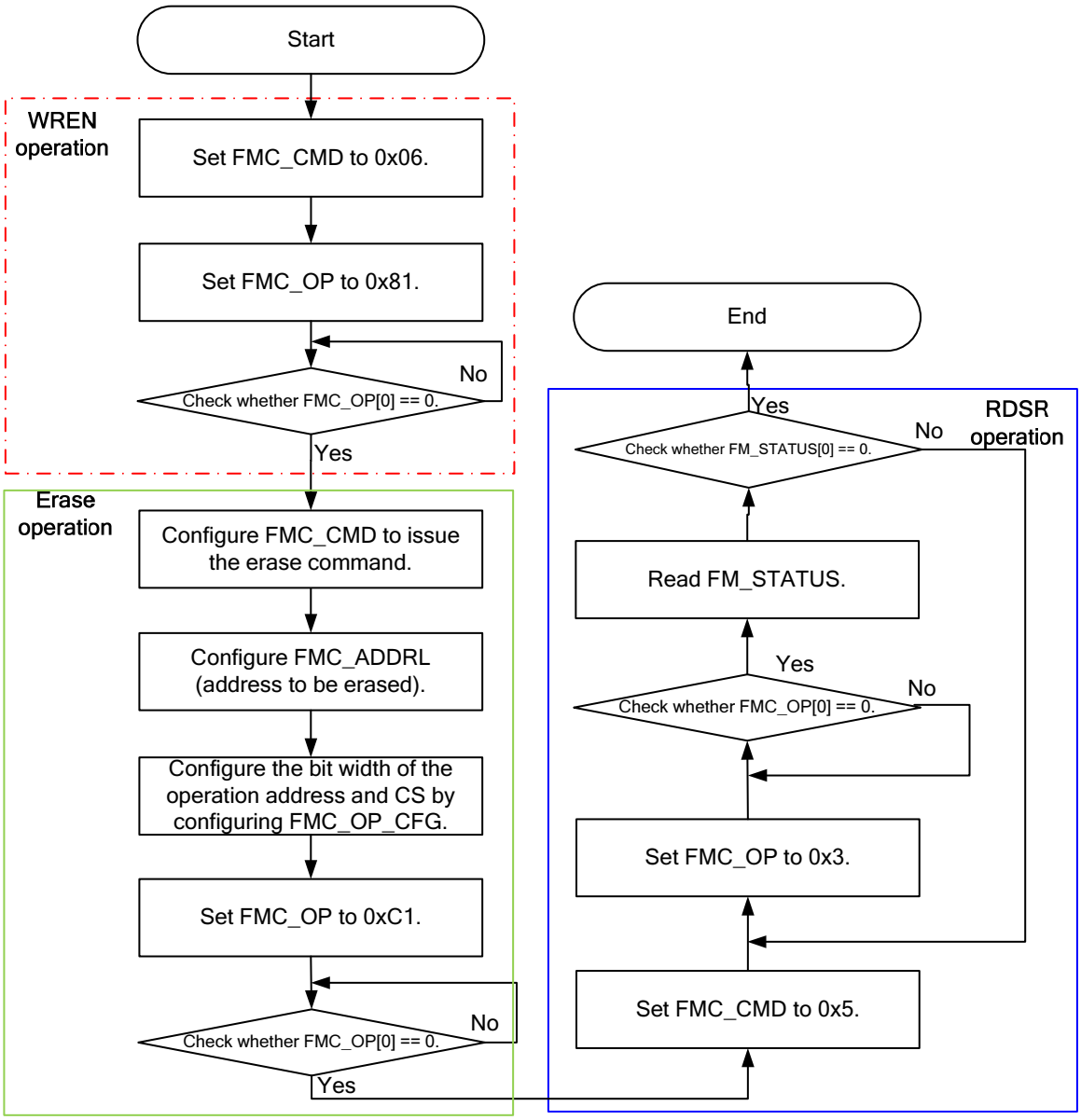
NOTE

For details about the command for changing the address mode of the SPI NOR flash, see the related flash manual.

4.2.4.5 Erase Operation Process

The data in the flash memory must be erased before the program operation is performed. Note that the write enable (WREN) operation must be performed before the erase operation. Figure 4-12 shows an example of erase operation process in query mode.

Figure 4-12 Example of erase operation process



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NOTE

- Figure 4-12 shows the erase operation process in query mode. If the interrupt mode is used, check whether FMC_INT[op_done_init] is 1; if yes, the operation is complete.
- During the erase operation, configure the erase instruction and operation address according to the manual of the corresponding flash.

4.2.4.6 Process of Reading Data in Internal DMA Mode

To read data in internal DMA mode, perform the following steps:

Step 1 Start a DMA read operation if FMC_OP_CTRL bit[0] is 0 and the DMA read operation is required.

Step 2 Configure the operation address of the flash memory by configuring FMC_ADDRL and FMC_ADDRH. For the SPI NOR flash, only FMC_ADDRL needs to be configured.

Step 3 Configure the start address for storing data in the DDR by configuring FMC_DMA_SADDR_D0 and FMC_DMA_SADDR_OOB.

- For the SPI NOR flash, FMC_DMA_SADDR_OOB does not need to be configured.
- For the SPI NAND flash, if only the OOB is read, only FMC_DMA_SADDR_OOB needs to be configured.

Step 4 Configure the length of data to be read for the SPI NOR flash and the length of the spare area data to be transferred for the SPI NAND flash in ECC0 mode by configuring FMC_DMA_LEN. Skip this step for other operations on the SPI NAND flash.

Step 5 Configure FMC_OP_CFG based on the requirements of the issued read command.

- Configure FMC_OP_CFG[dummy_num] based on the number of dummy cycles in the read timing.
- Configure FMC_OP_CFG[mem_if_type] based on the SPI required for the read operation.
- Configure FMC_OP_CFG[fm_cs] to select the CS to be operated.

Step 6 Set bit[0] of FMC_OP_CTRL to 1 to issue the flash read command.

- Set FMC_OP_CTRL[rw_op] to 0 to select the DMA read operation.
- For the SPI NAND flash, if only the OOB is read, FMC_OP_CTRL[rd_op_sel] needs to be configured.
- Configure FMC_OP_CTRL[rd_opcode] based on the instruction of the flash read operation.

Step 7 Check whether the read operation is complete by querying FMC_OP_CTRL bit[0] in query mode and FMC_INT[op_done_int] in interrupt mode. If FMC_OP_CTRL bit[0] or FMC_INT[op_done_int] is 1, the read operation is complete, and data is written to the DDR.

----End

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NOTE

- For the SPI NAND flash in ECC0 mode, the length configured in FMC_DMA_LEN must be 4-byte-aligned.
- For the SPI NAND flash, the start address for storing data in the DDR must be 4-byte-aligned.

4.2.4.7 Process of Writing Data in Internal DMA Mode

To write data in internal DMA mode, perform the following steps:

Step 1 Start a DMA write operation if FMC_OP_CTRL bit[0] is 0 and the DMA write operation is required.

Step 2 Configure the operation address of the flash memory by configuring FMC_ADDRL and FMC_ADDRH. For the SPI NOR flash, only FMC_ADDRL needs to be configured.

Step 3 Configure the start address for transferring data from the DDR by configuring FMC_DMA_SADDR_D0 and FMC_DMA_SADDR_OOB. For the SPI NOR flash, FMC_DMA_SADDR_OOB does not need to be configured.

Step 4 Configure the length of data to be transferred for the SPI NOR flash and the length of the spare area data to be transferred for the SPI NAND flash in ECC0 mode by configuring FMC_DMA_LEN. Skip this step for other operations on the SPI NAND flash.

Step 5 Configure FMC_OP_CFG based on the requirements of the issued write command.

- Configure FMC_OP_CFG[mem_if_type] based on the SPI required for the write operation.
- Configure FMC_OP_CFG[fm_cs] to select the CS to be operated.

Step 6 Configure FMC_OP_CTRL to issue corresponding commands.

- Set FMC_OP_CTRL[rw_op] to 1 to select the DMA write operation.
- Configure FMC_OP_CTRL[wr_opcode] based on the instruction of the flash program operation.

Step 7 Check whether the write operation is complete by querying FMC_OP_CTRL bit[0] in query mode and FMC_INT[op_done_int] in interrupt mode. If FMC_OP_CTRL bit[0] is 0 or FMC_INT[op_done_int] is 1, the write operation is complete, and data is written to the flash memory.

----End

NOTE

- For the SPI NAND flash in ECC0 mode, the length configured in FMC_DMA_LEN must be 4-byte-aligned.
- For the SPI NAND flash, the start address for storing data in the DDR must be 4-byte-aligned.

4.2.4.8 Notes

Note the following:

- You must reset the SPI NAND flash before use or after exceptions occur.
- You are advised not to configure registers when FMC_OP_CTRL[dma_op_ready] or FMC_OP[reg_op_start] is 1, indicating that the controller is performing an operation. Otherwise, the operation may become abnormal.

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4.2.5 Data Storage Structure

The data storage structure of the SPI NAND flash is different from that of the driver software because the ECC needs to be inserted and the bad block flag must be in the default position specified by the vendor. The following describes the format of data prepared by software and the data storage structure in the components.

Non-ECC Mode

When the FMC does not need to perform the ECC:

- During write operations, the FMC transparently writes the data in buffers to the SPI NAND flash without processing.
- During read operations, the FMC writes data read from the SPI NAND flash to the internal buffers without processing.
- The number of bytes of the read and written data is set by the FMC_DATA_NUM register.
- The non-ECC mode is used when the ID is read or features are configured and obtained.

4.2.5.2 Data Storage Structure of the Driver Software

2 KB Page Size

Figure 4-13 shows the data structure of the driver (2 KB page size).

Figure 4-13 Data structure of the driver (2 KB page size)



BB: bad block, 2 bytes
CTRL: 30-byte control area for software

4 KB Page Size

Figure 4-14 shows the data structure of the driver (4 KB page size).

Figure 4-14 Data structure of the driver (4 KB page size)



BB: bad block, 2 bytes
CTRL: 30-byte control area for software

4.2.5.3 8-Bit ECC Mode

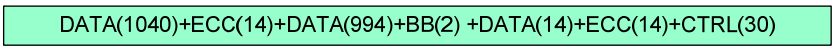
2 KB Page Size

Figure 4-15 shows the data structure of the SPI NAND flash in 8-bit ECC mode (2 KB page size). The software valid data is divided into two 1040-byte data segments, and the ECC is calculated for each data segment. When being written to the SPI NAND flash, the data is

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automatically stored in two groups in the alternate form of 1040-byte data+14-byte ECC. The bad block information is stored in the first two bytes of the SPI NAND flash spare area.

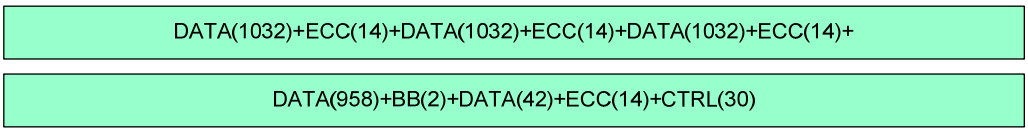
Figure 4-15 Data structure of the SPI NAND flash in 8-bit ECC mode (2 KB page size)



4 KB Page Size

Figure 4-16 shows the data structure of the SPI NAND flash in 8-bit ECC mode (4 KB page size). The software valid data is divided into four 1032-byte data segments, and the ECC is calculated for each data segment. Data written to the SPI NAND flash is automatically stored in three groups in the alternate form of 1032-byte data+14-byte ECC. For the last group (1032-byte data+14-byte ECC), the bad block flag is inserted in byte 4096 of the SPI NAND flash.

Figure 4-16 Data structure of the SPI NAND flash in 8-bit ECC mode (4 KB page size)

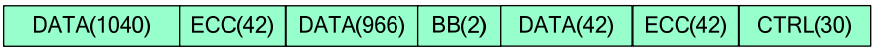


4.2.5.4 24-Bit ECC Mode

2 KB Page Size

Figure 4-17 shows the data structure of the SPI NAND flash in 24-bit ECC mode (2 KB page size). The bad block flag is inserted in byte 2048 of the SPI NAND flash.

Figure 4-17 Data structure of the SPI NAND flash in 24-bit ECC mode (2 KB page size)

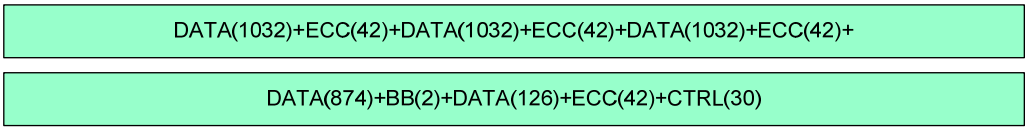


4 KB Page Size

Figure 4-18 shows the data structure of the SPI NAND flash in 24-bit ECC mode (4 KB page size). The data is divided into four 1024 bytes+8 bytes data segments, and a 42-byte ECC is calculated for each data segment. When being written to the SPI NAND flash, data and ECC in the first three data segments are stored in the alternative form. For the last data segment (1024 bytes+8 bytes+42 bytes), the bad block flag is inserted in byte 4096 of the SPI NAND flash.

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Figure 4-18 Data structure of the SPI NAND flash in 24-bit ECC mode (4 KB page size)



4.2.6 ECC Mode Selection

The ECC IP used by the controller processes data based on 1 KB data block. Therefore, the ECC performance of the controller is described in the format of *n* bits/1 KB, for example, 8 bits/1 KB or 24 bits/1 KB. To ensure component reliability, the recommended ECC performance of 1 bit/512 bytes, 4 bits/512 bytes, or 8 bits/512 bytes is provided in the component data sheet. The ECC performance of 8 bits/1 KB is equivalent to that of 4 bits/512 bytes. When you select the ECC mode for the controller, note the following:

- The priority of the ECC performance of the controller is higher than or equal to that of the recommended ECC performance.

  For example, if the recommended ECC performance is 1 bit/512 bytes or 4 bits/512 bytes, the controller can select the ECC mode with the performance of 8 bits/1 KB.
- The component page size must be greater than or equal to the required storage size of the ECC mode for the controller.

  For example, if the component page size is (2 KB + 64 bytes), the controller cannot select the ECC mode with the performance of 24 bits/1 KB because the required storage size of this ECC mode is (2 KB + 116 bytes). For details about the required storage size for the ECC mode, see section 4.2.5 "Data Storage Structure."

4.2.7 FMC Registers

4.2.7.1 Register Summary

Table 4-10 describes FMC registers.

Table 4-10 Summary of FMC registers (base address of 0x1001_0000)

Offset Address	Register	Description	Page
0x0000	FMC_CFG	Component configuration register	4-205
0x0004	GLOBAL_CFG	Global configuration register	4-206
0x0008	TIMING_SPI_CFG	SPI timing configuration register	4-206
0x0018	FMC_INT	Interrupt status register	4-207
0x001C	FMC_INT_EN	Interrupt enable register	4-208
0x0020	FMC_INT_CLR	Interrupt clear register	4-209
0x0024	FMC_CMD	Command word configuration register	4-210
0x0028	FMC_ADDRH	Upper-byte component address configuration register	4-211

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Offset Address	Register	Description	Page
0x002C	FMC_ADDRL	Lower-byte component address configuration register	4-211
0x0030	FMC_OP_CFG	Operation configuration register	4-211
0x0038	FMC_DATA_NUM	Data length register	4-212
0x003C	FMC_OP	Operation register	4-213
0x0040	FMC_DMA_LEN	DMA operation length register	4-214
0x0048	FMC_DMA_AHB_CTRL	DMA AHB bus control register	4-215
0x004C	FMC_DMA_SADDR_D0	DDR start address register 0 for DMA operations	4-215
0x005C	FMC_DMA_SADDR_OOB	DDR OOB information storage start address register for DMA operations	4-216
0x0068	FMC_OP_CTRL	DMA operation control register	4-216
0x006C	FMC_TIMEOUT_WR	Write operation timeout register	4-217
0x0070	FMC_OP_PARA	Data segment ID register for reading/writing to the SPI NAND flash in DMA mode	4-217
0x0074	FMC_BOOT_SET	Boot setting register	4-218
0x0078	FMC_LP_CTRL	Low-power control register	4-219
0x00A8	FMC_ERR_THD	ECC alarm threshold register	4-219
0x00AC	FMC_STATUS	Component operating status register	4-220
0x00B4	FMC_MEM_CTRL	SRAM pin configuration register	4-220
0x00BC	VERSION	Version register	4-221
0x00C0	FMC_ERR_NUM0_BUF0	SPI NAND flash error correction information 0 statistics register for the first buffer operation	4-221
0x00D0	FMC_ERR_ALARM_ADDRH	Upper-byte ECC alarm flash address register	4-222
0x00D4	FMC_ERR_ALARM_ADDRL	Lower-byte ECC alarm flash address register	4-222
0x00D8	FMC_ECC_INVALID_ADDRH	Upper-byte ECC uncorrectable address register	4-223
0x00DC	FMC_ECC_INVALID_ADDRL	Lower-byte ECC uncorrectable address register	4-223

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4.2.7.2 Register Description

FMC_CFG

FMC_CFG is a component configuration register.

Offset Address	Register Name	Total Reset Value
0x0000	FMC_CFG	0x0000_1820

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					spi_nor_addr_mode	block_size		ecc_type			page_size		flash_sel		op_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10]	RW	spi_nor_addr_mode	SPI address mode (valid only for the SPI NOR flash) 0 (default): 3-byte address mode 1: 4-byte address mode
[9:8]	RW	block_size	Block size of the SPI NAND flash 00: 64 pages Other values: reserved
[7:5]	RW	ecc_type	ECC type of the controller 000: no ECC 001: 8-bit ECC 011: 24-bit ECC Other values: reserved
[4:3]	RW	page_size	Page size of the SPI NAND flash 00: 2 KB page size 01: 4 KB page size Other values: reserved
[2:1]	RW	flash_sel	Flash type select 00: SPI NOR flash 01: SPI NAND flash Other values: reserved The reset value is determined by the pin.

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[0]	RW	op_mode	FMC operation mode 0: boot mode 1: normal mode

GLOBAL_CFG

GLOBAL_CFG is a global configuration register.

Offset Address	Register Name	Total Reset Value
0x0004	GLOBAL_CFG	0x0000_0040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									wp_en				reserved				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6]	RW	wp_en	Write protection enable for the WP pin. When this bit is enabled, the chip outputs 0 to the WP pin. 0: disabled 1: enabled
[5:0]	RO	reserved	Reserved

TIMING_SPI_CFG

TIMING_SPI_CFG is an SPI timing configuration register.

Offset Address	Register Name	Total Reset Value
0x0008	TIMING_SPI_CFG	0x0000_066F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				tcsh				tcss				tshsl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1	0	1	1	1	1

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:8]	RW	tcsh	CS hold time 0x0–0x7: (n + 1) interface clock cycles (n = 0, 1, 2, …, 7)

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[7:4]	RW	tcss	CS setup time 0x0–0x7: (*n* + 1) interface clock cycles (*n* = 0, 1, 2, …, 7)
[3:0]	RW	tshsl	CS deselect time. It is equal to the interval between two flash operations. 0x0–0xF: (*n* + 1) interface clock cycles (*n* = 0, 1, 2, …, 15)

FMC_INT

FMC_INT is an interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0018	FMC_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								ahb_op_int	reserved	dma_err_int	err_alarm_int	err_inval_int	err_val_int	op_fail_int	op_done_int
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RO	ahb_op_int	CPU read/write internal buffer interrupt enable when the FMC is reading/writing data from/to the flash memory 0: No interrupt is generated. 1: An interrupt is generated.
[6]	RO	reserved	Reserved
[5]	RO	dma_err_int	DMA transfer bus error interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[4]	RO	err_alarm_int	ECC alarm interrupt. An interrupt is generated when the number of error bits reaches the preset threshold. 0: No interrupt is generated. 1: An interrupt is generated.

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[3]	RO	err_inval_int	Uncorrectable ECC error interrupt In 8-bit ECC mode, if errors occur in eight or more bits of the checked 1024-byte data, an interrupt is generated. In 24-bit ECC mode, if errors occur in 24 or more bits of the checked 1024-byte data, an interrupt is generated. 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	err_val_int	Correctable ECC error interrupt In 8-bit ECC mode, if errors occur in one to eight bits of the checked 1024-byte data, an interrupt is generated. In 24-bit ECC mode, if errors occur in one to 24 bits of the checked 1024-byte data, an interrupt is generated. 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	op_fail_int	Programming (timeout) failure interrupt in DMA mode 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	op_done_int	Current controller operation completion interrupt. This bit is automatically cleared when the operation register is written. 0: No interrupt is generated. 1: An interrupt is generated.

FMC_INT_EN

FMC_INT_EN is an interrupt enable register.

Offset Address	Register Name	Total Reset Value
0x001C	FMC_INT_EN	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5	4	3	2	1	0
Name	reserved						ahb_op_int_en	reserved	dma_err_int_en	err_alarm_int_en	err_inval_int_en	err_val_int_en	op_fail_int_en	op_done_int_en
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved

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[7]	RW	ahb_op_int_en	CPU read/write internal buffer error interrupt enable when the FMC is reading/writing data from/to the flash memory 0: disabled 1: enabled
[6]	RW	reserved	Reserved
[5]	RW	dma_err_int_en	DMA transfer bus error interrupt enable 0: disabled 1: enabled
[4]	RW	err_alarm_int_en	ECC alarm interrupt enable. An interrupt is generated when the number of error bits reaches the threshold 0: disabled 1: enabled
[3]	RW	err_inval_int_en	ECC uncorrectable error interrupt enable 0: disabled 1: enabled
[2]	RW	err_val_int_en	ECC correctable error interrupt enable 0: disabled 1: enabled
[1]	RW	op_fail_int_en	Programming operation failure interrupt enable in DMA mode 0: disabled 1: enabled
[0]	RW	op_done_int_en	Current operation completion interrupt enable of the FMC 0: disabled 1: enabled

FMC_INT_CLR

FMC_INT_CLR is an interrupt clear register.

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Offset Address	Register Name	Total Reset Value
0x0020	FMC_INT_CLR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								ahb_op_int_clr	reserved	dma_err_int_clr	err_alarm_int_clr	err_inval_int_clr	err_val_int_clr	op_fail_int_clr	op_done_int_clr
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	WC	ahb_op_int_clr	ahb_op_err interrupt clear. Writing 1 clears the interrupt.
[6]	RO	reserved	Reserved
[5]	WC	dma_err_int_clr	DMA transfer bus error interrupt clear. Writing 1 clears the interrupt.
[4]	WC	err_alarm_int_clr	err_alarm interrupt clear. Writing 1 clears the interrupt.
[3]	WC	err_inval_int_clr	err_invalid interrupt clear. Writing 1 clears the interrupt.
[2]	WC	err_val_int_clr	err_valid interrupt clear. Writing 1 clears the interrupt.
[1]	WC	op_fail_int_clr	op_fail interrupt clear. Writing 1 clears the interrupt.
[0]	WC	op_done_int_clr	op_done interrupt clear. Writing 1 clears the interrupt.

FMC_CMD

FMC_CMD is a command word configuration register.

Offset Address	Register Name	Total Reset Value
0x0024	FMC_CMD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								cmd1							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	cmd1	Command sent to the SPI NOR/SPI NAND flash by the controller

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FMC_ADDRH

FMC_ADDRH is an upper-byte component address configuration register.

Offset Address	Register Name	Total Reset Value
0x0028	FMC_ADDRH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								addrh							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	addrh	Upper-byte of the operation address for the SPI NAND flash

FMC_ADDRL

FMC_ADDRL is a lower-byte component address configuration register.

Offset Address	Register Name	Total Reset Value
0x002C	FMC_ADDRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	addrl																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	addrl	Lower-byte of the operation address for the flash memory, and component address for the SPI NOR flash

FMC_OP_CFG

FMC_OP_CFG is an operation configuration register.

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Offset Address	Register Name	Total Reset Value
0x0030	FMC_OP_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				fm_cs	force_cs_en	mem_if_type			addr_num			dummy_num			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11]	RW	fm_cs	CS corresponding to the flash to be operated used for register operation and invalid for DMA operations 0: CS0 1: CS1
[10]	RW	force_cs_en	CS forcible pull-down enable 0: disabled 1: enabled
[9:7]	RW	mem_if_type	SPI flash interface type select (read operation) 000: standard SPI 001: dual-output/dual-input SPI 010: dual I/O SPI 011: quad-output/quad-input SPI 100: quad I/O SPI 101−111: reserved
[6:4]	RW	addr_num	Number of bytes of the address sent to the flash
[3:0]	RW	dummy_num	Number of bytes to be operated for dummy_en (one byte is equivalent to two clock cycles in 4-wire mode, four clock cycles in 2-wire mode, or eight clock cycles in 1-wire mode)

FMC_DATA_NUM

FMC_DATA_NUM is a data length register.

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Offset Address	Register Name	Total Reset Value
0x0038	FMC_DATA_NUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				op_data_num											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:0]	RW	op_data_num	Length of data to be processed during one operation. This register needs to be configured when there is data transfer and does not need to be configured for the DMA operations. This register is valid for the SPI NAND flash only in ECC0 mode.

FMC_OP

FMC_OP is an operation register.

Offset Address	Register Name	Total Reset Value
0x003C	FMC_OP	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																							dummy_en	cmd1_en	addr_en	write_data_en	reserved			read_data_en	read_status_en	reg_op_start
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:9]	RO	reserved	Reserved
[8]	RW	dummy_en	Dummy byte transfer enable after the address operation is enabled 0: disabled 1: enabled
[7]	RW	cmd1_en	Enable for transmitting command 1 to the flash 0: disabled 1: enabled
[6]	RW	addr_en	Enable for writing the operation address to the flash 0: disabled 1: enabled

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[5]	RW	write_data_en	Enable for writing data to the flash 0: disabled 1: enabled Note: read_data_en and write_data_en cannot be 1 at the same time.
[4:3]	RO	reserved	Reserved
[2]	RW	read_data_en	Enable for reading data from the flash 0: disabled 1: enabled Note: read_data_en and write_data_en cannot be 1 at the same time.
[1]	RW	read_status_en	Read component status register enable 0: disabled 1: enabled When this bit is 1, the value of the FMC read component status register is written in the FMC_STATUS[fm_status] instead of the internal butter.
[0]	RWSC	reg_op_start	Controller status when commands are issued by configuring the FMC_OP register 0: The controller is ready. This bit can be set only to 1 when software issues the command, indicating that the logic is enabled. 1: The controller is busy. This bit is automatically set to 0 after the operation is complete, indicating that the logic is complete.

FMC_DMA_LEN

FMC_DMA_LEN is a DMA operation length register.

Offset Address	Register Name	Total Reset Value
0x0040	FMC_DMA_LEN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				dma_len																											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:0]	RW	dma_len	Data transfer length for DMA operations, in byte. This register is used to configure the spare area length of the SPI NAND flash in ECC0 mode and length of the data to be read from or written to the SPI NOR flash in DMC mode.

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FMC_DMA_AHB_CTRL

FMC_DMA_AHB_CTRL is a DMA AHB bus control register.

Offset Address	Register Name	Total Reset Value
0x0048	FMC_DMA_AHB_CTRL	0x0000_0007

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													burst16_en	burst8_en	burst4_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	burst16_en	Burst16 enable 0: disabled 1: enabled
[1]	RW	burst8_en	Burst8 enable 0: disabled 1: enabled
[0]	RW	burst4_en	Burst4 enable 0: disabled 1: enabled

FMC_DMA_SADDR_D0

FMC_DMA_SADDR_D0 is DDR start address register 0 for DMA operations.

Offset Address	Register Name	Total Reset Value
0x004C	FMC_DMA_SADDR_D0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dma_mem_saddr_d0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dma_mem_saddr_d0	For the SPI NOR flash, this register indicates the DDR start address register for DMA operations. For the SPI NAND flash, this register indicates the DDR operation data base address register for DMA operations.

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FMC_DMA_SADDR_OOB

FMC_DMA_SADDR_OOB is a DDR OOB information storage start address register for DMA operations.

Offset Address	Register Name	Total Reset Value
0x005C	FMC_DMA_SADDR_OOB	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dma_mem_saddr_oob																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dma_mem_saddr_oob	DDR base address for the OOB area that stores the data to be read or written

FMC_OP_CTRL

FMC_OP_CTRL is an DMA operation control register.

Offset Address	Register Name	Total Reset Value
0x0068	FMC_OP_CTRL	0x0003_0200

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								rd_opcode								wr_opcode								reserved				rd_op_sel		reserved		rw_op	dma_op_ready
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	rd_opcode	SPI NAND/SPI NOR flash, DMA read command, non-bus mode (FAST_READ/READ/DUAL_READ)
[15:8]	RW	wr_opcode	SPI NAND/SPI NOR flash, DMA write command
[7:6]	RO	reserved	Reserved
[5:4]	RW	rd_op_sel	Area of data to be read 00: The whole page is read. 01: Only the OOB is read. 1x: reserved
[3:2]	RO	reserved	Reserved

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[1]	RW	rw_op	DMA write/read mode select 0: DMA read mode 1: DMA write mode
[0]	RWSC	dma_op_ready	Controller status when commands are issued by configuring the FMC_OP_CTRL register 0: The controller is ready. This bit can be set only to 1 when software issues the command, indicating that the logic is enabled. 1: The controller is busy. This bit is automatically set to 0 after the operation is complete, indicating that the logic is complete.

FMC_TIMEOUT_WR

FMC_TIMEOUT_WR is a write operation timeout register.

Offset Address	Register Name	Total Reset Value
0x006C	FMC_TIMEOUT_WR	0x00FF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								timeout_wr																							
Reset	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	timeout_wr	Program operation busy wait time, timeout period. The timeout period is in the unit of one interface clock cycle for the SPI NAND/SPI NOR flash.

FMC_OP_PARA

FMC_OP_PARA is a data segment ID register for reading/writing to the SPI NAND flash in DMA mode.

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Offset Address	Register Name	Total Reset Value
0x0070	FMC_OP_PARA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														rd_oob_only	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	rd_oob_only	Only the sector that stores the OOB information is read when the flash read command is issued.
[0]	RO	reserved	Reserved

FMC_BOOT_SET

FMC_BOOT_SET is a boot setting register.

Offset Address	Register Name	Total Reset Value
0x0074	FMC_BOOT_SET	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											bb_err	reserved		boot_quad_mode	boot_page0_cfg
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	RW	bb_err	Boot failure information 0: bootable 1: boot failure
[3:2]	RO	reserved	Reserved

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[1]	RW	boot_quad_mode	Whether to use the 4-wire boot mode for the SPI NAND flash. The reset value is determined by the pin. 0: no. The 1-wire boot mode is used. 1: yes
[0]	RW	boot_page0_cfg	Boot mode select 0: default boot mode 1: boot mode without configuring pins

FMC_LP_CTRL

FMC_LP_CTRL is a low-power control register.

Offset Address	Register Name	Total Reset Value
0x0078	FMC_LP_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															clk_gate_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	clk_gate_en	Clock gating select. If the clock gating is enabled, the low-power design is used and the module working clocks are disabled by logic. 0: All clocks are enabled. 1: Clocks are disabled according to the low-power design.

FMC_ERR_THD

FMC_ERR_THD is an ECC alarm threshold register.

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Offset Address	Register Name	Total Reset Value
0x00A8	FMC_ERR_THD	0x0000_00FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								fmc_err_thd							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	fmc_err_thd	ECC alarm threshold. When the number of error bits reaches the threshold, an ECC alarm interrupt is triggered. **NOTE** • If an uncorrectable ECC error occurs, an uncorrectable interrupt is reported regardless of the register value. • When the register value is set to 0 or 1, an alarm interrupt is reported as long as an error occurs. • When the configured value exceeds the number of correctable error bits, no alarm interrupt is reported regardless of the number of error bits. Only the correctable error interrupt and uncorrectable error interrupt are generated.

FMC_STATUS

FMC_STATUS is a component operating status register.

Offset Address	Register Name	Total Reset Value
0x00AC	FMC_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								fm_status							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RO	fm_status	Component operating status

FMC_MEM_CTRL

FMC_MEM_CTRL is an SRAM pin configuration register.

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Offset Address	Register Name	Total Reset Value
0x00B4	FMC_MEM_CTRL	0x0000_5222

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																reserved	tp_mem_ret1n	tp_mem_emab		tp_mem_emaa				reserved		sp_mem_ret1n	sp_mem_emaw			sp_mem_ema			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	0	0	0	1	0	0	0	1	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RW	reserved	Reserved and configurable
[14]	RW	tp_mem_ret1n	RET1N pin configuration of the dual-port SRAM
[13:11]	RW	tp_mem_emab	EMAB pin configuration of the dual-port SRAM
[10:8]	RW	tp_mem_emaa	EMAA pin configuration of the dual-port SRAM
[7:6]	RW	reserved	Reserved and configurable
[5]	RW	sp_mem_ret1n	RET1N pin configuration of the single-port SRAM
[4:3]	RW	sp_mem_emaw	EMAW pin configuration of the single-port SRAM
[2:0]	RW	sp_mem_ema	EMA pin configuration of the single-port SRAM

VERSION

VERSION is a version register.

Offset Address	Register Name	Total Reset Value
0x00BC	VERSION	0x0000_0100

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	version																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	version	FMC V100

FMC_ERR_NUM0_BUF0

FMC_ERR_NUM0_BUF0 is an SPI NAND flash error correction information 0 statistics register for the first buffer operation.

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Offset Address	Register Name	Total Reset Value
0x00C0	FMC_ERR_NUM0_BUF0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	err_num0_buf0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	err_num0_buf0	Number of error bits in the 4 KB data when the page size is 2 KB bit[31:24]: number of error bits in the fourth 1 KB data bit[23:16]: number of error bits in the third 1 KB data bit[15:8]: number of error bits in the second 1 KB data bit[7:0]: number of error bits in the first 1 KB data

FMC_ERR_ALARM_ADDRH

FMC_ERR_ALARM_ADDRH is an upper-byte ECC alarm flash address register.

Offset Address	Register Name	Total Reset Value
0x00D0	FMC_ERR_ALARM_ADDRH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								fmc_err_alarm_addrh							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RO	fmc_err_alarm_addrh	Upper bytes of the operation address of the flash that triggers the last interrupt when an ECC alarm interrupt is generated

FMC_ERR_ALARM_ADDRL

FMC_ERR_ALARM_ADDRL is a lower-byte ECC alarm flash address register.

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Offset Address	Register Name	Total Reset Value
0x00D4	FMC_ERR_ALARM_ADDRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fmc_err_alarm_addrl																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	fmc_err_alarm_addrl	Lower bytes of the operation address of the flash that triggers the last interrupt when an ECC alarm interrupt is generated

FMC_ECC_INVALID_ADDRH

FMC_ECC_INVALID_ADDRH is an upper-byte ECC uncorrectable address register.

Offset Address	Register Name	Total Reset Value
0x00D8	FMC_ECC_INVALID_ADDRH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								fmc_ecc_invalid_addrh							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RO	fmc_ecc_invalid_addrh	Upper bytes of the operation address of the flash that triggers the last interrupt when an uncorrectable ECC error occurs

FMC_ECC_INVALID_ADDRL

FMC_ECC_INVALID_ADDRL is a lower-byte ECC uncorrectable address register.

Offset Address	Register Name	Total Reset Value
0x00DC	FMC_ECC_INVALID_ADDRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fmc_ecc_invalid_addrl																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	fmc_ecc_invalid_addrl	Lower bytes of the operation address of the flash that triggers the last interrupt when an uncorrectable ECC error occurs

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5 ETH

5.1 Overview

The Ethernet (ETH) module provides an ETH module interface that is used to receive data or transmit data through the network interface at a speed of 10 Mbit/s or 100 Mbit/s. This module also supports half-duplex or full-duplex operating mode and provides the reduced media-independent interface (RMII). With the eight configurable DMAC address filter tables, the ETH module filters input frames received through the network interface, limiting the traffic of the CPU port to protect the CPU against heavy traffic.

5.2 Function Description

The ETH module has the following features:

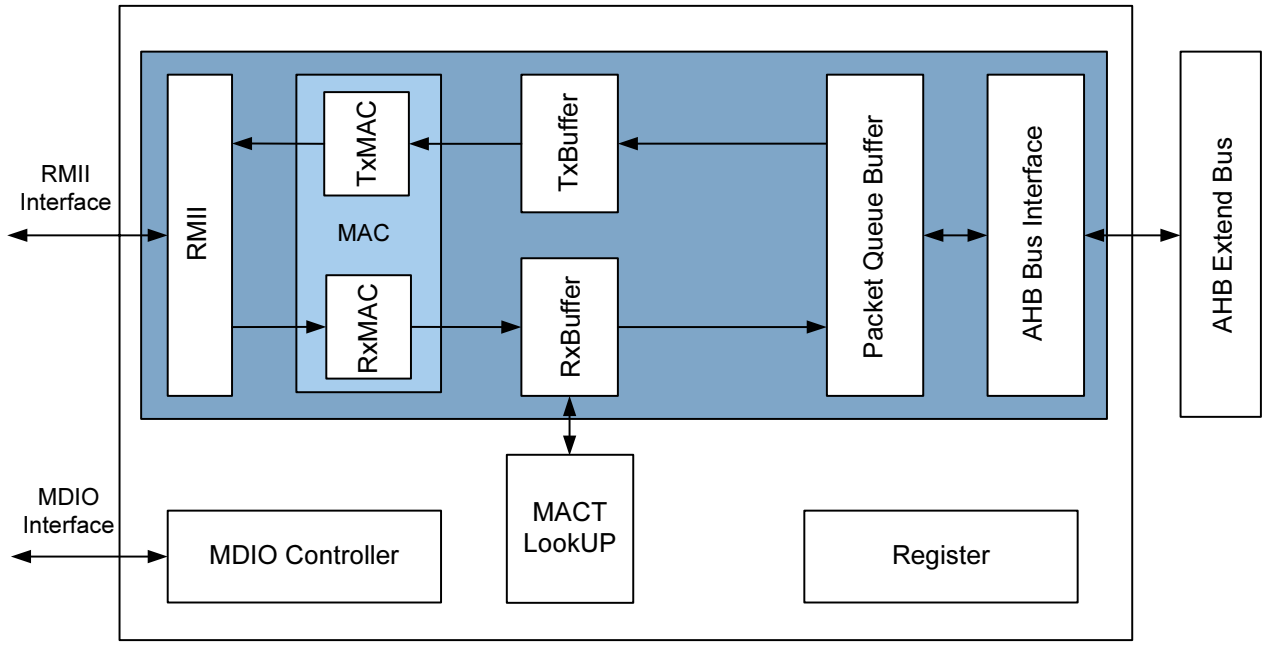
- Supports one ETH module interface.
- Supports the rate of 10 Mbit/s or 100 Mbit/s.
- Supports full-duplex or half-duplex operating mode.
- Supports the RMII.
- Supports collision back-off and retransmission and late collision in half-duplex mode.
- Supports the transmission of flow control frames in full-duplex mode.
- Supports detection of frame length validity and the discarding of extra-long and extra-short frames.
- Implements cyclic redundancy check (CRC) on the input frames. The frames with CRC errors are discarded.
- Implements CRC check on the output frames
- Supports short-frame stuffing.
- Supports the loopback to external in full-duplex mode.
- Supports adaptation.
- Provides the management data input/output (MDIO) interfaces with configurable frequency.
- Provides 64 frame management queues for both data receive (RX) and data transmit (TX).
- Provides traffic limit to prevent the CPU against traffic attack.
- Supports the count of received frames and transmitted frames.

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- Supports eight configurable DMAC address filter tables.
- Controls whether to forward or discard broadcast frames, multicast frames, and unicast frames.
- Supports the scatter gather (SG) function.
- Supports the checksum offload engine (COE) check and offload engine functions in the RX and TX directions.
- Supports Transmission Control Protocol (TCP) segmentation offload (TSO).
- Supports User Datagram Protocol (UDP) fragmentation offload (UFO).

Figure 5-1 shows the logic block diagram of the ETH module.

Figure 5-1 Logic block diagram of the ETH module



5.3 Signal Description

Table 5-1 and Table 5-2 list signals of the ETH interface.

Table 5-1 MDIO interface signals

Signal	Direction	Description	Pin
MDCK	O	Clock output of MDIO interface	MDCK
MDIO	I/O	Input/output signal of the MDIO interface	MDIO

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Table 5-2 MII interface signals

Signal	Direction	Description	Pin
RMII_REF_CLK	I/O	Uplink RMII reference clock	RMII_CLK
RMII_TXD[1:0]	O	RMII TX data	RMII_TXD1−RMII_TXD0
RMII_TXEN	O	RMII TX data valid	RMII_TX_EN
RMII_RXD[1:0]	I	RMII RX data	RMII_RXD1−RMII_RXD0
RMII_CRS_DV	I	RMII RX data valid	RMII_RX_DV

5.4 Operating Mode

5.4.1 Process of Receiving Frames

During initialization, software needs to perform the following operations:

- Software needs to request a certain number of buffers. The number is equal to the RX queue depths and the size of each buffer is 2 KB. Then, software writes the header addresses of the buffers to the frame RX queue one by one. The times of the write operation is equal to the configured RX queue depth.
- The configured buffers should not be released during frame receiving. If the configured header address is not a word aligned address, the byte address corresponding to the header address must be a writable address.

The CPU performs the following steps when it is informed that a frame needs to be received:

Step 1 Read the frame descriptor (including the start address and frame length of the RX frame) in the register UD_GLB_IQFRM_DES.

Step 2 Process the data and write 1 to clear GLB_IRQ_RAW[iraw_rx_up] (indicating that the CPU completes the frame receiving).

----End

After receiving a frame of data, software needs to re-apply for a buffer of 2 KB and re-write the header address to the frame descriptor of the current RX queue. Otherwise, the available depth of RX queues equals to the number of buffers assigned to the CPU rather than the value configured by the CPU.

Table 5-3 describes the data structure of the frame descriptor received by the CPU.

Table 5-3 Data structure of the frame descriptor received by the CPU

Bits	Name	Description
[63:32]	rxfrm_saddr	Start address for receiving frames.

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Bits	Name	Description
[31:18]	reserved	Reserved.
[17:12]	fd_in_addr	Relative address of the frame to be received in the input queue (IQ). It serves as the index (0 to iq_len - 1) of the absolute addresses for storing frames.
[11:0]	fd_in_len	Length of the frame to be received in the IQ.

NOTE

The length of the RX queue can be obtained by querying UD_GLB_ADDRQ_STAT.

The CPU can receive a frame in interrupt or query mode.

1. Receiving a frame in interrupt mode

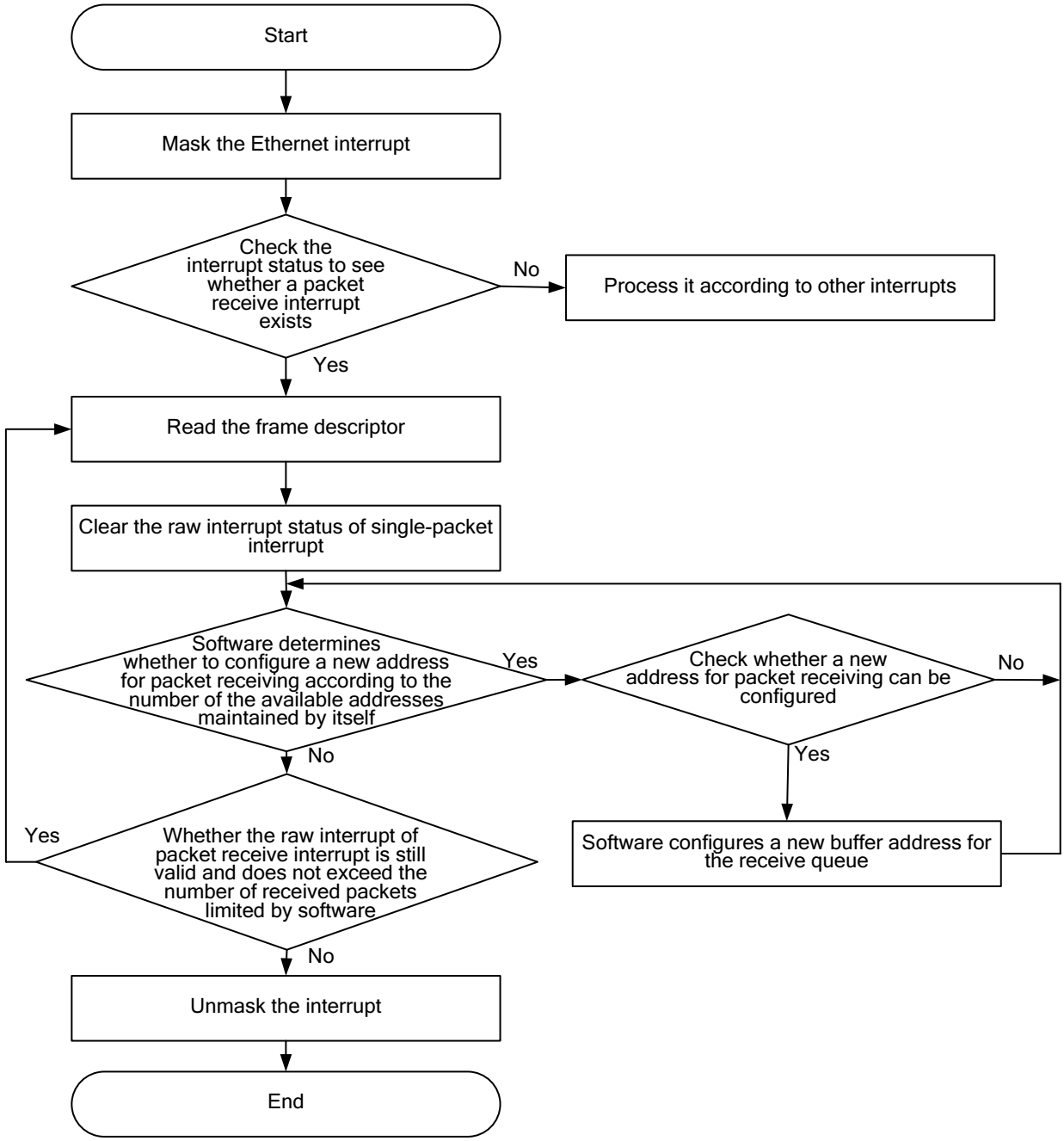
   When the CPU enables the frame RX interrupt, depending on the frames to be received, hardware generates frame RX interrupts (single-packet interrupt and multi-packet interrupt) int_rx_up and int_rxd_up.

   int_rx_up indicates that an interrupt is reported each time a packet is received.

   int_rxd_up indicates that an interrupt is reported each time a number of specified packets are received. Figure 5-2 shows the process of receiving a frame in interrupt mode.

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Figure 5-2 Process of receiving a frame in interrupt mode

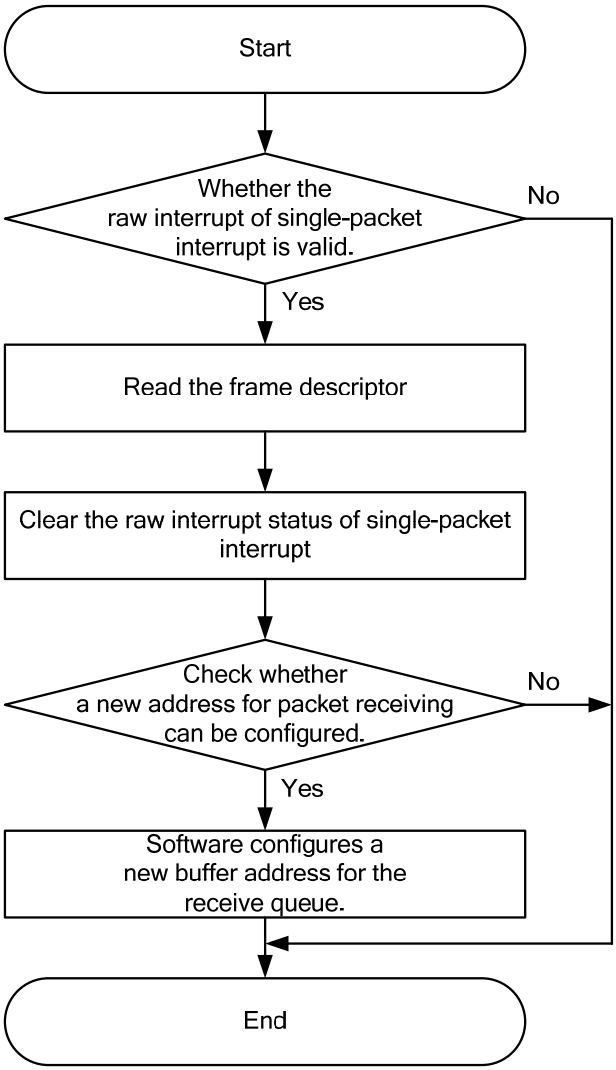


2. Receiving a frame in query mode

   In this mode, the CPU does not enable the frame RX interrupt bit, namely, GLB_IRQ_ENA[ien_rx_up], but automatically queries GLB_IRQ_RAW[iraw_rx_up]. If GLB_IRQ_RAW[iraw_rx_up] is 1, it indicates that there is a frame to be received by the CPU. Figure 5-3 shows the process of receiving a frame in query mode.

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Figure 5-3 Process of receiving a frame in query mode



5.4.2 Process of Transmitting a Frame

When a frame is transmitted, the CPU checks whether the current queue has any available space. If the space is sufficient, the CPU writes the header address of the buffer and then the length of the TX frame to the frame descriptor of the TX queue. The frame length trigger hardware for writing the TX frame writes the header address and frame length of the TX frame to the TX queue. Each time after a write is performed on the register, a data packet is transmitted. Therefore, software must control the write to the frame length register so that the frame length register is not written arbitrarily.

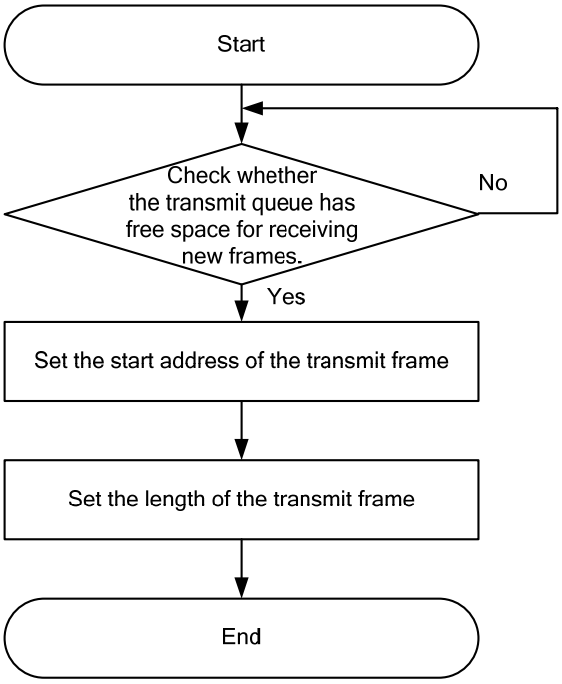
The frame format is as follows:

Destination MAC	Source MAC	Type	Data	FCS

Figure 5-4 shows the process of transmitting a frame by the CPU.

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Figure 5-4 Process of transmitting a frame by the CPU



When the frame transmitted by the CPU is buffered in the SDRAM, the frame descriptor is not included. The frame descriptor is written to UD_GLB_EQ_ADDR and UD_GLB_EQFRM_LEN to notify the ETH module of adding the frame (descriptor) to the queue. Table 5-4 describes the data structure of the frame descriptor transmitted by the CPU.

Table 5-4 Data structure of the frame descriptor transmitted by the CPU

Bits	Name	Description
[63:32]	start_addr_eq	Header address of a frame to be transmitted
[31]	tso_flag	Whether to implement TSO 0: no 1: yes
[30]	vlan_flag	Whether the current packet has the virtual local area network (VLAN) flag 0: no 1: yes
[29]	ip_version	IP protocol version of the current packet 0: IPv4 1: IPv6
[28]	protocol_type	Transport layer protocol type of the current packet 0: TCP 1: UDP

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Bits	Name	Description
[27]	tx_coe_flag	Whether to implement COE in the TX direction 0: no 1: yes
[26]	sg_flag	Whether to implement SG 0: no 1: yes
[25:24]	reserved	Reserved
[23:20]	ip_hdr_len	IP header length of the current packet, in the unit of word (four bytes) The IP header length is 5−15 words for an IPv4 packet and 10 words for an IPv6 packet.
[19:16]	protocol_hdr_len	TCP header length for a TCP packet, in the unit of word (four bytes), ranging from 5 words to 15 words; UDP header length for a UDP packet, in the unit of word (four bytes), 2 words
[15:11]	nfrags_num	Number of nfrags in the SKB of the current descriptor indicator. The maximum number is 16.
[10:0]	length	Maximum segment size (MSS) of the packet slice for the TSO packet, or total frame length for the SG or bypass packet

Note: The frames whose **length** is less than 20 bytes or greater than 1,900 bytes are discarded. In other words, the allowed range is from 20 bytes to 1900 bytes.

NOTE

The usage of the TX queues for the current CPU can be obtained by querying UD_GLB_ADDRQ_STAT.

The CPU can transmit a frame in interrupt or query mode.

1. Transmitting a frame in interrupt mode

   The CPU enables the nonempty-to-empty interrupt (int_freeeq_up) of the TX queue of the ETH module and allows the interrupt to be notified to the CPU. If the TX queue of the ETH module changes from nonempty to empty, it indicates the ETH module can transmit a frame. Then, hardware generates an interrupt to notify the CPU of transmitting the frame.

   If software needs to transmit a frame but the current TX queue is full, software enables the interrupt. After the TX queue is empty, an interrupt is generated to instruct software to transmit the waiting frame. Software uses the interrupt to send a group of frames at a time and then releases the buffers for storing the previously transmitted group of frames when the interrupt is valid.

2. Transmitting a frame in query mode

   Software queries the count of the TX frames. If the count is less than the configured depth of TX queue, it notifies the ETH module of the to-be transmitted frame directly. At the same time, it creates a corresponding TX frame index table whose content is the

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header address of the frame that is written to the TX queue of the Ethernet MAC. After transmitting a frame, the ETH module notifies the CPU of releasing the corresponding TX buffer through the address of the TX queue. After that, the CPU queries the corresponding TX buffer through the address of the TX queue and releases the buffer.

5.4.3 Interrupt Management

Interrupt Status Register

This register indicates the generated interrupt type. For details, see GLB_IRQ_STAT in section 5.6.3 "Description of the Global Control Registers."

Interrupt Enable Register

This register controls whether to generate the related interrupts. For details, see GLB_IRQ_ENA in section 5.6.3 "Description of the Global Control Registers." If an interrupt is enabled, the interrupt status is written to the related interrupt status register.

Raw Interrupt Status Register

This register can read the raw interrupt of a type and transmit it to the CPU. For details, see GLB_IRQ_RAW in section 5.6.3 "Description of the Global Control Registers." To clear the interrupt status, the raw interrupt of the interrupt must be cleared. After the raw interrupt is cleared, the interrupt status is cleared automatically.

5.4.4 Traffic Control

When the number of frames received at a certain interval exceeds the upper limit configured by software, the subsequently received frames are selectively discarded. Through the configuration of UD_GLB_FC_DROPCTRL, the broadcast frames, multicast frames, or unicast frames are discarded if the traffic limit is exceeded. The traffic limit is configured through UD_GLB_FC_RXLIMIT.

Software configures the time interval of traffic restriction through UD_GLB_FC_TIMECTRL. For a 10-bit time interval register, the time slot can be set to up to 1,023. A 17-bit counter is used for counting the time slots of the main clock. The default count is 100,000. For a 100-MHz main clock, the time slot is 1 ms. Software can configure the upper traffic limit of a 20-bit register. If the traffic limit is set to 0, it indicates that traffic is not limited.

5.4.5 Typical Application

Clock Gating

NOTE
When the ETH module is not used, its clocks can be disabled to reduce the power consumption.

To disable the ETH clocks, perform the following steps:

Step 1 Disable the link status of the ETH interface so that the ETH module cannot transmit or RX packets.

Step 2 Clear the RX queues of ETH interface so that the ETH module cannot report the packet RX interrupt.

Step 3 Software delivers a logic command to reset the ETH module and holds the reset status.

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Step 4 Set PERI_CRG59 [eth_cken] to 0 to disable the ETH clocks.

----End

To enable the ETH clocks, perform the following steps:

Step 1 Hold the reset status. Set PERI_CRG59 [eth_cken] to 1 to enable the ETH clocks.

Step 2 Set PERI_CRG59 [hrst_eth_s] to 0 to clear the reset status.

Step 3 Enable the link status of the ETH interface to ensure that the ETH module works properly.

----End

Soft Reset

To perform global soft reset on the ETH module, perform the following steps:

Step 1 Disable the link status of the ETH interface and the packet RX interrupt so that software cannot receive or transmit packets.

Step 2 After processing the current received and transmitted packets on the ETH interface, software clears the RX and TX queues and keeps the queue length the same as the value before soft reset. That is, the count values of the related pointers and queues return to 0.

Step 3 Set PERI_CRG59 [hrst_eth_s] to 1 to deliver the soft reset command for the ETH module.

Step 4 Set PERI_CRG59 [hrst_eth_s] to 0 to clear the soft reset on the ETH module.

Step 5 If packets need to be transmitted and received again, software also needs to initialize the RX and TX queues of the ETH interface.

Step 6 Enable the link status of the ETH interface to ensure that the ETH module works properly.

----End

Initialization

To initialize the ETH interface, perform the following steps:

Step 1 Configure the mode for obtaining the interface status.

The status of the ETH interface can be that of the PHY chip by means of auto-adaption or can be configured by the software. During initialization, select the mode for obtaining the interface status by configuring UD_MAC_PORTSEL[stat_ctrl]:

- If UD_MAC_PORTSEL[stat_ctrl] is set to 1, it indicates that the status of the ETH interface is configured by the software. In this case, go to Step 2.
- If UD_MAC_PORTSEL[stat_ctrl] is set to 0, it indicates that the status of the ETH interface is that of the PHY chip by means of auto-adaption. In this case, go to Step 3.

During reset, software configures the working status of the ETH interface.

Step 2 Configure the working status of the PHY chip.

- If UD_MAC_PORTSEL is set to 1, software needs to configure the rate, connection status, and duplex status in UD_MAC_PORTSET according to the actual application environment, and configures the information to the related registers of the PHY chip.

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- The ETH module provides an MDIO interface to implement the read/write control for the PHY chip. During software operation, the MDIO interface writes the address of the PHY chip, the address of the register, and related control information to the MDIO_RWCTRL register. When MDIO_RWCTRL[finish] is 1, it indicates the read/write operation on the PHY chip has been finished by hardware. For details about configuration information, see the data sheet related to the PHY chip.

After the configuration is complete, go to Step 4.

Step 3 Configure the working status in auto-adaption mode.

If UD_MAC_PORTSEL[stat_ctrl] is set to 0, you must specify the rate of the PHY chip, duplex mode, address of the connection register, and offset addresses of the registers of such status bits. The information is configured through

UD_MDIO_ANEG_CTRL.

Step 4 Set the depth of RX and TX queues.

Set the RX queue depth and TX queue depth in the register UD_GLB_QLEN_SET:

- The RX queue depth indicates the maximum number of buffered frames when data is received.
- The TX queue depth indicates the maximum number of buffered frames when data is transmitted.

The RX and TX queues share 64 management spaces, so the sum of the RX queue depth and the TX queue depth cannot exceed 64. Additionally, either the RX queue depth or the TX queue depth must be more than or equal to 1. If the sum exceeds 64, the depth of the RX queue remains unchanged and the depth of the TX queue is changed to 64 minus the depth of the RX queue.

Software can set the multi-packet interrupt configuration register. By configuring the register UD_GLB_IRQN_SET[int_frm_cnt], software controls how many packets need to be received before a multi-packet interrupt is reported. In addition, software can set the aging time register UD_GLB_IRQN_SET[int_timer].

Step 5 Initialize the buffer of the RX frame queue.

After the reset, software needs to apply for a certain number of buffers, and the number is equal to the configured depth of RX queues. The size of each buffer is 2 KB. Then, software writes the header addresses of the buffers to the RX queue. The number of times of the write operation must be equal to the configured depth of RX queues.

Step 6 Execute the soft reset on the ETH module.

Through the soft reset, the logic circuits and frame management queues inside the ETH module are reset, so that the ETH module returns to the initial status. However, the registers inside the ETH module keep the original values. After clearing the reset, software re-applies for a packet RX buffer and initializes the RX queue. Otherwise, the ETH module cannot receive network packets.

NOTE

After the soft reset of the ETH module, the registers configured by software remain unchanged. For details about these registers, see the register description.

----End

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Process of Receiving a Frame in Interrupt Mode

To receive a frame in interrupt mode, perform the following steps:

Step 1 Mask the Ethernet interrupt after entering the interrupt handling program.

Step 2 View the interrupt status bit GLB_IRQ_STAT[int_rx_up] to check whether a frame RX interrupt occurs. If a frame RX interrupt occurs, go to Step 2. Otherwise, continue to process other Ethernet interrupts.

Step 3 Read the frame descriptor UD_GLB_IQFRM_DES. Read the frame data of the related length (fd_in_len) according to the header address of the frame that corresponds to fd_in_addr.

Step 4 Return the raw interrupt signal bit GLB_IRQ_RAW[iraw_rx_up] of the single-packet interrupt to 0 and notify hardware of the completion of packet receiving.

Step 5 According to the number of the remaining available addresses maintained by itself, software determines whether to configure a new address for packet receiving. If a new address for packet receiving does not need to be configured, go to Step 7. Receiving a packet is complete.

Step 6 Read UD_GLB_QSTAT[cpu_addr_in_rdy] to check whether a new address for packet receiving can be configured. If a new address for packet receiving cannot be configured, return to Step 4.

Step 7 Software allocates a new buffer address to the RX queue through the register UD_GLB_IQ_ADDR. Return to Step 4.

Step 8 Read and check the raw interrupt signal bit GLB_IRQ_RAW[iraw_rx_up] of the single-packet interrupt. If the bit is valid and software can continue to receive packets (the upper limit of received packets is not exceeded), return to Step 2.

Step 9 Unmask the Ethernet interrupt.

----End

Process of Receiving a Frame in Query Mode

If the frame RX interrupt GLB_IRQ_ENA[ien_cpu_rx] is disabled, the CPU automatically queries the raw interrupt signal bit GLB_IRQ_RAW[iraw_rx_up] of the single-packet interrupt. If the bit is set to 1, it indicates that a frame needs to be received.

Receive a frame in query mode as follows:

Step 1 Read the raw interrupt signal bit GLB_IRQ_RAW[iraw_rx_up] of the single-packet interrupt. If this bit is invalid, the process ends.

Step 2 Read the frame descriptor UD_GLB_IQFRM_DES. Read the frame data of the related length (fd_in_len) according to the header address of the frame that corresponds to fd_in_addr.

Step 3 Write 1 to clear the raw interrupt signal bit GLB_IRQ_RAW[iraw_rx_up] of the single-packet interrupt and notify hardware of the completion of packet receiving.

Step 4 Read UD_GLB_QSTAT[cpu_addr_in_rdy] to check whether a new address for packet receiving can be configured. If a new address for packet receiving cannot be configured, the process ends.

Step 5 Software allocates a new buffer address to the RX queue through the register UD_GLB_IQ_ADDR.

----End

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Process of Transmitting a Frame

Transmit a frame as follows:

Step 1 Read UD_GLB_ADDRQ_STAT[eq_in_rdy] and check whether the TX queue of the ETH module has free space for receiving new TX frames. If the TX queue of the ETH module has no free space, continue to wait and query for a free space.

Step 2 Configure the header address UD_GLB_EQ_ADDR of the frame to be transmitted.

Step 3 Configure the length UD_GLB_EQFRM_LEN of the frame to be transmitted. The configuration for transmitting a frame is complete.

----End

5.5 Register Summary

MDIO Control Registers

Table 5-5 describes the MDIO control registers.

Table 5-5 Summary of the MDIO control registers (base address: 0x1009_0000)

Offset Address	Name	Description	Page
0x1100	MDIO_RWCTRL	MDIO command word register	5-18
0x1104	MDIO_RO_DATA	MDIO read data register	5-19

MAC Control Registers

Table 5-6 describes the MAC control registers.

Table 5-6 Summary of the MAC controller registers (base address: 0x1009_0000)

Offset Address	Name	Description	Page
0x0200	UD_MAC_PORTSEL	Port working status control register	5-23
0x0204	UD_MAC_RO_STAT	Port status register	5-23
0x0208	UD_MAC_PORTSET	Port working status configuration register	5-24
0x020C	UD_MAC_STAT_CHANGE	Port status change indicator register for the MAC	5-25
0x0210	UD_MAC_SET	MAC function configuration register	5-26

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Global Control Registers

Table 5-7 describes the Ethernet global control registers.

Table 5-7 Summary of the global control registers (base address: 0x1009_0000)

Offset Address	Name	Description	Page
0x1300	GLB_HOSTMAC_L32	Lower 32-bit register for the local MAC address	5-31
0x1304	GLB_HOSTMAC_H16	Upper 16-bit register for the local MAC address	5-32
0x1308	GLB_SOFT_RESET	Internal soft reset register	5-32
0x1310	GLB_FWCTRL	Forward control register	5-33
0x1314	GLB_MACTCTRL	MAC filter table control register	5-33
0x1318	GLB_ENDIAN_MOD	Endian control register	5-34
0x1330	GLB_IRQ_STAT	Interrupt status register	5-35
0x1334	GLB_IRQ_ENA	Interrupt enable register	5-37
0x1338	GLB_IRQ_RAW	Raw interrupt register	5-39
0x1400	GLB_MAC0_L32	MAC filter 0	5-40
0x1404	GLB_MAC0_H16	MAC filter 0	5-40
0x1408	GLB_MAC1_L32	MAC filter 1	5-41
0x140C	GLB_MAC1_H16	MAC filter 1	5-41
0x1410	GLB_MAC2_L32	MAC filter 2	5-42
0x1414	GLB_MAC2_H16	MAC filter 2	5-42
0x1418	GLB_MAC3_L32	MAC filter 3	5-43
0x141C	GLB_MAC3_H16	MAC filter 3	5-43
0x1420	GLB_MAC4_L32	MAC filter 4	5-44
0x1424	GLB_MAC4_H16	MAC filter 4	5-45
0x1428	GLB_MAC5_L32	MAC filter 5	5-45
0x142C	GLB_MAC5_H16	MAC filter 5	5-46
0x1430	GLB_MAC6_L32	MAC filter 6	5-46
0x1434	GLB_MAC6_H16	MAC filter 6	5-47
0x1438	GLB_MAC7_L32	MAC filter 7	5-47
0x143C	GLB_MAC7_H16	MAC filter 7	5-48

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Offset Address	Name	Description	Page
0x0340	UD_GLB_IRQN_SET	Multi-packet interrupt configuration register	5-48
0x0344	UD_GLB_QLEN_SET	Queue length configuration register	5-49
0x0348	UD_GLB_FC_LEVEL	Traffic control register	5-50
0x034C	UD_GLB_CAUSE	Cause register for the CPU to which the packet is transmitted	5-50
0x0350	UD_GLB_RXFRM_SADDR	RX frame start address register	5-51
0x0354	UD_GLB_IQFRM_DES	RX frame descriptor register	5-51
0x0358	UD_GLB_IQ_ADDR	RX frame header address register	5-52
0x035C	UD_GLB_BFC_STAT	Counter for traffic control status of forward buffer and aging time of multi-packet interrupt	5-52
0x0360	UD_GLB_EQ_ADDR	TX queue header address register	5-53
0x0364	UD_GLB_EQFRM_LEN	TX queue frame length configuration register	5-54
0x0368	UD_GLB_QSTAT	Queue status register	5-54
0x036C	UD_GLB_ADDRQ_STAT	Address queue status register	5-55
0x0370	UD_GLB_FC_TIMECTRL	Traffic control time configuration register	5-56
0x0374	UD_GLB_FC_RXLIMIT	Traffic control limit configuration register	5-57
0x0378	UD_GLB_FC_DROPCTRL	Packet drop control register for traffic control	5-57
0x0380	UD_GLB_RX_COE_EN	RX COE enable register	5-58

Statistics Counter Control Registers

Table 5-8 describes the statistics counter control registers.

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Table 5-8 Summary of the statistics counter control registers (base address: 0x1009_0000)

Offset Address	Name	Description	Page
0x0584	UD_STS_PORTCNT	Port status counter	5-58
0x05A0	UD_PORT2CPU_PKTS	Register for the total number of packets received by the CPU from the uplink or downlink port	5-59
0x05A4	UD_CPU2IQ_ADDRCNT	Register for the count of configuring packet receiving address queue by the CPU	5-60
0x05A8	UD_RX_IRQCNT	Register for the count of reporting single-packet interrupt by the uplink or downlink port	5-60
0x05AC	UD_CPU2EQ_PKTS	Register for the total number of packets transmitted by the CPU to the uplink or downlink port	5-60

Statistics Result Registers

Table 5-9 describes the statistics result registers.

Table 5-9 Summary of the statistics result registers (base address: 0x1009_0000)

Offset Address	Name	Description	Page
0x0600	UD_RX_DVCNT	RXDV rising edge count register	5-61
0x0604	UD_RX_OCTS	Register for the total number of received bytes	5-61
0x0608	UD_RX_RIGHTOCTS	Register for the total number of bytes of received correct packets	5-62
0x060C	UD_HOSTMAC_PKTS	Register for the number of packets matching the local MAC address	5-62
0x0610	UD_RX_RIGHTPKTS	Register for the total number of packets received by the port	5-63
0x0614	UD_RX_BROADPKTS	Register for the number of correct broadcast packets	5-63
0x0618	UD_RX_MULTPKTS	Register for the number of correct multicast packets	5-63

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Offset Address	Name	Description	Page
0x061C	UD_RX_UNIPKTS	Register for the number of correct unicast packets	5-64
0x0620	UD_RX_ERRPKTS	Register for the total number of incorrect packets	5-64
0x0624	UD_RX_CRCERR_PKTS	Register for the count of CRC errors	5-64
0x0628	UD_RX_LENERR_PKTS	Register for the number of packets with invalid length	5-65
0x062C	UD_RX_OCRCERR_PKTS	Register for the number of packets with odd nibbles and CRC errors	5-65
0x0630	UD_RX_PAUSE_PKTS	Register for the number of received pause packets	5-66
0x0634	UD_RF_OVERCNT	Register for the count of RXFIFO overflow events	5-66
0x0638	UD_FLUX_TOL_IPKTS	Register for the total number of received packets allowed by the traffic limit	5-66
0x063C	UD_FLUX_TOL_DPKTS	Register for the total number of packets discarded due to traffic limit	5-67
0x064C	UD_MN2CPU_PKTS	Register for the number of packets not forwarded to the CPU port due to MAC limit	5-67
0x0780	UD_TX_PKTS	Register for the total number of packets transmitted successfully	5-67
0x0784	UD_TX_BROADPKTS	Register for the number of broadcast packets transmitted successfully	5-68
0x0788	UD_TX_MULTPKTS	Register for the number of multicast packets transmitted successfully	5-68
0x078C	UD_TX_UNIPKTS	Register for the number of unicast packets transmitted successfully	5-69
0x0790	UD_TX_OCTS	Register for the total number of transmitted bytes	5-69
0x0794	UD_TX_PAUSE_PKTS	Register for the number of transmitted pause frames	5-69

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Offset Address	Name	Description	Page
0x0798	UD_TX_RETRYCNT	Register for the total count of retransmission	5-70
0x079C	UD_TX_COLCNT	Register for the total count of collisions	5-70
0x07A0	UD_TX_LC_PKTS	Register for the number of packets with late collision	5-70
0x07A4	UD_TX_COLOK_PKTS	Register for the number of packets transmitted successfully with collisions	5-71
0x07A8	UD_TX_RETRY15_PKTS	Register for the number of packets discarded due to more than 15 times of retransmission	5-71
0x07AC	UD_TX_RETRYN_PKTS	Register for the number of packets with the count of collisions being equal to the threshold	5-71

5.6 Register Description

5.6.1 Description of the MDIO Control Registers

MDIO_RWCTRL

MDIO_RWCTRL is an MDIO command word register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x1100	MDIO_RWCTRL	0x0000_8000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																finish	reserved	rw	phy_exaddr					frq_dv				phy_inaddr			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	cpu_data_in	Data used by the MDIO module to perform write operation on the PHY chip During write operation, the CPU first writes the 16-bit data to be written to the MDIO to this register.

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[15]	RW	finish	PHY read/write operation complete 0: not complete 1: complete When the read/write operation is required for the second time, the CPU must clear this bit first.
[14]	RO	reserved	Reserved
[13]	RW	rw	PHY read or write access control 0: read operation 1: write operation
[12:8]	RW	phy_exaddr	Physical address of the external PHY chip One MDIO can perform read/write operation on multiple external PHY chips. Each PHY chip has one corresponding address. When the MDIO connects to only one external PHY chip, this bit is equivalent to UD_MDIO_PHYADDR[phy0_addr] or UD_MDIO_PHYADDR[phy1_addr].
[7:5]	RW	frq_dv	Frequency division factor for the MDC (the MDIO interface clock) when the MDIO performs the read/write operation on external PHY chips Take the frequency 100 MHz of the main clock as an example to describe the matching relations between frq_dv and MDC frequency. 000: The frequency of the working main clock is divided by 50 and the obtained frequency is 2 MHz. 001: The frequency of the working main clock is divided by 100 and the obtained frequency is 1 MHz. 010: The frequency of the working main clock is divided by 200 and the obtained frequency is 500 kHz. 011: The frequency of the working main clock is divided by 400 and the obtained frequency is 250 kHz. 100: The frequency of the working main clock is divided by 800 and the obtained frequency is 125 kHz. 101: The frequency of the working main clock is divided by 1600 and the obtained frequency is 62.5 kHz. 110: The frequency of the working main clock is divided by 3200 and the obtained frequency is 31.25 kHz. 111: The frequency of the working main clock is divided by 6400 and the obtained frequency is 15.625 kHz.
[4:0]	RW	phy_inaddr	Internal register address of the external PHY chip. This address is presented by a 5-bit binary number.

MDIO_RO_DATA

MDIO_RO_DATAMDIO is a read data register. The register does not support soft reset.

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Offset Address	Register Name	Total Reset Value
0x1104	MDIO_RO_DATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																cpu_data_out															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	cpu_data_out	Data register used by the MDIO module to perform read operation on the PHY chip. The MDIO module first writes the 16-bit data read from the PHY chip to this register.

UD_MDIO_PHYADDR

UD_MDIO_PHYADDR is a PHY physical address register. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0108	UD_MDIO_PHYADDR	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												phy_addr			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	phy_addr	Physical address of the external PHY chip

UD_MDIO_RO_STAT

UD_MDIO_RO_STAT is a PHY status register. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x010C	UD_MDIO_RO_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													speed_mdio2mac	link_mdio2mac	duplex_mdio2mac

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RO	speed_mdio2mac	Port speed working status obtained from the MDIO interface, which is in either 10 Mbit/s or 100 Mbit/s working mode 0: 10 Mbit/s mode 1: 100 Mbit/s mode
[1]	RO	link_mdio2mac	Port link status obtained from the MDIO interface 0: No link exists. 1: A link exists.
[0]	RO	duplex_mdio2mac	Port duplex working status obtained from the MDIO interface 0: half-duplex 1: full-duplex

UD_MDIO_ANEG_CTRL

UD_MDIO_ANEG_CTRLPHY is an offset address configuration register for the PHY status. The register does not support soft reset.

NOTE

The PHY speed status is indicated by bit[14] of the register with the address of 17, internal_addr_speed is set to 0x11 and speed_index is set to 0xE. In this case, the ETH module reads the bit value as the current working speed mode of the PHY through the MDIO interface.

Offset Address	Register Name	Total Reset Value
0x0110	UD_MDIO_ANEG_CTRL	0x0463_1EA9

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				internal_addr_speed								internal_addr_link				internal_addr_duplex				speed_index				link_index				duplex_index			
Reset	0	0	0	0	0	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	1	1	0	1	0	1	0	1	0	0	1

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:22]	RW	internal_addr_speed	Address of the register in the PHY chip to store the status information (speed). The default value is set according to Intel 9785.

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[21:17]	RW	internal_addr_link	Address of the register in the PHY chip to store the status information (link). The default value is set according to Intel 9785.
[16:12]	RW	internal_addr_duplex	Address of the register in the PHY chip to store the status information (duplex). The default value is set according to Intel 9785.
[11:8]	RW	speed_index	Offset address in the PHY status register that is used to store the speed information. The default value is set according to Intel 9785.
[7:4]	RW	link_index	Offset address in the PHY status register that is used to store the link information. The default value is set according to Intel 9785.
[3:0]	RW	duplex_index	Offset address in the PHY status register that is used to store the duplex information. The default value is set according to Intel 9785.

UD_MDIO_IRQENA

UD_MDIO_IRQENA is a scan mask register for MDIO status changes. The register does not support soft reset.

NOTE

- If the status information about the PHY chip connecting to the port cannot be scanned and obtained by configuring
- UD_MDIO_ANEG_CTRL, you can scan the PHY status register by using MDIO_RWCTRL to check whether the port status changes and generate an interrupt to notify software of processing the interrupt.
- link_partner status change refers to the change of any bit of link, speed, and duplex for the PHY status.

Offset Address	Register Name	Total Reset Value
0x0114	UD_MDIO_IRQENA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												link_partner_ch_mask		speed_ch_mask		link_ch_mask		duplex_ch_mask	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RW	link_partner_ch_mask	Port link partner status scan change interrupt mask 0: mask 1: unmask

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[2]	RW	speed_ch_mask	Port speed mode scan change interrupt mask 0: mask 1: unmask
[1]	RW	link_ch_mask	Port link mode scan change interrupt mask 0: mask 1: unmask
[0]	RW	duplex_ch_mask	Port duplex mode scan change interrupt mask 0: mask 1: unmask

5.6.2 Description of the MAC Control Registers

MAC control registers are registers for port control. When the port status is valid, after configuring MAC control registers, you need to perform one soft reset on them.

UD_MAC_PORTSEL

UD_MAC_PORTSEL is a port working status control register. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0200	UD_MAC_PORTSEL	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															stat_ctrl
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	stat_ctrl	Port working status information select control register 0: Use the status information obtained from the MDIO interface. 1: Use the status information set by the CPU.

UD_MAC_RO_STAT

UD_MAC_RO_STAT is a port status register. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0204	UD_MAC_RO_STAT	0x0000_0000

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Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													peed_stat	link_stat	duplex_stat
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RO	speed_stat	Port current speed mode 0: 10 Mbit/s mode 1: 100 Mbit/s mode
[1]	RO	link_stat	Port current link status 0: No link exists. 1: A link exists.
[0]	RO	duplex_stat	Port current duplex status 0: half-duplex 1: full-duplex

UD_MAC_PORTSET

UD_MAC_PORTSET is a port working status configuration register. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0208	UD_MAC_PORTSET	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													speed_stat_dio	link_stat_dio	duplex_stat_dio
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	speed_stat_dio	Port speed mode set by the CPU 0: 10 Mbit/s mode 1: 100 Mbit/s mode

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[1]	RW	link_stat_dio	Port link status set by the CPU 0: No link exists. 1: A link exists.
[0]	RW	duplex_stat_dio	Port duplex mode set by the CPU 0: half-duplex 1: full-duplex

UD_MAC_STAT_CHANGE

UD_MAC_STAT_CHANGE is a port status change indicator register. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x020C	UD_MAC_STAT_CHANGE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													speed_stat_ch	link_stat_ch	duplex_stat_ch
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	WC	speed_stat_ch	Port speed mode change indicator 0: No change occurs. 1: A change occurs. Writing 1 clears this register.
[1]	WC	link_stat_ch	Port link status change indicator 0: No change occurs. 1: A change occurs. Writing 1 clears this register.
[0]	WC	duplex_stat_ch	Port duplex mode change indicator 0: No change occurs. 1: A change occurs. Writing 1 clears this register.

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UD_MAC_SET

UD_MAC_SET is a MAC function configuration register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0210	UD_MAC_SET	0x2027_55EE

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		add_pad_en	crcgen_dis	cntr_rdclr_en	cntr_clr_all	cntr_roll_dis	colthreshold				in_loop_en	ex_loop_en	pause_en	rx_shframe_en	rx_min_thr					len_max											
Reset	0	0	1	0	0	0	0	0	0	0	1	0	0	1	1	1	0	1	0	1	0	1	0	1	1	1	1	0	1	1	1	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29]	RW	add_pad_en	Port auto add PAD enable during transmission 0: disabled 1: enabled
[28]	RW	crcgen_dis	Port CRC generation disable control 0: CRC is recalculated for the output frame. 1: CRC is not recalculated for the output frame.
[27]	RW	cntr_rdclr_en	Port statistics counter read clear enable 0: disabled 1: enabled
[26]	RW	cntr_clr_all	Port statistics counter clear control 0: not clear 1: clear **NOTE** If cntr_clr_all is set to 1, the next clear all operation can be performed only after this bit is set to 0 and then to 1.
[25]	RW	cntr_roll_dis	Port statistics acyclic counter enable 0: disabled 1: enabled
[24:21]	RW	colthreshold	Port collision count statistics threshold The default value is 0x1, which indicates the count of frames with one collision.

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[20]	RW	in_loop_en	Port loopback to internal enable 0: disabled 1: enabled NOTE Loopback to internal enable and loopback to external enable cannot be configured at the same time. When the network interface is in normal state, you need to perform soft reset on the module after loopback to internal enable is configured instead of loopback to external enable and vice versa.
[19]	RW	ex_loop_en	Port loopback to external enable 0: disabled 1: enabled Note: Loopback to internal enable and loopback to external enable cannot be configured at the same time. When the network interface is in normal state, you need to perform soft reset on the module after loopback to internal enable is configured instead of loopback to external enable and vice versa.
[18]	RW	pause_en	Port pause frame TX enable 0: disabled 1: enabled
[17]	RW	rx_shframe_en	Port short frame RX enable 0: disabled 1: enabled NOTE If rx_shframe_en is set to 1, the minimum RX frame length allowed by the port is that set by rx_min_thr. If rx_shframe_en is set to 0, the minimum RX frame length allowed by the port is 64 bytes (including CRC) by default.
[16:11]	RW	rx_min_thr	Minimum RX frame length allowed by the port The value range is from 42 bytes to 63 bytes. The default value is 42 bytes. NOTE If rx_min_thr is set to a value smaller than 42, 42 is used instead of the value.
[10:0]	RW	len_max	Maximum RX frame length allowed by the port. The default value is 1518 bytes. The value is in a range of 1518 bytes to 1535 bytes. NOTE If len_max is set to a value greater than 2000, 2000 is used instead of the value. If len_max is set to a value smaller than 256, 256 is used instead of the value.

UD_MAC_EEE_INT

UD_MAC_EEE_INT is an EEE raw interrupt register.

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Offset Address	Register Name	Total Reset Value
0x0480	UD_MAC_EEE_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											tx_entry_start	rx_leave_lpi	rx_entry_lpi	tx_leave_lpi	tx_entry_lpi
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	RO	tx_entry_start	Raw interrupt that allows the PHY to enter the low-power idle (LPI) state in the ETH TX direction
[3]	RO	rx_leave_lpi	Raw interrupt that allows the PHY to exit the LPI state in the RX direction
[2]	RO	rx_entry_lpi	Raw interrupt that allows the PHY to enter the LPI state in the RX direction
[1]	RO	tx_leave_lpi	Raw interrupt that allows the PHY to exit the LPI state in the TX direction
[0]	RO	tx_entry_lpi	Raw interrupt that allows the PHY to enter the LPI state in the TX direction

UD_MAC_EEE_INTEN

UD_MAC_EEE_INTEN is an EEE interrupt enable register.

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Offset Address	Register Name	Total Reset Value
0x0484	UD_MAC_EEE_INTEN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						tx_entry_start_msk	rx_leave_lpi_msk	rx_entry_lpi_msk	tx_leave_lpi_msk	tx_entry_lpi_msk	tx_entry_start_en	rx_leave_lpi_en	rx_entry_lpi_en	tx_leave_lpi_en	tx_entry_lpi_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	tx_entry_start_msk	Mask of the raw interrupt that allows the PHY to enter the LPI state in the ETH TX direction
[8]	RW	rx_leave_lpi_msk	Mask of the raw interrupt that allows the PHY to exit the LPI state in the RX direction
[7]	RW	rx_entry_lpi_msk	Mask of the raw interrupt that allows the PHY to enter the LPI state in the RX direction
[6]	RW	tx_leave_lpi_msk	Mask of the raw interrupt that allows the PHY to exit the LPI state in the TX direction
[5]	RW	tx_entry_lpi_msk	Mask of the raw interrupt that allows the PHY to enter the LPI state in the TX direction
[4]	RO	tx_entry_start_en	Enable for the raw interrupt that allows the PHY to enter the LPI state in the ETH TX direction
[3]	RO	rx_leave_lpi_en	Enable for the raw interrupt that allows the PHY to exit the LPI state in the RX direction
[2]	RO	rx_entry_lpi_en	Enable for the raw interrupt that allows the PHY to enter the LPI state in the RX direction
[1]	RO	tx_leave_lpi_en	Enable for the raw interrupt that allows the PHY to exit the LPI state in the TX direction
[0]	RO	tx_entry_lpi_en	Enable for the raw interrupt that allows the PHY to enter the LPI state in the TX direction

UD_MAC_EEE_ENA

UD_MAC_EEE_ENA is an EEE enable register.

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Offset Address	Register Name	Total Reset Value
0x0488	UD_MAC_EEE_ENA	0x00F4_2400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	eee_ls_timer																												reserved		eee_assert	eee_enable
Reset	0	0	0	0	0	0	0	0	1	1	1	1	0	1	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RW	eee_ls_timer	LS timer
[3:2]	RW	reserved	Reserved
[1]	RW	eee_assert	EEE LPI state enable
[0]	RW	eee_enable	EEE enable

UD_MAC_EEE_TIMER

UD_MAC_EEE_TIMER is a timer register required for the EEE function.

Offset Address	Register Name	Total Reset Value
0x048C	UD_MAC_EEE_TIMER	0x001E_2710

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tx_wk_timer																lpi_cond_timer															
Reset	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1	0	0	1	1	1	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	tx_wk_timer	TX_WK_TIMER
[15:0]	RW	lpi_cond_timer	LPI_COND_TIMER

UD_MAC_EEE_LINK_STATUS

UD_MAC_EEE_LINK_STATUS an ETH port link status register dedicated for the EEE function.

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Offset Address	Register Name	Total Reset Value
0x0490	UD_MAC_EEE_LINK_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															phy_link_status
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RW	reserved	Reserved
[0]	RW	phy_link_status	PHY link status

UD_MAC_EEE_CLK_CNT

UD_MAC_EEE_CLK_CNT is an EEE clock unit counter register.

Offset Address	Register Name	Total Reset Value
0x0494	UD_MAC_EEE_CLK_CNT	0x0000_0063

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	eee_clk_cnt																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1

Bits	Access	Name	Description
[31:0]	RW	eee_clk_cnt	EEE clock unit counter

5.6.3 Description of the Global Control Registers

GLB_HOSTMAC_L32

GLB_HOSTMAC_L32 is a lower 32-bit register for the local MAC address.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x1300	GLB_HOSTMAC_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	local_mac																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	local_mac	Lower 32 bits of the local MAC address

GLB_HOSTMAC_H16

GLB_HOSTMAC_H16 is an upper 16-bit register for the local MAC address.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x1304	GLB_HOSTMAC_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																local_mac[47:32]															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	local_mac[47:32]	Upper 16 bits of the local MAC address

GLB_SOFT_RESET

GLB_SOFT_RESET is an internal soft reset register.

The register does not support soft reset.

NOTE

The time for each soft reset must remain for more than 2 ms.

Offset Address	Register Name	Total Reset Value
0x1308	GLB_SOFT_RESET	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															soft_reset
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	RW	soft_reset	Internal soft reset 0: not reset 1: reset In soft reset state, this bit must be set to 0 to clear soft reset.

GLB_FWCTRL

GLB_FWCTRL is a forward control register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x1310	GLB_FWCTRL	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								fwall2cpu_up	reserved	fw2cpu_ena_up	reserved				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	fwall2cpu_up	Whether to forcibly forward all valid input frames to the CPU port 0: no 1: yes
[6]	RO	reserved	Reserved
[5]	RW	fw2cpu_ena_up	Function enable of forwarding the input frames to the CPU port 0: disabled 1: enabled
[4:0]	RO	reserved	Reserved

GLB_MACTCTRL

GLB_MACTCTRL is a MAC filter table control register.

The register does not support soft reset.

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NOTE

- If the highest byte of the destination MAC address is even, the frame is a unicast frame.
- If the highest byte of the destination MAC address is odd, the frame is a multicast frame.
- If all bytes of the destination MAC address are 0xFF, the frame is a broadcast frame.

Offset Address	Register Name	Total Reset Value
0x1314	GLB_MACTCTRL	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								mact_ena_up	reserved	broad2cpu_up	reserved	multi2cpu_up	reserved	uni2cpu_up	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	mact_ena_up	Enable bit of all MAC filters of the port 0: disabled (no MAC filter is used) 1: enabled (MAC filters are used)
[6]	RO	reserved	Reserved
[5]	RW	broad2cpu_up	Whether to forward the input broadcast frames to the CPU port 0: no 1: yes
[4]	RO	reserved	Reserved
[3]	RW	multi2cpu_up	Whether to forward the input multicast frames that are not listed in the filter table to the CPU port 0: no 1: yes
[2]	RO	reserved	Reserved
[1]	RW	uni2cpu_up	Whether to forward the input unicast frames that are not listed in the filter table to the CPU port 0: no 1: yes
[0]	RO	reserved	Reserved

GLB_ENDIAN_MOD

GLB_ENDIAN_MOD is an endian control register.

The register does not support soft reset.

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Offset Address	Register Name	Total Reset Value
0x1318	GLB_ENDIAN_MOD	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														in_edian	out_edian
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	in_edian	RX packet write SDRAM endian configuration 0: big-endian mode 1: little-endian mode Data consists of bytes
[0]	RW	out_edian	TX packet read SDRAM endian configuration 0: big-endian mode 1: little-endian mode

GLB_IRQ_STAT

GLB_IRQ_STAT is an interrupt status register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x1330	GLB_IRQ_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			int_mdio_finish	reserved				int_rxd_up	int_freeeq_up	int_stat_up	int_duplex_up	int_speed_up	int_link_up	int_tx_up	int_rx_up
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RO	int_mdio_finish	Raw interrupt status indicating whether the MDIO interface completes the operation required by the CPU 0: not completed 1: Completed and an interrupt is generated.

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			After this interrupt is generated, software determines whether the MDIO completes the operation by querying MDIO_RWCTRL[finish].
[11:8]	RO	reserved	Reserved
[7]	RO	int_rxd_up	Interrupt status (multi-packet interrupt) for frames on the port to be received by the CPU 0: The interrupt is invalid. 1: The interrupt is valid. There are frames to be received by the CPU in the RX queue. After this interrupt is generated, software determines whether there are frames to be received by querying GLB_IRQ_RAW[iraw_rxd_up].
[6]	RO	int_freeeq_up	Interrupt status indicates that the status of the port output queue is changed from nonempty to empty, that is, the status of the TX queue buffer is changed from nonempty to empty, that is, the status of the TX queue buffer is changed from nonempty to empty. In this case, the CPU can write a group of new frames to be transmitted. 0: No interrupt is generated. 1: An interrupt is generated. After this interrupt is generated, software determines whether the current TX queue is empty by querying UD_GLB_ADDRQ_STAT[eq_cnt]. If the current TX queue is not empty, it indicates that the interrupt is invalid.
[5]	RO	int_stat_up	Interrupt status for port status changes, indicating that an interrupt is generated when the MDIO obtains the speed change, duplex mode change, and link status change of the PHY chip in auto-adaption mode. 0: The interrupt is invalid. 1: The interrupt is valid. The port status changes. After this interrupt is generated, software determines which status changes according to the configuration of UD_MDIO_IRQENA.
[4]	RO	int_duplex_up	Interrupt status for port duplex mode changes 0: The interrupt is invalid. 1: The interrupt is valid. The duplex mode changes. After this interrupt is generated, software determines whether the duplex mode changes by querying UD_MAC_STAT_CHANGE[duplex_stat_ch].
[3]	RO	int_speed_up	Interrupt status for port speed mode changes 0: The interrupt is invalid. 1: The interrupt is valid. The speed mode changes. After this interrupt is generated, software determines whether the speed mode changes by querying UD_MAC_STAT_CHANGE[speed_stat_ch].
[2]	RO	int_link_up	Interrupt status for port link status changes 0: The interrupt is invalid.

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			1: The interrupt is valid. The link status changes. After this interrupt is generated, software determines whether the link status changes by querying UD_MAC_STAT_CHANGE[link_stat_ch].
[1]	RO	int_tx_up	Interrupt status for the completion of transmitting a frame from the CPU by the port 0: not completed 1: Completed and an interrupt is generated. After this interrupt is generated, software determines whether to release the buffer of the TX frames by querying the current TX queue address eq_out_index in UD_GLB_QSTAT.
[0]	RO	int_rx_up	Interrupt status for frames on the port to be received by the CPU 0: The interrupt is invalid. 1: The interrupt is valid. There are frames to be received by the CPU in the RX queue. After this interrupt program is started, software determines whether frames are received by querying the GLB_IRQ_RAW[iraw_rxd_up] signal.

GLB_IRQ_ENA

GLB_IRQ_ENA is an interrupt enable register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x1334	GLB_IRQ_ENA	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19	18	17 16	15 14 13 12	11 10 9 8	7	6	5	4	3	2	1	0
Name	reserved			ien_all	ien_up	reserved	ien_mdio_finish	reserved	ien_rxd_up	ien_freeeq_up	ien_stat_up	ien_duplex_up	ien_speed_up	ien_link_up	ien_tx_up	ien_rx_up
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0 0 0	0 0 0 0	0 0 0 0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19]	RW	ien_all	All interrupts enable 0: disabled (none of the interrupt can be reported) 1: enabled (all interrupts are reported according to the configuration)

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[18]	RW	ien_up	All uplink port interrupts enable 0: disabled (none of the uplink port interrupt can be reported) 1: enabled (all uplink port interrupts are reported according to the configuration)
[17:13]	RO	reserved	Reserved
[12]	RW	ien_mdio_finish	Indicator enable for the MDIO to complete the operation required by the CPU 0: disabled 1: enabled
[11:8]	RO	reserved	Reserved
[7]	RW	ien_rxd_up	Interrupt enable (multi-packet interrupt) for a frame (frames) on the uplink port to be received by the CPU 0: disabled 1: enabled
[6]	RW	ien_freeeq_up	Interrupt signal enable for the TX queue of the uplink port to change from nonempty to empty 0: disabled 1: enabled
[5]	RW	ien_stat_up	Interrupt signal enable for uplink port status changes 0: disabled 1: enabled
[4]	RW	ien_duplex_up	Interrupt enable for uplink port duplex mode changes 0: disabled 1: enabled
[3]	RW	ien_speed_up	Interrupt enable for uplink port speed mode changes 0: disabled 1: enabled
[2]	RW	ien_link_up	Interrupt enable for uplink port link status changes 0: disabled 1: enabled
[1]	RW	ien_tx_up	Indicator enable for the completion of transmitting a frame from the CPU by the uplink port 0: disabled 1: enabled
[0]	RW	ien_rx_up	Interrupt enable for frames on the uplink port to be received by the CPU 0: disabled 1: enabled

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GLB_IRQ_RAW

GLB_IRQ_RAW is a raw interrupt register. The register does not support soft reset. Writing 1 clears this register.

Offset Address	Register Name	Total Reset Value
0x1338	GLB_IRQ_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			iraw_mdio_finish	reserved				iraw_rxd_up	iraw_freeeq_up	iraw_stat_up	iraw_duplex_up	iraw_speed_up	iraw_link_up	iraw_tx_up	iraw_rx_up
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	WC	iraw_mdio_finish	Raw interrupt status for the MDIO to complete the operation required by the CPU 0: No interrupt is generated. 1: An interrupt is generated.
[11:8]	RO	reserved	Reserved
[7]	WC	iraw_rxd_up	Raw interrupt status (multi-packet interrupt) for a frame (frames) on the uplink port to be received by the CPU 0: No interrupt is generated. 1: An interrupt is generated.
[6]	WC	iraw_freeeq_up	Raw interrupt status for the TX queue of the uplink port to change from nonempty to empty, indicating that the TX queue buffer changes from nonempty to empty and the CPU can write a group of new frames to be transmitted. 0: No interrupt is generated. 1: An interrupt is generated.
[5]	WC	iraw_stat_up	Raw interrupt status for uplink port status changes, indicating that an interrupt is generated when the MDIO obtains the speed change, duplex mode change, and link status change of the PHY chip in auto-adaption mode. 0: No interrupt is generated. 1: An interrupt is generated.
[4]	WC	iraw_duplex_up	Raw interrupt status for uplink port duplex mode changes 0: No interrupt is generated. 1: An interrupt is generated.

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[3]	WC	iraw_speed_up	Raw interrupt status for uplink port speed mode changes 0: The interrupt is invalid. 1: The interrupt is valid. The speed mode changes. Writing 1 clears this register.
[2]	WC	iraw_link_up	Raw interrupt status for uplink port link status changes 0: No interrupt is generated. 1: An interrupt is generated.
[1]	WC	iraw_tx_up	Raw interrupt status for the completion of transmitting a frame from the CPU by the uplink port 0: No interrupt is generated. 1: An interrupt is generated.
[0]	WC	iraw_rx_up	Raw interrupt status for frames on the uplink port to be received by the CPU 0: No interrupt is generated. 1: An interrupt is generated.

GLB_MAC0_L32

GLB_MAC0_L32 is a lower 32-bit register for the filter table MAC0.

Offset Address	Register Name	Total Reset Value
0x1400	GLB_MAC0_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac0	Lower 32 bits of the filter table MAC0

GLB_MAC0_H16

GLB_MAC0_H16 is an upper 16-bit register for the filter table MAC0.

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Offset Address	Register Name	Total Reset Value
0x1404	GLB_MAC0_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up	reserved			mac0_up	reserved	flt_mac0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac0_up	Control for setting this filter to be used by the uplink port 0: The uplink port does not use this filter. 1: The uplink port uses this filter.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac0	Upper 16 bits of the filter table MAC0

GLB_MAC1_L32

GLB_MAC1_L32 is a lower 32-bit register for the filter table MAC1.

Offset Address	Register Name	Total Reset Value
0x1408	GLB_MAC1_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac1	Lower 32 bits of the filter table MAC1

GLB_MAC1_H16

GLB_MAC1_H16 is an upper 16-bit register for the filter table MAC1.

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Offset Address	Register Name	Total Reset Value
0x140C	GLB_MAC1_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up	reserved			mac1_up	reserved	flt_mac1															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac1_up	Control for setting this filter to be used by the uplink port 0: The uplink port does not use this filter. 1: The uplink port uses this filter.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac1	Upper 16 bits of the filter table MAC1

GLB_MAC2_L32

GLB_MAC2_L32 is a lower 32-bit register for the filter table MAC2.

Offset Address	Register Name	Total Reset Value
0x1410	GLB_MAC2_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac2	Lower 32 bits of the filter table MAC2

GLB_MAC2_H16

GLB_MAC2_H16 is an upper 16-bit register for the filter table MAC2.

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Offset Address	Register Name	Total Reset Value
0x1414	GLB_MAC2_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up	reserved			mac2_up	reserved	flt_mac2															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac2_up	Control for setting this filter to be used by the uplink port 0: The uplink port does not use this filter. 1: The uplink port uses this filter.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac2	Upper 16 bits of the filter table MAC2

GLB_MAC3_L32

GLB_MAC3_L32 is a lower 32-bit register for the filter table MAC3.

Offset Address	Register Name	Total Reset Value
0x1418	GLB_MAC3_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac3																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac3	Lower 32 bits of the filter table MAC3

GLB_MAC3_H16

GLB_MAC3_H16 is an upper 16-bit register for the filter table MAC3.

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Offset Address	Register Name	Total Reset Value
0x141C	GLB_MAC3_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up	reserved			mac3_up	reserved	flt_mac3															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac3_up	Whether to forward the frames received by the downlink port that match this filter to the CPU port when the downlink port enables this filter 0: Do not forward the frames. 1: Forward the frames.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac3	Upper 16 bits of the filter table MAC3

GLB_MAC4_L32

GLB_MAC4_L32 is a lower 32-bit register for the filter table MAC4.

Offset Address	Register Name	Total Reset Value
0x1420	GLB_MAC4_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac4																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac4	Lower 32 bits of the filter table MAC4

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GLB_MAC4_H16

GLB_MAC4_H16 is an upper 16-bit register for the filter table MAC4.

Offset Address	Register Name	Total Reset Value
0x1424	GLB_MAC4_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up		reserved				mac4_up		reserved		flt_mac4											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac4_up	Control for setting this filter to be used by the uplink port 0: The uplink port does not use this filter. 1: The uplink port uses this filter.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac4	Upper 16 bits of the filter table MAC4

GLB_MAC5_L32

GLB_MAC5_L32 is a lower 32-bit register for the filter table MAC5.

Offset Address	Register Name	Total Reset Value
0x1428	GLB_MAC5_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac5																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac5	Lower 32 bits of the filter table MAC5

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GLB_MAC5_H16

GLB_MAC5_H16 is an upper 16-bit register for the filter table MAC5.

Offset Address	Register Name	Total Reset Value
0x142C	GLB_MAC5_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up	reserved			mac5_up	reserved	flt_mac5															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac5_up	Control for setting this filter to be used by the uplink port 0: The uplink port does not use this filter. 1: The uplink port uses this filter.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac5	Upper 16 bits of the filter table MAC5

GLB_MAC6_L32

GLB_MAC6_L32 is a lower 32-bit register for the filter table MAC6.

Offset Address	Register Name	Total Reset Value
0x1430	GLB_MAC6_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac6																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac6	Lower 32 bits of the filter table MAC6

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GLB_MAC6_H16

GLB_MAC6_H16 is an upper 16-bit register for the filter table MAC6.

Offset Address	Register Name	Total Reset Value
0x1434	GLB_MAC6_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up	reserved			mac6_up	reserved	flt_mac6															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac6_up	Control for setting this filter to be used by the uplink port 0: The uplink port does not use this filter. 1: The uplink port uses this filter.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac6	Upper 16 bits of the filter table MAC6

GLB_MAC7_L32

GLB_MAC7_L32 is a lower 32-bit register for the filter table MAC7.

Offset Address	Register Name	Total Reset Value
0x1438	GLB_MAC7_L32	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flt_mac7																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	flt_mac7	Lower 32 bits of the filter table MAC7

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GLB_MAC7_H16

GLB_MAC7_H16 is an upper 16-bit register for the filter table MAC7.

Offset Address	Register Name	Total Reset Value
0x143C	GLB_MAC7_H16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										fw2cpu_up	reserved			mac7_up	reserved	flt_mac7															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	fw2cpu_up	Whether to forward the frames received by the uplink port that match this filter to the CPU port when the uplink port enables this filter 0: no 1: yes
[20:18]	RO	reserved	Reserved
[17]	RW	mac7_up	Control for setting this filter to be used by the uplink port 0: The uplink port does not use this filter. 1: The uplink port uses this filter.
[16]	RO	reserved	Reserved
[15:0]	RW	flt_mac7	Upper 16 bits of the filter table MAC7

UD_GLB_IRQN_SET

UD_GLB_IRQN_SET is a multi-packet interrupt configuration register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0340	UD_GLB_IRQN_SET	0x0800_003A

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				int_frm_cnt				reserved								age_timer															
Reset	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved

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[28:24]	RW	int_frm_cnt	These bits are used to set the multi-packet interrupt function. That is, how many packets must be received before a multi-packet interrupt can be reported. NOTE The minimum value of int_frm_cnt can be set to 1. In this case, multi-packet interrupt is equivalent to single-packet interrupt.
[23:16]	RO	reserved	Reserved
[15:0]	RW	age_timer	After the multi-packet interrupt function is enabled, if the number of received packets cannot reach the specified number of packets required for reporting the multi-packet interrupt after a period, this period is defined as the aging time for generating the multi-packet interrupt. NOTE age_timer is counted in the unit of the main clock cycle divided by 256.

UD_GLB_QLEN_SET

UD_GLB_QLEN_SET is a queue length configuration register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0344	UD_GLB_QLEN_SET	0x0000_2020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				iq_len				reserved				eq_len			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	iq_len	RX (packet RX) queue length configuration NOTE iq_len cannot be set to 0. Otherwise, it is forcibly set to 1. The sum of the set values of iq_len and eq_len cannot be greater than 64. Otherwise, the value (non-zero) of iq_len is firstly assigned and the value of eq_len is calculated by the formula: 64 – iq_len.
[7:6]	RO	reserved	Reserved
[5:0]	RW	eq_len	TX (packet TX) queue length configuration NOTE eq_len cannot be set to 0. Otherwise, it is forcibly set to 1.

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UD_GLB_FC_LEVEL

UD_GLB_FC_LEVEL is a traffic control register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0348	UD_GLB_FC_LEVEL	0x3018_0508

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																	qlimit_ena	qlimit_up						reserved				qlimit_down			
Reset	0	0	1	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14]	RW	qlimit_ena	Traffic control enable for RX queue 0: Disabled (do not transmit the traffic control message according to the status of RX queue). 1: Enabled (transmit the traffic control message according to the status of RX queue).
[13:8]	RW	qlimit_up	Upper limit of traffic control for RX queue. When the free space of the RX queue is less than the upper limit, if traffic control for RX queue is enabled, the traffic control message is transmitted to the peer end. NOTE • If the upper limit qlimit_up is set to 0, the RX queue fails to enter the traffic control status. • The upper limit qlimit_up must be greater than the lower limit qlimit_down.
[7:6]	RO	reserved	Reserved
[5:0]	RW	qlimit_down	Lower limit of traffic control for RX queue. When the free space of the RX queue is equal to or greater than the upper limit, if the RX queue is in traffic control state, the current traffic control is stopped.

UD_GLB_CAUSE

UD_GLB_CAUSE is a cause register for the CPU to which the packet is transmitted.

The register does not support soft reset.

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Offset Address	Register Name	Total Reset Value
0x034C	UD_GLB_CAUSE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														mact_cause	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RO	mact_cause	Packet matching result types by querying the MAC table 000: forced forwarding 001: Packet whose destination MAC address is the local MAC address. 010: broadcast packet 011: packet matching the MAC table 100: multicast packet not matching the MAC table 101: unicast packet not matching the MAC table Others: reserved

UD_GLB_RXFRM_SADDR

UD_GLB_RXFRM_SADDR is an RX frame start address register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0350	UD_GLB_RXFRM_SADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rxfrm_saddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	rxfrm_saddr	Start address of the RX frame

UD_GLB_IQFRM_DES

UD_GLB_IQFRM_DES is an RX frame descriptor register.

The register does not support soft reset.

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Offset Address	Register Name	Total Reset Value
0x0354	UD_GLB_IQFRM_DES	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																fd_in_addr				fd_in_len												
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17:12]	RO	fd_in_addr	Relative address of the first frame to be received in the input queue (IQ). It serves as the index (0 to iq_len-1) of the absolute address for storing the frames.
[11:0]	RO	fd_in_len	Length of the frame to be received in the RX queue

UD_GLB_IQ_ADDR

UD_GLB_IQ_ADDR is an RX frame header address register.

The register does not support soft reset.

NOTE

If the address assigned by software is not word aligned, the logic writes data according to the word aligned address. In this case, the previously written data is invalid. For example, if the configured header address of a frame is 0xF000_8002 (non-word-aligned address), the logic writes 0x00 or other data to both the 0xF000_8000 and 0xF000_8001 addresses. Then, the logic writes the first byte (valid data) of the RX frame to the 0xF000_8002 address, writes the second byte (valid data) of the RX frame to the 0xF000_8003 address. Subsequent data is written to the buffer in sequence. If the configured header address of the RX frame is other non-word-aligned address, the logic writes data in a similar way.

Offset Address	Register Name	Total Reset Value
0x0358	UD_GLB_IQ_ADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	startaddr_iq																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	startaddr_iq	Header address (configured by the CPU) of the storage space corresponding to the RX frame. The RX frame requests the bus according to this address.

UD_GLB_BFC_STAT

UD_GLB_BFC_STAT is a counter for traffic control status of forward buffer and aging time of multi-packet interrupt.

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The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x035C	UD_GLB_BFC_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timerover_cnt																flowctrl_cnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	timerover_cnt	Register for the count of multi-packet interrupt aging time counter overflow events (the count reaches the configured value) **NOTE** If the value of timerover_cnt is too large in a unit of time, it indicates that UD_GLB_IRQN_SET[int_frm_cnt] is set improperly. Multi-packet interrupt is triggered by the aging time. Therefore, the configured value must be reduced.
[15:0]	RO	flowctrl_cnt	Register for the count of the forward buffer of the uplink or downlink port entering the traffic control status **NOTE** If the value of flowctrl_cnt is too large in a unit of time, it indicates that UD_GLB_FC_LEVEL[blimit_up] or UD_GLB_FC_LEVEL[blimit_down] is set to a too small value, or the external network condition is worsened. In this case, the configured value may be reduced.

UD_GLB_EQ_ADDR

UD_GLB_EQ_ADDR is a TX queue header address register.

The register does not support soft reset.

NOTE
If the header address of the TX frame is not word aligned, the logic reads data according to the word aligned address. In this case, the previously read data is invalid and discarded. For example, if the configured header address of the TX frame is 0xF000_8102 (non-word-aligned address), the logic directly discards the byte data read from the 0xF000_8100 and 0xF000_8101 addresses. Then, the logic considers the data read from the 0xF000_8102 address as the first byte (valid data) of the TX frame and considers the data read from the 0xF000_8103 address as the second byte (valid data) of the TX frame. All subsequent data is valid (until the data of the specified frame length is read). If the configured header address of the TX frame is other non-word-aligned address, the logic reads data in a similar way.

Offset Address	Register Name	Total Reset Value
0x0360	UD_GLB_EQ_ADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	add_fd_addr_out																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description

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[31:0]	RW	add_fd_addr_out	Header address of the TX frame added by the CPU to the TX queue

UD_GLB_EQFRM_LEN

UD_GLB_EQFRM_LEN is a TX queue frame length configuration register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0364	UD_GLB_EQFRM_LEN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								add_fd_len_out							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10:0]	RW	add_fd_len_out	Length of the TX frame added by the CPU to the TX queue Configure this register to trigger hardware so that software can write the header address and length of the TX frame to the TX queue for transmission. When transmitting a frame, software must write the header address of the frame before the length of the frame. Note: The frames whose add_fd_len_out is less than 20 bytes or greater than 1600 bytes are discarded. In other words, the allowed range is from 20 bytes to 1600 bytes.

UD_GLB_QSTAT

UD_GLB_QSTAT is a queue status register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0368	UD_GLB_QSTAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				iq_in_index				reserved				cpuw_index				reserved				eq_in_index				reserved				eq_out_index			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved

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[29:24]	RO	iq_in_index	RX index of the RX (packet RX) queue
[23:22]	RO	reserved	Reserved
[21:16]	RO	cpuw_index	RX index of frame header address of the RX (packet RX) queue
[15:14]	RO	reserved	Reserved
[13:8]	RO	eq_in_index	RX index of frame descriptor of the TX (packet TX) queue
[7:6]	RO	reserved	Reserved
[5:0]	RO	eq_out_index	TX index of frame descriptor of the TX (packet TX) queue

UD_GLB_ADDRQ_STAT

UD_GLB_ADDRQ_STAT is an address queue status register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x036C	UD_GLB_ADDRQ_STAT	0x0300_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved						cpuaddr_in_rdy	eq_in_rdy	reserved				cpu_cnt				reserved				iq_cnt				reserved				eq_cnt			
Reset	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25]	RO	cpuaddr_in_rdy	Whether the CPU can configure the frame header address of the RX queue 0: The CPU cannot configure the frame header address of the RX queue. 1: The CPU can configure the frame header address of the RX queue. **NOTE** The values of cpuaddr_in_rdy and eq_in_rdy are set to 0 during reset. The values, however, are set to 1 by the circuit immediately after reset. In other words, after reset, the iq address queue and eq descriptor queue are configurable.

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[24]	RO	eq_in_rdy	Whether the CPU can configure the frame descriptor (header address and length) of the TX queue 0: The CPU cannot configure the frame descriptor (header address and length) of the TX queue. 1: The CPU can configure the frame descriptor (header address and length) of the TX queue.
[23:22]	RO	reserved	Reserved
[21:16]	RO	cpu_cnt	Header address count for available frames assigned by the CPU to the RX queue
[15:14]	RO	reserved	Reserved
[13:8]	RO	iq_cnt	Used length of the RX queue (0 to iq_len)
[7:6]	RO	reserved	Reserved
[5:0]	RO	eq_cnt	Used length of the TX queue (0 to eq_len)

UD_GLB_FC_TIMECTRL

UD_GLB_FC_TIMECTRL is a traffic control time configuration register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0370	UD_GLB_FC_TIMECTRL	0x07FF_86A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								flux_timer_cfg												flux_timer_inter											
Reset	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:17]	RW	flux_timer_cfg	Traffic limit time interval counter, which is used to count the frequency division clock generated by flux_timer_inter. If this counter is set to 0, traffic limit is not performed.
[16:0]	RW	flux_timer_inter	Traffic limit time granularity counter, which is used to count the ETH working clock (for details, see the PERI_CRG59 register). The default field value is 100000. For example, if the default ETH working clock is 54 MHz and the time granularity is 1 ms, this field is set to 54000, which is calculated as follows: 1 ms/(1/54 MHz) = [1 x 10^(-3)]/[(1/54) x 10^(-6)]=54 x 10^3=54000

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UD_GLB_FC_RXLIMIT

UD_GLB_FC_RXLIMIT is a traffic control limit configuration register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0374	UD_GLB_FC_RXLIMIT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												flux_cfg																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:0]	RW	flux_cfg	Traffic limit upper threshold register. This group of bits is used to limit the number of frames received by software in the traffic limit time interval. The frames received after the configured upper threshold is exceeded are selectively discarded or received according to the configuration. When this group of bits is all set to 0, it indicates that traffic limit is not performed.

UD_GLB_FC_DROPCTRL

UD_GLB_FC_DROPCTRL is a packet drop control register for traffic limit.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0378	UD_GLB_FC_DROPCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													flux_	flux_	broa
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	flux_uni	Whether unicast packets are discarded when the upper threshold of traffic limit is exceeded 0: Do not discard unicast packets. 1: Discard unicast packets.
[1]	RW	flux_multi	Whether multicast packets are discarded when the upper threshold of traffic limit is exceeded 0: Do not discard multicast packets. 1: Discard multicast packets.

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[0]	RW	flux_broad	Whether broadcast packets are discarded when the upper threshold of traffic limit is exceeded 0: Do not discard broadcast packets. 1: Discard broadcast packets.

UD_GLB_RX_COE_EN

UD_GLB_RX_COE_EN is an RX COE enable register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0380	UD_GLB_RX_COE_EN	0x8000_E000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rx_coe_en	reserved															iphrd_drop	pro_drop	I6udp_drop	reserved												
Reset	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	rx_coe_en	COE enable in the RX direction
[30:16]	RO	reserved	Reserved
[15]	RW	iphdr_drop	Packet discarding enable when a checksum error occurs in the IPv4 IP header 0: The packet is transparently transmitted. 1: The packet is discarded.
[14]	RW	pro_drop	TCP/UDP packet discarding enable when a checksum error occurs 0: The packet is transparently transmitted. 1: The packet is discarded.
[13]	RW	i6udp_drop	Discarding enable for the invalid IPv6 UDP packet when the checksum domain is 0 0: The packet is transparently transmitted. 1: The packet is discarded.
[12:0]	RO	reserved	Reserved

5.6.4 Description of the Statistics Counter Control Registers

UD_STS_PORTCNT

UD_STS_PORTCNT is a port status counter.

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The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0584	UD_STS_PORTCNT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	rxsof_cnt	rxeof_cnt	rxcrcok_cnt	rxcrcbad_cnt	txsof_cnt	txeof_cnt	txcrcok_cnt	txcrcbad_cnt
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	rxsof_cnt	Count of the frame headers received by the port
[27:24]	RO	rxeof_cnt	Count of the frame trailers received by the port
[23:20]	RO	rxcrcok_cnt	Count of the frames without CRC errors received by the port
[19:16]	RO	rxcrcbad_cnt	Count of the frames with CRC errors received by the port
[15:12]	RO	txsof_cnt	Count of the frame headers transmitted by the port
[11:8]	RO	txeof_cnt	Count of the frame trailers transmitted by the port
[7:4]	RO	txcrcok_cnt	Count of the frames without CRC errors transmitted by the port
[3:0]	RO	txcrcbad_cnt	Count of the frames with CRC errors transmitted by the port

UD_PORT2CPU_PKTS

UD_PORT2CPU_PKTS is a register for the total number of packets received by the CPU from the uplink or downlink port.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x05A0	UD_PORT2CPU_PKTS	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				pkts_cpu			
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	WC	pkts_cpu	Total number of the packets received by the CPU port from the uplink or downlink port. Writing 0 clears this register. Writing 1 has no effect.

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UD_CPU2IQ_ADDRCNT

UD_CPU2IQ_ADDRCNT is a register for the count of configuring packet receiving address queue by the CPU.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x05A4	UD_CPU2IQ_ADDRCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																addr_cpu															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	WC	addr_cpu	Count of configuring packet receiving address queue by the CPU successfully. Writing 0 clears this register. Writing 1 has no effect.

UD_RX_IRQCNT

UD_RX_IRQCNT is a register for the count of reporting single-packet interrupt by the uplink or downlink port.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x05A8	UD_RX_IRQCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																pkts_port															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	WC	pkts_port	Count of the frame RX interrupts reported by the uplink or downlink port. Writing 0 clears this register. Writing 1 has no effect.

UD_CPU2EQ_PKTS

UD_CPU2EQ_PKTS is a register for the total number of packets transmitted by the CPU to the uplink or downlink port.

The register does not support soft reset.

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Offset Address	Register Name	Total Reset Value
0x05AC	UD_CPU2EQ_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																pkts_cpu2tx															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	WC	pkts_cpu2tx	Total number of packets transmitted by the CPU to the uplink or downlink port. Writing 0 clears this register. Writing 1 has no effect.

5.6.5 Description of the Statistics Result Registers

Statistics result registers can be configured in two modes: read only and read clear. If UD_MAC_SET[cntr_rdclr_en] is set to 1, it indicates the read clear mode. If UD_MAC_SET[cntr_rdclr_en] is set to 0, it indicates the read only mode. The following registers are described only in read only mode.

UD_RX_DVCNT

UD_RX_DVCNT is an RXDV rising edge count register.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0600	UD_rx_DVCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rxdvrise																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	rxdvrise	Count of all RXDV rising edges

UD_RX_OCTS

UD_RX_OCTS is a register for the total number of bytes received.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0604	UD_RX_OCTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

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Name	ifinoctets																											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	ifinoctets	Count of all received bytes, including the bytes in correct frames, error frames, and preambles. The frames without valid start of frame delimiters (SFDs) are not counted.

UD_RX_RIGHTOCTS

UD_RX_RIGHTOCTS is a register for the total number of bytes of received correct packets.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0608	UD_RX_RIGHTOCTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	octets_rx																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	octets_rx	Count of the received bytes, including the bytes in correct frames and error frames but excluding the bytes in preambles. The frames without valid SFDs are not counted.

UD_HOSTMAC_PKTS

UD_HOSTMAC_PKTS is a register for the number of packets matching the local MAC address.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x060C	UD_HOSTMAC_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	local_mac_match																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	local_mac_match	Count of correct RX frames whose destination MAC address is the same as the local MAC address, excluding short frames, long frames, frames with CRC errors, pause frames, and error TX frames.

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UD_RX_RIGHTPKTS

UD_RX_RIGHTPKTS is a register for the total number of packets received by the port.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0610	UD_RX_RIGHTPKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	pkts																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	pkts	Count of all frames

UD_RX_BROADPKTS

UD_RX_BROADPKTS is a register for the number of correct broadcast packets.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0614	UD_RX_broadpkts	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	broadcastpkts																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	broadcastpkts	Count of broadcast frames with valid length and without CRC errors, excluding pause frames and error TX frames

UD_RX_MULTPKTS

UD_RX_MULTPKTS is a register for the number of correct multicast packets.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0618	UD_RX_multpkts	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	multicastpkts																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	multicastpkts	Count of multicast frames with valid length and without CRC errors, excluding pause frames and error TX frames

UD_RX_UNIPKTS

UD_RX_UNIPKTS is a register for the number of correct unicast packets.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x061C	UD_RX_UNIpkts	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ifinucastpkts																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	ifinucastpkts	Count of unicast frames with valid length and without CRC errors, excluding pause frames and error TX frames

UD_RX_ERRPKTS

UD_RX_ERRPKTS is a register for the total number of incorrect packets.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0620	UD_RX_ERRPKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ifinerrors																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	ifinerrors	Count of all error frames, including frames with CRC errors, short frames, long frames, and error TX frames

UD_RX_CRCERR_PKTS

UD_RX_CRCERR_PKTS is a register for the count of CRC errors.

The register does not support soft reset.

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Offset Address	Register Name	Total Reset Value
0x0624	UD_RX_crcerr_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	crcerr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	crcerr	Count of RX frames with valid length (non short and long frames) but with CRC or alignment errors

UD_RX_LENERR_PKTS

UD_RX_LENERR_PKTS is a register for the number of packets with invalid length.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0628	UD_RX_LENERR_pkts	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	abnormalsizepkts																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	abnormalsizepkts	Count of frames (short frames and long frames) with invalid length (less than the set minimum valid length or greater than the set maximum valid length)

UD_RX_OCRCERR_PKTS

UD_RX_OCRCERR_PKTS is a register for the number of packets with odd nibbles and CRC errors.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x062C	UD_RX_OCRCERR_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dot3alignmenterr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dot3alignmenterr	Received frames with odd nibbles and CRC errors

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UD_RX_PAUSE_PKTS

UD_RX_PAUSE_PKTS is a register for the number of received pause packets.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0630	UD_RX_PAUSE_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dot3pause																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dot3pause	Count of received pause frames

UD_RF_OVERCNT

UD_RF_OVERCNT is a register for the count of RXFIFO overflow events.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0634	UD_RF_OVERCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dropevents																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dropevents	Accumulative count of RXFIFO overflow events during the reception of frames

UD_FLUX_TOL_IPKTS

UD_FLUX_TOL_IPKTS is a register for the total number of received packets allowed by the traffic limit.

The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0638	UD_flux_TOL_IPKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flux_frame_cnt																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	flux_frame_cnt	Total count of correct RX frames allowed by the traffic limit, excluding short frames, long frames, frames with CRC errors, pause frames, and error TX frames

UD_FLUX_TOL_DPKTS

UD_FLUX_TOL_DPKTS is a register for the total number of packets discarded due to traffic limit. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x063C	UD_flux_TOL_DPKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	flux_drop_cnt																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	flux_drop_cnt	Count of correct frames discarded due to traffic limit, excluding short frames, long frames, frames with CRC errors, pause frames, and error TX frames

UD_MN2CPU_PKTS

UD_MN2CPU_PKTS is a register for the number of packets not forwarded to the CPU port due to MAC limit. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x064C	UD_MN2CPU_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mac_not2cpu_pkts																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	mac_not2cpu_pkts	Number of packets not forwarded to the CPU port due to MAC limit

UD_TX_PKTS

UD_TX_PKTS is a register for the total number of packets transmitted successfully. The register does not support soft reset.

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Offset Address	Register Name	Total Reset Value
0x0780	UD_TX_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	pkts_tx																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	pkts_tx	Count of all configured TX frames, excluding the frames discarded due to timeout and the TX frames whose length of UD_GLB_EQFRM_LEN is not within valid range

UD_TX_BROADPKTS

UD_TX_BROADPKTS is a register for the number of broadcast packets transmitted successfully. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0784	UD_TX_BROADPKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	broadcastpkts_tx																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	broadcastpkts_tx	Count of broadcast frames transmitted successfully (excluding retransmission)

UD_TX_MULTPKTS

UD_TX_MULTPKTS is a register for the number of multicast packets transmitted successfully. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0788	UD_TX_MULTPKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	multicastpkts_tx																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	multicastpkts_tx	Count of multicast frames transmitted successfully (excluding retransmission)

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UD_TX_UNIPKTS

UD_TX_UNIPKTS is a register for the number of unicast packets transmitted successfully. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x078C	UD_TX_UNIPKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ifoutucastpkts_tx																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	ifoutucastpkts_tx	Count of unicast frames transmitted successfully (excluding retransmission)

UD_TX_OCTS

UD_TX_OCTS is a register for the total number of transmitted bytes. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0790	UD_TX_OCTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	octets_tx																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	octets_tx	Total count of transmitted bytes, including the bytes of retransmit frames, correct frames, and error frames, but excluding the preamble bytes

UD_TX_PAUSE_PKTS

UD_TX_PAUSE_PKTS is a register for the number of transmitted pause frames. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0794	UD_TX_PAUSE_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dot3outpause																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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Bits	Access	Name	Description
[31:0]	RO	dot3outpause	Count of transmitted pause frames

UD_TX_RETRYCNT

UD_TX_RETRYCNT is a register for the total count of retransmission. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x0798	UD_TX_RETRYCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	retry_times_tx																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	retry_times_tx	Total count of retransmissions of TX frames

UD_TX_COLCNT

UD_TX_COLCNT is a register for the total count of collisions. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x079C	UD_TX_COLCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	collisions																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	collisions	Count of collisions

UD_TX_LC_PKTS

UD_TX_LC_PKTS is a register for the number of packets with late collision. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x07A0	UD_TX_LC_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dot3latecol																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dot3latecol	Count of packets with late collision

UD_TX_COLOK_PKTS

UD_TX_COLOK_PKTS is a register for the number of packets transmitted successfully with collisions. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x07A4	UD_TX_COLOK_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dot3col_ok																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dot3col_ok	Count of packets transmitted successfully with collisions

UD_TX_RETRY15_PKTS

UD_TX_RETRY15_PKTS is a register for the number of packets discarded due to more than 15 times of retransmission. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x07A8	UD_TX_RETRY15_PKTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dot3excessivecol																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dot3excessivecol	Count of the packets discarded due to more than 15 times of retransmission

UD_TX_RETRYN_PKTS

UD_TX_RETRYN_PKTS is a register for the number of packets with the count of collisions being equal to the threshold. The register does not support soft reset.

Offset Address	Register Name	Total Reset Value
0x07AC	UD_TX_RETRYN_PKTS	0x0000_0000

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Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dot3colcnt																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	dot3colcnt	Count of packets with the count of collisions being equal to the threshold. This register is set by UD_MAC_SET[colthreshold]

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6 Video Encoder

6.1 Introduction

The encoding unit consists of the video encoding units AVC and JPEG encoder (JPGE). The AVC supports H.264 encoding. and the JPGE supports JPEG encoding.

6.2 AVC

6.2.1 Overview

The AVC is a CODEC that supports H.264 protocol and performs encoding by using hardware. It features low CPU usage, low bus bandwidth, short delay, and low power consumption.

6.2.2 Features

The VENC has the following features:

- ITU-T H.264 high profile/main profile/baseline profile@L5.1 encoding
  - Motion compensation with 1/2 or 1/4 pixel precision
  - Four subblock types of 16x16, 16x8, 8x16, and 8x8 for inter-prediction
  - Prediction modes of intra4x4, intra8x8, and intra16x16
  - Trans4x4 and Trans8x8
  - Context-based adaptive binary arithmetic coding (CABAC) and context-based adaptive variable-length coding (CAVLC) entropy encoding
  - De-blocking filtering
  - IPCM encoding
- SVC time-domain layering (SVC-T)
- Input picture format of semi-planar YCbCr4:2:0
- H.264 video encoding

  Hi3518E V200:
  - 720p@30 fps+VGA@30 fps+QVGA@30 fps
  - 1080p@20 fps+D1@20 fps+1080p@1 fps JPEG snapshot, maximum performance of 1080p@20 fps for a single channel

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Hi3518E V201:

- 720P@30 fps+QVGA@30 fps
- 960P@25 fps+D1@25 fps

Hi3516C V200: 1080p@30 fps+VGA@30 fps

- Configurable picture resolutions
  - Minimum picture resolution: 160 x 64
  - Maximum picture resolution: 1920 x 1920
- Step of the picture width or height: 2
- Region of interest (ROI) encoding
  - A maximum of eight ROIs
  - ROI encoding control (enabled or disabled)
- On-screen display (OSD) encoding protection

  OSD encoding protection control (enabled or disabled)
- OSD front-end overlaying
  - OSD overlaying before encoding for a maximum of eight regions
  - OSD overlaying anywhere (the maximum OSD size is the picture size)
  - 129-level alpha blending
  - OSD overlaying control
- Color-to-gray encoding
- Two bit rate control modes: constant bit rate (CBR) and variable bit rate (VBR)
- Fixed QP encoding
- Output bit rate ranging from 2 kbit/s to 100 Mbit/s

6.2.3 Function Description

Figure 6-1 shows the encoding functional block diagram of the AVC.

Based on related protocols and algorithms, the AVC supports motion estimation, inter-prediction, intra-prediction, motion vector prediction, transform/quantization, inverse transform/inverse quantization, variable length code (VLC) encoding, stream generation, and de-blocking filtering. The video firmware (FMW) controls the bit rate and handles interrupts.

Before the AVC is enabled for video encoding, the software allocates three types of buffers for the AVC in the external DDR SDRAM:

- Input picture buffer

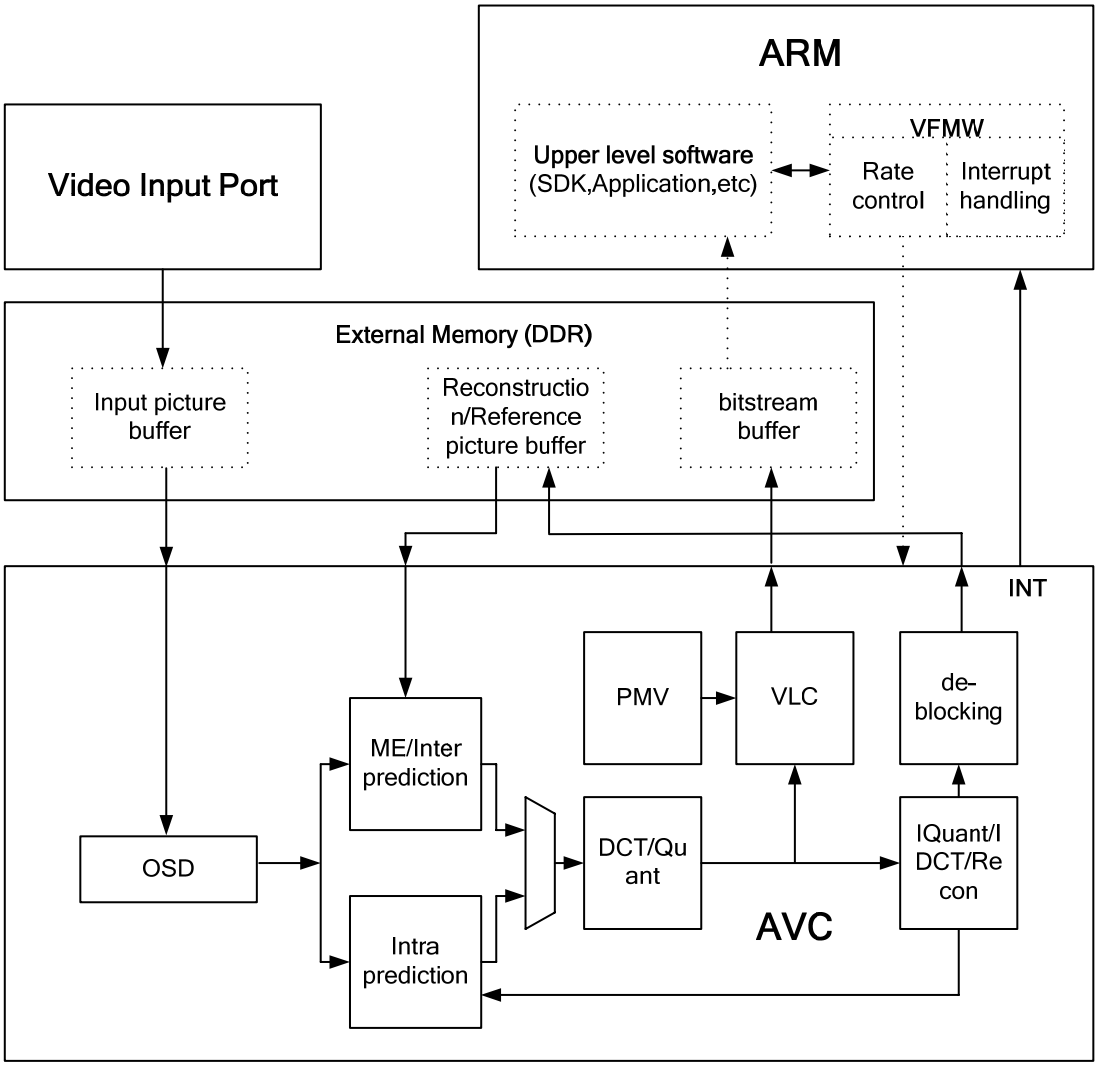
  The AVC reads the source pictures to be encoded from this buffer during encoding. This buffer is typically written by the VICAP module.
- Reconstruction/reference picture buffer

  The AVC writes reconstruction pictures to this buffer during encoding. These reconstruction pictures are used as the reference pictures of subsequent pictures. During the encoding of P frames, reference pictures are read from this buffer.
- Stream buffer

  This buffer stores encoded streams. The AVC writes streams to this buffer during encoding. This buffer is read by software.

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Figure 6-1 Encoding functional block diagram of the AVC



6.3 JPGE

6.3.1 Overview

The JPGE provides high-performance encoding performance by using hardware. It supports the snapshot of pictures with 8K x 8K resolution or HD MJPEG encoding.

6.3.2 Features

The JPGE has the following features:

- ISO/IEC 10918-1 (CCITT T.81) baseline process (DCT sequential) encoding
- Microprogrammed control unit (MCU) using the interleaved sequence
- Multiple input picture formats
  - Planar YCbCr4:2:0

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  - Planar YCbCr4:2:2
  - Planar YCbCr4:4:4
  - Semi-planar YCbCr4:2:0
  - Semi-planar YCbCr4:2:2
  - Package YUYV
- JPEG encoding
  - 2-megapixel@8 fps
- Configurable picture resolutions
  - Minimum picture resolution: 32 x 32
  - Maximum picture resolution: 8192 x 8192
- Picture width or height step of 4
- Configurable quantization tables

  An independent quantization table for the Y component, Cb component, and Cr component respectively
- OSD front-end overlaying
  - OSD overlaying before encoding for a maximum of eight regions
  - OSD overlapping with the maximum size of the source picture and within the picture position range
  - 129-level alpha blending
  - OSD overlaying control

6.3.3 Function Description

Figure 6-2 shows the functional block diagram of the JPGE.

Based on the protocols that require a large number of operands, the JPGE supports OSD, level shift, discrete cosine transform (DCT), quantization, scanning, VLC encoding, and stream generation. The VFMW configures quantization tables and handles interrupts.

Before the JPGE is enabled for video encoding, the software allocates two types of buffers for the JPGE in the external DDR SDRAM:

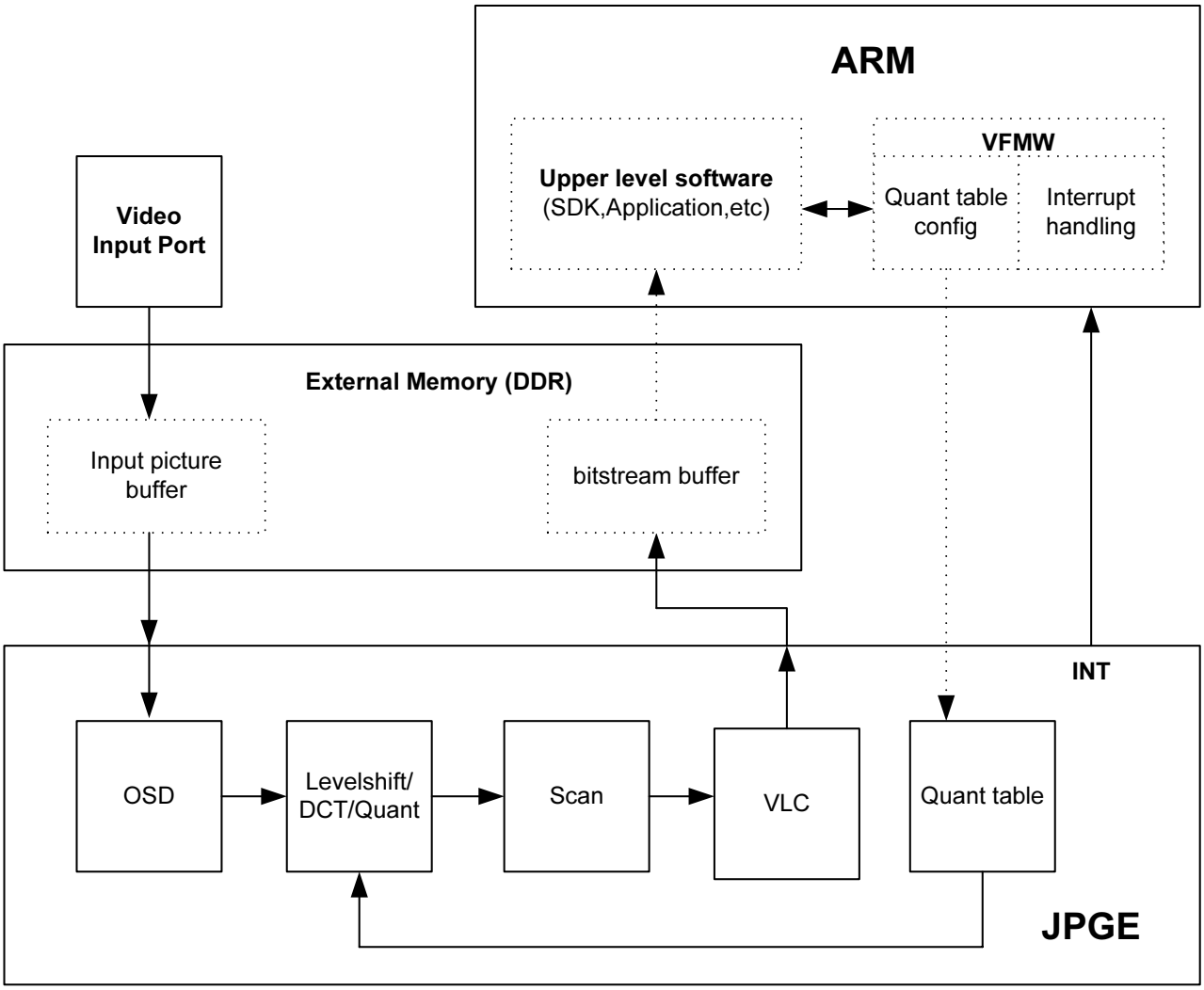
- Input picture buffer

  The JPGE reads the source pictures to be encoded from this buffer during encoding. This buffer is generally written by the VICAP module.
- Stream buffer

  This buffer stores encoded streams. The JPGE writes streams to this buffer during encoding. This buffer is read by software.

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Figure 6-2 Encoding functional block diagram of the JPGE



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7 Video and Graphics Processing

7.1 TDE

7.1.1 Overview

The two-dimensional engine (TDE) draws graphics using hardware. This significantly reduces the CPU usage and improves the utilization of the memory bandwidth. The TDE reads and writes the information about the bitmap data, filtering/scaling coefficients, parameters of linked list nodes, and linked list over the advanced eXtensible bus (AXI) master interface. By using the advanced peripheral bus (AHB) interface, the TDE obtains the CPU register configuration information.

The graphics data interface includes two channels. Their functions are as follows:

- Channel 1 implements direct copy and direct filling during a single-source operation.
- Channel 2 implements complicated functions such as scaling during a single-source operation.
- Channel 1 works with channel 2 to implement color blending and process the images in macroblock format.

7.1.2 Function Description

The TDE has the following features:

- Channel 1 supports the formats of ARGB4444, ARGB1555, ARGB8888, RGB444, RGB565, RGB888, ARGB8565, CLUT1, CLUT4, CLUT8, A1, A8, halfword.
- Channel 2 supports the formats of ARGB4444, ARGB1555, ARGB8888, RGB444, RGB565, RGB888, ARGB8565, CLUT1, CLUT4, CLUT8, A1, A8.
- The output image supports the formats of ARGB4444, ARGB1555, ARGB8888, RGB444, RGB565, RGB888, ARGB8565, CLUT1, CLUT4, CLUT8, A1, A8, byte, halfword.
- Supports only the little-endian system.
- Allows the formats of source bitmap 1, source bitmap 2, and output bitmaps to be configured separately.
- Supports gamma correction and adjustable contrast and luminance.
- Supports direct copy.
- Supports direct filling.

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- Supports 2D resize.
- Supports clip.
- Supports alpha blending.
- Supports colorkey.
- Supports clip mask.
- Supports inverse scanning.
- Provides status interrupts.

7.2 VPSS

7.2.1 Overview

The video processing subsystem (VPSS) implements video processing. It supports Gaussian 3D adaptive noise reduction (NR) as well as video covering, cropping, scaling, and single-component luminance processing.

The VPSS has the following features:

- Processing of video sources with 2048-pixel width using a single frame in online or offline mode
- A maximum of four video output channels including three encoding channels and one display channel
- Eight video regions covered by any filled quadrilaterals for each of the three output encoding channels. Eight sets of covering coordinates are fixed.
- Video cropping four three output channels (two encoding channels and one display channel)
- Video data compression
- Register configuration using the linked list in offline mode and register configuration using the APB in online mode
- Span of 4 KB boundary
- Input or output data format of semi-planar 420/422
- Outstanding configuration
- Memory low-power mode by using clock gating
- Low delay
- Single-component luminance processing

7.2.2 Features

The following describes the major features:

- Gaussian NR: The NR module removes the Gaussian noises from pictures by configuring parameters. Then pictures become smooth and the encoding bit rate is reduced.
- Scaling: Low-frequency filtering is supported when the input and output resolution are different. At most 15 times zoom out and 2 times zoom in are supported. Both zoom in and zoom out are supported in offline mode, and only zoom out is supported in online mode.

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7.3 VGS

7.3.1 Overview

The video graphics system (VGS) implements video and graphics processing. The functions include OSD overlaying, luminance region statistics, video cropping, video covering, rotation, and distortion correction.

The VGS has the following features:

- Processing of video sources with the 2048-pixel width by using a single frame in offline mode
- One video output channel
- Overlaying of the OSD and video in one region
- OSD input format of ARGB1555, ARGB4444, or ARGB8888
- Configuration of register linked lists
- Span of 4 KB boundary
- Linear storage for input/output data
- Video data compression
- Rotation by 90º or 270º
- Distortion correction for the picture with the maximum resolution of 2048 x 2048
- One video region covered by any filled or hollow quadrilateral
- Video cropping
- Outstanding configuration
- Memory low-power mode by using clock gating
- Statistics of the region luminance sum
- Single-component luminance processing

7.3.2 Features

The following describes the major features:

- Video covering: A video region is covered by any filled or hollow quadrilateral.
- Scaling: Low-frequency filtering is supported when the input and output resolution are different. At most 15.5 times zoom out and 16 times zoom in are supported.

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Tables

Table 8-1 Summary of IVE registers (base address: 0x206A_0000)....8-15

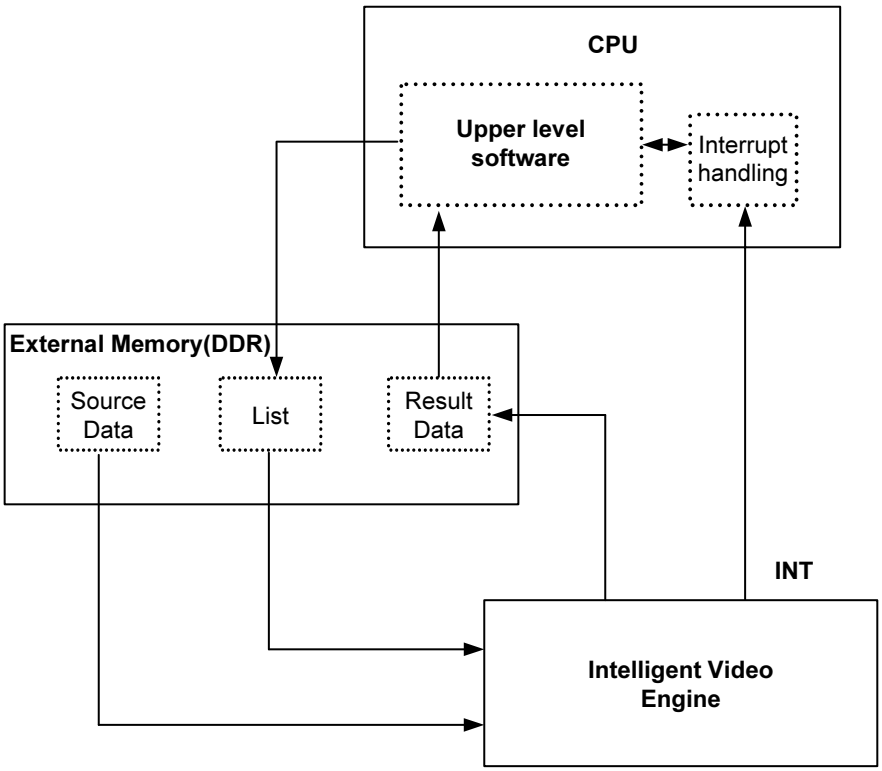
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8 IVE

8.1 Overview

The intelligent video engine (IVE) module is used to accelerate hardware processing. It provides a series of basic calculation functions and some time-consuming functions used in the intelligent analysis algorithm.

Figure 8-1 Position of the IVE in the system



8.2 Function Description

The IVE has the following features:

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- DMA: Supports direct copying, alternative copying, and memory filling.
- Filter: Supports 5x5 template filtering.
- Color space conversion (CSC): Supports the color space conversion of YUV2RGB, YUV2HSV, YUV2LAB, and RGB2YUV.
- FilterAndCSC: Combines 5x5 template filtering and CSC.
- Sobel: Supports the Sobel-like gradient calculation of 5x5 template.
- MagAndAng/Canny: Supports 5x5 template calculation and Canny edge extraction.
- Erode: Supports 5x5 template erode.
- Dilate: Supports 5x5 template dilate.
- Thresh\Thresh_S16\Thresh_U16: Supports image thresh processing.
- And\Or\Xor: Supports the AND, OR, or XOR operation on two images.
- Add\Sub: Deals with the weighted plus and minus of two images.
- Integ: Supports the integral calculation.
- Hist: Supports the histogram statistics.
- Map: Assigns values to images by using 256-level mapping.
- 16BitTo8Bit: Performs linear conversion from 16-bit data to 8-bit data.
- OrdStatFilter: Supports the sequence statistic filtering, including the median filtering, maximum filtering, and minimum filtering.
- NCC: Calculates the correlation coefficients of two images with the same size.
- CCL: Marks connected regions.
- NormGrad: Performs a normalized gradient calculation.
- Sum of absolute difference (SAD): Supports the calculation of the accumulated sum of the absolute pixel differences corresponding to two images by block.
- Supports separate soft reset.
- Supports 128-bit AXI bus and 32-bit APB.
- Supports linked-list interrupt, node interrupt, and timeout interrupt.
- Supports input formats includingSP 400, semi-planar 420 (SP420), semi-planar 422 (SP422), package, and planar.
- Supports output formats including SP400, SP420, SP422, package, and planar.
- Supports non-16-byte-aligned read and write addresses for some operators.

8.3 Operating Mode

8.3.1 Input and Output Data Formats

The w and h in Figure 8-2 to Figure 8-12 indicate the width and height of images in pixels. Unless otherwise specified, the sequence for storing data is the same as that in the little endian system. The following sections use word as the storage unit. In actual applications, the data alignment format varies with operators. The input and output format for operators described in section 8.3.2 "Supported Functions" are also different.

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Figure 8-2 SP422 storage format

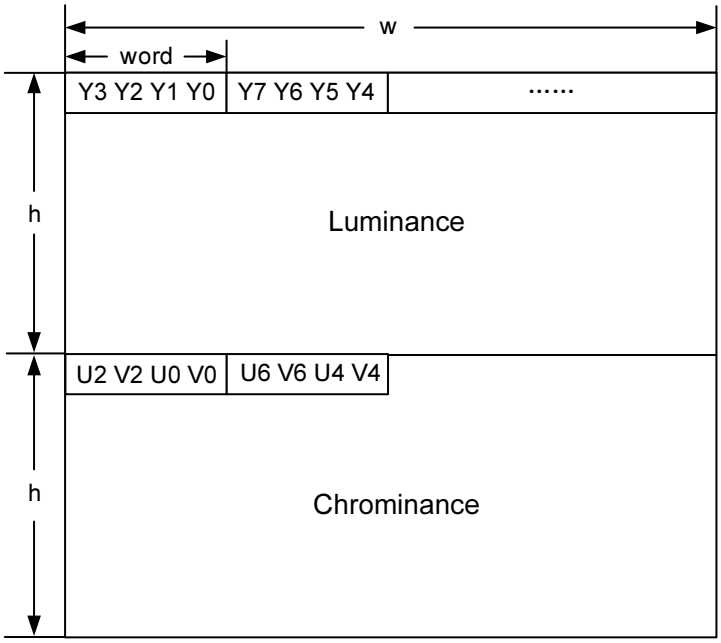
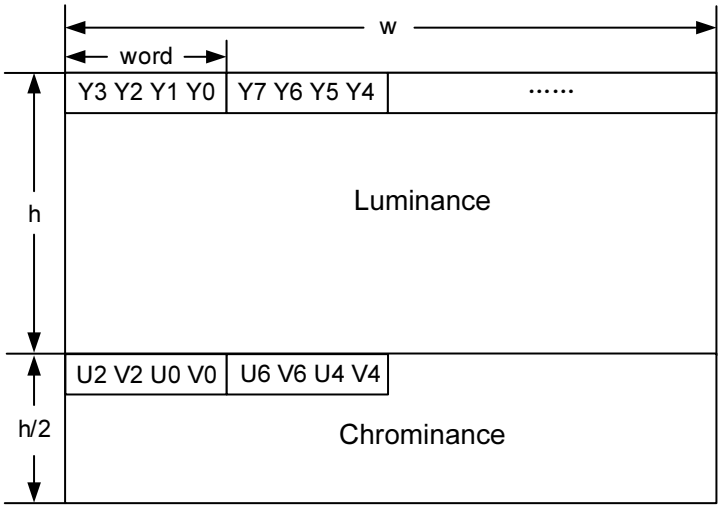


Figure 8-3 SP420 storage format



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Figure 8-4 Storage format of the 8-bit single component data in the memory

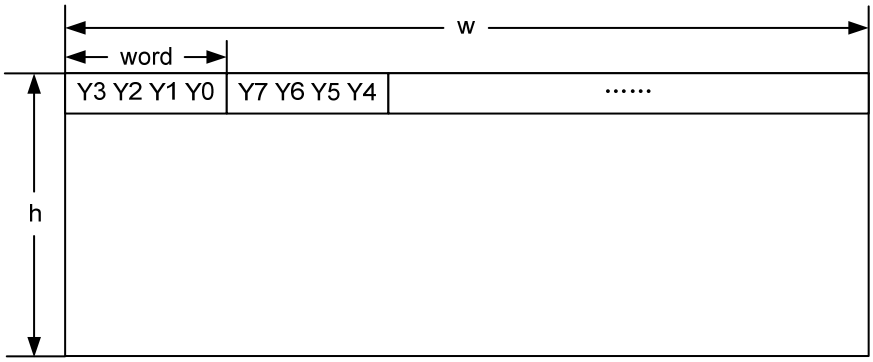
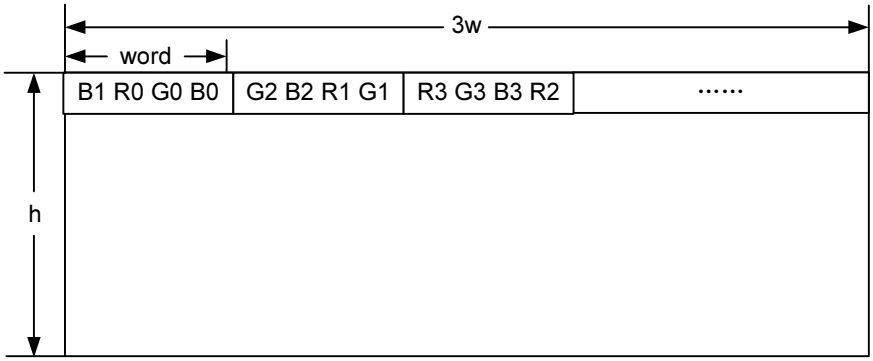


Figure 8-5 Package storage format



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Figure 8-6 Planar storage format

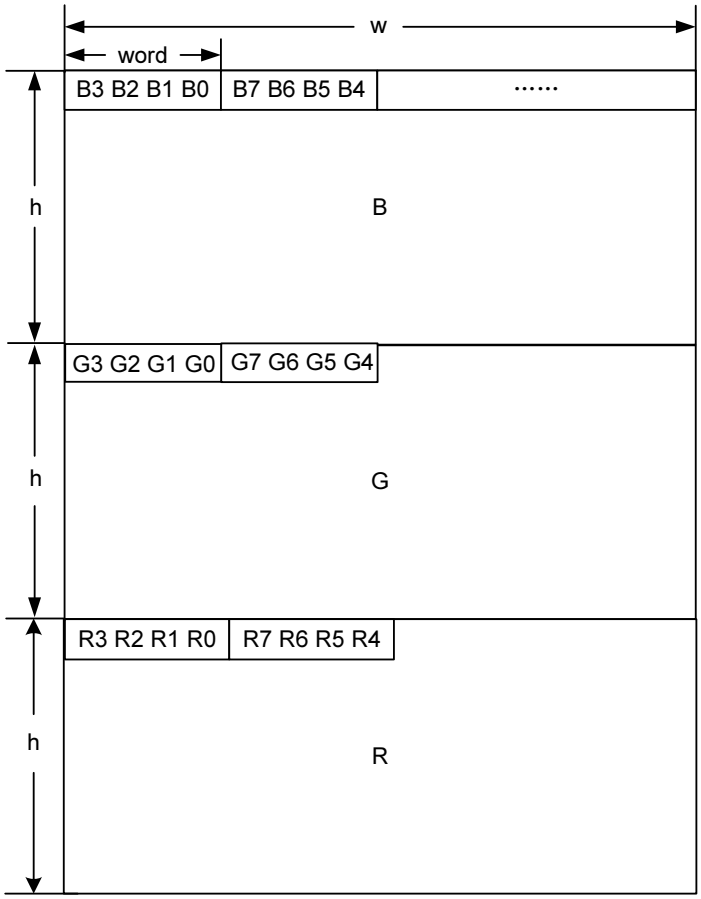
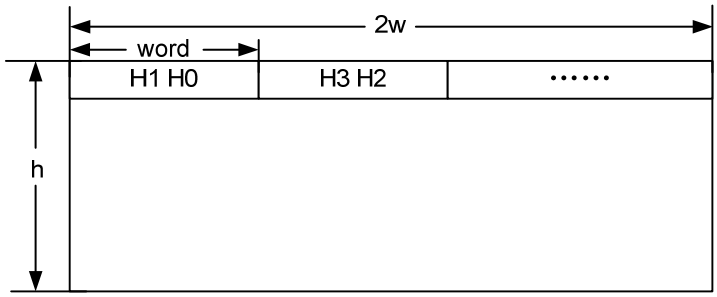


Figure 8-7 Storage format of the 16-bit single component data in the memory



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Figure 8-8 Storage format of the 32-bit single component data in the memory

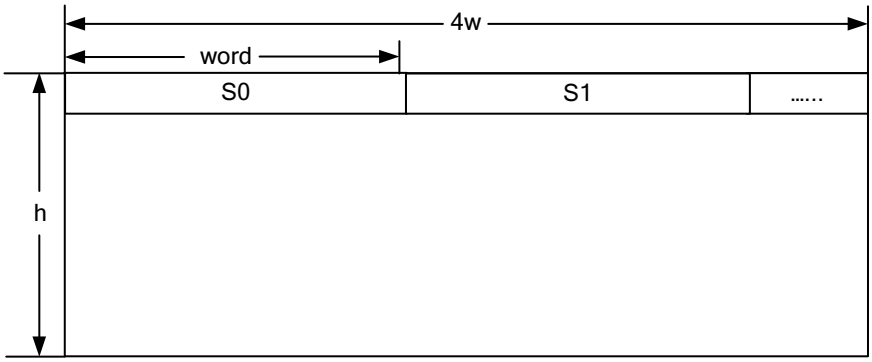


Figure 8-9 Storage format of the 64-bit single component data in the memory

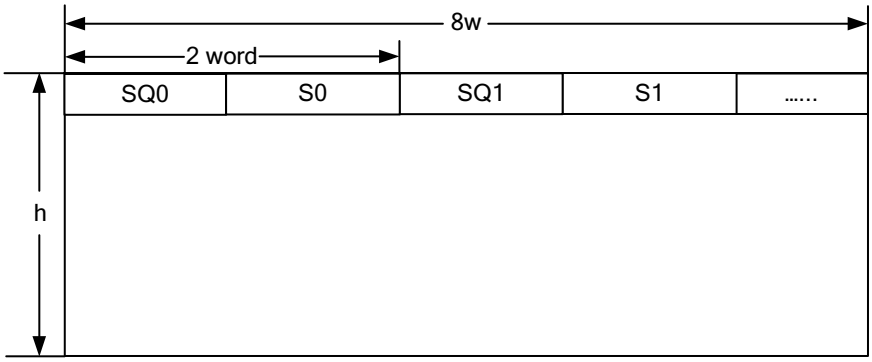


Figure 8-10 Storage format of NCC output data in the memory

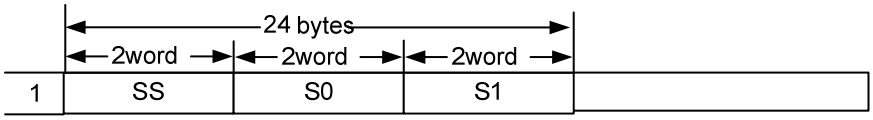
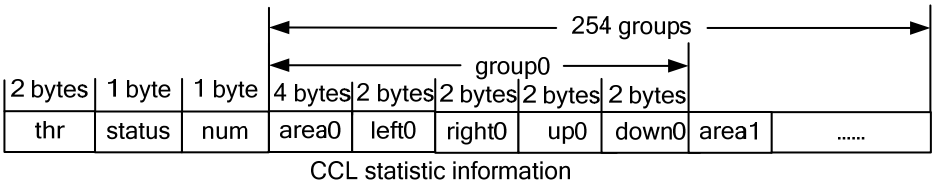


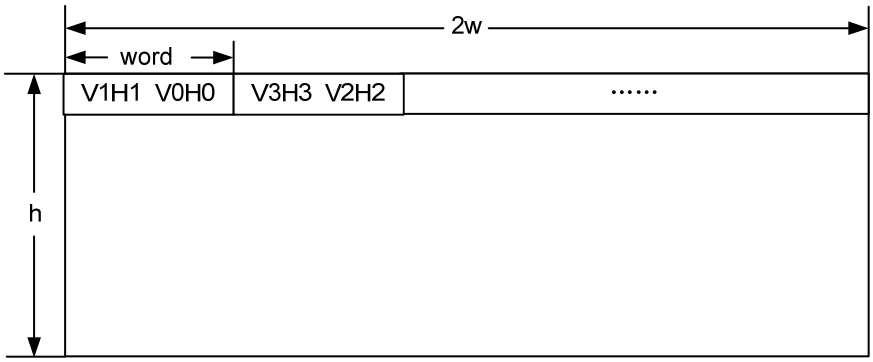
Figure 8-11 Storage format of CCL statistics in the memory (in sequence)



CCL statistic information

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Figure 8-12 Storage format of 16-bit interleaving data (in the horizontal and vertical directions) in the memory



8.3.2 Supported Functions

NOTE

For details about the resolutions supported by operators described in this section, see the *HiIVE API Reference*.

DMA

1. Direct copy mode

This mode allows you to quickly transfer the data of rectangle regions. In this mode, the source data is transferred to the destination region through the internal fast channel, and directly overwrites the data of the destination region.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats: The input and output are 8-bit single component data, as shown in Figure 8-4.

2. Alternative copy mode

This mode allows you to transfer the data of rectangle regions indirectly. In this mode, the source data with a specified size is transferred to the destination region at a specified distance in horizontal and vertical directions.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats: The input and output are 8-bit single component data, as shown in Figure 8-4.
- Others: The width of the source data must be an integral multiple of the distance.

3. Memset mode (3 bytes)

This mode allows you to set the memory in rectangle regions, and fill the destination region by three bytes.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.

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- Input and output formats: No input data is available. The output is 8-bit single component data, as shown in Figure 8-4.

4. Memset mode (8 bytes)

This mode allows you to set the memory in rectangle regions, with 8 byte as the unit for filling the target region.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats: No input data is available. The output is 8-bit single component data, as shown in Figure 8-4.

Filter

The filter function allows the output of the source image after filtering it based on the 5x5 template.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- The following lists the three input and output formats:
  - Both the input and output are 8-bit single component data, as shown in Figure 8-4.
  - Both the input and output are SP420 data, as shown in Figure 8-3.
  - Both the input and output are SP422 data, as shown in Figure 8-2.

CSC

The CSC among YUV2RGB, YUV2HSV, YUV2LAB, and RGB2YUV is supported.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats:
  - SP420 > package; SP420 > planar
  - SP422 > package; SP422 > planar
  - Package > SP420; package > SP422
  - planar > SP420; planar > SP422

  Figure 8-3 and Figure 8-2 show the SP420 format and SP422 format respectively.

  Figure 8-5 shows the package format.

  Figure 8-6 shows the planar format.

FilterAndCSC

The FilterAndCSC function indicates that the YUV SP420/SP422 images are filtered based on the 5x5 template and then the color space is converted into YUV2RGB for output.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats:
  - SP420 > package; SP420 > planar
  - SP422 > package; SP422 > planar

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Figure 8-3 and Figure 8-2 show the SP420 format and SP422 format respectively. Figure 8-5 shows the package format. Figure 8-6 shows the planar format.

Sobel

This function implements vertical and horizontal Sobel filtering based on the 5x5 template.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is an 8-bit single component image, as shown in Figure 8-4.
  - Only H or V is outputted, as shown in Figure 8-7.
  - Both H and V are outputted, as shown in Figure 8-7.

MagAndAng

The MagAndAng indicates that gradient magnitude and angle are calculated. It supports the thresh operation for TO_ZERO.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is the 8-bit single component image, as shown in Figure 8-5.
  - The output is the 16-bit single component amplitude image, as shown in Figure 8-7.
  - The output is the 8-bit single component angle image, as shown in Figure 8-4.

Dilate

This function implements dilation of a binary image based on the 5x5 template.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats: The input and output are 8-bit single component data, as shown in Figure 8-4.

Erode

This function implements erosion of a binary image based on the 5x5 template.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats: The input and output are 8-bit single component data, as shown in Figure 8-4.

Thresh

This function performs thresh processing for an image using specified thresh. Eight modes are available.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats: The input and output are 8-bit single component data, as shown in Figure 8-4.

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And

This function implements the calculation for source data 2 and 1, and outputs the data to the destination region.

- Resolution: The two inputted images should be of the same resolution.
- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is an 8-bit single component data of source 1.
  - The input is an 8-bit single component data of source 2.
  - The output is the 8-bit single component destination data, as shown in Figure 8-4.

Sub

This function performs the subtraction of the data from sources 2 and 1, and outputs the data to the destination region.

- Resolution: The two input images should be of the same resolution.
- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats:
  - An 8-bit single component minuend.
  - An 8-bit single component subtrahend.
  - The 8-bit single component destination data, as shown in Figure 8-4

Or

This function performs the Or calculation of the data from sources 2 and 1, and outputs the data to the destination region.

- Resolution: The two inputted images should be of the same resolution.
- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The two input images are 8-bit single component images.
  - The output is the 8-bit single component image, as shown in Figure 8-4.

Integral

This function performs the calculation of integrogram and square sum integrogram, and supports the output of sum integrogram, square sum integrogram, and the combination of the sum integrogram and square sum integrogram (the accumulated component sum occupied the lower 28-bit, and the accumulated component square sum occupied the upper 36-bit).

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is the 8-bit single component image, as shown in Figure 8-4.
  - As shown in Figure 8-8, if a sum integrogram is outputted, the output is a 32-bit single component image. As shown in Figure 8-9, if a square sum integrogram is

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outputted or is outputted by combining with a sum integrogram, the output is a 64-bit single component image.

Histogram

This function performs the 256-segment histogram statistics. The input is a single component, and the output is the statistics of a 32 bit wide 256-segment histogram.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is the 8-bit single component image, as shown in Figure 8-4.
  - The output is the 32-bit single component data of the statistic result, as shown in Figure 8-8.

Thresh_S16

This function implements thresh processing of signed 16-bit data to signed 8-bit data. It supports four comparison modes.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is the 16-bit single component data, as shown in Figure 8-7.
  - The output is the converted 8-bit single component data, as shown in Figure 8-4.

Thresh_U16

This function implements thresh processing of unsigned 16-bit data to unsigned 8-bit data. It supports two comparison modes.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is the 16-bit single component data, as shown in Figure 8-7.
  - The output is the converted 8-bit single component data, as shown in Figure 8-4.

16BitTo8Bit

This function implements linear conversion of 16-bit data to 8-bit data. It supports four comparison modes.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats:
  - The input is the 16-bit data, as shown in Figure 8-7.
  - The output is the converted 8-bit data, as shown in Figure 8-4.

OrdStatFilter

Supports median filtering, maximum value filtering, and minimum filtering in sequence based on a 3x3 template.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output formats: Both the input and output are 8-bit single component images, as shown in Figure 8-4.

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Map

This function outputs new data by mapping the source data to the 256-segment unsigned 8-bit mapping configured by the user.

- Address alignment mode: Both the input and output must be byte-aligned, and the mapping and all the strides must be 16-byte-aligned.
- Input and output data formats:
  - The input is the 8-bit single component data, as shown in Figure 8-4.
  - The input is the 8-bit wide mapping with a fixed entry length of 256.
  - The output is the 8-bit single component after mapping, as shown Figure 8-4.

Add

This function implements the sum of the weighted values of two gray images. The weight of the two images can be configured independently.

- Resolution: The two inputted images should be of the same resolution.
- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output data formats:
  - The input is the data of two 8-bit single component images, as shown in Figure 8-4.
  - The output is the sum of data of 8-bit single component image, as shown Figure 8-4.

Xor

This function performs the Xor operation for two binary image data.

- Resolution: The two inputted images should be of the same resolution.
- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output data formats:
  - The input is the data of two 8-bit single component images, as shown in Figure 8-4.
  - The output is the 8-bit single component data after performing the Xor operation, as shown in Figure 8-4.

NCC

This function calculates the cross correlation coefficient of two gray scale images of the same resolution.

- Resolution: The two inputted images should be of the same resolution.
- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output data formats:
  - The input is the data of two 8-bit single component images, as shown in Figure 8-4.
  - The output contains three types of data, namely the accumulative product of two images, accumulative product of square sum of image 1, and the accumulative product of square sum of image 2 (in sequence), as shown in Figure 8-10.

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CCL

This function marks eight connected regions.

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output data formats:
  - The input is an 8-bit single component image, as shown in Figure 8-4.
  - The 16-bit output data indicates the minimum area of the output valid connected components.
  - The 8-bit output data indicates whether the area of the detected connected components is greater than the threshold.
  - The output is the coordinate and area of the circumscribed rectangle for each connected component, as shown in Figure 8-11. The original image is changed to an 8-bit single component image after the labeling of the connected components, as shown in Figure 8-4.

CannyHysEdge

This function performs the threshold processing for the magnetic hysteresis, limited non-maximum values, outputted strong and weak edge image and the strong edge coordinate detected by the Canny edge:

- Address alignment mode: Input and output strides must be 16-byte-aligned.
- Input and output data formats:
  - The input is an 8-bit single component angle image, as shown in Figure 8-4.
  - The input is a 16-bit single component amplitude image, as shown in Figure 8-7.
  - The output is an 8-bit single edge mark image, as shown in Figure 8-4.
  - The output is a 32-bit single component data stack, as shown in Figure 8-8.
  - The output is the number of the 32-bit stack.

NormGrad

This function performs the normalized gradient calculation. The gradient components are all normalized to eight bits.

- Address alignment mode: The input and output addresses must be byte-aligned, and the input and output strides must be 16-byte-aligned.
- Input and output data formats:
  - The input is the 8-bit single component data, as shown in Figure 8-4.
  - When out_fmt is set to 0x00, 0x01, or 0x02, the output is the 8-bit single component data, as shown in Figure 8-4. When out_fmt is set to 0x03, the output is the 16-bit interleaving data of H and V, as shown in Figure 8-12.

SAD

This function calculates the accumulated sum of the absolute pixel differences corresponding to two images by block.

- Resolution: The width and height of the image must be an integral multiple of the block size.

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- Address alignment mode: The input and output addresses and strides must be byte-aligned.
- Input and output data formats:
  - The inputs are two frames of 8-bit single-component pictures with the same resolution, as shown in Figure 8-4.
  - The results of dst1 and dst2 depend on the configured value of out_fmt:

    If out_fmt is set to **0x00**, dst1indicates the SAD value (16-bit single component) and dst2 indicates the binary image (8-bit single component), as shown in Figure 8-7 and Figure 8-4 respectively.

    If out_fmt is set to **0x01**, dst1 indicates the SAD value (8-bit single component) and dst2 indicates the thresh image (8-bit single component), as shown in Figure 8-4.

    If out_fmt is set to **0x02**, dst1 indicates the SAD value (16-bit single component), as shown in Figure 8-7.

    If out_fmt is set to **0x03**, dst1 indicates the SAD value (8-bit single component), as shown in Figure 8-4.

    If out_fmt is set to **0x04**, dst1 indicates the thresh result (8-bit single component), as shown in Figure 8-4.
- Linked list configuration information
  - src1, input address of the source image
  - src2, input address of the source image
  - dst1, output address of the target image
  - (Optional) dst2, output address of the target image. dst2 needs to be configured when out_fmt is set to **0x00** or **0x01**.
  - src1_stride
  - src2_stride
  - dst1_stride
  - dst2_stride
  - Image width, which must be an integral multiple of the block size
  - Image height, which must be an integral multiple of the block size
  - op_type, which must be set to **0x2E**
  - op_mode

    0x00: The block size is 4 x 4.

    0x01: The block size is 8 x 8.

    0x02: The block size is 16 x 16.
  - out_fmt

    0x00: outputs the 16-bit SAD value and thresh result.

    0x01: outputs the 8-bit SAD value and thresh result.

    0x02: outputs the 16-bit SAD value.

    0x03: outputs the 8-bit SAD value.

    0x04: outputs the thresh result.
  - mask0, the smaller value between the thresh results
  - mask1, the larger value between the thresh results
  - thresh_16bit_1, thresh threshold

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8.4 Register Summary

Table 8-1 describes IVE registers.

Table 8-1 Summary of IVE registers (base address: 0x206A_0000)

Offset Address	Register	Description
0x0000	IVE_START	IVE start configuration register
0x0004	INT_EN	IVE interrupt enable register
0x0008	INT_RW	IVE clear interrupt register
0x000C	INT_STATUS	IVE interrupt status register
0x0010	LIST_POINTER	Linked list start address register
0x0014	IVE_STATUS	IVE working status register
0x0018	IVE_TASK_ID	ID register for the tasks processed by the IVE at the previous time
0x0030	NODE_CLK_VALUE	Register for the number of cycles consumed by the preceding node
0x0034	CLK_GT_EN	Clock gating of the IVE internal module
0x0040	NODE_DONE_CNT	Register for the number of processed nodes
0x0044	LIST_DONE_CNT	Register for the number of processed linked lists
0x0048	INT_DONE_CNT	Count register for the node interrupts reported by the current linked list
0x004C	INT_RD_CNT	Count register for the interrupts that have been successfully read by the current linked list
0x0050	INT_WR_CNT	Count register for the interrupts that have been cleared by writing the current linked list
0x0054	AXI_INFO	Read and write outstanding number register
0x0060	IVE_OVER_TIME_THR	Overtime interrupt threshold register
0x0064	IVE_OVER_TIME_CNT	Register for the dynamic counting number of the operator working cycle
0x0068	CHAIN_STAT_CLK_VALUE	Register for the number of cycles consumed by the preceding linked list
0x006C	CHAIN_CYCLE_CNT	Register for the counting number of the current linked list cycles

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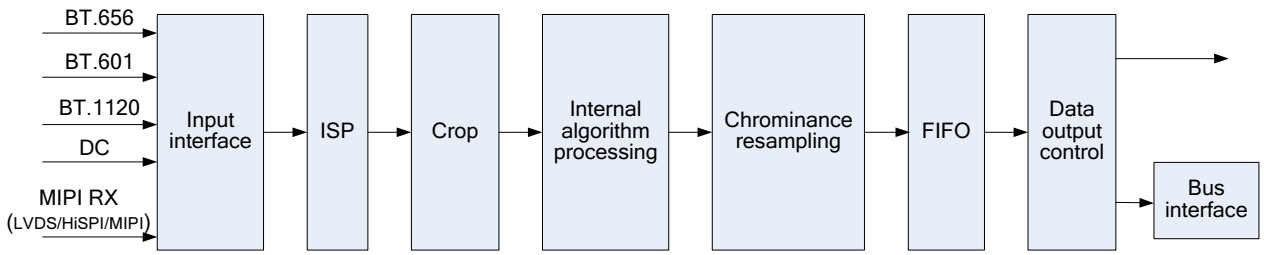
9 Video Interfaces

9.1 VICAP

9.1.1 Overview

The video capture (VICAP) module receives video data over the BT.656 interface, BT.601 interface, BT.1120 interface, digital camera (DC) interface, or mobile industry processor interface receiver (MIPI RX), and stores the data in the specified addresses of the memory or transmits the data to the video processing subsystem (VPSS). Note that the MIPI RX includes the MIPI, low-voltage differential signal (LVDS), and high-speed serial pixel interface (HiSPI). The VICAP module has a built-in image signal processor (ISP) that can directly receive external raw data (bayer RGB data). Figure 9-1 shows the functional block diagram of the VICAP module.

Figure 9-1 Functional block diagram of the VICAP module



9.1.2 Features

The VICAP module has the following features:

- Maximum input resolution of 1920 x 1080
- One external interface with maximum 16-bit width (maximum 14-bit width for the RAW data)
- One internal interface and 1-channel video processing in interlaced or progressive input mode
- BT.656, BT.601, BT.1120, DC, MIPI, LVDS, and HiSPI timings
- SMPTE293M or ITU-R BT.1358 timing (480p/576p)
- Flash trigger

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- Shutter trigger
- Built-in ISP
- Data obtained in a specified window
- Data output to the DDR in offline mode or to the VPSS in online mode
- Output storage modes
  - SPYCbCr 4:2:2 mode
  - SPYCbCr 4:2:0 mode

9.1.3 Function Description

9.1.3.1 Typical Applications

The VICAP module captures video data in multiple input timings and stores the captured data in the DDR or transmits data to the VPSS online. By using different function modes configured by the system, the VICAP module can be connected to different external VI interfaces, supporting multiple external input devices.

The VICAP module has 21 pins, one clock, two external sync signals (multiplexed with other pins), 16 data lines, one shutter trigger signal, and one camera flash trigger signal.

It has one port and one channel. The port parses the timings of an interconnected chip, and the channel processes one video signal.

The VICAP module supports the 1-channel 1080p typical input.

9.1.3.2 Function Implementation

ITU-R BT.656 YCbCr4:2:2

1. Horizontal timing

Based on the ITU-R BT.656 protocol, sync signals are included in data streams. The special bytes start of active video (SAV) and end of active video (EAV) indicate the start and end of the active line data respectively. In video data streams, the header of the timing reference code indicates that the following byte is SAV or EAV. The timing reference code consists of FF 00 00. FF and 00 indicate the reserved bytes of the image encoding data, that is, non-image data. Table 9-1 shows the format of the ITU-R BT.656 line data.

Table 9-1 Format of the ITU-R BT.656 YCbCr 4:2:2 line data

Timing Reference Code				Line Blanking Region					Timing Reference Code				YCbCr 4:2:2 with 720 Active Pixels						
FF	00	00	EAV	80	10	…	80	10	FF	00	00	SAV	Cb0	Y0	Cr0	Y1	…	Cr718	Y719

The difference between the SAV and EAV depends on the special bit H. Both the SAV and EAV include the vertical blanking bit V and field indicator bit F. For details about the SAV and EAV, see Table 9-2.

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Table 9-2 Formats of the SAV and EAV

Bit[7]	Bit[6] (F)	Bit[5] (V)	Bit[4] (H)	Bit[3:0] (P3–P0)
Fixed value 1	Field indicator bit 1st field: F = 0 2nd field: F = 1	Vertical blanking bit VBI: V = 1 Active video: V = 0	SAV: H = 0 EAV: H = 1	Check bits

The ITU-R BT.656 protocol defines valid SAV and EAV by using eight valid reserved bits. Four check bits can be used to correct 1-bit error and detect 2-bit errors. Table 9-3 describes the valid SAV and EAV values.

Table 9-3 Valid SAV and EAV values

Encoding	Binary Value	Field Number	Vertical Blanking Interval or Not
SAV	10000000	1	None
EAV	10011101	1	None
SAV	10101011	1	Yes
EAV	10110110	1	Yes
SAV	11000111	2	None
EAV	11011010	2	None
SAV	11101100	2	Yes
EAV	11110001	2	Yes

The four valid reserved bits (P0, P1, P2, and P3) provide the error correction function. They are determined by the F, V, and H bits. See Table 9-4.

Table 9-4 ITU-R BT.656 error-correcting codes

F	V	H	P3	P2	P1	P0
0	0	0	0	0	0	0
0	0	1	1	1	0	1
0	1	0	1	0	1	1
0	1	1	0	1	1	0
1	0	0	0	1	1	1
1	0	1	1	0	1	0
1	1	0	1	1	0	0
1	1	1	0	0	0	1

where

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P0 = F^V^H

P1 = F^V

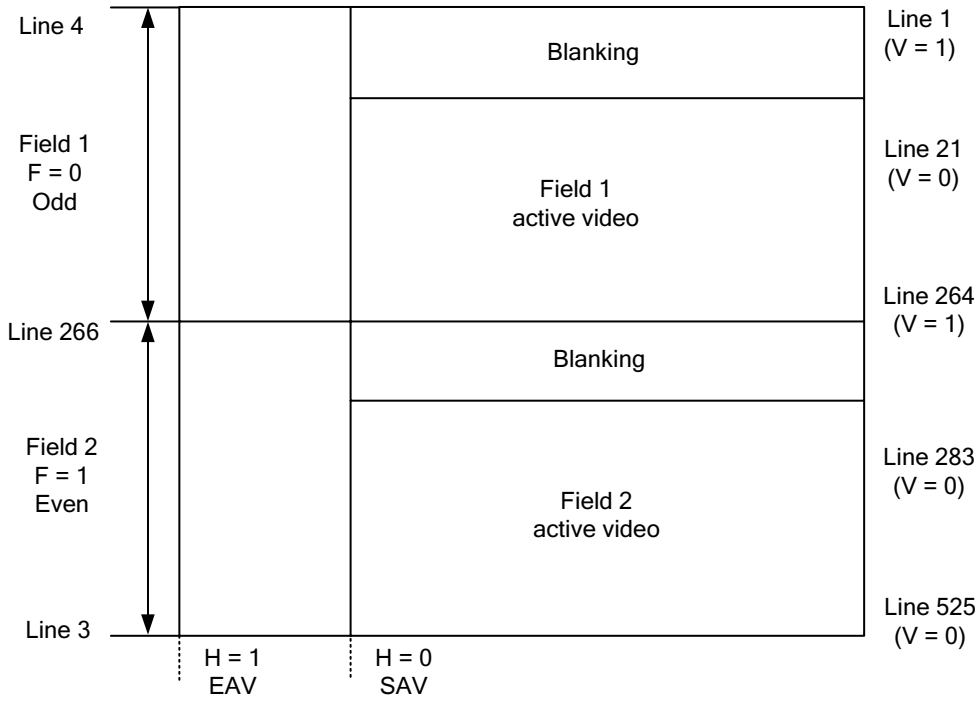
P2 = F^H

P3 = V^H

2. Vertical timing

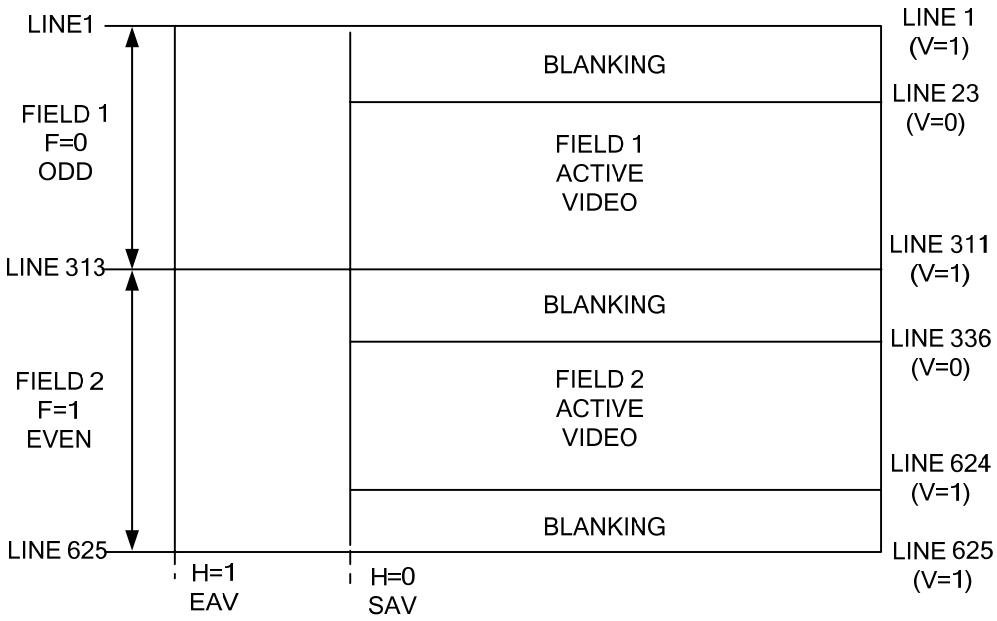
The positions of the vertical timings are determined by bit F and bit V of the timing reference codes SAV and EAV. Figure 9-2 shows the vertical timing of the 525-line 60 field/s video system, and Figure 9-3 shows the vertical timing of the 625-line 50 field/s video system.

Figure 9-2 Vertical timing of the 525-line 60 field/s video system



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Figure 9-3 Vertical timing of the 625-line 50 field/s video system



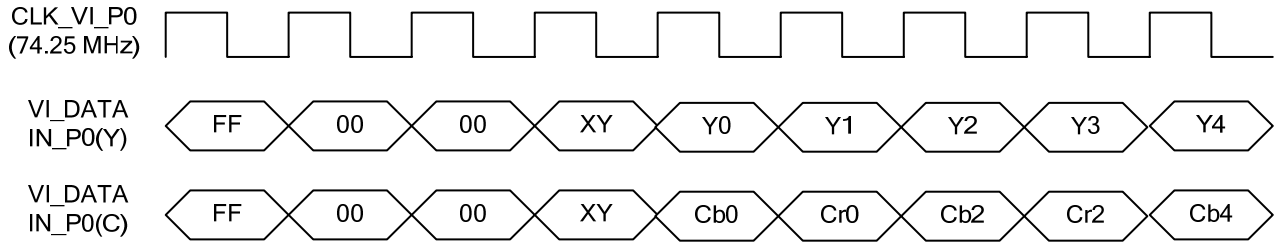
NOTE

The VICAP module identifies vertical timings based on SAV and EAV regardless of the lines.

BT.1120 (HD) Interface Timings

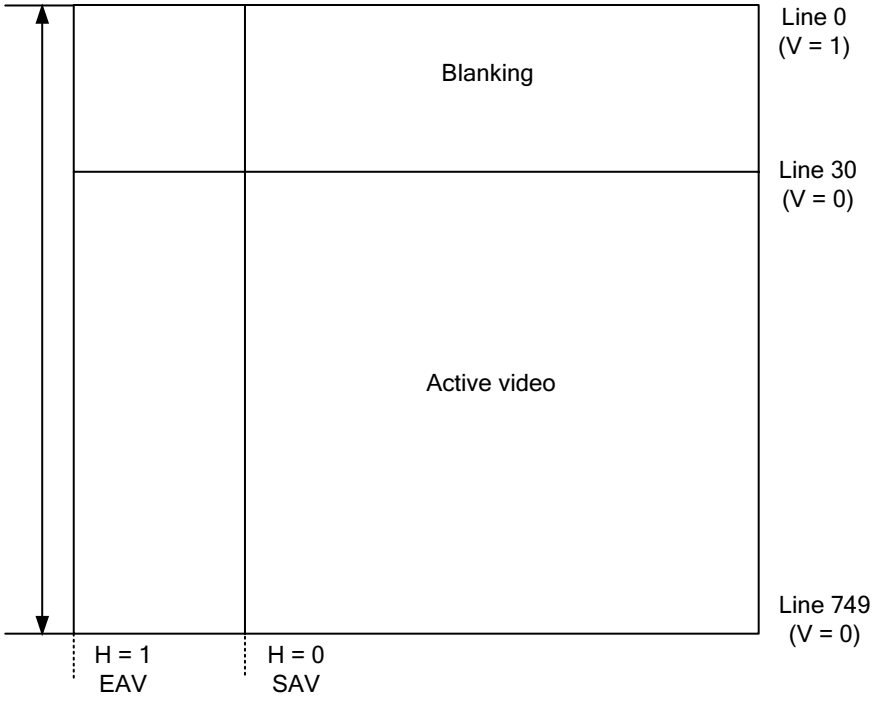
The VICAP module supports the HD interface timings with separated Y/C inputs. In this case, two data ports are required. One is used to transfer luminance and the other one is used to transfer chrominance, as shown in Figure 9-4 and Figure 9-5.

Figure 9-4 Horizontal input timing of the HD interface



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Figure 9-5 Vertical input timing of the HD interface

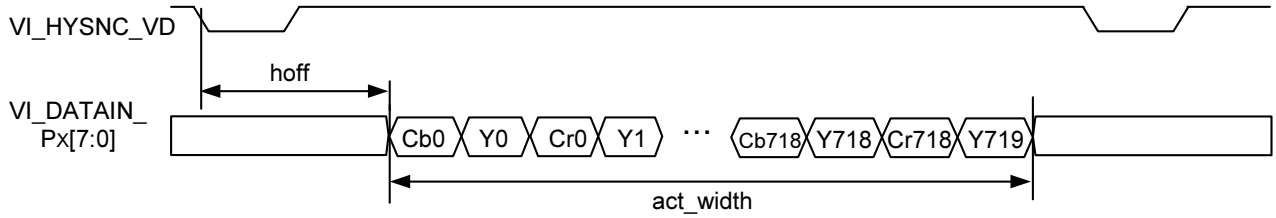


ITU-R BT.601 YCbCr4:2:2

1. Horizontal timing

The horizontal pulse indicates the start of a new line, as shown in Figure 9-6. After hoff clock cycles, an input signal passes the line front blanking region and enters the line active data region. The value of hoff is 244 in the NTSC 525-line system or 264 in the PAL 625-line system. After act_width clock cycles, the input signal passes the line active data region and enters the line back blanking region. The value of act_width can be configured, and the typical value is 720 or 704. In addition, the horizontal sync polarity is configurable.

Figure 9-6 ITU-R BT.601 horizontal timing



2. Vertical timings

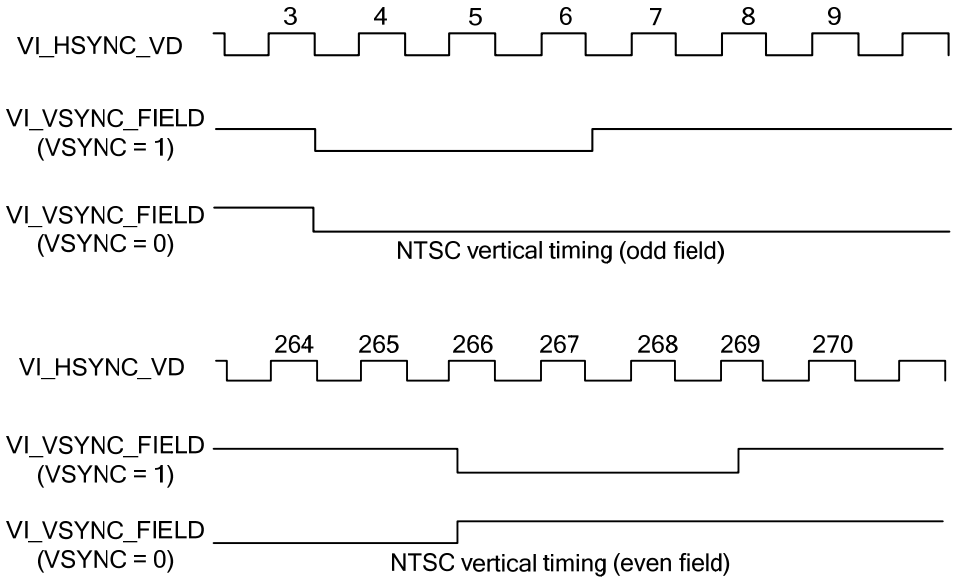
Based on the ITU-R BT.601 recommendation, the signals VSYNC and FIELD are vertical sync signals. The VSYNC pulse or FIELD transition indicates the start of the odd field or even field. The VICAP module supports two types of vertical sync.

Figure 9-7 shows VI vertical sync timings in the NTSC standard (525 lines), and Figure 9-8 shows VI vertical sync timings in the PAL standard (625 lines). VI_HSYNC_VD indicates the

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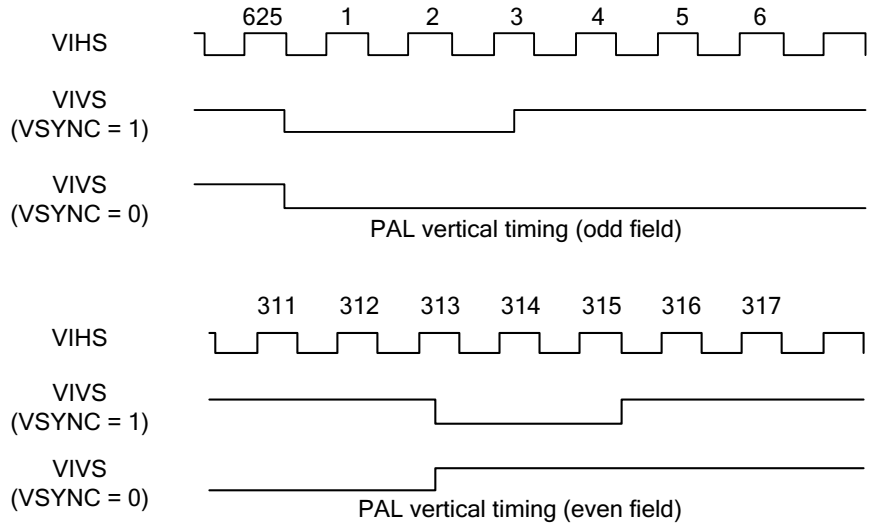
horizontal sync pulse, and VI_VSYNC_FIELD indicates the vertical sync pulse when VSYNC is 1 or field sync signal when VSYNC is 0.

Figure 9-7 Vertical sync timings in the NTSC standard



In NTSC interlaced scanning mode, the level of the vertical sync signal in field 1 becomes low in the start position of line 4, retains low for three consecutive lines, and then becomes high in the start position of line 7. The VICAP module receives 240-line data from line 22 to line 261. The level of the vertical sync signal in field 2 becomes low in the middle of line 266, retains low for three consecutive lines, and then becomes high in the middle of line 269. The VICAP module receives 240-line data from line 285 to line 524.

Figure 9-8 Vertical sync timings in the PAL standard



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In PAL interlaced scanning mode, the level of the vertical sync signal in field 1 becomes low in the start position of line 1, retains low for 2.5 consecutive lines, and then becomes high in the middle of line 3. The VICAP module receives 288-line data from line 24 to line 310. The level of the vertical sync signal in field 2 becomes low in the middle of line 313, retains low for 2.5 consecutive lines, and then becomes high in the start position of line 316. The VICAP module receives 288-line data from line 336 to line 623.

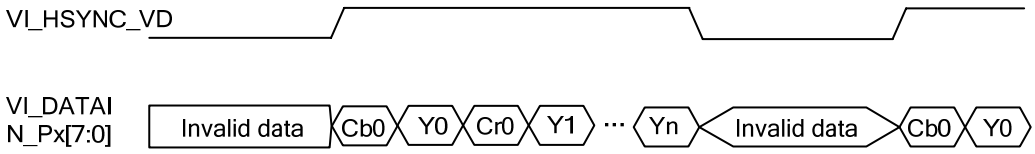
The preceding timings are typical BT.601 vertical timings. The number of lines from the start of the field to the active line, the number of field active lines, and vertical sync polarity can be configured.

DC Interface Timings

1. Horizontal timing

When a DC is connected to the VICAP module, VI_HSYNC_VD is the data valid signal. The polarity of this signal is configurable. Figure 9-9 shows the DC horizontal timing.

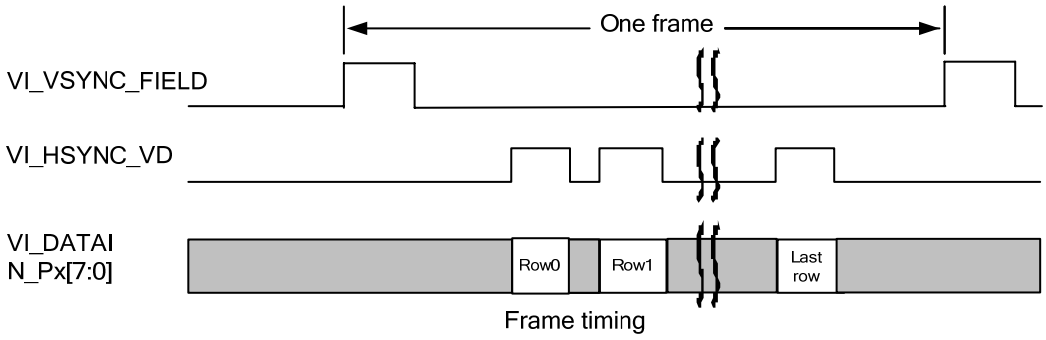
Figure 9-9 DC horizontal timing



2. Vertical timing

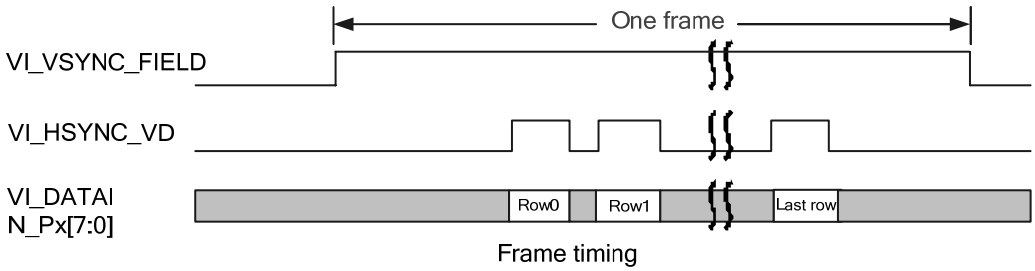
The VICAP module supports the vertical pulse timing and vertical line active timing, as shown in Figure 9-10 and Figure 9-11. The vertical sync polarity is configurable.

Figure 9-10 DC vertical pulse timing



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Figure 9-11 DC vertical line active timing

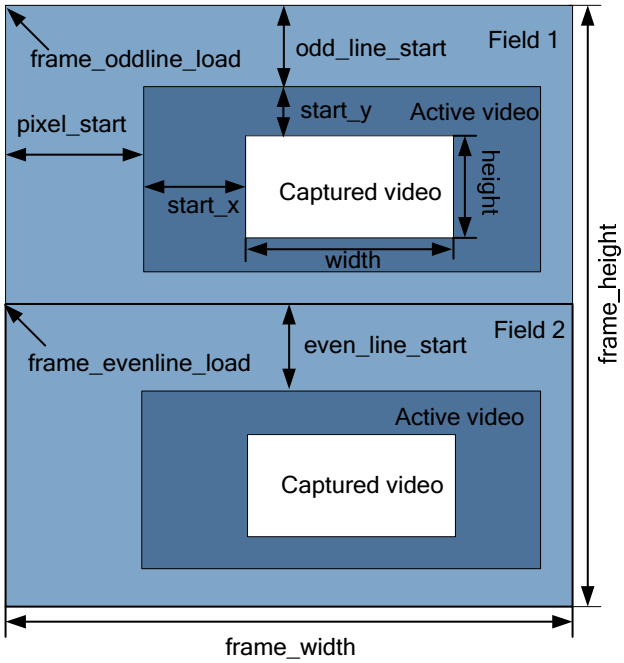


The preceding two timings are the same from the aspect of internal processing of the VICAP module. To be specific, the VICAP module considers the start of a frame after it detects a rising edge or a falling edge, and then detects the data active signal to check whether the current data is active.

9.1.3.3 Image Cropping

As shown in Figure 9-12, an active video starts from the end of the horizontal blanking region and vertical blanking region. The actual view range, however, is within the active video range. That is, compared with the boundary of the active video, the boundary of the actual view is shrunk, which avoids the boundary effect.

Figure 9-12 Relationship between the active video area and the horizontal/vertical blanking regions



9.1.3.4 Image Storage Modes

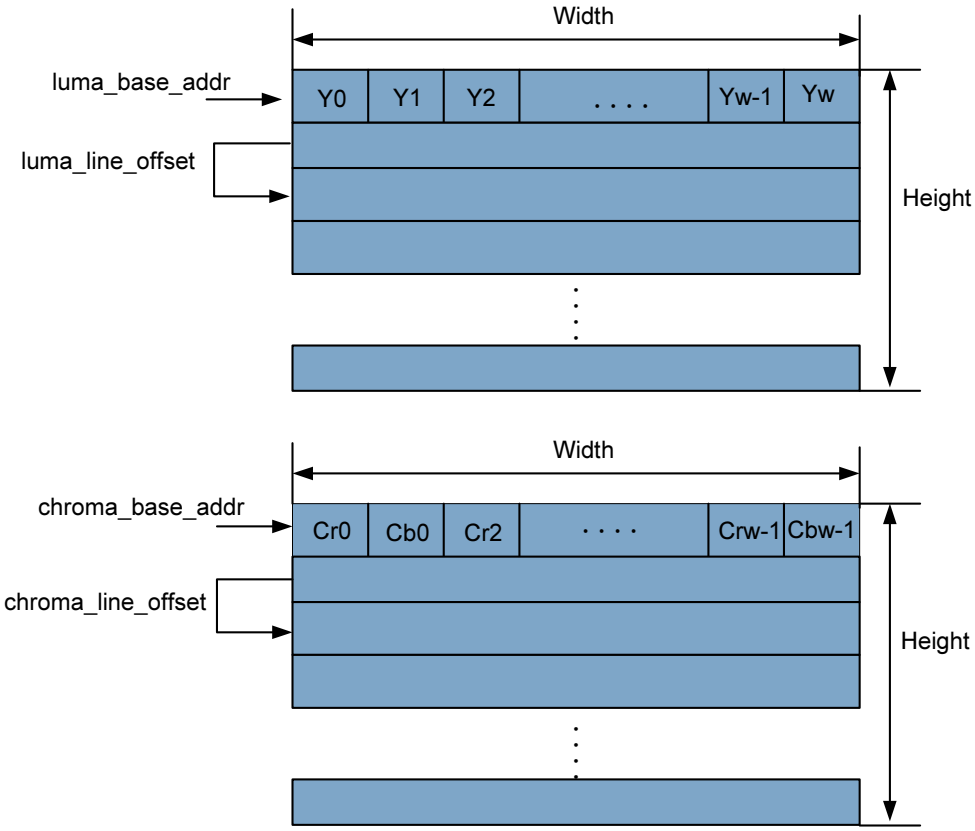
The supported image storage modes include:

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- Semi-planar YCbCr storage mode

  After setting a view area, the system stores the read data in semi-planar mode. That is, the luminance component and the chrominance component are stored in the luminance space and chrominance space of the DDR respectively.
- For one line, the luminance component and chrominance component are stored separately and consecutively.
- For two consecutive lines, the components are stored based on the offset parameter. This parameter defines the storage stride between the starts of two lines. The storage positions in the DDR of the luminance component and chrominance component are specified by the start address base_addr. Figure 9-13 shows the mode of storing the YCbCr4:2:2 data captured by the VICAP module.

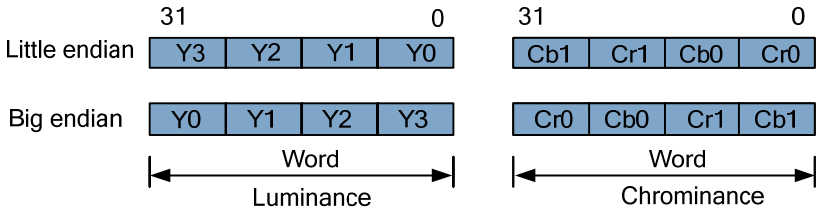
Figure 9-13 YCbCr4:2:2 storage mode



In the DDR, data is stored by word (32 bits). Four 8-bit pixels constitute a 32-bit word in big endian mode or little endian mode. Figure 9-14 shows the big endian and little endian storage modes of the luminance and chrominance components.

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Figure 9-14 Big endian and little endian storage modes



The VICAP module supports only the DDR that stores data in little endian mode.

9.1.4 Operating Mode

9.1.4.1 reg_newer Function

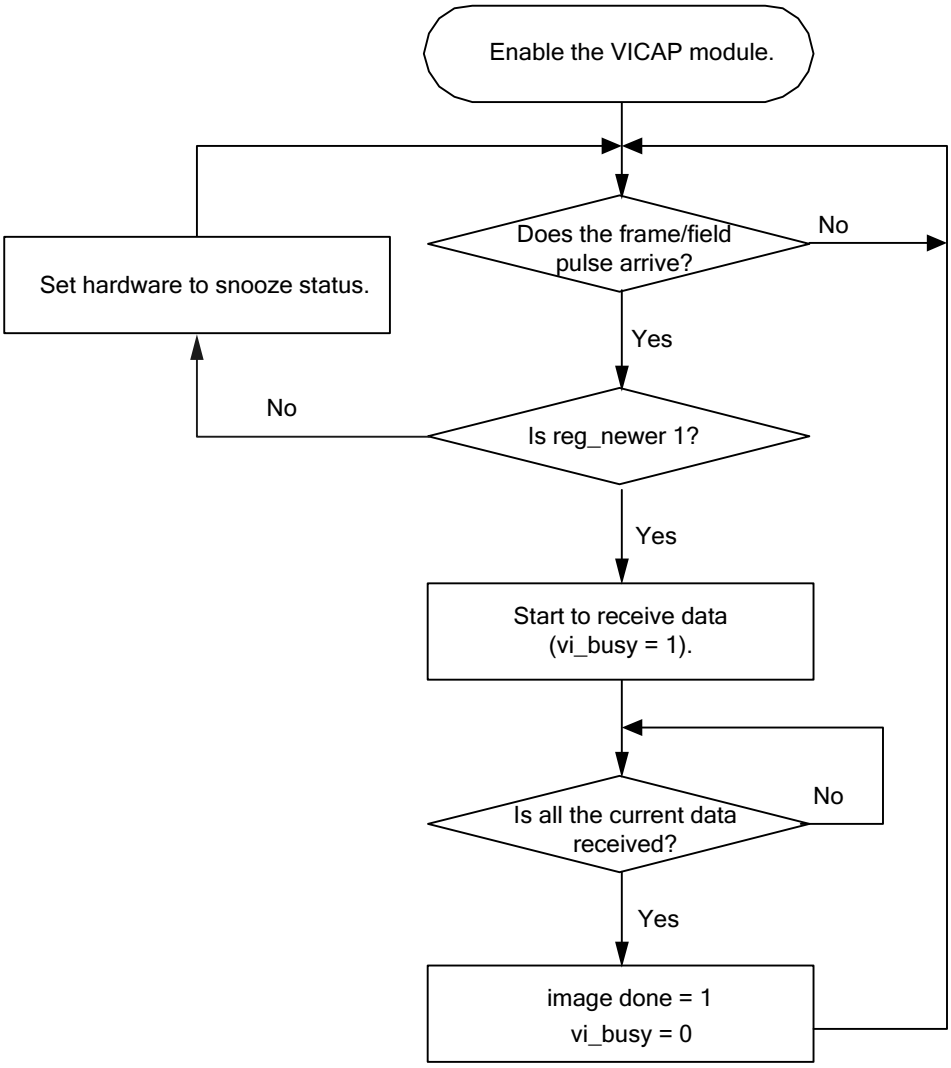
- Before enabling a channel of the VICAP module, the software must perform the following operations:
  - Configures the VICAP attribute register.
  - Writes 1 to the reg_newer bit to inform the VICAP module that the current register is ready.
- After the VICAP module is enabled, the VICAP logic starts to work. When a field or a frame arrives:
  - If reg_newer is 0, the VICAP module does not receive data. However, it sets the hardware status to snooze and waits for the next field or frame.
  - If reg_newer is 1, the VICAP module starts receiving data, generates the register update interrupt (reg_update_int), and sets the hardware to busy state.
- After all the current data is received, the busy status of the hardware is cleared. When the next field or frame arrives:
  - If reg_newer is 0, the VICAP module does not receive the data of the next field or frame.
  - If reg_newer is 1, the VICAP module receives the data of the next field or frame.

9.1.4.2 VICAP Hardware Workflow

Figure 9-15 shows the VICAP hardware workflow.

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Figure 9-15 VICAP hardware workflow



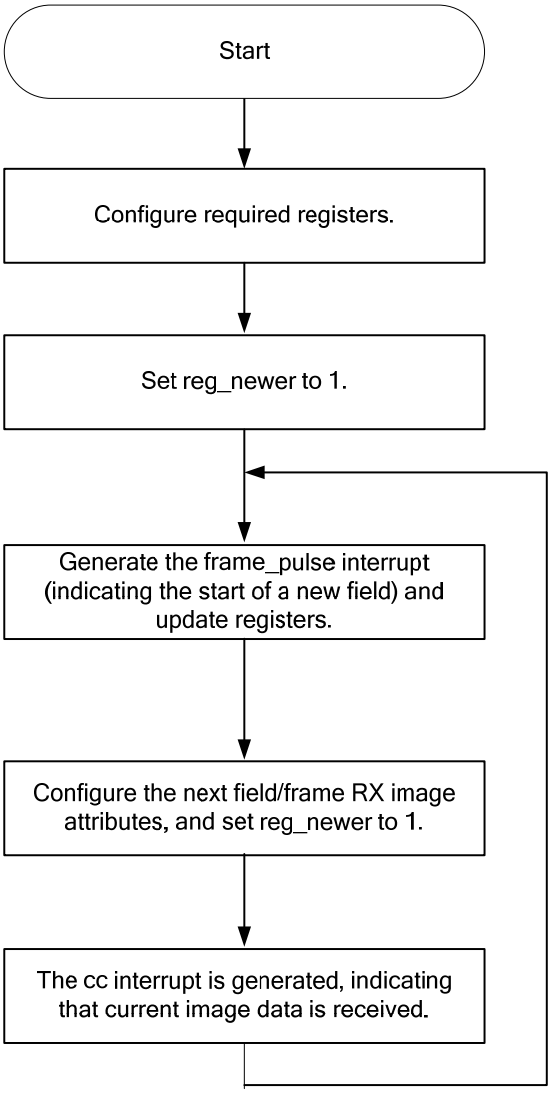
Each time after the VICAP module receives the data of a field or frame, it checks the reg_newer bit when the next field or frame arrives. If reg_newer is 1, software has updated or checked corresponding VICAP registers. In this case, the VICAP module automatically loads the register values configured by the software to the working register (this register is inaccessible to software), clears reg_newer, and starts to receive the data of the next field or frame. If reg_newer is 0, the VICAP module starts to receive data only when reg_newer is 1 and a new field or frame arrives.

9.1.4.3 Software Configuration Workflow

Figure 9-16 shows the software configuration process in interrupt mode.

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Figure 9-16 Software Configuration



When the BT.656, BT.1120, MIPI Rx, or DC interface is used, timing registers do not need to be configured. If the BT.601 interface is used, timing registers including vertical sync registers and horizontal sync registers need to be configured.

9.1.5 Register Summary

Table 9-5 describes VICAP registers.

Table 9-5 Summary of VICAP registers (base address: 0x2058_0000)

Offset Address	Register	Description	Page
0x0000	WK_MODE	Global operating mode configuration register	9-17
0x0010	AXI_CFG	Bus configuration register	9-18

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Offset Address	Register	Description	Page
0x0014	MAC_CFG	MAC configuration register	9-19
0x0040	DES_SEL	DES input data selection register	9-20
0x0050	ISP_SEL	ISP input data selection register	9-20
0x0060	CHN_MODE	SRC0/DES0 bus read/write rate control mode register	9-21
0x0070	BUF_MODE	LINE_BUF input data select register	9-22
0x00E0	APB_TIMEOUT	APB timeout register	9-22
0x00F0	VICAP_INT	Interrupt status register	9-22
0x00F8	VICAP_INT_MASK	Interrupt mask register	9-23
0x0100	PT_INTF_MOD	Port mode register	9-24
0x0110	PT_OFFSET0	Component 0 offset register	9-25
0x0114	PT_OFFSET1	Component 1 offset register	9-25
0x0118	PT_OFFSET2	Component 2 offset register	9-26
0x0120	PT_BT656_CFG	BT.656 configuration register	9-26
0x0130	PT_UNIFY_TIMING_CFG	Timing configuration register	9-28
0x0134	PT_GEN_TIMING_CFG	Timing recovery module configuration register	9-29
0x0140	PT_UNIFY_DATA_CFG	Data configuration register	9-30
0x0150	PT_YUV444_CFG	YUV444 configuration register	9-31
0x0180	PT_INTF_HFB	Horizontal front blanking width register	9-32
0x0184	PT_INTF_HACT	Horizontal active width register	9-32
0x0188	PT_INTF_HBB	Horizontal back blanking width register	9-33
0x018C	PT_INTF_VFB	Vertical front blanking width register	9-33
0x0190	PT_INTF_VACT	Vertical active width register	9-33
0x0194	PT_INTF_VBB	Vertical back blanking width register	9-34
0x0198	PT_INTF_VBFB	Vertical bottom front blanking width register	9-34
0x019C	PT_INTF_VBACT	Vertical bottom active width register	9-35

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Offset Address	Register	Description	Page
0x01A0	PT_INTF_VBBB	Vertical bottom back blanking width register	9-35
0x01A4	PT_ID_CFG	ID configuration register	9-35
0x01B0	PT_FLASH_CFG	Flash configuration register	9-36
0x01C0	PT_FLASH_CYC0	Flash timing 0 width register	9-37
0x01C4	PT_FLASH_CYC1	Flash timing 1 width register	9-38
0x01D0	PT_SHUTTER_CYC0	Shutter timing 0 width register	9-38
0x01D4	PT_SHUTTER_CYC1	Shutter timing 1 width register	9-39
0x01D8	PT_SHUTTER_CYC2	Shutter timing 2 width register	9-39
0x01DC	PT_SHUTTER_CYC3	Shutter timing 3 width register	9-39
0x01E0	PT_STATUS	Port status register	9-40
0x01E4	PT_BT656_STATUS	BT656 status register	9-40
0x01EC	PT_SIZE	Input picture size register	9-41
0x01F0	PT_INT	Port interrupt status register	9-41
0x01F8	PT_INT_MASK	Port interrupt mask register	9-42
0x0200	DES_CTRL	DES control register	9-43
0x0204	DES_REG_NEWER	Capture control register	9-43
0x0210	DES_ID_CFG	DES RX ID selection register	9-44
0x0220	DES_CROP_CFG	Crop enable register	9-44
0x0230	DES_CROP0_START	Crop start position register for region 0	9-45
0x0234	DES_CROP0_SIZE	Crop size configuration register for region 0	9-45
0x0270	DES_CFG	DES configuration register	9-46
0x0284	DES_SIZE	DES storage size register	9-46
0x0290	DES_FADDR	DES storage base address register	9-47
0x0294	DES_STRIDE	DES storage line stride register	9-47

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Offset Address	Register	Description	Page
0x029C	DES_BUF_CFG	DES bus buffer control register	9-47
0x02F0	DES_INT	DES raw interrupt register	9-48
0x02F8	DES_MASK	DES interrupt mask register	9-49
0x0500	LINE_BUF_CFG	LINE_BUF control register	9-50
0x0504	LINE_BUF_REG_NEWER	LINE_BUF updating register	9-50
0x0510	LINE_BUF_ID_CFG	LINE_BUF RX ID selection register	9-51
0x0520	LINE_BUF_CROP_CFG	Crop enable register	9-51
0x0530	LINE_BUF_CROP0_START	Crop start position register for region 0	9-52
0x0534	LINE_BUF_CROP0_SIZE	Crop size configuration register for region 0	9-52
0x0550	LINE_BUF_WIDTH	LINE_BUF picture width register	9-53
0x0554	LINE_BUF_HEIGHT	LINE_BUF picture height register	9-53
0x05F0	LINE_BUF_INT	LINE_BUF raw interrupt register	9-54
0x05F8	LINE_BUF_MASK	LINE_BUF interrupt mask register	9-55
0x0700	SRC_CTRL	SRC control register	9-55
0x0704	SRC_REG_NEWER	Capture control register	9-56
0x0770	SRC_CFG	SRC configuration register	9-57
0x0784	SRC_SIZE	SRC storage size register	9-57
0x0790	SRC_FADDR	SRC storage base address register	9-58
0x0794	SRC_STRIDE	SRC storage line stride register	9-58
0x079C	SRC_BUF_CFG	SRC bus buffer control register	9-58
0x07F0	SRC_INT	SRC raw interrupt register	9-59
0x07F8	SRC_MASK	SRC interrupt mask register	9-60
0x1004	CH_REG_NEWER	Capture control register	9-61
0x1010	CH_ADAPTER_CFG	Timing adaptation register	9-61

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Offset Address	Register	Description	Page
0x1080	CH_WCH_Y_CFG	Y component configuration register for the WCH module	9-62
0x1084	CH_WCH_Y_SIZE	Y component storage size register for the WCH module	9-63
0x1090	CH_WCH_Y_FADDR	Y component storage base address register for the WCH module	9-64
0x1094	CH_WCH_Y_HADDR	Y component header information storage base address register for the WCH module	9-64
0x1098	CH_WCH_Y_STRIDE	Y component line offset register for the WCH module	9-64
0x10A0	CH_WCH_C_CFG	C component configuration register for the WCH module	9-65
0x10A4	CH_WCH_C_SIZE	C component storage size register for the WCH module	9-66
0x10B0	CH_WCH_C_FADDR	C component storage base address register for the WCH module	9-66
0x10B4	CH_WCH_C_HADDR	C component header information storage base address register for the WCH module	9-67
0x10B8	CH_WCH_C_STRIDE	C component line offset register for the WCH module	9-67
0x10F0	CH_INT	Channel raw interrupt register	9-68
0x10F8	CH_INT_MASK	Channel interrupt mask register	9-69

9.1.6 Register Description

WK_MODE

WK_MODE is a global operating mode configuration register.

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Offset Address	Register Name	Total Reset Value
0x0000	WK_MODE	0x0000_9210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ros_mem_ema				rft_mem_emab				rft_mem_emaa				rfs_mem_emaw		rfs_mem_ema				reserved		reserved		power_mode	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16:14]	RW	ros_mem_ema	Speed adjustment configuration for the Q end of the read-only memory
[13:11]	RW	rft_mem_emab	Speed adjustment configuration for the D end of the dual-port memory
[10:8]	RW	rft_mem_emaa	Speed adjustment configuration for the Q end of the dual-port memory
[7:6]	RW	rfs_mem_emaw	Speed adjustment configuration for the D end of the single-port memory
[5:3]	RW	rfs_mem_ema	Speed adjustment configuration for the Q end of the single-port memory
[2]	RO	reserved	Reserved
[1]	RO	reserved	Reserved
[0]	RW	power_mode	Clock mode 0: The low-power mode is disabled. 1: The low-power mode is enabled.

AXI_CFG

AXI_CFG is a bus configuration register.

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Offset Address	Register Name	Total Reset Value
0x0010	AXI_CFG	0x1010_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																r_outstanding				reserved				w_outstanding				reserved			
Reset	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:12]	RW	r_outstanding	Number of read request outstandings Value range: [1, 8]
[11:8]	RO	reserved	Reserved
[7:4]	RW	w_outstanding	Number of write request outstandings Value range: [1, 8]
[3:0]	RO	reserved	Reserved

MAC_CFG

MAC_CFG is an MAC configuration register.

Offset Address	Register Name	Total Reset Value
0x0014	MAC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																r_req_prio								w_req_prio							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	r_req_prio	Read request arbitration priority 0: low priority 1: high priority bit[8]: reserved bit[9]: SRC1 read request priority bit[10]: reserved bit[11]: reserved bit[12]: reserved bit[13]: reserved bit[14]: reserved bit[15]: reserved

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[7:0]	RW	w_req_prio	Write request arbitration priority 0: low priority 1: high priority bit[0]: DES0 write request priority bit[1]: reserved bit[2]: reserved bit[3]: reserved bit[4]: reserved bit[5]: reserved bit[6]: reserved bit[7]: reserved

DES_SEL

DES_SEL is a DES input data selection register.

Offset Address	Register Name	Total Reset Value
0x0040	DES_SEL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														des_sel	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	des_sel	DES input data select 000: The DES connects to the port. 001: The DES connects to LINE_BUF. 010: The DES connects to the FPN. 011: reserved 100: The DES connects to the IR output. Other values: reserved

ISP_SEL

ISP_SEL is an ISP input data selection register.

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Offset Address	Register Name	Total Reset Value
0x0050	ISP_SEL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														isp_sel	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	isp_sel	ISP input data select 00: The ISP connects to LINE_BUF. 01: The ISP connects to SRC0. 10: The ISP connects to SRC1. 11: reserved

CHN_MODE

CHN_MODE is an SRC0/DES0 bus read/write rate control mode register.

Offset Address	Register Name	Total Reset Value
0x0060	CHN_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														chn_mode	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	chn_mode	SRC0/DES0 bus read/write rate control mode (used when SRC0 shares the buffer with DES0) 00: no rate control 01: The rate of SRC0 is controlled to ensure that it is lower than the DES0 write rate (read after write) 10: The rate of DES0 is controlled to ensure that it is lower than the SRC0 read rate (write after read) 11: reserved

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BUF_MODE

BUF_MODE is a LINE_BUF input data select register.

Offset Address	Register Name	Total Reset Value
0x0070	BUF_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															buf_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	buf_mode	Input data mode of LINE_BUF 0: raw data 1: YUV422

APB_TIMEOUT

APB_TIMEOUT is an APB timeout register.

Offset Address	Register Name	Total Reset Value
0x00E0	APB_TIMEOUT	0x8000_0180

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved															timeout															
Reset	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	Timeout enable 0: disabled 1: enabled
[30:16]	RO	reserved	Reserved
[15:0]	RW	timeout	Timeout threshold (in APB clock)

VICAP_INT

VICAP_INT is an interrupt status register.

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Offset Address	Register Name	Total Reset Value
0x00F0	VICAP_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															int_isp	reserved										int_buf	int_src1	reserved	int_des0	int_ch	int_pt
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RO	int_isp	ISP interrupt status 0: No interrupt is generated. 1: An interrupt is generated.
[15:6]	RO	reserved	Reserved
[5]	RO	int_buf	LINE_BUF interrupt status 0: No interrupt is generated. 1: An interrupt is generated.
[4]	RO	int_src1	SRC1 interrupt status 0: No interrupt is generated. 1: An interrupt is generated.
[3]	RO	reserved	Reserved
[2]	RO	int_des0	DES0 interrupt status 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	int_ch	Channel interrupt status 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	int_pt	Port interrupt status 0: No interrupt is generated. 1: An interrupt is generated.

VICAP_INT_MASK

VICAP_INT_MASK is an interrupt mask register.

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Offset Address	Register Name	Total Reset Value
0x00F8	VICAP_INT_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															int_isp	reserved										int_buf	int_src1	reserved	int_des0	int_ch	int_pt
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RW	int_isp	ISP interrupt enable 0: masked 1: enabled
[15:6]	RO	reserved	Reserved
[5]	RW	int_buf	LINE_BUF interrupt enable 0: masked 1: enabled
[4]	RW	int_src1	SRC1 interrupt enable 0: masked 1: enabled
[3]	RO	reserved	Reserved
[2]	RW	int_des0	DES0 interrupt enable 0: masked 1: enabled
[1]	RW	int_ch	Channel interrupt enable 0: masked 1: enabled
[0]	RW	int_pt	Port interrupt enable 0: masked 1: enabled

PT_INTF_MOD

PT_INTF_MOD is a port mode register.

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Offset Address	Register Name	Total Reset Value
0x0100	PT_INTF_MOD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																														mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	Port enable 0: disabled 1: enabled
[30:1]	RO	reserved	Reserved
[0]	RW	mode	Timing mode 0: external sync 1: BT.656

PT_OFFSET0

PT_OFFSET0 is a component 0 offset register.

Offset Address	Register Name	Total Reset Value
0x0110	PT_OFFSET0	0xFFF0_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mask																rev	reserved											offset			
Reset	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	mask	Component 0 mask
[15]	RW	rev	Whether the data line is reversed 0: no 1: yes
[14:6]	RO	reserved	Reserved
[5:0]	RW	offset	Component 0 offset

PT_OFFSET1

PT_OFFSET1 is a component 1 offset register.

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Offset Address	Register Name	Total Reset Value
0x0114	PT_OFFSET1	0xFFF0_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mask																rev	reserved											offset			
Reset	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	mask	Component 1 mask
[15]	RW	rev	Whether the data line is reversed 0: no 1: yes
[14:6]	RO	reserved	Reserved
[5:0]	RW	offset	Component 1 offset

PT_OFFSET2

PT_OFFSET2 is a component 2 offset register.

Offset Address	Register Name	Total Reset Value
0x0118	PT_OFFSET2	0xFFF0_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mask																rev	reserved											offset			
Reset	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	mask	Component 2 mask
[15]	RW	rev	Whether the data line is reversed 0: no 1: yes
[14:6]	RO	reserved	Reserved
[5:0]	RW	offset	Component 2 offset

PT_BT656_CFG

PT_BT656_CFG is a BT.656 configuration register.

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Offset Address	Register Name	Total Reset Value
0x0120	PT_BT656_CFG	0x0000_0303

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																				field_inv	vsync_inv	hsync_inv	reserved				mode			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31]	RW	enable	BT.656 enable 0: disabled 1: enabled
[30:11]	RO	reserved	Reserved
[10]	RW	field_inv	Reverse control 0: not reversed 1: reversed
[9]	RW	vsync_inv	Reverse control 0: not reversed 1: reversed
[8]	RW	hsync_inv	Reverse control 0: not reversed 1: reversed
[7:4]	RO	reserved	Reserved
[3:0]	RW	mode	Mode select mode[0] 0: hsync is not an active signal. 1: hsync is an active signal. mode[1] 0: The hsync output is active low. 1: The hsync output is active high. mode[3:2] 00: Component 0 is parsed. 01: Component 1 is parsed. 10: Component 2 is parsed. 11: reserved

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PT_UNIFY_TIMING_CFG

PT_UNIFY_TIMING_CFG is a timing configuration register.

Offset Address	Register Name	Total Reset Value
0x0130	PT_UNIFY_TIMING_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					field_inv	field_sel		reserved				vsync_mode		vsync_inv	vsync_sel	reserved	hsync_mode		hsync_and		hsync_inv	hsync_sel		reserved				de_inv	de_sel		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RW	field_inv	Field inverse (level-1 field processing) 0: not inverted 1: inverted
[25:24]	RW	field_sel	Field source select (level-0 field processing) 00: input field 01: input vsync 10: detected based on the relationship between vsync and hsync 11: fixed at 0
[23:21]	RO	reserved	Reserved
[20:19]	RW	vsync_mode	vsync processing mode (level-2 vsync processing) 00: not processed 01: detect the rising edge 10: detect the rising edge and falling edge 11: reserved
[18]	RW	vsync_inv	vsync inversion (level-1 vsync processing) 0: not inverted 1: inverted
[17:16]	RW	vsync_sel	vsync source select (level-0 vsync processing) 00: input vsync 01: input field 10: fixed at 0 11: reserved
[15]	RO	reserved	Reserved

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[14:13]	RW	hsync_mode	hsync processing mode (level-3 hsync processing) 0: not processed 1: detect the rising edge
[12:11]	RW	hsync_and	Whether hsync is operated with the result of level-1 vsync processing (level-2 hsync processing) 00: not processed 01: ANDed 10: exclusive ORed 11: reserved
[10]	RW	hsync_inv	hsync inversion (level-1 hsync processing) 0: not inverted 1: inverted
[9:8]	RW	hsync_sel	hsync source select (level-0 hsync processing) 00: input hsync 01: input de 10: fixed at 0 11: reserved
[7:3]	RO	reserved	Reserved
[2]	RW	de_inv	de inversion (level-1 de processing) 0: not inverted 1: inverted
[1:0]	RW	de_sel	de source select (level-0 de processing) 00: input de 01: result of level-2 hsync processing 10: fixed at 1 11: fixed at 0

PT_GEN_TIMING_CFG

PT_GEN_TIMING_CFG is a timing recovery module configuration register.

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Offset Address	Register Name	Total Reset Value
0x0134	PT_GEN_TIMING_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	mode	reserved																											vsync_mode	hsync_mode	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	Timing recovery enable (timings are recovered based on timing parameters) 0: disabled 1: enabled
[30]	RW	mode	Timing recovery mode (timings are recovered based on timing parameters) 0: Timings are generated based on the input valid signal of PT. 1: Timings are internally calculated and generated.
[29:3]	RO	reserved	Reserved
[2]	RW	vsync_mode	vsync recovery 0: not recovered 1: recovered
[1]	RW	hsync_mode	hsync recovery 0: not recovered 1: recovered
[0]	RO	reserved	Reserved

PT_UNIFY_DATA_CFG

PT_UNIFY_DATA_CFG is a data configuration register.

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Offset Address	Register Name	Total Reset Value
0x0140	PT_UNIFY_DATA_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																											uv_seq	yc_seq	comp_num	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	Data separation enable 0: disabled 1: enabled
[30:4]	RO	reserved	Reserved
[3]	RW	uv_seq	CbCr sequence 0: CbCr 1: CrCb
[2]	RW	yc_seq	YC sequence 0: CY 1: YC
[1:0]	RW	comp_num	Data component select 00: component 1 01: component 2 10: component 3 11: reserved

PT_YUV444_CFG

PT_YUV444_CFG is a YUV444 configuration register.

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Offset Address	Register Name	Total Reset Value
0x0150	PT_YUV444_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	YUV enable for converting YUV422 signals into YUV444 signals 0: disabled 1: enabled
[30:0]	RO	reserved	Reserved

PT_INTF_HFB

PT_INTF_HFB is a horizontal front blanking width register.

Offset Address	Register Name	Total Reset Value
0x0180	PT_INTF_HFB	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																hfb															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	hfb	Horizontal front blanking width

PT_INTF_HACT

PT_INTF_HACT is a horizontal active width register.

Offset Address	Register Name	Total Reset Value
0x0184	PT_INTF_HACT	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	hact																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	hact	Horizontal active width (in clock cycle)

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PT_INTF_HBB

PT_INTF_HBB is a horizontal back blanking width register.

Offset Address	Register Name	Total Reset Value
0x0188	PT_INTF_HBB	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																hbb															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	hbb	Horizontal back blanking width

PT_INTF_VFB

PT_INTF_VFB is a vertical front blanking width register.

Offset Address	Register Name	Total Reset Value
0x018C	PT_INTF_VFB	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																vfb															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	vfb	Vertical front blanking width

PT_INTF_VACT

PT_INTF_VACT is a vertical active width register.

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Offset Address	Register Name	Total Reset Value
0x0190	PT_INTF_VACT	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																vact															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	vact	Vertical active width

PT_INTF_VBB

PT_INTF_VBB is a vertical back blanking width register.

Offset Address	Register Name	Total Reset Value
0x0194	PT_INTF_VBB	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																vbb															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	vbb	Vertical back blanking width

PT_INTF_VBFB

PT_INTF_VBFB is a vertical bottom front blanking width register.

Offset Address	Register Name	Total Reset Value
0x0198	PT_INTF_VBFB	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																vbfb															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	vbfb	Vertical bottom front blanking width

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PT_INTF_VBACT

PT_INTF_VBACT is a vertical bottom active width register.

Offset Address	Register Name	Total Reset Value
0x019C	PT_INTF_VBACT	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																vbact															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	vbact	Vertical bottom active width

PT_INTF_VBBB

PT_INTF_VBBB is a vertical bottom back blanking width register.

Offset Address	Register Name	Total Reset Value
0x01A0	PT_INTF_VBBB	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																vbbb															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	vbbb	Vertical bottom back blanking width

PT_ID_CFG

PT_ID_CFG is an ID configuration register.

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Offset Address	Register Name	Total Reset Value
0x01A4	PT_ID_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	mode	reset	reserved																							id_det		id_max			id		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	ID generation enable 0: disabled 1: enabled
[30]	RW	mode	ID generation mode 0: automatic mode (the ID automatically increases for each frame) 1: non-automatic mode (the ID depends on the configured value)
[29]	WO	reset	In automatic mode, setting this field to 1 restores the ID of the next frame to the initial value. This field is automatically cleared after restoration.
[28:6]	RO	reserved	Reserved
[5:4]	RW	id_det	ID of the frame whose width and height are detected
[3:2]	RW	id_max	Maximum ID in automatic mode
[1:0]	RW	id	Initial ID in automatic mode or generated ID in non-automatic mode

PT_FLASH_CFG

PT_FLASH_CFG is a flash configuration register.

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Offset Address	Register Name	Total Reset Value
0x01B0	PT_FLASH_CFG	0x1000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	id_sel				reserved									shutter_times	shutter_phase	shutter_en	reserved															flash_phase	flash_en
Reset	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	id_sel	Trigger frame ID 0001: triggered by using the start interrupt of the frame whose ID is 0 0010: triggered by using the start interrupt of the frame whose ID is 1 0100: triggered by using the start interrupt of the frame whose ID is 2 1000: triggered by using the start interrupt of the frame whose ID is 3
[27:19]	RO	reserved	Reserved
[18]	RW	shutter_times	Shutter pulse times 0: twice 1: once
[17]	RW	shutter_phase	Shutter signal reverse 0: not reversed 1: reversed
[16]	WO	shutter_en	Shutter trigger enable. This field is automatically cleared.
[15:2]	RO	reserved	Reserved
[1]	RW	flash_phase	Flash signal reverse 0: not reversed 1: reversed
[0]	WO	flash_en	Flash trigger enable. This field is automatically cleared.

PT_FLASH_CYC0

PT_FLASH_CYC0 is a flash timing 0 width register.

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Offset Address	Register Name	Total Reset Value
0x01C0	PT_FLASH_CYC0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cyc																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cyc	Duration of flash timing 0 (in PT clock). The configured value is the actual value minus 1.

PT_FLASH_CYC1

PT_FLASH_CYC1 is a flash timing 1 width register.

Offset Address	Register Name	Total Reset Value
0x01C4	PT_FLASH_CYC1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cyc																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cyc	Duration of flash timing 1 (in PT clock). The configured value is the actual value minus 1.

PT_SHUTTER_CYC0

PT_SHUTTER_CYC0 is a shutter timing 0 width register.

Offset Address	Register Name	Total Reset Value
0x01D0	PT_SHUTTER_CYC0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cyc																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cyc	Duration of shutter timing 0 (in PT clock). The configured value is the actual value minus 1.

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PT_SHUTTER_CYC1

PT_SHUTTER_CYC1 is a shutter timing 1 width register.

Offset Address	Register Name	Total Reset Value
0x01D4	PT_SHUTTER_CYC1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cyc																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cyc	Duration of shutter timing 1 (in PT clock). The configured value is the actual value minus 1.

PT_SHUTTER_CYC2

PT_SHUTTER_CYC2 is a shutter timing 2 width register.

Offset Address	Register Name	Total Reset Value
0x01D8	PT_SHUTTER_CYC2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cyc																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cyc	Duration of shutter timing 2 (in PT clock). The configured value is the actual value minus 1.

PT_SHUTTER_CYC3

PT_SHUTTER_CYC3 is a shutter timing 3 width register.

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Offset Address	Register Name	Total Reset Value
0x01DC	PT_SHUTTER_CYC3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cyc																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cyc	Duration of shutter timing 3 (in PT clock). The configured value is the actual value minus 1.

PT_STATUS

PT_STATUS is a port status register.

Offset Address	Register Name	Total Reset Value
0x01E0	PT_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										id		field	vysnc	hsync	de
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5:4]	RO	id	Port output ID
[3]	RO	field	Field output over the port
[2]	RO	vysnc	vysnc signal output over the port
[1]	RO	hsync	hsync signal over the port
[0]	RO	de	de signal output over the port

PT_BT656_STATUS

PT_BT656_STATUS is a BT.656 status register.

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Offset Address	Register Name	Total Reset Value
0x01E4	PT_BT656_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																seav								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RO	seav	Sync code
[7:0]	RO	reserved	Reserved

PT_SIZE

PT_SIZE is an input picture size register.

Offset Address	Register Name	Total Reset Value
0x01EC	PT_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	height																width															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	height	Picture height
[15:0]	RO	width	Picture width

PT_INT

PT_INT is a channel interrupt status register.

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Offset Address	Register Name	Total Reset Value
0x01F0	PT_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													height_err	width_err	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	WC	height_err	Status of the picture height change interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[1]	WC	width_err	Status of the picture width change interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[0]	WC	fstart	Status of the field/frame start interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.

PT_INT_MASK

PT_INT_MASK is a port interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x01F8	PT_INT_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													height_err	width_err	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved

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[2]	RW	height_err	Picture height change interrupt enable 0: masked 1: enabled
[1]	RW	width_err	Picture width change interrupt enable 0: masked 1: enabled
[0]	RW	fstart	Frame/Field start interrupt enable 0: masked 1: enabled

DES_CTRL

DES_CTRL is a DES control register.

Offset Address	Register Name	Total Reset Value
0x0200	DES_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																							bit_width							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	DES enable 0: disabled 1: enabled
[30:6]	RO	reserved	Reserved
[5:0]	RW	bit_width	Data width 0x08: 8 bits 0x0A: 10 bits 0x0C: 12 bits 0x0E: 14 bits 0x10: 16 bits Other values: reserved

DES_REG_NEWER

DES_REG_NEWER is a capture control register.

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Offset Address	Register Name	Total Reset Value
0x0204	DES_REG_NEWER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															reg_newer
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	reg_newer	DES update. This bit is automatically cleared for each frame.

DES_ID_CFG

DES_ID_CFG is a DES RX ID selection register (instant register).

Offset Address	Register Name	Total Reset Value
0x0210	DES_ID_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												id_sel			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	id_sel	ID of the data received by the DES 0: Data with this ID is filtered. 1: Data with this ID is received. bit[0]: DES RX enable for data with the ID 0 bit[1]: DES RX enable for data with the ID 1 bit[2]: DES RX enable for data with the ID 2 bit[3]: DES RX enable for data with the ID 3

DES_CROP_CFG

DES_CROP_CFG is a crop enable register.

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Offset Address	Register Name	Total Reset Value
0x0220	DES_CROP_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															n0_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	n0_en	Region 0 enable 0: disabled 1: enabled

DES_CROP0_START

DES_CROP0_START is a crop start position register for region 0.

Offset Address	Register Name	Total Reset Value
0x0230	DES_CROP0_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				y_start												reserved				x_start											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	y_start	ID of the line from which pictures start to be captured
[15:13]	RO	reserved	Reserved
[12:0]	RW	x_start	ID of the pixel from which pictures start to be captured

DES_CROP0_SIZE

DES_CROP0_SIZE is a crop size configuration register for region 0.

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Offset Address	Register Name	Total Reset Value
0x0234	DES_CROP0_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height (in line) of the obtained picture. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Width (in pixel) of the obtained picture. The configured value is the actual value minus 1.

DES_CFG

DES_CFG is a DES configuration register.

Offset Address	Register Name	Total Reset Value
0x0270	DES_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cmp_en	k_coef_range							reserved																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	cmp_en	DES channel frame compress enable 0: disabled 1: enabled
[30:27]	RW	k_coef_range	Frame compression rate of the DES channel. The value range is [7:12]. The typical value is the data bit width minus 1. The configured value is 12 when the data bit width is 14 bits.
[26:0]	RO	reserved	Reserved

DES_SIZE

DES_SIZE is a DES storage size register.

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Offset Address	Register Name	Total Reset Value
0x0284	DES_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height (in line) of the stored picture. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Width (in pixel) of the stored picture. The configured value is the actual value minus 1.

DES_FADDR

DES_FADDR is a DES storage base address register.

Offset Address	Register Name	Total Reset Value
0x0290	DES_FADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	faddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	faddr	DES storage base address

DES_STRIDE

DES_STRIDE is a DES storage line stride register.

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Offset Address	Register Name	Total Reset Value
0x0294	DES_STRIDE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																stride															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	stride	Line stride for the DES channel frame

DES_BUF_CFG

DES_BUF_CFG is a DES bus buffer control register.

Offset Address	Register Name	Total Reset Value
0x029C	DES_BUF_CFG	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																buf_thd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	buf_thd	Bus buffer depth

DES_INT

DES_INT is a DES raw interrupt register.

Offset Address	Register Name	Total Reset Value
0x02F0	DES_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											update_cfg	field_throw	buf_ovf	cc_int	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved

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[4]	WC	update_cfg	Status of the register update interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[3]	WC	field_throw	Status of the field/frame loss interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[2]	WC	buf_ovf	Status of the internal FIFO overflow error interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[1]	WC	cc_int	Status of the capture completion interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[0]	WC	fstart	Status of the field/frame start interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.

DES_MASK

DES_MASK is a DES interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x02F8	DES_MASK	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3	2	1	0
Name	reserved							update_cfg	field_throw	buf_ovf	cc_int	fstart
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	RW	update_cfg	Register update interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.

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[3]	RW	field_throw	Field/Frame loss interrupt enable 0: masked 1: enabled
[2]	RW	buf_ovf	Internal FIFO overflow error interrupt enable 0: masked 1: enabled
[1]	RW	cc_int	Capture completion interrupt enable 0: masked 1: enabled
[0]	RW	fstart	Field/Frame start interrupt enable 0: masked 1: enabled

LINE_BUF_CFG

LINE_BUF_CFG is a LINE_BUF control register.

Offset Address	Register Name	Total Reset Value
0x0500	LINE_BUF_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																														vsync_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	LINE_BUF enable 0: disabled 1: enabled
[30:1]	RO	reserved	Reserved
[0]	RW	vsync_mode	0: Both data and the vsync signal are discarded during frame discarding. 1: Only data is discarded during frame discarding.

LINE_BUF_REG_NEWER

LINE_BUF_REG_NEWER is a LINE_BUF update register.

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Offset Address	Register Name	Total Reset Value
0x0504	LINE_BUF_REG_NEWER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															reg_newer
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	WO	reg_newer	LINE_BUF update. This bit is automatically cleared for each frame.

LINE_BUF_ID_CFG

LINE_BUF_ID_CFG is a LINE_BUF RX ID selection register (instant register).

Offset Address	Register Name	Total Reset Value
0x0510	LINE_BUF_ID_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												id_sel			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	id_sel	ID of the data received by the LINE_BUF 0: Data with this ID is filtered. 1: Data with this ID is received. bit[0]: LINE_BUF RX enable for data with the ID 0 bit[1]: LINE_BUF RX enable for data with the ID 1 bit[2]: LINE_BUF RX enable for data with the ID 2 bit[3]: LINE_BUF RX enable for data with the ID 3

LINE_BUF_CROP_CFG

LINE_BUF_CROP_CFG is a crop register.

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Offset Address	Register Name	Total Reset Value
0x0520	LINE_BUF_CROP_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															n0_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	n0_en	Region 0 enable 0: disabled 1: enabled

LINE_BUF_CROP0_START

LINE_BUF_CROP0_START is a crop start position register for region 0.

Offset Address	Register Name	Total Reset Value
0x0530	LINE_BUF_CROP0_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				y_start												reserved				x_start											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	y_start	ID of the line from which pictures start to be captured
[15:13]	RO	reserved	Reserved
[12:0]	RW	x_start	ID of the pixel from which pictures start to be captured

LINE_BUF_CROP0_SIZE

LINE_BUF_CROP0_SIZE is a crop size configuration register for region 0.

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Offset Address	Register Name	Total Reset Value
0x0534	LINE_BUF_CROP0_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height (in line) of the obtained picture. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Width (in pixel) of the obtained picture. The configured value is the actual value minus 1.

LINE_BUF_WIDTH

LINE_BUF_WIDTH is a LINE_BUF picture width register.

Offset Address	Register Name	Total Reset Value
0x0550	LINE_BUF_WIDTH	0x077F_077F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				line_buf_thd												reserved				line_buf_width											
Reset	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	line_buf_thd	Threshold for starting the LINE_BUF RAM read operation
[15:12]	RO	reserved	Reserved
[11:0]	RW	line_buf_width	LINE_BUF picture width

LINE_BUF_HEIGHT

LINE_BUF_HEIGHT is a LINE_BUF picture height register.

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Offset Address	Register Name	Total Reset Value
0x0554	LINE_BUF_HEIGHT	0x0000_0437

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				line_buf_height											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	1

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	line_buf_height	LINE_BUF picture height

LINE_BUF_INT

LINE_BUF_INT is a LINE_BUF raw interrupt register.

Offset Address	Register Name	Total Reset Value
0x05F0	LINE_BUF_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											update_cfg	field_throw	reserved	reserved	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	WC	update_cfg	Status of the register update interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[3]	WC	field_throw	Status of the field/frame loss interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	reserved	Reserved
[1]	RO	reserved	Reserved

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[0]	WC	fstart	Status of the field/frame start interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.

LINE_BUF_MASK

LINE_BUF_MASK is a LINE_BUF interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x05F8	LINE_BUF_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											update_cfg	field_throw	reserved	reserved	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	RW	update_cfg	Register update interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.
[3]	RW	field_throw	Field/Frame loss interrupt enable 0: masked 1: enabled
[2]	RO	reserved	Reserved
[1]	RO	reserved	Reserved
[0]	RW	fstart	Field/Frame start interrupt enable 0: masked 1: enabled

SRC_CTRL

SRC_CTRL is an SRC control register.

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Offset Address	Register Name	Total Reset Value
0x0700	SRC_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																							bit_width							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	SRC enable 0: disabled 1: enabled 0x700−0x7FF: address space of the SRC1 register
[30:6]	RO	reserved	Reserved
[5:0]	RW	bit_width	Data bit width 0x08: 8 bits 0x0A: 10 bits 0x0C: 12 bits 0x0E: 14 bits 0x10: 16 bits Other values: reserved

SRC_REG_NEWER

SRC_REG_NEWER is a capture control register.

Offset Address	Register Name	Total Reset Value
0x0704	SRC_REG_NEWER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															reg_newer
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	reg_newer	SRC update. This bit is automatically cleared for each frame.

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SRC_CFG

SRC_CFG is a configuration register.

Offset Address	Register Name	Total Reset Value
0x0770	SRC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dcmp_en	k_coef_range							reserved																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	dcmp_en	SRC read channel frame compress enable 0: disabled 1: enabled
[30:27]	RW	k_coef_range	Frame compression rate of the SRC channel. The value range is [7:12]. The configuration during decompression must be the same as the that during data compression. The typical value is the bit width minus 1. The configured value is 12 when the bit width is 14 bits.
[26:0]	RO	reserved	Reserved

SRC_SIZE

SRC_SIZE is an SRC storage size register.

Offset Address	Register Name	Total Reset Value
0x0784	SRC_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height (in line) of the stored picture. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved

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[12:0]	RW	width	Width (in pixel) of the stored picture. The configured value is the actual value minus 1.

SRC_FADDR

SRC_FADDR is an SRC storage base address register.

Offset Address	Register Name	Total Reset Value
0x0790	SRC_FADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	faddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	faddr	SRC storage base address

SRC_STRIDE

SRC_STRIDE is an SRC storage line stride register.

Offset Address	Register Name	Total Reset Value
0x0794	SRC_STRIDE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																stride															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	stride	Line stride for the SRC channel frame

SRC_BUF_CFG

SRC_BUF_CFG is an SRC bus buffer control register.

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Offset Address	Register Name	Total Reset Value
0x079C	SRC_BUF_CFG	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																buf_thd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	buf_thd	Depth of the bus buffer

SRC_INT

SRC_INT is an SRC raw interrupt register.

Offset Address	Register Name	Total Reset Value
0x07F0	SRC_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										dcmp_wrong	update_cfg	field_throw	buf_ovf	cc_int	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5]	WC	dcmp_wrong	Status of the decompress error interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[4]	WC	update_cfg	Status of the register update interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[3]	WC	field_throw	Status of the field/frame loss interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.

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[2]	WC	buf_ovf	Status of the internal FIFO overflow error interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[1]	WC	cc_int	Status of the capture completion interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[0]	WC	fstart	Status of the field/frame start interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.

SRC_MASK

SRC_MASK is an SRC interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x07F8	SRC_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										dcmp_wrong	update_cfg	field_throw	buf_ovf	cc_int	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5]	RW	dcmp_wrong	Decompress error interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.
[4]	RW	update_cfg	Register update interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.
[3]	RW	field_throw	Field/Frame loss interrupt enable 0: masked 1: enabled
[2]	RW	buf_ovf	Internal FIFO overflow error interrupt enable 0: masked 1: enabled

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[1]	RW	cc_int	Capture completion interrupt enable 0: masked 1: enabled
[0]	RW	fstart	Field/Frame start interrupt enable 0: masked 1: enabled

CH_REG_NEWER

CH_REG_NEWER is a capture control register.

Offset Address	Register Name	Total Reset Value
0x1004	CH_REG_NEWER	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								reg_newer
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	reg_newer	Channel update. This bit is automatically cleared for each frame.

CH_ADAPTER_CFG

CH_ADAPTER_CFG is a timing adaptation register.

Offset Address	Register Name	Total Reset Value
0x1010	CH_ADAPTER_CFG	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2	1	0
Name	reserved								vsync_mode	hsync_mode
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved

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[1]	RW	vsync_mode	Whether to perform delay processing on the vsync signal The delay processing is performed on the vsync signal only when both sharpen and uvnr are enabled. 0: not processed 1: processed
[0]	RW	hsync_mode	Whether to perform delay processing on the hsync signal 0: not processed 1: processed

CH_WCH_Y_CFG

CH_WCH_Y_CFG is a Y component configuration register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x1080	CH_WCH_Y_CFG	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved											interleave	bfield	bit_width		reserved								flip	mirror	head_tword	cmp_mode	cmp_en			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31]	RW	enable	CH write channel enable 0: disabled 1: enabled
[30:20]	RO	reserved	Reserved
[19]	RW	interleave	Progressive/Interlaced flag 0: The output data is progressive or interlaced and does not need to be interleaved as frames for storing. 1: The output data interlaced and needs to be interleaved as frames for storing.
[18]	RW	bfield	Top/Bottom field flag. This field is used when the output data interlaced and needs to be interleaved as frames for storing. 0: top field 1: bottom field
[17:16]	RW	bit_width	Data width 00: 8 bits 10: 16 bits Other values: reserved

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[15:5]	RO	reserved	Reserved
[4]	RW	flip	Channel flip enable 0: disabled 1: enabled
[3]	RW	mirror	Channel mirror enable 0: disabled 1: enabled
[2]	RW	head_tword	Header information stride 0: 128 bits (128 bits/line) 1: 256 bits (256 bits/line)
[1]	RW	cmp_mode	Compressed segment length 0: 128 pixels 1: 256 pixels
[0]	RW	cmp_en	Compression enable 0: disabled 1: enabled

CH_WCH_Y_SIZE

CH_WCH_Y_SIZE is a Y component storage size register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x1084	CH_WCH_Y_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height (in line) of the stored picture. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Width (in pixel) of the stored picture. The configured value is the actual value minus 1.

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CH_WCH_Y_FADDR

CH_WCH_Y_FADDR is a Y component storage base address register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x1090	CH_WCH_Y_FADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	faddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	faddr	Base address for storing the Y component

CH_WCH_Y_HADDR

CH_WCH_Y_HADDR is a Y component header information storage base address register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x1094	CH_WCH_Y_HADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	haddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	haddr	Base address for storing the Y component header information

CH_WCH_Y_STRIDE

CH_WCH_Y_STRIDE is a Y component line offset register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x1098	CH_WCH_Y_STRIDE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																stride															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	stride	Stride (in byte) for storing the Y component

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CH_WCH_C_CFG

CH_WCH_C_CFG is a C component configuration register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x10A0	CH_WCH_C_CFG	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved											interleave	bfield	bit_width		reserved											flip	mirror	head_tword	cmp_mode	cmp_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31]	RW	enable	CH write channel enable 0: disabled 1: enabled
[30:20]	RO	reserved	Reserved
[19]	RW	interleave	Progressive/Interlaced flag 0: The output data is progressive or interlaced and does not need to be interleaved as frames for storing. 1: The output data interlaced and needs to be interleaved as frames for storing.
[18]	RW	bfield	Top/Bottom field flag. This field is used when the output data interlaced and needs to be interleaved as frames for storing. 0: top field 1: bottom field
[17:16]	RW	bit_width	Data width 00: 8 bits 10: 16 bits Other values: reserved
[15:5]	RO	reserved	Reserved
[4]	RW	flip	Channel flip enable 0: disabled 1: enabled
[3]	RW	mirror	Channel mirror enable 0: disabled 1: enabled

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[2]	RW	head_tword	Header information stride 0: 128 bits (128 bits/line) 1: 256 bits (256 bits/line)
[1]	RW	cmp_mode	Compressed segment length 0: 128 pixels 1: 256 pixels
[0]	RW	cmp_en	Compression enable 0: disabled 1: enabled

CH_WCH_C_SIZE

CH_WCH_C_SIZE is a C component storage size register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x10A4	CH_WCH_C_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height (in line) of the stored picture. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Width (in pixel) of the stored picture. The configured value is the actual value minus 1.

CH_WCH_C_FADDR

CH_WCH_C_FADDR is a C component storage base address register for the WCH module.

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Offset Address	Register Name	Total Reset Value
0x10B0	CH_WCH_C_FADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	faddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	faddr	Base address for storing the C component

CH_WCH_C_HADDR

CH_WCH_C_HADDR is a C component header information storage base address register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x10B4	CH_WCH_C_HADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	haddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	haddr	Base address for storing the C component header information

CH_WCH_C_STRIDE

CH_WCH_C_STRIDE is a C component line offset register for the WCH module.

Offset Address	Register Name	Total Reset Value
0x10B8	CH_WCH_C_STRIDE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																stride															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	stride	Stride (in byte) for storing the C component

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CH_INT

CH_INT is a channel raw interrupt register.

Offset Address	Register Name	Total Reset Value
0x10F0	CH_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																fstart_dly	reserved											update_cfg	field_throw	buf_ovf	cc_int	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	WC	fstart_dly	Status of the delayed field/frame start interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[14:5]	RO	reserved	Reserved
[4]	WC	update_cfg	Status of the register update interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[3]	WC	field_throw	Status of the field/frame loss interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[2]	WC	buf_ovf	Status of the internal FIFO overflow error interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[1]	WC	cc_int	Status of the capture completion interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[0]	WC	fstart	Status of the field/frame start interrupt. Writing 1 clears the interrupt. 0: No interrupt is generated. 1: An interrupt is generated.

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CH_INT_MASK

CH_INT_MASK is a channel interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x10F8	CH_INT_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																fstart_dly	reserved										update_cfg	field_throw	buf_ovf	cc_int	fstart
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RW	fstart_dly	Delayed field/frame start interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.
[14:5]	RO	reserved	Reserved
[4]	RW	update_cfg	Register update interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.
[3]	RW	field_throw	Field/Frame loss interrupt enable 0: masked 1: enabled
[2]	RW	buf_ovf	Internal FIFO overflow error interrupt enable 0: masked 1: enabled
[1]	RW	cc_int	Capture completion interrupt enable 0: masked 1: enabled
[0]	RW	fstart	Field/Frame start interrupt enable 0: masked 1: enabled

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9.2 VDP

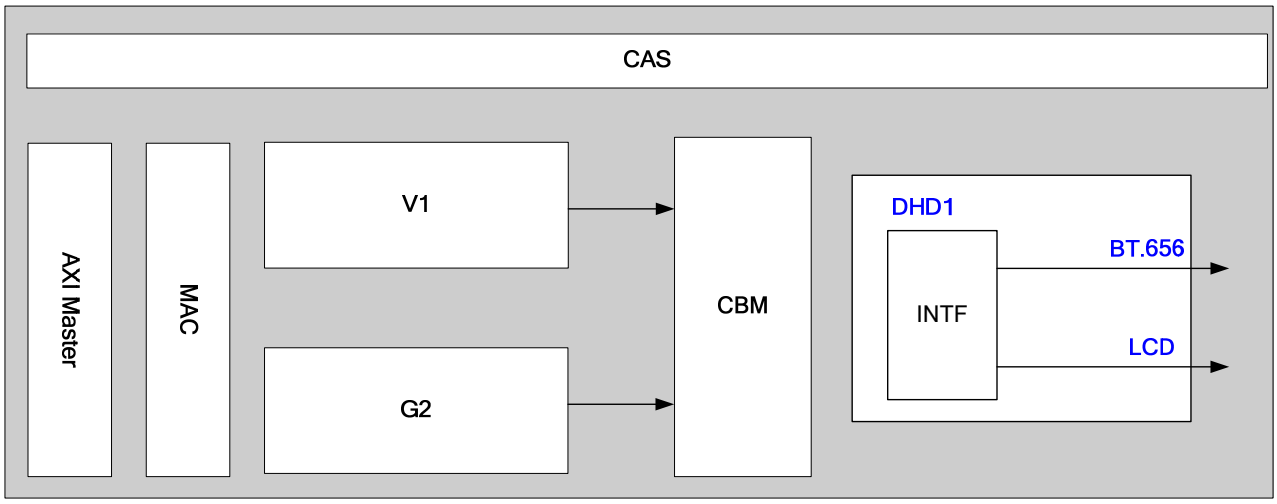
9.2.1 Overview

The video display processor (VDP) reads video and graphics data from the memory, overlays the data at the video and graphics layers, and transmits the data through the display channel.

9.2.2 Architecture

Figure 9-17 shows the block diagram of the VDP.

Figure 9-17 Block diagram of the VDP



The following describes the major concepts:

- Surface: data path of bus inputs. The surface reads and processes the bus data of a layer. The surfaces include the video layer V1 and graphics layer G2.
- Display channel: It includes the HD display channel DHD1.
- Crossbar mixer (CBM): It supports overlapping of the video layer and graphics layer.
- Memory access controller (MAC): Each module reads data from the memory over the AXI bus. The MAC arbitrates the requests from all surfaces.
- Control and status (CAS): It configures registers over the APB and reports the status of other modules to the CPU.

The VDP registers are classified into:

- Global registers

  Include the bus configuration registers, interrupt registers, and version register.
- Surface registers

  Include the video layer configuration registers and graphics layer configuration registers.
- Display channel registers

  Include the DHD1 configuration registers.

The VDP has the following features:

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- Digital output interfaces. The SD output interfaces support ITU-R BT.656 outputs and LCD 565 serial outputs.
- Video layer (V1)
- Graphics layer (G2)
- Overlay
- One video channel. It has a separate vertical timing interrupt (indicating the end of a field/frame) and a low bandwidth interrupt.

9.2.3 Operating Mode

9.2.3.1 Clock Configurations

The VDP clock source can be set to either of the following clocks:

Interface output clock source 2: VPLL

For details, see the description of the PERI_CRG13 for 3.2.2 section.

9.2.3.2 Reset

The VDP supports a hard reset signal and a soft reset signal.

CAUTION

Before soft-resetting the AXI bus, do as follows:

- Disable all layers.
- Configure the bus reset request after the next field/frame interrupt is detected.

9.2.3.3 Bus Configurations

AXI Master

The VDP provides one master interface to improve bus access efficiency.

The VDP supports the AXI master. The data read/write requests of V1 and G2 can be implemented by using the master.

APB Register Configuration

The VDP registers are read and written over the APB interface. The base address for VDP registers is 0x205C_0000, the register addressing space is 64 KB, and the address offset range is 0x0000–0xFFFF.

Outstanding Configuration

The depth of the AXI master outstanding ranges from 0 to 7. When the outstanding depth is set to 0, the AXI master does not operate the bus.

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9.2.3.4 Digital Output Interfaces

The VDP supports two digital interface output modes: ITU-R BT.656 and LCD RGB.

9.2.3.5 Interrupts

VDP interrupts are classified into:

- Vertical timing interrupt
- Low bandwidth interrupt

Vertical Timing Interrupt

The position of generating the vertical timing interrupt can be configured.

- The VDP supports one vertical timing interrupt, indicating the end of a frame or field.
- The interrupt can be generated by frame or field.
- In progressive display mode, the vertical timing interrupt must be generated by frame.
- In interlaced display mode, the vertical timing interrupt can be generated by frame or field. You are advised to set the interrupt generation mode to generation by field for HD and generation by frame for SD.
- Interrupt mask can be configured.
- The threshold for the vertical timing interrupt can be configured.
- Each interrupt source can be separately enabled or disabled. Writing 1 clears the interrupt.

Low Bandwidth Interrupt

The VDP allows the low bandwidth status to be reported by using an interrupt.

- The VDP supports one low bandwidth interrupt, indicating low bandwidth of a frame or field.
- The interrupt can be masked.
- Each interrupt source can be separately enabled or disabled. Writing 1 clears the interrupt.

9.2.4 Function Description

9.2.4.1 Video Layer

Features

The video layer has the following features:

- Input pixel formats of semi-planar420 and semi-planar422
- Minimum input resolution of 32 x 32 and maximum input resolution of 720 x 576
- Minimum output resolution of 32 x 32 and maximum output resolution of 720 x 576
- A multiple of 2 for the input horizontal resolution
- A multiplier of 4 for the interlaced 420 input vertical resolution and a multiple of 2 for the other input vertical resolutions
- Interlaced mode and progressive mode
- User-defined source luminance and chrominance start addresses, uncompressed data source (2-byte-aligned), and compressed data source (16-byte-aligned)

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- User-defined source luminance and chrominance stride, 16-byte-aligned
- Vertical chrominance up sampling, replication mode
- Horizontal chrominance up sampling, replication mode
- Color space conversion (CSC) and adjustment of contrast, hue, and saturation
- User-defined display position, anywhere on the screen
- User-defined global alpha ranging from 0 to 255

420-422 (Vertical Chrominance Up Sampling)

When YUV420 data is input, 2x vertical scaling must be performed on the chrominance to convert the data format into YUV422. V1 converts the data format in replication mode.

422-444 (Horizontal Chrominance Up Sampling)

Horizontal chrominance up sampling converts the YUV422 data format into YUV444 in replication mode.

CSC

- CSC between YCbCr BT601 and BT709
- Adjustment of luminance, hue, and saturation

9.2.4.2 Graphics Layer

Features

The graphics layer has the following features:

- Input pixel formats of ARGB1555 and ARGB4444
- Minimum input resolution of 32 x 32 and maximum input resolution of 720 x 576
- Minimum output resolution of 32 x 32 and maximum output resolution of 720 x 576
- A multiple of 2 for the input horizontal or vertical resolution
- Interlaced mode and progressive mode
- Frame update and field update
- User-defined source start address, 128-bit-aligned (16-byte-aligned)
- User-defined source stride, 128-bit-aligned (16-byte-aligned)
- CSC
- User-defined display position, anywhere on the screen
- User-defined global alpha ranging from 0 to 255
- Enabling or disabling of the pixel alpha, and alpha0 or alpha1 for the picture in ARGB1555 format with pixel alpha
- Colorkey processing
- Premultiplication

CSC

CSC between RGB2YCbCr601 and RGB2YCbCr709 is supported.

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Alpha Processing

The alpha value sources of the graphics layer are as follows:

- Pixel alpha value: overlay attribute of a pixel
- Global alpha value: overlay attributes of a layer

There is a special pixel alpha value. In RGB1555 format, the alpha value is only one bit. This bit is the index of the alpha value but not the actual alpha value. The index value is used to select the actual alpha value from the alpha register. When the index value is 0, alpha0 is selected; when the index value is other values, alpha1 is selected.

Colorkey

- The key mode is configurable. The pixels inside or outside the range can be selected for key processing.
- The key bit can be masked.

Premultiplication

Input data can be premultiplied data. In this case, the colorkey function must be disabled.

9.2.4.3 Overlaying

The VDP supports overlaying of V1 and G2.

Features

- Bottom-to-top overlaying
- User-defined overlaying background color
- User-defined priority of the overlaid surface

9.2.4.4 Display Channel

Video Channel Features

- DHD1 can function as an HD or SD output channel.
- Only one output timing is supported in an application scenario. HD and SD timings cannot be output at the same time.
- The following typical output timings are supported: 576p, 576i.

Timing Configuration

The VDP output interfaces support typical and non-typical timings to connect to different interfaces.

CAUTION

Interfaces must be disabled before timing parameters are configured and be enabled after configuration.

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9.2.4.5 SD Output Interface BT.656

Features

- 10 bits to 8 bits dither
- YCbCr444-to-YCbCr422 horizontal chrominance down sampling (DFIR)
- Clipping for data clamp (Y 16−235 and C 16−240 according to the interface protocol)
- Typical output timings of NTSC and PAL

9.2.4.6 LCD Output Interface RGB

Features

- 8-bit RGB output with adjustable bit sequence
- RGB serial output in the sequence of R, G, B. The lower eight bits are output.
- Maximum 27 MHz output clock

9.2.5 Register Summary

Table 9-6 describes VDP registers.

Table 9-6 Summary of VDP registers (base address: 0x205C_0000)

Offset Address	Register	Description	Page
0x0000	VOCTRL	VO control register	9-79
0x0004	VOINTSTA	VO interrupt status register (read-only)	9-80
0x0008	VOMSKINTSTA	VO masked interrupt status register	9-81
0x000C	VOINTMSK	VDP interrupt mask register	9-82
0x0010	VDPVERSION1	VDP version register 1	9-84
0x0014	VDPVERSION2	VDP version register 2	9-84
0x0034	VOAXICTRL	VO AXI bus configuration register	9-84
0x0100	VO_MUX	VO interface multiplexing register	9-85
0x1000	V1_CTRL	V1 configuration register (non-instant register)	9-85
0x1004	V1_UPD	V1 channel update enable register	9-87
0x1028	V1_IRESO	V1 input resolution register (non-instant register)	9-87
0x102C	V1_ORESO	V1 output resolution register (non-instant register)	9-88
0x1038	V1_CBMPARA	V1 overlay parameter register (non-instant register)	9-88

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Offset Address	Register	Description	Page
0x1044	V1_CPOS	Horizontal crop coordinate register for the source picture	9-88
0x1060	V1_DFPOS	V1 surface start position (in the display window) register (non-instant register)	9-89
0x1064	V1_DLPOS	V1 surface end position (in the display window) register (non-instant register)	9-89
0x1068	V1_VFPOS	V1 surface content start position (in the display window) register (non-instant register)	9-90
0x106C	V1_VLPOS	V1 surface content end position (in the display window) register (non-instant register)	9-90
0x1070	V1_BK	V1 background color register	9-91
0x1074	V1_ALPHA	V1 background filling color alpha register	9-91
0x1080	V1_CSC_IDC	V1 CSC input DC component register (instant register)	9-92
0x1084	V1_CSC_ODC	V1 CSC output DC component register (instant register)	9-92
0x1088	V1_CSC_IODC	V1 CSC input/output DC component register (instant register)	9-93
0x108C	V1_CSC_P0	V1 CSC parameter 0 register (instant register)	9-93
0x1090	V1_CSC_P1	V1 CSC parameter 1 register (instant register)	9-94
0x1094	V1_CSC_P2	V1 CSC parameter 2 register (instant register)	9-95
0x1098	V1_CSC_P3	V1 CSC parameter 3 register (instant register)	9-95
0x109C	V1_CSC_P4	V1 CSC parameter 4 register (instant register)	9-96
0x1180	V1_IFIRCOEF01	IFIR filtering coefficient 0/1 register	9-96
0x1184	V1_IFIRCOEF23	IFIR filtering coefficient 2/3 register	9-96
0x1188	V1_IFIRCOEF45	IFIR filtering coefficient 4/5 register	9-97
0x118C	V1_IFIRCOEF67	IFIR filtering coefficient 6/7 register	9-97
0x1204	V1_P0LADDR	V1 luminance address register	9-98
0x1208	V1_P0CADDR	V1 chrominance address register	9-98

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Offset Address	Register	Description	Page
0x120C	V1_P0STRIDE	V1 stride register	9-99
0x7000	G2_CTRL	G2 configuration register	9-99
0x7004	G2_UPD	G2 update enable register	9-100
0x7010	G2_ADDR	G2 address register	9-101
0x701C	G2_STRIDE	G2 stride register	9-101
0x7020	G2_IRESO	G2 input resolution register (non-instant register)	9-102
0x7024	G2_SFPOS	G2 surface read data start position (in the source bitmap) register (non-instant register)	9-102
0x7030	G2_CBMPARA	Overlaying parameter register (non-instant register)	9-103
0x7034	G2_CKEYMAX	Maximum colorkey value register (non-instant register)	9-104
0x7038	G2_CKEYMIN	Minimum colorkey value register (non-instant register)	9-104
0x703C	G2_CMASK	Colorkey mask value register (non-instant register)	9-105
0x7040	G2_PARAADDR	G2 LUT coefficient address register	9-105
0x7044	G2_PARAUP	G2 LUT coefficient update register	9-106
0x7080	G2_DFPOS	G2 surface start position (in the display window) register (non-instant register)	9-106
0x7084	G2_DLPOS	G2 surface end position (in the display window) register (non-instant register)	9-107
0x7088	G2_VFPOS	G2 surface content start position (in the display window) register (non-instant register)	9-107
0x708C	G2_VLPOS	G2 surface content end position (in the display window) register (non-instant register)	9-108
0x7090	G2_BK	G2 background color register	9-108
0x7094	G2_ALPHA	G2 background filling color alpha register	9-109
0x70C0	G2_CSC_IDC	G2 CSC input DC component register (instant register)	9-109
0x70C4	G2_CSC_ODC	G2 CSC output DC component register (instant register)	9-110

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Offset Address	Register	Description	Page
0x70C8	G2_CSC_IODC	G2 CSC input/output DC component register (instant register)	9-111
0x70CC	G2_CSC_P0	G2 CSC parameter 0 register (instant register)	9-112
0x70D0	G2_CSC_P1	G2 CSC parameter 1 register (instant register)	9-112
0x70D4	G2_CSC_P2	G2 CSC parameter 2 register (instant register)	9-113
0x70D8	G2_CSC_P3	G2 CSC parameter 3 register (instant register)	9-113
0x70DC	G2_CSC_P4	G2 CSC parameter 4 register (instant register)	9-114
0xB420	CBM_BKG2	CBM mixer 2 overlay background color register	9-114
0xB428	CBM_MIX2	CBM mixer 2 priority configuration register	9-115
0xC400	DHD1_CTRL	DSD1 global control register	9-115
0xC404	DHD1_VSYNC	Top field vertical sync timing register in interlaced output mode or frame vertical sync timing register in progressive output mode	9-117
0xC408	DHD1_HSYNC1	Horizontal sync configuration register in interlaced or progressive output mode	9-117
0xC40C	DHD1_HSYNC2	Horizontal sync configuration register in interlaced or progressive output mode (non-instant register)	9-118
0xC410	DHD1_VPLUS	Bottom field vertical sync timing register in interlaced output mode (non-instant register)	9-118
0xC414	DHD1_PWR	Sync signal pulse width register (non-instant register)	9-119
0xC418	DHD1_VTTHD3	Vertical timing threshold register (instant register)	9-119
0xC41C	DHD1_VTTHD	Vertical timing threshold register (instant register)	9-120
0xC4B0	DHD1_PARATHD	PARA coefficient update pointer threshold register	9-121
0xC4C0	DHD1_START_POS	DHD1 start signal start position register	9-121

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Offset Address	Register	Description	Page
0xC4F0	DHD1_STATE	DHD1 status register	9-121
0xD210	BT_CLIP0_L	BT.656 clip lowest threshold register (instant register)	9-122
0xD214	BT_CLIP0_H	BT.656 clip highest threshold register (instant register)	9-123
0xD280	BT_DITHER0_CTRL	BT.656 dither control register	9-123
0xD284	BT_DITHER0_COEF0	BT.656 dither coefficient 0 register	9-124
0xD288	BT_DITHER0_COEF1	BT.656 dither coefficient 1 register	9-125
0xD400	LCD_CTRL	LCD control register	9-125
0xD408	LCD_SYNC_INV	Sync signal polarity configuration register in LCD external sync timing input mode	9-126

9.2.6 Register Description

VOCTRL

VOCTRL is a VO control register. It is used to configure the arbitration mode of surface bus requests.

Offset Address	Register Name	Total Reset Value
0x0000	VOCTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	vo_ck_gt_en	reserved																											m0_arb_mode			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	vo_ck_gt_en	VDP clock gating enable 0: disabled 1: enabled
[30:4]	RO	reserved	Reserved

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[3:0]	RW	m0_arb_mode	Arbitration mode of data requests of internal surface buses of VO MAC0 0000: polling 0001: graphics layer first Other values: reserved

VOINTSTA

VOINTSTA is a VO interrupt status register (read-only).

Offset Address	Register Name	Total Reset Value
0x0004	VOINTSTA	0x0000_0044

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												v1_dc_err_int	v1_dl_err_int	reserved					vdac0_unload_int	reserved		reserved		reserved			dhd1uf_int	reserved			dhd1vtthd1_int	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19]	RO	v1_dc_err_int	V1 chrominance decompression error interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[18]	RO	v1_dl_err_int	V1 luminance decompression error interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[17:13]	RO	reserved	Reserved
[12]	RO	vdac0_unload_int	DAC0 offload interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[11]	RO	reserved	Reserved
[10]	RO	reserved	Reserved
[9]	RO	reserved	Reserved
[8]	RO	reserved	Reserved

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[7]	RO	dhd1uf_int	DHD1 low-bandwidth alarm interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[6]	RO	reserved	Reserved
[5]	RO	reserved	Reserved
[4]	RO	dhd1vtthd1_int	DHD1 vertical timing interrupt 1 0: An interrupt is reported. 1: No interrupt is reported.
[3]	RO	reserved	Reserved
[2]	RO	reserved	Reserved
[1]	RO	reserved	Reserved
[0]	RO	reserved	Reserved

VOMSKINTSTA

VOMSKINTSTA is a VO masked interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0008	VOMSKINTSTA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												v1_dc_err_int	v1_dl_err_int	reserved					vdac0_unload_int	reserved	reserved	reserved	reserved	dhd1uf_clr	reserved	reserved	dhd1vtthd1_clr	reserved	reserved	reserved	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19]	WC	v1_dc_err_int	V1 chrominance decompression error interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[18]	WC	v1_dl_err_int	V1 luminance decompression error interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[17:13]	RO	reserved	Reserved

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[12]	WC	vdac0_unload_int	DAC0 offload interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[11]	RO	reserved	Reserved
[10]	RO	reserved	Reserved
[9]	RO	reserved	Reserved
[8]	RO	reserved	Reserved
[7]	WC	dhd1uf_clr	DHD1 low-bandwidth alarm interrupt 0: An interrupt is reported. 1: No interrupt is reported.
[6]	RO	reserved	Reserved
[5]	RO	reserved	Reserved
[4]	WC	dhd1vtthd1_clr	DHD1 vertical timing interrupt 1 0: An interrupt is reported. 1: No interrupt is reported.
[3]	RO	reserved	Reserved
[2]	RO	reserved	Reserved
[1]	RO	reserved	Reserved
[0]	RO	reserved	Reserved

VOINTMSK

VOINTMSK is a VDP interrupt mask register.

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Offset Address	Register Name	Total Reset Value
0x000C	VOINTMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												v1_dc_err_int	v1_dl_err_int	reserved					vdac0_unload_intmsk	reserved				dhd1uf_intmsk	reserved		reserved	dhd1vtthd1_intmsk	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19]	RW	v1_dc_err_int	Mask for the V1 chrominance decompression error interrupt 0: masked 1: enabled
[18]	RW	v1_dl_err_int	Mask for the V1 luminance decompression error interrupt 0: masked 1: enabled
[17:13]	RO	reserved	Reserved
[12]	RW	vdac0_unload_intmsk	Mask for the DAC0 offload interrupt 0: masked 1: enabled
[11:8]	RO	reserved	Reserved
[7]	RW	dhd1uf_intmsk	Mask for the DHD1 low-bandwidth alarm interrupt 0: masked 1: enabled
[6]	RO	reserved	Reserved
[5]	RO	reserved	Reserved
[4]	RW	dhd1vtthd1_intmsk	Mask for DHD1 vertical timing interrupt 1 0: masked 1: enabled
[3:0]	RO	reserved	Reserved

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VDPVERSION1

VDPVERSION1 is VDP version register 1.

Offset Address	Register Name	Total Reset Value
0x0010	VDPVERSION1	0x7675_6F76

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	vdpversion1																															
Reset	0	1	1	1	0	1	1	0	0	1	1	1	0	1	0	1	0	1	1	0	1	1	1	1	0	1	1	1	0	1	1	0

Bits	Access	Name	Description
[31:0]	RO	vdpversion1	VDP version 1

VDPVERSION2

VDPVERSION2 is VDP version register 2.

Offset Address	Register Name	Total Reset Value
0x0014	VDPVERSION2	0x3031_3134

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	vdpversion2																															
Reset	0	0	1	1	0	0	0	0	0	0	1	1	0	0	0	1	0	0	1	1	0	0	0	1	0	0	1	1	0	1	0	0

Bits	Access	Name	Description
[31:0]	RO	vdpversion2	VDP version 2

VOAXICTRL

VOAXICTRL is a VO AXI bus configuration register.

Offset Address	Register Name	Total Reset Value
0x0034	VOAXICTRL	0x0111_0111

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												m0_outstd_rid0			
Reset	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	m0_outstd_rid0	Read ID0 outstanding of AXI master 0. The value ranges from 0 to 7.

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VO_MUX

VO_MUX is a VO interface multiplexing register.

Offset Address	Register Name	Total Reset Value
0x0100	VO_MUX	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	digital_sel				reserved																											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	digital_sel	Digital interface data select. The default value is 0. 1: BT.656 Other values: reserved
[27:0]	RO	reserved	Reserved

V1_CTRL

V1_CTRL is a V1 configuration register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x1000	V1_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	surface_en	reserved			reserved				crop_en	reserved	precharge_en	precharge_mode	ifir_mode		vup_mode	reserved	lm_rmode		chm_rmode		reserved				dcmp_en	time_out			reserved	ifmt		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	surface_en	Surface enable (non-instant) 0: disabled 1: enabled
[30:28]	RO	reserved	Reserved
[27]	RO	reserved	Reserved

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[26:24]	RO	reserved	Reserved
[23]	RW	crop_en	Crop enable (non-instant) 0: disabled 1: enabled
[22]	RO	reserved	Reserved
[21]	RW	precharge_en	Pre-charge enable. This field is valid when the precharge_mode field is 1 (software mode). 0: disabled 1: enabled
[20]	RW	precharge_mode	Pre-charge mode of the FDR FIFO 0: hardware mode. Hardware automatically enters the pre-charge status on the frame trailer. 1: software mode. Software controls whether to enter or exit the pre-charge status.
[19:18]	RW	ifir_mode	Horizontal chrominance IFIR mode 00: reserved 01: chrominance IFIR replication mode 10: bilinear interpolation 11: 16-order half-band filtering
[17]	RW	vup_mode	Surface coefficient update mode 0: update by frame 1: update by field
[16]	RO	reserved	Reserved
[15:14]	RW	lm_rmode	Luminance read mode 00: The read mode is bound to the interface. 01: The frame buffer data is read in progressive mode. 10: The top field is read in interlaced mode. 11: The bottom field is read in interlaced mode.
[13:12]	RW	chm_rmode	Chrominance read mode 00: The read mode is bound to the interface. 01: The frame buffer data is read in progressive mode. 10: The top field is read in interlaced mode. 11: The bottom field is read in interlaced mode.
[11:9]	RO	reserved	Reserved
[8]	RW	dcmp_en	Decompression enable
[7:5]	RW	time_out	Timeout period
[4]	RO	reserved	Reserved

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[3:0]	RW	ifmt	Format of the input data 0x1: SPYCbCr400 0x3: SPYCbCr420 0x4: SPYCbCr422 Other values: reserved

V1_UPD

V1_UPD is a V1 channel update enable register.

Offset Address	Register Name	Total Reset Value
0x1004	V1_UPD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															regup
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	WC	regup	Surface register update. After the registers at this layer are configured, the registers are updated when the value 1 is written to this bit. After the registers are updated, this bit is automatically cleared by the hardware.

V1_IRESO

V1_IRESO is a V1 input resolution register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x1028	V1_IRESO	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								ih												iw											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	ih	Height (in line). The configured value is the actual height minus 1. The frame height is referenced and its unit is line.
[11:0]	RW	iw	Width (in pixel). The configured value is the actual width minus 1.

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V1_ORESO

V1_ORESO is a V1 output resolution register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x102C	V1_ORESO	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								oh												ow											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	oh	Height (in line). The configured value is the actual height minus 1. The frame height is referenced and its unit is line.
[11:0]	RW	ow	Width (in pixel). The configured value is the actual width minus 1.

V1_CBMPARA

V1_CBMPARA is a V1 overlay parameter register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x1038	V1_CBMPARA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								galpha							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	galpha	Overlay global alpha value. The value ranges from 0 to 255. The value 255 indicates opaque, and the value 0 indicates completely transparent.

V1_CPOS

V1_CPOS is a horizontal crop coordinate register for the source picture.

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Offset Address	Register Name	Total Reset Value
0x1044	V1_CPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								src_xlpos												src_xfpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	src_xlpos	End coordinates for horizontally cropping the source picture
[11:0]	RW	src_xfpos	Start coordinates for horizontally cropping the source picture

V1_DFPOS

V1_DFPOS is a V1 surface start position (in the display window) register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x1060	V1_DFPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								disp_yfpos												disp_xfpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	disp_yfpos	Start coordinates of the display column The frame height is referenced and its unit is line.
[11:0]	RW	disp_xfpos	Start coordinates of the display row

V1_DLPOS

V1_DLPOS is a V1 surface end position (in the display window) register (non-instant register).

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Offset Address	Register Name	Total Reset Value
0x1064	V1_DLPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								disp_ylpos												disp_xlpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	disp_ylpos	End coordinates of the display column The frame height is referenced and its unit is line.
[11:0]	RW	disp_xlpos	End coordinates of the display row

V1_VFPOS

V1_VFPOS is a V1 surface content start position (in the display window) register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x1068	V1_VFPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								video_yfpos												video_xfpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	video_yfpos	Start coordinates of the video column The frame height is referenced and its unit is line.
[11:0]	RW	video_xfpos	Start coordinates of the video row

V1_VLPOS

V1_VLPOS is a V1 surface content end position (in the display window) register (non-instant register).

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Offset Address	Register Name	Total Reset Value
0x106C	V1_VLPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								video_ylpos												video_xlpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	video_ylpos	End coordinates of the video column The frame height is referenced and its unit is line.
[11:0]	RW	video_xlpos	End coordinates of the video row

V1_BK

V1_BK is a V1 background color register.

Offset Address	Register Name	Total Reset Value
0x1070	V1_BK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		vbk_y										vbk_cb								vbk_cr											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	vbk_y	Y component
[19:10]	RW	vbk_cb	Cb component
[9:0]	RW	vbk_cr	Cr component

V1_ALPHA

V1_ALPHA is a V1 background filling color alpha register.

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Offset Address	Register Name	Total Reset Value
0x1074	V1_ALPHA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								vbk_alpha							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	vbk_alpha	Level 0–255 of the background filling color of the video layer

V1_CSC_IDC

V1_CSC_IDC is a V1 CSC input DC component register (instant register).

Offset Address	Register Name	Total Reset Value
0x1080	V1_CSC_IDC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved									csc_en			cscidc1												cscidc0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22]	RW	csc_en	CSC enable 0: disabled 1: enabled
[21:11]	RW	cscidc1	DC parameter of the input U/G component. The MSB is the signed bit. The value is expressed as a two's complement.
[10:0]	RW	cscidc0	DC parameter of the input V/B component. The MSB is the signed bit. The value is expressed as a two's complement.

V1_CSC_ODC

V1_CSC_ODC is a V1 CSC output DC component register (instant register).

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Offset Address	Register Name	Total Reset Value
0x1084	V1_CSC_ODC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								csc_sign_mode	cscodc1															cscodc0								
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22]	RW	csc_sign_mode	CSC output mode 0: The CSC output is a 10-bit unsigned number. 1: The CSC output is a 12-bit signed number.
[21:11]	RW	cscodc1	DC parameter of the output U/G component. The MSB is the signed bit. The value is expressed as a two's complement.
[10:0]	RW	cscodc0	DC parameter of the output Y/R component. The MSB is the signed bit. The value is expressed as a two's complement.

V1_CSC_IODC

V1_CSC_IODC is a V1 CSC input/output DC component register (instant register).

Offset Address	Register Name	Total Reset Value
0x1088	V1_CSC_IODC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cscodc2												cscidc2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:11]	RW	cscodc2	DC parameter of the output V/B component. The MSB is the signed bit. The value is expressed as a two's complement.
[10:0]	RW	cscidc2	DC parameter of the input Y/R component. The MSB is the signed bit. The value is expressed as a two's complement.

V1_CSC_P0

V1_CSC_P0 is a V1 CSC parameter 0 register (instant register).

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Offset Address	Register Name	Total Reset Value
0x108C	V1_CSC_P0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	cscp01															reserved	cscp00														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RW	cscp01	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved
[14:0]	RW	cscp00	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

V1_CSC_P1

V1_CSC_P1 is a V1 CSC parameter 1 register (instant register).

Offset Address	Register Name	Total Reset Value
0x1090	V1_CSC_P1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	cscp10															reserved	cscp02														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RW	cscp10	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved
[14:0]	RW	cscp02	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

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V1_CSC_P2

V1_CSC_P2 is a V1 CSC parameter 2 register (instant register).

Offset Address	Register Name	Total Reset Value
0x1094	V1_CSC_P2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cscp12												reserved				cscp11											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RW	cscp12	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved
[14:0]	RW	cscp11	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

V1_CSC_P3

V1_CSC_P3 is a V1 CSC parameter 3 register (instant register).

Offset Address	Register Name	Total Reset Value
0x1098	V1_CSC_P3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cscp21												reserved				cscp20											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RW	cscp21	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved

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[14:0]	RW	cscp20	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

V1_CSC_P4

V1_CSC_P4 is a V1 CSC parameter 4 register (instant register).

Offset Address	Register Name	Total Reset Value
0x109C	V1_CSC_P4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																cscp22															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14:0]	RW	cscp22	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

V1_IFIRCOEF01

V1_IFIRCOEF01 is an IFIR filtering coefficient 0/1 register.

Offset Address	Register Name	Total Reset Value
0x1180	V1_IFIRCOEF01	0x000F_03F5

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								coef1								reserved								coef0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	1	1	1	1	1	1	0	1	0	1

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	coef1	IFIR filtering coefficient 1
[15:10]	RO	reserved	Reserved
[9:0]	RW	coef0	IFIR filtering coefficient 0

V1_IFIRCOEF23

V1_IFIRCOEF23 is an IFIR filtering coefficient 2/3 register.

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Offset Address	Register Name	Total Reset Value
0x1184	V1_IFIRCOEF23	0x001C_03EC

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								coef3								reserved								coef2							
Reset	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	1	0	1	1	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	coef3	IFIR filtering coefficient 3
[15:10]	RO	reserved	Reserved
[9:0]	RW	coef2	IFIR filtering coefficient 2

V1_IFIRCOEF45

V1_IFIRCOEF45 is an IFIR filtering coefficient 4/5 register.

Offset Address	Register Name	Total Reset Value
0x1188	V1_IFIRCOEF45	0x003D_03D8

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								coef5								reserved								coef4							
Reset	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	1	0	0	0	0	0	0	1	1	1	1	0	1	1	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	coef5	IFIR filtering coefficient 5
[15:10]	RO	reserved	Reserved
[9:0]	RW	coef4	IFIR filtering coefficient 4

V1_IFIRCOEF67

V1_IFIRCOEF67 is an IFIR filtering coefficient 6/7 register.

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Offset Address	Register Name	Total Reset Value
0x118C	V1_IFIRCOEF67	0x014A_0395

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								coef7								reserved								coef6							
Reset	0	0	0	0	0	0	0	1	0	1	0	0	1	0	1	0	0	0	0	0	0	0	1	1	1	0	0	1	0	1	0	1

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	coef7	IFIR filtering coefficient 7
[15:10]	RO	reserved	Reserved
[9:0]	RW	coef6	IFIR filtering coefficient 6

V1_P0LADDR

V1_P0LADDR is a V1 luminance address register.

Offset Address	Register Name	Total Reset Value
0x1204	V1_P0LADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	surface_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	surface_addr	Luminance start address for the video layer

V1_P0CADDR

V1_P0CADDR is a V1 chrominance address register.

Offset Address	Register Name	Total Reset Value
0x1208	V1_P0CADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	surface_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	surface_addr	Chrominance start address for the video layer

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V1_P0STRIDE

V1_P0STRIDE is a V1 stride register.

Offset Address	Register Name	Total Reset Value
0x120C	V1_P0STRIDE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	surface_cstride																surface_stride															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	surface_cstride	Stride of the chrominance buffer of the video layer (valid in semi-planar format), 128-bit alignment
[15:0]	RW	surface_stride	Stride of the video layer buffer (luminance stride in semi-planar format), 128-bit alignment

G2_CTRL

G2_CTR is a G2 configuration register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x7000	G2_CTRL	0x4000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	surface_en	reserved			upd_mode	read_mode	reserved				precharge_en	precharge_mode	reserved										bitext		ifmt							
Reset	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	surface_en	Surface enable (non-instant) 0: disabled 1: enabled
[30:28]	RO	reserved	Reserved
[27]	RW	upd_mode	Update mode 0: update by fame 1: update by field

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[26]	RW	read_mode	Data read mode 0: The read mode is automatically selected based on the interface read mode. That is, the progressive read mode is selected in progressive display mode, and the interlaced read mode is selected in interlaced display mode. 1: The progressive read mode is selected forcibly.
[25:22]	RO	reserved	Reserved
[21]	RW	precharge_en	Pre-charge enable. This field is valid when the precharge_mode field is 1 (software mode). 0: disabled 1: enabled
[20]	RW	precharge_mode	Pre-charge mode of the FDR FIFO 0: hardware mode. Hardware automatically enters the pre-charge status on the frame trailer. 1: software mode. Software controls whether to enter or exit the pre-charge status.
[19:10]	RO	reserved	Reserved
[9:8]	RW	bitext	Bit extend mode of the input bitmap 00: extend 0s to lower bits 10: extend the value of the MSB to a lower bit 11: extend the values of upper bits to lower bits Other values: reserved
[7:0]	RW	ifmt	Format of the input data 0x48: ARGB4444 0x49: ARGB1555 0x69: AYCbCr8888 Other values: reserved

G2_UPD

G2_UPD is a G2 update enable register.

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Offset Address	Register Name	Total Reset Value
0x7004	G2_UPD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															regup
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	regup	Surface register update. After the registers at this layer are configured, the registers are updated when the value 1 is written to this bit. After updates, this bit is automatically cleared by the hardware.

G2_ADDR

G2_ADDR is a G2 address register. When the horizontal pixel offsets, the address is calculated by following the description of G2_SFPOS.

Offset Address	Register Name	Total Reset Value
0x7010	G2_ADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	surface_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	surface_addr	Address for the surface frame buffer

G2_STRIDE

G2_STRIDE is a G2 stride register.

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Offset Address	Register Name	Total Reset Value
0x701C	G2_STRIDE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																surface_stride															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	surface_stride	Frame buffer stride

G2_IRESO

G2_IRESO is a G2 input resolution register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x7020	G2_IRESO	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								ih												iw											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	ih	Height (in line). The configured value is the actual height minus 1. Note: In interlaced output mode, the actual layer height must be an even number. There is no such limitation in progressive output mode.
[11:0]	RW	iw	Width (in pixel). The configured value is the actual width minus 1. Note: The actual layer width must be an even number.

G2_SFPOS

G2_SFPOS is a G2 surface read data start position (in the source bitmap) register (non-instant register)

If the read position is not 128-bit-word-aligned, this register indicates the number of pixels required for completing a 128-bit word. Note: The start position offset must be less than or equal to a 128-bit word. The exceeded part is expressed by an address.

The details are as follows:

The following assumes that the start address for the graphics layer of the source picture is addr_ori, the address configured for the logical physical layer is addr_offset, the graphics

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layer offset is offsetp pixels, and the data format of the graphics layer is bpp (for example, the bpp for ARGB8888 is 32). The offset is calculated as follows:

G2ADDR = addr_offset = addr_ori + int (offsetp x bpp/128)

G2SFPOS = offset%128

where int means the rounding operation, and % means the modulo operation.

Offset Address	Register Name	Total Reset Value
0x7024	G2_SFPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								src_xfpos							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6:0]	RW	src_xfpos	Start horizontal coordinate of the source picture. The value 0 indicates the first pixel of a line.

G2_CBMPARA

G2_CBMPARA is a G2 overlay parameter register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x7030	G2_CBMPARA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																key_mode	key_en	premult_en	palpha_en	reserved				galpha							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RW	key_mode	Colorkey mode 0: The color is regarded as the colorkey when the following condition is met: Keymin ≤ Pixel ≤ Keymax 1: The color is regarded as the colorkey when either of the following conditions is met: Pixel ≤ Keymin or Pixel ≥ Keymax
[14]	RW	key_en	Colorkey enable 0: disabled 1: enabled

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[13]	RW	premult_en	Whether the input bitmap is a premultiplied bitmap 0: non-premultiplied bitmap 1: premultiplied bitmap
[12]	RW	palpha_en	Pixel alpha enable 0: disabled 1: enabled
[11:8]	RO	reserved	Reserved
[7:0]	RW	galpha	Overlay global alpha value. The value ranges from 0 to 255. The value 255 indicates opaque, and the value 0 indicates completely transparent.

G2_CKEYMAX

G2_CKEYMAX is a maximum colorkey value register (non-instant register).

Offset Address	Register Name	Total Reset Value
0x7034	G2_CKEYMAX	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	va0								keyr_max								keyg_max								keyb_max							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	va0	Alpha0 value. When the data format is alphaRGB1555 and the alpha value is 0, the alpha0 value is used.
[23:16]	RW	keyr_max	Maximum value of the colorkey R component
[15:8]	RW	keyg_max	Maximum value of the colorkey G component
[7:0]	RW	keyb_max	Maximum value of the colorkey B component

G2_CKEYMIN

G2_CKEYMIN is a minimum colorkey value register (non-instant register).

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Offset Address	Register Name	Total Reset Value
0x7038	G2_CKEYMIN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	va1								keyr_min								keyg_min								keyb_min							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	va1	Alpha1 value. When the data format is alphaRGB1555 and the alpha value is 1, the alpha1 value is used.
[23:16]	RW	keyr_min	Minimum value of the colorkey R component
[15:8]	RW	keyg_min	Minimum value of the colorkey G component
[7:0]	RW	keyb_min	Minimum value of the colorkey B component

G2_CMASK

G2_CMASK is a colorkey mask register (non-instant register).

The value 1 indicates that a pixel bit is retained during colorkey comparison; the value 0 indicates that a pixel bit is forcibly set to 0 during color comparison no matter whether the bit is 0 or 1.

Offset Address	Register Name	Total Reset Value
0x703C	G2_CMASK	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								kmsk_r								kmsk_g								kmsk_b							
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	kmsk_r	R component of the masked colorkey
[15:8]	RW	kmsk_g	G component of the masked colorkey
[7:0]	RW	kmsk_b	B component of the masked colorkey

G2_PARAADDR

G2_PARAADDR is a G2 LUT coefficient address register.

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Offset Address	Register Name	Total Reset Value
0x7040	G2_PARAADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	surface_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	surface_addr	Surface LUT coefficient address

G2_PARAUP

G2_PARAUP is a G2 LUT coefficient update register.

Offset Address	Register Name	Total Reset Value
0x7044	G2_PARAUP	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															gdc_paraup
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	gdc_paraup	LUT coefficient update 0: not updated 1: updated

G2_DFPOS

G2_DFPOS is a G2 surface start position (in the display window) register (non-instant register). The position unit is pixel.

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Offset Address	Register Name	Total Reset Value
0x7080	G2_DFPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								disp_yfpos												disp_xfpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	disp_yfpos	Column start coordinates
[11:0]	RW	disp_xfpos	Row start coordinates

G2_DLPOS

G2_DLPOS is a G2 surface end position (in the display window) register (non-instant register). The position unit is pixel.

Offset Address	Register Name	Total Reset Value
0x7084	G2_DLPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								disp_ylpos												disp_xlpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	disp_ylpos	Column end coordinates
[11:0]	RW	disp_xlpos	Row end coordinates

G2_VFPOS

G2_VFPOS is a G2 surface content start position (in the display window) register (non-instant register). The position unit is pixel.

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Offset Address	Register Name	Total Reset Value
0x7088	G2_VFPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								video_yfpos												video_xfpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	video_yfpos	Start coordinates of the graphics column The frame height is referenced and its unit is line.
[11:0]	RW	video_xfpos	Start coordinates of the graphics row

G2_VLPOS

G2_VLPOS is a G2 surface content end position (in the display window) register (non-instant register). The position unit is pixel.

Offset Address	Register Name	Total Reset Value
0x708C	G2_VLPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								video_ylpos												video_xlpos											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:12]	RW	video_ylpos	End coordinates of the graphics column The frame height is referenced and its unit is line.
[11:0]	RW	video_xlpos	End coordinates of the graphics row

G2_BK

G2_BK is a G2 background color register.

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Offset Address	Register Name	Total Reset Value
0x7090	G2_BK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				vbk_y								vbk_cb								vbk_cr											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	vbk_y	Y component
[19:10]	RW	vbk_cb	Cb component
[9:0]	RW	vbk_cr	Cr component

G2_ALPHA

G2_ALPHA is a G2 background filling color alpha register.

Offset Address	Register Name	Total Reset Value
0x7094	G2_ALPHA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								vbk_alpha							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	vbk_alpha	Level 0–128 of the background filling color of the graphics layer

G2_CSC_IDC

G2_CSC_IDC is a G2 CSC input DC register (instant register).

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Offset Address	Register Name	Total Reset Value
0x70C0	G2_CSC_IDC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								csc_mode				csc_en	cscidc1											cscidc0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:23]	RW	csc_mode	This field is valid only when the G2 CSC coefficients are fixed. 000: reserved 001: YUV2YUV 010: YUV601_RGB 011: YUV709_RGB 100: YUV2YUV_709_601 101: YUV2YUV_601_709 110: RGB2YUV_601 111: RGB2YUV_709
[22]	RW	csc_en	CSC enable 0: disabled 1: enabled
[21:11]	RO	cscidc1	DC parameter of the input U/G component. The MSB is the signed bit. The value is expressed as a two's complement.
[10:0]	RO	cscidc0	DC parameter of the input V/B component. The MSB is the signed bit. The value is expressed as a two's complement.

G2_CSC_ODC

G2_CSC_ODC is a G2 CSC output DC component register (instant register).

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Offset Address	Register Name	Total Reset Value
0x70C4	G2_CSC_ODC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved									csc_sign_mode			cscodc1												cscodc0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22]	RW	csc_sign_mode	CSC output mode 0: The CSC output is a 10-bit unsigned number. 1: The CSC output is a 12-bit signed number.
[21:11]	RO	cscodc1	DC parameter of the output U/G component. The MSB is the signed bit. The value is expressed as a two's complement.
[10:0]	RO	cscodc0	DC parameter of the output Y/R component. The MSB is the signed bit. The value is expressed as a two's complement.

G2_CSC_IODC

G2_CSC_IODC is a G2 CSC input/output DC component register (instant register).

Offset Address	Register Name	Total Reset Value
0x70C8	G2_CSC_IODC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												cscodc2												cscidc2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:11]	RO	cscodc2	DC parameter of the output V/B component. The MSB is the signed bit. The value is expressed as a two's complement.
[10:0]	RO	cscidc2	DC parameter of the input Y/R component. The MSB is the signed bit. The value is expressed as a two's complement.

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G2_CSC_P0

G2_CSC_P0 is a G2 CSC parameter 0 register (instant register).

Offset Address	Register Name	Total Reset Value
0x70CC	G2_CSC_P0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	cscp01															reserved	cscp00														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RO	cscp01	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved
[14:0]	RO	cscp00	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

G2_CSC_P1

G2_CSC_P1 is a G2 CSC parameter 1 register (instant register).

Offset Address	Register Name	Total Reset Value
0x70D0	G2_CSC_P1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	cscp10															reserved	cscp02														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RO	cscp10	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved

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[14:0]	RO	cscp02	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

G2_CSC_P2

G2_CSC_P2 is a G2 CSC parameter 2 register (instant register).

Offset Address	Register Name	Total Reset Value
0x70D4	G2_CSC_P2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cscp12												reserved				cscp11											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RO	cscp12	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved
[14:0]	RO	cscp11	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

G2_CSC_P3

G2_CSC_P3 is a G2 CSC parameter 3 register (instant register).

Offset Address	Register Name	Total Reset Value
0x70D8	G2_CSC_P3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cscp21												reserved				cscp20											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved

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[30:16]	RO	cscp21	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.
[15]	RO	reserved	Reserved
[14:0]	RO	cscp20	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

G2_CSC_P4

G2_CSC_P4 is a G2 CSC parameter 4 register (instant register).

Offset Address	Register Name	Total Reset Value
0x70DC	G2_CSC_P4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				cscp22											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14:0]	RO	cscp22	5.10 data format that consists of a 1-bit sign part, a 4-bit integer part, and an 8-bit decimal part. The value is expressed as a two's complement.

CBM_BKG2

CBM_BKG2 is a CBM mixer 2 overlay background color register.

Offset Address	Register Name	Total Reset Value
0xB420	CBM_BKG2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cbm_bkgy2								cbm_bkgcb2								cbm_bkgcr2											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	cbm_bkgy2	Overlay background color of CBM mixer 2, Y component

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[19:10]	RW	cbm_bkgcb2	Overlay background color of CBM mixer 2, Cb component
[9:0]	RW	cbm_bkgcr2	Overlay background color of CBM mixer 2, Cr component

CBM_MIX2

CBM_MIX2 is a CBM mixer 2 priority configuration register (instant register). The register configuration takes effect only when the vsync signal is valid. mixer_prio_x is overly layer priority *x*.

Offset Address	Register Name	Total Reset Value
0xB428	CBM_MIX2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								mixer_prio1				mixer_prio0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:4]	RW	mixer_prio1	Drive layer of CBM mixer 2 with priority 1 0000: no drive layer 1001: V1 1010: G2 Other values: reserved
[3:0]	RW	mixer_prio0	Drive layer of CBM mixer 2 with priority 0 0000: no drive layer 1001: V1 1010: G2 Other values: reserved

DHD1_CTRL

DHD1_CTRL is a DHD1 global control register.

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Offset Address	Register Name	Total Reset Value
0xC400	DHD1_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	intf_en	cbar_en	cbar_sel	reserved																		precharge_en	precharge_mode	reserved				iop	reserved			regup
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	intf_en	Display interface enable (instant). Data is output over the interface only when this field is enabled. 0: disabled 1: enabled
[30]	RW	cbar_en	Color bar enable. The color bar is output over the interface when this field is enabled. 0: disabled 1: enabled
[29]	RW	cbar_sel	Color space select for the output color bar (instant) 0: VGA 1: YPbPr
[28:11]	RO	reserved	Reserved.
[10]	RW	precharge_en	Pre-charge enable. This field is valid when the precharge_mode field is 1 (software mode). 0: disabled 1: enabled
[9]	RW	precharge_mode	Pre-charge mode of AFIFO 0: hardware mode. Hardware automatically enters the pre-charge status in the vertical blanking region. 1: software mode. Software controls whether to enter or exit the pre-charge status.
[8:5]	RO	reserved	Reserved
[4]	RW	iop	Display mode (non-instant) 0: interlaced display 1: progressive display
[3:1]	RO	reserved	Reserved

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[0]	WC	regup	Surface register update. After the registers at this layer are configured, the registers are updated when the value 1 is written to this bit. After the registers are updated, this bit is automatically cleared by the hardware.

DHD1_VSYNC

DHD1_VSYNC is a top field vertical sync timing register in interlaced output mode or frame vertical sync timing register in progressive output mode. This register is a non-instant register.

Offset Address	Register Name	Total Reset Value
0xC404	DHD1_VSYNC	0x0011_321B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				vfb								vbb								vact											
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	1	1	0	0	1	0	0	0	0	1	1	0	1	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:20]	RW	vfb	In interlaced output mode: top vertical front blanking (TVFB) In progressive output mode: vertical front blanking (VFB)
[19:12]	RW	vbb	In interlaced output mode: top vertical back blanking (TVBB) In progressive output mode: vertical back blanking (VBB)+vertical pulse width (VPW)
[11:0]	RW	vact	In interlaced output mode: height of an active picture on the top field In progressive output mode: height of an active picture in a frame. The configured value is the actual value minus 1.

DHD1_HSYNC1

DHD1_HSYNC1 is a horizontal sync configuration register in interlaced or progressive output mode (non-instant register).

Offset Address	Register Name	Total Reset Value
0xC408	DHD1_HSYNC1	0x00BF_077F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	hbb																hact															
Reset	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:16]	RW	hbb	Horizontal back blanking (HBB), in pixel

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[15:0]	RW	hact	Number of horizontal pixels in an active region

DHD1_HSYNC2

DHD1_HSYNC2 is a horizontal sync configuration register in interlaced or progressive output mode (non-instant register).

Offset Address	Register Name	Total Reset Value
0xC40C	DHD1_HSYNC2	0x0000_020F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	hmid																hfb															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	1	1

Bits	Access	Name	Description
[31:16]	RW	hmid	Bottom vertical sync active pixel (active region)
[15:0]	RW	hfb	HFB, in pixel

DHD1_VPLUS

DHD1_VPLUS is a bottom field vertical sync timing in interlaced output mode (non-instant register).

Offset Address	Register Name	Total Reset Value
0xC410	DHD1_VPLUS	0x0021_321B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				bvfb								bvbb								bvact											
Reset	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	1	1	0	0	1	0	0	0	0	1	1	0	1	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:20]	RW	bvfb	Bottom vertical front blanking (BVFB) in interlaced output mode
[19:12]	RW	bvbb	Bottom vertical back blanking (BVBB)+VPW in interlaced output mode
[11:0]	RW	bvact	Height of an active picture in the bottom field in interlaced output mode The configured value is the actual value minus 1.

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DHD1_PWR

DHD1_PWR is a sync signal pulse width register (non-instant register).

Offset Address	Register Name	Total Reset Value
0xC414	DHD1_PWR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								vpw								hpw															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	vpw	VPW. The configured value is the actual value minus 1.
[15:0]	RW	hpw	Horizontal pulse width (HPW). The configured value is the actual value minus 1.

DHD1_VTTHD3

DHD1_VTTHD3 is a vertical timing threshold register (instant register). It can be used to set two thresholds for generating two interrupts separately.

Offset Address	Register Name	Total Reset Value
0xC418	DHD1_VTTHD3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	thd4_mode	reserved							vtmgthd4								thd3_mode	reserved							vtmgthd3							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	thd4_mode	Mode of generating threshold 4 0: frame mode. The threshold count is by frame. 1: field mode. The threshold count is by field during interlaced displaying.
[30:29]	RO	reserved	Reserved
[28:16]	RW	vtmgthd4	Vertical timing threshold 4 for simultaneously triggering DHD and DSD. The start time of the DSD interface timing is later than that of the DHD interface timing and is equal to the number of lines configured in vtmgthd4.

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[15]	RW	thd3_mode	Mode of generating threshold 3 0: frame mode. The threshold count is by frame. 1: field mode. The threshold count is by field during interlaced displaying.
[14:13]	RO	reserved	Reserved
[12:0]	RW	vtmgthd3	Vertical timing threshold 3. When the vertical timing counter reaches this threshold, the VOINTSTA[dhdvtthd3_int] interrupt is triggered.

DHD1_VTTHD

DHD1_VTTHD is a vertical timing threshold register (instant register). It can be used to set two thresholds for generating two interrupts separately.

Offset Address	Register Name	Total Reset Value
0xC41C	DHD1_VTTHD	0x0001_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	thd2_mode	reserved			vtmgthd2												thd1_mode	reserved			vtmgthd1											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31]	RW	thd2_mode	Mode of generating threshold 2 0: frame mode. The threshold count is by frame. 1: field mode. The threshold count is by field during interlaced displaying.
[30:29]	RO	reserved	Reserved
[28:16]	RW	vtmgthd2	Vertical timing threshold 2. When the vertical timing counter reaches this threshold, the VOINTSTA[dhdvtthd2_int] interrupt is triggered.
[15]	RW	thd1_mode	Mode of generating threshold 1 0: frame mode. The threshold count is by frame. 1: field mode. The threshold count is by field during interlaced displaying.
[14:13]	RO	reserved	Reserved
[12:0]	RW	vtmgthd1	Vertical timing threshold 1. When the vertical timing counter reaches this threshold, the VOINTSTA[dhdvtthd_int1] interrupt is triggered.

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DHD1_PARATHD

DHD1_PARATHDis a PARA coefficient update point threshold register.

Offset Address	Register Name	Total Reset Value
0xC4B0	DHD1_PARATHD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dfs_en	reserved																							para_thd							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	dfs_en	DFS enable 0: disabled 1: enabled
[30:6]	RW	reserved	Reserved
[5:0]	RW	para_thd	PARA coefficient update point threshold. It indicates the line after the vertical back blanking where the update point occurs. This field is valid only after DFS is enabled.

DHD1_START_POS

DHD1_START_POS is a DHD1 start signal start position register.

Offset Address	Register Name	Total Reset Value
0xC4C0	DHD1_START_POS	0x0000_0805

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																timing_start_pos								start_pos							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	1

Bits	Access	Name	Description
[31:16]	RW	reserved	Reserved
[15:8]	RW	timing_start_pos	Number of lines before the active region where the timing_gen state machine starts to work
[7:0]	RW	start_pos	Number of lines before the active region where the start signal is generated in online mode

DHD1_STATE

DHD1_STATE is a DHD1 status register.

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Offset Address	Register Name	Total Reset Value
0xC4F0	DHD1_STATE	0x0000_0006

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								count_int								vcnt														bottom_field	vblank	vback_blank
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RO	count_int	DHD1 interrupt count. The count value is increased by 1 each time a vertical timing interrupt is reported.
[15:3]	RO	vcnt	Active display line count of DHD1
[2]	RO	bottom_field	DHD1 top/bottom field indicator 0: top field 1: bottom field
[1]	RO	vblank	DHD1 blanking region indicator 0: active region 1: blanking region
[0]	RO	vback_blank	DHD1 back blanking region indicator 0: non-back blanking region 1: blanking region

BT_CLIP0_L

BT_CLIP0_L is a BT.656 clip lowest threshold register (instant register).

Offset Address	Register Name	Total Reset Value
0xD210	BT_CLIP0_L	0x0100_4010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		clip_cl2										clip_cl1								clip_cl0											
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved

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[29:20]	RW	clip_cl2	Lowest threshold Y/R of component 2, unsigned integer
[19:10]	RW	clip_cl1	Lowest threshold Cb/G of component 1, unsigned integer
[9:0]	RW	clip_cl0	Lowest threshold Cr/B of component 0, unsigned integer

BT_CLIP0_H

BT_CLIP0_H is a BT.656 clip highest threshold register (instant register). For example, the output data needs to be clipped in BT.656 output mode.

Offset Address	Register Name	Total Reset Value
0xD214	BT_CLIP0_H	0x0EB3_C0F0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				clip_ch2												clip_ch1								clip_ch0							
Reset	0	0	0	0	1	1	1	0	1	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	1	1	1	1	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	clip_ch2	Highest threshold Y/R of component 2, unsigned integer
[19:10]	RW	clip_ch1	Highest threshold Cb/G of component 1, unsigned integer
[9:0]	RW	clip_ch0	Highest threshold Cr/B of component 0, unsigned integer

BT_DITHER0_CTRL

BT_DITHER0_CTRL is a BT.656 dither control register.

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Offset Address	Register Name	Total Reset Value
0xD280	BT_DITHER0_CTRL	0x2000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dither_md				reserved																											
Reset	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RW	dither_md	Dither mode select 000: 12-bit inputs, 10-bit outputs, no dithering, and direct bit truncation 001: 12-bit inputs, 10-bit outputs, and time-domain dithering 010: 12-bit inputs, 10-bit outputs, and spatial-domain dithering 011: 12-bit inputs, 8-bit outputs, and time-domain and spatial-domain dithering 100: 12-bit inputs, 10-bit outputs, and round off 101: 12-bit inputs, 8-bit outputs, and round off Other values: reserved
[28:0]	RO	reserved	Reserved

BT_DITHER0_COEF0

BT_DITHER0_COEF0 is BT.656 dither coefficient register 0.

Offset Address	Register Name	Total Reset Value
0xD284	BT_DITHER0_COEF0	0xDD66_4400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dither_coef3								dither_coef2								dither_coef1								dither_coef0							
Reset	1	1	0	1	1	1	0	1	0	1	1	0	0	1	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dither_coef3	Coefficient 3 for time-domain dithering
[23:16]	RW	dither_coef2	Coefficient 2 for time-domain dithering
[15:8]	RW	dither_coef1	Coefficient 1 for time-domain dithering
[7:0]	RW	dither_coef0	Coefficient 0 for time-domain dithering

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BT_DITHER0_COEF1

BT_DITHER0_COEF1 is BT.656 dither coefficient register 1.

Offset Address	Register Name	Total Reset Value
0xD288	BT_DITHER0_COEF1	0xDD66_4400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dither_coef7								dither_coef6								dither_coef5								dither_coef4							
Reset	1	1	0	1	1	1	0	1	0	1	1	0	0	1	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dither_coef7	Coefficient 7 for time-domain dithering
[23:16]	RW	dither_coef6	Coefficient 6 for time-domain dithering
[15:8]	RW	dither_coef5	Coefficient 5 for time-domain dithering
[7:0]	RW	dither_coef4	Coefficient 4 for time-domain dithering

LCD_CTRL

LCD_CTRL is an LCD control register.

Offset Address	Register Name	Total Reset Value
0xD400	LCD_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		lcd_serial_mode	lcd_serial_perd	lcd_parallel_order	lcd_data_inv	reserved																									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	-	reserved	Reserved
[29]	RW	lcd_serial_mode	LCD mode 0: parallel mode 1: serial mode
[28]	RW	lcd_serial_perd	Number of output single-pixel clock cycles in serial LCD mode 0: 3 1: 4

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[27]	RW	lcd_parallel_order	LCD parallel output sequence 0: RGB 1: BGR
[26]	RW	lcd_data_inv	Inversion of the LCD output bits 0: Bits 15−0 are from upper bits to lower bits. 1: Bits 0−15 are from upper bits to lower bits.
[25:0]	-	reserved	Reserved

LCD_SYNC_INV

LCD_SYNC_INV is a sync signal polarity configuration register in LCD external sync timing input mode. The setting of this register takes effect immediately after configuration, that is, the polarity of the corresponding sync signal is immediately affected.

Offset Address	Register Name	Total Reset Value
0xD408	LCD_SYNC_INV	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												f_inv	vs_inv	hs_inv	dv_inv
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	-	reserved	Reserved
[3]	RW	f_inv	Output phase inversion enable for the odd/even field indicator signal (instant) 0: disabled 1: enabled
[2]	RW	vs_inv	Output phase inversion enable for the VS pulse (instant) 0: disabled 1: enabled
[1]	RW	hs_inv	Output phase inversion enable for the HS pulse (instant) 0: disabled 1: enabled
[0]	RW	dv_inv	Output phase inversion enable for the data validity signal (instant) 0: disabled 1: enabled

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9.3 MIPI RX

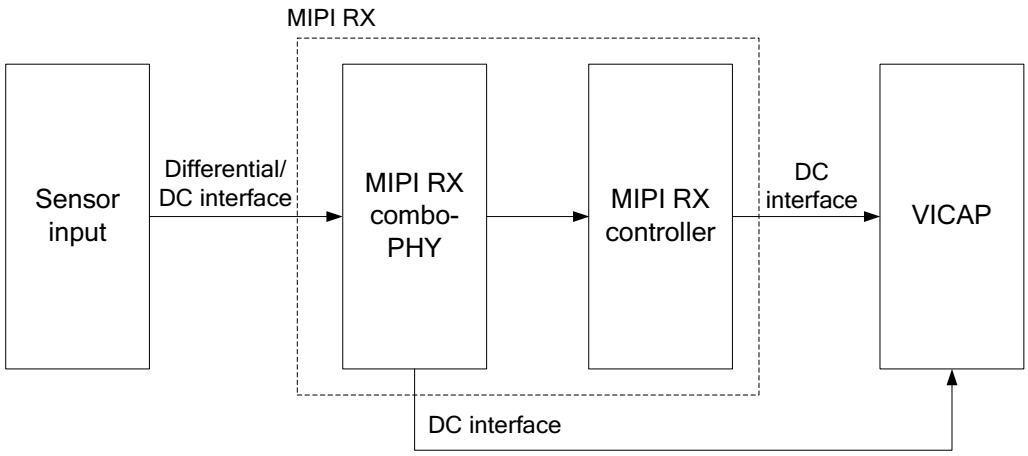
9.3.1 Overview

The mobile industry processor interface receiver (MIPI RX) receives low-voltage differential signals (LVDSs) that carry raw video data (Bayer RGB data), converts the serial differential signals into the digital camera (DC) timing, and transfers it to the downstream video capture (VICAP) module.

The MIPI RX supports serial video signal inputs such as the MIPI D-PHY, LVDS, and high-speed serial pixel interface (HiSPI). The serial video interfaces provide more bandwidth to enhance transmission stability. The MIPI RX is also compatible with the DC video interface, supports the 3.3 V/1.8 V DC parallel input, and requires a less number of chip pins while providing better compatibility.

The MIPI RX consists of the combo-PHY and controller. Figure 9-18 shows the MIPI RX workflow and its position in the system.

Figure 9-18 MIPI RX workflow and its position in the system



9.3.2 Features

The MIPI RX has the following features:

- 1/2/3/4-lane MIPI D-PHY interface, up to 1 Gbit/s per lane
- 1/2/4-lane LVDS. sub-LVDS, or HiSPI interface, up to 1 Gbit/s per lane
- CMOS parallel 1.8 V/3.3 V input
- RAW8, RAW10, RAW12, or RAW14 data type parsing
- User-defined data types in MIPI mode
- Endian mode configuration for LVDS or HiSPI mode pixel or synchronization code
- Configuration of channel quantity and sequence
- Configurable width, height, and start position of the output picture

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9.3.3 Function Description

9.3.3.1 Typical Applications

The MIPI RX is a collection unit that supports multiple differential video input interfaces. It uses the combo-PHY to receive data through the MIPI, LVDS, sub-LVDS, HiSPI, or DC interface. Depending on the functional mode configuration, the MIPI RX allows data transmission at different speeds and resolutions and supports multiple external input devices.

The MIPI RX supports a maximum of one link with four lanes for MIPI input and a maximum of one link with four lanes for LVDS input. The input pins can be multiplexed to provide single-end DC or BT.1120 channel input, thereby providing better compatibility with less chip pins.

Table 9-7 lists the interfaces supported by the MIPI RX.

Table 9-7 Interfaces supported by the MIPI RX

Interface Type	Common Mode Voltage	Differential Mode Voltage	Maximum Clock Frequency	Maximum Data Rate per Lane
D-PHY	200 mV	200 mV	500 MHz	1 Gbit/s
Sub-LVDS	900 mV	150 mV	500 MHz	1 Gbit/s
LVDS	1.25 V	350 mV	500 MHz	1 Gbit/s
HiSPI (HiVCM)	900 mV	280 mV	500 MHz	1 Gbit/s
HiSPI (SLVS)	200 mV	200 mV	500 MHz	1 Gbit/s
CMOS Parallel	1.8 V	1.8 V	150 MHz	150 Mbit/s
CMOS Parallel	3.3 V	3.3 V	150 MHz	150 Mbit/s

The MIPI RX supports the following typical input modes:

- 720P (1280 x 720) @30 fps
- 1080p (1920 x 1080) @60 fps
- 3M (2048 x 1536) @30 fps

9.3.3.2 Function Implementation

MIPI Video Data Input

MIPI specifications are developed and maintained by different working groups and cover diverse application requirements for use in different fields. The MIPI RX supports the D-PHY and CSI-2. D-PHY stipulates the rules of transmission at the physical layer, and CSI-2 stipulates the format and protocol of camera output data packets.

(1) D-PHY

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D-PHY is a high-speed physical layer standard released by the MIPI Alliance. The standard stipulates the physical specifications and transmission protocols at the physical layer for hosts and peripherals. D-PHY uses the LVDS technology with 200 mV source synchronization, and each channel has a frequency from 80 MHz to 1 GHz. Typically, each clock supports a maximum of four data channels, which provide a total transmission rate of 4 Gbit/s. D-PHY can work either in low-power (LP) or high-speed (HS) mode.

(2) CSI-2

CSI-2 is a camera data protocol that stipulates the data packet format used for communication between hosts and peripherals.

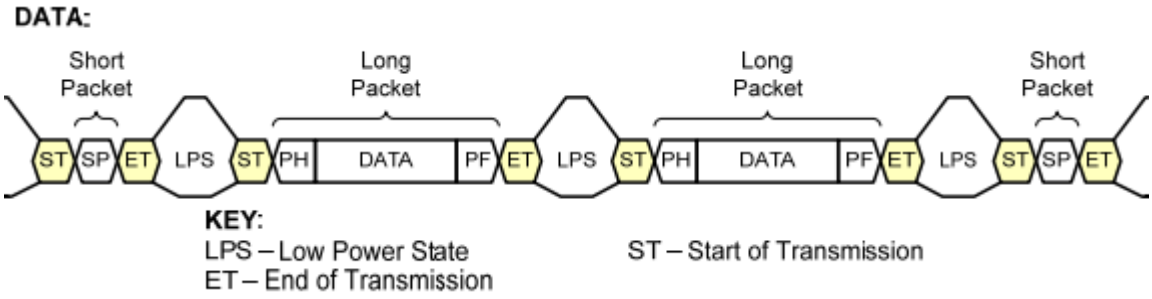
CSI-2 supports image application in different pixel formats, and the smallest unit for data transmission is byte. An appropriate number of data channels can be selected to attain enhanced CSI-2 performance. CSI-2 specifies how the TX end packetizes pixel data into bytes and how multiple data channels are allocated and managed. Byte data is organized in data packets, which are transmitted between the start of transmission (SoT) and the end of transmission (EoT). The RX end parses the data packets based on the applicable protocol and restores the pixel data.

The MIPI RX can parse pixel data in RAW8, RAW10, RAW12, RAW14 or User Defined format.

CSI-2 data packets can be long or short and contain parity codes for parity check and error correction.

Both long and short packets are transmitted between the SoT and EoT. D-PHY works in LP mode in gaps of data transmission. Figure 9-19 shows the mechanism for transmitting CSI-2 data packets. PH and PF mean packet header and packet footer, respectively.

Figure 9-19 CSI-2 data packet transmission mechanism



Long packets are used to transmit effective pixel data in five parts: Data ID, Word Count, ECC, Payload, and Checksum.

Data ID comprises Virtual Channel and Data Type. Virtual Channel controls the channel used for transmission and can specify channel multiplexing so that different channels will transmit different data. Data Type specifies the type of data.

Word Count indicates the amount of data to be received by the RX end.

ECC is 8-bit error correction code that can be used to detect and correct errors in Data Type and Word Count.

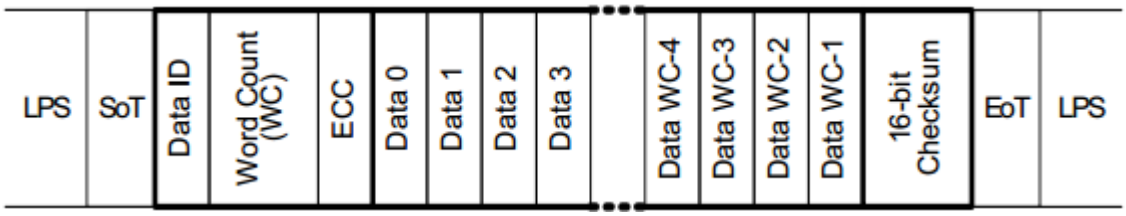
Payload is the pixel data to be transmitted.

Checksum is generated by a linear feedback shift register and is used to check payload data.

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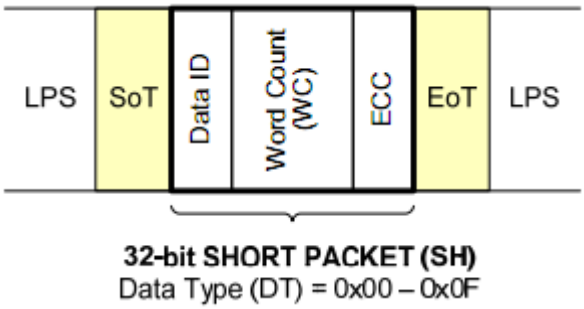
Figure 9-20 shows the format of a long packet.

Figure 9-20 CSI-2 long packet format



Short packets are used to transmit synchronization information in three parts: Data ID, Word Count, and ECC. Figure 9-21 shows its format.

Figure 9-21 CSI-2 short packet format



LVDS Video Data Input

The LVDS is widely used in front-end cameras. It identifies data of the blanking region and active region by synchronization code.

The MIPI RX controller receives parallel data from the HiSPI, PHY, splits and combines the data, and extracts the synchronization code to obtain the pixel data.

In LVDS transmission mode, frame or line synchronization signals are integrated into data streams. In data streams, SOF and EOF indicate the start and end of a frame, whereas SOL and EOL indicate the start and end of a line. In a data stream, each of SOF, EOF, SOL, and EOL consists of four fields, and the bit width and pixel data of each field are consistent. The first three fields are fixed reference codes, and the fourth field is used to identify the start of end of a frame or line. Table 9-8 illustrates the LVDS synchronization code format.

Table 9-8 LVDS synchronization code format

Field	Bit Width	Sync Code			
		SOL/SAV (Valid Line)	EOL/EAV (Valid Line)	SOF/SAV (Invalid Line)	EOF/EAV (Invalid Line)
1st code	8bit	FFh	FFh	FFh	FFh
	10-bit	3FFh	3FFh	3FFh	3FFh

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Field	Bit Width	Sync Code			
		SOL/SAV (Valid Line)	EOL/EAV (Valid Line)	SOF/SAV (Invalid Line)	EOF/EAV (Invalid Line)
	12-bit	FFFh	FFFh	FFFh	FFFh
	14-bit	3FFFh	3FFFh	3FFFh	3FFFh
2nd code	8bit	00h	00h	00h	00h
	10-bit	000h	000h	000h	000h
	12-bit	000h	000h	000h	000h
	14-bit	0000h	0000h	0000h	0000h
3rd code	8bit	00h	00h	00h	00h
	10-bit	000h	000h	000h	000h
	12-bit	000h	000h	000h	000h
	14-bit	0000h	0000h	0000h	0000h
4th code	8bit	XXh	XXh	XXh	XXh
	10-bit	XXXh	XXXh	XXXh	XXXh
	12-bit	XXXh	XXXh	XXXh	XXXh
	14-bit	XXXXh	XXXXh	XXXXh	XXXXh

NOTE

The first three fields of the synchronization code are fixed, and the fourth field identifies the start or end of a frame or line. The value of the fourth field is specified by image sensor vendors and therefore it varies according to vendors. Table 10-9 describes an implementation mode. Table 9-9 describes an implementation mode.

Table 9-9 Sample of the fourth field of the LVDS synchronization code

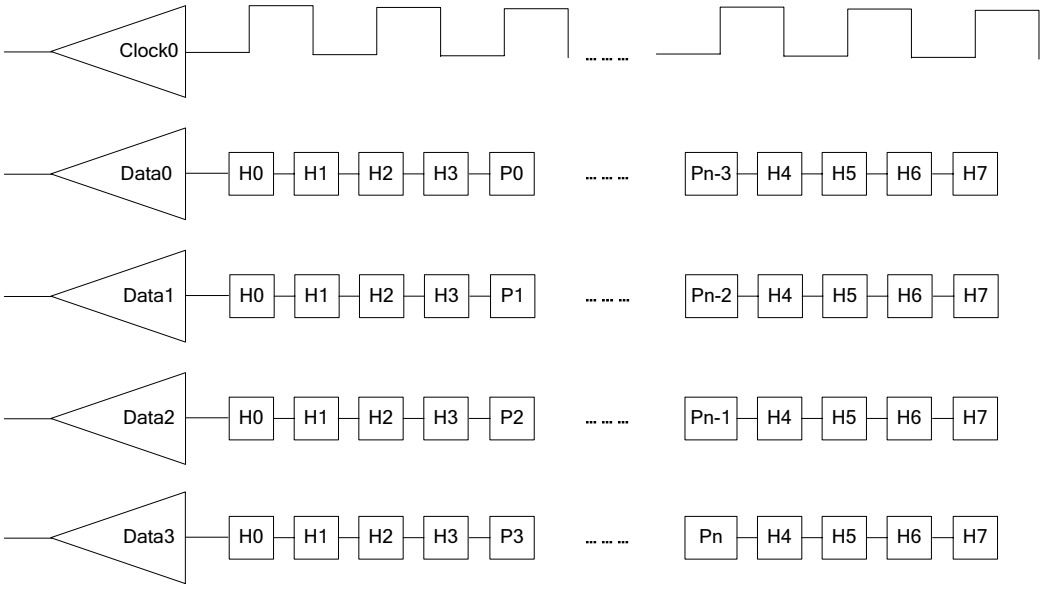
Field	Bit Width	Sync code			
		SAV(Valid line)	EAV(Valid line)	SAV(Invalid line)	EAV(Invalid line)
4th code	8bit	C7h	DAh	ECh	F1h
	10bit	200h	274h	2ACh	2D8h
	12bit	800h	9D0h	AB0h	B60h
	14bit	2000h	2740h	2AC0h	2D80h

Figure 9-22 shows how LVDS synchronization information and pixel data are transmitted on each channel. H indicates the synchronization code, and P indicates pixel. The bit widths of H and P are consistent with that of a single output pixel of the image sensor. In each data

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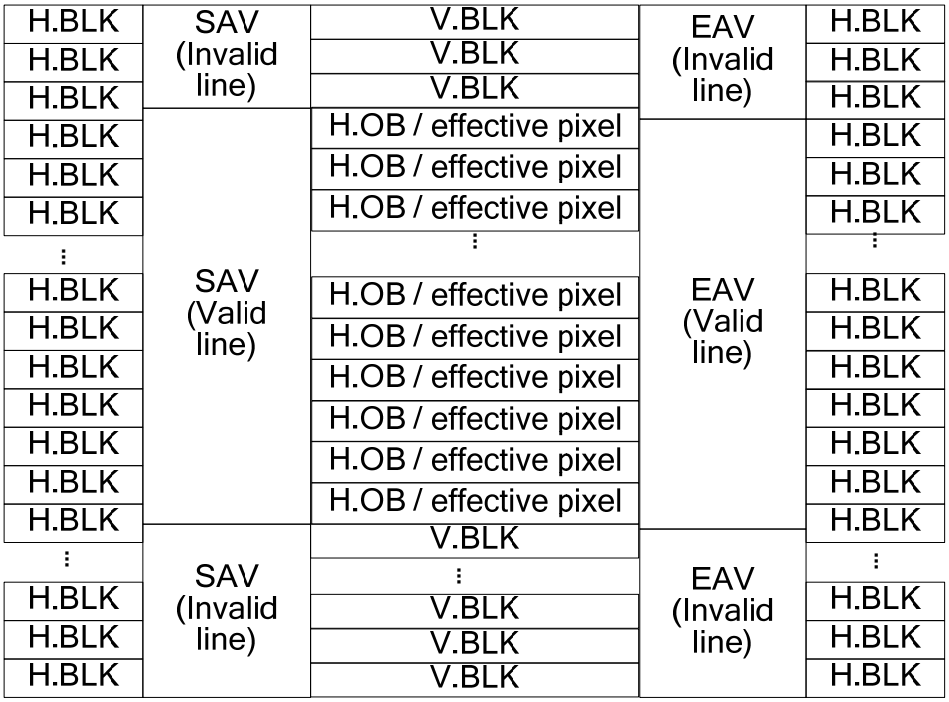
channel, the synchronization code with four-pixel bit width is transmitted before pixel data. The distribution of pixel data is related to the number of channels. Data is transmitted in series, and the endian mode is configurable in the MIPI RX.

Figure 9-22 LVDS data transmission mode



Two LVDS synchronization modes are available. In one mode, SAV (Invalid) and EAV (Invalid) identify invalid data of the blanking region whereas SAV (Valid) and EAV (Valid) identify pixel data of the active region. Figure 9-23 shows this synchronization mode.

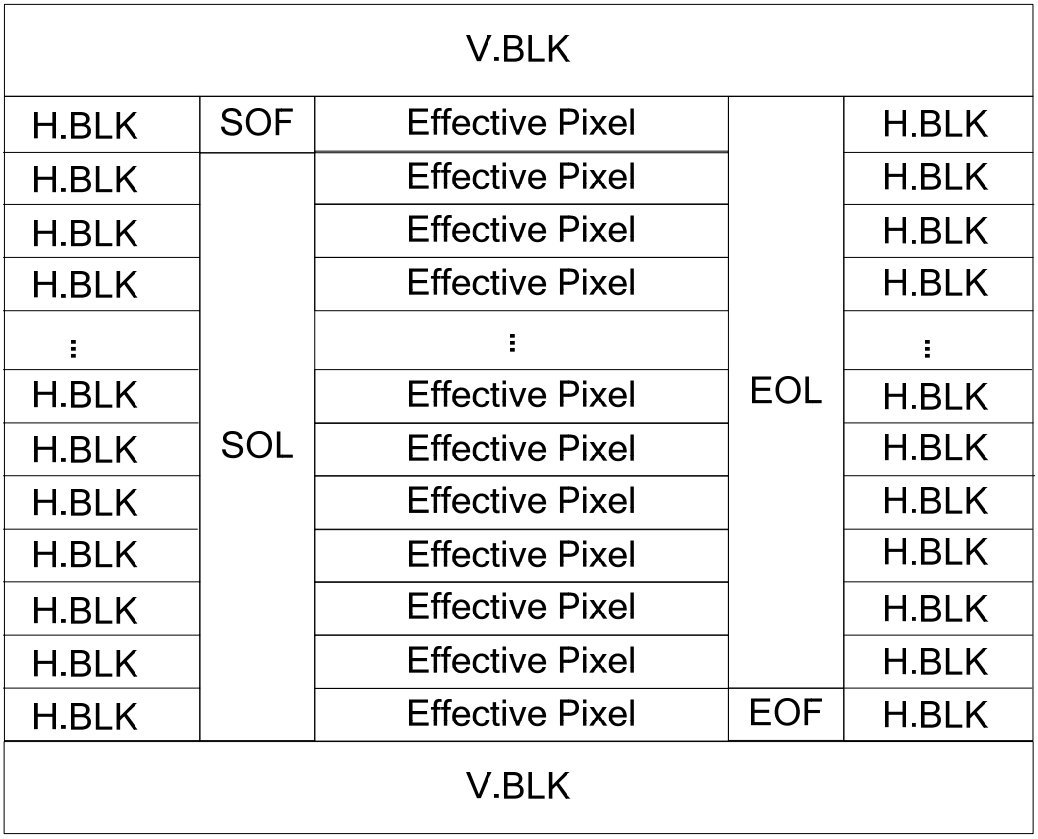
Figure 9-23 LVDS synchronization mode 1



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In the other mode, SOF identifies the start of the first line in the active region, EOF identifies the end of the last line in the active region, and SOL and EOL identify the start and end of the other lines in the active region. Figure 9-24 shows this synchronization mode.

Figure 9-24 LVDS synchronization mode 2



The sub-LVDS is a differential signal technology with ultra-low voltage swing, whose common or differential mode voltage is lower than the LVDS. The sub-LVDS uses the same format for data transmission as the LVDS and can be considered as one type of LVDS.

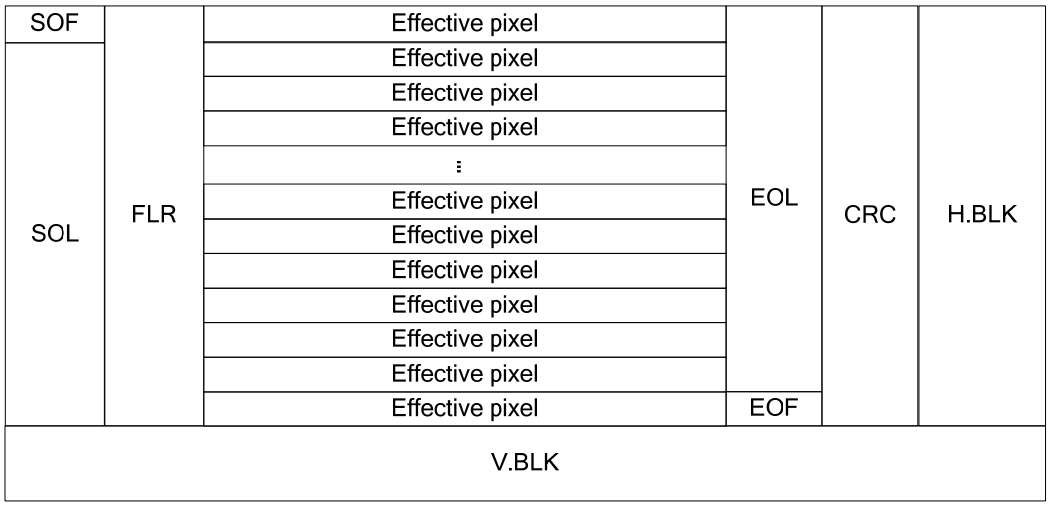
HiSPI Video Data Input

HiSPI, formulated by Aptina, includes the HiSPI physical layer protocol and HiSPI specification. The HiSPI physical layer protocol stipulates the electrical specifications and timing parameters, whereas the HiSPI specification defines the data packetizing modes.

The HiSPI specification stipulates two physical layer electric standards (HiVCM and SLVS) and four data transmission modes (Packetized-SP, Streaming-SP, Streaming-S, and ActiveStart-SP8). The Packetized-SP mode is similar to the LVDS in data packetizing mode, except that the checksum (CRC) and filler code (FLR) can be added to data packets in Packetized-SP mode. Figure 9-25 shows the data format in HiSPI Packetized-SP mode.

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Figure 9-25 HiSPI Packetized-SP data format



The MIPI RX supports the HiVCM and SLVS electric standards and supports the Packetized-SP mode (does not process the CRC and FLR data), meaning that the MIPI RX parses data in the same way for the HiSPI and LVDS.

9.3.4 Operating Mode of the MIPI RX Controller

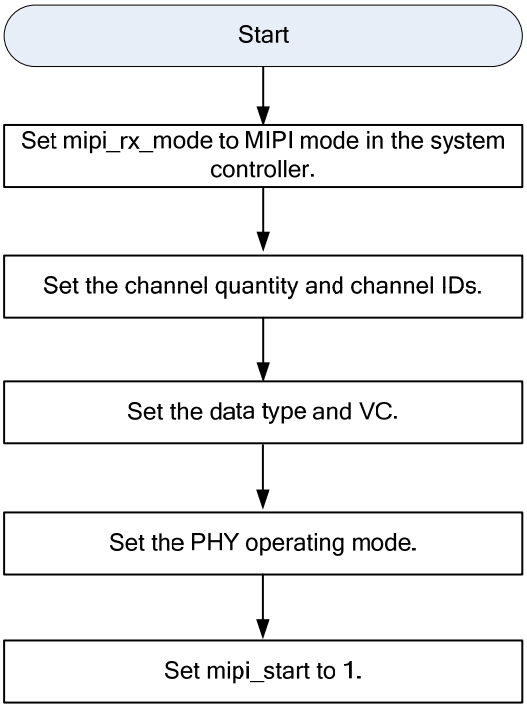
The MIPI RX controller supports the MIPI, LVDS, and HiSPI modes. The software configuration in each mode contains two parts: controller and PHY. All register configurations simultaneously take effect when MIPI_START has been set to 1.

9.3.4.1 MIPI Mode Configuration Process

In MIPI mode, configure the channel quantity, data type for data transmission. Frame or line synchronization information in MIPI mode is included in a short packet. The controller parses the short packet and restores the pixel data. Figure 9-26 shows the software configuration process in MIPI mode.

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Figure 9-26 MIPI RX configuration process in MIPI mode



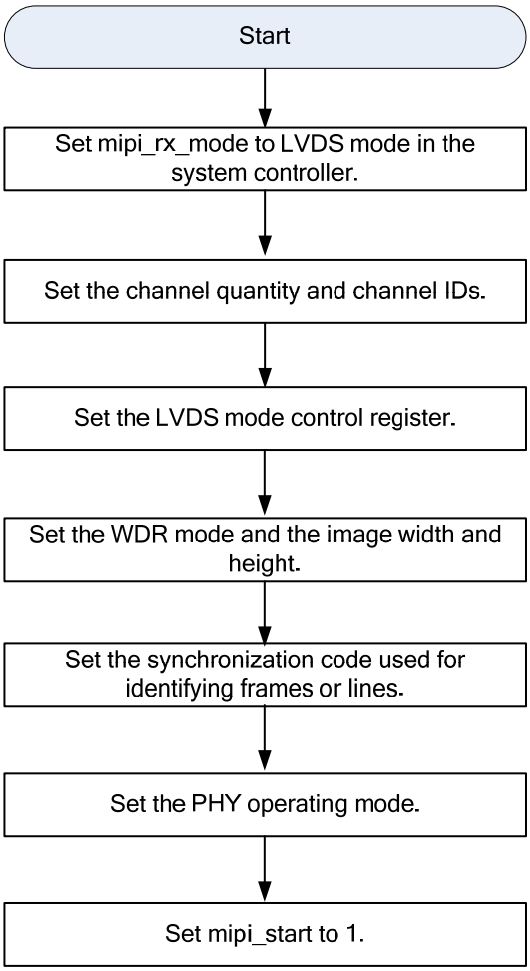
9.3.4.2 LVDS and HiSPI Mode Configuration Process

In LVDS mode, configure the type of raw data, endian mode, synchronization mode and image size registers. In LVDS mode, a synchronization code is used to identify frame or line synchronization information. Depending on the type of raw data, the synchronization code can be 8, 10, 12, or 14 bits. Configuration methods are the same for HiSPI and LVDS modes.

Figure 9-27 shows the software configuration process in LVDS and HiSPI modes.

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Figure 9-27 MIPI RX configuration process in LVDS and HiSPI modes



9.3.5 MIPI RX Register Summary

Table 9-10 describes the MIPI RX registers.

Table 9-10 Summary of MIPI RX registers (base address: 0x2068_0000)

Offset Address	Register	Description	Page
0x0000	MIPI_START	MIPI configuration enable register	9-138
0x0004	MIPI_CTRL_MODE	MIPI controller configuration register	9-139
0x000C	LVDS_IMGSIZE	MIPI image size register	9-140
0x0014	LVDS_CTRL_MODE	LVDS operating mode configuration register	9-140
0x0018	MIPI_USERDEF_DT	Pixel bit width configuration register for the user-defined data type in MIPI mode	9-142

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Offset Address	Register	Description	Page
0x001C	USER_DEF	User-defined data type enable register in MIPI mode	9-143
0x0020	CROP START	Crop start point horizontal/vertical coordinate register	9-144
0x0024	CTRL_MODE	Operating mode enable register	9-144
0x1000	INT_STAT_LINK0	Link 0 interrupt status register	9-145
0x1004	INT_MASK_LINK0	Link 0 interrupt mask register	9-147
0x1008	INT_RAW_LINK0	Link 0 raw interrupt status register	9-149
0x100C	INT_CLR_LINK0	Link 0 interrupt clear register	9-151
0x1010	LANE_EN_LINK0	Link 0 lane enable configuration register	9-153
0x1014	LANE_ID_LINK0	Link 0 lane priority configuration register	9-154
0x1024	PHY_EN_LINK0	Link 0 PHY channel enable register	9-154
0x1028	PHY_CFG_LINK0	Link 0 PHY mode configuration register	9-156
0x1030	MIPI_LANES_NUM	MIPI data channel quantity register	9-157
0x1038	MIPI_DI_1	MIPI controller data ID 1 register	9-158
0x103C	MIPI_DI_2	MIPI controller data ID 2 register	9-159
0x1040	MIPI_INTR1_RAW	MIPI raw interrupt 1 register	9-160
0x1044	MIPI_INTR2_RAW	MIPI raw interrupt 2 register	9-161
0x1048	MIPI_INTR1_MASK	MIPI interrupt 1 mask register	9-162
0x104C	MIPI_INTR2_MASK	MIPI interrupt 2 mask register	9-165
0x1060	MIPI_TIMEOUT	Link 0 MIPI timeout register	9-167
0x1064	MIPI_FSM_PARAM	Link 0 MIPI time parameter register	9-168
0x1068	MIPI_FSM_CKEN	MIPI FSM clock gating register	9-168
0x1080	LINK0_LANE0_SOF_01	SOF synchronization code configuration register for lane 0 in LVDS or HiSPI mode	9-169
0x1088	LINK0_LANE0_EOF_01	EOF synchronization code configuration register for lane 0 in LVDS or HiSPI mode	9-170
0x1090	LINK0_LANE0_SOL_01	SOL synchronization code configuration register for lane 0 in LVDS or HiSPI mode.	9-170
0x1098	LINK0_LANE0_EOL_01	EOL synchronization code configuration register for lane 0 in LVDS or HiSPI mode	9-171
0x10A0	LINK0_LANE1_SOF_01	SOF synchronization code configuration register for lane 1 in LVDS or HiSPI mode	9-171

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Offset Address	Register	Description	Page
0x10A8	LINK0_LANE1_EOF_01	EOF synchronization code configuration register for lane 1 in LVDS or HiSPI mode	9-172
0x10B0	LINK0_LANE1_SOL_01	SOL synchronization code configuration register for lane 1 in LVDS or HiSPI mode.	9-172
0x10B8	LINK0_LANE1_EOL_01	EOL synchronization code configuration register for lane 1 in LVDS or HiSPI mode	9-173
0x10C0	LINK0_LANE2_SOF_01	SOF synchronization code configuration register for lane 2 in LVDS or HiSPI mode	9-173
0x10C8	LINK0_LANE2_EOF_01	EOF synchronization code configuration register for lane 2 in LVDS or HiSPI mode	9-174
0x10D0	LINK0_LANE2_SOL_01	SOL synchronization code configuration register for lane 2 in LVDS or HiSPI mode.	9-174
0x10D8	LINK0_LANE2_EOL_01	EOL synchronization code configuration register for lane 2 in LVDS or HiSPI mode	9-175
0x10E0	LINK0_LANE3_SOF_01	SOF synchronization code configuration register for lane 3 in LVDS or HiSPI mode	9-175
0x10E8	LINK0_LANE3_EOF_01	EOF synchronization code configuration register for lane 3 in LVDS or HiSPI mode	9-176
0x10F0	LINK0_LANE3_SOL_01	SOL synchronization code configuration register for lane 3 in LVDS or HiSPI mode.	9-176
0x10F8	LINK0_LANE3_EOL_01	EOL synchronization code configuration register for lane 3 in LVDS or HiSPI mode	9-177

9.3.6 MIPI Register Description

MIPI_START

MIPI_START is an MIPI configuration enable register.

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Offset Address	Register Name	Total Reset Value
0x0000	MIPI_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															mipi_start
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	mipi_start	Configuration enable 0: disabled 1: Enable the APB clock domain configuration signal and convert it into the MIPI working clock domain output.

MIPI_CTRL_MODE

MIPI_CTRL_MODE is an MIPI controller configuration register.

Offset Address	Register Name	Total Reset Value
0x0004	MIPI_CTRL_MODE	0x0000_0222

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					emab			reserved	emaa			reserved		ret1n	mipi_mem_ck_gt
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10:8]	RW	emab	memory emab
[7]	RW	reserved	Reserved
[6:4]	RW	emaa	memory emaa
[3:2]	RO	reserved	Reserved
[1]	RW	ret1n	memory ret1n

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[0]	RW	mipi_mem_ck_gt	MIPI memory clock gating enable 0: disabled 1: enabled

LVDS_IMGSIZE

LVDS_IMGSIZE is an MIPI image size register.

Offset Address	Register Name	Total Reset Value
0x000C	LVDS_IMGSIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				lvds_imgheight												reserved				lvds_imgwidth											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	lvds_imgheight	Image height. The value is the actual image height minus 1, and the value range is [0, 8191].
[15:13]	RO	reserved	Reserved
[12:0]	RW	lvds_imgwidth	Image width. The value is the actual image width minus 1, and the value range is [0, 8191].

LVDS_CTRL_MODE

LVDS_CTRL_MODE is an LVDS operating mode configuration register.

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Offset Address	Register Name	Total Reset Value
0x0014	LVDS_CTRL_MODE	0x0000_0310

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			lvds_link_seq	reserved		lvds_code_big_endian	lvds_pix_big_endian	reserved			lvds_raw_type		reserved			li_en	lvds_sync_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	lvds_link_seq	Sequence of two links in LVDS mode 0: synchronous transmission mode for two links of the normal sensor 1: asynchronous transmission mode for two links of the Panasonic sensor
[11:10]	RW	reserved	Reserved
[9]	RW	lvds_code_big_endian	Serial bit transmission sequence for the synchronization code (sync_code) of raw data to be transmitted in LVDS or HiSPI mode 0: LSB. Least significant bits are transmitted first. The serial data sequence of the synchronization code (sync_code) received is bit 0, bit 1, …, bit 11. 1: MSB. Most significant bits are transmitted first. The serial data sequence of the synchronization code (sync_code) received is bit 11, bit 10, …, bit 0.
[8]	RW	lvds_pix_big_endian	Serial bit transmission sequence for the valid pixels of raw data to be transmitted in LVDS or HiSPI mode 0: LSB. Least significant bits are transmitted first. The serial data sequence of the valid pixels received is bit 0, bit 1, …, bit 11. 1: MSB. Most significant bits are transmitted first. The serial data sequence of the valid pixels received is bit 11, bit 10, …, bit 0.
[7]	RO	reserved	Reserved
[6:4]	RW	lvds_raw_type	Type of the RAW data to be transmitted in LVDS or HiSPI mode 001: 8-bit RAW data 010: 10-bit RAW data 011: 12-bit RAW data 100: 14-bit RAW data Other values: reserved

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[3:2]	RO	reserved	Reserved
[1]	RW	li_en	Sony line information detection enable 1'b0: disabled 1'b1: enabled
[0]	RW	lvds_sync_mode	Frame or line synchronization mode in LVDS mode 0: SOF, EOF, SOL, or EOL mode 1: SAV-EAV mode SAV-EAV mode is widely applied in SONY sensor. In this mode, set InValid SAV/EAV in LINKm_LANEn_SOF_xx or LINKm_LANEn_EOF_xx, and set Valid SAV/EAV in LINKm_LANEn_SOL_xx or LINKm_LANEn_SOL_xx. Here, *m* is 0 or 1, and *n* is 0, 1, 2, or 3. In HiSPI mode, this field must be set to 0.

MIPI_USERDEF_DT

MIPI_USERDEF_DT is a pixel bit width configuration register for the user-defined data type in MIPI mode.

Offset Address	Register Name	Total Reset Value
0x0018	MIPI_USERDEF_DT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				user_def3_dt	reserved / user_def2_dt	reserved / user_def1_dt	reserved / user_def0_dt
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14:12]	RW	user_def3_dt	Bit width of the transferred pixel when the data type is user_def3 000: 8 bits 001: 10 bits 010: 12 bits 011: 14 bits Other values: reserved
[11]	RO	reserved	Reserved

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[10:8]	RW	user_def2_dt	Bit width of the transferred pixel when the data type is user_def2 000: 8 bits 001: 10 bits 010: 12 bits 011: 14 bits Other values: reserved
[7]	RO	reserved	Reserved
[6:4]	RW	user_def1_dt	Bit width of the transferred pixel when the data type is user_def1 000: 8 bits 001: 10 bits 010: 12 bits 011: 14 bits Other values: reserved
[3]	RO	reserved	Reserved
[2:0]	RW	user_def0_dt	Bit width of the transferred pixel when the data type is user_def0 000: 8 bits 001: 10 bits 010: 12 bits 011: 14 bits Other values: reserved

USER_DEF

USER_DEF is a user-defined data type enable register in MIPI mode.

Offset Address	Register Name	Total Reset Value
0x001C	USER_DEF	0x1036_3534

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				user_def3				reserved				user_def2				reserved				user_def1				reserved				user_def0			
Reset	0	0	0	1	0	0	0	0	0	0	1	1	0	1	1	0	0	0	1	1	0	1	0	1	0	0	1	1	0	1	0	0

Bits	Access	Name	Description
[31:30]	RW	reserved	Reserved
[29:24]	RW	user_def3	User-defined data type 3
[23:22]	RW	reserved	Reserved
[21:16]	RW	user_def2	User-defined data type 2

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[15:14]	RO	reserved	Reserved
[13:8]	RW	user_def1	User-defined data type 1
[7:6]	RO	reserved	Reserved
[5:0]	RW	user_def0	User-defined data type 0

CROP START

CROP START is a crop start point horizontal/vertical coordinate register.

Offset Address	Register Name	Total Reset Value
0x0020	CROP START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				start_y												reserved				start_x											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	start_y	Vertical coordinate of the crop start point Value range: [0, 8191]
[15:13]	RO	reserved	Reserved
[12:0]	RW	start_x	Horizontal coordinate of the crop start point Value range: [0, 8191]

CTRL_MODE

CTRL_MODE is an operating mode enable register.

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Offset Address	Register Name	Total Reset Value
0x0024	CTRL_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												crop_en	user_def_en	reserved	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RW	crop_en	Crop enable 0: disabled 1: enabled
[2]	RW	user_def_en	User-defined mode enable 0: disabled 1: enabled
[1]	RO	reserved	Reserved
[0]	RO	reserved	Reserved

INT_STAT_LINK0

INT_STAT_LINK0 is a link 0 interrupt status register.

Offset Address	Register Name	Total Reset Value
0x1000	INT_STAT_LINK0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										vsync_err_stat	hsync_err_stat	lane3_sync_err_stat	lane2_sync_err_stat	lane1_sync_err_stat	lane0_sync_err_stat	reserved				timeout_d3_stat	timeout_d2_stat	timeout_d1_stat	timeout_d0_stat	timeout_ck_stat	reserved			escape_d3_stat	escape_d2_stat	escape_d1_stat	escape_d0_stat	escape_ck_stat
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved

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[21]	RO	vsync_err_stat	SOF or EOF signal async interrupt status for each lane on link 0 0: No interrupt is generated. 1: An interrupt is generated.
[20]	RO	hsync_err_stat	SOL or EOL signal async interrupt status for each lane on link 0 0: No interrupt is generated. 1: An interrupt is generated.
[19]	RO	lane3_sync_err_stat	EOL or EOF signal detection failure interrupt status for lane 3 on link 0 in LVDS mode 0: No interrupt is generated. 1: An interrupt is generated.
[18]	RO	lane2_sync_err_stat	EOL or EOF signal detection failure interrupt status for lane 2 on link 0 in LVDS mode 0: No interrupt is generated. 1: An interrupt is generated.
[17]	RO	lane1_sync_err_stat	EOL or EOF signal detection failure interrupt status for lane 1 on link 0 in LVDS mode 0: No interrupt is generated. 1: An interrupt is generated.
[16]	RO	lane0_sync_err_stat	EOL or EOF signal detection failure interrupt status for lane 0 on link 0 in LVDS mode 0: No interrupt is generated. 1: An interrupt is generated.
[15:13]	RO	reserved	Reserved
[12]	RO	timeout_d3_stat	FSM timeout interrupt status for data lane 3 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[11]	RO	timeout_d2_stat	FSM timeout interrupt status for data lane 2 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[10]	RO	timeout_d1_stat	FSM timeout interrupt status for data lane 1 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[9]	RO	timeout_d0_stat	FSM timeout interrupt status for data lane 0 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.

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[8]	RO	timeout_ck_stat	FSM timeout interrupt status for the clock lane on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[7:5]	RO	reserved	Reserved
[4]	RO	escape_d3_stat	Escape sequence interrupt status for data lane 3 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[3]	RO	escape_d2_stat	Escape sequence interrupt status for data lane 2 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	escape_d1_stat	Escape sequence interrupt status for data lane 1 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	escape_d0_stat	Escape sequence interrupt status for data lane 0 on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	escape_ck_stat	Escape sequence interrupt status for the clock lane on link 0 in MIPI mode 0: No interrupt is generated. 1: An interrupt is generated.

INT_MASK_LINK0

INT_MASK_LINK0 is a link 0 interrupt mask register.

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Offset Address	Register Name	Total Reset Value
0x1004	INT_MASK_LINK0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										vsync_err_mask	hsync_err_mask	lane3_sync_err_mask	lane2_sync_err_mask	lane1_sync_err_mask	lane0_sync_err_mask	reserved			timeout_d3_mask	timeout_d2_mask	timeout_d1_mask	timeout_d0_mask	timeout_ck_mask	reserved			escape_d3_mask	escape_d2_mask	escape_d1_mask	escape_d0_mask	escape_ck_mask
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	vsync_err_mask	SOF or EOF signal async raw interrupt mask for each lane on link 0 0: masked 1: enabled
[20]	RW	hsync_err_mask	SOL or EOL signal async raw interrupt mask for each lane on link 0 0: masked 1: enabled
[19]	RW	lane3_sync_err_mask	EOL or EOF signal detection failure raw interrupt mask for lane 3 on link 0 0: masked 1: enabled
[18]	RW	lane2_sync_err_mask	EOL or EOF signal detection failure raw interrupt mask for lane 2 on link 0 0: masked 1: enabled
[17]	RW	lane1_sync_err_mask	EOL or EOF signal detection failure raw interrupt mask for lane 1 on link 0 0: masked 1: enabled
[16]	RW	lane0_sync_err_mask	EOL or EOF signal detection failure raw interrupt mask for lane 0 on link 0 0: masked 1: enabled
[15:13]	RO	reserved	Reserved

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[12]	RW	timeout_d3_mask	FSM timeout interrupt mask for data lane 3 on link 0 0: masked 1: enabled
[11]	RW	timeout_d2_mask	FSM timeout interrupt mask for data lane 2 on link 0 0: masked 1: enabled
[10]	RW	timeout_d1_mask	FSM timeout interrupt mask for data lane 1 on link 0 0: masked 1: enabled
[9]	RW	timeout_d0_mask	FSM timeout interrupt mask for data lane 0 on link 0 0: masked 1: enabled
[8]	RW	timeout_ck_mask	FSM timeout interrupt mask for the clock lane on link 0 0: masked 1: enabled
[7:5]	RO	reserved	Reserved
[4]	RW	escape_d3_mask	Escape interrupt mask for data lane 3 on link 0 0: masked 1: enabled
[3]	RW	escape_d2_mask	Escape interrupt mask for data lane 2 on link 0 0: masked 1: enabled
[2]	RW	escape_d1_mask	Escape interrupt mask for data lane 1 on link 0 0: masked 1: enabled
[1]	RW	escape_d0_mask	Escape interrupt mask for data lane 0 on link 0 0: masked 1: enabled
[0]	RW	escape_ck_mask	Escape interrupt mask for the clock lane on link 0 0: masked 1: enabled

INT_RAW_LINK0

INT_RAW_LINK0 is a link 0 raw interrupt status register.

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Offset Address	Register Name	Total Reset Value
0x1008	INT_RAW_LINK0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										vsync_err_raw	hsync_err_raw	lane3_sync_err_raw	lane2_sync_err_raw	lane1_sync_err_raw	lane0_sync_err_raw	reserved			timeout_d3_raw	timeout_d2_raw	timeout_d1_raw	timeout_d0_raw	timeout_ck_raw	reserved			escape_d3_raw	escape_d2_raw	escape_d1_raw	escape_d0_raw	escape_ck_raw
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RO	vsync_err_raw	Raw SOF or EOF signal async interrupt status for each lane on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[20]	RO	hsync_err_raw	Raw SOL or EOL signal async interrupt status for each lane on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[19]	RO	lane3_sync_err_raw	Raw EOL or EOF signal detection failure interrupt status for lane 3 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[18]	RO	lane2_sync_err_raw	Raw EOL or EOF signal detection failure interrupt status for lane 2 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[17]	RO	lane1_sync_err_raw	Raw EOL or EOF signal detection failure interrupt status for lane 1 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[16]	RO	lane0_sync_err_raw	Raw EOL or EOF signal detection failure interrupt status for lane 0 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[15:13]	RO	reserved	Reserved

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[12]	RO	timeout_d3_raw	Raw FSM timeout interrupt status for data lane 3 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[11]	RO	timeout_d2_raw	Raw FSM timeout interrupt status for data lane 2 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[10]	RO	timeout_d1_raw	Raw FSM timeout interrupt status for data lane 1 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[9]	RO	timeout_d0_raw	Raw FSM timeout interrupt status for data lane 0 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[8]	RO	timeout_ck_raw	Raw FSM timeout interrupt status for the clock lane on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[7:5]	RO	reserved	Reserved
[4]	RO	escape_d3_raw	Raw escape sequence interrupt status for data lane 3 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[3]	RO	escape_d2_raw	Raw escape sequence interrupt status for data lane 2 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[2]	RO	escape_d1_raw	Raw escape sequence interrupt status for data lane 1 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[1]	RO	escape_d0_raw	Raw escape sequence interrupt status for data lane 0 on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[0]	RO	escape_ck_raw	Raw escape sequence interrupt status for the clock lane on link 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.

INT_CLR_LINK0

INT_CLR_LINK0 is a link 0 interrupt clear register.

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Offset Address	Register Name	Total Reset Value
0x100C	INT_CLR_LINK0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										vsync_err_clr	hsync_err_clr	lane3_sync_err_clr	lane2_sync_err_clr	lane1_sync_err_clr	lane0_sync_err_clr	reserved			timeout_d3_clr	timeout_d2_clr	timeout_d1_clr	timeout_d0_clr	timeout_ck_clr	reserved			escape_d3_clr	escape_d2_clr	escape_d1_clr	escape_d0_clr	escape_ck_clr
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21]	RW	vsync_err_clr	SOF or EOF signal async interrupt clear for each lane on link 0 Writing 1 clears this interrupt.
[20]	RW	hsync_err_clr	SOL or EOL signal async interrupt clear for each lane on link 0 Writing 1 clears this interrupt.
[19]	RW	lane3_sync_err_clr	EOL or EOF signal detection failure interrupt clear for lane 3 on link 0 Writing 1 clears this interrupt.
[18]	RW	lane2_sync_err_clr	EOL or EOF signal detection failure interrupt clear for lane 2 on link 0 Writing 1 clears this interrupt.
[17]	RW	lane1_sync_err_clr	EOL or EOF signal detection failure interrupt clear for lane 1 on link 0 Writing 1 clears this interrupt.
[16]	RW	lane0_sync_err_clr	EOL or EOF signal detection failure interrupt clear for lane 0 on link 0 Writing 1 clears this interrupt.
[15:13]	RO	reserved	Reserved
[12]	RW	timeout_d3_clr	FSM timeout interrupt clear for data lane 3 on link 0 Writing 1 clears this interrupt.
[11]	RW	timeout_d2_clr	FSM timeout interrupt clear for data lane 2 on link 0 Writing 1 clears this interrupt.
[10]	RW	timeout_d1_clr	FSM timeout interrupt clear for data lane 1 on link 0 Writing 1 clears this interrupt.
[9]	RW	timeout_d0_clr	FSM timeout interrupt clear for data lane 0 on link 0 Writing 1 clears this interrupt.

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[8]	RW	timeout_ck_clr	FSM timeout interrupt clear for the clock lane on link 0 Writing 1 clears this interrupt.
[7:5]	RO	reserved	Reserved
[4]	RW	escape_d3_clr	Escape interrupt clear for data lane 3 on link 0 Writing 1 clears this interrupt.
[3]	RW	escape_d2_clr	Escape interrupt clear for data lane 2 on link 0 Writing 1 clears this interrupt.
[2]	RW	escape_d1_clr	Escape interrupt clear for data lane 1 on link 0 Writing 1 clears this interrupt.
[1]	RW	escape_d0_clr	Escape interrupt clear for data lane 0 on link 0 Writing 1 clears this interrupt.
[0]	RW	escape_ck_clr	Escape interrupt clear for the clock lane on link 0 Writing 1 clears this interrupt.

LANE_EN_LINK0

LANE_EN_LINK0 is a link 0 lane enable configuration register.

Offset Address	Register Name	Total Reset Value
0x1010	LANE_EN_LINK0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			lane3_en	reserved			lane2_en	reserved			lane1_en	reserved			lane0_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	lane3_en	Lane 3 enable for link 0 0: disabled 1: enabled
[11:9]	RO	reserved	Reserved
[8]	RW	lane2_en	Lane 2 enable for link 0 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved

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[4]	RW	lane1_en	Lane 1 enable for link 0 0: disabled 1: enabled
[3:1]	RO	reserved	Reserved
[0]	RW	lane0_en	Lane 0 enable for link 0 0: disabled 1: enabled

LANE_ID_LINK0

LANE_ID_LINK0 is a link 0 lane priority configuration register.

Offset Address	Register Name	Total Reset Value
0x1014	LANE_ID_LINK0	0x0000_3210

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																		lane3_id		reserved		lane2_id		reserved		lane1_id		reserved		lane0_id	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:12]	RW	lane3_id	Lane 3 priority for link 0. The value ranges from 0 to 3.
[11:10]	RO	reserved	Reserved
[9:8]	RW	lane2_id	Lane 2 priority for link 0. The value ranges from 0 to 3.
[7:6]	RO	reserved	Reserved
[5:4]	RW	lane1_id	Lane 1 priority for link 0. The value ranges from 0 to 3.
[3:2]	RO	reserved	Reserved
[1:0]	RW	lane0_id	Lane 0 priority for link 0. The value ranges from 0 to 3.

PHY_EN_LINK0

PHY_EN_LINK0 is a link 0 PHY channel enable register.

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Offset Address	Register Name	Total Reset Value
0x1024	PHY_EN_LINK0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								phy_te_d3en	phy_te_d2en	phy_te_d1en	phy_te_d0en	reserved			phy_te_cken	phy_hs_d3en	phy_hs_d2en	phy_hs_d1en	phy_hs_d0en	reserved											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23]	RW	phy_te_d3en	Termination enable for data lane 3 on link 0 0: disabled 1: enabled
[22]	RW	phy_te_d2en	Termination enable for data lane 2 on link 0 0: disabled 1: enabled
[21]	RW	phy_te_d1en	Termination enable for data lane 1 on link 0 0: disabled 1: enabled
[20]	RW	phy_te_d0en	Termination enable for data lane 0 on link 0 0: disabled 1: enabled
[19:17]	RO	reserved	Reserved
[16]	RW	phy_te_cken	Termination enable for the clock lane on link 0 0: disabled 1: enabled
[15]	RW	phy_hs_d3en	High-speed mode enable for data lane 3 on link 0 0: disabled 1: enabled
[14]	RW	phy_hs_d2en	High-speed mode enable for data lane 2 on link 0 0: disabled 1: enabled
[13]	RW	phy_hs_d1en	High-speed mode enable for data lane 1 on link 0 0: disabled 1: enabled

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[12]	RW	phy_hs_d0en	High-speed mode enable for data lane 0 on link 0 0: disabled 1: enabled
[11:0]	RO	reserved	Reserved

PHY_CFG_LINK0

PHY_CFG_LINK0 is a link 0 PHY mode configuration register.

Offset Address	Register Name	Total Reset Value
0x1028	PHY_CFG_LINK0	0x6A1F_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		phy_rg_sa_en		phy_rg_it_dt		phy_rg_it_ck		phy_rg_it_lp		reserved		phy_da_ulpb_ck		phy_da_ulpb		reserved			phy_rg_en_lp	reserved			phy_rg_en_cmos	reserved			phy_rg_en_clk	phy_rg_en_d			
Reset	0		1		1		0		1		0		0		0		1		1		1		1		0		0		0		0	

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30]	RW	phy_rg_sa_en	SA enable (function tuning)
[29:28]	RW	phy_rg_it_dt	High-speed mode current adjustment for data channels
[27:26]	RW	phy_rg_it_ck	High-speed mode current adjustment for the clock channel
[25:24]	RW	phy_rg_it_lp	Low-power mode current adjustment
[23:21]	RO	reserved	Reserved
[20]	RW	phy_da_ulpb_ck	Ultra-low-power mode enable for the clock lane, active low 0: enable 1: disable
[19:16]	RW	phy_da_ulpb	Ultra-low-power mode enable for data lanes, active low. Each bit corresponds to one data lane. Bit[16] corresponds to data lane 0. Bit[17] corresponds to data lane 1. Bit[18] corresponds to data lane 2. Bit[19] corresponds to data lane 3 0: enable 1: disable
[15:13]	RO	reserved	Reserved

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[12]	RW	phy_rg_en_lp	LP mode enable, active high In LVDS mode, the common mode voltage may trigger an internal LP-related circuit. Therefore, this signal must be disabled in LVDS mode. 0: disabled 1: enabled
[11:9]	RO	reserved	Reserved
[8]	RW	phy_rg_en_cmos	CMOS mode enable, active high 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved.
[4]	RW	phy_rg_en_clk	Clock lane enable, active high 0: disabled 1: enabled
[3:0]	RW	phy_rg_en_d	Data lane enable, active high 0: disabled 1: enabled

MIPI_LANES_NUM

MIPI_LANES_NUM is an MIPI data channel quantity register.

Offset Address	Register Name	Total Reset Value
0x1030	MIPI_LANES_NUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														mipi_lane_num	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	1	Number of enabled data channels 00: one data channel 01: two data channels 10: three data channels 11: four data channels

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MIPI_DI_1

MIPI_DI_1 is an MIPI controller data with ID 1 register.

Offset Address	Register Name	Total Reset Value
0x1038	MIPI_DI_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	di3_vc		di3_dt						di2_vc		di2_dt						di1_vc		di1_dt						di0_vc		di0_dt					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RW	di3_vc	Virtual channel of data with ID 3
[29:24]	RW	di3_dt	Type of data with ID 3 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved
[23:22]	RW	di2_vc	Virtual channel of data with ID 2
[21:16]	RW	di2_dt	Type of data with ID 2 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved
[15:14]	RW	di1_vc	Virtual channel of data with ID 1
[13:8]	RW	di1_dt	Type of data with ID 1 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved
[7:6]	RW	di0_vc	Virtual channel of data with ID 0
[5:0]	RW	di0_dt	Type of data with ID 0 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved

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MIPI_DI_2

MIPI_DI_2 is an MIPI controller data with ID 2 register.

Offset Address	Register Name	Total Reset Value
0x103C	MIPI_DI_2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	di7_vc		di7_dt						di6_vc		di6_dt						di5_vc		di5_dt						di4_vc		di4_dt					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RW	di7_vc	Virtual channel of data with ID 7
[29:24]	RW	di7_dt	Type of data with ID 7 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved
[23:22]	RW	di6_vc	Virtual channel of data with ID 6
[21:16]	RW	di6_dt	Type of data with ID 6 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved
[15:14]	RW	di5_vc	Virtual channel of data with ID 5
[13:8]	RW	di5_dt	Type of data with ID 5 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved
[7:6]	RW	di4_vc	Virtual channel of data with ID 4
[5:0]	RW	di4_dt	Type of data with ID 4 0x2B: RAW10 0x2C: RAW12 0x2D: RAW14 Other values: reserved

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MIPI_INTR1_RAW

MIPI_INTR1_RAW is an MIPI raw interrupt 1 register.

Offset Address	Register Name	Total Reset Value
0x1040	MIPI_INTR1_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved			intr_ecc_double	intr_crc_vc3	intr_crc_vc2	intr_crc_vc1	intr_crc_vc0	intr_l_seq_di3	intr_l_seq_di2	intr_l_seq_di1	intr_l_seq_di0	intr_l_bndy_di3	intr_l_bndy_di2	intr_l_bndy_di1	intr_l_bndy_di0	intr_f_crc_vc3	intr_f_crc_vc2	intr_f_crc_vc1	intr_f_crc_vc0	intr_f_seq_vc3	intr_f_seq_vc2	intr_f_seq_vc1	intr_f_seq_vc0	intr_f_bndry_vc3	intr_f_bndry_vc2	intr_f_bndry_vc1	intr_f_bndry_vc0	reserved	reserved	reserved	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28]	RC	intr_ecc_double	The header has at least two errors, which cannot be fixed by ECC. This field is cleared on a read.
[27]	RC	intr_crc_vc3	VC3 data CRC error. This field is cleared on a read.
[26]	RC	intr_crc_vc2	VC2 data CRC error. This field is cleared on a read.
[25]	RC	intr_crc_vc1	VC1 data CRC error. This field is cleared on a read.
[24]	RC	intr_crc_vc0	VC0 data CRC error. This field is cleared on a read.
[23]	RC	intr_l_seq_di3	VC3 and DT3 line sequence error. This field is cleared on a read.
[22]	RC	intr_l_seq_di2	VC2 and DT2 line sequence error. This field is cleared on a read.
[21]	RC	intr_l_seq_di1	VC1 and DT1 line sequence error. This field is cleared on a read.
[20]	RC	intr_l_seq_di0	VC0 and DT0 line sequence error. This field is cleared on a read.
[19]	RC	intr_l_bndy_di3	SOL and EOL mismatch for VC3 and DT3. This field is cleared on a read.
[18]	RC	intr_l_bndy_di2	SOL and EOL mismatch for VC2 and DT2. This field is cleared on a read.
[17]	RC	intr_l_bndy_di1	SOL and EOL mismatch for VC1 and DT1. This field is cleared on a read.
[16]	RC	intr_l_bndy_di0	SOL and EOL mismatch for VC0 and DT0. This field is cleared on a read.
[15]	RC	intr_f_crc_vc3	In the last frame of data, VC3 channel data has at least one CRC error. This field is cleared on a read.
[14]	RC	intr_f_crc_vc2	In the last frame of data, VC2 channel data has at least one CRC error. This field is cleared on a read.

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[13]	RC	intr_f_crc_vc1	In the last frame of data, VC1 channel data has at least one CRC error. This field is cleared on a read.
[12]	RC	intr_f_crc_vc0	In the last frame of data, VC0 channel data has at least one CRC error. This field is cleared on a read.
[11]	RC	intr_f_seq_vc3	VC3 frame sequence error. This field is cleared on a read.
[10]	RC	intr_f_seq_vc2	VC2 frame sequence error. This field is cleared on a read.
[9]	RC	intr_f_seq_vc1	VC1 frame sequence error. This field is cleared on a read.
[8]	RC	intr_f_seq_vc0	VC0 frame sequence error. This field is cleared on a read.
[7]	RC	intr_f_bndry_vc3	SOF and EOF mismatch for VC3. This field is cleared on a read.
[6]	RC	intr_f_bndry_vc2	SOF and EOF mismatch for VC2. This field is cleared on a read.
[5]	RC	intr_f_bndry_vc1	SOF and EOF mismatch for VC1. This field is cleared on a read.
[4]	RC	intr_f_bndry_vc0	SOF and EOF mismatch for VC0. This field is cleared on a read.
[3]	RO	reserved	Reserved
[2]	RO	reserved	Reserved
[1]	RO	reserved	Reserved
[0]	RO	reserved	Reserved

MIPI_INTR2_RAW

MIPI_INTR2_RAW is an MIPI raw interrupt 2 register.

Offset Address	Register Name	Total Reset Value
0x1044	MIPI_INTR2_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								intr_1_seq_di7	intr_1_seq_di6	intr_1_seq_di5	intr_1_seq_di4	intr_1_bndry_di7	intr_1_bndry_di6	intr_1_bndry_di5	intr_1_bndry_di4	intr_err_id_vc3	intr_err_id_vc2	intr_err_id_vc1	intr_err_id_vc0	intr_vc3_ecc_corrected	intr_vc2_ecc_corrected	intr_vc1_ecc_corrected	intr_vc0_ecc_corrected	reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23]	RC	intr_1_seq_di7	VC 7 and DT 7 line sequence error. This field is cleared on a read.

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[22]	RC	intr_l_seq_di6	VC 6 and DT 6 line sequence error. This field is cleared on a read.
[21]	RC	intr_l_seq_di5	VC 5 and DT 5 line sequence error. This field is cleared on a read.
[20]	RC	intr_l_seq_di4	VC 4 and DT 4 line sequence error. This field is cleared on a read.
[19]	RC	intr_l_bndry_di7	SOL and EOL mismatch for VC 7 and DT 7. This field is cleared on a read.
[18]	RC	intr_l_bndry_di6	SOL and EOL mismatch for VC 6 and DT 6. This field is cleared on a read.
[17]	RC	intr_l_bndry_di5	SOL and EOL mismatch for VC 5 and DT 5. This field is cleared on a read.
[16]	RC	intr_l_bndry_di4	SOL and EOL mismatch for VC 4 and DT 4. This field is cleared on a read.
[15]	RC	intr_err_id_vc3	VC3 channel data type is not supported. This field is cleared on a read.
[14]	RC	intr_err_id_vc2	VC2 channel data type is not supported. This field is cleared on a read.
[13]	RC	intr_err_id_vc1	VC1 channel data type is not supported. This field is cleared on a read.
[12]	RC	intr_err_id_vc0	VC0 channel data type is not supported. This field is cleared on a read.
[11]	RC	intr_vc3_ecc_corrected	VC3 channel header error, fixed by ECC. This field is cleared on a read.
[10]	RC	intr_vc2_ecc_corrected	VC2 channel header error, fixed by ECC. This field is cleared on a read.
[9]	RC	intr_vc1_ecc_corrected	VC1 channel header error, fixed by ECC. This field is cleared on a read.
[8]	RC	intr_vc0_ecc_corrected	VC0 channel header error, fixed by ECC. This field is cleared on a read.
[7:0]	RO	reserved	Reserved

MIPI_INTR1_MASK

MIPI_INTR1_MASK is an MIPI interrupt 1 mask register.

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Offset Address	Register Name	Total Reset Value
0x1048	MIPI_INTR1_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved			mask_ecc_double	mask_crc_vc3	mask_crc_vc2	mask_crc_vc1	mask_crc_vc0	mask_l_seq_di3	mask_l_seq_di2	mask_l_seq_di1	mask_l_seq_di0	mask_l_bndy_di3	mask_l_bndy_di2	mask_l_bndy_di1	mask_l_bndy_di0	mask_f_crc_vc3	mask_f_crc_vc2	mask_f_crc_vc1	mask_f_crc_vc0	mask_f_seq_vc3	mask_f_seq_vc2	mask_f_seq_vc1	mask_f_seq_vc0	mask_f_bndry_vc3	mask_f_bndry_vc2	mask_f_bndry_vc1	mask_f_bndry_vc0	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28]	RW	mask_ecc_double	Ecc_double interrupt mask, corresponding to the relevant bit in the MIPI_INTR1_RAW register 0: masked 1: not masked
[27]	RW	mask_crc_vc3	Crc_vc3 interrupt mask 0: masked 1: not masked
[26]	RW	mask_crc_vc2	Crc_vc2 interrupt mask 0: masked 1: not masked
[25]	RW	mask_crc_vc1	Crc_vc1 interrupt mask 0: masked 1: not masked
[24]	RW	mask_crc_vc0	Crc_vc0 interrupt mask 0: masked 1: not masked
[23]	RW	mask_l_seq_di3	L_seq_di3 interrupt mask 0: masked 1: not masked
[22]	RW	mask_l_seq_di2	L_seq_di2 interrupt mask 0: masked 1: not masked
[21]	RW	mask_l_seq_di1	L_seq_di1 interrupt mask 0: masked 1: not masked

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[20]	RW	mask_l_seq_di0	L_seq_di0 interrupt mask 0: masked 1: not masked
[19]	RW	mask_l_bndy_di3	L_bndy_di3 interrupt mask 0: masked 1: not masked
[18]	RW	mask_l_bndy_di2	L_bndy_di2 interrupt mask 0: masked 1: not masked
[17]	RW	mask_l_bndy_di1	L_bndy_di1 interrupt mask 0: masked 1: not masked
[16]	RW	mask_l_bndy_di0	L_bndy_di0 interrupt mask 0: masked 1: not masked
[15]	RW	mask_f_crc_vc3	F_crc_vc3 interrupt mask 0: masked 1: not masked
[14]	RW	mask_f_crc_vc2	F_crc_vc2 interrupt mask 0: masked 1: not masked
[13]	RW	mask_f_crc_vc1	F_crc_vc1 interrupt mask 0: masked 1: not masked
[12]	RW	mask_f_crc_vc0	F_crc_vc0 interrupt mask 0: masked 1: not masked
[11]	RW	mask_f_seq_vc3	F_seq_vc3 interrupt mask 0: masked 1: not masked
[10]	RW	mask_f_seq_vc2	F_seq_vc2 interrupt mask 0: masked 1: not masked
[9]	RW	mask_f_seq_vc1	F_seq_vc1 interrupt mask 0: masked 1: not masked

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[8]	RW	mask_f_seq_vc0	F_seq_vc0 interrupt mask 0: masked 1: not masked
[7]	RW	mask_f_bndry_vc3	F_bndry_vc3 interrupt mask 0: masked 1: not masked
[6]	RW	mask_f_bndry_vc2	F_bndry_vc2 interrupt mask 0: masked 1: not masked
[5]	RW	mask_f_bndry_vc1	F_bndry_vc1 interrupt mask 0: masked 1: not masked
[4]	RW	mask_f_bndry_vc0	F_bndry_vc0 interrupt mask 0: masked 1: not masked
[3:0]	RO	reserved	Reserved

MIPI_INTR2_MASK

MIPI_INTR2_MASK is an MIPI interrupt 2 mask register.

Offset Address	Register Name	Total Reset Value
0x104C	MIPI_INTR2_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								intr_1_seq_di7	intr_1_seq_di6	intr_1_seq_di5	intr_1_seq_di4	intr_1_bndry_di7	intr_1_bndry_di6	intr_1_bndry_di5	intr_1_bndry_di4	intr_err_id_vc3	intr_err_id_vc2	intr_err_id_vc1	intr_err_id_vc0	intr_vc3_ecc_corrected	intr_vc2_ecc_corrected	intr_vc1_ecc_corrected	intr_vc0_ecc_corrected	reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23]	RW	intr_1_seq_di7	L_seq_di7 interrupt mask, corresponding to the relevant bit in the MIPI_INTR2_RAW register 0: masked 1: not masked

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[22]	RW	intr_l_seq_di6	L_seq_di6 interrupt mask 0: masked 1: not masked
[21]	RW	intr_l_seq_di5	L_seq_di5 interrupt mask 0: masked 1: not masked
[20]	RW	intr_l_seq_di4	L_seq_di4 interrupt mask 0: masked 1: not masked
[19]	RW	intr_l_bndry_di7	L_bndry_di7 interrupt mask 0: masked 1: not masked
[18]	RW	intr_l_bndry_di6	L_bndry_di6 interrupt mask 0: masked 1: not masked
[17]	RW	intr_l_bndry_di5	L_bndry_di5 interrupt mask 0: masked 1: not masked
[16]	RW	intr_l_bndry_di4	L_bndry_di4 interrupt mask 0: masked 1: not masked
[15]	RW	intr_err_id_vc3	Err_id_vc3 interrupt mask 0: masked 1: not masked
[14]	RW	intr_err_id_vc2	Err_id_vc2 interrupt mask 0: masked 1: not masked
[13]	RW	intr_err_id_vc1	Err_id_vc1 interrupt mask 0: masked 1: not masked
[12]	RW	intr_err_id_vc0	Err_id_vc0 interrupt mask 0: masked 1: not masked
[11]	RW	intr_vc3_ecc_corrected	Vc3_ecc_corrected interrupt mask 0: masked 1: not masked

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[10]	RW	intr_vc2_ecc_corrected	Vc2_ecc_corrected interrupt mask 0: masked 1: not masked
[9]	RW	intr_vc1_ecc_corrected	Vc1_ecc_corrected interrupt mask 0: masked 1: not masked
[8]	RW	intr_vc0_ecc_corrected	Vc0_ecc_corrected interrupt mask 0: masked 1: not masked
[7:0]	RO	reserved	Reserved

MIPI_TIMEOUT

MIPI_TIMEOUT is a link 0 MIPI timeout register.

Offset Address	Register Name	Total Reset Value
0x1060	MIPI_TIMEOUT	0x004C_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timeout_en_d	timeout_en_ck	reserved						timeout_param																							
Reset	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31]	RW	timeout_en_d	FSM timeout detection enable for data lanes on link 0 0: disabled 1: enabled
[30]	RW	timeout_en_ck	FSM timeout detection enable for the clock lane on link 0 When in the blanking region, the sensor's clock lane may be still in HS mode or enters the LP mode, depending on the features and configuration of the sensor. If the clock lane is still in HS mode, the clock lane timeout detection must be disabled. 0: disabled 1: enabled
[29:24]	RO	reserved	Reserved
[23:0]	RW	timeout_param	FSM timeout threshold for the link 0 PHYCIL module

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MIPI_FSM_PARAM

MIPI_FSM_PARAM is a link 0 MIPI time parameter register.

Offset Address	Register Name	Total Reset Value
0x1064	MIPI_FSM_PARAM	0x1D0C_000D

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cyc_clk_hs0				reserved				cyc_clk_lp00				reserved				cyc_data_hs_trail				reserved				cyc_data_lp00			
Reset	0	0	0	1	1	1	0	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:24]	RW	cyc_clk_hs0	FSM HS-0 status duration for the clock lane on link 0 The configured value is the actual value minus 1. 0 indicates that the status lasts for one cycle.
[23:22]	RO	reserved	Reserved
[21:16]	RW	cyc_clk_lp00	FSM LP-00 status duration for the clock lane on link 0 The configured value is the actual value minus 1. 0 indicates that the status lasts for one cycle.
[15:14]	RO	reserved	Reserved
[13:8]	RW	cyc_data_hs_trail	FSM HS-TRAIL status duration for data lanes on link 0 The configured value is the actual value minus 1. 0 indicates that the status lasts for one cycle.
[7:6]	RO	reserved	Reserved
[5:0]	RW	cyc_data_lp00	FSM LP-00 status duration for data lanes on link 0 The configured value is the actual value minus 1. 0 indicates that the status lasts for one cycle.

MIPI_FSM_CKEN

MIPI_FSM_CKEN is an MIPI FSM clock gating register.

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Offset Address	Register Name	Total Reset Value
0x1068	MIPI_FSM_CKEN	0x0000_00F1

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								mipi_d3_cken	mipi_d2_cken	mipi_d1_cken	mipi_d0_cken	reserved			mipi_ck_cken
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1

Bits	Access	Name	Description
[31:8]	RW	reserved	Reserved
[7]	RW	mipi_d3_cken	FSM clock enable for data lane 3 on link 0 0: disabled 1: enabled
[6]	RW	mipi_d2_cken	FSM clock enable for data lane 2 on link 0 0: disabled 1: enabled
[5]	RW	mipi_d1_cken	FSM clock enable for data lane 1 on link 0 0: disabled 1: enabled
[4]	RW	mipi_d0_cken	FSM clock enable for data lane 0 on link 0 0: disabled 1: enabled
[3:1]	RO	reserved	Reserved
[0]	RW	mipi_ck_cken	FSM clock enable for the clock lane on link 0 0: disabled 1: enabled

LINK0_LANE0_SOF_01

LINK0_LANE0_SOF_01 is an SOF synchronization code configuration register for lane 0 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x1080	LINK0_LANE0_SOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane0_sof_1																lane0_sof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane0_sof_1	SOF synchronization code of frame 1 for lane 0 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane0_sof_0	SOF synchronization code of frame 0 for lane 0 in LVDS or HiSPI mode In WDR mode, the value is the SOF of frame 0. In linear mode, the value is the SOF.

LINK0_LANE0_EOF_01

LINK0_LANE0_EOF_01 is an EOF synchronization code configuration register for lane 0 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x1088	LINK0_LANE0_EOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane0_eof_1																lane0_eof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane0_eof_1	EOF synchronization code of frame 1 for lane 0 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane0_eof_0	EOF synchronization code of frame 0 for lane 0 in LVDS or HiSPI mode In WDR mode, the value is the EOF of frame 0. In linear mode, the value is the EOF.

LINK0_LANE0_SOL_01

LINK0_LANE0_SOL_01 is an SOL synchronization code configuration register for lane 0 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x1090	LINK0_LANE0_SOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane0_sol_1																lane0_sol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane0_sol_1	SOL synchronization code of frame 1 for lane 0 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane0_sol_0	SOL synchronization code of frame 0 for lane 0 in LVDS or HiSPI mode In WDR mode, the value is the SOL of frame 0. In linear mode, the value is the SOL.

LINK0_LANE0_EOL_01

LINK0_LANE0_EOL_01 is an EOL synchronization code configuration register for lane 0 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x1098	LINK0_LANE0_EOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane0_eol_1																lane0_eol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane0_eol_1	EOL synchronization code of frame 1 for lane 0 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane0_eol_0	EOL synchronization code of frame 0 for lane 0 in LVDS or HiSPI mode In WDR mode, the value is the EOL of frame 0. In linear mode, the value is the EOL.

LINK0_LANE1_SOF_01

LINK0_LANE1_SOF_01 is an SOF synchronization code configuration register for lane 1 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x10A0	LINK0_LANE1_SOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane1_sof_1																lane1_sof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane1_sof_1	SOF synchronization code of frame 1 for lane 1 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane1_sof_0	SOF synchronization code of frame 0 for lane 1 in LVDS or HiSPI mode In WDR mode, the value is the SOF of frame 0. In linear mode, the value is the SOF.

LINK0_LANE1_EOF_01

LINK0_LANE1_EOF_01 is an EOF synchronization code configuration register for lane 1 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x10A8	LINK0_LANE1_EOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane1_eof_1																lane1_eof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane1_eof_1	EOF synchronization code of frame 1 for lane 1 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane1_eof_0	EOF synchronization code of frame 0 for lane 1 in LVDS or HiSPI mode In WDR mode, the value is the EOF of frame 0. In linear mode, the value is the EOF.

LINK0_LANE1_SOL_01

LINK0_LANE1_SOL_01 is an SOL synchronization code configuration register for lane 1 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x10B0	LINK0_LANE1_SOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane1_sol_1																lane1_sol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane1_sol_1	SOL synchronization code of frame 1 for lane 1 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane1_sol_0	SOL synchronization code of frame 0 for lane 1 in LVDS or HiSPI mode In WDR mode, the value is the SOL of frame 0. In linear mode, the value is the SOL.

LINK0_LANE1_EOL_01

LINK0_LANE1_EOL_01 is an EOL synchronization code configuration register for lane 1 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x10B8	LINK0_LANE1_EOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane1_eol_1																lane1_eol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane1_eol_1	EOL synchronization code of frame 1 for lane 1 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane1_eol_0	EOL synchronization code of frame 0 for lane 1 in LVDS or HiSPI mode In WDR mode, the value is the EOL of frame 0. In linear mode, the value is the EOL.

LINK0_LANE2_SOF_01

LINK0_LANE2_SOF_01 is an SOF synchronization code configuration register for lane 2 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x10C0	LINK0_LANE2_SOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane2_sof_1																lane2_sof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane2_sof_1	SOF synchronization code of frame 1 for lane 2 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane2_sof_0	SOF synchronization code of frame 0 for lane 2 in LVDS or HiSPI mode In WDR mode, the value is the SOF of frame 0. In linear mode, the value is the SOF.

LINK0_LANE2_EOF_01

LINK0_LANE2_EOF_01 is an EOF synchronization code configuration register for lane 2 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x10C8	LINK0_LANE2_EOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane2_eof_1																lane2_eof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane2_eof_1	EOF synchronization code of frame 1 for lane 2 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane2_eof_0	EOF synchronization code of frame 0 for lane 2 in LVDS or HiSPI mode In WDR mode, the value is the EOF of frame 0. In linear mode, the value is the EOF.

LINK0_LANE2_SOL_01

LINK0_LANE2_SOL_01 is an SOL synchronization code configuration register for lane 2 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x10D0	LINK0_LANE2_SOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane2_sol_1																lane2_sol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane2_sol_1	SOL synchronization code of frame 1 for lane 2 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane2_sol_0	SOL synchronization code of frame 0 for lane 2 in LVDS or HiSPI mode In WDR mode, the value is the SOL of frame 0. In linear mode, the value is the SOL.

LINK0_LANE2_EOL_01

LINK0_LANE2_EOL_01 is an EOL synchronization code configuration register for lane 2 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x10D8	LINK0_LANE2_EOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane2_eol_1																lane2_eol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane2_eol_1	EOL synchronization code of frame 1 for lane 2 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane2_eol_0	EOL synchronization code of frame 0 for lane 2 in LVDS or HiSPI mode In WDR mode, the value is the EOL of frame 0. In linear mode, the value is the EOL.

LINK0_LANE3_SOF_01

LINK0_LANE3_SOF_01 is an SOF synchronization code configuration register for lane 3 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x10E0	LINK0_LANE3_SOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane3_sof_1																lane3_sof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane3_sof_1	SOF synchronization code of frame 1 for lane 3 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane3_sof_0	SOF synchronization code of frame 0 for lane 3 in LVDS or HiSPI mode In WDR mode, the value is the SOF of frame 0. In linear mode, the value is the SOF.

LINK0_LANE3_EOF_01

LINK0_LANE3_EOF_01 is an EOF synchronization code configuration register for lane 3 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x10E8	LINK0_LANE3_EOF_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane3_eof_1																lane3_eof_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane3_eof_1	EOF synchronization code of frame 1 for lane 3 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane3_eof_0	EOF synchronization code of frame 0 for lane 3 in LVDS or HiSPI mode In WDR mode, the value is the EOF of frame 0. In linear mode, the value is the EOF.

LINK0_LANE3_SOL_01

LINK0_LANE3_SOL_01 is an SOL synchronization code configuration register for lane 3 in LVDS or HiSPI mode.

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Offset Address	Register Name	Total Reset Value
0x10F0	LINK0_LANE3_SOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane3_sol_1																lane3_sol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane3_sol_1	SOL synchronization code of frame 1 for lane 3 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane3_sol_0	SOL synchronization code of frame 0 for lane 3 in LVDS or HiSPI mode In WDR mode, the value is the SOL of frame 0. In linear mode, the value is the SOL.

LINK0_LANE3_EOL_01

LINK0_LANE3_EOL_01 is an EOL synchronization code configuration register for lane 3 in LVDS or HiSPI mode.

Offset Address	Register Name	Total Reset Value
0x10F8	LINK0_LANE3_EOL_01	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lane3_eol_1																lane3_eol_0															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	lane3_eol_1	EOL synchronization code of frame 1 for lane 3 in LVDS or HiSPI mode The field is configurable only in WDR mode, and the configuration is invalid in linear mode.
[15:0]	RW	lane3_eol_0	EOL synchronization code of frame 0 for lane 3 in LVDS or HiSPI mode In WDR mode, the value is the EOL of frame 0. In linear mode, the value is the EOL.

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10 ISP

10.1 Introduction

The image signal processor (ISP) module of Hi3518E V20X/Hi3516C V200 supports standard sensor picture data processing. The ISP module provides basic functions such as automatic white balance (AWB), automatic exposure (AE), demosaic, defect pixel correction (DPC), and lens shading correction (LSC) as well as advanced functions such as sensor built-in wide dynamic range (WDR), dynamic range compression (DRC), and denoising. The picture processing functions supported by the ISP module of Hi3518E V20X/Hi3516C V200 are as follows:

- Black level correction (BLC)
- Static and dynamic DPC and defect pixel cluster correction
- Bayer denoising
- Fixed pattern noise (FPN) removal
- Advanced demosaic
- Gamma correction
- DRC
- Sensor built-in WDR
- AWB
- AE
- Automatic focus (AF)
- 3A (AE, AF, AWB) statistics output
- Lens shading correction
- Lens distortion correction (not supported by Hi3518E V201)
- Picture sharpening
- Digital image stabilization (DIS)
- Automatic anti-fog
- Color management and enhancement

The processing capability of the ISP module is as follows:

- Maximum 14-bit bayer data input
- Maximum picture resolution of 2048 x 1208
- Minimum picture resolution of 480 x 240

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- Maximum frame rate of 30 fps for the 1920 x 1080 resolution
- Minimum horizontal blanking region of 64 pixels
- Minimum vertical blanking region of 24 lines

10.2 Overview

Functional Block Diagram

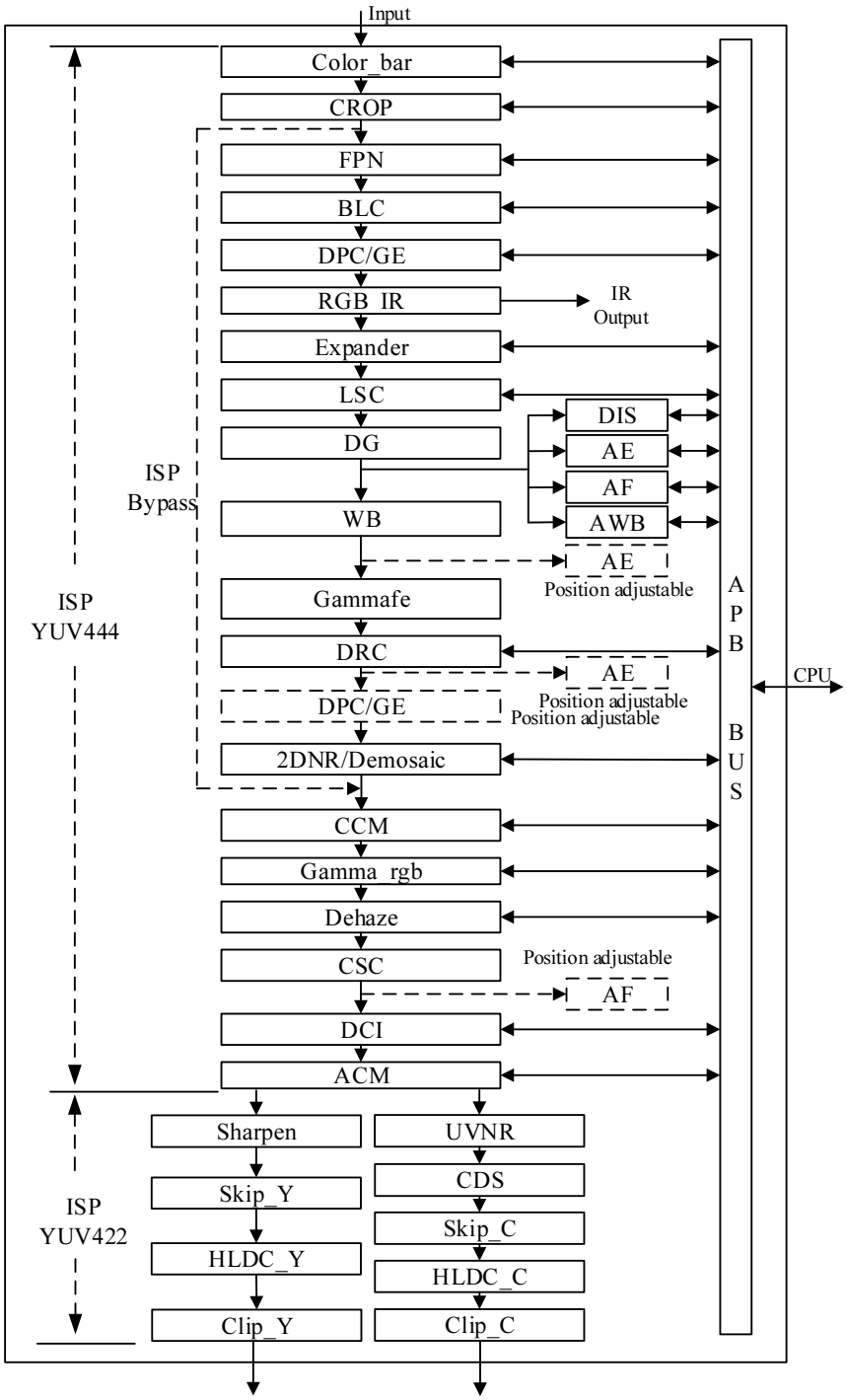
The base address of ISP registers is 0x20580000. The actual address of an ISP register is the base address plus the corresponding offset address. Figure 10-1 shows the functional block diagram of the ISP module. The YUV444 in this document indicates the set of the bayer field, RGB field, and YUV444 field.

NOTE

In this document, *U**.* and *S**.* represent the unsigned number and the signed number respectively. For example, U8.8 indicates that the data is an unsigned number consisting of an 8-bit integral part and an 8-bit decimal part. S8.8 indicates that the data is a signed number consisting of an 8-bit integral part (including a sign bit) and an 8-bit decimal part.

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Figure 10-1 Functional block diagram of the ISP module



Operating Mode

The ISP module supports the following operating modes:

- Maximum 14-bit raw RGB bayer data input

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When the input data width is less than 14 bits, the upper bits need to be aligned and the lower bits need to be stuffed with 0s. In this mode, any sequence of the R, Gr, Gb, and B components is supported, and the sequence can be changed by configuring ISP_YUV444_CTRL_F and ISP_YUV422_CTRL_F.

- Infrared (IR) mode

  In this mode, ISP_SKIP_C_CFG in the skip module needs to be set to **0x00000000** so that the C (U and V) component is discarded and the ISP module outputs only the Y component picture data, and the UVNR enable register ISP_UVNR_CFG needs to be set to **0**.

- External ISP mode

  In this mode, the ISP module supports gamma, dehaze, color space conversion (CSC), AF, automatic color management (ACM), dynamic contrast improvement (DCI), sharpening, UVNR (NR, CDS), and LDC_H processing.

  Table 10-1 describes the key parameters in external ISP mode.

Table 10-1 Key parameters in external ISP mode

Parameter	Description
BUF_MODE (0x20580070)	Register in the video capture (VICAP) module, indicating the line_buf input data mode 0: raw data 1: YUV422
ISP_ISPBYPASS	ISP bypass register 0: The embedded ISP is used. 1: ISP bypass. The modules from the FPN module to demosaic module are bypassed.

In external ISP mode, ISP_ISPBYPASS needs to be set to **0x1** to mask the raw field module (from the FPN module to the 2DNR/demosaic module), and BUF_MODE in the VICAP module needs to be set to **0x1** so that the VICAP line_buf input data is in YUV422 format.

- Adjustable module position

  The positions of the AE, AF, and DCG (DPC/GE) modules are adjustable.

  Table 10-2 describes key module position adjustment parameters.

Table 10-2 Key module position adjustment parameters

ISP_MODULE_POS	Description
af_sel	AF position adjustment register 0: The AF module is after the digital gain (DG) module. 1: The AF module is in the Y field after the CSC module.

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ISP_MODULE_POS	Description
ae_sel	AE position adjustment register 00: The AE module is after the DG module. 01: The AE module is after the white balance (WB) module. 10: The AE module is after the DRC module.
dcg_sel	DPC/GE position adjustment register 0: The DPC/GE module is after the BLC module. 1: The DPC/GE module is after the DRC module.

Memory Read and Write Solutions

Table 10-3 describes the memory address mapping of each module.

Table 10-3 Memory address mapping of each module

Range of the Memory Offset Address	Module to Which the Memory Belongs
0x22088–0x220A4	AE
0x22188–0x2218C	AWB
0x22288–0x2228C	AF
0x24880–0x2489C	2DNR (NR_LSC)
0x25080–0x2508C	DPC
0x25188–0x2518C	GE
0x25280–0x2528C	FPN
0x25880–0x2589C	LSC
0x26280–0x262AC	DRC
0x27080–0x270BC	Dehaze
0x27188–0x2719C	DIS
0x27380–0x27384	Gamma
0x27480–0x2749C	Gamma FE
0x27580–0x2758C	Compander

The ISP memory is indirectly read or written. Within the range of the memory address segments, the read data register (RDATA), read address register (RADDR), write data register (WDATA), and write address register (WADDR) are provided.

To read the memory, perform the following steps:

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Step 1 Write the start address of the data to be read in the memory to the read address register (RADDR).

Step 2 Read the read data register (RDATA) continuously. The address automatically increases by 1 in the logic each time the register is read.

----End

To write to the memory, perform the following steps:

Step 1 Write the start address of data in the memory to the write address register (WADDR).

Step 2 Write data to the write data register (WADDR) continuously. The address automatically increases by 1 in the logic each time the register is written.

----End

10.3 Interrupt System

Function Description

The ISP module has 13 hardware interrupt events. Table 10-4 describes the 13 interrupt events and their corresponding positions in the indicator register with the address of 0x205A00F0.

Table 10-4 Interrupt indicator register

Offset Address	Event	Bit	Description
0x200F0	reserved	31:15	Reserved
	update_cfg_yuv444	14	Status of the ISP YUV444 register update interrupt
	cfg_loss_yuv444	13	Status of the ISP YUV444 register configuration loss interrupt
	update_cfg_yuv422	12	Status of the ISP YUV422 register update interrupt
	field_throw_yuv422	11	Status of the ISP YUV422 field/frame loss interrupt
	int_disstat	10	DIS current frame statistics completion interrupt
	int_fstart_yuv422	9	ISP YUV422 frame start interrupt indicator
	reserved	8	Reserved
	reserved	7	Reserved
	int_afstat	6	AF current frame statistics transmit (TX) completion interrupt

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Offset Address	Event	Bit	Description
	int_aestat	5	AE current frame statistics TX completion interrupt
	int_awbstat	4	AWB current frame statistics TX completion interrupt
	int_user_yuv444	3	Configurable trigger position interrupt. The trigger position is specified by ISP_INT_TIMING_CTRL with the address of 0x205A0090. The interrupt can be triggered at any position from the frame start position to the frame end position with clock cycle precision.
	int_fstart2_yuv444	2	ISP YUV444 frame start interrupt indicator 2
	acm_para_finish	1	Status of the ACM lookup table load completion interrupt
	int_fstart_yuv444	0	ISP YUV444 frame start interrupt indicator

NOTE

Writing 1 clears the corresponding bit. If the bit value is 0, no interrupt is generated. If the bit value is 1, an interrupt is generated.

The meanings of the preceding events are described as follows:

- int_fstart_yuv444 and int_fstart2_yuv444 are the frame start interrupts of ISP YUV444 and int_fstart_yuv422 is the frame start interrupt of ISP YUV422, indicating the start of a frame. int_fstart_yuv444 and int_fstart2_yuv444 are the same interrupt.
- acm_para_finish indicates the status of the ACM lookup table load (from the DDR) completion interrupt.
- int_user_yuv444 is a user configurable interrupt. The trigger position is specified by ISP_INT_TIMING_CTRL with the address of 0x205A0090. The interrupt can be triggered at any position from the frame start position to the frame end position with clock cycle precision.
- update_cfg_yuv444 indicates the status of the ISP YUV444 register update interrupt. When this bit is pulled up, the register configured in the previous frame takes effect in the current frame.
- cfg_loss_yuv444 indicates the status of the ISP YUV444 register configuration loss interrupt. When this bit is pulled up, the register configured in the previous frame does not take effect in the current frame and exception handling is required for software.
- update_cfg_yuv422 indicates the status of the ISP YUV422 register update interrupt. When this bit is pulled up, the register configured in the previous frame takes effect in the current frame.
- field_throw_yuv422 indicates the status of the ISP YUV422 field/frame loss interrupt. When this bit is pulled up, the register configured in the previous frame does not take effect in the current frame, which results in picture data loss of the current frame. In this case, exception handling is required for software.

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Interrupt Timing

The positions of many interrupts are determined by the ISP module switch and register configuration. Figure 10-2 shows the position of each interrupt. Note that the interrupt position is arranged based on the time sequence in a frame. Table 10-5 describes the mapping between the interrupt ID and the interrupt event shown in Figure 10-2, and the number of lines from each interrupt to the rising edge of FSTART_I.

Figure 10-2 Interrupt timing

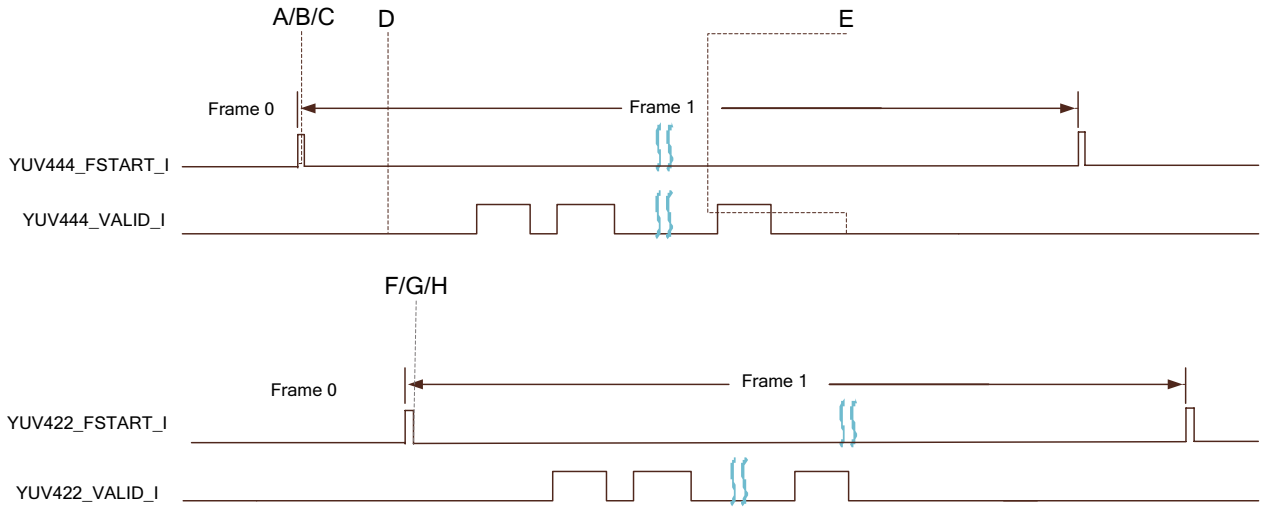


Table 10-5 Mapping between the interrupt ID and the interrupt event

No.	Interrupt Event	Number of Lines from the Interrupt to the Rising Edge of FSTART_I	Description
A	int_fstart_yuv444 int_fstart2_yuv444	0	-
B	update_cfg_yuv444	0	This event indicates whether the configuration in frame 0 takes effect in frame 1.
C	cfg_loss_yuv444	0	This event indicates whether the configuration in frame 0 is lost in frame 1.
D	acm_para_finish	N/A	-
E	int_user_yuv444	N/A	The trigger position is specified by ISP_INT_TIMING_CTRL with the address of 0x205A0090. The interrupt can be triggered at any position from the frame start position to the frame end position with clock cycle

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No.	Interrupt Event	Number of Lines from the Interrupt to the Rising Edge of FSTART_I	Description
			precision.
F	int_fstart_yuv422	0	-
G	update_cfg_yuv422	0	This event indicates whether the configuration in frame 0 takes effect in frame 1.
H	field_throw_yuv422	0	This event indicates whether the configuration in frame 0 is lost in frame 1.

10.4 Module Functions

Color_bar (Offset Address Segment: 0x28100−0x281FF)

The color_bar module supports the following five picture types:

- Pure color picture configured by ISP_COLORBAR_RBACKGND[rbackgnd], ISP_COLORBAR_GBACKGND[gbackgnd], and ISP_COLORBAR_BBACKGND[bbackgnd]

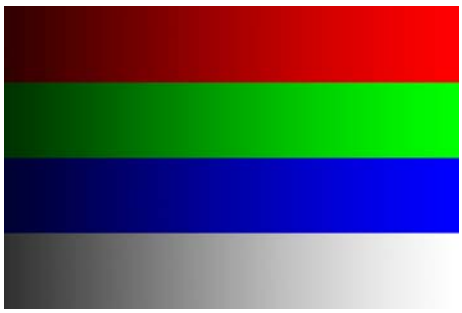
Figure 10-3 Pure color picture



- Picture consisting of four horizontal color stripes. The luminance from left to right is variable and the start value and increment are configurable.

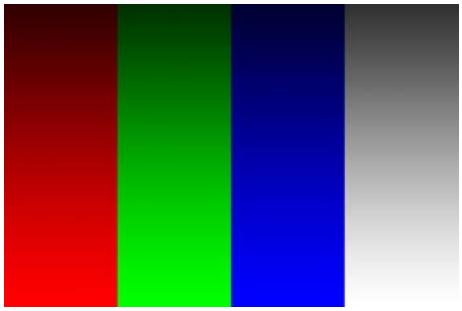
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Figure 10-4 Picture consisting of four horizontal color stripes



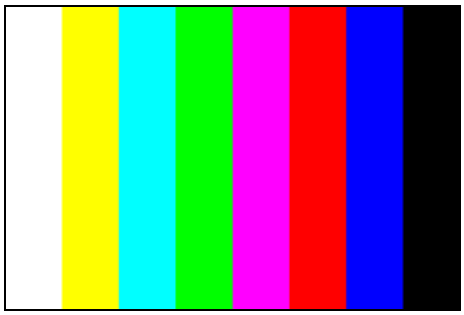
- Picture consisting of four vertical color stripes. The luminance from top to bottom is variable and the start value and increment are configurable.

Figure 10-5 Picture consisting of four vertical color stripes



- Picture consisting of eight vertical color stripes. The luminance is configured by ISP_COLORBAR_RBACKGND[rbackgnd], ISP_COLORBAR_GBACKGND[gbackgnd], and ISP_COLORBAR_BBACKGND[bbackgnd].

Figure 10-6 Picture consisting of eight vertical color stripes



- Picture with a rectangular pure color target area on the pure color background. The background color is configured by ISP_COLORBAR_RBACKGND[rbackgnd], ISP_COLORBAR_GBACKGND[gbackgnd], and ISP_COLORBAR_BBACKGND[bbackgnd]. The color of the target area is configured by ISP_COLORBAR_RFOREGND[rforegnd],

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ISP_COLORBAR_GFOREGND[gforegnd], and
ISP_COLORBAR_BFOREGND[bforegnd].

Figure 10-7 Picture with a rectangular pure color target area on the pure color background

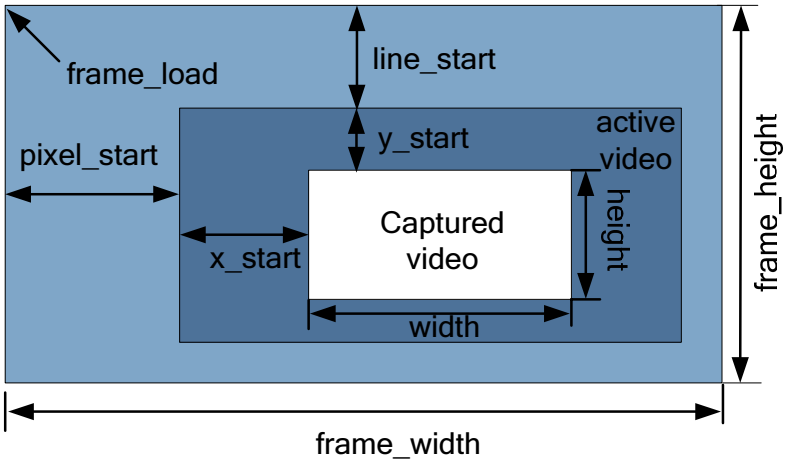


Crop (Offset Address Segment: 0x31100−0x31114)

The crop module crops the input picture. The actual displayed view area always falls within the valid video range. That is, compared with the valid video area, the actual displayed view area shrinks, which avoids the edge effect.

As shown in Figure 10-8, the crop module crops the valid picture area by configuring ISP_CROP0_START[y_start], ISP_CROP0_START[x_start], ISP_CROP0_SIZE[height], and ISP_CROP0_SIZE[width].

Figure 10-8 Relationship between the valid picture area and the horizontal/vertical blanking region



FPN (Offset Address Segment: 0x25200−0x252FF)

The FPN module corrects the sensor input picture based on the calibrated black frame or black line to remove the FPNs of the sensor.

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The FPN module supports calibration and correction in frame mode and line mode. This module needs to be enabled if the FPNs of the connected sensor are obvious, and does not need to be enabled if the the FPNs of the connected sensor are not obvious.

BLC (Offset Address Segment: 0x21000−0x210FF)

The BLC module provides the sensor-related BLC function, and configures the offsets of the R, Gr, Gb, and B components.

DPC (Offset Address Segment: 0x25000−0x250FF)

The DPC module detects and corrects static and dynamic defect pixels.

You can detect the static defect pixels of the sensor by using software and hardware, and store the coordinates of all the defect pixels in the external memory. Then the ISP corrects the defect pixels based on the coordinates. This module supports the detection and correction of at most 2048 static defect pixels.

This module automatically detects and corrects dynamic defect pixels in real time based on the configured threshold, and supports defect pixel cluster correction within 2 x 2 pixels.

The static defect pixel correction can be replaced with the dynamic defect pixel correction. However, to avoid definition degradation caused by misjudgment during dynamic detection, you are advised to first use the static defect pixel correction function and then set harsh dynamic defect pixel detection and correction conditions to implement defect pixel correction by combining static and dynamic defect pixel correction.

GE (Offset Address Segment: 0x25100−0x251FF)

The GE module equalizes the Gb and Gr channels when imbalance occurs and improves the picture quality in some scenarios. The GE statistics are zoned Gr/Gb average value statistics.

This module supports a maximum of 17 x 15 blocks and a minimum of 1 x 1 block. Each block can output the average values of the Gr and Gb components (average R/Gr/Gb/B). Table 10-6 describes the mapping between the address and read data of bloc statistics. *m* indicates the number of horizontal blocks and *n* indicates the number of vertical blocks.

Table 10-6 GE zone statistics

Memory Address	Zone	Memory Read Data		
		[31:28]	[27:14]	[13:0]
0	0	0	Average Gr	Average Gb
1	1	0	Average Gr	Average Gb
2	2	0	Average Gr	Average Gb
3	3	0	Average Gr	Average Gb
…				
m x n − 3	m x n − 3	0	Average Gr	Average Gb
m x n − 2	m x n − 2	0	Average Gr	Average Gb

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Memory Address	Zone	Memory Read Data		
		[31:28]	[27:14]	[13:0]
m x n − 1	m x n − 1	0	Average Gr	Average Gb

RGB_IR (Offset Address Segment: 0x23100−0x231FF)

The RGB_IR module is used to adapt the RGBIR sensor, convert the RGBIR pattern into the RGGB pattern, and restore the color to normal. Based on the position of the IR point and the G value in RGBIR pattern, this module obtains the G value corresponding to the IR position through direction decision and interpolation. According to the spectral characteristics, the IR component is removed from the RGB component to obtain the RGGB value.

Expander (Offset Address Segment: 0x27500−0x275FF)

The expander module is enabled only in sensor built-in WDR mode. This module decompresses the sensor built-in compressed non-linear data into linear data. The used lookup table (LUT) contains 129 points with 15-bit precision. The points between two nodes can be obtained through interpolation. The generated LUT depends on the corresponding sensor model.

LSC (Offset Address Segment: 0x25800−0x258FF)

The LSC module implements lens shading correction. Because of the optical characteristics of the lens, the luminous intensity of the edge area is lower than that of the central area in a sensor picture. Therefore, gain compensation based on pixel position is required.

This module provides the gains of four components (R, Gr, Gb, and B), and each gain is represented by a 13-bit unsigned number (3-bit integral part and 10-bit decimal part). The picture is divided into 16 x 16 blocks that are symmetric in the layout but differ in the block size. The closer the block is to the picture center, the larger the block size; the closer the block is to the four edges, the smaller the block size. The gains of the window vertex are provided. Other values are obtained through interpolation. The gain obtained after interpolation is multiplied by the corresponding pixel point value, and the compensated picture data is output.

DG (Offset Address Segment: 0x21400−0x214FF)

The DG module provides the digital gain, and configures the offsets of the R, Gr, Gb, and B components (U8.8 precision).

AE (Offset Address Segment: 0x22000−0x220FF)

The AE module collects AE statistics. Software adjusts the sensor based on statistics to implement AE. The AE statistics includes the global zoned weighted R/Gr/Gb/B average value statistics, global weighted R/Gr/Gb/B average value, and 256-segment histogram statistics.

- AE zoned statistics

  The AE module supports a maximum of 17 x 15 blocks and a minimum of 1 x 1 block. Each block can output the average values of the R, Gr, Gb, and B components (average R/Gr/Gb/B). Table 10-7 describes the mapping between the address and read data of

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block statistics. *m* indicates the number of horizontal blocks and *n* indicates the number of vertical blocks.

Table 10-7 AE zone statistics

Memory Address	Zone	Memory Read Data			
		[31:24]	[23:16]	[15:8]	[7:0]
0	0	Average R	Average Gr	Average Gb	Average B
1	1	Average R	Average Gr	Average Gb	Average B
2	2	Average R	Average Gr	Average Gb	Average B
3	3	Average R	Average Gr	Average Gb	Average B
…					
m x n − 3	m x n − 3	Average R	Average Gr	Average Gb	Average B
m x n − 2	m x n − 2	Average R	Average Gr	Average Gb	Average B
m x n − 1	m x n − 1	Average R	Average Gr	Average Gb	Average B

- AE global statistics

  The AE global statistics are basically the same as the AE zoned statistics. The AE module provides four types of global statistics. For details, see the ISP_AE_TOTAL_R_AVER, ISP_AE_TOTAL_GR_AVER, ISP_AE_TOTAL_GB_AVER, and ISP_AE_TOTAL_B_AVER registers. The physical meanings of the four types of global statistics are the same as those of the zoned statistics.

- Weighted AE histogram statistics

  The AE module provides the 256-segment histogram statistics. Table 10-8 describes the mapping between the address and read data of the histogram statistics for zone 0–255.

Table 10-8 AE histogram statistics

Memory Address	Zone	Memory Read Data	
		[31:28]	[27:0]
0	0	0	Number of pixels whose value is 0
1	1	0	Number of pixels whose value is 1
2	2	0	Number of pixels whose value is 2
3	3	0	Number of pixels whose value is 3
…			
254	254	0	Number of pixels whose value is 254
255	255	0	Number of pixels whose value is 255

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AF (Offset Address Segment: 0x22200−0x222FF)

The AF module collects statistics on picture definition evaluation information and implements AF. The number of picture blocks is configurable. This module supports a maximum of 17 x 15 blocks. The minimum block size is 32 x 32 and the maximum block size is 511 x 511. The definition evaluation information about each zone is provided.

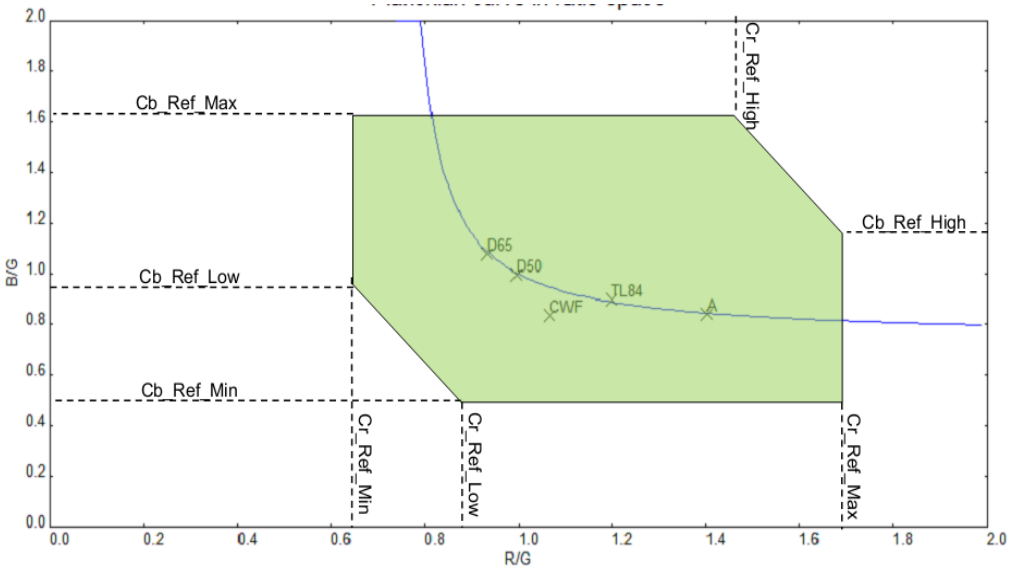
AWB (Offset Address Segment: 0x22100−0x221FF)

The AWB statistics includes the global statistics and zoned statistics.

- Global statistics: average values of the R, G, and B components for an entire picture, number of valid statistical points, and number of statistical points that exceed the limit
- Zoned statistics: average values of the R, G, and B components for each block (a maximum of 17 x 15 blocks are supported), number of valid statistical points, and number of statistical points that exceed the limit

  The valid AWB statistical points must meet the configured RGB upper limit and lower limit requirements described in ISP_AWB_THD_MIN and ISP_AWB_THD_MAX. In addition, the pixel point color must fall within the range of the configured color space of the valid pixel. As shown in Figure 10-9, the green area indicates the color range of the valid pixel.

Figure 10-9 Color space of valid AWB pixel



The valid pixel area of AWB statistics is a hexagon. The left, right, top, and bottom boundaries are determined by **cr_ref_min**, **cr_ref_max**, **cb_ref_min**, and **cb_ref_max**. The four parameters indicate the maximum and minimum values of R/G and B/G. The two hypotenuses of the valid area are determined by configuring the slope and intercept. ISP_AWB_TOP_K and ISP_AWB_TOP_B determine the hypotenuse on the upper right part, and ISP_AWB_BOT_K and ISP_AWB_BOT_B determine the hypotenuse on the lower left part.

- AWB zoned statistics

  The AWB module supports a maximum of 17 x 15 blocks. Each block can output the average values of the R, G, and B components (average R/G/B). The width of the pixel

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points involved in statistics is 12 bits. If the width of the input picture data is greater than 12 bits, the pixel data needs to be truncated based on the size configured in ISP_AWB_BITMOVE. There are three types of statistics on the statistical points of a block: number of valid statistical points (Count All), number of statistical points that is above the upper threshold (Count Max), and number of statistical points that is below the lower threshold (Count Min). The preceding statistics are normalized to 16 bits based on the picture size. Table 10-9 describes the mapping between the address and read data of block statistics.

Table 10-9 AWB zoned statistics

Memory Address	Zone	Memory Read Data	
		[31:16]	[15:0]
0	0	Average G	Average R
1	0	Count All	Average B
2	0	Count Max	Count Min
3	1	Average G	Average R
…			
3 x *m* x *n* − 3	*m* x *n*	Average G	Average R
3 x *m* x *n* − 2	*m* x *n*	Count All	Average B
3 x *m* x *n* − 1	*m* x *n*	Count Max	Count Min
0	0	Average G	Average R

- AWB global statistics

  The AWB global statistics are basically the same as the AWB zoned statistics. The AWB module provides six types of global statistics. For details, see the ISP_AWB_AVG_R, ISP_AWB_AVG_G, ISP_AWB_AVG_B, ISP_AWB_CNT_ALL, ISP_AWB_CNT_MIN, and ISP_AWB_CNT_MAX registers. The physical meanings of the six types of global statistics are the same as those of the zoned statistics.

DIS (Offset Address Segment: 0x27100−0x271FF)

The DIS module compares the two frames, finds the difference between them, and calculates the offset between frames. This module outputs the horizontal offset Offset_X and vertical offset Offset_Y. The statistics of nine windows are used for horizontal and vertical offset calculation. Each window outputs three result offsets, detail delta, and sum of absolute difference (SAD). The offset (in pixel) ranges from −64 to +64. Software calculates the Offset_X and Offset_Y of the entire picture based on the statistics of nine windows. The back-end module controls the position of the cropping window and crops the picture based on Offset_X and Offset_Y to eliminate picture jitter.

WB (Offset Address Segment: 0x21100−0x211FF)

The WB module provides the white balance function, and configures the gains of the R, Gr, Gb, and B components (U4.8 precision).

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Gamma FE (Offset Address Segment: 0x27400−0x274FF)

The gamma FE module is enabled only in WDR mode and works with the DRC module. This module implements gamma mapping and compresses composite data in WDR mode. The data on the gamma FE curve is represented by two LUTs. LUT0 contains 33 points with 16-bit precision and LUT1 contains 257 points with 16-bit precision.

DRC (Offset Address Segment: 0x26200−0x262FF)

The DRC module adjusts the display dynamic range of the picture so that the picture display effect on the display device is consistent with the picture observed by human eyes.

2DNR (Offset Address Segment: 0x24000−0x240FF & 0x24800−0x248FF)

The 2DNR module implements picture denoising during bayer-to-RGB picture conversion to retain details while removing noises. This module can remove noises for the sensor based on the noise model provided by the user.

The denoising strength is adaptively adjusted based on the picture detail, which is determined by **ISP_2DNR_LIMIT**. A smaller parameter value indicates greater NR strength. In addition, because the LSC function enhances noises, the 2NR module is associated with the LSC module. When the 2NR module is used, the NR_LSC module needs to be enabled and the corresponding LSC parameters need to be configured based on those in the LSC module.

Demosaic (Offset Address Segment: 0x23000−0x230FF)

The demosaic module converts the raw picture in bayer format into the RGB picture.

This module analyzes the characteristics of the internal picture edge, uses pixel relevance interpolation, and effectively suppresses the anti-false color while ensuring high resolution and definition.

CCM (Offset Address Segment: 0x21200−0x212FF)

The color correction matrix (CCM) module implements linear correction of the color space by using the standard 3 x 3 matrix and vector offset.

The firmware dynamically calculates the CCM coefficients of the current picture based on the several pre-calculated groups of CCM coefficients and picture color temperature. The firmware can also use the CCM to dynamically adjust the saturation based on the luminance of the current picture.

$$\begin{pmatrix} R \\ G \\ B \end{pmatrix} = \begin{pmatrix} coef00 & coef01 & coef02 \\ coef10 & coef11 & coef12 \\ coef20 & coef21 & coef22 \end{pmatrix} \times \begin{pmatrix} R \\ G \\ B \end{pmatrix} + \begin{pmatrix} out_dc0 \\ out_dc1 \\ out_dc2 \end{pmatrix}$$

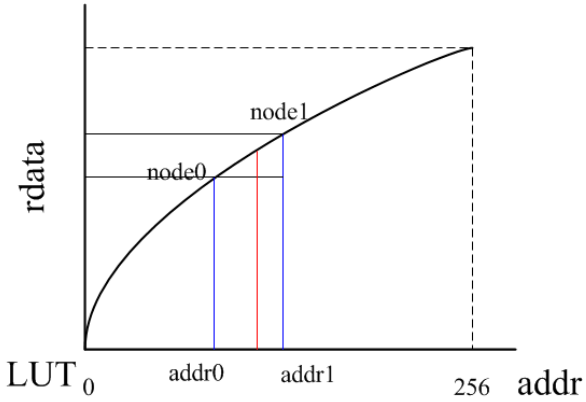
In the preceding formula, R, G, and B indicate the input data, and coef is a 3 x 3 configurable matrix coefficient. The coefficients are 15-bit numbers in S5.10 format.

Gamma_RGB (Offset Address Segment: 0x27300−0x273FF)

The gamma-RGB module applies to the three color channels (R, G, and B) and outputs the gamma-RGB adjustment result. This module adjusts the luminance based on the gamma curve. The gamma curve is typically configured to match BT.709 or sRGB. The three color channels (R, G, and B) are adjusted according to the gamma curve.

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Figure 10-10 Gamma curve



The gamma curve consists of 257 nodes (node 0 to node 256). The curve value corresponding to each node is a 12-bit unsigned number. The points between two nodes can be obtained through interpolation. The curve value of node 0 (gamma[0]) is 0 and the curve value of node 256 (gamma[256]) is 0xFFF.

Dehaze (Offset Address Segment: 0x27000−0x270FF)

The dehaze module provides the powerful zoned anti-fog function and improves video contrast and definition in the haze scenario. The picture can be divided into multiple regions by configuring ISP_DEHAZE_BLK_NUM. A maximum of 16 x 16 regions are supported.

This module analyzes the picture characteristics of each region, obtains the contrast specifications of each region, and implements pixel enhancement in each region. The enhancement strength of a pixel depends on the contrast specifications of the region this pixel belongs to and those of the neighboring regions.

CSC (Offset Address Segment: 0x21300−0x213FF)

The CSC module converts the {R, G, B} input into {Y, U, V} by using the standard 3 x 3 matrix and vector offset, and implements saturation adjustment and color correction by dynamically adjusting the matrix coefficient.

$$\begin{pmatrix} Y \\ U \\ V \end{pmatrix} = \begin{pmatrix} coef00 & coef01 & coef02 \\ coef10 & coef11 & coef12 \\ coef20 & coef21 & coef22 \end{pmatrix} \times \begin{pmatrix} R \\ G \\ B \end{pmatrix} + \begin{pmatrix} out_dc0 \\ out_dc1 \\ out_dc2 \end{pmatrix}$$

The parameters can be changed based on format conversion requirements.

DCI (Offset Address Segment: 0x31C00−0x31DFF)

The DCI module dynamically improves the picture contrast, and automatically adjusts picture luminance and contrast based on the picture luminance statistics to resolve issues (for example, the picture is too dark or too bright, and the contrast is too high or too low) so that excellent display effect can be achieved on the display device.

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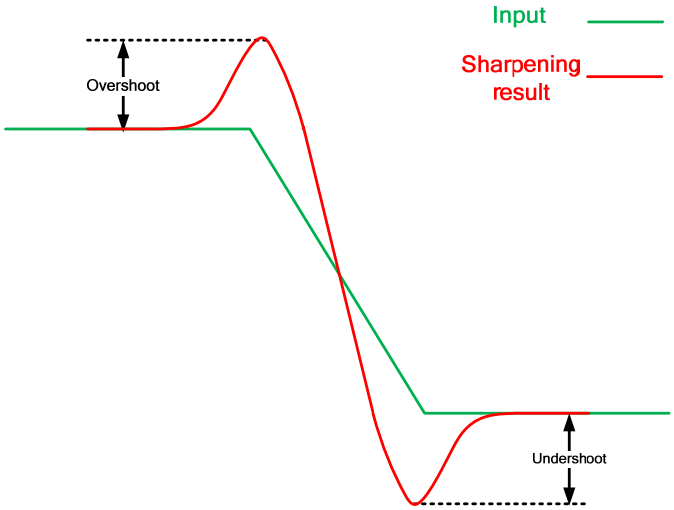
ACM (Offset Address Segment: 0x31400−0x31420)

The ACM implements automatic color management, corrects hue offset during video transfer, compensates for saturation loss during video transfer and the performance difference between color gamuts, and enhances specific colors such as the complexion, green, and blue based on user preference and style.

Sharpen (Offset Address Segment: 0x24200−0x242FF)

The sharpen module implements picture sharpening to improve picture definition. The sharpening strength can be adjusted by configuring the control parameters **edge_amt** and **sharp_amt**. Note that noises may be amplified if the sharpening strength is too high. Software adjusts parameter configurations based on the ISO value to achieve a balance between the definition and noise suppression while improving the visual effect of the picture.

Figure 10-11 Overshoot and undershoot occurrence during sharpening



In addition, white borders and black borders may appear due to the large positive and negative edge amplification caused by sharpening, as shown in Figure 10-11. The white borders and black borders can be suppressed by adjusting the overshoot and undershoot control parameters.

UVNR (Offset Address Segment: 0x21600−0x216FF)

The UVNR module is used to remove the chrominance noises of the picture. This module implements denoising only on the chrominance components of the picture and has no effect on the picture luminance. In scenarios with low ISO values, median chrominance filtering is implemented. In scenarios with low ISO values, dual-edge chrominance filtering is implemented. Software selects the filtering policy based on the ISO value, which effectively removes small chrominance noises and weakens clumpy low-frequency chrominance noises.

CDS (Offset Address Segment: 0x21600−0x216FF)

The CDS module converts the YUV444 into YUV422 or YUV420, and implements multi-order horizontal chrominance filtering. When the YUV420 output is required, this module implements average down sampling in the vertical direction of the chrominance. The visual

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loss caused by picture conversion can be minimized by setting the filtering parameters to appropriate values.

HLDC (Offset Address Segment: 0x32300−0x32324)

The HLDC module corrects the distortion caused by the lens.

This module implements correction by using the following parameters:

- **ratio** indicates the distortion degree. Its value range is [0, 1000]. It is a percentage number with a decimal bit. For example, if **ratio** is set to **100**, its physical meaning is 10.0%.
- **ldc_norcoef** is used for normalizing r. Theoretically the value range of r is [0.0, 1.0]. However, the r of the actual picture is measured by pixel. For example, the resolution of the D1 picture is 720 x 576, and the resolution of the 720p picture is 1280 x 720. As a result, the r value calculated by using the picture width and height needs to be normalized. The floating point number cannot be processed by logic. Therefore, 12-bit decimal precision is used when a floating point number is converted into a fixed point number for the maximum distance algorithm (1.0). That is, the value of r is changed from the actual picture size to 4096 and then involved in subsequent LDC calculation. **ldc_norcoef** is the coefficient for completing the preceding calculation.
- Scale. The range of the picture after LDC is only part of the view of the unprocessed picture. To obtain a picture with a larger view, the horizontal scaling coefficient and vertical scaling coefficient are required for implementing horizontal and vertical scaling during LDC.

10.5 Register Summary

Table 10-10 describes ISP registers.

Table 10-10 Summary of ISP registers (base address: 0x2058_0000)

Offset Address	Register	Description	Page
0x20080	ISP_VERSION	ISP version register	10-44
0x20088	ISP_MODULE_POS	ISP internal module position select register	10-44
0x2008C	ISP_CH_SWITCH	ISP input switch register	10-45
0x20090	ISP_INT_TIMING_CTRL	ISP adjustable interrupt trigger time configuration register	10-46
0x20094	ISP_ISPBYPASS	ISP bypass register	10-47
0x200F0	ISP_INT	ISP interrupt indicator register	10-47
0x200F8	ISP_INT_MASK	ISP interrupt mask register	10-49
0x201E0	ISP_YUV444_CTRL_F	ISP general update control register	10-51
0x201E4	ISP_YUV444_CTRL_I	ISP immediate update control register	10-51

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Offset Address	Register	Description	Page
0x201E8	ISP_YUV444_TIMING_CFG	Output timing configuration register	10-52
0x201EC	ISP_YUV444_REG_UPDATE	Register update register	10-52
0x201F0	ISP_YUV444_SIZE	Picture width/height register	10-53
0x202E0	ISP_YUV422_CTRL_F	ISP general update control register	10-53
0x202E4	ISP_YUV422_CTRL_I	ISP immediate update control register	10-54
0x202E8	ISP_YUV422_TIMING_CFG	Output timing configuration register	10-54
0x202EC	ISP_YUV422_REG_UPDATE	Register update register	10-55
0x202F0	ISP_YUV422_SIZE	Picture width/height register	10-55
0x21000	ISP_BLC_CFG	BLC control register	10-56
0x21010	ISP_BLC_OFFSET1	BLC offset register 1	10-56
0x21014	ISP_BLC_OFFSET2	BLC offset register 2	10-57
0x21100	ISP_WB_CFG	WB control register	10-57
0x21110	ISP_WB_GAIN1	WB gain (R & Gr) register	10-58
0x21114	ISP_WB_GAIN2	WB gain (B & Gb) register	10-58
0x21200	ISP_CCM_CFG	CCM enable register	10-59
0x21210	ISP_CCM_COEF0	CCM coefficient register 0	10-59
0x21214	ISP_CCM_COEF1	CCM coefficient register 1	10-60
0x21218	ISP_CCM_COEF2	CCM coefficient register 2	10-61
0x2121C	ISP_CCM_COEF3	CCM coefficient register 3	10-61
0x21220	ISP_CCM_COEF4	CCM coefficient register 4	10-62
0x21230	ISP_CCM_IN_DC0	CCM input direct current (DC) component register 0	10-63
0x21234	ISP_CCM_IN_DC1	CCM input DC component register 1	10-63
0x21240	ISP_CCM_OUT_DC0	CCM output DC component register 0	10-63
0x21244	ISP_CCM_OUT_DC1	CCM output DC component register 1	10-64
0x21300	ISP_CSC_CFG	CSC enable register	10-64

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Offset Address	Register	Description	Page
0x21310	ISP_CSC_COEF0	CSC coefficient register 0	10-65
0x21314	ISP_CSC_COEF1	CSC coefficient register 1	10-65
0x21318	ISP_CSC_COEF2	CSC coefficient register 2	10-66
0x2131C	ISP_CSC_COEF3	CSC coefficient register 3	10-67
0x21320	ISP_CSC_COEF4	CSC coefficient register 4	10-67
0x21330	ISP_CSC_IN_DC0	CSC input DC component register 0	10-68
0x21334	ISP_CSC_IN_DC1	CSC input DC component register 1	10-68
0x21340	ISP_CSC_OUT_DC0	CSC output DC component register 0	10-69
0x21344	ISP_CSC_OUT_DC1	CSC output DC component register 1	10-69
0x21400	ISP_DG_CFG	DG control register	10-69
0x21410	ISP_DG_GAIN1	DG gain (R & Gr) register	10-70
0x21414	ISP_DG_GAIN2	DG gain (B & Gb) register	10-70
0x21600	ISP_UVNR_CFG	UVNR control register	10-71
0x21614	ISP_HCDS_SIZE	Horizontal chrominance down sampling input size configuration register	10-72
0x21618	ISP_DSUS_MODE	Horizontal chrominance down sampling output size configuration register	10-72
0x2161C	ISP_HCDS_COEF0	Horizontal chrominance down sampling coefficient register 0	10-73
0x21620	ISP_HCDS_COEF1	Horizontal chrominance down sampling coefficient register 1	10-74
0x21624	ISP_HCDS_COEF2	Horizontal chrominance down sampling coefficient register 2	10-74
0x21628	ISP_VCDS_COEF	Vertical chrominance down sampling coefficient register	10-75
0x2162C	ISP_MIDF_SIZE	Median filtering input picture chrominance width/height register	10-75
0x21630	ISP_NR_SIZE	NR input picture chrominance width/height register	10-76
0x21634	ISP_UVNR_DIFFTHRES0	Difference threshold register	10-77
0x21638	ISP_UVNR_DIFFTHRES1	Difference threshold register	10-77

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Offset Address	Register	Description	Page
0x2163C	ISP_UVNR_RATIO0	Weight limit register	10-78
0x21640	ISP_UVNR_RATIO1	Weight limit register	10-78
0x21644	ISP_UVNR_PARA	NR blending ratio and shift register	10-79
0x22000	ISP_AE_CFG	AE enable register	10-80
0x22010	ISP_AE_ZONE	AE block configuration register	10-80
0x22014	ISP_AE_SKIP_CRG	AE point select configuration register	10-80
0x22018	ISP_AE_TOTAL_STAT	Selected point total number register in a picture frame during 256-segment histogram statistics	10-81
0x2201C	ISP_AE_COUNT_STAT	Selected point total weight register in a picture frame during 256-segment histogram statistics	10-82
0x22020	ISP_AE_TOTAL_R_AVER	R component average value register in a picture frame during average value statistics	10-82
0x22024	ISP_AE_TOTAL_GR_AVER	Gr component average value register in a picture frame during average value statistics	10-83
0x22028	ISP_AE_TOTAL_GB_AVER	Gb component average value register in a picture frame during average value statistics	10-83
0x2202C	ISP_AE_TOTAL_B_AVER	B component average value register in a picture frame during average value statistics	10-83
0x22040	ISP_AE_BITMOVE	AE pixel shift value register	10-84
0x22088	ISP_AE_MEM_HIST_RADDR	AE histogram statistics read address register	10-84
0x2208C	ISP_AE_MEM_HIST_RDATA	AE histogram statistics read data register	10-85
0x22098	ISP_AE_MEM_AVER_RADDR	AE average value statistics read address register	10-85
0x2209C	ISP_AE_MEM_AVER_RDATA	AE average value statistics read data register	10-85
0x220A0	ISP_AE_MEM_WEIGHT_WADDR	Zone weight register	10-86
0x220A4	ISP_AE_MEM_WEIGHT_WDATA	Zone weight register	10-86

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Offset Address	Register	Description	Page
0x22100	ISP_AWB_CFG	AWB enable register	10-86
0x22110	ISP_AWB_ZONE	AWB block configuration register	10-87
0x22118	ISP_AWB_THD_MIN	AWB minimum RGB field value register	10-87
0x2211C	ISP_AWB_THD_MAX	AWB maximum RGB field value register	10-88
0x22120	ISP_AWB_CR_MM	AWB reference maximum/minimum R/G value register	10-88
0x22124	ISP_AWB_CB_MM	AWB reference maximum/minimum B/G value register	10-88
0x22130	ISP_AWB_TOP_K	AWB gray area reference range top hypotenuse slope register	10-89
0x22134	ISP_AWB_TOP_B	AWB gray area reference range top hypotenuse intercept register	10-89
0x22138	ISP_AWB_BOT_K	AWB gray area reference range bottom hypotenuse slope register	10-90
0x2213C	ISP_AWB_BOT_B	AWB gray area reference range bottom hypotenuse intercept register	10-90
0x22140	ISP_AWB_AVG_R	R component average value register of an entire picture	10-91
0x22144	ISP_AWB_AVG_G	G component average value register of an entire picture	10-91
0x22148	ISP_AWB_AVG_B	B component average value register of an entire picture	10-91
0x2214C	ISP_AWB_CNT_ALL	Statistical point number register of an entire picture	10-92
0x22150	ISP_AWB_CNT_MIN	Number register for points below the minimum threshold in an entire picture	10-92
0x22154	ISP_AWB_CNT_MAX	Number register for points above the maximum threshold in an entire picture	10-93
0x22188	ISP_AWB_STAT_RADDR	AWB statistics read address register	10-93
0x2218C	ISP_AWB_STAT_RDATA	AWB statistics read return data register	10-94
0x22200	ISP_AF_CFG	AF control register	10-94

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Offset Address	Register	Description	Page
0x22210	ISP_AF_ZONE	AF block configuration register	10-95
0x22214	ISP_AF_CROP_START	AF picture crop start coordinate register	10-96
0x22218	ISP_AF_CROP_SIZE	AF picture crop size register	10-97
0x2221C	ISP_AF_MEAN_THRES	AF median filtering threshold register	10-97
0x22220	ISP_AF_IIRG0	AF IIR filtering parameter register 0	10-97
0x22224	ISP_AF_IIRG1	AF IIR filtering parameter register 1	10-98
0x22228	ISP_AF_IIRG2	AF IIR filtering parameter register 2	10-99
0x2222C	ISP_AF_IIRG3	AF IIR filtering parameter register 3	10-99
0x22230	ISP_AF_IIRG4	AF IIR filtering parameter register 4	10-100
0x22234	ISP_AF_IIRG5	AF IIR filtering parameter register 5	10-100
0x22238	ISP_AF_IIRG6	AF IIR filtering parameter register 6	10-101
0x22240	ISP_AF_SHIFT	AF IIR filtering shift parameter register	10-101
0x22250	ISP_AF_FIRH0	AF FIR filtering parameter register 0	10-102
0x22254	ISP_AF_FIRH1	AF FIR filtering parameter register 1	10-103
0x22258	ISP_AF_FIRH2	AF FIR filtering parameter register 2	10-103
0x2225C	ISP_AF_FIRH3	AF FIR filtering parameter register 3	10-104
0x22260	ISP_AF_FIRH4	AF FIR filtering parameter register 4	10-105
0x22270	ISP_AF_IIRTHRE	AF IIR filtering threshold register	10-105
0x22274	ISP_AF_FIRTHRE	AF FIR filtering threshold register	10-106
0x22278	ISP_AF_ACC_SHIFT	AF cumulative statistics shift register	10-106
0x2227C	ISP_AF_CNT_SHIFT	AF count statistics shift register	10-107
0x22288	ISP_AF_STAT_IND_RADDR	Statistics indirect read address register for AF blocks	10-107
0x2228C	ISP_AF_STAT_IND_RDATA	Statistics indirect read data register for AF blocks	10-107
0x23000	ISP_DEMOSAIC_CFG	Demosaic control register	10-109

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0x2811C	ISP_COLORBAR_BBACKGND	Color bar background color register	10-241
0x28120	ISP_COLORBAR_RFOREGND	Color bar foreground color register	10-241
0x28124	ISP_COLORBAR_GFOREGND	Color bar foreground color register	10-241
0x28128	ISP_COLORBAR_BFOREGND	Color bar foreground color register	10-242
0x2812C	ISP_COLORBAR_INIT	Color bar initial value register	10-242
0x28130	ISP_COLORBAR_GRAD	Color bar increasing value register	10-242
0x28134	ISP_COLORBAR_POS1	Color bar rectangle coordinate register	10-243
0x28138	ISP_COLORBAR_POS2	Color bar rectangle coordinate register	10-243
0x2813C	ISP_COLORBAR_RGGB	RGGB sequence register	10-244
0x28200	ISP_YUV444SUM_CFG	YUV444 sum control register	10-244
0x28210	ISP_CROP_DATA0_0	Data 0 sum lower 32 bits register	10-245
0x28214	ISP_CROP_DATA0_1	Data 0 sum upper 32 bits register	10-245
0x28218	ISP_CROP_DATA1_0	Data 1 sum lower 32 bits register	10-246
0x2821C	ISP_CROP_DATA1_1	Data 1 sum upper 32 bits register	10-246
0x28220	ISP_CROP_DATA2_0	Data 2 sum lower 32 bits register	10-246
0x28224	ISP_CROP_DATA2_1	Data 2 sum upper 32 bits register	10-247
0x28228	ISP_CROP_DATAFPN_0	Data FPN sum lower 32 bits register	10-247
0x2822C	ISP_CROP_DATAFPN_1	Data FPN sum upper 32 bits register	10-247
0x28230	ISP_WB_DATA0_0	Data 0 sum lower 32 bits register	10-248

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Offset Address	Register	Description	Page
0x28234	ISP_WB_DATA0_1	Data 0 sum upper 32 bits register	10-248
0x28300	ISP_YUV422SUM_CFG	FPGA sum control register	10-248
0x28310	ISP_CLIP_Y_DATA0_0	Y component data sum lower 32 bits register	10-249
0x28314	ISP_CLIP_Y_DATA0_1	Y component data sum upper 32 bits register	10-249
0x28318	ISP_CLIP_C_DATA0_0	C component data sum lower 32 bits register	10-249
0x2831C	ISP_CLIP_C_DATA0_1	C component data sum upper 32 bits register	10-250
0x28320	ISP_YUV422_SWITCH_DATA0_0	Data 0 sum lower 32 bits register	10-250
0x28324	ISP_YUV422_SWITCH_DATA0_1	Data 0 sum upper 32 bits register	10-250
0x28328	ISP_YUV422_SWITCH_DATA1_0	Data 1 sum lower 32 bits register	10-251
0x2832C	ISP_YUV422_SWITCH_DATA1_1	Data 1 sum upper 32 bits register	10-251
0x28330	ISP_YUV422_SWITCH_DATA2_0	Data 2 sum lower 32 bits register	10-251
0x28334	ISP_YUV422_SWITCH_DATA2_1	Data 2 sum upper 32 bits register	10-252
0x31010	ISP_ADAPTER444_CFG	YUV444 timing adaptation register	10-252
0x31020	ISP_SUM_Y	Input picture luminance statistics register	10-253
0x31100	ISP_CROP_CFG	Crop enable register	10-253
0x31110	ISP_CROP0_START	Crop start position register for region 0	10-253
0x31114	ISP_CROP0_SIZE	Crop size configuration register for region 0	10-254
0x31400	ISP_ACM_CTRL	ACM control register	10-254
0x31404	ISP_ACM_ADJ	ACM processed pixel change register	10-255
0x31410	ISP_ACM_SIZE	ACM processed picture size register	10-256
0x31414	ISP_ACM_PARA_ADDR	ACM coefficient configuration loading address register	10-257

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Offset Address	Register	Description	Page
0x31420	ISP_ACM_PARA_UP	ACM coefficient configuration update register	10-257
0x31C00	ISP_VPDCICTRL	DCI control register	10-257
0x31C04	ISP_VPDCIHPOS	DCI algorithm horizontal adjustment region register	10-259
0x31C08	ISP_VPDCIVPOS	DCI algorithm vertical adjustment region register	10-260
0x31C0C	ISP_VPDCIHISBLD	DCI histogram statistics weighted coefficient register	10-260
0x31C10	ISP_VPDCIHISOFT	DCI histogram statistics offset register	10-261
0x31C14	ISP_VPDCIHISCOR	DCI histogram coring register	10-261
0x31C18	ISP_VPDCIMERBLD	DCI adjustment unit blend value register	10-261
0x31C1C	ISP_VPDCIADJWGT	DCI manually configured curve weight register	10-262
0x31C20	ISP_VPDCICLIP0	DCI curve 0 weight range register	10-262
0x31C24	ISP_VPDCICLIP1	DCI curve 1 weight range register	10-263
0x31C28	ISP_VPDCICLIP2	DCI curve 2 weight range register	10-263
0x31C2C	ISP_VPDCIGLBGAIN	DCI luminance adjustment unit global gain register	10-264
0x31C30	ISP_VPDCIPOSTHR0	DCI adjustment unit threshold register 0 during positive adjustment	10-264
0x31C34	ISP_VPDCIPOSTHR1	DCI adjustment unit threshold register 1 during positive adjustment	10-265
0x31C38	ISP_VPDCIPOSGAIN0	DCI adjustment unit gain register 0 during positive adjustment	10-265
0x31C3C	ISP_VPDCIPOSGAIN1	DCI adjustment unit gain register 1 during positive adjustment	10-266
0x31C40	ISP_VPDCIPOSSLP0	DCI adjustment unit slope register 0 during positive adjustment	10-266
0x31C44	ISP_VPDCIPOSSLP1	DCI adjustment unit slope register 1 during positive adjustment	10-267
0x31C48	ISP_VPDCIPOSSLP2	DCI adjustment unit slope register 2during positive adjustment	10-267

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Offset Address	Register	Description	Page
0x31C4C	ISP_VPDCINEGTHR0	DCI adjustment unit threshold register 0 during negative adjustment	10-268
0x31C50	ISP_VPDCINEGTHR1	DCI adjustment unit threshold register 1 during negative adjustment	10-268
0x31C54	ISP_VPDCINEGGAIN0	DCI adjustment unit gain register 0 during negative adjustment	10-269
0x31C58	ISP_VPDCINEGGAIN1	DCI adjustment unit gain register 1 during negative adjustment	10-270
0x31C5C	ISP_VPDCINEGSLP0	DCI adjustment unit slope register 0 during negative adjustment	10-270
0x31C60	ISP_VPDCINEGSLP1	DCI adjustment unit slope register 1 during negative adjustment	10-271
0x31C64	ISP_VPDCINEGSLP2	DCI adjustment unit slope register 2 during negative adjustment	10-271
0x32000	ISP_SWITCH422_CTRL	Frame discarding module control register	10-272
0x32010	ISP_ADAPTER422_CFG	YUV422 timing adaptation register	10-272
0x32100	ISP_SKIP_Y_CFG	Y skip configuration register	10-273
0x32110	ISP_SKIP_C_CFG	C skip configuration register	10-273
0x32200	ISP_CLIP_Y_CFG	Luminance clamping configuration register	10-273
0x32204	ISP_CLIP_C_CFG	Chrominance clamping configuration register	10-274
0x32300	ISP_HLDC_CFG	HLDC enable register	10-274
0x32310	ISP_HLDC_SIZE	HLDC luminance size configuration register	10-275
0x32320	ISP_HLDC_CENTER	HLDC center coordinate parameter register	10-276
0x32324	ISP_HLDC_PARA	HLDC configuration parameter register	10-276

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10.6 Register Description

ISP_VERSION

ISP_VERSION is an ISP version register.

Offset Address	Register Name	Total Reset Value
0x20080	ISP_VERSION	0x0010_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	version												build				sn															
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	version	ISP version 0x1: V100 0x2: V200 0x3: V300 The same rule applies to other values.
[19:16]	RO	build	Subversion number 0x1: B01 0x2: B02 The same rule applies to other values.
[15:0]	RO	sn	Sequence number of the ISP version

ISP_MODULE_POS

ISP_MODULE_POS is an ISP internal module position select register.

Offset Address	Register Name	Total Reset Value
0x20088	ISP_MODULE_POS	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											reserved	af_sel		ae_sel		dcg_sel
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4]	RO	reserved	Reserved

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[3]	RW	af_sel	AF position adjustment register 0: The AF module is after the DG module. 1: The AF module is in the Y field after the CSC module.
[2:1]	RW	ae_sel	AE position adjustment register 00: The AE module is after the DG module. 01: The AE module is after the WB module. 10: The AE module is after the DRC module.
[0]	RW	dcg_sel	DPC/GE position adjustment register 0: The DPC/GE module is after the BLC module. 1: The DPC/GE module is after the DRC module.

ISP_CH_SWITCH

ISP_CH_SWITCH is an ISP input switch register.

Offset Address	Register Name	Total Reset Value
0x2008C	ISP_CH_SWITCH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										isp_ch_switch_4		reserved		isp_ch_switch_3		reserved		isp_ch_switch_2		reserved		isp_ch_switch_1		reserved		isp_ch_switch_0		reserved			isp_ch_switch_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22:20]	RW	isp_ch_switch_4	ISP FPN black frame input data select 000: channel 0 input 001: channel 1 input 010: channel 2 input 011: channel 3 input Other values: FPN black frame input
[19]	RO	reserved	Reserved

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[18:16]	RW	isp_ch_switch_3	Input data select for ISP channel 3 000: channel 0 input 001: channel 1 input 010: channel 2 input 100: FPN black frame input Other values: channel 3 input
[15]	RO	reserved	Reserved
[14:12]	RW	isp_ch_switch_2	Input data select for ISP channel 2 000: channel 0 input 001: channel 1 input 011: channel 3 input 100: FPN black frame input Other values: channel 2 input
[11]	RO	reserved	Reserved
[10:8]	RW	isp_ch_switch_1	Input data select for ISP channel 1 000: channel 0 input 010: channel 2 input 011: channel 3 input 100: FPN black frame input Other values: channel 1 input
[7]	RO	reserved	Reserved
[6:4]	RW	isp_ch_switch_0	Input data select for ISP channel 0 001: channel 1 input 010: channel 2 input 011: channel 3 input 100: FPN black frame input Other values: channel 0 input
[3:1]	RO	reserved	Reserved
[0]	RW	isp_ch_switch_en	ISP input switch enable

ISP_INT_TIMING_CTRL

ISP_INT_TIMING_CTRL is an ISP adjustable interrupt trigger time configuration register.

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Offset Address	Register Name	Total Reset Value
0x20090	ISP_INT_TIMING_CTRL	0x0025_4310

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	timing_cnt																															
Reset	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	1	0	0	0	0	1	1	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	timing_cnt	Adjustable interrupt trigger time

ISP_ISPBYPASS

ISP_ISPBYPASS is an ISP bypass register.

Offset Address	Register Name	Total Reset Value
0x20094	ISP_ISPBYPASS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															isp_bypass
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	isp_bypass	ISP bypass register 0: The embedded ISP is used. 1: The ISP is bypassed and data is transparently transmitted from the FPN module to the 2DNR/demosaic module

ISP_INT

ISP_INT is an ISP interrupt indicator register.

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Offset Address	Register Name	Total Reset Value
0x200F0	ISP_INT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																	update_cfg_yuv444	cfg_loss_yuv444	update_cfg_yuv422	field_throw_yuv422	int_disstat	int_fstart_yuv422	reserved	reserved	int_afstat	int_aestat	int_awbstat	int_user_yuv444	int_fstart2_yuv444	acm_para_finish	int_fstart_yuv444
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14]	WC	update_cfg_yuv444	ISP YUV444 register update interrupt status. Writing 1 clears this interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[13]	WC	cfg_loss_yuv444	ISP YUV444 register configuration loss interrupt status. Writing 1 clears this interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[12]	WC	update_cfg_yuv422	ISP YUV422 register update interrupt status. Writing 1 clears this interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[11]	WC	field_throw_yuv422	ISP YUV422 field/frame loss interrupt status. Writing 1 clears this interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[10]	WC	int_disstat	DIS current frame statistics completion interrupt
[9]	WC	int_fstart_yuv422	ISP YUV422 module frame start interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[8]	RO	reserved	Reserved
[7]	RO	reserved	Reserved
[6]	WC	int_afstat	AF current frame statistics TX completion interrupt 0: No interrupt is generated. 1: An interrupt is generated.

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[5]	WC	int_aestat	AE current frame statistics TX completion interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[4]	WC	int_awbstat	AWB current frame statistics TX completion interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[3]	WC	int_user_yuv444	ISP YUV444 configurable trigger position interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[2]	WC	int_fstart2_yuv444	ISP YUV444 frame start interrupt indicator 2. Writing 1 clears this interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[1]	WC	acm_para_finish	ACM lookup table load completion interrupt. Writing 1 clears this interrupt. 0: No interrupt is generated. 1: An interrupt is generated.
[0]	WC	int_fstart_yuv444	ISP YUV444 frame start interrupt indicator. Writing 1 clears this interrupt. 0: No interrupt is generated. 1: An interrupt is generated.

ISP_INT_MASK

ISP_INT_MASK is an ISP interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x200F8	ISP_INT_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																update_cfg_yuv444	cfg_loss_yuv444	update_cfg_yuv422	field_throw_yuv422	int_disstat	int_fstart_yuv422	reserved	reserved	int_afstat	int_aestat	int_awbstat	int_user_yuv444	int_fstart2_yuv444	acm_para_finish	int_fstart_yuv444	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved

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[14]	RW	update_cfg_yuv444	ISP YUV444 register update interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.
[13]	RW	cfg_loss_yuv444	ISP YUV444 configuration loss interrupt enable 0: masked 1: enabled
[12]	RW	update_cfg_yuv422	ISP YUV422 register update interrupt enable 0: No interrupt is generated. 1: An interrupt is generated.
[11]	RW	field_throw_yuv422	ISP YUV422 field/frame loss interrupt enable 0: masked 1: enabled
[10]	RW	int_disstat	DIS current frame statistics completion interrupt mask 0: masked 1: enabled
[9]	RW	int_fstart_yuv422	ISP YUV422 module frame start interrupt mask 0: masked 1: enabled
[8]	RO	reserved	Reserved
[7]	RO	reserved	Reserved
[6]	RW	int_afstat	AF current frame statistics TX completion interrupt mask 0: masked 1: enabled
[5]	RW	int_aestat	AE current frame statistics TX completion interrupt mask 0: masked 1: enabled
[4]	RW	int_awbstat	AWB current frame statistics TX completion interrupt mask 0: masked 1: enabled
[3]	RW	int_user_yuv444	ISP YUV444 configurable trigger position interrupt mask 0: masked 1: enabled
[2]	RW	int_fstart2_yuv444	ISP YUV444 frame start interrupt mask 2 0: masked 1: enabled

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[1]	RW	acm_para_finish	ACM lookup table load completion interrupt mask 0: masked 1: enabled
[0]	RW	int_fstart_yuv444	ISP YUV444 frame start interrupt mask 0: masked 1: enabled

ISP_YUV444_CTRL_F

ISP_YUV444_CTRL_F is an ISP general update control register.

Offset Address	Register Name	Total Reset Value
0x201E0	ISP_YUV444_CTRL_F	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														rggb_cfg	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	rggb_cfg	ISP YUV444 RGGB start 00: R Gr Gb B 01: Gr R B Gb 10: Gb B R Gr 11: B Gb Gr R

ISP_YUV444_CTRL_I

ISP_YUV444_CTRL_I is an ISP immediate update control register.

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Offset Address	Register Name	Total Reset Value
0x201E4	ISP_YUV444_CTRL_I	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								update_mode
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	update_mode	ISP YUV444 register update mode 0: ISP_YUV444_REG_UPDATE update mode 1: frame update mode

ISP_YUV444_TIMING_CFG

ISP_YUV444_TIMING_CFG is an output timing configuration register.

Offset Address	Register Name	Total Reset Value
0x201E8	ISP_YUV444_TIMING_CFG	0x0000_0080

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved					cpi_fix_timing_stat			reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	1 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:1]	RW	cpi_fix_timing_stat	ISP YUV444 manual timing parameter configuration. The length of the generated line blanking region is configured.
[0]	RO	reserved	Reserved

ISP_YUV444_REG_UPDATE

ISP_YUV444_REG_UPDATE is a register update register.

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Offset Address	Register Name	Total Reset Value
0x201EC	ISP_YUV444_REG_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															update
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	update	ISP YUV444 update register. This bit is automatically cleared for each frame.

ISP_YUV444_SIZE

ISP_YUV444_SIZE is a picture width/height register.

Offset Address	Register Name	Total Reset Value
0x201F0	ISP_YUV444_SIZE	0x0437_077F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	height	ISP YUV444 picture height. The configured value is the actual height minus 1. For example, if the actual picture height is 1080, this field needs to be set to **1079**.
[15:12]	RO	reserved	Reserved
[11:0]	RW	width	ISP YUV444 picture width. The configured value is the actual width minus 1. For example, if the actual picture width is 1920, this field needs to be set to **1919**.

ISP_YUV422_CTRL_F

ISP_YUV422_CTRL_F is an ISP general update control register.

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Offset Address	Register Name	Total Reset Value
0x202E0	ISP_YUV422_CTRL_F	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														rggb_cfg	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	rggb_cfg	ISP YUV422 RGGB start 00: R Gr Gb B 01: Gr R B Gb 10: Gb B R Gr 11: B Gb Gr R

ISP_YUV422_CTRL_I

ISP_YUV422_CTRL_I is an ISP immediate update control register.

Offset Address	Register Name	Total Reset Value
0x202E4	ISP_YUV422_CTRL_I	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															update_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	update_mode	ISP YUV422 register update mode 0: ISP_YUV422_REG_UPDATE update mode 1: frame update mode

ISP_YUV422_TIMING_CFG

ISP_YUV422_TIMING_CFG is an output timing configuration register.

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Offset Address	Register Name	Total Reset Value
0x202E8	ISP_YUV422_TIMING_CFG	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				cpi_fix_timing_stat											reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:1]	RW	cpi_fix_timing_stat	ISP YUV422 manual timing parameter configuration. The length of the generated line blanking region is configured.
[0]	RO	reserved	Reserved

ISP_YUV422_REG_UPDATE

ISP_YUV422_REG_UPDATE is a register update register.

Offset Address	Register Name	Total Reset Value
0x202EC	ISP_YUV422_REG_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															update
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	update	ISP YUV422 update register. This bit is automatically cleared for each frame.

ISP_YUV422_SIZE

ISP_YUV422_SIZE is a picture width/height register.

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Offset Address	Register Name	Total Reset Value
0x202F0	ISP_YUV422_SIZE	0x0437_077F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	height	ISP YUV422 picture height. The configured value is the actual height minus 1. For example, if the actual picture height is 1080, this field needs to be set to **1079**.
[15:12]	RO	reserved	Reserved
[11:0]	RW	width	ISP YUV422 picture width. The configured value is the actual width minus 1. For example, if the actual picture width is 1920, this field needs to be set to **1919**.

ISP_BLC_CFG

ISP_BLC_CFG is a BLC control register.

Offset Address	Register Name	Total Reset Value
0x21000	ISP_BLC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	BLC enable 0: disabled 1: enabled

ISP_BLC_OFFSET1

ISP_BLC_OFFSET1 is BLC offset register 1.

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Offset Address	Register Name	Total Reset Value
0x21010	ISP_BLC_OFFSET1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ofsr												reserved				ofsgr											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:16]	RW	ofsr	BLC offset of the R component, S14
[15:14]	RO	reserved	Reserved
[13:0]	RW	ofsgr	BLC offset of the Gr component, S14

ISP_BLC_OFFSET2

ISP_BLC_OFFSET2 is BLC offset register 2.

Offset Address	Register Name	Total Reset Value
0x21014	ISP_BLC_OFFSET2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ofsb												reserved				ofsgb											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:16]	RW	ofsb	BLC offset of the B component, S14
[15:14]	RO	reserved	Reserved
[13:0]	RW	ofsgb	BLC offset of the Gb component, S14

ISP_WB_CFG

ISP_WB_CFG is a WB control register.

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Offset Address	Register Name	Total Reset Value
0x21100	ISP_WB_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	WB enable 0: disabled 1: enabled

ISP_WB_GAIN1

ISP_WB_GAIN1 is a WB gain (R & Gr) register.

Offset Address	Register Name	Total Reset Value
0x21110	ISP_WB_GAIN1	0x0100_0100

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				rgain												reserved				grgain											
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	rgain	WB R gain, U4.8
[15:12]	RO	reserved	Reserved
[11:0]	RW	grgain	WB Gr gain, U4.8

ISP_WB_GAIN2

ISP_WB_GAIN2 is a WB gain (B & Gb) register.

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Offset Address	Register Name	Total Reset Value
0x21114	ISP_WB_GAIN2	0x0100_0100

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				bgain												reserved				gbgain											
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	bgain	WB B gain, U4.8
[15:12]	RO	reserved	Reserved
[11:0]	RW	gbgain	WB Gb gain, U4.8

ISP_CCM_CFG

ISP_CCM_CFG is a CCM enable register.

Offset Address	Register Name	Total Reset Value
0x21200	ISP_CCM_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	CCM enable 0: disabled 1: enabled

ISP_CCM_COEF0

ISP_CCM_COEF0 is CCM coefficient register 0.

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Offset Address	Register Name	Total Reset Value
0x21210	ISP_CCM_COEF0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef01															reserved	coef00															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef01	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef00	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

ISP_CCM_COEF1

ISP_CCM_COEF1 is CCM coefficient register 1.

Offset Address	Register Name	Total Reset Value
0x21214	ISP_CCM_COEF1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef10															reserved	coef02															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef10	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef02	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.

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[0]	RO	reserved	Reserved

ISP_CCM_COEF2

ISP_CCM_COEF2 is CCM coefficient register 2.

Offset Address	Register Name	Total Reset Value
0x21218	ISP_CCM_COEF2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef12															reserved	coef11															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef12	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef11	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

ISP_CCM_COEF3

ISP_CCM_COEF3 is CCM coefficient register 3.

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Offset Address	Register Name	Total Reset Value
0x2121C	ISP_CCM_COEF3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef21															reserved	coef20															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef21	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef20	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

ISP_CCM_COEF4

ISP_CCM_COEF4 is CCM coefficient register 4.

Offset Address	Register Name	Total Reset Value
0x21220	ISP_CCM_COEF4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															reserved	coef22															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RO	reserved	Reserved
[15:1]	RW	coef22	CCM coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

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ISP_CCM_IN_DC0

ISP_CCM_IN_DC0 is CCM input DC component register 0.

Offset Address	Register Name	Total Reset Value
0x21230	ISP_CCM_IN_DC0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	in_dc1												reserved				in_dc0								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RW	in_dc1	DC offset of the input G component, signed integer
[20:16]	RO	reserved	Reserved
[15:5]	RW	in_dc0	DC offset of the input R component, signed integer
[4:0]	RO	reserved	Reserved

ISP_CCM_IN_DC1

ISP_CCM_IN_DC1 is CCM input DC component register 1.

Offset Address	Register Name	Total Reset Value
0x21234	ISP_CCM_IN_DC1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												reserved				in_dc2								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RO	reserved	Reserved
[15:5]	RW	in_dc2	DC offset of the input B component, signed integer
[4:0]	RO	reserved	Reserved

ISP_CCM_OUT_DC0

ISP_CCM_OUT_DC0 is CCM output DC component register 0.

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Offset Address	Register Name	Total Reset Value
0x21240	ISP_CCM_OUT_DC0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	out_dc1												reserved				out_dc0								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RW	out_dc1	DC offset of the output U component, signed integer
[20:16]	RO	reserved	Reserved
[15:5]	RW	out_dc0	DC offset of the output Y component, signed integer
[4:0]	RO	reserved	Reserved

ISP_CCM_OUT_DC1

ISP_CCM_OUT_DC1 is CCM output DC component register 1.

Offset Address	Register Name	Total Reset Value
0x21244	ISP_CCM_OUT_DC1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												reserved				out_dc2								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RO	reserved	Reserved
[15:5]	RW	out_dc2	DC offset of the output V component, signed integer
[4:0]	RO	reserved	Reserved

ISP_CSC_CFG

ISP_CSC_CFG is a CSC enable register.

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Offset Address	Register Name	Total Reset Value
0x21300	ISP_CSC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	CSC enable 0: disabled 1: enabled

ISP_CSC_COEF0

ISP_CSC_COEF0 is CSC coefficient register 0.

Offset Address	Register Name	Total Reset Value
0x21310	ISP_CSC_COEF0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef01															reserved	coef00															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef01	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef00	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

ISP_CSC_COEF1

ISP_CSC_COEF1 is CSC coefficient register 1.

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Offset Address	Register Name	Total Reset Value
0x21314	ISP_CSC_COEF1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef10															reserved	coef02															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef10	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef02	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

ISP_CSC_COEF2

ISP_CSC_COEF2 is CSC coefficient register 2.

Offset Address	Register Name	Total Reset Value
0x21318	ISP_CSC_COEF2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef12															reserved	coef11															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef12	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef11	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.

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[0]	RO	reserved	Reserved

ISP_CSC_COEF3

ISP_CSC_COEF3 is CSC coefficient register 3.

Offset Address	Register Name	Total Reset Value
0x2131C	ISP_CSC_COEF3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	coef21															reserved	coef20															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	coef21	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[16]	RO	reserved	Reserved
[15:1]	RW	coef20	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

ISP_CSC_COEF4

ISP_CSC_COEF4 is CSC coefficient register 4.

Offset Address	Register Name	Total Reset Value
0x21320	ISP_CSC_COEF4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															reserved	coef22															reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved

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[16]	RO	reserved	Reserved
[15:1]	RW	coef22	CSC coefficient Its format is S5.10. The value is a signed number consisting of a 4-bit integral part and a 10-bit decimal part.
[0]	RO	reserved	Reserved

ISP_CSC_IN_DC0

ISP_CSC_IN_DC0 is CSC input DC component register 0.

Offset Address	Register Name	Total Reset Value
0x21330	ISP_CSC_IN_DC0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	in_dc1												reserved				in_dc0								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RW	in_dc1	DC offset of the input G component, signed integer
[20:16]	RO	reserved	Reserved
[15:5]	RW	in_dc0	DC offset of the input R component, signed integer
[4:0]	RO	reserved	Reserved

ISP_CSC_IN_DC1

ISP_CSC_IN_DC1 is CSC input DC component register 1.

Offset Address	Register Name	Total Reset Value
0x21334	ISP_CSC_IN_DC1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												reserved				in_dc2								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RO	reserved	Reserved
[15:5]	RW	in_dc2	DC offset of the input B component, signed integer
[4:0]	RO	reserved	Reserved

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ISP_CSC_OUT_DC0

ISP_CSC_OUT_DC0 is CSC output DC component register 0.

Offset Address	Register Name	Total Reset Value
0x21340	ISP_CSC_OUT_DC0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	out_dc1												reserved				out_dc0												reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RW	out_dc1	DC offset of the output U component, signed integer
[20:16]	RO	reserved	Reserved
[15:5]	RW	out_dc0	DC offset of the output Y component, signed integer
[4:0]	RO	reserved	Reserved

ISP_CSC_OUT_DC1

ISP_CSC_OUT_DC1 is CSC output DC component register 1.

Offset Address	Register Name	Total Reset Value
0x21344	ISP_CSC_OUT_DC1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												reserved				out_dc2												reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RO	reserved	Reserved
[15:5]	RW	out_dc2	DC offset of the output V component, signed integer
[4:0]	RO	reserved	Reserved

ISP_DG_CFG

ISP_DG_CFG is a DG control register.

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Offset Address	Register Name	Total Reset Value
0x21400	ISP_DG_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	DG enable 0: disabled 1: enabled

ISP_DG_GAIN1

ISP_DG_GAIN1 is a DG gain (R & Gr) register.

Offset Address	Register Name	Total Reset Value
0x21410	ISP_DG_GAIN1	0x0100_0100

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rgain																grgain															
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	rgain	DG R gain, U8.8
[15:0]	RW	grgain	DG Gr gain, U8.8

ISP_DG_GAIN2

ISP_DG_GAIN2 is a DG gain (B & Gb) register.

Offset Address	Register Name	Total Reset Value
0x21414	ISP_DG_GAIN2	0x0100_0100

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bgain																gbgain															
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	bgain	DG B gain, U8.8

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[15:0]	RW	gbgain	DG Gb gain, U8.8

ISP_UVNR_CFG

ISP_UVNR_CFG is a UVNR control register.

Offset Address	Register Name	Total Reset Value
0x21600	ISP_UVNR_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			uvnr_sel	reserved			nr_en	reserved			midf_en	uv2c_mode	uv2c_en	vcds_en	hcds_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	uvnr_sel	UVNR denoising channel select 0: MIDF channel 1: UNVR channel
[11:9]	RO	reserved	Reserved
[8]	RW	nr_en	UVNR denoising enable 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved
[4]	RW	midf_en	MIDFILTER enable 0: disabled 1: enabled
[3]	RW	uv2c_mode	UV2C bypass mode 0: The U component data is output. 1: The V component data is output.
[2]	RW	uv2c_en	UV2C enable for converting the internal dual-component (U and V) data into the single C component data 0: disabled 1: enabled
[1]	RW	vcds_en	Vertical chrominance down sampling enable 0: disabled 1: enabled

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[0]	RW	hcds_en	Horizontal chrominance down sampling enable 0: disabled 1: enabled

ISP_HCDS_SIZE

ISP_HCDS_SIZE is a horizontal chrominance down sampling input size configuration register.

Offset Address	Register Name	Total Reset Value
0x21614	ISP_HCDS_SIZE	0x077F_077F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				width_out												reserved				width_in											
Reset	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	width_out	U/V line width (in pixel) of the output picture during horizontal chrominance down sampling. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width_in	U/V line width (in pixel) of the input picture. The configured value is the actual value minus 1.

ISP_DSUS_MODE

ISP_DSUS_MODE is a horizontal chrominance down sampling output size configuration register.

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Offset Address	Register Name	Total Reset Value
0x21618	ISP_DSUS_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													vcus_mode	vcds_mode	hcds_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	vcus_mode	Vertical UV up sampling multiple 0: 2 1: 4
[1]	RW	vcds_mode	Vertical UV down sampling multiple 0: 2 1: 4
[0]	RW	hcds_mode	Horizontal UV down sampling multiple 0: 2 1: 4

ISP_HCDS_COEF0

ISP_HCDS_COEF0 is horizontal chrominance down sampling coefficient register 0.

Offset Address	Register Name	Total Reset Value
0x2161C	ISP_HCDS_COEF0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		coef2										coef1								coef0											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	coef2	Horizontal chrominance scaling and filtering coefficient 2
[19:10]	RW	coef1	Horizontal chrominance scaling and filtering coefficient 1

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[9:0]	RW	coef0	Horizontal chrominance scaling and filtering coefficient 0, 10-bit signed number Horizontal scaling contains eight taps, and the sum of the eight coefficients is 512.

ISP_HCDS_COEF1

ISP_HCDS_COEF1 is horizontal chrominance down sampling coefficient register 1.

Offset Address	Register Name	Total Reset Value
0x21620	ISP_HCDS_COEF1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				coef5								coef4								coef3											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	coef5	Horizontal chrominance scaling and filtering coefficient 5
[19:10]	RW	coef4	Horizontal chrominance scaling and filtering coefficient 4
[9:0]	RW	coef3	Horizontal chrominance scaling and filtering coefficient 3

ISP_HCDS_COEF2

ISP_HCDS_COEF2 is horizontal chrominance down sampling coefficient register 2.

Offset Address	Register Name	Total Reset Value
0x21624	ISP_HCDS_COEF2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												coef7								coef6											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:10]	RW	coef7	Horizontal chrominance scaling and filtering coefficient 7
[9:0]	RW	coef6	Horizontal chrominance scaling and filtering coefficient 6

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ISP_VCDS_COEF

ISP_VCDS_COEF is a vertical chrominance down sampling coefficient register.

Offset Address	Register Name	Total Reset Value
0x21628	ISP_VCDS_COEF	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				coef3				reserved				coef2				reserved				coef1				reserved				coef0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:24]	RW	coef3	Down sampling coefficient 3. The value meanings are the same as those of coefficient 0.
[23:19]	RO	reserved	Reserved
[18:16]	RW	coef2	Down sampling coefficient 2. The value meanings are the same as those of coefficient 0.
[15:11]	RO	reserved	Reserved
[10:8]	RW	coef1	Down sampling coefficient 1. The value meanings are the same as those of coefficient 0.
[7:3]	RO	reserved	Reserved
[2:0]	RW	coef0	Down sampling coefficient 0 0x0: multiplied by 0x1 0x1: multiplied by 0x2 0x2: multiplied by 0x4 0x3: multiplied by 0x8 0x4: multiplied by 0x10 0x5: multiplied by 0x20 0x6: multiplied by 0x0 0x7: multiplied by 0x0 (invalid)

ISP_MIDF_SIZE

ISP_MIDF_SIZE is a median filtering input picture chrominance width/height register.

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Offset Address	Register Name	Total Reset Value
0x2162C	ISP_MIDF_SIZE	0x0437_077F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Picture height after vertical chrominance down sampling. The configured value is the actual height minus 1. For example, if the actual picture height is 720, this field needs to be set to **719**.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Picture width after vertical chrominance down sampling (UV-to-C conversion). The configured value is the actual width minus 1. For example, if the actual picture width is 1280, this field needs to be set to **1279**.

ISP_NR_SIZE

ISP_NR_SIZE is an NR input picture chrominance width/height register.

Offset Address	Register Name	Total Reset Value
0x21630	ISP_NR_SIZE	0x0437_077F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Picture height after vertical chrominance down sampling. The configured value is the actual height minus 1. For example, if the actual picture height is 720, this field needs to be set to **719**.
[15:13]	RO	reserved	Reserved

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[12:0]	RW	width	Picture width after vertical chrominance down sampling (UV-to-C conversion). The configured value is the actual width minus 1. For example, if the actual picture width is 1280, this field needs to be set to **1279**.

ISP_UVNR_DIFFTHRES0

ISP_UVNR_DIFFTHRES0 is a difference threshold register.

Offset Address	Register Name	Total Reset Value
0x21634	ISP_UVNR_DIFFTHRES0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	diffthres4								diffthres3								diffthres2								diffthres1							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	diffthres4	Threshold of the difference with the distance of 4, 8-bit unsigned number
[23:16]	RW	diffthres3	Threshold of the difference with the distance of 3, 8-bit unsigned number
[15:8]	RW	diffthres2	Threshold of the difference with the distance of 2, 8-bit unsigned number
[7:0]	RW	diffthres1	Threshold of the difference with the distance of 1, 8-bit unsigned number

ISP_UVNR_DIFFTHRES1

ISP_UVNR_DIFFTHRES1 is a difference threshold register.

Offset Address	Register Name	Total Reset Value
0x21638	ISP_UVNR_DIFFTHRES1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																diffthres6								diffthres5							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	diffthres6	Threshold of the difference with the distance of 6, 8-bit unsigned number

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[7:0]	RW	diffthres5	Threshold of the difference with the distance of 5, 8-bit unsigned number

ISP_UVNR_RATIO0

ISP_UVNR_RATIO0 is a weight limit register.

Offset Address	Register Name	Total Reset Value
0x2163C	ISP_UVNR_RATIO0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	ratiolimit3							reserved	ratiolimit2							reserved	ratiolimit1							reserved	ratiolimit0						
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:24]	RW	ratiolimit3	Weight upper limit for distance 3, 7-bit unsigned number
[23]	RO	reserved	Reserved
[22:16]	RW	ratiolimit2	Weight upper limit for distance 2, 7-bit unsigned number
[15]	RO	reserved	Reserved
[14:8]	RW	ratiolimit1	Weight upper limit for distance 1, 7-bit unsigned number
[7]	RO	reserved	Reserved
[6:0]	RW	ratiolimit0	Weight upper limit for distance 0, 7-bit unsigned number

ISP_UVNR_RATIO1

ISP_UVNR_RATIO1 is a weight limit register.

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Offset Address	Register Name	Total Reset Value
0x21640	ISP_UVNR_RATIO1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								ratiolimit6								reserved	ratiolimit5							reserved	ratiolimit4						
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22:16]	RW	ratiolimit6	Weight upper limit for distance 6, 7-bit unsigned number
[15]	RO	reserved	Reserved
[14:8]	RW	ratiolimit5	Weight upper limit for distance 5, 7-bit unsigned number
[7]	RO	reserved	Reserved
[6:0]	RW	ratiolimit4	Weight upper limit for distance 4, 7-bit unsigned number

ISP_UVNR_PARA

ISP_UVNR_PARA is an NR blending ratio and shift register.

Offset Address	Register Name	Total Reset Value
0x21644	ISP_UVNR_PARA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								limit								reserved						diffshift		reserved				blendratio			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22:16]	RW	limit	Limit for the coring function. The value range is 0−127.
[15:10]	RO	reserved	Reserved
[9:8]	RW	diffshift	2-bit unsigned number. The value is 0, 1, 2, or 3.
[7:6]	RO	reserved	Reserved
[5:0]	RW	blendratio	Blending ratio, 6-bit unsigned number. The value range is 0−32.

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ISP_AE_CFG

ISP_AE_CFG is an AE enable register.

Offset Address	Register Name	Total Reset Value
0x22000	ISP_AE_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	AE enable 0: disabled 1: enabled

ISP_AE_ZONE

ISP_AE_ZONE is an AE block configuration register.

Offset Address	Register Name	Total Reset Value
0x22010	ISP_AE_ZONE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				vnum				reserved				hnum				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12:8]	RW	vnum	Number of vertical blocks
[7:5]	RO	reserved	Reserved
[4:0]	RW	hnum	Number of horizontal blocks

ISP_AE_SKIP_CRG

ISP_AE_SKIP_CRG is an AE point select configuration register.

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Offset Address	Register Name	Total Reset Value
0x22014	ISP_AE_SKIP_CRG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								offset_y		skip_y		offset_x		skip_x	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	offset_y	0: count from column 0 1: count from column 1
[6:4]	RW	skip_y	Vertical jump point configuration for statistical points 0x0: Each pixel point is collected for statistics. 0x1: Statistics are collected on every two pixels. 0x2: Statistics are collected on every three pixels. 0x3: Statistics are collected on every four pixels. 0x4: Statistics are collected on every five pixels. 0x5: Statistics are collected on every eight pixels. 0x6 or larger values: Statistics are collected on every nine pixels.
[3]	RW	offset_x	0: count from column 0 1: count from column 1
[2:0]	RW	skip_x	Horizontal jump point configuration for statistical points 0x0: Statistics are collected on every two pixels. 0x1: Statistics are collected on every three pixels. 0x2: Statistics are collected on every four pixels. 0x3: Statistics are collected on every five pixels. 0x4: Statistics are collected on every eight pixels. 0x5 or larger values: Statistics are collected on every nine pixels.

ISP_AE_TOTAL_STAT

ISP_AE_TOTAL_STAT is a selected point total number register in a picture frame during 256-segment histogram statistics.

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Offset Address	Register Name	Total Reset Value
0x22018	ISP_AE_TOTAL_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								total_pixels																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RO	total_pixels	Number of selected points in a picture frame during 256-segment histogram statistics

ISP_AE_COUNT_STAT

ISP_AE_COUNT_STAT is a selected point total weight register in a picture frame during 256-segment histogram statistics.

Offset Address	Register Name	Total Reset Value
0x2201C	ISP_AE_COUNT_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				count_pixels																											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:0]	RO	count_pixels	Total weight of selected points in a picture frame during 256-segment histogram statistics

ISP_AE_TOTAL_R_AVER

ISP_AE_TOTAL_R_AVER is an R component average value register in a picture frame during average value statistics.

Offset Address	Register Name	Total Reset Value
0x22020	ISP_AE_TOTAL_R_AVER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	total_r_aver																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	total_r_aver	Global average value of the R component

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ISP_AE_TOTAL_GR_AVER

ISP_AE_TOTAL_GR_AVER is a Gr component average value register in a picture frame during average value statistics.

Offset Address	Register Name	Total Reset Value
0x22024	ISP_AE_TOTAL_GR_AVER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	total_gr_aver																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	total_gr_aver	Global average value of the Gr component

ISP_AE_TOTAL_GB_AVER

ISP_AE_TOTAL_GB_AVER is a Gb component average value register in a picture frame during average value statistics.

Offset Address	Register Name	Total Reset Value
0x22028	ISP_AE_TOTAL_GB_AVER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	total_gb_aver																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	total_gb_aver	Global average value of the Gb component

ISP_AE_TOTAL_B_AVER

ISP_AE_TOTAL_B_AVER is a B component average value register in a picture frame during average value statistics.

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Offset Address	Register Name	Total Reset Value
0x2202C	ISP_AE_TOTAL_B_AVER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	total_b_aver																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	total_b_aver	Global average value of the B component

ISP_AE_BITMOVE

ISP_AE_BITMOVE is an AE pixel shift value register.

Offset Address	Register Name	Total Reset Value
0x22040	ISP_AE_BITMOVE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																							gamma_en	reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:9]	RO	reserved	Reserved
[8]	RW	gamma_en	AE gamma enable
[7:0]	RO	reserved	Reserved

ISP_AE_MEM_HIST_RADDR

ISP_AE_MEM_HIST_RADDR is an AE histogram statistics read address register.

Offset Address	Register Name	Total Reset Value
0x22088	ISP_AE_MEM_HIST_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	hist_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	hist_raddr	Histogram statistics read address

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ISP_AE_MEM_HIST_RDATA

ISP_AE_MEM_HIST_RDATA is an AE histogram statistics read data register.

Offset Address	Register Name	Total Reset Value
0x2208C	ISP_AE_MEM_HIST_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	hist_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	hist_rdata	Histogram statistics read data

ISP_AE_MEM_AVER_RADDR

ISP_AE_MEM_AVER_RADDR is an AE average value statistics read address register.

Offset Address	Register Name	Total Reset Value
0x22098	ISP_AE_MEM_AVER_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	aver_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	aver_raddr	Read address of the average value statistics

ISP_AE_MEM_AVER_RDATA

ISP_AE_MEM_AVER_RDATA is an AE average value statistics read data register.

Offset Address	Register Name	Total Reset Value
0x2209C	ISP_AE_MEM_AVER_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	aver_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	aver_rdata	Read data of the average value statistics

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ISP_AE_MEM_WEIGHT_WADDR

ISP_AE_MEM_WEIGHT_WADDR is a zone weight register.

Offset Address	Register Name	Total Reset Value
0x220A0	ISP_AE_MEM_WEIGHT_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wei_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	wei_waddr	Weight table write address

ISP_AE_MEM_WEIGHT_WDATA

ISP_AE_MEM_WEIGHT_WDATA is a zone weight register.

Offset Address	Register Name	Total Reset Value
0x220A4	ISP_AE_MEM_WEIGHT_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wei_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	wei_wdata	Weight table write data

ISP_AWB_CFG

ISP_AWB_CFG is an AWB enable register.

Offset Address	Register Name	Total Reset Value
0x22100	ISP_AWB_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															work_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	RW	work_en	AWB enable 1: enabled 0: disabled

ISP_AWB_ZONE

ISP_AWB_ZONE is an AWB block configuration register.

Offset Address	Register Name	Total Reset Value
0x22110	ISP_AWB_ZONE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				vnum				reserved				hnum			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12:8]	RW	vnum	Number of vertical blocks
[7:5]	RO	reserved	Reserved
[4:0]	RW	hnum	Number of horizontal blocks

ISP_AWB_THD_MIN

ISP_AWB_THD_MIN is an AWB minimum RGB field value register.

Offset Address	Register Name	Total Reset Value
0x22118	ISP_AWB_THD_MIN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				threshold_min											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	threshold_min	Minimum RGB field value during AWB statistics

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ISP_AWB_THD_MAX

ISP_AWB_THD_MAX is an AWB maximum RGB field value register.

Offset Address	Register Name	Total Reset Value
0x2211C	ISP_AWB_THD_MAX	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				threshold_max											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	threshold_max	Maximum RGB field value during AWB statistics

ISP_AWB_CR_MM

ISP_AWB_CR_MM is an AWB reference maximum/minimum R/G value register.

Offset Address	Register Name	Total Reset Value
0x22120	ISP_AWB_CR_MM	0x0040_01FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cr_ref_min												reserved				cr_ref_max											
Reset	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	cr_ref_min	AWB reference minimum R/G value, 4.8-bit unsigned value consisting of a 4-bit integral part and an 8-bit decimal part
[15:12]	RO	reserved	Reserved
[11:0]	RW	cr_ref_max	AWB reference maximum R/G value, 4.8-bit unsigned value consisting of a 4-bit integral part and an 8-bit decimal part

ISP_AWB_CB_MM

ISP_AWB_CB_MM is an AWB reference maximum/minimum B/G value register.

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Offset Address	Register Name	Total Reset Value
0x22124	ISP_AWB_CB_MM	0x0040_01FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cb_ref_min												reserved				cb_ref_max											
Reset	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	cb_ref_min	AWB reference minimum B/G value, 4.8-bit unsigned value consisting of a 4-bit integral part and an 8-bit decimal part
[15:12]	RO	reserved	Reserved
[11:0]	RW	cb_ref_max	AWB reference maximum B/G value, 4.8-bit unsigned value consisting of a 4-bit integral part and an 8-bit decimal part

ISP_AWB_TOP_K

ISP_AWB_TOP_K is an AWB gray area reference range top hypotenuse slope register.

Offset Address	Register Name	Total Reset Value
0x22130	ISP_AWB_TOP_K	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								awb_top_k																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:0]	RW	awb_top_k	Hypotenuse slope of the AWB gray area reference range, 13.12-bit signed number, configured by using complementary codes

ISP_AWB_TOP_B

ISP_AWB_TOP_B is an AWB gray area reference range top hypotenuse intercept register.

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Offset Address	Register Name	Total Reset Value
0x22134	ISP_AWB_TOP_B	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								awb_top_b																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	awb_top_b	Hypotenuse intercept of the AWB gray area reference range, 16.8-bit unsigned number

ISP_AWB_BOT_K

ISP_AWB_BOT_K is an AWB gray area reference range bottom hypotenuse slope register.

Offset Address	Register Name	Total Reset Value
0x22138	ISP_AWB_BOT_K	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								awb_bot_k																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:0]	RW	awb_bot_k	Hypotenuse slope of the AWB gray area reference range, 13.12-bit signed number, configured by using complementary codes

ISP_AWB_BOT_B

ISP_AWB_BOT_B is an AWB gray area reference range bottom hypotenuse intercept register.

Offset Address	Register Name	Total Reset Value
0x2213C	ISP_AWB_BOT_B	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								awb_bot_b																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved

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[23:0]	RW	awb_bot_b	Hypotenuse intercept of the AWB gray area reference range, 16.8-bit unsigned number

ISP_AWB_AVG_R

ISP_AWB_AVG_R is an R component average value register of an entire picture.

Offset Address	Register Name	Total Reset Value
0x22140	ISP_AWB_AVG_R	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				avg_r											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RO	avg_r	Average value of the R component for an entire picture

ISP_AWB_AVG_G

ISP_AWB_AVG_G is a G component average value register of an entire picture.

Offset Address	Register Name	Total Reset Value
0x22144	ISP_AWB_AVG_G	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				avg_g											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RO	avg_g	Average value of the G component for an entire picture

ISP_AWB_AVG_B

ISP_AWB_AVG_B is a B component average value register of an entire picture.

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Offset Address	Register Name	Total Reset Value
0x22148	ISP_AWB_AVG_B	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				avg_b											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RO	avg_b	Average value of the B component for an entire picture

ISP_AWB_CNT_ALL

ISP_AWB_CNT_ALL is a statistical point number register of an entire picture.

Offset Address	Register Name	Total Reset Value
0x2214C	ISP_AWB_CNT_ALL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																count_all															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	count_all	Number of statistical points in an entire picture

ISP_AWB_CNT_MIN

ISP_AWB_CNT_MIN is a number register for points below the minimum threshold in an entire picture.

Offset Address	Register Name	Total Reset Value
0x22150	ISP_AWB_CNT_MIN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																count_min															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	count_min	Number of points below the minimum threshold in an entire picture

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ISP_AWB_CNT_MAX

ISP_AWB_CNT_MAX is a number register for points above the maximum threshold in an entire picture.

Offset Address	Register Name	Total Reset Value
0x22154	ISP_AWB_CNT_MAX	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																count_max															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	count_max	Number of points above the maximum threshold in an entire picture

ISP_AWB_STAT_RADDR

ISP_AWB_STAT_RADDR is an AWB statistics read address register.

Offset Address	Register Name	Total Reset Value
0x22188	ISP_AWB_STAT_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						stat_raddr									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:0]	RW	stat_raddr	AWB statistics address The sequence is as follows: 0x0: AVG G+AVG R for block 0 0x1: Count ALL+AVG B for block 0 0x2: Count Max+Count Min for block 0 0x3: AVG G+AVG R for block 1 … The same rule applies to other values. There are 255 blocks, and the address value range is 0−754.

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ISP_AWB_STAT_RDATA

ISP_AWB_STAT_RDATA is an AWB statistics read return data register.

Offset Address	Register Name	Total Reset Value
0x2218C	ISP_AWB_STAT_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	stat_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	stat_rdata	Read return value of AWB statistics. Each statistical value occupies 16 bits.

ISP_AF_CFG

ISP_AF_CFG is an AF control register.

Offset Address	Register Name	Total Reset Value
0x22200	ISP_AF_CFG	0x0000_01DA

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ck_gt_en	reserved															bayer_mode	raw_mode	sqrt_en	mean_en	lpf_en	crop_en	data_mode	squ_mode	peak_mode	iir1_en2	iir1_en1	iir1_en0	iir0_en2	iir0_en1	iir0_en0	en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0	1	0

Bits	Access	Name	Description
[31]	RW	ck_gt_en	AF clock gating enable
[30:16]	RO	reserved	Reserved
[15]	RW	bayer_mode	AF raw format 0: RGGB or BGGR 1: GRBG or RBRG
[14]	RW	raw_mode	AF data format 0: YUV data 1: raw data
[13]	RW	sqrt_en	AF gamma enable
[12]	RW	mean_en	AF median filtering enable
[11]	RW	lpf_en	AF low-pass filter enable

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[10]	RW	crop_en	AF crop enable
[9]	RW	data_mode	AF statistical mode 0: five groups of values 1: nine groups of values
[8]	RW	squ_mode	Square mode of AF statistics 0: Statistics are directly accumulated. 1: Statistics are accumulated after the square operation.
[7]	RW	peak_mode	Peak value mode of AF statistics 0: All values of a block are selected for statistics. 1: The maximum value in each line of a block is selected for statistics.
[6]	RW	iir1_en2	Enable for IIR2 among the three cascaded IIRs in group 2 0: disabled 1: enabled
[5]	RW	iir1_en1	Enable for IIR1 among the three cascaded IIRs in group 2 0: disabled 1: enabled
[4]	RW	iir1_en0	Enable for IIR0 among the three cascaded IIRs in group 2 0: disabled 1: enabled
[3]	RW	iir0_en2	Enable for IIR2 among the three cascaded IIRs in group 1 0: disabled 1: enabled
[2]	RW	iir0_en1	Enable for IIR1 among the three cascaded IIRs in group 1 0: disabled 1: enabled
[1]	RW	iir0_en0	Enable for IIR0 among the three cascaded IIRs in group 1 0: disabled 1: enabled
[0]	RW	en	AF enable 0: disabled 1: enabled

ISP_AF_ZONE

ISP_AF_ZONE is an AF block configuration register.

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Offset Address	Register Name	Total Reset Value
0x22210	ISP_AF_ZONE	0x0000_0F11

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				vnum				reserved				hnum			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1	0	0	0	1

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12:8]	RW	vnum	Number of vertical AF blocks (15 blocks at most) Blocks are in the CH_AF_CROP_SIZE region when crop_en is 1. Blocks are in the CH_AF_SIZE region when crop_en is 0.
[7:5]	RO	reserved	Reserved
[4:0]	RW	hnum	Number of horizontal AF blocks (17 blocks at most) Blocks are in the CH_AF_CROP_SIZE region when crop_en is 1. Blocks are in the CH_AF_SIZE region when crop_en is 0.

ISP_AF_CROP_START

ISP_AF_CROP_START is an AF picture crop start coordinate register.

Offset Address	Register Name	Total Reset Value
0x22214	ISP_AF_CROP_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				pos_y												reserved				pos_x											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	pos_y	Vertical coordinate of the crop start point
[15:13]	RO	reserved	Reserved
[12:0]	RW	pos_x	Horizontal coordinate of the crop start point

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ISP_AF_CROP_SIZE

ISP_AF_CROP_SIZE is an AF picture crop size register.

Offset Address	Register Name	Total Reset Value
0x22218	ISP_AF_CROP_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				vsize												reserved				hsize											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	vsize	Picture height after AF cropping NOTE The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	hsize	Picture width after AF cropping NOTE The configured value is the actual value minus 1.

ISP_AF_MEAN_THRES

ISP_AF_MEAN_THRES is an AF median filtering threshold register.

Offset Address	Register Name	Total Reset Value
0x2221C	ISP_AF_MEAN_THRES	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																mean_thres															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	mean_thres	Threshold for AF median filtering

ISP_AF_IIRG0

ISP_AF_IIRG0 is AF IIR filtering parameter register 0.

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Offset Address	Register Name	Total Reset Value
0x22220	ISP_AF_IIRG0	0x00A1_00A0

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								iirg0_1								reserved								iirg0_0							
Reset	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	iirg0_1	g0 gain of AF IIR filtering group 2 NOTE This field affects the configured value of ISP_AF_SHIFT.
[15:8]	RO	reserved	Reserved
[7:0]	RW	iirg0_0	g0 gain of AF IIR filtering group 1 NOTE This field affects the configured value of ISP_AF_SHIFT.

ISP_AF_IIRG1

ISP_AF_IIRG1 is AF IIR filtering parameter register 1.

Offset Address	Register Name	Total Reset Value
0x22224	ISP_AF_IIRG1	0x0244_01BC

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								iirg1_1								reserved								iirg1_0							
Reset	0	0	0	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	1	1	0	1	1	1	1	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	iirg1_1	g1 gain of AF IIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −512.
[15:10]	RO	reserved	Reserved
[9:0]	RW	iirg1_0	g1 gain of AF IIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −512.

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ISP_AF_IIRG2

ISP_AF_IIRG2 is AF IIR filtering parameter register 2.

Offset Address	Register Name	Total Reset Value
0x22228	ISP_AF_IIRG2	0x0328_0328

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								iirg2_1								reserved								iirg2_0							
Reset	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	iirg2_1	g2 gain of AF IIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −512.
[15:10]	RO	reserved	Reserved
[9:0]	RW	iirg2_0	g2 gain of AF IIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −512.

ISP_AF_IIRG3

ISP_AF_IIRG3 is AF IIR filtering parameter register 3.

Offset Address	Register Name	Total Reset Value
0x2222C	ISP_AF_IIRG3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								iirg3_1								reserved								iirg3_0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	iirg3_1	g3 gain of AF IIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −512.
[15:10]	RO	reserved	Reserved

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[9:0]	RW	iirg3_0	g3 gain of AF IIR filtering group 1 **NOTE** The value is a signed number and cannot be the negative boundary value −512.

ISP_AF_IIRG4

ISP_AF_IIRG4 is AF IIR filtering parameter register 4.

Offset Address	Register Name	Total Reset Value
0x22230	ISP_AF_IIRG4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								iirg4_1								reserved								iirg4_0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	iirg4_1	g4 gain of AF IIR filtering group 2 **NOTE** The value is a signed number and cannot be the negative boundary value −512.
[15:10]	RO	reserved	Reserved
[9:0]	RW	iirg4_0	g4 gain of AF IIR filtering group 1 **NOTE** The value is a signed number and cannot be the negative boundary value −512.

ISP_AF_IIRG5

ISP_AF_IIRG5 is AF IIR filtering parameter register 5.

Offset Address	Register Name	Total Reset Value
0x22234	ISP_AF_IIRG5	0x0284_017C

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								iirg5_1								reserved								iirg5_0							
Reset	0	0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved

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[25:16]	RW	iirg5_1	g5 gain of AF IIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −512.
[15:10]	RO	reserved	Reserved
[9:0]	RW	iirg5_0	g5 gain of AF IIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −512.

ISP_AF_IIRG6

ISP_AF_IIRG6 is AF IIR filtering parameter register 6.

Offset Address	Register Name	Total Reset Value
0x22238	ISP_AF_IIRG6	0x033C_033C

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								iirg6_1								reserved								iirg6_0							
Reset	0	0	0	0	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	1	1	0	0	1	1	1	1	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	iirg6_1	g6 gain of AF IIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −512.
[15:10]	RO	reserved	Reserved
[9:0]	RW	iirg6_0	g6 gain of AF IIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −512.

ISP_AF_SHIFT

ISP_AF_SHIFT is an AF IIR filtering shift parameter register.

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Offset Address	Register Name	Total Reset Value
0x22240	ISP_AF_SHIFT	0x0027_1027

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	iirshift1_3	iirshift1_2	iirshift1_1	iirshift1_0	iirshift0_3	iirshift0_2	iirshift0_1	iirshift0_0
Reset	0 0 0 0	0 0 0 0	0 0 1 0	0 1 1 1	0 0 0 1	0 0 0 0	0 0 1 0	0 1 1 1

Bits	Access	Name	Description
[31:28]	RW	iirshift1_3	Shift adjustment for IIR2 in AF IIR filtering group 2
[27:24]	RW	iirshift1_2	Shift adjustment for IIR1 in AF IIR filtering group 2
[23:20]	RW	iirshift1_1	Shift adjustment for IIR0 in AF IIR filtering group 2
[19:16]	RW	iirshift1_0	Input shift adjustment for AF IIR filtering group 2 Note the following: iirg0_1 ≥ 128, iirshift0_0 ≥ 4 iirg0_1 ≥ 64, iirshift0_0 ≥ 3 iirg0_1 ≥ 32, iirshift0_0 ≥ 2 iirg0_1 ≥ 16, iirshift0_0 ≥ 1
[15:12]	RW	iirshift0_3	Shift adjustment for IIR2 in AF IIR filtering group 1
[11:8]	RW	iirshift0_2	Shift adjustment for IIR1 in AF IIR filtering group 1
[7:4]	RW	iirshift0_1	Shift adjustment for IIR0 in AF IIR filtering group 1
[3:0]	RW	iirshift0_0	Input shift adjustment for AF IIR filtering group 1 Note the following: iirg0_0 ≥ 128, iirshift0_0 ≥ 4 iirg0_0 ≥ 64, iirshift0_0 ≥ 3 iirg0_0 ≥ 32, iirshift0_0 ≥ 2 iirg0_0 ≥ 16, iirshift0_0 ≥ 1

ISP_AF_FIRH0

ISP_AF_FIRH0 is AF FIR filtering parameter register 0.

Offset Address	Register Name	Total Reset Value
0x22250	ISP_AF_FIRH0	0x0030_0030

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved			firh0_1	reserved			firh0_0
Reset	0 0 0 0	0 0 0 0	0 0 1 1	0 0 0 0	0 0 0 0	0 0 0 0	0 0 1 1	0 0 0 0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved

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[21:16]	RW	firh0_1	h0 gain of AF FIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −32.
[15:6]	RO	reserved	Reserved
[5:0]	RW	firh0_0	h0 gain of AF FIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −32.

ISP_AF_FIRH1

ISP_AF_FIRH1 is AF FIR filtering parameter register 1.

Offset Address	Register Name	Total Reset Value
0x22254	ISP_AF_FIRH1	0x0015_002B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												firh1_1				reserved												firh1_0			
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:16]	RW	firh1_1	h1 gain of AF FIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −32.
[15:6]	RO	reserved	Reserved
[5:0]	RW	firh1_0	h1 gain of AF FIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −32.

ISP_AF_FIRH2

ISP_AF_FIRH2 is AF FIR filtering parameter register 2.

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Offset Address	Register Name	Total Reset Value
0x22258	ISP_AF_FIRH2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												firh2_1				reserved												firh2_0			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:16]	RW	firh2_1	h2 gain of AF FIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −32.
[15:6]	RO	reserved	Reserved
[5:0]	RW	firh2_0	h2 gain of AF FIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −32.

ISP_AF_FIRH3

ISP_AF_FIRH3 is AF FIR filtering parameter register 3.

Offset Address	Register Name	Total Reset Value
0x2225C	ISP_AF_FIRH3	0x002B_0015

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												firh3_1				reserved												firh3_0			
Reset	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:16]	RW	firh3_1	h3 gain of AF FIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −32.
[15:6]	RO	reserved	Reserved
[5:0]	RW	firh3_0	h3 gain of AF FIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −32.

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ISP_AF_FIRH4

ISP_AF_FIRH4 is AF FIR filtering parameter register 4.

Offset Address	Register Name	Total Reset Value
0x22260	ISP_AF_FIRH4	0x0010_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												firh4_1				reserved												firh4_0			
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:16]	RW	firh4_1	h4 gain of AF FIR filtering group 2 NOTE The value is a signed number and cannot be the negative boundary value −32.
[15:6]	RO	reserved	Reserved
[5:0]	RW	firh4_0	h4 gain of AF FIR filtering group 1 NOTE The value is a signed number and cannot be the negative boundary value −32.

ISP_AF_IIRTHRE

ISP_AF_IIRTHRE is an AF IIR filtering threshold register.

Offset Address	Register Name	Total Reset Value
0x22270	ISP_AF_IIRTHRE	0x0014_0064

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								thre2_x								reserved				thre1_x											
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	thre2_x	Statistics threshold of AF IIR filtering group 2
[15:11]	RO	reserved	Reserved
[10:0]	RW	thre1_x	Statistics threshold of AF IIR filtering group 1

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ISP_AF_FIRTHRE

ISP_AF_FIRTHRE is an AF FIR filtering threshold register.

Offset Address	Register Name	Total Reset Value
0x22274	ISP_AF_FIRTHRE	0x000A_0032

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												thre2_y								reserved								thre1_y			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	thre2_y	Statistics threshold of AF FIR filtering group 2
[15:11]	RO	reserved	Reserved
[10:0]	RW	thre1_y	Statistics threshold of AF FIR filtering group 1

ISP_AF_ACC_SHIFT

ISP_AF_ACC_SHIFT is an AF cumulative statistics shift register.

Offset Address	Register Name	Total Reset Value
0x22278	ISP_AF_ACC_SHIFT	0x0002_0200

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												acc_shift_y				acc_shift2_v				acc_shift1_v				acc_shift2_h				acc_shift1_h			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:16]	RW	acc_shift_y	Luminance Y statistics shift (0−15). Y in CH_AF_STAT_RDATA is shifted.
[15:12]	RW	acc_shift2_v	Shift of vertical FIR filtering statistics in group 2 (0−15). V2 in CH_AF_STAT_RDATA is shifted.
[11:8]	RW	acc_shift1_v	Shift of vertical FIR filtering statistics in group 1 (0−15). V1 in CH_AF_STAT_RDATA is shifted.
[7:4]	RW	acc_shift2_h	Shift of horizontal IIR filtering statistics in group 2 (0−15). H2 in CH_AF_STAT_RDATA is shifted.
[3:0]	RW	acc_shift1_h	Shift of horizontal IIR filtering statistics in group 1 (0−15). H1 in CH_AF_STAT_RDATA is shifted.

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ISP_AF_CNT_SHIFT

ISP_AF_CNT_SHIFT is an AF count statistics shift register.

Offset Address	Register Name	Total Reset Value
0x2227C	ISP_AF_CNT_SHIFT	0x0000_0200

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																cnt_shift2_v				cnt_shift1_v				cnt_shift2_h				cnt_shift1_h			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:12]	RW	cnt_shift2_v	Shift of the vertical FIR filtering statistics in group 2 (0−15). vcnt2 in CH_AF_STAT_RDATA is shifted.
[11:8]	RW	cnt_shift1_v	Shift of the vertical FIR filtering statistics in group 1 (0−15). vcnt1 in CH_AF_STAT_RDATA is shifted.
[7:4]	RW	cnt_shift2_h	Shift of the horizontal IIR filtering statistics in group 2 (0−15). hcnt2 in CH_AF_STAT_RDATA is shifted.
[3:0]	RW	cnt_shift1_h	Shift of the horizontal IIR filtering statistics in group 1 (0−15). hcnt1 in CH_AF_STAT_RDATA is shifted.

ISP_AF_STAT_IND_RADDR

ISP_AF_STAT_IND_RADDR is a statistics indirect read address register for AF blocks.

Offset Address	Register Name	Total Reset Value
0x22288	ISP_AF_STAT_IND_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	af_stat_ind_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	af_stat_ind_raddr	Statistics indirect read address register for AF blocks

ISP_AF_STAT_IND_RDATA

ISP_AF_STAT_IND_RDATA is a statistics indirect read data register for AF blocks.

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Offset Address	Register Name	Total Reset Value
0x2228C	ISP_AF_STAT_IND_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	af_stat_ind_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	af_stat_ind_rdata	Statistics indirect read data register for AF blocks V1[16], H1[16], V2[16], H2[16], Y[16], hcnt1, hcnt2, vcnt1, and vcnt2 information about each block is collected for statistics. The following is the information for each address: When data_mode is 0: 0x00: {V1 (16 bits), H1 (16 bits)} of block 1 0x04: {V2 (16 bits), H2 (16 bits)} of block 1 0x08: Y (16 bits) of block 1 … The same rule applies to other values. When data_mode is 1: 0x00: {Y_h (6 bits), V1 (13 bits), H1 (13 bits)} of block 1 0x04: {Y_l (6 bits), V2 (13 bits), H2 (13 bits)} of block 1 0x08: {vcnt2 (8 bits), vcnt1 (8 bits), hcnt2 (8 bits), cnt1 (8 bits)} of block 1 … The same rule applies to other values. Note the following: V1 indicates the vertical FIR filtering statistics of group 1. V2 indicates the vertical FIR filtering statistics of group 2. H1 indicates the horizontal IIR filtering statistics of group 1. H2 indicates the horizontal IIR filtering statistics of group 2. Y indicates luminance statistics. vcnt1 indicates the number of times that the FIR filtering value is above the threshold. vcnt2 indicates the number of times that the FIR filtering value is above the threshold. hcnt1 indicates the number of times that the IIR filtering value is above the threshold. hcnt2 indicates the number of times that the IIR filtering value is above the threshold.

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ISP_DEMOSAIC_CFG

ISP_DEMOSAIC_CFG is a demosaic control register.

Offset Address	Register Name	Total Reset Value
0x23000	ISP_DEMOSAIC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	Demosaic enable 0: disabled 1: enabled

ISP_DEMOSAIC_CFG1

ISP_DEMOSAIC_CFG1 is a demosaic internal algorithm enable register.

Offset Address	Register Name	Total Reset Value
0x23004	ISP_DEMOSAIC_CFG1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															fcr_en	reserved															ahd_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RW	fcr_en	Demosaic FCR enable 0: disabled 1: enabled
[15:1]	RO	reserved	Reserved
[0]	RW	ahd_en	Demosaic AHD enable 0: disabled 1: enabled

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ISP_DEMOSAIC_COEF0

ISP_DEMOSAIC_COEF0 is a demosaic parameter register.

Offset Address	Register Name	Total Reset Value
0x23010	ISP_DEMOSAIC_COEF0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								bld_limit1								reserved								bld_limit2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	bld_limit1	Demosaic gradient blending threshold (U8.0)
[15:8]	RO	reserved	Reserved
[7:0]	RW	bld_limit2	Demosaic gradient blending threshold (U8.0)

ISP_DEMOSAIC_COEF1

ISP_DEMOSAIC_COEF1 is a demosaic parameter register.

Offset Address	Register Name	Total Reset Value
0x23014	ISP_DEMOSAIC_COEF1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								bld_ratio1								reserved								bld_ratio2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	bld_ratio1	Demosaic gradient blending threshold (U9.0)
[15:9]	RO	reserved	Reserved
[8:0]	RW	bld_ratio2	Demosaic gradient blending threshold (U9.0)

ISP_DEMOSAIC_COEF2

ISP_DEMOSAIC_COEF2 is a demosaic parameter register.

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Offset Address	Register Name	Total Reset Value
0x23018	ISP_DEMOSAIC_COEF2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				fcr_limit1												reserved				fcr_limit2											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	fcr_limit1	Demosaic FCR threshold (U12.0)
[15:12]	RO	reserved	Reserved
[11:0]	RW	fcr_limit2	Demosaic FCR threshold (U12.0)

ISP_DEMOSAIC_COEF3

ISP_DEMOSAIC_COEF3 is a demosaic parameter register.

Offset Address	Register Name	Total Reset Value
0x2301C	ISP_DEMOSAIC_COEF3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								ahd_par1								reserved								ahd_par2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	ahd_par1	Demosaic AHD coefficient (U10.0)
[15:8]	RO	reserved	Reserved
[7:0]	RW	ahd_par2	Demosaic AHD coefficient (U8.0)

ISP_DEMOSAIC_COEF4

ISP_DEMOSAIC_COEF4 is a demosaic parameter register.

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Offset Address	Register Name	Total Reset Value
0x23020	ISP_DEMOSAIC_COEF4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				fcr_thr												reserved												fcr_gain			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	fcr_thr	Demosaic FCR threshold (U12.0)
[15:5]	RO	reserved	Reserved
[4:0]	RW	fcr_gain	Demosaic FCR gain (U5.0)

ISP_DEMOSAIC_COEF5

ISP_DEMOSAIC_COEF5 is a demosaic parameter register.

Offset Address	Register Name	Total Reset Value
0x23024	ISP_DEMOSAIC_COEF5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								hv_ratio							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	hv_ratio	Direction ratio (U8.0)

ISP_DEMOSAIC_COEF6

ISP_DEMOSAIC_COEF6 is a demosaic parameter register.

Offset Address	Register Name	Total Reset Value
0x23028	ISP_DEMOSAIC_COEF6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										fcr_ratio					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved

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[5:0]	RW	fcr_ratio	FCR adjustment ratio (U6.0)

ISP_DEMOSAIC_SEL

ISP_DEMOSAIC_SEL is a demosaic threshold register.

Offset Address	Register Name	Total Reset Value
0x23030	ISP_DEMOSAIC_SEL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														hv_sel	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	hv_sel	Direction select (U2.0)

ISP_RGBIR_CFG

ISP_RGBIR_CFG is an RGBIR enable register.

Offset Address	Register Name	Total Reset Value
0x23100	ISP_RGBIR_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	RGBIR enable 0: disabled 1: enabled

ISP_RGBIR_CFG0

ISP_RGBIR_CFG0 is RGBIR enable register 0.

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Offset Address	Register Name	Total Reset Value
0x23104	ISP_RGBIR_CFG0	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												ir_out_en	ir_filter_en	blc_off_en	ir_remove_en	reserved														auto_gain_en	manu_gain_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19]	RW	ir_out_en	RGBIR IR component output enable 0: disabled 1: enabled
[18]	RW	ir_filter_en	RGBIR IR interpolation filtering enable 0: disabled 1: enabled
[17]	RW	blc_off_en	RGBIR de-BLC component enable 0: disabled 1: enabled
[16]	RW	ir_remove_en	RGBIR de-IR component enable 0: disabled 1: enabled
[15:2]	RO	reserved	Reserved
[1]	RW	auto_gain_en	RGBIR automatic gain enable 0: disabled 1: enabled
[0]	RW	manu_gain_en	RGBIR manual gain enable 0: disabled 1: enabled

ISP_RGBIR_CTRL

ISP_RGBIR_CTRL is an RGBIR control register.

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Offset Address	Register Name	Total Reset Value
0x23110	ISP_RGBIR_CTRL	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															ir_position
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	ir_position	1: The IR uses the Gb channel. 0: The IR uses the Gr channel.

ISP_RGBIR_MANU_GAIN

ISP_RGBIR_MANU_GAIN is an RGBIR manual gain register.

Offset Address	Register Name	Total Reset Value
0x23120	ISP_RGBIR_MANU_GAIN	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						manu_gain									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:0]	RW	manu_gain	RGBIR manual gain

ISP_RGBIR_MIN_G

ISP_RGBIR_MIN_G is an RGBIR G component minimum value register.

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Offset Address	Register Name	Total Reset Value
0x23130	ISP_RGBIR_MIN_G	0x0000_03FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				min_g											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	min_g	Minimum value of the G component

ISP_RGBIR_MAX_RB

ISP_RGBIR_MAX_RB is an RGBIR R/B component maximum value register.

Offset Address	Register Name	Total Reset Value
0x23134	ISP_RGBIR_MAX_RB	0x0000_03FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				max_rb											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	max_rb	Maximum value of the R/B component

ISP_RGBIR_BLACK_LEVEL

ISP_RGBIR_BLACK_LEVEL is an RGBIR black level register.

Offset Address	Register Name	Total Reset Value
0x23138	ISP_RGBIR_BLACK_LEVEL	0x0000_03FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				black_level											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	black_level	RGBIR black level

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ISP_RGBIR_F0

ISP_RGBIR_F0 is RGBIR filtering coefficient register 0.

Offset Address	Register Name	Total Reset Value
0x23140	ISP_RGBIR_F0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				coef02				coef01				coef00			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:8]	RW	coef02	RGBIR filtering coefficient
[7:4]	RW	coef01	RGBIR filtering coefficient
[3:0]	RW	coef00	RGBIR filtering coefficient

ISP_RGBIR_F1

ISP_RGBIR_F1 is RGBIR filtering coefficient register 1.

Offset Address	Register Name	Total Reset Value
0x23144	ISP_RGBIR_F1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				coef12				coef11				coef10			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:8]	RW	coef12	RGBIR filtering coefficient
[7:4]	RW	coef11	RGBIR filtering coefficient
[3:0]	RW	coef10	RGBIR filtering coefficient

ISP_RGBIR_F2

ISP_RGBIR_F2 is RGBIR filtering coefficient register 2.

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Offset Address	Register Name	Total Reset Value
0x23148	ISP_RGBIR_F2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				coef22				coef21				coef20			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:8]	RW	coef22	RGBIR filtering coefficient
[7:4]	RW	coef21	RGBIR filtering coefficient
[3:0]	RW	coef20	RGBIR filtering coefficient

ISP_RGBIR_THRE

ISP_RGBIR_THRE is an RGBIR threshold register.

Offset Address	Register Name	Total Reset Value
0x23150	ISP_RGBIR_THRE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								thre2								reserved								thre1							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	thre2	RGBIR direction determining threshold 2
[15:8]	RO	reserved	Reserved
[7:0]	RW	thre1	RGBIR direction determining threshold 1

ISP_RGBIR_EXP_VALUE

ISP_RGBIR_EXP_VALUE is an RGBIR exposure value register.

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Offset Address	Register Name	Total Reset Value
0x23160	ISP_RGBIR_EXP_VALUE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				exp_value											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	exp_value	RGBIR overexposure threshold

ISP_RGBIR_GAIN_USE

ISP_RGBIR_GAIN_USE is an RGBIR gain use value register.

Offset Address	Register Name	Total Reset Value
0x23164	ISP_RGBIR_GAIN_USE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				gain_use											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RO	gain_use	RGBIR gain_use statistics

ISP_RGBIR_EXP_CNT

ISP_RGBIR_EXP_CNT is an RGBIR overexposure statistics register.

Offset Address	Register Name	Total Reset Value
0x23168	ISP_RGBIR_EXP_CNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												exp_cnt																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:0]	RO	exp_cnt	Number of times that RGBIR overexposure occurs

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ISP_RGBIR_SCALE_B

ISP_RGBIR_SCALE_B is an RGBIR IR filtering B point coefficient register.

Offset Address	Register Name	Total Reset Value
0x23170	ISP_RGBIR_SCALE_B	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				b_b								b_g								b_r											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	b_b	B component coefficient for the B point during RGBIR matrix operation, 8-bit signed decimal
[19:10]	RW	b_g	G component coefficient for the B point during RGBIR matrix operation, 8-bit signed decimal
[9:0]	RW	b_r	R component coefficient for the B point during RGBIR matrix operation, 8-bit signed decimal

ISP_RGBIR_SCALE_G

ISP_RGBIR_SCALE_G is an RGBIR IR filtering G point coefficient register.

Offset Address	Register Name	Total Reset Value
0x23174	ISP_RGBIR_SCALE_G	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				g_b								g_g								g_r											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	g_b	B component coefficient for the G point during RGBIR matrix operation, 8-bit signed decimal
[19:10]	RW	g_g	G component coefficient for the G point during RGBIR matrix operation, 8-bit signed decimal

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[9:0]	RW	g_r	R component coefficient for the G point during RGBIR matrix operation, 8-bit signed decimal

ISP_RGBIR_SCALE_R

ISP_RGBIR_SCALE_R is an RGBIR IR filtering R point coefficient register.

Offset Address	Register Name	Total Reset Value
0x23178	ISP_RGBIR_SCALE_R	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				r_b								r_g								r_r											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	r_b	B component coefficient for the R point during RGBIR matrix operation, 8-bit signed decimal
[19:10]	RW	r_g	G component coefficient for the R point during RGBIR matrix operation, 8-bit signed decimal
[9:0]	RW	r_r	R component coefficient for the R point during RGBIR matrix operation, 8-bit signed decimal

ISP_RGBIR_SCALE_IR

ISP_RGBIR_SCALE_IR is an RGBIR IR filtering IR coefficient register.

Offset Address	Register Name	Total Reset Value
0x2317C	ISP_RGBIR_SCALE_IR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				b_ir								g_ir								r_ir											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved

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[29:20]	RW	b_ir	IR component coefficient for the B point during RGBIR matrix operation, 8-bit signed decimal
[19:10]	RW	g_ir	IR component coefficient for the G point during RGBIR matrix operation, 8-bit signed decimal
[9:0]	RW	r_ir	IR component coefficient for the R point during RGBIR matrix operation, 8-bit signed decimal

ISP_RGBIR_SCALE_C

ISP_RGBIR_SCALE_C is an RGBIR IR filtering C coefficient register.

Offset Address	Register Name	Total Reset Value
0x23180	ISP_RGBIR_SCALE_C	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				b_c								g_c								r_c											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	b_c	C component coefficient for the B point during RGBIR matrix operation, 8-bit signed decimal
[19:10]	RW	g_c	C component coefficient for the G point during RGBIR matrix operation, 8-bit signed decimal
[9:0]	RW	r_c	C component coefficient for the R point during RGBIR matrix operation, 8-bit signed decimal

ISP_2DNR_CFG

ISP_AWB_CFG is a 2DNR enable register.

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Offset Address	Register Name	Total Reset Value
0x24000	ISP_2DNR_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														luma_nr_en	en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	luma_nr_en	2DNR luminance denoising enable 0: disabled 1: enabled
[0]	RW	en	2DNR enable 0: disabled 1: enabled

ISP_2DNR_FCR

ISP_2DNR_FCR is a 2DNR anti-false color configuration register.

Offset Address	Register Name	Total Reset Value
0x24010	ISP_2DNR_FCR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								falsecolorenable	falsecolorshift							falsecolorgain								falsecolorlimit							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24]	RW	falsecolorenable	Anti-false color enable
[23:16]	RW	falsecolorshift	False color offset
[15:8]	RW	falsecolorgain	False color gain

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[7:0]	RW	falsecolorlimit	False color limit

ISP_2DNR_LUMANR

ISP_2DNR_LUMANR is a luminance denoising gain register.

Offset Address	Register Name	Total Reset Value
0x24014	ISP_2DNR_LUMANR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											luma_nr_gain				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RW	luma_nr_gain	Luminance denoising gain. The value range is [0, 16].

ISP_2DNR_LUMA

ISP_2DNR_LUMA is a current point luminance limit register.

Offset Address	Register Name	Total Reset Value
0x24018	ISP_2DNR_LUMA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				luma_low												reserved				luma_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:16]	RW	luma_low	Luminance lower limit of the current point. The bit width is the same as that of the bayer data. The maximum width is 14 bits.
[15:14]	RO	reserved	Reserved
[13:0]	RW	luma_high	Luminance upper limit of the current point. The bit width is the same as that of the bayer data. The maximum width is 14 bits.

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ISP_2DNR_LIMIT

ISP_2DNR_LIMIT is a 2DNR & demosaic blending ratio range limit register.

Offset Address	Register Name	Total Reset Value
0x2401C	ISP_2DNR_LIMIT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved		limit_low		reserved		limit_high	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	limit_low	Lower limit of the blending ratio, unsigned 9.0 number. The value range is [0, 0x100].
[15:9]	RO	reserved	Reserved
[8:0]	RW	limit_high	Upper limit of the blending ratio, unsigned 9.0 number. The value range is [0, 0x100].

ISP_2DNR_THRES0

ISP_2DNR_THRES0 is blending ratio linear range threshold register 0.

Offset Address	Register Name	Total Reset Value
0x24020	ISP_2DNR_THRES0	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved		thres_low		reserved		thres_high	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES1

ISP_2DNR_THRES1 is blending ratio linear range threshold register 1.

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Offset Address	Register Name	Total Reset Value
0x24024	ISP_2DNR_THRES1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES2

ISP_2DNR_THRES2 is blending ratio linear range threshold register 2.

Offset Address	Register Name	Total Reset Value
0x24028	ISP_2DNR_THRES2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES3

ISP_2DNR_THRES3 is blending ratio linear range threshold register 3.

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Offset Address	Register Name	Total Reset Value
0x2402C	ISP_2DNR_THRES3	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES4

ISP_2DNR_THRES4 is blending ratio linear range threshold register 4.

Offset Address	Register Name	Total Reset Value
0x24030	ISP_2DNR_THRES4	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES5

ISP_2DNR_THRES5 is blending ratio linear range threshold register 5.

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Offset Address	Register Name	Total Reset Value
0x24034	ISP_2DNR_THRES5	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES6

ISP_2DNR_THRES6 is blending ratio linear range threshold register 6.

Offset Address	Register Name	Total Reset Value
0x24038	ISP_2DNR_THRES6	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES7

ISP_2DNR_THRES7 is blending ratio linear range threshold register 7.

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Offset Address	Register Name	Total Reset Value
0x2403C	ISP_2DNR_THRES7	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES8

ISP_2DNR_THRES8 is blending ratio linear range threshold register 8.

Offset Address	Register Name	Total Reset Value
0x24040	ISP_2DNR_THRES8	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES9

ISP_2DNR_THRES9 is blending ratio linear range threshold register 9.

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Offset Address	Register Name	Total Reset Value
0x24044	ISP_2DNR_THRES9	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES10

ISP_2DNR_THRES10 is blending ratio linear range threshold register 10.

Offset Address	Register Name	Total Reset Value
0x24048	ISP_2DNR_THRES10	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	thres_low			reserved	thres_high		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES11

ISP_2DNR_THRES11 is blending ratio linear range threshold register 11.

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Offset Address	Register Name	Total Reset Value
0x2404C	ISP_2DNR_THRES11	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES12

ISP_2DNR_THRES12 is blending ratio linear range threshold register 12.

Offset Address	Register Name	Total Reset Value
0x24050	ISP_2DNR_THRES12	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES13

ISP_2DNR_THRES13 is blending ratio linear range threshold register 13.

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Offset Address	Register Name	Total Reset Value
0x24054	ISP_2DNR_THRES13	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES14

ISP_2DNR_THRES14 is blending ratio linear range threshold register 14.

Offset Address	Register Name	Total Reset Value
0x24058	ISP_2DNR_THRES14	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES15

ISP_2DNR_THRES15 is blending ratio linear range threshold register 15.

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Offset Address	Register Name	Total Reset Value
0x2405C	ISP_2DNR_THRES15	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_2DNR_THRES16

ISP_2DNR_THRES16 is blending ratio linear range threshold register 16.

Offset Address	Register Name	Total Reset Value
0x24060	ISP_2DNR_THRES16	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				thres_low												reserved				thres_high											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	thres_low	Lower limit of the blending ratio linear range
[15:12]	RO	reserved	Reserved
[11:0]	RW	thres_high	Upper limit of the blending ratio linear range

ISP_SHARPEN_CFG

ISP_SHARPEN_CFG is a sharpen control register.

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Offset Address	Register Name	Total Reset Value
0x24200	ISP_SHARPEN_CFG	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							en
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	Sharpen enable 0: disabled 1: enabled

ISP_SHARPEN_AMT

ISP_SHARPEN_AMT is a sharpen strength register.

Offset Address	Register Name	Total Reset Value
0x24210	ISP_SHARPEN_AMT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	edge_amt			reserved	sharp_amt		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	edge_amt	Edge enhancement strength (u, 12.0). (u, 12.0) indicates that the value is an unsigned number consisting of a 12-bit integral part and a 0-bit decimal part. The same rule applies to the meanings of other fields of sharpen registers.
[15:12]	RO	reserved	Reserved
[11:0]	RW	sharp_amt	Detail enhancement strength (u, 12.0)

ISP_SHARPEN_THD1

ISP_SHARPEN_THD1 is sharpen threshold register 1.

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Offset Address	Register Name	Total Reset Value
0x24214	ISP_SHARPEN_THD1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								edge_thd1								reserved								sharp_thd1							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	edge_thd1	Edge enhancement threshold (u, 10.0)
[15:10]	RO	reserved	Reserved
[9:0]	RW	sharp_thd1	Detail enhancement threshold (u, 10.0)

ISP_SHARPEN_COEF

ISP_SHARPEN_COEF is a sharpen strength coefficient register.

Offset Address	Register Name	Total Reset Value
0x24218	ISP_SHARPEN_COEF	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				edge_coef												reserved				sharp_coef											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	edge_coef	Edge enhancement coefficient (u, 12.0)
[15:12]	RO	reserved	Reserved
[11:0]	RW	sharp_coef	Detail enhancement coefficient (u, 12.0)

ISP_SHARPEN_SHOOTAMT

ISP_SHARPEN_SHOOTAMT is a sharpen shoot AMT control register.

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Offset Address	Register Name	Total Reset Value
0x2421C	ISP_SHARPEN_SHOOTAMT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								over_amt								reserved								under_amt							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	over_amt	Overshoot control coefficient (u, 8.0)
[15:8]	RO	reserved	Reserved
[7:0]	RW	under_amt	Undershoot control coefficient (u, 8.0)

ISP_SHARPEN_MID0

ISP_SHARPEN_MID0 is a sharpen IF filtering coefficient register.

Offset Address	Register Name	Total Reset Value
0x24230	ISP_SHARPEN_MID0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								mid_tmp02								mid_tmp01								mid_tmp00							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	mid_tmp02	IF filtering coefficient 02 (s, 8.0). (s, 8.0) indicates that the value is a signed number consisting of an 8-bit integral part and a 0-bit decimal part. The same rule applies to the meanings of other fields of sharpen registers.
[15:8]	RW	mid_tmp01	IF filtering coefficient 01 (s, 8.0)
[7:0]	RW	mid_tmp00	IF filtering coefficient 00 (s, 8.0)

ISP_SHARPEN_MID1

ISP_SHARPEN_MID1 is a sharpen IF filtering coefficient register.

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Offset Address	Register Name	Total Reset Value
0x24234	ISP_SHARPEN_MID1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								mid_tmp12								mid_tmp11								mid_tmp10							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	mid_tmp12	IF filtering coefficient 12 (s, 8.0)
[15:8]	RW	mid_tmp11	IF filtering coefficient 11 (s, 8.0)
[7:0]	RW	mid_tmp10	IF filtering coefficient 10 (s, 8.0)

ISP_SHARPEN_MID2

ISP_SHARPEN_MID2 is a sharpen IF filtering coefficient register.

Offset Address	Register Name	Total Reset Value
0x24238	ISP_SHARPEN_MID2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								mid_tmp22								mid_tmp21								mid_tmp20							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	mid_tmp22	IF filtering coefficient 22 (s, 8.0)
[15:8]	RW	mid_tmp21	IF filtering coefficient 21 (s, 8.0)
[7:0]	RW	mid_tmp20	IF filtering coefficient 20 (s, 8.0)

ISP_SHARPEN_LINETHD

ISP_SHARPEN_LINETHD is a sharpen line detection threshold register.

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Offset Address	Register Name	Total Reset Value
0x2423C	ISP_SHARPEN_LINETHD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								line_thd2								reserved								line_thd1							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	line_thd2	Line detection threshold 2 (u, 8.0)
[15:8]	RO	reserved	Reserved
[7:0]	RW	line_thd1	Line detection threshold 1 (u, 8.0)

ISP_SHARPEN_THD2

ISP_SHARPEN_THD2 is sharpen threshold register 2.

Offset Address	Register Name	Total Reset Value
0x24240	ISP_SHARPEN_THD2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								edge_thd2								reserved								sharp_thd2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	edge_thd2	Edge enhancement threshold (u, 10.0)
[15:10]	RO	reserved	Reserved
[9:0]	RW	sharp_thd2	Detail enhancement threshold (u, 10.0)

ISP_SHARPEN_SHOOTTHD

ISP_SHARPEN_SHOOTTHD is a sharpen shoot THD control register.

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Offset Address	Register Name	Total Reset Value
0x24244	ISP_SHARPEN_SHOOTTHD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	over_thd2								over_thd1								under_thd2								under_thd1							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	over_thd2	Overshoot control coefficient 2 (u, 8.0)
[23:16]	RW	over_thd1	Overshoot control coefficient 1 (u, 8.0)
[15:8]	RW	under_thd2	Undershoot control coefficient 2 (u, 8.0)
[7:0]	RW	under_thd1	Undershoot control coefficient 1 (u, 8.0)

ISP_SHARPEN_SHOOTCOEF

ISP_SHARPEN_SHOOTCOEF is a sharpen shoot COEF control register.

Offset Address	Register Name	Total Reset Value
0x24248	ISP_SHARPEN_SHOOTCOEF	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								over_coef								reserved				under_coef											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	over_coef	Overshoot control coefficient (u, 1.8)
[15:9]	RO	reserved	Reserved
[8:0]	RW	under_coef	Undershoot control coefficient (u, 1.8)

ISP_SHARPEN_ENPIXSEL

ISP_SHARPEN_ENPIXSEL is a sharpen enpixsel control register.

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Offset Address	Register Name	Total Reset Value
0x2424C	ISP_SHARPEN_ENPIXSEL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															enpixsel
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	enpixsel	Overshoot suppression 0: The original value is selected. 1: The maximum/minimum field value is selected.

ISP_NR_LSC_CFG

ISP_NR_LSC_CFG is an NR LSC enable register.

Offset Address	Register Name	Total Reset Value
0x24800	ISP_NR_LSC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	LSC enable 0: disabled 1: enabled

ISP_NR_LSC_WINNUM

ISP_NR_LSC_WINNUM is an NR LSC picture segmentation window quantity configuration register.

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Offset Address	Register Name	Total Reset Value
0x24810	ISP_NR_LSC_WINNUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												numv				reserved								numh							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	numv	Number of windows obtained after the picture is vertically segmented. The configured value is the actual value minus 1.
[15:5]	RO	reserved	Reserved
[4:0]	RW	numh	Number of windows obtained after the picture is horizontally segmented. The configured value is the actual value minus 1.

ISP_NR_LSC_WINX1

ISP_NR_LSC_WINX1 is an NR LSC interpolation horizontal first window width information register.

Offset Address	Register Name	Total Reset Value
0x24814	ISP_NR_LSC_WINX1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx1												reserved				deltax1											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx1	Reciprocal value of the width of the LSC interpolation horizontal first window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax1	Width of the LSC interpolation horizontal first window. The configured value is the actual value minus 1.

ISP_NR_LSC_WINX2

ISP_NR_LSC_WINX2 is an NR LSC interpolation horizontal second window width information register.

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Offset Address	Register Name	Total Reset Value
0x24818	ISP_NR_LSC_WINX2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx2												reserved				deltax2											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx2	Reciprocal value of the width of the LSC interpolation horizontal second window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax2	Width of the LSC interpolation horizontal second window

ISP_NR_LSC_WINX3

ISP_NR_LSC_WINX3 is an NR LSC interpolation horizontal third window width information register.

Offset Address	Register Name	Total Reset Value
0x2481C	ISP_NR_LSC_WINX3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx3												reserved				deltax3											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx3	Reciprocal value of the width of the LSC interpolation horizontal third window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax3	Width of the LSC interpolation horizontal third window

ISP_NR_LSC_WINX4

ISP_NR_LSC_WINX4 is an NR LSC interpolation horizontal fourth window width information register.

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Offset Address	Register Name	Total Reset Value
0x24820	ISP_NR_LSC_WINX4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx4												reserved				deltax4											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx4	Reciprocal value of the width of the LSC interpolation horizontal fourth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax4	Width of the LSC interpolation horizontal fourth window

ISP_NR_LSC_WINX5

ISP_NR_LSC_WINX5 is an NR LSC interpolation horizontal fifth window width information register.

Offset Address	Register Name	Total Reset Value
0x24824	ISP_NR_LSC_WINX5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx5												reserved				deltax5											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx5	Reciprocal value of the width of the LSC interpolation horizontal fifth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax5	Width of the LSC interpolation horizontal fifth window

ISP_NR_LSC_WINX6

ISP_NR_LSC_WINX6 is an NR LSC interpolation horizontal sixth window width information register.

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Offset Address	Register Name	Total Reset Value
0x24828	ISP_NR_LSC_WINX6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx6												reserved				deltax6											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx6	Reciprocal value of the width of the LSC interpolation horizontal sixth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax6	Width of the LSC interpolation horizontal sixth window

ISP_NR_LSC_WINX7

ISP_NR_LSC_WINX7 is an NR LSC interpolation horizontal seventh window width information register.

Offset Address	Register Name	Total Reset Value
0x2482C	ISP_NR_LSC_WINX7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx7												reserved				deltax7											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx7	Reciprocal value of the width of the LSC interpolation horizontal seventh window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax7	Width of the LSC interpolation horizontal seventh window

ISP_NR_LSC_WINX8

ISP_NR_LSC_WINX8 is an NR LSC interpolation horizontal eighth window width information register.

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Offset Address	Register Name	Total Reset Value
0x24830	ISP_NR_LSC_WINX8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx8												reserved				deltax8											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx8	Reciprocal value of the width of the LSC interpolation horizontal eighth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax8	Width of the LSC interpolation horizontal eighth window

ISP_NR_LSC_WINY1

ISP_NR_LSC_WINY1 is an NR LSC interpolation vertical first window width information register.

Offset Address	Register Name	Total Reset Value
0x24834	ISP_NR_LSC_WINY1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy1												reserved				deltay1											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy1	Reciprocal value of the width of the LSC interpolation vertical first window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay1	Width of the LSC interpolation vertical first window

ISP_NR_LSC_WINY2

ISP_NR_LSC_WINY2 is an NR LSC interpolation vertical second window width information register.

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Offset Address	Register Name	Total Reset Value
0x24838	ISP_NR_LSC_WINY2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy2												reserved								deltay2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy2	Reciprocal value of the width of the LSC interpolation vertical second window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay2	Width of the LSC interpolation vertical second window

ISP_NR_LSC_WINY3

ISP_NR_LSC_WINY3 is an NR LSC interpolation vertical third window width information register.

Offset Address	Register Name	Total Reset Value
0x2483C	ISP_NR_LSC_WINY3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy3												reserved								deltay3							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy3	Reciprocal value of the width of the LSC interpolation vertical third window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay3	Width of the LSC interpolation vertical third window

ISP_NR_LSC_WINY4

ISP_NR_LSC_WINY4 is an NR LSC interpolation vertical fourth window width information register.

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Offset Address	Register Name	Total Reset Value
0x24840	ISP_NR_LSC_WINY4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy4												reserved								deltay4							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy4	Reciprocal value of the width of the LSC interpolation vertical fourth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay4	Width of the LSC interpolation vertical fourth window

ISP_NR_LSC_WINY5

ISP_NR_LSC_WINY5 is an NR LSC interpolation vertical fifth window width information register.

Offset Address	Register Name	Total Reset Value
0x24844	ISP_NR_LSC_WINY5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy5												reserved								deltay5							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy5	Reciprocal value of the width of the LSC interpolation vertical fifth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay5	Width of the LSC interpolation vertical fifth window

ISP_NR_LSC_WINY6

ISP_NR_LSC_WINY6 is an NR LSC interpolation vertical sixth window width information register.

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Offset Address	Register Name	Total Reset Value
0x24848	ISP_NR_LSC_WINY6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy6												reserved								deltay6							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy6	Reciprocal value of the width of the LSC interpolation vertical sixth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay6	Width of the LSC interpolation vertical sixth window

ISP_NR_LSC_WINY7

ISP_NR_LSC_WINY7 is an NR LSC interpolation vertical seventh window width information register.

Offset Address	Register Name	Total Reset Value
0x2484C	ISP_NR_LSC_WINY7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy7												reserved								deltay7							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy7	Reciprocal value of the width of the LSC interpolation vertical seventh window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay7	Width of the LSC interpolation vertical seventh window

ISP_NR_LSC_WINY8

ISP_NR_LSC_WINY8 is an NR LSC interpolation vertical eighth window width information register.

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Offset Address	Register Name	Total Reset Value
0x24850	ISP_NR_LSC_WINY8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy8												reserved				deltay8											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy8	Reciprocal value of the width of the LSC interpolation vertical eighth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay8	Width of the LSC interpolation vertical eighth window

ISP_NR_LSC_GRRGAIN_WADDR

ISP_NR_LSC_GRRGAIN_WADDR is an NR LSC interpolation R/Gr channel gain write address register.

Offset Address	Register Name	Total Reset Value
0x24880	ISP_NR_LSC_GRRGAIN_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rgain_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rgain_waddr	Write address of the LSC interpolation R/Gr channel gain

ISP_NR_LSC_GRRGAIN_WDATA

ISP_NR_LSC_GRRGAIN_WDATA is an NR LSC interpolation R/Gr channel gain write data register.

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Offset Address	Register Name	Total Reset Value
0x24884	ISP_NR_LSC_GRRGAIN_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rgain_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rgain_wdata	Write data of the LSC interpolation R/Gr channel gain

ISP_NR_LSC_GRRGAIN_RADDR

ISP_NR_LSC_GRRGAIN_RADDR is an NR LSC interpolation R/Gr channel gain read address register.

Offset Address	Register Name	Total Reset Value
0x24888	ISP_NR_LSC_GRRGAIN_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rgain_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rgain_raddr	Read address of the LSC interpolation R/Gr channel gain

ISP_NR_LSC_GRRGAIN_RDATA

ISP_NR_LSC_GRRGAIN_RDATA is an NR LSC interpolation R/Gr channel gain read data register.

Offset Address	Register Name	Total Reset Value
0x2488C	ISP_NR_LSC_GRRGAIN_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	grgain_rdata	Read data of the LSC interpolation R/Gr channel gain

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ISP_NR_LSC_GBBGAIN_WADDR

ISP_NR_LSC_GBBGAIN_WADDR is an NR LSC interpolation B/Gb channel gain write address register.

Offset Address	Register Name	Total Reset Value
0x24890	ISP_NR_LSC_GBBGAIN_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	grgain_waddr	Write address of the LSC interpolation B/Gb channel gain

ISP_NR_LSC_GBBGAIN_WDATA

ISP_NR_LSC_GBBGAIN_WDATA is an NR LSC interpolation B/Gb channel gain write data register.

Offset Address	Register Name	Total Reset Value
0x24894	ISP_NR_LSC_GBBGAIN_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	grgain_wdata	Write data of the LSC interpolation B/Gb channel gain

ISP_NR_LSC_GBBGAIN_RADDR

ISP_NR_LSC_GBBGAIN_RADDR is an NR LSC interpolation B/Gb channel gain read address register.

Offset Address	Register Name	Total Reset Value
0x24898	ISP_NR_LSC_GBBGAIN_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	grgain_raddr	Read address of the LSC interpolation B/Gb channel gain

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ISP_NR_LSC_GBBGAIN_RDATA

ISP_NR_LSC_GBBGAIN_RDATA is an NR LSC interpolation B/Gb channel gain read data register.

Offset Address	Register Name	Total Reset Value
0x2489C	ISP_NR_LSC_GBBGAIN_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	grgain_rdata	Read data of the LSC interpolation B/Gb channel gain

ISP_DPC_CFG

ISP_DPC_CFG is a DPC control register.

Offset Address	Register Name	Total Reset Value
0x25000	ISP_DPC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	DPC enable 0: disabled 1: enabled

ISP_DPC_ALPHA

ISP_DPC_ALPHA is a DPC blending ratio register.

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Offset Address	Register Name	Total Reset Value
0x25008	ISP_DPC_ALPHA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	alpha0_rb								alpha0_g								alpha1_rb								alpha1_g							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	alpha0_rb	Blending ratio of the R/B component 5-point median filtering result to the 9-point median filtering result
[23:16]	RW	alpha0_g	Blending ratio of the G component 5-point median filtering result to the 9-point median filtering result
[15:8]	RW	alpha1_rb	Blending ratio of the R/B component input to the median filtering result
[7:0]	RW	alpha1_g	Blending ratio of the G component input to the median filtering result

ISP_DPC_MODE

ISP_DPC_MODE is a DPC mode register.

Offset Address	Register Name	Total Reset Value
0x25010	ISP_DPC_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						ir_channel	ir_position	dp_hightlight_en	dpt_det_sel	bpt_cor_en	grayscale_mode	bpt_update	six_det_en	cor_en	det_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	ir_channel	Format select 0: bayer format 1: RGBIR format
[8]	RW	ir_position	Channel select 0: The IR uses the Gb channel. 1: The IR uses the Gr channel.

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[7]	RW	dp_hightlight_en	Defect pixel highlight enable 0: disabled 1: enabled
[6]	RW	dpt_det_sel	Defect pixel table detection select 0: The bright dots are detected. 1: The dark dots are detected.
[5]	RW	bpt_cor_en	Defect pixel table correction enable 0: disabled 1: enabled
[4]	RW	grayscale_mode	Mode of the gray scale image 0: raw data 1: gray scale image
[3]	RW	bpt_update	Defect pixel table calibration enable 0: disabled 1: enabled
[2]	RW	six_det_en	Maximum/Minimum value detection, used only during defect pixel table calibration 0: disabled 1: enabled
[1]	RW	cor_en	Dynamic correction enable 0: disabled 1: enabled
[0]	RW	det_en	Dynamic detection enable 0: disabled 1: enabled

ISP_DPC_OUTPUT_MODE

ISP_DPC_OUTPUT_MODE is a dynamic correction mode register.

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Offset Address	Register Name	Total Reset Value
0x25014	ISP_DPC_OUTPUT_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												stage1_incl_rb_3x3	stage1_incl_g_3x3	stage1_incl_rb_center	stage1_incl_gr_center
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RW	stage1_incl_rb_3x3	Whether the R/B point correction uses the 9-point median 0: no 1: yes
[2]	RW	stage1_incl_g_3x3	Whether the G point correction uses the 9-point median 0: no 1: yes
[1]	RW	stage1_incl_rb_center	Whether the R/B point correction contains the central point 0: no 1: yes
[0]	RW	stage1_incl_gr_center	Whether the G point correction contains the central point 0: no 1: yes

ISP_DPC_SET_USE

ISP_DPC_SET_USE is a dynamic detection level register.

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Offset Address	Register Name	Total Reset Value
0x25018	ISP_DPC_SET_USE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												stage1_use_fix_set	stage1_use_set3	stage1_use_set2	stage1_use_set1
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RW	stage1_use_fix_set	Whether other detection levels are valid 0: no 1: yes
[2]	RW	stage1_use_set3	Whether detection level 3 is valid 0: no 1: yes
[1]	RW	stage1_use_set2	Whether detection level 2 is valid 0: no 1: yes
[0]	RW	stage1_use_set1	Whether detection level 1 is valid 0: no 1: yes

ISP_DPC_METHODS_SET_1

ISP_DPC_METHODS_SET_1 is detection switch register 1.

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Offset Address	Register Name	Total Reset Value
0x2501C	ISP_DPC_METHODS_SET_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			rg_red_blue1_enable	rnd_red_blue1_enable	ro_red_blue1_enable	lc_red_blue1_enable	pg_red_blue1_enable	reserved				rg_green1_enable	rnd_green1_enable	ro_green1_enable	lc_green1_enable	pg_green1_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	rg_red_blue1_enable	R/B point level 1 gradient sorting detection enable 0: disabled 1: enabled
[11]	RW	rnd_red_blue1_enable	R/B point level 1 adjacent difference sorting detection enable 0: disabled 1: enabled
[10]	RW	ro_red_blue1_enable	R/B point level 1 sorting detection enable 0: disabled 1: enabled
[9]	RW	lc_red_blue1_enable	R/B point level 1 linear detection enable 0: disabled 1: enabled
[8]	RW	pg_red_blue1_enable	R/B point level 1 peak value gradient detection enable 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved
[4]	RW	rg_green1_enable	G point level 1 gradient sorting detection enable 0: disabled 1: enabled
[3]	RW	rnd_green1_enable	G point level 1 adjacent difference sorting detection enable 0: disabled 1: enabled

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[2]	RW	ro_green1_enable	G point level 1 sorting detection enable 0: disabled 1: enabled
[1]	RW	lc_green1_enable	G point level 1 linear detection enable 0: disabled 1: enabled
[0]	RW	pg_green1_enable	G point level 1 peak value gradient detection enable 0: disabled 1: enabled

ISP_DPC_METHODS_SET_2

ISP_DPC_METHODS_SET_2 is detection switch register 2.

Offset Address	Register Name	Total Reset Value
0x25020	ISP_DPC_METHODS_SET_2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			rg_red_blue2_enable	rnd_red_blue2_enable	ro_red_blue2_enable	lc_red_blue2_enable	pg_red_blue2_enable	reserved			rg_green2_enable	rnd_green2_enable	ro_green2_enable	lc_green2_enable	pg_green2_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	rg_red_blue2_enable	R/B point level 2 gradient sorting detection enable 0: disabled 1: enabled
[11]	RW	rnd_red_blue2_enable	R/B point level 2 adjacent difference sorting detection enable 0: disabled 1: enabled
[10]	RW	ro_red_blue2_enable	R/B point level 2 sorting detection enable 0: disabled 1: enabled

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[9]	RW	lc_red_blue2_enable	R/B point level 2 linear detection enable 0: disabled 1: enabled
[8]	RW	pg_red_blue2_enable	R/B point level 2 peak value gradient detection enable 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved
[4]	RW	rg_green2_enable	G point level 2 gradient sorting detection enable 0: disabled 1: enabled
[3]	RW	rnd_green2_enable	G point level 2 adjacent difference sorting detection enable 0: disabled 1: enabled
[2]	RW	ro_green2_enable	G point level 2 sorting detection enable 0: disabled 1: enabled
[1]	RW	lc_green2_enable	G point level 2 linear detection enable 0: disabled 1: enabled
[0]	RW	pg_green2_enable	G point level 2 peak value gradient detection enable 0: disabled 1: enabled

ISP_DPC_METHODS_SET_3

ISP_DPC_METHODS_SET_3 is detection switch register 3.

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Offset Address	Register Name	Total Reset Value
0x25024	ISP_DPC_METHODS_SET_3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			rg_red_blue3_enable	rnd_red_blue3_enable	ro_red_blue3_enable	lc_red_blue3_enable	pg_red_blue3_enable	reserved				rg_green3_enable	rnd_green3_enable	ro_green3_enable	lc_green3_enable	pg_green3_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	rg_red_blue3_enable	R/B point level 3 gradient sorting detection enable 0: disabled 1: enabled
[11]	RW	rnd_red_blue3_enable	R/B point level 3 adjacent difference sorting detection enable 0: disabled 1: enabled
[10]	RW	ro_red_blue3_enable	R/B point level 3 sorting detection enable 0: disabled 1: enabled
[9]	RW	lc_red_blue3_enable	R/B point level 3 linear detection enable 0: disabled 1: enabled
[8]	RW	pg_red_blue3_enable	R/B point level 3 peak value gradient detection enable 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved
[4]	RW	rg_green3_enable	G point level 3 gradient sorting detection enable 0: disabled 1: enabled
[3]	RW	rnd_green3_enable	G point level 3 adjacent difference sorting detection enable 0: disabled 1: enabled

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[2]	RW	ro_green3_enable	G point level 3 sorting detection enable 0: disabled 1: enabled
[1]	RW	lc_green3_enable	G point level 3 linear detection enable 0: disabled 1: enabled
[0]	RW	pg_green3_enable	G point level 3 peak value gradient detection enable 0: disabled 1: enabled

ISP_DPC_LINE_THRESH_1

ISP_DPC_LINE_THRESH_1 is a linear detection parameter 1 register.

Offset Address	Register Name	Total Reset Value
0x25028	ISP_DPC_LINE_THRESH_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																line_thr1_rb								line_thr1_g							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	line_thr1_rb	R/B point detection parameter
[7:0]	RW	line_thr1_g	G point detection parameter

ISP_DPC_LINE_MAD_FAC_1

ISP_DPC_LINE_MAD_FAC_1 is a linear detection parameter 1 register.

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Offset Address	Register Name	Total Reset Value
0x2502C	ISP_DPC_LINE_MAD_FAC_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				line_mad_fac_1_rb				reserved				line_mad_fac_1_g			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	line_mad_fac_1_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	line_mad_fac_1_g	G point detection parameter

ISP_DPC_PG_FAC_1

ISP_DPC_PG_FAC_1 is a peak gradient parameter 1 register.

Offset Address	Register Name	Total Reset Value
0x25030	ISP_DPC_PG_FAC_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				pg_fac_1_rb				reserved				pg_fac_1_g			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	pg_fac_1_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	pg_fac_1_g	G point detection parameter

ISP_DPC_RND_THRESH_1

ISP_DPC_RND_THRESH_1 is an adjacent difference value sorting parameter 1 register.

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Offset Address	Register Name	Total Reset Value
0x25034	ISP_DPC_RND_THRESH_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rnd_thr1_rb								rnd_thr1_g							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	rnd_thr1_rb	R/B point detection parameter
[7:0]	RW	rnd_thr1_g	G point detection parameter

ISP_DPC_RG_FAC_1

ISP_DPC_RG_FAC_1 is a gradient sorting parameter 1 register.

Offset Address	Register Name	Total Reset Value
0x25038	ISP_DPC_RG_FAC_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				rg_fac_1_rb				reserved				rg_fac_1_g			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	rg_fac_1_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	rg_fac_1_g	G point detection parameter

ISP_DPC_LINE_THRESH_2

ISP_DPC_LINE_THRESH_2 is a linear detection parameter 2 register.

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Offset Address	Register Name	Total Reset Value
0x2503C	ISP_DPC_LINE_THRESH_2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																line_thr2_rb								line_thr2_g							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	line_thr2_rb	R/B point detection parameter
[7:0]	RW	line_thr2_g	G point detection parameter

ISP_DPC_LINE_MAD_FAC_2

ISP_DPC_LINE_MAD_FAC_2 is a linear detection parameter 2 register.

Offset Address	Register Name	Total Reset Value
0x25040	ISP_DPC_LINE_MAD_FAC_2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				line_mad_fac_2_rb				reserved				line_mad_fac_2_g			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	line_mad_fac_2_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	line_mad_fac_2_g	G point detection parameter

ISP_DPC_PG_FAC_2

ISP_DPC_PG_FAC_2 is a peak gradient parameter 2 register.

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Offset Address	Register Name	Total Reset Value
0x25044	ISP_DPC_PG_FAC_2	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				pg_fac_2_rb		reserved	pg_fac_2_g
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	pg_fac_2_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	pg_fac_2_g	G point detection parameter

ISP_DPC_RND_THRESH_2

ISP_DPC_RND_THRESH_2 is an adjacent difference value sorting parameter 2 register.

Offset Address	Register Name	Total Reset Value
0x25048	ISP_DPC_RND_THRESH_2	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				rnd_thr2_rb		rnd_thr2_g	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	rnd_thr2_rb	R/B point detection parameter
[7:0]	RW	rnd_thr2_g	G point detection parameter

ISP_DPC_RG_FAC_2

ISP_DPC_RG_FAC_2 is a gradient sorting parameter 2 register.

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Offset Address	Register Name	Total Reset Value
0x2504C	ISP_DPC_RG_FAC_2	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				rg_fac_2_rb		reserved	rg_fac_2_g
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	rg_fac_2_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	rg_fac_2_g	G point detection parameter

ISP_DPC_LINE_THRESH_3

ISP_DPC_LINE_THRESH_3 is a linear detection parameter 3 register.

Offset Address	Register Name	Total Reset Value
0x25050	ISP_DPC_LINE_THRESH_3	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				line_thr3_rb		line_thr3_g	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	line_thr3_rb	R/B point detection parameter
[7:0]	RW	line_thr3_g	G point detection parameter

ISP_DPC_LINE_MAD_FAC_3

ISP_DPC_LINE_MAD_FAC_3 is a linear detection parameter 3 register.

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Offset Address	Register Name	Total Reset Value
0x25054	ISP_DPC_LINE_MAD_FAC_3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																		line_mad_fac_3_rb								reserved				line_mad_fac_3_g			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	line_mad_fac_3_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	line_mad_fac_3_g	G point detection parameter

ISP_DPC_PG_FAC_3

ISP_DPC_PG_FAC_3 is a peak gradient parameter 3 register.

Offset Address	Register Name	Total Reset Value
0x25058	ISP_DPC_PG_FAC_3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																		pg_fac_3_rb								reserved				pg_fac_3_g			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	pg_fac_3_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	pg_fac_3_g	G point detection parameter

ISP_DPC_RND_THRESH_3

ISP_DPC_RND_THRESH_3 is an adjacent difference value sorting parameter 3 register.

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Offset Address	Register Name	Total Reset Value
0x2505C	ISP_DPC_RND_THRESH_3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rnd_thr3_rb								rnd_thr3_g							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	rnd_thr3_rb	R/B point detection parameter
[7:0]	RW	rnd_thr3_g	G point detection parameter

ISP_DPC_RG_FAC_3

ISP_DPC_RG_FAC_3 is a gradient sorting parameter 3 register.

Offset Address	Register Name	Total Reset Value
0x25060	ISP_DPC_RG_FAC_3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				rg_fac_3_rb				reserved				rg_fac_3_g			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	rg_fac_3_rb	R/B point detection parameter
[7:6]	RO	reserved	Reserved
[5:0]	RW	rg_fac_3_g	G point detection parameter

ISP_DPC_RO_LIMITS

ISP_DPC_RO_LIMITS is a sorting parameter 3 register.

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Offset Address	Register Name	Total Reset Value
0x25064	ISP_DPC_RO_LIMITS	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10	9 8	7 6	5 4	3 2	1 0
Name	reserved					ro_3_rb	ro_3_g	ro_2_rb	ro_2_g	ro_1_rb	ro_1_g
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0	0 0	0 0	0 0	0 0	0 0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:10]	RW	ro_3_rb	R/B point detection parameter
[9:8]	RW	ro_3_g	G point detection parameter
[7:6]	RW	ro_2_rb	R/B point detection parameter
[5:4]	RW	ro_2_g	G point detection parameter
[3:2]	RW	ro_1_rb	R/B point detection parameter
[1:0]	RW	ro_1_g	G point detection parameter

ISP_DPC_RND_OFFS

ISP_DPC_RND_OFFS is an adjacent difference value sorting parameter 3 register.

Offset Address	Register Name	Total Reset Value
0x25068	ISP_DPC_RND_OFFS	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10	9 8	7 6	5 4	3 2	1 0
Name	reserved					rnd_offs_3_rb	rnd_offs_3_g	rnd_offs_2_rb	rnd_offs_2_g	rnd_offs_1_rb	rnd_offs_1_g
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0	0 0	0 0	0 0	0 0	0 0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:10]	RW	rnd_offs_3_rb	R/B point detection parameter
[9:8]	RW	rnd_offs_3_g	G point detection parameter
[7:6]	RW	rnd_offs_2_rb	R/B point detection parameter

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[5:4]	RW	rnd_offs_2_g	G point detection parameter
[3:2]	RW	rnd_offs_1_rb	R/B point detection parameter
[1:0]	RW	rnd_offs_1_g	G point detection parameter

ISP_DPC_BPT_CTRL

ISP_DPC_BPT_CTRL is a defect pixel table control register.

Offset Address	Register Name	Total Reset Value
0x2506C	ISP_DPC_BPT_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bp_data																reserved				rb_3x3	g_3x3	bpt_incl_rb_center	bpt_incl_green_center	bpt_use_fix_set	bpt_use_set_3	bpt_use_set_2	bpt_use_set_1	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	bp_data	Defect pixel highlight value
[15:12]	RO	reserved	Reserved
[11]	RW	rb_3x3	Whether the R/B point correction uses the 9-point median 0: no 1: yes
[10]	RW	g_3x3	Whether the G point correction uses the 9-point median 0: no 1: yes
[9]	RW	bpt_incl_rb_center	Whether the R/B point correction contains the central point 0: no 1: yes
[8]	RW	bpt_incl_green_center	Whether the G point correction contains the central point 0: no 1: yes

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[7]	RW	bpt_use_fix_set	Whether other detection levels are valid 0: no 1: yes
[6]	RW	bpt_use_set_3	Whether detection level 3 is valid 0: no 1: yes
[5]	RW	bpt_use_set_2	Whether detection level 2 is valid 0: no 1: yes
[4]	RW	bpt_use_set_1	Whether detection level 1 is valid 0: no 1: yes
[3:0]	RO	reserved	Reserved

ISP_DPC_BPT_NUMBER

ISP_DPC_BPT_NUMBER is a defect pixel number register.

Offset Address	Register Name	Total Reset Value
0x25070	ISP_DPC_BPT_NUMBER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				bpt_number											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	bpt_number	Number of defect pixels. The value range is 0−2048.

ISP_DPC_BPT_CALIB_NUMBER

ISP_DPC_BPT_CALIB_NUMBER is a calibrated defect pixel number register.

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Offset Address	Register Name	Total Reset Value
0x25074	ISP_DPC_BPT_CALIB_NUMBER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				bpt_calib_number											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RO	bpt_calib_number	Number of calibrated defect pixels. The value range is 0–2048.

ISP_DPC_BPT_THRD

ISP_DPC_BPT_THRD is a calibrated defect pixel parameter register.

Offset Address	Register Name	Total Reset Value
0x2507C	ISP_DPC_BPT_THRD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	abs_hot_thresh								dev_hot_thresh								abs_dead_thresh								dev_dead_thresh							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	abs_hot_thresh	Absolute threshold for the bright defect pixel (U8.0). The current pixel is not detected if its value is less than this threshold.
[23:16]	RW	dev_hot_thresh	Offset threshold percentage for the bright defect pixel (U8.0). If the ratio of the absolute value of the difference between the value of the current pixel and the average value of its neighboring pixels to the value of the current pixel is less than or equal to this threshold, the current pixel is considered a non-defect pixel. The field value indicates a percentage number, and the value 128 indicates 100%.
[15:8]	RW	abs_dead_thresh	Absolute threshold for the dark defect pixel (U8.0). The current pixel is considered a non-defect pixel if its value is greater than this threshold.
[7:0]	RW	dev_dead_thresh	Offset threshold percentage for the dark defect pixel (U8.0). If the ratio of the absolute value of the difference between the value of the current pixel and the average value of its neighboring pixels to the average value of its neighboring pixels is less than or equal to this threshold, the current pixel is considered a non-defect pixel. The field value indicates a percentage number, and the value 128 indicates 100%.

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ISP_DPC_BPT_WADDR

ISP_DPC_BPT_WADDR is a defect pixel table write address register.

Offset Address	Register Name	Total Reset Value
0x25080	ISP_DPC_BPT_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bpt_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	bpt_waddr	Write address of the defect pixel table (bit [10:0])

ISP_DPC_BPT_WDATA

ISP_DPC_BPT_WDATA is a defect pixel table write data register.

Offset Address	Register Name	Total Reset Value
0x25084	ISP_DPC_BPT_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bpt_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	bpt_wdata	Write data of the defect pixel table. Bit [24:0] indicate the defect pixel coordinate.

ISP_DPC_BPT_RADDR

ISP_DPC_BPT_RADDR is a defect pixel table read address register.

Offset Address	Register Name	Total Reset Value
0x25088	ISP_DPC_BPT_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bpt_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	bpt_raddr	Read address of the defect pixel table (bit [10:0])

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ISP_DPC_BPT_RDATA

ISP_DPC_BPT_RDATA is a defect pixel table read data register.

Offset Address	Register Name	Total Reset Value
0x2508C	ISP_DPC_BPT_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bpt_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	bpt_rdata	Read data of the defect pixel table. Bit [24:0] indicate the defect pixel coordinate.

ISP_GE_CFG

ISP_GE_CFG is a GE control register.

Offset Address	Register Name	Total Reset Value
0x25100	ISP_GE_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																							isp_ge	reserved				isp_ge_aver	reserved				en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:9]	RO	reserved	Reserved
[8]	RW	isp_ge	GE correction enable 0: disabled 1: enabled
[7:5]	RO	reserved	Reserved
[4]	RW	isp_ge_aver	GE_AVER enable 0: disabled 1: enabled
[3:1]	RO	reserved	Reserved

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[0]	RW	en	GE enable 0: disabled 1: enabled

ISP_GE_CT_TH1

ISP_GE_CT_TH1 is a GE configuration register.

Offset Address	Register Name	Total Reset Value
0x25110	ISP_GE_CT_TH1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				ct_th1											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:0]	RW	ct_th1	Threshold

ISP_GE_CT_TH2

ISP_GE_CT_TH2 is a GE configuration register.

Offset Address	Register Name	Total Reset Value
0x25114	ISP_GE_CT_TH2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				ct_th2											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:0]	RW	ct_th2	Threshold

ISP_GE_CT_TH3

ISP_GE_CT_TH3 is a GE configuration register.

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Offset Address	Register Name	Total Reset Value
0x25118	ISP_GE_CT_TH3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				ct_th3											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:0]	RW	ct_th3	Threshold

ISP_GE_CT_SLOPE1

ISP_GE_CT_SLOPE1 is a GE configuration register.

Offset Address	Register Name	Total Reset Value
0x2511C	ISP_GE_CT_SLOPE1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												ct_slope1			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	ct_slope1	0x0: 1 0x1: 2 0x2: 4 0x3: 8 … This field can be set to **14** at most, and the value 14 indicates 2^{14}.

ISP_GE_CT_SLOPE2

ISP_GE_CT_SLOPE2 is a GE configuration register.

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Offset Address	Register Name	Total Reset Value
0x25120	ISP_GE_CT_SLOPE2	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved							ct_slope2
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	ct_slope2	0x0: 1 0x1: 2 0x2: 4 0x3: 8 … This field can be set to **14** at most, and the value 14 indicates 2^{14}.

ISP_GE_ZONE

ISP_GE_ZONE is a GE block configuration register.

Offset Address	Register Name	Total Reset Value
0x25130	ISP_GE_ZONE	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved					vnum	reserved	hnum
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12:8]	RW	vnum	Number of vertical blocks
[7:5]	RO	reserved	Reserved
[4:0]	RW	hnum	Number of horizontal blocks

ISP_GE_MEM_AVER_RADDR

ISP_GE_MEM_AVER_RADDR is a GE average value statistics read address register.

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Offset Address	Register Name	Total Reset Value
0x25188	ISP_GE_MEM_AVER_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	aver_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	aver_raddr	Read address of the average value statistics

ISP_GE_MEM_AVER_RDATA

ISP_GE_MEM_AVER_RDATA is a GE average value statistics read data register.

Offset Address	Register Name	Total Reset Value
0x2518C	ISP_GE_MEM_AVER_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	aver_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	aver_rdata	Read data of the average value statistics

ISP_FPN_CFG

ISP_FPN_CFG is an FPN configuration register.

Offset Address	Register Name	Total Reset Value
0x25200	ISP_FPN_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						line_frame	calib_corr	reserved							enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved

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[9]	RW	line_frame	FPN frame/line mode 0: frame mode 1: line mode
[8]	RW	calib_corr	FPN correction/calibration mode 0: correction mode 1: calibration mode
[7:1]	RO	reserved	Reserved
[0]	RW	enable	FPN enable 0: disabled 1: enabled

ISP_FPN_CALIB_START

ISP_FPN_CALIB_START is an FPN calibration start signal register.

Offset Address	Register Name	Total Reset Value
0x25204	ISP_FPN_CALIB_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															calib_start
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	calib_start	FPN calibration start. This bit is automatically cleared.

ISP_FPN_CORR_CFG

ISP_FPN_CORR_CFG is an FPN correction enable register.

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Offset Address	Register Name	Total Reset Value
0x25208	ISP_FPN_CORR_CFG	0x0000_000F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												correct3_en		correct2_en		correct1_en		correct0_en	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		1		1		1	

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	RW	correct3_en	FPN correction enable for channel 3 0: disabled 1: enabled
[2]	RW	correct2_en	FPN correction enable for channel 2 0: disabled 1: enabled
[1]	RW	correct1_en	FPN correction enable for channel 1 0: disabled 1: enabled
[0]	RW	correct0_en	FPN correction enable for channel 0 0: disabled 1: enabled

ISP_FPN_STAT

ISP_FPN_STAT is an FPN calibration status register.

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Offset Address	Register Name	Total Reset Value
0x2520C	ISP_FPN_STAT	0x1FFF_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				hcnt												reserved				vcnt				reserved							busy
Reset	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RO	hcnt	Number of lines
[15:14]	RO	reserved	Reserved
[13:8]	RO	vcnt	Number of calibrated frames
[7:1]	RO	reserved	Reserved
[0]	RO	busy	FPN calibration status 0: idle 1: being calibrated

ISP_FPN_WHITE_LEVEL

ISP_FPN_WHITE_LEVEL is an FPN calibration white point configuration register.

Offset Address	Register Name	Total Reset Value
0x25210	ISP_FPN_WHITE_LEVEL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				white_level											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:0]	RW	white_level	FPN calibration white point configuration

ISP_FPN_DIVCOEF

ISP_FPN_DIVCOEF is an FPN calibration division coefficient register.

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Offset Address	Register Name	Total Reset Value
0x25218	ISP_FPN_DIVCOEF	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				divcoef											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	divcoef	FPN calibration division coefficient

ISP_FPN_FRAMELOG2

ISP_FPN_FRAMELOG2 is an FPN calibrated frame number register.

Offset Address	Register Name	Total Reset Value
0x2521C	ISP_FPN_FRAMELOG2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														cpi_fpn_framelog2	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	cpi_fpn_framelog2	Number of FPN calibrated frames. The configured value is obtained by taking the logarithm of the number of frames to 2.

ISP_FPN_SUM0

ISP_FPN_SUM0 is an FPN calibration accumulated sum lower bits register.

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Offset Address	Register Name	Total Reset Value
0x25220	ISP_FPN_SUM0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sum																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sum	Lower bits of the FPN calibration accumulated sum

ISP_FPN_SUM1

ISP_FPN_SUM1 is an FPN calibration accumulated sum upper bits register.

Offset Address	Register Name	Total Reset Value
0x25224	ISP_FPN_SUM1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sum																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sum	Upper bits of the FPN calibration accumulated sum

ISP_FPN_CORR0

ISP_FPN_CORR0 is FPN correction configuration register 0.

Offset Address	Register Name	Total Reset Value
0x25230	ISP_FPN_CORR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	strength																reserved				offset											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	strength	FPN correction strength, U8.8
[15:12]	RO	reserved	Reserved
[11:0]	RW	offset	FPN correction bias

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ISP_FPN_CORR1

ISP_FPN_CORR1 is FPN correction configuration register 1.

Offset Address	Register Name	Total Reset Value
0x25234	ISP_FPN_CORR1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	strength																reserved				offset											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	strength	FPN correction strength, U8.8
[15:12]	RO	reserved	Reserved
[11:0]	RW	offset	FPN correction bias

ISP_FPN_CORR2

ISP_FPN_CORR2 is FPN correction configuration register 2.

Offset Address	Register Name	Total Reset Value
0x25238	ISP_FPN_CORR2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	strength																reserved				offset											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	strength	FPN correction strength, U8.8
[15:12]	RO	reserved	Reserved
[11:0]	RW	offset	FPN correction bias

ISP_FPN_CORR3

ISP_FPN_CORR3 is FPN correction configuration register 3.

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Offset Address	Register Name	Total Reset Value
0x2523C	ISP_FPN_CORR3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	strength																reserved				offset											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	strength	FPN correction strength, U8.8
[15:12]	RO	reserved	Reserved
[11:0]	RW	offset	FPN correction bias

ISP_FPN_SHIFT

ISP_FPN_SHIFT is an FPN shift configuration register.

Offset Address	Register Name	Total Reset Value
0x25240	ISP_FPN_SHIFT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				frame_calib_shift				reserved				out_shift				reserved				in_shift				reserved				fpn_shift			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:24]	RW	frame_calib_shift	Number of left shift bits for the last output frame during frame calibration
[23:20]	RO	reserved	Reserved
[19:16]	RW	out_shift	Number of left shift bits for the output
[15:12]	RO	reserved	Reserved
[11:8]	RW	in_shift	Number of right shift bits for the input
[7:4]	RO	reserved	Reserved
[3:0]	RW	fpn_shift	Number of right shift bits for the input FPN

ISP_FPN_MAX_O

ISP_FPN_MAX_O is an FPN maximum output value register.

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Offset Address	Register Name	Total Reset Value
0x25250	ISP_FPN_MAX_O	0x0000_3FFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																max_o															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	max_o	Maximum value of the FPN output

ISP_FPN_OVERFLOWTHR

ISP_FPN_OVERFLOWTHR is an FPN correction threshold register.

Offset Address	Register Name	Total Reset Value
0x25254	ISP_FPN_OVERFLOWTHR	0x0000_3FFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				overflowthr											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:0]	RW	overflowthr	FPN correction threshold

ISP_FPN_LINE_WADDR

ISP_FPN_LINE_WADDR is an FPN line mode black line write address register.

Offset Address	Register Name	Total Reset Value
0x25280	ISP_FPN_LINE_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fpn_line_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	fpn_line_waddr	Write address of the black line in FPN line mode

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ISP_FPN_LINE_WDATA

ISP_FPN_LINE_WDATA is an FPN line mode black line write data register.

Offset Address	Register Name	Total Reset Value
0x25284	ISP_FPN_LINE_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fpn_line_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	fpn_line_wdata	Write data of the black line in FPN line mode

ISP_FPN_LINE_RADDR

ISP_FPN_LINE_RADDR is an FPN line mode black line read address register.

Offset Address	Register Name	Total Reset Value
0x25288	ISP_FPN_LINE_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fpn_line_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	fpn_line_raddr	Read address of the black line in FPN line mode

ISP_FPN_LINE_RDATA

ISP_FPN_LINE_RDATA is an FPN line mode black line read data register.

Offset Address	Register Name	Total Reset Value
0x2528C	ISP_FPN_LINE_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	fpn_line_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	fpn_line_rdata	Read data of the black line in FPN line mode

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ISP_LSC_CFG

ISP_LSC_CFG is an LSC enable register.

Offset Address	Register Name	Total Reset Value
0x25800	ISP_LSC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	LSC enable 0: disabled 1: enabled

ISP_LSC_WINNUM

ISP_LSC_WINNUM is an LSC picture segmentation window quantity configuration register.

Offset Address	Register Name	Total Reset Value
0x25810	ISP_LSC_WINNUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												numv				reserved								numh							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	numv	Number of windows obtained after the picture is vertically segmented. The configured value is the actual value minus 1.
[15:5]	RO	reserved	Reserved
[4:0]	RW	numh	Number of windows obtained after the picture is horizontally segmented. The configured value is the actual value minus 1.

ISP_LSC_WINX1

ISP_LSC_WINX1 is an LSC interpolation horizontal first window width information register.

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Offset Address	Register Name	Total Reset Value
0x25814	ISP_LSC_WINX1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx1												reserved				deltax1											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx1	Reciprocal value of the width of the LSC interpolation horizontal first window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax1	Width of the LSC interpolation horizontal first window. The configured value is the actual value minus 1.

ISP_LSC_WINX2

ISP_LSC_WINX2 is an LSC interpolation horizontal second window width information register.

Offset Address	Register Name	Total Reset Value
0x25818	ISP_LSC_WINX2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx2												reserved				deltax2											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx2	Reciprocal value of the width of the LSC interpolation horizontal second window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax2	Width of the LSC interpolation horizontal second window

ISP_LSC_WINX3

ISP_LSC_WINX3 is an LSC interpolation horizontal third window width information register.

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Offset Address	Register Name	Total Reset Value
0x2581C	ISP_LSC_WINX3	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	invx3			reserved		deltax3	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx3	Reciprocal value of the width of the LSC interpolation horizontal third window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax3	Width of the LSC interpolation horizontal third window

ISP_LSC_WINX4

ISP_LSC_WINX4 is an LSC interpolation horizontal fourth window width information register.

Offset Address	Register Name	Total Reset Value
0x25820	ISP_LSC_WINX4	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	invx4			reserved		deltax4	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx4	Reciprocal value of the width of the LSC interpolation horizontal fourth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax4	Width of the LSC interpolation horizontal fourth window

ISP_LSC_WINX5

ISP_LSC_WINX5 is an LSC interpolation horizontal fifth window width information register.

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Offset Address	Register Name	Total Reset Value
0x25824	ISP_LSC_WINX5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx5												reserved				deltax5											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx5	Reciprocal value of the width of the LSC interpolation horizontal fifth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax5	Width of the LSC interpolation horizontal fifth window

ISP_LSC_WINX6

ISP_LSC_WINX6 is an LSC interpolation horizontal sixth window width information register.

Offset Address	Register Name	Total Reset Value
0x25828	ISP_LSC_WINX6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invx6												reserved				deltax6											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx6	Reciprocal value of the width of the LSC interpolation horizontal sixth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax6	Width of the LSC interpolation horizontal sixth window

ISP_LSC_WINX7

ISP_LSC_WINX7 is an LSC interpolation horizontal seventh window width information register.

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Offset Address	Register Name	Total Reset Value
0x2582C	ISP_LSC_WINX7	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	invx7			reserved		deltax7	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx7	Reciprocal value of the width of the LSC interpolation horizontal seventh window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax7	Width of the LSC interpolation horizontal seventh window

ISP_LSC_WINX8

ISP_LSC_WINX8 is an LSC interpolation horizontal eighth window width information register.

Offset Address	Register Name	Total Reset Value
0x25830	ISP_LSC_WINX8	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	invx8			reserved		deltax8	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invx8	Reciprocal value of the width of the LSC interpolation horizontal eighth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltax8	Width of the LSC interpolation horizontal eighth window

ISP_LSC_WINY1

ISP_LSC_WINY1 is an LSC interpolation vertical first window width information register.

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Offset Address	Register Name	Total Reset Value
0x25834	ISP_LSC_WINY1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	invy1			reserved		deltay1	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy1	Reciprocal value of the width of the LSC interpolation vertical first window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay1	Width of the LSC interpolation vertical first window

ISP_LSC_WINY2

ISP_LSC_WINY2 is an LSC interpolation vertical second window width information register.

Offset Address	Register Name	Total Reset Value
0x25838	ISP_LSC_WINY2	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	invy2			reserved		deltay2	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy2	Reciprocal value of the width of the LSC interpolation vertical second window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay2	Width of the LSC interpolation vertical second window

ISP_LSC_WINY3

ISP_LSC_WINY3 is an LSC interpolation vertical third window width information register.

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Offset Address	Register Name	Total Reset Value
0x2583C	ISP_LSC_WINY3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy3												reserved				deltay3											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy3	Reciprocal value of the width of the LSC interpolation vertical third window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay3	Width of the LSC interpolation vertical third window

ISP_LSC_WINY4

ISP_LSC_WINY4 is an LSC interpolation vertical fourth window width information register.

Offset Address	Register Name	Total Reset Value
0x25840	ISP_LSC_WINY4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy4												reserved				deltay4											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy4	Reciprocal value of the width of the LSC interpolation vertical fourth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay4	Width of the LSC interpolation vertical fourth window

ISP_LSC_WINY5

ISP_LSC_WINY5 is an LSC interpolation vertical fifth window width information register.

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Offset Address	Register Name	Total Reset Value
0x25844	ISP_LSC_WINY5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy5												reserved				deltay5											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy5	Reciprocal value of the width of the LSC interpolation vertical fifth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay5	Width of the LSC interpolation vertical fifth window

ISP_LSC_WINY6

ISP_LSC_WINY6 is an LSC interpolation vertical sixth window width information register.

Offset Address	Register Name	Total Reset Value
0x25848	ISP_LSC_WINY6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy6												reserved				deltay6											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy6	Reciprocal value of the width of the LSC interpolation vertical sixth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay6	Width of the LSC interpolation vertical sixth window

ISP_LSC_WINY7

ISP_LSC_WINY7 is an LSC interpolation vertical seventh window width information register.

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Offset Address	Register Name	Total Reset Value
0x2584C	ISP_LSC_WINY7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy7												reserved				deltay7											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy7	Reciprocal value of the width of the LSC interpolation vertical seventh window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay7	Width of the LSC interpolation vertical seventh window

ISP_LSC_WINY8

ISP_LSC_WINY8 is an LSC interpolation vertical eighth window width information register.

Offset Address	Register Name	Total Reset Value
0x25850	ISP_LSC_WINY8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				invy8												reserved				deltay8											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27:16]	RW	invy8	Reciprocal value of the width of the LSC interpolation vertical eighth window
[15:10]	RO	reserved	Reserved
[9:0]	RW	deltay8	Width of the LSC interpolation vertical eighth window

ISP_LSC_GRRGAIN_WADDR

ISP_LSC_GRRGAIN_WADDR is an LSC interpolation R/Gr channel gain write address register.

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Offset Address	Register Name	Total Reset Value
0x25880	ISP_LSC_GRRGAIN_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rgain_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rgain_waddr	Write address of the LSC interpolation R/Gr channel gain

ISP_LSC_GRRGAIN_WDATA

ISP_LSC_GRRGAIN_WDATA is an LSC interpolation R/Gr channel gain write data register.

Offset Address	Register Name	Total Reset Value
0x25884	ISP_LSC_GRRGAIN_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rgain_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rgain_wdata	Write data of the LSC interpolation R/Gr channel gain

ISP_LSC_GRRGAIN_RADDR

ISP_LSC_GRRGAIN_RADDR is an LSC interpolation R/Gr channel gain read address register.

Offset Address	Register Name	Total Reset Value
0x25888	ISP_LSC_GRRGAIN_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rgain_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rgain_raddr	Read address of the LSC interpolation R/Gr channel gain

ISP_LSC_GRRGAIN_RDATA

ISP_LSC_GRRGAIN_RDATA is an LSC interpolation R/Gr channel gain read data register.

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Offset Address	Register Name	Total Reset Value
0x2588C	ISP_LSC_GRRGAIN_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	grgain_rdata	Read data of the LSC interpolation R/Gr channel gain

ISP_LSC_GBBGAIN_WADDR

ISP_LSC_GBBGAIN_WADDR is an LSC interpolation B/Gb channel gain write address register.

Offset Address	Register Name	Total Reset Value
0x25890	ISP_LSC_GBBGAIN_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	grgain_waddr	Write address of the LSC interpolation B/Gb channel gain

ISP_LSC_GBBGAIN_WDATA

ISP_LSC_GBBGAIN_WDATA is an LSC interpolation B/Gb channel gain write data register.

Offset Address	Register Name	Total Reset Value
0x25894	ISP_LSC_GBBGAIN_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	grgain_wdata	Write data of the LSC interpolation B/Gb channel gain

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ISP_LSC_GBBGAIN_RADDR

ISP_LSC_GBBGAIN_RADDR is an LSC interpolation B/Gb channel gain read address register.

Offset Address	Register Name	Total Reset Value
0x25898	ISP_LSC_GBBGAIN_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	grgain_raddr	Read address of the LSC interpolation B/Gb channel gain

ISP_LSC_GBBGAIN_RDATA

ISP_LSC_GBBGAIN_RDATA is an LSC interpolation B/Gb channel gain read data register.

Offset Address	Register Name	Total Reset Value
0x2589C	ISP_LSC_GBBGAIN_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	grgain_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	grgain_rdata	Read data of the LSC interpolation B/Gb channel gain

ISP_DRC_CFG

ISP_DRC_CFG is a DRC configuration register.

Offset Address	Register Name	Total Reset Value
0x26200	ISP_DRC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															drc_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	RW	drc_en	DRC module enable 0: disabled 1: enabled

ISP_DRC_ZONE

ISP_DRC_ZONE is a DRC zone configuration register.

Offset Address	Register Name	Total Reset Value
0x26210	ISP_DRC_ZONE	0x0000_1220

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				vnum				reserved				hnum			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:8]	RW	vnum	DRC vertical zone configuration NOTE 2 ≤ vnum ≤ 24; vnum x hnum ≤ 768
[7:6]	RO	reserved	Reserved
[5:0]	RW	hnum	DRC horizontal zone configuration NOTE 2 ≤ hnum ≤ 40; vnum x hnum ≤ 768

ISP_DRC_ZONE_SIZE

ISP_DRC_ZONE_SIZE is a DRC block size register.

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Offset Address	Register Name	Total Reset Value
0x26214	ISP_DRC_ZONE_SIZE	0x003B_003B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				chk_y				vsize								reserved				chk_x				hsize							
Reset	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	1

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:24]	RW	chk_y	The block sizes are unequal. chk_y = vsize%vnum
[23:16]	RW	vsize	Height of each DRC block NOTE The configured value is the actual value minus 1.
[15:14]	RO	reserved	Reserved
[13:8]	RW	chk_x	The block sizes are unequal. chk_x = hsize%hnum
[7:0]	RW	hsize	Width of each DRC block NOTE The configured value is the actual value minus 1.

ISP_DRC_ZONE_DIV0

ISP_DRC_ZONE_DIV0 is a DRC block division coefficient register.

Offset Address	Register Name	Total Reset Value
0x26218	ISP_DRC_ZONE_DIV0	0x0222_0222

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				div_y												reserved				div_x											
Reset	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RW	div_y	Reciprocal value of the height of each DRC block div_y = 32768/vsize

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[15]	RO	reserved	Reserved
[14:0]	RW	div_x	Reciprocal value of the width of each DRC block div_x = 32768/hsize

ISP_DRC_ZONE_DIV1

ISP_DRC_ZONE_DIV1 is a DRC block division coefficient register.

Offset Address	Register Name	Total Reset Value
0x2621C	ISP_DRC_ZONE_DIV1	0x0222_0222

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				div_y												reserved				div_x											
Reset	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:16]	RW	div_y	Reciprocal value of the height of each DRC block div_y = 32768/vsize
[15]	RO	reserved	Reserved
[14:0]	RW	div_x	Reciprocal value of the width of each DRC block div_x = 32768/hsize

ISP_DRC_BIN

ISP_DRC_BIN is a DRC block histogram segment number register.

Offset Address	Register Name	Total Reset Value
0x26220	ISP_DRC_BIN	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												bin			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved

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[4:0]	RW	bin	Number of required histogram segments for each DRC block NOTE bin ≥ 8

ISP_DRC_BIN_DIV

ISP_DRC_BIN_DIV is a DRC histogram division coefficient register.

Offset Address	Register Name	Total Reset Value
0x26224	ISP_DRC_BIN_DIV	0x0000_0800

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				div_z											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14:0]	RW	div_z	Reciprocal value of the number of layers of each DRC block div_z = 32768/bin

ISP_DRC_BIN_SCALE

ISP_DRC_BIN_SCALE is a DRC histogram scaling ratio register.

Offset Address	Register Name	Total Reset Value
0x26228	ISP_DRC_BIN_SCALE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														bin_scale	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RW	bin_scale	Histogram statistics weight NOTE The fewer the pixels in a block, the greater the weight.

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ISP_DRC_RANGE_FLT_COEF

ISP_DRC_RANGE_FLT_COEF is a DRC value domain filtering coefficient register.

Offset Address	Register Name	Total Reset Value
0x2622C	ISP_DRC_RANGE_FLT_COEF	0x0000_0004

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												range_flt_coef			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	range_flt_coef	The value range is 0−8.

ISP_DRC_EXPOSURE

ISP_DRC_EXPOSURE is a DRC current frame exposure value register.

Offset Address	Register Name	Total Reset Value
0x26230	ISP_DRC_EXPOSURE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																exposure															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	exposure	Exposure value of the current DRC frame, for compensating bright-to-dark switching

ISP_DRC_LOCAL_EDGE_LMT

ISP_DRC_LOCAL_EDGE_LMT is a DRC local limit register.

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Offset Address	Register Name	Total Reset Value
0x26234	ISP_DRC_LOCAL_EDGE_LMT	0x0000_009B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								local_edge_lmt							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	1	1

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	local_edge_lmt	Limit on the input/output luminance difference of the dual-edge filter

ISP_DRC_STRENGTH

ISP_DRC_STRENGTH is a DRC strength adjustment register.

Offset Address	Register Name	Total Reset Value
0x26238	ISP_DRC_STRENGTH	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								strength							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	strength	DRC strength

ISP_DRC_DETAIL_GAIN

ISP_DRC_DETAIL_GAIN is a DRC detail gain adjustment register.

Offset Address	Register Name	Total Reset Value
0x2623C	ISP_DRC_DETAIL_GAIN	0x0060_8010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								detail_mixing_thres								detail_mixing_dark								detail_mixing_bright							
Reset	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	detail_mixing_thres	Detail enhancement threshold

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[15:8]	RW	detail_mixing_dark	Enhancement strength of the detail in dark regions
[7:0]	RW	detail_mixing_bright	Enhancement strength of the detail in bright regions

ISP_DRC_DARK_GAIN_LMT_Y

ISP_DRC_DARK_GAIN_LMT_Y is a DRC dark region luminance limit register.

Offset Address	Register Name	Total Reset Value
0x26240	ISP_DRC_DARK_GAIN_LMT_Y	0x0000_86CE

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sft2				val2				sft1				val1			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	1	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:14]	RW	sft2	Limit on the luminance gain of the dark region
[13:10]	RW	val2	Limit on the luminance gain of the dark region
[9:7]	RW	sft1	Limit on the luminance gain of the dark region
[6:0]	RW	val1	Limit on the luminance gain of the dark region

ISP_DRC_DARK_GAIN_LMT_C

ISP_DRC_DARK_GAIN_LMT_C is a DRC dark region chrominance limit register.

Offset Address	Register Name	Total Reset Value
0x26244	ISP_DRC_DARK_GAIN_LMT_C	0x0000_86CE

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sft2				val2				sft1				val1			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	0	1	1	0	0	1	1	1	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:14]	RW	sft2	Limit on the chrominance gain of the dark region
[13:10]	RW	val2	Limit on the chrominance gain of the dark region

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[9:7]	RW	sft1	Limit on the chrominance gain of the dark region
[6:0]	RW	val1	Limit on the chrominance gain of the dark region

ISP_DRC_BRIGHT_GAIN_LMT

ISP_DRC_BRIGHT_GAIN_LMT is a DRC bright region limit register.

Offset Address	Register Name	Total Reset Value
0x26248	ISP_DRC_BRIGHT_GAIN_LMT	0x0000_0238

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						sft		val							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	1	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:7]	RW	sft	Limit on the bright region gain
[6:0]	RW	val	Limit on the bright region gain

ISP_DRC_FLT_STRENGTH

ISP_DRC_FLT_STRENGTH is a DRC filtering strength register.

Offset Address	Register Name	Total Reset Value
0x26250	ISP_DRC_FLT_STRENGTH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								var_range				var_spatial			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:4]	RW	var_range	Value domain strength of dual-edge filtering
[3:0]	RW	var_spatial	Spatial domain strength of dual-edge filtering

ISP_DRC_LUT_MIX_CRTL

ISP_DRC_LUT_MIX_CRTL is a DRC TMLUT mix register.

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Offset Address	Register Name	Total Reset Value
0x26254	ISP_DRC_LUT_MIX_CRTL	0x0000_00FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								lut_mix_ctrl							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	lut_mix_ctrl	Mix ratio of LUT0 to LUT1

ISP_DRC_GAIN_CLIP

ISP_DRC_GAIN_CLIP is a DRC gain limit register.

Offset Address	Register Name	Total Reset Value
0x26258	ISP_DRC_GAIN_CLIP	0x0000_00B8

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								gain_clip_knee				gain_clip_step			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:4]	RW	gain_clip_knee	Limit start point of the bright region gain
[3:0]	RW	gain_clip_step	Limit range of the bright region gain

ISP_DRC_STAT_IND_WADDR0

ISP_DRC_STAT_IND_WADDR0 is a DRC tone mapping table indirect write address 0 register.

Offset Address	Register Name	Total Reset Value
0x26280	ISP_DRC_STAT_IND_WADDR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_waddr0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	drc_stat_ind_waddr0	Tone mapping entry 0

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ISP_DRC_STAT_IND_WDATA0

ISP_DRC_STAT_IND_WDATA0 is a DRC tone mapping table indirect write data 0 register.

Offset Address	Register Name	Total Reset Value
0x26284	ISP_DRC_STAT_IND_WDATA0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_wdata0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	drc_stat_ind_wdata0	Tone mapping entry 0

ISP_DRC_STAT_IND_RADDR0

ISP_DRC_STAT_IND_RADDR0 is a DRC tone mapping table indirect read address 0 register.

Offset Address	Register Name	Total Reset Value
0x26288	ISP_DRC_STAT_IND_RADDR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_raddr0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	drc_stat_ind_raddr0	Tone mapping entry 0

ISP_DRC_STAT_IND_RDATA0

ISP_DRC_STAT_IND_RDATA0 is a DRC tone mapping table indirect read data 0 register.

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Offset Address	Register Name	Total Reset Value
0x2628C	ISP_DRC_STAT_IND_RDATA0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_rdata0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	drc_stat_ind_rdata0	Tone mapping entry 0

ISP_DRC_STAT_IND_WADDR1

ISP_DRC_STAT_IND_WADDR1 is a DRC tone mapping table indirect write address 1 register.

Offset Address	Register Name	Total Reset Value
0x26290	ISP_DRC_STAT_IND_WADDR1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_waddr1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	drc_stat_ind_waddr1	Tone mapping entry 1

ISP_DRC_STAT_IND_WDATA1

ISP_DRC_STAT_IND_WDATA1 is a DRC tone mapping table indirect write data 1 register.

Offset Address	Register Name	Total Reset Value
0x26294	ISP_DRC_STAT_IND_WDATA1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_wdata1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	drc_stat_ind_wdata1	Tone mapping entry 1

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ISP_DRC_STAT_IND_RADDR1

ISP_DRC_STAT_IND_RADDR1 is a DRC tone mapping table indirect read address 1 register.

Offset Address	Register Name	Total Reset Value
0x26298	ISP_DRC_STAT_IND_RADDR1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_raddr1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	drc_stat_ind_raddr1	Tone mapping entry 1

ISP_DRC_STAT_IND_RDATA1

ISP_DRC_STAT_IND_RDATA1 is a DRC tone mapping table indirect read data 1 register.

Offset Address	Register Name	Total Reset Value
0x2629C	ISP_DRC_STAT_IND_RDATA1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_rdata1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	drc_stat_ind_rdata1	Tone mapping entry 1

ISP_DRC_STAT_IND_WADDR2

ISP_DRC_STAT_IND_WADDR2 is a DRC color correct table indirect write address 2 register.

Offset Address	Register Name	Total Reset Value
0x262A0	ISP_DRC_STAT_IND_WADDR2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_waddr2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	drc_stat_ind_waddr2	Color correct entry

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ISP_DRC_STAT_IND_WDATA2

ISP_DRC_STAT_IND_WDATA2 is a DRC color correct table indirect write data 2 register.

Offset Address	Register Name	Total Reset Value
0x262A4	ISP_DRC_STAT_IND_WDATA2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_wdata2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	drc_stat_ind_wdata2	Color correct entry

ISP_DRC_STAT_IND_RADDR2

ISP_DRC_STAT_IND_RADDR2 is a DRC color correct table indirect read address 2 register.

Offset Address	Register Name	Total Reset Value
0x262A8	ISP_DRC_STAT_IND_RADDR2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_raddr2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	drc_stat_ind_raddr2	Color correct entry

ISP_DRC_STAT_IND_RDATA2

ISP_DRC_STAT_IND_RDATA2 is a DRC color correct table indirect read data 2 register.

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Offset Address	Register Name	Total Reset Value
0x262AC	ISP_DRC_STAT_IND_RDATA2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	drc_stat_ind_rdata2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	drc_stat_ind_rdata2	Color correct entry

ISP_DEHAZE_CFG

ISP_DEHAZE_CFG is a dehaze enable register.

Offset Address	Register Name	Total Reset Value
0x27000	ISP_DEHAZE_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	Dehaze enable 0: disabled 1: enabled

ISP_DEHAZE_PRE_UPDATE

ISP_DEHAZE_PRE_UPDATE is a dehaze statistical result update register.

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Offset Address	Register Name	Total Reset Value
0x27010	ISP_DEHAZE_PRE_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															pre_update
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	pre_update	Dehaze statistical result update. This bit is automatically cleared for each frame.

ISP_DEHAZE_BLK_SIZE

ISP_DEHAZE_BLK_SIZE is a dehaze block size configuration register.

Offset Address	Register Name	Total Reset Value
0x27014	ISP_DEHAZE_BLK_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								block_sizeh								reserved				block_sizev											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	block_sizeh	Horizontal block size. The configured value is the actual value minus 1.
[15:9]	RO	reserved	Reserved
[8:0]	RW	block_sizev	Vertical block size. The configured value is the actual value minus 1.

ISP_DEHAZE_BLK_NUM

ISP_DEHAZE_BLK_NUM is a dehaze block quantity configuration register.

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Offset Address	Register Name	Total Reset Value
0x27018	ISP_DEHAZE_BLK_NUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												block_numh				reserved												block_numv			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	block_numh	Horizontal block size. The configured value is the actual value minus 1.
[15:5]	RO	reserved	Reserved
[4:0]	RW	block_numv	Vertical block size. The configured value is the actual value minus 1.

ISP_DEHAZE_DC_SIZE

ISP_DEHAZE_DC_SIZE is a dehaze bilinear interpolation point quantity configuration register.

Offset Address	Register Name	Total Reset Value
0x2701C	ISP_DEHAZE_DC_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						dc_numh						dc_numv			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:5]	RW	dc_numh	Number of bilinear interpolation horizontal points. The configured value is the actual value minus 1.
[4:0]	RW	dc_numv	Number of bilinear interpolation vertical points. The configured value is the actual value minus 1.

ISP_DEHAZE_X

ISP_DEHAZE_X is a configuration register for the horizontal phase difference between the pixels of the dehaze enlarged picture.

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Offset Address	Register Name	Total Reset Value
0x27020	ISP_DEHAZE_X	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												phase_x																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:0]	RW	phase_x	Horizontal phase difference between the pixels of the enlarged picture, unsigned integer

ISP_DEHAZE_Y

ISP_DEHAZE_Y is a configuration register for the vertical phase difference between the pixels of the dehaze enlarged picture.

Offset Address	Register Name	Total Reset Value
0x27024	ISP_DEHAZE_Y	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												phase_y																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:0]	RW	phase_y	Vertical phase difference between the pixels of the enlarged picture, unsigned integer

ISP_DEHAZE_STAT_MODE

ISP_DEHAZE_STAT_MODE is a dehaze statistics module control register.

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Offset Address	Register Name	Total Reset Value
0x27028	ISP_DEHAZE_STAT_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													max_mode	min_mode	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	max_mode	Maximum value statistics control 0: Mean filtering is performed on RGB components, and then the RGB components corresponding to the maximum RGB sum of each window are counted. 1: The RGB components corresponding to the maximum RGB sum of each window are counted.
[1:0]	RW	min_mode	Minimum value statistics control 00: The minimum value among RGB components is selected for mean filtering, and the minimum value of each window is counted. 01: The minimum value among RGB components is selected, and the minimum value of each window is counted. 10: The minimum value of the RGB mean filtering results is selected for mean filtering, and the minimum value of each window is counted. Other values: reserved

ISP_DEHAZE_NEG_MODE

ISP_DEHAZE_NEG_MODE is a dehaze inversion control register.

Offset Address	Register Name	Total Reset Value
0x2702C	ISP_DEHAZE_NEG_MODE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															neg_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	RW	neg_mode	Inversion mode control 0: normal mode 1: inversion mode

ISP_DEHAZE_AIR

ISP_DEHAZE_AIR is a dehaze airglow configuration register.

Offset Address	Register Name	Total Reset Value
0x27030	ISP_DEHAZE_AIR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								air_r								air_g								air_b							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:16]	RW	air_r	Airglow A corresponding to the R channel
[15:8]	RW	air_g	Airglow A corresponding to the G channel
[7:0]	RW	air_b	Airglow A corresponding to the B channel

ISP_DEHAZE_THLD

ISP_DEHAZE_THLD is a dehaze T threshold coefficient configuration register.

Offset Address	Register Name	Total Reset Value
0x27034	ISP_DEHAZE_THLD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								thld							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	thld	T threshold coefficient, unsigned integer

ISP_DEHAZE_GSTRTH

ISP_DEHAZE_GSTRTH is a dehaze global strength coefficient register.

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Offset Address	Register Name	Total Reset Value
0x27038	ISP_DEHAZE_GSTRTH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								gstrth							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	gstrth	G calculation coefficient, unsigned integer

ISP_DEHAZE_BLTHLD

ISP_DEHAZE_BLTHLD is a dehaze minimum value filtering threshold coefficient configuration register.

Offset Address	Register Name	Total Reset Value
0x2703C	ISP_DEHAZE_BLTHLD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								blthld							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	blthld	Minimum value filtering threshold coefficient, unsigned integer

ISP_DIS_CFG

ISP_DIS_CFG is a DIS enable register.

Offset Address	Register Name	Total Reset Value
0x27100	ISP_DIS_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															work_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	RW	work_en	DIS enable 0: disabled 1: enabled

ISP_DIS_BLK

ISP_DIS_BLK is a block configuration register.

Offset Address	Register Name	Total Reset Value
0x27104	ISP_DIS_BLK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												srch_range				srch_range_0	reserved							blk_size							blk_size_0
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RO	reserved	Reserved
[22:17]	RW	srch_range	Upper six bits of the search range. The value range is 4−32.
[16]	RO	srch_range_0	Least significant bit (LSB) of the search range. This bit is fixed at 0.
[15:8]	RO	reserved	Reserved
[7:1]	RW	blk_size	Upper seven bits of the side length of the square block. The value range is 8−64.
[0]	RO	blk_size_0	LSB of the side length of the square block. This bit is fixed at 0.

ISP_DIS_V0POS

ISP_DIS_V0POS is a vertical PRJ0 block address register.

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Offset Address	Register Name	Total Reset Value
0x27110	ISP_DIS_V0POS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the vertical PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the vertical PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_V4POS

ISP_DIS_V4POS is a vertical PRJ4 block address register.

Offset Address	Register Name	Total Reset Value
0x27114	ISP_DIS_V4POS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the vertical PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the vertical PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_V8POS

ISP_DIS_V8POS is a vertical PRJ8 block address register.

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Offset Address	Register Name	Total Reset Value
0x27118	ISP_DIS_V8POS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the vertical PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the vertical PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_V0POSE

ISP_DIS_V0POSE is a vertical PRJ0 block address register.

Offset Address	Register Name	Total Reset Value
0x27120	ISP_DIS_V0POSE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the end point of the vertical PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the end point of the vertical PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_V4POSE

ISP_DIS_V4POSE is a vertical PRJ4 block address register.

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Offset Address	Register Name	Total Reset Value
0x27124	ISP_DIS_V4POSE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the end point of the vertical PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the end point of the vertical PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_V8POSE

ISP_DIS_V8POSE is a vertical PRJ8 block address register.

Offset Address	Register Name	Total Reset Value
0x27128	ISP_DIS_V8POSE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the end point of the vertical PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the end point of the vertical PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_H0POS

ISP_DIS_H0POS is a horizontal PRJ0 block address register.

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Offset Address	Register Name	Total Reset Value
0x27130	ISP_DIS_H0POS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the horizontal PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the horizontal PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_H4POS

ISP_DIS_H4POS is a horizontal PRJ4 block address register.

Offset Address	Register Name	Total Reset Value
0x27134	ISP_DIS_H4POS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the horizontal PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the horizontal PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_H8POS

ISP_DIS_H8POS is a horizontal PRJ8 block address register.

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Offset Address	Register Name	Total Reset Value
0x27138	ISP_DIS_H8POS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the horizontal PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the horizontal PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_H0POSE

ISP_DIS_H0POSE is a horizontal PRJ0 block address register.

Offset Address	Register Name	Total Reset Value
0x27140	ISP_DIS_H0POSE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the end point of the horizontal PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the end point of the horizontal PRJ0 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_H4POSE

ISP_DIS_H4POSE is a horizontal PRJ4 block address register.

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Offset Address	Register Name	Total Reset Value
0x27144	ISP_DIS_H4POSE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the end point of the horizontal PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the end point of the horizontal PRJ4 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_H8POSE

ISP_DIS_H8POSE is a horizontal PRJ8 block address register.

Offset Address	Register Name	Total Reset Value
0x27148	ISP_DIS_H8POSE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				ver												reserved				hor											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:16]	RW	ver	Vertical coordinate of the end point of the horizontal PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.
[15:11]	RO	reserved	Reserved
[10:0]	RW	hor	Horizontal coordinate of the end point of the horizontal PRJ8 block. The LSB of the coordinate value is excluded, and one bit is subtracted.

ISP_DIS_GAMMA_EN

ISP_DIS_GAMMA_EN is a gamma enable register.

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Offset Address	Register Name	Total Reset Value
0x27154	ISP_DIS_GAMMA_EN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															gamma_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	gamma_en	Gamma enable 0: disabled 1: enabled

ISP_DIS_H_STAT_RADDR

ISP_DIS_H_STAT_RADDR is a horizontal statistics read address register.

Offset Address	Register Name	Total Reset Value
0x27188	ISP_DIS_H_STAT_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	h_stat_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	h_stat_raddr	h_delta, h_sad, and h_mv information about each block. Each address contains only one piece of information. For example: 0x00: h_delta[14:0] of block 0 0x01: h_sad[21:0] of block 0 0x02: h_mv[7:0] of block 0 … The same rule applies to other values.

ISP_DIS_H_STAT_RDATA

ISP_DIS_H_STAT_RDATA is a horizontal statistics read data register.

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Offset Address	Register Name	Total Reset Value
0x2718C	ISP_DIS_H_STAT_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	h_stat_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	h_stat_rdata	Horizontal statistics

ISP_DIS_V_STAT_RADDR

ISP_DIS_V_STAT_RADDR is a vertical statistics read address register.

Offset Address	Register Name	Total Reset Value
0x27198	ISP_DIS_V_STAT_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	v_stat_raddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	v_stat_raddr	v_delta, v_sad, and v_mv information about each block. Each address contains only one piece of information. For example: 0x00: v_delta[14:0] of block 0 0x01: v_sad[21:0] of block 0 0x02: v_mv[7:0] of block 0 … The same rule applies to other values.

ISP_DIS_V_STAT_RDATA

ISP_DIS_V_STAT_RDATA is a vertical statistics read data register.

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Offset Address	Register Name	Total Reset Value
0x2719C	ISP_DIS_V_STAT_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	v_stat_rdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	v_stat_rdata	Vertical statistics

ISP_GAMMA_CFG

ISP_GAMMA_CFG is a gamma control register.

Offset Address	Register Name	Total Reset Value
0x27300	ISP_GAMMA_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	Gamma enable 0: disabled 1: enabled

ISP_GAMMA_LUT_UPDATE

ISP_GAMMA_LUT_UPDATE is a gamma lookup table update register.

Offset Address	Register Name	Total Reset Value
0x27310	ISP_GAMMA_LUT_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															update
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	RW	update	Gamma lookup table update. This bit is automatically cleared for each frame.

ISP_GAMMA_WADDR

ISP_GAMMA_WADDR is a gamma lookup table write address register.

Offset Address	Register Name	Total Reset Value
0x27380	ISP_GAMMA_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_waddr	Write address of the 257-entry gamma lookup table

ISP_GAMMA_WDATA

ISP_GAMMA_WDATA is a gamma lookup table write data register.

Offset Address	Register Name	Total Reset Value
0x27384	ISP_GAMMA_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_wdata	Write data of the 257-entry gamma lookup table

ISP_GAMMAFE_CFG

ISP_GAMMAFE_CFG is a gamma FE configuration register.

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Offset Address	Register Name	Total Reset Value
0x27400	ISP_GAMMAFE_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	Gamma enable 0: disabled 1: enabled

ISP_GAMMAFE_CTRL_BITW

ISP_GAMMAFE_CTRL_BITW is a data bit width configuration register.

Offset Address	Register Name	Total Reset Value
0x27410	ISP_GAMMAFE_CTRL_BITW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												bitw_in				reserved												bitw_out			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:16]	RW	bitw_in	Bit width of the gamma FE input data 0xE: 14 bits 0x10: 16 bits 0x12: 18 bits 0x14: 20 bits Other values: reserved
[15:5]	RO	reserved	Reserved
[4:0]	RW	bitw_out	Bit width of the gamma FE output data 0xC: 12 bits 0xE: 14 bits 0x10: 16 bits Other values: reserved

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ISP_GAMMAFE_MIX_CTRL

ISP_GAMMAFE_MIX_CTRL is a gamma FE mix control register.

Offset Address	Register Name	Total Reset Value
0x27414	ISP_GAMMAFE_MIX_CTRL	0x0000_00FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								mix_ctrl							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7:0]	RW	mix_ctrl	Output data blending ratio controlled by the gamma FE 0x0: Data is output after direct bit truncation. 0xFF: The interpolation result is output. Other values: Data is blended according to a linear ratio.

ISP_GAMMAFE_WADDR0

ISP_GAMMAFE_WADDR0 is indirect write address register 0.

Offset Address	Register Name	Total Reset Value
0x27480	ISP_GAMMAFE_WADDR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_waddr0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_waddr0	Write address of the 257-entry gamma FE LUT0

ISP_GAMMAFE_WDATA0

ISP_GAMMAFE_WDATA0 is indirect write data register 0.

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Offset Address	Register Name	Total Reset Value
0x27484	ISP_GAMMAFE_WDATA0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_wdata0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_wdata0	Write data of the 257-entry gamma FE LUT0

ISP_GAMMAFE_RADDR0

ISP_GAMMAFE_RADDR0 is indirect read address register 0.

Offset Address	Register Name	Total Reset Value
0x27488	ISP_GAMMAFE_RADDR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_waddr0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_waddr0	Read address of the 257-entry gamma FE LUT0

ISP_GAMMAFE_RDATA0

ISP_GAMMAFE_RDATA0 is indirect read data register 0.

Offset Address	Register Name	Total Reset Value
0x2748C	ISP_GAMMAFE_RDATA0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_wdata0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	gamma_wdata0	Read data of the 257-entry gamma FE LUT0

ISP_GAMMAFE_WADDR1

ISP_GAMMAFE_WADDR1 is indirect write address register 1.

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Offset Address	Register Name	Total Reset Value
0x27490	ISP_GAMMAFE_WADDR1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_waddr1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_waddr1	Write address of the 33-entry gamma FE LUT1

ISP_GAMMAFE_WDATA1

ISP_GAMMAFE_WDATA1 is indirect write data register 1.

Offset Address	Register Name	Total Reset Value
0x27494	ISP_GAMMAFE_WDATA1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_wdata1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_wdata1	Write data of the 33-entry gamma FE LUT1

ISP_GAMMAFE_RADDR1

ISP_GAMMAFE_RADDR1 is indirect read address register 1.

Offset Address	Register Name	Total Reset Value
0x27498	ISP_GAMMAFE_RADDR1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_waddr1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	gamma_waddr1	Read address of the 33-entry gamma FE LUT1

ISP_GAMMAFE_RDATA1

ISP_GAMMAFE_RDATA1 is indirect read data register 1.

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Offset Address	Register Name	Total Reset Value
0x2749C	ISP_GAMMAFE_RDATA1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gamma_wdata1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	gamma_wdata1	Read data of the 33-entry gamma FE LUT1

ISP_EXPANDER_CFG

ISP_EXPANDER_CFG is an expander configuration register.

Offset Address	Register Name	Total Reset Value
0x27500	ISP_EXPANDER_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	Expander enable 0: disabled 1: enabled

ISP_EXPANDER_BITW

ISP_EXPANDER_BITW is a data bit width configuration register.

Offset Address	Register Name	Total Reset Value
0x27510	ISP_EXPANDER_BITW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												bitw_in				reserved								bitw_out							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved

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[20:16]	RW	bitw_in	Bit width of the expander input data 0xC: 12 bits 0xE: 14 bits Other values: reserved
[15:5]	RO	reserved	Reserved
[4:0]	RW	bitw_out	Bit width of the expander output data 0xE: 14 bits 0x10: 16 bits 0x12: 18 bits 0x14: 20 bits Other values: reserved

ISP_EXPANDER_OFFSETR

ISP_EXPANDER_OFFSETR is an R channel data offset configuration register.

Offset Address	Register Name	Total Reset Value
0x27514	ISP_EXPANDER_OFFSETR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												offset_r																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:0]	RW	offset_r	R channel data offset configuration

ISP_EXPANDER_OFFSETGR

ISP_EXPANDER_OFFSETGR is a Gr channel data offset configuration register.

Offset Address	Register Name	Total Reset Value
0x27518	ISP_EXPANDER_OFFSETGR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												offset_gr																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved

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[19:0]	RW	offset_gr	Gr channel data offset configuration

ISP_EXPANDER_OFFSETGB

ISP_EXPANDER_OFFSETGB is a Gb channel data offset configuration register.

Offset Address	Register Name	Total Reset Value
0x2751C	ISP_EXPANDER_OFFSETGB	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												offset_gb																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:0]	RW	offset_gb	Gb channel data offset configuration

ISP_EXPANDER_OFFSETB

ISP_EXPANDER_OFFSETB is a B channel data offset configuration register.

Offset Address	Register Name	Total Reset Value
0x27520	ISP_EXPANDER_OFFSETB	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												offset_b																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RO	reserved	Reserved
[19:0]	RW	offset_b	B channel data offset configuration

ISP_EXPANDER_WADDR

ISP_EXPANDER_WADDR is an indirect write address register.

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Offset Address	Register Name	Total Reset Value
0x27580	ISP_EXPANDER_WADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	expander_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	expander_waddr	Write address of the 129-entry expander LUT

ISP_EXPANDER_WDATA

ISP_EXPANDER_WDATA is indirect write data register.

Offset Address	Register Name	Total Reset Value
0x27584	ISP_EXPANDER_WDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	expander_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	expander_wdata	Write data of the 129-entry expander LUT

ISP_EXPANDER_RADDR

ISP_EXPANDER_RADDR is an indirect read address register.

Offset Address	Register Name	Total Reset Value
0x27588	ISP_EXPANDER_RADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	expander_waddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	expander_waddr	Read address of the 129-entry expander LUT

ISP_EXPANDER_RDATA

ISP_EXPANDER_RDATA is indirect read data register.

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Offset Address	Register Name	Total Reset Value
0x2758C	ISP_EXPANDER_RDATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	expander_wdata																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	expander_wdata	Read data of the 129-entry expander LUT

ISP_COLORBAR_CFG

ISP_COLORBAR_CFG is a color bar control register.

Offset Address	Register Name	Total Reset Value
0x28100	ISP_COLORBAR_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	Color bar enable 0: disabled 1: enabled

ISP_COLORBAR_PATTERN

ISP_COLORBAR_PATTERN is a color bar type register.

Offset Address	Register Name	Total Reset Value
0x28110	ISP_COLORBAR_PATTERN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														pattern	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved

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[2:0]	RW	pattern	Color bar type 000: pure color picture 001: picture consisting of four horizontal color stripes 010: picture consisting of four vertical color stripes 011: picture consisting of eight vertical color stripes 100: A rectangular pure color target area is generated on the pure color background. Other values: reserved

ISP_COLORBAR_RBACKGND

ISP_COLORBAR_RBACKGND is a color bar background color register.

Offset Address	Register Name	Total Reset Value
0x28114	ISP_COLORBAR_RBACKGND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rbackgnd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	rbackgnd	Background color of the R component

ISP_COLORBAR_GBACKGND

ISP_COLORBAR_GBACKGND is a color bar background color register.

Offset Address	Register Name	Total Reset Value
0x28118	ISP_COLORBAR_GBACKGND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																gbackgnd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	gbackgnd	Background color of the G component

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ISP_COLORBAR_BBACKGND

ISP_COLORBAR_BBACKGND is a color bar background color register.

Offset Address	Register Name	Total Reset Value
0x2811C	ISP_COLORBAR_BBACKGND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																bbackgnd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	bbackgnd	Background color of the B component

ISP_COLORBAR_RFOREGND

ISP_COLORBAR_RFOREGND is a color bar foreground color register.

Offset Address	Register Name	Total Reset Value
0x28120	ISP_COLORBAR_RFOREGND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rforegnd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	rforegnd	Foreground color of the R component

ISP_COLORBAR_GFOREGND

ISP_COLORBAR_GFOREGND is a color bar foreground color register.

Offset Address	Register Name	Total Reset Value
0x28124	ISP_COLORBAR_GFOREGND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																gforegnd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved

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[15:0]	RW	gforegnd	Foreground color of the G component

ISP_COLORBAR_BFOREGND

ISP_COLORBAR_BFOREGND is a color bar foreground color register.

Offset Address	Register Name	Total Reset Value
0x28128	ISP_COLORBAR_BFOREGND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																bforegnd															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	bforegnd	Foreground color of the B component

ISP_COLORBAR_INIT

ISP_COLORBAR_INIT is a color bar initial value register.

Offset Address	Register Name	Total Reset Value
0x2812C	ISP_COLORBAR_INIT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rgbinit															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	rgbinit	Initial value of the R/G/B component

ISP_COLORBAR_GRAD

ISP_COLORBAR_GRAD is a color bar increasing value register.

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Offset Address	Register Name	Total Reset Value
0x28130	ISP_COLORBAR_GRAD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rgbgrad															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	rgbgrad	Increasing value of the R/G/B component

ISP_COLORBAR_POS1

ISP_COLORBAR_POS1 is a color bar rectangle coordinate register.

Offset Address	Register Name	Total Reset Value
0x28134	ISP_COLORBAR_POS1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				pos_y1												reserved				pos_x1											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	pos_y1	Vertical start coordinate (y1) of the rectangle
[15:13]	RO	reserved	Reserved
[12:0]	RW	pos_x1	Horizontal start coordinate (x1) of the rectangle

ISP_COLORBAR_POS2

ISP_COLORBAR_POS2 is a color bar rectangle coordinate register.

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Offset Address	Register Name	Total Reset Value
0x28138	ISP_COLORBAR_POS2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				pos_y2												reserved				pos_x2											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	pos_y2	Vertical end coordinate (y2) of the rectangle
[15:13]	RO	reserved	Reserved
[12:0]	RW	pos_x2	Horizontal end coordinate (x2) of the rectangle

ISP_COLORBAR_RGGB

ISP_COLORBAR_RGGB is an RGGB sequence register.

Offset Address	Register Name	Total Reset Value
0x2813C	ISP_COLORBAR_RGGB	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														rggb	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	rggb	RGGB sequence 00: R Gr Gb B 01: Gr R B Gb 10: Gb B R Gr 11: B Gb Gr R

ISP_YUV444SUM_CFG

ISP_YUV444SUM_CFG is a YUV444 sum control register.

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Offset Address	Register Name	Total Reset Value
0x28200	ISP_YUV444SUM_CFG	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	en	FPGA sum enable 0: disabled 1: enabled

ISP_CROP_DATA0_0

ISP_CROP_DATA0_0 is a data 0 sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28210	ISP_CROP_DATA0_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data0_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data0_0	Lower 32 bits of the data 0 sum

ISP_CROP_DATA0_1

ISP_CROP_DATA0_1 is a data 0 sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28214	ISP_CROP_DATA0_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data0_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data0_1	Upper 32 bits of the data 0 sum

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ISP_CROP_DATA1_0

ISP_CROP_DATA1_0 is a data 1 sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28218	ISP_CROP_DATA1_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data1_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data1_0	Lower 32 bits of the data 1 sum

ISP_CROP_DATA1_1

ISP_CROP_DATA1_1 is a data 1 sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x2821C	ISP_CROP_DATA1_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data1_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data1_1	Upper 32 bits of the data 1 sum

ISP_CROP_DATA2_0

ISP_CROP_DATA2_0 is a data 2 sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28220	ISP_CROP_DATA2_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data2_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data2_0	Lower 32 bits of the data 2 sum

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ISP_CROP_DATA2_1

ISP_CROP_DATA2_1 is a data 2 sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28224	ISP_CROP_DATA2_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data2_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data2_1	Upper 32 bits of the data 2 sum

ISP_CROP_DATAFPN_0

ISP_CROP_DATAFPN_0 is a data FPN sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28228	ISP_CROP_DATAFPN_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	datafpn_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	datafpn_0	Lower 32 bits of the data FPN sum

ISP_CROP_DATAFPN_1

ISP_CROP_DATAFPN_1 is a data FPN sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x2822C	ISP_CROP_DATAFPN_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	datafpn_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	datafpn_1	Upper 32 bits of the data FPN sum

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ISP_WB_DATA0_0

ISP_WB_DATA0_0 is a data 0 sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28230	ISP_WB_DATA0_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data0_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data0_0	Lower 32 bits of the data 0 sum

ISP_WB_DATA0_1

ISP_WB_DATA0_1 is a data 0 sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28234	ISP_WB_DATA0_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data0_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data0_1	Upper 32 bits of the data 0 sum

ISP_YUV422SUM_CFG

ISP_YUV422SUM_CFG is an FPGA sum control register.

Offset Address	Register Name	Total Reset Value
0x28300	ISP_YUV422SUM_CFG	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved

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[0]	RW	en	FPGA sum enable 0: disabled 1: enabled

ISP_CLIP_Y_DATA0_0

ISP_CLIP_Y_DATA0_0 is a Y component data sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28310	ISP_CLIP_Y_DATA0_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data_y_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data_y_0	Lower 32 bits of the Y component data sum

ISP_CLIP_Y_DATA0_1

ISP_CLIP_Y_DATA0_1 is a Y component data sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28314	ISP_CLIP_Y_DATA0_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data_y_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data_y_1	Upper 32 bits of the Y component data sum

ISP_CLIP_C_DATA0_0

ISP_CLIP_C_DATA0_0 is a C component data sum lower 32 bits register.

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Offset Address	Register Name	Total Reset Value
0x28318	ISP_CLIP_C_DATA0_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data_c_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data_c_0	Lower 32 bits of the C component data sum

ISP_CLIP_C_DATA0_1

ISP_CLIP_C_DATA0_1 is a C component data sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x2831C	ISP_CLIP_C_DATA0_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data_c_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data_c_1	Upper 32 bits of the C component data sum

ISP_YUV422_SWITCH_DATA0_0

ISP_YUV422_SWITCH_DATA0_0 is a data 0 sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28320	ISP_YUV422_SWITCH_DATA0_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data0_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data0_0	Lower 32 bits of the data 0 sum

ISP_YUV422_SWITCH_DATA0_1

ISP_YUV422_SWITCH_DATA0_1 is a data 0 sum upper 32 bits register.

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Offset Address	Register Name	Total Reset Value
0x28324	ISP_YUV422_SWITCH_DATA0_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data0_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data0_1	Upper 32 bits of the data 0 sum

ISP_YUV422_SWITCH_DATA1_0

ISP_YUV422_SWITCH_DATA1_0 is a data 1 sum lower 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28328	ISP_YUV422_SWITCH_DATA1_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data1_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data1_0	Lower 32 bits of the data 1 sum

ISP_YUV422_SWITCH_DATA1_1

ISP_YUV422_SWITCH_DATA1_1 is a data 1 sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x2832C	ISP_YUV422_SWITCH_DATA1_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data1_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data1_1	Upper 32 bits of the data 1 sum

ISP_YUV422_SWITCH_DATA2_0

ISP_YUV422_SWITCH_DATA2_0 is a data 2 sum lower 32 bits register.

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Offset Address	Register Name	Total Reset Value
0x28330	ISP_YUV422_SWITCH_DATA2_0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data2_0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data2_0	Lower 32 bits of the data 2 sum

ISP_YUV422_SWITCH_DATA2_1

ISP_YUV422_SWITCH_DATA2_1 is a data 2 sum upper 32 bits register.

Offset Address	Register Name	Total Reset Value
0x28334	ISP_YUV422_SWITCH_DATA2_1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data2_1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	data2_1	Upper 32 bits of the data 2 sum

ISP_ADAPTER444_CFG

ISP_ADAPTER444_CFG is a YUV444 timing adaptation register.

Offset Address	Register Name	Total Reset Value
0x31010	ISP_ADAPTER444_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														vsync_mode	hsync_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved

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[1]	RW	vsync_mode	Whether to perform delay processing on the vsync signal 0: no 1: yes
[0]	RW	hsync_mode	Whether to perform delay processing on the hsync signal 0: no 1: yes

ISP_SUM_Y

ISP_SUM_Y is an input picture luminance statistics register.

Offset Address	Register Name	Total Reset Value
0x31020	ISP_SUM_Y	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sum_y																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sum_y	Luminance statistics

ISP_CROP_CFG

ISP_CROP_CFG is a crop enable register.

Offset Address	Register Name	Total Reset Value
0x31100	ISP_CROP_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															n0_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	n0_en	Region 0 enable

ISP_CROP0_START

ISP_CROP0_START is a crop start position register for region 0.

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Offset Address	Register Name	Total Reset Value
0x31110	ISP_CROP0_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				y_start												reserved				x_start											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	y_start	ID of the line from which the picture starts to be captured
[15:13]	RO	reserved	Reserved
[12:0]	RW	x_start	ID of the pixel from which the picture starts to be captured

ISP_CROP0_SIZE

ISP_CROP0_SIZE is a crop size configuration register for region 0.

Offset Address	Register Name	Total Reset Value
0x31114	ISP_CROP0_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height (in line) of the obtained picture. The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Width (in pixel) of the obtained picture. The configured value is the actual value minus 1.

ISP_ACM_CTRL

ISP_ACM_CTRL is an ACM control register.

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Offset Address	Register Name	Total Reset Value
0x31400	ISP_ACM_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	acm_en	acm_dbg_en	acm_stretch	acm_cliprange	acm_cliporwrap	reserved																			acm_cbcrthr							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	acm_en	ACM enable 0: disabled (data output bypassed) 1: enabled (implementing luminance and contrast adjustment)
[30]	RW	acm_dbg_en	ACM debugging enable. When acm_dbg_en is valid, the original picture is displayed on the left part of the screen, and the picture processed by ABC is displayed on the right part of the screen.
[29]	RW	acm_stretch	Input pixel range 0: Y 64−940, C 64−960 1: Y 0−1023, C 0−1023
[28]	RW	acm_cliprange	Output pixel range 0: Y 64−940, C 64−960 1: Y 0−1023, C 0−1023
[27]	RW	acm_cliporwrap	Hue component value limit after ACM 0: The hue component is wrapped around to fall within [0, 1023]. 1: The hue component is clipped to fall within [0, 1023].
[26:8]	RO	reserved	Reserved
[7:0]	RW	acm_cbcrthr	ACM algorithm adjustment enable threshold. It is a 9-bit unsigned number and ranges from 0 to 255.

ISP_ACM_ADJ

ISP_ACM_ADJ is an ACM processed pixel change register.

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Offset Address	Register Name	Total Reset Value
0x31404	ISP_ACM_ADJ	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				acm_gain0								acm_gain1												acm_gain2							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:20]	RW	acm_gain0	Control coefficient of the luminance adjustment range. The value range is 0−512.
[19:10]	RW	acm_gain1	Control coefficient of the hue adjustment range. The value range is 0−512.
[9:0]	RW	acm_gain2	Control coefficient of the saturation adjustment range. The value range is 0−512.

ISP_ACM_SIZE

ISP_ACM_SIZE is an ACM processed picture size register.

Offset Address	Register Name	Total Reset Value
0x31410	ISP_ACM_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Picture height (in pixel). The configured value is the actual value minus 1.
[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Picture width (in pixel). The configured value is the actual value minus 1.

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ISP_ACM_PARA_ADDR

ISP_ACM_PARA_ADDR is an ACM coefficient configuration loading address register.

Offset Address	Register Name	Total Reset Value
0x31414	ISP_ACM_PARA_ADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	acm_para_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	acm_para_addr	Address for loading ACM coefficients

ISP_ACM_PARA_UP

ISP_ACM_PARA_UP is an ACM coefficient configuration update register.

Offset Address	Register Name	Total Reset Value
0x31420	ISP_ACM_PARA_UP	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															para_up
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	para_up	Parameter update. This bit is automatically cleared for each frame.

ISP_VPDCICTRL

ISP_VPDCICTRL is a DCI control register.

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Offset Address	Register Name	Total Reset Value
0x31C00	ISP_VPDCICTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_en	dci_dbg_en	dci_scene_flg	dci_man_adj0	dci_man_adj1	dci_man_adj2	dci_cbcrcmp_en	dci_cbcrsta_en	dci_in_range	dci_out_range	dci_shift_ctrl		dci_histlpf_en	reserved																		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	dci_en	DCI enable 0: disabled (data output bypassed) 1: enabled (implementing luminance and contrast adjustment)
[30]	RW	dci_dbg_en	DCI demo mode enable 0: disabled 1: The original picture is displayed on the left part of the screen, and the picture processed by the DCI is displayed on the right part of the screen.
[29]	RW	dci_scene_flg	DCI scenario switch enable 0: disabled 1: enabled
[28]	RW	dci_man_adj0	Adjustment mode of curve 0 0: automatic mode. The result calculated by hardware is used. 1: manual mode. The result configured by software is used.
[27]	RW	dci_man_adj1	Adjustment mode of curve 1 0: automatic mode. The result calculated by hardware is used. 1: manual mode. The result configured by software is used.
[26]	RW	dci_man_adj2	Adjustment mode of curve 2 0: automatic mode. The result calculated by hardware is used. 1: manual mode. The result configured by software is used.
[25]	RW	dci_cbcrcmp_en	DCI chrominance compensation enable 0: The DCI algorithm does not adjust the chrominance component. 1: The DCI algorithm adjusts the chrominance component.

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[24]	RW	dci_cbcrsta_en	Histogram statistics select 0: Histogram statistics are collected only on the luminance component. 1: Histogram statistics are collected on the weighted Y, Cb, and Cr components.
[23]	RW	dci_in_range	Input range 0: limited range (Y ranges from 64 to 940 and C ranges from 64 to 960 for the input 10-bit pixel data) 1: full range (Y ranges from 0 to 1023 and C ranges from 0 to 1023 for the input 10-bit pixel data)
[22]	RW	dci_out_range	Output range 0: limited range (Y ranges from 64 to 940 and C ranges from 64 to 960 for the output 10-bit pixel data) 1: full range (Y ranges from 0 to 1023 and C ranges from 0 to 1023 for the output 10-bit pixel data)
[21:20]	RW	dci_shift_ctrl	DCI histogram statistics shift control 00: shifted rightwards by 6 bits 01: shifted rightwards by 7 bits 10: shifted rightwards by 8 bits 11: shifted rightwards by 9 bits
[19]	RW	dci_histlpf_en	Histogram low-pass filtering enable 0: disabled 1: enabled
[18:0]	RO	reserved	Reserved

ISP_VPDCIHPOS

ISP_VPDCIHPOS is a DCI algorithm horizontal adjustment region register.

Offset Address	Register Name	Total Reset Value
0x31C04	ISP_VPDCIHPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_hstart												dci_hend												reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RW	dci_hstart	Start value of the DCI algorithm horizontal active region
[19:8]	RW	dci_hend	End value of the DCI algorithm horizontal active region
[7:0]	RO	reserved	Reserved

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ISP_VPDCIVPOS

ISP_VPDCIVPOS is a DCI algorithm vertical adjustment region register.

Offset Address	Register Name	Total Reset Value
0x31C08	ISP_VPDCIVPOS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_vstart												dci_vend												reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:20]	RW	dci_vstart	Start value of the DCI algorithm vertical active region
[19:8]	RW	dci_vend	End value of the DCI algorithm vertical active region
[7:0]	RO	reserved	Reserved

ISP_VPDCIHISBLD

ISP_VPDCIHISBLD is a DCI histogram statistics weighted coefficient register.

Offset Address	Register Name	Total Reset Value
0x31C0C	ISP_VPDCIHISBLD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_cbcrsta_y								dci_cbcrsta_cb								dci_cbcrsta_cr								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dci_cbcrsta_y	Weighted coefficient of the Y component when histogram statistics are added. It is an 8-bit signed number and the most significant bit (MSB) is the sign bit.
[23:16]	RW	dci_cbcrsta_cb	Weighted coefficient of the Cb component when histogram statistics are added. It is an 8-bit signed number and the MSB is the sign bit.
[15:8]	RW	dci_cbcrsta_cr	Weighted coefficient of the Cr component when histogram statistics are added. It is an 8-bit signed number and the MSB is the sign bit.
[7:0]	RO	reserved	Reserved

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ISP_VPDCIHISOFT

ISP_VPDCIHISOFT is a DCI histogram statistics offset register.

Offset Address	Register Name	Total Reset Value
0x31C10	ISP_VPDCIHISOFT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_cbcrsta_oft												reserved																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:23]	RW	dci_cbcrsta_oft	Offset when histogram statistics are added. It is a 9-bit signed number and the MSB is the sign bit.
[22:0]	RO	reserved	Reserved

ISP_VPDCIHISCOR

ISP_VPDCIHISCOR is a DCI histogram coring register.

Offset Address	Register Name	Total Reset Value
0x31C14	ISP_VPDCIHISCOR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_histcor_thr0								dci_histcor_thr1								dci_histcor_thr2								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dci_histcor_thr0	Coring threshold for histogram 0
[23:16]	RW	dci_histcor_thr1	Coring threshold for histogram 1
[15:8]	RW	dci_histcor_thr2	Coring threshold for histogram 2
[7:0]	RO	reserved	Reserved

ISP_VPDCIMERBLD

ISP_VPDCIMERBLD is a DCI adjustment unit blend value register.

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Offset Address	Register Name	Total Reset Value
0x31C18	ISP_VPDCIMERBLD	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_metrc_abld0	dci_metrc_abld1	dci_metrc_abld2	dci_hist_abld	dci_org_abld	reserved		
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:26]	RW	dci_metrc_abld0	Weighted alpha blend value of the current frame for metric 0
[25:20]	RW	dci_metrc_abld1	Weighted alpha blend value of the current frame for metric 1
[19:14]	RW	dci_metrc_abld2	Weighted alpha blend value of the current frame for metric 2
[13:8]	RW	dci_hist_abld	Weighted alpha blend value of the current histogram and the previous histogram
[7:2]	RW	dci_org_abld	Alpha blend value when the luminance component is adjusted
[1:0]	RO	reserved	Reserved

ISP_VPDCIADJWGT

ISP_VPDCIADJWGT is a DCI manually configured curve weight register.

Offset Address	Register Name	Total Reset Value
0x31C1C	ISP_VPDCIADJWGT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_man_adjwgt0		dci_man_adjwgt1		dci_man_adjwgt2		reserved	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:24]	RW	dci_man_adjwgt0	Weighted value of curve 0 in manual mode
[23:16]	RW	dci_man_adjwgt1	Weighted value of curve 1 in manual mode
[15:8]	RW	dci_man_adjwgt2	Weighted value of curve 2 in manual mode
[7:0]	RO	reserved	Reserved

ISP_VPDCICLIP0

ISP_VPDCICLIP0 is a DCI curve 0 weight range register.

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Offset Address	Register Name	Total Reset Value
0x31C20	ISP_VPDCICLIP0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_wgt_cliplow0								dci_wgt_cliphigh0								reserved															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dci_wgt_cliplow0	Lower limit of the weighted value of curve 0
[23:16]	RW	dci_wgt_cliphigh0	Upper limit of the weighted value of curve 0
[15:0]	RO	reserved	Reserved

ISP_VPDCICLIP1

ISP_VPDCICLIP1 is a DCI curve 1 weight range register.

Offset Address	Register Name	Total Reset Value
0x31C24	ISP_VPDCICLIP1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_wgt_cliplow1								dci_wgt_cliphigh1								reserved															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dci_wgt_cliplow1	Lower limit of the weighted value of curve 1
[23:16]	RW	dci_wgt_cliphigh1	Upper limit of the weighted value of curve 1
[15:0]	RO	reserved	Reserved

ISP_VPDCICLIP2

ISP_VPDCICLIP2 is a DCI curve 2 weight range register.

Offset Address	Register Name	Total Reset Value
0x31C28	ISP_VPDCICLIP2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_wgt_cliplow2								dci_wgt_cliphigh2								reserved															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dci_wgt_cliplow2	Lower limit of the weighted value of curve 2

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[23:16]	RW	dci_wgt_cliphigh2	Upper limit of the weighted value of curve 2
[15:0]	RO	reserved	Reserved

ISP_VPDCIGLBGAIN

ISP_VPDCIGLBGAIN is a DCI luminance adjustment unit global gain register.

Offset Address	Register Name	Total Reset Value
0x31C2C	ISP_VPDCIGLBGAIN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_glb_gain0								dci_glb_gain1								dci_glb_gain2								reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RW	dci_glb_gain0	Global gain 0 when the luminance component is adjusted
[25:20]	RW	dci_glb_gain1	Global gain 1 when the luminance component is adjusted
[19:14]	RW	dci_glb_gain2	Global gain 2 when the luminance component is adjusted
[13:0]	RO	reserved	Reserved

ISP_VPDCIPOSTHR0

ISP_VPDCIPOSTHR0 is DCI adjustment unit threshold register 0 during positive adjustment.

Offset Address	Register Name	Total Reset Value
0x31C30	ISP_VPDCIPOSTHR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_gainpos_thr1								dci_gainpos_thr2								dci_gainpos_thr3								dci_gainpos_thr4							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dci_gainpos_thr1	Threshold 1 for luminance Y when the picture is adjusted from dark to bright
[23:16]	RW	dci_gainpos_thr2	Threshold 2 for luminance Y when the picture is adjusted from dark to bright
[15:8]	RW	dci_gainpos_thr3	Threshold 3 for luminance Y when the picture is adjusted from dark to bright
[7:0]	RW	dci_gainpos_thr4	Threshold 4 for luminance Y when the picture is adjusted from dark to bright

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ISP_VPDCIPOSTHR1

ISP_VPDCIPOSTHR1 is DCI adjustment unit threshold register 1 during positive adjustment.

Offset Address	Register Name	Total Reset Value
0x31C34	ISP_VPDCIPOSTHR1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainpos_thr5		dci_gainpos_thr6		dci_gainpos_thr7		reserved	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:24]	RW	dci_gainpos_thr5	Threshold 5 for luminance Y when the picture is adjusted from dark to bright
[23:16]	RW	dci_gainpos_thr6	Threshold 6 for luminance Y when the picture is adjusted from dark to bright
[15:8]	RW	dci_gainpos_thr7	Threshold 7 for luminance Y when the picture is adjusted from dark to bright
[7:0]	RO	reserved	Reserved

ISP_VPDCIPOSGAIN0

ISP_VPDCIPOSGAIN0 is DCI adjustment unit gain register 0 during positive adjustment

Offset Address	Register Name	Total Reset Value
0x31C38	ISP_VPDCIPOSGAIN0	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainpos_cbcr0	dci_gainpos_cbcr1	dci_gainpos_cbcr2	dci_gainpos_cbcr3	dci_gainpos_cbcr4	dci_gainpos_cbcr5	dci_gainpos_cbcr6	dci_gainpos_cbcr7
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RW	dci_gainpos_cbcr0	Chrominance compensation gain of threshold 0 when the picture is adjusted from dark to bright
[27:24]	RW	dci_gainpos_cbcr1	Chrominance compensation gain of threshold 1 when the picture is adjusted from dark to bright
[23:20]	RW	dci_gainpos_cbcr2	Chrominance compensation gain of threshold 2 when the picture is adjusted from dark to bright
[19:16]	RW	dci_gainpos_cbcr3	Chrominance compensation gain of threshold 3 when the picture is adjusted from dark to bright

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[15:12]	RW	dci_gainpos_cbcr4	Chrominance compensation gain of threshold 4 when the picture is adjusted from dark to bright
[11:8]	RW	dci_gainpos_cbcr5	Chrominance compensation gain of threshold 5 when the picture is adjusted from dark to bright
[7:4]	RW	dci_gainpos_cbcr6	Chrominance compensation gain of threshold 6 when the picture is adjusted from dark to bright
[3:0]	RW	dci_gainpos_cbcr7	Chrominance compensation gain of threshold 7 when the picture is adjusted from dark to bright

ISP_VPDCIPOSGAIN1

ISP_VPDCIPOSGAIN1 is DCI adjustment unit gain register 1 during positive adjustment.

Offset Address	Register Name	Total Reset Value
0x31C3C	ISP_VPDCIPOSGAIN1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainpos_cbcr8	reserved						
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RW	dci_gainpos_cbcr8	Chrominance compensation gain of threshold 8 when the picture is adjusted from dark to bright
[27:0]	RO	reserved	Reserved

ISP_VPDCIPOSSLP0

ISP_VPDCIPOSSLP0 is DCI adjustment unit slope register 0 during positive adjustment.

Offset Address	Register Name	Total Reset Value
0x31C40	ISP_VPDCIPOSSLP0	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainpos_slp0			dci_gainpos_slp1		dci_gainpos_slp2		reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:22]	RW	dci_gainpos_slp0	Slope of region 1 when the picture is adjusted from dark to bright, 10-bit signed number

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[21:12]	RW	dci_gainpos_slp1	Slope of region 2 when the picture is adjusted from dark to bright, 10-bit signed number
[11:2]	RW	dci_gainpos_slp2	Slope of region 3 when the picture is adjusted from dark to bright, 10-bit signed number
[1:0]	RO	reserved	Reserved

ISP_VPDCIPOSSLP1

ISP_VPDCIPOSSLP1 is DCI adjustment unit slope register 1 during positive adjustment.

Offset Address	Register Name	Total Reset Value
0x31C44	ISP_VPDCIPOSSLP1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainpos_slp3		dci_gainpos_slp4		dci_gainpos_slp5			reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:22]	RW	dci_gainpos_slp3	Slope of region 4 when the picture is adjusted from dark to bright, 10-bit signed number
[21:12]	RW	dci_gainpos_slp4	Slope of region 5 when the picture is adjusted from dark to bright, 10-bit signed number
[11:2]	RW	dci_gainpos_slp5	Slope of region 6 when the picture is adjusted from dark to bright, 10-bit signed number
[1:0]	RO	reserved	Reserved

ISP_VPDCIPOSSLP2

ISP_VPDCIPOSSLP2 is DCI adjustment unit slope register 2 during positive adjustment.

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Offset Address	Register Name	Total Reset Value
0x31C48	ISP_VPDCIPOSSLP2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_gainpos_slp6												dci_gainpos_slp7								reserved											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RW	dci_gainpos_slp6	Slope of region 7 when the picture is adjusted from dark to bright, 10-bit signed number
[21:12]	RW	dci_gainpos_slp7	Slope of region 8 when the picture is adjusted from dark to bright, 10-bit signed number
[11:0]	RO	reserved	Reserved

ISP_VPDCINEGTHR0

ISP_VPDCINEGTHR0 is DCI adjustment unit threshold register 0 during negative adjustment.

Offset Address	Register Name	Total Reset Value
0x31C4C	ISP_VPDCINEGTHR0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_gainneg_thr1								dci_gainneg_thr2								dci_gainneg_thr3								dci_gainneg_thr4							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	dci_gainneg_thr1	Threshold 1 for luminance Y when the picture is adjusted from bright to dark
[23:16]	RW	dci_gainneg_thr2	Threshold 2 for luminance Y when the picture is adjusted from bright to dark
[15:8]	RW	dci_gainneg_thr3	Threshold 3 for luminance Y when the picture is adjusted from bright to dark
[7:0]	RW	dci_gainneg_thr4	Threshold 4 for luminance Y when the picture is adjusted from bright to dark

ISP_VPDCINEGTHR1

ISP_VPDCINEGTHR1 is DCI adjustment unit threshold register 1 during negative adjustment.

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Offset Address	Register Name	Total Reset Value
0x31C50	ISP_VPDCINEGTHR1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainneg_thr5		dci_gainneg_thr6		dci_gainneg_thr7		reserved	
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:24]	RW	dci_gainneg_thr5	Threshold 5 for luminance Y when the picture is adjusted from bright to dark
[23:16]	RW	dci_gainneg_thr6	Threshold 6 for luminance Y when the picture is adjusted from bright to dark
[15:8]	RW	dci_gainneg_thr7	Threshold 7 for luminance Y when the picture is adjusted from bright to dark
[7:0]	RO	reserved	Reserved

ISP_VPDCINEGGAIN0

ISP_VPDCINEGGAIN0 is DCI adjustment unit gain register 0 during negative adjustment.

Offset Address	Register Name	Total Reset Value
0x31C54	ISP_VPDCINEGGAIN0	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainneg_cbcr0	dci_gainneg_cbcr1	dci_gainneg_cbcr2	dci_gainneg_cbcr3	dci_gainneg_cbcr4	dci_gainneg_cbcr5	dci_gainneg_cbcr6	dci_gainneg_cbcr7
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:28]	RW	dci_gainneg_cbcr0	Chrominance compensation gain of threshold 0 when the picture is adjusted from bright to dark
[27:24]	RW	dci_gainneg_cbcr1	Chrominance compensation gain of threshold 1 when the picture is adjusted from bright to dark
[23:20]	RW	dci_gainneg_cbcr2	Chrominance compensation gain of threshold 2 when the picture is adjusted from bright to dark
[19:16]	RW	dci_gainneg_cbcr3	Chrominance compensation gain of threshold 3 when the picture is adjusted from bright to dark
[15:12]	RW	dci_gainneg_cbcr4	Chrominance compensation gain of threshold 4 when the picture is adjusted from bright to dark
[11:8]	RW	dci_gainneg_cbcr5	Chrominance compensation gain of threshold 5 when the picture is adjusted from bright to dark

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[7:4]	RW	dci_gainneg_cbcr6	Chrominance compensation gain of threshold 6 when the picture is adjusted from bright to dark
[3:0]	RW	dci_gainneg_cbcr7	Chrominance compensation gain of threshold 7 when the picture is adjusted from bright to dark

ISP_VPDCINEGGAIN1

ISP_VPDCINEGGAIN1 is DCI adjustment unit gain register 1 during negative adjustment.

Offset Address	Register Name	Total Reset Value
0x31C58	ISP_VPDCINEGGAIN1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_gainneg_cbcr8				reserved																											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	dci_gainneg_cbcr8	Chrominance compensation gain of threshold 8 when the picture is adjusted from bright to dark
[27:0]	RO	reserved	Reserved

ISP_VPDCINEGSLP0

ISP_VPDCINEGSLP0 is DCI adjustment unit slope register 0 during negative adjustment.

Offset Address	Register Name	Total Reset Value
0x31C5C	ISP_VPDCINEGSLP0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dci_gainneg_slp0												dci_gainneg_slp1								dci_gainneg_slp2										reserved	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RW	dci_gainneg_slp0	Slope of region 1 when the picture is adjusted from bright to dark, 10-bit signed number
[21:12]	RW	dci_gainneg_slp1	Slope of region 2 when the picture is adjusted from bright to dark, 10-bit signed number
[11:2]	RW	dci_gainneg_slp2	Slope of region 3 when the picture is adjusted from bright to dark, 10-bit signed number

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[1:0]	RO	reserved	Reserved

ISP_VPDCINEGSLP1

ISP_VPDCINEGSLP1 is DCI adjustment unit slope register 1 during negative adjustment.

Offset Address	Register Name	Total Reset Value
0x31C60	ISP_VPDCINEGSLP1	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainneg_slp3		dci_gainneg_slp4		dci_gainneg_slp5			reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:22]	RW	dci_gainneg_slp3	Slope of region 4 when the picture is adjusted from bright to dark, 10-bit signed number
[21:12]	RW	dci_gainneg_slp4	Slope of region 5 when the picture is adjusted from bright to dark, 10-bit signed number
[11:2]	RW	dci_gainneg_slp5	Slope of region 6 when the picture is adjusted from bright to dark, 10-bit signed number
[1:0]	RO	reserved	Reserved

ISP_VPDCINEGSLP2

ISP_VPDCINEGSLP2 is DCI adjustment unit slope register 2 during negative adjustment.

Offset Address	Register Name	Total Reset Value
0x31C64	ISP_VPDCINEGSLP2	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	dci_gainneg_slp6		dci_gainneg_slp7		reserved			
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:22]	RW	dci_gainneg_slp6	Slope of region 7 when the picture is adjusted from bright to dark, 10-bit signed number
[21:12]	RW	dci_gainneg_slp7	Slope of region 8 when the picture is adjusted from bright to dark, 10-bit signed number
[11:0]	RO	reserved	Reserved

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ISP_SWITCH422_CTRL

ISP_SWITCH422_CTRL is a frame discarding module control register.

Offset Address	Register Name	Total Reset Value
0x32000	ISP_SWITCH422_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	enable	reserved																														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	enable	Frame discarding control 0: Frames are discarded. 1: ISP_YUV422_REG_UPDATE controls whether to discard frames.
[30:0]	RO	reserved	Reserved

ISP_ADAPTER422_CFG

ISP_ADAPTER422_CFG is a YUV422 timing adaptation register.

Offset Address	Register Name	Total Reset Value
0x32010	ISP_ADAPTER422_CFG	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														vsync_mode	hsync_mode
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	vsync_mode	Whether to perform delay processing on the vsync signal 0: no 1: yes

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[0]	RW	hsync_mode	Whether to perform delay processing on the hsync signal 0: no 1: yes

ISP_SKIP_Y_CFG

ISP_SKIP_Y_CFG is a Y skip configuration register.

Offset Address	Register Name	Total Reset Value
0x32100	ISP_SKIP_Y_CFG	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	skip_cfg																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:0]	RW	skip_cfg	Skip configuration

ISP_SKIP_C_CFG

ISP_SKIP_C_CFG is a C skip configuration register.

Offset Address	Register Name	Total Reset Value
0x32110	ISP_SKIP_C_CFG	0xFFFF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	skip_cfg																															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:0]	RW	skip_cfg	Skip configuration

ISP_CLIP_Y_CFG

ISP_CLIP_Y_CFG is a luminance clamping configuration register.

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Offset Address	Register Name	Total Reset Value
0x32200	ISP_CLIP_Y_CFG	0xFFFF_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	max																min															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	max	Maximum luminance clamping value. When the data bit width is less than 16 bits, the upper bits are valid.
[15:0]	RW	min	Minimum luminance clamping value. When the data bit width is less than 16 bits, the upper bits are valid.

ISP_CLIP_C_CFG

ISP_CLIP_C_CFG is a chrominance clamping configuration register.

Offset Address	Register Name	Total Reset Value
0x32204	ISP_CLIP_C_CFG	0xFFFF_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	max																min															
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	max	Maximum chrominance clamping value. When the data bit width is less than 16 bits, the upper bits are valid.
[15:0]	RW	min	Minimum chrominance clamping value. When the data bit width is less than 16 bits, the upper bits are valid.

ISP_HLDC_CFG

ISP_HLDC_CFG is an HLDC enable register.

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Offset Address	Register Name	Total Reset Value
0x32300	ISP_HLDC_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	hldc_l_en	hldc_c_en	reserved																													hldc_in_b422
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	hldc_l_en	HLDC luminance enable 0: disabled 1: enabled
[30]	RW	hldc_c_en	HLDC chrominance enable 0: disabled 1: enabled
[29:1]	RO	reserved	Reserved
[0]	RW	hldc_in_b422	Format of data processed by the HLDC 0: yuv420 1: yuv422

ISP_HLDC_SIZE

ISP_HLDC_SIZE is an HLDC luminance size configuration register.

Offset Address	Register Name	Total Reset Value
0x32310	ISP_HLDC_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				height												reserved				width											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved
[28:16]	RW	height	Height of a picture luminance column (in pixel). The configured value is the actual value minus 1.

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[15:13]	RO	reserved	Reserved
[12:0]	RW	width	Width of a picture luminance row (in pixel). The configured value is the actual value minus 1.

ISP_HLDC_CENTER

ISP_HLDC_CENTER is an HLDC center coordinate parameter register.

Offset Address	Register Name	Total Reset Value
0x32320	ISP_HLDC_CENTER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ldc_norcoef								ldc_y_center												ldc_x_center											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RW	ldc_norcoef	Normalization parameter
[23:12]	RW	ldc_y_center	Vertical central coordinate
[11:0]	RW	ldc_x_center	Horizontal central coordinate

ISP_HLDC_PARA

ISP_HLDC_PARA is an HLDC configuration parameter register.

Offset Address	Register Name	Total Reset Value
0x32324	ISP_HLDC_PARA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												ldc_black_mode	shift_right			ratio								scale							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved

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[21]	RW	ldc_black_mode	Mode select 0: normal mode 1: black border mode
[20:16]	RW	shift_right	Number of right shift bits for the denominator
[15:7]	RW	ratio	Distortion coefficient
[6:0]	RW	scale	Scaling ratio

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Tables

Table 11-1 Summary of AIAO registers (base address: 0x2065_0000)..... 11-7

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11 Audio Interfaces

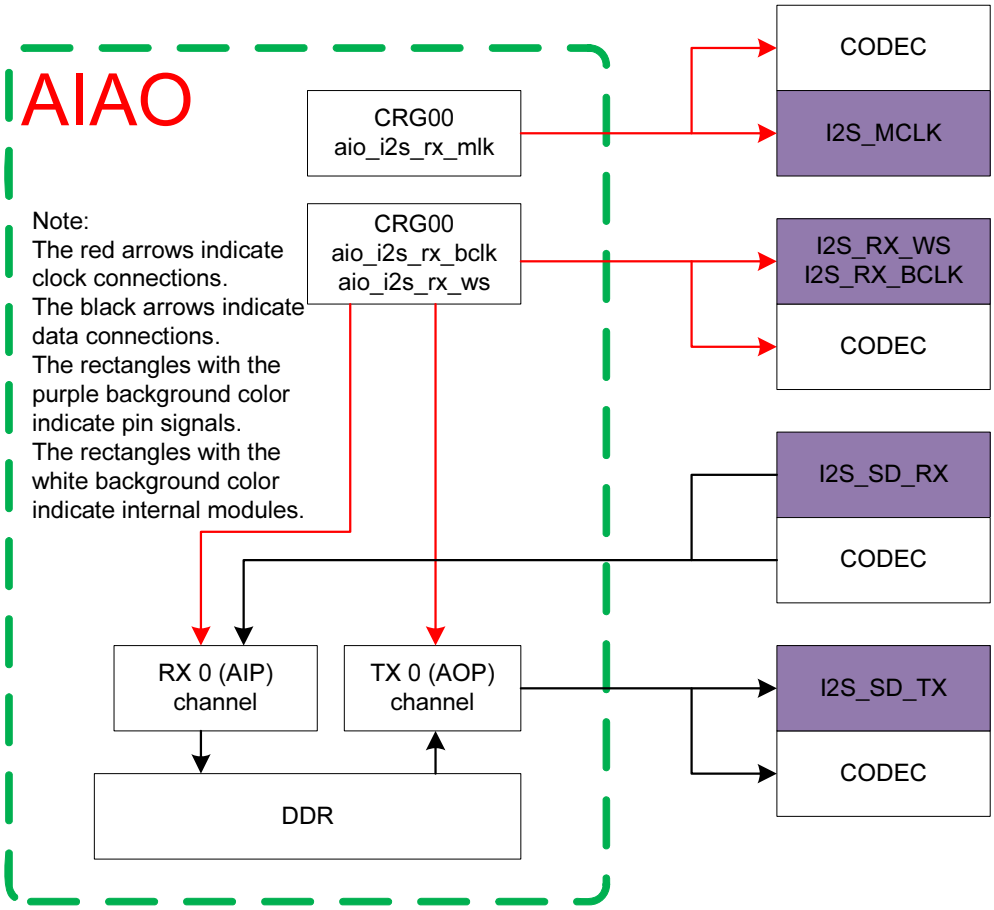
11.1 AIAO

11.1.1 Overview

The audio input/output (AIAO) interface is used to connect to the off-chip audio coder/decoder (CODEC) to input and output audio data, implementing the recording, talkback, and playback functions. Hi3518E V20*X*/Hi3516C V200 has an integrated AIAO interface that includes one audio input port (AIP) and one audio output port (AOP) and supported stereo inputs and outputs. Figure 11-1 shows the block diagram of the Hi3518E V20*X* AIAO.

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Figure 11-1 Block diagram of the Hi3518E V20*X* AIAO



11.1.2 Features

The AIAO interface supports the pulse code modulation (PCM) mode and inter-IC sound (I^2S) mode. The AIAO interface reads data from or writes data to the memory in DMA mode.

PCM Interfaces

The PCM interfaces have the following features:

- Transmit or receive the 16-bit linear PCM code from a channel in master mode.
- Support only short pulse sync signals in frame sync signals (the duration of those sync signals is one clock cycle). The PCM interfaces can work in both the standard mode and customized mode.
- Separately enable or disable input (AIP) and output (AOP).
- Support DMA for inputs (AIP) and outputs (AOP). The AIAO reads data from and writes data to a cyclic buffer created by using software. The cyclic buffer size and threshold are adjustable.

I²S Interfaces

The I^2S interfaces have the following features:

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- Transmit or receive 16- or 24-bit stereo data in master mode.
- Support the sampling rate ranging from 8 kHz to 192 kHz.
- Separately enables or disables input (AIP) and output (AOP).
- Support DMA for inputs (AIP) and outputs (AOP). The AIAO reads data from and writes data to a cyclic buffer created by using software. The cyclic buffer size and threshold are adjustable.

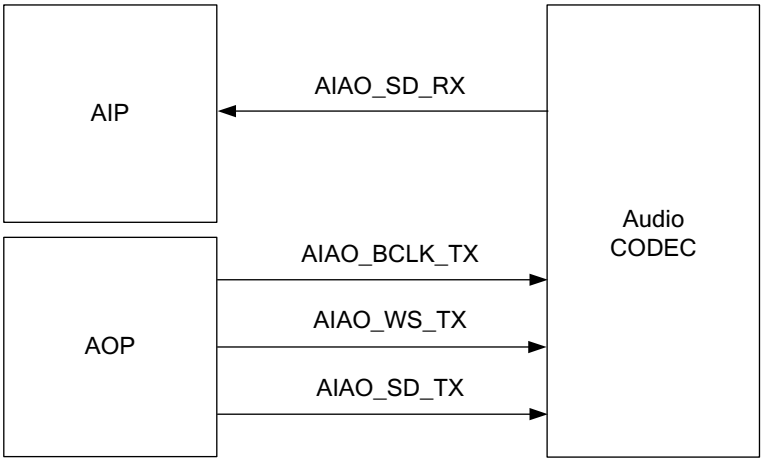
11.1.3 Function Description

Typical Application

Hi3518E V20*X*/Hi3516C V200 provides one AIP and one AOP. Their functions are as follows:

- The AIP supports data reception in PCM or I^2S mode.
- The AOP supports music playback in PCM or I^2S mode.
- The AIP supports interconnection with the internal audio CODEC in I^2S master mode or interconnection with the external analog-to-digital converter (ADC) in PCM/I^2S master mode to implement voice sampling.
- The AOP supports interconnection with the internal audio CODEC in I^2S master mode or interconnection with the external digital-to-analog converter (DAC) in PCM/I^2S master mode to implement music playback.

Figure 11-2 Connection diagram over the 4-line I^2S/PCM interface in master mode



Function Principle

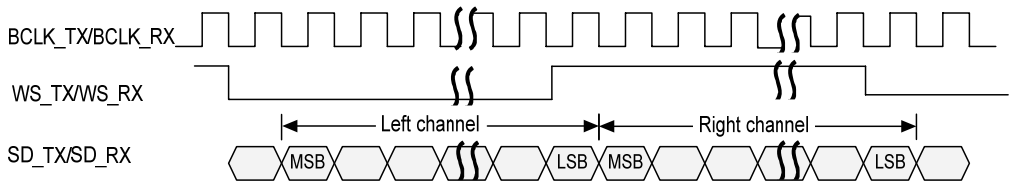
The AIP receives the audio data after the analog-to-digital (AD) conversion performed by the internal audio CODEC or external ADC over the I^2S or PCM interface, and stores the data into the cyclic FIFO created for the AIP. Then, the CPU fetches the data and stores it. In this way, audio recording is complete.

AOP reads audio data from the cyclic buffer and sends the audio data to the interconnected audio CODEC over the I^2S or PCM interface at a specified sampling rate. The internal audio CODEC or external ADC performs digital-to-analog (DA) conversion on the audio data and plays the audio.

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Figure 11-3 shows the supported I²S interface timing when the external I²S interface is connected.

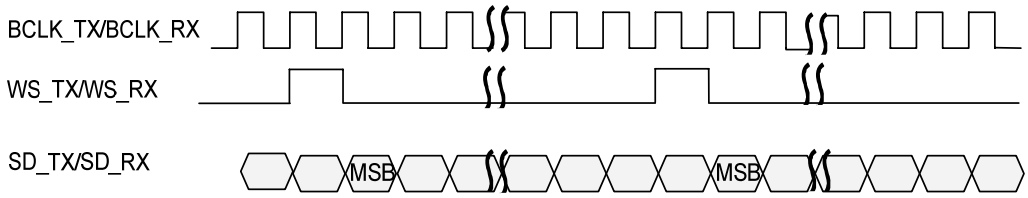
Figure 11-3 I²S interface timing



The I²S interface transmits data in most significant bit (MSB) first mode. Data and WS signals are sent at the falling edge of the BCLK and sampled at the rising edge of the BCLK. The data is delayed by one BCLK cycle compared with WS signals.

When the external PCM interface is connected, the PCM standard timing and data left-aligned timing are supported. Figure 11-4 shows the timing of the PCM interface in standard mode.

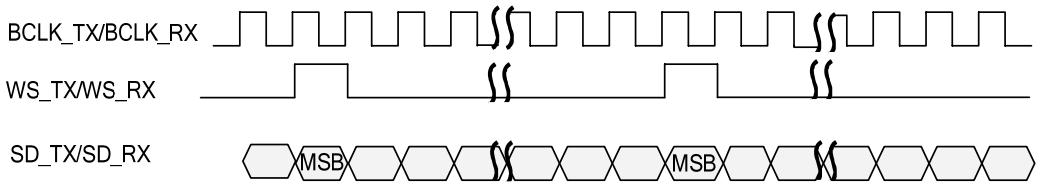
Figure 11-4 Timing of the PCM interface in standard mode



The PCM interface transmits data in MSB first mode. Data and WS signals are sent at the rising edge of the BCLK and sampled at the falling edge of the BCLK. In standard mode, the data is delayed by one BCLK cycle compared with WS signals.

Figure 11-5 shows the timing of the PCM interface in customized mode.

Figure 11-5 Timing of the PCM interface in customized mode



In customized mode, data and WS pulses start to be sent when they are in the same cycle.

11.1.4 Operating Mode

Channel Multiplexing Configuration

As there is only one AIP/AOP, the internal audio CODEC and the external ADC/DAC connected to the I²S/PCM interface cannot work simultaneously.

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If the system MISC registers i2s_pad_enable and audio_enable are set to 1 and 0 respectively, the AIP/AOP connects to the external I^2S/PCM pin. If i2s_pad_enable and audio_enable are set to 0 and 1 respectively, the AIP/AOP connects to the internal audio CODEC.

Clock Gating and Configuration

Before enabling the AIAO interface for audio recording or playback, you must enable clock gating of the related channels (AIP/AOP). To enable clock gating, perform the following steps:

Step 1 Set the system CRG register PERI_CRG35 to 0x2 to deassert the reset on the AIAO and enable clock gating.

Step 2 Configure the AIAO registers I2S_CRG_CFG0_00 and I2S_CRG_CFG1_00 to select appropriate frequency division coefficients. Note that the MCLK/WS frequency division ratio in the i2s1_fs_sel register for the audio CODEC must be consistent with the configuration in this step when the internal audio CODEC is used.

----End

Soft Reset

The two internal channels (AIP and AOP) of the AIAO module support separate soft reset. When the AIAO module is reset, the two channels are reset at the same time.

Recording Process

The following example assumes that the audio channels are stereo channels in I^2S mode, the sampling rate is 48 kHz, the sampling precision is 16 bits, the AIAO PLL clock source is a 1188 MHz clock, and the AIAO clock of the system controller is enabled. To record data, perform the following steps:

Step 1 Set I2S_CRG_CFG0_00 to 0x152EF0 to set the MCLK output frequency to 12.288 MHz.

Step 2 Set I2S_CRG_CFG1_00 to 0x00000133 to enable the clock, set the BCLK to the divide-by-4 clock of the MCLK, and set the FCLK to the divide-by-64 clock of the BCLK. The FCLK is 48 kHz.

Step 3 Set RX_IF_ATTRI to 0xE4800014 to set the operating mode of the RX channel to I2S stereo mode and sampling precision to 16 bits.

Step 4 Set RX_BUFF_SADDR to the start address of the allocated DDR (for example, 0x00000100), set RX_BUFF_SIZE to the size of the allocated DDR buffer (for example, 0x0000f000), set RX_BUFF_WPTR and RX_BUFF_RPTR to 0x0 to initialize the write and read pointers, and set RX_TRANS_SIZE to the data transfer length (for example, 0x00000f00).

Step 5 Configure RX_INT_ENA to enable the corresponding interrupt of the RX channel as required. For example, set RX_INT_ENA to 0x00000001 to enable the trans_int interrupt.

Step 6 Set RX_DSP_CTRL to 0x10000000 to enable the RX channel. Then the RX channel starts recording.

Step 7 Read RX_BUFF_WPTR and RX_BUFF_RPTR to check the empty/full status of the cyclic buffer and the valid data amount.

Ensure that data is fetched before the cyclic buffer is full and the updated read address for the cyclic buffer is written to RX_BUFF_RPTR. Otherwise, overflow occurs in the cyclic buffer and the audio is discontinuous.

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Step 8 After recording, write 0x00000000 to RX_DSP_CTRL and query RX_DSP_CTRL until its value is 0x20000000, which indicates that the RX channel stops working.

----End

CAUTION

Configure the AIP clock before starting the AIP, ensuring that AIAO_BCLK_RX and AIAO_WS_RX are normal.

Playback Process

The following assumes that the audio channels are stereo channels in I^2S mode, the sampling rate is 48 kHz, the sampling precision is 16 bits, the AIAO PLL clock source is a 1188 MHz clock, and the AIAO clock of the system controller is enabled. To play data, perform the following steps:

Step 1 Set I2S_CRG_CFG0_00 to 0x152EF0 to set the MCLK output frequency to 12.288 MHz.

Step 2 Set I2S_CRG_CFG1_00 to 0x00000133 to enable the clock, set the BCLK to the divide-by-4 clock of the MCLK, and set the FCLK to the divide-by-64 clock of the BCLK. The FCLK is 48 kHz.

Step 3 Set TX_IF_ATTRI to 0xE4000014 to set the operating mode of the RX channel to I^2S stereo mode and sampling precision to 16 bits.

Step 4 Set TX_BUFF_SADDR to the start address for the allocated buffer (for example, 0x00000100), set TX_BUFF_SIZE to the size of the allocated buffer, set TX_BUFF_WPTR and TX_BUFF_RPTR to 0x0, and set TX_TRANS_SIZE to the data transfer length. For details, see step 5 in the "Recording Process" section.

Step 5 Configure TX_INT_ENA to enable the corresponding interrupt of the RX channel as required. For example, set TX_INT_ENA to 0x00000001 to enable the trans_int interrupt.

Step 6 Set TX_DSP_CTRL to 0x10000000 to enable the playback channel.

Step 7 Read TX_BUFF_WPTR and TX_BUFF_RPTR to check the empty/full status of the cyclic buffer and the valid data amount.

Ensure that new audio data is stuffed before the cyclic buffer is empty and the updated write address for the cyclic buffer is written to TX_BUFF_WPTR. Otherwise, an underflow occurs in the cyclic buffer and the audio is discontinuous.

Step 8 After playback, write 0x00000000 to TX_DSP_CTRL to stop the playback channel and query TX_DSP_CTRL until its value is 0x20000000, which indicates that the playback channel stops working.

----End

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CAUTION

- Configure AOP clock before starting AOP, ensuring that AIAO_BCLK_RX and AIAO_WS_RX are normal.
- Ensure that the minimum available space of the AOP cyclic buffer is 32 bytes when writing data to the cyclic buffer and updating TX_BUFF_WPTR.

11.1.5 Register Summary

Table 11-1 describes AIAO registers.

Table 11-1 Summary of AIAO registers (base address: 0x2065_0000)

Offset Address	Register	Description	Page
0x0000	AIAO_INT_ENA	AIAO interrupt enable register	11-8
0x0004	AIAO_INT_STATUS	AIAO interrupt status register	11-9
0x0008	AIAO_INT_RAW	AIAO raw interrupt register	11-10
0x0100	I2S_CRG_CFG0_00	I^2S00 CRG configuration register 0	11-10
0x0104	I2S_CRG_CFG1_00	I^2S00 CRG configuration register 1	11-11
0x0140	I2S_CRG_CFG0_08	I^2S08 CRG configuration register 0	11-12
0x0144	I2S_CRG_CFG1_08	I^2S08 CRG configuration register 1	11-11
0x1000	RX_IF_ATTRI	Interface attribute configuration register for the RX channel	11-14
0x1004	RX_DSP_CTRL	RX channel control register	11-15
0x1080	RX_BUFF_SADDR	DDR buffer start address register for the RX channel	11-16
0x1084	RX_BUFF_SIZE	DDR buffer size register for the RX channel	11-17
0x1088	RX_BUFF_WPTR	DDR buffer write address register for the RX channel	11-17
0x108C	RX_BUFF_RPTR	DDR buffer read address register for the RX channel	11-17
0x1090	RX_BUFF_ALFULL_TH	DDR buffer almost full threshold register for the RX channel	11-18
0x1094	RX_TRANS_SIZE	Data transfer length register for the RX channel	11-18
0x10A0	RX_INT_ENA	Interrupt enable register for the RX channel	11-19

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Offset Address	Register	Description	Page
0x10A4	RX_INT_RAW	Raw interrupt register for the RX channel	11-20
0x10A8	RX_INT_STATUS	Interrupt status register for the RX channel	11-21
0x10AC	RX_INT_CLR	Interrupt clear register for the RX channel	11-22
0x2000	TX_IF_ATTRI	Interface attribute configuration register for the TX channel	11-23
0x2004	TX_DSP_CTRL	TX channel control register	11-24
0x2080	TX_BUFF_SADDR	DDR buffer start address register for the TX channel	11-26
0x2084	TX_BUFF_SIZE	DDR buffer size register for the TX channel	11-26
0x2088	TX_BUFF_WPTR	DDR buffer write address register for the TX channel	11-27
0x208C	TX_BUFF_RPTR	DDR buffer read address register for the TX channel	11-27
0x2090	TX_BUFF_ALEMPTY_TH	DDR buffer almost empty threshold register for the TX channel	11-28
0x2094	TX_TRANS_SIZE	Data transfer length register for the TX channel	11-28
0x20A0	TX_INT_ENA	Interrupt enable register for the TX channel	11-29
0x20A4	TX_INT_RAW	Raw interrupt register for the TX channel	11-30
0x20A8	TX_INT_STATUS	Interrupt status register for the TX channel	11-31
0x20AC	TX_INT_CLR	Interrupt clear register for the TX channel	11-33

11.1.6 Register Description

AIAO_INT_ENA

AIAO_INT_ENA is an AIAO interrupt enable register.

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Offset Address	Register Name	Total Reset Value
0x0000	AIAO_INT_ENA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															tx_ch0_int_ena	reserved															rx_ch0_int_ena
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RW	reserved	Reserved
[16]	RW	tx_ch0_int_ena	TX channel 0 interrupt enable 0: disabled 1: enabled
[15:1]	RW	reserved	Reserved
[0]	RW	rx_ch0_int_ena	RX channel 0 interrupt enable 0: disabled 1: enabled

AIAO_INT_STATUS

AIAO_INT_STATUS is an AIAO interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0004	AIAO_INT_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															tx_ch0_int_status	reserved															rx_ch0_int_status
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RO	tx_ch0_int_status	Interrupt status of TX channel 0 0: No interrupt is generated. 1: An interrupt is generated.

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[15:1]	RO	reserved	Reserved
[0]	RO	rx_ch0_int_status	Interrupt status of RX channel 0 0: No interrupt is generated. 1: An interrupt is generated.

AIAO_INT_RAW

AIAO_INT_RAW is an AIAO raw interrupt register.

Offset Address	Register Name	Total Reset Value
0x0008	AIAO_INT_RAW	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved			tx_ch0_int_raw	reserved			rx_ch0_int_raw
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved
[16]	RO	tx_ch0_int_raw	Raw interrupt of TX channel 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[15:1]	RO	reserved	Reserved
[0]	RO	rx_ch0_int_raw	Raw interrupt of RX channel 0 0: No raw interrupt is generated. 1: A raw interrupt is generated.

I2S_CRG_CFG0_00

I2S_CRG_CFG0_00 is I^2S00 CRG configuration register 0.

Offset Address	Register Name	Total Reset Value
0x0100	I2S_CRG_CFG0_00	0x00AA_AAAA

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved				aiao_mclk_div			

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Reset	0	0	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26:0]	RW	aiao_mclk_div	Configured value of the frequency division clock of the MCLK. The configured value is calculated as follows: (Target MCLK frequency/PLL frequency of the MCLK clock source) x 2^27. For details about the PLL frequency of the MCLK clock source, see section 3.2 "Clock" in chapter 3 "System" in the *Hi3518E V20X/Hi3516C V200 Economical HD IP Camera SoC Data Sheet*.

I2S_CRG_CFG1_00

I2S_CRG_CFG1_00 is I^{2}S00 CRG configuration register 1.

Offset Address	Register Name	Total Reset Value
0x0104	I2S_CRG_CFG1_00	0x0000_0131

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						aiao_srst_req	aiao_cken	reserved				aiao_fsclk_div			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	aiao_srst_req	Independent soft reset request of the RX 0 channel 0: deassert reset 1: reset
[8]	RW	aiao_cken	MCLK/BCLK/WS clock gating for CRG00 0: disabled 1: enabled
[7]	RO	reserved	Reserved

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[6:4]	RW	aiao_fsclk_div	Frequency division relationship between the bit clock BCLK and the sampling clock FS 000: The FS is obtained by dividing the BCLK by 16. 001: The FS is obtained by dividing the BCLK by 32. 010: The FS is obtained by dividing the BCLK by 48. 011: The FS is obtained by dividing the BCLK by 64. 100: The FS is obtained by dividing the BCLK by 128. 101: The FS is obtained by dividing the BCLK by 256. Other values: The FS is obtained by dividing the BCLK by 8.
[3:0]	RW	aiao_bclk_div	Frequency division relationship between the main clock MCLK and the bit clock BCLK 0000: The BCLK is obtained by dividing the MCLK by 1. 0001: The BCLK is obtained by dividing the MCLK by 3. 0010: The BCLK is obtained by dividing the MCLK by 2. 0011: The BCLK is obtained by dividing the MCLK by 4. 0100: The BCLK is obtained by dividing the MCLK by 6. 0101: The BCLK is obtained by dividing the MCLK by 8. 0110: The BCLK is obtained by dividing the MCLK by 12. 0111: The BCLK is obtained by dividing the MCLK by 16. 1000: The BCLK is obtained by dividing the MCLK by 24. 1001: The BCLK is obtained by dividing the MCLK by 32. 1010: The BCLK is obtained by dividing the MCLK by 48. 1011: The BCLK is obtained by dividing the MCLK by 64. Other values: The BCLK is obtained by dividing the MCLK by 8.

I2S_CRG_CFG0_08

I2S_CRG_CFG0_08 is I^{2}S08 CRG configuration register 0.

Offset Address	Register Name	Total Reset Value
0x0140	I2S_CRG_CFG0_08	0x00AA_AAAA

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				aiao_mclk_div																											
Reset	0	0	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved

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[26:0]	RW	aiao_mclk_div	Configured value of the frequency division clock of the MCLK. The configured value is calculated as follows: (Target MCLK frequency/Frequency of the MCLK clock source) x 2^27. The default frequency of the MCLK clock source (PLL clock) is 1188 MHz. For details about the PLL clock frequency, see section 3.2 "Clock" in chapter 3 "System" in the *Hi3518E V20X/Hi3516C V200 Economical HD IP Camera SoC Data Sheet*.

I2S_CRG_CFG1_08

I2S_CRG_CFG1_08 is I^{2}S08 CRG configuration register 1.

Offset Address	Register Name	Total Reset Value
0x0144	I2S_CRG_CFG1_08	0x0000_0131

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						aiao_srst_req	aiao_cken	reserved				aiao_fsclk_div				aiao_bclk_div			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0	1

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	aiao_srst_req	Independent soft reset request of the TX 0 channel 0: deassert reset 1: reset
[8]	RW	aiao_cken	MCLK/BCLK/WS clock gating for CRG08 0: disabled 1: enabled
[7]	RO	reserved	Reserved
[6:4]	RW	aiao_fsclk_div	Frequency division relationship between the bit clock BCLK and the sampling clock FS 000: The FS is obtained by dividing the BCLK by 16. 001: The FS is obtained by dividing the BCLK by 32. 010: The FS is obtained by dividing the BCLK by 48. 011: The FS is obtained by dividing the BCLK by 64. 100: The FS is obtained by dividing the BCLK by 128. 101: The FS is obtained by dividing the BCLK by 256. Other values: The FS is obtained by dividing the BCLK by 8.

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[3:0]	RW	aiao_bclk_div	Frequency division relationship between the main clock MCLK and the bit clock BCLK 0x0: The BCLK is obtained by dividing the MCLK by 1. 0x1: The BCLK is obtained by dividing the MCLK by 3. 0x2: The BCLK is obtained by dividing the MCLK by 2. 0x3: The BCLK is obtained by dividing the MCLK by 4. 0x4: The BCLK is obtained by dividing the MCLK by 6. 0x5: The BCLK is obtained by dividing the MCLK by 8. 0x6: The BCLK is obtained by dividing the MCLK by 12. 0x7: The BCLK is obtained by dividing the MCLK by 16. 0x8: The BCLK is obtained by dividing the MCLK by 24. 0x9: The BCLK is obtained by dividing the MCLK by 32. 0xA: The BCLK is obtained by dividing the MCLK by 48. 0xB: The BCLK is obtained by dividing the MCLK by 64. Other values: The BCLK is obtained by dividing the MCLK by 8.

RX_IF_ATTRI

RX_IF_ATTRI is an interface attribute configuration register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x1000	RX_IF_ATTRI	0xE400_0004

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19	18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2	1 0
Name	reserved		rx_sd_source_sel	reserved	rx_trackmode	rx_sd_offset		reserved	rx_ch_num	rx_i2s_precision	rx_mode
Reset	1 1 1 0	0 1 0 0	0 0 0 0	0	0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 1	0 0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:20]	RW	rx_sd_source_sel	This bit is set to 0x1000 in normal mode.
[19]	RO	reserved	Reserved
[18:16]	RW	rx_trackmode	Audio-left and audio-right channel mode in I^2S mode 000: The sound is not processed. 001: The sounds in two channels are audio-left channel sounds. 010: The sounds in two channels are audio-right channel sounds. 011: The sounds in two channels are exchanged. 100: The sounds of two channels are added and then output.

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			101: The audio-left channel is muted, and the sound of the audio-right channel is from the original audio-right channel. 110: The audio-right channel is muted, and the sound of the audio-left channel is from the original audio-left channel. 111: The audio-left and audio-right channels are muted. **NOTE** rx_trackmode is still valid in 1-channel RX mode.
[15:8]	RW	rx_sd_offset	PCM timing mode 0x0: left-aligned mode 0x1: standard mode Other values: reserved
[7]	RW	reserved	Reserved
[6:4]	RW	rx_ch_num	Number of RX channels 00: 1 01: 2 Other values: reserved
[3:2]	RW	rx_i2s_precision	Data sampling precision I^2S mode: 01: 16 bits 10: 24 bits Other values: reserved PCM mode: 01: 16 bits Other values: reserved
[1:0]	RW	rx_mode	Interface mode of the RX channel 00: I^2S mode 01: PCM mode Other values: reserved

RX_DSP_CTRL

RX_DSP_CTRL is an RX channel control register.

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Offset Address	Register Name	Total Reset Value
0x1004	RX_DSP_CTRL	0x2000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		rx_disable_done	rx_enable	bypass_en	reserved																										
Reset	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29]	RO	rx_disable_done	RX channel disable completion identifier 0: not complete 0: complete
[28]	RW	rx_enable	RX channel start/stop control bit 0: stop 1: start
[27]	RW	bypass_en	Data processing disable bit. The control function still takes effect after data processing is disabled. 0: Data processing is implemented normally. 1: Operations such as trackmode are not implemented.
[26:0]	RO	reserved	Reserved

RX_BUFF_SADDR

RX_BUFF_SADDR is a DDR buffer start address register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x1080	RX_BUFF_SADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rx_buff_saddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	rx_buff_saddr	Start address for the DDR buffer of RX channel 0. Its unit is byte. **NOTE** The start address must be 128x8-bit-aligned.

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RX_BUFF_SIZE

RX_BUFF_SIZE is a DDR buffer size register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x1084	RX_BUFF_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								rx_buff_size																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	rx_buff_size	DDR buffer size of the RX channel. Its unit is byte. **NOTE** The buffer size must be an integral multiple of 128 bytes.

RX_BUFF_WPTR

RX_BUFF_WPTR is a DDR buffer write address register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x1088	RX_BUFF_WPTR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								rx_buff_wptr																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	rx_buff_wptr	Write address for the DDR buffer of the RX channel. Its unit is byte. **NOTE** • The write address in the RX direction is maintained by the logic and is the offset address relative to the start address for the DDR buffer. • The write address must be 128x2-bit-aligned.

RX_BUFF_RPTR

RX_BUFF_RPTR is a DDR buffer read address register for the RX channel.

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Offset Address	Register Name	Total Reset Value
0x108C	RX_BUFF_RPTR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								rx_buff_rptr																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	rx_buff_rptr	Read address for the DDR buffer of the RX channel. Its unit is byte. **NOTE** • The read address in the RX direction is maintained by the logic and is the offset address relative to the start address for the DDR buffer. • The software operation is performed by byte, and the hardware operation is performed based on 128 x 2-bit alignment.

RX_BUFF_ALFULL_TH

RX_BUFF_ALFULL_TH is a DDR buffer almost full threshold register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x1090	RX_BUFF_ALFULL_TH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								rx_buff_alfull_th																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	rx_buff_alfull_th	Almost full threshold for the DDR buffer of the RX channel. Its unit is byte. If the available space of the DDR buffer is below the almost full threshold, the almost full raw interrupt is generated. **NOTE** If the rx_alfull_int interrupt is used, the field value must be an integral multiple of 16 bytes and greater than or equal to 0x40.

RX_TRANS_SIZE

RX_TRANS_SIZE is data transfer length register for the RX interrupt.

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Offset Address	Register Name	Total Reset Value
0x1094	RX_TRANS_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								rx_trans_size																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	rx_trans_size	After the RX channel receives the audio data with the length of rx_trans_size (in byte), a transfer completion interrupt is generated.

RX_INT_ENA

RX_INT_ENA is an interrupt enable register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x10A0	RX_INT_ENA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								rx_if_full_lost_int_ena	reserved	rx_stop_int_ena	reserved	rx_bfifo_full_int_ena	rx_alfull_int_ena	rx_full_int_ena	rx_trans_int_ena
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	rx_if_full_lost_int_ena	Interface data full loss raw interrupt enable for the RX channel 0: disabled 1: enabled
[6]	RO	reserved	Reserved
[5]	RW	rx_stop_int_ena	Stop interrupt enable for the RX channel 0: disabled 1: enabled
[4]	RO	reserved	Reserved

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[3]	RW	rx_bfifo_full_int_ena	Bus fifo overflow interrupt enable for the RX channel 0: disabled 1: enabled
[2]	RW	rx_alfull_int_ena	DDR buffer almost full interrupt enable for the RX channel 0: disabled 1: enabled
[1]	RW	rx_full_int_ena	DDR buffer full interrupt for the RX channel 0: disabled 1: enabled
[0]	RW	rx_trans_int_ena	Transfer completion interrupt enable for the RX channel 0: disabled 1: enabled

RX_INT_RAW

RX_INT_RAW is a raw interrupt register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x10A4	RX_INT_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								rx_if_full_lost_int_raw	reserved	rx_stop_int_raw	reserved	rx_bfifo_full_int_raw	rx_alfull_int_raw	rx_full_int_raw	rx_trans_int_raw
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RO	rx_if_full_lost_int_raw	Data full lost raw interrupt of the RX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[6]	RO	reserved	Reserved
[5]	RO	rx_stop_int_raw	Stop raw interrupt of the RX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[4]	RO	reserved	Reserved

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[3]	RO	rx_bfifo_full_int_raw	Bus FIFO overflow raw interrupt of the RX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[2]	RO	rx_alfull_int_raw	DDR buffer almost full raw interrupt of the RX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[1]	RO	rx_full_int_raw	DDR buffer full raw interrupt of the RX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[0]	RO	rx_trans_int_raw	Transfer completion raw interrupt of the RX interrupt 0: No raw interrupt is generated. 1: A raw interrupt is generated.

RX_INT_STATUS

RX_INT_STATUS is an interrupt status register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x10A8	RX_INT_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								tx_if_full_lost_int_status	reserved	rx_stop_int_status	reserved	rx_bfifo_full_int_status		rx_alfull_int_status		rx_full_int_status		rx_trans_int_status	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RO	tx_if_full_lost_int_status	Status of the data full loss interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.
[6]	RO	reserved	Reserved
[5]	RO	rx_stop_int_status	Status of the stop interrupt of the RX interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[4]	RO	reserved	Reserved

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[3]	RO	rx_bfifo_full_int_status	Status of the bus fifo overflow interrupt of the RC channel 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	rx_alfull_int_status	Status of the DDR buffer almost full interrupt of the RX channel 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	rx_full_int_status	Status of the DDR buffer full interrupt of the RX interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	rx_trans_int_status	Status of the transfer completion interrupt of the RX channel 0: No interrupt is generated. 1: An interrupt is generated.

RX_INT_CLR

RX_INT_CLR is an interrupt clear register for the RX channel.

Offset Address	Register Name	Total Reset Value
0x10AC	RX_INT_CLR	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7	6	5	4	3	2	1	0
Name	reserved						tx_if_full_lost_int_clear	reserved	rx_stop_int_clear	reserved	rx_bfifo_full_int_clear	rx_alfull_int_clear	rx_full_int_clear	rx_trans_int_clear
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	WC	tx_if_full_lost_int_clear	Data full loss interrupt clear for the TX channel 0: not cleared 1: cleared
[6]	RO	reserved	Reserved
[5]	WC	rx_stop_int_clear	Stop interrupt clear for the RX channel 0: not cleared 1: cleared
[4]	RO	reserved	Reserved

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[3]	WC	rx_bfifo_full_int_clear	Bus fifo overflow interrupt clear for the RX channel 0: not cleared 1: cleared
[2]	WC	rx_alfull_int_clear	DDR buffer almost full interrupt clear for the RX channel 0: not cleared 1: cleared
[1]	WC	rx_full_int_clear	DDR buffer full interrupt clear for the RX channel 0: not cleared 1: cleared
[0]	WO	rx_trans_int_clear	Transfer completion interrupt clear for the RX interrupt 0: not cleared 1: cleared

TX_IF_ATTRI

TX_IF_ATTRI is an interface attribute configuration register for the TX channel.

Offset Address	Register Name	Total Reset Value
0x2000	TX_IF_ATTRI	0xE400_0004

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved			tx_trackmode	tx_sd_offset		reserved	tx_ch_num	tx_i2s_precision	tx_mode
Reset	1 1 1 0	0 1 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 1 0 0

Bits	Access	Name	Description
[31:19]	RO	reserved	Reserved
[18:16]	RW	tx_trackmode	Audio-left and audio-right channel mode in I^2S mode 000: The sound is not processed. 001: The sounds in two channels are audio-left channel sounds. 010: The sounds in two channels are audio-right channel sounds. 011: The sounds in two channels are exchanged. 100: The sounds of two channels are added and then output. 101: The audio-left channel is muted, and the sound of the audio-right channel is from the original audio-right channel. 110: The audio-right channel is muted, and the sound of the audio-left channel is from the original audio-left channel. 111: The audio-left and audio-right channels are muted.

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			NOTE tx_trackmode is still valid in 1-channel TX mode.
[15:8]	RW	tx_sd_offset	PCM timing mode 0x0: left-aligned mode 0x1: standard mode Other values: reserved
[7:6]	RO	reserved	Reserved
[5:4]	RW	tx_ch_num	Select of the TX channel 00: Data is transmitted by one audio channel. 01: Data is transmitted by two audio channels as stereo. Other values: reserved
[3:2]	RW	tx_i2s_precision	Data sampling precision I^2S mode: 01: 16 bits 10: 24 bits Other values: reserved PCM mode: 01: 16 bits Other values: reserved
[1:0]	RW	tx_mode	Interface mode of the TX channel 00: I^2S mode 01: PCM mode Other values: reserved

TX_DSP_CTRL

TX_DSP_CTRL is a TX channel control register.

Offset Address	Register Name	Total Reset Value
0x2004	TX_DSP_CTRL	0x2000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		tx_disable_done	tx_enable	bypass_en	reserved			fade_out_rate				fade_in_rate				reserved	volume							reserved						mute_fade_en	mute_en

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Reset	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29]	RO	tx_disable_done	TX channel disable completion identifier 0: not complete 1: complete
[28]	RW	tx_enable	TX channel start/stop control bit 0: stop 1: start
[27]	RW	bypass_en	Data processing disable bit. The control function still takes effect after data processing is disabled. 0: Data processing is implemented normally. 1: Operations such as trackmode and volume processing are not implemented.
[26:24]	RO	reserved	Reserved
[23:20]	RW	fade_out_rate	Fade-out rate 0000: The fade-out rate changes every one sampling point. 0001: The fade-out rate changes every two sampling points. 0110: The fade-out rate changes every four sampling points. 0011: The fade-out rate changes every eight sampling points. 0100: The fade-out rate changes every 16 sampling points. 0101: The fade-out rate changes every 32 sampling points. 0110: The fade-out rate changes every 64 sampling points. 0111: The fade-out rate changes every 128 sampling points. Other values: reserved
[19:16]	RW	fade_in_rate	Fade-in rate 0000: The fade-in rate changes every one sampling point. 0001: The fade-in rate changes every two sampling points. 0010: The fade-in rate changes every four sampling points. 0011: The fade-in rate changes every eight sampling points. 0100: The fade-in rate changes every 16 sampling points. 0101: The fade-in rate changes every 32 sampling points. 0110: The fade-in rate changes every 64 sampling points. 0111: The fade-in rate changes every 128 sampling points. Other values: reserved
[15]	RO	reserved	Reserved

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[14:8]	RW	volume	Volume. The volume is incremented by 1 dB when the field value is incremented by 1. 0x00−0x28: mute 0x29: −80 dB 0x2A: −79 dB … 0x75: −4 dB 0x77: −2 dB 0x79: 0 dB 0x7B: +2 dB 0x7D: +4 dB 0x7F: +6 dB (maximum value)
[7:2]	RO	reserved	Reserved.
[1]	RW	mute_fade_en	Mute fade-in/fade-out control 0: disabled 1: enabled
[0]	RW	mute_en	Mute control 0: not muted 1: muted

TX_BUFF_SADDR

TX_BUFF_SADDR is a DDR buffer start address register for the TX channel.

Offset Address	Register Name	Total Reset Value
0x2080	TX_BUFF_SADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tx_buff_saddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	tx_buff_saddr	Start address for the DDR buffer of the TX channel. Its unit is byte. **NOTE** The start address must be 128x8-bit-aligned.

TX_BUFF_SIZE

TX_BUFF_SIZE is a DDR buffer size register for the TX channel.

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Offset Address	Register Name	Total Reset Value
0x2084	TX_BUFF_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								tx_buff_size																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	tx_buff_size	DDR buffer size of the TX channel. Its unit is byte. **NOTE** The buffer size must be an integral multiple of 128 bytes.

TX_BUFF_WPTR

TX_BUFF_WPTR is a DDR buffer write address register for the TX channel.

Offset Address	Register Name	Total Reset Value
0x2088	TX_BUFF_WPTR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								tx_buff_wptr																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	tx_buff_wptr	Write address for the DDR buffer of the TX channel **NOTE** • The write address in the TX direction is maintained by the logic and is the offset address relative to the start address for the DDR buffer. • The available space of the TX buffer must be greater than or equal to 128 bytes. • The software operation is performed by byte, and the hardware operation is performed by 128 x 2 bytes.

TX_BUFF_RPTR

TX_BUFF_RPTR is a DDR buffer read address register for the TX channel.

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Offset Address	Register Name	Total Reset Value
0x208C	TX_BUFF_RPTR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								tx_buff_rptr																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	tx_buff_rptr	Read address for the DDR buffer of the TX channel **NOTE** • The read address in the TX direction is maintained by the logic and is the offset address relative to the start address for the DDR buffer. • The address must be 128x2-bit-aligned.

TX_BUFF_ALEMPTY_TH

TX_BUFF_ALEMPTY_TH is a DDR buffer almost empty threshold register for the TX channel.

Offset Address	Register Name	Total Reset Value
0x2090	TX_BUFF_ALEMPTY_TH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								tx_buff_alempty_th																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	tx_buff_alempty_th	Almost empty threshold for the DDR buffer of the TX channel. Its unit is byte. If the available space of the DDR buffer is below the almost empty threshold, the almost empty raw interrupt is generated. **NOTE** If the tx_alempty_int interrupt is used, the field value must be an integral multiple of 16 bytes and greater than or equal to 0x20.

TX_TRANS_SIZE

TX_TRANS_SIZE is a data transfer length register for the TX channel.

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Offset Address	Register Name	Total Reset Value
0x2094	TX_TRANS_SIZE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								tx_trans_size																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	tx_trans_size	After the TX channel transmits the audio data with the length of tx_trans_size (in byte), a transfer completion interrupt is generated.

TX_INT_ENA

TX_INT_ENA is an interrupt enable register for the TX channel.

Offset Address	Register Name	Total Reset Value
0x20A0	TX_INT_ENA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								tx_dat_break_int_ena	tx_mfade_int_ena	tx_stop_int_ena	tx_ififo_empty_int_ena	tx_bfifo_empty_int_ena	tx_alempty_int_ena	tx_empty_int_ena	tx_trans_int_ena
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RW	tx_dat_break_int_ena	Interface data break interrupt enable for the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[6]	RW	tx_mfade_int_ena	Mute fade-in/fade-output completion interrupt enable for the TX channel 0: disabled 1: enabled

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[5]	RW	tx_stop_int_ena	Stop interrupt enable for the TX channel 0: disabled 1: enabled
[4]	RW	tx_ififo_empty_int_ena	Interface fifo underflow interrupt enable for the TX channel 0: disabled 1: enabled
[3]	RW	tx_bfifo_empty_int_ena	Bus fifo underflow interrupt enable for the TX channel 0: disabled 1: enabled
[2]	RW	tx_alempty_int_ena	DDR buffer almost empty interrupt enable for the TX channel 0: disabled 1: enabled
[1]	RW	tx_empty_int_ena	DDR buffer empty interrupt for the TX channel 0: disabled 1: enabled
[0]	RW	tx_trans_int_ena	Transfer completion interrupt enable for the TX channel 0: disabled 1: enabled

TX_INT_RAW

TX_INT_RAW is a raw interrupt register for the TX channel.

Offset Address	Register Name	Total Reset Value
0x20A4	TX_INT_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								tx_dat_break_int_raw	tx_mfade_int_raw	tx_stop_int_raw	tx_ififo_empty_int_raw	tx_bfifo_empty_int_raw	tx_alempty_int_raw	tx_empty_int_raw	tx_trans_int_raw
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved

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[7]	RO	tx_dat_break_int_raw	Interface data break raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[6]	RO	tx_mfade_int_raw	Mute fade-in/fade-output completion raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[5]	RO	tx_stop_int_raw	Stop raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[4]	RO	tx_ififo_empty_int_raw	Interface fifo underflow raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[3]	RO	tx_bfifo_empty_int_raw	Bus fifo underflow raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[2]	RO	tx_alempty_int_raw	DDR buffer almost empty raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[1]	RO	tx_empty_int_raw	DDR buffer empty raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[0]	RO	tx_trans_int_raw	Transfer completion raw interrupt of the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.

TX_INT_STATUS

TX_INT_STATUS is an interrupt status register for the TX channel.

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Offset Address	Register Name	Total Reset Value
0x20A8	TX_INT_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								tx_dat_break_int_status	tx_mfade_int_status	tx_stop_int_status	tx_ififo_empty_int_status	tx_bfifo_empty_int_status	tx_alempty_int_status	tx_empty_int_status	tx_trans_int_status
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RO	tx_dat_break_int_status	Status of the interface data break interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.
[6]	RO	tx_mfade_int_status	Status of the mute fade-in/fade-output completion interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.
[5]	RO	tx_stop_int_status	Status of the stop interrupt of the TX interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[4]	RO	tx_ififo_empty_int_status	Status of the interface FIFO underflow interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.
[3]	RO	tx_bfifo_empty_int_status	Status of the bus FIFO underflow interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	tx_alempty_int_status	DDR buffer almost empty interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	tx_empty_int_status	Status of the DDR buffer empty interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	tx_trans_int_status	Status of the transfer completion interrupt of the TX channel 0: No interrupt is generated. 1: An interrupt is generated.

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TX_INT_CLR

TX_INT_CLR is an interrupt clear register for the TX channel.

Offset Address	Register Name	Total Reset Value
0x20AC	TX_INT_CLR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																								tx_dat_break_int_clear	tx_mfade_int_clear	tx_stop_int_clear	tx_ififo_empty_int_clear	tx_bfifo_empty_int_clear	tx_alempty_int_clear	tx_empty_int_clear	tx_trans_int_clear
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	reserved	Reserved
[7]	RO	tx_dat_break_int_clear	Interface data break interrupt clear for the TX channel 0: No raw interrupt is generated. 1: A raw interrupt is generated.
[6]	WC	tx_mfade_int_clear	Mute fade-in/fade-output completion interrupt clear for the TX channel 0: not cleared 1: cleared
[5]	WC	tx_stop_int_clear	Stop interrupt clear for the TX channel 0: not cleared 1: cleared
[4]	WC	tx_ififo_empty_int_clear	Interface fifo underflow interrupt clear for the TX channel 0: not cleared 1: cleared
[3]	WC	tx_bfifo_empty_int_clear	Bus fifo underflow interrupt clear for the TX channel 0: not cleared 1: cleared
[2]	WC	tx_alempty_int_clear	DDR buffer almost empty interrupt clear for the TX channel 0: not cleared 1: cleared

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[1]	WC	tx_empty_int_clear	DDR buffer empty interrupt clear for the RX channel 0: not cleared 1: cleared
[0]	WC	tx_trans_int_clear	Transfer completion interrupt clear for the TX channel 0: not cleared 1: cleared

11.2 Audio CODEC

11.2.1 Overview

Hi3518E V20X/Hi3516C V200 is integrated with a high-performance audio CODEC: high-quality stereo playback DAC (96 dB DR A-weighted), single-ended line out; high-quality stereo recording ADC (90 dB DR A-weighted), one single-ended stereo input and one single-ended differential input; −1.5 dB to 50 dB MIC input, and gain control at 2 dB step (facilitating digital AGC). Hi3518E V20X/Hi3516C V200 supports two I^2S data interfaces, 8 kHz and 192 kHz standard sampling rates that can be used simultaneously, and digital audio mixing.

11.2.2 Features

The audio CODEC module has the following features:

- 96 dBA DR stereo DAC
- Single-ended stereo line out
- DAC digital volume control range: –121 dB to +6 dB, at 1 dB step
- 90 dBA DR stereo ADC
- Analog volume control range of ADC channel: –1.5 dB to 50 dB, at 2 dB step
- Digital volume control range of ADC channel: –96 dB to 30 dB, at 1 dB step
- One single-ended stereo input
- One single-ended differential input
- Internal MIC biasing
- Master and slave I^2S data interfaces, supporting 24 bits, 20 bits, 18 bits, and 16 bits, in binary format
- Audio sampling frequencies: 48 kHz, 44.1 kHz, and 32 kHz

  The sampling frequencies of each series are as follows:
  - The 32 kHz sampling frequencies series include 8 kHz, 16 kHz, 32 kHz, 64 kHz, and 128 kHz.
  - The 44.1 kHz sampling frequencies series include 11.025 kHz, 22.05 kHz, 44.1 kHz, 88.2 kHz, and 176.4 kHz.
  - The 48 kHz sampling frequencies series include 12 kHz, 24 kHz, 48 kHz, 96 kHz, and 192 kHz.
- The jitter tolerance of the analog clock CLKIN is 50 ps RMS.

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11.2.3 Function Description

The audio CODEC module provides the recording and playing function. In the case of audio recording, analog signals are input from the MIC or line-in end, the gain is amplified at the analog part, and the signals converted into digital signals and output from the I^2S interface. Stereo recording is supported. In the case of playback, audio signals are input through the I^2S interface, converted into analog signals by the DAC, and then output. Stereo playing is supported.

Recording Mode

In recording mode, the MIC or line-in signals are input from the analog input end, processed by using the programmable gain method, converted by the ADC, filtered and volume-adjusted at the digital part, and finally output from the I^2S interface. Then, the entire recording procedure is complete.

The procedure is as follows:

Step 1 Power on the audio CODEC module, and the reference voltage works normally one second later.

Step 2 Configure the registers according to register descriptions.

Step 3 Input analog audio signals to start recording. Signals are then output from the I^2S interface.

----End

Playing Mode

In playing mode, audio signals are transmitted from the I^2S interface to the DAC digital part for filtering and volume control, filtering is performed at the analog part, and finally output by the line-out end.

The procedure is as follows:

Step 1 Power on the audio CODEC module, and the reference voltage works normally one second later.

Step 2 Configure the registers according to register descriptions.

Step 3 Transmit signals from the I^2S interface, and then output analog audio signals from the line-out end.

----End

11.2.4 Register Summary

The audio CODEC is controlled by using the registers of the system controller with the base address 0x2012_0000. The offset addresses of these registers are 0x00C4, 0x00C8, 0x00CC, 0x00D0, 0x00D4 and 0x00D8. That is, no interface is provided for controlling the audio CODEC.

11.2.5 Register Description

For details about the registers related to the audio CODEC, see section 3.5.5.1 "Register Summary" and section 3.5.5.2 "Register Description."

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12 Peripherals

12.1 I²C Controller

12.1.1 Overview

The inter-integrated circuit (I^2C) module is used to read or write to the slave device connected on the I^2C bus through the CPU. When writing data to and reading data from the slave device, the CPU configures the I^2C configuration register over the bus and transmits the control information and the data to be used to the I^2C data communication register. After parsing the commands, the I^2C module transmits the data of the data channel register to the slave device over I^2C bus and notifies the CPU of the final status to the CPU by using interrupts after transmitting the data. The CPU reads data from the slave device in the similar way.

12.1.2 Function Description

The I^2C module has the following features:

- Hi3518E V20X/Hi3516C V200 provides the master I^2C interface. The frequency of the I^2C working reference clock is 24 MHz.
- Supports the bus arbitration in the case of multiple master devices.
- Supports clock synchronization and bit and byte waiting.
- Supports 7-bit standard address and 10-bit extended address.
- Supports standard mode (100 kbit/s) and high-speed mode (400 kbit/s).
- Supports general call and start byte.
- CBUS components are not supported.
- Supports the DMA operation.

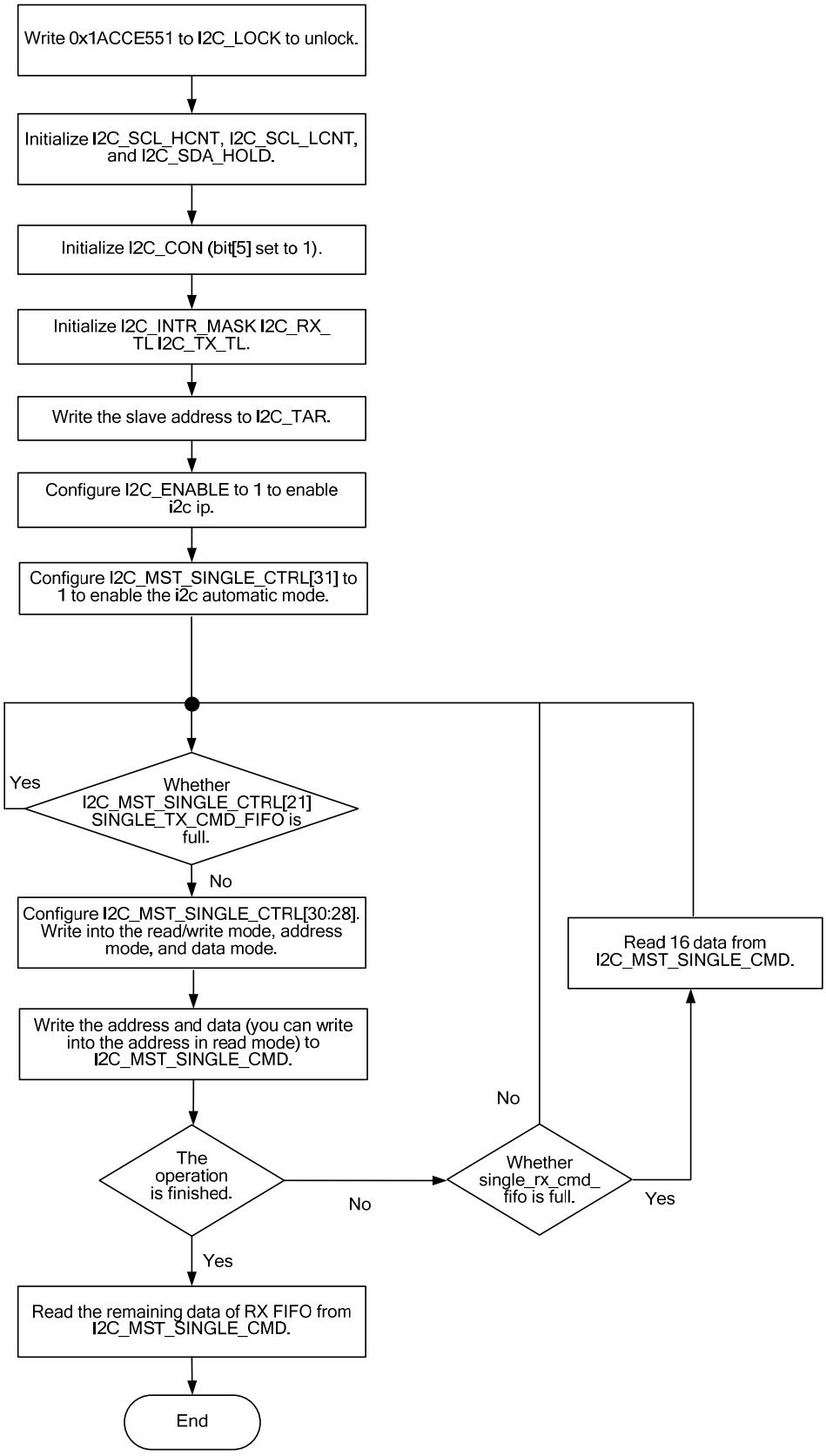
12.1.3 Operating Mode

Process of Receiving and Transmitting Data by a Single Operation through the I²C Master

Figure 12-1 shows the process of receiving data by a single operation through the I^2C master.

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Figure 12-1 Process of receiving data by a single operation through the I²C master

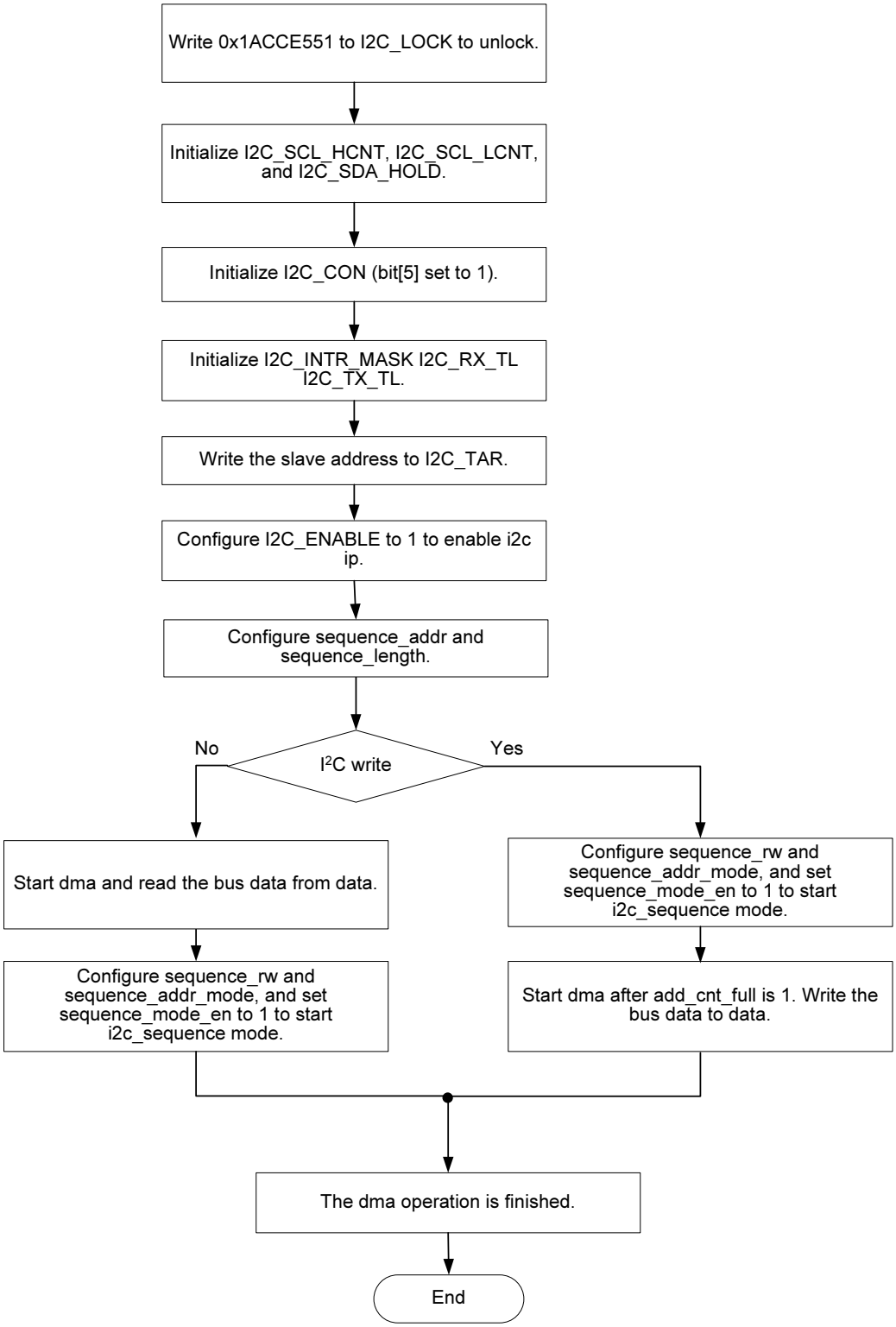


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Process of Continuously Receiving and Transmitting Data by the I²C Master

Figure 12-2 shows process of continuously receiving and transmitting data by the I²C master.

Figure 12-2 Process of continuously receiving and transmitting data by the I²C master



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12.1.4 Register Summary

Hi3518E V20X/Hi3516C V200 contains three I²C modules and Table 12-1 describes I²C registers.

Table 12-1 Summary of I²C registers (base addresses 0x200D_0000, 0x2024_0000, and 0x2025_0000 for I²C0, I²C1, and I²C2 respectively.)

Offset Address	Register	Description	Page
0x000	I2C_CON	I²C control register	12-5
0x004	I2C_TAR	I²C access slave device address register	12-6
0x010	I2C_DATA_CMD	I²C data operation register	12-6
0x01C	I2C_SCL_HCNT	I2C_SCL high level configuration register	12-7
0x020	I2C_SCL_LCNT	I2C_SCL low level configuration register	12-8
0x02C	I2C_INTR_STAT	I²C masked interrupt status register	12-8
0x030	I2C_INTR_MASK	I²C interrupt mask register	12-10
0x034	I2C_INTR_RAW	I²C raw interrupt status register	12-11
0x038	I2C_RX_TL	RX_FIFO threshold configuration register	12-13
0x03C	I2C_TX_TL	TX_FIFO threshold configuration register	12-14
0x040	I2C_CLR_INTR	I²C interrupt clear register	12-14
0x06C	I2C_ENABLE	I²C enable register	12-14
0x070	I2C_STATUS	I²C status register	12-15
0x074	I2C_TXFLR	TX_FIFO valid data indicator register	12-16
0x078	I2C_RXFLR	RX_FIFO valid data indicator register	12-16
0x07C	I2C_SDA_HOLD	SDA hold time register	12-17
0x080	I2C_TX_ABRT_SRC	I²C TX failure interrupt source register	12-17
0x088	I2C_DMA_CR	I²C DMA interface control register	12-20
0x08C	I2C_DMA_TDLR	TX_FIFO DMA operation threshold register	12-20
0x090	I2C_DMA_RDLR	RX_FIFO DMA operation threshold register	12-21
0x0A0	I2C_SCL_SWITCH	I²C anti-suspend enable register	12-21

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Offset Address	Register	Description	Page
0x0A4	I2C_SCL_SIM	I^{2}C anti-suspend analog register	12-21
0x0AC	I2C_LOCK	I^{2}C lock register	12-22
0x00B0	I2C_MST_SINGLE_CTRL	I2C_MST_SINGLE_CTRL register	12-22
0x00B4	I2C_MST_SINGLE_CMD	I2C_MST_SINGLE_CMD register	12-25
0x00B8	I2C_SEQUENCE_CMD0	I2C_SEQUENCE_CMD0 register	12-25
0x00BC	I2C_SEQUENCE_CMD1	I2C_SEQUENCE_CMD1 register	12-27
0x00C0	I2C_SEQUENCE_CMD2	I2C_SEQUENCE_CMD2 register	12-27

12.1.5 Register Description

I2C_CON

I2C_CON is an I^{2}C control register.

Offset Address	Register Name	Total Reset Value
0x000	I2C_CON	0x0000_0065

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved						restart_en	reserved
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 1 1 0	0 1 0 1

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5]	RW	restart_en	Restart command TX enable in master mode 0: disabled 1: enabled **NOTE** If the Restart function is disabled, the following functions are not supported: 1. Sending the start bit 2. High-speed mode 3. Read operation in 10-bit addressing mode 4. Combined addressing mode
[4:0]	RO	reserved	Reserved

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I2C_TAR

I2C_TAR is an I^{2}C access slave device address register.

Offset Address	Register Name	Total Reset Value
0x004	I2C_TAR	0x0000_002C

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																			master_10bit	special	gc_or_start	i2c_tar									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0

Bits	Access	Name	Description
[31:13]	RO	reserved	Reserved
[12]	RW	master_10bit	Address length in master mode 0: 7 bits 1: 10 bits
[11]	RW	special	General call or start byte function enable 0: disabled 1: enabled
[10]	RW	gc_or_start	I^{2}C command to be executed when the Special bit is 1 0: general call command (After the general call command is sent, you can perform only write operations. If you perform read operations, tx_abort interrupt is triggered. For details about the definition.) 1: start byte command
[9:0]	RW	i2c_tar	Slave address that is accessed by the I^{2}C when I^{2}C acts as a master device **NOTE** If the address length of the slave device is set to 7 bits, only bit [6:0] are valid.

I2C_DATA_CMD

I2C_DATA_CMD is an I^{2}C data operation register.

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Offset Address	Register Name	Total Reset Value
0x010	I2C_DATA_CMD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																							reserved	data							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:9]	RO	reserved	Reserved
[8]	RO	reserved	Reserved
[7:0]	RW	data	Data to be transmitted or received through the I^2C bus Read: The data received from the I^2C bus is read. Write: The written data is transmitted to the I^2C.

I2C_SCL_HCNT

I2C_SCL_HCNT is an I2C_SCL high level configuration register.

Offset Address	Register Name	Total Reset Value
0x01C	I2C_SCL_HCNT	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																i2c_scl_hcnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	i2c_scl_hcnt	High-level time of the SCL clock. The timing unit is I^2C working reference clock (24 MHz is used as an example). It is recommended that the field value be 1/2 of the SCL clock cycle in standard mode and be 0.36 of the SCL clock in fast mode. That is, i2c_scl_hcnt is calculated as follows: In standard mode: i2c_scl_hcnt = (fi2c/fSCL) x 0.5 In fast mode: i2c_scl_hcnt = (fi2c/fSCL) x 0.36 For example, if the 400 kHz SCL clock works in fast mode, i2c_scl_hcnt is 22 (0.36 x 24 MHz/400 kHz). **NOTE** 1. The field must be configured to obtain the appropriate I/O timing before data is transmitted through the I2C bus. 2. You can perform write operations only when the I2C interface is disabled (I2C_ENABLE = 0).

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I2C_SCL_LCNT

I2C_SCL_LCNT is an I2C_SCL low level configuration register.

Offset Address	Register Name	Total Reset Value
0x020	I2C_SCL_LCNT	0x0000_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																i2c_scl_lcnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	i2c_scl_lcnt	Low-level time of the SCL clock. The timing unit is I^2C working reference clock (24 MHz is used as an example). It is recommended that the field value be 1/2 of the SCL clock cycle in standard mode and be 0.64 of the SCL clock in fast mode. That is, i2c_scl_lcnt is calculated as follows: In standard mode: i2c_scl_lcnt = (fi2c/fSCL) x 0.5 In fast mode: i2c_scl_lcnt = (fi2c/fSCL) x 0.64 For example, if the 400 kHz SCL clock works in fast mode, i2c_scl_lcnt is 38 (0.64 x 24 MHz/400 kHz). **NOTE** 1. The field must be correctly configured to obtain the appropriate I/O timing before data is transmitted through the I^2C bus. 2. You can perform write operations only when the I^2C interface is disabled (I2C_ENABLE = 0).

I2C_INTR_STAT

I2C_INTR_STAT is an interrupt status register after the I^2C is masked.

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Offset Address	Register Name	Total Reset Value
0x02C	I2C_INTR_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				gen_call	start_det	stop_det	activity	reserved	tx_abrt	reserved	tx_empty	tx_over	rx_full	rx_over	rx_under
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11]	RO	gen_call	Status of the interrupt indicating that a general call request is received 0: No interrupt is generated. 1: An interrupt is generated.
[10]	RO	start_det	Status of the start detect interrupt indicating the I^2C bus interface has the START or RESTART condition 0: No interrupt is generated. 1: An interrupt is generated.
[9]	RO	stop_det	Status of the start detect interrupt indicating the I^2C bus interface has the STOP condition 0: No interrupt is generated. 1: An interrupt is generated.
[8]	RO	activity	Activity interrupt status. The I^2C activity status is recorded until the interrupt is cleared. 0: No interrupt is generated. 1: An interrupt is generated.
[7]	RO	reserved	Reserved
[6]	RO	tx_abrt	TX abort interrupt status 0: No interrupt is generated. 1: An interrupt is generated.
[5]	RO	reserved	Reserved
[4]	RO	tx_empty	Status of the interrupt indicating that TX_FIFO reaches or is below the threshold 0: No interrupt is generated. 1: An interrupt is generated.
[3]	RO	tx_over	TX_FIFO overflow interrupt status 0: No interrupt is generated. 1: An interrupt is generated.

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[2]	RO	rx_full	Status of the interrupt indicating that RX_FIFO reaches or is above the threshold 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	rx_over	RX_FIFO overflow interrupt status 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	rx_under	Read data overflow (that is, the CPU reads the FIFO when RX_FIFO is empty) interrupt status 0: No interrupt is generated. 1: An interrupt is generated.

I2C_INTR_MASK

I2C_INTR_MASK is an I^2C interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x030	I2C_INTR_MASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				gen_call_mask	start_det_mask	stop_det	activity	reserved	tx_abrt	reserved	tx_empty	tx_over	rx_full	rx_over	rx_under
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11]	RW	gen_call_mask	Interrupt mask when a general call request is received 0: not masked 1: masked
[10]	RW	start_det_mask	Start detect interrupt mask 0: not masked 1: masked
[9]	RW	stop_det	Stop detect interrupt mask 0: not masked 1: masked
[8]	RW	activity	Activity interrupt mask 0: not masked 1: masked

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[7]	RO	reserved	Reserved
[6]	RW	tx_abrt	TX abort interrupt mask 0: not masked 1: masked
[5]	RO	reserved	Reserved
[4]	RW	tx_empty	Interrupt mask when TX_FIFO reaches or is below the threshold 0: not masked 1: masked
[3]	RW	tx_over	TX_FIFO overflow interrupt mask 0: not masked 1: masked
[2]	RW	rx_full	Interrupt mask when RX_FIFO reaches or is below the threshold 0: not masked 1: masked
[1]	RW	rx_over	RX_FIFO overflow interrupt mask 0: not masked 1: masked
[0]	RW	rx_under	Interrupt mask when read data overflows (The CPU reads the FIFO when RX_FIFO is empty.) 0: not masked 1: masked

I2C_INTR_RAW

I2C_INTR_RAW is an I^{2}C raw interrupt status register.

Offset Address	Register Name	Total Reset Value
0x034	I2C_INTR_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				gen_call	start_det	stop_det	activity	reserved	tx_abrt	reserved	tx_empty	tx_over	rx_full	rx_over	rx_under
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved

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[11]	RO	gen_call	Interrupt status when a general call request is received 0: No interrupt is generated. 1: The slave receives the general call. **NOTE** This bit is set to 1 only when the general call request is received and confirmed. This value is retained unless being cleared.
[10]	RO	start_det	Start detect interrupt status, indicating whether the I^2C bus interface has the START or RESTART condition 0: No interrupt is generated. 1: The START or RESTART condition is detected.
[9]	RO	stop_det	Stop detect interrupt status, indicating whether the I^2C bus interface has the STOP condition 0: No interrupt is generated. 1: The STOP condition is detected.
[8]	RO	activity	Activity interrupt status. The I^2C activity status is recorded until the interrupt is cleared. 0: idle 1: busy
[7]	RO	reserved	Reserved
[6]	RO	tx_abrt	TX abort interrupt status 0: No interrupt is generated. 1: An interrupt is generated. **NOTE** When this bit is set to 1, the I2C_TX_ABRT_SRC register indicates the reason for sending the interrupt. If this bit is set to a value, the I^2C clears TX_FIFO. TX_FIFO remains in this state until the interrupt is cleared. After the interrupt is cleared, TX_FIFO reads more data over the APB interface.
[5]	RO	reserved	Reserved
[4]	RO	tx_empty	Interrupt status when TX_FIFO reaches or is below the threshold 0: No interrupt is generated. 1: An interrupt is generated. **NOTE** When data amount of TX_FIFO is below the threshold, this interrupt is automatically cleared. If the I^2C is disabled, TX_FIFO is cleared.
[3]	RO	tx_over	TX_FIFO overflow interrupt status 0: No interrupt is generated. 1: An interrupt is generated.

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[2]	RO	rx_full	Interrupt status when RX_FIFO reaches or is below the threshold 0: No interrupt is generated. 1: An interrupt is generated. **NOTE** This bit is set to 1 when RX_FIFO reaches or is below the threshold set by I2C_RX_TL. When RX_FIFO is below the threshold, this interrupt is automatically cleared. If the I^2C is disabled, RX_FIFO and this interrupt are automatically cleared.
[1]	RO	rx_over	RX_FIFO overflow interrupt status 0: No interrupt is generated. 1: An interrupt is generated. **NOTE** This bit is set to 1 if RX_FIFO is full but still receives bytes from the external I^2C device. The I^2C responds to data received over the I^2C bus. However, bytes received after RX_FIFO is full are lost.
[0]	RO	rx_under	Interrupt status when read data overflows (The CPU reads the FIFO when RX_FIFO is empty.) 0: No interrupt is generated. 1: An interrupt is generated.

I2C_RX_TL

I2C_RX_TL is an RX_FIFO threshold configuration register.

Offset Address	Register Name	Total Reset Value
0x038	I2C_RX_TL	0x0000_0003

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												rx_tl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	rx_tl	RX_FIFO threshold **NOTE** Valid values: 0–15. The actual value is the configured value plus 1. An rx_full interrupt is triggered when the number of data items in RX_FIFO is greater than or equal to the value. The upper threshold is configured as the RX_FIFO depth by default if the configured value is greater than the RX_FIFO depth.

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I2C_TX_TL

I2C_TX_TL is a TX_FIFO threshold configuration register.

Offset Address	Register Name	Total Reset Value
0x03C	I2C_TX_TL	0x0000_0004

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												tx_tl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	tx_tl	TX_FIFO threshold **NOTE** Valid values: 0–15. The actual value is equal to the configured value. A tx_empty interrupt is triggered when the amount of data in TX_FIFO is less than or equal to the value. The upper threshold is configured as the TX_FIFO depth by default if the configured value is greater than the TX_FIFO depth.

I2C_CLR_INTR

I2C_CLR_INTR is an I^2C interrupt clear register.

Offset Address	Register Name	Total Reset Value
0x040	I2C_CLR_INTR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															clr_intr
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	WO	clr_intr	Combined interrupts, all independent interrupts, and the I2C_TX_ABRT_SRC register are cleared when 1 is written to this bit.

I2C_ENABLE

I2C_ENABLE is an I^2C enable register.

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Offset Address	Register Name	Total Reset Value
0x06C	I2C_ENABLE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	enable	I^2C enable 0: disabled 1: enabled

I2C_STATUS

I2C_STATUS is an I^2C status register.

Offset Address	Register Name	Total Reset Value
0x070	I2C_STATUS	0x0000_0006

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									reserved	mst_activity	rx_fifo_full	rx_fifo_nempty	tx_fifo_nempty	tx_fifo_full	i2c_activity
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6]	RO	reserved	Reserved
[5]	RO	mst_activity	I^2C master activity status 0: The master enters the idle state, and the I^2C is deactivated. 1: The master does not enter the idle state, and the I^2C is activated.
[4]	RO	rx_fifo_full	RX_FIFO full flag 0: not full 1: full

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[3]	RO	rx_fifo_nempty	RX_FIFO non-empty flag 0: empty 1: non-empty
[2]	RO	tx_fifo_nempty	TX_FIFO empty flag 0: non-empty 1: empty
[1]	RO	tx_fifo_full	TX_FIFO non-full flag 0: full 1: not full
[0]	RO	i2c_activity	I^2C operating status 0: idle 1: active

I2C_TXFLR

I2C_TXFLR is a TX_FIFO valid data indicator register.

Offset Address	Register Name	Total Reset Value
0x074	I2C_TXFLR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											tx_flr				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RO	tx_flr	Number of valid data items in TX_FIFO The register is cleared when or a tx_abort interrupt is generated.

I2C_RXFLR

I2C_RXFLR is an RX_FIFO valid data indicator register.

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Offset Address	Register Name	Total Reset Value
0x078	I2C_RXFLR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												rx_flr			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved
[4:0]	RO	rx_flr	Number of valid data items in RX_FIFO The register is cleared when a tx_abort interrupt is generated.

I2C_SDA_HOLD

I2C_SDA_HOLD is an SDA hold time register.

Offset Address	Register Name	Total Reset Value
0x07C	I2C_SDA_HOLD	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sda_hold_fs															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	sda_hold_fs	SDA hold time. The timing unit is I^2C working reference clock. It is recommended that sda_hold_fs be set to I2C_SCL_LCNT[i2c_fs_lcnt]/2. **NOTE** This bit indicates the delay from the SCL falling edge to SDA change. The SDA changes after the SCL falling edge and the delay (value of sda_hold_fs x cycle of i2c_clk). This bit must be set based on the speed mode.

I2C_TX_ABRT_SRC

I2C_TX_ABRT_SRC is an I^2C TX failure interrupt source register.

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Offset Address	Register Name	Total Reset Value
0x080	I2C_TX_ABRT_SRC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																reserved	reserved	reserved	arb_lost	reserved	abrt_10b_rd_norstrt	abrt_sbyte_norstrt	reserved	abrt_sbyte_ackdet	reserved	abrt_gcall_read	abrt_gcall_noack	abrt_txdata_noack	abrt_10addr2_noack	abrt_10addr1_noack	abrt_7b_addr_noack
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RO	reserved	Reserved
[14]	RO	reserved	Reserved
[13]	RO	reserved	Reserved
[12]	RO	arb_lost	Whether an error occurs after the master loses the bus control permission 0: No error occurs. 1: An error occurs.
[11]	RO	reserved	Reserved
[10]	RO	abrt_10b_rd_norstrt	Whether an error occurs after the restart is disabled (I2C_CON[restart_en] set to 0) and the master attempts to initiate a read operation in 10-bit mode 0: No error occurs. 1: An error occurs.
[9]	RO	abrt_sbyte_norstrt	Whether an error occurs after the restart is disabled (I2C_CON[restart_en] set to 0) and a user attempts to initiate a start operation 0: No error occurs. 1: An error occurs.
[8]	RO	reserved	Reserved
[7]	RO	abrt_sbyte_ackdet	Whether an error occurs when the master sends a start command and receives a response 0: No error occurs. 1: An error occurs. (The slave does not need to respond to the start command.)
[6]	RO	reserved	Reserved

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[5]	RO	abrt_gcall_read	tx_abort generation cause 0: tx_abort is triggered due to a reason not defined by the bit. 1: The master sends a common request, but the user initiates a read operation after the request.
[4]	RO	abrt_gcall_noack	tx_abort generation cause 0: tx_abort is triggered due to a reason not defined by the bit. 1: The master sends a common request but does not receive a response from the slave over the bus.
[3]	RO	abrt_txdata_noack	tx_abort generation cause 0: tx_abort is triggered due to a reason not defined by the bit. 1: After the master sends an address and receives a response from the slave, the master sends data to the slave but does not receive any response.
[2]	RO	abrt_10addr2_noack	tx_abort generation cause 0: tx_abort is triggered due to a reason not defined by the bit. 1: The master sends address bit[10:8] the second time but does not receive any response from the slave when the master operates in 10-bit mode.
[1]	RO	abrt_10addr1_noack	tx_abort generation cause 0: tx_abort is triggered due to a reason not defined by the bit. 1: The master sends address bit[7:0] the first time but does not receive any response from the slave when the master operates in 10-bit mode.
[0]	RO	abrt_7b_addr_noack	tx_abort generation cause 0: tx_abort is triggered due to a reason not defined by the bit. 1: The master sends an address but does not receive the response from any slave on the bus when the master operates in 7-bit mode.

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I2C_DMA_CR

I2C_DMA_CR is an I^2C DMA interface control register.

Offset Address	Register Name	Total Reset Value
0x088	I2C_DMA_CR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														rdmae	tdmae
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	rdmae	DMA enable for the TX FIFO 0: disabled 1: enabled
[0]	RW	tdmae	DMA enable for the RX FIFO 0: disabled 1: enabled

I2C_DMA_TDLR

I2C_DMA_TDLR is a TX_FIFO DMA operation threshold register.

Offset Address	Register Name	Total Reset Value
0x08C	I2C_DMA_TDLR	0x0000_0004

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												dma_txtl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	dma_txtl	TX FIFO DMA threshold. When the number of data items of the TX FIFO is less than or equal to this value, a DMA operation request is sent. The DMA moves data to the TX FIFO.

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I2C_DMA_RDLR

I2C_DMA_RDLR is an RX_FIFO DMA operation threshold register.

Offset Address	Register Name	Total Reset Value
0x090	I2C_DMA_RDLR	0x0000_0004

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												dma_rxtl			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	dma_rxtl	RX FIFO DMA threshold. When the number of data items of the TX FIFO is greater than or equal to this value, a DMA operation request is sent. The DMA moves data of the RX FIFO to the specified destination. The actual value is equal to the configured value plus 1.

I2C_SCL_SWITCH

I2C_SCL_SWITCH is an I^2C anti-suspend enable register.

Offset Address	Register Name	Total Reset Value
0x0A0	I2C_SCL_SWITCH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															scl_switch
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	scl_switch	Master simulate SCL signal enable 0: disabled 1: enabled

I2C_SCL_SIM

I2C_SCL_SIM is an I^2C anti-suspend analog register.

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Offset Address	Register Name	Total Reset Value
0x0A4	I2C_SCL_SIM	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															scl_sim
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	scl_sim	SCL enable signals simulated by the master 0: The output SCL is 0. 1: The output SCL is 1.

I2C_LOCK

I2C_LOCK is an I^2C lock register.

Offset Address	Register Name	Total Reset Value
0x0AC	I2C_LOCK	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															i2c_lock
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	i2c_lock	I^2C lock register. To unlock the register, write 1ACCE551 to the register. To lock the register, write other values. When this register is read: 0: The register is unlocked, and all registers can be configured. 1: The register is locked, and only this register can be configured.

I2C_MST_SINGLE_CTRL

I2C_MST_SINGLE_CTRL is an I2C_MST_SINGLE_CTRL register.

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Offset Address	Register Name	Total Reset Value
0x00B0	I2C_MST_SINGLE_CTRL	0x0030_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	single_model_en	single_model	single_addr_model	single_data_model	single_mst_tx_abrt_clr	single_tx_cmd_fifo_over_clr	single_rx_cmd_fifo_under_clr	single_rx_cmd_fifo_over_clr	single_mst_tx_abrt	single_tx_cmd_fifo_over	single_tx_cmd_fifo_not_full	single_tx_cmd_fifo_empty	reserved				single_tx_cmd_fifo_cnt				single_rx_cmd_fifo_under	single_rx_cmd_fifo_over	single_rx_cmd_fifo_full	single_rx_cmd_fifo_not_empty	reserved				single_rx_cmd_fifo_cnt			
Reset	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	single_model_en	I^2C master single operation enable control 0: disabled 1: enabled
[30]	RW	single_model	I^2C master single operation mode control 0: write 1: read
[29]	RW	single_addr_model	I^2C master single operation address mode control 0: The slave register has an 8-bit address width. 1: The slave register has a 16-bit address width.
[28]	RW	single_data_model	I^2C master single operation data mode control 0: The slave register has an 8-bit data width. 1: The slave register has a 16-bit data width.
[27]	WC	single_mst_tx_abrt_clr	Transmission failure indicator clear in single-operation mode
[26]	WC	single_tx_cmd_fifo_over_clr	SINGLE_TX_FIFO overflow indicator clear
[25]	WC	single_rx_cmd_fifo_under_clr	SINGLE_RX_FIFO read data overflow indicator clear
[24]	WC	single_rx_cmd_fifo_over_clr	SINGLE_RX_FIFO overflow indicator clear
[23]	RO	single_mst_tx_abrt	Transmission failure indicator in single-operation mode 0: No transmission failure occurs. 1: A transmission failure occurs.

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[22]	RO	single_tx_cmd_fifo_over	SINGLE_TX_FIFO overflow indicator 0: Data in SINGLE_TX_FIFO does not overflow. 1: Data in SINGLE_TX_FIFO overflows.
[21]	RO	single_tx_cmd_fifo_not_full	SINGLE_TX_FIFO non-full indicator 0: SINGLE_TX_FIFO is full. The FIFO has at least 16 data items. 1: SINGLE_TX_FIFO is not full. The FIFO has less than 16 data items.
[20]	RO	single_tx_cmd_fifo_empty	SINGLE_TX_FIFO empty indicator 0: There is data in the FIFO. 1: There is no data in the FIFO.
[19:17]	RO	reserved	Reserved
[16:12]	RO	single_tx_cmd_fifo_cnt	Number of valid data items in SINGLE_TX_FIFO This register is cleared when the I^2C master single operation is disabled (I2C_MST_SINGLE_CTRL bit[31] set to 0).
[11]	RO	single_rx_cmd_fifo_under	Overflow status when read data overflows (The CPU reads the FIFO when SINGLE_RX_FIFO is empty.) 0: No read data overflows. 1: Read data overflows.
[10]	RO	single_rx_cmd_fifo_over	SINGLE_RX_FIFO overflow status 0: Data in SINGLE_RX_FIFO does not overflow. 1: Data in SINGLE_RX_FIFO overflows.
[9]	RO	single_rx_cmd_fifo_full	SINGLE_RX_FIFO full indicator 0: SINGLE_RX_FIFO is not full. The FIFO has less than 16 data items. 1: SINGLE_RX_FIFO is full. The FIFO has at least 16 data items.
[8]	RO	single_rx_cmd_fifo_not_empty	SINGLE_RX_FIFO non-empty indicator 0: There is no data in the FIFO. 1: There is data in the FIFO.
[7:5]	RO	reserved	Reserved
[4:0]	RO	single_rx_cmd_fifo_cnt	Number of valid data items in SINGLE_RX_FIFO This register is cleared when the I^2C master single operation is disabled (I2C_MST_SINGLE_CTRL bit[31] set to 0).

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I2C_MST_SINGLE_CMD

I2C_MST_SINGLE_CMD is an I2C_MST_SINGLE_CMD register.

Offset Address	Register Name	Total Reset Value
0x00B4	I2C_MST_SINGLE_CMD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	single_addr																single_data															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	WO	single_addr	Address of the slave register accessed using the I^2C master single operation **NOTE** The lower 8 bits are sent first, and the upper 8 bits are sent at last.
[15:0]	RW	single_data	Data of the slave register accessed using the I^2C master single operation **NOTE** The lower 8 bits are sent first, and the upper 8 bits are sent at last.

I2C_SEQUENCE_CMD0

I2C_SEQUENCE_CMD0 is an I2C_SEQUENCE_CMD0 register.

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Offset Address	Register Name	Total Reset Value
0x00B8	I2C_SEQUENCE_CMD0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sequential_mode_en	sequential_rw	sequential_addr_mode		reserved			reserved	reserved	add_cnt_full	sequential_busy	sequential_finish	sequential_tx_fifo_cnt																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	sequential_mode_en	Sequential mode enable (This function must be used together with the DMA.) 0: disabled 1: enabled
[30]	RW	sequential_rw	Operation type in sequential mode 0: read 1: write
[29:28]	RW	sequential_addr_mode	Start address type in sequential mode 00: 8 bits 01: 16 bits 10: 24 bits 11: 32 bits
[27]	RW	reserved	Reserved
[26:24]	RO	reserved	Reserved
[23]	RW	reserved	Reserved
[22]	RO	add_cnt_full	Start address transmission completion indicator, indicating that the DMA can be started
[21]	RO	sequential_busy	Operation busy indicator in sequential mode
[20]	RO	sequential_finish	Operation completion indicator in sequential mode
[19:0]	RO	sequential_tx_fifo_cnt	Number of completed operations in sequential mode

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I2C_SEQUENCE_CMD1

I2C_SEQUENCE_CMD1 is an I2C_SEQUENCE_CMD1 register.

Offset Address	Register Name	Total Reset Value
0x00BC	I2C_SEQUENCE_CMD1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sequential_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sequential_addr	Start address in sequential mode **NOTE** The lower 8 bits are sent first, and the upper 8 bits are sent at last.

I2C_SEQUENCE_CMD2

I2C_SEQUENCE_CMD2 is an I2C_SEQUENCE_CMD2 register.

Offset Address	Register Name	Total Reset Value
0x00C0	I2C_SEQUENCE_CMD2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sequential_length																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sequential_length	Sequential mode length The configured value is the actual value minus 1.

12.2 UART

12.2.1 Overview

The universal asynchronous receiver transmitter (UART) is an asynchronous serial communication interface. It performs serial-to-parallel conversion on the data received from peripherals and transmits the converted data to the internal bus. It also performs parallel-to-serial conversion on the data that is transmitted to peripherals. The UART is mainly used to interconnect the Hi3518E V20X/Hi3516C V200 with the UART of an external chip so that the two chips can communicate with each other.

Hi3518E V20X/Hi3516C V200 provides three UART units (UART0 to UART2).

- UART0: a two -wire UART for debugging

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- UART1: a four-wire UART for controlling the pan, tile, and zoom (PTZ).
- UART2: a two-wire UART for generating alarms or connecting to a universal UART device

12.2.2 Features

The UART unit has the following features:

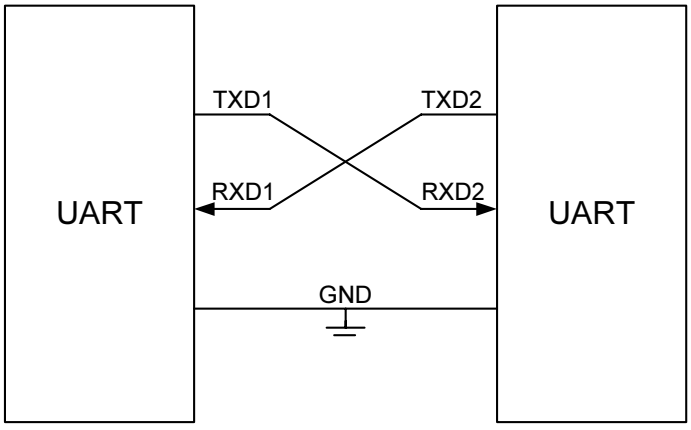
- Supports 64x8-bit transmit first-in, first-out (FIFO) and 64x12-bit RX FIFO.
- Supports programmable widths for the data bit and stop bit. The width of the data bit can be set to 5 bits, 6 bits, 7 bits, or 8 bits and width of the stop bit can be set to 1 bit or 2 bits by programming.
- Supports parity check or no check.
- Supports the programmable transfer rate.
- Supports RX FIFO interrupts, TX FIFO interrupts, RX timeout interrupts, and error interrupts.
- Supports the query of the raw interrupt status and the masked interrupt status.
- Allows the UART or the TX/RX function of the UART to be disabled by programming to reduce power consumption.
- Allows the UART clock to be disabled to reduce power consumption.
- Supports DMA operations.

12.2.3 Function Description

Application Block Diagram

Figure 12-3 shows the typical application of the UART.

Figure 12-3 Typical application block diagram of the UART



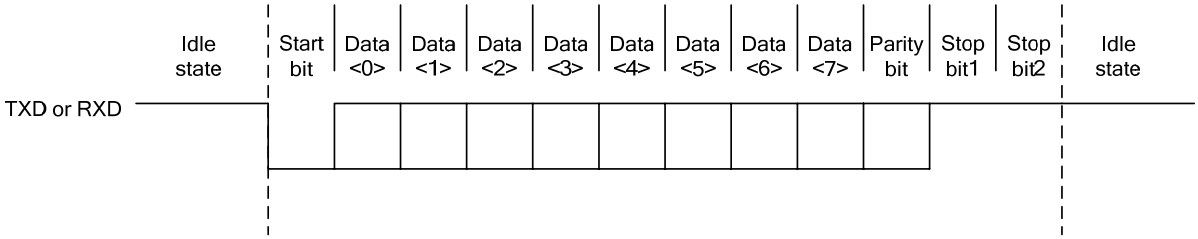
The UART serves as an asynchronous bidirectional serial bus. Through the UARTs connected by two data lines, a simplified and effective data transfer mode is implemented.

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Function Principle

A frame transfer of the UART involves the start signal, data signal, parity bit, and stop signal, as shown in Figure 12-4. The data frame is output from the TXD end of one UART and then is input to the RXD end of the other UART.

Figure 12-4 Frame format of the UART



The definitions of the start signal, data signal, parity bit, and stop signal are as follows:

- Start signal (start bit)

  It is the start flag of a data frame. According to the UART protocol, the low level of the TXD signal indicates the start of a data frame. When the UART does not transmit data, the level must remain high.

- Data signal (data bit)

  The data bit width can be set to 5 bits, 6 bits, 7 bits, or 8 bits according to the requirements in different applications.

- Parity bit

  It is a 1-bit error correction signal. The UART parity bit can be an odd parity bit, even parity bit, or stick parity bit. The UART can enable and disable the parity bit. For details, see the description of the

  UART_LCR_H register.

- Stop signal (stop bit)

  It is the stop bit of a data frame. The stop bit width can be set to 1 bit or 2 bits. The high level of TXD indicates the end of the data frame.

12.2.4 Operating Mode

12.2.4.1 Baud Rate Configuration

The operating baud rate of the UART can be set by configuring the registers UART_IBRD and UART_FBRD. The baud rate is calculated as follows:

Current baud rate = Frequency of the UART reference clock/(16 x Clock divider). The frequency of the UART reference clock is 24 MHz by default.

The clock frequency divider consists of the integral part and the fractional part that correspond to UART_IBRD and UART_FBRD respectively.

For example, assume that the frequency of the UART reference clock is 24 MHz. If UART_IBRD is set to 0x1E and UART_FBRD is set to 0x00, the current baud rate is calculated as follows: 24/(16 x 30) = 0.005 Mbit/s

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The typical UART baud rates are 9600 bit/s, 14,400 bit/s, 19,200 bit/s, 38,400 bit/s, 57,600 bit/s, 76,800 bit/s, 115,200 bit/s, 230,400 bit/s, and 460,800 bit/s.

The following examples show how to calculate the clock divider and how to configure the clock divider register:

If the required baud rate is 230,400 bit/s and the frequency of the UART reference clock is 24 MHz, the clock divider is calculated as follows: $(24x10^6)/(16 \times 230,400) = 6.5104$. The integral part IBRD is 6 and the fractional part FBRD is 0.5104.

To calculate the value of the 6-bit UART_FBRD register, do as follows: calculate the value of *m* by using the following formula: $m = integer (FBRD \times 2^n + 0.5) = (0.5104 \times 2^6 + 0.5) = 33$(n = the width of UART_FBRD). Then set UART_IBRD to 0x0006 and set UART_FBRD to 0x21.

If the fractional part of the frequency divider is set to 33, the actual divisor of the baud rate is 6 + 33/64 = 6.5156, the baud rate is (24 x 16)/(16 x 605156) = 230,216.7107, and the error rate is (230,216.7107 – 230,400)/230,400 x 100 = -0.07956%.

The maximum error rate is 1/64 x 100 = 1.56% when the 6-bit UART_FBRD is used. If the value of *m* is 1, the total error rate is greater than 64 clock cycles.

12.2.4.2 Soft Reset

The UART controller can be separately reset by configuring the CRG registers.

- The UART0 controller can be separately reset by setting PERI_CRG57[uart0_srst_req] to 1.
- The UART1 controller can be separately reset by setting PPERI_CRG57[uart1_srst_req] to 1.
- The UART2 controller can be separately reset by setting PERI_CRG57[uart2_srst_req] to 1.

NOTE

For details about reset registers, see section 3.2.7 "Register Description."

After reset, the configuration registers are restored to default values. Therefore, these registers must be initialized again.

12.2.4.3 Data Transfer in Interrupt or Query Mode

Initialization

The initialization is implemented as follows:

Step 1 Write 0 to UART_CR bit[0] to disable the UART.

Step 2 Write to UART_IBRD and UART_FBRD to configure the transfer rate.

Step 3 Configure UART_CR and UART_LCR_H to set the UART operating mode.

Step 4 Configure UART_IFLS to set the thresholds of TX and RX FIFOs.

Step 5 If the driver runs in interrupt mode, set UART_IMSC to enable the corresponding interrupt; if the driver runs in query mode, disable the generation of corresponding interrupts.

Step 6 Write 1 to UART_CR bit[0] to enable the UART.

----End

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Data Transmission

To transmit data, perform the following steps:

Step 1 Write the data to be transmitted to UART_DR and start data transmission.

Step 2 In query mode, check the TX_FIFO status by reading UART_FR bit[5] during the continuous data transmission. According to the TX_FIFO status, determine whether to transmit data to TX_FIFO. In interrupt mode, check the TX_FIFO status by reading the corresponding interrupt status bits and then determine whether to transmit data to TX_FIFO.

Step 3 Check whether the UART transmits all data by reading UART_FR bit[7]. If UART_FR bit[7] is 1, the UART transmits all data.

----End

Data Reception

To receive data, perform the following steps:

- In query mode, detect the RX_FIFO status by reading UART_FR[rxfe] during data reception and then determine whether to read data from the RX_FIFO according to the RX_FIFO status.
- In interrupt mode, determine whether to read data from RX_FIFO according to corresponding interrupt status bits.

12.2.4.4 Data Transfer in DMA Mode

Initialization

To initialize the UART, perform the following steps:

Step 1 Write 0 to UART_CR[uarten] to disable the UART.

Step 2 Write values to UART_IBRD and UART_FBRD to configure the data transfer rate.

Step 3 Configure UART_CR and UART_LCR_H to set the UART operating mode.

Step 4 Configure UART_IFLS to set the TX and RX FIFO thresholds.

Step 5 If the driver runs in interrupt mode, set UART_IMSC to enable corresponding interrupts. If the driver runs in query mode, disable the generation of corresponding interrupts.

Step 6 Write 1 to UART_CR[uarten] to enable the UART.

----End

Data Transfer

To transmit data, perform the following steps (the following is an example using the DMA to transmit data):

Step 1 Configure the DMA data channel, including the transfer source address, transfer destination address, number of data items to be transferred, and transfer type. For details, see the description in xx "DMAC."

Step 2 Set UART_DMACR to 0x2 to enable the DMA transfer function of the UART.

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Step 3 Check whether the data is transferred completely based on the interrupt report status of the DMA. If all data is transferred, disable the DMA transfer function of the UART.

----End

Data Reception

To receive data, perform the following steps (the following is an example using the DMA to receive data):

Step 1 Configure the DMA data channel, including data transfer source and destination addresses, data receive area address, number of data items to be transferred, and transfer type.

Step 2 Set UART_DMACR to 0x1 to enable the DMA receive function of the UART.

Step 3 Check whether the data is received completely by querying the DMA status. If all data is received, disable the DMA receive function of the UART.

----End

12.2.5 Register Summary

Hi3518E V20X/Hi3516C V200 provides three UART units: UART0, UART1 and UART2. Their base addresses are as follows:

- The base address of UART0 registers is 0x2008_0000.
- The base address of UART1 registers is 0x2009_0000.
- The base address of UART2 registers is 0x200A_0000.

Table 12-2 describes the UART registers.

Table 12-2 Summary of UART registers

Offset Address	Register	Description	Page
0x000	UART_DR	Data register	12-33
0x004	UART_RSR	Receive status register or error clear register	12-34
0x018	UART_FR	Flag register	12-35
0x024	UART_IBRD	Integral baud rate register	12-36
0x028	UART_FBRD	Fractional baud rate register	12-36
0x02C	UART_LCR_H	Line control register	12-37
0x030	UART_CR	Control register	12-38
0x034	UART_IFLS	Interrupt FIFO threshold select register	12-40
0x038	UART_IMSC	Interrupt mask register	12-41
0x03C	UART_RIS	Raw interrupt status register	12-42
0x040	UART_MIS	Masked interrupt status register	12-43

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Offset Address	Register	Description	Page
0x044	UART_ICR	Interrupt clear register	12-44
0x048	UART_DMACR	DMA control register	12-45

12.2.6 Register Description

UART_DR

UART_DR is a UART data register that stores the received data and the data to be transmitted. The receive status can be queried by reading this register.

Offset Address	Register Name	Total Reset Value
0x000	UART_DR	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				oe	be	pe	fe	data							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:12]	RO	reserved	Reserved
[11]	RO	oe	Overflow error 0: No overflow error occurs. 1: An overflow error occurs. That is, a data segment is received when the RX FIFO is full.
[10]	RO	be	Break error 0: No break error occurs. 1: A break error occurs. That is, the time of RX data input signal keeping low is longer than a full word transfer. A full word consists of a start bit, a data bit, a parity bit, and a stop bit.
[9]	RO	pe	Parity error 0: No parity error occurs. 1: A parity error occurs.
[8]	RO	fe	Frame error 0: No frame error occurs. 1: A frame error (namely, stop bit error) occurs.
[7:0]	RW	data	Data received and to be transmitted

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UART_RSR

UART_RSR is a receive status register or error clear register.

- It acts as the receive status register when being read.
- It acts as the error clear register when being written.

You can query the receive status by reading UART_DR. The status information about the break, frame, and parity read from UART_DR has priority over that read from UART_RSR. That is, the status read from UART_DR changes faster than that read from UART_RSR.

UART_RSR is reset when any value is written to it.

Offset Address	Register Name	Total Reset Value
0x004	UART_RSR	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved				oe	be	pe	fe
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:4]	RO	reserved	Reserved
[3]	RW	oe	Overflow error 0: No overflow error occurs. 1: An overflow error occurs. When the FIFO is full, the next data segment cannot be written to the FIFO and an overflow occurs in the shift register. Therefore, the contents in the FIFO are valid. In this case, the CPU must read the data immediately to spare the FIFO.
[2]	RW	be	Break error 0: No break error occurs. 1: A break error occurs. A break error occurs when the time of the RX data signal keeping low is longer than a full word transfer. A full word consists of a start bit, a data bit, a parity bit, and a stop bit.
[1]	RW	pe	Parity error 0: No parity error occurs. 1: A parity error occurs when the received data is checked. In FIFO mode, the error is associated with the data at the top of the FIFO.
[0]	RW	fe	Frame error 0: No frame error occurs. 1: The stop bit of the received data is incorrect. The valid stop bit is 1.

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UART_FR

UART_FR is a UART flag register.

Offset Address	Register Name	Total Reset Value
0x018	UART_FR	0x0012

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								txfe	rxff	txff	rxfe	busy	reserved		
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0

Bits	Access	Name	Description
[15:8]	RO	reserved	Reserved
[7]	RO	txfe	The definition of the bit is determined by the status of UART_LCR_H [fen]. If UART_LCR_H [fen] is 0, this bit is set to 1 when the TX holding register is empty. If UART_LCR_H [fen] is 1, this bit is set to 1 when the TX FIFO is empty.
[6]	RO	rxff	The definition of the bit is determined by the status of UART_LCR_H [fen]. If UART_LCR_H [fen] is 0, this bit is set to 1 when the RX holding register is full. If UART_LCR_H [fen] is 1, this bit is set to 1 when the RX FIFO is full.
[5]	RO	txff	The definition of the bit is determined by the status of UART_LCR_H [fen]. If UART_LCR_H [fen] is 0, this bit is set to 1 when the TX holding register is full. If UART_LCR_H [fen] is 1, this bit is set to 1 when the TX FIFO is full.
[4]	RO	rxfe	The definition of the bit is determined by the status of UART_LCR_H [fen]. If UART_LCR_H [fen] is 0, this bit is set to 1 when the RX holding register is empty. If UART_LCR_H [fen] is 1, this bit is set to 1 when the RX FIFO is empty.
[3]	RO	busy	UART busy/idle status 0: The UART is idle or data transmission is complete. 1: The UART is busy in transmitting data. If the bit is set to 1, the status is kept until the entire byte (including all stop bits) is transmitted from the shift register. Regardless of whether the UART is enabled, this bit is set to 1 when the TX FIFO is not empty.
[2:0]	RO	reserved	Reserved

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UART_IBRD

UART_IBRD is an integral baud rate register.

Offset Address	Register Name	Total Reset Value
0x024	UART_IBRD	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	baud divint															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:0]	RW	baud divint	Clock frequency divider corresponding to the integral part of the baud rate. All bits are cleared after reset.

UART_FBRD

UART_FBRD is a fractional baud rate register.

CAUTION

- The values of UART_IBRD and UART_FBRD can be updated only after the current data is transmitted and received completely.
- The minimum clock frequency divider is 1 and the maximum divider is 65,535 (216 - 1). That is, UART_IBRD cannot be 0 and UART_FBRD is ignored if UART_IBRD is 0. Similarly, if UART_IBRD is equal to 65,535 (0xFFFF), UART_IBRD must be 0. If UART_FBRD is greater than 0, the data fails to be transmitted or received.

Offset Address	Register Name	Total Reset Value
0x028	UART_FBRD	0x00

Bit	7	6	5	4	3	2	1	0
Name	reserved		baud divfrac					
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:6]	RO	reserved	Reserved
[5:0]	RW	band divfrac	Clock frequency divider corresponding to the fractional part of the baud rate. All bits are cleared after reset.

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UART_LCR_H

UART_LCR_H is a line control register. The registers UART_LCR_H, UART_IBRD, and UART_FBRD are combined to form a 30-bit register. If UART_IBRD and UART_FBRD are updated, UART_LCR_H must be updated at the same time.

Offset Address	Register Name	Total Reset Value
0x02C	UART_LCR_H	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								sps	wlen		fen	stp2	eps	pen	brk
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:8]	RO	reserved	Reserved
[7]	RW	sps	Parity select When bit 1, bit 2, and bit 7 of this register are set to 1, the parity bit is 0 during transmission and detection. When bit 1 and bit 7 of this register are set to 1 and bit 2 is set to 0, the parity bit is 1 during transmission and detection. When bit 1, bit 2, and bit 7 are cleared, the stick parity bit is disabled.
[6:5]	RW	wlen	Count of bits in a transmitted or received frame 00: 5 bits 01: 6 bits 10: 7 bits 11: 8 bits
[4]	RW	fen	TX/RX FIFO enable 0: disabled 1: enabled
[3]	RW	stp2	2-bit stop bit at the end of a transmitted frame 0: There is no 2-bit stop bit at the end of the transmitted frame. 1: There is a 2-bit stop bit at the end of the transmitted frame. The RX logic does not check for the 2-bit stop bit during data reception.
[2]	RW	eps	Parity select during data transmission and reception 0: The odd parity is generated or checked during data transmission and reception. 1: The even parity is generated or checked during data transmission and reception. When UART_LCR_H[fen] is 0, this bit becomes invalid.
[1]	RW	pen	Parity enable 0: The parity is disabled.

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			1: The parity is generated on the TX side and checked on the RX side.
[0]	RW	brk	Break transmit 0: invalid 1: After the current data transmission is complete, UTXD outputs low level continuously. **NOTE** This bit must retain 1 during the period of at least two full frames to ensure the break command is executed properly. In general, the bit must be set to 0.

UART_CR

UART_CR is a UART control register.

To configure UART_CR, perform the following steps:

Step 1 Write 0 to UART_CR[uarten] to disable the UART.

Step 2 Wait until the current data transmission or reception is complete.

Step 3 Clear UART_LCR_H[fen].

Step 4 Configure UART_CR.

Step 5 Write 1 to UART_CR[uarten] to enable the UART.

----End

Offset Address	Register Name	Total Reset Value
0x030	UART_CR	0x0300

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ctsen	rtsen	reserved		rts	dtr	rxe	txe	lbe	reserved						uarten
Reset	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15]	RW	ctsen	CTS hardware flow control enable 0: disabled 1: enabled. Data is transmitted only when the nUARTCTS signal is valid.
[14]	RW	rtsen	RTS hardware flow control enable 0: disabled 1: enabled. The data reception request is raised only when the RX FIFO has available space.
[13:12]	RO	reserved	Reserved
[11]	RW	rts	Request transmit

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			This bit is the inversion of the status output signal nUARTRTS of the UART modem. 0: The output signal retains. 1: When this bit is set to 1, the output signal is 0.
[10]	RW	dtr	Data transmit ready This bit is the inversion of the status output signal nUARTDTR of the UART modem. 0: The output signal retains. 1: When this bit is set to 1, the output signal is 0.
[9]	RW	rxe	UART receive enable 0: disabled 1: enabled If the UART is disabled during data reception, the current data reception is stopped abnormally.
[8]	RW	txe	UART transmit enable 0: disabled 1: enabled If the UART is disabled during data transmission, the current data transmission is stopped abnormally.
[7]	RW	lbe	Loopback enable 0: disabled 1: UARTTXD is looped back to UARTRXD.
[6:1]	RO	reserved	Reserved
[0]	RW	uarten	UART enable 0: disabled 1: enabled If the UART is disabled during data reception and transmission, the data transfer is stopped abnormally.

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UART_IFLS

UART_IFLS is an interrupt FIFO threshold select register. It is used to set a threshold for triggering a FIFO interrupt (UART_TXINTR or UART_RXINTR).

Offset Address	Register Name	Total Reset Value
0x034	UART_IFLS	0x0012

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										rxiflsel			txiflsel		
Reset	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0

Bits	Access	Name	Description
[15:6]	RO	reserved	Reserved
[5:3]	RW	rxiflsel	RX interrupt FIFO threshold select. An RX interrupt is triggered when any of the following conditions is met: 000: RX FIFO ≥ 1/8 full 001: RX FIFO ≥ 1/4 full 010: RX FIFO ≥ 1/2 full 011: RX FIFO ≥ 3/4 full 100: RX FIFO ≥ 7/8 full 101–111: reserved
[2:0]	RW	txiflsel	TX interrupt FIFO threshold select. A TX interrupt is triggered when any of the following conditions is met: 000: TX FIFO ≤ 1/8 full 001: TX FIFO ≤ 1/4 full 011: TX FIFO ≤ 3/4 full 010: TX FIFO ≤ 1/2 full 100: TX FIFO ≤ 7/8 full 101–111: reserved

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UART_IMSC

UART_IMSC is an interrupt mask register. It is used to mask interrupts.

Offset Address	Register Name	Total Reset Value
0x038	UART_IMSC	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					oeim	beim	peim	feim	rtim	txim	rxim	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:11]	RO	reserved	Reserved
[10]	RW	oeim	Mask status of the overflow error interrupt 0: masked 1: not masked
[9]	RW	beim	Mask status of the break error interrupt 0: masked 1: not masked
[8]	RW	peim	Mask status of the parity interrupt 0: masked 1: not masked
[7]	RW	feim	Mask status of the frame error interrupt 0: masked 1: not masked
[6]	RW	rtim	Mask status of the RX timeout interrupt 0: masked 1: not masked
[5]	RW	txim	Mask status of the TX interrupt 0: masked 1: not masked
[4]	RW	rxim	Mask status of the RX interrupt 0: masked 1: not masked
[3:0]	RO	reserved	Reserved

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UART_RIS

UART_RIS is a raw interrupt status register. The contents of this register are not affected by the UART_IMSC register.

Offset Address	Register Name	Total Reset Value
0x03C	UART_RIS	0x0002

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					oeris	beris	peris	feris	rtris	txris	rxris	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[15:11]	RO	reserved	Reserved
[10]	RO	oeris	Status of the raw overflow error interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[9]	RO	beris	Status of the raw break error interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[8]	RO	peris	Status of the raw parity interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[7]	RO	feris	Status of the raw error interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[6]	RO	rtris	Status of the raw RX timeout interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[5]	RO	txris	Status of the raw TX interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[4]	RO	rxris	Status of the raw RX interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[3:0]	RO	reserved	Reserved

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UART_MIS

UART_MIS is a masked interrupt status register. The contents of this register are the results obtained after the raw interrupt status is ANDed with the interrupt mask status.

Offset Address	Register Name	Total Reset Value
0x040	UART_MIS	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					oemis	bemis	pemis	femis	rtmis	txmis	rxmis	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:11]	RO	reserved	Reserved
[10]	RO	oemis	Status of the masked overflow error interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[9]	RO	bemis	Status of the masked break error interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[8]	RO	pemis	Status of the masked parity interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[7]	RO	femis	Status of the masked error interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[6]	RO	rtmis	Status of the masked RX timeout interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[5]	RO	txmis	Status of the masked TX interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[4]	RO	rxmis	Status of the masked RX interrupt 0: No interrupt is generated. 1: An interrupt is generated.
[3:0]	RO	reserved	Reserved

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UART_ICR

UART_ICR is an interrupt clear register. When 1 is written to it, the corresponding interrupt is cleared. Writing 0 has no effect.

Offset Address	Register Name	Total Reset Value
0x044	UART_ICR	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved					oeic	beic	peic	feic	rtic	txic	rxic	reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:11]	RO	reserved	Reserved
[10]	WO	oeic	Overflow error interrupt clear 0: invalid 1: cleared
[9]	WO	beic	Break error interrupt clear 0: invalid 1: cleared
[8]	WO	peic	Parity interrupt clear 0: invalid 1: cleared
[7]	WO	feic	Error interrupt clear 0: invalid 1: cleared
[6]	WO	rtic	RX timeout interrupt clear 0: invalid 1: cleared
[5]	WO	txic	TX interrupt clear 0: invalid 1: cleared
[4]	WO	rxic	RX interrupt clear 0: invalid 1: cleared
[3:0]	RO	reserved	Reserved

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UART_DMACR

UART_DMACR is a DMA control register. It is used to enable or disable the DMAs of the TX and RX FIFOs.

Offset Address	Register Name	Total Reset Value
0x048	UART_DMACR	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved													dmaonerr	txdmae	rxdmae
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:3]	RO	reserved	Reserved
[2]	RW	dmaonerr	DMA enable for the RX channel when the UART error interrupt (UARTEINTR) occurs. 0: When UARTEINTR is valid, the DMA output request (UARTRXDMASREQ or UARRTXDMABREQ) of the RX channel is valid. 1: When UARTEINTR is valid, the DMA output request (UARTRXDMASREQ or UARRTXDMABREQ) of the RX channel is invalid.
[1]	RW	txdmae	TX FIFO DMA enable 0: disabled 1: enable
[0]	RW	rxdmae	RX FIFO DMA enable 0: disabled 1: enable

12.3 SPI

12.3.1 Overview

The serial peripheral interface (SPI) controller implements serial-to-parallel conversion and parallel-to-serial conversion, and serves as a master to communicate with peripherals in synchronous and serial modes. The SPI controller supports three types of peripheral interfaces including the SPI, TI synchronous serial interface, and MicroWire interface.

12.3.2 Features

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CAUTION

Hi3518E V20X/Hi3516C V200 has two serial peripheral interfaces (SPIs). SPI0 supports a chip select (CS), whereas SPI1 supports two CSs.

Hi3518E V20X/Hi3516C V200 SPIs are master interfaces. The working reference clock is the APB bus clock. SPI_CLK output by the SPI supports the maximum working frequency of 24.75 MHz.

The SPI controller has the following features:

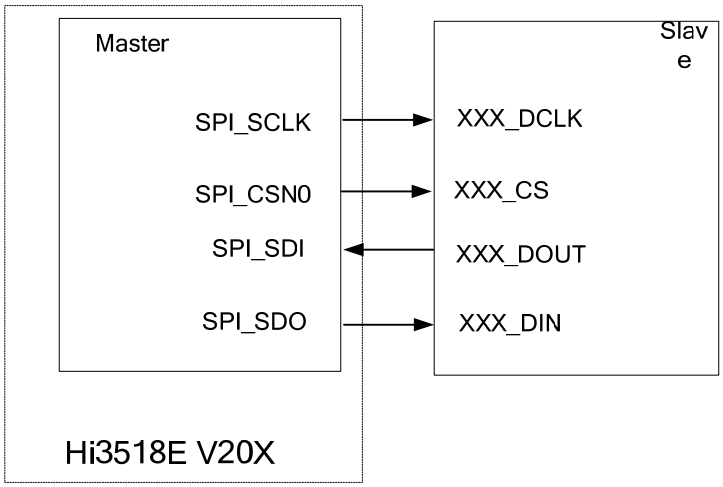
- Supports programmable frequency of the interface clock.
- Supports two separate FIFOs. One acts as an RX FIFO and the other one acts as a TX FIFO. Each of them is 16-bit wide and 256-location deep.
- Supports programmable serial data frame length: 4 bits to 16 bits.
- Provides internal loopback test mode.
- Supports the direct memory access (DMA) operation.
- Supports three types of peripheral interfaces including the SPI, MicroWire interface, and TI synchronous serial interface.
- Supports SPI in full-duplex mode and configurable clock polarity and phase.
- Supports MicroWire in half-duplex mode.
- Supports TI synchronous serial interface in full-duplex mode.

12.3.3 Function Description

Typical Application

Figure 12-5 shows the application block diagram when the SPI is connected to a slave device. The default CS pin SPI_CSN0 is used.

Figure 12-5 Application block diagram when the SPI is connected to a single slave device



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12.3.4 Peripheral Bus Timings

The meanings of the abbreviations and acronyms in Figure 12-6 to Figure 12-13 are as follows:

- MSB: most significant bit
- LSB: least significant bit
- Q: Q is an undefined signal

SPI

NOTE

SPO indicates the polarity of SPICLKOUT and SPH indicates the phase of SPICLKOUT. They correspond to SPICR0 bit[7:6].

1. **SPO = 0, SPH = 0**

Figure 12-6 shows the SPI single frame format.

Figure 12-6 SPI single frame format (SPO = 0, SPH = 0)

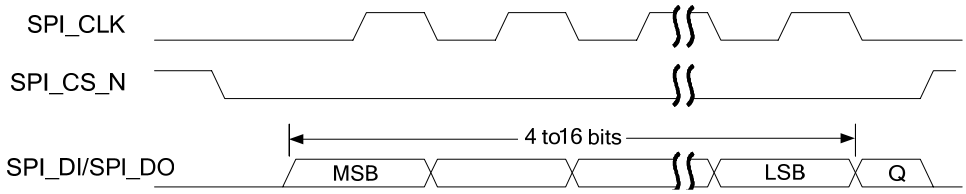
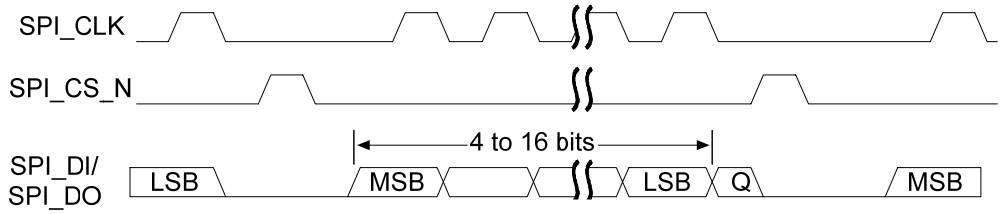


Figure 12-7 shows the SPI continuous frame format.

Figure 12-7 SPI continuous frame format (SPO = 0, SPH = 0)



When the SPI is idle in this mode:

- The SPI_CLK signal is set to low.
- The SPI_CS_N signal is set to high.
- The TX data line SPI_DO is forced to low.

When the SPI is enabled and valid data is ready in the TX FIFO, data transfer starts if the SPI_CS_N signal is set to low. The data from the slave device is transferred to the RX data line SIP_DI of the master device immediately. Half SPI_CLK clock cycle later, the valid master data is transmitted to SPI_DO. At this time, both the master and slave data become valid. The SPI_CLK pin changes to high level in the next half SPI_CLK clock cycle. Then,

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data is captured on the rising edge and is transmitted on the falling edge of the SPI_CLK clock.

If a single word is transferred, SPI_CS_N is restored to high level one SPI_CLK clock later after the last bit is captured.

For continuous transfer, the SPI_CS_N signal must pull up the SPI_CLK clock by one clock cycle between the transfers of two words. When SPH is 0, the slave select pin fixes the data of the internal serial device register. Therefore, the master device must pull up the SPI_CS_N signal between the transfers of two words in continuous transfer. When the continuous transfer ends, SPI_CS_N is restored to high level one SPI_CLK clock cycle later after the last bit is captured.

2. **SPO = 0, SPH = 1**

Figure 12-8 shows the SPI single frame format.

Figure 12-8 SPI single frame format (SPO = 0, SPH = 1)

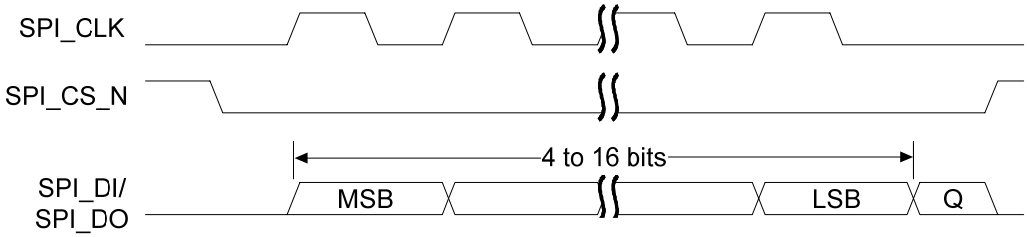
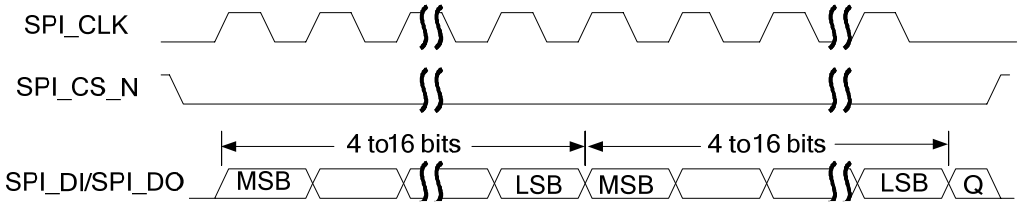


Figure 12-9 shows the SPI continuous frame format.

Figure 12-9 SPI continuous frame format (SPO = 0, SPH = 1)



When the SPI is idle in this mode:

- The SPI_CLK signal is set to low.
- The SPI_CS_N signal is set to high.
- The TX data line SPI_DO is forced to low.

When the SPI is enabled and valid data is ready in the TX FIFO, data transfer start if the SPI_CS_N signal is set to low. The master and slave data become valid on their respective transfer line half SPI_CLK clock cycle later. In addition, SPI_CLK becomes valid from the first rising edge. Then, data is captured on the falling edge and is transmitted on the rising edge of the SPI_CLK clock.

If a single word is transferred, SPI_CS_N is restored to high level one SPI_CLK clock later after the last bit is captured.

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For a continuous transfer, SPI_CS_N remains low between the transfers of two words. When the continuous transfer ends, SPI_CS_N is restored to high level one SPI_CLK clock cycle later after the last bit is captured.

3. **SPO = 1, SPH = 0**

Figure 12-10 shows the SPI single frame format.

Figure 12-10 SPI single frame format (SPO = 1, SPH = 0)

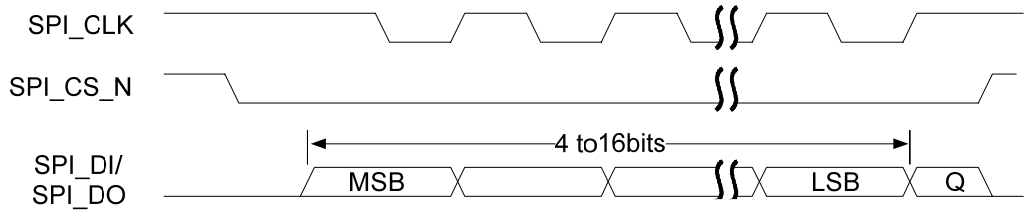
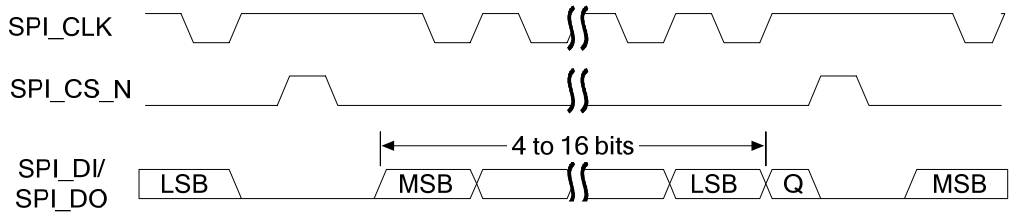


Figure 12-11 shows the SPI continuous frame format.

Figure 12-11 SPI continuous frame format (SPO = 1, SPH = 0)



When the SPI is idle in this mode:

- The SPI_CLK signal is set to high.
- The SPI_CS_N signal is set to high.
- The TX data line SPI_DO is forced to low.

When the SPI is enabled and valid data is ready in the TX FIFO, data transfer starts if the SPI_CS_N signal is set to low. The data from the slave device is transferred to the RX data line SPI_DI of the master device immediately. Half SPI_CLK clock cycle later, the valid master data is transmitted to SPI_DO. After another half SPI_CLK clock cycle, the SPI_CLK master pin is set to low. Then, data is captured on the falling edge and is transmitted on the rising edge of the SPI_CLK clock.

If a single word is transferred, SPI_CS_N is restored to high level after one SPI_CLK clock since the data of the last bit is captured.

For a continuous transfer, the SPI_CS_N signal must be pulled up between the transfers of two words. This is because that when SPH is 0, the slave select pin fixes the data of the internal serial device register. SPI_CS_N is restored to high level one SPI_CLK clock later after the last bit is captured.

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4. **SPO = 1, SPH = 1**

Figure 12-12 shows the SPI single frame format.

Figure 12-12 SPI single frame format (SPO = 1, SPH = 1)

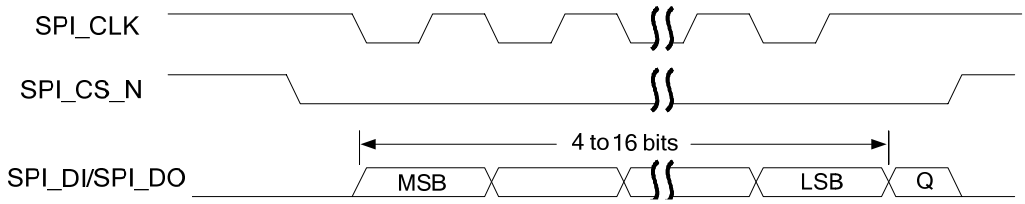
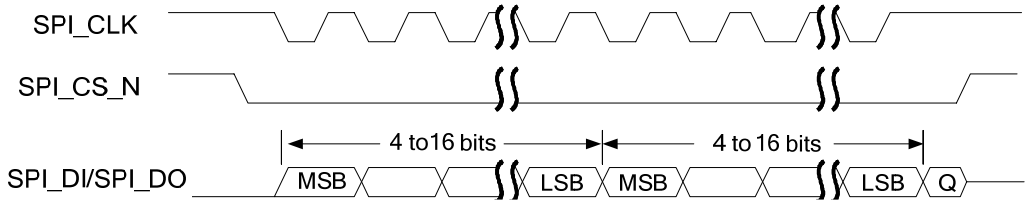


Figure 12-13 shows the SPI continuous frame format.

Figure 12-13 SPI continuous frame format (SPO = 1, SPH = 1)



When the SPI is idle in this mode:

- The SPI_CLK signal is set to high.
- The SPI_CS_N signal is set to high.
- The TX data line SPI_DO is forced to low.

When the SPI is enabled and valid data is ready in the TX FIFO, data transfer starts if the SPI_CS_N signal is set to low. Half SPI_CLK clock cycle later, the master data and slave data are valid on respective transfer line. In addition, SPI_CLK becomes valid from a falling edge of SPI_CLK. Then, data is captured on the rising edge and is transmitted on the falling edge of the SPI_CLK clock. If a single word is being transmitted, SPI_CS_N is restored to high level one SPI_CLK clock cycle later after the last bit is captured.

For a continuous transfer, the SPI_CS_N signal remains low. SPI_CS_N is restored to high level one SPI_CLK clock cycle after the last bit is captured. For a continuous transfer, SPI_CS_N remains low during transfer. The method of ending data transfer is the same as that in single transfer mode.

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5. **Interface timings**

Figure 12-14 SPI timings

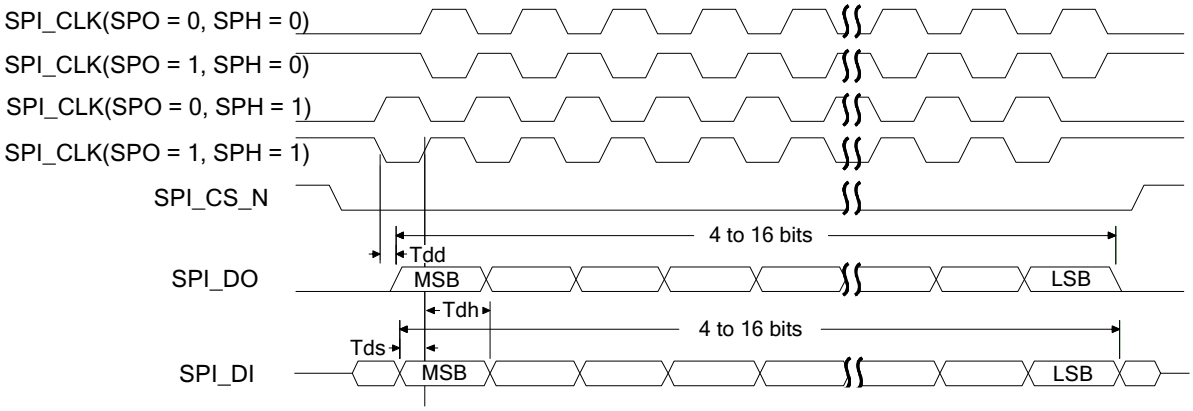


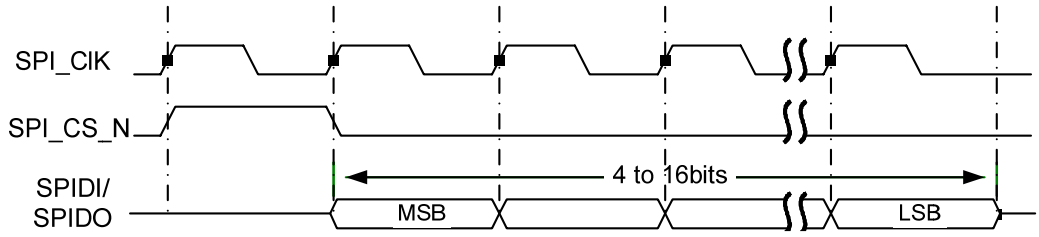
Table 12-3 Timing parameters of the SPI

Parameter	Description	Min	Max	Unit
Tdd	Output data delay	–3.5	5	ns
Tds	Input control signal setup time	23	None	ns
Tdh	Input control signal hold time	0	None	ns

TI Synchronous Serial Interface

Figure 12-15 shows the TI synchronous serial single frame format.

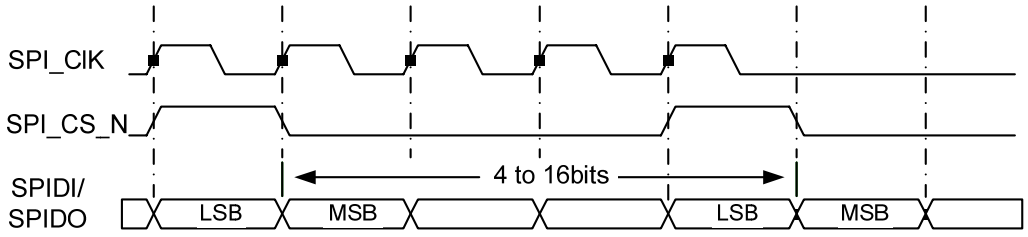
Figure 12-15 TI synchronous serial single frame format



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Figure 12-16 shows the TI synchronous serial continuous frame format.

Figure 12-16 TI synchronous serial continuous frame format



When the SPI is idle in this mode:

- The SPI_CK signal is set to low.
- The SPI_CS_N signal is set to low.
- The transfer data line SPI_DO retains high impedance.

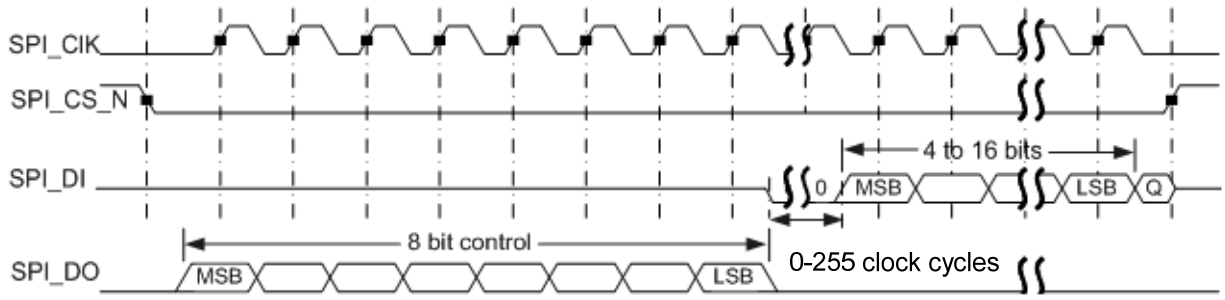
If there is data in the TX FIFO, SPI_CS_N generates a high-level pulse in one SPI_CK clock cycle. Then, the data to be transmitted is transferred from the TX FIFO to the TX logic serial shift register. In addition, the MSBs of the data frames with the length of 4–16 bits are shifted and output from SPI_DO on the next rising edge of the SPI_CK clock. Similarly, the MSB of the data received from the external serial slave device is shifted and input from the SPI_DI pin.

The SPI and off-chip serial device stores the data in the serial shift register on the falling edge of the SPI_CK clock. The RX serial register transmits the data to the RX FIFO on the rising edge of the first SPI_CK clock after receiving the LSB.

National Semiconductor Microwire Interface

Figure 12-17 shows the national semiconductor microwire single frame format.

Figure 12-17 National semiconductor microwire single frame format

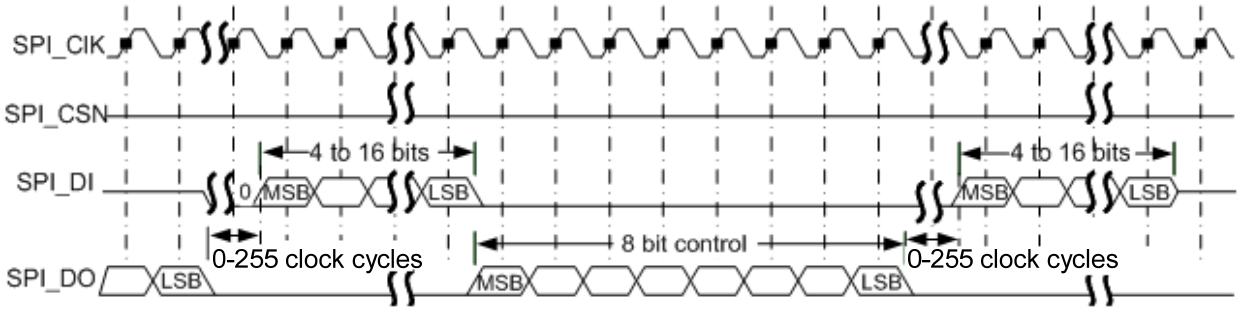


0 to 255 clock cycles can be delayed between the end of SPI_DO LSB and the start of SPI_DI MSB.

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Figure 12-18 shows the national semiconductor microwire continuous frame format.

Figure 12-18 National semiconductor microwire continuous frame format



0 to 255 clock cycles can be delayed between the end of SPI_DO LSB and the start of SPI_DI MSB.

The microwire format is similar to the SPI format because both of them use the technology of transferring master-slave information. The only difference is that the SPI works in full-duplex mode and the microwire interface works in half-duplex mode. Before the SPI transmits serial data to the external chip, an 8-bit control word needs to be added. In this process, the SPI does not receive any data. After data transfer is complete, the external chip decodes the received data. One clock cycle later after 8-bit control information is transmitted, the slave device starts to respond to the required data. The returned data length is 4 bits to 16 bits, and the length of the entire frame is 13 bits to 25 bits.

When the SPI is idle in this mode:

- The SPI_CLK signal is set to low.
- The SPI_CS_N signal is set to high level.
- The TX data signal SPI_DO is forced to low level.

Writing one control byte to the TX FIFO starts a data transfer. The data transfer is triggered on a falling edge of SPI_CS_N. The data of the TX FIFO is sent to the serial shift register. The MSB of the 8-bit control frame is transmitted to the SPI_DO pin. During frame transfer, SPI_CS_N remains low. Whereas SPI_DI retains high impedance.

The off-chip serial slave device latches the data to the serial shift register on each rising edge of the SPI_CLK clock. After the slave device latches the data of the last bit, the slave device starts to decode the received data in the next clock cycle. Then, the slave device provides the data required by the SPI. Each bit is written to SPI_DI on the falling edge of the SPI_CK clock. For a single data transfer, SPI_CS_N is pulled up at the end of the frame one clock cycle after the last bit is written to the RX serial register. In this way, the RX data is transmitted to the RX FIFO.

The start and end for a continuous data transfer are the same as those for a signal data transfer. During the continuous data transfer, SPI_CS_N retains low and the data transferred is continuous. The control word of the next frame is adjacent to the LSB of the previous frame. When the LSB of the frame is latched to the SPI, each received value is originated from the receive shift register on the falling edge of the SPI_CLK clock.

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12.3.5 Operating Mode

Working Modes

The SPI working modes include the data transmission in the interrupt or query mode and the data transmission in the DMA mode.

Clock and Reset

The frequency of the output SPI clock is calculated as follows:

Fsspclkout = Fsspclk/[CPSDVR x (1 + SCR)]

Fsspclk is the working reference clock of the SPI and its value is 1/4 of the bus clock (49.5MHz).

For details about CPSDVR and SCR, query the corresponding registers.

The SPI of Hi3518E V20X/Hi3516C V200 supports separately soft reset, which is controlled by the PERI_CRG57 bit[6:5] register. To disable the soft reset on the SPI, write 0 to corresponding bits. To enable soft-reset on the SPI, write 1 to corresponding bits. The default value is 0 when the SPI is powered on.

Interrupts

The SPI has five interrupts. Four of them have separate interrupts sources and maskable and active high.

- SPIRXINTR

  RX FIFO interrupt request. When there are four or more valid data segments in the RX FIFO, the interrupt is set to 1.

- SPITXINTR

  TX FIFO interrupt request. When there are four or less valid data segments in the TX FIFO, the interrupt is set to 1.

- SPIRORINTR

  RX overrun interrupt request. When the FIFO is full and new data is written to the FIFO, the FIFO is overrun and the interrupt is set to 1. In this case, the data is written to the RX shift register rather than the FIFO.

- SPIRTINTR

  RX timeout interrupt request. When the RX FIFO is not empty and the SPI is idle for more than a fixed 32-bit cycle, the interrupt is set to 1.

  It indicates that data in the RX FIFO needs to be transmitted. When the RX FIFO is empty or new data is transmitted to SPIRXD, the interrupt is cleared. The interrupt can also be cleared by writing to the SPIICR[RTIC] register.

- SPIINTR

  Combined interrupt. This interrupt is obtained by performing an OR operation on the preceding four interrupts. If any of the preceding four interrupts is set to 1 and enabled, SPIINTR is set to 1.

For details about how to connect the SPIINTR of SPI, see "Interrupts" in this section.

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Initialization

The initialization is implemented as follows:

Step 1 Write 0 to SPICR1[sse] to disable the SPI.

Step 2 Write to SPICR0 to set the parameters such as the frame format and transfer data bit width.

Step 3 Configure SPICPSR to set the required clock divider.

Step 4 In the interrupt mode, configure SPIIMSC to enable the corresponding interrupts or disable the generation of corresponding interrupts in query or DMA mode.

Step 5 Configure SPITXFIFOCR and SPIRXFIFOCR in interrupt or DMA mode.

Step 6 Configure SPIDMACR to enable the DMA function of the SPI in DMA mode.

----End

Data Transfer in Query Mode

The full status of the TX or RX FIFO is not considered, because the depth of the TX FIFO or RX FIFO is 512.

Perform the following steps:

Step 1 Write 1 to SPICR1[sse] to enable the SPI.

Step 2 Write the data to be transmitted to SPIDR continuously.

Step 3 Poll SPISR until SPISR[BSY] is 0, SPISR[TFE] is 1, and SPISR[RNE] is 1. If SPISR[BSY] is 0, the bus is busy; if SPISR[TFE] is 1, the TX FIFO is empty; if SPISR[RNE] is 1, the RX FIFO is not empty.

Step 4 Read data until the RX FIFO is empty. You can check whether the RX FIFO is empty by querying SPISR[RNE].

CAUTION

The SPI/Microwire has full-duplex features. That is, a data segment is received after a data segment is transmitted. Even if only data is transmitted, the RX FIFO must be cleared.

Step 5 Write 0 to SPICR1[sse] to disable the SPI.

----End

Data Transfer in Interrupt Mode

Perform the following steps:

Step 1 Write 1 to SPICR1[sse] to disable the SPI.

Step 2 Write the data to be transmitted to SPIDR continuously.

Step 3 Wait for the interrupt SPIRXINTR and read data in cyclic mode until all data is read.

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Note that the full-duplex features of the SPI/Microwire. That is, a data segment is received after a data segment is transmitted. Even if only data is transmitted, the RX FIFO must be cleared.

Step 4 Write 0 to SPICR1[sse] to disable the SPI.

----End

Data Transfer in DMA Mode

Perform the following steps:

Step 1 Obtain a DMAC channel.

Step 2 Write 1 to SPICR1[sse] to disable the SPI.

Step 3 Transmit data.

1. Configure the parameters of the configuration register and control register related to the DMAC channel.
2. Start the DMAC and respond to the DMA request of the SPI TX FIFO for the data transfer.
3. Check whether all data is transmitted by viewing the DMA interrupt. If all data is transmitted, disable the DMA transmit function of the SPI.

Step 4 Receive data

1. Configure the parameters of the configuration register and control register related to the DMAC channel.
2. Start the DMAC and respond to the DMA request of the SSP RX FIFO for the data transfer.
3. Check whether all data is received by viewing the DMA interrupt. If all data is received, disable the DMA receive function of the SPI.

Step 5 Write 0 to SPICR1[sse] to disable the SPI.

----End

12.3.6 Register Summary

Table 12-4 describes the registers.

- The base address of SPI registers is 0x200C_0000.
- The base address of SPI registers is 0x200E_0000.

Table 12-4 Summary of SPI registers

Offset Address	Register	Description	Page
0x000	SPICR0	Control register 0	12-57
0x004	SPICR1	Control register 1	12-58
0x008	SPIDR	Data register	12-59
0x00C	SPISR	Status register	12-60

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Offset Address	Register	Description	Page
0x010	SPICPSR	Clock divider register	12-60
0x014	SPIIMSC	Interrupt mask register	12-61
0x018	SPIRIS	Raw interrupt status register	12-62
0x01C	SPIMIS	Masked interrupt status register	12-62
0x020	SPIICR	Interrupt clear register	12-63
0x024	SPIDMACR	DMA control register	12-63
0x028	SPITXFIFOCR	TX FIFO control register	12-64
0x02C	SPIRXFIFOCR	RX FIFO control register	12-65

12.3.7 Register Description

SPICR0

SPICR0 is SPI control register 0.

Offset Address	Register Name	Total Reset Value
0x000	SPICR0	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SCR								SPH	SPO	FRF		DSS			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:8]	RW	SCR	Serial clock rate, ranging from 0 to 255. The value of the SCR is used to generate the TX and RX bit rates of the SPI. The formula is as follows: FSSPCLK/(CPSDVSR (1 + SCR). CPSDVSR is an even ranging from 2 to 254, and it is configured by SPICPSR.
[7]	RW	SPH	SPICLKOUT phase. For details, see section 12.3.4 "Peripheral Bus Timings."
[6]	RW	SPO	SPICLKOUT polarity. For details, see section 12.3.4 "Peripheral Bus Timings."
[5:4]	RW	FRF	Frame format select. 00: Motorola SPI frame format 01: TI synchronous serial frame format 10: National microwire frame format 11: reserved
[3:0]	RW	DSS	Data width

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			0011: 4 bits 1000: 9 bits 1101: 14 bits 0100: 5 bits 1001: 10 bits 1110: 15 bits 0101: 6 bits 1010: 11 bits 1111: 16 bits 0110: 7 bits 1011: 12 bits 0111: 8 bits 1100: 13 bits Other values: reserved

SPICR1

SPICR1 is SPI control register 1.

Offset Address	Register Name	Total Reset Value
0x004	SPICR1	0x7F00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	WaitEn	WaitVal							reserved	mode_altasens	reserved	BigEnd	reserved	MS	SSE	LBM
Reset	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15]	RW	WaitEn	Wait enable. This bit is valid when the SPICR0[FRF] is set to the national microwire frame format. 0: disabled 1: enabled
[14:8]	RW	WaitVal	Number of waiting beats between read and write when in the national microwire frame format. When WaitEn is 1 and the frame format is national microwire, WaitVal is valid.
[7]	RW	reserved	Reserved

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[6]	RW	mode_altasens	0: The CS signal is automatically generated by the chip logic based on the selected timing. 1: The CS signal is controlled by the SPI enable when the Motorola SPI frame format is used. If the SPI is enabled, the CS signal is pulled down; otherwise, the CS signal is pulled up.
[5]	RO	reserved	Reserved
[4]	RW	BigEnd	Data endian mode 0: little endian 1: big endian
[3]	RW	reserved	Reserved
[2]	RW	MS	Master or slave mode. This bit can be changed only when the SPI is disabled. 0: master mode (default value) 1: reserved
[1]	RW	SSE	SPI enable 0: disabled 1: enabled
[0]	RW	LBM	Loopback mode 0: A normal serial port operation is enabled. 1: The output of the TX serial shift register is connected to the input of the RX serial shift register.

SPIDR

SPIDR is a data register.

Offset Address	Register Name	Total Reset Value
0x008	SPIDR	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DATA															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:0]	RW	DATA	TX or RX FIFO Read: RX FIFO Write: TX FIFO If the data is less than 16 bits, the data must be aligned to the right. The TX logic ignores the unused upper bits, and the RX logic automatically aligns the data to the right.

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SPISR

SPISR is a status register.

Offset Address	Register Name	Total Reset Value
0x00C	SPISR	0x0003

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved											BSY	RFF	RNE	TNF	TFE
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

Bits	Access	Name	Description
[15:5]	RW	reserved	Reserved
[4]	RW	BSY	SPI busy flag 0: idle 1: busy
[3]	RW	RFF	Whether the RX FIFO is full 0: not full 1: full
[2]	RW	RNE	Whether the RX FIFO is not empty 0: empty 1: not empty
[1]	RW	TNF	Whether the TX FIFO is not full 0: full 1: not full
[0]	RW	TFE	Whether the TX FIFO is empty 0: not empty 1: empty

SPICPSR

SPICPSR is a clock divider register.

Offset Address	Register Name	Total Reset Value
0x010	SPICPSR	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								CPSDVSR							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:8]	RW	reserved	Reserved

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[7:0]	RW	CPSDVSR	Clock divider. The value must be an even ranging from 2 to 254. It depends on the frequency of the input clock SPICLK. The LSB is read as 0.

SPIIMSC

SCIIMSC is an interrupt mask register. The value 0 indicates an interrupt is masked and the value1 indicates an interrupt is not masked.

Offset Address	Register Name	Total Reset Value
0x014	SPIIMSC	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												TXIM	RXIM	RTIM	RORIM
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:4]	RW	reserved	Reserved
[3]	RW	TXIM	TX FIFO interrupt mask 0: The interrupt is masked when only half of or less data is left in the TX FIFO. 1: The interrupt is not masked when only half of or less data is left in the TX FIFO.
[2]	RW	RXIM	RX FIFO interrupt mask 0: The interrupt is masked when only half of or less data is left in the RX FIFO. 1: The interrupt is not masked when only half of or less data is left in the RX FIFO.
[1]	RW	RTIM	RX timeout interrupt mask 0: masked 1: not masked
[0]	RW	RORIM	RX overflow interrupt mask 0: masked 1: not masked When the value is 1, the hardware stream control function is enabled. That is, when the RX FIFO is full, the SPI stops transmitting data.

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SPIRIS

SPIRIS is a raw interrupt status register. The value 0 indicates no interrupts are generated, and the value 1 indicates interrupts are generated.

Offset Address	Register Name	Total Reset Value
0x018	SPIRIS	0x0008

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												TXRIS	RXRIS	RTRIS	RORRIS
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

Bits	Access	Name	Description
[15:4]	RO	reserved	Reserved
[3]	RO	TXRIS	Raw TX FIFO interrupt status
[2]	RO	RXRIS	Raw RX FIFO interrupt status
[1]	RO	RTRIS	Raw RX timeout interrupt status
[0]	RO	RORRIS	Raw RX overflow interrupt status

SPIMIS

SPIMIS is a masked interrupt status register. The value 0 indicates no interrupts are generated, and the value 1 indicates interrupts are generated.

Offset Address	Register Name	Total Reset Value
0x01C	SPIMIS	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												TXMIS	RXMIS	RTMIS	RORMIS
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:4]	RO	reserved	Reserved
[3]	RO	TXMIS	Status of the masked TX FIFO interrupt
[2]	RO	RXMIS	Status of the masked RX FIFO interrupt
[1]	RO	RTMIS	Status of the masked RX timeout interrupt
[0]	RO	RORMIS	Status of the masked RX overflow interrupt

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SPIICR

SCIICR is an interrupt clear register. Writing 1 clears an interrupt, and writing 0 has no effect.

Offset Address	Register Name	Total Reset Value
0x020	SPIICR	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														RTIC	RORIC
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:2]	RO	reserved	Reserved
[1]	RO	RTIC	RX timeout interrupt clear
[0]	RO	RORIC	RX overflow interrupt clear

SPIDMACR

SCIDMACR is a DMA control register.

Offset Address	Register Name	Total Reset Value
0x024	SPIDMACR	0x0000

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														TXDMAE	RXDMAE
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[15:2]	WO	reserved	Reserved
[1]	WO	TXDMAE	DMA TX FIFO enable 0: disabled 1: enabled
[0]	WO	RXDMAE	DMA RX FIFO enable 0: disabled 1: enabled

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SPITXFIFOCR

SPITXFIFOCR is a TX FIFO control register.

Offset Address	Register Name	Total Reset Value
0x028	SPITXFIFOCR	0x0009

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										TXINTSize			DMATXBRSize		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1

Bits	Access	Name	Description
[15:6]	RW	reserved	Reserved
[5:3]	RW	TXINTSize	Threshold for generating the TX FIFO request interrupt. That is, when the number of data segments in the TX FIFO is less than or equal to the value of TXINTSize, TXRIS is valid. 000: 1 001: 4 010: 8 011: 16 100: 32 101: 64 110: 64 111: 64
[2:0]	RW	DMATXBRSize	Threshold for generating the TX FIFO request DMA transfer burst. That is, when the number of data segments in the TX FIFO is less than or equal to the configured value (256 – DMATXBRSize), DMATXBREQ is valid. The width of the TX FIFO is 16 bits. 000: 1 001: 4 010: 8 011: 16 100: 32 101: 64 110: 128 111: 128

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SPIRXFIFOCR

SPIRXFIFOCR is an RX FIFO control register.

Offset Address	Register Name	Total Reset Value
0x02C	SPIRXFIFOCR	0x0009

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved										RXINTSize			DMARXBRSize		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1

Bits	Access	Name	Description
[15:6]	RW	reserved	Reserved
[5:3]	RW	RXINTSize	Threshold for generating the RX FIFO request interrupt. That is, when the number of data segments in the TX FIFO is less than or equal to the configured value (256 – RXINTSize), RXRIS is valid. The width of the RX FIFO is 16 bits. 000: 1 001: 4 010: 8 011: 16 100: 32 101: 64 110: 64 111: 64
[2:0]	RW	DMARXBRSize	Burst transfer threshold. When this threshold is reached, the RX FIFO asks the DMA to perform a burst transfer. That is, when number of data segments in the TX FIFO is less than or equal to the value of DMARXBRSize, DMARXBREQ is valid. 000: 1 001: 4 010: 8 011: 16 100: 32 101: 64 110: 128 111: 224

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12.4 MMC/SD/SDIO Controller

12.4.1 Function Description

Functional Block Diagram

The MMC/SD/SDIO controller (MMC controller for short) controls the read/write operations on the secure digital (SD) card and multimedia card (MMC), and supports various extended devices such as Bluetooth and Wi-Fi devices based on the secure digital input/output (SDIO) protocol. Hi3518E V20X/Hi3516C V200 provides two MMC controllers for controlling the SD card, MMC, and SDIO device.

The MMC0 controller controls the devices that comply with the following protocols:

- Secure Digital Memory (SD mem-version 3.0)
- Secure Digital I/O (SDIO-version 3.0)
- MultiMediaCard (MMC-version 5.0)

The MMC1 controller controls the devices that comply with the following protocols:

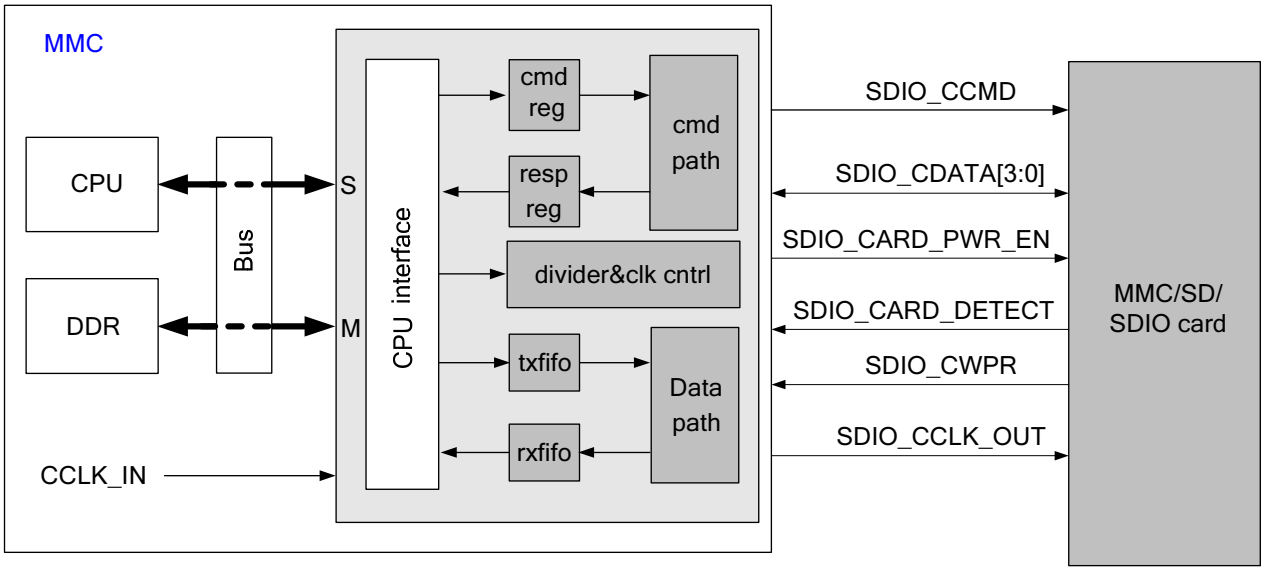
- Secure Digital Memory (SD mem-version 2.0)
- Secure Digital I/O (SDIO-version 2.0)

NOTE

The devices controlled by the two MMC controllers comply with different protocols. In addition, the MMC0 controller can connect to the embedded multimedia card (eMMC) and SD card and support MMC-version 5.0 (the kernel version must be 3.16 or later), while the MMC1 controller can connect to only one SD 2.0 card.

Figure 12-19 shows the functional block diagram of the MMC controller.

Figure 12-19 Functional block diagram of the MMC controller



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NOTE

S indicates slave interface and M indicates master interface.

Connected to the system through the internal bus, the MMC controller consists of the following parts:

- Command path

  It is used to transmit commands and receive responses.
- Data path

  It is used to perform data read and write operations by working with the command path.
- Interface clock control unit

  It is used to change the frequency of the interface clock as required and enable or disable the interface clock. SDIO_CLK_OUT can be generated by dividing CCLK_IN.

The MMC controller has the following features:

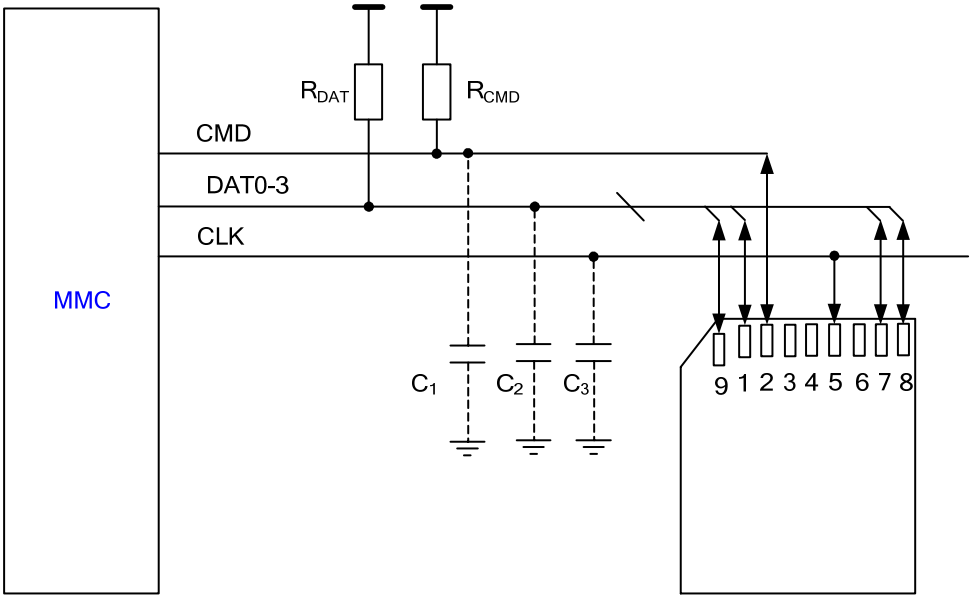
- Transfers data by using the internal direct memory access controller (IDMAC).
- Supports one TX FIFO and one RX FIFO. The depth of each FIFO is 256 words.
- Supports configurable burst size during DMA transfer and configurable FIFO threshold.
- Supports FIFO overflow and underflow interrupts to avoid errors during data transfer.
- Supports the cyclic redundancy check (CRC) generation and check for data and commands.
- Supports programmable frequency of the interface clock.
- Disables the MMC clock and interface clock in low-power mode.
- Supports 1-bit, 4-bit or 8-bit data width based on the connected component.
- Reads or writes data blocks with the size of 1 byte to 65,535 bytes.
- Reads/writes stream data from/to the MMC card.
- Supports the suspend operation, resume operation, and read wait operation on the SDIO card.
- Supports 4-bit and 8-bit DDR SDRAMs, and the GO_PRE_IDLE_STATE command for MMC 4.41.
- Supports hardware reset for eMMC 4.41.
- Supports 4-bit and 8-bit HS200 (1.8 V voltage), CMD21, CMD49, and the voltage of 3.3 V or 1.8 V for eMMC 4.5.
- Supports the HS400 mode for eMMC 5.0.

Typical Application

Figure 12-20 shows the typical application circuit of the MMC controller.

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Figure 12-20 Typical application circuit of the MMC controller



The MMC controller exchanges commands and data with the connected card over a clock signal line, a bidirectional command signal line, and four bidirectional data signal lines. Both the command signal line and data signal lines work in pull-up mode. Table 12-5 describes the pull-up resistors and the load capacitance of each signal line.

Table 12-5 Load parameters of signal lines

Parameter	Min.	Max.	Description
R_{DAT} R_{CMD}	10 kΩ	100 kΩ	Pull-up resistors
Load capacitance Cx	None	30 pF	$C_x = C_{mmchost} + C_{bus} + C_{card}$The maximum capacitance of each card (C_{card}) is 10 pF; therefore, the sum of $C_{mmchost}$ and C_{bus} must be less than 30 pF.
Inductance of each signal line	None	16 nH	Fpp ≤ 20 MHz

CAUTION

Besides the signal lines in Figure 12-20, the card slot also provides the mechanical write protection signal and card detection signal. However, the signal interfaces are not provided in Figure 12-20.

Commands and Responses

All interactions between the MMC controller and the card, including initializing the card, reading/writing to registers, querying the status, and transferring data, are implemented by using commands.

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A MMC command is a segment of 48-bit serial data consisting of a start bit, a transmission bit, a command sequence, a CRC bit, and an end bit. After receiving a command, the card returns a 48-bit or a 136-bit response based on the command type.

Figure 12-21 Format of an MMC command

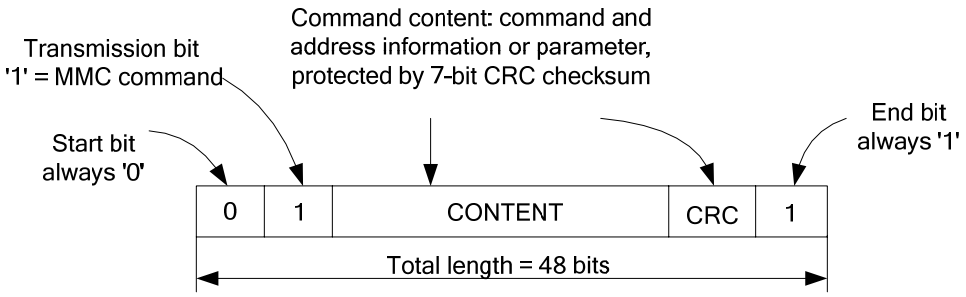
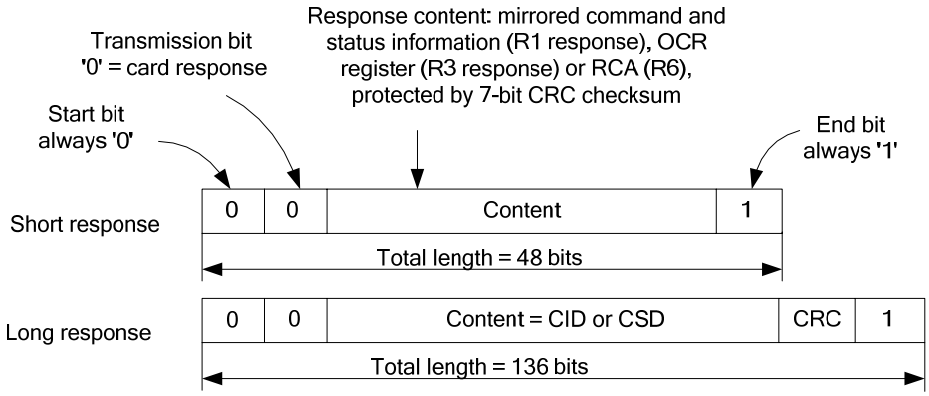


Figure 12-22 Format of the response to an MMC command



Commands are classified into the following two types based on whether data is transferred:

- Non-data transfer command

  The MMC controller transmits/receives commands to/from the card through the command signal line.

- Data transfer command

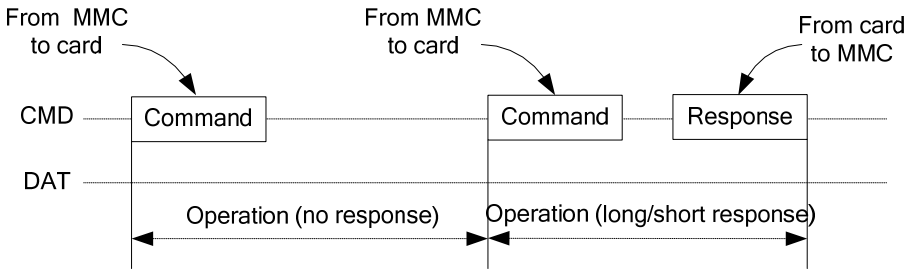
  Besides the interaction on the command line, data is also transferred through the data lines DAT0 to DAT3.

1. Non-data transfer command

Figure 12-23 shows the operations between the MMC controller and the card by running a non-data transfer command.

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Figure 12-23 Operations by running an MMC non-data transfer command



2. Data transfer command

The MMC supports the following data transfer commands:

- Stream data read/write command

  Only the MMC card supports the stream data read/write command. In this case, only the data line DAT0 is used for data transfer, and no CRC check is performed.

- Single-block read/write command

  After this command is executed, one single data block is read and written each time. No stop command is required for stopping each data transfer.

- Multi-block read/write command
  - Predefined block count mode

    Before the multi-block read/write command is executed, the block count command is transmitted to specify the number of data blocks to be transferred.
  - Open-ended mode

    After a read/write command is transmitted, a stop command is required for stopping data transfer at the end of data transfer.

  The difference between the two modes lies on how the MMC controller notifies the card of the end of each data transfer. The SD card supports only the open-ended mode, whereas the MMC supports both modes.

  The multi-block read/write command for the SDIO card is different from the preceding two modes. To be specific, the command parameter contains the number of data blocks to be transferred when the read/write command is transmitted.

Responses are classified into the following three types:

- Command without response

  For example, the card reset command.

- Short response command

  For example, the data transfer command and card status query command.

- Long response command

  This type of commands are used to read only the information about the card identification (CID) and card specific data (CSD) registers of a card.

Data Transfer

The single-block read/write command and the multi-block read/write command are widely used during data transfer. The data block during the data transfer of the SD card and MMC is 512 bytes; the block size during the data transfer of the SDIO card can be customized

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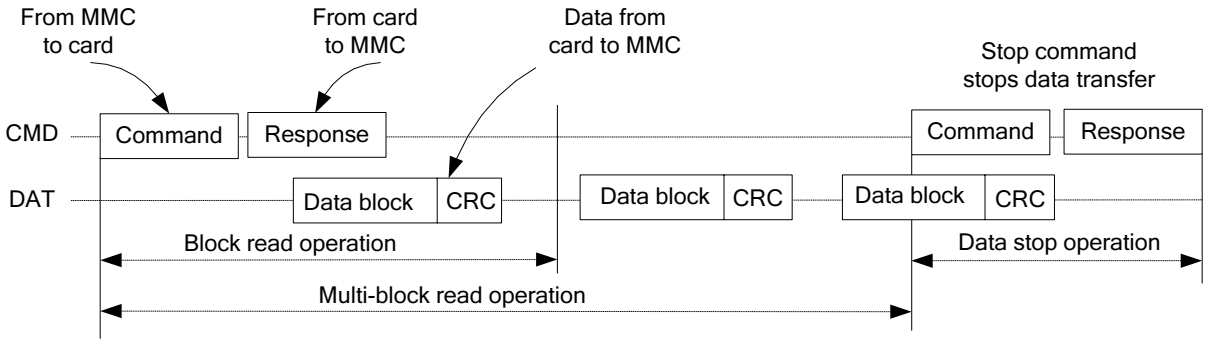
NOTE

During the data transfer by using the block read/write command, the total data amount to be transferred must be an integral multiple of the block size.

All data transfer commands are short response commands with data transferred through data lines. Figure 12-24 and Figure 12-25 show the relationships among the commands, responses, and timings of data lines.

1. Single-block and multi-block read operations

Figure 12-24 Single-block and multi-block read operations

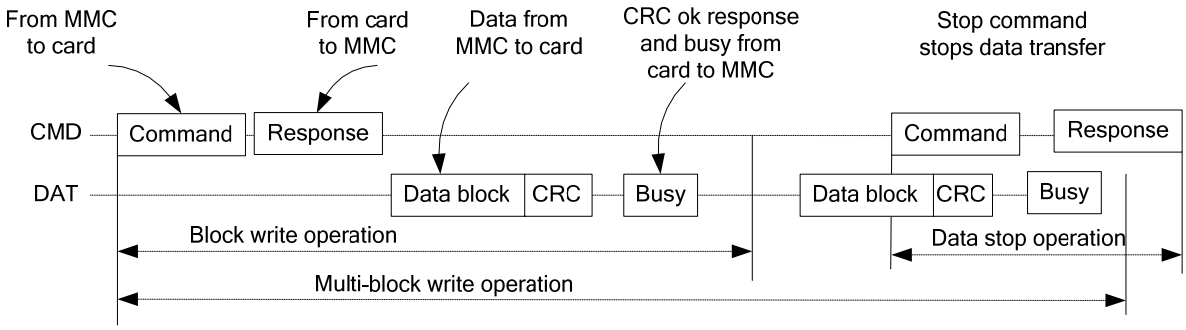


The MMC controller transmits a single-block or multi-block command to a card. When a response is being received, data is received by blocks. Each data block contains a CRC check bit for ensuring the integrity of the transferred data.

In a single-block read operation, the data transfer is completed after the MMC controller receives a data block. In a multi-block read operation, the MMC controller needs to transmit a stop command to end the data transfer after receiving multiple data blocks only in open-ended mode.

2. Single-block and multi-block read operations

Figure 12-25 Single-block and multi-block write operations



The MMC controller transmits a single-block or multi-block command to a card. After receiving a response, the MMC controller starts to transmit data to the card by blocks. Each data block contains a CRC bit. The card performs CRC on each data block and transfers the CRC status to the MMC controller. This ensures that data is transferred properly.

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In a single-block write operation, the data transfer is completed after the MMC controller transmits a data block. In a multi-block write operation, the MMC controller needs to transmit a stop command to end the data transfer after transmitting multiple data blocks only in open-ended mode. After a write operation, the card may be busy in programming the flash memory. The MMC controller can perform the next operation on the card only after it confirms that the card is not busy by querying the status of the signal line DAT0.

3. Data transfer format

   During the block read/write operations, the 1- or 4-bit data line can be used to transfer data between the MMC controller and the card. Before a data transfer command is transmitted, the data transfer widths of the MMC controller and card must be the same (1 bit or 4 bits). You can set the data bit width of the MMC controller by configuring MMC_CTYPE and set the data bit width of the card by transmitting the corresponding command.

   Figure 12-26 shows the data transfer format in 1-bit mode, and Figure 12-27 shows the data transfer format in 4-bit mode.

Figure 12-26 Data transfer format in 1-bit mode

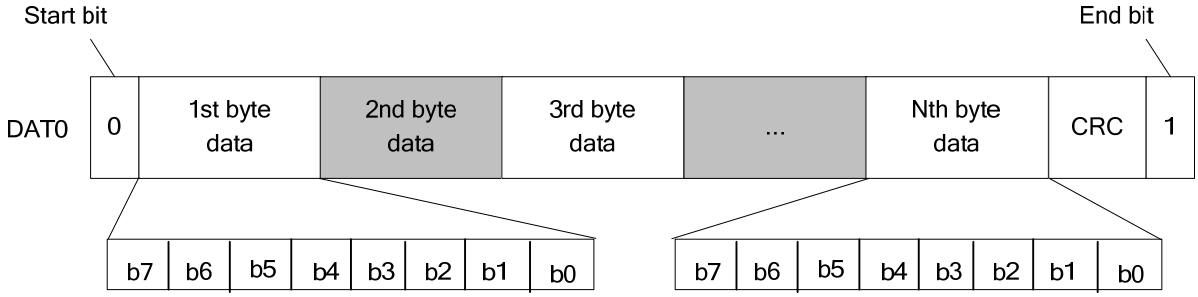
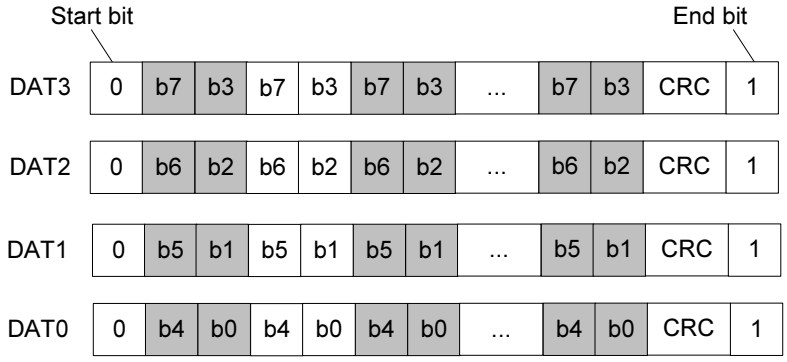


Figure 12-27 Data transfer format in 4-bit mode



Speed Mode and Voltage Switching Supported by SD3.0

The MMC controller supports SD 3.0 ultra high speed (UHS-1) and allows the voltage to be switched in SD mode. Table 12-6 describes the relationship between the transmission modes and voltages.

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Table 12-6 Transmission mode

Mode	Input Clock (MHz)	Clock on the Card Side	Maximum Data Bit Width	Voltage
SDR104	100 MHz	100 MHz	4 bits	1.8 V
SDR50	100 MHz	100 MHz	4 bits	1.8 V
DDR50	50 MHz	50 MHz	4 bits	1.8 V
SDR25	50 MHz	50 MHz	4 bits	1.8 V
SDR12	25 MHz	25 MHz	4 bits	1.8 V
HS	50 MHz	50 MHz	4 bits	3.3 V
DS	25 MHz	25 MHz	4 bits	3.3 V
MMC HS	50 MHz	50 MHz	4 bits	3.3 V

12.4.2 Application Notes

NOTE

For details about clock reset registers, see section 3.2.7 "Register Description."

Clock Gating

When the software completes the current command or data transfer and does not start a new data transfer, the SDIO_CCLK_OUT clock can be disabled if the MMC controller is idle.

Perform the following steps:

Step 1 Read MMC_STATUS.

Step 2 If both MMC_STATUS[Command fsm_states] and MMC_STATUS[data_state_mc_busy] are 0, write 0 to MMC_CTRL to mask the MMC interrupt and enable the DMA request, and go to Step 3. If any of [Command fsm_states] and [data_state_mc_busy] is not 0, wait with delay, and go to Step 1.

Step 3 Write 0 to PERI_CRG49 bit[9] to disable the SDIO0 clock, and write 0 to PERI_CRG49 bit[1] to disable the SDIO1 clock.

NOTE

To enable the SDIO0 clock, write 1 to PERI_CRG49 bit[9]. To enable the SDIO1 clock, write 1 to PERI_CRG49 bit[1].

----End

Soft Reset

If the MMC controller cannot be restored to the idle state due to data transfer exceptions, you can write 1 to PERI_CRG49 bit[8] to soft reset SDIO0, and write 1 to PERI_CRG49 bit[0] to soft reset SDIO1. In addition, you can query MMC_STATUS[Data_busy] to check whether the MMC controller is idle.

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Before using the MMC controller, you are advised to soft-reset the MMC controller after hot-swapping the card.

Configuring the Working Clock

Before using the MMC controller, you need to configure the frequency of its working clock. SDIO0 by configuring PERI_CRG49 bit[11:10], and configure SDIO1 by setting PERI_CRG49 bit[3:2].

Configuring the Interface Clock

Clock frequencies vary according to the MMCs complying with different protocols and the status of MMCs. The MMC controller provides an internal even frequency divider that generates appropriate interface clocks by dividing the frequency of the working clock. The relationship between the frequencies of the working clock CCLK_IN and the interface clock SDIO_CCLK_OUT of the MMC controller is as follows:

$F_{SDIO_CCLK_OUT} = F_{CCLK_IN}/(2 \times clk_divider)$

Where, clk_divider is the value of MMC_CLKDIV[clk_divider]. Clock frequencies vary according to card types. The maximum value of $F_{SDIO_CCLK_OUT}$ is 100 MHz.

Before changing the clock frequency of an MMC, ensure that no data or command is being transferred. In addition, to avoid the occurrence of glitches in the output clock of the MMC, you need to perform the following steps:

Step 1 Disable the interface clock.

Set MMC_CLKENA to 0x0000_0000, set MMC_CMD[Start_cmd], MMC_CMD[Update_clk_regs_only], and MMC_CMD[Wait_prvdata_complete] to 1, and wait until MMC_CMD[Start_cmd] is cleared automatically.

Step 2 Set the clock divider.

Configure MMC_CLKDIV based on the required clock frequency, set MMC_CMD[Start_cmd] and MMC_CMD[Update_clk_regs_only] to 1, and wait until MMC_CMD[Start_cmd] is cleared automatically.

Step 3 Enable the interface clock again.

Set MMC_CLKENA to 0x0000_0001, set MMC_CMD[Start_cmd] and MMC_CMD[Update_clk_regs_only] to 1, and wait until MMC_CMD[Start_cmd] is cleared automatically.

----End

CAUTION

The values of MMC_CMD and MMC_CMD are loaded only after MMC_CLKDIV[Start_cmd] and MMC_CLKENA[Update_clk_only] are set to 1. After the values are loaded successfully, the MMC controller clears MMC_CMD[Start_cmd] automatically. If a command is being executed, a hardware locked error (HLE) interrupt is generated. In this case, you need to clear the interrupt, and then transmit a command again.

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When a command is being executed or data is being transferred, the clock parameter values of the card cannot be changed.

Initialization

Before commands and data are exchanged between a card and the MMC controller, the MMC controller needs to be initialized. Perform the following steps:

Step 1 Configure the frequency of the working clock of the MMC controller. For details, see "Working Clock Configuration" in section 12.4.2 "Application Notes."

Step 2 Soft-reset the MMC controller after the card is powered on and the command and data signal lines are pulled up and become stable. For details, see "Soft Reset" in section 12.4.2 "Application Notes."

Step 3 Clear interrupts. Set all bits of MMC_RINTSTS bit[15:0] to 1 to clear raw interrupt status bits.

Step 4 Configure MMC_INTMASK. Set all bits of MMC_INTMASK bit[15:0] to 1 to enable all interrupt sources.

If data is transferred in DMA mode, set MMC_INTMASK bit[4] and MMC_INTMASK bit[5] to 0 to mask the TX/RX FIFO data request interrupts.

Step 5 Set MMC_CTRL[Int_enable] to 1 to enable the MMC interrupt.

Step 6 Configure the timeout parameter register MMC_TMOUT.

Step 7 Configure the FIFO parameter register MMC_FIFOTH.

----End

After the preceding steps, the interface clock can be configured and commands can be transmitted to the card.

Non-Data Transfer Command

If the MMC controller receives a response (correct response, error response, or timeout response) after transmitting a command, the MMC controller sets MMC_RINTSTS bit[2] to 1. Short responses are stored in MMC_RESP0, and long responses are stored in MMC_RESP0 to MMC_RESP3. MMC_RESP3 bit[31] is the MSB, and MMC_RESP0 bit[0] is the LSB. After a command is transmitted, its error status is reflected by the response command and the corresponding error bit of MMC_RINTSTS.

To transmit a non-data transfer command, perform the following steps:

Step 1 Set the corresponding command parameters of MMC_CMDARG.

Step 2 Configure the command register MMC_CMD based on the description in Table 12-7.

Step 3 Wait until the MMC controller runs the command. If the command is executed, the MMC controller clears MMC_CMD[Start_cmd] automatically.

Step 4 Check whether an HLE interrupt is generated by MMC_RINTSTS bit[12].

Step 5 Wait until the command is executed. If the MMC controller receives a response (correct response, error response, or timeout response), the MMC controller sets MMC_RINTSTS bit[2] to 1. This indicates that the command is executed.

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Step 6 Check whether there is any response exception and read the response value if necessary.

You can check the response timeout error, response CRC error, and response error by reading MMC_RINTSTS bit[8], MMC_RINTSTS bit[6], and MMC_RINTSTS bit[1] respectively.

----End

CAUTION

The values of MMC_CMD, MMC_CMDARG, MMC_BYTCNT, and MMC_BLKSIZ are loaded only after MMC_CMDARG[Start_cmd] is set to 1 and MMC_CMD[Update_clock_registers_only] is set to 0. After the values are loaded successfully, the MMC controller clears MMC_CMD[Start_cmd] automatically.

If other commands are being executed, an HLE interrupt is generated. In this case, you need to perform the preceding operations again. When a non-data transfer command is executed, the values of MMC_BYTCNT and MMC_BLKSIZ are ignored.

Table 12-7 Default values of MMC_CMD when a non-data transfer command is executed

Parameter	Value	Description
Start_cmd	1	Indicates that a command starts to be transmitted.
Update_clock_registers_only	0	Indicates a non-clock parameter update command.
data_transfer_expected	0	Indicates a non-data transfer command.
card_number	0	-
cmd_index	Cmd index	Command index
send_initialization	0	Indicates that this bit is set to 1 when a command is a card reset command such as CMD0.
stop_abort_cmd	0	Indicates that this bit is set to 1 when a command is a data transfer stop command such as CMD12.
response_length	0	Indicates that this bit is set to 1 when the response is a long response.
response_expect	1	Indicates that this bit is set to 0 when a command is not responded, such as CMD0, CMD4, or CMD15.
Wait_prvdata_complete	1 or 0	Indicates that the MMC controller must wait until the current data transfer command is executed before transmitting another command. You are advised to set this bit to the fixed value 1 except that the command is used for querying the card status during data transfer or stopping the current data transfer.

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Parameter	Value	Description
Check_response_crc	1 or 0	Indicates whether the MMC controller checks the responded CRC bit.

Reading a Single Data Block or Multiple Data Blocks

To read a single data block or multiple data blocks, perform the following steps:

Step 1 Write 1 to MMC_CTRL[fifo_reset] to reset the FIFO pointer, and query and wait until this bit is cleared automatically.

Step 2 Write the number of bytes to be transmitted to MMC_BYTCNT.

Step 3 Write the block size to MMC_BLKSIZ.

Step 4 Write the start address for reading data to MMC_CMDARG.

Step 5 Configure MMC_CMD according to the description in Table 12-8.

For the SD card or MMC, you need to read a single data block by running CMD17, and read multiple data blocks by running CMD18; for the SDIO card, you need to read a single data block or multiple data blocks by running CMD53.

The MMC controller starts to run commands after MMC_CMD is written. After commands are transferred to the bus, the cmd_done interrupt is generated.

Step 6 Check the values of MMC_RINTSTS bit[5] and MMC_RINTSTS bit[10]. If any or both of them are 1, read the data in the FIFO by reading MMC_DATA. This ensures that the MMC controller can receive the subsequent data. In addition, check data error interrupts, that is, check the values of MMC_RINTSTS bit[7], MMC_RINTSTS bit[9], MMC_RINTSTS bit[13], and MMC_RINTSTS bit[15]. In this case, you can transmit a stop command to stop the data transfer by using the software.

Step 7 If MMC_RINTSTS bit[3] is 1, data transfer is complete. In this case, read the remaining data in the FIFO by reading MMC_DATA.

Step 8 If MMC_CMD[Send_auto_stop] is set to 1 during the command execution, the MMC controller automatically transmits a stop command to stop the data transfer. For details, see "Configuration for Using the Auto-Stop Function" in section 12.4.2 "Application Notes."

----End

Table 12-8 Default values of MMC_CMD when a single data block or multiple data blocks are read

Parameter	Value	Description
Start_cmd	1	Indicates that a command starts to be transmitted.
Update_clock_registers_only	0	Indicates a non-clock parameter update command.
card_number	0	-
send_initialization	0	Indicates that this bit is set to 1 when a

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Parameter	Value	Description
		command is a card reset command such as CMD0.
stop_abort_cmd	0	Indicates that this bit is set to 1 when a command is a data transfer stop command such as CMD12.
send_auto_stop	0 or 1	For details, see "Configuration for Using the Auto-Stop Function" in section 12.4.2 "Application Notes."
transfer_mode	0	Indicates block transfer.
read/write	0	Indicates that data is read from the card.
response_length	0	Indicates that all responses to data commands are short responses.
data_transfer_expected	1	Indicates a data transfer command.
response_expect	1	Indicates that this bit is set to 0 when a command is not responded, such as CMD0, CMD4, or CMD15.
cmd_index	Cmd index	Indicates the command index.
Wait_prvdata_complete	1 or 0	Indicates that the master device must wait until the current data transfer command is executed before transmitting another command. You are advised to set this bit to the fixed value 1 except that the command is used for querying the card status or stopping the current data transfer.
Check_response_crc	1 or 0	Indicates whether the MMC controller checks the responded CRC bit.

Writing a Single Data Block or Multiple Data Blocks

To write a single data block or multiple data blocks, perform the following steps:

Step 1 Write 1 to MMC_CTRL[fifo_reset] to reset the FIFO pointer, and query and wait until this bit is cleared automatically.

Step 2 Write the size of the data to be transmitted to MMC_BYTCNT.

Step 3 Write the block size to MMC_BLKSIZ.

Step 4 Write the start address for writing data to MMC_CMDARG.

Step 5 Write data to the FIFO by writing to MMC_DATA. The FIFO needs to be filled with data completely at the very beginning.

Step 6 Configure MMC_CMD according to the description in Table 12-9.

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For the SD card or MMC, you need to write a single data block by running CMD24, and write multiple data blocks by running CMD25; for the SDIO card, you need to write a single data block or multiple data blocks by running CMD53.

Step 7 Check the values of MMC_RINTSTS bit[4] and MMC_RINTSTS bit[10]. If any or both of them are 1, fill the FIFO with data by writing to MMC_DATA. In addition, check data error interrupts, that is, check the values of MMC_RINTSTS bit[7], MMC_RINTSTS bit[9], MMC_RINTSTS bit[13], and MMC_RINTSTS bit[15]. If necessary, you can transmit a stop command to stop the data transfer by using the software. If MMC_RINTSTS bit[3] is 1, data transfer is complete.

Step 8 If MMC_CMD[Send_auto_stop] is set to 1 during the command execution, the MMC controller automatically transmits a stop command to stop the data transfer. For details, see "Configuration for Using the Auto-Stop Function" in section 12.4.2 "Application Notes."

Step 9 Query and wait until the value of MMC_STATUS[data_busy] is changed from 1 to 0.

----End

Table 12-9 Default values of MMC_CMD when a single data block or multiple data blocks are written

Parameter	Value	Description
Start_cmd	1	Indicates that a command starts to be transmitted.
Update_clock_registers_only	0	Indicates a non-clock parameter update command.
card_number	0	-
send_initialization	0	Indicates that this bit is set to 1 when a command is a card reset command such as CMD0.
stop_abort_cmd	0	Indicates that this bit is set to 1 when a command is a data transfer stop command such as CMD12.
send_auto_stop	0 or 1	For details, see "Configuration for Using the Auto-Stop Function" in section 12.4.2 "Application Notes."
transfer_mode	0	Indicates block transfer.
read_write	1	Indicates that data is written to the card.
response_length	0	Indicates that all responses to data commands are short responses.
data_transfer_expected	1	Indicates a data transfer command.
response_expect	1	Indicates that this bit is set to 0 when a command is not responded, such as CMD0, CMD4, or CMD15.
cmd_index	Cmd index	-
Wait_prvdata_complete	1 or 0	Indicates that the master device must wait until the current data transfer command is executed before transmitting another command. You are advised to

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Parameter	Value	Description
		set this bit to the fixed value 1 except that the command is used for querying the card status or stopping the current data transfer.
Check_response_crc	1 or 0	Indicates whether the MMC controller checks the responded CRC bit.

Reading/Writing Stream Data

The modes of reading/writing stream data are the same as those of reading/writing data blocks, except that MMC_CMD[Transfer_mode] needs to be set to 1. During a stream data transfer, the auto-stop function is required.

Transferring Data by Using the IDMAC

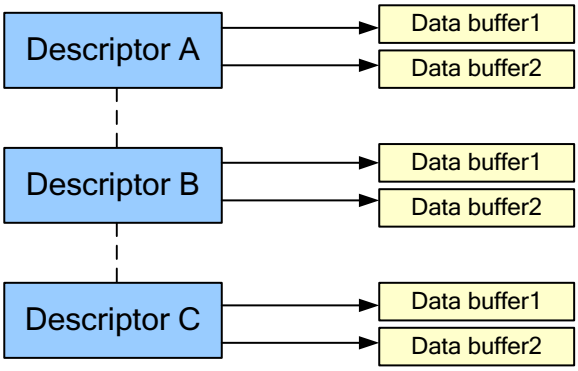
The MMC controller has an embedded IDMAC that transfers data from the original address to the destination address based on the specified descriptor.

Descriptor

The IDMAC supports the following two types of descriptors:

- Dual-buffer descriptor. For this type of descriptor, the span between two descriptors is determined by the DSL bit of MMC_BMOD. Figure 12-28 shows the structure of the dual-buffer descriptor.

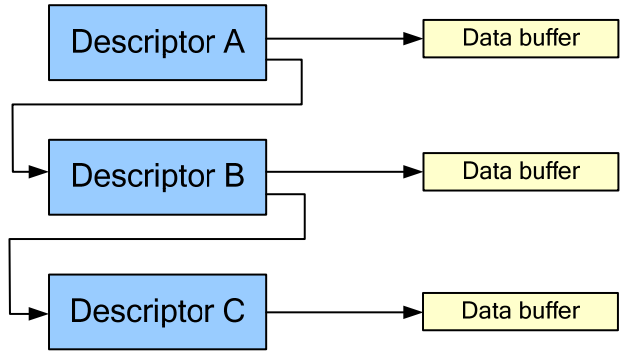
Figure 12-28 Structure of the dual-buffer descriptor



- Linked descriptor. For this type of descriptor, each descriptor points to a unique buffer and the next descriptor. Figure 12-29 shows the structure of the linked descriptor.

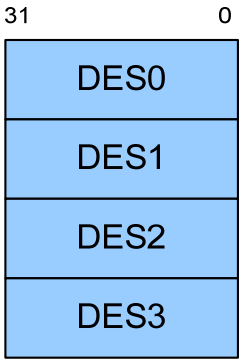
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Figure 12-29 Structure of the linked descriptor



Each descriptor must be word-aligned, and each descriptor contains 16-byte control and status information. Figure 12-30 shows the internal structure of 32-bit descriptors.

Figure 12-30 Internal structure of 32-bit descriptors



DES0 is used to protect the control and status information. Table 12-10 describes the definition of each bit.

Table 12-10 Definition of each bit of DES0

Bits	Register	Description
31	OWN	Indicates the attributes of the descriptor. 0: The descriptor belongs to the CPU. 1: The descriptor belongs to the IDMAC. After data is transferred by using the IDMAC, the IDMAC clears this bit.
30	CES	Indicates the error status when a card is read. 0: No error occurs. 1: An error occurs.
29:6	RES	Reserved
5	ER	Indicates the link end of the descriptor.

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Bits	Register	Description
		0: The descriptor is not the last one on the link. 1: The descriptor is the last one on the link. This bit is valid only for the dual-buffer descriptor.
4	CH	Indicates the definition of the second address in DES3. 0: The second address in DES3 is the address of the second buffer. 1: The second address in DES3 is the address of the next descriptor. When this bit is 1, DES1[25:13] must be 0.
3	FS	Indicates that the descriptor contains the first data buffer when this bit is 1. If the size of the first data buffer is 0, the next descriptor contains the start data.
2	LD	Indicates that the descriptor points to the last data buffer when this bit is 1.
1	DIC	Indicates that the data transfer completion interrupt is masked when this bit is 1.
0	RES	Reserved

DES1 is used to specify the buffer size. Table 12-11 describes the definition of each bit of DES1.

Table 12-11 Definition of each bit of DES1

Bits	Register	Description
31: 26	RES	Reserved
25: 13	BS2	Indicates the number of bytes in the second data buffer. This value must be an integral multiple of 4. This bit is invalid when DES0[4] is 1.
12: 0	BS1	Indicates the number of bytes in the first data buffer. This value must be an integral multiple of 4.

DES2 is the address pointer of the first data buffer. Table 12-12 describes the definition of each bit of DES2.

Table 12-12 Definition of each bit of DES2

Bits	Register	Description
31: 0	BAP1	Indicates the physical address of the first data buffer. The address must be word-aligned.

DES3 indicates the second address. Table 12-13 describes the definition of each bit of DES3.

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Table 12-13 Definition of each bit of DES3

Bits	Register	Description
31: 0	BAP2	Indicates the physical address of the second data buffer when dual-buffer descriptors are used, or indicates the physical address of the next descriptor when DES0[4] is 1.

Initialization

Perform the following steps:

Step 1 Configure MMC_BMOD to set bus parameters.

Step 2 Configure MMC_IDINTEN to mask unnecessary registers.

Step 3 Create TX/RX descriptor linked lists, configure MMC_DBADDR, and set the start address.

Step 4 The IDMAC attempts to obtain descriptors from the descriptor linked lists.

----End

Transmission

Perform the following steps:

Step 1 Create the descriptors DES0 to DES3 by using the CPU, set DES0 bit[31] (OWN bit) to 1, and provide data buffers.

Step 2 Write data commands to MMC_CMD.

Step 3 Set TX_Wmark by using MMC_FIFOTH.

Step 4 The IDMAC obtains descriptors and check whether the value of the OWN bit is 1. If the OWN bit is not 1, wait until the CPU releases descriptors. During this process, the IDMAC enters the suspend state. Therefore, the CPU needs to configure MMC_PLDMND to enable the IDMAC to obtain descriptors again.

Step 5 The IDMAC transfers data from data buffers to the internal FIFO of the MMC controller when the OWN bit is 1.

Step 6 If the interrupts are enabled, the corresponding bit of the IDMAC status register MMC_IDSTS is updated and the OWN bit is cleared after data transfer.

----End

Reception

Perform the following steps:

Step 1 Create the descriptors DES0 to DES3 by using the CPU, and set the DES0 bit[31] (OWN bit) to 1.

Step 2 Write read commands to MMC_CMD.

Step 3 Set RX_WMark by using MMC_FIFOTH.

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Step 4 The IDMAC obtains descriptors and check whether the value of the OWN bit is 1. If the OWN bit is not 1, wait until the CPU releases descriptors. During this process, the IDMAC enters the suspend state. Therefore, the CPU needs to configure MMC_PLDMND to enable the IDMAC to obtain descriptors again.

Step 5 The IDMAC transfers data from the internal FIFO of the MMC controller to the external data buffers when the OWN bit is 1.

Step 6 If the interrupts are enabled, the corresponding bit of the IDMAC status register MMC_IDSTS is updated and the OWN bit is cleared after data transfer.

----End

Auto-Stop Function Configuration

When multiple data blocks are read or written, a stop command is required to stop each data transfer. The stop command can be transmitted in non-data transfer command mode or by using the auto-stop function.

The auto-stop function is applicable in the following scenarios:

- SD card

  Multi-block read/write operation by running CMD18 or CMD25
- MMC
  - Stream data read/write operation
  - Multi-block read/write operation in open-ended mode by running CMD18 or CMD25

Before using the auto-stop function of the MMC controller, you are advised to perform the followings steps:

Step 1 Set MMC_CMD[Send_auto_stop] to 1 during the block transfer command operation.

Step 2 After data transfer, the MMC controller automatically transmits a stop command to enable the card to restore to the corresponding state.

Step 3 Read MMC_RINTSTS[auto_cmd_done] to check whether the stop command is executed. The response is stored in MMC_RESP1.

----End

Stopping or Aborting the Data Transfer

The stop command is used to interrupt the data transfer between the MMC controller and the card, whereas the abort command is used to interrupt the I/O data transfer only in SDIO_IOONLY or SDIO_COMBO mode.

The two commands are used as follows:

- Stop command

  This command can be transmitted at any time during data transfer, because this command is used to stop the data transfer. In this case, you need to set MMC_CMD bit[5:0] to CMD12, MMC_CMD bit[14] to 1, and MMC_CMD bit[13] to 0.
- Abort command

  This command is available only for SDIO_IOONLY or SDIO_COMBO. To abort the data transfer, you need to configure the CCCR[ASx] register of the SDIO card by running the CMD52 command.

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Suspend and Resume Operations

An SDIO card can store the data of a maximum of seven functional devices. The MMC controller can suspend the data transfer of a device by performing the suspend operation. Then the SD interface bus is available for another device with higher priority. After the device with higher priority transfers data, the MMC controller can resume the suspended data transfer of the previous device.

The suspend and resume operations are implemented by configuring the corresponding bits of the CCCR register of the SDIO card. The CCCR register is written or read by running the CMD52 command.

To implement a suspend operation, perform the following steps:

Step 1 Query the SBS bit of the CCCR register to check whether the SDIO card supports suspend and resume operations.

Step 2 Query the FSx and bus status (BS) bits of the CCCR register to check whether the functional device to be suspended is transferring data.

If the BS bit is 1, the device specified by the FSx bit is transferring data.

Step 3 Set the bus release (BR) bit of the CCCR register to 1 to suspend the current data transfer.

Step 4 Check whether the BS and the BR bits of the CCCR register are cleared.

The BS bit retains 1 when the data bus is being used. The BR bit retains 1 before the bus is released completely. When both the BR and BS bits are 0, the data transfer of the selected functional device is suspended.

Step 5 If the current read operation is suspended, MMC_CTRL[Abort_read_data] needs to be set to 1 to restart the data transfer function of the MMC controller after the suspend operation. Then MMC_CTRL[Abort_read_data] is cleared automatically.

Step 6 Read MMC_TCBCNT to query the number of transferred bytes.

----End

To implement a resume operation, perform the following steps:

Step 1 Query the transfer status of the card to check whether the bus is idle.

Step 2 If the card is disconnected, run the CMD7 command to select it. The card status can be queried by running the CMD52 or CMD53 command.

Step 3 Check whether the device to be resumed is ready for data transfer by querying the RF bit of the CCCR register. If the RF bit is 1, the device is ready for data transfer.

Step 4 Run the CMD52 command to write the device ID to the FS bit of the CCCR register to resume the data transfer, and enable the MMC controller to enter the data transfer state (that is, write the block size to MMC_BLKSIZ, and write the amount of remaining data to be transferred to MMC_BYTCNT).

For details about the configuration of MMC_CMDARG, see Table 12-14. The configuration of MMC_CMD is similar to the configuration during block transfer.

Step 5 The data transfer is resumed after the CMD52 command is transmitted successfully. Read the resume data flag (DF) bit of the SDIO device. If this bit is 1, data transfer starts when the transfer function is resumed. If this bit is 0, no data is ready for transferring.

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Step 6 If the DF bit is 0, the MMC generates a data timeout error interrupt a period of time later during data read.

----End

Table 12-14 Reference configuration of MMC_CMDARG when the resume operation is performed

MMC_CMDARG	Value	Description
Bit[31]	1	Indicates the read/write flag.
Bit[30:28]	0	Indicates the ID of a functional device that accesses the CCCR register.
Bit[27]	1	Indicates the real-time flag, that is, write-to-read.
Bit[26]	None	-
Bit[25:9]	0x0D	Indicates the register address.
Bit[8]	None	-
Bit[7:0]	ID of the functional device that is resumed	Indicates write data.

CAUTION

The MMC controller cannot be woken up after the system enters the low-power mode.

Read Wait Operation

The read wait operation is performed to suspend the data transfer of the device that is using the SDIO card. The MMC controller determines the duration of pausing the data transfer.

To implement a read wait operation, perform the following steps:

Step 1 Check whether the card supports the read wait operation.

You can read the SRW bit of the CCCR register by running the CMD52 command. If the SRW bit is 1, all the devices supporting the card support the read wait operation.

Step 2 Set MMC_CTRL[Read_wait] to 1.

Step 3 If you want to resume the data transfer, clear MMC_CTRL[Read_wait].

----End

12.4.3 Register Summary

Table 12-15 describes the MMC registers.

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Table 12-15 Summary of MMC registers (the base address for SDIO0 is 0x1002_0000 and for SDIO1 is 0x1003_0000)

Offset Address	Register	Description	Page
0x0000	MMC_CTRL	MMC control register	12-88
0x0004	MMC_PWREN	Power_en control register	12-90
0x0008	MMC_CLKDIV	Clock divider register	12-90
0x000C	MMC_CLKSRC	Clock source select register of the SD card	12-91
0x0010	MMC_CLKENA	Clock enable register	12-91
0x0014	MMC_TMOUT	Timeout register	12-92
0x0018	MMC_CTYPE	Card type register	12-93
0x001C	MMC_BLKSIZ	Block size configuration register	12-93
0x0020	MMC_BYTCNT	Block transfer count register	12-94
0x0024	MMC_INTMASK	Interrupt mask register	12-94
0x0028	MMC_CMDARG	Command parameter register	12-95
0x002C	MMC_CMD	Command register	12-96
0x0030	MMC_RESP0	Response register 0	12-99
0x0034	MMC_RESP1	Response register 1	12-99
0x0038	MMC_RESP2	Response register 2	12-99
0x003C	MMC_RESP3	Response register 3	12-100
0x0040	MMC_MINTSTS	Masked interrupt status register	12-100
0x0044	MMC_RINTSTS	Raw interrupt status register	12-101
0x0048	MMC_STATUS	Status register	12-102
0x004C	MMC_FIFOTH	FIFO threshold register	12-104
0x0050	MMC_CDETECT	Card detection register	12-105
0x0054	MMC_WRTPRT	Card write protection register	12-105
0x005C	MMC_TCBCNT	Count of bytes transmitted to the card register	12-106
0x0060	MMC_TBBCNT	Count of bytes transmitted from the bus interface unit (BIU) FIFO register	12-106
0x0064	MMC_DEBNCE	Dejitter count register	12-107
0x0074	MMC_UHS_REG	UHS-1 register	12-107
0x0078	MMC_CARD_RSTN	eMMC reset control register	12-108

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Offset Address	Register	Description	Page
0x0080	MMC_BMOD	Bus mode register	12-108
0x0084	MMC_PLDMND	Poll demand register	12-109
0x0088	MMC_DBADDR	Base address register of the descriptor linked list	12-110
0x008C	MMC_IDSTS	IDMAC status register	12-110
0x0090	MMC_IDINTEN	IDMAC interrupt enable register	12-111
0x0094	MMC_DSCADDR	Address register of the current descriptor	12-113
0x0098	MMC_BUFADDR	Address register of the current data buffer	12-113
0x0100	MMC_CARDTHRCTL	Threshold control register	12-117
0x0108	MMC_UHS_REG_EXT	UHS extension register	12-115
0x010C	MMC_DDR_REG	eMMC 4.5 DDR start bit detection control register	12-116
0x0110	MMC_ENABLE_SHIFT	Phase shift register	12-117
0x0200	MMC_DATA	Data register (entrance address of the FIFO)	12-117

12.4.4 Register Description

MMC_CTRL

MMC_CTRL is a MMC control register.

Offset Address	Register Name	Total Reset Value
0x0000	MMC_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved						use_internal_dmac	reserved																abort_read_data	send_irq_response	read_wait	reserved	int_enable	reserved	dma_reset	fifo_reset	controller_reset
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved

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[25]	RW	Use_internal_dmac	Whether to transfer data by using the IDMAC 0: The CPU transfers data by using the slave interface. 1: The CPU transfers data by using the IDMAC.
[24:9]	RO	reserved	Reserved
[8]	RW	Abort_read_data	Whether to abort the data transfer during data read 0: invalid 1: After transmitting a suspend command during data read, the software polls the card to check when suspend occurs. After the suspend occurs, the software sets the bit to 1. This enables the data transfer state machine to be restored to idle state for the next block transfer. After the state machine restores to the idle status, this bit is cleared automatically.
[7]	RW	Send_irq_response	TX interrupt response control 0: invalid 1: An interrupt request (IRQ) response is transmitted automatically. After the response is transmitted, this bit is cleared automatically. To wait for the interrupt generated by the MMC controller, the host transmits the CMD40 command and waits for the interrupt response from the MMC controller. If you do not want the host to keep in the interrupt wait state, set this bit to 1 and transmit the CMD40 command to restore the host to the idle state.
[6]	RW	Read_wait	Read wait control 0: disabled 1: enabled This bit is valid only for the SDIO card that supports the read wait function.
[5]	RO	reserved	Reserved.
[4]	RW	Int_enable	Global interrupt enable 0: disabled 1: enabled The interrupt output is valid only when this bit is valid and the interrupt source is enabled.
[3]	RO	reserved	Reserved
[2]	RW	Dma_reset	Soft reset control for the IDMAC 0: invalid 1: reset the internal DMA interface This bit is automatically reset after two AHB clock cycles.

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[1]	RW	Fifo_reset	Soft reset control for the internal FIFO 0: invalid 1: reset the FIFO pointer This bit is reset automatically after the reset operation is complete.
[0]	RW	Controller_reset	Soft reset control for the controller 0: invalid 1: reset the MMC/SD/SDIO host module

MMC_PWREN

MMC_PWREN is a Power_en control register.

Offset Address	Register Name	Total Reset Value
0x0004	MMC_PWREN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														Power_enable	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1:0]	RW	Power_enable	Power control Bit 1 and bit 0 control the eMMC power and SDIO power respectively. The following takes bit 0 as an example: 0: power off 1: power on This value is used to drive the SDIOn_CARD_POWER_EN (n = 0 or 1) pin.

MMC_CLKDIV

MMC_CLKDIV is a clock divider register that shows the ratio of the frequency of the module output clock to the frequency of the input clock. For example, if the module input clock is 40 MHz and the register value is set to 1, the output clock is 20 MHz.

The clock divider is 2 x N. If the value of *N* is 0x0, it indicates no frequency division (2 x 0 = 0); if the value of *N* is 0x1, the frequency is divided by 2; if the value of *N* is 0xFF, the frequency is divided by 510.

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Offset Address	Register Name	Total Reset Value
0x0008	MMC_CLKDIV	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																clk_divider1								clk_divider0							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:8]	RW	clk_divider1	eMMC clock divider. The clock divider is 2 x *n*. If the value of *n* is 0, it indicates no frequency division; if the value of *n* is 1, the frequency is divided by 2; if the value of *n* is 0xFF, the frequency is divided by 510.
[7:0]	RW	clk_divider0	SDIO clock divider. The clock divider is 2 x *n*. If the value of *n* is 0, it indicates no frequency division; if the value of *n* is 1, the frequency is divided by 2; if the value of *n* is 0xFF, the frequency is divided by 510.

MMC_CLKSRC

MMC_CLKSRC is a clock source select register of the SD card.

Offset Address	Register Name	Total Reset Value
0x000C	MMC_CLKSRC	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												clk_source			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3:0]	RW	clk_source	This bit must be set to 4.

MMC_CLKENA

MMC_CLKENA is a clock enable register.

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Offset Address	Register Name	Total Reset Value
0x0010	MMC_CLKENA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														cclk_low_power		reserved														cclk_enable	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17:16]	RW	cclk_low_power	Card low-power control for disabling the card clock Bit 17 and bit 16 indicate the low-power control for eMMC and SDIO respectively. The following takes bit 16 as an example: 0: non lower-power mode 1: low-power mode When the card is idle, the card clock is disabled. This function is used only for the MMC and SD card, because the card clock of the SDIO card must be retained for detecting interrupts.
[15:2]	RO	reserved	Reserved
[1:0]	RW	cclk_enable	Card clock enable Bit 1 and bit 0 indicate the clock enable for eMMC and SDIO respectively. The following takes bit 0 as an example: 0: disabled 1: enabled

MMC_TMOUT

MMC_TMOUT is a timeout register.

Offset Address	Register Name	Total Reset Value
0x0014	MMC_TMOUT	0xFFFF_FF40

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data_timeout																								response_timeout							
Reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RW	data_timeout	Timeout during the data transfer of the card, in unit of the mmc_clk cycle of the card. The timeout is also the data starvation timeout of the CPU

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[7:0]	RW	response_timeout	Response timeout, in unit of the mmc_clk cycle of the card.

MMC_CTYPE

MMC_CTYPE is a card type register.

Offset Address	Register Name	Total Reset Value
0x0018	MMC_CTYPE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														card_width		reserved														card_width1	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17:16]	RW	card_width_0	Bus width of the card Bit 17 and bit 16 indicate the eMMC bus width and SDIO bus width respectively. The following takes bit 16 as an example: This field must be set to 0 (non 8-bit mode). The bus width of the card is set to 1-bit mode or 4-bit mode, which depends on the value of bit[0].
[15:2]	RO	reserved	Reserved
[1:0]	RW	card_width_1	Bus width of the card Bit 1 and bit 0 indicate the eMMC bus width and SDIO bus width respectively. The following takes bit 0 as an example: 0: 1-bit mode 1: 4-bit mode

MMC_BLKSIZ

MMC_BLKSIZ is a block size configuration register.

Offset Address	Register Name	Total Reset Value
0x001C	MMC_BLKSIZ	0x0000_0200

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																block_size															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	block_size	Block size. The initial size of each block is 512 bytes.

MMC_BYTCNT

MMC_BYTCNT is a block transfer count register.

Offset Address	Register Name	Total Reset Value
0x0020	MMC_BYTCNT	0x0000_0200

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Byte_count																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	Byte_count	Number of transferred bytes. The number must be an integral multiple of the block size. If the data transfer is not block transfer, this register must be set to 0. In this case, the software needs to transmit a stop or an abort command to control the data transfer.

MMC_INTMASK

MMC_INTMASK is an interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x0024	MMC_INTMASK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															sdio_int_mask	Int_mask															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved

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[17:16]	RW	sdio_int_mask	Interrupt mask control Bit 17 and bit 16 indicate the interrupt mask control for eMMC and SDIO respectively. The following takes bit 16 as an example: 0: masked 1: enabled
[15:0]	RW	int_mask	Interrupt mask 0: masked 1: enabled Bit[15]: end-bit error (read)/write no CRC (EBE) Bit[14]: auto command done (ACD) Bit[13]: start-bit error (SBE) Bit[12]: hardware locked write error (HLE) Bit[11]: FIFO underrun/overrun error (FRUN) Bit[10]: data starvation-by-host timeout (HTO) Bit[9]: data read timeout (DTO) Bit[8]: response timeout (RTO) Bit[7]: data CRC error (DCRC) Bit[6]: response CRC error (RCRC) Bit[5]: RX FIFO data request (RXDR) Bit[4]: TX FIFO data request (TXDR) Bit[3]: data transfer over (DTO) Bit[2]: command done (CD) Bit[1]: response error (RE) Bit[0]: card detect (CD)

MMC_CMDARG

MMC_CMDARG is a command parameter register.

Offset Address	Register Name	Total Reset Value
0x0028	MMC_CMDARG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cmd_arg																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cmd_arg	Command parameter that is transferred to the card. The command parameter is related to the protocol, and each command corresponds to a command parameter.

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MMC_CMD

MMC_CMD is a command register.

Offset Address	Register Name	Total Reset Value
0x002C	MMC_CMD	0x2000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	start_cmd	reserved	use_hold_reg	volt_switch	boot_mode	disable_boot	expect_boot_ack	enable_boot	reserved		update_clock_registes_only	card_number					send_initialization	stop_abort_cmd	wait_prvdata_complete	send_auto_stop	transfer_mode	read_write	data_transfer_expected	check_response_crc	response_length	response_expect	cmd_index					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	start_cmd	Start control 0: do not start 1: start a command After the command is transferred to the card interface unit (CIU), this bit is cleared. The CPU does not allow modifications to this register. Otherwise, an HLE interrupt is generated. After transmitting a command, the CPU queries this bit, and transmits the next command when the value of this bit changes to 0.
[30]	RO	reserved	Reserved
[29]	RW	use_hold_reg	Whether to use the hold register 0: The CMD and DATA signals transmitted to the MMC do not pass through the hold register. 1: The CMD and DATA signals transmitted to the MMC pass through the hold register.
[28]	RW	volt_switch	Voltage switching enable 0: disabled 1: enabled
[27]	RW	boot_mode	Boot mode 0: boot mode 1: alternative boot mode

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[26]	RW	disable_boot	Boot disable If the software enables this bit and the Start_cmd bit at the same time, the controller aborts the boot operation. The Enable_boot and Disable_boot bits cannot be enabled at the same time.
[25]	RW	expect_boot_ack	Boot response enable If the software enables this bit and the Enable_boot bit at the same time, the controller detects the boot response signal in "0–1–0" sequence.
[24]	RW	enable_boot	Boot enable This bit is available when the boot mode is mandatory. If the software enables this bit and the Start_cmd bit, the controller pulls down the CMD signal to start the boot process. The Enable_boot and Disable_boot bits cannot be enabled at the same time.
[23:22]	RO	reserved	Reserved
[21]	RW	update_clock_registers_only	Automatic update 0: The normal command sequence is used. That is, the values of MMC_CMD, MMC_CMDARG, MMC_TMOUT, MMC_CTYPE, MMC_BLKSIZ, and MMC_BYTCNT are transferred from the BIU to the CIU. The CIU uses the new values of registers when running new commands. 1: No command is transmitted, and only the clock register values in the card clock domain are updated. The values of MMC_CLKDIV, MMC_CLKSRC, and MMC_CLKENA are transferred to the card clock domain. The card clock can be changed (frequency change and clock enable) even no command is transmitted. Each time the card clock is changed; this bit must be set to 1. In this case, no command is transmitted to the card, and no command done interrupt is generated.
[20:16]	RW	card_number	Serial number of the card that is being used
[15]	RW	send_initialization	Whether to transmit the initialization sequence 0: do not transmit the initialization sequence before transmitting the Send_initialization command (high level in 80 clock cycles) 1: transmit the initialization sequence before transmitting the Send_initialization command When a card is powered on, the initialization sequence must be transmitted for initialization before any command is transmitted. That is, this bit must be set to 1.
[14]	RW	stop_abort_cmd	Whether to transmit the stop/abort command when data is being transferred in open-ended mode or in fixed length mode. 0: do not transmit the stop/abort command 1: transmit the stop/abort command for stopping the current data transfer

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[13]	RW	wait_prvdata_complete	Whether to transmit a command immediately 0: transmit a command immediately even though the previous data transfer is not complete 1: transmit a command only when the previous data transfer is complete The typical value is 0. If this bit is set to 0, the transfer status can be read during data transfer, and the interrupt transfer is supported.
[12]	RW	send_auto_stop	Whether to transmit the stop command 0: do not transmit the stop command after data transfer 1: transmit the stop command after data transfer In non-data transfer mode, this bit is ignored.
[11]	RW	transfer_mode	Transfer mode 0: block transfer mode 1: stream transfer mode In non-data transfer mode, this bit is ignored.
[10]	RW	read_write	Read/write control 0: read data from the card 1: write data to the card In non-data transfer mode, this bit is ignored.
[9]	RW	data_transfer_expected	Data transfer indicator 0: No data is output from the card. 1: Data is output from the card.
[8]	RW	check_response_crc_	Whether to perform the CRC check 0: Do not check the CRC response. 1: Check the CRC response. No valid CRC is returned when some commands are responded. To prevent the host from performing CRC, the software needs to disable this function based on related commands.
[7]	RW	response_length	Response length 0: Short responses are output from the card 1: Long responses are output from the card The length of a long response is 128 bits, whereas the length of a short response is 32 bits.
[6]	RW	response_expect	Whether to output response 0: No response is output from the card. 1: Responses are output from the card.
[5:0]	RW	cmd_index	Command index

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MMC_RESP0

MMC_RESP0 is response register 0.

Offset Address	Register Name	Total Reset Value
0x0030	MMC_RESP0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	response0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	response0	Bit[31:0] of a response

MMC_RESP1

MMC_RESP1 is response register 1.

Offset Address	Register Name	Total Reset Value
0x0034	MMC_RESP1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	response1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	response1	Bit[63:32] of a long response After the CIU transmits an auto-stop command, the corresponding response is stored in this register, and the response to the previous command is still stored in MMC_RESP0. The auto-stop command is available only during data transfer, and the corresponding response is always a short response.

MMC_RESP2

MMC_RESP2 is response register 2.

Offset Address	Register Name	Total Reset Value
0x0038	MMC_RESP2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	response2																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	response2	Bit[95:64] of a long response

MMC_RESP3

MMC_RESP3 is response register 3.

Offset Address	Register Name	Total Reset Value
0x003C	MMC_RESP3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	response3																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	response3	Bit[127:96] of a long response

MMC_MINTSTS

MMC_MINTSTS is a masked interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0040	MMC_MINTSTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sdio_interrupt	int_status														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17:16]	RO	sdio_interrupt	Interrupt mask status The SDIO interrupt is valid only when MMC_INTMASK [sdio_int_mask] is enabled. Bit 17 and bit 16 indicate the interrupt mask control for eMMC and SDIO respectively. The following takes bit 16 as an example: 0: No SDIO interrupt is output from the card. 1: An SDIO interrupt is output from the card.

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[15:0]	RO	int_status	Status of each interrupt Bit[15]: end-bit error (read)/write no CRC (EBE) Bit[14]: auto command done (ACD) Bit[13]: start-bit error (SBE) Bit[12]: hardware locked write error (HLE) Bit[11]: FIFO underrun/overrun error (FRUN) Bit[10]: data starvation by the host timeout (HTO) Bit[9]: data read timeout (DTO) Bit[8]: response timeout (RTO) Bit[7]: data CRC error (DCRC) Bit[6]: response CRC error (RCRC) Bit[5]: RX FIFO data request (RXDR) Bit[4]: TX FIFO data request (TXDR) Bit[3]: data transfer over (DTO) Bit[2]: command done (CD) Bit[1]: response error (RE) Bit[0]: card detect (CD)

MMC_RINTSTS

MMC_RINTSTS is a raw interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0044	MMC_RINTSTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved															sdio_interrupt	int_status															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17:16]	RW	sdio_interrupt	Raw status of an interrupt Bit 17 and bit 16 control the eMMC interrupts and SDIO interrupts respectively. The following takes bit 16 as an example: 0: No SDIO interrupt is output from the card. 1: An SDIO interrupt is output from the card. The value of the interrupt status bit is independent of the interrupt mask status.

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[15:0]	RW	int_status	Raw status of each interrupt. Writing 1 clears the bits, and writing 0 has no effect. The value of the interrupt status bit is independent of the interrupt mask status. Bit[15]: end-bit error (read)/write no CRC (EBE) Bit[14]: auto command done (ACD) Bit[13]: start-bit error (SBE) Bit[12]: hardware locked write error (HLE) Bit[11]: FIFO underrun/overrun error (FRUN) Bit[10]: data starvation by the host timeout (HTO) Bit[9]: data read timeout (DRTO)/Boot Data Start (BDS) Bit[8]: response timeout (RTO)/Boot Ack Received (BAR) Bit[7]: data CRC error (DCRC) Bit[6]: response CRC error (RCRC) Bit[5]: RX FIFO data request (RXDR) Bit[4]: TX FIFO data request (TXDR) Bit[3]: data transfer over (DTO) Bit[2]: command done (CD) Bit[1]: response error (RE) Bit[0]: card detect (CD)

MMC_STATUS

MMC_STATUS is a status register.

Offset Address	Register Name	Total Reset Value
0x0048	MMC_STATUS	0x0000_0106

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				fifo_count												response_index				data_state_mc_busy		data_busy		data_3_status		commandfsm_states				fifo_full		fifo_empty		fifo_tx_watermark		fifo_rx_watermark	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0

Bits	Access	Name	Description
[31:30]	RO	reserved	Reserved
[29:17]	RO	fifo_count	FIFO count
[16:11]	RO	response_index	Serial number of the previous response, including the response to the auto-stop command

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[10]	RO	data_state_mc_busy	0: The data TX/RX state machine is idle. 1: The data TX/RX state machine is busy.
[9]	RO	data_busy	This field value depends on the reverse level of DATA0. 0: idle (DATA0 is high level) 1: busy (DATA0 is low level)
[8]	RO	data_3_status	Card detection signal. This field value depends on the level of DATA3. 0: No card is inserted (DATA3 is low level). 1: A card is inserted (DATA3 is high level).
[7:4]	RO	commandfsm_states	Status of the command state machine 0x0: idle 0x1: send init sequence 0x2: Tx cmd start bit 0x3: Tx cmd tx bit 0x4: Tx cmd index + arg 0x5: Tx cmd crc7 0x6: Tx cmd end bit 0x7: Rx resp start bit 0x8: Rx resp IRQ response 0x9: Rx resp tx bit 0xA: Rx resp cmd idx 0xB: Rx resp data 0xC: Rx resp crc7 0xD: Rx resp end bit 0xE: Cmd path wait NCC 0xF: wait, CMD-to-response turnaround
[3]	RO	fifo_full	FIFO full flag 0: empty 1: full
[2]	RO	fifo_empty	FIFO empty flag 0: not empty 1: empty
[1]	RO	fifo_tx_watermark	Whether the FIFO reaches the TX threshold 0: no 1: yes
[0]	RO	fifo_rx_watermark	Whether the FIFO reaches the RX threshold 0: no 1: yes

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MMC_FIFOTH

MMC_FIFOTH is a FIFO threshold register.

Offset Address	Register Name	Total Reset Value
0x004C	MMC_FIFOTH	0x60FF_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	Multiple_Transaction_Size			rx_wmark												reserved				tx_wmark											
Reset	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:28]	RW	Multiple_Transaction_Size	Transferred burst length 000: 1 001: 4 010: 8 011: 16 100: 32 101: 64 110: 128 111: 256
[27:16]	RW	rx_wmark	FIFO threshold during data read. If the data amount in the FIFO is above the threshold, a DMA request is enabled. After a data transfer, a DMA request is raised for transferring the remaining data no matter whether the threshold is reached. In non-DMA mode, the RXDR interrupt is enabled. If the data amount in the FIFO is not above the threshold after a data transfer, no interrupt is generated. In this case, the software needs to read the remaining data by querying the DTD interrupt. If a data transfer is complete in DMA mode, the DMA raises a single transfer request to read data until the DTD interrupt is generated even though the remaining data amount is below the threshold. Restriction: RX_WMark ≤ FIFO_DEPTH – 2 Recommendation: A request is raised when the value of [(FIFO_DEPTH/2) – 1] is above the threshold.
[15:12]	RW	reserved	Reserved

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[11:0]	RW	tx_wmark	FIFO threshold during data transmission. If the data amount in the FIFO is below the threshold, a DMA request is enabled. After a data transfer, a DMA request is raised for transferring the remaining data no matter whether the threshold is reached. In non-DMA mode, the RXDR interrupt is enabled. If the data amount in the FIFO is not above the threshold after a data transfer, no interrupt is generated. In this case, the software needs to read the remaining data by querying the DTD interrupt. If a data transfer is complete in DMA mode, the DMA raises a single transfer request to read data until the DTD interrupt is generated even though the remaining data amount is below the threshold. Restriction: TX_WMark ≤ FIFO_DEPTH – 2 Recommendation: This field must be less than or equal to FIFO_DEPTH/2.

MMC_CDETECT

MMC_CDETECT is a card detection register.

Offset Address	Register Name	Total Reset Value
0x0050	MMC_CDETECT	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															card_detect_n
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	card_detect_n	Card detection signal The value depends on the SDIO_CARD_DETECT pin.

MMC_WRTPRT

MMC_WRTPRT is a card write protection register.

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Offset Address	Register Name	Total Reset Value
0x0054	MMC_WRTPRT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															write_protect
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	write_protect	Card write protection signal The value depends on the SDIO_CWPR pin.

MMC_TCBCNT

MMC_TCBCNT is a count of bytes transmitted to the card register.

Offset Address	Register Name	Total Reset Value
0x005C	MMC_TCBCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	trans_card_byte_count																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	trans_card_byte_count	Count of bytes transmitted from the CIU to the card When this register is accessed through a 32-bit AHB, the 32-bit data needs to be read at a time. This avoids the read-coherency error.

MMC_TBBCNT

MMC_TBBCNT is a count of bytes transmitted from the BIU FIFO register.

Offset Address	Register Name	Total Reset Value
0x0060	MMC_TBBCNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	trans_fifo_byte_count																															

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	trans_fifo_byte_count	Count of bytes transferred between the CPU/DMA and BIU FIFO When this register is accessed through a 32-bit AHB, the 32-bit data needs to be read at a time. This avoids the read-coherency error.

MMC_DEBNCE

MMC_DEBNCE is a dejitter count register.

Offset Address	Register Name	Total Reset Value
0x0064	MMC_DEBNCE	0x00FF_FFFF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								debounce_count																							
Reset	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:0]	RW	debounce_count	Number of bus clock cycles used by the dejitter filter logic. Generally, the dejitter duration ranges from 5 ms to 25 ms.

MMC_UHS_REG

MMC_UHS_REG is a UHS-1 register.

Offset Address	Register Name	Total Reset Value
0x0074	MMC_UHS_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														ddr_reg		reserved														volt_reg	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RW	reserved	Reserved

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[17:16]	RW	ddr_reg	DDR mode control Bit 17 and bit 16 control the DDR modes of eMMC and SDIO respectively. The following takes bit 16 as an example: 0: non-DDR mode 1: DDR mode
[15:2]	RO	reserved	Reserved
[1:0]	RW	volt_reg	Voltage mode control Bit 1 and bit 0 control the voltage modes of eMMC and SDIO respectively. The following takes bit 0 as an example: 0: 3.3 V 1: 1.8 V

MMC_CARD_RSTN

MMC_CARD_RSTN is an eMMC reset control register.

Offset Address	Register Name	Total Reset Value
0x0078	MMC_CARD_RSTN	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														card_reset	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RW	card_reset	eMMC reset. The control pin is CARD_RESET. 0: reset 1: deassert reset
[0]	RW	reserved	Reserved

MMC_BMOD

MMC_BMOD is a bus mode register.

Offset Address	Register Name	Total Reset Value
0x0080	MMC_BMOD	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

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Name	reserved																				pbl			de	dsl						fb	swr
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10:8]	RW	pbl	Length of the IDMAC burst transfer 000: 1 001: 4 010: 8 011: 16 100: 32 101: 64 110: 128 111: 256
[7]	RW	de	IDMAC enable 0: disabled 1: enabled
[6:2]	RW	dsl	Span between two descriptors, that is, number of words between two non-linked descriptors. This bit is valid only for the dual-buffer descriptor.
[1]	RW	fb	Fixed burst length type 0: single and INCR burst types 1: single, INCR4, INCR8, and INCR16 burst types
[0]	RW	swr	Soft reset control for the internal register of the IDMAC 0: not reset 1: reset After reset, this bit is automatically cleared one clock cycle later.

MMC_PLDMND

MMC_PLDMND is a poll demand register.

Offset Address	Register Name	Total Reset Value
0x0084	MMC_PLDMND	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	pd																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	pd	If DES0[OWN] is 0, the IDMAC enters the suspend state. The

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			CPU can write any value to this register to enable the IDMAC to obtain descriptors again.

MMC_DBADDR

MMC_DBADDR is a base address register of the descriptor linked list.

Offset Address	Register Name	Total Reset Value
0x0088	MMC_DBADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sdl																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sdl	Start address of the descriptor linked list, that is, the base address of the first descriptor

MMC_IDSTS

MMC_IDSTS is an IDMAC status register.

Offset Address	Register Name	Total Reset Value
0x008C	MMC_IDSTS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																fsm				eb				ais	nis	reserved				ces	du	reserved				fbe	ri	ti
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved

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[16:13]	RO	fsm	Current state of the IDMAC state machine 0x0: DMA_IDLE 0x1: DMA_SUSPEND 0x2: DESC_RD 0x3: DESC_CHK 0x4: DMA_RD_REQ_WAIT 0x5: DMA_WR_REQ_WAIT 0x6: DMA_RD 0x7: DMA_WR 0x8: DESC_CLOSE Other values: reserved
[12:10]	RW	eb	Bus error type 001: The transmit operation is aborted. 010: The receive operation is aborted. Other values: reserved
[9]	RW	ais	Abnormal total interrupt. The value of this bit is obtained after the values of FBE, DU, and CES bits are ORed. Writing 1 clears this bit.
[8]	RW	nis	Normal total interrupt. The value of this bit is obtained after the values of the TI and RI bits are ORed. Writing 1 clears this bit.
[7:6]	RO	reserved	Reserved
[5]	RW	ces	Card error indicator. This bit indicates the card status when data is being received.
[4]	RW	du	Descriptor invalid interrupt. When DES0[OWN] is 0, this bit is set to 1. Writing 1 clears this bit.
[3]	RO	reserved	Reserved
[2]	RW	fbe	Fatal bus error interrupt. If this bit is set to 1, the IDMAC stops all accesses through the bus. Writing 1 clears this bit.
[1]	RW	ri	Receive done interrupt. This bit indicates that the data of a descriptor is received. Writing 1 clears this bit.
[0]	RW	ti	Transmit done interrupt. This bit indicates that a descriptor finishes transmitting data. Writing 1 clears this bit.

MMC_IDINTEN

MMC_IDINTEN is an IDMAC interrupt enable register.

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Offset Address	Register Name	Total Reset Value
0x0090	MMC_IDINTEN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						ai	ni	reserved		ces	du	reserved	fbe	ri	ti
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9]	RW	ai	Abnormal interrupt enable 0: disabled 1: The FBE, DU, and CES interrupts are enabled.
[8]	RW	ni	Normal interrupt enable 0: disabled 1: The TI and RI interrupts are enabled.
[7:6]	RO	reserved	Reserved
[5]	RW	ces	Card error interrupt enable 0: disabled 1: enabled
[4]	RW	du	Descriptor invalid interrupt enable 0: disabled 1: enabled
[3]	RO	reserved	Reserved
[2]	RW	fbe	Fatal bus error interrupt enable 0: disabled 1: enabled
[1]	RW	ri	RX interrupt enable 0: disabled 1: enabled
[0]	RW	ti	TX interrupt enable 0: disabled 1: enabled

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MMC_DSCADDR

MMC_DSCADDR is an address register of the current descriptor.

Offset Address	Register Name	Total Reset Value
0x0094	MMC_DSCADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	had																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	had	Descriptor pointer. The value is automatically refreshed during data transfer. The register points to the start address of the descriptor that will be used by the IDMAC.

MMC_BUFADDR

MMC_BUFADDR is an address register of the current data buffer.

Offset Address	Register Name	Total Reset Value
0x0098	MMC_BUFADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	hba																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	hba	Data buffer pointer. The value is automatically refreshed during data transfer. The register points to the start address of the data buffer that is being used by the IDMAC.

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MMC_CARDTHRCTL

MMC_CARDTHRCTL is a threshold control register.

Offset Address	Register Name	Total Reset Value
0x0100	MMC_CARDTHRCTL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				cardrdthreshold												reserved													cardwrthr_en	BsyClrIntEn	cardrdthr_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RW	reserved	Reserved
[27:16]	RW	cardrdthreshold	Read threshold The maximum value is 512.
[15:3]	RW	reserved	Reserved
[2]	RW	cardwrthr_en	Write threshold enable 0: disabled 1: enabled
[1]	RW	BsyClrIntEn	Busy clear interrupt enable 0: disabled 1: enabled
[0]	RW	cardrdthr_en	Read threshold enable 0: disabled 1: enabled

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MMC_UHS_REG_EXT

MMC_UHS_REG_EXT is a UHS extension register.

Offset Address	Register Name	Total Reset Value
0x0108	MMC_UHS_REG_EXT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved						clk_drv_phase_ctrl		reserved				clk_smpl_phase_ctrl				reserved															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

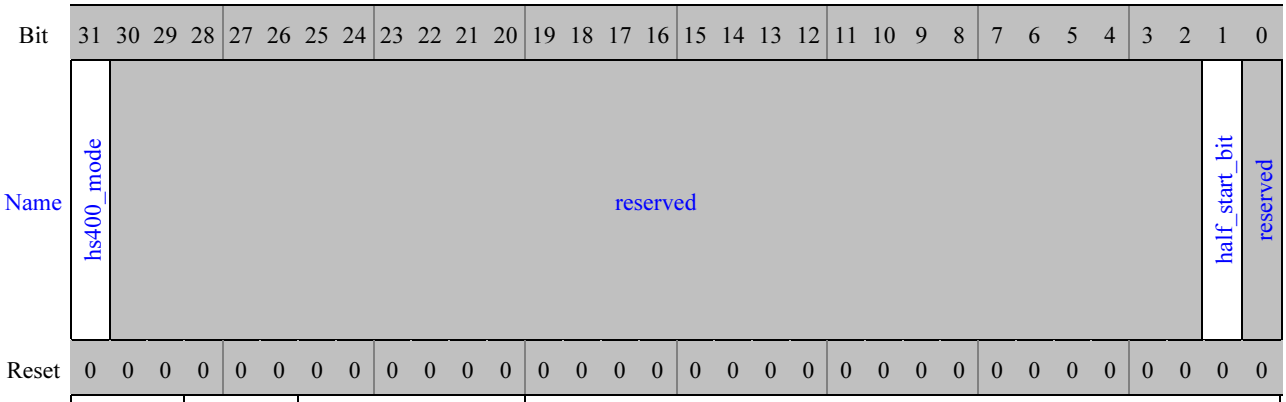
Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:23]	RW	clk_drv_phase_ctrl	clk_in_drv clock phase control 000: 0° 001: 45° 010: 90° 011: 135° 100: 180° 101: 225° 110: 270° 111: 315°
[22:19]	RO	reserved	Reserved
[18:16]	RW	clk_smpl_phase_ctrl	clk_in_sample clock phase control 000: 0° 001: 45° 010: 90° 011: 135° 100: 180° 101: 225° 110: 270° 111: 315°
[15:0]	RO	reserved	Reserved

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MMC_DDR_REG

MMC_DDR_REG is an eMMC 4.5 DDR start bit detection control register.

Offset Address	Register Name	Total Reset Value
0x010C	MMC_DDR_REG	0x0000_0000



Bits	Access	Name	Description
[31]	RW	hs400_mode	HS400 mode control 0: non-HS400 mode 1: HS400 mode
[30:2]	RO	reserved	Reserved
[1]	RW	half_start_bit	For the eMMC4.5 component, when this bit is set to 1'b1, RINTSTS[15] is set to 1 when CMD12 is received in DDR mode.
[0]	RO	reserved	Reserved

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MMC_ENABLE_SHIFT

MMC_ENABLE_SHIFT is a phase shift register.

Offset Address	Register Name	Total Reset Value
0x0110	MMC_ENABLE_SHIFT	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2	1 0
Name	reserved							Enable Phase Shift Register_1	Enable Phase Shift Register_0
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0	0 0

Bits	Access	Name	Description
[31:4]	RW	reserved	Reserved
[3:2]	RW	Enable Phase Shift Register_1	eMMC phase shift control 00: default phase shift 01: Phase shift on the next rising edge 10: Phase shift on the next falling edge 11: Reserved
[1:0]	RW	Enable Phase Shift Register_0	SDIO phase shift control 00: default phase shift 01: Phase shift on the next rising edge 10: Phase shift on the next falling edge 11: Reserved

MMC_DATA

MMC_DATA is a data register (entrance address of the FIFO). The FIFO overflow must be read first before the data is read from or written to the FIFO.

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Offset Address	Register Name	Total Reset Value
0x0200	MMC_DATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	data	Address for reading/writing to the FIFO. If the address ranges from 0x200 to 0x200 + FIFO_DEPTH, the FIFO is selected.

12.5 IR Interface

12.5.1 Overview

The infrared (IR) module receives data over the IR interface.

12.5.2 Features

The IR module has the following features:

- Allows you to disable the IR receive module by using software.
- Supports two operating modes:
  - Mode 0: supports decoding in four formats (including NEC with simple repeat code, NEC with full repeat code, SONY, and TC9012), error detection on received data, and IR wakeup.
  - Mode 1: supports the symbol level width detection in any data format.
- In mode 0, supports the RX data overflow interrupt, RX data frame format error interrupt, RX data frame interrupt, key release interrupt, and combined interrupt.
- In mode 1, supports the RX symbol overflow interrupt, RX symbol interrupt, symbol timeout interrupt, and combined interrupt.
- Supports the query of the raw interrupt status and the masked interrupt status.
- Supports interrupt clear and mask (write to clear).
- Supports IR wakeup.
- Supports the reference clock frequency ranging from 1 MHz to 128 MHz and controls the clock frequency divider by software programming, enabling the frequency of the working clock to be prescaled to 1 MHz.

12.5.3 Function Description

The IR module receives infrared signals transmitted by the infrared remote control, decodes the signals, and then transmits the decoded signals to the ARM system. The ARM system performs corresponding operations according to the received codes, which implements expected functions. The IR module connects to the APB of the ARM subsystem. When the chip is in the low-power state (the CPU is at a low frequency), the IR module generates an

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interrupt after receiving a complete frame, and transmits the interrupt to the CPU. In this way, the IR wake function is implemented.

The analysis of the signals transmitted by various infrared remote controls shows that the lead codes in the infrared commands vary according to remote controls. In addition, the subsequent control commands and the bits of command codes are also different. This is because infrared remote controls are not designed based on a unified infrared remote control standard. The basic encoding principles, however, are the same. That is, the pulses with different periods and duty ratios are used to represent 0 and 1. The duty ratios and pulse cycles may vary according to remote controls. Based on preceding differences, the code formats of the infrared data are classified into NEC with simple repeat code, NEC with full repeat code, TC9012 code, and SONY code.

Table 12-16 to Table 12-18 describe the code formats of the received infrared data.

NOTE

The values in Table 12-16 to Table 12-18 are tested before and may not be accurate. Therefore, they are only for reference.

Table 12-16 Code formats of the received infrared data (NEC with simple repeat code)

Data Format		NEC with Simple Repeat Code			
		uPD6121G	D6121/BU5777/D1913	LC7461M-C13	AEHA
Lead code (10 µs)	LEAD_S	900	900	900	337.6
	LEAD_E	450	450	450	168.8
Bit0 (10 µs)	B0_L	56	56	56	42.2
	B0_H	56	56	56	42.2
Bit1 (10 µs)	B1_L	56	56	56	42.2
	B1_H	169	169	169	126.6
Simple repeat code (10 µs)	SLEAD_S	900	900	900	337.6
	SLEAD_E	225	225	225	337.6
Burst (10 µs)		55	55	55	42.2
Frame length (10 µs)		108,00	10,800	10,800	8,777.6 – 12,828.8
Valid data bit		32	32	42	48

Table 12-17 Code formats of the received infrared data (NEC with full repeat code)

Data Format		NEC with Full Repeat Code						
		uPD6121G	LC7461M-C13	MN6024-C5D6	MN6014-C6D6	MATNEW	MN6030	PANASONIC
Lead code	LEAD_S	900	900	337.6	349.2	348.8	349	352

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Data Format		NEC with Full Repeat Code						
		uPD6121G	LC7461M-C13	MN6024-C5D6	MN6014-C6D6	MATNEW	MN6030	PANASONIC
(10 µs)	LEAD_E	450	450	337.6	349.2	374.4	349	352
Bit 0 (10 µs)	B0_L	56	56	84.4	87.3	43.6	87.3	88
	B0_H	56	56	84.4	87.3	43.6	87.3	88
Bit 1 (10 µs)	B1_L	56	56	84.4	87.3	43.6	87.3	88
	B1_H	169	169	253.2	174.6	130.8	261.9	264
Simple repeat code (10 µs)	SLEAD_S	None	None	None	None	None	None	None
	SLEAD_E							
Burst (10 µs)		55	55	84.4	87.3	43.6	87.3	88
Frame length (10 µs)		10,800	10,800	10,130	10,470	12,413.6 – 16,594.4	10,500	10,400
Valid data bit		32	42	22	24	48	22	22

Table 12-18 Code formats of the received infrared data (TC9012 code and SONY code)

Data Format		TC9012	SONY			
		TC9012F/9243	SONY-D7C5	SONY-D7C6	SONY-D7C8	SONY-D7C13
Lead code (10 µs)	LEAD_S	450	240	240	240	240
	LEAD_E	450	60	60	60	60
Bit0 (10 µs)	B0_L	56	60	60	60	60
	B0_H	56	60	60	60	60
Bit1 (10 µs)	B1_L	56	120	120	120	120
	B1_H	169	60	60	60	60
Simple repeat code (10 µs)	SLEAD_S	None	None	None	None	None
	SLEAD_E					
Burst (10 µs)		56	None	None	None	None
Frame length (10 µs)		10,800	4500	4500	4500	4500
Valid data bit		32	12	13	15	20

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12.5.3.1 NEC with Simple Repeat Code

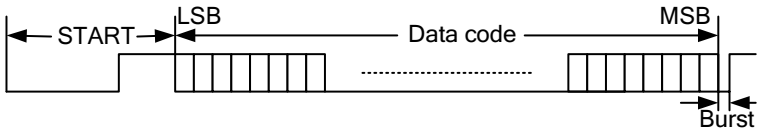
Frame Format

The NEC with simple repeat code consists of the following:

- Start code (lead code): Consists of a start code (low level) and an end code (high level).
- Data code: The valid bits and the definition of each bit depend on specific code format. During data code reception, its LSB is received first.
- Burst signal: It is used to receive the last data bit.

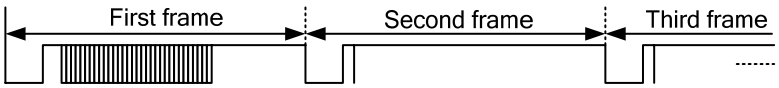
Figure 12-31 shows the frame format of transmitting a single NEC with simple repeat code.

Figure 12-31 Frame format for transmitting a single NEC with simple repeat code



If a complete data frame is received after the key is held down for more than one frame length, the subsequently received data frame consists only of a simple lead code and a burst signal. The lead code also consists of a start code (low level) and an end code (high level). Figure 12-32 shows the frame format for transmitting NEC with simple repeat codes by pressing the key continuously.

Figure 12-32 Frame format for transmitting NEC with simple repeat codes by pressing the key continuously



Code Format

Figure 12-33 shows the definitions of bit0 and bit1 of the NEC with simple repeat code.

Figure 12-33 Definitions of bit0 and bit1 of the NEC with simple repeat code

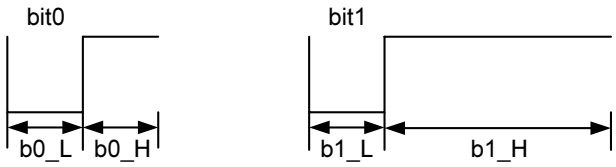


Figure 12-34 shows the format for transmitting a single NEC with simple repeat code. Figure 12-35 shows the format for transmitting consecutive NEC with simple repeat codes.

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Figure 12-34 Format for transmitting a single NEC with simple repeat code

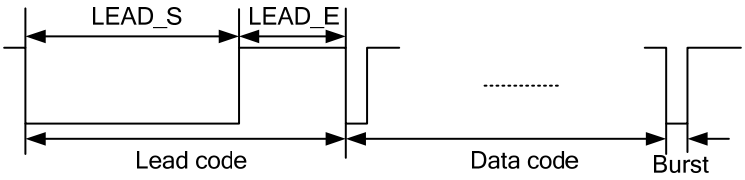
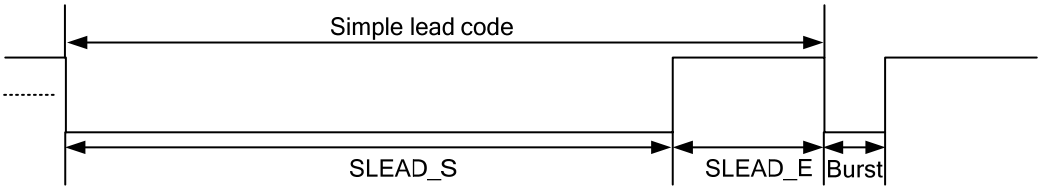


Figure 12-35 Code format for transmitting consecutive NEC with simple repeat codes



NOTE

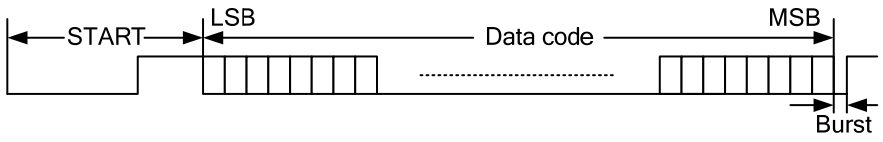
- The pulse width of the high and low levels and the frame length depend on specific code formats. See Table 12-16 to Table 12-18.
- The frame length must be less than or equal to 160 ms. Otherwise, the simple lead code cannot be identified.

12.5.3.2 NEC with Full Repeat Code

Frame Format

The data format of the NEC with full repeat code consists of the following parts: START (lead code), data code, and burst. START (lead code): Consists of a start code (low level) and an end code (high level). Data code: The valid bits and the definition of each bit are determined by the specific code format. During data code reception, its LSB is received first. Burst signal: It is used to receive the last data bit. Figure 12-36 shows the frame format for transmitting a single NEC with full repeat code.

Figure 12-36 Frame format for transmitting a single NEC with full repeat code



If a complete data frame (first frame) is received after the key is held down for more than one frame length, the subsequently received data frame is still a complete data frame. That is, the first frame is transmitted repeatedly based on the frame length. Figure 12-37 shows the frame format for transmitting NEC with full repeat codes by pressing the key continuously.

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Figure 12-37 Frame format for transmitting NEC with full repeat codes by pressing the key continuously

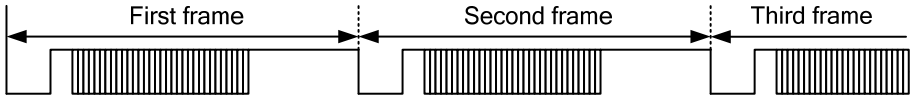


Figure 12-36 and Figure 12-37 show that the only difference between the NEC with simple repeat code and the NEC with full repeat code is the format of the repeat frame. For the NEC with simple repeat code, the simple lead code is transmitted; for the NEC with full repeat code, the complete frame is transmitted. That is, the first frame and the repeat frame are the same.

Code Format

Figure 12-38 shows the definitions of bit0 and bit1 of the NEC with full repeat code.

Figure 12-38 Definitions of bit0 and bit1 of the NEC with full repeat code

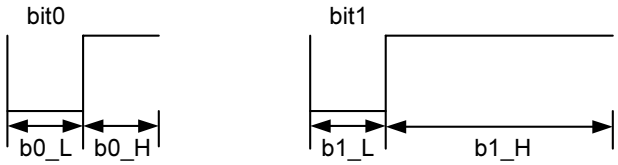
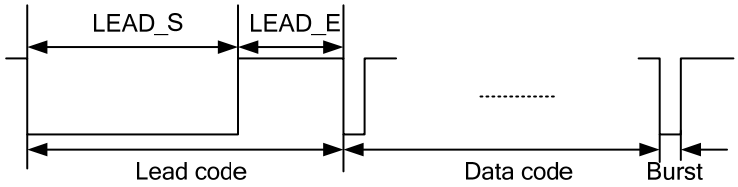


Figure 12-39 shows the format for transmitting a single NEC with full repeat code.

Figure 12-39 Format for transmitting a single NEC with full repeat code



NOTE

The pulse width of the high and low levels and the frame length depend on specific code formats. See Table 12-16 to Table 12-18.

12.5.3.3 TC9012 Code

Frame Format

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CAUTION

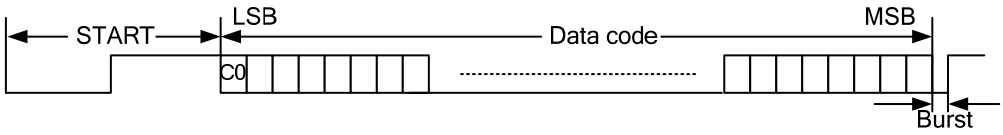
According to the features of the TC9012 code, the first bit of all key codes must be all 1s or all 0s. Otherwise, unnecessary frames are generated when the key are pressed continuously.

The TC9012 code consists of the following parts:

- Start code (lead code): Consists of a start code (low level) and an end code (high level).
- Data code: The valid bits and the definition of each bit depend on the specific code pattern. During data code reception, its LSB is received first.
- Burst signal: It is used to receive the last data bit.

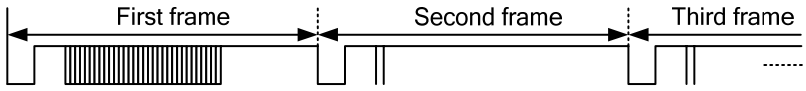
Figure 12-40 shows the frame format for transmitting a single TC9012 code.

Figure 12-40 Frame format for transmitting a single TC9012 code



When a complete data frame is received after the key is held down for more than one frame length, the subsequently received data frame consists of a lead code, a data bit, and a burst signal. The lead code also consists of a start code (low level) and an end code (high level). The data bit is the complement of the first data bit (C0) received in the previous frame. Figure 12-41 shows the frame format for transmitting TC9012 codes by pressing the key continuously.

Figure 12-41 Frame format for transmitting TC9012 codes by pressing the key continuously



Code Format

Figure 12-42 shows the definitions of bit0 and bit1 of the TC9012 code.

Figure 12-42 Definitions of bit0 and bit1 of the TC9012 code

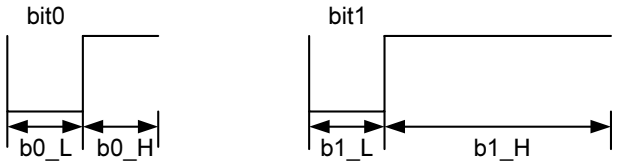


Figure 12-43 shows the format for transmitting a single TC9012 code.

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Figure 12-43 Format for transmitting a single TC9012 code

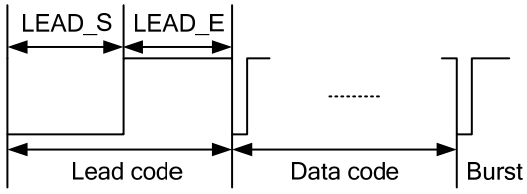


Figure 12-44 shows the format for transmitting consecutive TC9012 codes when C0 is 1.

Figure 12-44 Format for transmitting consecutive TC9012 codes (C0 = 1)

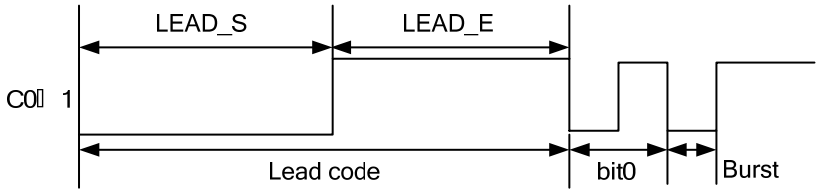
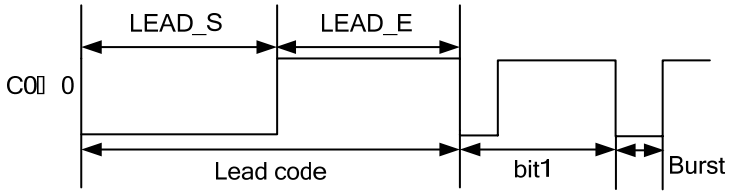


Figure 12-45 shows the format for transmitting consecutive TC9012 codes when C0 is 0.

Figure 12-45 Format for transmitting consecutiveTC9012 codes (C0 = 0)



NOTE

The pulse width of the high level and low level and the frame length depend on specific code patterns. For details, see Table 12-16 to Table 12-18. In addition, the frame length must be less than or equal to 160 ms. Otherwise, the repeat frame cannot be identified.

12.5.3.4 SONY Code

Frame Format

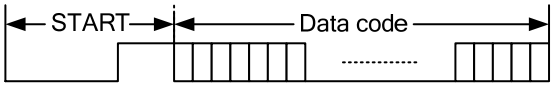
A SONY code consists of the following parts:

- Start code (lead code): Consists of a start code (low level) and an end code (high level).
- Data code: The valid bits and the definition of each bit are determined by the specific code pattern.

During reception, the LSB is received first. Figure 12-46 shows the frame format for transmitting a single SONY code.

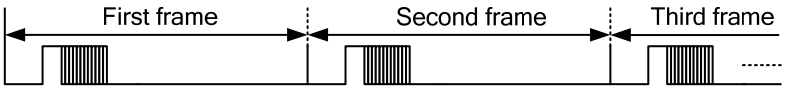
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Figure 12-46 Frame format for transmitting a single SONY code



When a complete data frame is received after the key is pressed for more than one frame length, the subsequently received data frame is also a complete data frame. Figure 12-47 shows the frame format for continuously transmitting SONY codes by pressing the key.

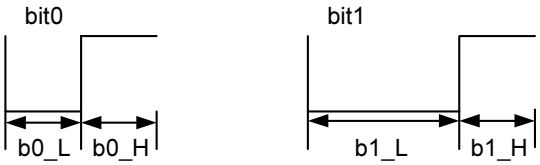
Figure 12-47 Frame format for continuously transmitting SONY codes by pressing the key



Code Format

Figure 12-48 shows the definitions of bit0 and bit1 of a SONY code.

Figure 12-48 Definitions of bit0 and bit1 of a SONY code



NOTE

The pulse width of the high level and low level and the frame length depend on specific code patterns. For details, see Table 12-16 to Table 12-18.

12.5.4 Operating Mode

Soft Reset

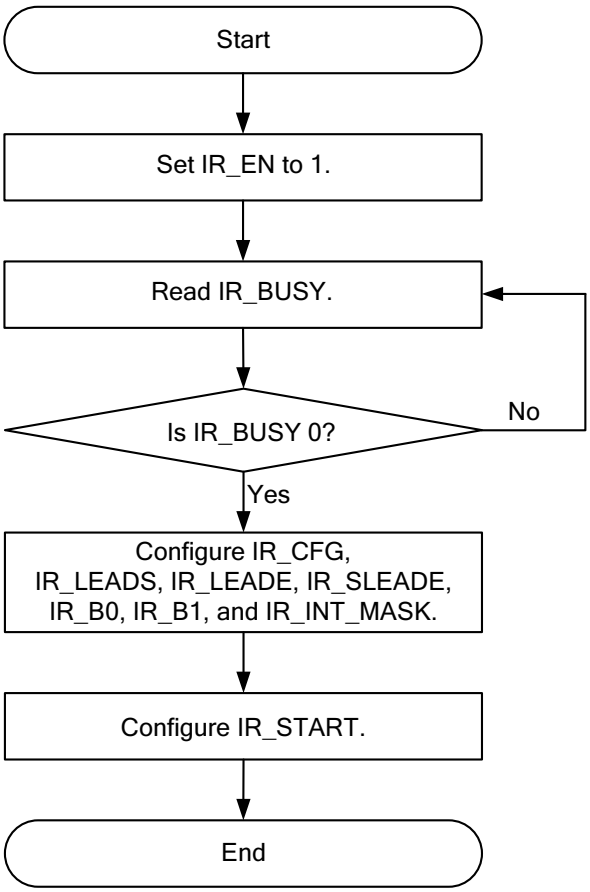
When PERI_CRG57[ir_srst_req] is set to 1, the IR module is soft-reset separately. After reset, each configuration register is restored its default value. Therefore, these registers must be reinitialized.

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Instances of Configuring Registers

Figure 12-49 shows the process of initializing the IR module.

Figure 12-49 Process of initializing the IR module



To initialize the IR module, perform the following steps:

Step 1 Select the address space of the IR module.

Step 2 Set IR_EN bit [0] to 1 to enable the IR receive module.

Step 3 Read IR_BUSY to check the current status of the IR module.

- If the value of IR_BUSY is 1, the IR module is busy. Then continue to read IR_BUSY. Note that configuring other control registers of the IR module by using software has no effect in this case.
- If the value of IR_BUSY is 0, the IR module is idle. Then go to Step 4.

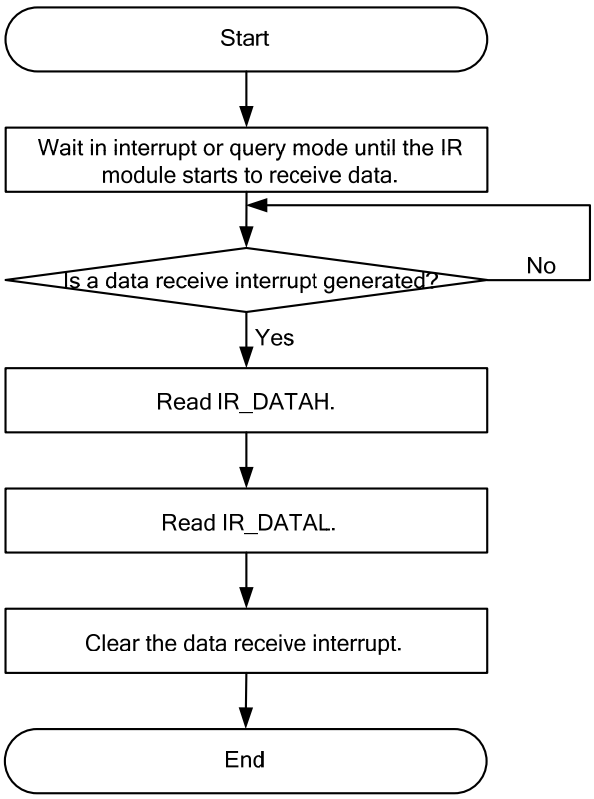
Step 4 Configure IR_CFG, IR_LEADS, IR_LEADE, IR_SLEADE, IR_B0, IR_B1, and IR_INT_MASK. Note: You can update corresponding registers as required. If the registers are not updated, the original values are retained.

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Step 5 Configure IR_STARTafter all IR control registers are configured. This is because IR_START is used to generate the start signal. If IR_START is configured, the IR module starts to receive infrared data based on the values of IR control registers.

----End

Figure 12-50 Process of reading the decoded data



To read the decoded data, perform the following steps:

Step 1 Select the address space of the IR module.

Step 2 Wait in interrupt or query mode until data frames are received.

- In interrupt mode, when the CPU receives an interrupt request signal from the IR module, read the value of IR_INT_STATUS[intms_rcv]. If the value is 1, the IR module receives a data frame. Then, go to Step 3. If the value is 0, repeat Step 2 to wait for an interrupt.
- In query mode, continuously read the value of IR_INT_STATUS[intrs_rcv] by using software or read the value at intervals. If the value is 1, the IR module receives a data frame. Then, go to Step 3. If the value is 0, the IR module does not receive any data frame. Then, repeat Step 2 to continue the query.

Step 3 Read IR_DATAH. If the number of data bits in one frame is less than or equal to 32, skip this step.

Step 4 Read IR_DATAL.

Step 5 Clear the data receive interrupt.

----End

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12.5.5 Register Summary

Table 12-19 describes IR registers.

Table 12-19 Summary of IR registers (base address: 0x2007_0000)

Offset Address	Register	Description	Page
0x000	IR_EN	IR receive enable control register	12-130
0x004	IR_CFG	IR configuration register	12-130
0x008	IR_LEADS	Lead code start bit margin configuration register (valid when IR_CFG[ir_mode] is 0 only)	12-132
0x00C	IR_LEADE	Lead code end bit margin configuration register (valid when IR_CFG[ir_mode] is 0 only)	12-133
0x010	IR_SLEADE	Simple lead code end bit margin configuration register (valid when IR_CFG[ir_mode] is 0 only)	12-134
0x014	IR_B0	Data 0 level judge margin configuration register (used only when IR_CFG[ir_mode] is 0)	12-135
0x018	IR_B1	Data 1 level judge margin configuration register (used only when IR_CFG[ir_mode] is 0)	12-136
0x01C	IR_BUSY	Configuration busy flag register	12-137
0x020	IR_DATAH	Upper 16-bit IR receive decoded data register (IR_CFG[ir_mode] = 0) or symbol count register in the symbol FIFO (IR_CFG[ir_mode] = 1)	12-138
0x024	IR_DATAL	Lower 32-bit IR receive decoded data register (IR_CFG[ir_mode] = 0) or IR received symbol width register (IR_CFG[ir_mode] = 1)	12-139
0x028	IR_INT_MASK	IR interrupt mask register	12-139
0x02C	IR_INT_STATUS	IR interrupt status register	12-141
0x030	IR_INT_CLR	IR interrupt clear register	12-143
0x034	IR_START	IR start configuration register	12-145

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12.5.6 Register Description

IR_EN

IR_EN is an IR receive enable control register.

CAUTION

Before configuring other registers, you must set IR_EN[ir_en] to 1 by using software. When IR_EN[ir_en] is 0, other registers are read-only and the read values are their reset values.

Offset Address	Register Name	Total Reset Value
0x000	IR_EN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															ir_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	ir_en	IR receive module enable 0: disabled 1: enabled

IR_CFG

IR_CFG is an IR configuration register.

CAUTION

Before configuring this register, you must set

IR_BUSY[ir_busy] to 0 and set IR_EN[ir_en] to 1. Otherwise, the original value is retained after configuration.

The reference clock frequency supported by the IR module ranges from 1 MHz to 128 MHz. The relationship between the frequency and the clock frequency divider ir_freq is as follows:

- When the reference clock frequency is 1 MHz, ir_freq must be set to 0x00.
- If the reference clock frequency is 128 MHz, ir_freq must be set to 0x7F.

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When the frequency of the IR reference clock is not an integer ranging from 1 MHz to 128 MHz, the clock frequency divider is rounded off. For example, if the reference clock frequency is 12.1 MHz, the clock frequency divider is 0x0B; if the reference clock frequency is 12.8 MHz, the clock frequency divider is 0x0C.

The relationship between the frequency offset and the count deviation is as follows: If the base frequency is f and the frequency variation is D_f, the frequency offset ratio is D_f/f. If the count deviation of the counter is Dcnt, and the judge level width is s (in μs), the count deviation Dcnt is calculated as follows: Dcnt = |0.1 x s x ratio|Therefore, when the clock has frequency offset, the valid range of the parameter value needs to be changed. If the frequency increases, the corresponding margin range is changed to [min + Dcnt, max + Dcnt]. Where, min and max indicate the margins without frequency offset. If the frequency decreases, the offset range is changed to [min – Dcnt, max – Dcnt]. Take the margin of the start bit in the lead code as an example. If the base frequency is 100 MHz, and the frequency increases by 0.1 MHz, then the ratio is 0.001 (0.1/100). Assume that s is 9000 μs. Dcnt is calculated as follows: Dcnt = |0.1 x 9,000 x 0.001| = 1. In this case, the margin range of ir_leads must be changed to [0x033D, 0x3CD].

Offset Address	Register Name	Total Reset Value
0x004	IR_CFG	0x3E80_1F0B

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ir_max_level_width																ir_format				ir_bits				ir_mode	ir_freq							
Reset	0	0	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0	0	1	0	1	1

Bits	Access	Name	Description
[31:16]	RW	ir_max_level_width	Invalid when IR_CFG[ir_mode] is 0. Indicates the maximum level width (in 10 μs) of a symbol whenIR_CFG[ir_mode] is 1. This width indicates the end of a symbol stream.
[15:14]	RW	ir_format	Indicates the data code format when IR_CFG[ir_mode] is 0. 00: NEC with simple repeat code 01: TC9012 code 10: NEC with full repeat code 11: SONY code For details about the relationship between code types and code formats, see Table 12-16 to Table 12-18. Indicates the symbol format when IR_CFG[ir_mode] is 1. bit[15]: reserved The definitions of bit[14] are as follows: 0: The symbol is from low to high, and the symbol stream ends at the high level. 1: The symbol is from high to low, and the symbol stream ends at the low level.
[13:8]	RW	ir_bits	Indicates the number of data bits in a frame when IR_CFG[ir_mode] is 0. 0x00–0x2F: 1–48 data bits in a frame 0x30–0x3F: reserved

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			If ir_bits is set to a value ranging from 0x30 to 0x3F by using software, the setting has no effect and the original value is retained. Indicates the symbol receive interrupt threshold when IR_CFG[ir_mode] is 1. bit[13]: reserved bit[12:8]: 0x0–0x1F. 0x0 indicates that an interrupt is reported when there is at least one symbol in the FIFO; 0x1F indicates that an interrupt is reported when there are at least 32 symbols in the FIFO, and so on.
[7]	RW	ir_mode	IR operating mode 0: The decoded complete data frames are output. 1: Only the symbol width is output.
[6:0]	RW	ir_freq	Frequency divider of the working clock 0x00–0x7F: correspond to the working clock divider 1–128 respectively.

IR_LEADS

IR_LEADS is a lead code start bit margin configuration register (valid only when IR_CFG[ir_mode] = 0).

CAUTION

Before setting this register, you must set IR_BUSY[ir_busy] to 0 and set Register Description[ir_en] to 1. Otherwise, the original value of the register and the reserved value are retained after setting.

The margin must be considered based on the typical value of the specific code type for accurately judging the start bit of the lead code. For details about the typical values of specified code types, see the values of LEAD_S in Table 12-16 to Table 12-18.

- For a pulse width whose typical value is greater than or equal to 400 (10 μs precision), the recommended margin is 8% of the typical value. Take the D6121 code as an example. If the typical value of LEAD_S is 900, the values of cnt_leads_min and cnt_leads_max are calculated as follows:

  cnt_leads_min = 900 x 92% = 828 = 0x33C cnt_leads_max = 900 x 108% = 972 = 0x3CC
- For a pulse width whose typical value is less than 400 (10 μs precision), the recommended margin is 20% of the typical value. Take the SONY-D7C5 as an example. If the typical value of LEAD_S is 240, the values of cnt_leads_min and cnt_leads_max are calculated as follows:

  cnt_leads_min = 240 x 80% = 192 = 0xC0 cnt_leads_max = 240 x 120% = 288 = 0x120

The basic configuration principle is as follows: cnt_leads_max is greater than or equal to cnt_leads_min, and cnt_leads_min is greater than cnt0_b_max and cnt1_b_max.

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Offset Address	Register Name	Total Reset Value
0x008	IR_LEADS	0x033C_03CC

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved								cnt_leads_min								reserved								cnt_leads_max							
Reset	0	0	0	0	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:16]	RW	cnt_leads_min	Minimum pulse width of the start bit of the lead code 0x000–0x007: reserved
[15:10]	RO	reserved	Reserved
[9:0]	RW	cnt_leads_max	Maximum pulse width of the start bit of the lead code 0x000–0x007: reserved

IR_LEADE

IR_LEADE is a lead code end bit margin configuration register (valid only when IR_CFG[ir_mode] = 0).

CAUTION

- Before setting this register, you must set IR_BUSY[ir_busy] to 0 and set IR_EN[ir_en] to 1. Otherwise, the original value of the register and the reserved value are retained after setting.
- For the NEC with simple repeat code, the margins of cnt_sleade and cnt_leade cannot be overlapped. Otherwise, when the actual count value is within the overlapped range, the simple lead code cannot be identified. As a result, a frame format error occurs.

The margin must be considered based on the typical value of the specific code type for accurately judging the end bit of the lead code. The margin is about 8% of the type value. For details about the typical values of specific code types, see the values of LEAD_E in Table 12-16 to Table 12-18.

- For the pulse width whose typical value is greater than or equal to 400 (10 μs precision), the recommended margin is 8% of the typical value. Take the D6121 code as an example. If the typical value of LEAD_E is 450, the values of cnt_leade_min and cnt_leade_max are calculated as follows:

  cnt_leade_min = 450 x 92% = 414 = 0x19E cnt_leade_max = 450 x 108% = 486 = 0x1E6
- For a pulse width whose typical value is less than 400 (10 μs precision), the recommended margin is 20% of the typical value. Take the SONY-D7C5 code as an example. If the typical value of LEAD_E is 60, the values of cnt_leade_min and cnt_leade_max are calculated as follows:

  cnt_leade_min = 60 x 80% = 48 = 0x030 cnt_leade_max = 60 x 120% = 72 = 0x048

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The basic configuration principle is as follows: cnt_leade_max is greater than or equal to cnt_leade_min.

Offset Address	Register Name	Total Reset Value
0x00C	IR_LEADE	0x019E_01E6

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								cnt_leade_min								reserved								cnt_leade_max							
Reset	0	0	0	0	0	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	cnt_leade_min	Minimum pulse width of the end bit of the lead code 0x000–0x007: reserved
[15:9]	RO	reserved	Reserved
[8:0]	RW	cnt_leade_max	Maximum pulse width of the end bit of the lead code 0x000–0x007: reserved

IR_SLEADE

IR_SLEADE is a simple lead code end bit margin configuration register (IR_CFG[ir_mode] = 0).

CAUTION

- Before setting this register, you must set IR_BUSY[ir_busy] to 0 and set IR_EN[ir_en] to 1. Otherwise, the original value of the register and the reserved value are retained after setting.
- For the NEC with simple repeat code, the margins of cnt_sleade and cnt_leade cannot be overlapped. Otherwise, when the actual count value is within the overlapped range, the simple lead code cannot be identified. As a result, a frame format error occurs.
- This register must be configured only for the NEC with simple repeat code.

The margin must be considered based on the typical value of the specific code type for accurately judging the end bit of the simple lead code. For details about the typical values of specific code types, see the values of SLEAD_E in Table 12-16 to Table 12-18.

- For a pulse width whose typical value is greater than or equal to 225 (10 µs precision), the recommended margin is 8% of the typical value. Take the D6121 code as an example. If the typical value of SLEAD_E is 225, the values of cnt_sleade_min and cnt_sleade_max are calculated as follows:

  cnt_sleade_min = 225 x 92% = 207 = 0xCF cnt_sleade_max = 225 x 108% = 243 = 0xF3
- For a pulse width whose typical value is less than 225 (10 µs precision), the recommended margin is 20% of the typical value. For example, if the typical value of

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SLEAD_E of a code type is 60, the values of cnt_sleade_min and cnt_sleade_max are calculated as follows:

cnt_sleade_min = 60 x 80% = 48 = 0x30 cnt_sleade_max = 60 x 120% = 72 = 0x48

The basic configuration principle is as follows: cnt_sleade_max is greater than or equal to cnt_sleade_min.

Offset Address	Register Name	Total Reset Value
0x010	IR_SLEADE	0x00CF_00F3

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								cnt_sleade_min								reserved								cnt_sleade_max							
Reset	0	0	0	0	0	0	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1	1

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	cnt_sleade_min	Minimum pulse width of the end bit of the simple lead code 0x000–0x007: reserved
[15:9]	RO	reserved	Reserved
[8:0]	RW	cnt_sleade_max	Maximum pulse width of the start bit of the simple lead code 0x000–0x007: reserved

IR_B0

IR_B0 is data 0 level judge margin configuration register (valid only when IR_CFG[ir_mode] = 0).

CAUTION

- Before setting this register, you must set IR_BUSY[ir_busy] to 0 and set IR_EN[ir_en] to 1. Otherwise, the original value of the register and the reserved value are retained after setting.
- For the preceding four code types, the margins for judging the levels of bit0 and bit1 cannot be overlapped. Otherwise, when the actual count value is within the overlapped range, bit1 cannot be identified and is regarded as bit0 by mistake.

A margin must be considered based on the typical value of the specific code type to accurately judging bit0. The margin is about 20% of the typical value.

- For details about the typical values of the NEC with full repeat code, NEC with simple repeat code, and TC9012 code, see the values of B0_H in Table 12-16 to Table 12-18. Take the D6121 code as an example. If the typical value of B0_H is 56 (10 μs precision), the values of cnt0_b_min and cnt0_b_max are calculated as follows:

  cnt0_b_min = 56 x 80% = 45 = 0x2D cnt0_b_max = 56 x 120% = 67 = 0x43

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- For details about the typical value of the SONY code, see the values of B0_L in Table 12-16 to Table 12-18. Take the SONY-D7C5 code as an example. If the typical value of B0_L is 60 (10 μs precision), the values of cnt0_b_min and cnt0_b_max are calculated as follows:

  cnt0_b_min = 60 x 80% = 48 = 0x30 cnt0_b_max = 60 x 120% = 72 = 0x48

The basic configuration principle is as follows: cnt0_b_max is greater than or equal to cnt0_b_min.

Offset Address	Register Name	Total Reset Value
0x014	IR_B0	0x002D_0043

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								cnt0_b_min								reserved								cnt0_b_max							
Reset	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	1

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	cnt0_b_min	Minimum pulse width of the level for judging bit0 0x000–0x007: reserved
[15:9]	RO	reserved	Reserved
[8:0]	RW	cnt0_b_max	Maximum pulse width of the level for judging bit0 0x000–0x007: reserved

IR_B1

IR_B1 is data 1 judge level margin configuration register (valid only when IR_CFG[ir_mode] = 0).

CAUTION

- Before setting this register, you must set IR_BUSY bit[0] to 0 and set IR_EN bit[0] to 1. Otherwise, the original value of the register and the reserved value are retained after setting.
- For the preceding four code types, the margins for judging the levels of bit0 and bit1 cannot be overlapped. Otherwise, when the actual count value is within the overlapped range, bit1 cannot be identified and is regarded as bit0 by mistake.

A margin must be considered based on the typical value of the specific code type for accurately judging bit1. The margin is about 20% of the typical value.

- For details about the typical values of the NEC with simple repeat code, NEC with simple repeat code, and TC9012 code, see the values of B1_H in Table 12-16 to Table 12-18. Take the D6121 code as an example. If the typical value of B1_H is 169 (10 μs precision), the values of cnt1_b_min and cnt1_b_max are calculated as follows:

  cnt1_b_min = 169 x 80% = 135 = 0x87 cnt1_b_max = 169 x 120% = 203 = 0xCB

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- For details about the typical value of the SONY code, see the values of B1_L in Table 12-16 to Table 12-18. Take the SONY-D7C5 code as an example. If the typical value of B1_L is 120 (10 μs precision), the values of cnt1_b_min and cnt1_b_max are calculated as follows:

  cnt1_b_min = 120 x 80% = 96 = 0x60 cnt1_b_max = 120 x 120% = 144 = 0x90

The basic configuration principle is as follows: cnt1_b_max is greater than or equal to cnt1_b_min.

Offset Address	Register Name	Total Reset Value
0x018	IR_B1	0x0087_00CB

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								cnt1_b_min								reserved								cnt1_b_max							
Reset	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	1	1	0	0	1	0	1	1

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:16]	RW	cnt1_b_min	Minimum pulse width of the level for judging bit1 0x000–0x007: reserved
[15:9]	RO	reserved	Reserved
[8:0]	RW	cnt1_b_max	Maximum pulse width of the level for judging bit1 0x000–0x007: reserved

IR_BUSY

IR_BUSY is a configuration busy flag register.

Offset Address	Register Name	Total Reset Value
0x01C	IR_BUSY	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															ir_busy
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	ir_busy	Busy status flag 0: idle state. In this case, software can configure data. 1: busy state. In this state, software cannot configure data.

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IR_DATAH

IR_DATAH is an upper 16-bit IR receive decoded data register (IR_CFG[ir_mode] = 0) or symbol count register in the symbol FIFO (IR_CFG[ir_mode] = 1)

The IR_DATAH register stores the upper 16 bits of the decoded data received by the IR, whereas IR_DATAL stores the lower 32 bits of the decoded data received by the IR. The data bits depend on the valid data bits in a frame and the specific code. For details, see the valid data bits in Table 12-16 to Table 12-18.

Data is stored as follows: The data is stored in IR_DATAH and IR_DATAL in sequence from MSB to LSB. That is, after IR_DATAL is full, the remaining data is stored in IR_DATAH. The unused upper bits are reserved. Data is read as follows: IR_DATAH and IR_DATAL are read in sequence.

The hardware receives all data bits without checking the definition of each data bit. The software is responsible for processing data bits.

Offset Address	Register Name	Total Reset Value
0x020	IR_DATAH	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ir_datah															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	ir_datah	Indicates the upper 16 bits of the decoded data received by the IR module when IR_CFG[ir_mode] is 0. Indicates the symbol count in the symbol FIFO when IR_CFG[ir_mode] is 1. bit[15:6]: reserved bit[5:0]: number of symbols in the symbol FIFO

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IR_DATAL

IR_DATAL is a lower 32-bit IR receive decoded data register (IR_CFG[ir_mode] = 0) or IR receive symbol width register (IR_CFG[ir_mode] = 1).

Offset Address	Register Name	Total Reset Value
0x024	IR_DATAL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ir_datal																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	ir_datal	Indicates the lower 32 bits of the decoded data received by the IR module when IR_CFG[ir_mode] is 0. Indicates the width of the symbol received by the IR module when IR_CFG[ir_mode] is 1. The definitions of bit[31:16] are as follows: Indicates the high-level width (a multiple of 10 μs) of the symbol received by the IR when the symbol level is from low to high. Indicates the low-level width (a multiple of 10 μs) of the symbol received by the IR module when the symbol level is from high to low. The definitions of bit[15:0] are as follows: Indicates the low-level width (a multiple of 10 μs) of the symbol received by the IR module when the symbol level is from low to high. Indicates the high-level width (a multiple of 10 μs) of the symbol received by the IR module when the symbol level is from high to low.

IR_INT_MASK

IR_INT_MASK is an IR interrupt mask register.

CAUTION

- Before setting this register, you must set IR_EN[ir_en] to 1. Otherwise, the original value of the register is retained after setting.
- If all interrupts are masked, the IR wake-up function is unavailable.
- When IR_CFG[ir_mode] is 0, IR_INT_MASK bit[3:0] are valid; when IR_CFG[ir_mode] is 1, IR_INT_MASK bit[18:16] are valid.

The definitions of the interrupts related to the register are as follows:

- RX data overflow interrupt

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If the CPU does not fetch the current frame and the next frame is already received, the next frame overwrites the current frame and a raw RX data overflow error interrupt is reported.

- RX data frame format error interrupt

  If the received data frame is not complete or the data pulse width does not meet the margin requirements, a raw RX frame format error interrupt is reported.

- RX data frame interrupt

  After a complete frame data is received, a raw RX data frame interrupt is reported.

- Key release detection interrupt

  For the NEC with simple repeat code and TC9012 code, if the start synchronous code is not detected again within 160 ms after the previously detected start synchronous code, or a valid data frame rather than a simple lead code is detected, a raw key release detection interrupt is reported. Both the NEC with full repeat code and the SONY code do not support the key release detection interrupt.

- RX symbol overflow interrupt

  If the symbol FIFO is full because the CPU does not fetch the data in time and the subsequent symbol is already received, a raw RX symbol overflow error interrupt is reported.

- RX symbol interrupt

  If a complete symbol is received and the symbol count of the symbol FIFO is above the threshold configured by IR_CFG[ir_bits], a raw RX symbol interrupt is reported.

- Symbol timeout interrupt

  If no new symbol interrupt is received during the period configured by IR_CFG[ir_max_level_width] after a valid symbol is received, a raw symbol timeout interrupt is reported.

The hardware does not identify the interrupt priority. An interrupt is generated if one or more masked interrupt sources are valid.

Offset Address	Register Name	Total Reset Value
0x028	IR_INT_MASK	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved			intm_overrun, intm_time_out, intm_symb_rcv	reserved			intm_release, intm_overflow, intm_frame_error, intm_rcv
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:19]	RO	reserved	Reserved
[18]	RW	intm_overrun	Symbol overflow interrupt mask when IR_CFG[ir_mode] is 1 0: not masked 1: masked

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[17]	RW	intm_time_out	Symbol timeout interrupt mask when IR_CFG[ir_mode] is 1 0: not masked 1: masked
[16]	RW	intm_symb_rcv	RX N symbol interrupt mask when IR_CFG[ir_mode] is 1 0: not masked 1: masked
[15:4]	RO	reserved	Reserved
[3]	RW	intm_release	Key release interrupt mask when IR_CFG[ir_mode] is 0 0: not masked 1: masked
[2]	RW	intm_overflow	RX data overflow interrupt mask when IR_CFG[ir_mode] is 0 0: not masked 1: masked
[1]	RW	intm_frame_error	RX data frame format error interrupt mask when IR_CFG[ir_mode] is 0 0: not masked 1: masked
[0]	RW	intm_rcv	RX data frame interrupt mask when IR_CFG[ir_mode] is 0 0: not masked 1: masked

IR_INT_STATUS

IR_INT_STATUS is an IR interrupt status register.

CAUTION

- When IR_CFG[ir_mode] is 0, IR_INT_STATUS bit[3:0] and IR_INT_STATUS bit[19:16] are valid.
- When IR_CFG[ir_mode] is 1, IR_INT_STATUS bit[10:8] and IR_INT_STATUS bit[26:24] are valid.

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Offset Address	Register Name	Total Reset Value
0x02C	IR_INT_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved					intms_overrun	intms_time_out	intms_symb_rcv	reserved				intms_release	intms_overflow	intms_frame_error	intms_rcv	reserved				intrs_overrun	intrs_time_out	intrs_symb_rcv		reserved				intrs_release	intrs_overflow	intrs_frame_error	intrs_rcv
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:27]	RO	reserved	Reserved
[26]	RO	intms_overrun	Masked symbol overflow interrupt status when IR_CFG[ir_mode] is 1 0: No interrupt is generated. 1: An interrupt is generated.
[25]	RO	intms_time_out	Masked symbol timeout interrupt status when IR_CFG[ir_mode] is 1 0: No interrupt is generated. 1: An interrupt is generated.
[24]	RO	intms_symb_rcv	Masked RX symbol interrupt status when IR_CFG[ir_mode] is 1 0: No interrupt is generated. 1: An interrupt is generated.
[23:20]	RO	reserved	Reserved
[19]	RO	intms_release	Masked key release interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.
[18]	RO	intms_overflow	Masked RX data overflow error interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.
[17]	RO	intms_frame_error	Masked RX data frame format error interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.
[16]	RO	intms_rcv	Masked RX data frame interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.
[15:11]	RO	reserved	Reserved

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[10]	RO	intrs_overrun	Raw symbol overflow interrupt status when IR_CFG[ir_mode] is 1 0: No interrupt is generated. 1: An interrupt is generated.
[9]	RO	intrs_time_out	Raw symbol timeout interrupt status when IR_CFG[ir_mode] is 1 0: No interrupt is generated. 1: An interrupt is generated.
[8]	RO	intrs_symb_rcv	Raw RX symbol interrupt status when IR_CFG[ir_mode] is 1 0: No interrupt is generated. 1: An interrupt is generated.
[7:4]	RO	reserved	Reserved
[3]	RO	intrs_release	Raw key release interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.
[2]	RO	intrs_overflow	Raw RX data overflow error interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.
[1]	RO	intrs_frame_error	Raw RX data frame format error interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.
[0]	RO	intrs_rcv	Raw RX data frame interrupt status when IR_CFG[ir_mode] is 0 0: No interrupt is generated. 1: An interrupt is generated.

IR_INT_CLR

IR_INT_CLR is an IR interrupt clear register.

CAUTION

- When IR_CFG[ir_mode] is 0, IR_INT_CLR bit[3:0] are valid.
- When IR_CFG[ir_mode] is 1, IR_INT_CLR bit[18:16] are valid.

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Offset Address	Register Name	Total Reset Value
0x030	IR_INT_CLR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved													intc_overrun	intc_time_out	intc_symb_rcv	reserved												intc_release	intc_overflow	intc_framerr	intc_rcv
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:19]	RO	reserved	Reserved
[18]	WC	intc_overrun	When IR_CFG[ir_mode] is 1, this bit indicates whether the symbol overflow interrupt request is cleared. 0: no effect 1: cleared
[17]	WC	intc_time_out	When IR_CFG[ir_mode] is 1, this bit indicates whether the symbol timeout interrupt request is cleared. 0: no effect 1: cleared
[16]	WC	intc_symb_rcv	When IR_CFG[ir_mode] is 1, this bit indicates whether the RX symbol interrupt request is cleared. 0: no effect 1: cleared
[15:4]	RO	reserved	Reserved
[3]	WC	intc_release	When IR_CFG[ir_mode] is 0, this bit indicates whether the key release interrupt request is cleared. 0: no effect 1: cleared
[2]	WC	intc_overflow	When IR_CFG[ir_mode] is 0, this bit indicates whether the RX data overflow error interrupt request is cleared. 0: no effect 1: cleared
[1]	WC	intc_framerr	When IR_CFG[ir_mode] is 0, this bit indicates whether the RX data frame format error interrupt request is cleared. 0: no effect 1: cleared

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[0]	WC	intc_rcv	When IR_CFG[ir_mode] is 0, this bit indicates whether the RX data frame interrupt request is cleared. 0: no effect 1: cleared If an RX data frame interrupt is generated and the software writes 1 to the bit without reading the data in IR_DATAL, the interrupt cannot be cleared.

IR_START

IR_START is an IR start configuration register.

After other registers are configured, IR_START can be started when any value is written to the corresponding address during the startup of the IR module.

Offset Address	Register Name	Total Reset Value
0x034	IR_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															ir_start
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	WO	ir_start	IR start configuration register

12.6 GPIO

12.6.1 Overview

Hi3518E V20X/Hi3516C V200 supports 9 groups of general purpose input/output (GPIO) pins, that is, GPIO0 to GPIO8. Each group of GPIO pins provides eight programmable input/output pins (GPIO8 has only two pins). Each pin can be configured as input or output and these pins are used to generate input signals or output signals for special purposes. As input, each GPIO pin can act as an interrupt source; as output, each GPIO pin can be set to 0 or 1 separately.

The GPIO can generate maskable interrupts based on the level or transition value. The general purpose input output interrupt (GPIOINTR) signal provides an indicator to the interrupt controller, indicating that an interrupt occurs.

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CAUTION

- For details about the number of GPIO pins and multiplexing relationship between GPIO pins and other pins, see chapter 2 "Hardware." For details about control modes, see section 2.3 "Pin Multiplexing Registers."
- For the multiplexed GPIO pins that are output by default, note that the pins that connect to the chip and the components must be input.
- The power of the pins SDIO0_CCLK_OUT, SDIO0_CCMD, and SDIO0_CDATA0–3 (corresponding to GPIO5_1, GPIO5_2, GPIO5_3, GPIO5_4, GPIO5_5, and GPIO5_6) are supplied by power switch in the chip. Therefore, when these pins function as GPIO pins, MISC_CTRL3[powerswitch_sel_mux] needs to be set to 0x1 so that power switch is controlled by the MISC_CTRL3 register. If these pins need to support the 3.3 V voltage, MISC_CTRL3[powerswitch_sel] needs to be set to 0x0. If these pins need to support the 1.8 V voltage, MISC_CTRL3[powerswitch_sel] needs to be set to 0x1 and the power pin DVDD18_SDIO needs to connect to the 1.8 V power.

12.6.2 Features

Each GPIO pin can be configured as input or output.

- As input, each GPIO pin can act as an interrupt source.
- As output, each GPIO can be set to 0 or 1 separately.

12.6.3 Operating Mode

Interface Reset

The GPIO is reset during chip power-on reset or system reset, and GPIO pins are in input state after being reset.

GPIO

Each pin can be configured as input or output. To configure a GPIO pin, perform the following steps:

Step 1 Enable the required GPIO pins by configuring corresponding pins according to the description of multiplexing registers.

Step 2 Configure the GPIO as input or output by using the GPIO_DIR register.

Step 3 When the GPIO pins are in input mode, read the GPIO_DATA register to check the input signal value. When the GPIO pins are in output mode, write the output value to the GPIO_DATA register to control the output level of the GPIO pins.

----End

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CAUTION

When the GPIO pins are in output mode, do not enable the GPIO interrupt. Otherwise, the GPIO interrupt will be generated when output signals meet interrupt generation conditions.

Interrupt Operation

The GPIO interrupt is controlled through seven registers (such as GPIO_IS). These registers enable you to select the interrupt source, interrupt edge (falling edge or rising edge), and interrupt trigger modes (level-sensitive mode or edge-sensitive mode). For details about the corresponding interrupt registers of the GPIO, see section 3.3 "Interrupt Systems."

When multiple interrupts are generated at the same time, these interrupts are combined into an interrupt and then reported. For details about the GPIO interrupt mapping, see section 3.3 "Interrupt Systems."

The GPIO_IS, GPIO_IBE, and GPIO_IEV registers determine the features of the interrupt source and interrupt trigger type.

GPIO_RIS and GPIO_MIS are used to read the raw interrupt status and masked interrupt status, respectively. GPIO_IEV controls the final report status of each interrupt. In addition, GPIO_IC is provided to clear the interrupt status.

Perform the following operations to configure GPIO pins to the interrupt mode:

Step 1 Select the edge-sensitive mode or level-sensitive mode by configuring the GPIO_IS register.

Step 2 Select the falling-/rising-edge-sensitive mode or high-/low-level-sensitive mode by configuring the GPIO_IEV register.

Step 3 If the edge-sensitive mode is selected, select single-edge-sensitive mode or dual-edge-sensitive mode by configuring the GPIO_IBE register.

Step 4 Write 0xFF to the GPIO_IC register to clear the interrupt.

Step 5 Set the GPIO_IE to 1 to enable the interrupt.

----End

CAUTION

Ensure that data in GPIO pins is stable during initialization to prevent pseudo interrupts.

The GPIO interrupts are controlled by seven registers. When one or more GPIO pins generate interrupts, a combined interrupt is output to the interrupt controller. The differences between the edge-sensitive mode and level-sensitive mode are as follows:

- Edge-sensitive mode: Software must clear this interrupt to enable superior interrupts.
- Level-sensitive mode: The external interrupt source must keep this level until the processor identifies this interrupt.

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12.6.4 Register Summary

Table 12-20 lists the base addresses for 9 groups of GPIO registers.

Table 12-20 Base addresses for 9 groups of GPIO registers

Register	Base Address
GPIO8	0x201C_0000
GPIO7	0x201B_0000
GPIO6	0x201A_0000
GPIO5	0x2019_0000
GPIO4	0x2018_0000
GPIO3	0x2017_0000
GPIO2	0x2016_0000
GPIO1	0x2015_0000
GPIO0	0x2014_0000

Table 12-21 describes the offset addresses and definitions of a group of internal GPIO registers. GPIO0 to GPIO8 also have the same internal GPIO registers.

NOTE

- Register address of GPIOn = GPIOn base address + Offset address of the register
- The value of n ranges from 0 to 8.

Table 12-21 Summary of GPIO registers

Offset Address	Register	Description	Page
0x000–0x3FC	GPIO_DATA	GPIO data register	12-149
0x400	GPIO_DIR	GPIO direction control register	12-149
0x404	GPIO_IS	GPIO interrupt trigger register	12-150
0x408	GPIO_IBE	GPIO interrupt dual-edge trigger register	12-150
0x40C	GPIO_IEV	GPIO interrupt trigger event register	12-151
0x410	GPIO_IE	GPIO interrupt mask register	12-151
0x414	GPIO_RIS	GPIO raw interrupt status register	12-152
0x418	GPIO_MIS	GPIO masked interrupt status register	12-152
0x41C	GPIO_IC	GPIO interrupt clear register	12-153

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12.6.5 Register Description

GPIO_DATA

GPIO_DATA is a GPIO data register. It is used to buffer the input or output data.

When the corresponding bit of the GPIO_DIR is configured as output, the values written to the GPIO_DATA register are sent to the corresponding pin (note that the pin multiplexing configuration must be correct). If the bit is configured as input, the value of the corresponding input pin is read.

CAUTION

If the corresponding bit of GPIO_DIR is configured as input, the pin value is returned after a valid read; if the corresponding bit is configured as output, the written value is returned after a valid read.

Through PADDR[9:2], the GPIO_DATA register masks the read and write operations on the register. The register corresponds to 256 address spaces. PADDR[9:2] corresponds to GPIO_DATA[7:0]. When the corresponding bit is high, it can be read or written. When the corresponding bit is low, no operations are supported. For example:

- If the address is 0x3FC (0b11_1111_1100), the operations on all the eight bits of GPIO_DATA bit[7:0] are valid.
- If the address is 0x200 (0b10_0000_0000), only the operation on GPIO_DATA bit[7] is valid.

Offset Address	Register Name	Total Reset Value
0x000–0x3FC	GPIO_DATA	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_data							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	gpio_data	Indicates the GPIO input data when the GPIO is configured as input; indicates the GPIO output data when the GPIO is configured as output. Each bit can be controlled separately. The register is used together with GPIO_DIR.

GPIO_DIR

GPIO_DIR is a GPIO direction control register. It is used to configure the direction of each GPIO pin.

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Offset Address	Register Name	Total Reset Value
0x400	GPIO_DIR	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_dir							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	gpio_dir	GPIO direction control register. Bit[7:0] correspond to GPIO_DATA [7:0] respectively. Each bit can be controlled separately. 0: input 1: output

GPIO_IS

GPIO_IS is a GPIO interrupt trigger register. It is used to configure the interrupt trigger mode.

Offset Address	Register Name	Total Reset Value
0x404	GPIO_IS	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_is							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	gpio_is	GPIO interrupt trigger control register. Bit[7:0] correspond to GPIO_DATA [7:0]. Each bit is controlled separately. 0: edge-sensitive mode 1: level-sensitive mode

GPIO_IBE

GPIO_IBE is a GPIO interrupt dual-edge trigger register. It is used to configure the edge trigger mode of each GPIO pin.

Offset Address	Register Name	Total Reset Value
0x408	GPIO_IBE	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_ibe							

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Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	gpio_ibe	GPIO interrupt edge control register. Bit[7:0] correspond to GPIO_DATA [7:0] respectively. Each bit is controlled independently. 0: single-edge-sensitive mode. The GPIO_IEV register controls whether the interrupt is rising-edge-sensitive or falling-edge-sensitive. 1: dual-edge-sensitive mode

GPIO_IEV

GPIO_IEV is a GPIO interrupt event register. It is used to configure the interrupt trigger event of each GPIO pin.

Offset Address	Register Name	Total Reset Value
0x40C	GPIO_IEV	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_iev							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	gpio_iev	GPIO interrupt trigger event register. Bit[7:0] correspond to GPIO_DATA [7:0]. Each bit is controlled separately. 0: falling-edge-sensitive mode or low-level-sensitive mode 1: rising-edge-sensitive mode or high-level-sensitive mode.

GPIO_IE

GPIO_IE is a GPIO interrupt mask register. It is used to mask GPIO interrupts.

Offset Address	Register Name	Total Reset Value
0x410	GPIO_IE	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_ie							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RW	gpio_ie	GPIO interrupt mask register. Bit[7:0] correspond to GPIO_DATA [7:0]. Each bit is controlled separately. 0: masked

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			1: not masked

GPIO_RIS

GPIO_RIS is a GPIO raw interrupt status register. It is used to query the raw interrupt status of each GPIO pin.

Offset Address	Register Name	Total Reset Value
0x414	GPIO_RIS	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_ris							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RO	gpio_ris	GPIO raw interrupt status register. Bit[7:0] correspond to GPIO_DATA [7:0], indicating the unmasked interrupt status. The status cannot be masked and controlled by the GPIO_IE register. 0: No interrupt occurs. 1: An interrupt is generated.

GPIO_MIS

GPIO_MIS is a GPIO masked interrupt status register. It is used to query the masked interrupt status of each GPIO pin.

Offset Address	Register Name	Total Reset Value
0x418	GPIO_MIS	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_mis							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	RO	gpio_mis	GPIO masked interrupt status register. Bit[7:0] correspond to GPIO_DATA [7:0], indicating the masked interrupt status. The status is controlled by the GPIO_IE register. 0: The interrupt is invalid. 1: The interrupt is valid.

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GPIO_IC

GPIO_IC is a GPIO interrupt clear register. It is used to clear the interrupts generated by GPIO pins and clear the GPIO_RIS and GPIO_MIS registers.

Offset Address	Register Name	Total Reset Value
0x41C	GPIO_IC	0x00

Bit	7	6	5	4	3	2	1	0
Name	gpio_ic							
Reset	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[7:0]	WC	gpio_ic	GPIO interrupt clear register. Bit[7:0] correspond to GPIO_DATA [7:0]. Each bit is controlled separately. 0: no effect 1: cleared

12.7 USB 2.0 Host

12.7.1 Overview

The USB 2.0 host controller supports the high-speed (480 Mbit/s), full-speed (12 Mbit/s), and low-speed (1.5 Mbit/s) data transfer modes. It complies with the USB 2.0, open host controller interface (OHCI) V1.0a, and enhanced host controller interface (EHCI) V1.0 protocols. The USB 2.0 host controller has a root hub. As a part of the USB system, the root hub is used to extend the USB port. By using in the controller, the following functions are achieved:

- Controls and processes data transfer.
- Parses data packets and packages data.
- Encodes and decodes the signals transmitted through the USB port.
- Provide interfaces (such as the interrupt vector interface) for the driver.

The USB 2.0 device controller supports high-speed (480 Mbit/s) and full-speed (12 Mbit/s) data transfer modes. It also supports host/device intelligent switching.

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12.7.2 Function Description

Logic Block Diagram

Figure 12-51 shows the logic block diagram of the USB 2.0 host controller.

Figure 12-51 Logic block diagram of the USB 2.0 host controller

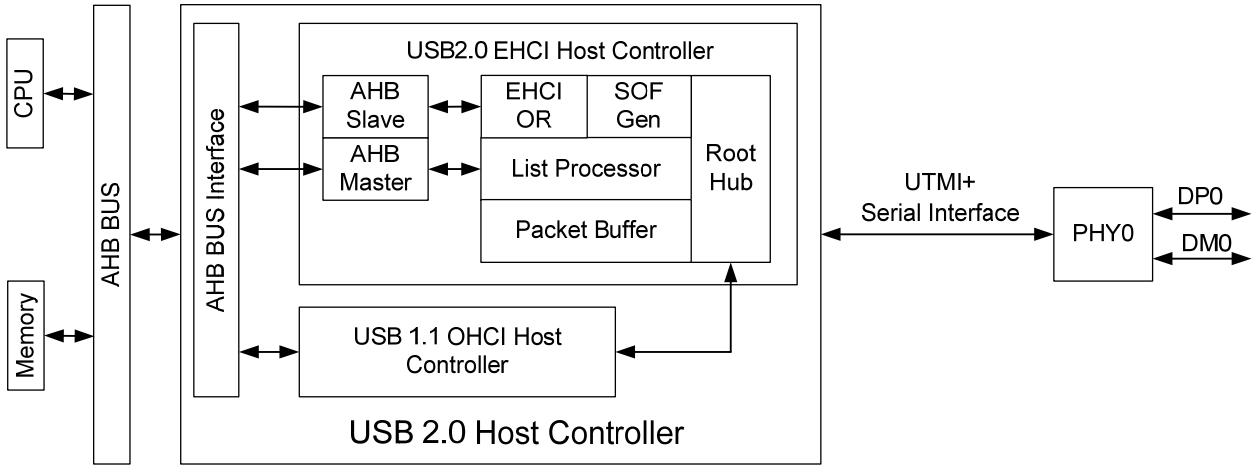
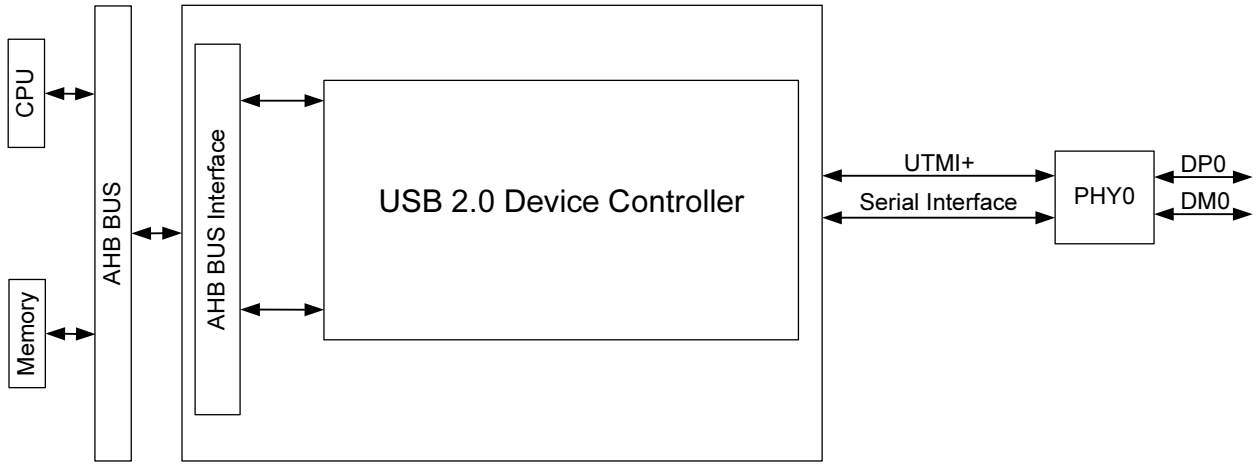


Figure 12-52 shows the logic block diagram of the USB 2.0 device.

Figure 12-52 Logic block diagram of the USB 2.0 device controller



NOTE

- UTMI: USB2.0 transceiver macrocell interface
- EHCI: enhanced host controller interface
- OHCI: open host controller interface

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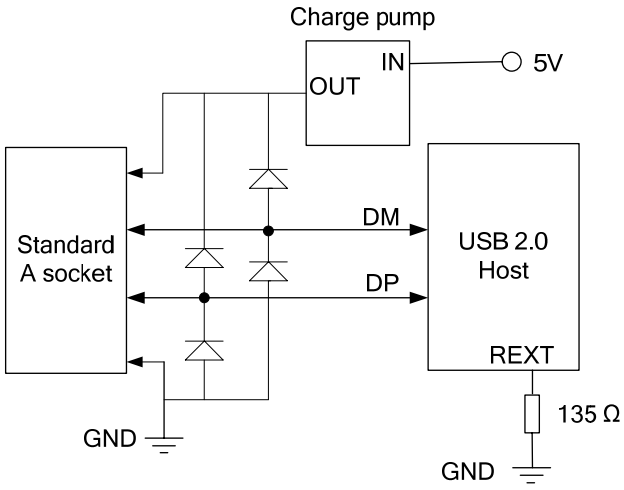
Typical Application

Figure 12-53 shows the reference design of the USB 2.0 host controller.

CAUTION

- The impedance of DP or DM is 45 Ω ±1% and no extra matched resistors are required.
- The REXT resistor is a ±1% resistor.
- If high-speed electro static discharge (ESD) components are used, 1 pF capacitors are recommended.

Figure 12-53 Reference design of the USB 2.0 host controller



Functions

The USB 2.0 host controller has the following features:

- Fully compatible with USB 2.0.
- Complies with OHCI V1.0a and EHCI V1.0 protocols.
- Supports high-speed, full-speed, and low-speed devices.
- Supports low-power solutions.
- Supports four basic data transfer modes including control transfer, bulk transfer, isochronous transfer, and interrupt transfer.
- Supports a maximum of 127 devices by using USB hubs

The USB 2.0 device controller has the following features:

- Complies with the USB 2.0 standard.
- Supports high-speed, full-speed, and low-speed devices.
- Supports four basic data transfer modes including control transfer, bulk transfer, isochronous transfer, and interrupt transfer.

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Working Principle

The USB 2.0 host controller supports the following four standard transfer modes:

- Control transfer

  This mode applies to the data transfer between endpoints 0 of USB host and the USB device. For the USB devices of specific models, other endpoints may be used. The control transfer is bidirectional and the transferred data amount is small. Depending on the device and transfer speed, 8-byte data, 16-byte data, 32-byte data, or 64-byte data can be transferred.

- Bulk transfer

  This mode applies to the data transfer in bulk when there is no limit on the bandwidth and time interval. This mode is the best choice when the transfer speed is very low and many data transfers are delayed. Bulk transfer can be performed after all other types of data transfers are complete. In bulk transfer mode, data is transferred between the USB Host and USB Device without errors by using an error detection and retransmission mechanism.

- Isochronous transfer

  This mode applies to the stream data transfer with strict time requirements and strong error tolerance or the instant data transfer at a constant transfer rate. This mode provides a fixed bandwidth and time interval.

- Interrupt transfer

  This mode applies to the data transfer when the data is scattered, unpredictable, and is of small volume. In this mode, the device is checked whether there is interrupt data to be sent at a fixed interval. The query frequency ranges from 1 ms to 255 ms and it depends on the device endpoint mode. The typical interrupt transfer mode is unidirectional and only input is available for the USB host.

12.7.3 Operating Mode

Host/Device Switching

You can configure the MISC_CTRL30[usb_chipid] to operate in host or device mode.

To enable the register to operate in host mode, set MISC_CTRL30[usb_chipid] to 0 and MISC_CTRL30[usb2_phy_dmpulldown] and MISC_CTRL30[usb2_phy_dppulldown] to 1.

To enable the register to operate in device mode, set MISC_CTRL30[usb_chipid] to 1 and MISC_CTRL30[usb2_phy_dmpulldown] and MISC_CTRL30[usb2_phy_dppulldown] to 0.

Pin Polarity Control

The valid polarity of USB_PWREN can be configured by setting MISC_CTRL30[usbpwr_p_ctrl].

The valid polarity of USB0_OVRCUR and USB1_OVRCUR can be configured by setting MISC_CTRL30[usbovr_p_ctrl].

Clock Gating

If the USB 2.0 host controller is not used, its clock can be disabled to reduce power consumption.

To disable the clock, perform the following steps:

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Step 1 Write 1 to PERI_CRG46 [usb_phy_port0_treq], PERI_CRG46[usb_phy_req], and PERI_CRG46[usb_ctrl_utmi0_req] respectively to reset the USB controller and PHY.

Step 2 Set PERI_CRG46 [usb_cken] to 0 to disable the clocks of the USB 2.0 host controller.

----End

To enable the clock, perform the following steps:

Step 1 Set PERI_CRG46 [usb_cken] to 1 to enable the clocks of the USB 2.0 host controller.

Step 2 Deassert the reset on the USB controller and PHY. For details, see "Reset Deassert"

----End

Reset Deassert

By default, the USB controller and PHY are reset after power-on. To deassert reset, perform the following steps:

Step 1 Delay at least 10 μs.

Step 2 Write 0 to PERI_CRG46[usb_phy_port0_treq] to deassert the soft reset on USB PHY port0.

Step 3 Write 0 to PERI_CRG46[usb_phy_req] and PERI_CRG46 [usb_otg_phy_srst_req] to deassert the reset on the USB PHY.

Step 4 After 250 μs delay, write 0 to PERI_CRG46[usb_ctrl_utmi0_req] to deassert the soft reset on USB controller port0.

Step 5 Write 0 to PERI_CRG46[usb_ahb_srst_req] to deassert the soft reset on the USB bus.

----End

Separately Resetting Ports During the Working Process

To reset port separately during the working process, perform the following steps:

Step 1 Write 1 to PERI_CRG46[usb_ctrl_utmi0_req] to soft-reset port0 of the USB controller.

Step 2 Write 1 to PERI_CRG46[usb_phy_port0_treq] and PERI_CRG46 [usb_otg_phy_srst_req] to soft-reset port0 of the USB PHY.

Step 3 After 200 μs delay, write 0 to PERI_CRG46[usb_phy_port0_treq] to deassert the reset on port0 of the USB PHY.

Step 4 Write 0 to PERI_CRG46[usb_ctrl_utmi0_req] to deassert the reset on port0 of the USB controller.

----End

Variables in the Offset Addresses for USB 2.0 Registers

Table 12-22 describes the value ranges and meanings of the variables in the offset addresses of registers.

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Table 12-22 Variables in the offset addresses for USB 2.0 registers

Variable	Value Range	Description
FIFO_num	0–14	Number of FIFOs
n	0–15	Endpoint *n*

12.7.4 Register Summary

USB HOST Register

Table 12-23 describes USB 2.0 registers.

Table 12-23 Summary of USB 2.0 registers (base address: 0x100B_0000)

Offset Address	Register	Description	Page
0x90	INTNREG00	Micro-frame length configuration register	12-161
0x94	INSNREG01	PBUF OUT/IN register	12-162
0x98	INSNREG02	PBUF depth register	12-162
0x9C	INSNREG03	Interrupt memory transfer enable register	12-162
0xA0	INTNREG04	Debug register	12-163
0xA4	INTNREG05	UTMI Vendor Control and status register	12-165
0xA8	INTNREG06	AHB error status register	12-165
0xAC	INTNREG07	AHB error address register	12-166

NOTE

The base address of the EHCI register is 0x100B_0000 and the base address of the OHCI register is 0x100A_0000. The base address of Table 12-23 is that of the EHCI register.

USB Device Register Overview

Table 12-24 describes the UBS device register overview.

Table 12-24 USB device register overview (base address: 0x1008_0000)

Offset Address	Name	Description	Page
0x0000	GOTGCTL	Device behavior control and status query register	12-166
0x0004	GOTGINT	Device interrupt generation indicator/clear register	12-169
0x0008	GAHBCFG	AHB configuration register	12-170

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Offset Address	Name	Description	Page
0x000C	GUSBCFG	USB configuration register	12-172
0x0010	GRSTCTL	Reset hardware feature register	12-175
0x0014	GINTSTS	System interrupt register	12-178
0x0018	GINTMSK	System interrupt mask register	12-181
0x001C	GRXSTSR	RX status debug read register	12-184
0x0020	GRXSTSP	RX status read&pop register	12-185
0x0024	GRXFSIZ	RX FIFO size configuration register	12-186
0x0028	GNPTXFSIZ	Non-periodic TX FIFO size configuration register	12-186
0x002C	GNPTXSTS	Non-periodic TX FIFO and non-periodic TX request query register	12-187
0x0030	GI2CCTL	I^2C access register	12-188
0x0034	GPVNDCTL	PHY vendor control register	12-189
0x0038	GGPIO	GPIO register	12-190
0x003C	GUID	User ID query register	12-190
0x0040	GSNPSID	Synopsys ID query register	12-191
0x0044	GHWCFG1	User hardware configuration register 1	12-191
0x0048	GHWCFG2	User hardware configuration register 2	12-192
0x004C	GHWCFG3	User hardware configuration register 3	12-194
0x0050	GHWCFG4	User hardware configuration register 4	12-195
0x0054	GLPMCFG	LPM configuration register	12-197
0x0058	GPWRDN	Power-down register	12-200
0x005C	GDFIFOCFG	DFIFO software configuration register	12-203
0x0060	GADPCTL	ADP timer control and status register	12-203
0x0100	HPTXFSIZ	TX FIFO configuration register	12-206
0x0104 + 0x0004 x FIFO_num	DPTXFSIZN	Device periodic TX FIFO-n size register	12-206
0x0104 + 0x0004 x FIFO_num	DIEPTXFN	Device IN endpoint TX FIFO size register	12-207
0x0800	DCFG	Device configuration register	12-207
0x0804	DCTL	Device control register	12-209
0x0808	DSTS	Device status register	12-210

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Offset Address	Name	Description	Page
0x0810	DIEPMSK	Common interrupt mask register for device IN endpoints	12-211
0x0814	DOEPMSK	Common interrupt mask register for device OUT endpoints	12-213
0x0818	DAINT	Interrupt register for all device endpoints	12-214
0x081C	DAINTMSK	Interrupt mask register for all endpoints	12-215
0x0820	DTKNQR1	Device IN token learning sequence queue read register 1	12-215
0x0824	DTKNQR2	Device IN token learning sequence queue read register 2	12-216
0x0830	DTKNQR3	Device IN token learning sequence queue read register 3	12-217
0x0834	DTKNQR4	Device IN token learning sequence queue read register 4	12-217
0x0828	DVBUSDIS	Device VBUS discharge time register	12-218
0x082C	DVBUSPULSE	Device VBUS pulsing time register	12-218
0x0830	DTHRCTL	Device threshold control register	12-218
0x0834	DIEPEMPMSK	FIFO empty interrupt mask register for device IN endpoints	12-220
0x0838	DEACHINT	Interrupt register for each device endpoint	12-220
0x083C	DEACHINTMSK	Interrupt register for each device endpoint	12-220
0x0840 + (0x0004 x *n*)	DIEPEACHMSKN	Interrupt register for each device IN endpoint-n	12-221
0x0880 + (0x0004 x *n*)	DOEPEACHMSKN	Interrupt register for each device OUT endpoint-n	12-222
0x0900	DIEPCTL0	Control register for device control IN endpoint 0	12-224
0x0B00	DOEPCTL0	Control register for device control OUT endpoint 0	12-226
0x0900 + (0x0020 x *n*)	DIEPCTLN	Device IN endpoint-n control register	12-228
0x0B00 + (0x0020 x *n*)	DOEPCTLN	Device OUT endpoint-n control register	12-231
0x0908 + (0x0020 x *n*)	DIEPINTn	Device IN endpoint-n interrupt register	12-233
0x0B08 +	DOEPINTn	Device OUT endpoint-n interrupt register	12-236

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Offset Address	Name	Description	Page
(0x0020 x *n*)			
0x0910	DIEPTSIZ0	Transfer size register for device IN endpoint 0	12-238
0x0B10	DOEPTSIZ0	Transfer size register for device OUT endpoint 0	12-239
0x0910 + (0x0020 x *n*)	DIEPTSIZn	Transfer size register for device IN Endpoint-n	12-240
0x0B10 + (0x0020 x *n*)	DOEPTSIZn	Transfer size register for device OUT endpoint-n	12-240
0x0914 + (0x0020 x *n*)	DIEPDMAN	Device IN endpoint-n DMA address register	12-241
0x0B14 + (0x0020 x *n*)	DOEPDMAN	Device OUT endpoint-n DMA address register	12-242
0x091C + (0x0020 x *n*)	DIEPDMABN	Device IN endpoint-n DMA buffer address register	12-242
0x0B1C + (0x0020 x *n*)	DOEPDMABN	Device OUT endpoint-n DMA buffer address register	12-243
0x0938	DTXFSTSn	Device IN endpoint TX FIFO status register	12-243

12.7.5 Register Description

12.7.5.1 USB HOST Register Description

INTNREG00

INTNREG00 is a micro-frame length configuration register.

Offset Address	Register Name	Total Reset Value
0x90	INTNREG00	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				val											en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13:1]	RW	val	Value of the micro-frame counter. This register is used only for emulation. In normal mode, the micro-frame length is 125 µs defined by the protocol. During emulation, you can change the micro-frame length by configuring this register as required to

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			reduce the emulation time.
[0]	RW	en	Micro-frame length register enable 0: disabled 1: enabled

INSNREG01

INSNREG01 is a PBUF out/in configuration register.

Offset Address	Register Name	Total Reset Value
0x94	INSNREG01	0x0020_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	out_threshold																in_threshold															
Reset	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	out_threshold	TX threshold. If the data amount in the PBUF is above the TX threshold, data starts to be transmitted. The depth is measured by 32 bits.
[15:0]	RW	in_threshold	RX threshold. If the data amount in the PBUF is above the RX threshold, data is read from the PBUF. The depth is measured by 32 bits.

INSNREG02

INSNREG02 is a PBUF depth configuration register.

Offset Address	Register Name	Total Reset Value
0x98	INSNREG02	0x0000_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				pbuf_depth											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RW	pbuf_depth	PBUF depth. The depth is measured by 32 bits.

INSNREG03

INSNREG03 is an interrupt memory transfer enable register.

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Offset Address	Register Name	Total Reset Value
0x9C	INSNREG03	0x0000_0001

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																	dis	ctrl		val			fetch	offset							brk_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14]	RW	dis	256 MHz clock check enable 0: disabled 1: enabled
[13]	RW	ctrl	Line state ignore during TESTSE0 NAK 0: disabled 1: enabled
[12:10]	RW	val	TX-TX turnaround delay addition
[9]	RW	fetch	Instruction fetching from the periodic frame list, valid only in CONFIG1 mode 0: The controller fetches instructions from the periodic linked list only in micro-frame 0. 1: The controller fetches instructions from the periodic linked list in every micro-frame.
[8:1]	RW	offset	Available time offset
[0]	RO	brk_en	Interrupt memory transfer enable 0: disabled 1: enabled

INTNREG04

INTNREG04 is a debug register.

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Offset Address	Register Name	Total Reset Value
0xA0	INTNREG04	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																										auto_en	nak_reldfix_en	reserved	scaledwn_enum_time	hccparam_en	hcsparam_en
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:6]	RO	reserved	Reserved
[5]	RW	auto_en	Automatic feature enable 0: enabled. The suspend signal is valid when the run/stop bit is reset by software, but the hchalted bit is not set. 1: disabled. The port is not suspended when software clears the run/stop bit. The default value is 0.
[4]	RW	nak_reldfix_en	NAK reload enable 0: enabled 1: disabled
[3]	RO	reserved	Reserved
[2]	RW	scaledwn_enum_time	Port enumeration time scale down enable 0: disabled 1: enabled
[1]	RW	hccparam_en	HCCPARAMS register write enable 0: disabled 1: enabled
[0]	RW	hcsparam_en	HCSPARAMS register write enable 0: disabled 1: enabled

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INTNREG05

INTNREG05 is a UTMI Vendor control and status register. It is used to read or write to the UTMI PHY.

Offset Address	Register Name	Total Reset Value
0xA4	INTNREG05	0x0000_1000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														vbusy		vport			vcontrol_loadm	reserved				reserved							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17]	RO	vbusy	The value 1 indicates that the hardware is writing data. This bit is cleared only when the process ends.
[16:13]	RW	vport	Port ID. It cannot exceed the supported number of ports.
[12]	RW	vcontrol_loadm	Load enable 0: enabled 1: disabled
[11:8]	RO	reserved	Reserved
[7:0]	RO	reserved	Reserved

INTNREG06

INTNREG06 is an AHB error status register.

Offset Address	Register Name	Total Reset Value
0xA8	INTNREG06	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	err_capture	reserved																			hbusrt_err				num_beat_err				num_beat_ok			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	err_capture	AHB error
[30:12]	RO	reserved	Reserved

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[11:9]	RO	hbusrt_err	hbrust value during a control transfer when an AHB error occurs
[8:4]	RO	num_beat_err	Number of beats during a burst transfer when an AHB error occurs. The maximum number of beats is 16. 0x00–0x10: valid 0x11–0x1F: reserved
[3:0]	RO	num_beat_ok	Number of completed beats during a burst transfer when an AHB error occurs

INTNREG07

INTNREG07 is an AHB error address register.

Offset Address	Register Name	Total Reset Value
0xAC	INTNREG07	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	err_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	err_addr	Address during a control transfer when an AHB error occurs

12.7.5.2 USB Device Register Description

GOTGCTL

GOTGCTL is a device behavior control and status query register.

Offset Address	Register Name	Total Reset Value
0x0000	GOTGCTL	0x04C1_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				chirpen	multvalidbc					reserved	otgver	bsesvld	asesvld	dbnctime	conidsts	reserved				devhnpen	hstsethnpen	hnpreq	hstnegscs	bvalidovval	bvalidoven	avalidovval	avalidoven	vbvalidovval	vbvalidoven	sesreq	sesreqscs
Reset	0	0	0	0	0	1	0	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved

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[27]	RW	chirpen	Chirp on enable 0: disabled 1: enabled **NOTE** This bit is valid only when OTG_BC_SUPPORT is 1. If OTG_BC_SUPPORT is not 1, this bit is reserved.
[26:22]	RO	multvalidbc	BC ACA input Bit[26]: rid_float Bit[25]: rid_gnd Bit[24]: rid_a Bit[23]: rid_b Bit[22]: rid_c
[21]	RO	reserved	Reserved
[20]	RW	otgver	Device version 0: Device version 1.3 1: Device version 2.0
[19]	RO	bsesvld	Transceiver status in device mode 0: B-session invalid 1: B-session valid
[18]	RO	asesvld	Transceiver status in host mode 0: A-session invalid 1: A-session valid
[17]	RO	dbnctime	Dejitter time 0: long dejitter time 1: short dejitter time
[16]	RO	conidsts	USB_ID status 0: The device works in A-device mode. 1: The device works in B-device mode.
[15:12]	RO	reserved	Reserved
[11]	RW	devhnpen	Device HNP enable 0: disabled 1: enabled
[10]	RW	hstsethnpen	Host HNP enable 0: disabled 1: enabled
[9]	RW	hnpreq	HNP request 0: no 1: yes

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[8]	RO	hstnegscs	Host negotiation indicator 0: Host negotiation fails. 1: Host negotiation is successful.
[7]	RW	bvalidovval	Bvalid setting 0: Bvalid = 0 1: Bvalid = 1 When GOTGCTL[bvalidoven] is 1, this bit is valid.
[6]	RW	bvalidoven	Bvalid signal overwrite enable 0: Bvalid can be overwritten. 1: Bvalid cannot be overwritten.
[5]	RW	avalidovval	Avalid setting 0: Avalid = 0 1: Avalid = 1 When GOTGCTL[avalidoven] is 1, this bit is valid.
[4]	RW	avalidoven	Avalid signal overwrite enable 0: Avalid can be overwritten. 1: Avalid cannot be overwritten.
[3]	RW	vbvalidovval	vbusvalid setting 0: vbusvalid = 0 1: vbusvalid = 1 When GOTGCTL[vbvalidoven] is 1, this bit is valid.
[2]	RW	vbvalidoven	vbusvalid signal overwrite enable. 0: vbusvalid can be overwritten. 1: vbusvalid cannot be overwritten.
[1]	RW	sesreq	Session request 0: no 1: yes
[0]	RO	sesreqscs	Session request status 0: failure 1: success

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GOTGINT

GOTGINT is a device interrupt generation indicator/clear register.

Offset Address	Register Name	Total Reset Value
0x0004	GOTGINT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved											multvalipchng	dbncedone	adevtoutchg	hstnegdet	reserved							hstnegsucstschng	sesreqsucstschng	reserved						sesenddet	reserved	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20]	RO	multvalipchng	An interrupt is generated when the value of at least one ACA pin changes. Setting this bit to 1 clears the interrupt. This bit is valid only when OTG_BC_SUPPORT is 1.
[19]	RO	dbncedone	An interrupt is generated when dejitter is successful. Setting this bit to 1 clears the interrupt. This bit is valid only when the HNP Capable or SRP Capable bit is set to 1.
[18]	RO	adevtoutchg	An interrupt is generated when waiting for the connection of device B times out. Setting this bit to 1 clears the interrupt.
[17]	RO	hstnegdet	An interrupt is generated when host negotiation is detected. Setting this bit to 1 clears the interrupt.
[16:10]	RO	reserved	Reserved
[9]	RO	hstnegsucstschng	An interrupt is generated when the host negotiation request fails or is successful. Setting this bit to 1 clears the interrupt.
[8]	RO	sesreqsucstschng	An interrupt is generated when the session request fails or is successful. Setting this bit to 1 clears the interrupt.
[7:3]	RO	reserved	Reserved
[2]	RO	sesenddet	An interrupt is generated when utmiotg_bvalid is deasserted. Setting this bit to 1 clears the interrupt.
[1:0]	RO	reserved	Reserved

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GAHBCFG

GAHBCFG is an AHB configuration register.

Offset Address	Register Name	Total Reset Value
0x0008	GAHBCFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								invdescendianness	ahbsingle	notialldmawrit	remmemsupp	reserved												nptxfemplvl	reserved	dmaen	hbstlen				glblintrmsk
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24]	RW	invdescendianness	Descriptor byte invert 0: The descriptor byte sequence is similar to the AHB master byte sequence. 1: When the AHB master byte sequence is big endian, the descriptor byte sequence is little endian. When the AHB master byte sequence is little endian, the descriptor byte sequence is big endian.
[23]	RW	ahbsingle	Transfer in DMA mode 0: The remaining data is transmitted based on the INCR burst size. 1: The remaining data is transmitted based on the single burst size.
[22]	RW	notialldmawrit	DMA write operation notification. This bit is valid only when GAHBCFG[remmemsupp] is 1.
[21]	RW	remmemsupp	Remote memory support 0: int_dma_req and int_dma_done are not set to **1**, and the controller core handles the XferComp interrupt immediately after the DMA write operation is complete. 1: int_dma_req is set to **1** when the HSOTG DMA starts writing data to an external memory; int_dma_done is set to **1** when the transfer of the controller core is complete, indicating that the HSOTG write operation is complete. In this case, the controller core waits until the system sys_dma_done signal becomes valid, and then starts the next DMA transfer.
[20:8]	RO	reserved	Reserved

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[7]	RW	nptxfemplvl	Empty status level of the non-periodic TX FIFO 0: The TxFEmp interrupt indicates that the TX FIFO at the IN endpoint is half empty. 1: The TxFEmp interrupt indicates that the TX FIFO at the IN endpoint is empty.
[6]	RO	reserved	Reserved
[5]	RW	dmaen	DMA mode enable 0: The core works in slave mode. 1: The core works in DMA mode.
[4:1]	RW	hbstlen	Burst length/type for both the external and internal DMA modes In external DMA mode: 0000: 1 word 0001: 4 words 0010: 8 words 0011: 16 words 0100: 32 words 0101: 64 words 0110: 128 words 0111: 256 words Other values: reserved In internal DMA mode: 0000: Single 0001: INCR 0011: INCR4 0101: INCR8 0111: INCR16 Other values: reserved
[0]	RW	glblintrmsk	Global interrupt mask 0: masked 1: not masked

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GUSBCFG

GUSBCFG is a USB configuration register.

Offset Address	Register Name	Total Reset Value
0x000C	GUSBCFG	0x0000_1400

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	corrupttxpkt	forcedevmode	forcehstmode	txenddelay	usbtrafctl	ic_usbcap	ulpi	indicator	complement	termseldlpulse	ulpiextvbusindicator	ulpiextvbusdrv	ulpiclksusm	ulpiautores	ulpifsls	otgi2csel	phylpwrclksel	reserved	usbtrdtim				hnpcap	srpcap	ddrsel	physel	fsintf	ulpi_utmi_sel	phyif	toutcal		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	WO	corrupttxpkt	Unexpected TX packet This bit is for debugging only and is always set to 0.
[30]	RW	forcedevmode	Forcible device mode 0: normal mode 1: forcible device mode
[29]	RW	forcehstmode	Forcible host mode 0: normal mode 1: forcible host mode
[28]	RW	txenddelay	TX end delay enable 0: disabled 1: enabled
[27]	RW	usbtrafctl	USB pull-up/pull-down resistor control (traffic signaling) 0: Pull-up and pull-down resistors are attached. 1: Pull-up and pull-down resistors are detached. **NOTE** This bit is valid only when OTG_ENABLE_IC_USB and USBCFG [ic_usbcap] are 1.
[26]	RO	ic_usbcap	IC_USB enable 0: The IC_USB PHY interface is not selected. 1: The IC_USB PHY interface is selected.
[25]	RW	ulpi	ULPI interface protection enable 0: enabled 1: disabled

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[24]	RW	indicator	Indicator pass through 0: The complementary output signal is determined by the vbusvalid comparator. 1: The complementary output signal is not determined by the vbusvalid comparator. When TG_HSPHY_INTERFACE is 1, this bit is read-only. When TG_HSPHY_INTERFACE is 0, this bit is reserved.
[23]	RW	complement	Indicator complement 0: The PHY does not invert the ExternalVbusIndicator signal. 1: The PHY inverts the ExternalVbusIndicator signal.
[22]	RW	termseldlpulse	SRP line pulse drive 0: utmi_txvalid drives the line pulse. 1: utmi_termsel drives the line pulse.
[21]	RW	ulpiextvbusindicator	ULPI external VBUS over-current indicator 0: The PHY uses the internal vbusvalid comparator. 1: The PHY uses the external vbusvalid comparator. This bit is valid only when OTG_HSPHY_INTERFACE is 2 or 3.
[20]	RW	ulpiextvbusdrv	ULPI external VBUS drive 0: The PHY uses the internal charge pump to drive VBUS. 1: The PHY uses the external power supply to drive VBUS. **NOTE** This bit is valid only when OTG_HSPHY_INTERFACE is 2 or 3.
[19]	RW	ulpiclksusm	ULPI clock suspend 0: The PHY disables the internal clock in suspend mode. 1: The PHY does not disable the internal clock in suspend mode. This bit is valid only when OTG_HSPHY_INTERFACE is 2 or 3.
[18]	RW	ulpiautores	ULPI automatic restore 0: The PHY does not use the automatic restore feature. 1: The PHY uses the automatic restore feature. **NOTE** This bit is valid only when OTG_HSPHY_INTERFACE is 2 or 3.
[17]	RW	ulpifsls	ULPI FS/LS select 0: ULPI interface 1: ULPI FS/LS serial interface **NOTE** Set GUSBCFG[ulpi_utmi_sel] to 1 before setting this bit.

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[16]	RW	otgi2csel	UTMI or I^{2}C interface select 0: UTMI USB 1.1 full-speed interface 1: I^{2}C interface **NOTE** This bit is valid only when OTG_I2C_INTERFACE is 2. Reading this bit returns 0.
[15]	RW	phylpwrclksel	PHY low-power clock select 0: internal 480 MHz PLL clock 1: external 48 MHz clock
[14]	RO	reserved	Reserved
[13:10]	RW	usbtrdtim	USB turnaround time 0101: This field is set to this value when the MAC interface is a 16-bit UTMI+ interface. 1001: This field is set to this value when the MAC interface is an 8-bit UTMI+ interface. Other values: reserved
[9]	RW	hnpcap	HNP enable 0: disabled 1: enabled
[8]	RW	srpcap	SRP enable 0: disabled 1: enabled
[7]	RW	ddrsel	ULPI DDR select 0: single-rate ULPI interface 1: double-rate ULPI interface This bit is valid only when OTG_HSPHY_INTERFACE is 2 or 3.
[6]	RW	physel	USB 2.0 high-speed PHY or USB 1.1 full-speed serial transceiver 0: USB 2.0 high-speed UTMI+ or ULPI PHY interface 1: USB 1.1 full-speed serial transceiver If the USB 1.1 full-speed serial transceiver is not selected, this bit is always 0 and read-only. If the USB 2.0 high-speed PHY is not selected, this bit is always 1 and read-only. If the preceding interface types are not selected (non-zero value), this bit is used to select the interface to be activated, and this bit can be read or written.

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[5]	RW	fsintf	Full-speed serial interface select 0: 6-pin unidirectional full-speed serial interface 1: 3-pin bidirectional full-speed serial interface If the USB 1.1 full-speed serial transceiver is not selected, this bit is always 0 and write-only. If the USB 1.1 full-speed interface is selected, this bit can be used to select the 3-pin or 6-pin interface and can be read or written.
[4]	RW	ulpi_utmi_sel	ULPI or UTMI select 0: UTMI+ interface 1: ULPI interface
[3]	RW	phyif	PHY interface 0: 8 bits 1: 16 bits
[2:0]	RW	toutcal	HS/FS timeout calibration High-speed operations: One 30-MHz PHY clock = 16 bit times One 60-MHz PHY clock = 8 bit times Full speed operation: One 30-MHz PHY clock = 0.4 bit times One 60-MHz PHY clock = 0.2 bit times One 48-MHz PHY clock = 0.25 bit times

GRSTCTL

GRSTCTL is a reset hardware feature register.

Offset Address	Register Name	Total Reset Value
0x0010	GRSTCTL	0x8000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ahbidle	dmareq	reserved																		txfnum						txfflsh	rxfflsh	intknqflsh		frmcntrrst	reserved	csftrst
Reset	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	ahbidle	Whether the AHB master state machine is idle 0: no 1: yes

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[30]	RO	dmareq	DMA request signal, indicating that the DMA request is being processed. This bit is only for debugging. 0: other cases 1: The DMA request is being processed.
[29:11]	RO	reserved	Reserved
[10:6]	RW	txfnum	TX FIFO ID, which must be updated by using the TX FIFO flush bit. This field cannot be overwritten before the controller core clears the TX FIFO flush bit. 0x0: In host mode, non-periodic TX FIFOs are flushed. During shared FIFO operations, non-periodic TX FIFOs are flushed in device mode. In dedicated FIFO mode, TX FIFO 0 is flushed in device mode. 0x1: In host mode, periodic TX FIFOs are flushed. During shared FIFO operations, periodic TX FIFO 1 is flushed in device mode. In dedicated FIFO mode, TX FIFO 1 is flushed in device mode. 0x2: During shared FIFO operations, periodic TX FIFO 2 is flushed in device mode. In dedicated FIFO mode, TX FIFO 2 is flushed in device mode. 0xF: During shared FIFO operations, periodic TX FIFO 15 is flushed in device mode. In dedicated FIFO mode, TX FIFO 15 is flushed in device mode. 0x10: All TX FIFOs are flushed in device or host mode.
[5]	R_WS_SC	txfflsh	TX FIFO flush This bit can be used to refresh one or all TX FIFOs but FIFOs cannot be refreshed during transactions. This bit can be written only after the application program confirms that the controller core does not read or write to the TX FIFO. It takes eight clock cycles to clear this bit, during which the application program must wait. 0: not flushed 1: flushed
[4]	RO	rxfflsh	RX FIFO flush This bit can be used to refresh all RX FIFOs but FIFOs cannot be refreshed during transactions. This bit can be written only after the application program confirms that the controller core does not read or write to the RX FIFO. It takes eight clock cycles to clear this bit, during which the application program must wait. 0: not flushed 1: flushed

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[3]	R_WS_SC	intknqflsh	In token sequence learning queue flush This bit is valid only when OTG_EN_DED_TX_FIFO is 0. 0: not flushed 1: flushed
[2]	RO	frmcntrrst	Host frame counter reset This bit is used to reset the frame number counter. When the frame number counter is reset, the ID of next transmitted SOF micro-frames is 0. 0: not reset 1: reset
[1]	RO	reserved	Reserved
[0]	RW	csftrst	Core soft interrupt • All interrupts and the CSR register excluding the following bits are cleared: – PCGCCTL[rstpdwnmodule] – PCGCCTL[gatehclk] – PCGCCTL[pwrclmp] – PCGCCTL[stoppphylpwrclkselclk] – GUSBCFG[phylpwrclksel] – GUSBCFG[ddrsel] – GUSBCFG[physel] – GUSBCFG[fsintf] – GUSBCFG[ulpi_utmi_sel] – GUSBCFG[phyif] – GUSBCFG[txenddelay] – GUSBCFG[termseldlpulse] – GUSBCFG[ulpiclksusm] – GUSBCFG[ulpiautores] – GUSBCFG[ulpifsls] – GGPIO – GPWRDN – GADPCTL – HCFG[fslspclksel] – DCFG[devspd] – DCTL[sftdiscon] • All module state machines are reset to the idle status, and all TX FIFOs and RX FIFOs are refreshed. • All transactions on the AHB master are interrupted as soon as possible, and all USB transactions are immediately interrupted. • When the hibernation or ADP feature is enabled, the PMU module is not soft-reset by the core.

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GINTSTS

GINTSTS is a system interrupt register.

Offset Address	Register Name	Total Reset Value
0x0014	GINTSTS	0x0800_0080

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wkupint	sessreqint	disconnint	conidstschng	lpm_int	ptxfemp	hchint	prtint	resetdet	fetsusp	incomplpincompisoout	incompisoin	oepint	iepint	epmis	rstrdoneint	eopf	isooutdrop	enumdone	usbrst	usbsusp	erlysusp	i2cint	ulpickinti2cckint	goutnakeff	ginnakeff	nptxfemp	rxflvl	sof	otgint	modemis	curmod
Reset	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RWSC	wkupint	Suspend/Resume detection interrupt. Setting this bit to 1 clears the interrupt.
[30]	RWSC	sessreqint	In host mode, an interrupt is generated when a device session request is detected. In device mode, an interrupt is generated when utmisrp_bvalid is 1. Setting this bit to 1 clears the interrupt.
[29]	RWSC	disconnint	An interrupt is generated when the system detects that the device is disconnected. Setting this bit to 1 clears the interrupt.
[28]	RWSC	conidstschng	An interrupt is generated when the slot ID changes. Setting this bit to 1 clears the interrupt.
[27]	RWSC	lpm_int	In device mode, an interrupt is generated when the device receives an LPM transaction and transmits a response indicating no errors. In host mode, an interrupt is generated when the device receives an LPM transaction and transmits a response indicating no errors, or when the host core completes the preset LPM transactions. **NOTE** This bit is valid only when LPMCapable or OTG_ENABLE_LPM is 1. Setting this bit to 1 clears the interrupt.
[26]	RO	ptxfemp	An interrupt is generated when the periodic TX FIFO is empty or half empty.
[25]	RO	hchint	In host mode, an interrupt is generated in a core channel. This bit is cleared by clearing the corresponding status bit.

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[24]	RO	prtint	In host mode, an interrupt is generated when the status of a port of the DWC_otg core changes. **NOTE** This bit is cleared by clearing the corresponding bit in the host port control and status register.
[23]	RWSC	resetdet	In device mode, an interrupt is generated when the device is suspended and the system detects that the USB module is reset in power-down module. In host mode, this interrupt is not generated. **NOTE** Setting this bit to 1 clears the interrupt.
[22]	RWSC	fetsusp	This interrupt is valid only in DMA mode. This interrupt indicates that the core stops obtaining data from IN endpoints.
[21]	RWSC	incomplpincompisoout	In host mode, an interrupt is generated if there are pending transactions. In device mode, an interrupt is generated when there are isochronous OUT transfers to be completed. **NOTE** Setting this bit to 1 clears the interrupt.
[20]	RWSC	incompisoin	An interrupt is generated when there are isochronous IN transfers to be completed. In scatter/gather DMA mode, this interrupt is not generated. **NOTE** Setting this bit to 1 clears the interrupt.
[19]	RO	oepint	In device mode, an interrupt is generated on an OUT endpoint. This interrupt is cleared by clearing the corresponding status bit in the DOEPINT*n* register.
[18]	RO	iepint	In device mode, an interrupt is generated on an IN endpoint. This interrupt is cleared by clearing the corresponding status bit in the DOEPINT*n* register.
[17]	RO	epmis	An interrupt is generated when endpoints mismatch. This bit is valid only during shared FIFO operations. **NOTE** Setting this bit to 1 clears the interrupt.
[16]	RWSC	rstrdoneint	An interrupt is caused by the restore command after hibernation. **NOTE** This bit is valid only when the hibernation function is enabled.
[15]	RWSC	eopf	Periodic frame end interrupt Setting this bit to 1 clears the interrupt.
[14]	RWSC	isooutdrop	Isochronous OUT packet drop interrupt **NOTE** Setting this bit to 1 clears the interrupt.

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[13]	RWSC	enumdone	Speed enumeration completion interrupt **NOTE** Setting this bit to 1 clears the interrupt.
[12]	RWSC	usbrst	USB reset detection interrupt **NOTE** Setting this bit to 1 clears the interrupt.
[11]	RWSC	usbsusp	USB suspend detection interrupt **NOTE** Setting this bit to 1 clears the interrupt.
[10]	RWSC	erlysusp	USB idle 3 ms detection interrupt **NOTE** Setting this bit to 1 clears the interrupt.
[9]	RWSC	i2cint	An interrupt is generated when I^2C access through the I^2C interface is complete. This bit is enabled only when OTG_I2C_INTERFACE is 1. **NOTE** Setting this bit to 1 clears the interrupt.
[8]	RWSC	ulpickinti2cckint	When the ULPI Carkit interrupt is generated, the core sets the interrupt bit to 1. This bit is valid only when OTG_ULPI_CARKIT is 1; otherwise, reading this bit returns 0. When the I^2C Carkit interrupt is generated, the core sets the interrupt bit to 1. This bit is valid only when OTG_I2C_INTERFACE is 1; otherwise, reading this bit returns 0. **NOTE** Setting this bit to 1 clears the interrupt.
[7]	RO	goutnakeff	The global OUT NAK bit in the device control register is set. This interrupt is cleared by clearing the global OUT NAK bit.
[6]	RO	ginnakeff	An interrupt is generated when the global non-periodic IN NAK is set in the device control register.
[5]	RO	nptxfemp	An interrupt is generated when the non-periodic TX FIFO is empty or half empty.
[4]	RO	rxflvl	There is at least a pending packet to be read by the RX FIFO.
[3]	RWSC	sof	In host mode, an SOF, micro-SOF, or keep-active is being transmitted. In device mode, an SOF token is received by the USB module.
[2]	RO	otgint	An OTG even is in progress. This interrupt is cleared by clearing the corresponding bit in the GOTGINT register.

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[1]	RWSC	modemis	Mode mismatch interrupt When the core works in device mode, a register in host mode needs to be accessed. When the core works in host mode, a register in device mode needs to be accessed. **NOTE** Setting this bit to **1** clears the interrupt. 0: invalid 1: valid
[0]	RO	curmod	Current operating mode 0: device mode 1: host mode

GINTMSK

GINTMSK is a system interrupt mask register.

Offset Address	Register Name	Total Reset Value
0x0018	GINTMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	wkupintmsk	sessreqintmsk	disconnintmsk	conidstschngmsk	lpm_intmsk	ptxfempmsk	hchintmsk	prtintmsk	resetdetmsk	fetsuspmsk	incomplpmskincompisooutmsk	incompisoinmsk	oepintmsk	iepintmsk	epmismsk	rstrdoneintmsk	eopfmsk	isooutdropmsk	enumdonemsk	usbrstmsk	usbsuspmsk	erlysuspmsk	i2cintmsk	ulpickintmsk	goutnakeffmsk	ginnakeffmsk	nptxfempmsk	rxflvlmsk	sofmsk	otgintmsk	modemismsk	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	wkupintmsk	Resume/Remote wakeup detection interrupt mask 0: masked 1: not masked
[30]	RW	sessreqintmsk	Session request/new session detection interrupt mask 0: masked 1: not masked
[29]	RW	disconnintmsk	Disconnect detection interrupt mask 0: masked 1: not masked

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[28]	RW	conidstschngmsk	Connector ID status change interrupt mask 0: masked 1: not masked
[27]	RW	lpm_intmsk	LPM transaction RX interrupt mask 0: masked 1: not masked
[26]	RW	ptxfempmsk	Periodic TX FIFO empty interrupt mask 0: masked 1: not masked
[25]	RW	hchintmsk	Host channel interrupt mask 0: masked 1: not masked
[24]	RW	prtintmsk	Host port interrupt mask 0: masked 1: not masked
[23]	RW	resetdetmsk	Reset detection interrupt mask 0: masked 1: not masked
[22]	RW	fetsuspmsk	Data fetch suspended interrupt mask 0: masked 1: not masked
[21]	RW	incomplpmskincompisooutmsk	Incomplete periodic transfer interrupt mask 0: masked 1: not masked Incomplete isochronous OUT transfer interrupt mask 0: masked 1: not masked
[20]	RW	incompisoinmsk	Incomplete isochronous in transfer interrupt mask 0: masked 1: not masked This bit can be enabled only when periodic endpoints are enabled in dedicated TX FIFO mode.
[19]	RW	oepintmsk	OUT endpoint interrupt mask 0: masked 1: not masked
[18]	RW	iepintmsk	IN endpoint interrupt mask 0: masked 1: not masked

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[17]	RW	epmismsk	Endpoint mismatch interrupt mask 0: masked 1: not masked
[16]	RW	rstrdoneintmsk	Restore done interrupt mask 0: masked 1: not masked This bit is valid only when the hibernation function is enabled (OTG_EN_PWROPT = 2).
[15]	RW	eopfmsk	End of periodic frame interrupt mask 0: masked 1: not masked
[14]	RW	isooutdropmsk	Device only isochronous OUT packet drop interrupt mask 0: masked 1: not masked
[13]	RW	enumdonemsk	Enumeration done interrupt mask 0: masked 1: not masked
[12]	RW	usbrstmsk	USB reset interrupt mask 0: masked 1: not masked
[11]	RW	usbsuspmsk	USB suspend interrupt mask 0: masked 1: not masked
[10]	RW	erlysuspmsk	Early suspend interrupt mask 0: masked 1: not masked
[9]	RW	i2cintmsk	I^2C interrupt mask 0: masked 1: not masked
[8]	RW	ulpickintmsk	ULPI Carkit interrupt mask 0: masked 1: not masked I^2C Carkit interrupt mask 0: masked 1: not masked
[7]	RW	goutnakeffmsk	Global OUT NAK effective interrupt mask 0: masked 1: not masked

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[6]	RW	ginnakeffmsk	Global non-periodic IN NAK effective interrupt mask 0: masked 1: not masked
[5]	RW	nptxfempmsk	Non-periodic TX FIFO empty interrupt mask 0: masked 1: not masked
[4]	RW	rxflvlmsk	RX FIFO non-empty interrupt mask 0: masked 1: not masked
[3]	RW	sofmsk	Start of (micro) frame interrupt mask 0: masked 1: not masked
[2]	RW	otgintmsk	Device interrupt mask 0: masked 1: not masked
[1]	RW	modemismsk	Mode mismatch interrupt mask 0: masked 1: not masked
[0]	RW	reserved	Reserved

GRXSTSR

GRXSTSR is an RX status debugging read register.

Offset Address	Register Name	Total Reset Value
0x001C	GRXSTSR	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved_devicemode	fn_devicemode		pktsts_devicemode	dpid_devicemode	bcnt_devicemode		chnum_devicemode
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:25]	RO	reserved_devicemode	Reserved
[24:21]	RO	fn_devicemode	Number of frames

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[20:17]	RO	pktsts_devicemode	Data packet status 0001: global OUT NAK (interrupt triggered) 0010: IN data packet received 0011: IN transfer completion (interrupt triggered) 0101: data toggle error (interrupt triggered) 0111: channel stop (interrupt triggered) Other values: reserved
[16:15]	RO	dpid_devicemode	Data PID of the received packet 00: DATA0 10: DATA1 01: DATA2 11: MDATA
[14:4]	RO	bcnt_devicemode	Byte size of the received IN data packets
[3:0]	RO	chnum_devicemode	Number of channels from which the current data packets are received

GRXSTSP

GRXSTSP is an RX status read&pop register.

Offset Address	Register Name	Total Reset Value
0x0020	GRXSTSP	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1 0
Name	reserved	fn_devicemode	pktsts_devicemode	dpid_devicemode	bcnt_devicemode			chnum_devicemode
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0

Bits	Access	Name	Description
[31:25]	RO	reserved	Reserved
[24:21]	RO	fn_devicemode	Number of frames

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[20:17]	RO	pktsts_devicemode	Data packet status 0001: global OUT NAK (interrupt triggered) 0010: IN data packet received 0011: IN transfer completion (interrupt triggered) 0101: data toggle error (interrupt triggered) 0111: channel stop (interrupt triggered) Other values: reserved
[16:15]	RO	dpid_devicemode	Data PID of the received packet 00: DATA0 10: DATA1 01: DATA2 11: MDATA
[14:4]	RO	bcnt_devicemode	Byte size of the received IN data packets
[3:0]	RO	chnum_devicemode	Number of channels from which the current data packets are received

GRXFSIZ

GRXFSIZ is an RX FIFO size configuration register.

Offset Address	Register Name	Total Reset Value
0x0024	GRXFSIZ	0x0000_0211

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																rxfdep															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	1

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	rxfdep	RX FIFO depth. Its value range is 16−32768.

GNPTXFSIZ

GNPTXFSIZ is a non-periodic TX FIFO size configuration register.

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Offset Address	Register Name	Total Reset Value
0x0028	GNPTXFSIZ	0x0100_0211

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ineptxf0dep																ineptxf0staddr															
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	1

Bits	Access	Name	Description
[31:16]	RW	ineptxf0dep	Depth of IN endpoint TX FIFO 0. Its value range is 16–32768.
[15:0]	RW	ineptxf0staddr	Start address for the TX RAM of IN endpoint FIFO 0

GNPTXSTS

GNPTXSTS is a non-periodic TX FIFO and non-periodic TX request query register.

Offset Address	Register Name	Total Reset Value
0x002C	GNPTXSTS	0x0008_0100

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	nptxqtop							nptxqspcavail								nptxfspcavail															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:24]	RO	nptxqtop	Non-periodic TX request queue top Bit[30:27]: number of channels or endpoints bit[26:25]: 00: IN/OUT token 01: zero-length transmit packet (device IN/host OUT) 10: PING/CSPLIT token 11: channel stop command Bit[24]: stop (the selected channel or endpoint is used for attempting)
[23:16]	RO	nptxqspcavail	Available space for the non-periodic TX request queue 0x0: The non-periodic TX request queue is full. 0x1: One address is available. 0x2: Two addresses are available. 0xn: n addresses ($0 \leq n \leq 8$) are available. Other values: reserved

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[15:0]	RO	nptxfspcavail	Total available space for the non-periodic TX request queue 0x0: The non-periodic TX FIFO is full. 0x1: One word is available. 0x2: Two words are available. 0xn: *n* words ($0 \le n \le 32768$) are available. 0x8000: 32768 words are available. Other values: reserved

GI2CCTL

GI2CCTL is an I^{2}C access register.

Offset Address	Register Name	Total Reset Value
0x0030	GI2CCTL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	bsydne	rw	reserved	i2cdatse0	i2cdevadr		i2csuspctl	ack	i2cen				addr				regaddr								rwdata							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RWSC	bsydne	I^{2}C busy/done 0: done 1: busy
[30]	RW	rw	Read/Write indicator 0: write 1: read
[29]	RW	reserved	Reserved
[28]	RW	i2cdatse0	USB mode of the FS interface 0: VP_VM USB mode 1: DAT_SE0 USB mode
[27:26]	RW	i2cdevadr	I^{2}C device address 00: 7'h2C 01: 7'h2D 10: 7'h2E 11: 7'h2F

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[25]	RW	i2csuspctl	I^2C suspend control 0: utmi_suspend_n is used. 1: The suspend bit in the PHY register is edited by using the I^2C write function.
[24]	RO	ack	I^2C ACK 0: NAK 1: ACK
[23]	RW	i2cen	I^2C transaction enable 0: disabled 1: enabled
[22:16]	RW	addr	I^2C slave address
[15:8]	RW	regaddr	I^2C register address
[7:0]	RW	rwdata	I^2C read//write data

GPVNDCTL

GPVNDCTL is a PHY vendor control register.

Offset Address	Register Name	Total Reset Value
0x0034	GPVNDCTL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	disulpidrvr	reserved			vstsdone	vstsbsy	newregreq	reserved		regwr			regaddr				vctrlextregaddr								regdata							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RWSC	disulpidrvr	ULPI interface enable 0: enabled 1: disabled **NOTE** This bit is valid only when OTG_ULPI_CARKIT is 1. Otherwise, reading this bit returns 0.
[30:28]	RO	reserved	Reserved
[27]	RWSC	vstsdone	Vendor control access (done) 0: The new register request is set. 1: Vendor control access ends.

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[26]	RO	vstsbsy	Vendor control access (busy) 0: Vendor control access ends. 1: Vendor control access is in progress.
[25]	RWSC	newregreq	New vendor control access 0: no access request 1: new vendor control access request
[24:23]	RO	reserved	Reserved
[22]	RW	regwr	Register read/write 0: read 1: write
[21:16]	RW	regaddr	Register access address
[15:8]	RW	vctrlextregaddr	Address for the UTMI+ vendor control register bit[15:12]: 4-bit parallel output bus addressing bit[11:8]: utmi_vcontrol[3:0]
[7:0]	RW	regdata	Register data This bit is valid after the Vstatus Done bit is set.

GGPIO

GGPIO is a GPIO register.

Offset Address	Register Name	Total Reset Value
0x0038	GGPIO	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gpo																gpi															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	gpo	General-purpose output, which is driven by gp_o[15:0] of the core
[15:0]	RW	gpi	General-purpose input, which reflects gp_i[15:0] of the core

GUID

GUID is a user ID register.

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Offset Address	Register Name	Total Reset Value
0x003C	GUID	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	userid																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	userid	User ID

GSNPSID

GSNPSID is a Synopsys ID query register.

Offset Address	Register Name	Total Reset Value
0x0040	GSNPSID	0x4F54_300A

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	synopsysid																															
Reset	0	1	0	0	1	1	1	1	0	1	0	1	0	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	0	1	0

Bits	Access	Name	Description
[31:0]	RW	synopsysid	ID of the current DWC_otg core

GHWCFG1

GHWCFG1 is user hardware configuration register 1.

Offset Address	Register Name	Total Reset Value
0x0044	GHWCFG1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	epdir																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	epdir	Endpoint direction The endpoint direction is expressed by two bits. Endpoint bit[31:30]: direction of endpoint 15 bit[29:28]: direction of endpoint 14 ...

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			bit[3:2]: direction of endpoint 1 bit[1:0]: direction of endpoint 0 (always bidirectional) Direction 00: bidirectional (IN and OUT) endpoint 01: IN endpoint 10: OUT endpoint 11: reserved

GHWCFG2

GHWCFG2 is user hardware configuration register 2.

Offset Address	Register Name	Total Reset Value
0x0048	GHWCFG2	0x2284_C850

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	otg_enable_ic_usb			tknqdepth			ptxqdepth		nptxqdepth		reserved	multiprocintrpt	dynfifosizing	periosupport	numhstchnl		numdeveps				fsphytype		hsphytype		singpnt	otgarch	otgmode					
Reset	0	0	1	0	0	0	1	0	1	0	0	0	0	1	0	0	1	1	0	0	1	0	0	0	0	1	0	1	0	0	0	0

Bits	Access	Name	Description
[31]	RO	otg_enable_ic_usb	IC_USB select enable 0: disabled 1: enabled
[30:26]	RO	tknqdepth	Depth of the IN token sequence learning queue in device mode The value range is 0−30.
[25:24]	RO	ptxqdepth	Depth of the periodic request queue in host mode 00: 2 01: 4 10: 8 11: 16
[23:22]	RO	nptxqdepth	Depth of the non-periodic request queue 00: 2 01: 4 10: 8 Other values: reserved

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[21]	RO	reserved	Reserved
[20]	RW	multiprocintrpt	Multi-processor interrupt enable 0: disabled 1: enabled
[19]	RO	dynfifosizing	FIFO size dynamic change enable 0: disabled 1: enabled
[18]	RO	periosupport	Periodic OUT channel support in host mode 0: not supported 1: supported
[17:14]	RO	numhstchnl	Number of host channels 0–15: The value 0 indicates one channel, and the value 15 indicates 16 channels.
[13:10]	RO	numdeveps	Number of device endpoints (excluding endpoint 0) The value range is 1−15.
[9:8]	RO	fsphytype	Type of the full-speed PHY interface 00: The full-speed interface is not supported. 01: The full-speed interface is supported. 10: FS pins share with UTMI+ pins. 11: FS pins share with ULPI pins.
[7:6]	RO	hsphytype	Type of the high-speed PHY interface 00: high-speed interface not supported 01: UTMI+ 10: ULPI 11: UTMI+ and ULPI
[5]	RO	singpnt	Point-to-point 0: multi-point application (hub and split supported) 1: single-point application (hub and split not supported)
[4:3]	RO	otgarch	Device architecture 00: slave only 01: external DMA 10: internal DMA Other values: reserved

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[2:0]	RO	otgmode	Operating mode 000: HNP- and SRP-capable OTG (host & device) 001: SRP-capable OTG (host & device) 010: non-HNP and non-SRP capable OTG (host & device) 011: SRP-capable device 100: non-OTG device 101: SRP-Capable host 110: non-OTG host Other values: reserved

GHWCFG3

GHWCFG3 is user hardware configuration register 3.

Offset Address	Register Name	Total Reset Value
0x004C	GHWCFG3	0x0501_54E8

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dfifodepth																lpmmode	bcsupport	hsicmode	adpsupport	rsttype	optfeature	vndctlsupt	i2cintsel	otgen	pktsizewidth			xfersizewidth			
Reset	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	1	0	1	0	1	0	1	0	0	1	1	1	0	1	0	0	0

Bits	Access	Name	Description
[31:16]	RO	dfifodepth	DFIFO depth. The value range is 32−32768.
[15]	RO	lpmmode	LPM working mode (configurable)
[14]	RO	bcsupport	Whether the HS device controller supports the battery charger 0: not supported 1: supported
[13]	RO	hsicmode	HSIC mode select 0: The HSIC-capable shares with the UTMI PHY interface 1: The HSIC mode is not selected.
[12]	RO	adpsupport	Whether the device controller has an ADP logic 0: no 1: yes
[11]	RO	rsttype	Reset mode of clock always blocks 0: asynchronous reset 1: synchronous reset

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[10]	RO	optfeature	Feature removal Features include the user ID register, GPIO interface ports, and SOF toggle and counter ports. 0: not removed 1: removed
[9]	RO	vndctlsupt	Vendor control interface support 0: not supported 1: supported
[8]	RO	i2cintsel	I^2C interface select 0: not selected 1: selected
[7]	RO	otgen	Device function enable 0: disabled 1: enabled
[6:4]	RO	pktsizewidth	Data packet width 000: 4 bits 001: 5 bits 010: 6 bits 011: 7 bits 100: 8 bits 101: 9 bits 110: 10 bits Other values: reserved
[3:0]	RO	xfersizewidth	Transfer width 0000: 11 bits 0001: 12 bits 1000: 19 bits Other values: reserved

GHWCFG4

GHWCFG4 is user hardware configuration register 4.

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Offset Address	Register Name	Total Reset Value
0x0050	GHWCFG4	0x4600_8020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	descdma	descdmaen	ineps				dedfifomode	sessendfltr	bvalidfltr	avalidfltr	vbusvalidfltr	iddgfltr	numctleps				phydatawidth		reserved						extndedhibernation	hibernation	ahbfreq	partialpwrdn	numdevperioeps			
Reset	0	1	0	0	0	1	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	descdma	Scatter/Gather DMA 0: no dynamic configuration 1: dynamic configuration
[30]	RO	descdmaen	PAD slew rate 0: fast 1: slow
[29:26]	RO	ineps	Number of IN endpoints in device mode 0: one IN endpoint 1: two IN endpoints ... 15: 16 IN endpoints
[25]	RO	dedfifomode	Dedicated TX FIFO enable for device IN endpoints 0: disabled 1: enabled
[24]	RO	sessendfltr	session_end filter enable 0: disabled 1: enabled
[23]	RO	bvalidfltr	b_valid filter enable 0: disabled 1: enabled
[22]	RO	avalidfltr	a_valid filter enable 0: disabled 1: enabled
[21]	RO	vbusvalidfltr	VBUS valid filter enable 0: disabled 1: enabled

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[20]	RO	iddgfltr	IDDIG filter enable 0: disabled 1: enabled
[19:16]	RO	numctleps	Number of controlled endpoints excluding endpoint 0 in device mode The value range is 0−15.
[15:14]	RO	phydatawidth	Data width of the UTMI+ PHY/ULPI-to-internal UTMI+ wrapper 00: 8 bits 01: 16 bits 10: 8 bit or 16 bits, which is configured by using software Other values: reserved
[13:8]	RO	reserved	Reserved
[7]	RO	extndedhibernation	Extended hibernation enable 0: disabled 1: enabled
[6]	RO	hibernation	Hibernation enable 0: disabled 1: enabled
[5]	RO	ahbfreq	Whether the minimum AHB frequency is lower than 60 MHz 0: no 1: yes
[4]	RO	partialpwrdn	Partial power down enable 0: disabled 1: enabled
[3:0]	RO	numdevperioeps	Number of periodic IN endpoints in device mode The value range is 0−15.

GLPMCFG

GLPMCFG is an LPM configuration register.

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Offset Address	Register Name	Total Reset Value
0x0054	GLPMCFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	invselhsic		hsiccon		rstrslpsts		enbesl		lpm_retrycnt_sts			sndlpm	lpm_retry_cnt			lpm_chnl_indx				l1resumeok	slpsts		corel1res		hird_thres				enblslpm	bremotewake	hird				appl1res	lpmcap
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	invselhsic	HSIC select (by using HSIC-invert) enable If if_sel_hsic is 1: InvSelHsic = 1: disabled InvSelHsic = 0: enabled If if_sel_hsic is 0: InvSelHsic = 1: enabled InvSelHsic = 0: disabled
[30]	RW	hsiccon	HSIC-connect Host mode: When this bit is set to 1, the bus HSIC is driven to idle status and waits for the HSIC connect sequence transmitted from the device. Device mode: When this bit is set to 1, the device waits for the HSIC idle line state on the bus. The HSIC connection starts after the idle line state occurs. **NOTE** This bit is valid only when OTG_ENABLE_HSIC is 1; otherwise, reading this bit returns 0.
[29]	RW	rstrslpsts	Restore SlpSts 0: The core enters the shallow sleep mode. 1: The core enters the deep sleep mode.
[28]	RW	enbesl	Best effort service latency (BESL) enable 0: disabled 1: enabled
[27:25]	RO	lpm_retrycnt_sts	LPM retry count This field is used to count the remaining retries.
[24]	RWSC	sndlpm	LPM transaction TX
[23:21]	RW	lpm_retry_cnt	LPM retry count statistics When a device provides error information, this field shows the extra LPM retries of the host before the device validity signal arrives.

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[20:17]	RW	lpm_chnl_indx	LPM channel index The number of LPM transaction channels is provided. The core automatically calculates the device address and number of endpoints based on the LPM channel index.
[16]	RO	l1resumeok	Resume from the sleep state 0: resumed 1: not resumed
[15]	RO	slpsts	Port sleep state 0: resume 1: sleep
[14:13]	RO	corel1res	LPM response In device mode, this field reflects the response to the LPM transaction. In host mode, this field indicates the handshake response from the device during LPM transaction. 11: ACK 10: NYET 01: STALL 00: error (no handshake response)
[12:8]	RW	hird_thres	BESL or HIRD threshold
[7]	RW	enblslpm	utmi_sleep_n enable 0: disabled 1: enabled
[6]	RW	bremotewake	Remote wake enable In host mode, the enable signal is transmitted from the wIndex field of the LPM transaction. In device mode (read-only), when the ACK, NYET, or STALL response is transmitted to the LPM transaction, this bit is updated together with the received LPM token, bRemoteWake, and bmAttribute.
[5:2]	RW	hird	When EnBESL is 0: Host-initiated resume period In host mode, this field is assigned by the LPM transaction. In device mode, this field is updated by LPM token HIRD bmAttribute. Sl. No HIRD[3:0] THIRD (µs) 1 4'b0000 50 2 4'b0001 125 3 4'b0010 200 4 4'b0011 275 5 4'b0100 350

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			6 4'b0101 425 7 4'b0110 500 8 4'b0111 575 9 4'b1000 650 10 4'b1001 725 11 4'b1010 800 12 4'b1011 875 13 4'b1100 950 14 4'b1101 1025 15 4'b1110 1100 16 4'b1111 1175 When EnBESL is 1: BESL In host mode, the BESL value is transmitted to the LPM transaction In device mode, this field is updated by LPM token BESL bmAttribute.
[1]	RW	appl1res	LPM response 0: NYET 1: ACK
[0]	RW	lpmcap	LPM enable 0: disabled 1: enabled

GPWRDN

GPWRDN is a power down register.

Offset Address	Register Name	Total Reset Value
0x0058	GPWRDN	0x1320_0010

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				multvalidbc				adpint	bsessvld	iddig	linestate	stschngintmsk	stschngint	srpdetectmsk	srpdetect	conndetmsk	connectdet	disconnectdetectmsk	disconnectdetect	resetdetmsk	resetdetected	linestagechangemsk	Instschng	disablevbus	pwrdnswtch	pwrdnrst_n	pwrdnclmp	restore	pmuactv	pmuintsel	
Reset	0	0	0	1	0	0	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bits	Access	Name	Description
[31:29]	RO	reserved	Reserved

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[28:24]	RO	multvalidbc	BC ACA input Bit 28 - rid_float Bit 27 - rid_gnd Bit 26 - rid_a Bit 25 - rid_b Bit 24 - rid_c
[23]	RWSC	adpint	An interrupt is generated when the ADP is in progress. Setting this bit to 1 clears this interrupt.
[22]	RO	bsessvld	B session validity 0: B-Valid = 0 1: B-valid = 1
[21]	RO	iddig	IDDIG signal status indicator Current operating mode 0: host mode 1: device mode
[20:19]	RO	linestate	Current line state indicator 00: DM = 0, DP = 0 01: DM = 0, DP = 1 10: DM = 1, DP = 0 11: undefined
[18]	RW	stschngintmsk	StsChng interrupt mask 0: masked 1: not masked
[17]	RWSC	stschngint	StsChng interrupt The bit indicates whether the status of the IDDIG or BSessVld signal changes. 0: not changed 1: changed
[16]	RW	srpdetectmsk	SRPDetect interrupt mask, valid only in host mode 0: not masked 1: masked
[15]	RWSC	srpdetect	SRP detection 0: not detected 1: detected
[14]	RW	conndetmsk	Connection detection interrupt mask 0: masked 1: not masked This bit is valid only when OTG_EN_PWROPT is 2.

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[13]	RO	connectdet	Connection detection 0: disconnected 1: connected This bit is valid only when OTG_EN_PWROPT is 2.
[12]	RW	disconnectdetectmsk	Disconnection detection interrupt mask 0: masked 1: not masked This bit is valid only when OTG_EN_PWROPT is 2.
[11]	RWSC	disconnectdetect	Disconnection detection 0: not disconnected 1: disconnected This bit is valid only when OTG_EN_PWROPT is 2.
[10]	RW	resetdetmsk	Reset detection interrupt mask 0: masked 1: not masked This bit is valid only when OTG_EN_PWROPT is 2.
[9]	RWSC	resetdetected	Reset detection 0: not reset 1: reset This bit is valid only when OTG_EN_PWROPT is 2.
[8]	RW	linestagechangemsk	Line state change interrupt mask 0: masked 1: not masked This bit is valid only when OTG_EN_PWROPT is 2.
[7]	RWSC	lnstschng	Line state change detection 0: not changed 1: changed This bit is valid only when GPWRDN[pmuactv] is 1 and OTG_EN_PWROPT is 2.
[6]	RW	disablevbus	Vbus disable In host mode: 0: PrtPwr is not 0. 1: PrtPwr is 0. In device mode: 0: The level of the bvalid signal is high. 1: The level of the bvalid signal is low. This bit is valid only when GPWRDN[pmuactv] is 1.

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[5]	RW	pwrdnswtch	Power down switch 0: on 1: off
[4]	RW	pwrdnrst_n	Power down reset 0: reset 1: not reset
[3]	RW	pwrdnclmp	Power down beat enable 0: disabled 1: enabled
[2]	RW	restore	Restore 0: DWC_otg is in normal mode. 1: DWC_otg is in restore mode. This bit is valid only when OTG_EN_PWROPT is 2.
[1]	RW	pmuactv	PMU enable 0: disabled 1: enabled
[0]	RW	pmuintsel	PMU interrupt select 0: internal DWC_otg_core interrupt 1: external DWC_otg_core interrupt

GDFIFOCFG

GDFIFOCFG is a DFIFO software configuration register.

Offset Address	Register Name	Total Reset Value
0x005C	GDFIFOCFG	0x0501_0511

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	epinfobaseaddr																gdfifocfg															
Reset	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0	0	1	0	0	0	1

Bits	Access	Name	Description
[31:16]	RW	epinfobaseaddr	Start address for the EP info controller
[15:0]	RW	gdfifocfg	DFIFO size, which is dynamically configured

GADPCTL

GADPCTL is an ADP timer control and status register.

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Offset Address	Register Name	Total Reset Value
0x0060	GADPCTL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved			ar		adptoutmsk	adpsnsintmsk	adpprbintmsk	adptoutint	adpsnsint	adpprbint	adpen	adpres	enasns	enaprb	rtim												prbper			prbdelta			prbdschg
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:29]	RW	reserved	Reserved
[28:27]	RW	ar	Access request 00: read/write 01: read-only 10: write-only 11: reserved
[26]	RW	adptoutmsk	ADP timeout interrupt mask 0: masked 1: not masked **NOTE** This bit is valid only when OTG_Ver is 1.
[25]	RW	adpsnsintmsk	ADP sense interrupt mask 0: masked 1: not masked **NOTE** This bit is valid only when OTG_Ver is 1.
[24]	RWC	adpprbintmsk	ADB probe interrupt mask 0: masked 1: not masked **NOTE** This bit is valid only when OTG_Ver 1.
[23]	RWC	adptoutint	ADP timeout interrupt Setting this bit to 1 clears this interrupt. **NOTE** This bit is valid only when OTG_Ver 1.
[22]	RWC	adpsnsint	ADP sense interrupt Setting this bit to 1 clears this interrupt. **NOTE** This bit is valid only when OTG_Ver 1.

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[21]	RWC	adpprbint	ADP probe interrupt Setting this bit to 1 clears this interrupt. **NOTE** This bit is valid only when OTG_Ver 1.
[20]	RW	adpen	ADP enable 0: disabled 1: enabled **NOTE** This bit is valid only when OTG_Ver 1.
[19]	RWC	adpres	ADP reset 0: not reset 1: reset **NOTE** This bit is valid only when OTG_Ver 1.
[18]	RW	enasns	Sense enable 0: disabled 1: enabled **NOTE** This bit is valid only when OTG_Ver 1.
[17]	RW	enaprb	Probe enable 0: disabled 1: enabled **NOTE** This bit is valid only when OTG_Ver 1.
[16:6]	RO	rtim	Ramp time The following is the mapping between the values and 32 kHz clock cycle: 0x000: 1 cycles 0x001: 2 cycles 0x002: 3 cycles ... 0x7FF: 2048 cycles
[5:4]	RW	prbper	Probe period 00: 0.625s to 0.925s (0.775s typically) 01: 1.25s to 1.85s (1.55s typically) 10: 1.9s to 2.6s (2.275s typically) 11: reserved

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[3:2]	RW	prbdelta	Probe delta The following is the mapping between the values and 32 kHz clock cycle: 00: 1 cycle 01: 2 cycles 10: 3 cycles 11: 4 cycles
[1:0]	RW	prbdschg	Probe discharge TADP_DSCHG time 00: 4 ms 01: 8 ms 10: 16 ms 11: 32 ms

HPTXFSIZ

HPTXFSIZ is a TX FIFO configuration register.

Offset Address	Register Name	Total Reset Value
0x0100	HPTXFSIZ	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	ptxfsize																ptxfstaddr															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	ptxfsize	Depth of the host periodic TX FIFO. Its value range is 16−32768.
[15:0]	RW	ptxfstaddr	Start address for the host periodic TX FIFO

DPTXFSIZN

DPTXFSIZN is a device periodic TX FIFO-*n* size register.

NOTE

This register is valid only in shared FIFO mode.

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Offset Address	Register Name	Total Reset Value
0x0104 + 0x0004 x FIFO_num (FIFO_num = 0–14)	DPTXFSIZN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dptxfsize																dptxfstaddr															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	dptxfsize	Size of the device periodic TX FIFO The value range is 4–768.
[15:0]	RW	dptxfstaddr	Start address for the device periodic TX FIFO RAM

DIEPTXFN

DIEPTXFN is a device IN endpoint TX FIFO size register.

NOTE

This register is valid only in dedicated FIFO mode.

Offset Address	Register Name	Total Reset Value
0x0104 + 0x0004 x FIFO_num (FIFO_num = 0–14)	DIEPTXFN	0x0300_0251

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	inepntxfdep																inepntxfstaddr															
Reset	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0	0	1

Bits	Access	Name	Description
[31:16]	RW	inepntxfdep	Depth of the IN endpoint TX FIFO The value range is 16–32768.
[15:0]	RW	inepntxfstaddr	Start address for the IN endpoint FIFO*n* TX RAM

DCFG

DCFG is a device configuration register.

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Offset Address	Register Name	Total Reset Value
0x0800	DCFG	0x8100_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	resvalid						perschintvl		descdma	epmiscnt					reserved				endevoutnak	perfrint	devaddr								ena32khzsusp	nzstsouthshk	devspd	
Reset	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RW	resvalid	Resume time control This bit is valid only when DCFG[ena32khzsusp] is 1.
[25:24]	RW	perschintvl	Ratio of the (micro) frames in scatter/gather DMA mode 00: 25% of (micro) frames 01: 50% of (micro) frames 10: 75% of (micro) frames 11: reserved
[23]	RW	descdma	Scatter/Gather DMA enable in device mode 0: disabled 1: enabled
[22:18]	RW	epmiscnt	IN endpoint mismatch statistics
[17:14]	RO	reserved	Reserved
[13]	RW	endevoutnak	Device OUT NAK enable 0: disabled 1: enabled
[12:11]	RW	perfrint	Periodic frame interval 00: 80% of the (micro) frame interval 01: 85% 10: 90% 11: 95%
[10:4]	RW	devaddr	Device address
[3]	RW	ena32khzsusp	32 kHz suspend mode enable 0: disabled 1: enabled
[2]	RW	nzstsouthshk	Non-zero-length status OUT handshake select

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[1:0]	RW	devspd	Device speed 00: high speed (30 MHz or 60 MHz USB 2.0 PHY clock) 01: full speed (30 MHz or 60 MHz USB 2.0 PHY clock) 10: low speed (6 MHz USB 1.1 transceiver clock) 11: full speed (48 MHz USB 1.1 transceiver clock)

DCTL

DCTL is a device control register.

Offset Address	Register Name	Total Reset Value
0x0804	DCTL	0x0000_0002

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved														encontonbna	nakonbble	ignrfrmnum	reserved				pwronprgdone	cgoutnak	sgoutnak	cgnpinnak	sgnpinnak	tstctl			goutnaksts	gnpinnaksts	sftdiscon	rmtwkupsig
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:18]	RO	reserved	Reserved
[17]	RW	encontonbna	Continue on BNA enable 0: disabled 1: enabled
[16]	RW	nakonbble	NAK on babble error 0: no operation 1: NAK configured on the received babble
[15]	RW	ignrfrmnum	Whether the number of frames on isochronous endpoints is ignored 0: not ignored 1: ignored
[14:12]	RO	reserved	Reserved
[11]	RW	pwronprgdone	Resume from the power-down mode 0: not woken up 1: woken up
[10]	WO	cgoutnak	Global OUT NAK clear 0: no operation 1: cleared

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[9]	WO	sgoutnak	Global OUT NAK 0: no operation 1: configured
[8]	WO	cgnpinnak	Global non-periodic IN NAK clear 0: no operation 1: cleared
[7]	WO	sgnpinnak	Global non-periodic IN NAK 0: no operation 1: configured
[6:4]	RW	tstctl	Test control 000: test mode disabled 001: Test_J mode 010: Test_K mode 011: Test_SE0_NAK mode 100: Test_Packet mode 101: Test_Force_Enable Other values: reserved
[3]	RO	goutnaksts	Global OUT NAK status 0: handshake TX 1: NAK handshake TX
[2]	RO	gnpinnaksts	Global non-periodic IN NAK status 0: handshake TX 1: NAK handshake TX
[1]	RW	sftdiscon	Soft disconnection 0: common mode 1: The core drives phy_opmode_o to 2'b01, and then triggers USB disconnection.
[0]	RW	rmtwkupsig	Remote wakeup signal TX 0: not woken up remotely 1: woken up remotely

DSTS

DSTS is a device status query register.

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Offset Address	Register Name	Total Reset Value
0x0808	DSTS	0x0007_FF02

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								devlnsts				soffn												reserved				errticerr		enumspd		suspsts	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31:24]	RO	reserved	Reserved
[23:22]	RO	devlnsts	Logic level of the current USB data line bit[23]: logic level of D+ bit[22]: logic level of D−
[21:8]	RO	soffn	Number of frames or micro frames in the received SOF packets
[7:4]	RO	reserved	Reserved
[3]	RO	errticerr	Irregular error indicator The controller core sets this bit to **1** when any irregular error (for example, phy_rxvalid_i is set to **1** and retains this value for 2 ms due to an PHY error) occurs in the UTMI+ interface. 0: No irregular error occurs in the UTMI+ interface. 1: An irregular error occurs in the UTMI+ interface.
[2:1]	RO	enumspd	Enumeration speed 00: high speed (30 MHz or 60 MHz PHY clock) 01: full speed (30 MHz or 60 MHz PHY clock) 10: low speed (6 MHz PHY clock). 11: full speed (48 MHz PHY clock).
[0]	RO	suspsts	Suspend status. When the suspend conditions are detected, this bit is 1.

DIEPMSK

DIEPMSK is a common interrupt mask register for device IN endpoints.

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Offset Address	Register Name	Total Reset Value
0x0810	DIEPMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																		nakmsk	reserved			bnainintrmsk	txfifoundrnmsk	reserved	inepnakeffmsk	intknepmismsk	intkntxfempmsk	timeoutmsk	ahberrmsk	epdisbldmsk	xfercomplmsk
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved
[13]	RW	nakmsk	NAK interrupt mask 0: masked 1: not masked
[12:10]	RO	reserved	Reserved
[9]	RW	bnainintrmsk	BNA interrupt mask 0: masked 1: not masked
[8]	RW	txfifoundrnmsk	FIFO underrun interrupt mask 0: masked 1: not masked
[7]	RO	reserved	Reserved
[6]	RW	inepnakeffmsk	IN endpoint NAK effective interrupt mask 0: masked 1: not masked
[5]	RW	intknepmismsk	IN token received with EP mismatch interrupt mask 0: masked 1: not masked
[4]	RW	intkntxfempmsk	IN token received when TX FIFO empty interrupt mask 0: masked 1: not masked
[3]	RW	timeoutmsk	Timeout interrupt mask 0: masked 1: not masked
[2]	RW	ahberrmsk	AHB error interrupt mask 0: masked 1: not masked

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[1]	RW	epdisbldmsk	Endpoint disabled interrupt mask 0: masked 1: not masked
[0]	RW	xfercomplmsk	Transfer completed interrupt mask 0: masked 1: not masked

DOEPMSK

DOEPMSK is a common interrupt mask register for device OUT endpoints.

Offset Address	Register Name	Total Reset Value
0x0814	DOEPMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																	nyetmsk	nakmsk	bbleerrmsk	reserved		bnaoutintrmsk	outpktermsk	reserved	back2backsetupmsk	stsphsercvdmsk	outtknepdismsk	setupmsk	ahberrmsk	epdisbldmsk	xfercomplmsk
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14]	RW	nyetmsk	NYET interrupt mask 0: masked 1: not masked
[13]	RW	nakmsk	NAK interrupt mask 0: masked 1: not masked
[12]	RW	bbleerrmsk	Babble error interrupt mask 0: masked 1: not masked
[11:10]	RO	reserved	Reserved
[9]	RW	bnaoutintrmsk	BNA interrupt mask 0: masked 1: not masked

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[8]	RW	outpkterrmsk	OUT packet error interrupt mask 0: masked 1: not masked
[7]	RO	reserved	Reserved
[6]	RW	back2backsetupmsk	Back-to-back setup packets received interrupt mask 0: masked 1: not masked
[5]	RW	stsphsercvdmsk	Status phase received interrupt mask 0: masked 1: not masked
[4]	RW	outtknepdismsk	W OUT token received when endpoint disabled interrupt mask 0: masked 1: not masked
[3]	RW	setupmsk	Setup phase done interrupt mask 0: masked 1: not masked
[2]	RW	ahberrmsk	AHB error interrupt mask 0: masked 1: not masked
[1]	RW	epdisbldmsk	Endpoint disabled interrupt mask 0: masked 1: not masked
[0]	RW	xfercomplmsk	Transfer completed interrupt mask 0: masked 1: not masked

DAINT

DAINT is an interrupt register for all device endpoints.

Offset Address	Register Name	Total Reset Value
0x0818	DAINT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	outepint																inepint															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	outepint	OUT endpoint interrupt

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			Bit 16: OUT endpoint 0 ... Bit 31: OUT endpoint 15
[15:0]	RO	inepint	IN endpoint interrupt Bit 0: IN endpoint 0 ... Bit 15: IN endpoint 15

DAINTMSK

DAINTMSK is an interrupt mask register for all endpoints.

Offset Address	Register Name	Total Reset Value
0x081C	DAINTMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	outepmsk																inepmsk															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	outepmsk	OUT endpoint interrupt mask Bit 16: OUT endpoint 0 ... Bit 31: OUT endpoint 15
[15:0]	RW	inepmsk	IN endpoint interrupt mask Bit 0: IN endpoint 0 ... Bit 15: IN endpoint 15

DTKNQR1

DTKNQR1 is device IN token sequence learning queue read register 1.

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Offset Address	Register Name	Total Reset Value
0x0820	DTKNQR1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	eptkn																								wrapbit	reserved			intknwptr			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:8]	RO	eptkn	Token endpoint ID Four bits are used to indicate the token endpoint ID. Bit[31:28]: endpoint ID of token 5 Bit[27:24]: endpoint ID of token 4 … Bit[15:12]: endpoint ID of token 1 Bit[11:8]: endpoint ID of token 0
[7]	RO	wrapbit	Wrapping bit 0: The learning queue of the IN token is cleared. 1: The write pointer is wrapped.
[6:5]	RO	reserved	Reserved
[4:0]	RO	intknwptr	IN token queue write pointer

DTKNQR2

DTKNQR2 is device IN token sequence learning queue read register 2.

Offset Address	Register Name	Total Reset Value
0x0824	DTKNQR2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	eptkn																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	eptkn	Token endpoint ID Four bits are used to indicate the token endpoint ID. Bit[31:28]: endpoint ID of token 13 Bit[27:24]: endpoint ID of token 12 … Bit[7:4]: endpoint ID of token 7

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			Bit[3:0]: endpoint ID of token 6

DTKNQR3

DTKNQR3 is device in token sequence learning queue read register 3.

Offset Address	Register Name	Total Reset Value
0x0830	DTKNQR3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	eptkn																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	eptkn	Token endpoint ID Four bits are used to indicate the token endpoint ID. Bit[31:28]: endpoint ID of token 21 Bit[27:24]: endpoint ID of token 20 … Bit[7:4]: endpoint ID of token 15 Bit[3:0]: endpoint ID of token 14

DTKNQR4

DTKNQR4 is device IN token sequence learning queue read register 4.

Offset Address	Register Name	Total Reset Value
0x0834	DTKNQR4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	eptkn																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	eptkn	Token endpoint ID Four bits are used to indicate the token endpoint ID. Bit[31:28]: endpoint ID of token 29 Bit[27:24]: endpoint ID of token 28 … Bit[7:4]: endpoint ID of token 23 Bit[3:0]: endpoint ID of token 22

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DVBUSDIS

DVBUSDIS is a device VBUS discharge time register.

Offset Address	Register Name	Total Reset Value
0x0828	DVBUSDIS	0x0000_17D7

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																dvbusdis															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	0	1	0	1	1	1

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	dvbusdis	Device VBUS discharge time

DVBUSPULSE

DVBUSPULSE is a device VBUS pulsing time register.

Offset Address	Register Name	Total Reset Value
0x082C	DVBUSPULSE	0x0000_05B8

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																				dvbuspulse											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	1	1	1	0	0	0

Bits	Access	Name	Description
[31:12]	RO	reserved	Reserved
[11:0]	RO	dvbuspulse	Device VBUS pulse time

DTHRCTL

DTHRCTL is a device threshold control register.

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Offset Address	Register Name	Total Reset Value
0x0830	DTHRCTL	0x0C10_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				arbprken	reserved	rxthrlen									rxthren	reserved				ahbthrratio				txthrlen						isothren	nonisothren
Reset	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:28]	RO	reserved	Reserved
[27]	RW	arbprken	Arbitrator parking enable 0: disabled 1: enabled
[26]	RO	reserved	Reserved
[25:17]	RW	rxthrlen	RX threshold length
[16]	RW	rxthren	RX threshold enable Some RX FIFO problems may be caused after the RX threshold is enabled, especially when there are RX and babble errors. 0: disabled 1: enabled
[15:13]	RO	reserved	Reserved
[12:11]	RW	ahbthrratio	AHB threshold ratio 00: AHB threshold = MAC threshold 01: AHB threshold = MAC threshold/2 10: AHB threshold = MAC threshold/4 11: AHB threshold = MAC threshold/8
[10:2]	RW	txthrlen	TX threshold length
[1]	RW	isothren	ISO IN endpoint threshold enable 0: disabled 1: enabled
[0]	RW	nonisothren	Non-ISO IN endpoint threshold enable 0: disabled 1: enabled

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DIEPEMPMSK

DIEPEMPMSK is a FIFO empty interrupt mask register for device IN endpoints.

Offset Address	Register Name	Total Reset Value
0x0834	DIEPEMPMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ineptxfempmsk															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	ineptxfempmsk	IN EP TX FIFO empty interrupt mask Bit 0: IN endpoint 0 ... Bit 15: IN endpoint 15

DEACHINT

DEACHINT is an interrupt register for each device endpoint.

Offset Address	Register Name	Total Reset Value
0x0838	DEACHINT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	echoutepint																echinepint															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	echoutepint	OUT endpoint interrupt Bit 16: OUT endpoint 0 ... Bit 31: OUT endpoint 15
[15:0]	RW	echinepint	IN endpoint interrupt Bit 15: IN endpoint 0 ... Bit 0: IN endpoint 15

DEACHINTMSK

DEACHINTMSK is an interrupt mask register for each device endpoint.

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Offset Address	Register Name	Total Reset Value
0x083C	DEACHINTMSK	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	echoutepmsk																echinepmsk															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	echoutepmsk	OUT endpoint interrupt mask Bit 16: OUT endpoint 0 ... Bit 31: OUT endpoint 15 0: masked 1: not masked
[15:0]	RW	echinepmsk	IN endpoint interrupt mask Bit 16: IN endpoint 0 ... Bit 31: IN endpoint 15 0: masked 1: not masked

DIEPEACHMSKN

DIEPEACHMSKN is an interrupt register for device IN endpoint *n*.

Offset Address	Register Name	Total Reset Value
0x0840 + (0x0004 x *n*) (*n* = 0–15)	DIEPEACHMSKN	0x0040_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																		nakmsk	reserved			bnainintrmsk	txfifoundrnmsk	reserved	inepnakeffmsk	intknepmismsk	intkntxfempmsk	timeoutmsk	ahberrmsk	epdisbldmsk	xfercomplmsk
Reset	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:14]	RO	reserved	Reserved

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[13]	RW	nakmsk	NAK interrupt mask 0: masked 1: not masked
[12:10]	RO	reserved	Reserved
[9]	RW	bnainintrmsk	BNA interrupt mask 0: masked 1: not masked
[8]	RW	txfifoundrnmsk	FIFO underrun interrupt mask 0: masked 1: not masked
[7]	RO	reserved	Reserved
[6]	RW	inepnakeffmsk	IN endpoint NAK effective interrupt mask 0: masked 1: not masked
[5]	RW	intknepmismsk	IN token received with EP mismatch interrupt mask 0: masked 1: not masked
[4]	RW	intkntxfempmsk	IN token received when TX FIFO empty interrupt mask 0: masked 1: not masked
[3]	RW	timeoutmsk	Timeout interrupt mask (non-isochronous endpoints) 0: masked 1: not masked
[2]	RW	ahberrmsk	AHB error interrupt mask 0: masked 1: not masked
[1]	RW	epdisbldmsk	Endpoint disable interrupt mask 0: masked 1: not masked
[0]	RW	xfercomplmsk	Transfer completion interrupt mask 0: masked 1: not masked

DOEPEACHMSKN

DOEPEACHMSKN is an interrupt register for device OUT endpoint *n*.

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Offset Address	Register Name	Total Reset Value
0x0880 + (0x0004 x *n*) (*n* = 0−15)	DOEPEACHMSKN	0x0080_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																	nyetmsk	nakmsk	bbleerrmsk	reserved		bnaoutintrmsk	outpkterrmsk	reserved	back2backsetup	reserved	outtknepdismsk	setupmsk	ahberrmsk	epdisbldmsk	xfercomplmsk
Reset	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:15]	RO	reserved	Reserved
[14]	RW	nyetmsk	NYET interrupt mask 0: masked 1: not masked
[13]	RW	nakmsk	NAK interrupt mask 0: masked 1: not masked
[12]	RW	bbleerrmsk	Babble error interrupt mask 0: masked 1: not masked
[11:10]	RO	reserved	Reserved
[9]	RW	bnaoutintrmsk	BNA interrupt mask 0: masked 1: not masked
[8]	RW	outpkterrmsk	OUT packet error interrupt mask 0: masked 1: not masked
[7]	RO	reserved	Reserved
[6]	RW	back2backsetup	Back-to-back setup packets received interrupt mask 0: masked 1: not masked
[5]	RO	reserved	Reserved
[4]	RW	outtknepdismsk	OUT token received when endpoint disabled interrupt mask 0: masked 1: not masked

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[3]	RW	setupmsk	Setup phase done interrupt mask 0: masked 1: not masked
[2]	RW	ahberrmsk	AHB error interrupt mask 0: masked 1: not masked
[1]	RW	epdisbldmsk	Endpoint disabled interrupt mask 0: masked 1: not masked
[0]	RW	xfercomplmsk	Transfer completion interrupt mask 0: masked 1: not masked

DIEPCTL0

DIEPCTL0 is a control register for device control IN endpoint 0.

Offset Address	Register Name	Total Reset Value
0x0900	DIEPCTL0	0x0000_8000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	epena	epdis	reserved		snak	cnak	txfnum				stall	reserved	eptype		naksts	reserved	usbactep	nextep			reserved								mps			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RWSC	epena	Endpoint enable When the IN endpoint works in scatter or gather DMA mode, this bit indicates that the descriptor structure and data buffer used for transmitting data have been set up. When the scatter or gather DMA mode is disabled, this bit indicates that data to be transmitted is ready. 0: invalid 1: valid
[30]	RWSC	epdis	Endpoint disable The application program sets this bit to **1** to end the data transmission of this endpoint. This bit is enabled only in DMA mode. 0: invalid 1: valid

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[29:28]	RO	reserved	Reserved
[27]	WO	snak	NAK configuration Writing to this bit sets the NAK bit of the endpoint to **1**. The application program can use this bit to control the NAK handshake of an endpoint during a transfer. The controller core can set this bit of an endpoint after this endpoint receives the setup packet. 0: invalid 1: valid
[26]	WO	cnak	NAK clear Writing to this bit clears the NAK bit of the endpoint. 0: invalid 1: valid
[25:22]	RW	txfnum	Number of TX FIFOs
[21]	RWSC	stall	Stall handshake The application program can set this bit. The controller core clears this bit after the endpoint receives the setup packet. 0: invalid 1: valid
[20]	RO	reserved	Reserved
[19:18]	RO	eptype	Endpoint type 00: control transfer endpoint 01: real-time transfer endpoint 10: bulk transfer endpoint 11: interrupt transfer endpoint
[17]	RO	naksts	NAK status 0: non-NAK handshake TX 1: NAK handshake TX
[16]	RO	reserved	Reserved
[15]	RO	usbactep	USB activation endpoint 0 This bit is always 1, indicating that control endpoint 0 is always valid. 0: invalid 1: valid
[14:11]	RO	nextep	Next endpoint This field specifies the ID of the next endpoint that receives data.
[10:2]	RO	reserved	Reserved

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[1:0]	RW	mps	Minimum packet size 00: 64 bytes 01: 32 bytes 10: 16 bytes 11: 8 bytes

DOEPCTL0

DOEPCTL0 is a control register for device control OUT endpoint 0.

Offset Address	Register Name	Total Reset Value
0x0B00	DOEPCTL0	0x0000_8000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	epena	epdis	reserved		snak	cnak	reserved				stall	snp	eptype		naksts	reserved	usbactep	reserved														mps
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RWSC	epena	Endpoint enable When the scatter/gather DMA is enabled: 0: The descriptor structure and the data RX function of the data buffer are disabled. 1: The descriptor structure and the data RX function of the data buffer are enabled. When the scatter/gather DMA is disabled: 0: The function of receiving data from the USB module to the memory is disabled. 1: The function of receiving data from the USB module to the memory is enabled.
[30]	RO	epdis	Endpoint disable The application program cannot disable OUT control transfer endpoint 0. 0: enabled 1: disabled
[29:28]	RO	reserved	Reserved

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[27]	WO	snak	NAK configuration Writing to this bit sets the NAK bit of the endpoint to **1**. The application program can use this bit to control the NAK handshake of an endpoint during a transfer. The controller core can set this bit of an endpoint after this endpoint receives the setup packet. 0: invalid 1: valid
[26]	WO	cnak	NAK clear Writing to this bit clears the NAK bit of the endpoint. 0: invalid 1: valid
[25:22]	RO	reserved	Reserved
[21]	RWSC	stall	Stall handshake The application program can set this bit. The controller core clears this bit after the endpoint receives the setup packet. 0: invalid 1: valid
[20]	RW	snp	Snooping mode This bit is used to set the endpoint to the snooping mode. In snooping mode, the controller core checks whether the OUT packet is correct after it sends the packet to the application memory. 0: invalid 1: valid
[19:18]	RO	eptype	Endpoint type 00: control transfer endpoint 01: real-time transfer endpoint 10: bulk transfer endpoint 11: interrupt transfer endpoint
[17]	RO	naksts	NAK status 0: non-NAK handshake TX 1: NAK handshake TX
[16]	RO	reserved	Reserved
[15]	RO	usbactep	USB activation endpoint 0 This bit is always 1, indicating that control endpoint 0 is always valid. 0: invalid 1: valid
[14:2]	RO	reserved	Reserved

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[1:0]	RW	mps	Minimum packet size 00: 64 bytes 01: 32 bytes 10: 16 bytes 11: 8 bytes

DIEPCTLN

DIEPCTLN is a control register for device IN endpoint *n*.

Offset Address: 0x0900 + (0x0020 x *n*) (*n* = 0−15)

Register Name: DIEPCTLN

Total Reset Value: 0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EPEna	EPDis	SetD1PID_SetOddFr	SetD0PID_SetEvenFr	SNAK	CNAK	TxFNum				Stall	Snp	EPType		NAKSts	DPID_EO_FrNum	USBActEP	NextEp			MPS											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RWSC	EPEna	Endpoint enable When the scatter or gather DMA is enabled: 0: The descriptor structure and the data TX function of the data buffer are disabled. 1: The descriptor structure and the data TX function of the data buffer are disabled. When the scatter/gather DMA is disabled: 0: The data on endpoints is not ready. 1: The data on endpoints is ready.
[30]	RWSC	EPDis	Endpoint disable This bit applies to the IN and OUT endpoints. The application program can set this bit to **1** to stop the data transmission/reception of an IN/OUT endpoint. 0: enabled 1: disabled

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[29]	WO	SetD1PID_SetOddFr	DATA1 PID/odd micro-frame configuration The data PID of the interrupt endpoint and bulk IN/OUT endpoint can be set to DATA1 by setting this bit to **1**. 0: invalid 1: valid For the real-time IN/OUT endpoint, the odd and even micro-frames can be set to odd micro-frames by setting this bit to **1**. 0: invalid 1: valid
[28]	WO	SetD0PID_SetEvenFr	DATA0 PID/even micro-frame configuration The data PID of the interrupt endpoint and bulk IN/OUT endpoint can be set to DATA0 by setting this bit to **1**. 0: invalid 1: valid For the real-time IN/OUT endpoint, the odd and even micro-frames can be set to even micro-frames by setting this bit to **1**. 0: invalid 1: valid
[27]	WO	SNAK	NAK configuration Writing to this bit sets the NAK bit of the endpoint to **1**. The application program can use this bit to control the NAK handshake of an endpoint during a transfer. For an OUT endpoint, this bit is set after the controller core receives the transfer completion interrupt or the OUT endpoint receives the setup packet. 0: invalid 1: valid
[26]	WO	CNAK	NAK clear Writing to this bit clears the NAK bit of an IN/OUT endpoint. 0: invalid 1: valid
[25:22]	RW	TxFNum	Number of TX FIFOs Shared FIFO operation: 0: non-periodic TX FIFO Other values: specified periodic TX FIFO number Dedicated FIFO operation: Number of FIFOs This field is valid only for IN endpoints.

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[21]	RW	Stall	Stall handshake (RW) This bit applies to the non-control transfer and non-real-time transfer IN and OUT endpoints. If this bit is set to **1**, the application program blocks the token packets transmitted from the USB host controller to the endpoint. 0: invalid 1: valid For the control endpoint (RWSC), this bit is set to **1** by the application program and cleared by the controller core only when the endpoint receives the setup token. 0: invalid 1: valid
[20]	RW	Snp	Snooping mode This bit is used to set the OUT endpoints to the snooping mode. In snooping mode, the controller core does not check whether the OUT packet is correct before sending the packet to the application memory. 0: invalid 1: valid
[19:18]	RW	EPType	Endpoint type 00: control transfer endpoint 01: real-time transfer endpoint 10: bulk transfer endpoint 11: interrupt transfer endpoint
[17]	RO	NAKSts	NAK status 0: non-NAK handshake TX 1: NAK handshake TX
[16]	RO	DPID_EO_FrNum	Endpoint data PID 0: DATA0 1: DATA1 Even/Odd frame Non-scatter/Gather DMA mode: 0: even frame 1: odd frame Scatter/Gather DMA mode: Reserved
[15]	RWSC	USBActEP	USB activation endpoint 0 This bit is always 1, indicating that control endpoint 0 is always valid. 0: invalid 1: valid

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[14:11]	RW	NextEp	Next endpoint This field specifies the ID of the next endpoint that receives data.
[10:0]	RW	MPS	Maximum packet size

DOEPCTLN

DOEPCTLN is a control register for device OUT endpoint *n*.

Offset Address	Register Name	Total Reset Value
0x0B00 + (0x0020 x *n*) (*n* = 0−15)	DOEPCTLN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	EPEna	EPDis	SetD1PID_SetOddFr	SetD0PID_SetEvenFr	SNAK	CNAK	TxFNum				Stall	Snp	EPType		NAKSts	DPID_EO_FrNum	USBActEP	NextEp			MPS											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RWSC	EPEna	Endpoint enable When the scatter or gather DMA is enabled: 0: The descriptor structure and the data RX function of the data buffer are disabled. 1: The descriptor structure and the data RX function of the data buffer are enabled. When the scatter/gather DMA is disabled: 0: The function of receiving data from the USB module to the memory is disabled. 1: The function of receiving data from the USB module to the memory is enabled.
[30]	RWSC	EPDis	Endpoint disable This bit applies to the IN and OUT endpoints. The application program can set this bit to **1** to stop the data transmission/reception of an IN/OUT endpoint. 0: enabled 1: disabled

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[29]	WO	SetD1PID_SetOddFr	DATA1 PID/odd micro-frame configuration The data PID of the interrupt endpoint and bulk IN/OUT endpoint can be set to DATA1 by setting this bit to **1**. 0: invalid 1: valid For the real-time IN/OUT endpoint, the odd and even micro-frames can be set to odd micro-frames by setting this bit to **1**. 0: invalid 1: valid
[28]	WO	SetD0PID_SetEvenFr	DATA0 PID/even micro-frame configuration The data PID of the interrupt endpoint and bulk IN/OUT endpoint can be set to DATA0 by setting this bit to **1**. 0: invalid 1: valid For the real-time IN/OUT endpoint, the odd and even micro-frames can be set to even micro-frames by setting this bit to **1**. 0: invalid 1: valid
[27]	WO	SNAK	NAK configuration Writing to this bit sets the NAK bit of the endpoint to **1**. The application program can use this bit to control the NAK handshake of an endpoint during a transfer. For an OUT endpoint, this bit is set after the controller core receives the transfer completion interrupt or the OUT endpoint receives the setup packet. 0: invalid 1: valid
[26]	WO	CNAK	NAK clear Writing to this bit clears the NAK bit of an IN/OUT endpoint. 0: invalid 1: valid
[25:22]	RW	TxFNum	Number of TX FIFOs Shared FIFO operation: 0: non-periodic TX FIFO Other values: specified periodic TX FIFO.number Dedicated FIFO operation: Number of FIFOs This field is valid only for IN endpoints.
[21]	RW	Stall	Stall handshake (RW) Endpoint control (RWSC)

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[20]	RW	Snp	Snooping mode This bit is used to set the OUT endpoint to the snooping mode. In snooping mode, the controller core does not check whether the OUT packet is correct before sending the packet to the application memory. 0: invalid 1: valid
[19:18]	RW	EPType	Endpoint type 00: control transfer endpoint 01: real-time transfer endpoint 10: bulk transfer endpoint 11: interrupt transfer endpoint
[17]	RO	NAKSts	NAK status 0: non-NAK handshake TX 1: NAK handshake TX
[16]	RO	DPID_EO_FrNum	Endpoint data PID 0: DATA0 1: DATA1 Even/Odd frame Non-scatter/Gather DMA mode: 0: even frame 1: odd frame Scatter/Gather DMA mode: Reserved
[15]	RWSC	USBActEP	USB activation endpoint This bit applies to the IN and OUT endpoints and indicates whether an endpoint is valid. 0: invalid 1: valid
[14:11]	RW	NextEp	Next endpoint This field specifies the ID of the next endpoint that receives data.
[10:0]	RW	MPS	Maximum packet size

DIEPINTn

DIEPINTn is an interrupt register for device IN endpoint *n*.

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Offset Address	Register Name	Total Reset Value
0x0908 + (0x0020 x *n*) (*n* = 0–15)	DIEPINTn	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																StupPktRcvd	NYETIntrpt	NAKIntrpt	BbleErrIntrpt	PktDrpSts	reserved	BNAIntr	TxfifoUndrn_OutPktErr	TxFEmp	INEPNakEff_Back2BackSETup	INTknEPMis_StsPhseRcvd	INTknTXFEmp_OUTTknEPdis	TimeOUT_SetUp	AHBErr	EPDisbld	XferCompl
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RW	reserved	Reserved
[15]	RWSC	StupPktRcvd	Whether setup packets are received 0: no 1: yes
[14]	RWSC	NYETIntrpt	NYET interrupt NYET response TX interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[13]	RWSC	NAKIntrpt	NAK interrupt NAK TX interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[12]	RWSC	BbleErrIntrpt	Babble error interrupt Endpoint received babble interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[11]	RWSC	PktDrpSts	Packet drop status interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[10]	RO	reserved	Reserved

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[9]	RWSC	BNAIntr	BNA interrupt An interrupt is generated when descriptor access is not ready. 0: An interrupt is cleared. 1: An interrupt is generated.
[8]	RWSC	TxfifoUndrn_OutPktErr	FIFO underrun interrupt An interrupt is generated when the FIFO underrun conditions are transmitted. The interrupt is valid when the following conditions are met: • OTG_EN_DED_TX_FIFO is 1. • Thresholding is enabled. OUT packet error An interrupt is generated when an overrun or CRC error is detected. The interrupt is valid when the following conditions are met: • OTG_EN_DED_TX_FIFO is 1. • Thresholding is enabled.
[7]	RO	TxFEmp	TX FIFO empty status interrupt An interrupt is generated when the TX FIFO is empty or half empty. **NOTE** This bit is valid only for IN endpoints. 0: An interrupt is cleared. 1: An interrupt is generated.
[6]	RWSC	INEPNakEff_Back2BackSETup	IN endpoint NAK effective interrupt An interrupt is generated when the IN endpoint bit is set. Back-to-back setup package received interrupt An interrupt is generated when the core receives more than three back-to-back setup packets. 0: An interrupt is cleared. 1: An interrupt is generated.
[5]	RWSC	INTknEPMis_StsPhseRcvd	An interrupt is generated when the IN tokens with mismatch EP are received. A control write pulse is received. 0: An interrupt is cleared. 1: A control write pulse is received.
[4]	RWSC	INTknTXFEmp_OUTTknEPdis	An interrupt is generated if IN tokens are received when the corresponding TX FIFO is empty. An interrupt is generated if OUT tokens are received when endpoints are disabled. 0: An interrupt is cleared. 1: An interrupt is generated.

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[3]	RWSC	TimeOUT_SetUp	Timeout interrupt An interrupt is generated when timeout conditions are detected. Setup phase done interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[2]	RWSC	AHBErr	AHB error interrupt An interrupt is generated when an AHB error occurs during AHB read/write operations in internal DMA mode. 0: An interrupt is cleared. 1: An interrupt is generated.
[1]	RWSC	EPDisbld	Endpoint disable interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[0]	RWSC	XferCompl	Transfer completion interrupt 0: An interrupt is cleared. 1: An interrupt is generated.

DOEPINTn

DOEPINTn is an interrupt register for device OUT endpoint *n*.

Offset Address	Register Name	Total Reset Value
0x0B08 + (0x0020 x *n*) (*n* = 0−15)	DOEPINTn	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																StupPktRcvd	NYETIntrpt	NAKIntrpt	BbleErrIntrpt	PktDrpSts	reserved	BNAIntr	TxfifoUndrn_OutPktErr	TxFEmp	INEPNakEff_Back2BackSETup	INTknEPMis_StsPhseRcvd	INTknTXFEmp_OUTTknEPdis	TimeOUT_SetUp	AHBErr	EPDisbld	XferCompl
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RWSC	StupPktRcvd	Whether setup packets are received 0: no 1: yes

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[14]	RWSC	NYETIntrpt	NYET interrupt NYET response TX interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[13]	RWSC	NAKIntrpt	NAK interrupt NAK TX interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[12]	RWSC	BbleErrIntrpt	Babble error interrupt Endpoint received babble interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[11]	RWSC	PktDrpSts	Packet drop status interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[10]	RO	reserved	Reserved
[9]	RWSC	BNAIntr	BNA interrupt An interrupt is generated when descriptor access is not ready. 0: An interrupt is cleared. 1: An interrupt is generated.
[8]	RWSC	TxfifoUndrn_OutPktErr	FIFO underrun interrupt An interrupt is generated when the FIFO underrun conditions are transmitted. The interrupt is valid when the following conditions are met: – OTG_EN_DED_TX_FIFO is 1. – Thresholding is enabled. OUT packet error An interrupt is generated when an overrun or CRC error is detected. The interrupt is valid when the following conditions are met: – OTG_EN_DED_TX_FIFO is 1. – Thresholding is enabled.
[7]	RO	TxFEmp	TX FIFO empty status interrupt An interrupt is generated when the TX FIFO is empty or half empty. This bit is valid only for IN endpoints. 0: An interrupt is cleared. 1: An interrupt is generated.

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[6]	RWSC	INEPNakEff_Back2BackSETup	IN endpoint NAK effective interrupt An interrupt is generated when the IN endpoint bit is set. Back-to-back setup package received interrupt An interrupt is generated when the core receives more than three back-to-back setup packets. 0: An interrupt is cleared. 1: An interrupt is generated.
[5]	RWSC	INTknEPMis_StsPhseRcvd	An interrupt is generated when the IN tokens with mismatch EP are received. A control write pulse is received. 0: An interrupt is cleared. 1: An interrupt is generated.
[4]	RWSC	INTknTXFEmp_OUTTknEPdis	An interrupt is generated if IN tokens are received when the corresponding TX FIFO is empty. An interrupt is generated if OUT tokens are received when endpoints are disabled. 0: An interrupt is cleared. 1: An interrupt is generated.
[3]	RWSC	TimeOUT_SetUp	Timeout interrupt An interrupt is generated when timeout conditions are detected. Setup phase done interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[2]	RWSC	AHBErr	AHB error interrupt An interrupt is generated when an AHB error occurs during AHB read/write operations in internal DMA mode. 0: An interrupt is cleared. 1: An interrupt is generated.
[1]	RWSC	EPDisbld	Endpoint disable interrupt 0: An interrupt is cleared. 1: An interrupt is generated.
[0]	RWSC	XferCompl	Transfer completion interrupt 0: An interrupt is cleared. 1: An interrupt is generated.

DIEPTSIZ0

DIEPTSIZ0 is a transfer size register for device IN endpoint 0.

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Offset Address	Register Name	Total Reset Value
0x0910	DIEPTSIZ0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												PktCnt				reserved								XferSize							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:21]	RO	reserved	Reserved
[20:19]	RW	PktCnt	Number of packets
[18:7]	RO	reserved	Reserved
[6:0]	RW	XferSize	Transferred data size

DOEPTSIZ0

DOEPTSIZ0 is a transfer size register for device OUT endpoint 0.

Offset Address	Register Name	Total Reset Value
0x0B10	DOEPTSIZ0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	SUPCnt			reserved								PktCnt				reserved								XferSize							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30:29]	RW	SUPCnt	Number of setup packets 00: 0 packet 01: 1 packet 10: 2 packets 11: 3 packets
[28:20]	RO	reserved	Reserved
[19]	RW	PktCnt	Number of packets
[18:7]	RO	reserved	Reserved
[6:0]	RW	XferSize	Transferred data size

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DIEPTSIZn

DIEPTSIZn is a transfer size register for device IN endpoint *n*.

Offset Address: 0x0910 + (0x0020 x *n*) (*n* = 0−15)

Register Name: DIEPTSIZn

Total Reset Value: 0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	mc_rxdpid_supcnt			PktCnt								XferSize																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	reserved	Reserved
[30:29]	RW	mc_rxdpid_supcnt	Number of packets (IN endpoint, RW) 00: 0 packet 01: 1 packet 10: 2 packets 11: 3 packets Number of packets (non-periodic IN endpoints, RO) This field is valid only in internal DMA mode. PID of the received data (isochronous OUT endpoints, RO) 00: DATA0 01: DATA2 10: DATA1 11: MDATA Number of setup packets (OUT endpoint, RW) 01: 1 packet 10: 2 packets 11: 3 packets
[28:19]	RW	PktCnt	Number of packets
[18:0]	RW	XferSize	Transferred data size

DOEPTSIZn

DOEPTSIZn is a transfer size register for device OUT endpoint *n*.

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Offset Address	Register Name	Total Reset Value
0x0B10 + (0x0020 x *n*) (*n* = 0−15)	DOEPTSIZn	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved	mc_rxdpid_supcnt			PktCnt												XferSize															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	reserved	Reserved
[30:29]	RW	mc_rxdpid_supcnt	Number of packets (IN endpoint, RW) 00: 0 packet 01: 1 packet 10: 2 packets 11: 3 packets Number of packets (non-periodic IN endpoints, RO) This field is valid only in internal DMA mode. PID of the received data (isochronous OUT endpoints, RO) 00: DATA0 01: DATA2 10: DATA1 11: MDATA Number of setup packets (OUT endpoint, RW) 01: 1 packet 10: 2 packets 11: 3 packets
[28:19]	RW	PktCnt	Number of packets
[18:0]	RW	XferSize	Transferred data size

DIEPDMAN

DIEPDMAN is a DMA address register for device IN endpoint *n*.

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Offset Address	Register Name	Total Reset Value
0x0914 + (0x0020 x *n*) (*n* = 0–15)	DIEPDMAN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dmaaddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dmaaddr	DMA address

DOEPDMAN

DOEPDMAN is a DMA address register for device OUT endpoint *n*.

Offset Address	Register Name	Total Reset Value
0x0B14 + (0x0020 x n) (*n* = 0–15)	DOEPDMAN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dmaaddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dmaaddr	DMA address

DIEPDMABN

DIEPDMABN is a DMA buffer address register for device IN endpoint *n*.

Offset Address	Register Name	Total Reset Value
0x091C + (0x0020 x *n*) (*n* = 0–15)	DIEPDMABN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DMABufferAddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	DMABufferAddr	DMA buffer address

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DOEPDMABN

DOEPDMABN is a DMA buffer address register for device OUT endpoint *n*.

Offset Address	Register Name	Total Reset Value
0x0B1C + (0x0020 x *n*) (*n* = 0−15)	DOEPDMABN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DMABufferAddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	DMABufferAddr	DMA buffer address

DTXFSTSn

DTXFSTSn is a TX FIFO status register for device IN endpoints.

Offset Address	Register Name	Total Reset Value
0x0938	DTXFSTSn	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																INEPTxFSpcAvail															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	INEPTxFSpcAvail	Available space for the IN endpoint TX FIFO 0x0: The endpoint TX FIFO is full. 0x1: One word is available. 0x2: Two words are available. 0xn: *n* words ($0 \leq n \leq 32768$) are available. 0x8000: 32768 words are available. Other values: reserved

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12.8 LSADC_CTRL

12.8.1 Overview

The successive approximation register analog-to-digital converter controller (LSADC_CTRL) provides an ADC controller interface to connect to the on-chip LSADC for driving the LSADC developed by HiSilicon and reporting key values.

12.8.2 Features

The LSADC has the following features:

- 3.3 V power voltage
- Maximum scanning frequency of 200000 times per second
- Four independent channels

The LSADC_CTRL has the following features:

- Single start, one channel scanned at a time, interrupt, query, and no deglitch
- Continuous scanning
  - Channels are polled based on ch_vld.
  - Continuous query is started based on Tscan (scanning interval).
  - Deglitch is performed based on Tglitch and the glitch validity is determined.
  - The LSADC conversion result and corresponding channel IDs are reported.
  - The key press interrupt and key change interrupt are reported.

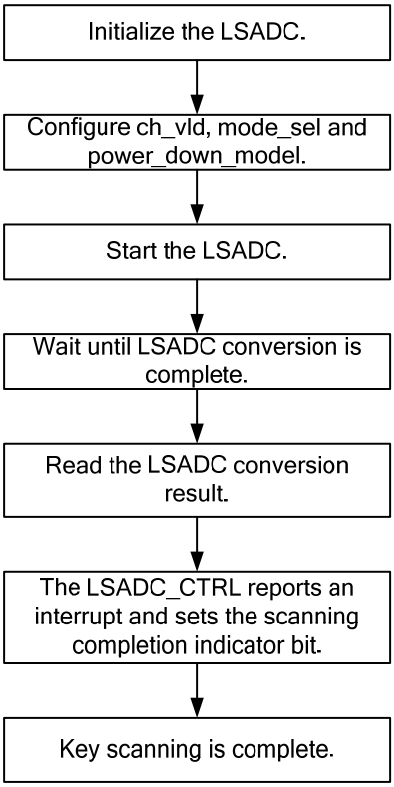
12.8.3 Operating Mode

Single Scanning Procedure

In single read mode (LSADC_CTRL0[model_sel] = 0), the CPU configures the ID of the channel to scanned, scanning mode, and key value mapping table and starts the LSADC to complete a channel scanning. The LSADC_CTRL obtains the ADC conversion value and sets the conversion completion indicator bit while reporting an interrupt. No deglitch is performed in single read mode.

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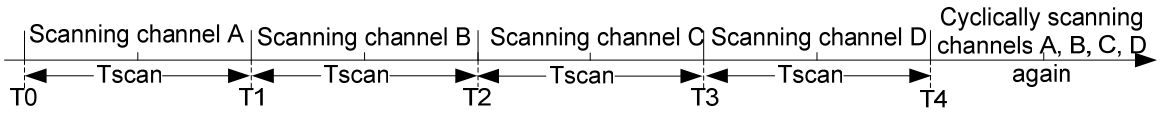
Figure 12-54 Single scanning procedure



Continuous Scanning Procedure

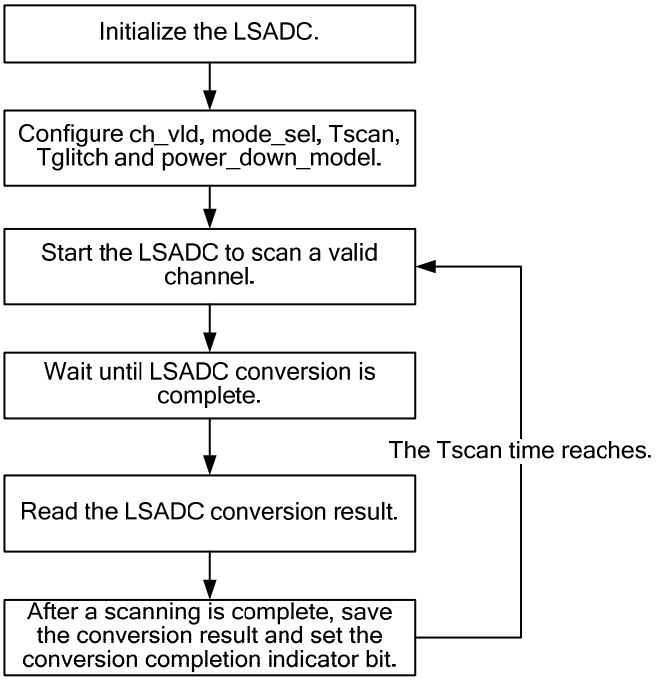
In continuous scanning mode (LSADC_CTRL0[model_sel] = 1), the CPU configures the continuous scanning interval (Tscan), glitch width (Tglitch), and valid channel ID (ch_vld) based on the application scenario, and starts the LSADC_CTRL. The LSADC_CTRL scans a valid channel within Tscan, and scans the next valid channel when the next scanning time reaches. After all valid channels are scanned, the LSADC_CTRL cyclically scans them again. See Figure 12-56. Figure 12-55 illustrates channel polling in continuous scanning mode.

Figure 12-55 Channel polling in continuous scanning mode



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Figure 12-56 Continuous scanning procedure



Deglitch Procedure

The deglitch circuit adopts the majority decision algorithm. In the deglitch window, if the levels sampled by the ADC are the levels when keys are pressed, the value is considered as a valid key value; otherwise, a glitch signal is considered.

In continuous mode, when valid scanning values are obtained after deglitch, a key press interrupt is reported, and the scanning completion indicator bits are set. As there are sampling errors, the precision of the LSADC result is 6 bits.

Hardware deglitch can be enabled or disabled as required.

12.8.4 Register Summary

Table 12-25 describes LSADC_CRTL registers.

Table 12-25 Summary of LSADC_CRTL registers (base address: 0x200B_0000)

Offset Address	Register	Description	Page
0x0000	LSADC_CTRL0	LSADC_CTRL configuration register	12-247
0x0004	LSADC_CTRL1	Deglitch configuration register	12-248
0x0008	LSADC_CTRL2	Scanning interval configuration register	12-249
0x000C	LSADC_CTRL3	LSADC data hold register	12-249
0x0010	LSADC_CTRL4	Interrupt enable register	12-249

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Offset Address	Register	Description	Page
0x0014	LSADC_CTRL5	Interrupt status register	12-250
0x0018	LSADC_CTRL6	Interrupt clear register	12-251
0x001C	LSADC_CTRL7	Start configuration register	12-251
0x0020	LSADC_CTRL8	Stop configuration register	12-252

12.8.5 Register Description

LSADC_CTRL0

LSADC_CTRL0 is an LSDAC_CTRL configuration register.

Offset Address	Register Name	Total Reset Value
0x0000	LSADC_CTRL0	0x0000_80FF

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lsadc_active_bit								lsadc_data_delta				reserved		deglitch_bypass	reserved	lsadc_reset	power_down_model	model_sel	reserved	ch_d_vld	ch_c_vld	ch_b_vld	ch_a_vld	lsadc_zero							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

Bits	Access	Name	Description
[31:24]	RW	lsadc_active_bit	LSADC conversion result precision. 8'b11111100: 6-bit precision 8'b11111000: 5-bit precision ……
[23:20]	RW	lsadc_data_delta	Error range of the LSADC conversion result. In continuous scanning mode, if the difference between two conversion results falls within the error range, the two results are considered as the same.
[19:18]	RO	reserved	Reserved
[17]	RW	deglitch_bypass	Deglitch bypass (used in continuous scanning mode) 0: enabled 1: bypassed
[16]	RO	reserved	Reserved

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[15]	RW	lsadc_reset	Whether the LSADC enters the reset status 0: The LSADC exits the reset status. 1: The LSADC enters the reset status.
[14]	RW	power_down_mode l	Whether power_down is supported 0: not supported 1: supported
[13]	RW	model_sel	Scanning mode of the LSADC_CTRL 0: single scanning mode 1: continuous scanning mode
[12]	RO	reserved	Reserved
[11]	RW	ch_d_vld	LSADC channel D validity 0: invalid 1: valid
[10]	RW	ch_c_vld	LSADC channel C validity 0: invalid 1: valid
[9]	RW	ch_b_vld	LSADC channel B validity 0: invalid 1: valid
[8]	RW	ch_a_vld	LSADC channel A validity 0: invalid 1: valid
[7:0]	RW	lsadc_zero	LSADC value when no key is pressed.

LSADC_CTRL1

LSADC_CTRL1 is a deglitch configuration register

Offset Address	Register Name	Total Reset Value
0x0004	LSADC_CTRL1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	glitch_sample																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	glitch_sample	Deglitch time window (in ms typically). When the LSADC conversion result is retained in the deglitch time window, the conversion result is considered as a valid value; otherwise, a glitch is considered. The deglitch time window cannot be 0 in continuous

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			scanning mode.

LSADC_CTRL2

LSADC_CTRL2 is a scanning interval configuration register.

Offset Address	Register Name	Total Reset Value
0x0008	LSADC_CTRL2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	time_scan																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	time_scan	Interval of scanning two channels in continuous scanning mode (in the unit of 0.333 ms). The interval must be greater than the LSADC conversion time, that is, the configured value must be greater than or equal to 14.

LSADC_CTRL3

LSADC_CTRL3 is an LSADC data hold register.

Offset Address	Register Name	Total Reset Value
0x000C	LSADC_CTRL3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	lsadc_data_ind								lsadc_data_inc								lsadc_data_inb								lsadc_data_ina							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	lsadc_data_ind	Scanned value of LSADC channel D.
[23:16]	RO	lsadc_data_inc	Scanned value of LSADC channel C.
[15:8]	RO	lsadc_data_inb	Scanned value of LSADC channel B.
[7:0]	RO	lsadc_data_ina	Scanned value of LSADC channel A.

LSADC_CTRL4

LSADC_CTRL4 is an interrupt enable register.

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Offset Address	Register Name	Total Reset Value
0x0010	LSADC_CTRL4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															int_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	int_enable	Scanned value valid interrupt enable 0: disabled 1: enabled

LSADC_CTRL5

LSADC_CTRL5 is an interrupt status register.

Offset Address	Register Name	Total Reset Value
0x0014	LSADC_CTRL5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																											lsadc_auto_busy	int_flag_ind	int_flag_inc	int_flag_inb	int_flag_ina
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:5]	RO	reserved	Reserved Writing to this field has no effect and reading this field returns 0.
[4]	RO	lsadc_auto_busy	LSADC busy indicator in automatic scanning mode 0: idle 1: busy
[3]	RO	int_flag_ind	Scanned value valid interrupt flag of channel D
[2]	RO	int_flag_inc	Scanned value valid interrupt flag of channel C
[1]	RO	int_flag_inb	Scanned value valid interrupt flag of channel B

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[0]	RO	int_flag_ina	Scanned value valid interrupt flag of channel A

LSADC_CTRL6

LSADC_CTRL6 is an interrupt clear register.

Offset Address	Register Name	Total Reset Value
0x0018	LSADC_CTRL6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																												clr_int_flag_ind	clr_int_flag_inc	clr_int_flag_inb	clr_int_flag_ina
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:4]	RO	reserved	Reserved
[3]	WO	clr_int_flag_ind	Channel D interrupt clear 0: not cleared 1: cleared
[2]	WO	clr_int_flag_inc	Channel C interrupt clear 0: not cleared 1: cleared
[1]	WO	clr_int_flag_inb	Channel B interrupt clear 0: not cleared 1: cleared
[0]	WO	clr_int_flag_ina	Channel A interrupt clear 0: not cleared 1: cleared

LSADC_CTRL7

LSADC_CTRL7 is a start configuration register.

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Offset Address	Register Name	Total Reset Value
0x001C	LSADC_CTRL7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	start																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	start	LSADC_CTRL start signal. Writing any value to this register starts the LSADC_CTRL.

LSADC_CTRL8

LSADC_CTRL8 is a stop configuration register.

Offset Address	Register Name	Total Reset Value
0x0020	LSADC_CTRL8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	stop																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	stop	Automatic scanning stop. In automatic scanning mode, writing any value to this register stops automatic scanning of the LSADC_CTRL. Automatic scanning is restarted only after the start bit is enabled.

12.9 PWM

12.9.1 Overview

Hi3518E V20X/Hi3516C V200 has four independent pulse width modulation (PWM) outputs in one group.

12.9.2 Features

Each PWM output has the following features:

- Supports the optional 3 MHz, 24 MHz or 50 MHz clock source.
- Provides an internal 26-bit counter with configurable output frequency. The maximum frequency of the output square waves is 25 MHz (50 MHz/2), and the minimum frequency of the output square waves is 0.045 Hz (3 MHz/67108863).
- Allows you to set the number of high-level beats (26 bits).

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- Provides a 10-bit internal counter and allows you to set the number of pulses. A maximum of 1023 pulses are supported. The number of output pulses can be fixed at a value and the always output mode is supported.
  - When pwmn_keep is set to 0 (PWM*n* indicates the PWM ID), PWM*n* outputs the square waves of a specific number, which depends on the setting of the pwmn_num register.
  - When pwmn_keep is set to 1, PWM*n* always outputs square waves.

12.9.3 Operating Mode

The internal working clock of the PWM module is 3 MHz, 24 MHz or 50 MHz. The following takes PWM0 as an example to configure 1-channel PWM output:

Step 1 Select an appropriate clock source, and calculate the number of required cycles and high-level beats.

Step 2 Write the corresponding values to PWM0_CFG0, PWM0_CFG1, and PWM0_CFG2.

Step 3 Set PWM0_CTRL bit[0] to 1 to enable the PWM output.

----End

For example, to output a 3 kHz waveform with 72.5% high levels and 10 pulses (the duty ratio is 72.5%), calculate the number of high levels as follows:

- If the 3 MHz clock is the source clock, Cycle = 3 MHz/1 kHz ≈ 1000 (0x00003E8 in hexadecimal)
- Number of high levels = Cycle x Duty ratio = 1000 x 72.5% = 725 (0x00002D5 in hexadecimal)

To output a desired waveform, perform the following steps:

Step 1 Write 0x2 to PERI_CRG14, select the 3 MHz clock as the PWM clock source, and enable the PWM clock.

Step 2 Read PWM0_STATE2 bit[10] until it is 0. This indicates that the PWM module is idle and can output square waves.

Step 3 Write 0x0000_03E8 to PWM0_CFG0.

Step 4 Write 0x0000_02D5 to PWM0_CFG1.

Step 5 Write 0x0000_000A to PWM0_CFG2.

Step 6 Write 0x1 to PWM0_CTRL.

NOTE

The following steps can be skipped. They are used to check whether the square wave is output as expected.

Step 7 Read PWM0_STATE2 bit[10] until it is 1. This indicates that the PWM module is outputting square waves.

Step 8 Read PWM0_STATE0 and compare the queried value with 0x0000_03E8.

Step 9 Read PWM0_STATE1 and compare the queried value with 0x0000_02D5.

Step 10 Read PWM0_STATE2 bit[9:0] and compare the queried value with 0x0A. If PWM0_STATE2 bit[10] is 1, the PWM module is outputting square waves. If PWM0_STATE2 bit[10] is 0, the configured number of square waves are output.

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----End

12.9.4 Register Summary

The registers of PWM*n* (*n* = 0−3) have the same functions but their base addresses are different. See Table 12-1.

Table 12-1 Base addresses for the registers of PWM*n*

Value of n	Base Address
0	0x2013_0000
1	0x2013_0020
2	0x2013_0040
3	0x2013_0060

Table 12-2 describes PWM0 registers as an example.

Table 12-2 Summary of PWM0 registers (base address: 0x2013_0000)

Offset Address	Register	Description	Page
0x0000	PWM0_CFG0	PWM0 configuration 0 register	12-254
0x0004	PWM0_CFG1	PWM0 configuration 1 register	12-255
0x0008	PWM0_CFG2	PWM0 configuration 2 register	12-255
0x000C	PWM0_CTRL	PWM0 control register	12-256
0x0010	PWM0_STATE0	PWM0 status 0 register	12-256
0x0014	PWM0_STATE1	PWM0 status 1 register	12-257
0x0018	PWM0_STATE2	PWM0 status 2 register	12-258

12.9.5 Register Description

PWM0_CFG0

PWM0_CFG0 is a PWM0 configuration 0 register.

Offset Address	Register Name	Total Reset Value
0x0000	PWM0_CFG0	0x0000_018F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								pwm0_period																							

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Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	1	1

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:0]	RW	pwm0_period	Number of cycles for PWM0 **NOTE** The configured value must be greater than or equal to 2.

PWM0_CFG1

PWM0_CFG1 is a PWM0 configuration 1 register.

Offset Address	Register Name	Total Reset Value
0x0004	PWM0_CFG1	0x0000_00C7

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								pwm0_duty																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	1	1

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:0]	RW	pwm0_duty	Number of level beats for PWM0. If the number is greater than or equal to the number of cycles, the output level is always high. **NOTE** The configured value must be greater than or equal to 1.

PWM0_CFG2

PWM0_CFG2 is a PWM0 configuration 2 register.

Offset Address	Register Name	Total Reset Value
0x0008	PWM0_CFG2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																						pwm0_num									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:10]	RO	reserved	Reserved
[9:0]	RW	pwm0_num	Number of square waves output by PWM0

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PWM0_CTRL

PWM0_CTRL is a PWM0 control register.

Offset Address	Register Name	Total Reset Value
0x000C	PWM0_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved																													pwm0_keep	pwm0_inv	pwm0_enable
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2]	RW	pwm0_keep	PWM output mode 0: PWM0 outputs a fixed number of square waves. 1: PWM0 always outputs square waves.
[1]	RW	pwm0_inv	PWM output control 0: PWM0 outputs square waves in normal mode. 1: PWM0 outputs square waves in inverted mode.
[0]	RW	pwm0_enable	PWM enable 0: disabled 1: enabled

PWM0_STATE0

PWM0_STATE0 is a PWM0 status 0 register.

Offset Address	Register Name	Total Reset Value
0x0010	PWM0_STATE0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								pwm0_period_st																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:0]	RO	pwm0_period_st	Number of count cycles for the internal module of PWM0

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PWM0_STATE1

PWM0_STATE1 is a PWM0 status 1 register.

Offset Address	Register Name	Total Reset Value
0x0014	PWM0_STATE1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved								pwm0_duty_st																							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:26]	RO	reserved	Reserved
[25:0]	RO	pwm0_duty_st	Number of high level beats for the internal module of PWM0

PWM0_STATE2

PWM0_STATE2 is a PWM0 status 2 register.

Offset Address	Register Name	Total Reset Value
0x0018	PWM0_STATE2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved												pwm0_cnt_st								pwm0_keep_st	pwm0_busy	pwm0_period_st									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	reserved	Reserved
[21:12]	RO	pwm0_cnt_st	Number of remaining square waves output by PWM0. This field is valid only when the following conditions are met: pwm0_busy = 1, pwm0_keep_st = 0
[11]	RO	pwm0_keep_st	Square wave output mode for the internal module of PWM0 0: PWM0 outputs a fixed number of square waves 1: PWM0 always outputs square waves.
[10]	RO	pwm0_busy	Working status of PWM0 0: Wave output is complete and PWM0 is idle. 1: PWM0 is outputting square waves.
[9:0]	RO	pwm0_period_st	Number of output square waves for the internal module of PWM0

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13 Security Modules

13.1 CIPHER

13.1.1 Overview

The cipher module supports data encryption and decryption by following the data encryption standard (DES), 3DES, or advanced encryption standard (AES) algorithm. The DES, 3DES, and AES algorithms are implemented according to FIPS46-3 and FIPS 197 standards. The DES/3DES and AES operating modes comply with FIPS-81 and NIST special 800-38a standards.

The cipher module can encrypt or decrypt a large amount of data effectively. In addition, it encrypts and decrypts one or more blocks at one time.

13.1.2 Features

The cipher module has the following features:

- Supports the AES key length of 128 bits, 192 bit, or 256 bits. If keys are configured by the key management module, the key length can be set only to 128 bits.
- Supports the DES key length of 64 bits. The values for bit 0, bit 8, bit 16, bit 24, bit 32, bit 40, bit 48, and bit 56 represent the parity check values for eight bytes respectively. The parity check values are not used during encryption or decryption.
- Supports 3-key and 2-key modes for the 3DES algorithm. If keys are configured by the key management module, only the 2-key mode is supported.
- Supports the operating modes of electronic codebook (ECB), cipher block chaining (CBC), 1-/8-/128-cipher feedback (CFB), 128-output feedback (OFB), and counter (CTR) for the AES algorithm. These operating modes comply with the NIST special 800-38a standard.
- Supports the operating modes of ECB, CBC, 1-/8-/64-CFB, and 1-/8-/64-OFB for the DES or 3DES algorithm. These operating modes comply with the FIPS-81 standard.
- Encrypts and decrypts one or more blocks at one time in ECB, CBC, CFB, OFB or CTR operating mode.
- Encrypts and decrypts one or more blocks at one time in CTR operating mode using the AES algorithm.
- Provides eight encryption/decryption keys (64 bits, 128 bits, 192 bits, or 256 bits) configured by the CPU.

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- Provides eight keys (fixed at 128 bits) configured by the key management module. The master CPU supports the write operation but not the read operation.
- Provides a single-block encryption/decryption channel and seven multi-block encryption/decryption channels. The single-block encryption/decryption channel can encrypt or decrypt a single block only at one time. In this case, the CPU writes data to the channel register and reads the results. For the multi-block encryption/decryption channel, the logic reads data from the DDR, and writes the encrypted or decrypted data to the DDR automatically.
- Supports weighted round robin policy for each channel. For a single-block channel, the weighted value is 1 by default; for a multi-block channel, the weighted value is configurable.
- Supports the same set of keys or different sets of keys for any channel.
- Keeps the data in the last incomplete block unprocessed when the data of the multi-block channels is not an integral multiple of encryption/decryption blocks.
- Supports data combination for multi-block channels. To be specific, if the remaining data in a linked list data block is insufficient to form an encryption/decryption block and the data block is not the last one to be processed, the remaining data is combined with the data in the next data block for encryption or decryption. This avoids data stuffing.
- Supports byte address for the multi-block encryption/decryption channel.
- Supports the multi-linked-list structure for the multi-block encryption/decryption channel and supports the combination of data from multiple linked lists. The linked list length is 20 bits. That is, the maximum data amount is 1 MB minus 1.
- Queries the interrupt status and masks and clears interrupts.
- Separately processes and controls interrupts for each channel.
- Supports multi-packet interrupts and aging interrupts.

13.1.3 Function Description

The operating modes of the DES, 3DES, and AES algorithms comply with the FIPS-81 and NIST special 800-38a standards. In the DES, 3DES, and AES algorithms, the ECB, CBC, and CFB operating modes are identical; however, the CTR (for the AES algorithm only) and OFB operating modes are slightly different.

3DES Algorithm

The 3DES algorithm supports both 3-key and 2-key operations. A 2-key operation can be regarded as a simplified 3-key operation. To be specific, key 3 is represented by key 1 in a 2-key operation.

Figure 13-1 shows the 3DES encryption of a 3-key operation and a 2-key operation.

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Figure 13-1 3DES encryption of a 3-key operation and a 2-key operation

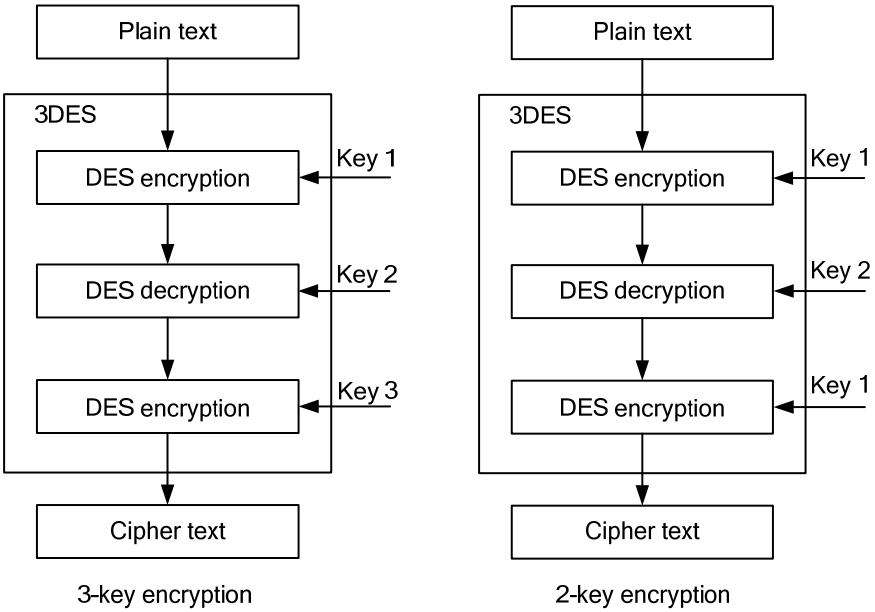
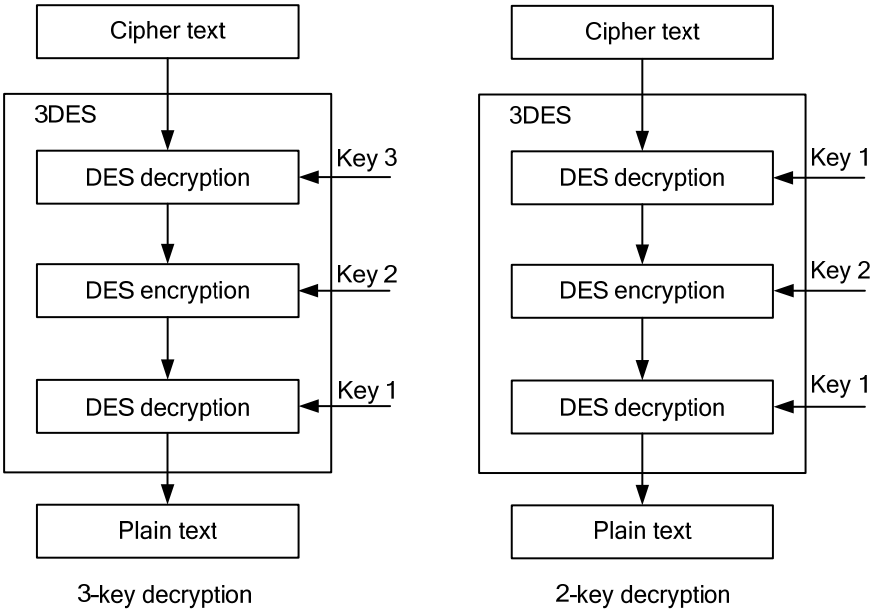


Figure 13-2 shows the 3DES decryption of a 3-key operation and a 2-key operation.

Figure 13-2 3DES decryption of a 3-key operation and a 2-key operation



ECB Mode

In ECB mode, encryption and decryption algorithms are directly applied to the block data. The operation of each block is independent. With this feature, the plain text encryption and cipher text decryption can be performed concurrently. Figure 13-3 shows the ECB mode of the AES and DES algorithms, and Figure 13-4 shows the ECB mode of the 3DES algorithm.

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Figure 13-3 ECB mode of the AES and DES algorithms

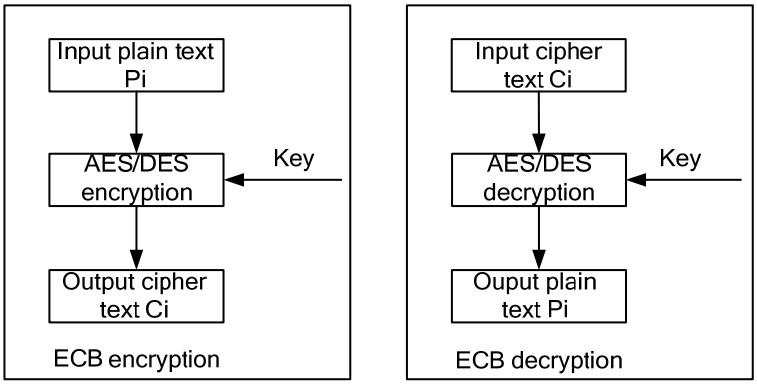
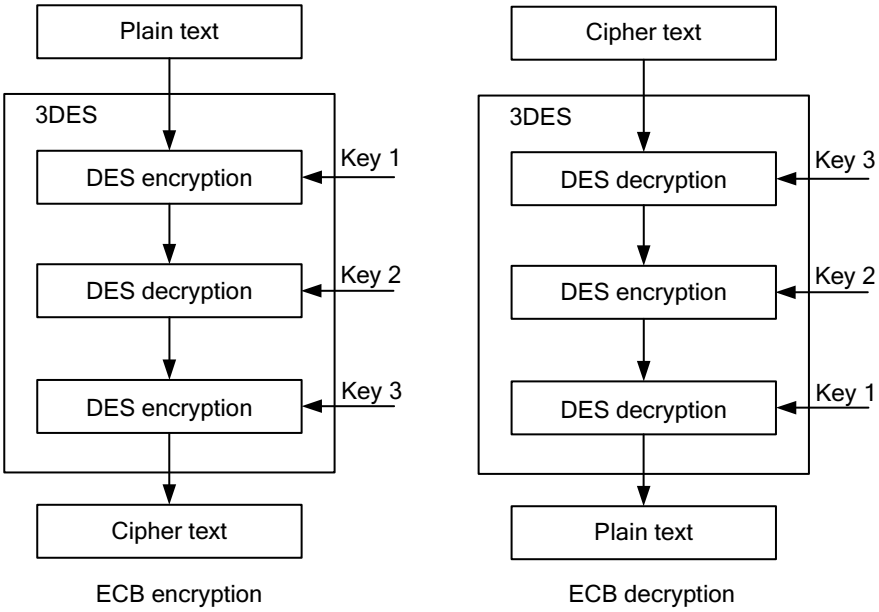


Figure 13-4 ECB mode of the 3DES algorithm

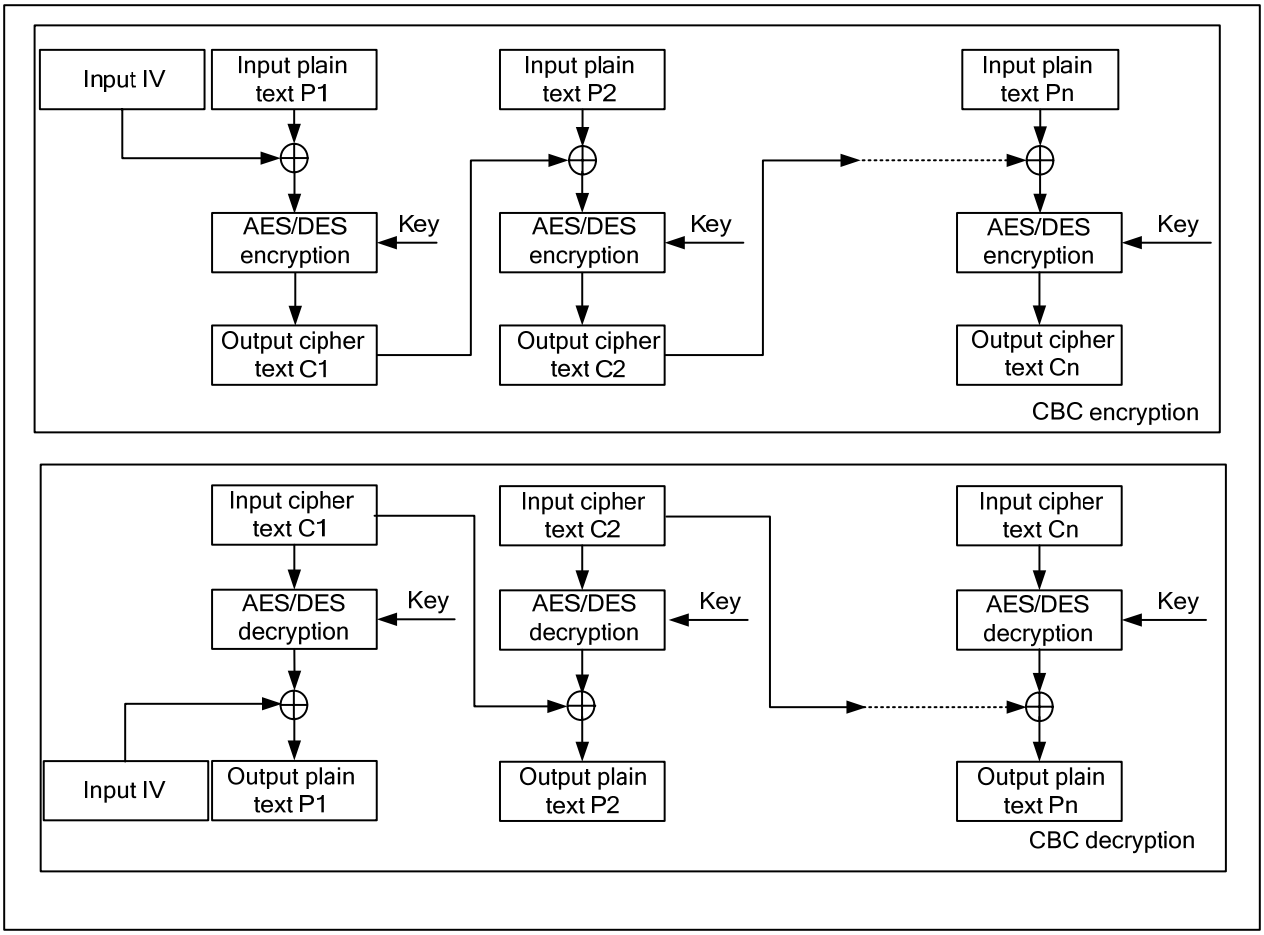


CBC Mode

In CBC mode, the encrypted input plain text block must be exclusively ORed with the input initialization vector (IV) before being encrypted. The encryption processing of each plain text block is related to the block processing result (cipher text) of the previous plain text. Therefore, encryption operations cannot be concurrently performed in CBC mode. The decryption operation, however, is independent of output plain text of the previous block. Therefore, decryption operations can be performed concurrently. Figure 13-5 shows the CBC mode of the AES and DES algorithms, and Figure 13-6 shows the CBC mode of the 3DES algorithm.

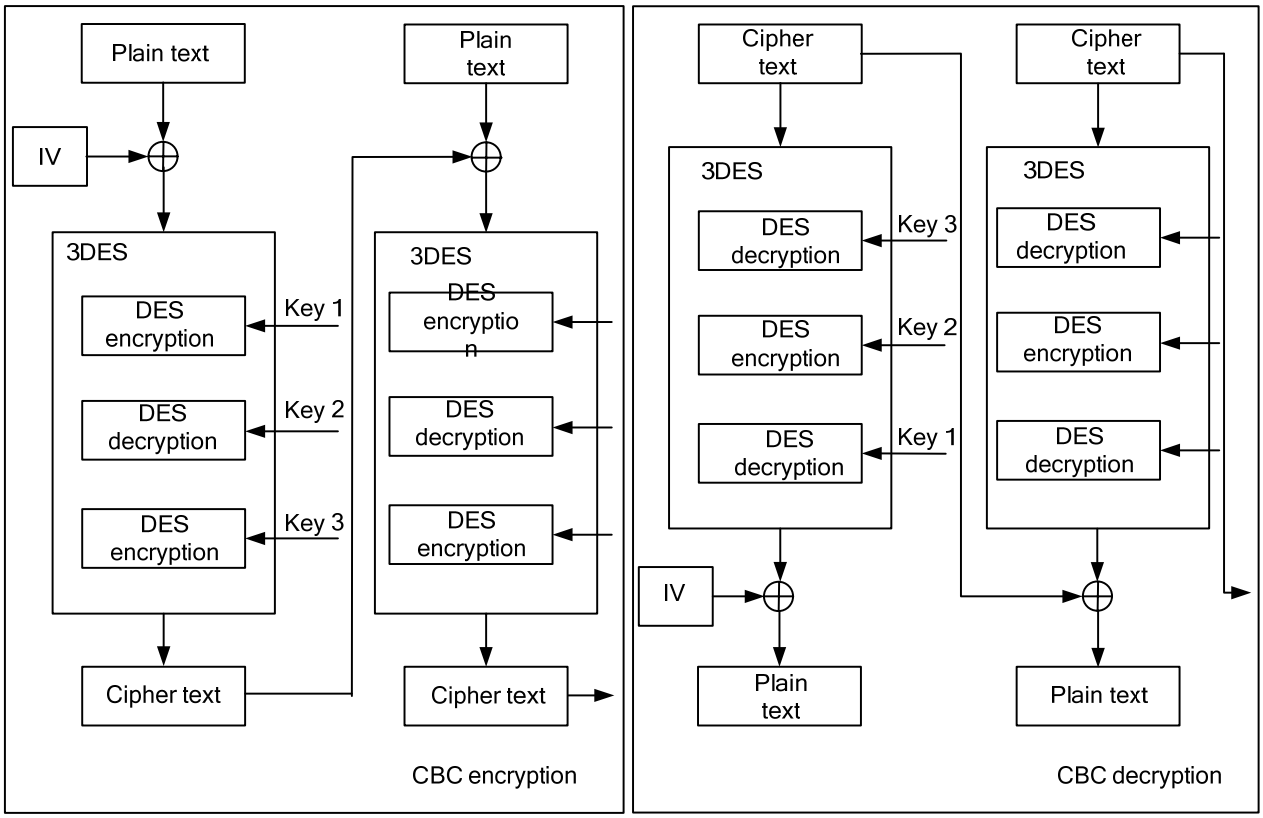
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Figure 13-5 CBC mode of the AES and DES algorithms



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Figure 13-6 CBC mode of the 3DES algorithm



CFB Mode

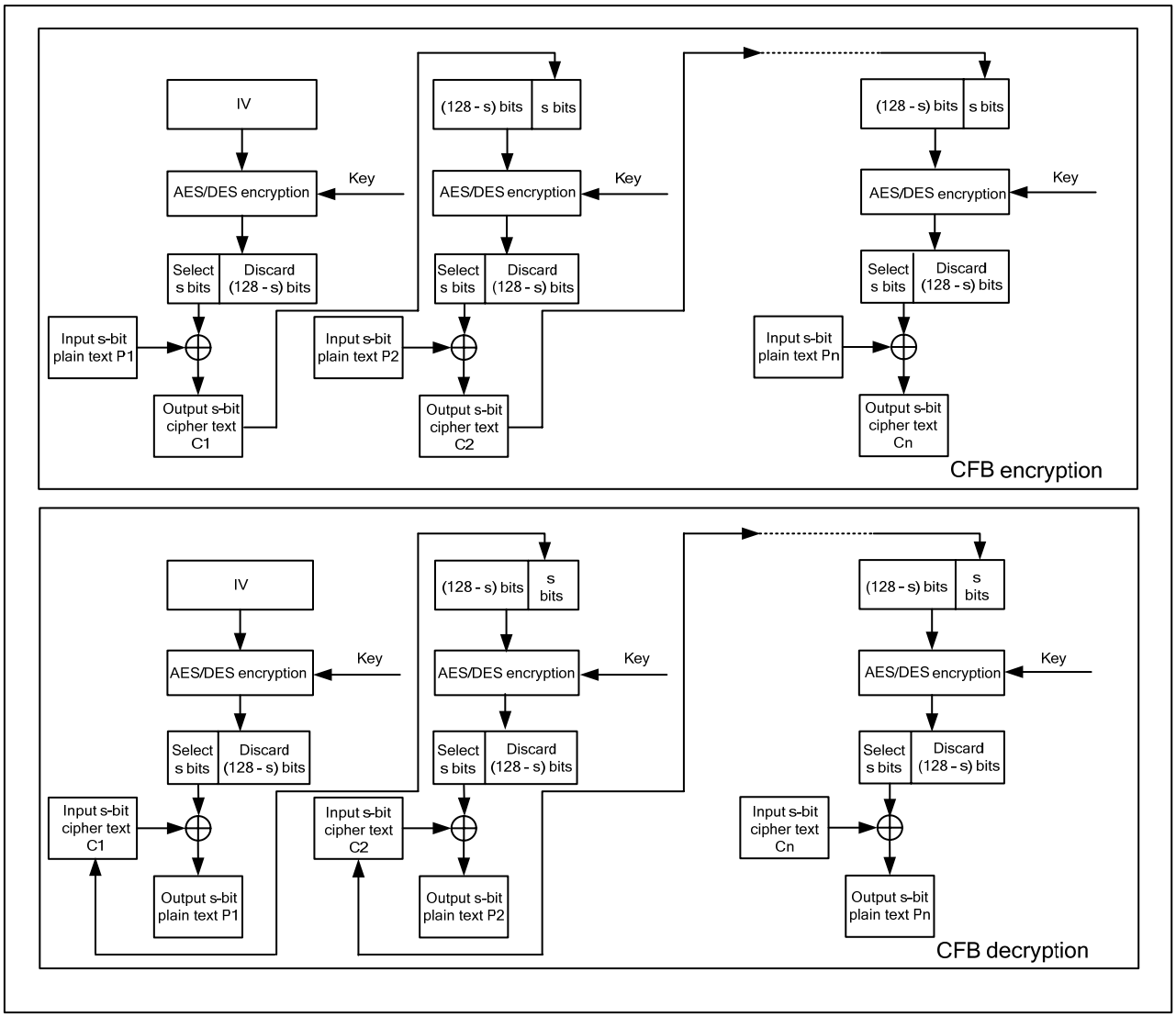
The CFB mode is used to convert a block cipher into a stream cipher. This mode is implemented by selecting the operation bits of the CFB. The shift operation bits are represented by the letter s. The value of s is as follows:

- 1 bit, 8 bits, or 64 bits for the DES or 3DES algorithm
- 1 bit, 8 bits, or 128 bits for the AES algorithm

Figure 13-7 shows the s-bit CFB mode of the AES and DES algorithms, and Figure 13-8 shows the s-bit CFB mode of the 3DES algorithm.

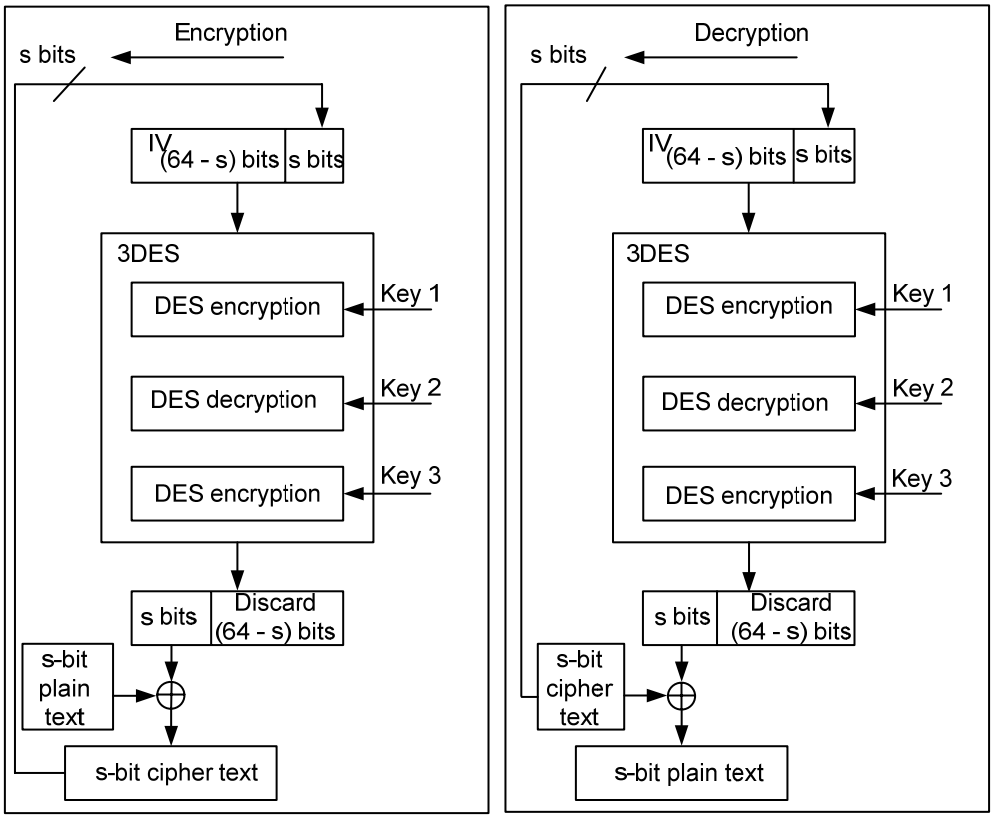
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Figure 13-7 S-bit CFB mode of the AES and DES algorithms



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Figure 13-8 S-bit CFB mode of the 3DES algorithm



OFB Mode

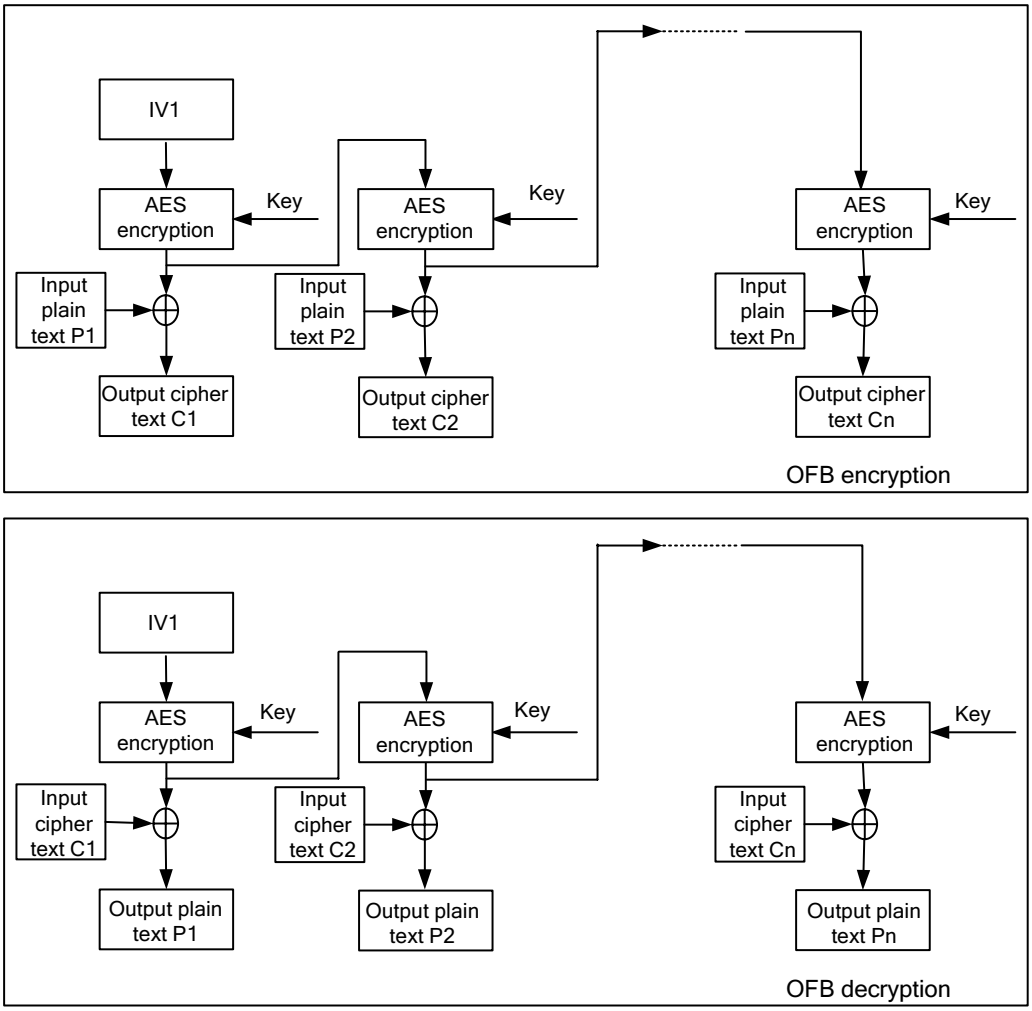
In OFB mode, IVs serve as the inputs during encryption. If a same key is used, different IVs must be used to ensure operation security. The value of the s bit is as follows:

- 1 bit, 8 bits, or 64 bits for the DES or 3DES algorithm
- 128 bits for the AES algorithm

Figure 13-9 shows the OFB mode of the AES algorithm.

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Figure 13-9 OFB mode of the AES algorithm



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Figure 13-10 shows the s-bit OFB mode of the DES algorithm.

Figure 13-10 S-bit OFB mode of the DES algorithm

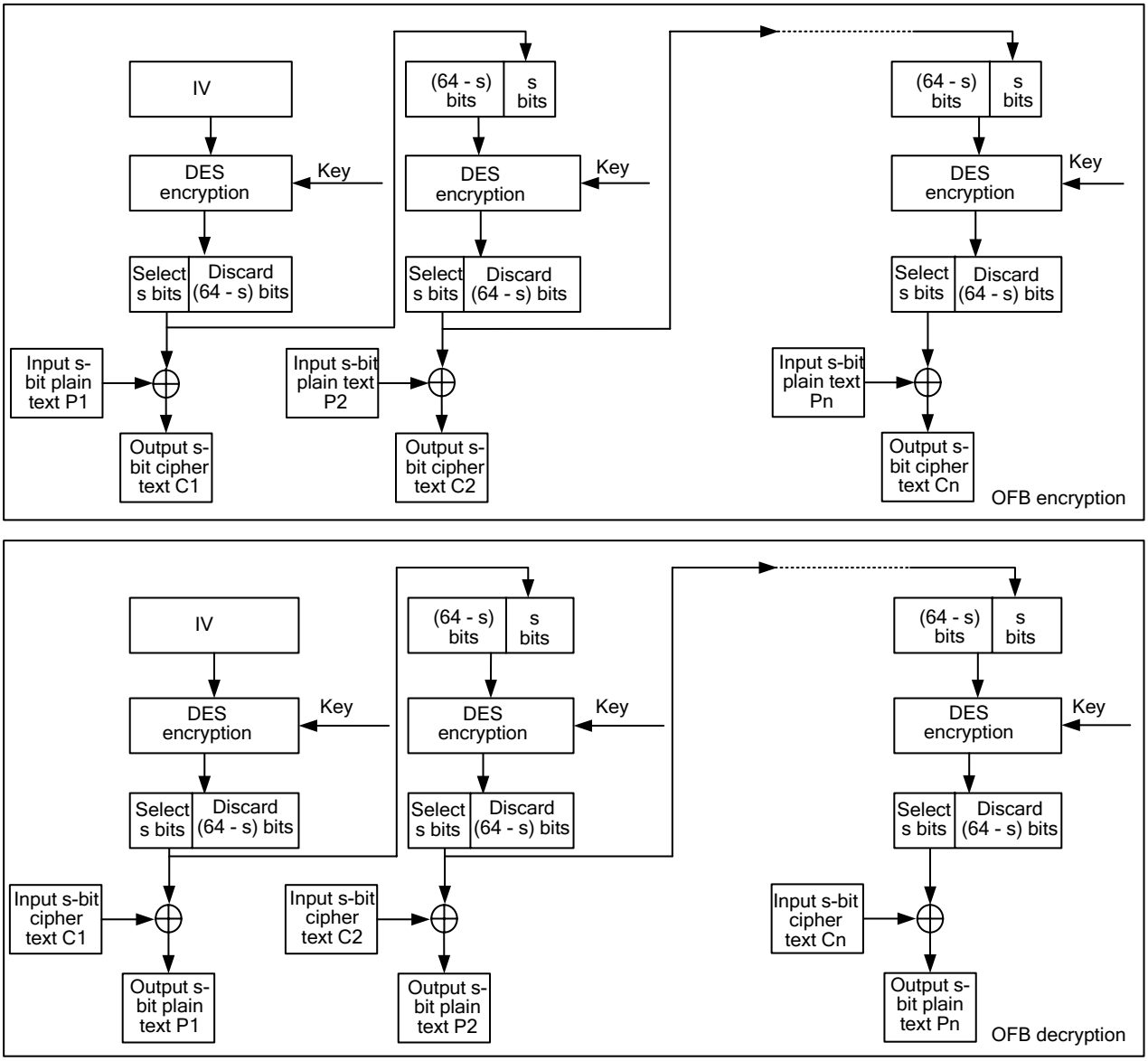
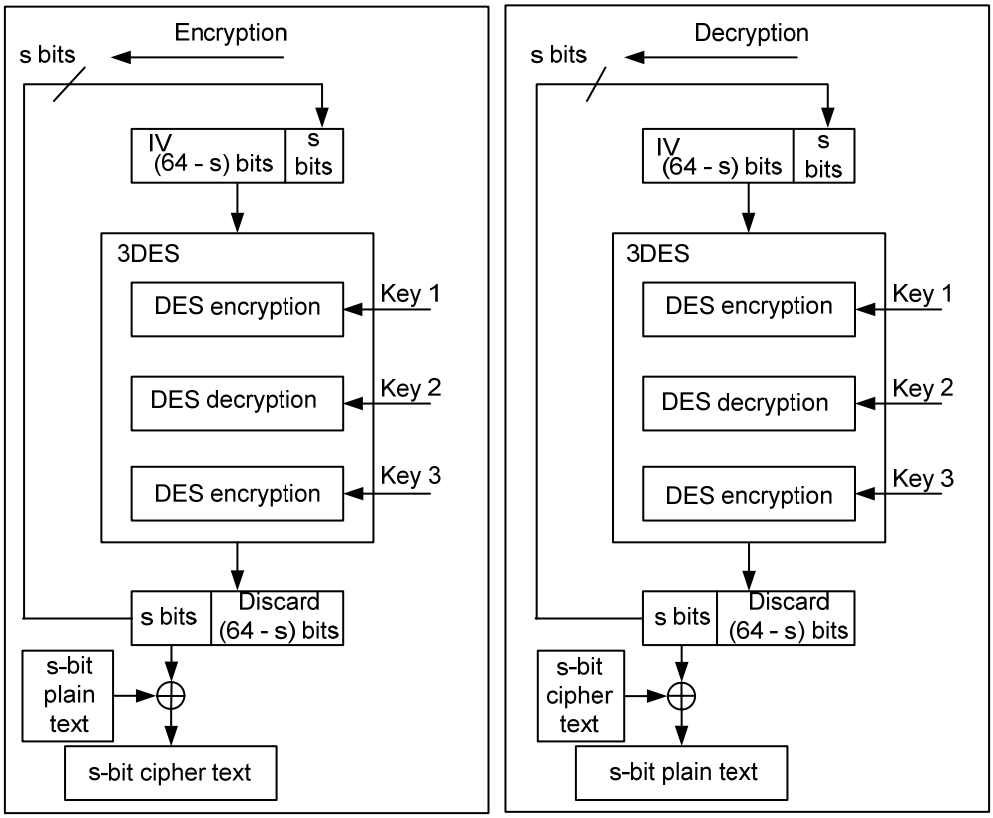


Figure 13-11 shows the s-bit OFB mode of the 3DES algorithm.

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Figure 13-11 S-bit OFB mode of the 3DES algorithm



CTR Mode

In CTR mode, different data segments are input to the cipher module by using the AES algorithm to ensure data security. Such data can be the count value CTRn. Therefore, CTRn determines the security of the CTR mode.

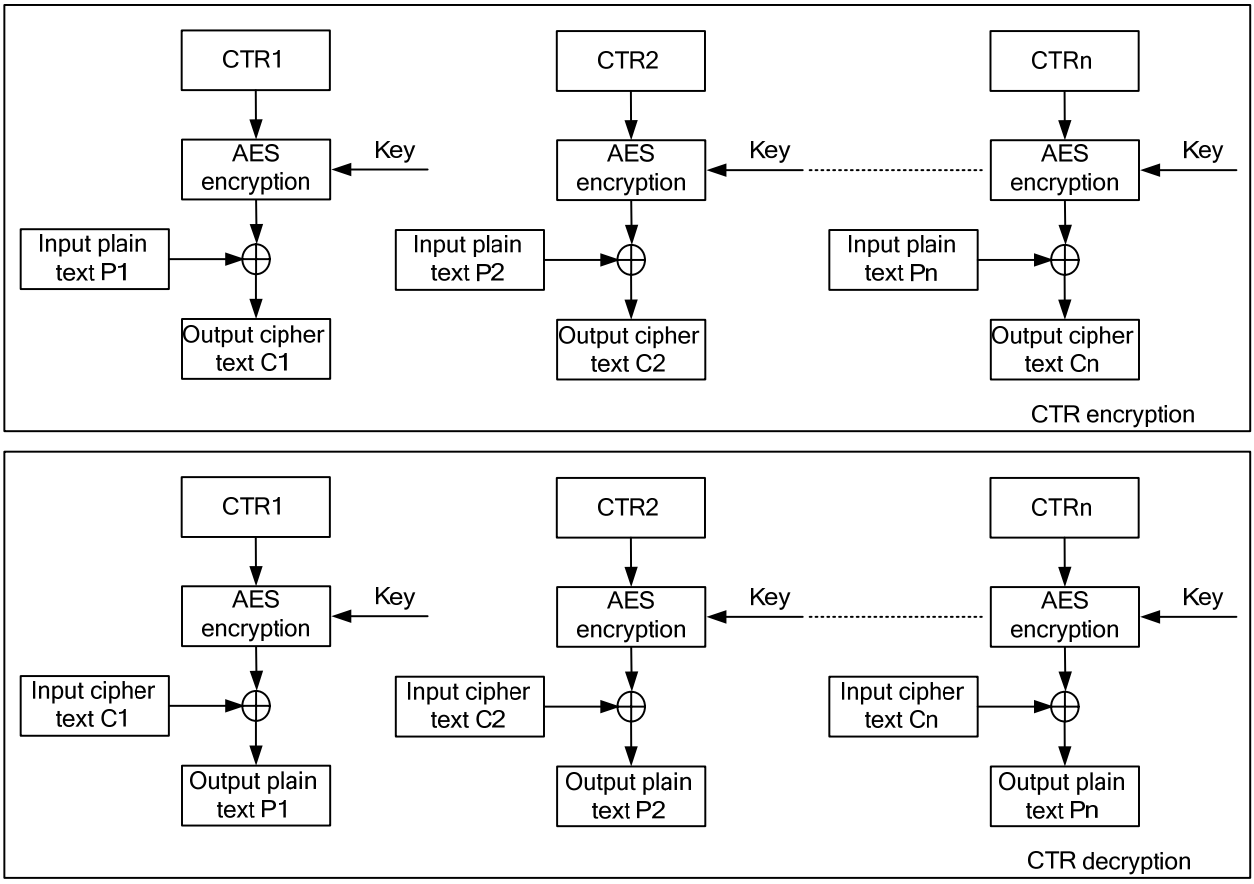
NOTE

CTRn is obtained by using the accumulation count mode.

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Figure 13-12 shows the CTR mode of the AES algorithm.

Figure 13-12 CTR mode of the AES algorithm



13.1.4 Operating Mode

Single-Block Operation Process of the Cipher Module

The cipher module provides channel 0 as the single-block encryption/decryption channel. A single-block operation is performed as follows:

Step 1 Check whether channel 0 is busy by querying the ch0_busy field of the configuration register CHAN0_CFG of channel 0. If channel 0 is not busy, configure data inputs and write related configuration information to the registers of channel 0.

Step 2 Write to the ch0_start field of CHAN0_CFG to enable channel 0 to start encryption or decryption.

Step 3 Check whether encryption and decryption of channel 0 are complete in either of following ways:

- Query the ch0_busy field. If ch0_busy indicates that channel 0 is not busy, encryption and decryption are complete.
- Check whether the interrupt of channel 0 is generated. If the interrupt is generated, encryption and decryption are complete.

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Step 4 Read the registers CHAN0_CIPHER_DOUT and CHAN0_CIPHER_IVOUT of channel 0.

----End

Multi-Block Operation Process of the Cipher Module

The cipher module provides seven multi-block encryption/decryption channels. The weighted value of each channel can be set using software based on its rate. These channels automatically read data from the DDR, and write the encrypted or decrypted data to the DDR.

A multi-block operation is performed as follows:

Step 1 Initialize the channels, including setting the depth, start address, number of multi-packet interrupts, aging interrupt time of the input and output queues of each channel, and setting the control register of each channel.

Step 2 When data needs to be encrypted or decrypted, query CHANn_IBUF_CNT. If the value of this register is smaller than the value of CHANn_IBUF_NUM, add the header of the data linked list corresponding to the data to be encrypted or decrypted to the input queue, and go to Step 4; otherwise, go to step 3.

Step 3 Enable the interrupt corresponding to the input queue channel, wait for the generation of the interrupt, read CHANn_IEMPTY_CNT, write to this register to clear the interrupt, and add new data to the input queue.

Step 4 Add the linked list header of the output buffer to the output queue.

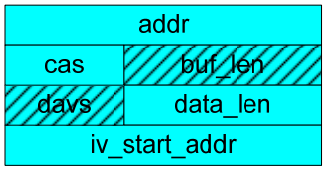
Step 5 Enable the interrupt corresponding to the output queue.

Step 6 Fetch data from the output queue, and write the number of currently received packets to CHANn_OFULL_CNT to clear the interrupt.

----End

Figure 13-13 shows the structure of the linked list header of a multi-block encryption/decryption channel.

Figure 13-13 Structure of the linked list header of a multi-block encryption/decryption channel



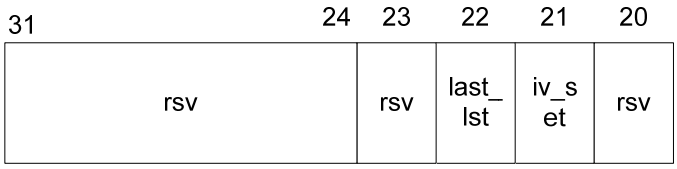
The field definitions are as follows:

- addr is the start address of the buffer pointer by the linked list header. The start address can be byte address.
- data_len is the length of the valid data pointed by the linked list header.
- cas is cipher control information.

Figure 13-14 shows the bits of cas.

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Figure 13-14 Bits of cas



- iv_set indicates that the IV of the data pointed by the current linked list header must be replaced. iv_start_addr indicates the start address of the IV in the DDR. This address must be aligned by word.
- last_lst indicates that the data pointed by the current linked list header is the last linked list of a data block. If the logic encounters an incomplete encryption/decryption block after processing the linked list, the logic writes the incomplete block to the output buffer without performing encryption and decryption.

Clock Gating

When no encryption is required and the cipher module is idle, the cipher clock can be disabled by configuring the registers of the system controller, which reduces power consumption.

Soft Reset

The cipher module can be soft-reset by configuring the registers of the system controller.

13.1.5 Register Summary

Table 13-1 describes cipher registers.

NOTE

The variable *n* in the offset addresses indicates the channel ID and ranges from 1 to 7.

Table 13-1 Summary of cipher registers (base address: 0x100C_0000)

Offset Address	Register	Description	Page
0x0000–0x000C	CHAN0_CIPHER_DOUT	Cipher output register for channel 0 (for single-block encryption/decryption)	13-16
0x0010–0x001C	CHAN0_CIPHER_IVOUT	Operation complete IV output register of the cipher module	13-16
0x0020–0x008C	CHAN_CIPHER_IVOUT	IV output register for channels 1–7	13-17
0x0090–0x018C	CIPHER_KEY	CPU configuration key register of the cipher module	13-18
0x1000	CHAN0_CIPHER_CTRL	Encryption/decryption control register for channel 0	13-19
0x1004–0x1010	CHAN0_CIPHER_IV	Cipher VI block input register for	13-21

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Offset Address	Register	Description	Page
	IN	channel 0	
0x1014–0x1020	CHAN0_CIPHER_DIN	128-bit block input register of the cipher module	13-22
0x1000 + *n* x 0x80	CHANn_IBUF_NUM	Input queue total depth register for channel n (n = 1–7) (linked list header count register)	13-23
0x1000 + *n* x 0x80 + 0x4	CHANn_IBUF_CNT	Pending data buffer count register for channel n in the input queue	13-23
0x1000 + *n* x 0x80 + 0x8	CHANn_IEMPTY_CNT	Processed data buffer count register for channel n in the input queue	13-24
0x1000 + *n* x 0x80 + 0xC	CHANn_INT_ICNTCFG	Input queue multi-packet interrupt threshold register for channel n	13-24
0x1000 + *n* x 0x80 + 0x10	CHANn_CIPHER_CTRL	Encryption/decryption control register for channel n	13-25
0x1000 + *n* x 0x80 + 0x14	CHANn_SRC_LST_SADDR	Input queue start address register for channel n	13-27
0x1000 + *n* x 0x80 + 0x18	CHANn_IAGE_TIMER	Input queue interrupt aging time configuration register for channel n	13-27
0x1000 + *n* x 0x80 + 0x3C	CHANn_OBUF_NUM	Output queue total depth register for channel n (linked list header count register)	13-28
0x1000 + *n* x 0x80 + 0x40	CHANn_OBUF_CNT	Pending data buffer count register for channel n in the output queue	13-28
0x1000 + *n* x 0x80 + 0x44	CHANn_OFULL_CNT	Processed data buffer count register for channel n in the output queue	13-29
0x1000 + *n* x 0x80 + 0x48	CHANn_INT_OCNTCFG	Output queue multi-packet interrupt threshold register for channel n	13-29
0x1000 + *n* x 0x80 + 0x4C	CHANn_DEST_LST_SADDR	Output queue start address register for channel n	13-29
0x1000 + *n* x 0x80 + 0x50	CHANn_OAGE_TIMER	Output queue interrupt aging time configuration register for channel n	13-30
0x1400	INT_STATUS	Interrupt status register	13-31
0x1404	INT_EN	Interrupt enable register	13-32
0x1408	INT_RAW	Raw interrupt status register	13-33
0x140C	RST_STATUS	Reset status indicator register	13-34
0x1410	CHAN0_CFG	Channel 0 configuration register	13-34

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13.1.6 Register Description

CHAN0_CIPHER_DOUT

CHAN0_CIPHER_DOUT is a cipher output register for channel 0 (for single-block encryption/decryption).

The data read from this register is the results of a single-block operation. The results of the AES algorithm are different from those of the DES or 3DES algorithm. The details are as follows:

- For the AES algorithm
  - If the 1-CFB mode is selected, the least significant bit (LSB) is valid, that is, CIPHER_DOUT bit[0] is valid.
  - If the 8-CFB mode is selected, lower eight bits are valid, that is, CIPHER_DOUT bit[7:0] is valid.
  - If the 128-CFB mode is selected, 128 bits are valid.
  - In other modes, 128 bits are valid.
- For the DES or 3DES algorithm
  - If the 1-CFB or 1-OFB mode is selected, the LSB is valid, that is, CIPHER_DOUT bit[0] is valid.
  - If the 8-CFB or 8-OFB mode is selected, lower eight bits are valid, that is, CIPHER_DOUT bit[7:0] is valid.
  - If the 64-CFB or 64-OFB mode is selected, lower 64 bits are valid, that is, CIPHER_DOUT bit[63:0] is valid.
  - In other modes, lower 64 bits are valid, that is, CIPHER_DOUT bit[63:0] is valid.

Offset Address	Register Name	Total Reset Value
0x0000–0x000C	CHAN0_CIPHER_DOUT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	chan0_cipher_dout																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	chan0_cipher_dout	128-bit block output of the cipher module. Each address maps to a 32-bit data segment. CIPHER_DOUT[31:0]: address 0x0000 CIPHER_DOUT[63:32]: address 0x0004 CIPHER_DOUT[95:64]: address 0x0008 CIPHER_DOUT[127:96]: address 0x000C

CHAN0_CIPHER_IVOUT

CHAN0_CIPHER_IVOUT is an operation complete IV output register of the cipher module.

Note the following points when reading this register:

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- This register can be ignored in ECB or CTR mode.
- If a single-block operation is performed, the data of this register is the vector output of the block. The data can be used as the vector input in the next block operation for the same data packet.
  - If the AES algorithm is selected, 128 bits are valid.
  - If the DES or 3DES algorithm is selected (CIPHER_CTRL[cipher_mode] = 0b00, 0b01, or 0b11), lower 64 bits are valid, that is, CIPHER_IVOUT bit[63:0] is valid.
- If a multi-block operation is performed, the data read from this register is the output vector of the last block operation.
  - If the AES algorithm is selected, 128 bits are valid.
  - If the DES or 3DES algorithm is selected, lower 64 bits are valid, that is, CIPHER_IVOUT bit[63:0] is valid.

Offset Address	Register Name	Total Reset Value
0x0010–0x001C	CHAN0_CIPHER_IVOUT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	chan0_cipher_ivout																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	chan0_cipher_ivout	Vector output after the operation of the cipher module is complete. It can be ignored in ECB or CTR mode. Each address maps to a 32-bit data segment. CIPHER_DOUT[31:0]: address 0x0010 CIPHER_IVOUT[63:32]: address 0x0014 CIPHER_IVOUT[95:64]: address 0x0018 CIPHER_IVOUT[127:96]: address 0x001C

CHAN_CIPHER_IVOUT

CHAN_CIPHER_IVOUT is an IV output register for channels 1–7.

Offset Address	Register Name	Total Reset Value
0x0020–0x008C	CHAN_CIPHER_IVOUT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	chan_cipher_ivout																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	chan_cipher_ivout	0x0020–0x002C: channel 1 0x0030–0x003C: channel 2 0x0040–0x004C: channel 3 0x0050–0x005C: channel 4

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			0x0060–0x006C: channel 5 0x0070–0x007C: channel 6 0x0080–0x008C: channel 7

CIPHER_KEY

CIPHER_KEY is a CPU configuration key register of the cipher module. The key is the configured value of the CPU, and the CPU can be read or written.

Note the following points when configuring this register:

- If the DES algorithm is selected, lower 64 bits are valid, that is, CIPHER_KEY[63:0] is valid.
- For the 3DES algorithm

  If a 3-key operation is performed (CIPHER_CTRL[key_length] = 0b00, 0b01, or 0b10), low 192 bits are valid.

  where

  - CIPHER_KEY bit[63:0] indicates key 1.
  - CIPHER_KEY bit[127:64] indicates key 2.
  - CIPHER_KEY bit[191:128] indicates key 3.

  If a 2-key operation is selected (CIPHER_CTRL[key_length] = 0b11), lower 128 bits are valid.

  where

  - CIPHER_KEY bit[63:0] indicates key 1.
  - CIPHER_KEY bit[127:64] indicates key 2.
- For the AES algorithm
  - If a 128-bit key operation is performed, lower 128 bits are valid, that is, CIPHER_KEY bit[127:0] is valid.
  - If a 192-bit key operation is performed, lower 192 bits are valid, that is, CIPHER_KEY bit[191:0] is valid.
  - If a 256-bit key operation is performed, 256 bits are valid.

The cipher module allows you to configure eight keys. Each channel can use one key, and multiple channels can share one key.

Offset Address	Register Name	Total Reset Value
0x0090–0x018C	CIPHER_KEY	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	cipher_key																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	cipher_key	Key input of the cipher module. Each address maps to a 32-bit data segment. CIPHER_KEY[31:0]: address 0x0090

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			CIPHER_KEY[63:32]: address 0x0094 CIPHER_KEY[95:64]: address 0x0098 CIPHER_KEY[127:96]: address 0x009C CIPHER_KEY[159: 128]: address 0x00A0 CIPHER_KEY[191:160]: address 0x00A4 CIPHER_KEY[223:192]: address 0x00A8 CIPHER_KEY[255:224]: address 0x00AC 0x0090–0x00AC: host_key0 0x00B0–0x00CC: host_key1 0x00D0–0x00EC: host_key2 0x00F0–0x010C: host_key3 0x0110–0x012C: host_key4 0x0130–0x014C: host_key5 0x0150–0x016C: host_key6 0x0170–0x018C: host_key7

CHAN0_CIPHER_CTRL

CHAN0_CIPHER_CTRL is an encryption/decryption control register for channel 0. Channel 0 is a single-block encryption/decryption channel.

Note the following points when configuring this register:

- Configure this register before configuring others registers of the cipher module.
- In the modes except the CFB mode of the AES algorithm, the CIPHER_CTRL[width] cannot be set to 01 or 10.
- In the modes except the CFB and OFB modes of the DES and 3DES algorithms, CIPHER_CTRL[width] cannot be set to 01 or 10.

Offset Address	Register Name	Total Reset Value
0x1000	CHAN0_CIPHER_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																key_adder		key_sel	reserved	reserved	key_length		ivin_sel	width		alg_sel		mode			decrypt
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:17]	RO	reserved	Reserved

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[16:14]	RW	key_adder	ID of the key used by the current channel 000: host_key0 001: host_key1 010: host_key2 011: host_key3 100: host_key4 101: host_key5 110: host_key6 111: host_key7
[13]	RW	key_sel	Key select 0: keys configured by the CPU 1: keys generated by the key management module
[12]	RO	reserved	Reserved
[11]	RO	reserved	Reserved
[10:9]	RW	key_length	Key length select For the AES algorithm: 00: 128 bits 01: 192 bits 10: 256 bits 11: 128 bits For the DES algorithm: 00: 3 keys 01: 3 keys 10: 3 keys 11: 2 keys
[8]	RW	ivin_sel	Input select of CIPHER_IVIN 0: do not configure CIPHER_IVIN 1: configure CIPHER_IVIN
[7:6]	RW	width	Bit width control For the DES or 3DES algorithm: 00: 64 bits 01: 8 bits 10: 1 bit 11: 64 bits For the AES algorithm: 00: 128 bits 01: 8 bits 10: 1 bit 11: 128 bits

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[5:4]	RW	alg_sel	Algorithm select 00: DES algorithm 01: 3DES algorithm 10: AES algorithm 11: DES algorithm
[3:1]	RW	mode	Operating mode select For the AES algorithm: 000: ECB mode 001: CBC mode 010: CFB mode 011: OFB mode 100: CTR mode Other values: ECB mode For the DES algorithm: 000: ECB mode 001: CBC mode 010: CFB mode 011: OFB mode Other values: ECB mode
[0]	RW	decrypt	Encryption/decryption select 0: encryption 1: decryption

CHAN0_CIPHER_IVIN

CHAN0_CIPHER_IVIN is a cipher VI block input register for channel 0.

Assume that channel 0 is selected for the single-block encryption/decryption and the selected mode is not ECB mode (CIPHER_CTRL[mode] = 0b001, 0b010, 0b011, or 0b100).

- If you do not want to configure the input vector (CIPHER_CTRL[ivin_sel] = 0b0), CIPHER_IVIN can be ignored.
- If you want to configure the input vector (CIPHER_CTRL[ivin_sel] = 0b1), CIPHER_IVIN needs to be configured. If the AES algorithm is selected (CIPHER_CTRL [alg_sel] = 0b10), CIPHER_IVIN bit[127:0] is valid. If the DES or 3DES algorithm is selected (CIPHER_CTRL[alg_sel] = 0b00, 0b01, or 0b11), lower 64 bits are valid, that is, CIPHER_IVIN bit[63:0] is valid.

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Offset Address	Register Name	Total Reset Value
0x1004–0x1010	CHAN0_CIPHER_IVIN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	chan0_cipher_ivin																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	chan0_cipher_ivin	128-bit IV of the cipher module for channel 0 or the data input from the counter. Each address maps to a 32-bit data segment. CIPHER_IVIN[31:0]: address 0x1004 CIPHER_IVIN[63:32]: address 0x1008 CIPHER_IVIN[95:64]: address 0x100C CIPHER_IVIN[127:96]: address 0x1010

CHAN0_CIPHER_DIN

CHAN0_CIPHER_DIN is a 128-bit block input register of the cipher module.

Note the following points when configuring this register:

- If channel 0 is selected for the single-block operation, this register needs to be configured.
- Assume that the AES algorithm (CIPHER_CTRL[alg_sel] = 0b10) is selected.
  - If the 1-CFB mode is selected, the LSB is valid, that is, CIPHER_DIN bit[0] is valid.
  - If the 8-CFB mode is selected, lower eight bits are valid, that is, CIPHER_DIN bit[7:0] is valid.
  - If the 128-CFB mode is selected, 128 bits are valid.
  - In other modes, 128 bits are valid.
- Assume that the DES or the 3DES algorithm (CIPHER_CTRL[alg_sel] = 0b00, 0b01, or 0b11) is selected.
  - If the 1-CFB or 1-OFB mode is selected, the LSB is valid, that is, CIPHER_DIN bit[0] is valid.
  - If the 8-CFB or 8-OFB mode is selected, lower eight bits are valid, that is, CIPHER_DIN bit[7:0] is valid.
  - If the 64-CFB or 64-OFB mode is selected, lower 64 bits are valid, that is, CIPHER_DIN bit[63:0] is valid.
  - In other modes, lower 64 bits are valid, that is, CIPHER_DIN bit[63:0] is valid.

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Offset Address	Register Name	Total Reset Value
0x1014–0x1020	CHAN0_CIPHER_DIN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	chan0_cipher_din																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	chan0_cipher_din	128-bit block input of the cipher module for channel 0. Each address maps to a 32-bit data segment. CIPHER_DIN[31:0]: address 0x1014 CIPHER_DIN[63:32]: address 0x1018 CIPHER_DIN[95:64]: address 0x101c CIPHER_DIN[127:96]: address 0x1020

CHANn_IBUF_NUM

CHANn_IBUF_NUM is an input queue total depth register for channel *n* (*n* = 1–7). This register can be used to configure the count of linked list headers.

Offset Address	Register Name	Total Reset Value
0x1000 + *n* x 0x80	CHANn_IBUF_NUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ibuf_num															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	ibuf_num	Input queue depth, that is, count of linked list headers configured for each channel

CHANn_IBUF_CNT

CHANn_IBUF_CNT is a pending data buffer count register for channel *n* in the input queue. When this register is written by using software, the logic adds the original value of the register and the written value. After the logic processes a buffer, the value of this register is decreased by 1.

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Offset Address	Register Name	Total Reset Value
0x1000 + *n* x 0x80 + 0x4	CHANn_IBUF_CNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ibuf_cnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	ibuf_cnt	Count of buffers to be processed in the input queue

CHANn_IEMPTY_CNT

CHANn_IEMPTY_CNT is a processed buffer count register for channel *n* in the input queue. When this register is written by software, the logic subtracts the written value from the original value of this register. After the logic processes a buffer, the value of this register is increased by 1.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x8	CHANn_IEMPTY_CNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																iempty_cnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	iempty_cnt	Count of processed buffers in the input queue

CHANn_INT_ICNTCFG

CHANn_INT_ICNTCFG is an input queue multi-packet interrupt threshold register for channel *n*. When the count of buffers in the input queue processed by the logic is above the threshold, an input queue interrupt is reported.

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Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0xC	CHANn_INT_ICNTCFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																int_icnt_cfg															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	int_icnt_cfg	Input queue multi-packet interrupt threshold

CHANn_CIPHER_CTRL

CHANn_CIPHER_CTRL is an encryption/decryption control register for channel *n*.

Note the following points when configuring this register:

- You must configure this register before performing encryption or decryption using the channel.
- In the modes other than the CFB mode of the AES algorithm, CIPHER_CTRL[width] cannot be set to 01 or 10.
- In the modes other than the CFB and OFB modes of the DES or 3DES algorithm, CIPHER_CTRL[width] cannot be set to 01 or 10.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x10	CHANn_CIPHER_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	weight										reserved						key_adder				key_sel	byte_seq	ts_vld		key_length			reserved	width		alg_sel		mode			decrypt
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:22]	RO	weight	Weighted value of the current channel, in the unit of 64 bytes
[21:17]	RO	reserved	Reserved
[16:14]	RW	key_adder	ID of the key used by the current channel. The key can be any one in the addresses 0–7.
[13]	RW	key_sel	Key select 0: keys configured by the CPU 1: keys generated by the key management module
[12:11]	RO	reserved	Reserved

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[10:9]	RW	key_length	Key length select For the AES algorithm: 00: 128 bits 01: 192 bits 10: 256 bits 11: 128 bits For the DES algorithm: 00: 3 keys 01: 3 keys 10: 3 keys 11: 2 keys
[8]	RO	reserved	Reserved
[7:6]	RW	width	Bit width control For the DES or 3DES algorithm: 00: 64 bits 01: 8 bits 10: 1 bit 11: 64 bits For the AES algorithm: 00: 128 bits 01: 8 bits 10: 1 bit 11: 128 bits
[5:4]	RW	alg_sel	Algorithm select 00: DES algorithm 01: 3DES algorithm 10: AES algorithm 11: DES algorithm

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[3:1]	RW	mode	Operating mode select For the AES algorithm: 000: ECB mode 001: CBC mode 010: CFB mode 011: OFB mode 100: CTR mode Other values: ECB mode For the DES algorithm: 000: ECB mode 001: CBC mode 010: CFB mode 011: OFB mode Other values: ECB mode
[0]	RW	decrypt	Encryption/decryption select 0: encryption 1: decryption

CHANn_SRC_LST_SADDR

CHANn_SRC_LST_SADDR is an input queue start address register for channel n. The address must be aligned by word.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x14	CHANn_SRC_LST_SADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	src_lst_saddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	src_lst_saddr	Start address for the input queue

CHANn_IAGE_TIMER

CHANn_IAGE_TIMER is an input queue interrupt aging time configuration register for channel *n*. If overflow occurs in the aging time counter and the count of processed buffers in the input queue is greater than 0, an input queue processing complete interrupt is reported.

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Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x18	CHANn_IAGE_TIMER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																iage_timer															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	iage_timer	Aging interrupt timer

CHANn_OBUF_NUM

CHANn_OBUF_NUM is an output queue total depth register for channel *n*. This register can be used to configure the count of linked list headers.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x3C	CHANn_OBUF_NUM	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																obuf_num															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	obuf_num	Total depth of the output queue

CHANn_OBUF_CNT

CHANn_OBUF_CNT is a pending data buffer count register for channel *n* in the output queue. When this register is written by using software, the logic adds the original value of the register and the written value. After the logic processes a buffer, the value of this register is decreased by 1.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x40	CHANn_OBUF_CNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																obuf_cnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved

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[15:0]	RW	obuf_cnt	Count of buffers to be processed in the output queue

CHANn_OFULL_CNT

CHANn_OFULL_CNT is a processed buffer count register for channel *n* in the output queue. When this register is written by software, the logic subtracts the written value from the original value of this register. After the logic processes a buffer, the value of this register is increased by 1.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x44	CHANn_OFULL_CNT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ofull_cnt															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	ofull_cnt	Count of processed buffers in the output queue

CHANn_INT_OCNTCFG

CHANn_INT_OCNTCFG is an output queue multi-packet interrupt threshold register for channel *n*. When the count of buffers in the output queue processed by the logic is above the threshold, an output queue interrupt is reported.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x48	CHANn_INT_OCNTCFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																int_ocnt_cfg															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	int_ocnt_cfg	Output queue multi-packet interrupt threshold

CHANn_DEST_LST_SADDR

CHANn_DEST_LST_SADDR is an output queue start address register for channel *n*. The address must be aligned by word.

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Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x4C	CHANn_DEST_LST_SADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dest_lst_saddr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dest_lst_saddr	Start address for the output queue

CHANn_OAGE_TIMER

CHANn_OAGE_TIMER is an output queue interrupt aging time configuration register for channel *n*. If overflow occurs in the aging time counter and the count of processed buffers in the output queue is greater than 0, an output queue processing complete interrupt is reported.

Offset Address	Register Name	Total Reset Value
0x1000 + n x 0x80 + 0x50	CHANn_OAGE_TIMER	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																oage_timer															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RW	oage_timer	Aging interrupt timer

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INT_STATUS

INT_STATUS is an interrupt status register.

Offset Address	Register Name	Total Reset Value
0x1400	INT_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ch7_ibuf_int	ch6_ibuf_int	ch5_ibuf_int	ch4_ibuf_int	ch3_ibuf_int	ch2_ibuf_int	ch1_ibuf_int	ch0_ibuf_int	ch7_obuf_int	ch6_obuf_int	ch5_obuf_int	ch4_obuf_int	ch3_obuf_int	ch2_obuf_int	ch1_obuf_int	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RO	ch7_ibuf_int	Input queue data interrupt for channel 7
[14]	RO	ch6_ibuf_int	Input queue data interrupt for channel 6
[13]	RO	ch5_ibuf_int	Input queue data interrupt for channel 5
[12]	RO	ch4_ibuf_int	Input queue data interrupt for channel 4
[11]	RO	ch3_ibuf_int	Input queue data interrupt for channel 3
[10]	RO	ch2_ibuf_int	Input queue data interrupt for channel 2
[9]	RO	ch1_ibuf_int	Input queue data interrupt for channel 1
[8]	RO	ch0_ibuf_int	Data processing complete interrupt for channel 0
[7]	RO	ch7_obuf_int	Output queue data interrupt for channel 7
[6]	RO	ch6_obuf_int	Output queue data interrupt for channel 6
[5]	RO	ch5_obuf_int	Output queue data interrupt for channel 5
[4]	RO	ch4_obuf_int	Output queue data interrupt for channel 4
[3]	RO	ch3_obuf_int	Output queue data interrupt for channel 3
[2]	RO	ch2_obuf_int	Output queue data interrupt for channel 2
[1]	RO	ch1_obuf_int	Output queue data interrupt for channel 1
[0]	RO	reserved	Reserved

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INT_EN

INT_EN is an interrupt enable register.

Offset Address	Register Name	Total Reset Value
0x1404	INT_EN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	int_en	reserved															ch7_ibuf_en	ch6_ibuf_en	ch5_ibuf_en	ch4_ibuf_en	ch3_ibuf_en	ch2_ibuf_en	ch1_ibuf_en	ch0_ibuf_en	ch7_obuf_en	ch6_obuf_en	ch5_obuf_en	ch4_obuf_en	ch3_obuf_en	ch2_obuf_en	ch1_obuf_en	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RW	int_en	Total interrupt enable for the cipher module
[30:16]	RO	reserved	Reserved
[15]	RW	ch7_ibuf_en	Input queue data interrupt enable for channel 7
[14]	RW	ch6_ibuf_en	Input queue data interrupt enable for channel 6
[13]	RW	ch5_ibuf_en	Input queue data interrupt enable for channel 5
[12]	RW	ch4_ibuf_en	Input queue data interrupt enable for channel 4
[11]	RW	ch3_ibuf_en	Input queue data interrupt enable for channel 3
[10]	RW	ch2_ibuf_en	Input queue data interrupt enable for channel 2
[9]	RW	ch1_ibuf_en	Input queue data interrupt enable for channel 1
[8]	RW	ch0_ibuf_en	Data processing complete interrupt enable for channel 0
[7]	RW	ch7_obuf_en	Output queue data interrupt enable for channel 7
[6]	RW	ch6_obuf_en	Output queue data interrupt enable for channel 6
[5]	RW	ch5_obuf_en	Output queue data interrupt enable for channel 5
[4]	RW	ch4_obuf_en	Output queue data interrupt enable for channel 4
[3]	RW	ch3_obuf_en	Output queue data interrupt enable for channel 3
[2]	RW	ch2_obuf_en	Output queue data interrupt enable for channel 2
[1]	RW	ch1_obuf_en	Output queue data interrupt enable for channel 1
[0]	RO	reserved	Reserved

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INT_RAW

INT_RAW is a raw interrupt status register.

Offset Address	Register Name	Total Reset Value
0x1408	INT_RAW	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																ch7_obuf_raw	ch6_obuf_raw	ch5_obuf_raw	ch4_obuf_raw	ch3_obuf_raw	ch2_obuf_raw	ch1_obuf_raw	ch0_ibuf_raw	ch7_ibuf_raw	ch6_ibuf_raw	ch5_ibuf_raw	ch4_ibuf_raw	ch3_ibuf_raw	ch2_ibuf_raw	ch1_ibuf_raw	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15]	RWC	ch7_obuf_raw	Raw output queue data interrupt for channel 7
[14]	RWC	ch6_obuf_raw	Raw output queue data interrupt for channel 6
[13]	RWC	ch5_obuf_raw	Raw output queue data interrupt for channel 5
[12]	RWC	ch4_obuf_raw	Raw output queue data interrupt for channel 4
[11]	RWC	ch3_obuf_raw	Raw output queue data interrupt for channel 3
[10]	RWC	ch2_obuf_raw	Raw output queue data interrupt for channel 2
[9]	RWC	ch1_obuf_raw	Raw output queue data interrupt for channel 1
[8]	RWC	ch0_ibuf_raw	Raw data processing complete interrupt for channel 0
[7]	RWC	ch7_ibuf_raw	Raw input queue data interrupt for channel 7
[6]	RWC	ch6_ibuf_raw	Raw input queue data interrupt for channel 6
[5]	RWC	ch5_ibuf_raw	Raw input queue data interrupt for channel 5
[4]	RWC	ch4_ibuf_raw	Raw input queue data interrupt for channel 4
[3]	RWC	ch3_ibuf_raw	Raw input queue data interrupt for channel 3
[2]	RWC	ch2_ibuf_raw	Raw input queue data interrupt for channel 2
[1]	RWC	ch1_ibuf_raw	Raw input queue data interrupt for channel 1
[0]	RO	reserved	Reserved

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RST_STATUS

RST_STATUS is a reset status indicator register.

Offset Address	Register Name	Total Reset Value
0x140C	RST_STATUS	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															rst_status
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	rst_status	Reset status indicator of the cipher module 0: The cipher module is being reset. 1: The cipher module is working properly.

CHAN0_CFG

CHAN0_CFG is channel 0 configuration register.

Offset Address	Register Name	Total Reset Value
0x1410	CHAN0_CFG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														ch0_busy	ch0_start
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RO	ch0_busy	Status signal of channel 0
[0]	RW	ch0_start	Encryption/Decryption start signal of channel 0

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13.2 HASH

13.2.1 Overview

The hash is a module for implementing the SHA1, SHA256, HMAC_SHA1, and HMAC_SHA256 algorithms. The SHA1 and SHA256 algorithms comply with the FIPS 180-2 standard. The HMAC_SHA1 and HMAC_SHA256 algorithms comply with the RFC2104 standard.

The hash module can be used to authenticate data integrity and construct digital signatures.

13.2.2 Features

The hash module has the following features:

- Supports the SHA1, SHA256, HMAC_SHA1, and HMAC_SHA256 algorithms.
- Supports the input data configured by the CPU and read operation in DMA mode.
- Supports block-aligned total length of the input data (padded by software), that is, a multiple of 64 bytes. The total length is configured by software and the maximum value is (2^{64} – 64) bytes. The message length after padding (by software) is used as the configured data length for the SHA1 and SHA256 algorithms, and the (padded message length + 64 bytes) is used as the configured data length for the HMAC_SHA1 and HMAC_SHA256 algorithms.
- Supports the 128-bit HMAC key configured by hardware (cipher hash key ctrl) or software.
- Supports configurable initial values for the SHA1 and SHA256 algorithms.

13.2.3 Function Description

The hash module is used to authenticate data integrity and construct digital signatures. The hash module can be set to the initial value update mode (the initial value needs to be configured in this mode) or non-initial value update mode by software. Then the message digest can be calculated by using the SHA1 or SHA256 algorithm. Software can start the HMAC_SHA1 or HMAC_SHA256 algorithm to calculate the MAC. The initial value cannot be configured.

13.2.4 Operating Mode

SHA1 Operation Process in CPU Mode

The CPU configures the input data. An SHA1 operation in CPU mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[read_ctrl] to 1 and HASH_CTRL[sha_sel], HASH_CTRL[hmac_flag], HASH_CTRL[hardkey_sel], HASH_CTRL[small_end_en], and HASH_CTRL[sha_init_update_en] to 0. Set HASH_CTRL[sha_init_update_en] to 1 if the initial value needs to be updated.

Step 4 Configure SHA_START[sha_start].

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Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DATA_IN[data_in] by using the CPU.

Step 7 Repeat step 4 and step 5 until all the data required for calculation is input.

Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT5[sha_out5]; if no, wait.

----End

SHA1 Operation Process in DMA Mode

The DMA reads the input data. An SHA1 operation in DMA mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[read_ctrl], HASH_CTRL[sha_sel], HASH_CTRL[hmac_flag], HASH_CTRL[hardkey_sel], HASH_CTRL[small_end_en], and HASH_CTRL[sha_init_update_en] to 0. Set HASH_CTRL[sha_init_update_en] to 1 if the initial value needs to be updated.

Step 4 Configure SHA_START[sha_start].

Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DMA_START_ADDR[dma_start_addr] by using the CPU.

Step 7 Configure DMA_LEN[dma_len] by using the CPU.

Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT5[sha_out5]; if no, wait.

----End

SHA256 Operation Process in CPU Mode

The CPU configures the input data. An SHA256 operation in CPU mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[read_ctrl] and HASH_CTRL[sha_sel] to 1 and HASH_CTRL[hmac_flag], HASH_CTRL[hardkey_sel], HASH_CTRL[small_end_en], and HASH_CTRL[sha_init_update_en] to 0. Set HASH_CTRL[sha_init_update_en] to 1 if the initial value needs to be updated.

Step 4 Configure SHA_START[sha_start].

Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DATA_IN[data_in] by using the CPU.

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Step 7 Repeat step 4 and step 5 until all the data required for calculation is input.

Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT8[sha_out8]; if no, wait.

----End

SHA256 Operation Process in DMA Mode

The DMA reads the input data. An SHA256 operation in DMA mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[sha_sel] to 1 and HASH_CTRL[read_ctrl], HASH_CTRL[hmac_flag], HASH_CTRL[hardkey_sel], HASH_CTRL[small_end_en], and HASH_CTRL[sha_init_update_en] to 0. Set HASH_CTRL[sha_init_update_en] to 1 if the initial value needs to be updated.

Step 4 Configure SHA_START[sha_start].

Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DMA_START_ADDR[dma_start_addr] by using the CPU.

Step 7 Configure DMA_LEN[dma_len] by using the CPU.

Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT8[sha_out8]; if no, wait.

----End

HMAC_SHA1 Operation Process in CPU Mode

The CPU configures the input data. An HMAC_SHA1 operation in CPU mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[sha_sel], HASH_CTRL[small_end_en], and HASH_CTRL[sha_init_update_en] to 0 and HASH_CTRL[read_ctrl], HASH_CTRL[hmac_flag], and HASH_CTRL[hardkey_sel] to 1 (the cipher hash key ctrl must first configure the key. If HASH_CTRL[hardkey_sel] is 0, the CPU must configure MCU_KEY0−3[mcu_key0−3]).

Step 4 Configure SHA_START[sha_start].

Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DATA_IN[data_in] by using the CPU.

Step 7 Repeat step 4 and step 5 until all the data required for calculation is input.

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Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT5[sha_out5]; if no, wait.

----End

HMAC_SHA1 Operation Process in DMA Mode

The DMA reads the input data. An HMAC_SHA1 operation in DMA mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[sha_sel], HASH_CTRL[small_end_en], HASH_CTRL[sha_init_update_en], HASH_CTRL[read_ctrl], HASH_CTRL[hmac_flag], and HASH_CTRL[hardkey_sel] to 0 (the cipher hash key ctrl must first configure the key. If HASH_CTRL[hardkey_sel] is 0, the CPU must configure MCU_KEY0−3[mcu_key0−3]).

Step 4 Configure SHA_START[sha_start].

Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DMA_START_ADDR[dma_start_addr] by using the CPU.

Step 7 Configure DMA_LEN[dma_len] by using the CPU.

Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT5[sha_out5]; if no, wait.

----End

HMAC_SHA256 Operation Process in CPU Mode

The CPU configures the input data. An HMAC_SHA256 operation in CPU mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[small_end_en] and HASH_CTRL[sha_init_update_en] to 0 and HASH_CTRL[read_ctrl], HASH_CTRL[sha_sel], HASH_CTRL[hmac_flag], and HASH_CTRL[hardkey_sel] to 1 (the cipher hash key ctrl must first configure the key. If HASH_CTRL[hardkey_sel] is 0, the CPU must configure MCU_KEY0−3[mcu_key0−3]).

Step 4 Configure SHA_START[sha_start].

Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DATA_IN[data_in] by using the CPU.

Step 7 Repeat step 5 and step 6 until all the data required for calculation is input.

Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT8[sha_out8]; if no, wait.

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----End

HMAC_SHA256 Operation Process in DMA Mode

The DMA reads the input data. An HMAC_SHA256 operation in DMA mode is performed as follows:

Step 1 Check whether HASH_STATUS[hash_rdy] is 1. If yes, there is no calculation currently; if no, wait until the calculation is complete.

Step 2 Configure TOTAL_LEN1[total_len1] and TOTAL_LEN2[total_len2]. The configured message length must be 64-byte-aligned.

Step 3 Set HASH_CTRL[sha_sel] to 1 and HASH_CTRL[read_ctrl], HASH_CTRL[hmac_flag], HASH_CTRL[small_end_en], HASH_CTRL[sha_init_update_en], and HASH_CTRL[hardkey_sel] to 0 (the cipher hash key ctrl must first configure the key. If HASH_CTRL[hardkey_sel] is 0, the CPU must configure MCU_KEY0−3[mcu_key0−3]).

Step 4 Configure SHA_START[sha_start].

Step 5 Check whether HASH_STATUS[rec_rdy] is 1. If yes, go to the next step; if no, wait.

Step 6 Configure DMA_START_ADDR[dma_start_addr] by using the CPU.

Step 7 Configure DMA_LEN[dma_len] by using the CPU.

Step 8 Check whether HASH_STATUS[hash_rdy] is 1. If yes, read the calculation results from SHA_OUT1[sha_out1] to SHA_OUT8[sha_out8]; if no, wait.

----End

Clock Gating

When the hash calculation is not required, you can disable the clock of the hash module to reduce power consumption by configuring the system controller register.

Soft Reset

You can soft-rest the hash module by configuring the system controller register.

13.2.5 Register Summary

Table 13-2 describes hash registers.

Table 13-2 Summary of hash registers (base address: 0x1007_0000)

Offset Address	Register	Description	Page
0x0	TOTAL_LEN1	Hash message total length register (lower 32 bits)	13-41
0x4	TOTAL_LEN2	Hash message total length register (upper 32 bits)	13-41
0x8	HASH_STATUS	Status register	13-41

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Offset Address	Register	Description	Page
0xC	HASH_CTRL	Control register	13-42
0x10	SHA_START	Calculation start register	13-43
0x14	DMA_START_ADDR	Read message start address register	13-44
0x18	DMA_LEN	DMA transfer length register	13-44
0x1C	DATA_IN	Input data register	13-45
0x20	REC_LEN1	RX message length register 1	13-45
0x24	REC_LEN2	RX message length register 2	13-46
0x30	SHA_OUT1	Hash output register 1	13-46
0x34	SHA_OUT2	Hash output register 2	13-46
0x38	SHA_OUT3	Hash output register 3	13-47
0x3C	SHA_OUT4	Hash output register 4	13-47
0x40	SHA_OUT5	Hash output register 5	13-48
0x44	SHA_OUT6	Hash output register 6	13-48
0x48	SHA_OUT7	Hash output register 7	13-48
0x4C	SHA_OUT8	Hash output register 8	13-49
0x70	MCU_KEY0	Key 0 (configured by MCU) register	13-49
0x74	MCU_KEY1	Key 1 (configured by MCU) register	13-49
0x78	MCU_KEY2	Key 2 (configured by MCU) register	13-50
0x7C	MCU_KEY3	Key 3 (configured by MCU) register	13-50
0x90	SHA_INIT1_UPDATE	SHA initial value register 1	13-51
0x94	SHA_INIT2_UPDATE	SHA initial value register 2	13-51
0x98	SHA_INIT3_UPDATE	SHA initial value register 3	13-51
0x9C	SHA_INIT4_UPDATE	SHA initial value register 4	13-52
0xA0	SHA_INIT5_UPDATE	SHA initial value register 5	13-52
0xA4	SHA_INIT6_UPDATE	SHA initial value register 6	13-52
0xA8	SHA_INIT7_UPDATE	SHA initial value register 7	13-53
0xAC	SHA_INIT8_UPDATE	SHA initial value register 8	13-53

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13.2.6 Register Description

TOTAL_LEN1

TOTAL_LEN1 is a hash message total length register (lower 32 bits).

Offset Address	Register Name	Total Reset Value
0x0	TOTAL_LEN1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	total_len1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	total_len1	SHA: lower bits of the padded message length (in byte) HMAC: lower bits of the length (in byte) of (i_key_pad + padded message)

TOTAL_LEN2

TOTAL_LEN2 is a hash message total length register (upper 32 bits).

Offset Address	Register Name	Total Reset Value
0x4	TOTAL_LEN2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	total_len2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	total_len2	SHA: upper bits of the padded message length (in byte) HMAC: upper bits of the length (in byte) of (i_key_pad + padded message)

HASH_STATUS

HASH_STATUS is an HASH status register.

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Offset Address	Register Name	Total Reset Value
0x8	HASH_STATUS	0x0000_000F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									error_state				rec_rdy	msg_rdy	dma_rdy	hash_rdy
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6]	RO	len_err	Length correctness flag 0: correct 1: incorrect
[5:4]	RO	error_state	AHB bus status err_state[4] 0: No error occurs on the AHB. 1: An error occurs on the AHB. err_state[5] 0: The AHB slave is idle. 1: The AHB slave is busy.
[3]	RO	rec_rdy	Data RX ready flag for the internal logic (reg_rdy = msg_rdy & dma_rdy) 0: not ready 1: ready
[2]	RO	msg_rdy	Block calculation debugging information 0: The CPU channel is not ready to receive data. 1: The CPU channel is ready to receive data.
[1]	RO	dma_rdy	DMA debugging information 0: A data block is not completely read in DMA mode. 1: A data block is completely read in DMA mode.
[0]	RO	hash_rdy	TOTAL_LEN calculation completion signal 0: The calculation is not complete. 1: The calculation is complete.

HASH_CTRL

HASH_CTRL is an HASH control register.

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Offset Address	Register Name	Total Reset Value
0xC	HASH_CTRL	0x0000_0020

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									sha_init_update_en	small_end_en	hardkey_sel	hmac_flag	sha_sel		read_ctrl
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6]	RW	sha_init_update_en	SHA initial value update enable. This bit is set to 0 for HMAC. 0: disabled. The default SHA initial value is used. 1: enabled. The value of sha_initn_update is used as the initial value when the SHA calculation starts.
[5]	RW	small_end_en	Word reverse order enable 0: big-endian mode 1: little-endian mode
[4]	RW	hardkey_sel	HMAC key select. This bit is set to 0 for the SHA. 0: kl_key (hardware key) 1: MCU key (software key)
[3]	RW	hmac_flag	Calculation mode select 0: SHA calculation 1: HMAC calculation
[2:1]	RW	sha_sel	Hash mode control 00: SHA1 mode 01: SHA256 mode Other values: reserved, Currently this field can be set to only 00 or 01.
[0]	RW	read_ctrl	Message transfer mode control 0: The message is read by the DMA. 1: The message is read by the CPU.

SHA_START

SHA_START is an SHA computing start register.

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Offset Address	Register Name	Total Reset Value
0x10	SHA_START	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															sha_start
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	sha_start	Hash calculation start signal 0: not start 1: start

DMA_START_ADDR

DMA_START_ADDR is a read message start address register.

Offset Address	Register Name	Total Reset Value
0x14	DMA_START_ADDR	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	dma_start_addr																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	dma_start_addr	Byte address. The address must be configured as word-aligned by software.

DMA_LEN

DMA_LEN is a DMA transfer length register.

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Offset Address	Register Name	Total Reset Value
0x18	DMA_LEN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	segment_len1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	segment_len1	DMA transfer length (in byte), which must be word-aligned After the length is configured, data is read and calculated automatically by the internal logic.

DATA_IN

DATA_IN is an SHA computing input data register.

Offset Address	Register Name	Total Reset Value
0x1C	DATA_IN	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	data_in																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	data_in	The input data of the message digest needs to be determined through SHA computing, and the input data is configured by the CPU. This register cannot be read when debug_disable is 1.

REC_LEN1

REC_LEN1 is RX message length register 1.

Offset Address	Register Name	Total Reset Value
0x20	REC_LEN1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	receive_byte_cnt																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	receive_byte_cnt	Number of bytes calculated by the hash module (64 bytes for each block). This register is cleared when the calculation starts.

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REC_LEN2

REC_LEN2 is RX message length register 2.

Offset Address	Register Name	Total Reset Value
0x24	REC_LEN2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	receive_byte_cnt																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	receive_byte_cnt	Number of bytes calculated by the hash module (64 bytes for each block). This register is cleared when the calculation starts.

SHA_OUT1

SHA_OUT1 is hash output result register 1.

Offset Address	Register Name	Total Reset Value
0x30	SHA_OUT1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out1	Bits 0−31 of the SHA1/SHA256 message digest output The reset terminal is removed, and the initial value is random.

SHA_OUT2

SHA_OUT2 is hash output result register 2.

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Offset Address	Register Name	Total Reset Value
0x34	SHA_OUT2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out2	Bits 32−63 of the SHA1/SHA256 message digest output The reset terminal is removed, and the initial value is random.

SHA_OUT3

SHA_OUT3 is hash output result register 3.

Offset Address	Register Name	Total Reset Value
0x38	SHA_OUT3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out3																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out3	Bits 64−95 of the SHA1/SHA256 message digest output The reset terminal is removed, and the initial value is random.

SHA_OUT4

SHA_OUT4 is hash output result register 4.

Offset Address	Register Name	Total Reset Value
0x3C	SHA_OUT4	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out4																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out4	Bits 96−127 of the SHA1/SHA256 message digest output The reset terminal is removed, and the initial value is random.

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SHA_OUT5

SHA_OUT5 is hash output result register 5.

Offset Address	Register Name	Total Reset Value
0x40	SHA_OUT5	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out5																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out5	Bits 128−159 of the SHA1/SHA256 message digest output The reset terminal is removed, and the initial value is random.

SHA_OUT6

SHA_OUT6 is hash output result register 6.

Offset Address	Register Name	Total Reset Value
0x44	SHA_OUT6	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out6																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out6	This register does not apply to SHA1. This field indicates bits 160−191 of the SHA256 message digest output.

SHA_OUT7

SHA_OUT7 is hash output result register 7.

Offset Address	Register Name	Total Reset Value
0x48	SHA_OUT7	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out7																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out7	This register does not apply to SHA1. This field indicates bits 192−223 of the SHA256 message digest output.

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SHA_OUT8

SHA_OUT8 is hash output result register 8.

Offset Address	Register Name	Total Reset Value
0x4C	SHA_OUT8	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_out8																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sha_out8	This register does not apply to SHA1. This field indicates bits 224−255 of the SHA256 message digest output.

MCU_KEY0

MCU_KEY0 is an HMAC input key register (bits 0−31, configured by the CPU).

Offset Address	Register Name	Total Reset Value
0x70	MCU_KEY0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mcu_key0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	mcu_key0	HMAC input key register (bits 0−31, configured by the CPU) This register cannot be read when debug_disable is 1.

MCU_KEY1

MCU_KEY1 is an HMAC input key register (bits 32−63, configured by the CPU).

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Offset Address	Register Name	Total Reset Value
0x74	MCU_KEY1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mcu_key1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	mcu_key1	HMAC input key register (bits 32−63, configured by the CPU) This register cannot be read when debug_disable is 1.

MCU_KEY2

MCU_KEY2 is an HMAC input key register (bits 64−95, configured by the CPU).

Offset Address	Register Name	Total Reset Value
0x78	MCU_KEY2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mcu_key2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	mcu_key2	HMAC input key register (bits 64−95, configured by the CPU) This register cannot be read when debug_disable is 1.

MCU_KEY3

MCU_KEY3 is an HMAC input key register (bits 96−127, configured by the CPU).

Offset Address	Register Name	Total Reset Value
0x7C	MCU_KEY3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	mcu_key3																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	mcu_key3	HMAC input key register (bits 96−127, configured by the CPU) This register cannot be read when debug_disable is 1.

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SHA_INIT1_UPDATE

SHA_INIT1_UPDATE is SHA initial value register 1.

Offset Address	Register Name	Total Reset Value
0x90	SHA_INIT1_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init1_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init1_update	SHA initial value. (bits 0−31)

SHA_INIT2_UPDATE

SHA_INIT2_UPDATE is SHA initial value register 2.

Offset Address	Register Name	Total Reset Value
0x94	SHA_INIT2_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init2_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init2_update	SHA initial value. (bits 32−63)

SHA_INIT3_UPDATE

SHA_INIT3_UPDATE is SHA initial value register 3.

Offset Address	Register Name	Total Reset Value
0x98	SHA_INIT3_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init3_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init3_update	SHA initial value. (bits 64−95)

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SHA_INIT4_UPDATE

SHA_INIT4_UPDATE is SHA initial value register 4.

Offset Address	Register Name	Total Reset Value
0x9C	SHA_INIT4_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init4_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init4_update	SHA initial value. (bits 96−127)

SHA_INIT5_UPDATE

SHA_INIT5_UPDATE is SHA initial value register 5.

Offset Address	Register Name	Total Reset Value
0xA0	SHA_INIT5_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init5_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init5_update	SHA initial value. (bits 128−159)

SHA_INIT6_UPDATE

SHA_INIT6_UPDATE is SHA initial value register 6.

Offset Address	Register Name	Total Reset Value
0xA4	SHA_INIT6_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init6_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init6_update	SHA initial value. (bits 160−191)

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SHA_INIT7_UPDATE

SHA_INIT7_UPDATE is SHA initial value register 7.

Offset Address	Register Name	Total Reset Value
0xA8	SHA_INIT7_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init7_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init7_update	SHA initial value. (bits 192−223)

SHA_INIT8_UPDATE

SHA_INIT8_UPDATE is SHA initial value register 8.

Offset Address	Register Name	Total Reset Value
0xAC	SHA_INIT8_UPDATE	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sha_init8_update																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RW	sha_init8_update	SHA initial value. (bits 224−255)

13.3 RNG_GEN

13.3.1 Overview

RNG_GEN is a module that generates true random numbers in compliance with the FIPS 140-1 random test standard.

13.3.2 Features

The RNG_GEN module has the following features:

- Generates true random numbers.
- Supports three random number sources that can be configured by the CPU.
- Disables the random number sources.

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13.3.3 Operating Mode

RNG_GEN Operation Process

The RNG_GEN operation process is as follows:

Step 1 Check whether RNG_STAT[rng_data_count] is greater than or equal to 1. If yes, go to the next step because there are random numbers; if no, wait.

Step 2 Read RNG_FIFO_DATA[rng_fifo_data].

Step 3 Repeat step 1 and step 2 until the required number of random numbers are read.

----End

13.3.4 Register Summary

Table 13-3 describes RNG_GEN registers.

Table 13-3 Summary of RNG_GEN registers (base address: 0x2028_0000)

Offset Address	Register	Description	Page
0x00	RNG_CTRL	RNG control register	13-54
0x04	RNG_FIFO_DATA	RNG FIFO data register	13-56
0x08	RNG_STAT	RNG status register	13-56

13.3.5 Register Description

RNG_CTRL

RNG_CTRL is an RNG control register.

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Offset Address	Register Name	Total Reset Value
0x00	RNG_CTRL	0x0000_1082

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved		low_osc_st1	low_osc_st0	reserved												post_process_depth								post_process_enable	reserved	drop_enable	filter_enable	mix_en	cleardata	osc_sel	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	1	0

Bits	Access	Name	Description
[31]	RO	reserved	Reserved
[30]	WC	low_osc_st1	Exception test of the low-frequency oscillator 0: The low-frequency oscillator works properly. 1: The output of the low-frequency oscillator is always 1, and writing this bit clears it.
[29]	WC	low_osc_st0	Exception test of the low-frequency oscillator 0: The low-frequency oscillator works properly. 1: The output of the low-frequency oscillator is always 1, and writing this bit clears it.
[28:16]	RO	reserved	Reserved
[15:8]	RW	post_process_depth	Post-processing mixing depth. A larger value indicates a better random effect and lower speed.
[7]	RW	post_process_enable	Pre-processing enable for the random number. If this function is enabled, the random number generation speed is decreased by 1/(post_process_depth + 1). 0: disabled 1: enabled
[6]	RO	reserved	Reserved
[5]	RW	drop_enable	Enable for discarding two-bit data when the values of the consecutive two bits are the same (00 or 11). If this function is enabled, the random number generation speed is decreased by 1/3. 0: disabled 1: enabled
[4]	RW	filter_enable	Pre-processing filtering enable for the random number. If this function is enabled, the random number generation speed is decreased by 1/4. 0: disabled 1: enabled

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[3]	RW	mix_en	Data mixing enable 0: disabled 1: enabled
[2]	WC	cleardata	Random number clear 0: not cleared 1: cleared
[1:0]	RW	osc_sel	Random number source select 00: RNG is disabled. 01: Random number source 1 is used. 10: Random number source 2 is used. 11: Random number source 3 is used.

RNG_FIFO_DATA

RNG_FIFO_DATA is an RNG FIFO data register.

Offset Address	Register Name	Total Reset Value
0x04	RNG_FIFO_DATA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	rng_fifo_data																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	rng_fifo_data	You can read the random number by directly reading this register.

RNG_STAT

RNG_STAT is an RNG status register.

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Offset Address	Register Name	Total Reset Value
0x08	RNG_STAT	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																													rng_data_count		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:3]	RO	reserved	Reserved
[2:0]	RO	rng_data_count	Number of 32-bit random numbers in the random number FIFO. The maximum value is 4.

13.4 RSA

13.4.1 Overview

The Rivest-Shamir-Adleman (RSA) is a public key encryption/decryption algorithm implemented through the modular exponentiation operation. The ciphertext is obtained as follows: $C = M^E \bmod N$. The plaintext is obtained as follows: $M=C^D \bmod N$. *M* indicates the plaintext, *C* indicates the ciphertext, *(N, E)* indicates the public key, and *(N, D)* indicates the private key.

The RSA module supports RSA 1024, RSA 2048, and RSA 4096 that comply with the PKCS#1 V1.5/2.1 standard.

The RSA module is used for data encryption/decryption and digital signature verification.

13.4.2 Features

The RSA module has the following features:

- Supports 1024-bit, 2048-bit, and 4096-bit keys.
- Supports the minimum performance of 10 operations per second (OPS) for the 2048-bit key.
- Supports key debugging by using CRC16.
- Clears the random access memories (RAMs) that store the internal RSA keys, packets, and results.

13.4.3 Function Description

The RSA module is used for data encryption/decryption and digital signature verification. The registers are configured by software. Then the RSA module is started to implement the

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modular exponentiation operation. The RSA module also uses CRC16 to calculate the key stored in the logic and outputs the result for checking whether the key is correct. The RSA module also clears the RAMs that store the internal keys, packets, and results.

13.4.4 Operating Mode

Encrypting/Decrypting Data by Using RSA

Data is encrypted/decrypted by using RSA as follows:

Step 1 Read and check whether SEC_RSA_BUSY_REG[sec_rsa_busy_reg] is 0. If yes, go to the next step; if no, wait.

Step 2 Set SEC_RSA_MOD_REG[sec_rsa_key_width] to 1 and SEC_RSA_MOD_REG[sec_rsa_mod_sel], SEC_RSA_MOD_REG[sec_rsa_data0_clr], SEC_RSA_MOD_REG[sec_rsa_data1_clr], and SEC_RSA_MOD_REG[sec_rsa_data2_clr] to 0. When SEC_RSA_MOD_REG[sec_rsa_key_width] is 0 or 3, RSA 1024 is used; when the field value is 1, RSA 2048 is used; when the field value is 2, RSA 4096 is used.

Step 3 Configure SEC_RSA_WPKT_REG[sec_rsa_wpkt_reg]. The configured packet length must be the same as the key bit width configured in SEC_RSA_MOD_REG[sec_rsa_key_width].

Step 4 Configure SEC_RSA_WSEC_REG[sec_rsa_wsec_reg]. Set the key parameter N and then E/D. The length of each parameter must be the same as the key bit width configured in SEC_RSA_MOD_REG[sec_rsa_key_width].

Step 5 Set SEC_RSA_START_REG[sec_rsa_start_reg] to 1.

Step 6 Read and check whether SEC_RSA_BUSY_REG[sec_rsa_busy_reg] is 0. If yes, go to the next step; if no, wait.

Step 7 Read SEC_RSA_RRSLT_REG[sec_rsa_rrslt_reg]. The length of the read result must be the same as the key bit width configured in SEC_RSA_MOD_REG[sec_rsa_key_width].

----End

Debugging the Key by Using CRC16

The key is debugged by using CRC16 as follows:

Step 1 Read and check whether SEC_RSA_BUSY_REG[sec_rsa_busy_reg] is 0. If yes, go to the next step; if no, wait.

Step 2 Set SEC_RSA_MOD_REG[sec_rsa_data0_clr], SEC_RSA_MOD_REG[sec_rsa_data1_clr], and SEC_RSA_MOD_REG[sec_rsa_data2_clr] to 0, SEC_RSA_MOD_REG[sec_rsa_mod_sel] to 3, and SEC_RSA_MOD_REG[sec_rsa_key_width] to 1. When SEC_RSA_MOD_REG[sec_rsa_key_width] is 0 or 3, RSA 1024 is used; when the field value is 1, RSA 2048 is used; when the field value is 2, RSA 4096 is used.

Step 3 Set SEC_RSA_START_REG[sec_rsa_start_reg] to 1.

Step 4 Read and check whether SEC_RSA_BUSY_REG[sec_rsa_busy_reg] is 0. If yes, go to the next step; if no, wait.

Step 5 Read the cyclic redundancy check (CRC) result from SEC_CRC16_REG[sec_rsa_crc16_dat].

----End

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Clearing the RAM

The RAM is cleared as follows:

Step 1 Read and check whether SEC_RSA_BUSY_REG[sec_rsa_busy_reg] is 0. If yes, go to the next step; if no, wait.

Step 2 Set SEC_RSA_MOD_REG[sec_rsa_mod_sel] to 2 and SEC_RSA_MOD_REG[sec_rsa_key_width], SEC_RSA_MOD_REG[sec_rsa_data0_clr], SEC_RSA_MOD_REG[sec_rsa_data1_clr], and SEC_RSA_MOD_REG[sec_rsa_data2_clr] to 1. If SEC_RSA_MOD_REG[sec_rsa_key_width] is 0 or 3, RSA 1024 is used; if the field value is 1, RSA 2048 is used; if the field value is 2, RSA 4096 is used. If SEC_RSA_MOD_REG[sec_rsa_data0_clr] is 1, the RAM that stores the RSA key is cleared; if the field value is 0, the RAM that stores the RSA key is not cleared. If SEC_RSA_MOD_REG[sec_rsa_data1_clr] is 1, the RAM that stores the RSA packet is cleared; if the field value is 0, the RAM that stores the RSA packet is not cleared. If SEC_RSA_MOD_REG[sec_rsa_data2_clr] is 1, the RAM that stores the RSA result is cleared; if the field value is 0, the RAM that stores the RSA result is not cleared.

Step 3 Set SEC_RSA_START_REG[sec_rsa_start_reg] to 1.

Step 4 Read and check whether SEC_RSA_BUSY_REG[sec_rsa_busy_reg] is 0. If yes, go to the next step; if no, wait.

Step 5 Read SEC_RSA_RPKT_REG[sec_rsa_rpkt_reg] to determine whether the RAM that stores the packet is cleared.

Step 6 Read SEC_RSA_RRSLT_REG[sec_rsa_rrslt_reg] to determine whether the RAM that stores the result is cleared.

Step 7 Check the key through CRC to determine whether the RAM that stores the key is cleared.

----End

13.4.5 Register Summary

Table 13-4 describes RSA registers.

Table 13-4 Summary of RSA registers (base address: 0x2026_0000)

Offset Address	Register	Description	Page
0x50	SEC_RSA_BUSY_REG	Busy status register for the SEC_RSA module	13-60
0x54	SEC_RSA_MOD_REG	Operating mode register for the SEC_RSA module	13-60
0x58	SEC_RSA_WSEC_REG	Key write operation register for the SEC_RSA module	13-61
0x5C	SEC_RSA_WPKT_REG	Packet write operation register for the SEC_RSA module	13-62
0x60	SEC_RSA_RPKT_REG	Packet read operation register for the SEC_RSA module	13-62

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Offset Address	Register	Description	Page
0x64	SEC_RSA_RRSLT_REG	Calculation result read operation register for the SEC_RSA module	13-62
0x68	SEC_RSA_START_REG	Modular exponentiation operation configuration start register for the SEC_RSA module	13-63
0x6C	SEC_RSA_ADDR_REG	RAM address register for the key, packet, and result of the SEC_RSA module	13-63
0x70	SEC_RSA_ERROR_REG	Error alarm status register for the SEC_RSA module	13-64
0x74	SEC_CRC16_REG	Key CRC result register for the SEC_RSA module	13-65

13.4.6 Register Description

SEC_RSA_BUSY_REG

SEC_RSA_BUSY_REG is a busy status register for the SEC_RSA module.

Offset Address	Register Name	Total Reset Value
0x50	SEC_RSA_BUSY_REG	0x0000_0000

Bit	31 30 29 28	27 26 25 24	23 22 21 20	19 18 17 16	15 14 13 12	11 10 9 8	7 6 5 4	3 2 1	0
Name	reserved								sec_rsa_busy_reg
Reset	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RO	sec_rsa_busy_reg	Busy status of the SEC_RSA module 0: idle 1: busy

SEC_RSA_MOD_REG

SEC_RSA_MOD_REG is an operating mode register for the SEC_RSA module.

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Offset Address	Register Name	Total Reset Value
0x54	SEC_RSA_MOD_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																									sec_rsa_data2_clr	sec_rsa_data1_clr	sec_rsa_data0_clr	sec_rsa_key_width		sec_rsa_mod_sel	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:7]	RO	reserved	Reserved
[6]	RW	sec_rsa_data2_clr	Clear enable for the RAM that stores the RSA result 0: not cleared 1: cleared
[5]	RW	sec_rsa_data1_clr	Clear enable for the RAM that stores the RSA packet 0: not cleared 1: cleared
[4]	RW	sec_rsa_data0_clr	Clear enable for the RAM that stores the RSA key 0: not cleared 1: cleared
[3:2]	RW	sec_rsa_key_width	Key bit width 00: 1024 bits 01: 2048 bits 10: 4096 bits 11: 1024 bits
[1:0]	RW	sec_rsa_mod_sel	Operating mode select 00: modular exponentiation encryption and decryption operation 01: The key is updated, and the modular exponentiation operation is not performed (this mode is not supported currently). 10: The RAM is cleared. 11: The key is checked by using CRC16.

SEC_RSA_WSEC_REG

SEC_RSA_WSEC_REG is a key write operation register for the SEC_RSA module.

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Offset Address	Register Name	Total Reset Value
0x58	SEC_RSA_WSEC_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sec_rsa_wsec_reg																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	sec_rsa_wsec_reg	Key address configured by the CPU

SEC_RSA_WPKT_REG

SEC_RSA_WPKT_REG is a packet write operation register for the SEC_RSA module.

Offset Address	Register Name	Total Reset Value
0x5C	SEC_RSA_WPKT_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sec_rsa_wpkt_reg																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	sec_rsa_wpkt_reg	Packet address configured by the CPU

SEC_RSA_RPKT_REG

SEC_RSA_RPKT_REG is a packet read operation register for the SEC_RSA module.

Offset Address	Register Name	Total Reset Value
0x60	SEC_RSA_RPKT_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sec_rsa_rpkt_reg																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sec_rsa_rpkt_reg	Packet address read by the CPU

SEC_RSA_RRSLT_REG

SEC_RSA_RRSLT_REG is a calculation result read operation register for the SEC_RSA module.

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Offset Address	Register Name	Total Reset Value
0x64	SEC_RSA_RRSLT_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sec_rsa_rrslt_reg																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	RO	sec_rsa_rrslt_reg	Calculation result read by the CPU

SEC_RSA_START_REG

SEC_RSA_START_REG is a modular exponentiation operation configuration start register for the SEC_RSA module.

Offset Address	Register Name	Total Reset Value
0x68	SEC_RSA_START_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																															sec_rsa_start_reg
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:1]	RO	reserved	Reserved
[0]	RW	sec_rsa_start_reg	CPU configuration start 0: not start 1: start

SEC_RSA_ADDR_REG

SEC_RSA_ADDR_REG is an RAM address register for the key, packet, and result of the SEC_RSA module.

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Offset Address	Register Name	Total Reset Value
0x6C	SEC_RSA_ADDR_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	sec_rsa_addere_rslt								sec_rsa_addre_pkt								sec_rsa_addre_d								sec_rsa_addr_n							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:24]	RO	sec_rsa_addere_rslt	Result RAM address The address offset is 4 bytes each time. That is, if the value of this field is 1, 4-byte data is stored. When the number of stored data segments reaches the maximum key bit width, the value of this field is 0 because the count value starts from 0.
[23:16]	RO	sec_rsa_addre_pkt	Packet RAM address The address offset is 4 bytes each time. That is, if the value of this field is 1, 4-byte data is stored or read. When the number of stored data segments reaches the maximum key bit width, the value of this field is 0 because the count value starts from 0.
[15:8]	RO	sec_rsa_addre_d	RAM address of the key parameter E/D The address offset is 4 bytes each time. That is, if the value of this field is 1, 4-byte data is stored. When the number of stored data segments reaches the maximum key bit width, the value of this field is 0 because the count value starts from 0.
[7:0]	RO	sec_rsa_addr_n	RAM address of the key parameter N The address offset is 4 bytes each time. That is, if the value of this field is 1, 4-byte data is stored. When the number of stored data segments reaches the maximum key bit width, the value of this field is 0 because the count value starts from 0.

SEC_RSA_ERROR_REG

SEC_RSA_ERROR_REG is an error alarm status register for the SEC_RSA module.

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Offset Address	Register Name	Total Reset Value
0x70	SEC_RSA_ERROR_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																														sec_rsa_err1	reserved
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:2]	RO	reserved	Reserved
[1]	RO	sec_rsa_err1	Alarm when the external input RSA key (N) is an even number (the odd/even feature of the key is not checked when the RAM that stores the key is cleared) 0: The external input key is normal. 1: The external input key (N) is an even number.
[0]	RO	reserved	Reserved

SEC_CRC16_REG

SEC_CRC16_REG is a key CRC result register for the SEC_RSA module.

Offset Address	Register Name	Total Reset Value
0x74	SEC_CRC16_REG	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																sec_rsa_crc16_dat																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:16]	RO	reserved	Reserved
[15:0]	RO	sec_rsa_crc16_dat	CRC result of the key

13.5 Cipher Hash Key Ctrl

13.5.1 Operating Mode

The cipher hash key control module has the following three working modes:

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- Burning the key to the one-time programming (OTP)
- Loading the key to the cipher
- Loading the key to the hash

13.5.1.1 Burning the Key to the OTP

Before burning, the data in the OTP is 0. After burning is enabled, the cipher hash key control module burns the corresponding macro unit to 1 based on the key value and OTP address of the key (KL_CTRL[otp_key_add]).

The operation process of burning the key to the OTP is as follows:

Step 1 Assign values to KL_WKEY0, KL_WKEY1, KL_WKEY2, and KL_WKEY3.

Step 2 Query the status register KL_STA, and wait until KL_STA[ctrl_rdy] is 1 and KL_STA[ctrl_busy0] and KL_STA[ctrl_busy1] are 0.

Step 3 Write the OTP address of the key to KL_CTRL[otp_key_add], and set KL_CTRL[otp_kd_mode] to 1 and KL_CTRL[cipher_kl_mode] and KL_CTRL[hash_kl_mode] to 0 to enable the mode of burning the key to the OTP.

Step 4 Write 1 to KL_CTRL[start] to start the operation.

Step 5 Query KL_STA[key_wt_finish]. If the value changes from 0 to 1, the burning is complete. Note that if KL_STA[key_wt_error] is 1, the burning fails and the key address has been burned before.

----End

13.5.1.2 Loading the Key to the Cipher

The operation process of loading the key to the cipher is as follows:

Step 1 Query the status register KL_STA, and wait until KL_STA[ctrl_rdy] is 1 and KL_STA[ctrl_busy0] and KL_STA[ctrl_busy1] are 0.

Step 2 Write the OTP address of the key to KL_CTRL[otp_key_add], write the key address for loading the key to the cipher to KL_CTRL[cipher_key_add], and set KL_CTRL[cipher_kl_mode] to 1 and KL_CTRL[otp_kd_mode] and KL_CTRL[hash_kl_mode] to 0 to enable the mode of loading the key to the cipher.

Step 3 Write 1 to KL_CTRL[start] to start the operation.

Step 4 Query KL_STA[cipher_kl_finish]. If the value changes from 0 to 1, the process of loading the key to the cipher is complete.

----End

13.5.1.3 Loading the Key to the Hash

The operation process of loading the key to the hash is as follows:

Step 1 Query the status register KL_STA, and wait until KL_STA[ctrl_rdy] is 1 and KL_STA[ctrl_busy0] and KL_STA[ctrl_busy1] are 0.

Step 2 Write the OTP address of the key to KL_CTRL[otp_key_add], and set KL_CTRL[hash_kl_mode] to 1 and KL_CTRL[cipher_kl_mode] and KL_CTRL[otp_kd_mode] to 0 to enable the mode of loading the key to the hash.

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Step 3 Write 1 to KL_CTRL[start] to start the operation.

Step 4 Query KL_STA[hash_key_read_busy]. If the value changes from 1 to 0, the process of loading the key to the hash is complete.

----End

13.5.2 Register Summary

Table 13-5 describes cipher hash key control registers.

Table 13-5 Summary of cipher hash key control registers (base address: 0x2010_0800)

Offset Address	Register	Description	Page
0x0000	KL_WKEY0	Key burning register 0	13-67
0x0004	KL_WKEY1	Key burning register 1	13-67
0x0008	KL_WKEY2	Key burning register 2	13-68
0x000C	KL_WKEY3	Key burning register 3	13-68
0x0010	KL_CTRL	Key control register	13-68
0x0014	KL_STA	Status indicator register	13-70

13.5.3 Register Description

KL_WKEY0

KL_WKEY0 is key burning register 0.

Offset Address	Register Name	Total Reset Value
0x0000	KL_WKEY0	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	key0																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	key0	Key[31:0] that are burnt to the OTP

KL_WKEY1

KL_WKEY1 is key burning register 1.

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Offset Address	Register Name	Total Reset Value
0x0004	KL_WKEY1	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	key1																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	key1	Key[63:32] that are burnt to the OTP

KL_WKEY2

KL_WKEY2 is key burning register 2.

Offset Address	Register Name	Total Reset Value
0x0008	KL_WKEY2	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	key2																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	key2	Key[95:64] that are burnt to the OTP

KL_WKEY3

KL_WKEY3 is key burning register 3.

Offset Address	Register Name	Total Reset Value
0x000C	KL_WKEY3	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	key3																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:0]	WO	key3	Key[127:96] that are burnt to the OTP

KL_CTRL

KL_CTRL is a key control register.

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Offset Address	Register Name	Total Reset Value
0x0010	KL_CTRL	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved																					cipher_key_add			reserved		otp_key_add		hash_kl_mode	otp_kd_mode	cipher_kl_mode	start
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31:11]	RO	reserved	Reserved
[10:8]	RW	cipher_key_add	Key address for loading the key to the cipher
[7:6]	RW	reserved	Reserved
[5:4]	RW	otp_key_add	Address of the key in the OTP for burning or loading the key
[3]	RW	hash_kl_mode	Enable for loading the key to the hash 0: disabled 1: enabled **NOTE** Only one of hash_kl_mode, otp_kd_mode, and cipher_kl_mode can be set to 1 at a time. That is, only one function takes effect at a time.
[2]	RW	otp_kd_mode	Enable for burning the key to the OTP 0: disabled 1: enabled **NOTE** Only one of hash_kl_mode, otp_kd_mode, and cipher_kl_mode can be set to 1 at a time. That is, only one function takes effect at a time.
[1]	RW	cipher_kl_mode	Enable for loading the key to the cipher 0: disabled 1: enabled **NOTE** Only one of hash_kl_mode, otp_kd_mode, and cipher_kl_mode can be set to 1 at a time. That is, only one function takes effect at a time.
[0]	RW	start	Operation start. Writing 1 to this bit starts the operation. Reading back this bit has no effect. **NOTE** A new operation can be initiated only when ctrl_busy1 and ctrl_busy0 are 0 and ctrl_rdy is 1.

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KL_STA

KL_STA is a status indicator register.

Offset Address	Register Name	Total Reset Value
0x0014	KL_STA	0x0000_0000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	key_wt_finish	key_wt_error	ctrl_busy1	ctrl_busy0	ctrl_rdy	reserved																									hash_key_read_busy	cipher_kl_finish
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Access	Name	Description
[31]	RO	key_wt_finish	Whether the operation of burning the key to the OTP is complete 0: not complete 1: complete
[30]	RO	key_wt_error	Whether an error occurs during the operation of burning the key to the OTP 0: No error occurs. The burning is successful. 1: An error occurs. The key address has been burnt before.
[29]	RO	ctrl_busy1	Controller busy signal 1. A new operation can be initiated only when ctrl_busy1 and ctrl_busy0 are 0 and ctrl_rdy is 1.
[28]	RO	ctrl_busy0	Controller busy signal 0. A new operation can be initiated only when ctrl_busy1 and ctrl_busy0 are 0 and ctrl_rdy is 1.
[27]	RO	ctrl_rdy	Controller ready signal. A new operation can be initiated only when ctrl_busy1 and ctrl_busy0 are 0 and ctrl_rdy is 1.
[26:2]	RO	reserved	Reserved
[1]	RO	hash_key_read_busy	Whether the operation of loading the key to the hash is complete 0: complete 1: not complete The hash encryption and decryption operations are started only after the loading operation is complete.
[0]	RO	cipher_kl_finish	Whether the operation of loading the key to the cipher is complete 0: not complete 1: complete The cipher encryption and decryption operations are started only after the loading operation is complete.

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Contents

A Ordering Information A-1

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A Ordering Information

Figure A-1 shows the Hi3518E V200 mark.

Figure A-1 Hi3518E V200 mark

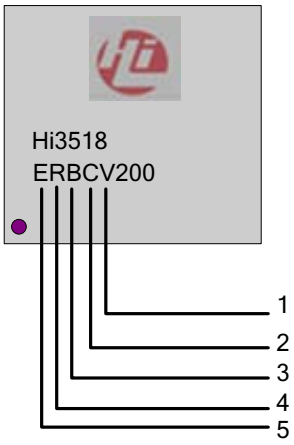
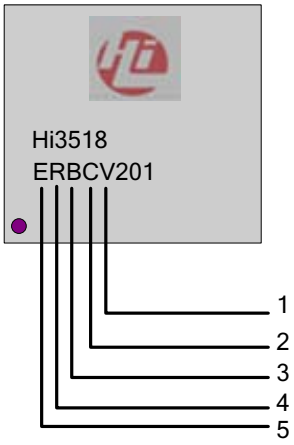


Figure A-2 shows the Hi3518E V201 mark.

Figure A-2 Hi3518E V201 mark



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Figure A-3 shows the Hi3516C V200 mark.

Figure A-3 Hi3516C V200 mark

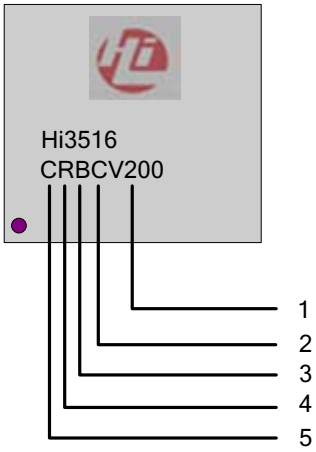


Table A-1 describes the Hi3518E V200/Hi3518E V201/Hi3516C V200 mark.

Table A-1 Meaning of the Hi3518E V200/Hi3518E V201/Hi3516C V200 mark

No.	Item	Description
1	Version number	It indicates the chip version number.
2	Temperature flag	The letter C stands for commercial. It indicates that the chip is for commercial use.
3	Package flag	The letter B indicates the ball grid array (BGA) package.
4	Environmental protection flag	The letter R stands for Restriction of Hazardous Substances (RoHS).
5	Product differentiation flag	It is the last digit of a chip name for differentiating one chip from another one.

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